SILVÈRE MARTIN-MICHIELLOT & ANTIGRAVITY

THE WAY OF THE NUMBER

A Constructive Odyssey, from the First Pebble to Advanced High School Mathematics

58 chapters of living history, motor manipulations, column algorithms, and crystal-clear logic

DEFINITIVE REFERENCE EDITION — 2026

CONSTRUCTIVIST PEDAGOGY • GEOMETRIC INTUITION • FORMAL RIGOR

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THE WAY OF THE NUMBER — A Constructive Odyssey, from the First Pebble to Advanced High School Mathematics

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“Philosophy is written in that great book which ever lies before our eyes — I mean the Universe —, but we cannot understand it if we do not first learn the language and the characters in which it is written. It is written in the language of mathematics.”

— Galileo Galilei, The Assayer (1623)

To all children and young minds who ever felt they were “not made for math”,
and to the parents and educators who guide them with patience, clarity, and joyful rigor.

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VOLUME I — THE AGE OF THE HAND AND THE SAVANNA

Ages 6–8 / Grades 1–2

From the awakening of prehistoric vision to early pebble manipulation, this foundational volume establishes number sense, place value, and the elementary operations.

Chapters of Volume I :

Chapter 1 : The Monkey's Eyes and the Bag of Pebbles
Chapter 2 : The Five-Fingered Hand and Bundles of Ten
Chapter 3 : The Shared Hunt and the Balance Scale of Justice
Chapter 4 : The Merchant's Abacus and the Traveling Carry
Chapter 5 : The Brick Wall and Magic Rectangles
Chapter 6 : The Pharaoh's Cubit and the Hourglass
Chapter 7 : The Shapes of the World and the Gardener's String

Facing: Hexagonal tessellation of honeycomb cells (Apis mellifera).

CHAPTER 1: The Monkey's Eyes and the Bag of Pebbles

> "Before learning to count with words and numerals, humans began by looking at the world with their eyes and touching the earth with their fingers. A pebble for a beast, a notch for a day: thus was born the greatest adventure of the human mind."

What You Are About to Discover

- Why your eyes instantly recognize 1, 2, or 3 objects in a flash without counting, but get confused when there are 5 or more (the secret of *subitizing*).
- The incredible visual memory of the chimpanzee Ayumu.
- How a prehistoric shepherd protected his flock without knowing a single number word, thanks to the **tally pebble** (one-to-one correspondence).
- What the magic terms truly mean: **as many as**, **more than**, **fewer than**.
- How to properly trace and recognize the ten digits from **0** to **9**.

---

1. The Problem: The Flock at Sunset

Close your eyes. Travel twenty thousand years back in time. There is no school, no book, no clock, no phone. The world is an immense savanna of golden grasses, deep forests, and rocky hills.

You are seven years old, and in your clan, you have a vital mission: every morning, you must lead the flock of sheep out of the cave to graze on the fresh hillside grass.

Yet there is a problem that makes your heart race: **in your clan's language, words for counting do not exist yet.** Nobody knows what "twelve," "twenty," or "fifty" mean. You do not know how many animals you have.

Evening arrives. The sky turns crimson, then deep twilight blue. Far off in the forest, the first wolves begin to howl. You herd the sheep into the cave at a gallop and pull shut the heavy thornwood gate.

You stand by the gate, anxious: *"Did they all make it inside? Or did one sheep get trapped in the brambles up on the hill, all alone in the dark?"*

If you cannot count, how can you be absolutely sure?

---

2. The Magic Glance: Subitizing

Before solving the shepherd's riddle, let us try an experiment with your own eyes.

Ask the person accompanying you to place two buttons, or two hazelnuts, on the table. Close your eyes. Open them for a quarter of a second—just a quick blink—and close them immediately.

How many objects did you see? You answer without a second's hesitation: **"Two!"**

Try it again with three buttons. You blink your eyes: you answer **"Three!"** right away. You did not need to point your finger and say: *"one… and another one…"*. Your brain photographed the three in a single chunk.

Now, ask them to scatter **eight buttons** across the table in random disarray. Close your eyes. Open them for a flash. Close them.

How many were there? This time, it is impossible to tell! You hesitate: *"Maybe six? Or nine?"* To be sure, you must point your finger at every single button and count slowly: *"1, 2, 3, 4, 5, 6, 7, and 8."*

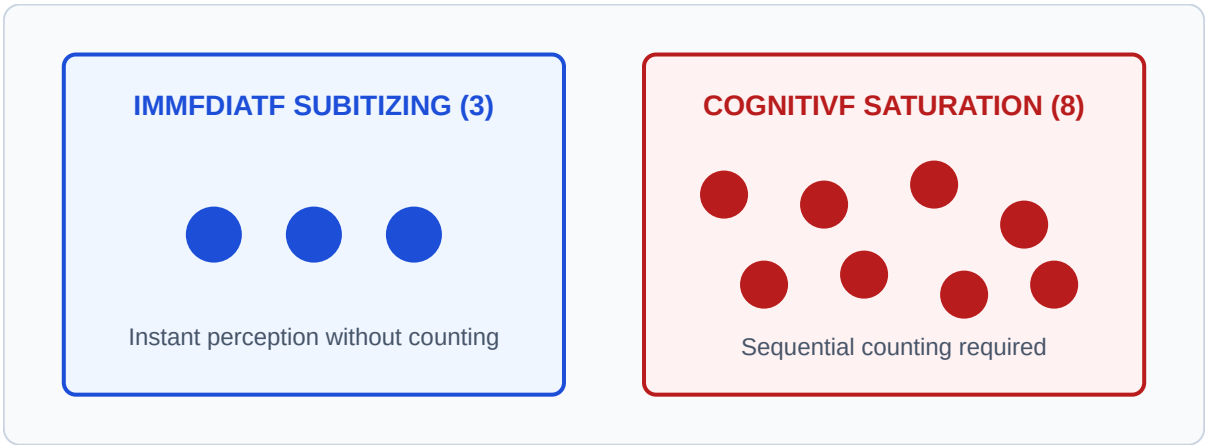

Figure 1: The subitizing phenomenon: immediate perception vs cognitive saturation

📐 Definition — Subitizing

Subitizing (from the Latin *subitus*, meaning "sudden, immediate") is our brain's capacity to perceive **at a single glance**, without counting one by one, the exact quantity of objects in a small collection. In humans, this magic glance works reliably up to **3** or **4** objects. Starting from **5**, our eyes become confused if the objects are scattered randomly.

💡 Did You Know? — Dice and Domino Patterns

Why do standard playing dice always arrange their dots in the same pattern? Look at the 5 on a die: four corner dots and one in the center. Thanks to this cross-shaped pattern, your eye immediately spots two at the top, two at the bottom, and one in the center: $2 + 2 + 1 = 5$! Humans invented these structured geometric arrangements to help their eyes recognize quantities larger than 3 without wasting time counting one by one.

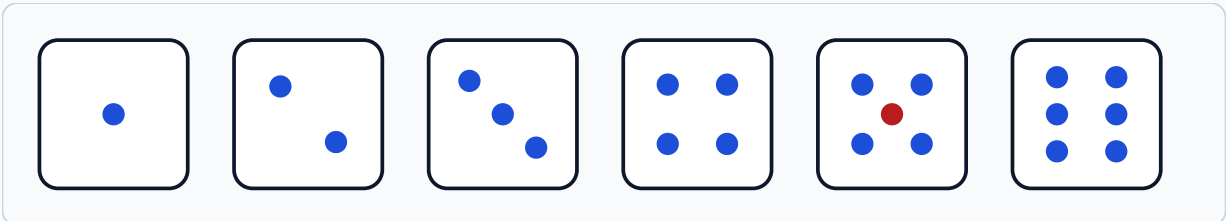

Figure 2: The six faces of the die: instant spatial recognition

🔍 ***True Story — The Lightning-Fast Glance of Ayumu the Chimpanzee*** *At Kyoto University in Japan, researcher Tetsuro Matsuzawa studied a young chimpanzee named **Ayumu**. On a touchscreen monitor, the digits 1 through 9 appear scattered randomly, then disappear under blank white squares in less than two-tenths of a second! An adult human cannot remember where the digits were. But Ayumu taps the white squares in the exact numerical order: 1, then 2, 3, 4, 5, 6, 7, 8, and 9, almost never making a mistake! Why does a chimpanzee possess this extraordinary ability? Because in the canopy, spotting in a split second the exact number of ripe fruits on a branch or the number of predators hidden among leaves is a matter of life or death.*

Yet if humans do not possess this superpower for large numbers, how did we manage to survive and build civilizations? We invented a trick far more powerful than the eye alone: **object manipulation**.

---

3. One-to-One Correspondence: The Bag of Pebbles

Let us return to our prehistoric shepherd. The sun rises, and the sheep are pawing at the gate. Here is the stroke of genius he invents:

1. He picks up an empty leather pouch and hangs it from his belt.
2. He opens the thornwood gate just enough to let the sheep pass **one by one**.
3. The first sheep walks out? The shepherd picks up a small pebble and drops it into his pouch: *Plop!*
4. The second sheep walks out? Another pebble goes into the pouch: *Plop!*
5. Every single time an animal crosses the threshold, **one pebble and only one** enters the bag.

When the last sheep bounds onto the hill, the shepherd pulls the drawstring closed. He still does not know the number of his sheep. Yet he knows something monumental: **there are in his bag exactly as many pebbles as there are living animals on the hill!** Every pebble is the secret twin of a sheep.

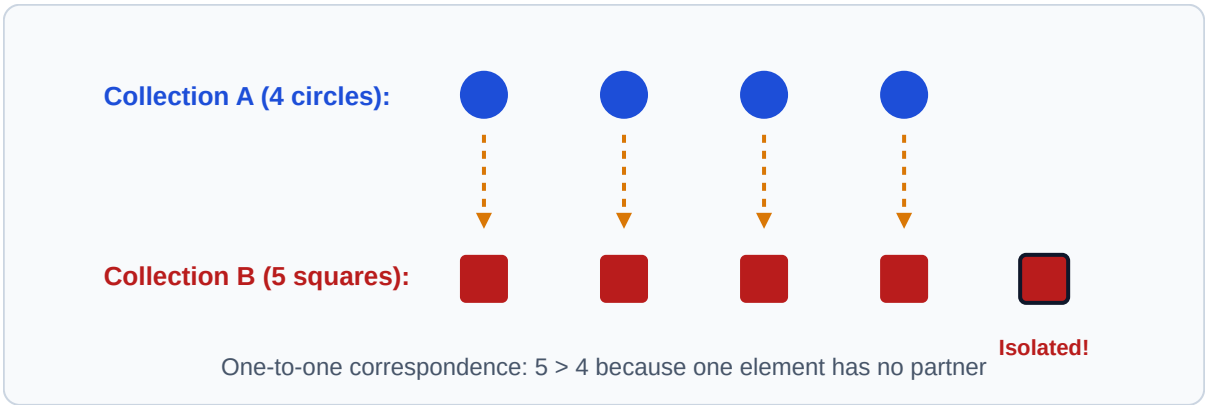

Figure 3: Term-to-term correspondence: direct comparison of two collections

At dusk, the sheep return. The shepherd stands at the cave entrance:

- For every animal that enters, he reaches into his pouch, pulls out one pebble, and sets it on the ground: *Clink!*

Look at what can happen:

- **The pouch is empty at the end?** Relief! There were **as many** sheep as there were pebbles. Everyone is safely accounted for.
- **One pebble remains in the pouch?** Alarm! This pebble never found its sheep. There are **fewer** sheep returned than pebbles: one sheep was left behind outside! The shepherd lights a torch and goes out to search.
- **Another sheep appears and the pouch is already empty?** A joyous surprise: a ewe gave birth to a lamb out on the meadow! There are **more** sheep than pebbles.

> 📐 **Definition — One-to-One Correspondence**
>
> Establishing a **one-to-one correspondence** (also called bijection or item-by-item pairing) means pairing every object of one group with one and only one object of another group, leaving none out and repeating none. This is the foundational principle of all mathematical calculation.

> 💡 **Did You Know? — Where Does the Word "Calculate" Come From?**
>
> In Latin, a small smooth pebble gathered from a stream bed was called a **calculus**. It is from that very pebble slipped into the prehistoric shepherd's pocket that our modern word comes: **to calculate**!

---

4. As Many As, More Than, Fewer Than

Before using written numerals, mathematicians use three fundamental phrases to compare two collections of objects:

- **As many as**: Every circle has found a partner square, and none are left over. Both groups contain the same quantity.

- **More than**: The group of squares has leftover items. There are **more squares than circles**.
- **Fewer than**: The group of circles ran out first. There are **fewer circles than squares**.

> ⚠️ **Watch Out! — The Illusion of Occupied Space**
>
> Look at these two rows of buttons:

Solved Example — Visual Comparison in Parallel Rows

> - *Row 1:*
> - *Row 2:*

> *The first row looks longer because the circles are spread wide apart! But if you connect each circle to a square with your finger, you discover that there are 4 circles and 6 squares: there are therefore **more** squares than circles, even though the circles occupy more space on the table. **In mathematics, it is not space that counts—it is quantity!***

---

5. From the Pebble to the Carved Tally: The Ishango Bone

The bag of pebbles has one major flaw: if your flock grows to 80 sheep, your pouch becomes as heavy as a boulder and tears your belt.

Our ancestors therefore devised a much lighter tool: **the tally stick**. Each time an animal passes, one strikes a sharp flint against a polished bone or flat piece of wood: *Scritch!*

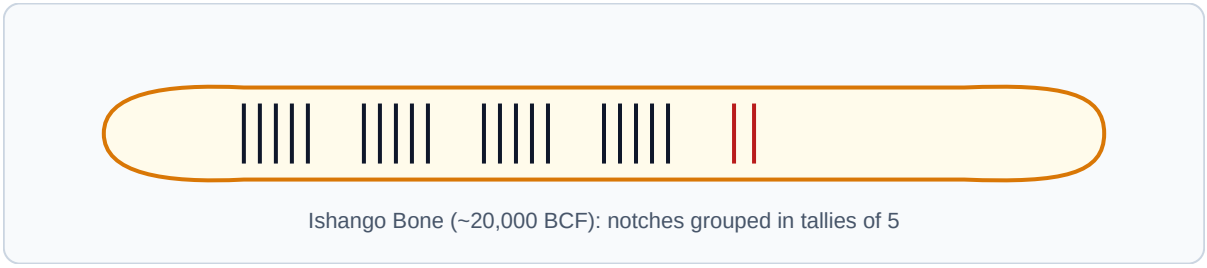

Figure 4: The Ishango bone: early tally notches grouped in bundles

> 🔍 ***True Story — The Ishango Tally Bone*** *On the shores of Lake Edward, between the Democratic Republic of the Congo and Uganda in Africa, archaeologists unearthed a fossilized baboon fibula over **20,000 years old**: the **Ishango bone**. It bears notches arranged into regular series. While early researchers interpreted these as counting tally marks in base 5 or 10, ethnomathematicians and paleoanthropologists point out that they might equally represent a lunar calendar, an arithmetic game, or a rhythmic notation. It remains one of humanity's oldest surviving testimonies of mathematical abstraction and the physical inscription of quantity.*

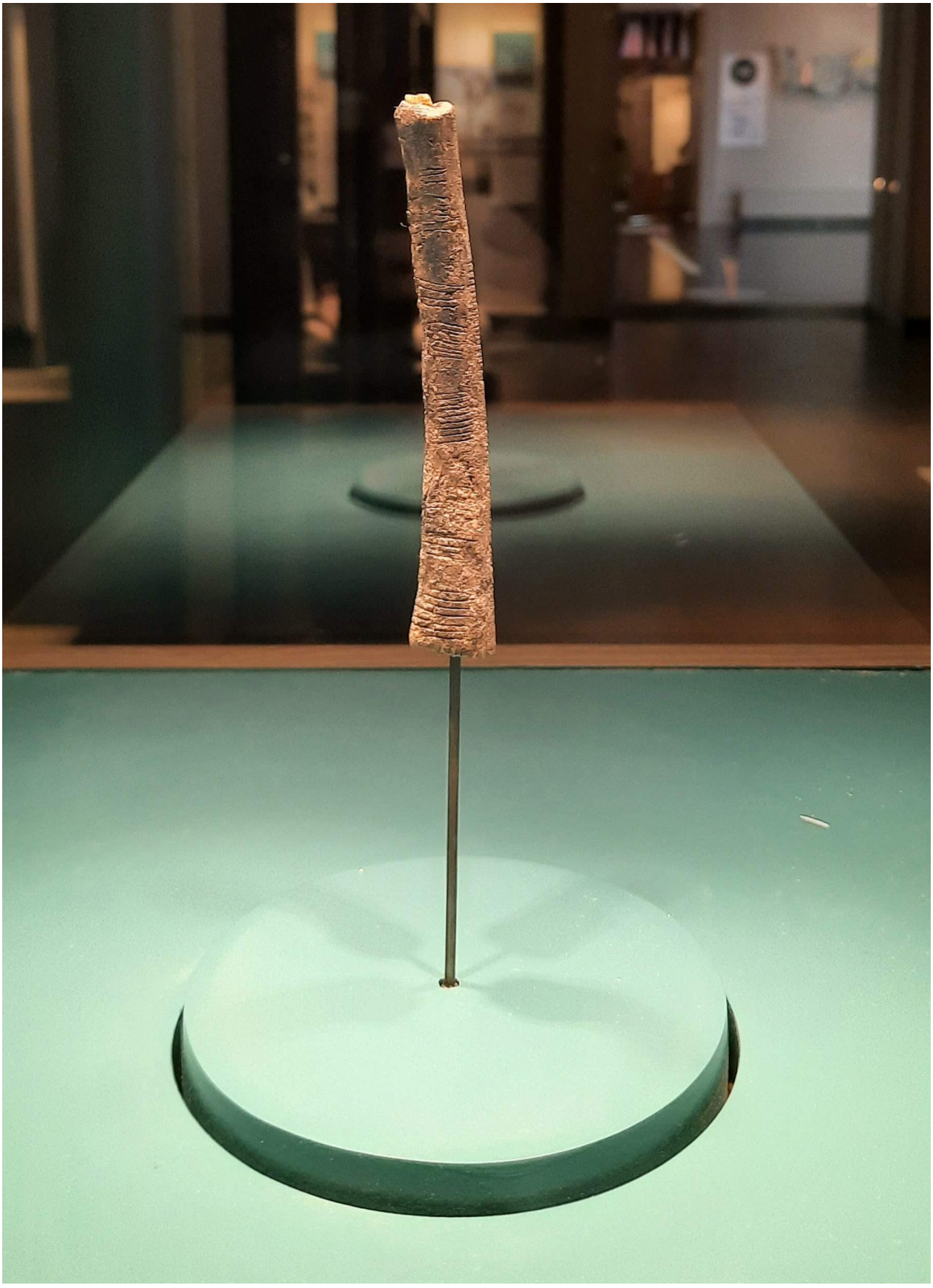

Figure 5: The fossilized Ishango bone (~20,000 BCE, Congo): humanity's earliest counting notches and numerical groupings

6. Writing the World: The Ten Digits from 0 to 9

Carving notches on bones still takes up substantial space. To work faster, humans eventually invented **symbols**—concise glyphs that can be drawn in seconds. It takes only ten of them to write all the numbers in the universe:

$$\mathbf{0},\quad \mathbf{1},\quad \mathbf{2},\quad \mathbf{3},\quad \mathbf{4},\quad \mathbf{5},\quad \mathbf{6},\quad \mathbf{7},\quad \mathbf{8},\quad \mathbf{9}$$

> **📐 Definition — Digit vs. Number**
>
> • A **digit** is an individual written character, like a letter of the alphabet. There are only 10: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
> • A **number** expresses a quantity or measure, like a word built from letters. *For example: using the digits 2 and 5, we write the number 25.*

How to Properly Draw the Ten Digits

Here is the secret to forming each digit cleanly without shaking, always starting from the correct position:

- **0 (Zero)**: Start at the top, curve down to the left as if making a large circle, touch the baseline, and sweep back up to close at the top.
- **1 (One)**: Start with a short stroke slanting up to the right, then drop straight down like a vertical fence post.
- **2 (Two)**: Start near the top, arch over to the right in a rounded dome, slant diagonally down to the bottom left, then sweep a flat baseline stroke straight to the right.
- **3 (Three)**: Draw a small rounded curve to the right, tuck in toward the middle without lifting your pencil, and sweep out a second rounded curve below.
- **4 (Four)**: Drop straight down, make a sharp right-angle turn to the right, lift your pencil, and cross through that horizontal line with a vertical stroke.
- **5 (Five)**: Start at the top, drop a short vertical stroke, sweep out a broad rounded belly to the right, then top it off with a flat horizontal cap.
- **6 (Six)**: Start at the top, curve down to the left as on a steep slide, and roll into a tight closed loop at the bottom.
- **7 (Seven)**: Draw a flat horizontal roof from left to right, slant down diagonally to the bottom left, and add a small crossbar across the middle.
- **8 (Eight)**: Do not stack two circles on top of each other! Trace a smooth winding "S" curve, then loop back up diagonally through the middle to close the loop at the top.
- **9 (Nine)**: Start by tracing a complete circle at the top (like a balloon), then bring the string straight down with a gentle hook curving to the left.

---

7. Five Solved Examples and Practical Workshops

Solved Example 1 — The Squirrels' Harvest

Problem: Three little squirrels gather hazelnuts. Hazelnut the squirrel found 3, Bushy found 5, and Chestnut found 3. Who has as many as whom? *Step-by-step resolution:* 1. Let us compare Hazelnut (3 nuts) and Chestnut (3 nuts). Let us perform item-by-item pairing:

- Hazelnut's 1st nut ↔ Chestnut's 1st nut
- 2nd nut ↔ 2nd nut
- 3rd nut ↔ 3rd nut

2. Every nut is paired, leaving none left over: **Hazelnut has as many hazelnuts as Chestnut**.
3. Bushy has 5: he has two more than Hazelnut and Chestnut. Bushy has **more than** the others.

Solved Example 2 — Setting the Table Without Counting

Practical Workshop: You must set the dinner table for 4 people. You have a box of forks. *The shepherd's method:* You do not count the forks in the box. You walk to each plate set around the table:

- In front of Mom's plate: place **one** fork.
- In front of Dad's plate: **one** fork.
- In front of your sister's plate: **one** fork.
- In front of your plate: **one** fork. *Conclusion:* Every plate has received one fork and only one. There are **as many** forks on the table as plates, without ever having needed to count!

Solved Example 3 — Comparing Two Spread-Out Collections

Problem: On a sheet of paper, 5 small red stars are clustered tightly together, while 4 large blue hearts are spaced widely apart. Are there more stars or more hearts? *Method:* Connect each heart to a star with a pencil line:

- Heart 1 → Star 1
- Heart 2 → Star 2
- Heart 3 → Star 3
- Heart 4 → Star 4 There remains **one red star** that has no partner heart. *Conclusion:* There are **more red stars than blue hearts**, even though the hearts occupy more physical space across the page.

Solved Example 4 — The Clothes Hanger Balance

Practical Workshop: Take a plastic clothes hanger. Clip 3 red clothespins on the left arm and 3 blue clothespins on the right arm. *Observation:* If you balance the hanger by its hook on your index finger, it stays perfectly horizontal! Both sides balance because there are **as many** clothespins on the left as on the right. If you add another red pin to the left, the left side dips down: there are **more pins on the left than on the right**.

Solved Example 5 — Spotting the Empty Set with Zero

Problem: In one nest, there are 3 chicks. In a second nest, there is 1 chick. In a third abandoned nest, there are no chicks at all. What digit do you write for each nest? *Resolution:*

- First nest: digit **3**.
- Second nest: digit **1**.
- Third nest: the nest is empty. To represent the absence of chicks, the null quantity, we write the digit **0**. *Conclusion:* Zero signifies that there is **nothing**, the container is empty!

Key Takeaways

1. **Subitizing** enables our visual cortex to recognize small quantities of 1, 2, or 3 objects instantly without counting one by one.
2. For larger quantities, we employ **one-to-one correspondence**: pairing every item in one set with one item in another set (like one pebble per sheep).
3. **As many as** means that both groups have identical quantities with none left over.
4. **More than** means that unmatched items remain in the first set; **fewer than** means items are missing.
5. Direct pairing eliminates perceptual traps caused by object size or spread-out spacing.
6. There are **10 digits**: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. They allow us to write all numbers. The digit **0** represents the empty quantity.

Practice Exercises

Level (Foundation & Fluency)

Exercise 1.1 Flash glance! Look at each of the three boxes below for a single blink. State the quantity of dots without pointing your finger:

- Box A:
- Box B:
- Box C:

Exercise 1.2 Horses and saddles. In a pasture, there are 4 horses. In the stable, there are 4 saddles. Are there enough saddles for every horse to receive one? State the exact mathematical phrase.

Exercise 1.3 Rabbits and carrots. On the table, you have 3 toy rabbits and 5 plastic carrots. Connect each rabbit to one carrot. Do any carrots remain? Are there more rabbits or more carrots?

Exercise 1.4 Tracing numerals. Take a sheet of lined paper and carefully write five times in a row: the digit **2**, the digit **4**, and the digit **7**.

Exercise 1.5 The empty nest. On a branch, there are 3 bird nests. The first contains 2 eggs, the second contains 4 eggs, the third contains nothing at all. Write the three digits representing the eggs in these nests.

Level (Reasoning & Comparison)

Exercise 1.6 The shepherd and the wolf. A shepherd has 6 pebbles in his pouch in the morning. At dusk, 5 sheep enter the fold, and the shepherd removes one pebble for each returned animal. 1. How many pebbles remain in the pouch? 2. Are all sheep safely inside? What must the shepherd do?

Exercise 1.7 The spread-out trap. Lola places 4 large pillows widely spaced across the living room rug. Tom places 5 small erasers huddled together on his desk. Lola claims: *"I have more items, because my line reaches all the way across the room!"* Is she right? Explain why using one-to-one correspondence.

Exercise 1.8 Birds on a wire. 5 sparrows are perched on an electric wire. 3 sparrows suddenly take flight. Are there now more birds on the wire or more birds in the sky?

Exercise 1.9 Birthday candles. For his birthday, Leo places 6 candles on his cake. His younger sister Maya has 4 candles on hers. 1. Who is older? 2. How many more candles does Leo have than Maya?

Exercise 1.10 Digit or number? In the following sentences, state whether the bold element is used as a **digit** (a writing symbol) or a **number** (a counted quantity):

1. "To write the number 38, I use the digits **3** and **8**."
2. "In this basket, there are **38** apples."
3. "My bib number for the race is **7**."

Level (Challenges & Puzzles)

Exercise 1.11 The greedy squirrels' feast. Four squirrels (Shadow, Rusty, Nutty, and Patch) each gathered acorns:

- Shadow gathered as many acorns as Rusty.
- Nutty gathered more acorns than Shadow.
- Patch gathered fewer acorns than Rusty.

Rank the four squirrels from the one with the most acorns to the one with the fewest.

Exercise 1.12 The mystery flock. A shepherd leads out his flock in the morning with his pouch of pebbles. During the day on the pasture, two ewes each give birth to a newborn lamb. At dusk, all animals (ewes and lambs) return to the fold, and the shepherd removes one pebble for every animal that passes the gate. What will happen when the last animal enters?

Exercise 1.13 The deceptive dominoes. Look at these three dominoes:

- Domino 1: (2 dots on the left, 3 dots on the right)
- Domino 2: (4 dots on the left, 1 dot on the right)
- Domino 3: (3 dots on the left, 3 dots on the right)

Two of these dominoes carry the exact same total number of dots. Which ones?

Exercise 1.14 The secret cave code. To unlock the secret cave door, one must punch in a code made of three distinct digits:

- The first digit is the number of eyes on your face.
- The second digit is the number of paws on a dog.
- The third digit is the digit signifying that nothing is there.

What is the three-digit secret code?

Exercise 1.15 The Ayumu home challenge. Take 5 playing cards numbered 1 through 5. Shuffle them and place them face down on the table. Have your mentor flip over 3 cards at random for 2 seconds, then cover them back up. Can you name the revealed cards from memory? Try again to beat your speed record!

Solutions

Solution 1.1 - Box A: **2**. - Box B: **1**. - Box C: **3**.

Your visual cortex recognized these small quantities via subitizing without counting.

Solution 1.2 Every horse receives one saddle and none remain left over. There are **as many** saddles as horses.

Solution 1.3 Pairing each rabbit with one carrot matches 3 rabbits to 3 carrots. 2 carrots remain alone. There are **more carrots than rabbits** (or fewer rabbits than carrots).

Solution 1.4 Verify with your mentor that your pencil started from the top for 2 and 7, and that the strokes of 4 meet at a crisp right angle.

Solution 1.5 The digits to write are: **2**, **4**, and **0** for the empty nest.

Solution 1.6 1. There were 6 pebbles. 5 were removed for the 5 returned sheep. Therefore **1 pebble** remains in the pouch ($6 - 5 = 1$). 2. No, not all sheep are safe: one sheep is missing out on the hill. The shepherd must take his torch and go search for it.

Solution 1.7 Lola is mistaken! Even though the pillows occupy more physical space, there are only 4 pillows, whereas Tom has 5 erasers. Pairing each pillow with an eraser leaves one eraser unmatched. There are **more erasers than pillows**. Occupied space does not alter numerical quantity.

Solution 1.8 There were 5 sparrows. 3 flew away into the sky, leaving 2 on the wire. Since 3 is greater than 2, there are **more birds in the sky than on the wire**.

Solution 1.9 1. Leo has 6 candles and Maya has 4. **Leo** is older because he has more candles. 2. Pairing 4 of Leo's candles with Maya's 4 leaves 2 candles. Leo has **2 more candles** than his sister ($6 - 4 = 2$).

Solution 1.10 1. **Digits** (building blocks used for notation). 2. A **number** (an actual counted quantity). 3. A **digit** used as an identifier (on a race bib or jersey, 7 serves as a nominal label to identify a runner, not as a measured quantity).

Solution 1.11 Analyzing the clues:

- Shadow and Rusty are tied with equal amounts.
- Nutty has more than Shadow, so Nutty ranks ahead of both.
- Patch has fewer than Rusty, so Patch ranks behind both.

Ranking from most to fewest: **Nutty**, then **Shadow and Rusty (tied)**, then **Patch**.

Solution 1.12 The shepherd started with one pebble per sheep. But two lambs were born! At dusk, there are 2 more animals than pebbles in the pouch. When all original pebbles have been removed, the bag will be empty while **2 lambs** are still standing outside the gate. The shepherd will realize with joy that his flock has expanded by two animals!

Solution 1.13 Counting the dots:

- Domino 1: $2 + 3 = 5$ dots.

- Domino 2: $4 + 1 = 5$ dots.
- Domino 3: $3 + 3 = 6$ dots. **Dominoes 1 and 2** carry the exact same total count (5 dots each).

Solution 1.14 - Eyes on your face: 2. - Paws on a dog: 4. - Null quantity (nothing): 0.

The secret cave code is: **2 - 4 - 0**.

Solution 1.15 Well done! The more you practice recognizing small quantities without counting on your fingers, the faster your brain develops its visual working memory, just like the young chimpanzee Ayumu.

The Question That Opens the Journey Ahead…

You are now able to track and verify any collection of objects using a pouch of pebbles or carved notches on a bone.

Yet imagine that your settlement expands. You no longer herd six sheep, but **three hundred**! And the miller has harvested thousands of wheat grains in his sacks.

If you had to carry three hundred pebbles in your pouch, it would weigh so much that you could barely walk! And if you tried to carve three thousand notches on a bone, it would snap in half.

How did humans learn to count very large numbers without collapsing under the weight of stones? The answer is magnificent: they simply looked down at **the ends of their arms**!

See you in **Chapter 2: The Five-Fingered Hand and Bundles of Ten**!

CHAPTER 2: The Five-Fingered Hand and Bundles of Ten

> "If we had had six fingers on each hand like creatures in fairy tales, the entire world would count in a completely different way. Human mathematics was born at the tips of our ten fingers."

What You Are About to Discover

- Why almost the entire planet counts in bundles of **10** (the decimal place-value system).
- Civilizations that also counted with their toes in bundles of **20** (vigesimal systems).
- The magic trade: exchanging 10 small pebbles for 1 cork marker or white seashell (the **ten**).
- How to construct **hundreds** and **thousands** using the egg carton model.
- The secret of **place value**: why the 1 in 14 does not mean the same thing at all as the 1 in 41!
- The indispensable role of **zero** to hold the empty column (the difference between 45 and 405).
- Walking along the number line from 0 to 100, and comparing numbers using the signs $<$ and $>$.

---

1. The Problem: A Pouch That Weighs a Ton

The flock of sheep has flourished across the seasons. New lambs were born in spring, and other shepherds have joined your settlement. This morning, you no longer have just seven or eight sheep to lead out.

You have **forty-five**!

You open the gate and begin your item-by-item pairing with pebbles in your leather pouch: *Plop, plop, plop…* By the fortieth sheep, your bag is heavy as a paving stone. It bumps against your knee, pulls down on your belt, and threatens to tear in two!

You then try replacing the pebbles with notches carved onto a wooden tally stick, like ancestral hunters:

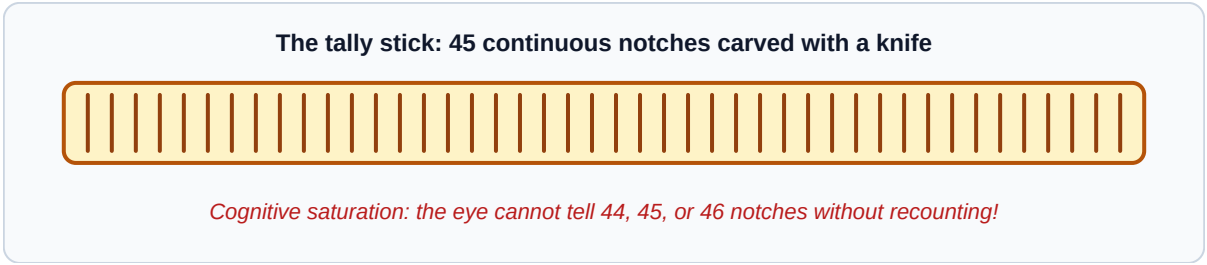

Figure 6: The tally stick: cognitive saturation of 45 continuous strokes

Look at this long, unbroken row of notches. Can you tell at a single glance whether there are 44, 45, or 46 marks? It is impossible! Your eyes blur, the lines seem to swim, and if a fly distracts you while you are pointing your finger, you are forced to start all over again from the very first mark.

To count large quantities without carrying a mountain of stones or straining your eyes, a new trick had to be invented. And that trick you have carried with you from the very first day of your life.

---

2. The Secret at the Ends of Your Arms: The Ten

Hold up both hands open before your eyes. Look at your fingers:

- On your left hand: 5 fingers (thumb, index, middle, ring, pinky).
- On your right hand: 5 fingers.

- Together: $5 + 5 = 10$ fingers!

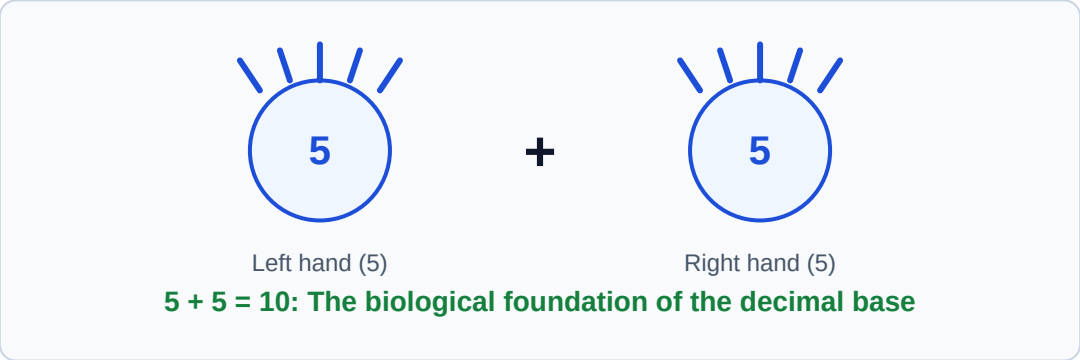

Figure 7: Two open hands: 5 + 5 = 10, foundation of the decimal base

One morning, watching the sheep stream past, a young shepherd has an epiphany. Along the trail, he has gathered ordinary small gray pebbles, but in his pocket, he carries several lightweight cork stoppers carved from the bark of an oak tree (or smooth white sea shells).

He invents a brilliant rule of exchange: 1. Each time an animal crosses the gate, he raises a finger: one, two, three, four, five, six, seven, eight, nine, **ten**! 2. All ten fingers are raised. His hands are full! He has no finger left for the eleventh sheep. 3. So, he drops all ten fingers at once. 4. And instead of keeping 10 heavy pebbles in his bag, he makes a trade: he places the 10 small pebbles into a ground basket, and **slips 1 single cork stopper into his pocket**!

This cork stopper is no ordinary pebble. It possesses a superpower: on its own, it whispers: *"I am worth ten small pebbles. I am worth ten sheep!"*

Figure 8: Regular bundling: 10 loose units become 1 bundled ten

The shepherd can now continue with free hands:

- Ten more sheep pass? A second cork stopper into the pocket.
- Ten more? A third cork stopper.
- Ten more? A fourth cork stopper.
- And for the final five sheep bringing up the rear? There are not ten of them; his hands are not full: he simply holds **5 small gray pebbles** in his palm.

When all 45 sheep have crossed the threshold, what does the shepherd have in his pocket?

- Only **4 cork stoppers** (worth $4 \times 10 = 40$),
- And **5 small gray pebbles** (worth 5).

In total: 9 small, featherlight items instead of 45 heavy stones that were bruising his hips!

📐 Definition — Unit (One) and Ten

- A **unit** (or **one**) is an isolated, individual item (1 pebble, 1 marble, 1 finger).
- A **ten** is a closed bundle of **10 units** grouped together ($1 \text{ ten} = 10 \text{ ones}$).

> 💡 **Did You Know? — The 10-Egg Carton**
>
> In your kitchen, look at a standard cardboard egg carton. It contains two rows of 5 compartments, exactly 10 spaces. One full, closed carton is **1 ten of eggs**. If you have 3 full cartons and 4 loose eggs on the counter, you know instantly that you have 34 eggs, without having to count them one by one!

---

3. Civilizations That Counted with Their Toes: Base 20

Not all civilizations across history stopped at 10 fingers.

In the warm climates of Mesoamerica thousands of years ago, the **Mayas** and **Aztecs** walked barefoot year-round. To count, they used their 10 fingers… plus their **10 toes**!

They therefore counted in bundles of **20**: this is known as a **vigesimal system** (from the Latin *vicesimus*, twentieth).

> 🔍 ***True Story — Why Does French Say "Quatre-vingts"?*** *Ancient Celts and Gauls also counted in bundles of twenty! That is why in France, instead of a single word for eighty, people still say:*
>
> $$\text{“quatre-vingts”} = 4 \quad \text{twenties} = 20 + 20 + 20 + 20 = 80$$
>
> *And for 90, they say “quatre-vingt-dix” ($4 \times 20 + 10$). Languages preserve living traces of those barefoot ancient hunter-gatherers!*

---

4. The Grand Hierarchy: Tens, Hundreds, and Thousands

What happens when a flock becomes enormous? Suppose you have collected ten cork stoppers in your pouch:

$$10 \quad \text{cork stoppers} = 10 \quad \text{tens} = 100 \quad \text{sheep!}$$

Your pocket of stoppers is now full. What do you do? You invent another trade:

- You exchange your **10 cork stoppers** for **1 golden pinecone**!
- This pinecone is called **one hundred**. It represents 100 sheep.

And if your settlement one day gathers ten golden pinecones?

$$10 \quad \text{hundreds} = 1\,000 \quad \text{sheep!}$$

You exchange them for **1 large painted river stone**: this is **one thousand**.

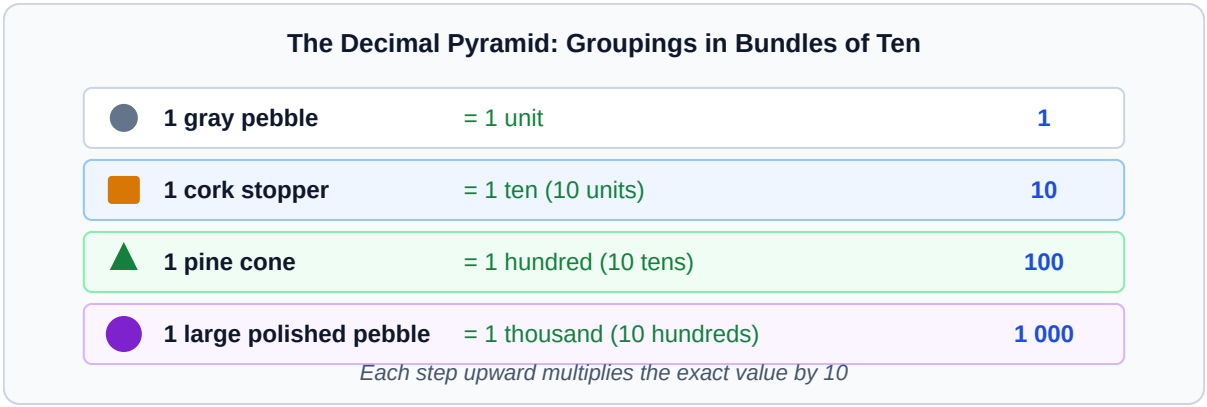

Figure 9: The decimal pyramid: units, tens, hundreds, thousands

Every time you accumulate 10 items of one level, you bundle them into a single item of the next level! This is the core of our **decimal place-value system** (from the Latin *decem*, meaning ten).

---

5. Place Value: Position Dictates Power!

To write these large quantities using only our digits 0 through 9 without drawing pinecones or corks all day long, ancient mathematicians discovered a master principle: **place value**.

Look carefully at these two numbers written with the exact same digits, 1 and 4:

$$\mathbf{14} \quad \text{and} \quad \mathbf{41}$$

Are they the same? Absolutely not!

- In **14**:
  - The rightmost digit (**4**) indicates loose units: 4 single pebbles.
  - The leftmost digit (**1**) indicates tens: 1 bundle of 10.
  - In total: $10 + 4 = 14$.
- In **41**:
  - The rightmost digit (**1**) indicates 1 small single pebble.
  - The leftmost digit (**4**) indicates 4 tens, that is, 4 bundles of 10 ($10 + 10 + 10 + 10 = 40$)!
  - In total: $40 + 1 = 41$.

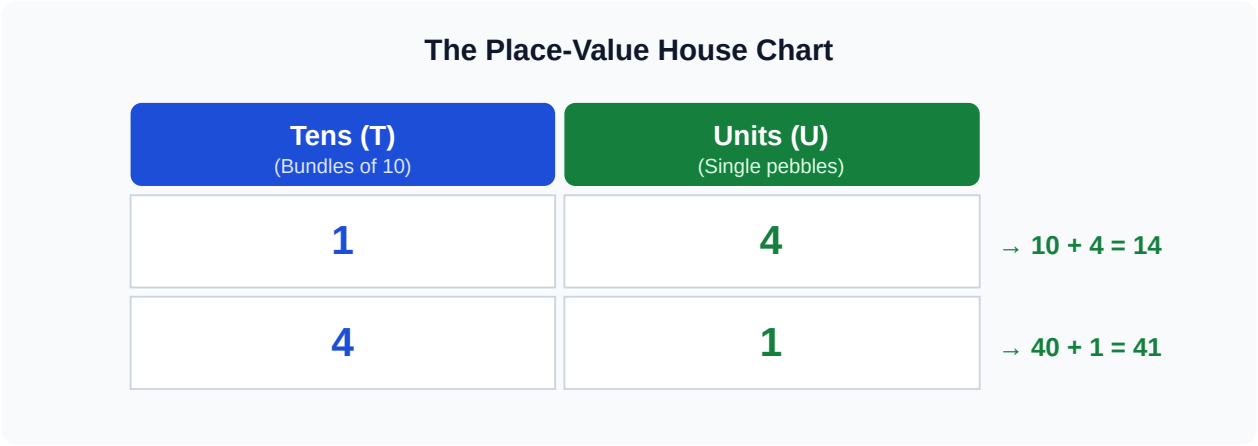

Figure 10: The place-value house chart: hundreds, tens, and units

> **Golden Rule — Place Value**
>
> In mathematics, the true value of a digit depends entirely upon the **place** (the column) it occupies within the number:
>
> - Furthest to the right: the **ones** (units) column (u).
> - Immediately to the left: the **tens** column (t).
> - Further to the left: the **hundreds** column (h).
> - Further still: the **thousands** column (th).

---

6. The Invisible Guardian: Zero Holding the Empty Place

Imagine you have in your collection box:

- **4 golden pinecones** (4 hundreds = 400),
- **0 cork stoppers** (no tens at all),
- **5 gray pebbles** (5 ones).

How do you write this number? If you simply write the 4 and the 5 side by side: **45**, anyone reading your note will interpret it as 4 tens and 5 ones, meaning 45 sheep! Four hundred sheep have vanished into thin air!

To ensure that the 4 remains anchored in the hundreds column, we must station a sentry in the empty tens column: **the digit zero (0)**.

$$405 = 4 \text{ hundreds, } 0 \text{ tens, and } 5 \text{ ones}$$

📐 Fundamental Rule — The Role of Zero as a Place Keeper

- Hundreds Tens Units
- 4 0 5 —> 405
- Zero prevents the 4 and 5 from collapsing together!

💡 Did You Know? — The Invention of Zero

Zero is one of the greatest conceptual breakthroughs in human history! It was perfected by mathematicians in **India** around 1,500 years ago. Its Sanskrit name was *shunya*, meaning "void" or "empty." It traveled into the Arabic world as *sifr* (which gave rise to both our English word "cipher" and the French word "chiffre").

⚠️ Watch Out! — The Missing Middle Zero

Two hundred three is never written 23! $23 = 2 \text{ tens and } 3 \text{ units}$. Two hundred three is written 203: there are 0 tens in the middle. Never let an empty column collapse without its sentry zero!

---

7. The Number Line and Inequality Signs: $<$ and $>$

To organize numbers like beads along a string, we draw a straight graduated axis: **the number line**. It originates at 0 on the left and extends toward the right, growing step by step:

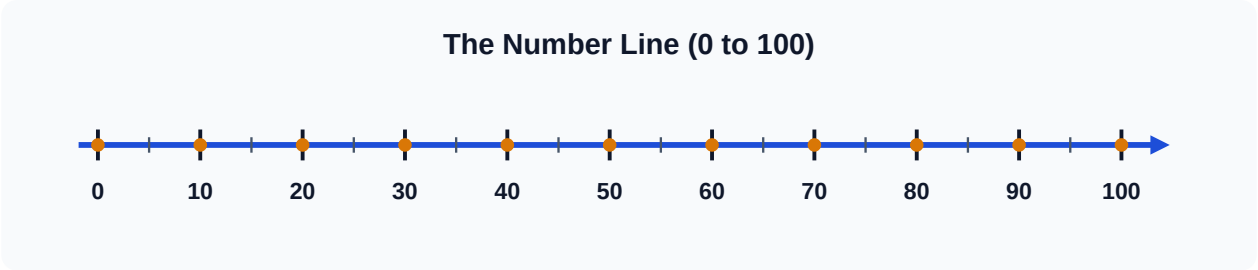

Figure 11: The directed number line from 0 to 100

The further a number is located to the right, the greater its value.

The Hungry Alligator: $<$ and $>$

To compare two numbers and declare which is greater, we use two convenient mathematical symbols:

- **The sign $<$** is read "**is less than**." *Example:* $3 < 8$ (3 is less than 8).
- **The sign $>$** is read "**is greater than**." *Example:* $9 > 4$ (9 is greater than 4).

🧭 Method — The Hungry Alligator

Imagine that the symbols $<$ and $>$ are the **wide-open jaws of a hungry alligator**. The alligator always snaps toward the larger feast (the greater number)! The narrow point points toward the smaller number.

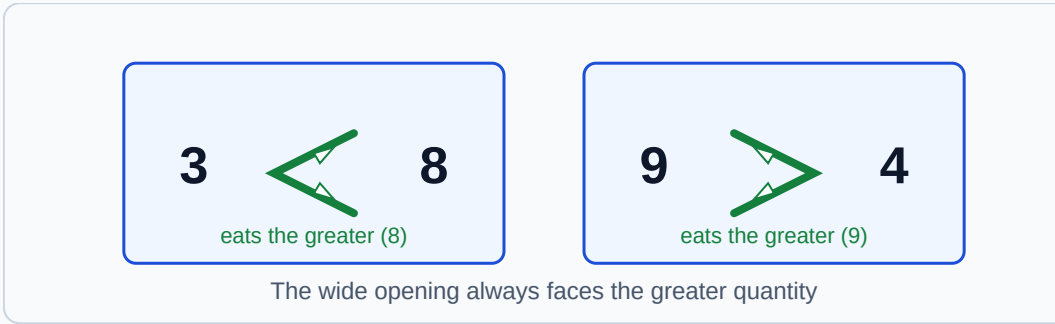

Figure 12: The comparison crocodile: jaws open toward the greater quantity

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8. Five Step-by-Step Solved Examples

Solved Example 1 — Decomposing a Number into Tens and Ones

Problem: Decompose the number **63** using 10-egg cartons and loose eggs. *Step-by-step resolution:* 1. In 63, the left digit is 6: it represents **6 tens**. $6 \text{ tens} = 6 \text{ cartons of } 10 = 60 \text{ eggs}$. 2. The right digit is 3: it represents **3 loose units**. 3. We express this in expanded form:

$$63 = 60 + 3 = (6 \times 10) + 3$$

Conclusion: 63 eggs means 6 full cartons plus 3 loose eggs on the side.

Solved Example 2 — Comparing 37 and 52

Problem: Which number is greater between 37 and 52? Express the comparison with the correct sign. *Method:*
1. Always inspect the **tens digit** first (the larger bundles):

- In 37, there are 3 tens (30).
- In 52, there are 5 tens (50).

2. Because $5 \text{ tens} > 3 \text{ tens}$, the number 52 contains more tens than 37. We do not even need to examine the units!
3. The alligator opens its jaws toward 52:

$$37 < 52 \qquad (37 \text{ is less than } 52)$$

Solved Example 3 — Comparing Numbers with Equal Tens: 74 and 79

Problem: Compare 74 and 79. *Method:* 1. Inspect the tens digits: 7 tens in 74, and 7 tens in 79. It is a tie! 2. Since the tens are identical, we break the tie using the **ones digit**:

- 74 has 4 ones.
- 79 has 9 ones.

3. Since $4 < 9$, we conclude:

$$74 < 79$$

Solved Example 4 — Rounding Bounds to the Nearest Ten

Problem: Bound the number **47** between the multiple of ten immediately preceding it and the one immediately following it. *Method:* 1. 47 has 4 tens and 7 ones. 2. The full ten immediately preceding 47 is **40** (4 tens). 3. The full ten immediately following 47 is the next multiple: **50** (5 tens). 4. We express the double inequality:

$$40 < 47 < 50$$

Solved Example 5 — Discovering the Mystery Number

Problem: "I am a number between 60 and 70. My units digit is double of 3. Who am I?" *Method:* 1. The number lies between 60 and 70: its tens digit must be **6**. 2. Its units digit is double of 3: $3 + 3 = 6$. The units digit is therefore **6**. 3. Combining 6 tens and 6 units yields: **66**.

Key Takeaways

1. Our numerical system is in **base ten** because we possess ten fingers on our two hands.
2. **1 ten = 10 units (ones)**. 1 hundred = 10 tens = 100 units. 1 thousand = 10 hundreds = 1,000 units.
3. **Place value**: A digit's value depends on its column. In 82, the 8 represents 80 (eight tens). In 28, the 8 represents 8 (eight ones).
4. The digit **zero (0)** is an indispensable placeholder: it reserves the space of an empty column (e.g., 306 contains 0 tens).
5. To compare two two-digit numbers:
   - First compare the **tens digit**. Whoever has more tens always wins!
   - If the tens digits are equal, compare the **units digit**.
6. Inequality symbols: $<$ (less than) and $>$ (greater than). The alligator's mouth always opens toward the greater number.

Practice Exercises

Level (Foundation & Fluency)

Exercise 2.1 How many eggs? Write as a standard two-digit number: 1. 3 full cartons of 10 eggs and 6 loose eggs. 2. 5 full cartons of 10 eggs and 1 loose egg. 3. 8 full cartons of 10 eggs and zero loose eggs.

Exercise 2.2 Tens or units? In the number **73**: 1. Which digit is in the units place? 2. Which digit is in the tens place? 3. What is the actual value of the digit 7? (7 or 70?)

Exercise 2.3 The alligator's mouth. Fill in with $<$ or $>$:

- $24 \quad ... \quad 42$
- $68 \quad ... \quad 61$
- $89 \quad ... \quad 90$
- $50 \quad ... \quad 15$

Exercise 2.4 Expanded form. Decompose these numbers as an addition ($25 = 20 + 5$):

- $48 = ... + ...$
- $72 = ... + ...$
- $36 = ... + ...$
- $94 = ... + ...$

Exercise 2.5 The number line. On a number line marked in tens, what number sits exactly halfway between 20 and 30? And between 70 and 80?

Level (Reasoning & Comparison)

Exercise 2.6 Sentry zero. Write in standard digits: 1. Five hundreds, three tens, and two units. 2. Seven hundreds, zero tens, and four units. 3. Two hundreds, six tens, and zero units. What would happen to the second number if you forgot to write zero?

Exercise 2.7 Largest and smallest. Using two number cards showing the digits **3** and **8**, create: 1. The smallest possible two-digit number. 2. The largest possible two-digit number. Compare them using the sign $<$.

Exercise 2.8 Ordering numbers. Arrange the following numbers from least to greatest using the sign $<$:

$$54 \quad ; \quad 19 \quad ; \quad 72 \quad ; \quad 45 \quad ; \quad 78 \quad ; \quad 9$$

Exercise 2.9 The marble merchant's count. A merchant packs marbles into bags of 10. He has 67 loose marbles. 1. How many full bags of 10 can he pack? 2. How many marbles remain outside the bags? 3. How many marbles does he need to fill one more bag?

Exercise 2.10 Rounding bounds. Identify the tens immediately preceding and following each number:

- $\ldots < 34 < \ldots$
- $\ldots < 58 < \ldots$
- $\ldots < 81 < \ldots$

Level (Challenges & Puzzles)

Exercise 2.11 Grandmother's mystery number. "My number has 2 digits. The sum of my two digits is 10 ($t + u = 10$). My tens digit is 7. Who am I?"

Exercise 2.12 The Mayan and Celtic challenge. 1. A Mayan elder counts with hands and toes. He says: "I have 3 twenties and 5 units." What number is that in our decimal system? 2. How would a speaker of traditional vigesimal French describe the number 85 in bundles of 20?

Exercise 2.13 Pirates and the gold chest. Pirates discover a chest containing 132 gold coins. The captain wants to pack them into pouches of 10 coins each. 1. How many full pouches of 10 coins can the pirates form? 2. How many gold coins will remain loose on the table?

Exercise 2.14 Reversed digits. Take a two-digit number whose tens digit is strictly greater than its units digit (e.g., 62). Reverse its digits (giving 26). Is the reversed number always strictly smaller than the original? Explain why in terms of place value.

Exercise 2.15 The tens lottery game. Find all possible two-digit numbers whose tens digit is **5**, and whose units digit is strictly greater than 6. Arrange them from least to greatest.

Solutions

Solution 2.1 1. 3 cartons of 10 and 6 eggs = **36**. 2. 5 cartons of 10 and 1 egg = **51**. 3. 8 cartons of 10 and 0 eggs = **80**.

Solution 2.2 1. Units digit is **3**. 2. Tens digit is **7**. 3. The 7 is in the tens column: it represents 7 bundles of 10, so its value is **70**.

Solution 2.3 - $24 < 42$ (2 tens is less than 4 tens). - $68 > 61$ (same tens digit 6, but $8 > 1$). - $89 < 90$ (8 tens is less than 9 tens). - $50 > 15$ (5 tens versus 1 ten).

Solution 2.4 - $48 = 40 + 8$ - $72 = 70 + 2$ - $36 = 30 + 6$ - $94 = 90 + 4$

Solution 2.5 - Halfway between 20 and 30 is **25** ($20 + 5$). - Halfway between 70 and 80 is **75** ($70 + 5$).

Solution 2.6 1. $500 + 30 + 2 = \mathbf{532}$. 2. $700 + 0 + 4 = \mathbf{704}$. 3. $200 + 60 + 0 = \mathbf{260}$. If zero were omitted in the second number, one would write 74, representing 7 tens and 4 units rather than 7 hundreds and 4 units. Zero is indispensable as a placeholder!

Solution 2.7 1. Smallest number: place smaller digit in tens: **38**. 2. Largest number: place larger digit in tens: **83**. Comparison: $38 < 83$.

Solution 2.8 Checking digit counts first (9 is a single digit, hence smallest), then tens:

$$9 < 19 < 45 < 54 < 72 < 78$$

Solution 2.9 1. In 67, there are 6 tens: he can fill **6 full bags** of 10 marbles. 2. **7 loose marbles** remain outside. 3. To reach a full 10 from 7, he needs $10 - 7 = \mathbf{3}$ more marbles.

Solution 2.10 - $30 < 34 < 40$ - $50 < 58 < 60$ - $80 < 81 < 90$

Solution 2.11 The tens digit is 7. We seek a units digit u such that $7 + u = 10$. The complement of 7 to 10 is 3 ($7 + 3 = 10$). The mystery number is **73**.

Solution 2.12 1. 3 twenties is $20 + 20 + 20 = 60$. Adding 5 units gives $60 + 5 = \mathbf{65}$. 2. 85 is $80 + 5$, which is 4 twenties and 5 ones: "**quatre-vingt-cinq**."

Solution 2.13 1. In 132, there are 13 full tens ($13 \times 10 = 130$). The pirates can fill **13 full pouches**. 2. **2 gold coins** will remain loose.

Solution 2.14 Yes, it is always true! Take a number where tens exceed units, such as 62 ($6 > 2$). Initially, it contains 6 tens (60). Upon reversing digits to 26, it contains only 2 tens (20). Since 2 tens is strictly less than 6 tens, the reversed number will **always be smaller** than the original.

Solution 2.15 The tens digit must be 5. The units digit must be strictly greater than 6, leaving choices 7, 8, or 9. The valid numbers are: **57**, **58**, and **59**. In ascending order: $57 < 58 < 59$.

The Question That Opens the Journey Ahead…

You are now able to bundle items into tens, construct hundreds, and compare large numbers thanks to place value.

Yet what happens when two clan hunters return at dusk carrying two different baskets filled with ripe berries? How do we **put their harvests together**? And what happens if one basket has a hole in the bottom and berries are lost along the trail?

To learn how to gather, take away, and share with absolute justice, we are about to construct a magical tool with two suspended wooden pans…

See you in **Chapter 3: The Shared Hunt and the Balance Scale of Justice**!

CHAPTER 3: The Shared Hunt and the Balance Scale of Justice

> "Nothing resembles justice more closely than a straight wooden beam suspended from its exact midpoint, whose two arms refuse to tilt in either direction. It is from this physical balance of hands that our most precious mathematical symbols were born."

What You Are About to Discover

- How to unite two separate harvests into one large basket using **addition (+)**.
- How to measure what has vanished, been eaten, or is missing using **subtraction (−)**.
- Why the equal sign = is not a calculator key meaning "give me the answer," but **a two-pan balance scale in absolute equilibrium**.
- The **friends of 10** (number bonds to 10 essential for lightning-fast mental math) and number bonds to 100.
- Magic mental math shortcuts (such as adding 9 by doing $+10-1$).
- How to solve authentic word problems: "how many remain?", "who has more?", "how many are missing?".

---

1. The Problem: The Evening Cauldron and the Night Thief

The sun dips below the forest edge. Hunters and foragers return to the clan's grand campfire.

You went to the wild orchard by the stream: in your woven wicker basket, you carry **4 crisp red apples**. Your older sister climbed the sunny rocky slopes: in her linen apron, she carries **3 plump yellow pears**.

Tonight, the entire clan prepares the feast. You pour your 4 apples into the large earthenware cooking pot: *Thump, thump, thump, thump!* Your sister steps up and pours her 3 pears right on top: *Thump, thump, thump!*

How many fruits are resting at the bottom of the pot? Your eyes and fingers register it immediately: **seven fruits**!

Addition combines two distinct collections into a single total

Figure 13: The addition pot: pooling two quantities together

This magnificent physical action, where two hands unite two separate piles into a single shared vessel, is **Addition**. To record this action without sketching fruit baskets all day, humans drew a small perpendicular cross: **the plus sign** +.

$$4 + 3 = 7$$

> 💡 **Did You Know? — Where Does the + Sign Come From?**
>
> In medieval times, merchants scribbled the Latin word *et* (meaning "and also"). Writing at high speed with quill pens, they slurred the strokes: the loop of the *e* and the crossbar of the *t* merged into a simple upright cross: +! The Latin root *ad-dere* means "to give in addition," which produced our modern word **addition**.

---

2. Order Does Not Matter: Commutativity

Imagine that your sister returned to camp first, exhausted from her climb. She is the one who pours her **3 pears** into the bottom of the pot. You arrive a bit later with your **4 apples**, pouring them on top.

Are there more fruits? Are there fewer? Not a single gram has changed! Inside the pot, there remain exactly **7 fruits**.

$$3 + 4 = 7$$

Whether we pour the apples first or the pears first, the total harvest in the pot is strictly identical:

$$4 + 3 = 3 + 4$$

📐 Property — Commutativity of Addition

In an addition, one can **reverse the order of the numbers** without altering the final sum. We say that addition is **commutative**. *Practical tip:* To mentally calculate $2 + 19$, it is much easier to flip the calculation in your mind and compute $19 + 2 = 21$!

---

3. The Night of the Fox: Subtraction

Yet life does not consist solely of joyful additions. Reserves dwindle, and thieves move in the shadows.

That evening, the fishermen brought back **7 fine fish** from the river. They were strung from a wooden bough above the glowing embers to smoke slowly overnight. The village falls peacefully asleep.

In the dead of night, a hungry red fox creeps forward on silent paws. Without a sound, it climbs a boulder, leaps onto the bough, snatches **2 fish** in its jaws, and bolts into the dark forest!

At dawn, you stretch your arms and look up at the smoking bough:

- There were 7 fish last evening.
- The fox carried off 2.
- How many fish remain for breakfast?

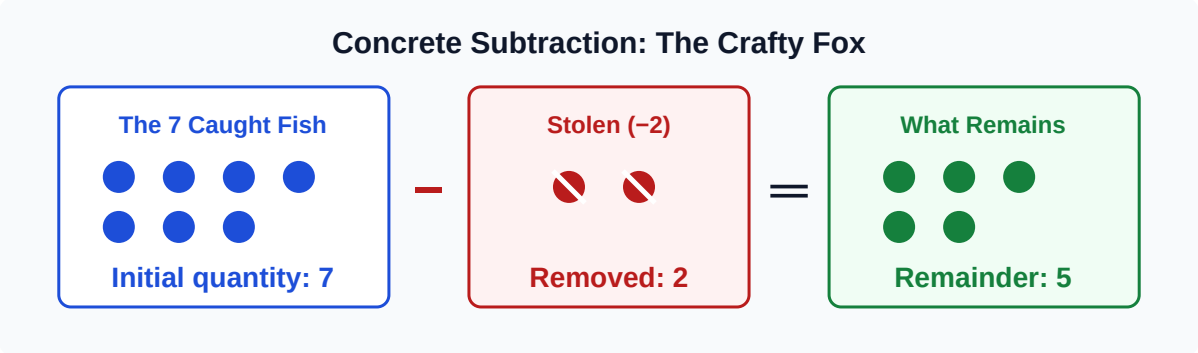


Figure 14: Concrete subtraction: 7 fish minus 2 snatched away

There are **5** remaining!

This physical gesture where material is removed, where a quantity is eaten, lost, or concealed, is **Subtraction**. To record it, we draw a flat horizontal stroke: **the minus sign −**.

$$7 - 2 = 5$$

📐 Definition — Subtraction and the Inverse Operation

Subtraction determines what remains after a portion is taken away, or what is needed to complete a whole. Subtraction is **the inverse operation of addition**: it undoes what addition constructed.

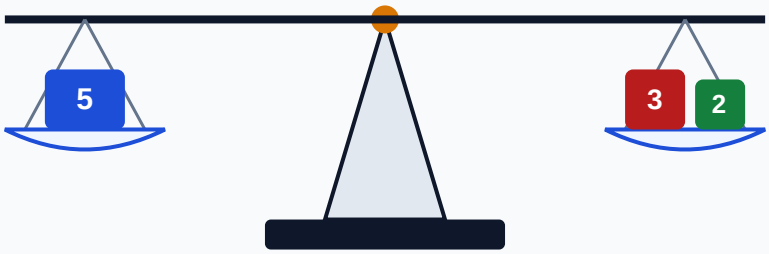
$$7 - 2 = 5 \qquad \text{because} \qquad 5 + 2 = 7$$

4. The Scale of Justice and the True Secret of the Sign $=$

Here lies one of the most foundational concepts in all of mathematics, from early childhood to rocket science!

Two traders argue vehemently in the market square. One says: *"My 5 bags of grain are worth as much as your 3 bags of flour and 2 bags of olives!"* The other shouts: *"False! You are trying to cheat me!"*

To establish truth, the village elder does not shout. He brings out **the Balance Scale of Justice**: a sturdy wooden beam suspended by a cord at its exact center, carrying two suspended pans on either side.



Perfect balance: the '=' sign asserts exact equality of both pans

Figure 15: The balance of justice: state of perfect equilibrium

The elder places the 5 bags of grain on the left pan. He places the 3 bags of flour and 2 bags of olives on the right pan. The wooden beam wavers… hesitates… and settles **perfectly level**. Both pans hang at the exact same height!

The elder declares: *"It is an equality!"*

📐 Golden Rule — The Equal Sign $=$ is a Balance Scale

The sign $=$ (equals) does **NOT** mean: "press the button to produce the answer." The sign $=$ means: **"WHAT IS ON THE LEFT WEIGHS EXACTLY THE SAME AS WHAT IS ON THE RIGHT."** Both sides are in absolute equilibrium!

One is fully entitled to write:

$$5 = 5$$

$$4 + 1 = 5$$

$$5 = 2 + 3$$

$$3 + 4 = 5 + 2 \qquad \text{(because both sides equal 7!)}$$

🔍 ***True Story — Robert Recorde's Twin Parallel Lines*** *In the year 1557, Welsh physician and mathematician* ***Robert Recorde*** *grew tired of repeatedly writing out* "is equal to" *in his books. He chose to draw two parallel horizontal line segments: $=$. Why twin parallel lines? He wrote:* "No two things can be more equal than a pair of parallel lines of the same length!"

⚠️ Watch Out! — The Run-On Equality Fallacy

Writing $3 + 4 = 7 + 2 = 9$ is **completely false**! Why? Because the first expression evaluates to 7, while the next evaluates to 9. The balance scale collapses: 7 is not equal to 9! Each state of equilibrium must be written independently:

$$3 + 4 = 7 \qquad \text{and then} \qquad 7 + 2 = 9$$

5. Number Bonds to 10: Mental Math Superpowers

Because we have ten fingers, the number **10** is the bedrock fortress of arithmetic. Mastering the pairs of numbers that sum to ten allows you to calculate mentally with extraordinary speed!

These are known as **the friends of 10** (or **number bonds to 10**):

> 💡 **Discovery & History — Friends of 10 and Complementary Pairs**
>
> - 0 + 10 = 10 (and 10 + 0 = 10)
> - 1 + 9 = 10 (and 9 + 1 = 10)
> - 2 + 8 = 10 (and 8 + 2 = 10)
> - 3 + 7 = 10 (and 7 + 3 = 10)
> - 4 + 6 = 10 (and 6 + 4 = 10)
> - 5 + 5 = 10 (the twin hands!)

> 🖐 **Your Turn — The Hanger and Clothespin Workshop**
>
> Take a plastic clothes hanger. Clip exactly **10 clothespins** onto it. Close your eyes. Have a partner slide an index card in between them: for example, leaving 4 pins visible on the left. Without peeking under the card, state how many pins are hidden on the right! It is the partner of 4 to make 10: there must be **6**!

Number Bonds to 100!

The exact same pattern holds when working with tens:

$$10 + 90 = 100 \qquad 20 + 80 = 100 \qquad 30 + 70 = 100 \qquad 50 + 50 = 100$$

---

6. Mental Calculation Strategies

Here are three mental shortcuts used by math practitioners:

1. **Bridging Through 10:** To compute $8 + 5$:
   - How much does 8 need to reach 10? It needs 2 (its partner to 10).
   - Split 5 into $2 + 3$.
   - Compute: $8 + 2 = 10$, leaving 3, so $10 + 3 = \mathbf{13}$!
2. **The $+9$ Strategy (Add 10, then subtract 1):** Adding 9 is almost adding 10—it is just one less! To compute $25 + 9$:
   - Jump forward by 10: $25 + 10 = 35$.
   - Take one small step back: $35 - 1 = \mathbf{34}$!
3. **The -9 Strategy (Subtract 10, then add 1):** To compute $43 - 9$:
   - Subtract an entire ten: $43 - 10 = 33$.
   - Since you subtracted one too many, return it: $33 + 1 = \mathbf{34}$!

---

7. Five Solved Examples: Categorizing Word Problems

In life and mathematics, problems are not bare equations: they are narratives. Here is how to select the right operation:

Solved Example 1 — Combining Collections ("How many in total?")

Problem: Leo has 6 blue marbles in his right pocket and 7 red marbles in his left pocket. How many marbles does he have in total? *Diagnostic Question:* Are we combining two separate collections into one? Yes! Combining = **addition**. *Calculation:*

$$6 + 7 = 6 + (4 + 3) = (6 + 4) + 3 = 10 + 3 = 13$$

Sentence Answer: Leo has **13 marbles** in total.

Solved Example 2 — Separation / Loss ("How many remain?")

Problem: A box contained 12 biscuits. Children ate 4 of them for a snack. How many biscuits remain in the box? *Diagnostic Question:* Did items disappear or get removed? Yes! Removal = **subtraction**. *Calculation:*

$$12 - 4 = 12 - 2 - 2 = 10 - 2 = 8$$

Sentence Answer: There are **8 biscuits** remaining in the box.

Solved Example 3 — Comparison ("How many more than…?")

Problem: Sophie picked 15 daisies. Her brother Hugo picked 9. How many more flowers did Sophie pick than her brother? *Classic Trap!* Many students see the word "more" and add! That is an error! We are not pooling their flowers together; we seek **the gap**, the difference between the two quantities. To find a difference, we use **subtraction**:

$$15 - 9 = 15 - 10 + 1 = 6$$

Sentence Answer: Sophie has **6 more daisies** than her brother.

Solved Example 4 — Missing Part ("How many more needed?")

Problem: A sticker album has 20 pages. Tom has already completed 14 pages. How many pages does he still need to complete his album? *Diagnostic Question:* How much is needed to bridge from 14 to 20? This is a missing-addend equation: $14 + \ldots = 20$. Finding the difference:

$$20 - 14 = 6$$

Sentence Answer: Tom needs **6 more pages** to finish his album.

Solved Example 5 — The Mystery Scale

Problem: On a balanced scale, the left pan holds a 10 kg weight. The right pan holds a 3 kg melon and a sealed mystery box. What is the mass of the mystery box? *Equilibrium equation:*

$$\text{Left Pan} = \text{Right Pan}$$

$$10 = 3 + \text{Box}$$

This is the bond of 3 to make 10!

$$\text{Box} = 10 - 3 = 7 \ \text{kg}$$

Sentence Answer: The mystery box has a mass of **7 kg**.

Key Takeaways

1. **Addition (+)** unites quantities. Number order can be swapped ($4 + 3 = 3 + 4$): addition is commutative.
2. **Subtraction (−)** removes a quantity, measures what remains, or computes a difference. It inverts addition.
3. **The equal sign =** is a two-pan balance scale: both sides must carry identical value ($6 + 2 = 5 + 3$).

4. **Number bonds to 10** must be second nature: $1+9$, $2+8$, $3+7$, $4+6$, $5+5$.
5. To add 9, add 10 and subtract 1 ($24+9=34-1=33$).
6. Choose the operation by visualizing the action: uniting ($+$), taking away ($-$), finding a difference or missing part ($-$).

Practice Exercises

Level (Foundation & Fluency)

Exercise 3.1 Number bonds to 10. Fill in the blanks without counting on fingers:

- $7+...=10$
- $4+...=10$
- $2+...=10$
- $9+...=10$
- $5+...=10$

Exercise 3.2 Is the scale level? State whether these statements are **True** (scale is balanced) or **False** (scale tilts): 1. $5+3=8$ 2. $6+4=7+3$ 3. $10-2=5+2$ 4. $4+4=10-2$

Exercise 3.3 The $+9$ shortcut. Calculate mentally by adding 10 and subtracting 1:

- $15+9=...$
- $34+9=...$
- $62+9=...$

Exercise 3.4 Farmyard chicks. In a barnyard, 8 yellow chicks peck at corn seeds. 5 black chicks run in to join them. How many chicks are in the barnyard now?

Exercise 3.5 Birthday slices. Mom prepared 16 slices of apple tart. By the end of the party, children have eaten 10 slices. How many slices remain on the platter?

Level (Reasoning & Problem Solving)

Exercise 3.6 Selecting the operation. For each scenario, state whether you must use **addition** or **subtraction**: 1. “Lucas has 8 marbles. He wins 4 at recess. How many does he have now?” 2. “Clara has 14 markers. 3 no longer work and she throws them out. How many markers remain?” 3. “Arthur is 12 years old and his cousin Lea is 7. What is the age difference between them?” 4. “Juliette has 8 trading cards. She needs 15 to complete her set. How many cards is she missing?”

Exercise 3.7 Balancing the scale. Find the value of the question mark that establishes equilibrium:

- $8+?=15$
- $?+6=14$
- $13-?=7$
- $20=12+?$

Exercise 3.8 Number bonds to 100. Find the complement to 100:

- $40+...=100$
- $70+...=100$
- $80+...=100$
- $25+...=100$ (Hint: bridge to 30 first!)

Exercise 3.9 Colored pencils. In her pencil case, Jade has 18 colored pencils. Her friend Manon has 11. How many more pencils does Jade have than Manon? Provide the equation and full sentence answer.

Exercise 3.10 Bridging through 10. Decompose the second term to bridge through 10: *Example:* $9+6=9+1+5=10+5=15$ - $8+7=8+...+...=...$ - $7+5=7+...+...=...$ - $6+6=6+...+...=...$

Level (Challenges & Puzzles)

Exercise 3.11 The three-number family. Using only the numbers **2**, **5**, and **7**, write two distinct additions and two distinct subtractions!

Exercise 3.12 The addition brick pyramid. In an addition pyramid, each brick equals the sum of the two bricks directly beneath it:

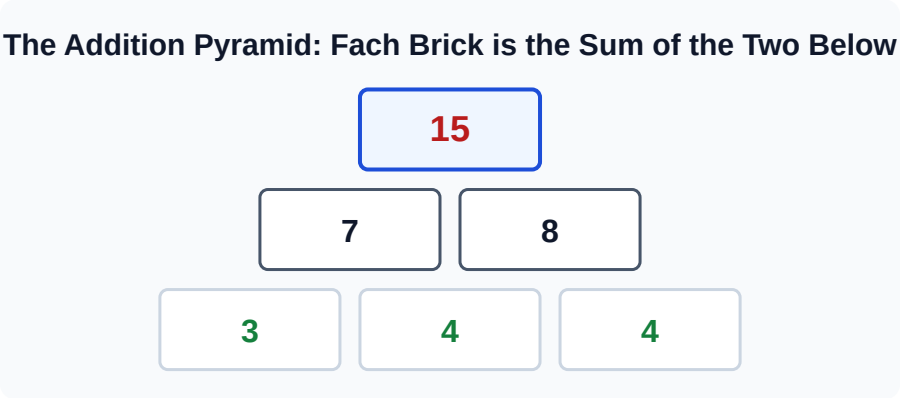


Figure 16: The addition pyramid: sum of underlying bricks

1. Verify the ground row: $3 + 4 = 7$ and $4 + 4 = 8$.
2. What number belongs in the peak brick?

Exercise 3.13 The prudent squirrel. A squirrel stored 17 hazelnuts in a hollow oak on Monday morning.

- On Tuesday, he eats 5.
- On Wednesday, he finds 8 fresh hazelnuts and stores them.
- On Thursday, a blue jay steals 4 hazelnuts.

How many hazelnuts remain in the tree on Thursday evening?

Exercise 3.14 Three-weight balance. On the left pan of a balance scale sit two weights: **6 kg** and **9 kg**. On the right pan sits an **8 kg** weight and a closed mystery box. How much must the box weigh for the beam to remain horizontal?

Exercise 3.15 Age puzzle. When her little brother Noah was born, Sarah was exactly 5 years old. Today, Noah celebrates his 8th birthday. How old is Sarah today? Provide the calculation.

Solutions

Solution 3.1 - $7 + \mathbf{3} = 10$ - $4 + \mathbf{6} = 10$ - $2 + \mathbf{8} = 10$ - $9 + \mathbf{1} = 10$ - $5 + \mathbf{5} = 10$

Solution 3.2 1. $5 + 3 = 8$: **True** ($8 = 8$). 2. $6 + 4 = 7 + 3$: **True** (both sides equal 10). 3. $10 - 2 = 5 + 2$: **False** ($8 \neq 7$, left side tilts downward). 4. $4 + 4 = 10 - 2$: **True** ($8 = 8$).

Solution 3.3 - $15 + 9 = 15 + 10 - 1 = 25 - 1 = \mathbf{24}$. - $34 + 9 = 34 + 10 - 1 = 44 - 1 = \mathbf{43}$. - $62 + 9 = 62 + 10 - 1 = 72 - 1 = \mathbf{71}$.

Solution 3.4 Uniting the two groups of chicks:

$$8 + 5 = 13$$

There are now **13 chicks** in the barnyard.

Solution 3.5 Removing the eaten portions:

$$16 - 10 = 6$$

There are **6 slices** remaining on the platter.

Solution 3.6 1. Lucas wins marbles: **addition** ($8 + 4 = 12$). 2. Clara discards broken markers: **subtraction** ($14 - 3 = 11$). 3. Finding an age difference: **subtraction** ($12 - 7 = 5$). 4. Finding a missing quantity to complete a set: **subtraction** ($15 - 8 = 7$).

Solution 3.7 - $8 + \mathbf{7} = 15$ - $\mathbf{8} + 6 = 14$ - $13 - \mathbf{6} = 7$ - $20 = 12 + \mathbf{8}$

Solution 3.8 - $40 + \mathbf{60} = 100$ - $70 + \mathbf{30} = 100$ - $80 + \mathbf{20} = 100$ - For 25: from 25 to 30 is 5; from 30 to 100 is 70. Thus $25 + \mathbf{75} = 100$.

Solution 3.9 We compute the difference between 18 and 11:

$$18 - 11 = 7$$

Jade has **7 more colored pencils** than Manon.

Solution 3.10 Step-by-step decomposition to bridge ten:

- $8 + 7 = 8 + \mathbf{2} + \mathbf{5} = 10 + 5 = \mathbf{15}$
- $7 + 5 = 7 + \mathbf{3} + \mathbf{2} = 10 + 2 = \mathbf{12}$
- $6 + 6 = 6 + \mathbf{4} + \mathbf{2} = 10 + 2 = \mathbf{12}$

Solution 3.11 With 2, 5, and 7:

- Additions: $2 + 5 = 7$ and $5 + 2 = 7$.
- Subtractions: $7 - 2 = 5$ and $7 - 5 = 2$.

They form a fact family!

Solution 3.12 The top brick is the sum of the two supporting bricks:

$$7 + 8 = 15$$

We must write **15** in the peak brick.

Solution 3.13 Tracking the hazelnuts step by step:

- Monday: 17 hazelnuts.
- Tuesday: $17 - 5 = 12$ hazelnuts.
- Wednesday: $12 + 8 = 20$ hazelnuts.
- Thursday: $20 - 4 = 16$ hazelnuts.

There are **16 hazelnuts** left in the tree on Thursday evening.

Solution 3.14 - Mass on left pan: $6 + 9 = 15$ kg. - The right pan must also equal 15 kg for balance. - We already have an 8 kg weight. - Mass of the box: $15 - 8 = \mathbf{7}$ **kg**.

Solution 3.15 Sarah is always 5 years older than her brother, as time flows at the exact same pace for both! Since Noah is 8:

$$8 + 5 = 13$$

Sarah is **13 years old** today.

The Question That Opens the Journey Ahead…

You are now able to maneuver with addition, subtraction, and balance scale equations for small numbers.

Yet imagine that the clan leader asks you to sum the agricultural yields of the entire valley: **47 bags of wheat** gathered in the northern plain, and **38 bags of wheat** harvested in the southern plain!

Doing this in your head or on ten fingers quickly becomes precarious: what should you do when loose units overflow past 10? How can one record the operation neatly in vertical columns on a slate without making errors?

Ancient merchants invented a wooden sliding frame, and a secret traveler known as **the carry**…

See you in **Chapter 4: The Merchant's Abacus and the Traveling Carry**!

CHAPTER 4: The Merchant's Abacus and the Traveling Carry

> "When numbers grow too large to hold in the palm of your hand, we build them a house with stories and columns. And when a room overflows, a guest climbs the secret staircase to knock at the neighboring door: this is the journey of the carry."

What You Are About to Discover

- The secret of the **abacus** and the partitioned shoe box (the Ones column and the Tens column).
- Setting up and executing **column addition without regrouping (no carry)**.
- The mystery of **addition with carrying**: when $7 + 8 = 15$, the ones compartment overflows! Why and how 1 brand-new ten travels to the top.
- Setting up and executing **column subtraction with borrowing (decomposing a ten)** by opening a sealed bundle of 10.
- The detective's superpower: **verifying calculations using the inverse operation** to be 100% certain of eliminating all mistakes.

---

1. The Problem: The Merchant's Heavy Cargo

We stand on the river wharves. A sturdy wooden river barge has just docked, laden with sacks of grain.

The dock foreman holds a clay tablet. He must tally the shipments from two separate agricultural estates:

- The Sun Estate brings **47 sacks of wheat**.
- The Moon Estate brings **38 sacks of barley**.

The city governor demands the grand total before nightfall. The foreman begins carving small notches on his slate: $1, 2, 3, 4\ldots$ It is far too slow! His vision blurs, the lines cross, and dockworkers shout with mounting impatience: *"How many sacks in total? We want to unload the ship!"*

To calculate without error, even with large quantities, ancient merchants did not line up loose stones on the dirt. They utilized an ingenious device: **the counting abacus**.

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2. The Calculating Machine: The Abacus and the Two-Column Box

Let us construct the earliest precursor to the computer! You only need a shoe box and a cardboard partition dividing it into two compartments:

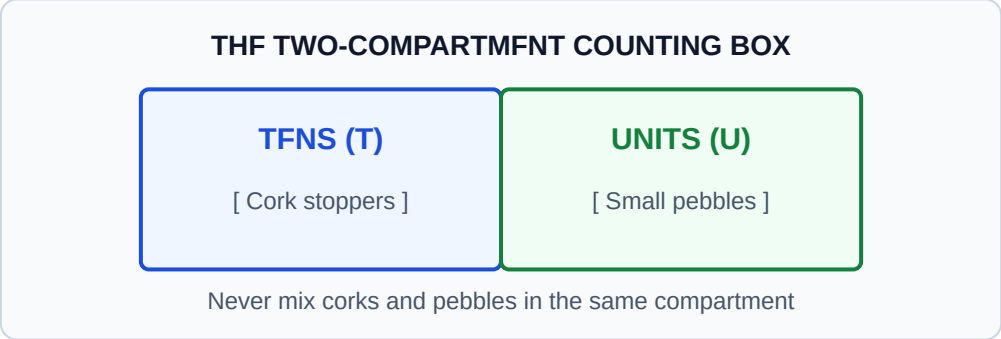

Figure 17: The merchant's abacus: units and tens compartments

Recall the foundational rule discovered in Chapter 2:

- In the **Ones (o)** compartment, we only place small single pebbles (1 pebble = 1 sack).
- In the **Tens (t)** compartment, we only place cork stoppers (1 stopper = 10 sacks).
- **Absolute Rule:** The moment **10 pebbles** accumulate in the ones compartment, the bin is completely full! We must immediately exchange those 10 pebbles for **1 cork stopper**, which shifts across into the tens compartment.

On grid paper, we draw this exact structure in vertical columns: this is known as **setting up a standard vertical algorithm**.

---

3. Vertical Addition Without Regrouping: Perfect Alignment

Let us begin with a gentle addition:

$$23 + 45$$

To set up this calculation on grid paper, follow the golden rule of alignment:

> **📐 Alignment Rule — Everyone in Their Column!**
>
> Write the numbers one directly above the other, aligning vertically:
>
> - **Ones under ones** (the right-hand column).
> - **Tens under tens** (the left-hand column).
> - Draw a straight horizontal line beneath: this line represents the equal sign $=$.
> - **ALWAYS begin by adding the ones column first** (from right to left)!

```
    t   o
    2   3      (2 tens and 3 ones)
+   4   5      (4 tens and 5 ones)
----------
```

1. **Step 1: Add the ones.** In the right-hand column: $3 + 5 = 8$. Write **8** below the line, in the ones column.
2. **Step 2: Add the tens.** In the left-hand column: $2 + 4 = 6$. Write **6** below the line, in the tens column.

```
    t   o
    2   3
+   4   5
----------
    6   8      The sum is 68!
```

Everything remained orderly: neither column reached 10, no pebbles overflowed.

---

4. Addition with Regrouping: The Traveling Carry

Now let us return to the barge cargo:

$$47 + 38$$

Set up the columns:

```
    t   o
    4   7
+   3   8
----------
```

1. **Inspect the ones column:** We unite 7 pebbles and 8 pebbles.

$$7 + 8 = 15$$

Alert! 15 is a two-digit number! We cannot write 15 in the ones box, because each column can hold only a single digit from 0 to 9.

2. **The magic trade:** What do 15 pebbles represent?

$$15 \text{ ones} = 1 \text{ ten} + 5 \text{ ones}$$

- The **5 ones** remain seated at the bottom of the column, below the line.
- The **1 brand-new ten** cannot stay in the ones bin: it packs its bag and **travels up to the top of the tens column**!

This traveling ten is called **the carry** (or regrouped ten). We write it as a small superscript numeral circled at the top of the tens column:

```
    (1) <----- The traveling carry!
     t   o
     4   7
+    3   8
----------
     5      (7 + 8 = 15: write 5, carry 1)
```

3. **Compute the tens column:** Do not forget to include the traveling newcomer!

$$1 \text{ (carry)} + 4 + 3 = 8 \text{ tens}$$

```
    (1)
     t   o
     4   7
+    3   8
----------
     8   5      The total is 85 sacks!
```

📐 Definition — The Carry (Regrouping)

In a vertical algorithm, a **carry** is a ten (or hundred) produced by an overflow of the preceding column, which travels to join the top of the next larger place-value column.

⚠️ Watch Out! — Forgetting the Carry

The most frequent student mistake is forgetting to add the carry when summing tens! To guarantee success: 1. Record the carry immediately the moment you write down the ones digit. 2. Form the habit of starting the tens addition **with the carry**: "1 + 4 = 5, and 5 + 3 = 8."

> 🔍 ***True Story — Blaise Pascal's Calculator (1642)*** *To assist his father, a royal tax collector who spent long nights adding endless ledgers of figures, 19-year-old French scientist Blaise Pascal invented the first mechanical calculator:* ***the Pascaline****. Inside, an ingenious sautoir gear mechanism automatically carried over a 1 to the neighboring wheel whenever a dial turned past 9!*

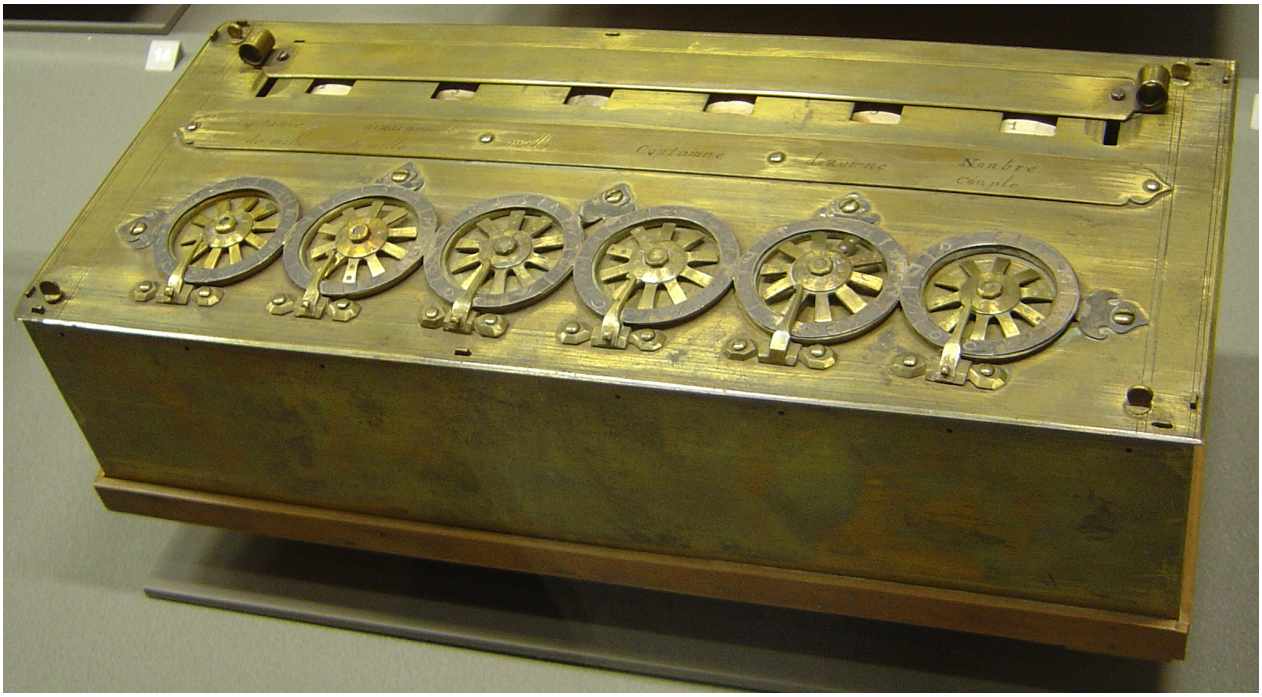

Figure 18: Blaise Pascal's arithmetic machine (Pascaline, 1642): the first mechanical automatic decimal carry mechanism

---

5. Vertical Subtraction: Unpacking a Ten

Now consider the reverse narrative: The barge initially contained **85 sacks**. Merchants sell **38 sacks** on the wharf. How many sacks remain on board?

$$85 - 38$$

Set up the subtraction vertically:

```
    t   o
    8   5

-   3   8
----------
```

As always, start with the ones column on the far right: We have 5 loose pebbles, and we must remove 8:

$$5 - 8 = \ ??$$

That is impossible! If you have 5 berries in your pocket, you cannot hand 8 to your friends!

What does the merchant do? He does not panic. He looks to the tens column, which holds 8 sealed bundles of 10 sacks.

The Decomposing Method (Borrowing / Unpacking):

1. The merchant takes **1 bundle of ten** from the tens store. There were 8 tens: now only **7 tens** remain! (Cross out the 8 and write 7 above).
2. He unties this bundle: it releases **10 fresh ones**.
3. He adds these 10 ones to the 5 already sitting in the ones column:

   $$10 + 5 = 15 \ \text{ ones!}$$

   (Place a small 1 beside the 5 to form 15).

```
      (7)  (15)
       t     o
      \8\   \5\

  -    3     8
  -------------
```

4. Now subtraction proceeds effortlessly!
   - In the ones column: $15 - 8 = \mathbf{7}$.
   - In the tens column: $7 - 3 = \mathbf{4}$.

```
      (7)  (15)
       t     o
      \8\   \5\

  -    3     8
  -------------
       4     7      47 sacks remain!
```

> 💡 ***Physical Action — Unpacking a Ten*** *Take 8 matchboxes filled with 10 matches each (80) and 5 loose matches (5). You must remove 8 matches. You cannot draw 8 from the 5 loose ones. You take* ***one closed matchbox****, open it, and dump its 10 matches onto the table with the 5 loose ones: you now have 15 loose matches and 7 closed boxes. You can remove 8 matches with complete ease!*

---

6. The Detective's Secret: Verifying with the Inverse Operation

How does a skilled mathematician confirm that an answer is accurate without waiting for an instructor's red pen? By performing **the inverse operation proof**!

- You calculated the subtraction: $85 - 38 = 47$.
- To verify, execute the reverse: take the 47 remaining sacks and add back the 38 sacks sold ($47 + 38$).
- If your work was flawless, you must recover **the exact 85 initial sacks**!

```
Executed subtraction:      Verification addition:
    8 5                        4 7

  - 3 8                      + 3 8
  -----                      -----
    4 7                        8 5   <--- Exactly 85 recovered: verified!
```

If the result does not yield 85, a calculation error occurred: you can track down the slip before anyone else spots it!

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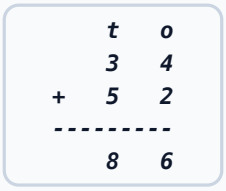
7. Five Step-by-Step Solved Examples

Solved Example 1 — Vertical Addition Without Carrying

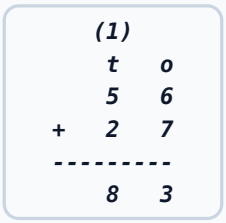
Problem: Calculate $34 + 52$. *Steps:*



- *Ones:* $4 + 2 = 6$.
- *Tens:* $3 + 5 = 8$. Answer: $34 + 52 = 86$.

Solved Example 2 — Vertical Addition with a Carry

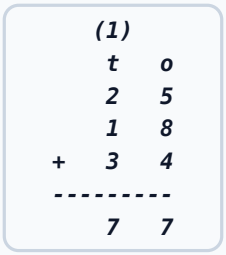
Problem: Calculate $56 + 27$. *Steps:*



- *Ones:* $6 + 7 = 13$. *Write 3 in ones; carry 1 to tens.*
- *Tens:* $1 \ \ (\text{carry}) + 5 + 2 = 8$. Answer: $56 + 27 = 83$.

Solved Example 3 — Three-Story Addition

Problem: Calculate $25 + 18 + 34$. *Steps:*



- *Ones:* $5 + 8 = 13$, *and* $13 + 4 = 17$. *Write 7, carry 1.*
- *Tens:* $1 \ \ (\text{carry}) + 2 + 1 + 3 = 7$. Answer: $25 + 18 + 34 = 77$.

Solved Example 4 — Vertical Subtraction Without Borrowing

Problem: Calculate $69 - 25$. *Steps:*

```
    t   o
    6   9

-   2   5
---------
    4   4
```

- *Ones:* $9 - 5 = 4$.
- *Tens:* $6 - 2 = 4$. Answer: $69 - 25 = 44$.

Solved Example 5 — Vertical Subtraction with Decomposing

Problem: Calculate $72 - 36$. *Steps:*

- Ones: $2 - 6$ is impossible!
- Borrow 1 ten from 7: 6 tens remain.
- Transfer 10 ones to 2: it becomes 12.

```
   (6)  (12)
    t    o
   \7\  \2\

-   3    6
-----------
    3    6
```

Fundamental Principle & Definition

- Ones: $12 - 6 = 6$.
- Tens: $6 - 3 = 3$. *Check by addition:* $36 + 36 = 72$. Flawless!

Key Takeaways

1. To execute vertical operations, align **ones under ones** and **tens under tens**. Compute from right to left.
2. **In addition**, if the sum of ones reaches 10 or greater, write down the ones digit and send the ten as a **carry** to the top of the tens column.
3. Always include the carry when summing the tens column.
4. **In subtraction**, if the top digit is smaller than the bottom digit, **decompose a ten**: the tens column loses 1, and the ones column gains 10.
5. Verify subtractions by adding the difference back to the subtracted amount.

Practice Exercises

Level (Foundation & Fluency)

Exercise 4.1 Compute without regrouping: - $21 + 35$ - $42 + 16$ - $63 + 24$

Exercise 4.2 Compute with regrouping: - $38 + 25$ - $47 + 36$ - $59 + 14$

Exercise 4.3 Compute subtractions without borrowing: - $57 - 23$ - $88 - 41$ - $76 - 35$

Exercise 4.4 The forgotten carry. Examine this calculation:

```
  4 8
+ 2 5
-----
  6 3
```

What error was committed? What is the correct answer?

Exercise 4.5 Fuel tank. A truck fuel tank held 24 liters. The driver adds 38 liters at the pump. How many liters are in the tank now?

Level (Reasoning & Problem Solving)

Exercise 4.6 Compute subtractions requiring decomposing a ten: - $53 - 28$ - $61 - 37$ - $84 - 49$

Exercise 4.7 The detective's verification. A classmate claims that $75 - 39 = 46$. Perform the inverse addition ($46 + 39$) to verify whether they are correct or mistaken.

Exercise 4.8 Crossing 100! Compute additions whose totals exceed 100:

- $65 + 47$
- $78 + 56$
- $89 + 35$

Exercise 4.9 Sheep from the fold. In an enclosure, there are 92 sheep. At dusk, the shepherd leads out 35 to drink at the river. How many sheep remain inside?

Exercise 4.10 Three-term addition. For school decorations, students inflated 28 red balloons, 34 yellow balloons, and 19 blue balloons. How many balloons did they inflate in total?

Level (Challenges & Puzzles)

Exercise 4.11 Mystery digits. Identify the digits hidden under the stars ($\star$):

```
    3  ?
+   ?  4
--------
    8  1
```

Exercise 4.12 Hidden subtraction digit. Identify the digit under the block ($\blacksquare$):

```
    6  4

-   2  ?
--------
    3  7
```

Exercise 4.13 Subtraction with zero in ones. Calculate:

$$80 - 36$$

Explain how you borrow a ten when the ones digit is zero.

Exercise 4.14 The bus route. A bus departs the terminal carrying 38 passengers.

- At stop 1, 15 board and 7 depart.
- At stop 2, 12 board and 9 depart.

How many passengers remain on the bus after stop 2?

Exercise 4.15 Giant abacus challenge. By choosing the digits of each number from **2**, **5**, and **8** (without repeating the same digit within a single number), form two two-digit numbers to obtain the largest possible sum. Compute that sum.

Solutions

Solution 4.1 - $21 + 35 = \mathbf{56}$. - $42 + 16 = \mathbf{58}$. - $63 + 24 = \mathbf{87}$.

Solution 4.2 - $38 + 25 = \mathbf{63}$ (carry 1). - $47 + 36 = \mathbf{83}$ (carry 1). - $59 + 14 = \mathbf{73}$ (carry 1).

Solution 4.3 - $57 - 23 = \mathbf{34}$. - $88 - 41 = \mathbf{47}$. - $76 - 35 = \mathbf{41}$.

Solution 4.4 The student calculated $8 + 5 = 13$, wrote down 3, but **forgot to add the carried 1** in the tens column! They computed $4 + 2 = 6$ instead of $1 + 4 + 2 = 7$. The correct sum is **73**.

Solution 4.5 $24 + 38 = \mathbf{62 \ \ liters}$.

Solution 4.6 - $53 - 28 = \mathbf{25}$ (borrow 1 ten: $13 - 8 = 5$, $4 - 2 = 2$). - $61 - 37 = \mathbf{24}$ (borrow 1 ten: $11 - 7 = 4$, $5 - 3 = 2$). - $84 - 49 = \mathbf{35}$ (borrow 1 ten: $14 - 9 = 5$, $7 - 4 = 3$).

Solution 4.7 $46 + 39 = 85 \neq 75$. The classmate is mistaken by one ten! The true difference is **36** ($36 + 39 = 75$).

Solution 4.8 - $65 + 47 = \mathbf{112}$. - $78 + 56 = \mathbf{134}$. - $89 + 35 = \mathbf{124}$.

Solution 4.9 $92 - 35 = \mathbf{57 \ \ sheep}$.

Solution 4.10 $28 + 34 + 19 = \mathbf{81 \ \ balloons}$ (carry 2).

Solution 4.11 - Ones: $\star + 4 = 11 \Longrightarrow \star = \mathbf{7}$. - Tens: $1 + 3 + \star = 8 \Longrightarrow \star = \mathbf{4}$.

The addition was $37 + 44 = 81$.

Solution 4.12 Borrowing 1 ten turns 4 into 14: $14 - \blacksquare = 7 \Longrightarrow \blacksquare = \mathbf{7}$. Verification: $5 - 2 = 3$. Flawless!

Solution 4.13 In $80 - 36$, 0 cannot yield 6. Borrow 1 ten from 8 (leaving 7). The 10 ones join 0 to make 10. $10 - 6 = 4$, and $7 - 3 = 4$. Result is **44**.

Solution 4.14 - Departure: 38. - Stop 1: $38 + 15 = 53$; $53 - 7 = 46$. - Stop 2: $46 + 12 = 58$; $58 - 9 = 49$. **49 passengers** remain.

Solution 4.15 To maximize the sum of two two-digit numbers, place the highest possible digit in the tens position for both numbers, which is **8**. For the units positions, assign the remaining largest available digits, **5** and **2**. The two numbers are **85** and **82**, giving the maximal sum: $85 + 82 = \mathbf{167}$. He holds **85** and **82**.

The Question That Opens the Journey Ahead…

You now master vertical addition and subtraction, carry overs, and borrowing.

Yet imagine a royal master mason constructing a massive brick rampart. The wall spans **6 bricks in length** and stands **4 bricks in height**. To calculate the total brick count, he could compute: $6 + 6 + 6 + 6$. But what if the rampart is 100 bricks long by 8 bricks tall? Repeated addition becomes an exhausting slog!

How did ancient builders devise a brilliant shortcut turning repeated additions into a dance of rectangular geometry?

See you in **Chapter 5: The Brick Wall and Magic Rectangles**!

CHAPTER 5: The Brick Wall and Magic Rectangles

> "Why count ten times what the eye can see in a single block? Arranging objects in regular rows gives our mind the superpower of multiplying space and time."

What You Are About to Discover

- Why counting 24 bricks one by one takes too long, and how **multiplication (×)** saves precious time.
- **Sensible commutativity**: why rotating a chocolate bar by a quarter turn (90°) changes neither its taste nor its number of squares ($4 \times 6 = 6 \times 4$).
- How to learn multiplication tables without tears thanks to geometric patterns and finger tricks (the magical nine-times table).
- **Doubles** and **halves**.
- Multiplying by 10 and 100 by sliding digits one place to the left.
- Your first **column multiplication** of a two-digit number by a single digit.

---

1. The Problem: The Storm Approaches the Brickyard

Centuries have rolled by. Humans no longer sleep in caves: they have built sturdy villages along riverbanks.

To construct a great granary capable of feeding the village throughout the winter, the masons molded clay bricks and set them out in the sun to dry.

Suddenly, a cold wind whips up. The blue sky turns dark gray, then ink-black: **a ferocious summer storm is bearing down!** The master mason waves his arms and shouts: *"We must cover the bricks with heavy tarpaulins before the downpour drenches them and turns them into mush! How many bricks are there on the drying yard? Give me the exact count, quickly!"*

You run toward the site. The bricks have been neatly aligned on the ground: There are **4 long rows**, and in each row, there are **6 bricks**.

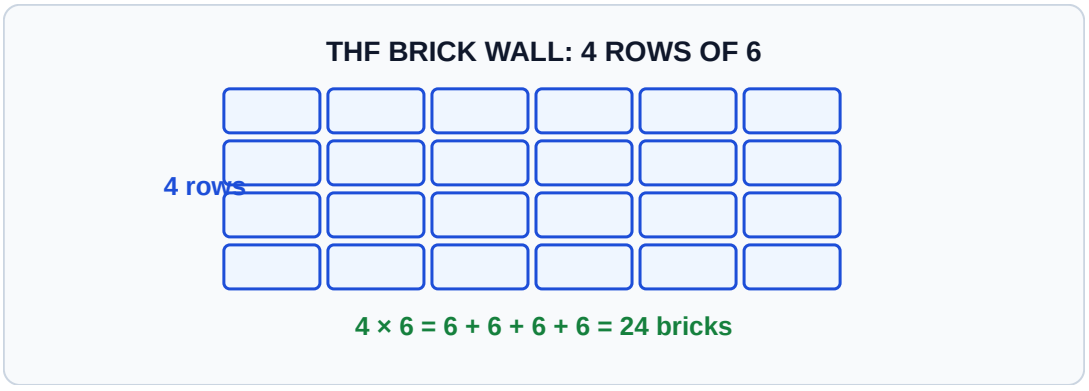


Figure 19: The brick wall: 4 rows of 6 make 24

If you start pointing your finger at every brick one by one: *"1, 2, 3, 4, 5, 6, 7…"* By the twelfth brick, the first flashes of lightning rip through the sky. By the fifteenth, the first heavy raindrops splatter in the dust. You will never finish counting in time before everything is ruined!

You could perform repeated addition:

$$6 + 6 + 6 + 6$$

$6 + 6 = 12$, then $12 + 6 = 18$, then $18 + 6 = 24$. That is already much faster. But ancient builders invented an even more lightning-fast shortcut: **Multiplication**.

---

2. The Giant Leap: Multiplication

Look at the shape traced by the bricks on the ground: this is no scattered mess, it is a magnificent **Rectangle**!

- Vertically (height): **4 rows**.
- Horizontally (width): **6 columns**.

Instead of repeating the addition "6 plus 6 plus 6 plus 6," we say: *"I have 4 rows of 6 bricks."* And we write this with a little cross: **the sign** $\times$ (times or multiplied by).

$$4 \times 6 = 24$$

This little $\times$ sign marks the crossing of two directions: the horizontal line meets the vertical column to weave a grid. **To multiply is to forge a rectangular surface in a flash!**

Definition — Multiplication

Multiplication is an operation that replaces repeated addition of the same number. The symbol $\times$ is read "**times**" or "**multiplied by**".

$$4 \times 6 = \underbrace{6 + 6 + 6 + 6}_{4 \text{ times the number } 6} = 24$$

The numbers being multiplied are called **factors**, and the result is called the **product**.

---

3. The Chocolate Bar: Commutativity at 90 Degrees

Take a chocolate bar in your kitchen. It has 4 rows of 6 squares. Lay it flat before you horizontally: you see **4 rows of 6 squares** (4×6).

Now, take the bar and rotate it by a quarter turn on the table (a right angle of $90°$).

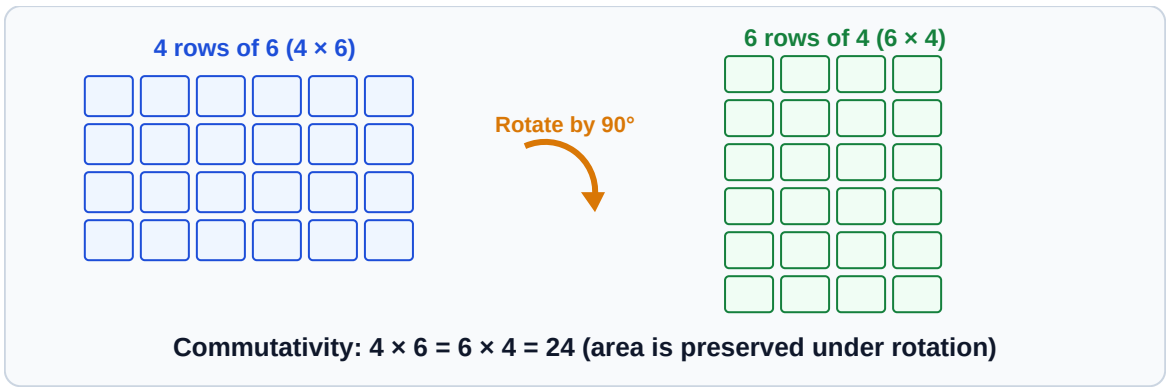


Figure 20: Commutativity by rotation: 4 × 6 = 6 × 4

What do you see now? You see **6 rows of 4 squares** (6×4).

Did a single square of chocolate melt away? No! Did anyone bite a piece off during the turn? No! The number of squares before your eyes is strictly the same:

$$4 \times 6 = 6 \times 4 = 24$$

📐 Property — Commutativity of Multiplication

We can **change the order of the factors** without changing the result. Turning a rectangle on its side never changes the number of objects it contains! *A wonderful consequence:* When learning your multiplication tables, you only have **half the work to do**! Once you know that $3 \times 8 = 24$, you already automatically know that $8 \times 3 = 24$.

---

4. Doubles and Halves

Before exploring all the tables, there are two words you hear every day: **double** and **half**.

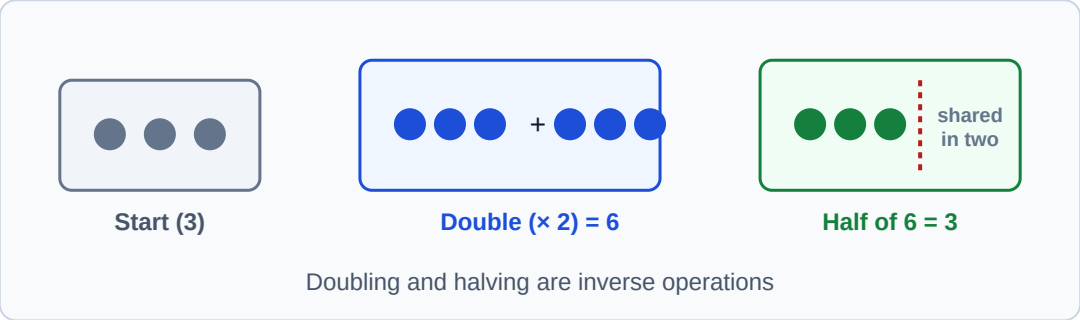


Figure 21: Double and half: bilateral multiplicative symmetry

- **The double** is taking an amount **twice** ($n \times 2$ or $n + n$).
  - The double of 4 is $4 + 4 = 8$.
  - The double of 10 is $10 + 10 = 20$.
- **The half** is sharing fairly into two strictly equal shares.
  - Half of 8 is 4.
  - Half of 20 is 10.

---

5. A Journey Through the Multiplication Tables

Memorizing tables by rote like a robot can feel dull. But when you look at their hidden patterns, each table has its own unique personality!

1. The 2-Times Table: Even Numbers

Multiplying by 2 simply means taking giant leaps of two:

$$2, \quad 4, \quad 6, \quad 8, \quad 10, \quad 12, \quad 14, \quad 16, \quad 18, \quad 20$$

All results are **even numbers**: they always end in 0, 2, 4, 6, or 8.

2. The 5-Times Table: The Dance of the Hand

Every hand has 5 fingers. So two hands make 10, three hands make 15, four hands make 20… The results of the 5-times table **always alternate ending in 5 or 0**:

$$5, \quad 10, \quad 15, \quad 20, \quad 25, \quad 30, \quad 35, \quad 40, \quad 45, \quad 50$$

3. The 10-Times Table: The Awakening of Zero

Multiplying by 10 turns each unit into a ten! The digit shifts one column to the left, and a 0 is placed in the units column:

$$3 \times 10 = 30 \qquad 7 \times 10 = 70 \qquad 12 \times 10 = 120$$

4. The Magical 9-Times Table: The Ten-Finger Trick!

Lay your ten fingers flat on the table before you. Your fingers are numbered from 1 to 10 from left to right (left thumb is 1, right little finger is 10).

> 🖐 ***The Finger Magic Trick for the 9-Times Table*** *Want to calculate* 9×4*? 1. Fold down your 4th finger (the ring finger of your left hand). 2. Look at your fingers:*
>
> - *To the left of the folded finger, there are* ***3 upright fingers****: these are the tens (****30****).*
> - *To the right of the folded finger, there are* ***6 upright fingers****: these are the units (****6****).*
>
> 3. *Put them together:* ***36****! And there it is:* $9 \times 4 = 36$*!* (Try it with 9×7: fold down the 7th finger. You have 6 fingers on the left and 3 on the right: 63! It never fails!)

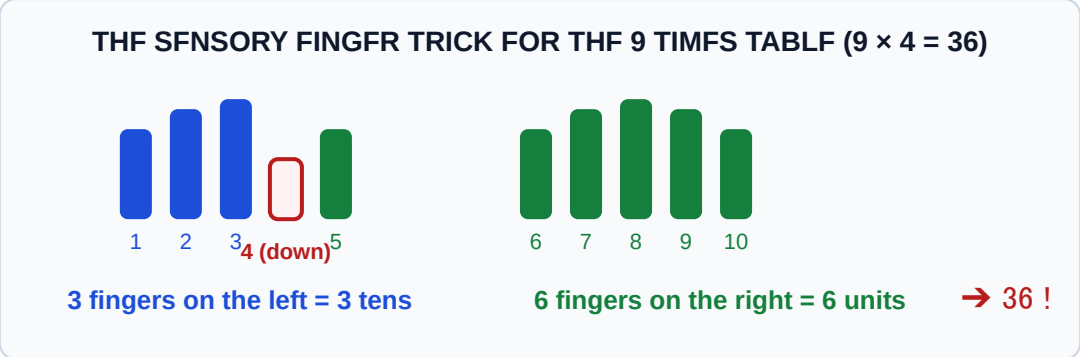


Figure 22: The finger trick for the 9 times table

---

6. Multiplying by 10, 100, and 1,000

What happens when we multiply a number by 10 or 100? Recall our place-value chart:

Centaines	Dizaines	Unités	Valeur décimale
		4	4
	4	0	$4 \times 10 = 40$
4	0	0	$4 \times 100 = 400$

- Multiplying by **10**: each unit becomes a ten. The number shifts **one place to the left**, and we append a **0** on the right.
- Multiplying by **100**: each unit becomes a hundred. The number shifts **two places to the left**, and we append **two zeros** (**00**) on the right.

$$7 \times 100 = 700 \qquad 25 \times 100 = 2,500$$

---

7. Column Multiplication: A Two-Digit Number by a Single Digit

Imagine you need to prepare packages for 3 classrooms. Each class needs **24 notebooks**. How many notebooks must you hand out?

$$24 \times 3$$

Let us set up the calculation in columns, just as with addition:

```
    t   u
    2   4
×       3
---------
```

1. **Step 1: The 3 multiplies the units.**

$$3 \times 4 \text{ units} = 12 \text{ units}$$

In 12, there are 2 units and 1 ten:

- Write the **2** under the line in the units column.

- Keep the **1 ten as a carry** in your mind or in a small circle atop the tens column.

```
    (1) <--- carry
     t   u
     2   4
×        3
---------
         2
```

2. **Step 2: The 3 multiplies the tens.**

   $3 \times 2 \text{ tens} = 6 \text{ tens}$

   Do not forget to add the carried ten:

   $6 + 1 = 7 \text{ tens}$

   Write the **7** in the tens column.

```
    (1)
     t   u
     2   4
×        3
---------
     7   2      The total is 72 notebooks!
```

> ⚠️ **Watch Out for the Trap! — Multiply first, add the carry after!**
>
> In the tens column, **never** perform addition before multiplication! If you do $(1 + 2) \times 3 = 3 \times 3 = 9$, that is false! The sacred rule: **multiply first ($3 \times 2 = 6$), and add the carry after ($6 + 1 = 7$)**.

7bis. The Great Shortcut: Long Multiplication with Two-Digit Multipliers

Once you master multiplying by a single digit, merchants invented an ingenious shortcut for multiplying by a **two-digit number** like 146×23.

This calculation is done in **two distinct tiers**:

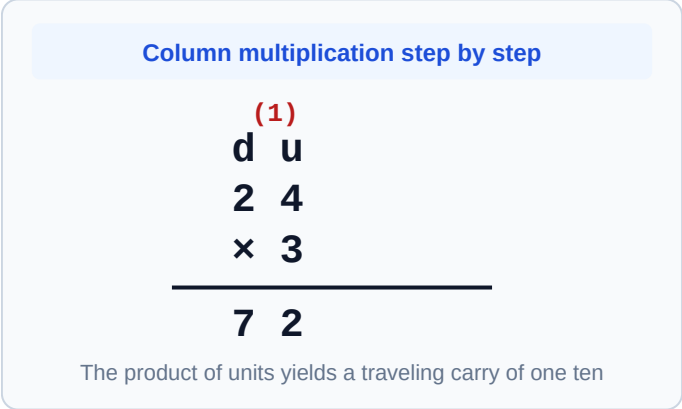


Figure 23: Two-stage column multiplication with regrouping (carry)

```
     1   4   6
×        2   3
-------------
     4   3   8      <-- Tier 1: 146 × 3 units
+2   9   2   0      <-- Tier 2: 146 × 2 tens (THE MANDATORY RED ZERO!)
-------------
 3   3   5   8      <-- Final sum of both tiers
```

> 🧭 **The Mechanical Algorithm in 4 Simple Gestures:**
>
> 1. Multiply
>
> the whole top number by the **units** digit of the bottom number ($146 \times 3 = 438$). 2. On the line below, place a **red zero** in the units column immediately (because we are now multiplying by tens!). 3. Multiply the whole top number by the **tens** digit of the bottom number ($146 \times 2 = 292$). 4. Draw a line and **add the two tiers**.

🛠️ The Boundary Case Laboratory: What Happens with ZERO?

(This section is isolated so you can explore it peacefully without confusing your initial practice.)

Once you understand that $4 \times 3 = 12$ bricks, let us ask two extreme questions at the borders of arithmetic:

1. How much is 4×0?

- **The thought experiment:** Imagine the foreman prepares a space 4 bricks wide, but decides to lay **zero rows** of bricks (he never even starts). How many bricks are on the ground? **None.**
- **The descending sequence:**
  - $4 \times 3 = 12$
  - $4 \times 2 = 8$ (subtract 4)
  - $4 \times 1 = 4$ (subtract 4)
  - $4 \times 0 = 0$ (subtract 4 again!)
- **The universal rule:** Any number multiplied by zero always yields **zero**:

$$a \times 0 = 0 \quad \text{and} \quad 0 \times a = 0$$

  In mathematics, we say that **zero is an absorbing element**: it swallows everything it touches!

2. And what about 0×0?

- You have prepared zero width (0), and you build zero rows (0).
- There is neither brick, nor wall, nor lot: the result is simply **0**!

⚡ The Practitioner's Scales: Mechanical Drills (Timed)

To master calculation, practice your fingers just as a pianist plays scales. Answer as quickly as possible without recounting one by one:

Challenge	Calculation	Your Answer	Self-Check
1	7×8	…….	
2	6×9	…….	
3	9×0	…….	
4	4×7	…….	
5	8×8	…….	
6	0×0	…….	
7	52×10	…….	
8	23×3 column	…….	
9	15×4 column	…….	
10	12×11	…….	

---

8. Five Step-by-Step Worked Examples

Worked Example 1 — Translating a Situation into Multiplication

Problem: A florist prepares 5 bouquets. In each bouquet, she puts 6 roses. How many roses does she use in total? *Method:* This is repeated addition: $6 + 6 + 6 + 6 + 6$. We write the multiplication: 5×6. From the 5-times table (or 6-times table): $5 \times 6 = 30$. *Answer:* The florist uses **30 roses**.

Worked Example 2 — Finding Doubles and Halves

Problem: Leo has 14 marbles. His older brother has twice as many (double), and his little sister has half as many. How many marbles do the brother and sister have? *Solution:*

- The brother has double: $14 \times 2 = 14 + 14 = \mathbf{28}$ **marbles**.
- The sister has half: split 14 into two equal parts ($7 + 7 = 14$). The sister has **7 marbles**.

Worked Example 3 — Multiplying by 10 and 100 Mentally

Problem: Calculate mentally: 8×10, 8×100, and 43×10. *Solution:*

- $8 \times 10 = \mathbf{80}$ (one zero appended).
- $8 \times 100 = \mathbf{800}$ (two zeros appended).
- $43 \times 10 = \mathbf{430}$ (shift 43 one place left, append 0).

Worked Example 4 — Column Multiplication without Regrouping

Problem: Set up and compute 32×3. *Step by step:*

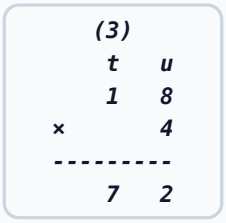
```
    t   u
    3   2
×       3
---------
    9   6
```

- *Units:* $3 \times 2 = 6$.
- *Tens:* $3 \times 3 = 9$. Result: $32 \times 3 = 96$.

Worked Example 5 — Column Multiplication with Regrouping (Carry)

Problem: Set up and compute 18×4. *Step by step:*

- Units: $4 \times 8 = 32$. Write down 2 and carry 3.
- Tens: $4 \times 1 = 4$, plus carry 3: $4 + 3 = 7$.



Result: $18 \times 4 = 72$.

Key Takeaways

1. **Multiplication ($\times$)** replaces repeated addition of the same quantity ($4 \times 6 = 6 + 6 + 6 + 6$).
2. It is **commutative**: you can reverse the order of factors ($4 \times 6 = 6 \times 4 = 24$). It is just like turning a rectangle or chocolate bar by a quarter turn!
3. **Double** means multiplying by 2; **half** means dividing into two equal shares.
4. To multiply by **10**, shift one column left (append one zero); to multiply by **100**, shift two columns left (append two zeros).
5. For long multiplication: multiply the units first, record any carry, then multiply the tens and add the carry at the end.

Exercises

Level (Practice and Apply)

Exercise 5.1 From Addition to Multiplication. Rewrite each addition as a multiplication, then find the result:

- $5 + 5 + 5 = \ldots \times \ldots = \ldots$
- $7 + 7 = \ldots \times \ldots = \ldots$
- $3 + 3 + 3 + 3 = \ldots \times \ldots = \ldots$
- $10 + 10 + 10 + 10 + 10 = \ldots \times \ldots = \ldots$

Exercise 5.2 Lightning Tables. Answer as fast as possible without using your fingers:

- $2 \times 7 = \ldots$
- $5 \times 4 = \ldots$
- $3 \times 6 = \ldots$
- $9 \times 2 = \ldots$
- $10 \times 8 = \ldots$

Exercise 5.3 Doubles and Halves. 1. What is the double of 6? of 9? of 15? 2. What is half of 10? of 16? of 40?

Exercise 5.4 Magical Zeros. Calculate mentally:

- $6 \times 10 = \ldots$

- $9 \times 100 = ...$
- $17 \times 10 = ...$
- $30 \times 10 = ...$

Exercise 5.5 The Chocolate Box. A box of delicious chocolates contains 3 rows of 8 chocolates each. How many chocolates are in the box?

Level (Think and Calculate)

Exercise 5.6 The 9-Times Table Trick. Use the ten-finger trick to find:

- 9×3
- 9×6
- 9×8

Check each time that the sum of the digits in your answer is always 9!

Exercise 5.7 Set up and calculate these multiplications without carries: - 21×4 - 33×2 - 42×2

Exercise 5.8 Set up and calculate these multiplications with carries: - 26×3 - 35×2 - 17×5

Exercise 5.9 Kitchen Floor Tiling. A kitchen floor is covered with a rectangular tile pattern: 7 tiles wide by 9 tiles long. 1. How many tiles are there in total? 2. If the tiler had laid them in the other orientation (9 wide by 7 long), how many tiles would there have been?

Exercise 5.10 The Biscuit Pack. A pack contains 4 pouches of 6 biscuits each. 1. How many biscuits are in one pack? 2. Mom buys 3 full packs for the school fair. How many biscuits did she buy in total?

Level (Challenges and Puzzles)

Exercise 5.11 The Mystery Rectangle. “I am a number in the 4-times table. I am also in the 6-times table. I am between 20 and 30. Who am I?”

Exercise 5.12 The Missing Digit Mystery. Find the digit hidden under the star:

```
   2  ?
×     4
--------
   9  6
```

Exercise 5.13 Farm Legs. In a pasture, there are 4 chickens (2 legs each) and 3 cows (4 legs each). How many legs are there in total in the pasture? Write the whole calculation in a single line.

Exercise 5.14 Mental Multiplication: Distributivity. To calculate 14×3 mentally, a champion breaks 14 down into $10 + 4$:

- $10 \times 3 = 30$
- $4 \times 3 = 12$
- $30 + 12 = 42$!

Use this trick to calculate mentally:

- 13×4
- 15×3

Exercise 5.15 The Chess Tournament. A chessboard has 8 rows of 8 square spaces alternating white and black. 1. How many squares are on the chessboard in total? 2. Since half the squares are black and half are white, how many black squares are on the board?

Solutions

Solution 5.1 - $5+5+5=3\times5=\mathbf{15}$. - $7+7=2\times7=\mathbf{14}$. - $3+3+3+3=4\times3=\mathbf{12}$. - $10+10+10+10+10=5\times10=\mathbf{50}$.

Solution 5.2 - $2\times7=\mathbf{14}$ - $5\times4=\mathbf{20}$ - $3\times6=\mathbf{18}$ - $9\times2=\mathbf{18}$ - $10\times8=\mathbf{80}$

Solution 5.3 1. Doubles: $6\times2=\mathbf{12}$; $9\times2=\mathbf{18}$; $15\times2=\mathbf{30}$. 2. Halves: half of 10 is $\mathbf{5}$; half of 16 is $\mathbf{8}$; half of 40 is $\mathbf{20}$.

Solution 5.4 - $6\times10=\mathbf{60}$ - $9\times100=\mathbf{900}$ - $17\times10=\mathbf{170}$ - $30\times10=\mathbf{300}$

Solution 5.5 $3\ \text{ rows of }\ 8=3\times8=\mathbf{24\ \ chocolates}$.

Solution 5.6 - $9\times3=\mathbf{27}$ (finger 3 folded: 2 tens and 7 units; $2+7=9$). - $9\times6=\mathbf{54}$ (finger 6 folded: 5 tens and 4 units; $5+4=9$). - $9\times8=\mathbf{72}$ (finger 8 folded: 7 tens and 2 units; $7+2=9$).

Solution 5.7 - 21×4: units $4\times1=4$, tens $4\times2=8\Longrightarrow\mathbf{84}$. - 33×2: units $2\times3=6$, tens $2\times3=6\Longrightarrow\mathbf{66}$. - 42×2: units $2\times2=4$, tens $2\times4=8\Longrightarrow\mathbf{84}$.

Solution 5.8 - 26×3: units $3\times6=18$ (write 8, carry 1); tens $3\times2=6+1=7\Longrightarrow\mathbf{78}$. - 35×2: units $2\times5=10$ (write 0, carry 1); tens $2\times3=6+1=7\Longrightarrow\mathbf{70}$. - 17×5: units $5\times7=35$ (write 5, carry 3); tens $5\times1=5+3=8\Longrightarrow\mathbf{85}$.

Solution 5.9 1. $7\times9=\mathbf{63\ \ tiles}$. 2. $9\times7=\mathbf{63\ \ tiles}$ as well, thanks to commutativity.

Solution 5.10 1. In one pack: $4\times6=\mathbf{24\ \ biscuits}$. 2. In 3 packs: $3\times24=3\times20+3\times4=60+12=\mathbf{72\ \ biscuits}$.

Solution 5.11 4-times table between 20 and 30: 24, 28. 6-times table between 20 and 30: 24. The mystery common number is **24** ($4\times6=24$).

Solution 5.12 In the units, $4\times\star$ must end in 6. In the 4-times table, $4\times4=16$ (ends in 6) or $4\times9=36$. Test $\star=4$: $4\times4=16$ (carry 1). Tens: $4\times2=8$, plus carry 1: $8+1=9$. This gives exactly 96! The hidden digit is $\star=\mathbf{4}$.

Solution 5.13 Chicken legs: $4\times2=8$. Cow legs: $3\times4=12$. Total: $(4\times2)+(3\times4)=8+12=\mathbf{20\ \ legs}$.

Solution 5.14 - For 13×4: $(10\times4)+(3\times4)=40+12=\mathbf{52}$. - For 15×3: $(10\times3)+(5\times3)=30+15=\mathbf{45}$.

Solution 5.15 1. $8\ \text{ rows of }\ 8=8\times8=\mathbf{64\ \ squares}$. 2. Half of 64 is half of 60 (30) plus half of 4 (2), giving **32 black squares** (and 32 white squares).

Solutions to the Practitioner's Scales:

1. $7\times8=\mathbf{56}$; 2. $6\times9=\mathbf{54}$; 3. $9\times0=\mathbf{0}$; 4. $4\times7=\mathbf{28}$; 5. $8\times8=\mathbf{64}$; 6. $0\times0=\mathbf{0}$; 7. $52\times10=\mathbf{520}$; 8. $23\times3=\mathbf{69}$; 9. $15\times4=\mathbf{60}$; 10. $12\times11=\mathbf{132}$.

The Question That Opens the Next Chapter…

You are now able to multiply space and calculate the number of bricks in a wall in the blink of an eye.

Yet when the masons want to build that wall, they need to know: **how long is this wall?** Builders in Ancient Egypt used their own bodies: the length of their forearm, foot, or stride. But the pharaoh had long arms, while his servant had short arms! How could they ensure that carved stone blocks would fit together seamlessly without ever breaking?

And how could they measure the passage of time or the weight of a sack of gold?

Join us in **Chapter 6: The Pharaoh's Cubit and the Hourglass**!

CHAPTER 6: The Pharaoh's Cubit and the Hourglass

> "To measure the world, early humans turned to what was closest at hand: their fingers, their feet, and their arms. But as long as a measurement changes with the size of the person measuring, houses collapse and merchants quarrel. A universal standard that no one can counterfeit had to be invented."

What You Are About to Discover

- How ancient peoples measured using their own bodies (the inch, the foot, the pace, the cubit) and why this led to gargantuan disputes!
- The invention of a single standard: the **meter (m)** and the **centimeter (cm)**.
- How to measure an object with a graduated ruler **without ever falling into the zero trap**.
- Weighing the world: the Roberval balance scale, the **gram (g)**, and the **kilogram (kg)**.
- Liquids and capacities: the **liter (L)**.
- Taming the flow of time: the calendar of seasons, the 12 months, and **reading time on an analog clock**.
- Everyday currency: coins, banknotes, and euro cents.

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1. The Problem: The Feud of the Egyptian Stonecutters

Under the scorching sun on the banks of the Nile, four thousand years ago, hundreds of workers are constructing a grand temple for the pharaoh.

The chief architect traces the blueprints upon a sheet of papyrus: *"I demand that each block of stone for the monumental gateway measure exactly **three cubits** in length!"*

Two stonecutters set to work in the quarry:

- The first is named Senned. He is a towering giant over two meters tall, with long, powerful arms like oak trunks.
- The second is named Bak. He is a slender, agile young man with short, delicate arms.

What is a **cubit** to them? It is the distance from the tip of the elbow to the tip of the middle finger (the longest finger of the hand)!

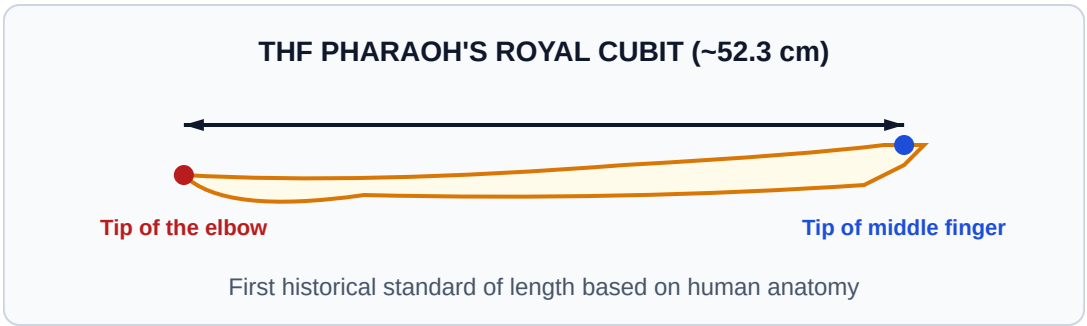

Figure 24: The pharaonic royal cubit: anatomical standard of measure

Each mason measures three cubits using his own arm, carves his limestone block from the white bedrock, and hauls it to the construction site.

When the architect attempts to assemble the lintel of the gate, disaster strikes! The stone carved by the giant Senned juts out by twenty centimeters, while the block carved by little Bak is far too short and crashes down into the dust! The two stonecutters seize each other by the collars, each accusing the other of cheating.

Why did this disaster occur? Because **the body of every human being is different**. If a giant's stride is wider than a child's stride, and if a grandfather's foot is longer than his grandson's foot, no one on Earth will ever be able to build a straight house!

---

2. From the Body to the Ruler: The Meter and the Centimeter

To put an end to endless disputes, the pharaoh decreed a royal law: *"Henceforth, there shall exist only one single cubit throughout all the land: **the cubit of my own arm**!"*

Royal sculptors carved this length into a rod of polished black granite, preserved within the sacred royal treasury. Wooden copies were crafted for every job site: this was a **standard of measurement**.

Centuries later, during the French Revolution, scientists sought something even grander: a standard belonging to no single monarch, but to all inhabitants of the Earth. They created **the Meter**.

💡 Discovery & History — Historical Anatomical Measurements

- The inch: The width of an adult's thumb (~ 2 to 3 cm)
- The foot: The length of a shod foot (~ 30 cm)
- The pace: A long adult walking stride (~ 1 meter)

📐 Definition — The Meter and the Centimeter

- **The meter (m)** is the official base unit of length. It is

roughly the width of a bedroom doorway, or the length of a long adult walking stride.

- To measure small objects (such as a pencil or an eraser), one meter is divided into 100 equal slices: each slice is called a **centimeter (cm)**.
- The prefix "centi-" means: *one hundredth*.

$$1 \text{ meter } (m) = 100 \text{ centimeters } (cm)$$

How to Measure with Your Ruler Without Falling into the Trap

Pick up your plastic school ruler.

⚠️ Watch Out for the Trap! — The plastic edge is not zero!

Many students press the plastic end of their ruler against the start of the object. That is an error! On most rulers, there is a small blank margin before the first graduation line. **To measure accurately, you must align the ZERO (0) mark directly with the very start of your object.** Then, you observe which number lands exactly at the end of the object.

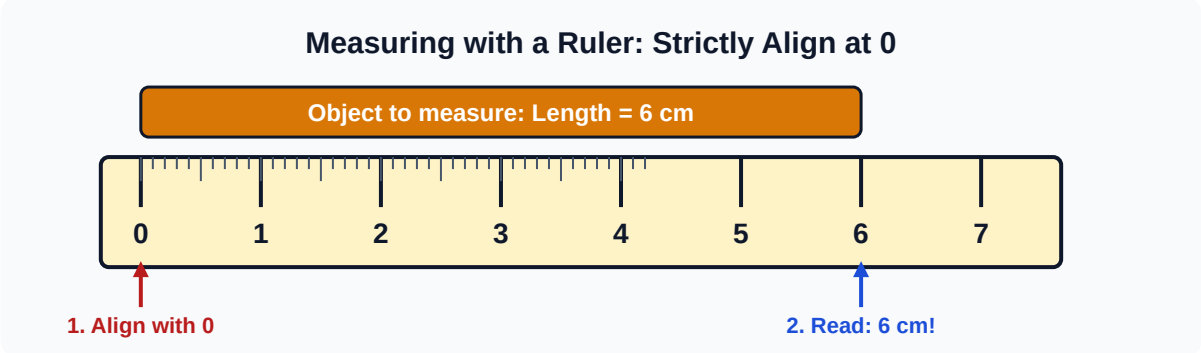

Figure 25: Ruler measurement: alignment at zero and direct reading

3. Mass and Weight: The Roberval Scale and the Kilogram

How heavy is your backpack? How heavy is a bird's feather?

To weigh objects, we use a two-pan balance perfected in 1669 by French mathematician and inventor **Gilles Personne de Roberval**. Unlike ancient hanging-beam balances where balance depended on the exact position of objects on the pan, Roberval's balance incorporates an ingenious articulated parallelogram mechanism concealed beneath the pans. Thanks to this system, both pans remain strictly horizontal throughout motion, ensuring exact measurement whether objects rest at the center or near the edge! On one pan, we place the object to be weighed (for instance, a ripe melon). On the other pan, we place calibrated metal **weights** until the pointer rests perfectly vertical in the center.

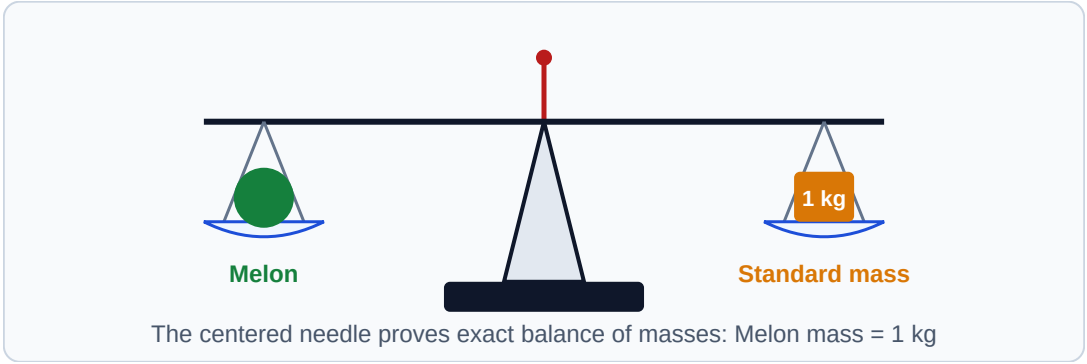

Figure 26: Weighing the melon: mass balance of 1 kilogram

Definition — The Kilogram and the Gram

- For heavy objects, we use the **kilogram (kg)**. A bag of flour or a carton of milk weighs approximately 1 kg.
- For very light objects, we divide a kilogram into 1,000 tiny parts: each part is a **gram (g)**.
- A paperclip or a kernel of corn weighs roughly 1 g.
- The prefix "kilo-" comes from Greek and means **thousand**.

$$1 \text{ kilogram } (kg) = 1,000 \text{ grams } (g)$$

Half a kilogram is 500 g.

4. Capacities: The Liter

How much water fills your bathtub? How much orange juice is in your cup? The volume of liquid that a container can hold is called its **capacity** or capacity volume.

Definition — The Liter (L)

The standard unit of liquid measurement is the **liter (L)**.

- A standard bottle of mineral water often holds 1 liter (or 1.5 L).
- A carton of vegetable soup contains 1 liter.
- A small soda can contains roughly a third of a liter (33 cL).
- A large bucket holds approximately 10 liters.
- A filled bathtub holds around 150 liters.

5. Taming the River of Time: From the Hourglass to the Clock

Unlike a rope lying motionless on the floor or a stone resting on a scale, **time never stops**. You cannot touch it with your fingers.

To track time, early humans began by watching the heavens: 1. **Day and Night**: The Earth spins on its axis. When the sun shines, it is day; when it sets, it is night. A full day lasts **24 hours**. 2. **The Week**: It spans **7 days** (Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday). 3. **The Month**: The time required for the Moon to cycle through its phases (roughly 30 or 31 days, except February with 28 or 29). 4. **The Year**: The time Earth takes to complete one full revolution around the Sun. It contains **12 months** (365 days) and journeys through the four seasons (spring, summer, autumn, winter).

Reading an Analog Clock

On a clock face, there are 12 numbers and two hands turning in clockwise rotation:

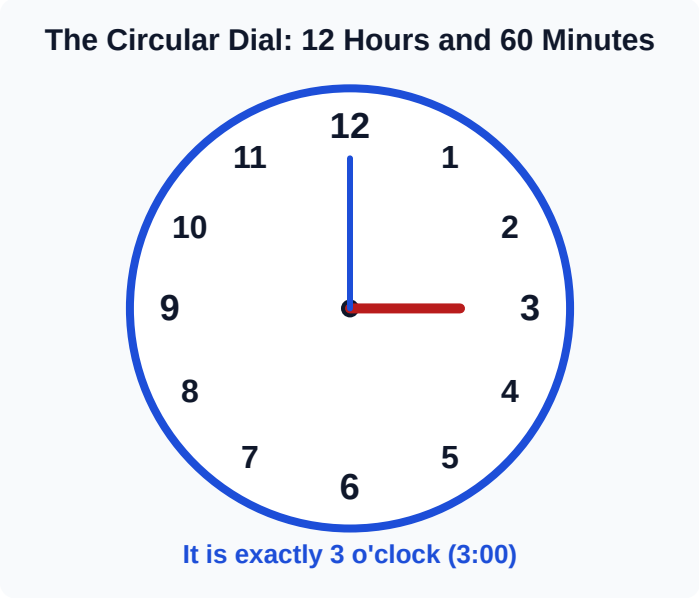

Figure 27: The circular dial: 12 hours and 60 minutes

- **The short hand (small and slow)**: This is the **hour hand**. It takes 12 hours to complete one revolution, gliding slowly from one numeral to the next.
- **The long hand (tall and fast)**: This is the **minute hand**. It completes a full lap in **60 minutes** (1 hour).

> **Method — Reading the Clock Step by Step**
>
> 1. **On the hour:** When the long hand points straight up at **12**, the short hand indicates the exact hour. *Example: short hand on 4, long hand on 12 → it is 4 o'clock (*$4:00$*).* 2. **Half past (30 minutes):** When the long hand points straight down at **6**, it has completed half a turn. We say "half past". *Example: short hand halfway between 4 and 5, long hand on 6 → 4:30.* 3. **Quarter past (15 minutes):** Long hand pointing right at **3** → quarter past ($4:15$). 4. **Quarter to (45 minutes):** Long hand pointing left at **9** → quarter to ($4:45$).
>
> $$1 \text{ hour } (h) = 60 \text{ minutes } (min)$$
>
> $$1 \text{ minute } (min) = 60 \text{ seconds } (s)$$

---

6. Money: Everyday Currency

To purchase bread from the baker or buy a book, we no longer trade goats or bushels of wheat: we use **currency**. In many European countries, the currency is the **Euro (€)**; in other places, the dollar ($), pound, or yen is used.

- **Coins:**
  - Copper coins: 1 cent, 2 cents, 5 cents.
  - Brass coins: 10 cents, 20 cents, 50 cents.
  - Bimetallic coins: **1 euro** and **2 euros**.
- **Paper Banknotes:**
  - 5 €, 10 €, 20 €, 50 €, 100 €, 200 €.

> 📐 ***Currency Standard — Cents and the Euro*** *Just like centimeters and meters:*
>
> $$1 \text{ Euro } (€) = 100 \text{ cents } (c)$$
>
> *To make one euro, you need one hundred cents!*

---

7. Five Step-by-Step Worked Examples

📐 Worked Example 1 — Measuring a Paintbrush with a Ruler

Problem: A paintbrush is laid beside a ruler. The handle begins at the 0 graduation line. The bristle tip touches mark 14. How long is the brush? *Solution:* Since the start aligns with 0, read the length directly from the end: the brush measures **14 cm**. *(If it started at mark 2 and ended at mark 14, it would measure* $14 - 2 = 12$ cm.*)*

Worked Example 2 — Comparing Lengths in Meters and Centimeters

Problem: A blue ribbon measures 1 m 25 cm. A red ribbon measures 140 cm. Which ribbon is longer? *Method:* Convert to the same unit (centimeters):

- Blue ribbon: $1 \text{ m} = 100$ cm, so $1 \text{ m } 25 \text{ cm} = 100 + 25 = 125$ cm.
- Red ribbon: 140 cm. Since $140 > 125$, the **red ribbon** is longer.

Worked Example 3 — Balancing a Basket on a Roberval Scale

Problem: A bag of apples is placed on the left pan of a balance scale. On the right pan, we place a 1 kg weight, a 500 g weight, and two 100 g weights. The scale balances perfectly. What is the mass of the bag of apples? *Solution:* Sum the weights:

- $1 \text{ kg} = 1,000$ g.
- Remaining weights: $500 + 100 + 100 = 700$ g.
- Total: $1,000 + 700 = 1,700$ g, which is **1 kg 700 g**.

Worked Example 4 — Duration of a Cartoon

Problem: A cartoon starts at $14:15$ and ends at $15:15$. How long did it last? *Solution:* The minutes match (15 min). Look at the hours: from 14:00 to 15:00, exactly **1 hour** (60 minutes) elapsed.

Worked Example 5 — Making Change at the Bakery

Problem: A loaf of bread costs 1 € 20 c. You hand the baker a 2 € coin. How much change will you receive? *Method:* Convert to cents:

- 2 € $= 200$ cents.
- Bread cost: 1 € 20 c $= 120$ cents.
- Subtract: $200 - 120 = 80$ cents. You receive **80 cents** in change (e.g. a 50c, 20c, and 10c coin).

Key Takeaways

1. To avoid misunderstandings, everyone uses **standardized units**:
   - Length: **meter (m)** and **centimeter (cm)** ($1\ \text{m} = 100\ \text{cm}$).
   - Mass: **kilogram (kg)** and **gram (g)** ($1\ \text{kg} = 1,000\ \text{g}$).
   - Capacity: **liter (L)**.
2. When measuring with a ruler, always align the **0** mark with the start of the object.
3. Time is structured into days (24 h), weeks (7 d), months (30 or 31 d), and years (12 mo).
4. On a clock: short hand indicates **hours**, long hand indicates **minutes**. $1\ \text{h} = 60\ \text{min}$.
5. $1\ \text{euro} = 100\ \text{cents}$.

Exercises

Level (Practice and Apply)

Exercise 6.1 Choose the correct unit (mm, cm, or m):
1. Length of a ladybug: 7 ...
2. Height of a living room door: 2 ...
3. Length of a ballpoint pen: 15 ...
4. Length of the school playground: 30 ...

Exercise 6.2 Basic Conversions. Fill in the blanks:
- $2\ \text{m} = ...\ \text{cm}$
- $4\ \text{m} = ...\ \text{cm}$
- $300\ \text{cm} = ...\ \text{m}$
- $1\ \text{kg} = ...\ \text{g}$
- $3\ \text{kg} = ...\ \text{g}$

Exercise 6.3 Choose the correct mass unit (g or kg): 1. A newborn kitten: 150 ... 2. A bag of cement: 25 ... 3. A chocolate bar: 100 ... 4. A schoolbag filled with textbooks: 3 ...

Exercise 6.4 Clock Reading. What time is it in each scenario? 1. Short hand on 8, long hand on 12. 2. Short hand between 2 and 3, long hand on 6. 3. Short hand on 11, long hand on 12.

Exercise 6.5 Counting Coins and Bills. How much money is in this piggy bank?
- One 10 € bill
- Two 2 € coins
- One 1 € coin
- Three 10-cent coins

Level (Think and Measure)

Exercise 6.6 The Broken Ruler Trap. A careless student has a snapped ruler with the zero end missing! He places an eraser starting at mark **3 cm**, and the eraser reaches mark **7 cm**. What is the true length of the eraser? Write out the calculation.

Exercise 6.7 The Wizard's Magic Potion. A cauldron already holds $3\ \text{liters}$ of spring water. The wizard pours in:
- 2 bottles of toad juice ($1\ \text{liter}$ each),
- 4 bottles of snail slime ($50\ \text{cL}$ / half-liter each).

How many liters of magic potion are now in the cauldron? *(Hint: 2 half-liters equal 1 full liter!)*

Exercise 6.8 Days of the Week. 1. If today is Tuesday, what day will it be in exactly 3 days? 2. What day was it the day before yesterday? 3. In a full week, how many school days are there if classes are held Monday, Tuesday, Thursday, and Friday?

Exercise 6.9 The Airline Bag Limit. To board a flight, luggage must not exceed $20\ \text{kg}$. Dad's suitcase weighs $14\ \text{kg}$. He wishes to pack a heavy dictionary ($3\ \text{kg}$) and canned goods ($4\ \text{kg}$). Will the suitcase be accepted without extra fees?

Exercise 6.10 Exact Change. A storybook costs $7\ €\ 50\ \text{c}$. Find a combination that pays this exact sum using the fewest possible coins and banknotes.

Level (Puzzles and Riddles)

Exercise 6.11 The Melting Candle Riddle. A candle is $24\ \text{cm}$ tall at 6:00 PM when lit. Each passing hour, it burns down by exactly $3\ \text{cm}$. 1. How tall will the candle be at 9:00 PM? 2. At what time will the candle be completely burned down (0 cm)?

Exercise 6.12 The Frog's Measured Leap. A tiny green frog hops $15\ \text{cm}$ per jump. She begins at the 0 mark of a $100\ \text{cm}$ meter stick. How many jumps must she take to leap past the $100\ \text{cm}$ finish line? At what centimeter mark will she land on her final jump?

Exercise 6.13 The Chimes of Midnight. An antique grandfather clock chimes once at 1 o'clock, twice at 2 o'clock, three times at 3 o'clock, and so forth up to twelve chimes at noon and midnight. How many chimes does the clock strike in total from 1:00 AM to 6:00 AM inclusive?

Exercise 6.14 The Riddle of Two Jugs. You are by a riverbank with two ungraduated buckets: a **3-liter** bucket and a **5-liter** bucket. How can you measure out exactly **4 liters** of water to bring back to the village?

Exercise 6.15 Coins in the piggy bank. Leo has a piggy bank full of 20-cent and 50-cent coins. He has exactly **4 coins in total**, and together they are worth **1 euro 40 cents** (140 c). How many 20-cent coins and how many 50-cent coins does he have?

Solutions

Solution 6.1 1. Ladybug: 7 $\boldsymbol{mm}$ (an insect is measured in millimeters!). 2. Door height: 2 $\boldsymbol{m}$. 3. Pen length: 15 $\boldsymbol{cm}$. 4. Playground: 30 $\boldsymbol{m}$.

Solution 6.2 Conversions:

- $2\ \text{m} = \mathbf{200}\ \text{cm}$
- $4\ \text{m} = \mathbf{400}\ \text{cm}$
- $300\ \text{cm} = \mathbf{3}\ \text{m}$
- $1\ \text{kg} = \mathbf{1\,000}\ \text{g}$
- $3\ \text{kg} = \mathbf{3\,000}\ \text{g}$

Solution 6.3 1. Kitten: 150 $\boldsymbol{g}$. 2. Cement bag: 25 $\boldsymbol{kg}$. 3. Chocolate bar: 100 $\boldsymbol{g}$. 4. Schoolbag: 3 $\boldsymbol{kg}$.

Solution 6.4 1. **8 o'clock sharp** ($8:00$). 2. **Half past two** ($2:30$). 3. **11 o'clock sharp** ($11:00$).

Solution 6.5 - Banknotes and euro coins: $10 + 2 + 2 + 1 = 15\ €$. - Cents: $3 \times 10 = 30\ \text{c}$.

Total: $\mathbf{15\ €\ 30\ cents}$.

Solution 6.6 The eraser spans from 3 cm to 7 cm:

$$7 - 3 = 4\ \text{cm}$$

The true length is $\mathbf{4\ cm}$.

Solution 6.7 - Water: 3 L. - Toad juice: 2 L. - Snail slime: 4 half-liters = 2 liters.

Total: $3 + 2 + 2 = \mathbf{7\ liters}$.

Solution 6.8 1. Tuesday $+3$ days (Wed, Thu, Fri): **Friday**. 2. Day before yesterday: **Sunday**. 3. School days: **4 days**.

Solution 6.9 Total mass:

$$14 + 3 + 4 = 21\ \text{kg}$$

Since $21 \text{ kg} > 20 \text{ kg}$, the bag exceeds the limit: **no**, it will not be accepted without an overweight fee or lightening.

Solution 6.10 To form $7 \text{ €} \ 50 \text{ c}$ with minimum pieces:

- One 5 € bill
- One 2 € coin ($5 + 2 = 7$ €)
- One 50-cent coin.

Total: **3 items** (one bill, two coins).

Solution 6.11 1. From 6:00 PM to 9:00 PM is 3 hours: burn is $3 \times 3 = 9 \text{ cm}$. Height remaining: $24 - 9 = \mathbf{15 \text{ cm}}$. 2. To burn all 24 cm at 3 cm/h requires $24 \div 3 = 8$ hours. $18:00 + 8 \text{ h} = \mathbf{2:00 \text{ AM}}$.

Solution 6.12 Leaps of 15 cm: 15, 30, 45, 60, 75, 90, 105. At the 6th jump, she is at 90 cm. On her **7th jump**, she clears the line, landing at $\mathbf{105 \text{ cm}}$. She needs 7 jumps.

Solution 6.13 Sum of chimes:

$$1 + 2 + 3 + 4 + 5 + 6 = 21 \text{ chimes}$$

Total: **21 chimes**.

Solution 6.14 1. Fill the 5-liter bucket completely. 2. Pour from the 5L bucket into the 3L bucket until full: exactly **2 liters** remain in the 5L bucket. 3. Empty the 3L bucket. 4. Pour the 2 liters from the 5L bucket into the 3L bucket (leaving space for 1 L). 5. Fill the 5L bucket completely again. 6. Pour into the 3L bucket until full (transferring exactly 1 L): exactly **4 liters** remain in the 5L bucket!

Solution 6.15 Testing combinations of 4 coins with 50 c and 20 c coins: 2 coins of 50 c and 2 coins of 20 c yield $2 \times 50 + 2 \times 20 = 100 + 40 = 140 \text{ c}$ (€1.40). Leo holds **two 50-cent coins** and **two 20-cent coins**.

The Question That Opens the Next Chapter…

You now know how to measure lengths, weigh objects, tell time, and make change with money.

Yet have you looked closely at the shapes of the stones forming houses? Some have four straight sides and crisp corners: rectangles and squares. Others are pointed like pyramids, circular like the sun, or cubic like a rolling die.

How did master builders learn to trace a flawless circle on the ground using only a simple string and two pegs? And how does a folded sheet of paper bring forth a magical butterfly?

Join us for the final grand journey of Volume I: **Chapter 7: The Shapes of the World and the Gardener's String**!

CHAPTER 7: The Shapes of the World and the Gardener's String

> "Look around you: the Earth does not speak solely in numbers, it speaks in shapes. Tree trunks are cylinders, honeycombs are hexagons, and river-polished stones are spheres. To trace a straight figure in the sand is to begin drawing the world."

What You Are About to Discover

- How to recognize plane figures all around you: the **square**, the **rectangle**, the **triangle**, and the **circle**.
- The secret of the gardener's string: tracing a flawless circle without a compass using two wooden pegs.
- Entering the three-dimensional world: solids (the **cube**, the **rectangular prism**, the **sphere**, the **cylinder**).
- The vocabulary of builders: **vertex**, **face**, and **edge**.
- Using a ruler and crafting a **folded-paper set square** to test right angles.
- The magic of the mirror and the paper butterfly: the **axis of symmetry**.
- Navigating a grid and programming a robot's path with step-by-step arrows.

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1. The Problem: The Queen's Flowerbed

Long ago, in a splendid kingdom of the Orient, the queen summoned her head gardener: *"I desire a new garden before the palace gates. But take heed! I will not tolerate a chaotic tangle of wild weeds. I want four magnificent parterres: - a flawless square parterre for my white rosebushes, - an elongated rectangular parterre for my crimson tulips, - a pointed triangular parterre for my fragrant herbs, - and a perfectly circular pool in the center, where my golden carp may swim!"*

The gardener bowed low. Yet upon stepping onto the vast stretch of bare earth, he was deeply troubled: How could he trace a perfect circle without a giant compass? How could he ensure a rectangle had sharp, true corners and did not look like a crushed, crooked box? And how could he direct the apprentices to dig the pathways in the right places without losing their way?

To accomplish this masterpiece, the gardener relied on the simplest, most brilliant tools in the history of geometry: **a piece of string, two wooden pegs, and a folded sheet of paper**.

---

2. Plane Figures: Polygons and Their Secrets

A flat (or plane) figure is a closed drawing that can be traced on a sheet of paper or on the ground without rising up into the air.

When a plane figure is bounded solely by straight line segments (straight sides drawn with a ruler), it is called a **polygon** (from the Greek *polus*, many, and *gônia*, angle or corner).

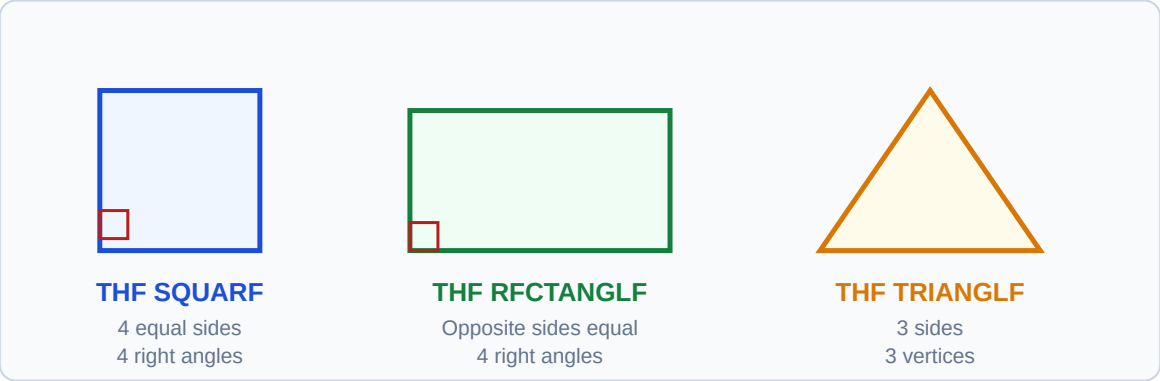

Figure 28: The three fundamental shapes of planar geometry

1. The Square

- It has **4 sides** that measure **exactly the same length**.
- It has **4 right angles** (crisp corners like the corner of a sheet of paper).
- It has **4 vertices** (pointed corners).

2. The Rectangle

- It has **4 sides**, but they are not all equal: it has 2 long lengths and 2 short widths.
- Opposite sides (the sides facing each other) are equal in length.
- It has **4 right angles** and **4 vertices**.

3. The Triangle

- It has **3 straight sides** and **3 vertices** (three angles).
- Its name means: "three angles".

---

3. The Gardener's String: Tracing a Perfect Circle

Unlike the square or the triangle, **the circle has no straight sides and no corners**! It is a closed, beautifully harmonious curved line.

How did the gardener trace the queen's great round pool?

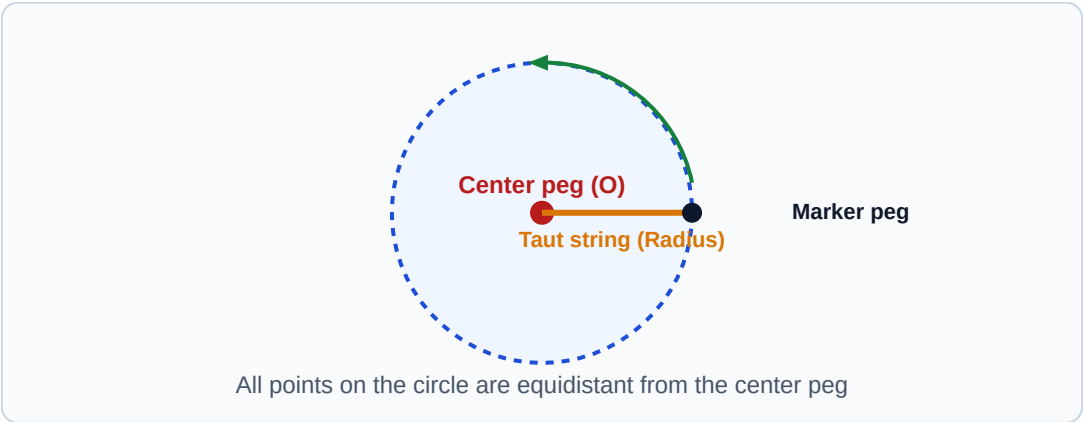

Figure 29: The gardener's string: drawing a perfect circle

1. He drives a sturdy **wooden peg straight into the ground at the center** of the future pool (this is the **center** of the circle).
2. He ties a strong cord securely to this peg.
3. To the other end of the cord, he fastens a second pointed peg (or a piece of chalk).
4. He pulls the string until it is **completely taut**.
5. Keeping the string tight without ever letting it slacken, he walks in a full circle around the center peg, letting the point scratch into the earth.

When he completes the loop, he has traced a **flawless circle**!

> **📐 Definition — Circle, Radius, and Diameter**
>
> - A **circle** is the collection of all points located **at the exact same distance** from a central point called the **center**.
> - The length of the taut string is called the **radius** of the circle.
> - The straight line passing through the center from one edge to the opposite edge is called the **diameter**. The diameter is always twice the radius ($2 \times \text{radius}$).

---

4. The Right-Angle Gauge: Crafting Your Magic Set Square

To verify that a wall rises vertically or that a table has square corners, carpenters use a triangular tool made of wood or metal: a **set square**. Yet you can craft an infallible set square in ten seconds using any scrap of torn paper!

> **🖐 Workshop — Make Your Own Folded-Paper Set Square**
>
> 1. Take
>
> any scrap piece of paper, even one with a jagged, crooked edge. 2. Fold it in half once, pressing firmly along the fold with your thumbnail. You now have a perfectly straight edge. 3. Now, fold the paper in half a second time, **bringing the first straight crease directly onto itself**, edge to edge! 4. Crease it firmly. 5. The corner formed by these two folds is a **flawless right angle of 90°**! Place this magical corner against the corner of your notebook, your desk, or a door frame: it fits to the millimeter!

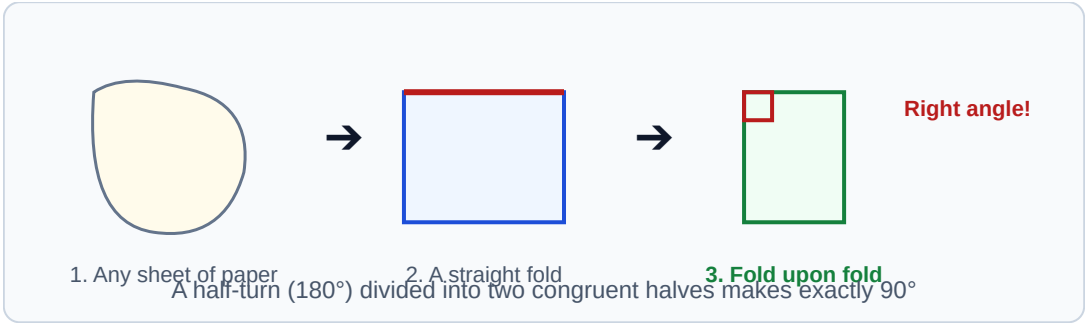

Figure 30: Folding a set square: the 90° right angle

---

5. Solids: The World in Three Dimensions

On paper, figures are flat. But in the physical world, objects possess thickness and depth: they take up volume in space. We call them **solids**.

Figure 31: Fundamental geometric solids: cube, rectangular prism, and cylinder

Let us learn the vocabulary of master builders:

- A **face**: a flat wall of the solid (the smooth surface upon which it can rest).
- An **edge**: the straight line segment where two faces meet (the sharp boundary).
- A **vertex**: the sharp corner point where several edges meet.

The Cube (like a game die)

- It has **6 square faces**, all strictly identical.
- It has **12 edges** of equal length.
- It has **8 sharp vertices**.

The Rectangular Prism (like a shoebox or brick)

- It has **6 rectangular faces** (which may sometimes include 2 square faces).
- It has **12 edges** and **8 vertices**.

The Sphere (like a marble, an orange, or a soccer ball)

- It has no flat faces, no straight edges, and no vertices. It rolls freely in every direction!

The Cylinder (like a paper towel roll or soup can)

- It has **2 circular flat faces** (top and bottom disks) and a curved side that lets it roll in a straight line.

---

6. The Magic Mirror: The Axis of Symmetry

Take a sheet of white paper. Place a drop of blue ink or watercolor in the center. Fold the sheet in half along a straight line immediately, press firmly with the palm of your hand, and unfold:

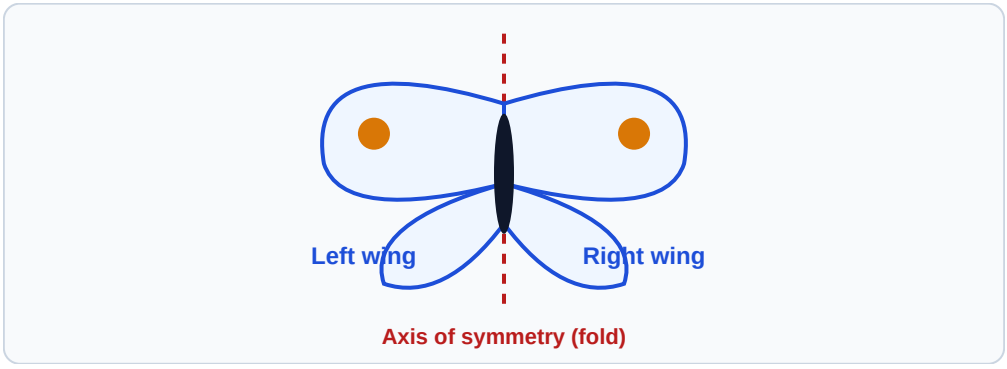

Figure 32: The butterfly and bilateral axial symmetry

You behold a magnificent **butterfly**! The right wing is the identical reflection of the left wing. If you refold the sheet, the two wings will overlap **exactly**, stroke for stroke, point for point.

📐 Definition — Axis of Symmetry

An **axis of symmetry** is a straight line that divides a figure into two halves that match up perfectly when folded. It mirrors what you would see if a flat mirror were placed upright along that line!

Countless things in nature display an axis of symmetry: your own face, a tree leaf, a ladybug, or a snowflake.

---

7. Grids and the Little Robot's Code

To dispatch an apprentice to plant seeds at an exact spot in the garden, the master divided the ground into regular squares: **a grid**.

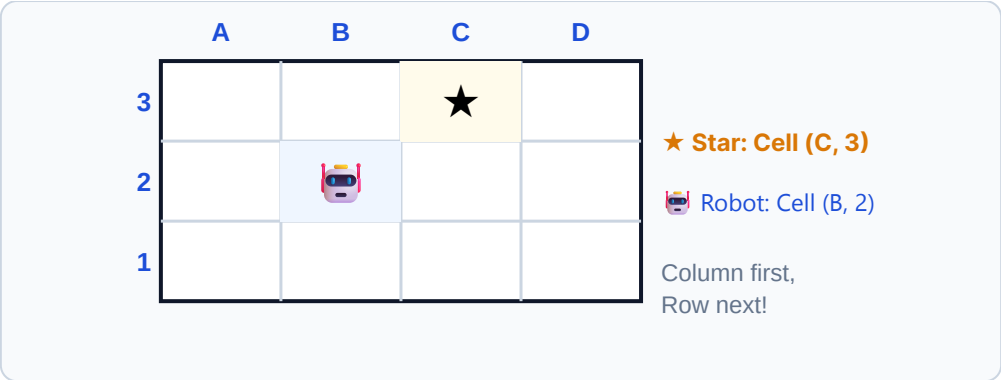

Figure 33: Plotting coordinates on a discrete Cartesian grid

To locate any square, we provide its address with two coordinates: 1. First the **column** (the letter: A, B, C, D). 2. Next the **row** (the number: 1, 2, 3). *Example:* The robot is in square **B2**. The golden star shines in square **C3**.

Coding Movement with Arrows:

To navigate the robot from square **B2** to the star at **C3**, we write a program using directional arrows:

- →: move one square to the right.
- ↑: move one square up.
- ←: move one square to the left.
- ↓: move one square down.

The shortest path from B2 to C3 is:

→ ↑

(One step right to land on C2, then one step up to reach C3!)

---

8. Five Step-by-Step Worked Examples

Worked Example 1 — Distinguishing a Square from Rectangles

Problem: Consider three shapes, each with 4 right angles:

- Figure A: width 5 cm, length 8 cm.
- Figure B: 4 sides, each measuring 6 cm.
- Figure C: width 3 cm, length 7 cm. Which figure is a square? *Solution:* A square must have **4 equal sides** in addition to 4 right angles. Only **Figure B** has 4 equal sides (6 cm). Figures A and C are non-square rectangles.

Worked Example 2 — Calculating Diameter from Radius

Problem: The gardener staked a 4-meter taut cord from the central peg to outline his pool. What is the total diameter of the pool? *Solution:* The radius is $4\ \text{m}$. The diameter spans straight through the center from edge to edge, equal to twice the radius:

$$\text{Diameter} = 4\ \text{m} + 4\ \text{m} = 2 \times 4\ \text{m} = \mathbf{8\ \text{meters}}$$

Worked Example 3 — Counting Features of a Solid

Problem: Look at a closed cardboard tissue box (a rectangular prism). How many stickers would you need to put one sticker on every vertex? *Solution:* Vertices are the sharp corners. There are 4 vertices on the top face and 4 on the bottom face:

$$4 + 4 = \mathbf{8\ \text{vertices}}$$

You would need **8 stickers**.

Worked Example 4 — Finding Axes of Symmetry of a Rectangle

Problem: How many axes of symmetry does an oblong (non-square) rectangle possess? *Practical Method:* Take a sheet of A4 printer paper. 1. Fold it in half horizontally: the two halves align. That is the first axis! 2. Fold it in half vertically: the two halves align. That is the second axis! 3. *What about the diagonal (corner to corner)?* Try folding corner to opposite corner: the edges jut out and do not match! The diagonal of an oblong rectangle is **not** an axis of symmetry. *Conclusion:* A non-square rectangle has **2 axes of symmetry**.

Worked Example 5 — Moving on a Grid

Problem: A treasure is located at square **D4**. A pirate starts at **A1**. Write the shortest directional code to find the treasure. *Solution:*

- From column A to column D: 3 steps right ($A \to B \to C \to D$): $\to\to\to$.
- From row 1 to row 4: 3 steps up ($1 \to 2 \to 3 \to 4$): $\uparrow\uparrow\uparrow$. *Complete Program:* $\to\ \ \to\ \ \to\ \ \uparrow\ \ \uparrow\ \ \uparrow$.

Key Takeaways

1. **A square** has 4 equal sides and 4 right angles. **A rectangle** has equal opposite sides and 4 right angles. **A triangle** has 3 sides and 3 vertices.
2. **A circle** is the set of points equidistant from the center. This distance is the **radius**; the **diameter** is twice the radius.
3. A **right angle ($90°$)** is verified with a set square or double-folded paper.
4. Solids have **faces** (flat surfaces), **edges** (meeting lines), and **vertices** (corner points).
   - Cube and rectangular prism: **6 faces**, **12 edges**, **8 vertices**.
5. A figure has an **axis of symmetry** if its two halves overlap perfectly when folded.
6. On a grid, a cell is identified by its **column** (letter) and **row** (number). Paths are coded using directional arrows.

Exercises

Level (Practice and Apply)

Exercise 7.1 Name each plane figure: 1. I have 3 sides and 3 vertices. Who am I? 2. I have 4 equal sides and 4 right angles. Who am I? 3. I have no straight sides and resemble a bicycle wheel. Who am I? 4. I have 4 right angles, 2 long equal sides, and 2 short equal sides. Who am I?

Exercise 7.2 Vertex, Face, or Edge? On a game die: 1. What do we call the flat square surface bearing the number 5? 2. What do we call the pointed corner where three faces meet? 3. What do we call the straight edge you can trace with your finger along the side?

Exercise 7.3 Everyday Solids. Match each household object to the solid it most resembles:

- A soup can → ...
- A tennis ball → ...
- A cereal box → ...
- A game die → ...

Exercise 7.4 Radius and Diameter. 1. A circle has a radius of $5\ \text{cm}$. What is its diameter? 2. A circle has a diameter of $12\ \text{cm}$. What is its radius?

Exercise 7.5 Locating Grid Squares. Examine the grid below:

Practical Application — Grid Treasure Hunt

	A	B	C
2		Apple	
1	Banana		Grapes

Give the coordinate (Letter, Number) of each fruit: apple, banana, grapes.

Level (Think and Build)

Exercise 7.6 Hunting Right Angles. Using your paper set square: 1. How many right angles does a right-angled triangle have? 2. How many right angles does a circle have? 3. How many right angles does a square have?

Exercise 7.7 True or False on Solids? Answer **True** or **False**: 1. A sphere has 1 flat face and 1 vertex. 2. A cube has more edges than vertices. 3. A cylinder can roll on its curved side and stand flat on its round bases. 4. All faces of a cube are squares.

Exercise 7.8 Symmetric Letters. Which of these capital letters have a vertical axis of symmetry (can be folded left-to-right)?

A B M T F

Exercise 7.9 Mouse Navigation Code. A little mouse starts at square **A1**. She follows this movement code:

↑ ↑ → → ↓

In which square does she end her journey?

Exercise 7.10 Cube Net. A cube has 6 square faces. If you unfold a cardboard box flat, how many squares are joined together to form the net of the cube?

Level (Puzzles and Riddles)

Exercise 7.11 Triangles in the Square. Take a square piece of paper. Cut straight across from corner to opposite corner along the diagonal. 1. How many pieces do you obtain? 2. What exact geometric shape is each piece?

Exercise 7.12 The Matchstick Challenge. With 12 equal matchsticks laid flat on a table: 1. Can you construct a square using all 12 matchsticks? How many matchsticks per side? 2. Can you construct a non-square rectangle with these 12 matchsticks? Give the number of matchsticks for each side.

Exercise 7.13 Cart Wheels. A cart has two large rear wheels of diameter 80 cm, and two small front wheels of diameter 40 cm. 1. What is the radius of a large wheel? 2. What is the radius of a small wheel? 3. When the cart moves forward, which wheel completes more revolutions to cover the same distance?

Exercise 7.14 The Foraging Bee. On a garden grid, a bee leaves her hive at **B1**. She flies to visit a rose at **D3**, then lavender at **B4**, and finally returns to her hive at **B1**. Write out the complete path in arrow steps.

Exercise 7.15 The Mystery Solid. "I have 5 faces in total: my base on the ground is a flat square, and my other 4 faces rising into the sky are triangles meeting at a single apex at the very top. Who am I?" How many edges and vertices do I have?

Solutions

Solution 7.1 1. A **triangle**. 2. A **square**. 3. A **circle**. 4. A **rectangle**.

Solution 7.2 1. The flat square surface is a **face**. 2. The pointed corner is a **vertex**. 3. The straight line boundary is an **edge**.

Solution 7.3 - Soup can $\rightarrow$ **Cylinder**. - Tennis ball $\rightarrow$ **Sphere**. - Cereal box $\rightarrow$ **Rectangular prism**. - Game die $\rightarrow$ **Cube**.

Solution 7.4 1. Radius 5 cm $\Longrightarrow$ $\text{Diameter} = 2 \times 5 = \mathbf{10 \text{ cm}}$. 2. Diameter 12 cm $\Longrightarrow$ $\text{Radius} = 12 \div 2 = \mathbf{6 \text{ cm}}$.

Solution 7.5 - Apple: **B2**. - Banana: **A1**. - Grapes: **C1**.

Solution 7.6 1. A right-angled triangle has **1 right angle**. 2. A circle has no straight sides, so **0 right angles**. 3. A square has **4 right angles**.

Solution 7.7 1. False (a sphere has neither flat faces nor vertices). 2. True (a cube has 12 edges and 8 vertices, and $12 > 8$). 3. True (it rolls on its curved wall and stands stably on its circular bases). 4. True (all 6 faces of a cube are identical squares).

Solution 7.8 - **A**: yes (vertical axis down the center). - **B**: no (it has a horizontal axis, not vertical). - **M**: yes (vertical axis). - **T**: yes (vertical axis). - **F**: no (no axis).

The letters are **A**, **M**, and **T**.

Solution 7.9 - Start at A1. - $\uparrow\uparrow$: moves to A2, then A3. - $\rightarrow\rightarrow$: moves to B3, then C3. - $\downarrow$: moves down to C2.

The mouse lands on **C2**.

Solution 7.10 A cube has 6 faces: its net requires exactly **6 squares**.

Solution 7.11 1. You obtain **2 pieces**. 2. Each piece is an **isosceles right-angled triangle** (one right angle and two equal sides).

Solution 7.12 1. Yes! $12 \div 4 = 3$. Use **3 matchsticks per side**. 2. Yes! Two lengths and two widths must sum to 12 ($L + w = 6$). For instance: **4 matchsticks for each length** and **2 matchsticks for each width** ($4 + 2 + 4 + 2 = 12$). Or 5 for length and 1 for width.

Solution 7.13 1. Large wheel radius: $80 \div 2 = \mathbf{40 \text{ cm}}$. 2. Small wheel radius: $40 \div 2 = \mathbf{20 \text{ cm}}$. 3. The **small wheel** turns more times! Because its circumference is smaller, it must rotate twice as fast to cover the same distance.

Solution 7.14 - From hive (B1) to rose (D3): 2 right, 2 up $\rightarrow \quad \rightarrow \quad \uparrow \quad \uparrow$. - From rose (D3) to lavender (B4): 2 left, 1 up $\leftarrow \quad \leftarrow \quad \uparrow$. - From lavender (B4) to hive (B1): 3 down $\downarrow \quad \downarrow \quad \downarrow$.

Solution 7.15 This solid is a square-based **pyramid** (like the great pyramids of Egypt).

- Number of edges: 4 base edges $+4$ rising edges $= \mathbf{8 \text{ edges}}$.
- Number of vertices: 4 base corners $+1$ top apex $= \mathbf{5 \text{ vertices}}$.

The End of Volume I and the Open Road to Volume II…

You have traveled an extraordinary journey!

You began like the prehistoric shepherd watching over his flock with pebbles in a pouch. You discovered the secrets of your ten fingers, learned to assemble bundles of tens and hundreds, weighed goods upon the balance scale of justice, aligned columns with the traveling carry, multiplied space with magic rectangles, measured the earth with the meter, and befriended the shapes of the world.

Yet soon, a new challenge will arise. When city merchants bring back sacks of 13 precious stones to divide among 4 warriors, the stones refuse to divide evenly: there is always one lonely pebble left at the bottom of the basket! And when a mother must share a single loaf of flatbread among 3 hungry children, stone must break and we must invent numbers that break into pieces…

Prepare to enter **the Age of the Market and Partitions** with **Volume II: The Rebel Stones and the Lonely Numbers**!

VOLUME II — THE SECRET OF SHARING AND THE MAGNITUDES OF THE WORLD

Ages 8–11 / Grades 3–5

From sharing flatbread to astronomical measurements, the student masters fractions, decimals, proportionality, and Euclidean geometry.

Chapters of Volume II :

Chapter 8 : The Rebel Stones and the Lonely Numbers
Chapter 9 : The Feast Remainders and the Game Teams
Chapter 10 : The Broken Galette and the Measuring Sieve
Chapter 11 : The Tenths of Simon Stevin
Chapter 12 : The Meter of the Revolution
Chapter 13 : Grandmother's Recipe and the Rule of Three
Chapter 14 : The Merchant's Slate and the Sand Pit

Facing: Logarithmic spiral of the Nautilus shell (Nautilus pompilius).

CHAPTER 8: The Rebel Stones and the Lonely Numbers

> "There are numbers that willingly accept being folded, organized into battalions, and sliced into rectangles. And there are rebel numbers, wild and untamed, that refuse to bend and stand like solitary stones in the desert. These wild stones are the atoms of all numbers."

What You Are About to Discover

- What **multiples** of a number are (the numbers you land on when always leaping by the same length).
- What **divisors** of a number are (row lengths that divide evenly with nothing left over).
- **Detective tricks** to know, without calculating long divisions, whether a number divides evenly by 2, 3, 5, 9, or 10.
- **Prime numbers**, those numbers that refuse to form rectangles, and why they are the bricks of all other numbers.
- The **Sieve of Eratosthenes**, a number-sorting machine invented more than 2,000 years ago.

---

1. The Problem: The Market Guards' Parade

You are now eight years old. The tent campsite has grown into a small town on the banks of a great river. Every morning, merchants arrive from distant oases with camels laden with spices, pottery, and woven wool.

To ensure no one steals from the market stalls, the governor has recruited a small troop of guards. Today is the great harvest festival: the guards must march in parade across the grand central square.

The captain commands **12 soldiers**. He shouts: *"Form a rectangle! I want perfect rows, without a single soldier sticking out!"*

The 12 soldiers obey with a smile:

- they form **2 rows of 6**: $2 \times 6 = 12$;
- then **3 rows of 4**: $3 \times 4 = 12$;
- and to pass through the city's narrow gate, a single **file of 12**: $1 \times 12 = 12$.

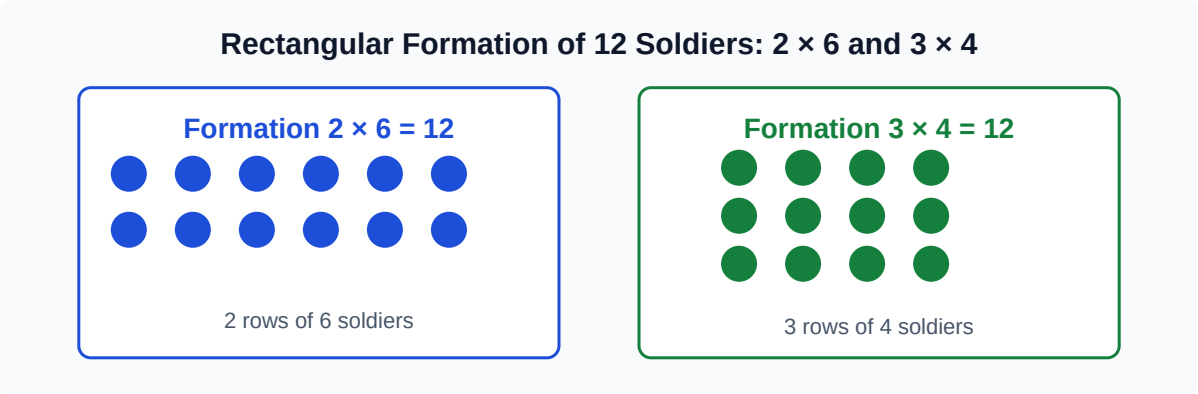

Figure 34: Rectangular formation of soldiers: 2×6 and 3×4

Everyone applauds: 12 is a **docile** number. It loves forming rectangles.

Suddenly, a messenger arrives at a full gallop: one of the soldiers must leave immediately to deliver an urgent message to the king. Only **11 soldiers** remain. The captain blows his whistle: *"Reform a rectangle!"*

- In rows of 2: five in front, five behind… and the eleventh finds himself **all alone** at the end!
- In rows of 3: three rows of three make 9… and **2 soldiers** are left stranded outside.
- In rows of 4: $4 + 4 = 8$, leaving **3** soldiers stranded.
- In rows of 5: $5 + 5 = 10$, leaving **1** soldier sticking out.

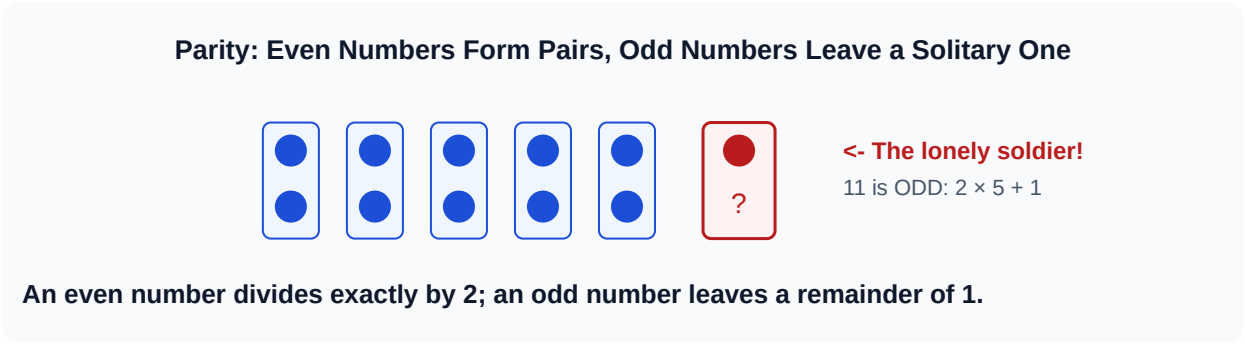

Figure 35: Parity: paired even numbers vs solitary odd soldier

The captain sweats under his bronze helmet. No matter how hard he tries: **with 11 soldiers, no solid rectangle is possible**, except for the long single line of 1 by 11.

It is neither the soldiers' fault nor the captain's. It is the number 11 itself that **refuses** to be broken into rows. Why are some numbers so flexible, while others are so stubborn? To understand this mystery, we must learn two crucial words: **multiple** and **divisor**.

> 🖐 ***Your Turn*** *— Take 12 small objects (buttons, beans, or coins). Form every possible rectangle. Then remove one object and try again with 11. Feel the difference with your own fingers!*

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2. Multiples: Always Leaping by the Same Length

Imagine a frog hopping along a long number line. It starts at 0 and always makes jumps of **3 steps**.

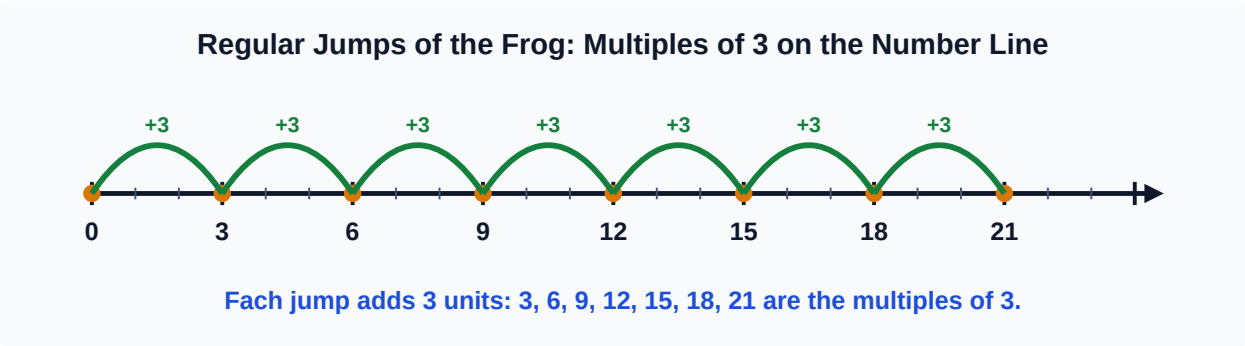

Figure 36: Regular jumps of 3 by 3 of the frog

The points where its feet land are: $0, 3, 6, 9, 12, 15, 18, 21, \ldots$ These are exactly the products of the **times-3 table**: 3×0, 3×1, 3×2, 3×3, and so on.

> 📐 **Definition — Multiple**
>
> A number is a **multiple of 3** if it belongs to the 3-times table, which means it can be written as $3 \times$ (a whole number). More generally, a number is a **multiple of *b*** if it can be written as $b \times$ (a whole number).

The three small dots "..." written at the end of a list read as "**and so on**". They mean: the list continues in exactly the same way, **never stopping**. For the frog can keep jumping as long as it likes: there are **infinitely many** multiples of 3!

> **Worked Example 1 — Is it a multiple?**
>
> *Question:* Is 42 a multiple of 6? What about 45? *Looking into the 6-times table:* $6 \times 7 = 42$. Yes, 42 is a multiple of 6. *For 45:* $6 \times 7 = 42$ and $6 \times 8 = 48$. 45 falls **between** the two; it is not in the table. 45 is **not** a multiple of 6. (The frog jumping by 6 leaps right over 45 without landing on it.)

Worked Example 2 — Common Multiples

Question: Which numbers up to 40 are multiples of both 4 and 6? Multiples of 4: 0, 4, 8, **12**, 16, 20, **24**, 28, 32, **36**, 40. Multiples of 6: 0, 6, **12**, 18, **24**, 30, **36**. The numbers present in both lists: **0, 12, 24, 36**. Two frogs, one jumping by 4s and the other by 6s, meet on these exact stepping stones!

---

3. Divisors: Row Lengths that Divide Evenly

Let us return to our soldiers. With 12 soldiers, we could form rows of 1, 2, 3, 4, 6, and 12 (since 4×3 and 6×2 also work by simply turning the rectangle). With 11, only rows of 1 and 11 worked.

📐 Definition — Divisor

A number b is a **divisor** of a if we can arrange a objects into rows of b **without a single object left over**. In other words: a is in the times table of b. We also say: "a **is divisible by** b" or "b **divides** a".

Notice how these two concepts go hand in hand, like two faces of the same coin:

$$12 = 3 \times 4 \quad \Rightarrow \quad \{12 \text{ is a multiple of } 3$$
$$3 \text{ is a divisor of } 12$$

The heavy arrow "⇒" reads as "**therefore**" or "**implies that**".

💡 Did You Know?

Later, in middle and high school, mathematicians will write "$3 \mid 12$" (a small vertical bar) to say "3 divides 12". Do not confuse this with a fraction bar, which is horizontal or slanted!

The Pair Method for Finding ALL Divisors

Every rectangle reveals **two** divisors at once: its width and its length. To make sure nothing is forgotten, we look for pairs systematically, starting with 1.

Worked Example 3 — All Divisors of 24

- $1 \times 24 = 24 \Longrightarrow$ divisors 1 and 24
- $2 \times 12 = 24 \Longrightarrow$ divisors 2 and 12
- $3 \times 8 = 24 \Longrightarrow$ divisors 3 and 8
- 4? $4 \times 6 = 24 \Longrightarrow$ divisors 4 and 6
- 5? $5 \times 4 = 20$, $5 \times 5 = 25$: no.
- 6? Already found with 4! The pairs begin to repeat in reverse: **we can stop here**.

The divisors of 24 are: **1, 2, 3, 4, 6, 8, 12, 24**. There are 8 of them, forming 4 distinct rectangles.

Worked Example 4 — Divisors of 36 (A Square Number)

1×36, 2×18, 3×12, 4×9, 6×6. Divisors: **1, 2, 3, 4, 6, 9, 12, 18, 36**. There are 9 divisors: an **odd** number of divisors! Why? Because the rectangle 6×6 is a **square**: the 6 serves as both width and length, so it is counted only once.

⚠ Pitfall Warning — Peculiar Cases

- **1 is a divisor of every whole number**: we can always form a single file (1×12, 1×11, etc.).
- **Every number is a divisor of itself** and a multiple of itself: $12 = 12 \times 1$.
- **0 is a multiple of every number**: $0 = 3 \times 0 = 7 \times 0$. Before jumping, the frog is already standing on 0!
- **Never divide by 0**: arrange 12 soldiers into rows of 0 soldiers? You could make empty rows for a thousand years without ever placing a single person. It is completely meaningless.
- Do not mix up the words: "3 is a **divisor** of 12", but "12 is a **multiple** of 3". The **smaller** divides, the **larger** is a multiple.

---

4. Detective Tricks: Divisibility Rules

At the market, a merchant has 3,471 dates. Can he divide them into bags of 3 without any leftover? Carrying out long division by hand would be tedious. Fortunately, mathematicians have discovered **divisibility rules**: clever clues that let you answer in the blink of an eye.

Rules for the Last Digit: 2, 5, and 10

Remember: a number like 3,471 consists of thousands, hundreds, tens, and units. But **a ten (10) can always be divided evenly by 2, 5, and 10**: $10 = 2 \times 5 = 5 \times 2 = 10 \times 1$. And hundreds and thousands are simply bundles of tens! Therefore, everything in the tens, hundreds, and thousands columns divides evenly without remainder. **Only the single units digit can cause a leftover.** That is why you only need to inspect the **last digit**.

📐 Divisibility Rules for 2, 5, and 10

- A number is divisible by **2** if its units digit is **0, 2, 4, 6, or 8**. We say it is **even**. Otherwise, it is **odd**.
- A number is divisible by **5** if its units digit is **0 or 5**.
- A number is divisible by **10** if its units digit is **0**.

Worked Example 5 — The Last Digit Speaks

- 3,476: ends in 6 $\Longrightarrow$ divisible by 2; not by 5; not by 10.
- 985: ends in 5 $\Longrightarrow$ divisible by 5; odd, so not by 2; not by 10.
- 7,230: ends in 0 $\Longrightarrow$ divisible by 2, by 5, **and** by 10.

The Magic Rules of Summing the Digits: 3 and 9

For 3 and 9, inspecting the last digit does not work at all! (13 ends in 3, but $13 = 3 \times 4 + 1$ is not divisible by 3.) We need a sharper insight.

Look closely:

$$10 = 9 + 1 \qquad 100 = 99 + 1 \qquad 1\,000 = 999 + 1$$

Now, 9, 99, and 999 divide perfectly into groups of 9 (and thus into groups of 3 as well, since $9 = 3 \times 3$). Each ten, each hundred, and each thousand therefore leaves **exactly 1 stray unit sticking out**.

Take 234: this is 2 hundreds, 3 tens, and 4 units.

- The 2 hundreds leave **2** stray units (1 each).
- The 3 tens leave **3** stray units.
- The 4 units are **4** stray units.

In total, the leftover stray pieces equal $2 + 3 + 4 = 9$ units: **the sum of the digits**! And 9 divides evenly by both 9 and 3. Therefore, 234 is divisible by both 9 and 3. (Check: $234 = 9 \times 26$.)

📐 Divisibility Rules for 3 and 9

- A number is divisible by **3** if the **sum of its digits**

is a multiple of 3.
- A number is divisible by **9** if the **sum of its digits** is a multiple of 9.

Worked Example 6 — The Merchant's Dates

3,471: $3 + 4 + 7 + 1 = 15$. 15 is in the 3-times table (3×5), so **3,471 is divisible by 3**: the merchant can pack his bags without any leftovers! ($3\,471 = 3 \times 1\,157$.) However, 15 is not in the 9-times table: 3,471 is **not** divisible by 9.

Worked Example 7 — When the Sum Is Still Large

98,769: $9 + 8 + 7 + 6 + 9 = 39$. Is 39 a multiple of 3? Repeat the trick: $3 + 9 = 12$, and $12 = 3 \times 4$. Yes! Therefore 98,769 is divisible by 3. By 9? 39 is not in the 9-times table ($9 \times 4 = 36$, $9 \times 5 = 45$): no.

⚠️ Pitfall Warning

- For 3 and 9, **never look only at the last digit**: 23 ends in 3

but is not divisible by 3 ($2 + 3 = 5$). 27 ends in 7 and is divisible by 3 ($2 + 7 = 9$).
- **Divisible by 9 $\Longrightarrow$ divisible by 3**, because $9 = 3 \times 3$. But the **reverse is false**: 12 is divisible by 3, but not by 9.
- **Divisible by 2 and by 5 $\Longrightarrow$ divisible by 10** (the last digit must be even AND be 0 or 5: it must be 0).
- An **odd** number is never divisible by 10, nor by 4, nor by 6 (because all these divisors are even).

🖐 ***Your Turn — The Detective Game*** *One player calls out a large number. The other player must announce as fast as possible which numbers among 2, 3, 5, 9, and 10 divide it evenly. Each correct answer scores one point; a mistake loses a point.*

---

5. Prime Numbers: The Atoms of Numbers

Let us return to our stubborn troop of soldiers. The number 11 has only **two** divisors: 1 and 11. It cannot form any true rectangle.

📐 Definition — Prime Number

A **prime number** is a whole number greater than or equal to 2 that has **exactly two distinct divisors**: 1 and itself. A whole number (greater than 1) that has more than two divisors is called **composite**: it can be composed into a rectangle.

The word "prime" comes from the Latin *primus*: "first, original, that which comes before all else". These are numbers that exist **before** all others, because every other number is built from them, as you will see.

Let us inspect the first few whole numbers:

Number	Possible Rectangles	Divisors	Prime?
1	1×1	1	**No** (only one divisor)
2	1×2	1, 2	Yes
3	1×3	1, 3	Yes
4	1×4, 2×2	1, 2, 4	No
5	1×5	1, 5	Yes
6	1×6, 2×3	1, 2, 3, 6	No
7	1×7	1, 7	Yes
8	1×8, 2×4	1, 2, 4, 8	No
9	1×9, 3×3	1, 3, 9	No
10	1×10, 2×5	1, 2, 5, 10	No
11	1×11	1, 11	Yes

⚠ Pitfall Warning — The Special Cases of 1 and 2

- **1 is not prime.** It has only **one** divisor (itself),

whereas the definition strictly demands exactly two. Mathematicians deliberately excluded 1: otherwise, the fundamental law "every number breaks down in a unique way into prime numbers" would collapse (one could multiply by 1 as many times as desired without changing the value).

- **2 is the only even prime number.** Every other even number is divisible by 2, meaning it has at least three divisors: 1, 2, and itself.
- **Odd does not mean prime!** 9, 15, 21, 25, 27 are odd… and composite ($9 = 3 \times 3$, $15 = 3 \times 5$, $21 = 3 \times 7$, etc.). Beware especially of **51** ($= 3 \times 17$) and **91** ($= 7 \times 13$): they look prime at first glance, but they are impostors!

Breaking a Number Down to Its Atoms: The Factor Tree

Take any composite number, for example **60**. Break it into two factors, then break down each factor as long as possible:

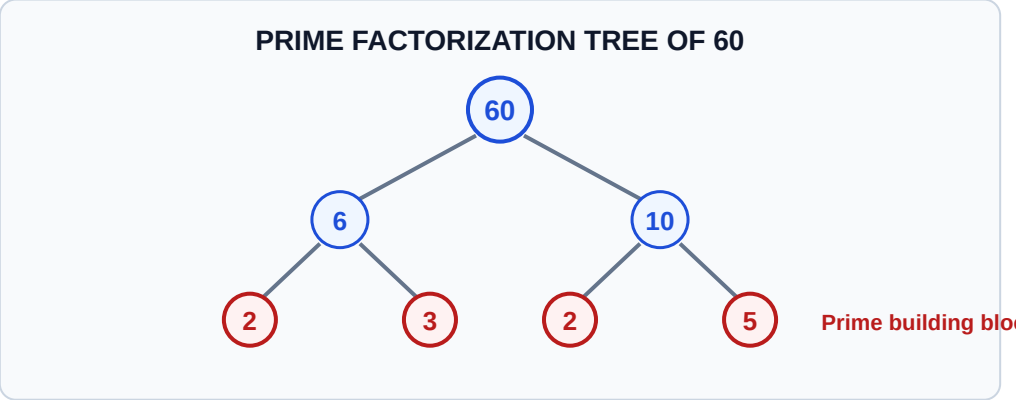

Figure 37: Prime factorization tree

$$60 = 2 \times 2 \times 3 \times 5$$

The numbers multiplied together are called **factors**. Here, all factors are prime numbers: we say we have **factored 60 into a product of prime factors**.

Worked Example 8 — Another Path for 60

What if we had started with $60 = 4 \times 15$? $4 = 2 \times 2$ and $15 = 3 \times 5$. We still obtain $60 = 2 \times 2 \times 3 \times 5$! No matter which path you take, **you always arrive at the exact same prime atoms** (only the order of factors might change). This is a profound law of arithmetic, already known to the ancient Greek mathematician **Euclid**, which you will prove formally in high school.

This is why prime numbers are called **the atoms of numbers**: just as all physical objects are built of atoms, **every whole number greater than 1 is uniquely built of prime numbers multiplied together**.

CHAPTER 9: The Feast Remainders and the Game Teams

> "To share is to distribute justice turn by turn. But when objects cannot be broken into pieces, what do we do with what remains in the center of the table? It is within the remainder that the whole truth of division lies hidden."

What You Are About to Discover

- Why division arises from two everyday questions: **fair sharing** (how many does each person get?) and **grouping** (how many full groups can we form?).
- The **four actors** of Euclidean division: the *dividend*, the *divisor*, the *quotient*, and the *remainder*.
- The **fundamental equation** $a = b \times q + r$ and the **inviolable golden rule**: the remainder must always be strictly less than the divisor ($r < b$).
- The **step-by-step long division method**, explained clearly without panic.
- How to **verify** your calculation in a single glance using multiplication.

---

1. The Problem: The Fourteen Sturgeons of the Great River

Morning mist floats over the slow waters of the great river. The village's heavy fishing dugout pulls up against the wooden logs of the landing dock. The fishermen leap into the cool water and heave a dripping net onto the bank: they have caught **14 large golden sturgeons**, glistening in the morning sun.

Four fishing families gather in the central square around a massive flat stone table. Each family has sent its representative. The 14 fish are lined up across the table, their eyes bright and scales sparkling.

The village elder steps forward with a solemn gaze: *"These fish must feed our children. But take heed: every family must receive exactly the same share. Not one bone more, not one scale less! Otherwise, jealousy will tear our village apart."*

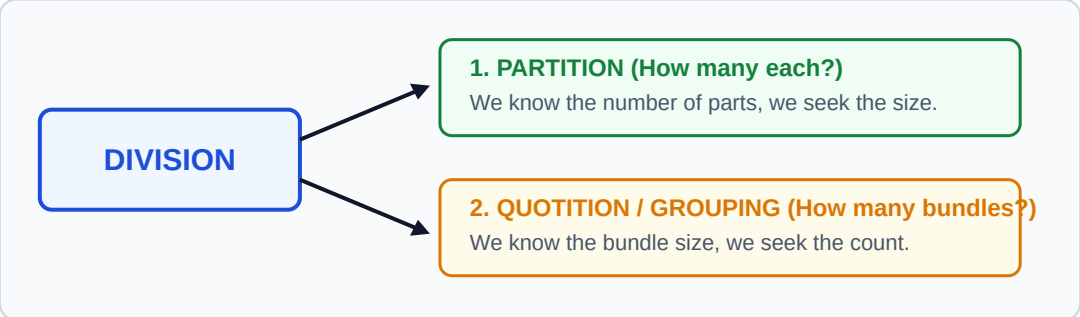

Figure 38: The two meanings of division: fair sharing and grouping

You are tasked with carrying out the division. How can you proceed so that no injustice is done?

You do not plunge into complicated formulas. You place your hands on the fish and perform a universal gesture—the same one children use when dealing cards in a game: **you distribute one fish at a time, round after round**.

Look at the table after the third round:

- Each family has received **3 large fish**. Everyone is treated completely equally.
- But in the center of the stone, **2 whole fish** remain.

Can you deal a fourth round? No! If you give one fish to Family A and one to Family B, Families C and D will protest loudly: *"Why them and not us? That is unjust!"*

Because these fish are fresh and must be salted whole to be preserved, they cannot be sliced into pieces today. Sharing whole objects has reached its limit.

You have just carried out a fundamental operation in mathematics: **Euclidean division** (integer division with remainder), named in honor of the ancient Greek scholar **Euclid**, who taught in Alexandria more than 2,300 years ago.

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2. The Four Actors of Sharing: The Anatomy of Division

Let us give precise names to the four quantities that played a part in this distribution:

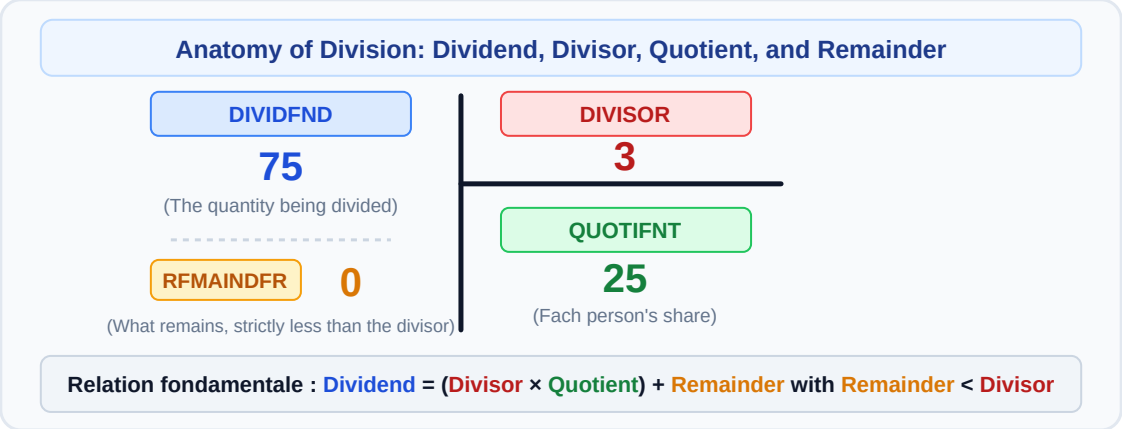

Figure 39: Anatomy of the Euclidean long division bracket

> **📐 Definition — The Four Roles of Euclidean Division**
>
> When we share an integer a into b equal parts: 1. The **dividend** (a) is the total quantity of items to be shared. 2. The **divisor** (b) is the number of equal shares to form (it can never be 0!). 3. The **quotient** (q) is the fair share received by each recipient. 4. The **remainder** (r) is what is left unshared at the end.

These four numbers are bound together by an essential equation that every mathematician keeps in mind:

$$a = (b \times q) + r$$

We read aloud: "**Dividend equals divisor times quotient plus remainder.**"

The Inviolable Golden Rule: The Condition on the Remainder

Imagine a clumsy distributor who says: *"I shared the 14 fish among the 4 families. I gave 2 fish to each, and I have 6 fish left on the table!"*

Look at the corresponding equality:

$$14 = (4 \times 2) + 6$$

In pure arithmetic, the sum is correct: $4 \times 2 = 8$, and $8 + 6 = 14$. Yet this distribution is **wrong**! Why? Because with the 6 fish left on the table, there were easily enough fish to give one more to each of the 4 families ($6 \geq 4$). The distributor simply stopped too early out of laziness!

> **⚠️ Pitfall Warning — The Inviolable Golden Rule**
>
> In any Euclidean division, **the remainder must ALWAYS be strictly less than the divisor**:
>
> $$r < b$$
>
> And naturally, the remainder cannot be negative: $0 \leq r < b$.
>
> If you divide by 4, the only possible remainders are: **0, 1, 2, or 3**. The moment your remainder is equal to or greater than 4, your quotient is too small: you can still distribute more!

When the remainder is **0**, we say that **the division has no remainder** (it divides evenly). The dividend is then an exact multiple of the divisor (as we explored in Chapter 8).

Worked Example 1 — Identifying the Four Actors

In the division of 37 by 5:

- In the 5-times table, find the largest multiple less than or equal to 37: $5 \times 7 = 35$, and $5 \times 8 = 40$ (too large).
- Thus: $37 = (5 \times 7) + 2$.
- The dividend is $a = 37$.
- The divisor is $b = 5$.
- The quotient is $q = 7$.
- The remainder is $r = 2$.
- Checking the golden rule: $2 < 5$. The condition is perfectly satisfied!

---

3. Two Meanings for One Operation: Sharing or Grouping?

Division actually answers two very distinct types of real-world questions. Learning to tell them apart is essential for solving word problems clearly.

Meaning 1: Fair Sharing (Dealing Cards)

- **What is known:** the total number of items (a) and the number of recipients (b).
- **What we seek:** the size of the share received by each person (q).
- *Example:* 35 candies shared among 4 children. Each child receives 8 candies, and 3 remain in the bag.

Meaning 2: Grouping or Capacity (Packing Boxes)

- **What is known:** the total number of items (a) and the capacity of one box (b).
- **What we seek:** how many full boxes can be packed (q).
- *Example:* A baker bakes 35 chocolate eggs. He packs them into boxes of 4 eggs. How many full boxes can he seal? He can seal 8 full boxes, leaving 3 eggs left over.

Worked Example 2 — Sharing or Grouping?

Situation A: A teacher has 28 notebooks for 7 students. How many notebooks does each student receive? **Answer:** This is **sharing** (we seek each student's share). $28 = 7 \times 4 + 0$. Each student receives 4 notebooks.

Situation B: A florist has 28 roses. She arranges them into bouquets of 6 roses. How many bouquets can she make? **Answer:** This is **grouping** (we package by groups of 6). $28 = 6 \times 4 + 4$. She makes 4 complete bouquets, and 4 roses remain.

---

4. Step-by-Step Long Division: Calm and Methodical

When numbers grow large, we can no longer distribute pebbles or fish one by one. We need a reliable mechanical method: **long division**.

Why this layout? Because it enables us to divide the number **chunk by chunk**, starting from the left (the highest place-value: hundreds, then tens, then units). It is the only standard arithmetic algorithm that works from left to right!

Detailed Case Study: Calculating $75 \div 3$

We wish to share 75 gold coins among 3 treasure chests. 75 is **7 tens and 5 units**.

```
 7 5  |  3
      |-------
      |
```

Step 1: Divide the tens. - Place your finger on the 7. In 7 tens, how many times does 3 fit? - In the 3-times table: $3 \times 2 = 6 \leq 7$. We can put **2** tens into each chest. - Write 2 in the quotient. - Subtract the distributed tens: $7 - 6 = 1$. There is **1 ten unshared**.

```
  7 5  |  3
       |
- 6   |-------
----  |  2
  1   |
```

Step 2: Bring down the next digit. - 1 ten alone cannot be shared among 3 chests. - Exchange it for 10 units and bring down the 5 beside it. This yields **15 units**.

```
  7 5  |  3
       |
- 6   |-------
----  |  2 5
  1 5 |
```

Step 3: Divide the units. - In 15 units, how many times does 3 fit? - In the 3-times table: $3 \times 5 = 15$ exactly! - Write 5 in the quotient, to the right of 2. - Subtract the distributed units: $15 - 15 = 0$.

```
  7 5  |  3
       |
- 6   |-------
----  |  2 5
  1 5 |
      |
- 1 5 |
-----  |
    0  |
```

Result: The quotient is **25**, and the remainder is **0**. Each chest receives 25 gold coins. The division divides evenly!

A Richer Case: Calculating $473 \div 4$

Let us see what happens when digits have non-zero remainders at each stage.

```
4 7 3  |  4
       |-------
       |
```

1. **Hundreds:** In 4, how many times 4? Exactly **1** time ($1 \times 4 = 4$). Remainder: $4 - 4 = 0$.
2. **Bring down 7 (tens):** In 7, how many times 4? Exactly **1** time ($1 \times 4 = 4$). Remainder: $7 - 4 = 3$.
3. **Bring down 3 (units):** We get 33. In 33, how many times 4? $4 \times 8 = 32$. It fits **8** times. Remainder: $33 - 32 = 1$.

```
  4 7 3  |  4
         |
- 4     |-------
----    |  1 1 8
  0 7   |
        |
-   4   |
-----   |
    3 3 |
        |
  - 3 2 |
  -----  |
      1  |
```

Result: $473 = (4 \times 118) + 1$. The quotient is 118, and the remainder is 1. The condition $1 < 4$ is fully respected!

🧭 Method — The 4 Magic Moves of Long Division

For each column in division, we repeat the same four moves: 1. **D**ivide (how many times does the divisor fit?) 2. **M**ultiply (calculate what was just distributed: $q \times b$) 3. **S**ubtract (find the remainder) 4. **B**ring down (drop down the next digit to form a new working number)

Remember the magic acronym: **D - M - S - B** (*Divide, Multiply, Subtract, Bring down*)!

---

5. Checking Your Division: The Foolproof Verification

In mathematics, you never finish a calculation without verifying your work. To check a Euclidean division, we perform the inverse operation: **we retrace our steps using multiplication**.

Worked Example 3 — Verifying $473 = 4 \times 118 + 1$

1.

Multiply the quotient by the divisor:

$$118 \times 4 = (100 \times 4) + (10 \times 4) + (8 \times 4) = 400 + 40 + 32 = 472$$

2. Add the remainder:

$$472 + 1 = 473$$

3. Do we get back the original dividend? **Yes, exactly 473!** The calculation is certified error-free.

⚠️ Pitfall Warning — The Trap of Zero in the Quotient

This is the most common mistake made by fourth-grade students! Let us divide $615 \div 3$:

- In 6, how many times 3? 2 times ($2 \times 3 = 6$). Remainder 0.
- Bring down the 1. In 1, how many times 3? **It fits 0 times!**
- Many students forget to write **0** in the quotient and immediately bring down the 5. They end up with a quotient of 25 instead of **205**! But $3 \times 25 = 75$, which is nowhere near 615! With the zero in place, $3 \times 205 = 615$. **Whenever you bring down a digit, you MUST write a digit in the quotient, even if that digit is zero!**

---

6. Additional Worked Examples

Worked Example 4 — Bags of Marbles

A child has 146 marbles. He wants to pack them into bags of 6 marbles each. *Question:* How many full bags will he get, and how many marbles will remain?

- In 1: 6 does not fit. Take 14.
- In 14, how many times 6? $6 \times 2 = 12$. Remainder: $14 - 12 = 2$.
- Bring down the 6: we have 26.
- In 26, how many times 6? $6 \times 4 = 24$. Remainder: $26 - 24 = 2$.
- Euclidean form: $146 = (6 \times 24) + 2$. **Conclusion:** He fills **24 full bags** and has **2 marbles left over**.

Worked Example 5 — Minibuses for Field Trips

83 people are going on a field trip. Each minibus can carry at most 9 passengers. *Question:* What is the minimum number of minibuses required to transport everyone?

- Perform the Euclidean division of 83 by 9: $83 = (9 \times 9) + 2$ (since $9 \times 9 = 81$).
- The quotient is 9, the remainder is 2.
- 9 minibuses will transport $9 \times 9 = 81$ people.
- But 2 people would be left standing on the curb! Will we leave them behind? Of course not! **Conclusion:** We must order **10 minibuses** (9 full minibuses and 1 minibus with only 2 passengers). *This example shows why we must always consider the concrete meaning of a problem: sometimes the answer is the quotient, and sometimes it is quotient + 1!*

5bis. The Challenge of Great Merchants: Two-Digit Divisors

In fourth and fifth grade, real-world problems often require dividing by a two-digit number, such as distributing $7,845 \text{ liters of oil}$ into barrels holding 23 liters.

Faced with a divisor like 23, no one has memorized the "23-times table"! Seasoned calculators use a foolproof tool: **the scratchpad mini-table**.

Calculating $7,845 \div 23$ Step by Step:

1. **The scratchpad mini-table:** Quickly calculate the first few multiples of 23:
   - $23 \times 1 = 23$
   - $23 \times 2 = 46$
   - $23 \times 3 = 69$
   - $23 \times 4 = 92$
2. **Set up long division:**
   - In 7, 23 does not fit. Take **78**.
   - In our mini-table: $23 \times 3 = 69 \leq 78$. Place **3** in the quotient.
   - Subtraction: $78 - 69 = 9$.
   - Bring down 4: we get **94**.
   - In our mini-table: $23 \times 4 = 92 \leq 94$. Place **4** in the quotient.
   - Subtraction: $94 - 92 = 2$.
   - Bring down 5: we get **25**.
   - In 25, how many times 23? It fits **1** time ($1 \times 23 = 23$). Place **1** in the quotient.
   - Final subtraction: $25 - 23 = 2$.

```
  7 8 4 5  |  2 3
           |
- 6 9      |-------
-----      |  3 4 1
   9 4     |
           |
  - 9 2    |
  -----    |
     2 5   |
           |
    - 2 3  |
    -----  |
       2   |
```

Result: $7,845 = (23 \times 341) + 2$ with $2 < 23$. Everything is validated!

🛠️ The Boundary Case Laboratory: What Happens with ZERO in Division?

(This section is isolated to analyze extreme edge cases without disrupting your standard calculations.)

Case A: Zero Divided by a Number ($0 \div 5$)

- **The scenario:** You have a completely empty basket (0 apples), and 5 children sit at the table. How many apples does each child receive?
- Each child receives **0 apples**.
- Verification: $5 \times 0 = 0$. Everything works smoothly!
- **Rule:** Zero divided by any non-zero number always equals **zero**:

$$\frac{0}{b} = 0 \quad (\text{for all} \quad b \neq 0)$$

Case B: The Absolute Prohibition: Dividing by ZERO ($5 \div 0$)

- Imagine trying to calculate $5 \div 0 = q$.
- If such a quotient existed, our verification formula would require:

$$0 \times q = 5$$

- But we established that **zero multiplied by any number is ALWAYS zero** ($0 \times q = 0$)! It is strictly impossible for it to equal 5.
- No number in the entire universe can satisfy this equation.
- **Absolute Rule of Mathematics: WE NEVER DIVIDE BY ZERO.** It is a mathematical black hole that destroys the meaning of equality.

⚡ The Practitioner's Drills: Timed Fluency Practice

Find the integer quotient q and remainder r mentally or on paper as swiftly as possible:

Challenge	Division	Quotient q	Remainder r	Verification ($a = bq + r$)
1	$37 \div 5$	.....	.....	.....
2	$50 \div 6$	.....	.....	.....
3	$72 \div 8$	.....	.....	.....
4	$43 \div 7$	.....	.....	.....
5	$0 \div 9$	.....	.....	.....
6	$612 \div 3$	.....	.....	.....
7	$85 \div 4$	.....	.....	.....
8	$105 \div 10$	.....	.....	.....
9	$490 \div 7$	.....	.....	.....
10	$150 \div 25$	.....	.....	.....

What You Need to Remember

1. **Euclidean division** is used to divide a whole quantity into equal parts or to count how many full groups can be formed.

2. **The 4 actors:**

$$\text{Dividend} = (\text{Divisor} \times \text{Quotient}) + \text{Remainder}$$

$$a = (b \times q) + r$$

3. **The mandatory golden rule:** The remainder must always be strictly less than the divisor:

$$0 \leq r < b$$

4. **Long division layout:** We process digits from left to right following the cycle: *Divide* → *Multiply* → *Subtract* → *Bring down*.
5. **Zero in the quotient:** Whenever you bring down a digit and cannot form a group, you must write a `0` in the quotient before bringing down the next digit.
6. **Verification:** Always verify using $(b \times q) + r$.

Exercises

Level (Direct Application)

1. Write the Euclidean equation $a = b \times q + r$ matching this statement: “When I share 29 cookies among 4 friends, each receives 7 and 1 remains.”
2. In the division of 43 by 6, the quotient is 7 and the remainder is 1. What is the dividend? What is the divisor?
3. Which of the following equalities represent a valid Euclidean division?
   a) $34 = 5 \times 6 + 4$
   b) $34 = 5 \times 5 + 9$
   c) $50 = 8 \times 6 + 2$
   d) $50 = 8 \times 5 + 10$
4. What are all the possible remainders when dividing by 6?
5. Calculate mentally (give quotient and remainder):
   a) $23 \div 5$
   b) $31 \div 4$
   c) $47 \div 9$
   d) $58 \div 8$
6. Set up and calculate: $84 \div 4$.
7. Set up and calculate: $95 \div 3$.

Level (Reflection & Technique)

8. Set up and calculate: $734 \div 5$. Verify your result using multiplication.
9. Watch out for zero! Set up and calculate: $816 \div 4$.
10. A florist received 135 tulips. She wants to make bouquets of 8 tulips each.
    a) How many full bouquets can she sell?
    b) How many more tulips does she need to make one additional bouquet?
11. A school has 172 students going to a museum. Each coach bus has 50 seats. How many buses must be booked?
12. In a division, the divisor is 7, the quotient is 13, and the remainder is 4. What was the dividend?
13. Solve the mystery number: “I am between 40 and 50. If you divide me by 6, the remainder is 3. If you divide me by 7, the remainder is 3. Who am I?”

Level (Expert Challenges)

14. A treasure of 925 gold coins must be shared equally among 8 sailors. The captain keeps the remainder for himself.
    a) How many coins does each sailor receive?
    b) How many coins does the captain pocket?
    c) Could the captain have pocketed 9 coins? Justify your answer.
15. A cyclist rides on a circular running track that is 400 meters long. He rides a total of 3,500 meters.
    a) How many full laps did he complete?
    b) How far from the starting line does he stop?

16. *The Calendar Riddle:* Today is Tuesday. What day of the week will it be in exactly 100 days? (Think about the divisor 7!).

Solutions

1. $29 = (4 \times 7) + 1$. **2.** The dividend is $a = 43$, the divisor is $b = 6$ (quotient $q = 7$, remainder $r = 1$). **3.** - a) **Yes**, because the remainder 4 is strictly less than the divisor 5 ($4 < 5$). - b) **No**, because the remainder 9 is greater than the divisor 5 ($9 \geq 5$). Another group could have been distributed: $34 = 5 \times 6 + 4$. - c) **Yes**, because $2 < 8$. - d) **No**, because the remainder 10 is greater than the divisor 8 ($10 \geq 8$).

4. The possible remainders are all whole numbers strictly less than 6: **0, 1, 2, 3, 4, and 5**. **5.** - a) $23 = 5 \times 4 + 3 \Longrightarrow q = 4, r = 3$. - b) $31 = 4 \times 7 + 3 \Longrightarrow q = 7, r = 3$. - c) $47 = 9 \times 5 + 2 \Longrightarrow q = 5, r = 2$. - d) $58 = 8 \times 7 + 2 \Longrightarrow q = 7, r = 2$.

6. In 8: $2 \times 4 = 8$ (remainder 0). Bring down 4: in 4, $1 \times 4 = 4$ (remainder 0). **Quotient = 21, Remainder = 0**. ($84 = 4 \times 21$). **7.** In 9: $3 \times 3 = 9$ (remainder 0). Bring down 5: in 5, $1 \times 3 = 3$ (remainder 2). **Quotient = 31, Remainder = 2**. ($95 = 3 \times 31 + 2$). **8.** - In 7: $1 \times 5 = 5$, remainder 2. - Bring down $3 \rightarrow 23$: $4 \times 5 = 20$, remainder 3. - Bring down $4 \rightarrow 34$: $6 \times 5 = 30$, remainder 4. - **Quotient = 146, Remainder = 4** ($4 < 5$). - *Check:* $146 \times 5 = 730$. And $730 + 4 = 734$. Perfect!

9. - In 8: $2 \times 4 = 8$, remainder 0. - Bring down 1: in 1, 4 fits **0 times**! Write **0** in the quotient. - Bring down $6 \rightarrow 16$: in 16, $4 \times 4 = 16$, remainder 0. - **Quotient = 204, Remainder = 0**. ($204 \times 4 = 816$).

10. a) $135 = 8 \times 16 + 7$. She can sell **16 full bouquets** (leaving 7 tulips). b) Since she has 7 tulips left and each bouquet requires 8, she needs only **1 more tulip** ($8 - 7 = 1$) to form another bouquet.

11. $172 = 50 \times 3 + 22$. 3 buses are not enough because 22 students would be left on the sidewalk. Therefore, **4 buses** must be booked. **12.** Apply the formula: $a = (b \times q) + r = (7 \times 13) + 4 = 91 + 4 = 95$. The dividend was **95**. **13.** Multiples of 6 plus 3 between 40 and 50: $6 \times 7 + 3 = 42 + 3 = \mathbf{45}$. Multiples of 7 plus 3: $7 \times 6 + 3 = 42 + 3 = \mathbf{45}$. The mystery number is **45**. **14.** a) $925 \div 8$:

- In 9: $1 \times 8 = 8$, remainder 1.
- Bring down $2 \rightarrow 12$: $1 \times 8 = 8$, remainder 4.
- Bring down $5 \rightarrow 45$: $5 \times 8 = 40$, remainder 5.

Each sailor receives **115 coins**. b) The captain pockets the remainder: **5 coins**. c) **No**, that is impossible! According to the golden rule, the remainder must always be strictly less than the divisor ($r < 8$). The maximum possible remainder is 7 coins. If 9 coins remained, the sailors would demand another coin each!

15. a) Division of 3,500 by 400: $3,500 = (400 \times 8) + 300$. He completed **8 full laps**. b) The remainder is 300: he stops **300 meters** past the starting line.

16. A week has 7 days. Divide 100 by 7: $100 = (7 \times 14) + 2$. 14 full weeks elapse (returning us to Tuesday). Then **2 extra days** remain. Two days after Tuesday: Wednesday (+1), then **Thursday** (+2). In 100 days, it will be **Thursday**!

Solutions to the Practitioner's Scales:

1. $37 = 5 \times \mathbf{7} + \mathbf{2}$ ($q = 7, r = 2$); 2. $50 = 6 \times \mathbf{8} + \mathbf{2}$ ($q = 8, r = 2$); 3. $72 = 8 \times \mathbf{9} + \mathbf{0}$ ($q = 9, r = 0$); 4. $43 = 7 \times \mathbf{6} + \mathbf{1}$ ($q = 6, r = 1$); 5. $0 = 9 \times \mathbf{0} + \mathbf{0}$ ($q = 0, r = 0$); 6. $612 = 3 \times \mathbf{204} + \mathbf{0}$ ($q = 204, r = 0$); 7. $85 = 4 \times \mathbf{21} + \mathbf{1}$ ($q = 21, r = 1$); 8. $105 = 10 \times \mathbf{10} + \mathbf{5}$ ($q = 10, r = 5$); 9. $490 = 7 \times \mathbf{70} + \mathbf{0}$ ($q = 70, r = 0$); 10. $150 = 25 \times \mathbf{6} + \mathbf{0}$ ($q = 6, r = 0$).

The Question that Opens What Lies Ahead

Look once more at our four fishing families and the 14 fish from the start of the chapter. Each family departed with 3 magnificent whole sturgeons. And **2 large fish** remained on the stone table.

The families are still hungry. The children look at these two remaining fish with longing. The village elder declares: *“These two fish must not be wasted! Since we cannot distribute them whole, we will draw the sharp knife and slice them into equal parts!”*

But when we break a whole object into pieces, how do we name these parts? How do we write with numbers a part that is smaller than 1? That is the great leap into the next chapter: **The Broken Galette and the Measuring Sieve**!

CHAPTER 10: The Broken Galette and the Measuring Sieve

> "Until now, our numbers counted whole, solid objects: pebbles, sheep, gold coins. But the real world is made of bread that is broken, cloth that is cut, and time that flows in fractions of hours. To measure the fractured world, the human mind had to invent the fraction."

What You Are About to Discover

- Why whole numbers are no longer enough when we must share things that do not divide evenly.
- The two-story anatomy of a fraction: the **denominator** (which names the size of the parts) and the **numerator** (which counts how many parts are taken).
- How to tell at a glance whether a fraction is **less than**, **equal to**, or **greater than 1**.
- How to decompose a large fraction into **whole units plus a remaining fractional part** (such as $\frac{7}{3} = 2 + \frac{1}{3}$).
- How to place fractions accurately on a **graduated number line**.
- Why $\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$: the secret of **equivalent fractions** revealed through paper folding.
- How to **add and subtract** fractions sharing the same denominator without making mistakes.

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1. The Problem: The Baker's Barley Galette

The mouth-watering aroma of warm, golden bread drifts from the mouth of the communal clay-brick oven. The baker pulls out a magnificent flat barley galette onto a large wooden board—round like a golden disk, steaming and crispy. Around the table, **four children** have bellies rumbling with anticipation. There is only one single galette for four hungry diners.

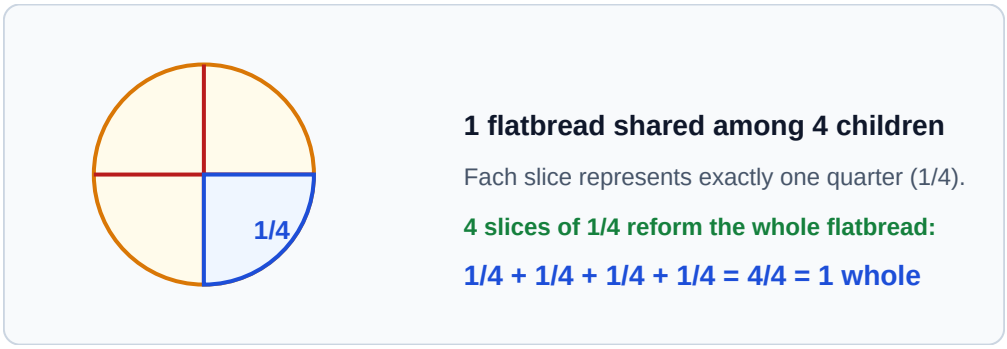

Figure 40: Sharing the flatbread: fractions of a disk

If the baker applied Euclidean division from the previous chapter:

$$1 = (4 \times 0) + 1$$

The quotient is 0, and the remainder is 1! The children would receive 0 galettes each, and the whole galette would sit on the table growing cold. That would be absurd and sorrowful!

The baker smiles, draws a sharp steel knife, and rests the blade right at the center of the galette: 1. *First straight cut:* the blade slices cleanly from edge to edge. Behold: **two identical half-galettes**. 2. *Second perpendicular cut:* the blade cuts across at a right angle. The galette is now divided into **four strictly equal quarters**.

Each child holds out a hand and receives their share. How much did each child receive? They did not receive 0 galettes. Nor did they receive 1 whole galette. They received **one part coming from a whole cut into 4 equal pieces**.

To record this new kind of quantity, humans invented a two-story notation, separated by a horizontal bar:

$$\frac{1}{4}$$

This notation is read: **“one fourth”** (or **“one quarter”**). The word comes from the Latin *fractio*, meaning “the act of breaking into pieces”. You have just stepped into the world of **fractions**.

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2. The Anatomy of a Fraction: The Sieve and the Harvest

A fraction is not an ordinary number resting on a line. It is an alliance between two integers playing two very distinct roles:

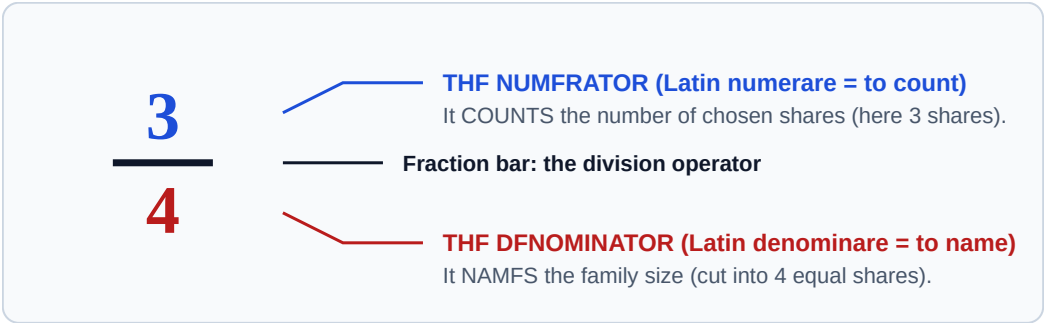

Figure 41: Anatomy of a fraction: role of numerator and denominator

Definition — Numerator and Denominator

In a fraction written as $\frac{a}{b}$ (with $b \neq 0$):

- The **denominator** (b, at the bottom) indicates how many **equal** parts the whole unit has been divided into. It is the size of the “sieve”.
- The **numerator** (a, at the top) indicates the number of parts chosen, eaten, or shaded. It is the “harvest”.

How to Read a Fraction Aloud?

- The denominator is read using ordinal numbers: $\frac{1}{5}$ is “one fifth”, $\frac{3}{7}$ is “three sevenths”, $\frac{7}{10}$ is “seven tenths”.
- Pay attention to the **historical exceptions**:
  - $\frac{1}{2}$ is read “one **half**”;
  - $\frac{1}{3}$ is read “one **third**”;
  - $\frac{1}{4}$ is read “one **quarter**” (or “one fourth”).

The Denominator’s Paradox: The Larger It Is, the Smaller the Piece!

Ask yourself this vital question: *Would you rather receive $\frac{1}{3}$ of an apple pie or $\frac{1}{8}$ of that exact same pie?*
A careless glance might suggest: *“8 is bigger than 3, so $\frac{1}{8}$ must be bigger!”* **That is a catastrophic mistake for a dessert lover!**

Think about the physical act of sharing:

- If you share a pie among **3 people** ($\frac{1}{3}$), you only need three slices: they are thick and hearty.
- If you must share the same pie among **8 people** ($\frac{1}{8}$), you have to make many more knife cuts: each slice becomes thin as paper!

$$\frac{1}{3} > \frac{1}{8}$$

⚠️ Pitfall Warning — The Denominator Rule

For fractions sharing the **same numerator** (for instance 1 on top): **The larger the denominator, the smaller the individual piece!**

$$\frac{1}{2} > \frac{1}{3} > \frac{1}{4} > \frac{1}{5} > \frac{1}{10} > \frac{1}{100}$$

💡 True History — The Scribes of Ancient Egypt (The Rhind Papyrus)

More than 3,500 years ago, the pharaohs' scribes almost exclusively recorded fractions with a **1** in the numerator ($\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{10}$). To express a quantity like $\frac{3}{4}$, they decomposed it into a sum of unit fractions: $\frac{1}{2} + \frac{1}{4}$. The famous **Rhind Papyrus** (preserved in the British Museum) contains dozens of division tables designed to fairly allocate bread and beer among the workers building the pyramids.

Figure 42: The Rhind Mathematical Papyrus (~1550 BC, Ancient Egypt): scribes' unit fraction decomposition tables

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3. Comparing a Fraction to One: Smaller, Equal, or Greater?

Look at a box of 6 chocolates. The whole unit is the complete box (6 chocolates).

💡 Discovery — The Complete Unit as a Fraction $6/6$

- o | o | o | o | o | o] <– 1 full box = 6/6

1. **If you eat 4 chocolates:** you have eaten $\frac{4}{6}$ of the box. The numerator (4) is smaller than the denominator (6). You have eaten **less than one whole box**: $\frac{4}{6} < 1$.
2. **If you eat all 6 chocolates:** you have eaten $\frac{6}{6}$ of the box. The numerator equals the denominator. You have eaten **exactly 1 whole box**: $\frac{6}{6} = 1$.

3. **If you are ravenous and eat 11 chocolates** (by opening a second box): You have eaten $\frac{11}{6}$ of a box. The numerator (11) is larger than the denominator (6). You have eaten **more than one whole box**: $\frac{11}{6} > 1$.

📐 Property — Position Relative to 1

For any fraction $\frac{a}{b}$:

- If the numerator is strictly less than the denominator ($a < b$), then $\frac{a}{b} < 1$.
- If the numerator is equal to the denominator ($a = b$), then $\frac{a}{b} = 1$.
- If the numerator is strictly greater than the denominator ($a > b$), then $\frac{a}{b} > 1$.

Worked Example 1 — Comparing to 1

Let us compare the following fractions to 1: $\frac{3}{8}$, $\frac{7}{7}$, $\frac{9}{4}$, $\frac{5}{6}$.

- For $\frac{3}{8}$: $3 < 8$, so $\frac{3}{8} < 1$.
- For $\frac{7}{7}$: $7 = 7$, so $\frac{7}{7} = 1$.
- For $\frac{9}{4}$: $9 > 4$, so $\frac{9}{4} > 1$.
- For $\frac{5}{6}$: $5 < 6$, so $\frac{5}{6} < 1$.

---

4. Decomposing a Fraction: The Bridge Between Division and Fractions

When a fraction is larger than 1, it hides complete whole units inside! Consider the fraction $\frac{7}{3}$ (seven thirds of a galette). Each whole galette contains 3 thirds ($\frac{3}{3} = 1$). How many whole galettes can we assemble with 7 thirds?
Recall Euclidean division of 7 by 3:

$$7 = (3 \times 2) + 1$$

In 7, there are **2 groups of 3**, with **1 left over**. This means that with 7 thirds, we can build **2 full galettes**, with **1 third remaining**!

$$\frac{7}{3} = \frac{3}{3} + \frac{3}{3} + \frac{1}{3} = 1 + 1 + \frac{1}{3} = 2 + \frac{1}{3}$$

🧭 Method — Writing a Fraction as “Whole Number + Fraction < 1”

To decompose an improper fraction $\frac{a}{b}$: 1. Carry out the Euclidean division of a by b: find quotient q and remainder r ($a = b \times q + r$). 2. Write directly:

$$\frac{a}{b} = q + \frac{r}{b}$$

Since $r < b$, the fraction $\frac{r}{b}$ is guaranteed to be strictly less than 1!

Worked Example 2 — Decomposing an Improper Fraction

Decompose $\frac{23}{5}$.

- Division of 23 by 5: $23 = (5 \times 4) + 3$ (quotient 4, remainder 3).
- We therefore have: $\frac{23}{5} = 4 + \frac{3}{5}$. This shows that $\frac{23}{5}$ lies strictly between 4 and 5: $4 < \frac{23}{5} < 5$.

---

5. Placing Fractions on a Graduated Number Line

A graduated number line is a measuring path. To place fractions upon it: 1. Identify the unit interval between 0 and 1 (and between 1 and 2, 2 and 3, etc.). 2. Look at the denominator: it indicates the number of equal sub-intervals to create within each whole unit. 3. Count steps forward according to the numerator.

Let us draw a line where each unit is partitioned into **4 equal parts** (quarters):

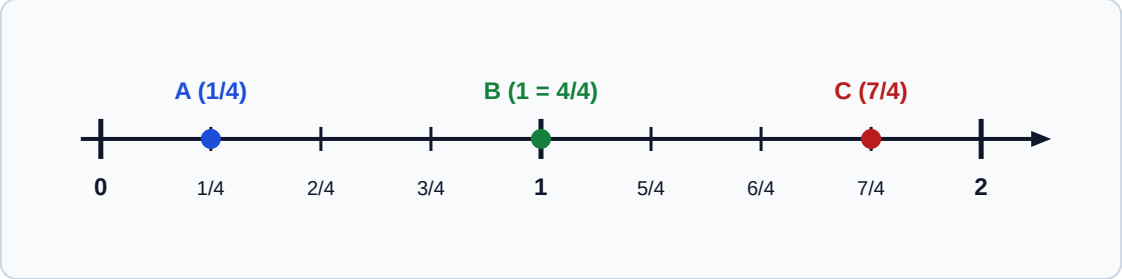

Figure 43: The number line graduated in fourths

- Point A is located 1 quarter from the start: it represents $\frac{1}{4}$.
- Point B is located right on 1: it represents $\frac{4}{4} = 1$.
- Point C is located 7 quarters from the start: it represents $\frac{7}{4} = 1 + \frac{3}{4}$.

Worked Example 3 — Reading and Placing on the Line

Question: On a number line where each unit is divided into 5 equal parts (fifths), where does $\frac{8}{5}$ lie? *Answer:* Start at 0 and move forward 8 tick marks of size $\frac{1}{5}$. Since $\frac{8}{5} = 1 + \frac{3}{5}$, the point is located between 1 and 2, exactly 3 ticks past 1.

---

6. Equivalent Fractions: The Secret of Paper Folding

Take a strip of white paper. Fold it exactly in half. Color one half blue. You have colored **1 part out of 2**: that is $\frac{1}{2}$. Now, without cutting anything or adding any ink, fold the folded strip in half once more. Unfold it. What do you notice? The whole strip is now divided into **4 equal sections**. And the blue part now contains **2 sections out of 4**!

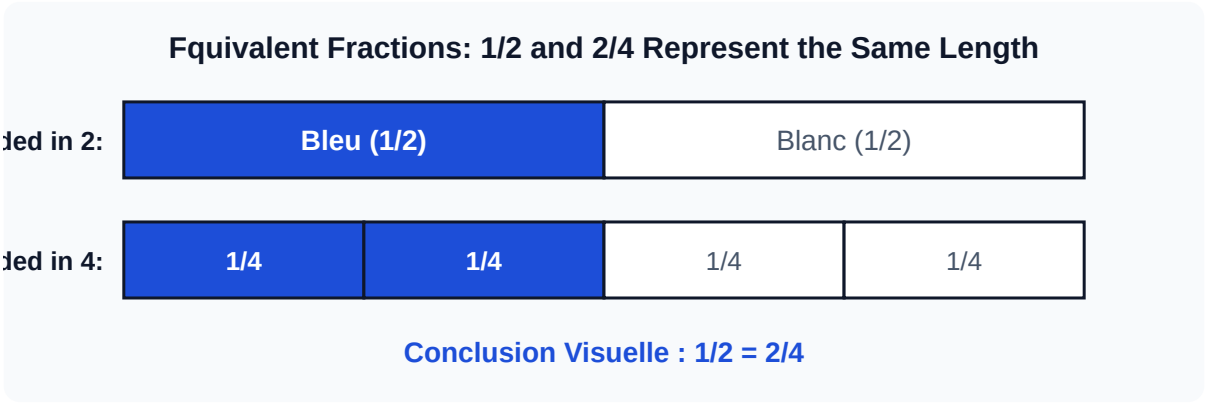

Figure 44: Equivalent fractions by strip division: 1/2 = 2/4

Did the amount of blue ink increase? No! Is the strip any longer? No! Only the fineness of the division changed. You multiplied the total number of slices by 2 (at the bottom), but you also multiplied the number of blue slices by 2 (at the top):

$$\frac{1}{2} = \frac{1 \times 2}{2 \times 2} = \frac{2}{4}$$

If you fold the strip in half a third time, you obtain:

$$\frac{2}{4} = \frac{2 \times 2}{4 \times 2} = \frac{4}{8}$$

All these fractions describe the exact **same quantity of matter**. We call them **equivalent fractions** (or equal fractions).

📐 The Golden Rule of Equivalent Fractions

You do not alter the value of a fraction if you **multiply** (or **divide**) both its numerator and denominator by the **same non-zero whole number**:

$$\frac{a}{b} = \frac{a \times k}{b \times k} \quad \text{and} \quad \frac{a}{b} = \frac{a \div k}{b \div k}$$

Worked Example 4 — Generating Equivalent Fractions

- Find a fraction equal to $\frac{3}{5}$ with a denominator of 20: To go from

5 to 20, we multiply by 4 ($5 \times 4 = 20$). We must also multiply the top by 4: $3 \times 4 = 12$.

$$\frac{3}{5} = \frac{3 \times 4}{5 \times 4} = \frac{12}{20}$$

- Simplify $\frac{6}{8}$: 6 and 8 are both divisible by 2:

$$\frac{6}{8} = \frac{6 \div 2}{8 \div 2} = \frac{3}{4}$$

---

7. Adding and Subtracting Fractions with the Same Denominator

Imagine a large cake cut into 8 equal slices (eighths).

- At lunch, Leo eats 2 slices: he eats $\frac{2}{8}$ of the cake.
- At afternoon snack, his sister Sarah eats 3 slices: she eats $\frac{3}{8}$ of the same cake.

How much cake did they eat altogether? They ate $2 \text{ slices} + 3 \text{ slices} = 5 \text{ slices}$ of size “eighth”. Therefore, they ate $\frac{5}{8}$ of the cake!

$$\frac{2}{8} + \frac{3}{8} = \frac{2+3}{8} = \frac{5}{8}$$

Solved Example — Dividing the Cake into Eighths

- Cake cut into 8 slices:
- Leo | Leo | Sarah | Sarah | Sarah | | |]
- 2/8 —> <——- 3/8 ——->
- 5/8 ————>

📐 Rule — Addition and Subtraction with the Same Denominator

To add (or subtract) two fractions that have the **same denominator**: 1. **Keep the same denominator** (do not touch the bottom!). 2. **Add (or subtract) the numerators** (compute only on the top).

$$\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c} \quad \text{and} \quad \frac{a}{c} - \frac{b}{c} = \frac{a-b}{c}$$

⚠ Pitfall Warning — The Forbidden Mistake

Never add denominators together!

$$\frac{1}{4}+\frac{2}{4}\neq\frac{1+2}{4+4}=\frac{3}{8}\quad\text{TOTALLY FALSE!}$$

If you combine 1 quarter of a pie and 2 quarters of a pie, you get 3 quarters of a pie ($\frac{3}{4}$), not 3 eighths! The size of the slice does not shrink when you place several on the same plate.

Worked Example 5 — Simple Calculations

- $A=\frac{3}{7}+\frac{2}{7}=\frac{3+2}{7}=\frac{5}{7}$
- $B=\frac{9}{10}-\frac{4}{10}=\frac{9-4}{10}=\frac{5}{10}=\frac{1}{2}$
- $C=1-\frac{3}{5}$: Rewrite the integer 1 as a fraction with denominator 5: $1=\frac{5}{5}$.

$$C=\frac{5}{5}-\frac{3}{5}=\frac{5-3}{5}=\frac{2}{5}$$

What You Need to Remember

1. **A fraction** $\frac{a}{b}$ represents an equal division: the bottom (b, denominator) gives the number of equal parts in the whole; the top (a, numerator) counts the parts taken.
2. **Vocabulary:** $\frac{1}{2}$ (one half), $\frac{1}{3}$ (one third), $\frac{1}{4}$ (one quarter / fourth), $\frac{a}{b}$ (a b-ths).
3. **Comparison to 1:**
   - $a<b\Longrightarrow\frac{a}{b}<1$
   - $a=b\Longrightarrow\frac{a}{b}=1$
   - $a>b\Longrightarrow\frac{a}{b}>1$
4. **Decomposition:** For $\frac{a}{b}>1$, perform Euclidean division $a=b\times q+r$, yielding $\frac{a}{b}=q+\frac{r}{b}$ (a whole number plus a proper fraction).
5. **Equivalent fractions:** Multiplying or dividing both top and bottom by the same non-zero number leaves the fraction's value unchanged ($\frac{1}{2}=\frac{2}{4}=\frac{5}{10}$).
6. **Addition / Subtraction:** With identical denominators, compute only across the numerators: $\frac{a}{c}+\frac{b}{c}=\frac{a+b}{c}$.

Exercises

Level (Direct Application)

1. Write each as a numerical fraction:
   a) three fifths
   b) seven eighths
   c) one half
   d) four thirds
2. In the fraction $\frac{5}{9}$:
   a) What is the numerator? What does it concretely represent?
   b) What is the denominator? What does it concretely represent?
3. State whether each fraction is less than 1, equal to 1, or greater than 1:

$$\frac{4}{7},\quad\frac{9}{9},\quad\frac{12}{5},\quad\frac{1}{10},\quad\frac{8}{3},\quad\frac{15}{15}$$

4. Arrange in ascending order (from smallest to largest):

$$\frac{1}{9},\quad \frac{1}{2},\quad \frac{1}{5},\quad \frac{1}{100},\quad \frac{1}{3}$$

5. Fill in the missing values to create equal fractions:
   a) $\frac{1}{3} = \frac{\dots}{6}$
   b) $\frac{2}{5} = \frac{\dots}{10}$
   c) $\frac{3}{4} = \frac{9}{\dots}$
   d) $\frac{10}{20} = \frac{1}{\dots}$
6. Compute each sum and difference:
   a) $\frac{2}{7} + \frac{3}{7}$
   b) $\frac{5}{9} - \frac{1}{9}$
   c) $\frac{4}{12} + \frac{7}{12}$
   d) $\frac{8}{10} - \frac{5}{10}$

Level (Reflection & Geometry)

7. On a graduated line, the unit interval is divided into 6 equal segments.
   a) Which fractions correspond to the points located 1 tick, 5 ticks, and 8 ticks from origin 0?
   b) Decompose the third point's fraction into "integer + fraction".
8. Decompose the following fractions into a whole number plus a proper fraction less than 1 (as in $\frac{7}{3} = 2 + \frac{1}{3}$):
   a) $\frac{11}{4}$
   b) $\frac{17}{5}$
   c) $\frac{26}{8}$
   d) $\frac{31}{6}$
9. Compute:
   a) $1 - \frac{2}{3}$
   b) $1 - \frac{5}{8}$
   c) $2 + \frac{3}{4}$ (write the final result as a single fraction)
10. A group of 5 friends orders pizzas cut into 8 slices each. In total, they eat 22 slices.
    a) What fraction of a pizza did they eat in total?
    b) How many whole pizzas were entirely consumed?
    c) How many slices remain on the final opened pizza?
11. True or False? Justify your answer in each case:
    a) $\frac{3}{4}$ is greater than $\frac{5}{4}$.
    b) $\frac{2}{3}$ is equal to $\frac{6}{9}$.
    c) $\frac{1}{2} + \frac{1}{2} = 1$.
    d) $\frac{3}{5} + \frac{1}{5} = \frac{4}{10}$.

Level (Surveyors' & Merchants' Challenges)

12. Three children each receive an identical chocolate bar made of 24 squares.
    - Manon eats $\frac{1}{2}$ of her bar.
    - Arthur eats $\frac{3}{8}$ of his bar.
    - Chloe eats $\frac{5}{12}$ of her bar. How many squares of chocolate did each child eat? Who ate the most?
13. Bound each fraction between two consecutive whole numbers:

$$\dots < \frac{14}{3} < \dots \qquad \dots < \frac{37}{5} < \dots \qquad \dots < \frac{49}{8} < \dots$$

14. Nadia runs a 1,200-meter cross-country trail.
    - At the first checkpoint, she has completed $\frac{2}{6}$ of the course.
    - At the second checkpoint, she has completed $\frac{5}{6}$ of the course.
    a) What fraction of the course remains to be run at the second checkpoint?
    b) How many meters does this remaining distance represent?

15. *The Magic Ribbon Challenge:* A ribbon measures 1 meter in length. We cut it into two equal halves. We take one of the two pieces and cut it in half again. We repeat this process two more times. What fraction of the original ribbon does the smallest resulting piece measure?

Solutions

1. a) $\frac{3}{5}$; b) $\frac{7}{8}$; c) $\frac{1}{2}$; d) $\frac{4}{3}$. **2.** a) The numerator is 5: it indicates that 5 parts have been taken. b) The denominator is 9: it indicates that the unit was divided into 9 equal parts. **3.** - Less than 1: $\frac{4}{7}$, $\frac{1}{10}$ (since the top is smaller than the bottom). - Equal to 1: $\frac{9}{9}$, $\frac{15}{15}$ (since top = bottom). - Greater than 1: $\frac{12}{5}$, $\frac{8}{3}$ (since the top is larger than the bottom).

4. Same numerator (1): the larger the denominator, the smaller the fraction.

$$\frac{1}{100} < \frac{1}{9} < \frac{1}{5} < \frac{1}{3} < \frac{1}{2}$$

5. a) $\frac{1}{3} = \frac{1\times2}{3\times2} = \mathbf{\frac{2}{6}}$. b) $\frac{2}{5} = \frac{2\times2}{5\times2} = \mathbf{\frac{4}{10}}$. c) $\frac{3}{4} = \frac{3\times3}{4\times3} = \mathbf{\frac{9}{12}}$. d) $\frac{10}{20} = \frac{10\div10}{20\div10} = \mathbf{\frac{1}{2}}$.

6. a) $\frac{2+3}{7} = \frac{5}{7}$; b) $\frac{5-1}{9} = \frac{4}{9}$; c) $\frac{4+7}{12} = \frac{11}{12}$; d) $\frac{8-5}{10} = \frac{3}{10}$.

7. a) 1 tick = $\frac{1}{6}$; 5 ticks = $\frac{5}{6}$; 8 ticks = $\frac{8}{6}$. b) $\frac{8}{6} = \frac{6}{6} + \frac{2}{6} = 1 + \frac{2}{6}$ (or $1 + \frac{1}{3}$).

8. a) $11 = 4 \times 2 + 3 \Longrightarrow \frac{11}{4} = 2 + \frac{3}{4}$. b) $17 = 5 \times 3 + 2 \Longrightarrow \frac{17}{5} = 3 + \frac{2}{5}$. c) $26 = 8 \times 3 + 2 \Longrightarrow \frac{26}{8} = 3 + \frac{2}{8}$ (or $3 + \frac{1}{4}$). d) $31 = 6 \times 5 + 1 \Longrightarrow \frac{31}{6} = 5 + \frac{1}{6}$.

9. a) $1 - \frac{2}{3} = \frac{3}{3} - \frac{2}{3} = \frac{1}{3}$. b) $1 - \frac{5}{8} = \frac{8}{8} - \frac{5}{8} = \frac{3}{8}$. c) $2 + \frac{3}{4} = \frac{8}{4} + \frac{3}{4} = \frac{11}{4}$.

10. a) One pizza = 8 slices. They ate $\mathbf{\frac{22}{8}}$ of a pizza. b) $22 = 8 \times 2 + 6 \Longrightarrow \frac{22}{8} = 2 + \frac{6}{8}$. Therefore **2 whole pizzas** were completely eaten. c) On the third pizza, they ate 6 out of 8 slices. Thus $8 - 6 =$ **2 slices** remain.

11. a) **False**, because $3 < 5$ with the same denominator, so $\frac{3}{4} < \frac{5}{4}$. b) **True**, because $\frac{2\times3}{3\times3} = \frac{6}{9}$. c) **True**, because $\frac{1+1}{2} = \frac{2}{2} = 1$. d) **False**, we never add denominators! $\frac{3}{5} + \frac{1}{5} = \frac{4}{5}$, not $\frac{4}{10}$.

12. - Manon: $\frac{1}{2}$ of 24 squares = $24 \div 2 =$ **12 squares**. - Arthur: 1 eighth is $24 \div 8 = 3$ squares. Thus 3 eighths are $3 \times 3 =$ **9 squares**. - Chloe: 1 twelfth is $24 \div 12 = 2$ squares. Thus 5 twelfths are $5 \times 2 =$ **10 squares**. **Manon** ate the most (12 squares).

13. - $14 = 3 \times 4 + 2 \Longrightarrow 4 < \frac{14}{3} < 5$. - $37 = 5 \times 7 + 2 \Longrightarrow 7 < \frac{37}{5} < 8$. - $49 = 8 \times 6 + 1 \Longrightarrow 6 < \frac{49}{8} < 7$.

14. a) The whole trail is $\frac{6}{6}$. At the second stop, she completed $\frac{5}{6}$. There remains $\frac{6}{6} - \frac{5}{6} = \frac{1}{6}$ of the trail. b) The full course is 1,200 m. $\frac{1}{6}$ of 1,200 m is $1,200 \div 6 =$ **200 meters**.

15. - 1st cut in half: pieces of $\frac{1}{2}$. - 2nd cut in half: pieces of $\frac{1}{4}$ (since $\frac{1}{2} \div 2 = \frac{1}{4}$). - 3rd cut in half: pieces of $\frac{1}{8}$. - 4th cut in half: pieces of $\frac{1}{16}$.

The smallest piece measures $\mathbf{\frac{1}{16}}$ **of a meter** (or 6.25 cm).

The Question that Opens What Lies Ahead

Thanks to fractions, you now know how to cut any galette into 3, 4, 7, or 12 equal parts. Yet Renaissance merchants and engineers faced a persistent, frustrating headache: To compare $\frac{3}{7}$ and $\frac{4}{9}$, or to add mixed thirds, fourths, and fifths, they constantly had to search for awkward common denominators.

In 1585, a brilliant Flemish engineer and mathematician named **Simon Stevin** cried out: *"Why burden ourselves with all these disparate denominators? Our number system already counts in bundles of 10: units, tens, hundreds. Let us simply extend this very scale below the number 1: tenths, hundredths, thousandths!"*

To separate the realm of whole items from the realm of decimal crumbs, all that was needed was a single magic punctuation mark: **the decimal point** (or comma). Prepare to discover decimal numbers in **Chapter 11: The Tenths of Simon Stevin**!

CHAPTER 11: The Tenths of Simon Stevin

> "Everyone admires the perfection of our number system that counts by tens, hundreds, and thousands. Why stop at the unit as if standing before an insurmountable wall? By crossing this threshold into tenths and hundredths, all measurements become as simple as whole numbers." — *Adapted from Simon Stevin,* The Decimal *(*De Thiende*, 1585).*

What You Are About to Discover

- The incredible stroke of genius by Flemish engineer **Simon Stevin** in 1585.
- What **decimal fractions** are ($\frac{1}{10}$, $\frac{1}{100}$, $\frac{1}{1\,000}$) and how they extend the place value chart.
- The role of **the decimal point** (or comma): the watertight border between whole units and decimal fractions.
- The classic pitfall: understanding why **3.2 is greater than 3.18**, even though 18 is greater than 2!
- **Bounding and interpolating** decimals: discovering that between any two numbers, there are always infinitely many other numbers.
- How to set up **addition and subtraction** of decimals by lining up decimal points like buttons on a shirt.
- How to multiply by 10, 100, and 1,000 in the blink of an eye by shifting place value ranks.

---

1. The Problem: The Nightmare of the Merchants of Bruges

The year is 1585 in the bustling Flemish city of Bruges. The canals are crowded with barges laden with grain, spices, barrels of beer, and rolls of fine Flemish cloth.

On the quays, merchants and tax collectors are tearing their hair out. To calculate the total cost of a shipment of wheat, a merchant must add:

- 14 whole florins,
- plus $\frac{3}{7}$ of a florin for cart transport,
- plus $\frac{5}{12}$ of a florin for the canal bridge toll,
- minus $\frac{1}{6}$ of a florin for a trade discount!

Every trader uses fractions with completely different denominators: thirds, sevenths, twelfths, sixteenths. They constantly have to find common denominators, convert fractions, simplify… Calculations drag on for hours, errors multiply, and bitter arguments erupt across the marketplace.

One man observes this chaos with calm clarity. His name is **Simon Stevin** (1548–1620). He is an engineer, mathematician, and quartermaster general of the Dutch army.

> 💡 **Discovery & History — Simon Stevin (1548—1620)**
>
> *"Why make complicated what can be simple? We count by tens, hundreds, and thousands. Let us do the exact same thing BELOW the unit!"*

Stevin asks an illuminating question: *"When we write 111, the first 1 represents 100, the second represents 10, and the third represents 1. Each rank is 10 times smaller than the one before it. Why stop abruptly at 1? Let us continue! The next rank will be 10 times smaller than 1, and the one after that 10 times smaller still!"*

In 1585, Stevin publishes a slim 36-page booklet written in Flemish entitled *De Thiende* ("The Tenth"), translated into French as *La Disme* and English as *The Art of Tenths* or *The Decimal*. This marks the official birth of **decimal numbers**.

---

2. From Sliced Cake to Place Value: Decimal Fractions

To understand the journey below the number 1, imagine a long measuring strip of length 1 meter.

MULTIPLICATION BY 10, 100, 1000: DFCIMAL POINT SHIFT

3,45 × 10	➔ shifts 1 rank to the right ➔	34,5
3,45 × 100	➔ shifts 2 ranks to the right ➔	345
3,45 × 1 000	➔ shifts 3 ranks to the right ➔	3 450

Figure 45: Operational shifting of the decimal point

1. **The Tenths:** Divide this meter into **10 equal sections**. Each piece is a decimal fraction:
$$\frac{1}{10} \text{ of a meter (one tenth of a meter, or } 1 \text{ decimeter)}$$
It takes 10 tenths to make 1 whole unit: $\frac{10}{10} = 1$.
2. **The Hundredths:** Take one of these tenths and divide it into **10 tiny sub-pieces**. Since there were 10 tenths, the whole meter is now partitioned into $10 \times 10 = 100$ small pieces! Each tiny piece is called a **hundredth**:
$$\frac{1}{100} \text{ of a meter (one hundredth of a meter, or } 1 \text{ centimeter)}$$
It takes 10 hundredths to build 1 tenth ($\frac{10}{100} = \frac{1}{10}$). And it takes 100 hundredths to form 1 whole unit ($\frac{100}{100} = 1$).
3. **The Thousandths:** Divide each hundredth into 10 even finer dust-like specks. The entire unit is now cut into $100 \times 10 = 1\,000$ parts! Each speck is called a **thousandth**:
$$\frac{1}{1\,000} \text{ of a meter (one thousandth of a meter, or } 1 \text{ millimeter)}$$

> **Definition — Decimal Fraction**
>
> A **decimal fraction** is any fraction whose denominator is $10, 100, 1\,000, 10\,000\ldots$ (a power of 10).

---

3. The Decimal Point: The Border Guard of Place Value

How can we write these fractions without drawing two-story fraction bars every time? Look at our grand place-value chart extended to the right:

…	Hundreds (h)	Tens (t)	Units (u)	.	Tenths ($\frac{1}{10}$)	Hundredths ($\frac{1}{100}$)	Thou-sandths ($\frac{1}{1\,000}$)
	100	10	1	.	0.1	0.01	0.001
			3	.	**4**	**7**	

Look at this number: **3.47**. It consists of two distinct territories:

📐 Definition — Decimal Number, Whole Part, and Decimal Part

A **decimal number** is a number that can be expressed with a finite number of digits after the decimal point.

- The **whole-number part** is located to the left of the decimal point (units, tens, hundreds…).
- The **decimal part** is located to the right of the decimal point (tenths, hundredths, thousandths…).
- The **decimal point** (or comma) is the border marking where the single units column ends.

How to Read 3.47 Aloud?

- In casual speech: people often say "three point four seven".
- But the precise mathematical reading is: "**3 units and 47 hundredths**" or "**3 units, 4 tenths, and 7 hundredths**".

$$3.47 = 3 + \frac{4}{10} + \frac{7}{100} = 3 + \frac{47}{100} = \frac{347}{100}$$

💡 Did You Know?

Simon Stevin did not yet use our modern decimal point! He drew tiny circled numbers above the digits: $3\text{ \textcircled{0} }4\text{ \textcircled{1} }7\text{ \textcircled{2} }$ to designate whole units, tenths, and hundredths. It was the Scottish mathematician **John Napier** (the inventor of logarithms) who popularized the decimal point and comma in 1617. Today, English-speaking countries use a dot (3.47), while France and many European nations use a decimal comma ($3,47$). The mathematical meaning is identical!

---

4. The Great Pitfall: Why 3.2 Is Greater Than 3.18!

This is the number-one pitfall in elementary school mathematics—one that almost every student stumbles into at least once!

Look at these two numbers:

$$3.2 \quad \text{and} \quad 3.18$$

An untrained eye glances at the decimal part and thinks: *"18 is bigger than 2, so 3.18 must be bigger than 3.2!"* **That is completely wrong!**

Why? Because you are comparing slices of completely different sizes!

- In 3.2, the 2 represents **2 tenths** of a cake (2 large, thick slices).
- In 3.18, the 1 represents **1 tenth** and the 8 represents **8 tiny hundredths**.

To compare fairly, we must express both in the same unit of measure (hundredths). Add a **trailing zero** at the end of 3.2:

$$3.2 = 3.20 \quad (3 \text{ whole units and } 20 \text{ hundredths})$$

Now compare:

- 3.20 is 3 cakes and **20 hundredths**.
- 3.18 is 3 cakes and **18 hundredths**.

Since $20 > 18$, it is crystal clear that:

$$\mathbf{3.2 > 3.18}$$

🧭 Method — Comparing Two Decimal Numbers Without Error

1.

First compare the **whole-number part**: $7.1 > 5.99$ because $7 > 5$. (No need to even look past the decimal point!). 2. If the whole parts are equal, compare the **tenths digit**: In $3.\mathbf{2}$ and $3.1\mathbf{8}$, the tenths digits are 2 versus 1. Since $2 > 1$, it follows that $3.2 > 3.18$. 3. If the tenths are equal, compare the **hundredths**, and so on. *Foolproof tip:* Pad the numbers with trailing zeros on the right so both have the same number of decimal places!

> ⚠️ **Pitfall Warning — Meaningful vs. Superfluous Zeros**
>
> - **Superfluous zeros:** You can freely add or remove zeros at the very end of the decimal part without changing the number's value:
>
> $$4.5 = 4.50 = 4.500$$
>
> - **Indispensable zeros:** A zero trapped between the decimal point and another non-zero digit can NEVER be removed:
>
> $$4.05 \neq 4.5$$
>
> 4.05 is 4 units and 5 hundredths. 4.5 is 4 units and 5 tenths (50 hundredths)! That is 10 times more crumb!

---

5. Interpolation and Bounding: The Infinite World Between Two Numbers

With whole numbers, between 3 and 4, there is **nothing**. No integer can slip between them. With decimals, the world becomes infinitely rich!

Look at a ruler between 3 and 4:

Between 3 and 4, we have interpolated 9 numbers at the tenths level: $3.1, 3.2, 3.3...$

And what happens between 3.1 and 3.2? Write them in hundredths: 3.10 and 3.20. Between 3.10 and 3.20, we can slip:

$$3.11 \quad ; \quad 3.12 \quad ; \quad 3.15 \quad ; \quad 3.19$$

And between 3.14 and 3.15? We can write 3.140 and 3.150, and insert thousandths like 3.145!

> 💡 **Did You Know?**
>
> Between any two decimal numbers, no matter how close they seem, you can ALWAYS find another decimal number (in fact, infinitely many!). There is never an empty gap that cannot be filled by zooming in further.

Worked Example 1 — Bounding and Interpolation

- *Bound 4.72 between consecutive whole numbers:* $4 < 4.72 < 5$.
- *Bound 4.72 to the nearest tenth:* $4.7 < 4.72 < 4.8$ (since $4.70 < 4.72 < 4.80$).
- *Insert a decimal between 6.3 and 6.4:* Think of 6.30 and 6.40. We can choose 6.35 (or 6.31 or 6.38).

---

6. Column Addition and Subtraction: The Shirt Buttons

To add or subtract decimal numbers, the golden rule is simple and absolute: **You must vertically align the decimal points directly under one another.**

Think of the decimal point as the line of buttons on a shirt: if one button is off by a single hole, the entire shirt sits crooked!

Worked Example: Calculate $43.7 + 8.56$

```
    t   u . tenths  hundredths
    1   1                 <-- carry digits
    4   3 .    7      0   <-- we pad with a 0 to balance columns
+       8 .    5      6
 ----------------------
    5   2 .    2      6
```

1. Align the decimal points vertically.

2. Add a trailing zero to 43.7 so both numbers have two decimal digits: 43.70.
3. Add column by column from right to left:
   - Hundredths: $0 + 6 = 6$.
   - Tenths: $7 + 5 = 12 \Longrightarrow$ write 2, carry 1 to the units column.
   - **Immediately place the decimal point in the answer, perfectly aligned!**
   - Units: $1 \quad (\text{carry}) + 3 + 8 = 12 \Longrightarrow$ write 2, carry 1 to the tens.
   - Tens: $1 \quad (\text{carry}) + 4 = 5$.
4. **Result:** 52.26.

Worked Example: Calculate $35.2 - 7.64$

```
      t   u . tenths  hundredths
      2  14     11      10   <-- borrowing / regrouping
      3   5 .    2       0   <-- this zero is MANDATORY!

-         7 .    6       4
-----------------------------
      2   7 .    5       6
```

> ⚠️ **Pitfall Warning — The Mandatory Zero in Subtraction**
>
> In $35.2 - 7.64$, you cannot simply say "nothing minus 4 is 4"! That is a grave error. You must write $35.\mathbf{20}$. Then calculate: 0 hundredths minus 4 hundredths cannot be done directly $\Longrightarrow$ regroup 1 tenth into 10 hundredths, and $10 - 4 = 6$!

---

7. Multiplying by 10, 100, and 1,000: The Great Place-Value Shift

What happens when you multiply a decimal number by 10? Remember: each unit becomes a ten, each tenth becomes a unit, and each hundredth becomes a tenth! Every digit jumps one place-value column to the left. On paper, this creates the visual appearance that **the decimal point shifts to the right**.

Worked Example 2 — Mental Fluency

- $0.6 \times 10 = 6$
- $2.38 \times 100 = 238$
- $4.7 \times 100 = 470$ (shift one place to make 47, then append a zero for the second place!)
- $0.005 \times 1\,000 = 5$

What You Need to Remember

1. **A decimal number** consists of a whole-number part and a decimal part separated by a decimal point.
2. **Place value:**
   - 1st digit after the point = **tenths** ($\frac{1}{10} = 0.1$);
   - 2nd digit = **hundredths** ($\frac{1}{100} = 0.01$);
   - 3rd digit = **thousandths** ($\frac{1}{1\,000} = 0.001$).
3. **Comparing decimals:** Always look at the whole-number part first. If equal, compare tenths, then hundredths. Pad with trailing zeros to level columns ($3.2 = 3.20 > 3.18$).

4. **Column addition/subtraction:** Line up decimal points vertically without fail. Remember to pad top numbers with zeros in subtraction when needed ($5 - 2.3 = 5.0 - 2.3$).
5. **Multiplying by 10, 100, 1,000:** Shift the decimal point 1, 2, or 3 places to the right, adding zeros as necessary.

Exercises

Level (Direct Application)

1. Write each decimal fraction in standard decimal notation:
   a) $\frac{3}{10}$
   b) $\frac{45}{100}$
   c) $\frac{7}{100}$
   d) $\frac{128}{10}$
   e) $\frac{2\,456}{1\,000}$
2. In the number 84.375:
   a) What is the tens digit?
   b) What is the tenths digit?
   c) What is the hundredths digit?
   d) What is the thousandths digit?
3. Cross out any superfluous zeros:
   $$045.20 \qquad 3.08 \qquad 7.500 \qquad 0.040 \qquad 120.0$$
4. Insert $<$, $>$, or $=$:
   a) $5.4 \ldots 5.19$
   b) $8.30 \ldots 8.3$
   c) $12.08 \ldots 12.8$
   d) $0.75 \ldots 0.705$
   e) $9.99 \ldots 10.1$
5. Calculate mentally:
   a) 4.3×10
   b) 0.75×100
   c) 12.8×100
   d) $0.006 \times 1\,000$
   e) 3.25×10

Level (Operations and Bounds)

6. Set up and calculate:
   a) $24.6 + 8.75$
   b) $135.08 + 48.9$
   c) $7 + 14.35$
7. Set up and calculate:
   a) $45.8 - 12.35$
   b) $82.4 - 37.86$
   c) $15 - 4.75$
8. Bound each decimal between consecutive whole numbers:
   $\ldots < 7.38 < \ldots$ $\quad \ldots < 0.84 < \ldots$ $\quad \ldots < 29.99 < \ldots$
9. Bound each decimal to the nearest tenth:
   $\ldots < 5.47 < \ldots$ $\quad \ldots < 12.83 < \ldots$ $\quad \ldots < 0.39 < \ldots$
10. Insert a decimal number between each pair:
    a) Between 4.6 and 4.7
    b) Between 8.15 and 8.16

c) Between 0.9 and 1

11. Arrange in ascending order:

$$4.25 \quad ; \quad 4.2 \quad ; \quad 4.05 \quad ; \quad 4.205 \quad ; \quad 4.5$$

Level (Real-World Word Problems)

12. To make a theater costume, a tailor buys:
    - a velvet remnant for $\$24.50$,
    - gold trim for $\$8.75$,
    - a box of buttons for $\$3.80$. She pays with a $\$50$ bill.

    a) What is the total cost of her purchases?

    b) How much change does the merchant return to her?

13. A runner is training for a marathon. On Monday, he runs $12.4 \ \text{km}$. On Wednesday, he runs $3.85 \ \text{km}$ more than on Monday. On Friday, he runs $15.2 \ \text{km}$. What total distance did he cover during the week?

14. A roll of wire fencing measures $25 \ \text{meters}$. A gardener cuts off $7.45 \ \text{m}$ to fence a vegetable garden, and another piece of $11.8 \ \text{m}$ to protect raspberry bushes. How much fencing remains on the roll?

15. *The Mystery of the Erased Bill:* An addition on a slate was partially wiped away by rain:

$$2\,?\,.4 + 6.7\,?\, = 35.18$$

    Find the two missing digits represented by the question marks.

16. *The Paper Stack Challenge:* A sheet of paper has a thickness of $0.1 \ \text{mm}$ (one tenth of a millimeter). If you stack 100 identical sheets, what is the height of the stack in millimeters? What is it in centimeters?

Solutions

1. a) 0.3; b) 0.45; c) 0.07; d) 12.8; e) 2.456. **2.** a) Tens: **8**; b) Tenths: **3**; c) Hundredths: **7**; d) Thousandths: **5**. **3.** - $045.20 \to \mathbf{45.2}$ - $3.08 \to$ **no superfluous zeros** (the zero is trapped) - $7.500 \to \mathbf{7.5}$ - $0.040 \to \mathbf{0.04}$ (only the final trailing zero is superfluous) - $120.0 \to \mathbf{120}$

4. a) $5.40 > 5.19 \implies 5.4 > 5.19$. b) $8.30 = 8.3$. c) $12.08 < 12.80 \implies 12.08 < 12.8$. d) $0.750 > 0.705 \implies 0.75 > 0.705$. e) $9 < 10 \implies 9.99 < 10.1$.

5. a) 43; b) 75; c) 1 280; d) 6; e) 32.5. **6.** a) $24.60 + 8.75 = \mathbf{33.35}$. b) $135.08 + 48.90 = \mathbf{183.98}$. c) $7.00 + 14.35 = \mathbf{21.35}$. **7.** a) $45.80 - 12.35 = \mathbf{33.45}$. b) $82.40 - 37.86 = \mathbf{44.54}$. c) $15.00 - 4.75 = \mathbf{10.25}$. **8.** - $7 < 7.38 < 8$ - $0 < 0.84 < 1$ - $29 < 29.99 < 30$

9. - $5.4 < 5.47 < 5.5$ (since $5.40 < 5.47 < 5.50$) - $12.8 < 12.83 < 12.9$ - $0.3 < 0.39 < 0.4$

10. a) Between 4.60 and 4.70: for example $\mathbf{4.65}$. b) Between 8.150 and 8.160: for example $\mathbf{8.155}$. c) Between 0.90 and 1.00: for example $\mathbf{0.95}$.

11. Align in thousandths: 4.250; 4.200; 4.050; 4.205; 4.500.

$$\mathbf{4.05 < 4.2 < 4.205 < 4.25 < 4.5}$$

12. a) Total cost: $24.50 + 8.75 + 3.80 = \mathbf{\$37.05}$. b) Change returned: $50.00 - 37.05 = \mathbf{\$12.95}$.

13. - Monday: $12.4 \ \text{km}$. - Wednesday: $12.4 + 3.85 = 16.25 \ \text{km}$. - Friday: $15.2 \ \text{km}$. - Total distance: $12.40 + 16.25 + 15.20 = \mathbf{43.85 \ km}$.

14. Total length cut: $7.45 + 11.80 = 19.25 \ \text{m}$. Remaining length: $25.00 - 19.25 = \mathbf{5.75 \ meters}$.

15. Analyze by column:

- Hundredths: $0 + ? = 8 \implies$ bottom hundredths digit is **8** (giving 6.78).
- Tenths: $4 + 7 = 11 \implies$ write 1, carry 1.
- Units: $1 \ (\text{carry}) + ? + 6 = 15 \implies ? = 8$. The first number is 28.4.
- Tens: $1 \ (\text{carry}) + 2 = 3$.

The full addition is $\mathbf{28.4 + 6.78 = 35.18}$. The missing digits were **8** and **8**.

16. - Height in mm: $0.1 \times 100 = \mathbf{10 \ mm}$. - Since $10 \ \text{mm} = 1 \ \text{cm}$, the stack measures exactly $\mathbf{1 \ centimeter}$ tall!

The Question that Opens What Lies Ahead

Thanks to Simon Stevin's decimals, humanity gained a flawless system for measuring continuous magnitudes. Yet until the end of the 18th century, every city and province in France used entirely different units: a league in Paris was not a league in Lyon, a pint varied from town to town, and the King's Foot was not the size of anyone else's shoe!

In 1789, the French Revolution broke out. Citizens and scholars resolved to unite the nation—and soon the entire world—around a single universal standard of measure drawn directly from the Earth itself: **the Meter**. And for all physical magnitudes, from length and mass to liquid volume, they embraced the elegance of the decimal system!

How can we measure the Earth, calculate the perimeter of a field, or find the area of a pasture without ever stumbling? Join us in **Chapter 12: The Meter of the Revolution**!

CHAPTER 12: The Meter of the Revolution

> "Nothing greater or more sublime has ever come from the hands of human beings than this metric system, which gives one and the same standard to all nations and for all time." — *Decree of the National Convention (1795).*

What You Are About to Discover

- The daring expedition of astronomers **Delambre and Méchain** during the French Revolution to measure planet Earth.
- The secret of universal prefixes: Greek to multiply (kilo-, hecto-, deca-) and Latin to divide (deci-, centi-, milli-).
- The three great families of measurement: **lengths** (m), **masses** (g, kg, t), and **capacities** (L).
- The fundamental distinction between **perimeter** (the boundary, in 1 dimension) and **area** (the covered surface, in 2 dimensions).
- How to compute the perimeter and area of rectangles and squares without ever confusing the two formulas.
- How to recognize a **right angle**, an **acute angle**, and an **obtuse angle** using a set square.

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1. The Problem: The Tower of Babel of Old Regime Measurements

Before 1789, France was not united in its measurements: it was a bewildering labyrinth! Every province, every town, and often every feudal estate had its own measuring units:

- In Paris, cloth was measured using the **Paris ell** (*aune*, about 1.18 meters). But in Rouen, the ell was shorter, and in Lyon, it was longer!
- To measure land, surveyors used the **King's Foot** (about 32.5 cm), but the foot in Bordeaux was 35.7 cm.
- To weigh grain, people used the **pound** (*livre*), but more than 50 different kinds of pounds existed across France!
- For liquids, the Saint-Denis **pint** was twice as large as the Paris pint.

Historians estimate that France possessed over **800 different names for measures** and nearly **250,000 different ways of calculating them**!

> 💡 **Discovery & History — The Chaos of Weights and Measures under the Ancien Régime**
>
> - Before 1789:
> - In Paris: 1 foot = 32.5 cm
> - In Bordeaux: 1 foot = 35.7 cm
> - In Strasbourg: 1 foot = 28.9 cm
> - Honest trade between cities was practically impossible!

Dishonest merchants took advantage of this confusion to cheat customers; peasants were ruined by feudal lords who secretly enlarged grain bushels to collect heavier taxes; nobody could agree.

In 1789, the French Revolution erupted. In their registers of grievances (*cahiers de doléances*), citizens demanded with one voice: *"One king, one law, one weight, and one measure!"*

---

2. Measuring the Earth: The Adventure of the Universal Meter

The scientists of the French Academy of Sciences (Lavoisier, Condorcet, Lagrange, Laplace) refused to base the new unit on the length of a king's foot or an emperor's arm: kings die, and human limbs vary from person to person. They sought a standard belonging to all humankind, drawn from nature itself: **planet Earth**!

💡 True History — The Epic Journey of Delambre and Méchain (1792—1798)

In 1792, in the midst of revolutionary turmoil and the threat of war, two French astronomers set out to measure the Earth:

- **Jean-Baptiste Delambre** traveled south from Dunkirk.
- **Pierre Méchain** traveled north from Barcelona. For six arduous years, across rugged mountains, violent storms, and arrests by suspicious revolutionary guards who mistook their surveying theodolites for spy equipment, they measured the distance between Dunkirk and Barcelona using triangulation.

From these measurements, scientists determined the exact total distance along the Earth's meridian arc from the North Pole to the Equator. They divided this immense distance into **ten million equal parts**. This ten-millionth part of the Earth's meridian quadrant became the new standard of length for humanity: **the Meter** (from the Greek *metron*, meaning "measure").

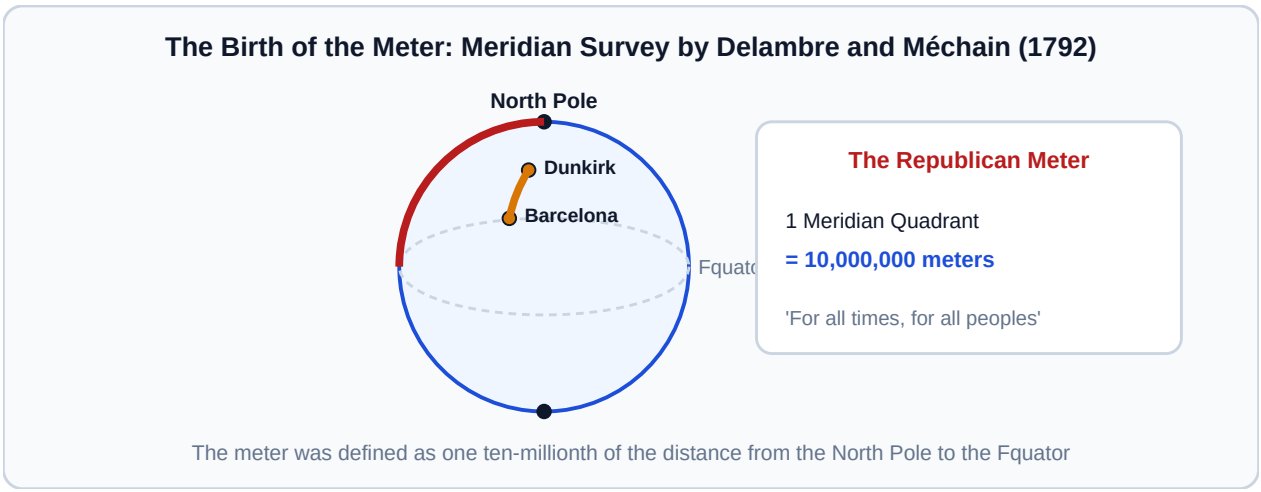

Figure 46: Measurement of Earth's meridian and the universal meter standard

The motto of this new universal standard was engraved upon the first official commemorative medals: **"For all times, for all peoples."**

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3. The Universal Chart: The Magic of Greek and Latin Prefixes

The scholars had a second stroke of genius: apply Simon Stevin's decimal place-value system to every physical measurement. Gone were the clumsy systems where 1 foot was 12 inches and 1 inch was 12 lines! From now on, **everything scales by tens**.

To name the units, they devised a brilliant linguistic convention:

- For units **larger** than the base unit, they borrowed words from **Greek** (the language of giants):
  - **Deca-** (from Greek *deka*) = $\times$ 10
  - **Hecto-** (from Greek *hekaton*) = $\times$ 100
  - **Kilo-** (from Greek *khilioi*) = $\times$ 1 000
- For units **smaller** than the base unit, they borrowed words from **Latin** (the language of division):
  - **Deci-** (from Latin *decimus*) = $\div$ 10 ($\frac{1}{10}$)
  - **Centi-** (from Latin *centesimus*) = $\div$ 100 ($\frac{1}{100}$)
  - **Milli-** (from Latin *millesimus*) = $\div$ 1 000 ($\frac{1}{1\,000}$)

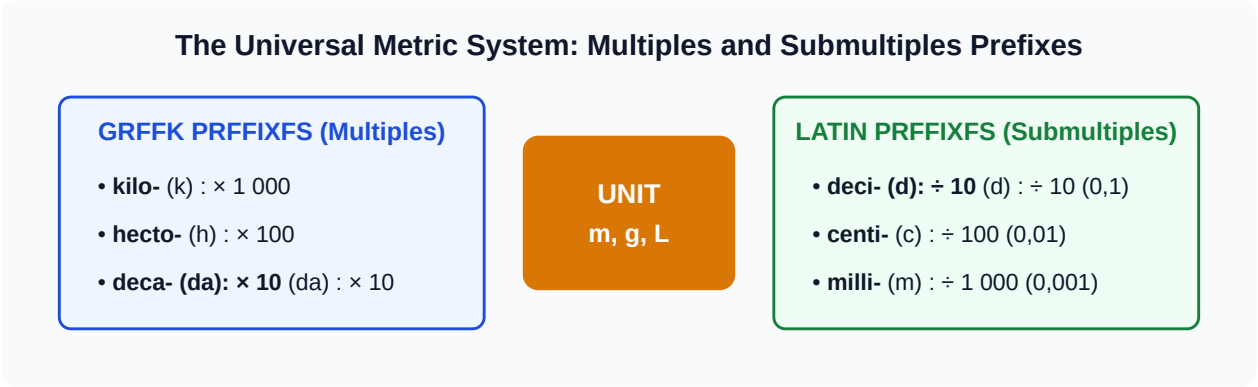


Figure 47: The universal metric system: multiple and submultiple prefixes

This exact same chart structure works **identically** for lengths, masses, and capacities!

1. Lengths (Base unit: the meter m)

Kilometer (km)	Hectometer (hm)	Decameter (dam)	Meter (m)	Decimeter (dm)	Centimeter (cm)	Millimeter (mm)
1 000 m	100 m	10 m	**1 m**	0.1 m	0.01 m	0.001 m

2. Capacities (Base unit: the liter L)

Hectoliter (hL)	Decaliter (daL)	Liter (L)	Deciliter (dL)	Centiliter (cL)	Milliliter (mL)
100 L	10 L	**1 L**	0.1 L	0.01 L	0.001 L

(Note: the kiloliter is rarely used in daily conversation; people usually refer to cubic meters).

3. Masses (Base units: the gram g and the kilogram kg)

Metric Ton (t)	Quintal (q)	—	Kilo-gram (kg)	Hec-togram (hg)	Deca-gram (dag)	Gram (g)	Deci-gram (dg)	Centi-gram (cg)	Mil-ligram (mg)
1 000 kg	100 kg		**1 kg**	100 g	10 g	**1 g**	0.1 g	0.01 g	0.001 g

⚠ Pitfall Warning — The Conversion Table Rules

1. Place only **one digit per column** in the conversion chart. 2. The **units digit** of the starting number must always sit in the column of the given unit! 3. *Example:* To enter 34.8 m, the units digit is 4. Place 4 in the m column, 3 in dam, and 8 in dm. To read this in centimeters: append zeros out to the cm column $\rightarrow$ 3 480 cm.

Worked Example 1 — Direct Conversions

- Convert 2.5 km to meters:

2.5 km = 2 500 m.

- Convert 750 mL to liters: 750 mL = 0.750 L = 0.75 L.
- Convert 3.4 metric tons to kilograms: 1 t = 1 000 kg $\Longrightarrow$ 3.4 t = 3 400 kg.

---

4. Length vs. Area: The String and the Tile (1D vs. 2D)

Here is one of the most common confusions in all of geometry: the difference between **perimeter** and **area**.

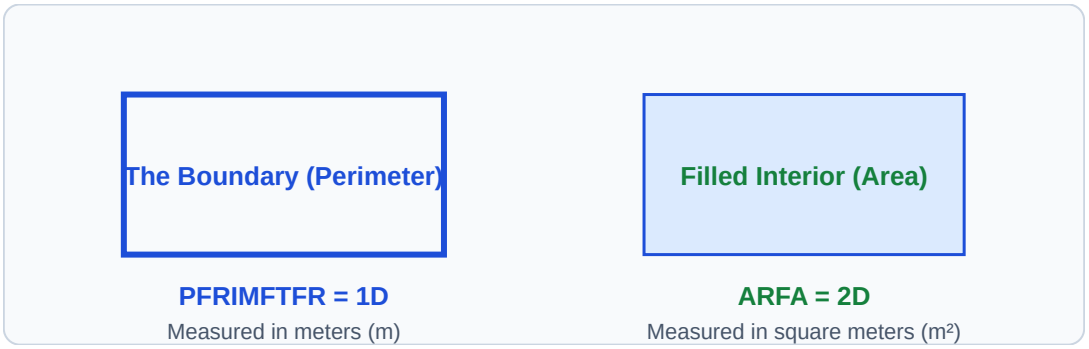


Figure 48: 1D Linear perimeter vs 2D surface area

- **Perimeter:** the total distance around the outside edge of a shape. Imagine an ant walking all the way around a field, or unrolling a spool of fence wire along the boundary. It is a **1-dimensional** measure (pure length).
- **Area (surface):** the total amount of 2D space enclosed within the boundary. Imagine laying down square floor tiles, or painting a wall. It is a **2-dimensional** measure ($\text{width} \times \text{length}$).

---

5. Calculating the Perimeter of a Polygon

To find the perimeter of any polygon, simply **add together the lengths of all its sides**.

> 📐 ***Common Perimeter Formulas***
> - ***Any polygon:*** $\mathcal{P} = \text{sum of all outer sides.}$
> - ***Rectangle with length L and width w:***
>
> $$\mathcal{P} = L + w + L + w = 2 \times (L + w)$$
>
> - ***Square with side length s:***
>
> $$\mathcal{P} = 4 \times s$$

Worked Example 2 — Perimeter Calculations

- A rectangle measures 8 cm long and 5 cm wide. $\mathcal{P} = 2 \times (8 + 5) = 2 \times 13 = 26$ cm.
- A square has side length 6 m. $\mathcal{P} = 4 \times 6 = 24$ m.

---

6. Measuring Area: From Grids to Square Centimeters (cm^2)

To measure surface area, we pave the surface with standard square tiles.

Consider a rectangle 5 cm long and 3 cm wide. Cover it completely with 1 cm^2 tiles:

To count the total number of square tiles, there is no need to count them one by one: We have **3 rows of 5 tiles each**!

$$\text{Area} = 5 \times 3 = 15 \text{ tiles of } 1 \text{ cm}^2 = 15 \text{ cm}^2$$

> 📐 ***Common Area Formulas***
> - ***Area of a rectangle:***
>
> $$\mathcal{A} = \text{Length} \times \text{Width} = L \times w$$
>
> - ***Area of a square:***
>
> $$\mathcal{A} = \text{side} \times \text{side} = s \times s$$

Worked Example 3 — Never Confuse Perimeter and Area!

Consider a square with side length $s = 4 \text{ cm}$.

- Its perimeter: $\mathcal{P} = 4 \times 4 = 16 \text{ cm}$. (Length of string around the edges).
- Its area: $\mathcal{A} = 4 \times 4 = 16 \text{ cm}^2$. (Number of square carpet tiles). Both numbers happen to be 16, but their units describe completely different things: one is in **centimeters**, the other in **square centimeters**! Now look at a square with side length 5 cm:
- $\mathcal{P} = 4 \times 5 = 20 \text{ cm}$.
- $\mathcal{A} = 5 \times 5 = 25 \text{ cm}^2$. Here, $25 \neq 20$!

---

7. Angles: Acute, Right, or Obtuse with a Set Square

When two straight lines meet at a point (the vertex), they form an opening called an **angle**. To inspect this opening, the fundamental reference tool of students and carpenters is the **set square** (or drafting triangle).

Acute Angle	Right Angle	Obtuse Angle
More closed than square	Fits square exactly	More open than square
Aperture $< 90°$	Aperture $= 90°$	Aperture $> 90°$

1. **Right angle ($90°$):** the angle of a perfect sheet of paper, the corner of a square room, or the corner of a set square. Its two rays are perpendicular.
2. **Acute angle:** a sharp, narrow angle tighter than a right angle (opening strictly between $0°$ and $90°$).
3. **Obtuse angle:** an open, wide angle broader than a right angle (opening strictly between $90°$ and $180°$).

🖐 Method — Checking Angles with a Set Square

1. Place the

vertex of the set square's right angle directly onto the angle's vertex. 2. Align one straight edge of the set square along one arm of the angle. 3. Observe the other arm:

- If it hides beneath the body of the set square: the angle is **acute** (smaller than $90°$).
- If it traces exactly along the second edge: the angle is **right** ($90°$).
- If it spreads outward beyond the set square: the angle is **obtuse** (greater than $90°$).

What You Need to Remember

1. **The metric system** is anchored on the **meter** (m), defined during the French Revolution as a decimal fraction of planet Earth.
2. **Prefixes:**
   - Kilo (1 000), Hecto (100), Deca (10) to multiply;
   - Deci ($\frac{1}{10}$), Centi ($\frac{1}{100}$), Milli ($\frac{1}{1\,000}$) to divide.
3. **Masses:** $1 \text{ kg} = 1\,000 \text{ g}$ and $1 \text{ metric ton (t)} = 1\,000 \text{ kg}$.
4. **Perimeter (1D):** Total boundary length.
   - Rectangle: $\mathcal{P} = 2 \times (L + w)$.
   - Square: $\mathcal{P} = 4 \times s$.
5. **Area (2D):** Total enclosed surface.
   - Rectangle: $\mathcal{A} = L \times w$.

- Square: $\mathcal{A} = s \times s$.

6. **Angles:**
   - Acute ($< 90^\circ$);
   - Right ($= 90^\circ$, corner of set square);
   - Obtuse ($> 90^\circ$).

Exercises

Level (Direct Application)

1. Convert into the requested unit:
   a) $4 \text{ m} = \ldots \text{ cm}$
   b) $3.5 \text{ km} = \ldots \text{ m}$
   c) $250 \text{ mm} = \ldots \text{ cm}$
   d) $12 \text{ dm} = \ldots \text{ m}$
2. Convert the following masses:
   a) $5 \text{ kg} = \ldots \text{ g}$
   b) $2.8 \text{ t} = \ldots \text{ kg}$
   c) $750 \text{ g} = \ldots \text{ kg}$
   d) $4\,200 \text{ mg} = \ldots \text{ g}$
3. Convert the following capacities:
   a) $2 \text{ L} = \ldots \text{ cL}$
   b) $1.5 \text{ L} = \ldots \text{ mL}$
   c) $33 \text{ cL} = \ldots \text{ L}$
   d) $25 \text{ dL} = \ldots \text{ L}$
4. Compute the perimeter of a rectangle whose length is 7 cm and width is 4 cm.
5. Compute the area of a square whose side measures 6 cm.
6. An angle is narrower than the corner of your set square. What is its name? What if it is wider?

Level (Geometry and Combined Conversions)

7. A rectangular field has a length of 45 m and a width of 20 m.
   a) Calculate its perimeter.
   b) The farmer encloses it with wire fence costing $\$8$ per meter. How much will the fencing cost?
   c) Calculate the area of this field in square meters (m^2).
8. Watch your units! A rectangle has a length of 1.2 m and a width of 40 cm.
   a) Express the length in centimeters.
   b) Calculate its perimeter in centimeters, then in meters.
   c) Calculate its area in square centimeters (cm^2).
9. A square has a perimeter of 36 cm.
   a) What is the length of each side?
   b) What is its area?
10. A bag of flour weighs 1 kg. A baker uses 350 g for brioche and 420 g for pancakes. How many grams of flour remain?
11. A water bottle holds 1.5 L. How many full glasses of 25 cL can be poured from this bottle?

Level (Surveying Challenges & Logic)

12. Two rectangles have the exact same area of 24 cm^2.
    - Rectangle A measures 6 cm by 4 cm.
    - Rectangle B measures 8 cm by 3 cm.

    a) Calculate the perimeter of Rectangle A.
    b) Calculate the perimeter of Rectangle B.

c) Do two shapes with the same area necessarily have the same perimeter? Explain your discovery!

13. A rectangular swimming pool measures $10\ \text{m}$ long by $5\ \text{m}$ wide. Around the pool, a paved walkway $1\ \text{meter}$ wide is constructed on all four sides.
    a) What are the length and width of the large outer rectangle formed by pool and walkway combined?
    b) Calculate the total area of the pool and walkway together.
    c) What is the area of the paved walkway alone?
14. An empty truck weighs $2.4\ \text{metric tons}$. It is loaded with 80 bags of cement weighing $35\ \text{kg}$ each. What is the total mass of the loaded truck, in metric tons?
15. *The Square Scaling Challenge:* If you double the side length of a square:
    a) By what factor is its perimeter multiplied?
    b) By what factor is its area multiplied? (Draw a square of side 2 cm and a square of side 4 cm to help visualize!).

Solutions

1. a) $4\ \text{m} = \mathbf{400\ cm}$; b) $3.5\ \text{km} = \mathbf{3\,500\ m}$; c) $250\ \text{mm} = \mathbf{25\ cm}$; d) $12\ \text{dm} = \mathbf{1.2\ m}$.

2. a) $5\ \text{kg} = \mathbf{5\,000\ g}$; b) $2.8\ \text{t} = \mathbf{2\,800\ kg}$; c) $750\ \text{g} = \mathbf{0.75\ kg}$; d) $4\,200\ \text{mg} = \mathbf{4.2\ g}$.

3. a) $2\ \text{L} = \mathbf{200\ cL}$; b) $1.5\ \text{L} = \mathbf{1\,500\ mL}$; c) $33\ \text{cL} = \mathbf{0.33\ L}$; d) $25\ \text{dL} = \mathbf{2.5\ L}$.

4. $\mathcal{P} = 2 \times (7 + 4) = 2 \times 11 = \mathbf{22\ cm}$. **5.** $\mathcal{A} = s \times s = 6 \times 6 = \mathbf{36\ cm^2}$. **6.** Sharper than a right angle = **acute angle**. Broader than a right angle = **obtuse angle**. **7.** a) $\mathcal{P} = 2 \times (45 + 20) = 2 \times 65 = \mathbf{130\ m}$. b) Fencing cost: $130 \times 8 = \mathbf{\$1\,040}$. c) Field area: $\mathcal{A} = 45 \times 20 = \mathbf{900\ m^2}$.

8. a) $1.2\ \text{m} = \mathbf{120\ cm}$. b) $\mathcal{P} = 2 \times (120 + 40) = 2 \times 160 = \mathbf{320\ cm = 3.2\ m}$. c) $\mathcal{A} = 120 \times 40 = \mathbf{4\,800\ cm^2}$.

9. a) Side length: $36 \div 4 = \mathbf{9\ cm}$. b) Square area: $9 \times 9 = \mathbf{81\ cm^2}$.

10. Flour used: $350 + 420 = 770\ \text{g}$. Since $1\ \text{kg} = 1\,000\ \text{g}$, there remains: $1\,000 - 770 = \mathbf{230\ g}$ of flour.

11. $1.5\ \text{L} = 150\ \text{cL}$. Number of glasses: $150 \div 25 = \mathbf{6\ full\ glasses}$.

12. a) Perimeter of A: $2 \times (6 + 4) = 2 \times 10 = \mathbf{20\ cm}$. b) Perimeter of B: $2 \times (8 + 3) = 2 \times 11 = \mathbf{22\ cm}$. c) **No!** Two shapes can encompass the exact same surface area (here $24\ \text{cm}^2$) while having perimeters of different lengths. The more elongated and narrow a rectangle is, the larger its perimeter for a given area.

13. a) The walkway adds $1\ \text{m}$ on each side:

- New length: $10 + 1 + 1 = \mathbf{12\ m}$.
- New width: $5 + 1 + 1 = \mathbf{7\ m}$.

b) Total combined area: $12 \times 7 = \mathbf{84\ m^2}$.

c) Pool area alone: $10 \times 5 = 50\ \text{m}^2$. Walkway area alone: $84 - 50 = \mathbf{34\ m^2}$.

14. Mass of cement bags: $80 \times 35 = 2\,800\ \text{kg} = 2.8\ \text{metric tons}$. Total mass of loaded truck: $2.4 + 2.8 = \mathbf{5.2\ metric\ tons}$.

15. Let a square have side length $s = 2\ \text{cm}$: $\mathcal{P}_1 = 4 \times 2 = 8\ \text{cm}$ and $\mathcal{A}_1 = 2 \times 2 = 4\ \text{cm}^2$. If we double the side ($s = 4\ \text{cm}$): $\mathcal{P}_2 = 4 \times 4 = 16\ \text{cm}$ and $\mathcal{A}_2 = 4 \times 4 = 16\ \text{cm}^2$. a) Perimeter is multiplied by **2** ($16 = 8 \times 2$). b) Area is multiplied by **4** ($16 = 4 \times 4$, because $2 \times 2 = 4$)! A square twice as wide contains 4 smaller squares inside.

The Question that Opens What Lies Ahead

You are now equipped with the metric system: you know how to weigh bags of flour, measure lengths of fencing, and calculate field areas. Yet in the kitchen of a great restaurant or the workshop of a master builder, another challenge arises constantly: If a chocolate cake recipe is written for **4 people**, how do you adjust the ingredients for **6 people** or **10 guests** without making the cake overly sweet or heavy?

When one quantity doubles, the other doubles as well: both magnitudes march in lockstep. How can we calculate these relationships swiftly without ever getting confused? Welcome to the realm of **Proportionality** in **Chapter 13: Grandmother's Recipe and the Rule of Three**!

CHAPTER 13: Grandmother's Recipe and the Rule of Three

> "If two loaves of bread cost six coins, how much will five loaves cost? As soon as the world advances at a steady, rhythmic pace, a single measurement is enough to unveil all the others. That is the secret art of proportionality."

What You Are About to Discover

- How to adapt any **cooking recipe** without ever spoiling its taste.
- How to recognize a genuine situation of **proportionality** and spot false ones (no, you will not be 10 feet tall at age 20!).
- How to organize information into a clean, crystal-clear **proportionality table**.
- The three classic calculation methods:
  1. The **column-jumping method** (doubling, tripling, adding columns).
  2. The **unit rate method** (finding the price of a single item).
  3. The **constant of proportionality** (the direct vertical multiplier).
- The secret behind the famous **"rule of three"**, the ancient merchants' shortcut explained without mystery.
- Navigating with **average speeds** (km/h or mph) and **map scales**.

---

1. The Problem: The Birthday Feast and the Cake for 4

Thomas's birthday is just around the corner. To treat his friends, his grandmother pulls out her treasured recipe notebook, stained with cocoa and dog-eared from years of love. She turns to her legendary pear chocolate fudge cake:

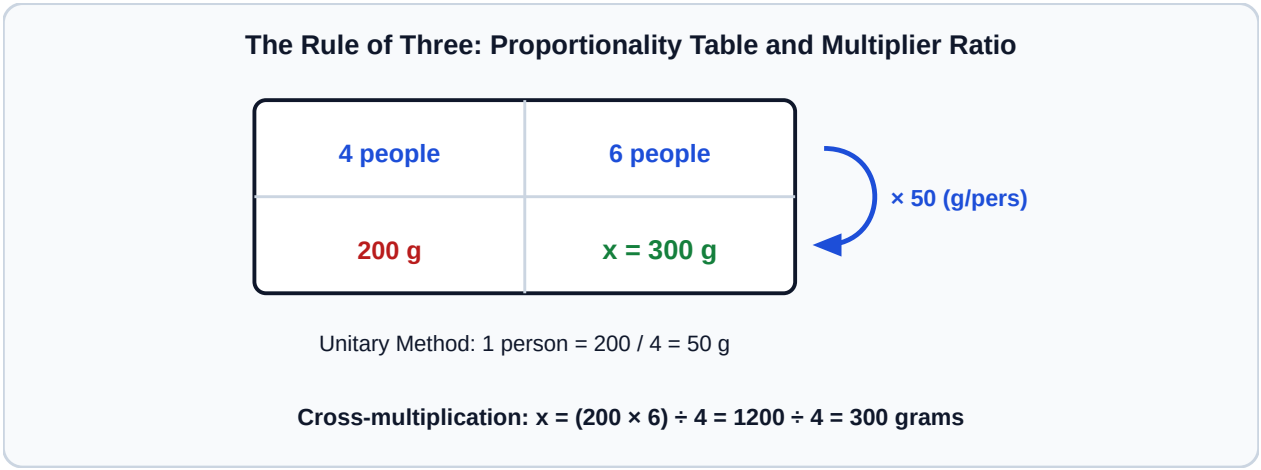


Figure 49: The proportionality table and the rule of three

Thomas jumps with delight: *"Grandma, this is wonderful! But there will be **6 people** at the table for my birthday: my parents, you, my two cousins, and me! And tomorrow, we're having a second party with **10 classmates**!"*

Grandmother smiles: *"If we simply toss in 6 eggs at random and 300 grams of sugar, our cake will turn out cloyingly sweet or terribly dry. We must respect **the right proportions**!"*

How can we calculate the exact amount of chocolate for 6 people without ruining Grandma's recipe?

Thomas takes a sheet of scrap paper and thinks it through:

- For 4 people, we need 200 g of chocolate.
- Therefore, for half as many (**2 people**), we need half as much: $200 \div 2 = 100$ g.
- And **6 people** is simply 4 people plus 2 people!

We can just add the corresponding amounts together:

$$200 \text{ g} + 100 \text{ g} = 300 \text{ g of chocolate!}$$

Thomas has just used, without realizing it, one of the most powerful tools in everyday mathematics: **proportionality**.

---

2. What Is Proportionality? The Doubling Test

In nature and daily life, many quantities are linked together. But they are not all linked in the same way!

> **📐 Definition — Two Proportional Quantities**
>
> Two quantities are **proportional** when the values of one are obtained by **always multiplying the values of the other by the exact same constant number**. This implies that:
> - If one quantity becomes **2 times larger**, the other becomes **2 times larger**.
> - If one quantity becomes **3 times larger**, the other becomes **3 times larger**.
> - If one quantity becomes **2 times smaller** (halved), the other becomes **2 times smaller**.

The Pitfall: Not Everything in Life Is Proportional!

Let us put three everyday situations to the **"doubling test"**:

1. **The Price of Apples:** 1 kg of apples costs \$3. 2 kg of apples cost \$6 (2×3). 4 kg of apples cost \$12 (4×3). → **This is a true proportional relationship!**
2. **A Child's Age and Height:** At age 5, Tom was 1.10 m tall. At age 10 (his age has doubled: $5 \times 2 = 10$), Tom is about 1.40 m tall. Is he twice as tall as 1.10 m, which would be 2.20 m? Absolutely not! And at age 20, will he stand 4.40 m tall? That would be a terrifying giant! → **Height is NOT proportional to age!** Human bodies grow quickly in early childhood, slow down, and stop growing altogether in adulthood.
3. **A Cinema Membership:** An annual membership card costs \$10, and then each movie ticket costs \$5.
   - For 1 movie: $\$10 + \$5 = \$15$.
   - For 2 movies: $\$10 + (2 \times \$5) = \$20$. The number of movies doubled (from 1 to 2), but the total cost did not double ($\$20 \neq 2 \times \$15 = \$30$). → Because of the fixed starting fee, this is **NOT direct proportionality**.

---

3. The Proportionality Table: Three Calculation Methods

To solve problems cleanly without getting tangled up, we arrange our numbers into a two-row table:

Grandeur 1 (nombre de cahiers)	2	4	6	10
Grandeur 2 (prix en euros)	5 €	10 €	15 €	25 €

There are **three reliable methods** for filling in missing entries in this table. Always pick the one that makes the numbers simplest to compute!

Method 1: Operating Across Columns (Multiplication and Addition)

In a proportionality table, you are allowed to:

- Multiply or divide an entire column by any number.
- Add (or subtract) two columns together!

Grandeur 1 (nombre de cahiers)	2	4	6	10
Grandeur 2 (prix en euros)	5 €	10 €	15 €	25 €

Method 2: Finding the Unit Rate (How Much for 1?)

This is the common-sense method favored by generations of shoppers: *"First find the value for **1 single item**, then multiply by the desired number of items!"*

🧭 Worked Example 1 — The Unit Rate Method

Problem: 3 identical pens cost $4.50. How much do 7 of these pens cost?

- **Step 1: Find the price of 1 pen.** $4.50 \div 3 = \$1.50$. (One pen costs $1.50).
- **Step 2: Calculate the cost for 7 pens.** $7 \times \$1.50 = \$10.50$. 7 pens cost $\mathbf{\$10.50}$.

Method 3: The Constant of Proportionality (The Vertical Bridge)

This is the multiplier that takes you directly from the top row to the bottom row in one step.

Nombre de gâteaux	3	5	8	× 4
Nombre de fraises	12	20	32	(Coefficient)

To find this multiplier, divide any number in the second row by its corresponding partner in the first row:

$$\text{Constant} = \frac{12}{3} = 4$$

Every cake requires 4 strawberries. For any order, simply **multiply by 4**! And to work backwards from strawberries to cakes, divide by 4.

---

4. The Legendary "Rule of Three" Explained

For centuries across European schools, pupils were taught **"the rule of three"**. Why that name? Because you are given **three known numbers**, and your goal is to uncover the **fourth**!

Consider the setup:

$$a \to b$$
$$c \to ?$$

Example: If 4 meters of fabric cost $28, how much do 9 meters cost? - The three known numbers are: 4, 28, and 9. - The unknown number is labeled ?.

How did ancient merchants solve this in a single concise line? They combined finding the unit rate and multiplying: 1. The price of 1 meter is: $\frac{28}{4} = \$7$. 2. The price of 9 meters is: $9 \times 7 = \$63$.

In a single expression:

$$? = \frac{28 \times 9}{4} = \frac{252}{4} = \$63$$

Multiply the two numbers located along the diagonal pair, then divide by the third remaining number! This is also known as **cross-multiplication**.

---

5. Two Classic Applications: Speed and Map Scales

1. Traveling at Constant Speed: Kilometers per Hour (km/h)

When a train or car travels at a steady speed of $\mathbf{80\ \ km/h}$ (80 kilometers per hour), it covers $80\ \ \text{km}$ in exactly $1\ \ \text{hour}$. Distance covered is directly proportional to elapsed travel time!

Let us build the table:

Time (h)	$1\ \ \text{h}$	$2\ \ \text{h}$	$3\ \ \text{h}$	$30\ \ \text{min}$ ($\frac{1}{2}\ \ \text{h}$)	$15\ \ \text{min}$ ($\frac{1}{4}\ \ \text{h}$)
Distance (km)	**80**	160	240	40	20

- In $2\ \ \text{h}$: twice as much time $\to 2 \times 80 = 160\ \ \text{km}$.
- In $30\ \ \text{min}$ (half an hour): half the distance $\to 80 \div 2 = 40\ \ \text{km}$.
- In $15\ \ \text{min}$ (a quarter of an hour): $80 \div 4 = 20\ \ \text{km}$.

Worked Example 2 — Road Trip

A car drives at an average speed of $90\ \text{km/h}$. What distance does it cover in $2\ \text{h}\ \ 30\ \text{min}$?

- In $2\ \text{h}$: $2 \times 90 = 180\ \text{km}$.
- In $30\ \text{min}$ (half-hour): $90 \div 2 = 45\ \text{km}$.
- Total distance: $180 + 45 = \mathbf{225\ \text{km}}$.

2. Geographic Maps and Architectural Models: Scales

On a road map or city floorplan, the real world is scaled down proportionally. If the scale says: $\mathbf{1\ \text{cm on the map} = 100\ \text{m in the real world}}$ This means all real-world distances are divided by $10\,000$ (since $100\ \text{m} = 10\,000\ \text{cm}$).

On the map (cm)	$1\ \text{cm}$	$2\ \text{cm}$	$5\ \text{cm}$	$10\ \text{cm}$
In reality	$100\ \text{m}$	$200\ \text{m}$	$500\ \text{m}$	$1\,000\ \text{m}\ (1\ \text{km})$

Worked Example 3 — Forest Hike

On a hiking trail map where $1\ \text{cm}$ represents $250\ \text{m}$, two mountain villages are linked by a trail measuring $8\ \text{cm}$ on the map. What is the actual distance between the two villages?

- $8 \times 250\ \text{m} = 2\,000\ \text{m} = \mathbf{2\ \text{km}}$.

What You Need to Remember

1. **Two quantities are proportional** if you can move from one to the other by multiplying by a fixed constant value (the **constant of proportionality**).
2. **Properties of proportionality tables:**
   - You can multiply or divide any column by a constant.
   - You can add or subtract columns together.
3. **The unit rate method:** Find the value for 1 unit (by division), then multiply to find the desired target quantity.
4. **The rule of three:** To solve $\frac{a}{b} = \frac{c}{?}$, calculate $? = \frac{b \times c}{a}$.
5. **Constant speed:** Distance is proportional to elapsed time ($\text{distance} = \text{speed} \times \text{time}$).
6. **Map scale:** Distances drawn on paper are proportional to physical distances on the ground.

Exercises

Level (Direct Application)

1. Which of the following situations represent true proportionality? Explain briefly:
   a) The number of baguettes purchased and the total price paid.
   b) A tree's age and its height in meters.
   c) The quantity of gasoline pumped and the total shown on the meter.
   d) The number of pages in a novel and the time required to read it.
2. To bake a cake for 4 people, 200 g of flour are required. How much flour is needed:
   a) For 8 people?
   b) For 2 people?
   c) For 10 people?
3. 5 packs of chewing gum cost \$6.
   a) What is the price of a single pack?

b) How much do 8 packs cost?

4. Complete this proportionality table (first identify the vertical multiplier):

Number of notebooks	3	5	8	12
Total cost ($)	6	…	…	…

5. A map scale indicates: $1 \text{ cm on the map} = 500 \text{ m on the ground}$. What real-world distance corresponds to:
   a) 2 cm on the map?
   b) 6 cm on the map?
   c) 10 cm on the map?

Level (Tables and Speed Calculations)

6. A painter uses 3 liters of paint to cover a wall of $24\ \text{m}^2$.
   a) What surface area can he paint with 1 liter of paint?
   b) What area can he cover with 7 liters?
   c) How many liters of paint does he need to cover a building facade of $96\ \text{m}^2$?
7. A cyclist rides at a constant speed of 20 km/h.
   a) How far does she ride in 2 hours?
   b) How far does she ride in 3 h 30 min?
   c) How long does she take to travel 10 km?
8. Complete this proportionality table using column properties (doubles, halves, additions):

Mass of apples (kg)	2	4	6	10	1
Total cost ($)	5	…	…	…	…

9. A farmer sells farm-fresh eggs at the market: 6 eggs cost $2.40. How much do:
   a) 12 eggs cost?
   b) 3 eggs cost?
   c) 15 eggs cost? (Use a clever addition of columns!).
10. A passenger train travels at a steady speed of 120 km/h.
    a) What distance does it cover in 1 hour?
    b) What distance does it cover in 15 minutes?
    c) What distance does it cover in 45 minutes?

Level (Expert Challenges)

11. *The Giant Cafeteria Recipe:* To make chocolate mousse for 6 children, a chef uses 9 eggs, 150 g butter, and 240 g dark chocolate. Today, he must prepare this mousse for all 90 children in the school! Calculate the exact quantity of each ingredient needed.
12. *The Leaky Faucet:* A dripping faucet loses 2 liters of clean water every 3 hours.
    a) How many liters of water are wasted in 24 hours (one full day)?
    b) How many liters are wasted over a 30-day month?
13. On a treasure map, the distance between Skull Rock and Pirate Cove is 12 cm. The map key indicates that 2 cm represents 150 meters in reality. What is the actual physical distance between Skull Rock and Pirate Cove?
14. Two supermarkets offer deals on the same brand of cookies:
    - Store A: a bundle of 3 boxes sells for $4.80.
    - Store B: a bundle of 5 boxes sells for $7.50. Which store offers the lower unit price per box?
15. *The Candle Challenge:* A cylindrical candle 18 cm tall burns down at a steady rate. After burning for 2 hours, its height is reduced to 14 cm.
    a) By how many centimeters does the candle shorten each hour?
    b) After how many total hours will the candle be completely burned away?

Solutions

1. a) **Yes**, with a fixed price per baguette, twice as many loaves cost twice as much. b) **No**, a tree does not grow at the same pace throughout its life (growth slows down and plateaus). c) **Yes**, the price per gallon or liter of gasoline is constant, so total cost is directly proportional to volume. d) **No**, reading speed varies with fatigue, difficulty of the text, and plot tension.

2. a) For 8 people (double 4): $200 \times 2 = \mathbf{400}\ \ \mathbf{g}$. b) For 2 people (half of 4): $200 \div 2 = \mathbf{100}\ \ \mathbf{g}$. c) For 10 people ($8 + 2$): $400 + 100 = \mathbf{500}\ \ \mathbf{g}$ of flour.

3. a) Price of 1 pack: $\$6 \div 5 = \mathbf{\$1.20}$. b) Price of 8 packs: $8 \times \$1.20 = \mathbf{\$9.60}$.

4. Multiplier constant: $6 \div 3 = 2$ (multiply top row by 2). - For 5 notebooks: $5 \times 2 = \mathbf{\$10}$. - For 8 notebooks: $8 \times 2 = \mathbf{\$16}$. - For 12 notebooks: $12 \times 2 = \mathbf{\$24}$.

5. a) $2 \times 500 = 1\,000\ \text{m} = \mathbf{1\ \ km}$. b) $6 \times 500 = 3\,000\ \text{m} = \mathbf{3\ \ km}$. c) $10 \times 500 = 5\,000\ \text{m} = \mathbf{5\ \ km}$.

6. a) With 1 liter: $24 \div 3 = \mathbf{8\ m^2}$. b) With 7 liters: $7 \times 8 = \mathbf{56\ m^2}$. c) For 96 m^2: $96 \div 8 = \mathbf{12\ \ liters}$ of paint.

7. a) In 2 h: $2 \times 20 = \mathbf{40\ \ km}$. b) In 3 h 30 min: 3 h $\rightarrow$ 60 km, 30 min $\rightarrow$ 10 km. Total = $\mathbf{70\ \ km}$. c) For 10 km (half of 20 km): half an hour, or $\mathbf{30\ \ minutes}$.

8. - For 4 kg (double of 2 kg): $\$5 \times 2 = \mathbf{\$10}$. - For 6 kg (2 kg + 4 kg): $\$5 + \$10 = \mathbf{\$15}$. - For 10 kg (4 kg + 6 kg): $\$10 + \$15 = \mathbf{\$25}$. - For 1 kg (half of 2 kg): $\$5 \div 2 = \mathbf{\$2.50}$.

9. a) 12 eggs (double 6): $\$2.40 \times 2 = \mathbf{\$4.80}$. b) 3 eggs (half of 6): $\$2.40 \div 2 = \mathbf{\$1.20}$. c) 15 eggs ($12 + 3$): $\$4.80 + \$1.20 = \mathbf{\$6.00}$.

10. a) In 1 h: $\mathbf{120\ \ km}$. b) In 15 min (a quarter hour): $120 \div 4 = \mathbf{30\ \ km}$. c) In 45 min ($3 \times 15$ min): $3 \times 30 = \mathbf{90\ \ km}$.

11. To scale from 6 children to 90 children: $90 \div 6 = 15$ times more children! Multiply all ingredients by 15:

- Eggs: $9 \times 15 = \mathbf{135\ \ eggs}$.
- Butter: $150 \times 15 = \mathbf{2\,250\ \ g = 2.25\ \ kg\ of\ butter}$.
- Dark chocolate: $240 \times 15 = \mathbf{3\,600\ \ g = 3.6\ \ kg\ of\ chocolate}$.

12. a) In 24 hours, there are $24 \div 3 = 8$ three-hour intervals. Water wasted: $8 \times 2 = \mathbf{16\ \ liters\ per\ day}$. b) In 30 days: $30 \times 16 = \mathbf{480\ \ liters}$ wasted!

13. If 2 cm represents 150 m, then 1 cm represents $150 \div 2 = 75$ m. Actual distance for 12 cm: $12 \times 75 = \mathbf{900\ \ meters}$.

14. - Store A: $\$4.80 \div 3 = \$1.60$ per box. - Store B: $\$7.50 \div 5 = \$1.50$ per box. **Store B** offers the better value ($\$1.50 < \$1.60$).

15. a) Over 2 hours, height drops from 18 cm to 14 cm, shortening by $18 - 14 = 4$ cm. In 1 hour, it shortens by: $4 \div 2 = \mathbf{2\ \ cm\ per\ hour}$. b) To consume the full 18 cm: $18 \div 2 = \mathbf{9\ \ hours}$ in total.

The Question that Opens What Lies Ahead

You have mastered the law of proportions: you can expand or shrink recipes, calculate travel times, and decode geographic maps. Yet in the ancient ledger books of traveling merchants, a mysterious shadow lingers.

Imagine an itinerant trader in a bustling winter market. In the morning, he begins his day with 5 silver coins. At midday, he purchases 8 silver coins worth of warm woolen blankets from a wholesaler, saying: *"Take my 5 coins now, and I will owe you the remaining 3 coins tonight!"* How much money remains in his pocket? Less than zero! He carries a debt of 3 coins. Meanwhile outside, the thermometer plunges below the freezing point of water: ice forms on the canal, and the mercury reads -5° C!

Do numbers exist beneath the ground level of zero? Numbers colder and deeper than nothingness? Step through the looking-glass into **Chapter 14: The Merchant's Slate and the Sand Pit**!

CHAPTER 14: The Merchant's Slate and the Sand Pit

> "To a naive mind, nothing exists beneath zero: an empty purse holds nothing, and counting halts there. But to the merchant balancing ledger books, to the mason digging deep foundations, or to the traveler caught in a winter blizzard, zero is merely a threshold. Beneath the surface begins the mirror-world of debts and depths."

What You Are About to Discover

- Why **zero is not pure nothingness**, but a reference level and a point of equilibrium.
- The physical model of the **sand heap and sand pit**: how matter and hollow cancel each other out ($(+1) + (-1) = 0$).
- Elevators, underground parking levels ($-1, -2$), depth below sea level, and **winter thermometers**.
- The **complete number line**: crossing the zero border and venturing leftward.
- How to compare negative numbers without stumbling: understanding why **-5 is smaller than -2** (the colder it is, the lower the number!).
- **Adding negative numbers**: filling pits and accumulating debts.
- Calculating **temperature intervals** and elevation differences.

---

1. The Problem: The Woolen Blanket and the Black Slate

Autumn draws to a close across the grand central square of the trading city. Northern winds bring the first white frosts, and camel caravans hasten to unload their bundles before the heavy snows set in.

A young river fisherman pauses before the stall of a cloth merchant from the high plateaus. The fisherman shivers in his worn linen tunic. He admires a magnificent alpine wool blanket—thick, soft, and warm.

The merchant announces the price: *"This blanket costs **5 silver coins**!"*

The fisherman unties his leather pouch and counts his coins onto the wooden table: *One… two…* and that is all! He has only **2 silver coins** in his pouch. He is short by 3 coins. With downcast eyes, he packs away his two coins and prepares to head back out into the freezing wind.

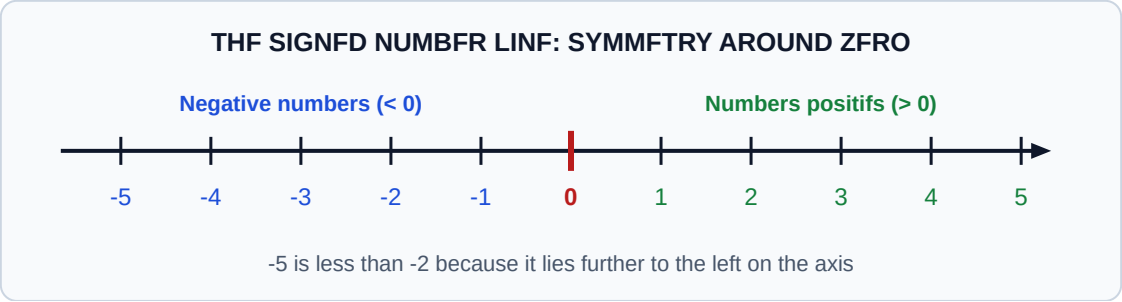


Figure 50: The signed number line of positive and negative integers

Seeing the fisherman's honest character, the merchant places a comforting hand on his shoulder: *"Take the blanket today so you do not freeze tonight. Tomorrow, after your net fishing, bring me the 3 coins that are missing!"*

The fisherman gives heartfelt thanks and walks away wrapped in the snug warmth of the blanket. The merchant then takes a black slate and a stick of white chalk to record the transaction in his ledger.

What should he write beside the fisherman's name?

- If he writes "0", that is false! Zero would mean the fisherman owes him nothing, that their account is settled.
- The fisherman does not have zero coins: **he holds a debt of 3 coins**.

- To return to a neutral purse of zero (0), he must first earn 3 silver coins and surrender them to the merchant without keeping any for himself.

On his slate, the merchant writes the digit 3 preceded by a small horizontal dash:

$$\mathbf{-3}$$

You have just met **negative numbers**.

---

2. The Heap and the Pit: Zero as Equilibrium

To understand clearly what a negative number means in the physical world, let us leave the bustling market and walk down to the river's sandy beach.

Look at the beach: the wind has smoothed the sand. The ground is level and flat. This is our starting baseline, our reference level: **Level Zero (0)**.

Take a small shovel: 1. Plunge the blade into the sand and lift out one shovelful. 2. Deposit this shovelful right next to the hole.

Look closely at what your single action created in the landscape:

- You built a **Heap** rising above the ground: a positive quantity, written $\mathbf{+1}$.
- At the exact same instant, unavoidably, you hollowed out a **Pit** sinking below the surface: an absence of sand, a debt of matter, written $\mathbf{-1}$.

Now, what happens if you kick the heap $+1$ back into the pit -1? The sand falls back into the cavity. The pit is filled completely. The beach becomes perfectly flat once more! The heap has vanished, the pit has vanished: only the zero level of the ground remains.

$$(+1) + (-1) = 0$$

> 💡 ***An Illuminating Truth — The Meaning of Zero*** *Zero is not an absolute void. It is* ***the mutual cancellation of two equal and opposite forces****: A positive quantity $(+1)$ and a negative quantity (-1) possess the exact same magnitude. When they meet, they balance out and yield zero.*

---

3. Negative Numbers All Around Us

In our daily lives, we encounter negative numbers continuously:

1. Thermometers and the Freezing of Water

On a Celsius thermometer (° C), zero (0° C) is not the absence of heat: it is the precise temperature at which liquid water turns to solid ice.

- Above zero: water is liquid ($+5$° C, $+20$° C).
- Below zero: water freezes, frost forms, snow falls (-2° C, -10° C).

2. Elevators and Underground Parking

In a tall building, the ground floor entrance is level **0**.

- Apartments rise upward: $+1, +2, +3$...
- Basements and parking garages sink beneath the earth: $\mathbf{-1}$ (first basement), $\mathbf{-2}$ (second basement).

3. Geographic Elevation

The Earth's zero level is **mean sea level**.

- Mont Blanc peaks at $+4\,807$ meters above sea level.
- The Dead Sea lies in a geological depression at $\mathbf{-430}$ **meters** below sea level!
- Submarines cruise at -100 m or -300 m.

---

4. The Complete Number Line: Through the Looking-Glass

Until now, your number line began at zero and stretched exclusively to the right, like a beam of sunlight. Now, the line extends **infinitely in both directions**, like an endless two-way highway!

📐 Definition — Directed (Signed) Numbers / Integers

The complete collection formed by positive numbers, zero, and negative numbers is known as the set of **signed numbers** (or directed numbers; for whole values, **integers**).

- **Positive numbers** carry a $+$ sign (which is often omitted: $+3 = 3$).
- **Negative numbers** must always carry a $-$ sign.
- The number **0** is the unique number that is simultaneously positive and negative: it is the origin and center of symmetry.

Two numbers located at the exact same distance from zero on opposite sides are called **opposite numbers**:

- The opposite of $+4$ is -4.
- The opposite of -7 is $+7$.
- The sum of two opposite numbers is always zero: $(+4) + (-4) = 0$.

---

5. Comparing Negative Numbers: The Thermometer Pitfall

Look carefully at this question: *Which number is smaller: -5 or -2?*

Many students hesitate and say: *"5 is greater than 2, so -5 must be greater than -2!"* **That is a classic mistake! The reality is exactly the reverse!**

Think of the thermometer:

- At -2° C, it is chilly; there is light frost.
- At -5° C, it is **even colder**; the mercury has plunged deeper!

Think of the debt ledger:

- Owing \$2 (-2) is a minor debt.
- Owing \$5 (-5) is a heavier debt: you have less wealth!

On the number line, the rule is universal and never changes: **The smaller number is always the one positioned further to the left!**

$$\mathbf{-5 < -2}$$

⚠️ Pitfall Warning — Rule for Comparing Negative Numbers

For any two negative numbers: **The larger number is the one closest to zero!**

$$-1 > -10 \qquad -3 > -8 \qquad -0.5 > -2$$

Every negative number is strictly less than 0 and less than every positive number:

$$-100 < +1$$

Worked Example 1 — Ordering Temperatures

Here are the morning temperatures recorded in five mountain villages: $+2^\circ$ C ; -4° C ; 0° C ; -8° C ; -1° C Let us arrange them from coldest to mildest (ascending order):

$$-8^\circ \text{ C} < -4^\circ \text{ C} < -1^\circ \text{ C} < 0^\circ \text{ C} < +2^\circ \text{ C}$$

---

6. Simple Additions with Negatives: Filling the Pits

How do we add signed numbers intuitively? Think of shovelfuls of sand or monetary balances!

Case 1: Two Gains or Two Debts with the Same Sign

- If you gain 3 coins, then gain 4 more: $(+3) + (+4) = +7$.
- If you dig a hole 3 scoops deep, then dig 2 scoops deeper from the bottom: you have a hole 5 scoops deep!

$$(-3) + (-2) = -5$$

Adding two debts together results in an even larger debt.

Case 2: A Pit and a Heap (Opposite Signs)

Imagine you have a hole 3 scoops deep (-3) and you arrive with a heap of 5 scoops ($+5$). You pour the sand into the hole:

- 3 scoops completely fill the hole back to level zero (0).
- The 2 leftover scoops form a small heap above the ground ($+2$)!

$$(-3) + (+5) = +2$$

What if you have a hole 6 scoops deep (-6) and only 2 scoops of sand ($+2$)? You pour in your 2 scoops: the hole is partly filled, but 4 scoops are still missing!

$$(-6) + (+2) = -4$$

> **Worked Example 2 — Elevator Movements**
>
> An elevator is on the 2nd basement level (-2). 1. It rides up 4 floors: $(-2) + 4 = +\mathbf{2}$ (it arrives on the 2nd floor). 2. From the 2nd floor ($+2$), it travels down 5 floors: $2 - 5 = -\mathbf{3}$ (it arrives on the 3rd basement level).

---

7. Intervals and Differences: Measuring Temperature Swings

What is the temperature difference between a freezing morning at -3° C and an afternoon at $+5^\circ$ C?

To find the gap, count the intervals on the thermometer:

- Rising from -3° C up to 0° C: you must climb **3 degrees**.
- Rising from 0° C up to $+5^\circ$ C: you must climb another **5 degrees**.
- Total temperature difference: $3 + 5 = \mathbf{8^\circ\ C}$!

> **Worked Example 3 — Elevation Difference**
>
> A seaside harbor sits at $+10$ m elevation, and an underwater sea cave sits at -45 m. What is the vertical distance between the harbor pier and the cave floor? Distance = 10 m (above water) $+ 45$ m (underwater) = $\mathbf{55}$ **meters**.

What You Need to Remember

1. **Negative numbers** carry a minus sign ($-$) and describe values beneath a reference baseline of zero (debts, temperatures below freezing, basement levels, ocean depths).
2. **Zero** is the origin and balance point where opposites neutralize one another: $(+a) + (-a) = 0$.
3. **On the number line:** Positive numbers sit to the right of 0; negative numbers sit to the left.
4. **Comparison:**
   - Every negative number is less than 0 and less than any positive number.

- Between two negative numbers, the greater one is **closest to zero** (furthest right on the axis): $-2 > -5$.

5. **Addition:**
   - Two negatives combine to form a larger negative: $(-3) + (-4) = -7$.
   - A positive and a negative partially cancel: $(-4) + 6 = +2$ and $(-7) + 3 = -4$.
6. **Temperature differences:** To measure the difference between a negative and a positive temperature, sum their distances from zero ($|-a| + b$).

Exercises

Level (Direct Application)

1. Write the signed number matching each description:
   a) It is 7 degrees below zero.
   b) A submarine dives 150 meters below sea level.
   c) A stone tower stands 85 meters tall.
   d) Ali owes 20 dollars to his sister.
   e) The thermometer shows 18 degrees above zero.
2. State the opposite of each number:
   a) $+5$
   b) -12
   c) -3.5
   d) 0
3. Insert $<$ or $>$:
   a) $-3 \ldots +2$
   b) $-6 \ldots -1$
   c) $-10 \ldots 0$
   d) $+4 \ldots -8$
   e) $-15 \ldots -20$
4. Arrange in ascending order (from smallest to largest):
$$+3 \quad ; \quad -7 \quad ; \quad 0 \quad ; \quad -2 \quad ; \quad +8 \quad ; \quad -5$$
5. Compute mentally:
   a) $(-2) + (-3)$
   b) $(+5) + (-2)$
   c) $(-7) + (+7)$
   d) $(-4) + (+6)$
   e) $(-8) + (+3)$

Level (Thermometers and Elevators)

6. A thermometer reads -4° C at 6:00 AM.
   - By noon, the temperature has risen by 9° C. What is the new temperature?
   - By midnight, it drops by 7° C. What is the midnight temperature?
7. In a department store:
   - Paul enters at the ground floor (0).
   - He rides down 2 levels to buy tools in the basement. What level is he on?
   - Next, he rides up 5 floors to visit the toy department. Which floor does he reach?
8. Fill in the blank to make each equation true:
   a) $(-5) + \ldots = 0$
   b) $(-3) + \ldots = +2$
   c) $(+4) + \ldots = -1$
   d) $(-6) + \ldots = -10$

9. Calculate the temperature difference between:
   a) -5° C and $+8^\circ$ C
   b) -12° C and -2° C
   c) -3° C and $+15^\circ$ C
10. A scuba diver is at a depth of -18 meters. She ascends 7 meters, then descends 4 meters. At what depth is she now located?
11. True or False? Justify your answer:
    a) The number -8 is greater than -3.
    b) Zero is greater than every negative number.
    c) The opposite of a negative number is always positive.
    d) Adding two negative numbers produces a positive number.

Level (Merchants' Accounts & History Challenges)

12. *The Bank Account:* Here are the transactions in an artisan's account across one week:
    - Monday morning: balance is $+\$120$.
    - Tuesday: pays a materials invoice of $\$150$.
    - Thursday: deposits a customer check for $\$80$.
    - Friday: pays workshop rent of $\$100$. Calculate his balance at the close of each day. Is his account in surplus or in debt (overdrawn) at week's end?
13. *Figures of Antiquity:* Years before the common era are often indexed with negative numbers.
    - Greek philosopher Socrates was born in year -470 and died in year -399.
    - Mathematician Euclid was born around -325 and died around -265.
    a) At what age did Socrates die?
    b) Who was born first: Socrates or Euclid?
    c) On the mathematical timeline, the origin is set at 0. How many years separate Socrates' birth from this origin 0? (Historical note: in the civil calendar, year 0 does not exist; dates skip directly from 1 BC to AD 1!).
14. *The Game of Red and Green Dice:* Two special dice are rolled:
    - The green die awards positive points ($+1$ to $+6$).
    - The red die inflicts negative points (-1 to -6). A player rolls a green 3 and a red 5. What is the player's score? Which rolls yield a net score of 0?
15. *The Mine Shaft Challenge:* The peak of a hill rises to $+340$ meters elevation. The entrance to a mine shaft sits on the hillside at $+180$ meters. The vertical shaft drops 250 meters straight down from the entrance. What is the elevation of the bottom of the mine shaft relative to sea level?

Solutions

1. a) -7° C; b) -150 m; c) $+85$ m; d) $-\$20$; e) $+18^\circ$ C. **2.** a) -5; b) $+12$; c) $+3.5$; d) 0 (0 is its own opposite). **3.** a) $-3 < +2$ (a negative is always less than a positive). b) $-6 < -1$ (-1 is closer to 0). c) $-10 < 0$. d) $+4 > -8$. e) $-15 > -20$ (-15 is closer to 0 than -20).

4. Ascending order:

$$\mathbf{-7 < -5 < -2 < 0 < +3 < +8}$$

5. a) $(-2) + (-3) = \mathbf{-5}$; b) $(+5) + (-2) = \mathbf{+3}$; c) $(-7) + (+7) = \mathbf{0}$; d) $(-4) + (+6) = \mathbf{+2}$; e) $(-8) + (+3) = \mathbf{-5}$.

6. - At noon: $(-4) + 9 = \mathbf{+5}^\circ$ **C**. - At midnight: $(+5) - 7 = \mathbf{-2}^\circ$ **C**.

7. - After going down 2 floors: he is on **level $\mathbf{-2}$** (2nd basement). - After riding up 5 floors: $(-2) + 5 = \mathbf{+3}$ (he arrives on the **3rd floor**).

8. a) $(-5) + \mathbf{5} = 0$; b) $(-3) + \mathbf{5} = +2$; c) $(+4) + (\mathbf{-5}) = -1$; d) $(-6) + (\mathbf{-4}) = -10$.

9. a) Between -5° C and $+8^\circ$ C: $5 + 8 = \mathbf{13^\circ\ C}$ difference. b) Between -12° C and -2° C: $12 - 2 = \mathbf{10^\circ\ C}$ difference. c) Between -3° C and $+15^\circ$ C: $3 + 15 = \mathbf{18^\circ\ C}$ difference.

10. $(-18) + 7 - 4 = -11 - 4 = \mathbf{-15\ \ meters}$ (she is 15 meters below sea level). **11.** a) **False**, $-8 < -3$ because -8 lies further to the left from zero. b) **True**, zero is to the right of every negative number. c) **True**, the opposite of $-x$ is $+x$. d) **False**, combining two debts yields an even greater debt (a negative number)! For instance $(-2) + (-3) = -5$.

12. - Monday: $+\$120$. - Tuesday: $120 - 150 = \mathbf{-\$30}$ (he has a \$30 deficit). - Thursday: $(-30) + 80 = \mathbf{+\$50}$ (back in surplus). - Friday: $50 - 100 = \mathbf{-\$50}$.

At the end of the week, his balance is $\mathbf{-\$50}$ (he is overdrawn by \$50).

13. a) From -470 to -399: $(-399) - (-470) = 470 - 399 = \mathbf{71\ \ years\ old}$. Socrates died at age 71. b) **Socrates** was born first (-470 is further back in time than -325). c) On the number line, the distance from -470 to origin 0 is $|-470 - 0| = \mathbf{470\ \ years}$. (In historical civil calendars, year 0 does not exist as historians jump directly from 1 BC to AD 1; however, on the continuous coordinate axis of mathematicians and astronomers, the distance to zero is exactly 470 units).

14. - Score: $(+3) + (-5) = \mathbf{-2}$. - To score 0, the green die and red die must show identical numbers (for instance green 1 and red 1, up to green 6 and red 6). Their magnitudes cancel: $(+x) + (-x) = 0$.

15. Elevation of the shaft bottom: $180 - 250 = \mathbf{-70\ \ meters}$. The bottom of the mine sits $\mathbf{70\ \ meters}$ **below sea level**.

Grand Review of Volume II: The Ground Conquered

As you close this second volume, look back upon the vast intellectual territory your mind has mastered between ages 8 and 11: 1. In **Chapter 8**, you investigated multiples, divisibility tests, and **prime numbers**, the indivisible atoms of arithmetic. 2. In **Chapter 9**, you mastered the justice of fair sharing through **Euclidean division**, the long division bracket, and the inviolable remainder rule ($r < b$). 3. In **Chapter 10**, you broke the bread and cake to tame **fractions**, the meaning of denominators, and the art of folding paper to find equivalent fractions. 4. In **Chapter 11**, following **Simon Stevin**, you extended decimal place value below the threshold of one using the decimal point. 5. In **Chapter 12**, you joined the scholars of the French Revolution to unite measurements under the **meter**, exploring perimeters, areas, and angles. 6. In **Chapter 13**, you mastered **proportionality**, grandmother's recipes, unit rates, and the rule of three. 7. And finally, in **Chapter 14**, you crossed the looking-glass into **negative numbers** and the complete bilateral number line.

The Question that Opens Volume III…

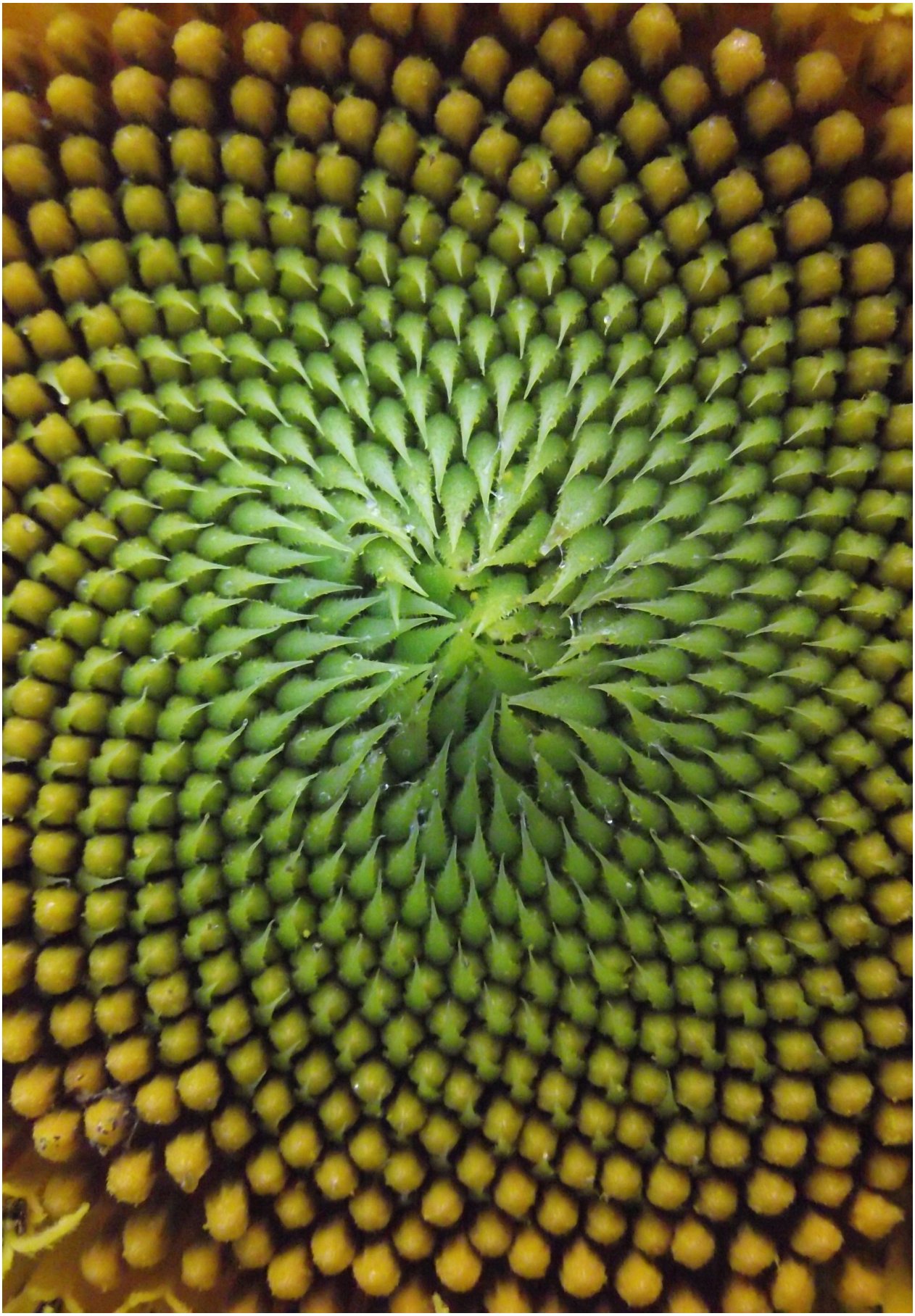
The trading city flourishes in peace along the river banks. Yet in early spring, the melting snows from distant mountains swell the waters beyond measure: **the great river bursts its banks in a catastrophic flood**. For weeks, the fertile valley vanishes beneath thick muddy waters.

When the floodwaters finally recede into the riverbed, a desolate scene greets the farming clans: Every hedge has been swept away. Every boundary stone marking the borders between fields has vanished under a thick, uniform layer of river silt. No one knows where their pasture begins or where their neighbor's land ends!

The governor summons the geometers: *"We must resurvey the entire plain, retrace every family's boundaries, and construct defensive flood dikes with true, perfect right angles of* 90°*!"*

How can one lay out a mathematically perfect right angle across a vast expanse of muddy fields, without plastic drafting squares, using nothing more than… **a simple closed loop of rope tied with thirteen equally spaced knots**?

Welcome to the epic quest of builders and surveyors in **Volume III: The Age of Surveyors and Master Builders (Ages 10–13 / Grades 5–8)**!



VOLUME III — THE STAR OF ALEXANDRIA AND EUCLIDEAN SPACE

Ages 11–13 / Grades 6–7

At the gates of middle school, deductive geometry organizes itself around Euclid's axioms, relative numbers, and planar symmetries.

Chapters of Volume III :

Chapter 15 : The Flooding of the Nile and the Thirteen-Knot Rope
Chapter 16 : The Potter's Wheel and Archimedes' Secret
Chapter 17 : The Shadow of the Pyramid and the Upright Staff
Chapter 18 : The Grain of Rice on the Chessboard
Chapter 19 : The Forbidden Secret of the Pythagoreans
Chapter 20 : Treasure Island and the Two Paths
Chapter 21 : The Tax Collector and the Hundred Pieces

Facing: Phyllotaxis and Fibonacci spirals of the Sunflower (Helianthus annuus).

CHAPTER 15: The Flooding of the Nile and the Thirteen-Knot Rope

> "When the floodwaters have washed everything away and mud covers the earth, man can no longer trust his eyes. He needs a taut cord between his fingers, and the geometry of the rope to make the right angle rise again."

What You Are About to Discover

- What an angle **truly** is (an opening, a sweep of the arm… and not a length!) and how to measure it with a **protractor**.
- The angle families: **acute**, **right**, **obtuse**, **straight**… and the words used to name them.
- How to construct **perpendicular** and **parallel** lines, using a set square, a compass… or a simple length of rope.
- Why the **sum of the angles of a triangle** is always 180° — and three distinct ways to prove it.
- **Special triangles** (isosceles, equilateral, right-angled) and their geometric secrets.
- The ancient mystery of the **thirteen-knot rope** that builds a flawless right angle on the ground.

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1. The Problem: The Mud of the Nile and the Wrath of the Farmers

You are now ten or eleven years old. Let us leave our small village behind and journey to the banks of a mighty river: the **Nile**, in Egypt, more than three thousand years ago.

Every year, in the heart of summer, an extraordinary event occurs. Far to the south, in the highlands of Ethiopia, torrential rains pour down. The waters of the Nile swell, burst from their banks, and submerge the entire valley. For weeks on end, the dry earth vanishes beneath a vast sea of brown water.

For the Egyptians, this **annual flood** (*akhet*) is a supreme blessing: the receding river leaves behind a layer of intensely fertile black mud, called **silt**. Thanks to this mineral-rich mud, wheat and barley flourish in abundance. The ancient Egyptians affectionately named their homeland *Kemet*, "the Black Land."

Yet when the waters recede in autumn, a nightmare begins for the villagers: **every single stone marker, every furrow and boundary peg separating the farm plots has been swept away, displaced, or buried beneath feet of thick mud.** The immense plain has become a uniform grey expanse devoid of the slightest landmark.

"My boundary reached all the way to the tall date palm!" shouts Ameni. *"Liar! It ended ten cubits further south!"* bellows Nakht, raising his hoe in fury.

To prevent bloody land disputes and restore order, the Pharaoh dispatches his most elite civil servants: the **royal surveyors**. Centuries later, Greek travelers would marvel at them and call them the **harpedonaptai**, which literally translates to "the rope-stretchers."

Their mission is formidable: they must lay out **rectangular** fields once more. And to construct a rectangle, one must produce **right angles**. In your notebook, you simply reach for your plastic set square. But across a muddy field two hundred meters wide, a small hand square is completely useless: if your angle is off by just a single fraction of a degree at the start, two hundred meters down the field your line will have drifted by more than three meters… cutting directly into your neighbor's property!

How can one produce a flawless right angle directly upon the open ground, without a metal drafting square, without a telescope, and without a protractor? Before answering this question, we must understand what an angle truly is.

> 💡 **Did You Know?**
>
> The word **geometry** originates from the ancient Greek *gê* ("the earth") and *metron* ("measure"). Geometry was born, quite literally, as "the measurement of the earth." The Greek historian Herodotus recorded that geometry was invented in Egypt out of the urgent practical necessity of re-measuring agricultural parcels after each Nile inundation.

---

2. What Is an Angle?

2.1 The Intuition with Your Own Body

Extend your right arm straight out in front of you, perfectly horizontal. Keep your shoulder pinned in place and slowly pivot your arm to your right, like the hand of a clock sweeping across a dial.

- At first, your arm has barely moved: the **opening** between your starting direction and your current direction is narrow.
- When your arm points straight out to your right side, it has made a **quarter-turn**: this is a right angle.
- If you keep pivoting until your arm points straight behind you, you have completed a **half-turn**.
- And if you pivot all the way back to the front, you have made a **full turn**.

This is the essence of an angle: **the amount of rotation between two directions that originate from the same common point**. Your shoulder acts as the **vertex**, and the two pointing directions are its two **sides** (rays).

2.2 Words, Notations, and Geometrical Symbols

Before proceeding further, let us recall the foundational notation of geometry:

Written Form	Pronunciation	Exact Mathematical Meaning
$[AB]$	"segment AB"	The straight line segment connecting A and B, including both endpoints
(AB)	"line AB"	The infinite straight line passing through A and B, extending without end in both directions
$[AB)$	"ray AB"	The ray starting at origin A, passing through B, and extending infinitely beyond B
AB	"AB" (no brackets)	The **length** of segment $[AB]$, which is a number (e.g., 5 cm)

> 📐 **Definition — Angle**
>
> An **angle** is the geometric figure formed by two rays sharing the same starting point. This shared starting point is the **vertex** of the angle; the two rays are its **sides**.

> 📐 **Fundamental Definition — The Geometric Anatomy of an Angle**
>
> - Side [BC)
> - ← The opening: this rotation is the angle
> - B A
> - Vertex Side [BA)

To denote this angle symbolically, we write:

$$\widehat{ABC} \quad \text{or} \quad \angle ABC$$

We read this aloud as **"angle ABC"**. The circumflex hat (or the $\angle$ sign) placed above the letters serves as a clear warning: *"I am not speaking of individual points, but of the rotational opening between these two rays."*

> ⚠️ **Warning: Common Pitfall!**
>
> In the notation $\widehat{ABC}$, **the middle letter must always designate the vertex**. $\widehat{ABC}$ and $\widehat{CBA}$ describe the exact same angle (vertex B), but $\widehat{BAC}$ is an entirely different angle whose vertex is A! When there is no risk of confusion, mathematicians often write simply $\hat{B}$ ("angle B").

2.3 An Angle Is NOT a Length

Here lies the most widespread misunderstanding among apprentice mathematicians. Examine these two angles:

> 💡 **Observation — Comparing the Aperture of an Angle's Rays**
>
> • Angle 1: Angle 2:

Which of the two angles is larger? Many beginners reflexively answer "the second one," simply because its printed rays are drawn much longer on the paper. **That is completely false!** The sides of an angle are rays: they extend infinitely into space, and we merely sketch a small portion of them. What truly counts is the **rotational opening**, never the length of the pencil strokes. The two angles shown above possess the exact same angular opening.

> 🖐 ***Try It Yourself*** *— Take two pencils and bind them firmly together at one end with a tight rubber band, like a pair of compasses. Open them slightly. Now replace them with two long meter sticks, keeping the exact same opening: the angle has not changed by a single hair, only the length of the arms has changed!*

---

3. Measuring an Angle: The Degree and the Protractor

3.1 Why 360 Degrees?

To quantify rotation precisely, we require a standard **unit of measurement**. Approximately four thousand years ago, the astronomers and mathematicians of Babylon divided a complete circular turn into **360 equal slices**. Each elementary slice is called a **degree**, symbolized by a small raised circle:

$$1^\circ \quad \text{reads "one degree"}$$

Why did the ancients choose 360 instead of a round decimal number like 100? For several compelling reasons:

- The Babylonians computed in **sexagesimal (base 60)**; our 60 minutes in an hour and 60 seconds in a minute stem directly from their system, and $360 = 6 \times 60$.
- The solar year lasts roughly 365 days; to the ancient sky-watchers, the Sun appeared to advance across the zodiac by approximately one degree each day.
- Above all, **360 is exceptionally divisible**. It can be cleanly split into equal integer parts by 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 18, 20, 24, 30, 36, 40, 45, 60, 72, 90, 120, and 180. A third of a turn, an eighth of a turn, a fifth of a turn—all yield whole integer degrees!

Rotation	Fractional Turn	Exact Degree Measure
Complete turn	1 full turn	360°
Half-turn	$\frac{1}{2}$ turn	180°
Quarter-turn	$\frac{1}{4}$ turn	90°
Eighth-turn	$\frac{1}{8}$ turn	45°
Twelfth-turn (one hour on a clock face)	$\frac{1}{12}$ turn	30°

3.2 Families of Angles

📐 Definition — Angle Classification

According to its measure θ, an angle is classified as:

- **Zero angle:** $\theta = 0^\circ$ (the two rays completely overlap).
- **Acute angle:** $0^\circ < \theta < 90^\circ$ (sharp, pointed, narrower than a quarter-turn).
- **Right angle:** $\theta = 90^\circ$ (the perpendicular corner).
- **Obtuse angle:** $90^\circ < \theta < 180^\circ$ (wide, blunt, opened wider than a quarter-turn).
- **Straight angle:** $\theta = 180^\circ$ (the two rays form a continuous straight line).
- **Reflex angle:** $180^\circ < \theta < 360^\circ$ (an angle opening beyond a straight line).
- **Full (perigon) angle:** $\theta = 360^\circ$ (one complete rotation).

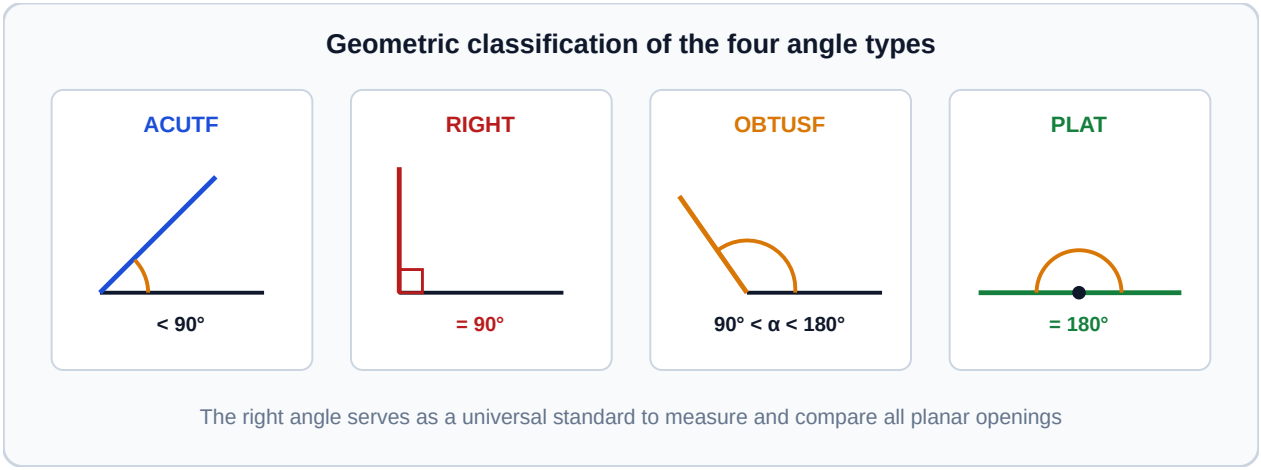

Figure 51: Classification of the four angle types

In geometric drawings, a right angle is always highlighted by a **small square** nestled in the corner: ⌞.

Mnemonic tip: **Acute** shares its Latin root with *needle* (sharp, piercing, narrow); **obtuse** shares its root with *blunt* (dull, wide open).

3.3 Mastering the Protractor

A standard protractor is a transparent semicircular tool graduated from 0° to 180°. It features **two concentric scales**: one reads from left to right, and the other reads from right to left. This dual scale is the source of nearly all beginner measurement blunders!

🧭 Method — Measuring an Angle $\widehat{ABC}$ with a Protractor

1. Place the **center point** of the protractor (the crosshair or small hole) directly over the **vertex** B. 2. Align the **baseline zero-mark** of the protractor perfectly along one of the rays, say $[BA)$. 3. Locate the scale **that starts at 0° on that chosen ray**, follow it around the rim, and read the number where the second ray $[BC)$ crosses the curved graduation. 4. **Perform a visual sanity check:** Look at the angle with your naked eye! Is it acute or obtuse? Your numerical reading must agree with physical reality.

⚠️ Warning: Double Scale Trap!

If an angle is visibly acute but your reading says "130°", you have read from the wrong concentric ring! The correct reading is $180^\circ - 130^\circ = 50^\circ$. Asking yourself *"Is it sharp or blunt?"* prevents 99% of mistakes.

Method — Constructing an Angle of Given Measure (e.g., $\widehat{xOy} = 65°$)

1. Draw a straight ray $[Ox)$ with a

ruler. 2. Place the protractor center on point O, aligning the zero-line with $[Ox)$. 3. Starting from $0°$ along $[Ox)$, count up to $65°$ and make a crisp pencil tick on the paper. 4. Remove the protractor and use your ruler to draw ray $[Oy)$ from vertex O through your tick mark. 5. Mark the angle with a small arc and label it $65°$.

Solved Examples

Solved Example 1 — Classifying Angles. Classify each of the following angular measures: $17°$; $90°$; $91°$; $179°$; $180°$; $45°$.

Solution: $17°$ and $45°$ are **acute** (strictly between $0°$ and $90°$). $90°$ is a **right** angle. $91°$ and $179°$ are **obtuse** (strictly between $90°$ and $180°$). $180°$ is a **straight** angle. *Remark:* Even though $91°$ looks almost identical to a right angle to the untrained eye, in geometry a single degree makes all the difference: it is strictly obtuse!

Solved Example 2 — Clock Hands. What angle is formed by the hands of an analog clock at 2:00? At 5:00?

Solution: A clock face represents a complete turn of $360°$ partitioned into 12 hourly increments. Between any two consecutive numbers, the angle is:

$$360° \div 12 = 30°$$

At 2:00, the minute hand points at 12 and the hour hand at 2. There are 2 hourly intervals:

$$2 \times 30° = 60° \quad (\text{acute angle})$$

At 5:00, there are 5 intervals:

$$5 \times 30° = 150° \quad (\text{obtuse angle})$$

Solved Example 3 — Reading a Protractor Without Confusion. Leah places her protractor over angle $\widehat{DEF}$ (vertex E, with ray $[ED)$ aligned with the zero mark on the right). Ray $[EF)$ passes through the marks 40 and 140. The angle is clearly wide open. What is its measure?

Solution: Ray $[ED)$ lies on the **right** zero; we must follow the scale starting at 0 on the right. Alternatively, using our visual check: the angle is visibly obtuse (wider than $90°$), so it cannot possibly be $40°$. Therefore, $\widehat{DEF} = 140°$.

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4. Perpendicular and Parallel Lines

4.1 Two Intersecting Lines

When two straight lines cross at a point, they form **four angles**:

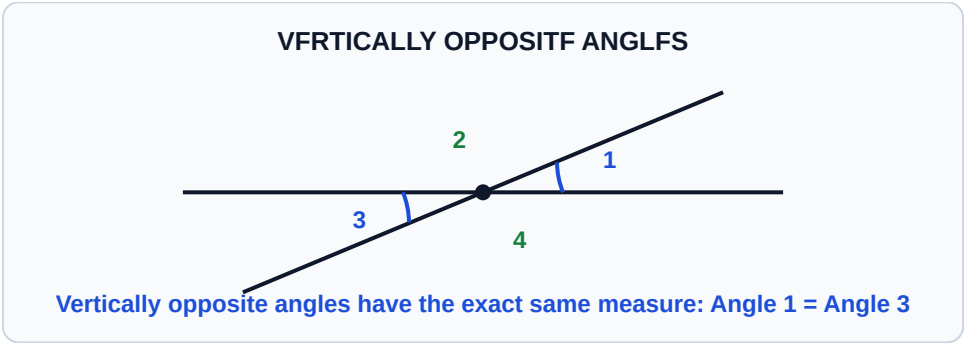

Figure 52: Vertically opposite angles and geometric equality

- Angles ① and ③ are called **vertically opposite angles** (or vertical angles): they sit face-to-face across the intersection, like the two blades of an open pair of scissors. They have **identical measures**.
- Angles ① and ② lie side-by-side along a straight line: their measures add up to $180°$. They are called **supplementary angles**.

📐 ***Vocabulary — Angle Pairs***

- *Two angles are **supplementary** if their sum equals 180°.*
- *Two angles are **complementary** if their sum equals 90°.*
- *Two angles are **adjacent** if they share the same vertex, have a common side, and lie on opposite sides of that common side.*

Why are vertically opposite angles strictly equal? Because ① + ② = 180° and ② + ③ = 180°. Both ① and ③ complete the exact same angle ② to reach a straight line (180°). Therefore:

$$\text{Angle } ① = 180^\circ - \text{Angle } ② = \text{Angle } ③$$

This is our very first formal **geometric proof**: we did not rely on physical measurements, but solely on deductive logical reasoning!

4.2 Perpendicular Lines

If just one of the four angles at an intersection is a right angle (90°), then all four angles are automatically right angles ($180^\circ - 90^\circ = 90^\circ$).

📐 **Definition — Perpendicular Lines**

Two lines are **perpendicular** if they intersect at a right angle (90°). We write:

$$(d_1) \perp (d_2)$$

and read "line d_1 is perpendicular to line d_2." The symbol $\perp$ is itself a graphic depiction of two perpendicular lines!

4.3 Parallel Lines

📐 **Definition — Parallel Lines**

Two lines lying in the same plane are **parallel** if they never intersect—no matter how far they are extended—or if they are identical (coincident). We write:

$$(d_1) \parallel (d_2) \quad \text{or} \quad (d_1) \,//\, (d_2)$$

and read "d_1 is parallel to d_2." The symbol depicts two side-by-side lines that never meet, like straight railway tracks.

Three fundamental properties govern these relationships in a Euclidean plane:

📐 ***Fundamental Euclidean Properties —***

- ***P1.** If two lines are perpendicular to the same third line, then they are parallel to each other.*
- ***P2.** If two lines are parallel, any third line perpendicular to one is also perpendicular to the other.*
- ***P3.** If two lines are parallel to the same third line, they are parallel to each other.*

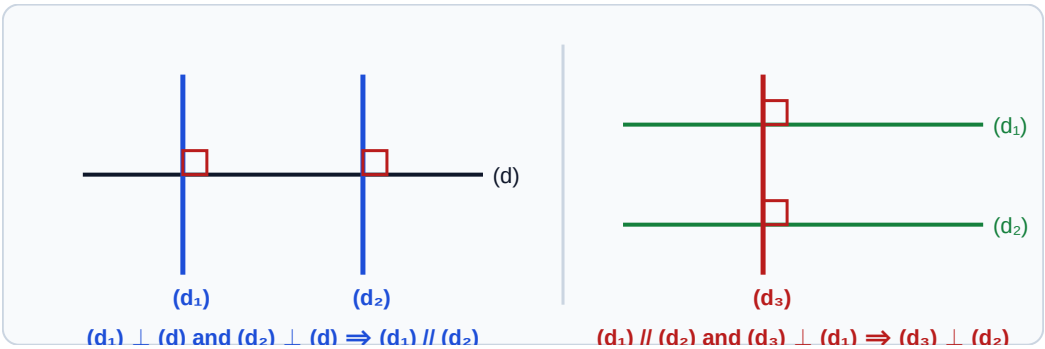

Figure 53: Fundamental parallel and perpendicular lines

The logical symbol ⇒ reads "therefore" or "implies": the truth of the statement on the left guarantees the truth of the statement on the right.

4.4 Construction with Tools

> 🖐 **Method — Drawing the Perpendicular to (d) Through Point A (Set Square)**
>
> Place one side of the set square's right angle firmly along line (d). Slide the set square along (d) until its other perpendicular side meets point A. Draw the line through A.

> 📐 **Method — Drawing the Parallel to (d) Through Point A (Ruler + Set Square)**
>
> First, draw the line (p) perpendicular to (d) passing through A. Then construct the line perpendicular to (p) passing through A. By property **P1**, this second line is strictly parallel to (d).

> 🧭 **Method — The Compass-Only Perpendicular (The Master Builder's Method)**
>
> 1. Place the compass point at A and strike an arc
>
> that cuts line (d) at two points, M and N. 2. With a wider opening, place the compass point at M, then at N, and strike two intersecting arcs at a point P (on the other side of (d)). 3. Draw line (AP): it is guaranteed to be perpendicular to (d). *Why?* Both A and P are equidistant from M and N, placing them on the **perpendicular bisector** of segment $[MN]$.

This compass method is precisely what Egyptian and medieval master builders performed on vast building sites using **two pegs and a piece of cord** functioning as a giant compass!

4.5 Distance from a Point to a Line

Imagine you are standing at point A in the middle of a field and wish to reach a straight paved road (d) as quickly as possible. Instinctively, you walk straight towards the road at a **perpendicular angle**.

> 📐 **Definition — Distance from a Point to a Line**
>
> The **distance from point A to line (d)** is the length of segment $[AH]$, where H is the foot of the perpendicular dropped from A onto (d). It represents the shortest possible path connecting A to (d).

Solved Example 4 — Angles Around an Intersection

Two straight lines intersect at O. One of the angles measures $38°$. Find the measures of the remaining three angles.

Solution: The vertically opposite angle also measures $38°$. The two adjacent angles are supplementary to $38°$:

$$180° - 38° = 142°$$

The four angles measure $38°$, $142°$, $38°$, and $142°$. *Verification:* $38° + 142° + 38° + 142° = 360°$ (one full rotation). ✓

---

5. The Sum of the Angles in Any Triangle: Always $180°$

Here stands one of the most stunning and universal truths of all geometry: take any triangle in the universe—whether microscopic or spanning light-years, razor-sharp or flat and obtuse—and measure its three internal angles. Add them together. **The sum will always be exactly $180°$!**

$$\hat{A} + \hat{B} + \hat{C} = 180°$$

Let us explore three distinct ways to convince ourselves of this eternal truth.

5.1 Proof 1: The Paper-Tear Experiment (Physical Intuition)

1. Take a sheet of blank paper and draw any triangle you like with a ruler.

2. Color each of the three corners with a different colored pencil: Blue at vertex A, Red at vertex B, and Amber at vertex C.
3. Carefully cut out the triangle, then tear off the three colored corners.
4. Arrange the three points together on a flat tabletop so their tips meet at a single common point, placing the colored edges snugly side-by-side:

The three angles fit together seamlessly to form a perfect straight edge! Since a straight edge represents a straight angle of 180°, their sum is demonstrably 180°.

5.2 Proof 2: The Euclidean Proof (Deductive Rigor)

To transform this sensory experiment into a watertight mathematical proof, we follow Euclid's luminous reasoning using parallel lines:

1. Given any triangle ABC, construct line (DE) passing through vertex A parallel to the opposite base (BC).
2. The three angles meeting at point A along the straight line (DE) form a straight angle:
$$\widehat{DAB} + \widehat{BAC} + \widehat{CAE} = 180^\circ$$
3. Because (DE) is parallel to (BC), line (AB) acts as a transversal line cutting two parallel lines:
   - Angle $\widehat{DAB}$ and angle $\widehat{ABC}$ are **alternate interior angles**, so they are equal: $\widehat{DAB} = \hat{B}$.
   - Similarly, angle $\widehat{CAE}$ and angle $\widehat{ACB}$ are alternate interior angles, so: $\widehat{CAE} = \hat{C}$.
4. Substituting these equalities into our sum yields:
$$\hat{B} + \hat{A} + \hat{C} = 180^\circ$$

   The proof is complete!

5.3 Proof 3: The Walker's Turn (Algorithmic / Robotic View)

Imagine a tiny explorer (or a robot) walking along the perimeter of triangle ABC:

- The walker starts at A, walks toward B, and at vertex B turns left by exterior angle $\theta_1 = 180^\circ - \hat{B}$.
- She walks to C and turns left by $\theta_2 = 180^\circ - \hat{C}$.
- She walks back to A and turns left to face her initial heading: $\theta_3 = 180^\circ - \hat{A}$.

Upon returning to A, the walker has made exactly **one full rotation** (360°) in the plane:

$$(180^\circ - \hat{A}) + (180^\circ - \hat{B}) + (180^\circ - \hat{C}) = 360^\circ$$

$$540^\circ - (\hat{A} + \hat{B} + \hat{C}) = 360^\circ$$

$$\hat{A} + \hat{B} + \hat{C} = 540^\circ - 360^\circ = 180^\circ$$

---

6. Special Triangles and the Secret of the Thirteen-Knot Rope

6.1 The Equilateral Triangle

An **equilateral triangle** has three sides of equal length. By symmetry, its three internal angles are also strictly equal:

$$\hat{A} = \hat{B} = \hat{C} = \frac{180^\circ}{3} = 60^\circ$$

Every equilateral triangle in the universe has three 60° angles.

6.2 The Isosceles Triangle

An **isosceles triangle** has two sides of equal length. The angles sitting opposite those equal sides (called the **base angles**) are strictly equal:

- If the apex angle is 40°, the two base angles share the remaining 140° equally:
$$\frac{180^\circ - 40^\circ}{2} = 70^\circ \quad \text{each}$$

6.3 The Right-Angled Triangle

A **right-angled triangle** has one right angle (90°). The other two acute angles must therefore absorb the remaining 90°:

$$\hat{A} + \hat{B} = 90^\circ$$

The two acute angles of a right triangle are always **complementary**. If one measures 35°, the other must measure $90^\circ - 35^\circ = 55^\circ$.

6.4 The Thirteen-Knot Rope: The Harpedonaptai's Secret Weapon

We can now resolve the mystery posed in our opening problem: how did the Egyptian rope-stretchers lay out a flawless right angle on the ground?

They used a single length of rope tied with **thirteen knots spaced at identical intervals**, and joined its ends to form a continuous closed loop:

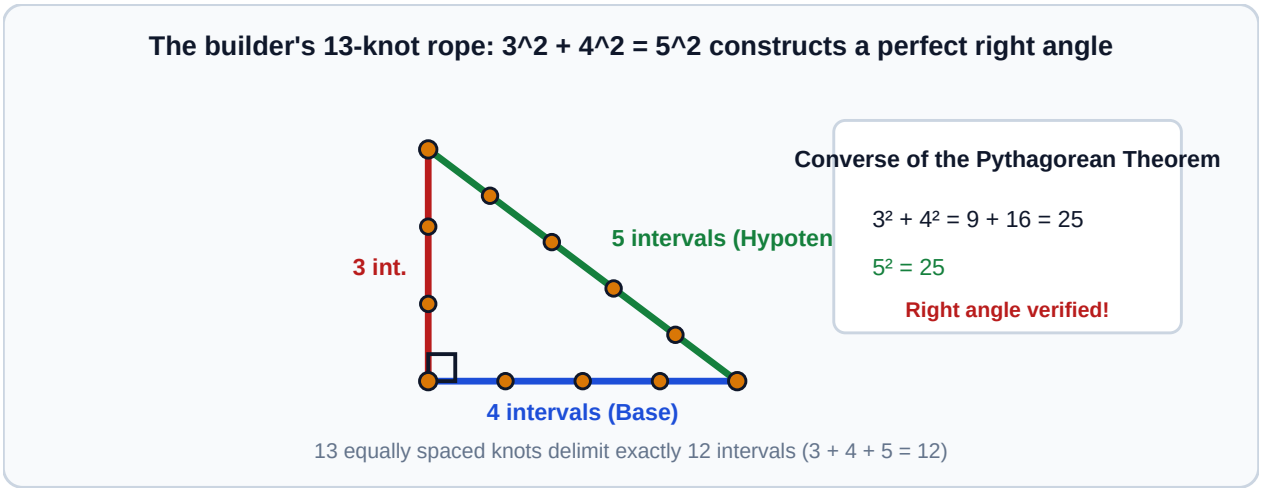

Figure 54: The Egyptian thirteen-knot rope and the 3-4-5 right triangle

Because knot 13 joins knot 1, the loop contains exactly **12 equal segments**. The surveyors drove three wooden stakes into the ground and pulled the rope tight around them, distributing the segments as follows:

- **3 segments** along the first side (from knot 1 to knot 4);
- **4 segments** along the second side (from knot 4 to knot 8);
- **5 segments** along the third side (from knot 8 back to knot 13/1).

$$3 + 4 + 5 = 12 \text{ segments}$$

When pulled taut, the angle formed between the side of 3 units and the side of 4 units is **guaranteed to be an absolute, mathematical right angle (90°)**! *Why does this work?* Because $3^2 + 4^2 = 9 + 16 = 25 = 5^2$. This is the famous $(3, 4, 5)$ **Pythagorean triple**, an algebraic harmony discovered empirically by Egyptian builders more than a millennium before Pythagoras was born!

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7. 🛠️ The Boundary Case Laboratory

To sharpen our mathematical thinking, let us test the boundaries of our rules where everyday intuition falters:

Case 1: Can a Triangle Have Two Right Angles?

Suppose a triangle had two right angles: $\hat{A} = 90^\circ$ and $\hat{B} = 90^\circ$. The sum of all three angles must equal 180°:

$$\hat{C} = 180^\circ - (90^\circ + 90^\circ) = 0^\circ$$

An angle of 0° means the two sides are parallel and never meet! In a Euclidean flat plane, **it is strictly impossible for a triangle to possess two right angles**.

> 🌎 ***Beyond Flat Space — Spherical Geometry:*** *However, on the curved surface of a sphere—such as planet Earth—lines of longitude meet the equator at 90° right angles, and they also meet at the North Pole! A triangle drawn from the North Pole down to the equator at 0° longitude, along the equator to 90° East longitude, and back up to the pole has **three right angles** ($90^\circ + 90^\circ + 90^\circ = 270^\circ$). The 180° theorem is true only in **flat Euclidean space**!*

Case 2: The Degenerate (Flat) Triangle

What happens if the vertex C is pushed down onto the base segment $[AB]$? The angle $\hat{C}$ opens all the way to $180°$, while $\hat{A}$ and $\hat{B}$ collapse to $0°$:

$$0° + 180° + 0° = 180°$$

The triangle flattens into a single straight line segment. It is called a **degenerate triangle**.

Case 3: Angles in Non-Coplanar Space

In three-dimensional space, two lines that never intersect are not necessarily parallel! Think of a bridge crossing high above a highway: the road below and the bridge above never meet, yet they point in different directions. Such lines are called **skew lines** (or non-coplanar lines). Parallelism requires the lines to lie within the **same flat plane**.

What You Must Remember

1. **Nature of an Angle:** An angle measures an opening of rotation between two rays sharing a common vertex. It does not depend on the length of the drawn strokes.
2. **Standard Units:** A full turn is $360°$; a straight angle is $180°$; a right angle is $90°$.
3. **Protractor Sanity Check:** Always verify whether the angle is acute ($< 90°$) or obtuse ($> 90°$) before recording your measurement.
4. **Opposite and Supplementary Angles:** Vertically opposite angles are equal. Adjacent angles along a straight line are supplementary (sum $= 180°$).
5. **Sum of Angles in a Triangle:** For any flat triangle, $\hat{A} + \hat{B} + \hat{C} = 180°$.
6. **Special Triangles:**
   - Equilateral: three equal sides, three $60°$ angles.
   - Isosceles: two equal sides, two equal base angles.
   - Right: one $90°$ angle, acute angles sum to $90°$.
7. **The Thirteen-Knot Rope:** A closed loop with 12 equal intervals formed into a $3 - 4 - 5$ triangle guarantees a perfect $90°$ corner.

Exercises

Level (Direct Application)

Exercise 15.1 Classify each of the following angle measures as acute, right, obtuse, or straight: $12°$, $90°$, $144°$, $89°$, $180°$, $93°$.

Exercise 15.2 In a triangle ABC, we measure $\hat{A} = 52°$ and $\hat{B} = 68°$. Calculate the measure of angle $\hat{C}$.

Exercise 15.3 One of the acute angles of a right-angled triangle measures $37°$. What is the measure of the second acute angle?

Exercise 15.4 An isosceles triangle has an apex angle measuring $50°$. Find the measures of its two base angles.

Exercise 15.5 Two straight lines cross at point O. One of the angles formed measures $72°$. Determine the measures of the other three angles.

Level (Reasoning and Geometric Deductions)

Exercise 15.6 In triangle DEF, angle $\hat{D}$ is three times as large as angle $\hat{E}$, and angle $\hat{F} = 40°$. Find the measures of angles $\hat{D}$ and $\hat{E}$.

Exercise 15.7 Can a triangle exist whose angles measure $45°$, $65°$, and $75°$? Explain your answer.

Exercise 15.8 In an isosceles triangle, one of the angles measures $100°$. 1. Can this $100°$ angle be one of the base angles? Why or why not? 2. What are the measures of the other two angles?

Exercise 15.9 A medieval mason wishes to check if the corner of a stone wall is square ($90°$). He measures $60\ \text{cm}$ along one wall and marks point A. He measures $80\ \text{cm}$ along the adjoining wall and marks point B. What distance must separate A and B for the corner to be a perfect right angle?

Exercise 15.10 Lines (d_1) and (d_2) are parallel. A transversal line crosses (d_1) at an angle of $64°$. What are the measures of all eight angles formed at the two intersections?

Level (Challenges and Investigations)

Exercise 15.11 Sum of Angles in Any Quadrilateral. 1. Draw any four-sided polygon (quadrilateral) $ABCD$. 2. Divide it into two triangles by drawing diagonal $[AC]$. 3. Deduce the exact sum of the four interior angles of any quadrilateral.

Exercise 15.12 The Star of Alexandria. Consider a regular five-pointed star. What is the sum of the five angles located at the five pointed tips of the star? *(Hint: relate the tip angles to the interior triangles of the star).*

Solutions

Solution 15.1 $12°$ and $89°$: acute. $90°$: right. $93°$ and $144°$: obtuse. $180°$: straight.

Solution 15.2 The sum of angles in triangle ABC is $180°$: $\hat{C} = 180° - (52° + 68°) = 180° - 120° = 60°$.

Solution 15.3 The acute angles of a right triangle are complementary: $\text{Second angle} = 90° - 37° = 53°$.

Solution 15.4 The base angles share the remaining degrees equally: $\text{Base angle} = \frac{180° - 50°}{2} = \frac{130°}{2} = 65°$ each.

Solution 15.5 The vertically opposite angle also measures $72°$. The two adjacent supplementary angles each measure $180° - 72° = 108°$. The four angles are $72°$, $108°$, $72°$, and $108°$.

Solution 15.6 We know that $\hat{D} + \hat{E} + \hat{F} = 180°$, so $\hat{D} + \hat{E} = 180° - 40° = 140°$. Since $\hat{D} = 3 \times \hat{E}$, we have $3 \times \hat{E} + \hat{E} = 4 \times \hat{E} = 140°$. Thus $\hat{E} = 140° \div 4 = 35°$, and $\hat{D} = 3 \times 35° = 105°$. Check: $105° + 35° + 40° = 180°$. ✓

Solution 15.7 Let us compute the sum: $45° + 65° + 75° = 185°$. Since $185° \neq 180°$, such a triangle **cannot exist** in Euclidean geometry!

Solution 15.8 1. No, because if one base angle were $100°$, the second base angle would also be $100°$, giving $100° + 100° = 200° > 180°$, which is impossible. 2. The $100°$ angle must be the apex angle. The two base angles are $\frac{180° - 100°}{2} = 40°$ each.

Solution 15.9 The mason is using a scaled version of the Egyptian $(3, 4, 5)$ triangle multiplied by 20: $3 \times 20 = 60\ \text{cm}$, $4 \times 20 = 80\ \text{cm}$. The hypotenuse must be $5 \times 20 = 100\ \text{cm} = 1\ \text{meter}$. If the distance between A and B is exactly $100\ \text{cm}$, the corner is perfectly square.

Solution 15.10 The transversal forms four angles of $64°$ and four supplementary angles of $180° - 64° = 116°$.

Solution 15.11 Diagonal $[AC]$ splits quadrilateral $ABCD$ into two non-overlapping triangles, $\triangle ABC$ and $\triangle ACD$. Each triangle has an angle sum of $180°$. The sum of all interior angles of the quadrilateral is therefore:

$$180° + 180° = 360°$$

Solution 15.12 By considering the central regular pentagon and the five isosceles triangles that form the points, each tip angle of a regular star measures $36°$. The sum of the five tip angles is $5 \times 36° = 180°$!

The Question That Opens What Follows

We now know how to master straight lines, perpendiculars, and the sharp angles of polygons. But what happens when the lines begin to curve and sweep smoothly through space without a single corner?

In the dark corner of an ancient artisan's workshop, a heavy wooden disc spins smoothly upon an iron pivot: the **potter's wheel**. How can one measure the curved rim of a circle when no ruler can bend around it without slipping? And what is this infinite, mysterious number that the great **Archimedes of Syracuse** tracked down between two vast armies of polygons? That is the secret of the circle and the number π, which we shall unveil in the next chapter.

CHAPTER 16: The Potter's Wheel and Archimedes' Secret

> "Every polygon built of right angles readily submits to a grid. But the gentle curve of the circle slips through the surveyor's fingers. To measure the perfect ring, one had to unroll the circumference upon the earth and discover that mysterious, unending number without repetition that the sages of Syracuse hunted down between two vast armies of polygons."

What You Are About to Discover

- The essential distinction between the **circle** (the boundary line) and the **disc** (the solid interior surface).
- The complete vocabulary of roundness: **center**, **radius**, **diameter**, **chord**, and **arc**.
- The rolling wheel experiment and the magic constant: why the ratio of perimeter to diameter always yields the same number $\pi \approx 3.14159...$
- The astonishing intellectual feat of **Archimedes of Syracuse** and his 96-sided polygon method.
- The two indispensable formulas: the circumference of the circle ($C = 2\pi r$) and the area of the disc ($A = \pi r^2$), proved visually by unrolling a disc into a near-rectangle!
- 🛠️ The Boundary Case Laboratory: extreme limits of circles, concentric rings, and the Earth-rope paradox.

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1. The Problem: The Chariot Wheel and the Clay Pot

You are eleven years old. After exploring the straight lines and right angles of the Egyptian surveyors in the previous chapter, let us step into the workshop of an ancient artisan.

In a sunlit corner of the potter's studio, a heavy wooden turntable spins smoothly upon an iron pivot. It is the **potter's wheel**. The craftsman drops a lump of wet grey clay onto the exact center of the spinning disc. His wet palms cup the earth. Under his thumbs, as if by enchantment, the shapeless lump blossoms into an exquisitely symmetrical bowl.

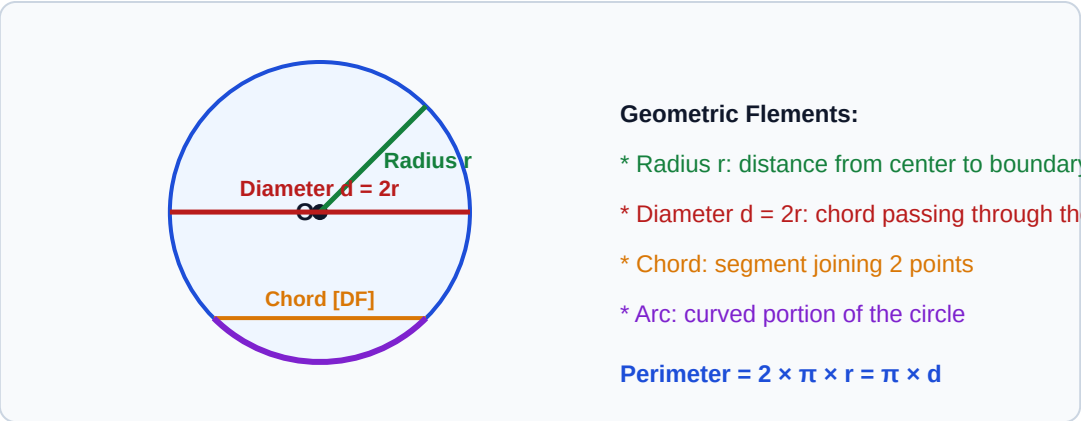

Figure 55: Key circle elements: radius, diameter, and chord

Every single point along the outer rim of the bowl spins at the exact same distance from the central pivot. If one side were even a fraction of an inch further from the center than another, the whirling vessel would wobble violently and collapse in an instant.

The next day, in the neighboring village, the blacksmith faces a far more punishing challenge. He must forge an **iron tire** to encase the heavy wooden wheel of a war chariot. This iron band must be forged straight on the anvil, heated white-hot, bent into a ring, and fitted around the wooden wheel with millimeter precision.

- If the blacksmith cuts the iron band two centimeters too short, it will never fit over the wooden rim.

- If he cuts it two centimeters too long, the iron hoop will rattle loosely against the wood, and the wheel will shatter against the very first rock on the road.

The smith measures the width of the wooden wheel: it measures exactly **1 meter straight across**, passing through the center. What exact length of straight iron must he cut from his stock?

He tries wrapping a thin cord around the curved rim of the wheel, but the cord sags, slips, and stretches. The frustrated craftsman grumbles: *"Why is it so easy to measure a square or a triangle with a rigid ruler, and so treacherous to measure the curved rim of a round wheel?"*

To answer this question, we must do what the ancient geometers did: unroll the circle upon the ground.

---

2. The Vocabulary of the Circle and the Disc

Before unrolling our wheel, let us establish precise terms. Mathematical language is rigorous; confusing these terms is the most frequent beginner trap.

2.1 Circle vs. Disc

Definition — Circle and Disc

- The **circle** of center O and radius r is the collection of

all points in the plane located at the exact same distance r from center O. The circle is **a closed curved line**—it is the hollow boundary (like a wedding ring, a hoop, or an iron tire).

- The **disc** of center O and radius r is the collection of all points in the plane located at a distance less than or equal to r from center O. The disc is **a solid filled surface** (like a coin, a dinner plate, or a pancake).

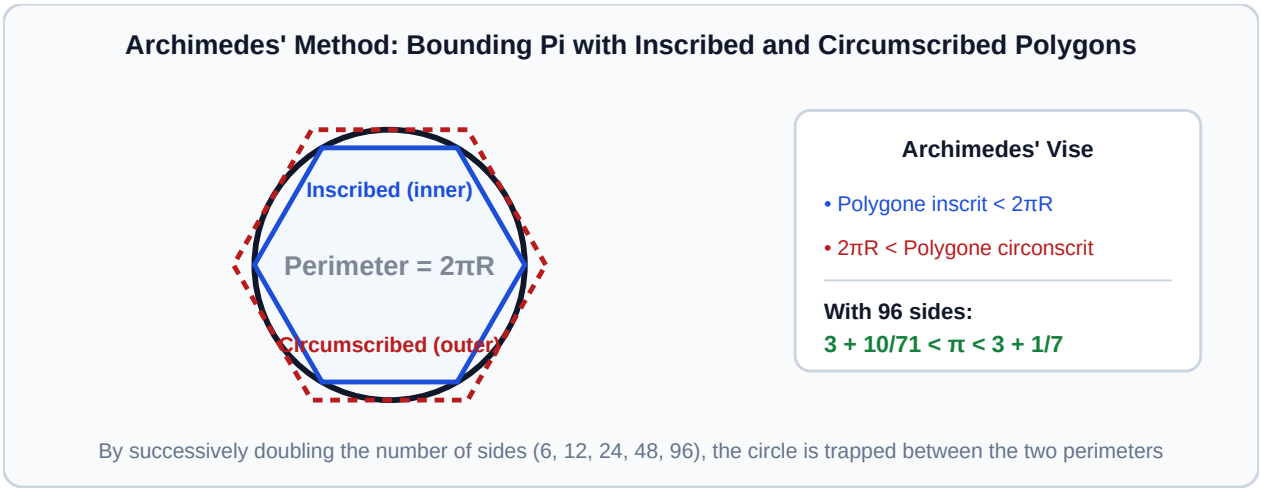

Figure 56: Archimedes' method: bounding Pi with regular polygons

2.2 Remarkable Segments and Curves

Mark a point O on a sheet of paper:

- The **center**: the fixed point O located at the exact midpoint of the figure.
- The **radius** (r): both the straight line segment $[OB]$ joining the center to any point on the circle, and the **numerical length** $r = OB$. All radii of the same circle have identical lengths.
- The **diameter** (d): a straight line segment $[AB']$ passing through center O and connecting two opposite points on the circle. Its length is denoted d. Since it consists of two radii placed end-to-end ($[AO]$ and $[OB']$), we have the fundamental relationship:

$$d = 2 \times r \quad \Leftrightarrow \quad r = \frac{d}{2}$$

- A **chord**: a line segment $[CD]$ connecting any two points on the circle without necessarily passing through the center. The diameter is simply the longest possible chord of the circle.
- An **arc**: a continuous portion of the curved circle between two points, denoted $\overset{\frown}{CD}$.

> 💡 **Did You Know?**
>
> The word *radius* comes from the Latin word for the spoke of a chariot wheel or a weaver's rod. The word *diameter* comes from the ancient Greek *dia* ("across, through") and *metron* ("measure"): it is the measure that cuts straight through the figure from one side to the other.

---

3. The Rolling Experiment and the Magic Constant

Let us return to the blacksmith and his chariot wheel. Consider an experiment you can easily perform at home using a coin, a roll of masking tape, or a circular tin can:

1. Take a felt-tip pen and draw a tiny red mark on the bottom rim of your circular tin can.
2. Place the can on a long sheet of blank paper taped to the floor, aligning the red mark with the 0 cm mark on your ruler.
3. Roll the can along the ruler in a straight line without slipping, until the red mark touches the ground again after completing **exactly one full revolution**.
4. Measure the distance traveled along the ruler: this represents the length of one full turn of the rim—the **perimeter** of the circle, historically known as the **circumference**.

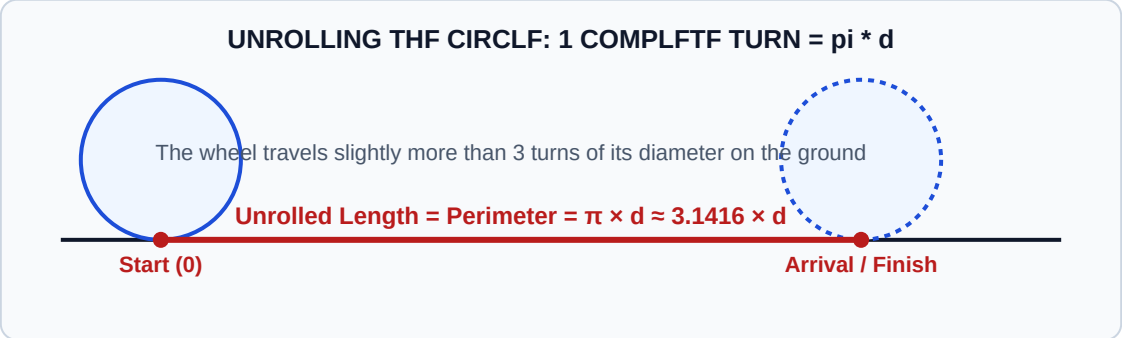

Figure 57: Unrolling the circle: geometric definition of Pi

Let us record careful measurements for round objects of widely varying sizes:

Object Measured	Diameter d	Measured Perimeter P	Ratio $\frac{P}{d}$ (division)
2-Euro coin	2.57 cm	8.07 cm	$\frac{8.07}{2.57} \approx 3.140$
Roll of tape	8.00 cm	25.13 cm	$\frac{25.13}{8.00} \approx 3.141$
Dinner plate	22.00 cm	69.10 cm	$\frac{69.10}{22.00} \approx 3.141$
Bicycle wheel	68.00 cm	213.60 cm	$\frac{213.60}{68.00} \approx 3.141$

Look closely at the final column: **the division $\frac{P}{d}$ always yields the exact same number: approximately 3.14!** Whether the object is as minuscule as the head of a pin or as colossal as the orbit of the Moon around the Earth, **the circumference of any circle is always just a little more than three times its diameter.**

This constant ratio does not depend on the size of the object: it is a universal mathematical constant of Euclidean geometry. Mathematicians denote it with the Greek letter π (pronounced *"pie"*).

> 📐 ***The Universal Constant π —*** *For every circle of circumference P and diameter d:*
>
> $$\frac{P}{d} = \pi \approx 3.1415926535...$$

Multiplying both sides by diameter d yields the famous formula for the **circumference of a circle**:

$$P = \pi \times d = 2 \times \pi \times r$$

Generations of students remember it by its rhythmic cadence: **"Two π r"**.

> 🔍 ***Historical Note — Where Did the Letter π Come From?*** *The letter π is the first letter of the Greek word* periphereia *(which gave us "periphery," meaning outer boundary). The Welsh mathematician William Jones was the first to use π to represent this ratio in 1706 in an educational textbook. However, it was the Swiss mathematical titan* ***Leonhard Euler*** *who popularized it worldwide starting in 1737 by adopting it in all his master treatises.*

---

4. Archimedes' Secret: Taming the Circle with Polygons

Ancient Babylonians and Egyptians already knew that the circumference was roughly three times the diameter. Egyptian scribes used an approximation near $\left(\frac{16}{9}\right)^2 \approx 3.16$, and the Old Testament (1 Kings 7:23) describes a great circular bronze basin measuring 10 cubits across and 30 cubits around ($\pi \approx 3$).

Yet no one knew how to calculate the true value of this number with mathematical certainty. Enter the greatest scientific mind of Antiquity: **Archimedes of Syracuse** (c. 287 – 212 BC).

4.1 The Stroke of Genius: The Polygon Vise

Archimedes reasoned: *"I cannot measure a curved line directly. However, I know how to calculate the perimeter of straight-sided figures: regular polygons!"*

Archimedes trapped the circle between two regular polygons like a vise: 1. An **inscribed polygon** (lying entirely **inside** the circle, whose vertices touch the circle). 2. A **circumscribed polygon** (lying entirely **outside** the circle, whose straight sides touch the circle).

Physical reality is undeniable:

- The path along the inner polygon is strictly shorter than the curve of the circle.
- The path along the outer polygon is strictly longer than the curve of the circle.

Therefore:

$$\text{Perimeter of Inscribed Polygon} < P_{\text{circle}} < \text{Perimeter of Circumscribed Polygon}$$

Dividing each term by diameter d, Archimedes trapped π within an unbreakable arithmetic interval!

4.2 The Climb to 96 Sides

Archimedes began with regular hexagons (6 sides), computing their perimeters using the Pythagorean theorem. Then, through a brilliant geometric formula of his own invention, he systematically **doubled** the number of sides:

- 12-sided polygons (dodecagons);
- 24-sided polygons;
- 48-sided polygons;
- And finally, polygons of **96 sides**!

Imagine the immense labor: at 96 sides, the inner and outer polygons hug the circular line so tightly that the human eye cannot distinguish the broken straight edges from the smooth curve.

After heroic manual computations involving square roots and fractions, Archimedes proved his historic inequality:

$$\frac{223}{71} < \pi < \frac{22}{7}$$

In decimal form:

$$3.140845... < \pi < 3.142857...$$

Thanks to Archimedes, humanity knew beyond doubt that π begins with $\mathbf{3.14}$. Archimedes' upper fraction, $\frac{\mathbf{22}}{\mathbf{7}}$, is so remarkably accurate (error under 0.04%) that master builders and engineers used it for mental calculations for over two thousand years!

---

5. The Area of the Disc: Turning a Circle into a Rectangle

We now know how to measure the outer boundary cord ($P = 2\pi r$). But what is the **surface area** of the disc? How many square meters of paint are required to coat a round tabletop?

The answer is given by a celebrated formula:

$$A = \pi \times r^2$$

Where does this formula come from? Why does the square of the radius (r^2) appear, multiplied by the exact same constant π? Here is a breathtaking visual demonstration, attributed to Leonardo da Vinci and anticipated by ancient Greek thinkers.

The Apple Pie Metaphor

Imagine a large circular disc of pastry of radius r: 1. Slice this disc into a large number of identical, razor-thin wedge-shaped slices (say 16 slices, like narrow arrowheads). 2. Separate the 16 slices and interlock them alternately, one pointing up, one pointing down:

Look at the reassembled figure: it astonishingly resembles a **rectangle** (or a near-perfect parallelogram)!

- What is the **height** of this rectangle? It is the length of one pastry wedge, which is the **radius r** of the circle.
- What is the **base** of this rectangle? The upper scalloped rim consists of half the wedges, and the bottom rim consists of the other half. The base is therefore **half the circumference of the circle**:

$$\text{Base} = \frac{1}{2} \times P = \frac{1}{2} \times (2\pi r) = \pi r$$

What is the area of this rectangle?

$$\text{Area} = \text{base} \times \text{height} = (\pi r) \times r = \pi r^2$$

If instead of 16 wedges, we slice the disc into 100, 1,000, or 1,000,000 infinitely thin slices, the tiny curved scallops flatten completely into straight lines: the shape becomes an exact mathematical rectangle of area πr^2.

> **Theorem — Area of a Disc**
>
> The area A of a disc of radius r (and diameter $d = 2r$) equals:
>
> $$A = \pi \times r^2 = \pi \times r \times r$$
>
> Expressed in terms of diameter d: $A = \pi \times \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$.

---

6. Beware of Pitfalls!

> **Warning: Deadly Traps!**
>
> 1. **Never confuse radius and diameter!** In any word problem, highlight immediately whether you are given the radius or the diameter. If a wheel has a diameter of $10\ \text{cm}$, its radius is $5\ \text{cm}$. If you blindly write $A = \pi \times 10^2$, your result will be four times too large! 2. **Never confuse perimeter and area!**
> - Perimeter is a **length** (in m, cm, mm): $P = 2\pi r$. The radius has power 1.
> - Area is a **surface** (in m^2, cm^2, mm^2): $A = \pi r^2$. The radius is squared.
>
> 3. **The Semicircle Perimeter Trap:** The perimeter of a closed semicircle is NOT simply half of $2\pi r$. You must add the straight flat edge that closes the shape—the diameter!
>
> $$P_{\text{semicircle}} = \pi r + 2r$$
>
> 4. **π is neither 3.14 nor $\frac{22}{7}$!** These are merely practical approximations. The decimal expansion of π continues infinitely without any repeating periodic pattern: $\pi = 3.141592653589793...$ (π is an irrational number). When a problem asks for the "exact value," you must leave the symbol π in your final answer (e.g., $16\pi\ \text{cm}^2$).

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7. Solved Examples

Solved Example 1 — Iron Hooping of a Wine Barrel

A cooper wishes to encase a round wooden barrel with an iron hoop. The barrel has a diameter of 80 cm. What minimum length of iron strap must he cut? (Use $\pi \approx 3.14$).

Solution: - We are looking for a perimeter. Given diameter $d = 80 \text{ cm}$. - Formula: $P = \pi \times d$. - Calculation: $P \approx 3.14 \times 80 = 251.2 \text{ cm}$.

He must cut an iron band of at least 251.2 cm (about 2.51 meters). ✓

Solved Example 2 — The Bicycle Odometer

A mountain bike tire has a radius of $r = 35 \text{ cm}$. 1. How far does the bicycle travel when the wheel completes exactly one full revolution? 2. How many full turns does the wheel make over a journey of 1.1 km?

Solution: 1. One revolution equals the wheel's circumference:

$$P = 2 \times \pi \times r = 2 \times \pi \times 35 = 70\pi \approx 219.91 \text{ cm} \approx 2.20 \text{ m}$$

2. Convert total distance into meters: $1.1 \text{ km} = 1,100 \text{ m}$.

$$\text{Number of turns} = \frac{\text{Total distance}}{\text{Circumference}} = \frac{1,100}{2.20} = 500 \text{ turns}$$

The wheel makes exactly 500 full rotations. ✓

Solved Example 3 — The Pizza Dilemma: A Feast of Geometry!

At a pizzeria, you face a hungry choice for the exact same price of $16:

- One **large pizza** with a diameter of 40 cm.
- Two **small pizzas** with a diameter of 28 cm each.

Which choice gives you more pizza to eat?

Solution: Calculate the edible surface area in both cases (watch the radii!):

- For the large pizza: diameter is 40 cm, so radius $r_1 = 20 \text{ cm}$.

$$A_1 = \pi \times r_1^2 = \pi \times 20^2 = 400\pi \approx 1,256.6 \text{ cm}^2$$

- For the two small pizzas: diameter is 28 cm, so radius $r_2 = 14 \text{ cm}$. Area of one small pizza: $A_{\text{small}} = \pi \times 14^2 = 196\pi \approx 615.75 \text{ cm}^2$. For two small pizzas:

$$A_2 = 2 \times 196\pi = 392\pi \approx 1,231.5 \text{ cm}^2$$

Conclusion: $400\pi > 392\pi$. The single large pizza provides more food than both small pizzas combined! The single large pizza is the smarter choice. ✓

Solved Example 4 — Finding the Radius from Perimeter or Area

1. A venerable oak tree has a trunk circumference measured at 314 cm. What is its diameter?
2. A circular plaza has an area of 706.5 m^2. What is its radius? (Use $\pi \approx 3.14$).

Solution: 1. Since $P = \pi \times d$:

$$d = \frac{P}{\pi} \approx \frac{314}{3.14} = 100 \text{ cm} = 1 \text{ meter}$$

2. Since $A = \pi \times r^2$:

$$r^2 = \frac{A}{\pi} \approx \frac{706.5}{3.14} = 225$$

What positive number squared equals 225? We find the square root: $15 \times 15 = 225$, so $r = 15 \text{ meters}$. ✓

Solved Example 5 — The Running Track and Circular Ring (Annulus)

A training stadium features a central circular grass lawn of radius $R_1 = 30 \text{ m}$, surrounded by a tartan running track of uniform width 4 m. What is the area of the tartan running track?

Solution: The track forms an **annulus** (a circular ring): the difference between the large outer disc and the small inner lawn disc.

- Inner radius: $R_1 = 30 \text{ m}$.
- Outer radius: $R_2 = 30 + 4 = 34 \text{ m}$.

- Large disc area: $A_{\text{large}} = \pi \times R_2^2 = \pi \times 34^2 = 1,156\pi \text{ m}^2$.
- Small disc area: $A_{\text{small}} = \pi \times R_1^2 = \pi \times 30^2 = 900\pi \text{ m}^2$.
- Area of the ring:

$$A_{\text{track}} = A_{\text{large}} - A_{\text{small}} = 1,156\pi - 900\pi = 256\pi \approx 256 \times 3.1416 \approx 804.25 \text{ m}^2$$

The track covers approximately 804.25 m^2. ✓

What You Must Remember

1. **Circle vs. Disc:**
   - The circle is the boundary line (length / perimeter).
   - The disc is the solid interior surface (area).
2. **Radius and Diameter:**
   - Diameter is twice the radius: $d = 2r \Leftrightarrow r = \frac{d}{2}$.
3. **The Constant π:**
   - It is the invariant ratio of perimeter to diameter: $\pi = \frac{P}{d} \approx 3.14159...$
   - Archimedes bounded it with 96-sided polygons: $\frac{223}{71} < \pi < \frac{22}{7}$.
4. **Fundamental Formulas:**
   - **Circumference:** $P = 2\pi r = \pi d$.
   - **Area of a Disc:** $A = \pi r^2 = \frac{\pi d^2}{4}$.
5. **Exact vs. Rounded Values:**
   - Keep the symbol π for exact mathematical values (e.g., $25\pi \text{ cm}^2$).
   - Replace π with 3.14 or calculator π for decimal approximations.

Exercises

Level (Direct Application)

Exercise 16.1 Complete the table below for different circles (use $\pi \approx 3.14$): | Circle | Radius r | Diameter d | Circumference P | |—|—|—|—| | A | 6 cm | ? | ? | | B | ? | 18 m | ? | | C | ? | ? | 31.4 dm |

Exercise 16.2 Calculate the exact area (leaving π), then round to the nearest tenth: 1. A disc of radius 4 cm. 2. A disc of diameter 14 m.

Exercise 16.3 A gardener installs wooden edging around a circular flowerbed of radius 3 m. What length of edging must he purchase?

Exercise 16.4 A dinner plate has a diameter of 24 cm. Calculate its area to the nearest cm^2.

Exercise 16.5 Give the exact perimeter of a closed semicircle of radius 5 cm, including its straight baseline.

Level (Reasoning and Geometry)

Exercise 16.6 A round dining table has a perimeter of 376.8 cm. 1. What is its radius? 2. What is its area in m^2?

Exercise 16.7 A goat is tethered to a stake in the middle of an open flat meadow by a rope 6 meters long. 1. What geometric shape can the goat graze? 2. Calculate the exact area of this grazing zone.

Exercise 16.8 The radius of a disc is doubled (from r to $2r$). 1. By what factor is its perimeter multiplied? 2. By what factor is its area multiplied? Justify using algebraic formulas.

Exercise 16.9 A sports field has the shape of a rectangle 100 m long and 60 m wide, capped at both ends by semicircular regions of diameter 60 m. 1. Calculate the total outer track perimeter. 2. Calculate the total surface area of the sports complex.

Exercise 16.10 A truck tire has an outer diameter of 90 cm. At cruising speed, the wheel rotates at 10 revolutions per second. How far does the truck travel in one minute? State the answer in meters, then convert to kilometers per hour (km/h).

Exercise 16.11 An audio CD has an outer diameter of 12 cm and a center spindle hole of diameter 1.5 cm. What surface area is available for recorded musical data?

Level (Challenges and In-Depth Investigations)

Exercise 16.12 Inscribed Square and Circle. From a square sheet of copper 20 cm on each side, a metalsmith cuts out the largest possible circular disc. 1. What is the diameter of this disc? 2. What area of copper is discarded as scrap from the four corners? 3. What percentage of the initial metal plate was wasted?

Exercise 16.13 The Lunes of Hippocrates of Chios. Let ABC be a right triangle at vertex A, with $AB = 4$ cm and $AC = 3$ cm. The hypotenuse is $BC = 5$ cm. On each of the three sides taken as diameters, draw a semicircle outward. The two crescent shapes formed above the hypotenuse are called *lunes*. Prove that the combined area of these two curved lunes is strictly equal to the area of the flat right triangle ABC!

Exercise 16.14 The Rope Around the Earth. Imagine a snug rope wrapped tightly around the equator of the Earth (assume Earth is a perfect sphere of circumference $C_1 = 40,000$ km). We now add **just 1 meter** to the length of this rope ($C_2 = 40,000$ km $+ 1$ m), and prop it up uniformly all around the globe so it forms a concentric circle. Can a mouse crawl under the elevated rope without ducking? A cat? A human being? *(Hint: calculate the clearance height $h = r_2 - r_1$. The result will astonish you!)*

Exercise 16.15 Archimedes' Hexagon Method. Consider a circle of radius $R = 1$. 1. Inscribe a regular hexagon inside this circle. How long are its 6 sides? Deduce the perimeter of this hexagon and the elementary lower bound for π. 2. Calculate the perimeter of a square circumscribed outside this circle. What upper bound for π does this produce? Between which two whole integers is π thus proven to lie?

Solutions

Solution 16.1 - Circle A: $d = 2 \times 6 = 12$ cm; $P = \pi \times 12 \approx 3.14 \times 12 = 37.68$ cm. - Circle B: $r = 18/2 = 9$ m; $P = \pi \times 18 \approx 3.14 \times 18 = 56.52$ m. - Circle C: $d = P/\pi \approx 31.4/3.14 = 10$ dm; $r = 10/2 = 5$ dm.

Solution 16.2 1. $r = 4$ cm: Exact Area $A = \pi \times 4^2 = 16\pi$ cm$^2 \approx 50.3$ cm^2. 2. $d = 14$ m $\Longrightarrow r = 7$ m: Exact Area $A = \pi \times 7^2 = 49\pi$ m$^2 \approx 153.9$ m^2.

Solution 16.3 $P = 2\pi r = 2 \times \pi \times 3 = 6\pi \approx 18.85$ m. He should buy about 18.9 m of edging.

Solution 16.4 $d = 24$ cm $\Longrightarrow r = 12$ cm. $A = \pi \times 12^2 = 144\pi \approx 452.39$ cm$^2 \approx 452$ cm^2.

Solution 16.5 The semicircle consists of the curved arc of length $\frac{1}{2}(2\pi r) = \pi r = 5\pi$ cm, plus the straight baseline diameter $d = 2 \times 5 = 10$ cm. Exact perimeter: $P = (5\pi + 10)$ cm ≈ 25.71 cm.

Solution 16.6 1. $P = 2\pi r \Longrightarrow r = \frac{P}{2\pi} \approx \frac{376.8}{2 \times 3.14} = \frac{376.8}{6.28} = 60$ cm. 2. $A = \pi r^2 = \pi \times 60^2 = 3,600\pi \approx 11,304$ cm$^2 = 1.1304$ m^2.

Solution 16.7 1. The accessible area is a solid disc centered at the stake with radius 6 m. 2. Exact area: $A = \pi \times 6^2 = 36\pi$ m$^2 \approx 113.10$ m^2.

Solution 16.8 1. New perimeter: $P' = 2\pi(2r) = 2(2\pi r) = 2P$. Perimeter is **multiplied by 2**. 2. New area: $A' = \pi(2r)^2 = \pi \times 4r^2 = 4(\pi r^2) = 4A$. Area is **multiplied by 4** (2^2)!

Solution 16.9 1. The perimeter consists of two straight lengths of 100 m plus two semicircular arcs of diameter 60 m (which form one complete circle of diameter 60 m): $P_{\text{total}} = 2 \times 100 + \pi \times 60 = 200 + 60\pi \approx 200 + 188.5 = 388.5$ m. 2. Total area is the central rectangle ($100 \times 60 = 6,000$ m^2) plus the complete disc formed by the two semicircles ($r = 30$ m): $A_{\text{disc}} = \pi \times 30^2 = 900\pi \approx 2,827.4$ m^2. $A_{\text{total}} = 6,000 + 900\pi \approx 8,827.4$ m^2.

Solution 16.10 Distance per turn: $P = \pi \times 0.90$ m ≈ 2.8274 m. In one second (10 turns): $10 \times 2.8274 = 28.274$ m/s. In one minute (60 s): $D = 28.274 \times 60 \approx 1,696.46$ m ≈ 1.70 km. Speed in km/h: $28.274 \times 3.6 \approx 101.8$ km/h.

Solution 16.11 Outer radius $R = 6$ cm, inner hole radius $r = 0.75$ cm. $A = \pi R^2 - \pi r^2 = \pi(6^2 - 0.75^2) = \pi(36 - 0.5625) = 35.4375\pi \approx 111.33$ cm^2.

Solution 16.12 1. The largest disc has a diameter equal to the width of the square: $d = 20 \text{ cm} \Longrightarrow r = 10 \text{ cm}$. 2. Area of square: $A_{\text{square}} = 20 \times 20 = 400 \text{ cm}^2$. Area of disc: $A_{\text{disc}} = \pi \times 10^2 = 100\pi \approx 314.16 \text{ cm}^2$. Scrap area: $400 - 100\pi \approx 85.84 \text{ cm}^2$. 3. Waste percentage: $\frac{85.84}{400} \approx 0.2146 = 21.46\%$. More than 21% of the raw material is lost!

Solution 16.13 Triangle ABC is right-angled at A; its area is $\frac{3\times4}{2} = 6 \text{ cm}^2$. The semicircles on the legs have radii $r_1 = 2$ and $r_2 = 1.5$. Their areas are $\frac{1}{2}\pi(2^2) = 2\pi$ and $\frac{1}{2}\pi(1.5^2) = 1.125\pi$. The large semicircle on hypotenuse BC has radius $R = 2.5$ and area $\frac{1}{2}\pi(2.5^2) = 3.125\pi$. Notice: $2\pi + 1.125\pi = 3.125\pi$! The sum of the two smaller semicircles equals the large semicircle. The total figure area is (Triangle + 2 small semicircles). Subtracting the large semicircle leaves the lunes: $\text{Area of lunes} = (\text{Area of triangle} + 3.125\pi) - 3.125\pi = \text{Area of triangle} = 6 \text{ cm}^2$. A curved shape whose area is strictly equal to a straight polygon!

Solution 16.14 Let R_E be Earth's radius and R_2 the radius of the lengthened rope. Initial circumference: $C_1 = 2\pi R_E$. Lengthened circumference: $C_2 = 2\pi R_2 = C_1 + 1 \text{ meter}$. Subtract: $2\pi R_2 - 2\pi R_E = 1 \Longrightarrow 2\pi(R_2 - R_E) = 1$. The clearance height $h = R_2 - R_E$ is:

$$h = \frac{1}{2\pi} \approx \frac{1}{6.283} \approx 0.159 \text{ m} \approx 16 \text{ cm !}$$

This clearance is entirely independent of the size of the sphere! Whether wrapped around the Earth, the Sun, or a baseball, adding 1 meter to the circumference lifts the rope by nearly 16 cm everywhere. A mouse, and even a large cat, can slip underneath with ease!

Solution 16.15 1. An inscribed regular hexagon splits into 6 equilateral triangles of side length equal to radius $R = 1$. Each side measures 1; perimeter is $6 \times 1 = 6$. Since diameter is 2, the ratio gives $\pi > \frac{6}{2} = 3$. 2. The circumscribed square has sides equal to the diameter, 2. Its perimeter is $4 \times 2 = 8$. The outer ratio gives $\pi < \frac{8}{2} = 4$. We obtain the clean integer bracket: $3 < \pi < 4$.

The Question That Opens What Follows

We now know how to measure round, curving shapes upon the ground. But what can we do when the object to be measured is neither a small wheel before our eyes nor a cord we can lay across the sand, but a towering mountain of stone: the **Great Pyramid of Giza**?

How did a Greek philosopher named **Thales**, armed with nothing more than a simple wooden stick planted in the scorching desert sand, deduce the exact height of the greatest tomb on Earth without ever climbing a single step? That is the secret of **geometric proportions and similar triangles**, which we shall master in the next chapter.

CHAPTER 17: The Shadow of the Pyramid and the Upright Staff

> "Why risk one's life attempting to scale a mountain of stone? When the sun illuminates the world, every creature casts upon the earth a faithful mirror of its own height. One must simply learn to read the shadow."

What You Are About to Discover

- The historic challenge of **Thales of Miletus** at the foot of the Great Pyramid of Giza.
- What an **enlargement** or **reduction** by scale factor k truly means.
- The astonishing geometric effect of factor k on our physical universe: why lengths scale by k, surface areas by k^2, and volumes by k^3.
- **Thales' Theorem** in its two classic configurations: nested triangles and the hourglass (butterfly).
- The equality of the three side ratios and the foolproof technique of cross-multiplication.
- The **Converse of Thales' Theorem** to prove beyond doubt that two lines are parallel.
- 🛠️ The Boundary Case Laboratory: extreme scaling limits, Galileo's giant bone paradox, and the pinhole camera (*camera obscura*).

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1. The Problem: The Pharaoh's Insolent Challenge

The desert wind whips up golden dust swirls across the Giza plateau in Egypt, twenty-six hundred years ago. Before you looms the Great Pyramid of Pharaoh Khufu: an artificial mountain of solid limestone soaring to a dizzying height.

Pharaoh Amasis gathers his high priests, court architects, and surveyors at the foot of the stone giant: *"Who among you is wise enough to determine the exact vertical height of this sacred tomb, from the bedrock of the earth to the gilded peak that touches the sky?"*

The master builders lower their eyes in embarrassment: *"Your Majesty, the four faces are polished smooth as glass and slope steeply toward the heavens. No man can climb with a surveyor's measuring chain without slipping and falling one hundred and forty meters to his death!"*

A foreign traveler, hailing from the Greek city of Miletus and cloaked in a simple woolen mantle, steps calmly forward. His name is **Thales**. Thales gazes up at the stone colossus, observes the blazing sun in the azure sky, and smiles: *"Your Majesty, I have no need to climb your pyramid. Hand me a simple wooden walking staff, and before sunset, I shall give you the exact height of the monument without ever lifting my feet from the ground!"*

The Egyptian courtiers scoff. How could an itinerant Greek merchant measure a stone mountain with a common piece of wood?

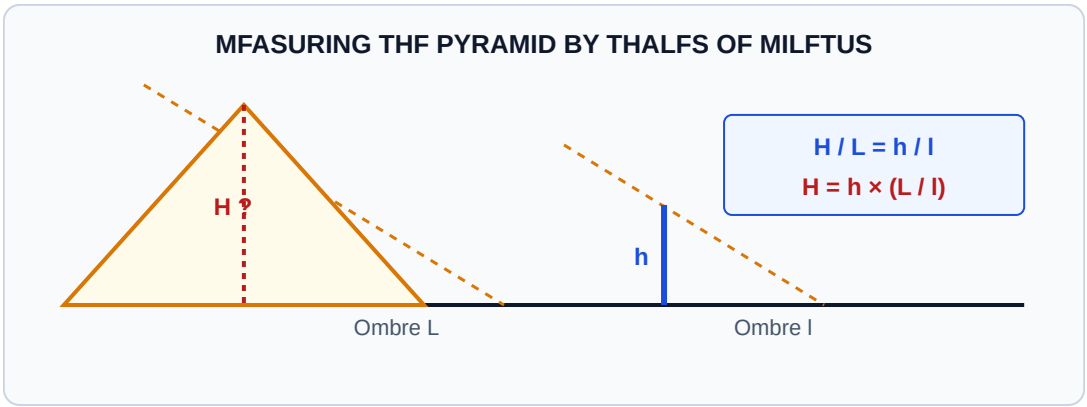

Figure 58: Right-triangle trigonometry: Cosine, Sine, and Tangent

2. From the Magic Instant to Universal Proportions

2.1 The First Insight: The Magic Hour

Thales sits in the warm sand and carefully observes the path of the sun.

- Early in the morning, the sun sits low on the horizon: shadows stretch out enormously across the plain.
- At noon, the sun reaches its zenith overhead: shadows shrink to tiny pools huddled under one's feet.
- Through the afternoon, the sun descends again, and the shadows begin to lengthen once more.

Thales drives his wooden staff vertically into the sand. The staff stands exactly **2 meters tall**. He marks a tick mark in the sand and tracks the tip of the staff's shadow:

- At 2:00 PM, the shadow measures $1.20 \ \text{m}$ (shorter than the staff).
- At 3:00 PM, it measures $1.60 \ \text{m}$.
- Suddenly, at **3:45 PM**, Thales measures the shadow again: **it measures precisely 2 meters!**

At that singular instant, a geometric miracle occurs:

$$\text{Length of Staff's Shadow} = \text{Actual Height of Staff} = 2 \ \text{meters}$$

The sunlight strikes the ground at an angle of exactly 45°, forming an **isosceles right-angled triangle** with the upright staff and the horizontal ground. Thales shouts to the servants: *"Measure the shadow of the pyramid right now!"* At that exact second, the length of the pyramid's shadow (measured from the vertical line below its apex to the far tip of its shadow) is **strictly equal to its true vertical height**!

> 💡 **Did You Know?**
>
> According to the Greek historian Plutarch, Pharaoh Amasis was so utterly astonished to see Thales determine the monument's height in just a few paces that he showered the philosopher with gifts and revered him as the wisest living mortal.

2.2 What If the Sun Won't Wait? The Power of Proportion

The "magic hour" technique is brilliant, but it has one glaring limitation: one must wait for the exact moment when the solar angle reaches 45°. What if a storm cloud hides the sun at 3:45 PM? What if you must measure a tall tree at 10:00 AM?

Thales realized a far deeper truth: **sunlight rays, arriving from an astronomical body 150 million kilometers away, reach the surface of the Earth along lines that are virtually parallel to one another.**

Therefore, at **any hour of the day**, the pyramid and its shadow form a triangle that has the **exact same geometric shape** as the triangle formed by the staff and its shadow! One triangle is simply an **enlargement** of the other.

The ratio between vertical height and cast shadow is **identical for all vertical objects standing under the sun at that same moment**:

$$\frac{\text{Pyramid Height}}{\text{Pyramid Shadow}} = \frac{\text{Staff Height}}{\text{Staff Shadow}} \Leftrightarrow \frac{H}{L} = \frac{h}{l}$$

Hence, the formula that delivers the height at any time without waiting:

$$H = L \times \frac{h}{l}$$

3. Enlargement, Reduction, and the Dimensional Trap

Before stating the theorem in its full mathematical generality, let us pause to examine what it truly means to "enlarge" or "reduce" a geometric shape. This is an area where human intuition is easily deceived!

3.1 The Scale Factor k

> **📐 Definition — Enlargement and Reduction**
>
> - Transforming a figure by multiplying **all its linear dimensions** by the same strictly positive number k is called:
> - An **enlargement** (or dilation) if $k > 1$.
> - A **reduction** (or contraction) if $0 < k < 1$.
> - The constant multiplier k is called the **scale factor** (or scale ratio).
> - In an enlargement or reduction, **the overall shape is strictly preserved**: angles remain completely unchanged, and parallel lines remain parallel.

3.2 The Effect on Lengths, Areas, and Volumes

Here is the golden rule that governs the geometry of our three-dimensional physical universe:

> **📐 Fundamental Scaling Theorem — The Powers of Factor k**
>
> When all linear dimensions of a figure or solid are multiplied by a scale factor k ($k > 0$):
>
> - **Lengths** are multiplied by k.
> - **Surface areas** are multiplied by k^2 ($k \times k$).
> - **Volumes** are multiplied by k^3 ($k \times k \times k$).

Why Squares and Cubes? The Visual Proof with Blocks

Take a small modeling-clay cube measuring 1 cm along each edge:

- Its edge length is 1 cm.
- The surface area of one square face is $1 \times 1 = 1 \text{ cm}^2$.
- Its solid volume is $1 \times 1 \times 1 = 1 \text{ cm}^3$.

Now perform an enlargement with scale factor $\boldsymbol{k = 2}$ (doubling all linear dimensions: edge $= 2 \text{ cm}$):

Look closely: 1. The edge length was multiplied by $\mathbf{2}$ (2 cm). 2. The area of a square face: it now contains $2 \times 2 = 4$ unit squares. It was multiplied by $2^2 = \mathbf{4}$! 3. The solid volume: to build the new cube of edge 2 cm, you need 4 unit cubes on the bottom tier and 4 unit cubes on the top tier, totaling $2 \times 2 \times 2 = 8$ unit cubes. The volume was multiplied by $2^3 = \mathbf{8}$!

If we tripled all linear dimensions ($k = 3$):

- Lengths would scale by $\times 3$.
- Surface areas would scale by $\times 3^2 = \mathbf{9}$.
- Volumes would scale by $\times 3^3 = \mathbf{27}$!

> **💡 Did You Know? — Why Giant Movie Monsters Cannot Exist**
>
> Galileo Galilei was the first to demonstrate this mathematical reality: if a human being were enlarged to 10 times his normal height ($k = 10$), he would be 10 times taller, but his body weight (which depends on volume) would multiply by $10^3 = \mathbf{1,000}$! However, the mechanical strength of his leg bones (which depends on their cross-sectional area) would only multiply by $10^2 = \mathbf{100}$. His bones would experience 10 times the normal stress: they would snap under his own weight at his very first step!

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4. Thales' Theorem in Triangles

Let us formulate this foundational theorem as it appears in standard Euclidean geometry.

4.1 The Two Canonical Configurations

Thales' Theorem applies whenever a straight line intersects two sides of a triangle while being strictly **parallel** to the third side. There are two geometric arrangements:

In both configurations:

- Lines (MB) and (NC) are secant lines intersecting at point A.
- Lines (MN) and (BC) are **strictly parallel**.

Triangle AMN is a scaled replica of triangle ABC: it represents a reduction (or enlargement) by some factor k. Their corresponding angles are equal, and all pairs of corresponding sides are proportional.

4.2 Formal Statement of Thales' Theorem

📐 Thales' Theorem (Intercept Theorem) —

Let (d) and (d') be two straight lines intersecting at point A. Let B and M be two points on (d), distinct from A. Let C and N be two points on (d'), distinct from A. If the lines (MN) and (BC) are parallel, then:

$$\frac{AM}{AB} = \frac{AN}{AC} = \frac{MN}{BC}$$

Each fraction compares a side of the smaller triangle to the corresponding side of the larger triangle:

$$\frac{\text{side of } \ AMN}{\text{side of } \ ABC} = \text{scale factor } \ k$$

🧭 Method — Writing the Equal Ratios Without Error

1.

Identify the **shared intersection vertex** (the point where the two secant lines cross, here A). 2. Write the first two ratios radiating outward from this shared vertex: $\frac{A\ldots}{A\ldots} = \frac{A\ldots}{A\ldots}$ (e.g., $\frac{AM}{AB} = \frac{AN}{AC}$). 3. For the third ratio, "ignore" vertex A and take the two remaining letters from the numerators, then denominators: $\frac{MN}{BC}$. 4. Check that all side lengths of the upper triangle (AM, AN, MN) reside exclusively in the **numerators**, and all side lengths of the lower triangle (AB, AC, BC) reside in the **denominators** (or vice versa—never mix them up!).

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5. The Converse of Thales' Theorem: Proving Parallelism

Thales' Theorem asserts:

$$\text{If lines are parallel} \Rightarrow \text{corresponding ratios are equal.}$$

What if we wish to travel in the opposite direction? Suppose we are on a construction site or surveying land, measure lengths on the ground, and need to prove with absolute certainty that two beams or property lines are **parallel**. That is the purpose of the **Converse of Thales' Theorem**.

⚠ ⚠ Critical Exam Trap — The Required Order of Points

Testing equality of length ratios $\frac{AM}{AB} = \frac{AN}{AC}$ is **not sufficient alone** to establish parallelism! Points must lie **in the exact same order** along both transversals (A, M, B and A, N, C). If M lies on the opposite side of A, equal ratios can occur without parallel lines!

📐 Converse of Thales' Theorem —

Let (AB) and (AC) be two lines intersecting at A. Let M be a point on line (AB) and N a point on line (AC). If: 1. The points A, M, B and A, N, C are aligned **in the exact same order**, 2. And $\frac{AM}{AB} = \frac{AN}{AC}$, then the lines (MN) and (BC) are **parallel**.

The Contrapositive: Proving Lines Are NOT Parallel

If after computing the two quotients, you find that they are not equal ($\frac{AM}{AB} \neq \frac{AN}{AC}$), you can immediately conclude via the **contrapositive of Thales' Theorem** that **the lines (MN) and (BC) are not parallel**.

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6. Beware of Pitfalls!

⚠ Warning: Classic Exam Traps!

1. **Inverting Numerator and Denominator:** Writing $\frac{AM}{AB} = \frac{AC}{AN}$ is a fatal error. You have placed "small over large" on the left, and "large over small" on the right. Any cross-multiplication will be completely wrong. 2. **Using the Intermediate Segment Instead of the Full Triangle Side:** In the nested configuration, the side of the larger triangle is AB. Some students mistakenly plug in the length MB. But MB is not the side of any triangle! Remember: $AB = AM + MB$. 3. **Omitting the Alignment Order Condition:** To apply the converse rigorously, you **must explicitly state**: *"Points A, M, B on one hand, and A, N, C on the other, lie in the same order along their respective lines."* 4. **Multiplying Areas by k Instead of k^2:** If a scale model is built at scale $\frac{1}{100}$, lengths are divided by 100, but the area of a surface on the model is divided by $100^2 = 10,000$!

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7. Solved Examples

Solved Example 1 — Measuring the Pyramid with Historical Data

Let us compute the height of the Great Pyramid using authentic measurements:

- The staff has height $h = 2$ m. Its shadow measures $l = 3$ m.
- The pyramid has a square base measuring 230 m on each side.
- The shadow cast by the pyramid's apex extends 74 m beyond the exterior edge of its base.

⚠ ⚠ The Historical Trap of the Pyramid's Shadow

Thales could not measure the shadow from the hidden center of the pyramid buried beneath stone! The shadow measured on the sand (74 m) only extended from the outer wall. To apply scale proportionality, Thales rigorously added half the base width ($\frac{230}{2} = 115$ m):

$$L_{\text{total}} = 115 + 74 = 189 \text{ m}.$$

Without this crucial geometric correction, his estimate would have undershot the real height by over 75 meters!

What is the vertical height H of the pyramid?

Solution: 1. Find total shadow length L, measured from directly below the summit (the center of the square base):

$$L = \text{half-base} + \text{exterior shadow} = \frac{230}{2} + 74 = 115 + 74 = 189 \text{ m}$$

2. Because the sun's rays are parallel, we set up Thales' proportion:

$$\frac{H}{h} = \frac{L}{l} \Longrightarrow \frac{H}{2} = \frac{189}{3}$$

3. Solve for H using cross-multiplication:

$$H = \frac{2 \times 189}{3} = 2 \times 63 = 126 \text{ meters}$$

The Great Pyramid stood at approximately 126 meters on that day (accounting for summit erosion). ✓

Solved Example 2 — Computing Unknown Side Lengths (Nested Configuration)

Lines (AB) and (AC) intersect at A. Given: $AM = 3$ cm, $AB = 7.5$ cm, $AC = 10$ cm, and $BC = 12$ cm. Lines (MN) and (BC) are parallel. Calculate lengths AN and MN.

Solution: - Standard formal presentation: In triangle ABC, point M lies on $[AB]$ and point N lies on $[AC]$. Since lines (MN) and (BC) are parallel, by **Thales' Theorem**:

$$\frac{AM}{AB} = \frac{AN}{AC} = \frac{MN}{BC}$$

- Substitute known numerical values:

$$\frac{3}{7.5} = \frac{AN}{10} = \frac{MN}{12}$$

- Solve for AN:

$$AN = \frac{3 \times 10}{7.5} = \frac{30}{7.5} = 4 \text{ cm}$$

- Solve for MN:

$$MN = \frac{3 \times 12}{7.5} = \frac{36}{7.5} = 4.8 \text{ cm}$$

The requested lengths are $AN = 4$ cm and $MN = 4.8$ cm. ✓

Solved Example 3 — The Impassable River (Hourglass Configuration)

A surveyor needs to determine the width AB of a rushing river without swimming across. Lines (AD) and (BC) cross at point O, with $(AB) \parallel (CD)$. Measurements on the accessible bank yield: $OB = 12$ m, $OC = 30$ m, and $CD = 45$ m. Calculate distance AB.

Solution: - Lines (AD) and (BC) intersect at O, with $(AB) \parallel (CD)$. - By **Thales' Theorem** in the hourglass configuration:

$$\frac{OA}{OD} = \frac{OB}{OC} = \frac{AB}{CD}$$

- Using the known ratio:

$$\frac{12}{30} = \frac{AB}{45}$$

- Cross-multiplication:

$$AB = \frac{12 \times 45}{30} = \frac{540}{30} = 18 \text{ meters}$$

The distance across the river is 18 meters. ✓

Solved Example 4 — Proving Parallelism with the Converse

Consider triangle ABC with $AB = 6$ cm and $AC = 8$ cm. On ray $[AB)$, mark point E such that $AE = 15$ cm. On ray $[AC)$, mark point F such that $AF = 20$ cm. Are lines (BC) and (EF) parallel?

Solution: - Compute both ratios separately:

$$\frac{AB}{AE} = \frac{6}{15} = \frac{2}{5} = 0.4$$

$$\frac{AC}{AF} = \frac{8}{20} = \frac{2}{5} = 0.4$$

- We observe that:

$$\frac{AB}{AE} = \frac{AC}{AF}$$

- Formal conclusion: Points A, B, E on one hand, and A, C, F on the other, lie in the same order along their lines. Furthermore, $\frac{AB}{AE} = \frac{AC}{AF}$. Therefore, by the **Converse of Thales' Theorem**, lines (BC) and (EF) are strictly parallel. ✓

Solved Example 5 — The Skyscraper Architectural Model

An architecture firm builds a scale model of a planned glass skyscraper at scale $\frac{1}{200}$. 1. The model stands 1.40 m tall. What will be the actual height of the skyscraper? 2. The model uses 0.35 m^2 of glass siding. What total area of glass will the actual tower require? 3. The interior volume of the model is 0.05 m^3. What will be the actual building volume?

Solution: Moving from the model to the real structure represents an enlargement by factor $K = 200$. 1. **Lengths ($K = 200$):**

$$\text{Actual Height} = 1.40 \times 200 = 280 \text{ meters}$$

2. **Areas ($K^2 = 200^2 = 40,000$):**

$$\text{Actual Glass Area} = 0.35 \times 40,000 = 14,000 \text{ m}^2$$

3. **Volumes ($K^3 = 200^3 = 8,000,000$):**

$$\text{Actual Volume} = 0.05 \times 8,000,000 = 400,000 \text{ m}^3$$

Notice the dimensional explosion: the real volume is eight million times greater than the model's! ✓

What You Must Remember

1. **Enlargement and Reduction:**
   - Scale factor $k > 1$ represents an enlargement; $0 < k < 1$ represents a reduction.
   - Dimensional consequences:
     - Lengths scale by k
     - Surface areas scale by k^2
     - Volumes scale by k^3
2. **Thales' Theorem (Length Calculations):**
   - **Hypothesis:** Two lines intersecting at A, cut by parallel lines (MN) and (BC).
   - **Conclusion:** $\frac{AM}{AB} = \frac{AN}{AC} = \frac{MN}{BC}$.
3. **Converse of Thales (Proof of Parallelism):**
   - **Hypothesis:** Points aligned in the same order on two intersecting lines, AND $\frac{AM}{AB} = \frac{AN}{AC}$.
   - **Conclusion:** Lines (MN) and (BC) are parallel.
   - If the ratios differ, the **contrapositive** proves the lines are not parallel.

Exercises

Level (Direct Application)

Exercise 17.1 Apply an enlargement of scale factor $k = 3$ to a triangle whose sides measure 4 cm, 5 cm, and 6 cm. What are the side lengths of the enlarged triangle?

Exercise 17.2 A square has an area of 25 cm^2. We reduce its side lengths by scale factor $k = 0.4$. 1. What is the side length of the original square? 2. What is the side length of the reduced square? 3. What is the area of the reduced square? Verify that $\text{Area}' = k^2 \times \text{Area}$.

Exercise 17.3 In triangle ABC, point $M \in [AB]$ and point $N \in [AC]$ with $(MN) \parallel (BC)$. Given $AM = 2 \text{ cm}$, $AB = 6 \text{ cm}$, and $AN = 3 \text{ cm}$. Calculate length AC.

Exercise 17.4 Hourglass configuration: lines (AB) and (CD) intersect at O, with $(AC) \parallel (BD)$. Given $OA = 5$ cm, $OB = 15$ cm, and $AC = 4$ cm. Calculate length BD.

Exercise 17.5 On line (d), $AM = 4$ cm and $AB = 10$ cm. On intersecting line (d'), $AN = 6$ cm and $AC = 15$ cm. Are the ratios $\frac{AM}{AB}$ and $\frac{AN}{AC}$ equal?

Level (Reasoning and Geometric Deductions)

Exercise 17.6 The Height of the Church Belfry. One sunny afternoon, a pedestrian 1.80 m tall casts a shadow of 2.40 m across a stone courtyard. At the exact same instant, the shadow cast by the church belfry measures 36 meters. What is the vertical height of the belfry?

Exercise 17.7 The Mid-Segment Trap! In triangle ABC, $AB = 12$ cm and $AC = 16$ cm. Point M lies on $[AB]$ such that $MB = 3$ cm. The parallel to (BC) through M intersects $[AC]$ at N. 1. Calculate length AM. 2. Calculate length AN, then length NC.

Exercise 17.8 The Wall Shelves. A carpenter mounts two horizontal wooden shelves $[MN]$ and $[BC]$ between two sloping struts that meet at apex A. Given: $AM = 45$ cm, $MB = 30$ cm, and shelf $MN = 60$ cm. Since both shelves are horizontal, $(MN) \parallel (BC)$. Calculate the length of the lower shelf $[BC]$.

Exercise 17.9 Parallelism Check. In triangle RST, $RS = 7$ cm and $RT = 9$ cm. Point P is placed on $[RS]$ such that $RP = 2.8$ cm, and point Q on $[RT]$ such that $RQ = 3.6$ cm. Are lines (PQ) and (ST) parallel? Write out your proof carefully.

Exercise 17.10 The Non-Parallel Counterexample. In triangle ABC, $AB = 5$ cm and $AC = 8$ cm. Point D lies on $[AB]$ with $AD = 2$ cm, and point E on $[AC]$ with $AE = 3$ cm. Are lines (DE) and (BC) parallel? Justify your answer.

Exercise 17.11 Painting a Billboard: The k^2 Factor. A commercial painter takes 2 hours and uses a 5-liter can of paint to cover a square billboard measuring 3 meters on each side. How many cans of paint and how many hours of labor will he require to paint a giant billboard of the same shape measuring 9 meters on each side (assuming he works at the same pace)?

Level (Challenges and Modeling)

Exercise 17.12 Surveying Across a Marsh. To measure the distance between two survey beacons A and B separated by a deep swamp, a surveyor selects an observation station S on dry ground from which both A and B are visible. He measures on dry land: $SA = 120$ m and $SB = 150$ m. He extends sightlines straight past S: On the ray opposite to $[SA)$, he marks point A' with $SA' = 40$ m. On the ray opposite to $[SB)$, he marks point B' with $SB' = 50$ m. He measures the distance between the pegs directly on the meadow: $A'B' = 36$ m. 1. Prove that line $(A'B')$ is parallel to line (AB). 2. Determine the inaccessible distance AB across the swamp.

Exercise 17.13 Slicing the Cake into Equal Areas. A baker bakes a triangular pastry ABC of vertical height 18 cm. He wishes to divide it into three portions of equal surface area using two horizontal cuts parallel to base $[BC]$, located at heights h_1 and h_2 below apex A. 1. By what scale factor is the surface area multiplied when moving from the small top triangle to the middle triangle? And to the full pastry? 2. At what exact vertical distances h_1 and h_2 from vertex A must the cuts be made?

Exercise 17.14 The Frustum of the Pyramid. Consider a regular square pyramid of base side 10 m and height 12 m. Its total volume is $V = \frac{1}{3} \times \text{Base} \times \text{Height} = \frac{1}{3} \times 100 \times 12 = 400$ m^3. The pyramid is sliced horizontally at mid-height (6 m from the summit) by a plane parallel to the base. 1. What is the reduction scale factor k of the smaller upper pyramid? 2. What is the volume of the small removed pyramid? 3. What is the volume of the remaining lower "frustum"? What fraction of the total volume does it represent?

Exercise 17.15 Thales in Optics: The Pinhole Camera (*Camera Obscura*). The earliest camera is a lightproof box with a tiny pinhole O in its front face. A tree 15 meters tall stands 30 meters in front of the pinhole. The depth of the camera box from the pinhole to the rear photographic plate is 20 centimeters. 1. Sketch a ray diagram showing light rays crossing at pinhole O. Is the projected image upright or upside-down? 2. What will be the vertical height of the tree's image formed on the photographic plate?

Solutions

Solution 17.1 Each side is multiplied by 3: $12\ \text{cm}$, $15\ \text{cm}$, and $18\ \text{cm}$.

Solution 17.2 1. Side of original square: $\sqrt{25} = 5\ \text{cm}$. 2. Side of reduced square: $5 \times 0.4 = 2\ \text{cm}$. 3. Area of reduced square: $2 \times 2 = 4\ \text{cm}^2$. Check: $k^2 \times \text{Area} = 0.4^2 \times 25 = 0.16 \times 25 = 4\ \text{cm}^2$. Exact!

Solution 17.3 By Thales' Theorem: $\frac{AM}{AB} = \frac{AN}{AC} \Longrightarrow \frac{2}{6} = \frac{3}{AC} \Longrightarrow AC = \frac{6\times 3}{2} = 9\ \text{cm}$.

Solution 17.4 Hourglass configuration: $\frac{OA}{OB} = \frac{AC}{BD} \Longrightarrow \frac{5}{15} = \frac{4}{BD} \Longrightarrow BD = \frac{15\times 4}{5} = 12\ \text{cm}$.

Solution 17.5 $\frac{AM}{AB} = \frac{4}{10} = 0.4$ and $\frac{AN}{AC} = \frac{6}{15} = 0.4$. Yes, both ratios are equal.

Solution 17.6 Sunlight parallelism: $\frac{H_{\text{belfry}}}{h_{\text{man}}} = \frac{L_{\text{belfry}}}{l_{\text{man}}} \Longrightarrow \frac{H}{1.80} = \frac{36}{2.40} \Longrightarrow H = \frac{1.80\times 36}{2.40} = 27\ \text{meters}$.

Solution 17.7 1. $AM = AB - MB = 12 - 3 = 9\ \text{cm}$. 2. Thales: $\frac{AM}{AB} = \frac{AN}{AC} \Longrightarrow \frac{9}{12} = \frac{AN}{16} \Longrightarrow AN = \frac{9\times 16}{12} = 12\ \text{cm}$. $NC = AC - AN = 16 - 12 = 4\ \text{cm}$.

Solution 17.8 Total leg $AB = AM + MB = 45 + 30 = 75\ \text{cm}$. Thales: $\frac{AM}{AB} = \frac{MN}{BC} \Longrightarrow \frac{45}{75} = \frac{60}{BC} \Longrightarrow BC = \frac{75\times 60}{45} = 100\ \text{cm} = 1\ \text{meter}$.

Solution 17.9 Compute separately: $\frac{RP}{RS} = \frac{2.8}{7} = 0.4$ and $\frac{RQ}{RT} = \frac{3.6}{9} = 0.4$. Points R, P, S and R, Q, T lie in the same order. Since $\frac{RP}{RS} = \frac{RQ}{RT}$, by the Converse of Thales' Theorem, lines (PQ) and (ST) are parallel.

Solution 17.10 $\frac{AD}{AB} = \frac{2}{5} = 0.40$ while $\frac{AE}{AC} = \frac{3}{8} = 0.375$. Since $\frac{AD}{AB} \neq \frac{AE}{AC}$, by the contrapositive of Thales' Theorem, lines (DE) and (BC) are not parallel.

Solution 17.11 Linear dimension scales by $k = \frac{9}{3} = 3$. Surface area scales by $k^2 = 3^2 = 9$.

- Paint: $5\ \text{L} \times 9 = 45\ \text{liters}$ (9 cans).
- Time: $2\ \text{h} \times 9 = 18\ \text{hours}$ of labor.

Solution 17.12 1. Points A', S, A and B', S, B are aligned in order. Ratios: $\frac{SA'}{SA} = \frac{40}{120} = \frac{1}{3}$ and $\frac{SB'}{SB} = \frac{50}{150} = \frac{1}{3}$. By the Converse of Thales' Theorem, $(A'B') \parallel (AB)$. 2. By Thales: $\frac{A'B'}{AB} = \frac{1}{3} \Longrightarrow AB = 3 \times 36 = 108\ \text{meters}$.

Solution 17.13 1. Let A be total area. Top portion $= \frac{1}{3}A$, middle triangle $= \frac{2}{3}A$. Area scale factors are $\frac{1}{3}$ and $\frac{2}{3}$. Linear scale factors are square roots of area ratios: $k_1 = \sqrt{1/3}$ and $k_2 = \sqrt{2/3}$. 2. $h_1 = 18 \times \frac{1}{\sqrt{3}} = 6\sqrt{3} \approx 10.39\ \text{cm}$. $h_2 = 18 \times \sqrt{\frac{2}{3}} = 6\sqrt{6} \approx 14.70\ \text{cm}$.

Solution 17.14 1. Cut at half-height ($6\ \text{m}$ of $12\ \text{m}$): $k = \frac{6}{12} = \frac{1}{2} = 0.5$. 2. Volume scales by $k^3 = (1/2)^3 = 1/8$. $V_{\text{small}} = \frac{1}{8} \times 400 = 50\ \text{m}^3$. 3. Volume of frustum: $V_{\text{frustum}} = 400 - 50 = 350\ \text{m}^3$. It accounts for $\frac{350}{400} = \frac{7}{8}$ (87.5%) of the pyramid's entire volume!

Solution 17.15 1. Light rays travel in straight lines through pinhole O: the ray from the treetop hits the bottom of the plate, and the ray from the base hits the top. The image is formed **upside-down (inverted)**. 2. Hourglass configuration: Object distance $D = 30\ \text{m} = 3,000\ \text{cm}$; Camera depth $d = 20\ \text{cm}$; Tree height $H = 15\ \text{m} = 1,500\ \text{cm}$. $\frac{h}{H} = \frac{d}{D} \Longrightarrow \frac{h}{1,500} = \frac{20}{3,000} = \frac{1}{150} \Longrightarrow h = \frac{1,500}{150} = 10\ \text{cm}$. The tree projects an inverted miniature image $10\ \text{cm}$ tall.

The Question That Opens What Follows

When we scale a 3D pyramid or cube, its volume multiplies by k^3: it explodes upward at tremendous speed. But what happens when we do not merely triple a dimension once, but repeat a multiplication dozens of times in succession?

It is the legend of an Eastern king who thought he was granting a humble handful of rice to the inventor of chess, only to discover his entire empire buried beneath a mountain of grain. Welcome to the dizzying universe of **exponents, order of operations, and scientific notation**!

CHAPTER 18: The Grain of Rice on the Chessboard

> "Addition is the slow step of the walker advancing stone by stone. Multiplication is the gallop of the horseman. But exponentiation is the dazzling ascent of the rocket: in a few beats of its wings, it outstrips the comprehension of kings and fills the entire cosmos."

What You Are About to Discover

- The ancient Indian legend of sage **Sissa** and the unstoppable avalanche of rice grains across the chessboard.
- The universal **order of operations** (PEMDAS): why without strict rules of the road, no two mathematicians would ever agree on a calculation.
- **Exponential notation**: how to compress giant repeated multiplications into a tiny raised exponent (a^n).
- The secrets of **powers of 10** and **negative exponents** ($10^{-n} = \frac{1}{10^n}$).
- The four golden rules for multiplying, dividing, and raising powers without ever getting tangled.
- **Scientific notation** and the exploration of scales, from the subatomic nucleus to the edge of the observable universe.
- 🛠️ The Boundary Case Laboratory: exponents of zero ($a^0 = 1$), the indeterminate mystery of 0^0, and Kasner's Googolplex.

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1. The Problem: The Reckless Promise of King Belkib

More than fifteen hundred years ago, in a magnificent palace in northern India, lived a powerful king named Belkib (or Shirham). Despite all his riches, armies, and palaces, the monarch suffered from profound boredom. No banquet, no dance performance, and no cavalry tournament could dispel his heavy melancholy.

One afternoon, a wise court scholar named **Sissa** appeared before the throne carrying a carved ebony box. From it, he drew a game board of 64 alternating black and white squares, accompanied by two armies of carved wooden figurines: kings, horses, towers, and foot soldiers. It was the game of chess (then called *Chaturanga*).

The king was instantly mesmerized. For weeks on end, he studied strategies, shielded his king, and sacrificed pawns. Cured of his sorrow, he summoned Sissa before the entire assembled court: *"Sissa, your invention is sublime! Name any reward you desire, be it half the gold in my royal treasuries or my grandest marble palace, and it shall be yours!"*

Sissa bowed with humble grace: *"Great King, I am merely a modest scholar. I seek neither gold coins nor glittering jewels. Place your chessboard before you: - Place **1 grain of rice** upon the first square. - Place **2 grains of rice** upon the second square. - Place **4 grains of rice** upon the third square. - And double the number of grains on each subsequent square until the sixty-fourth square is reached. Grant me simply these few grains of rice, and my heart will be content."*

1re case	2e case	3e case	4e case	…	64e case
1 grain	2 grains	4 grains	8 grains	…	2^{63} grains !

The king burst into hearty laughter: *"What foolish modesty! You could have asked for a royal fortune, and you ask for bird food! Fetch a sack of rice from the royal granaries and pay this humble man at once!"*

Yet that evening, the king's chief royal treasurer returned pale, trembling, and bowing to the marble floor: *"Your Majesty… we have emptied every granary in the palace, every silo in the imperial city, and we could not even cover the thirtieth square! To fulfill Sissa's request, every harvest produced across the entire Earth for over two thousand years would not suffice!"*

How could the simple act of doubling a quantity 63 times in succession produce a mountain of grain larger than planet Earth itself? That is the staggering power of **exponential growth and powers**.

---

2. The Order of Operations: Who Takes Command in a Calculation?

Before handling colossal powers, we need an unambiguous navigational compass. Consider this disarmingly simple arithmetic expression:

$$2 + 3 \times 5$$

If you pose this question to random people on the street:

- The first computes strictly from left to right: $2 + 3 = 5$, followed by $5 \times 5 = \mathbf{25}$.
- The second performs the multiplication first: $3 \times 5 = 15$, followed by $2 + 15 = \mathbf{17}$.

Who is correct? 25 or 17?

In nature and physical life, this is not a matter of subjective opinion! Imagine visiting the market: you purchase 1 apple for \$2, and 3 bags of sweets for \$5 each. How much do you owe? You pay $2 + 15 = \$17$, not \$25! Multiplication binds objects into indivisible packets.

To ensure that every engineer, scientist, and computer program across the globe computes the exact same value without ambiguity, mathematicians established an international code of the road: the **order of operations**.

2.1 The PEMDAS Hierarchy

Mathematicians memorize this operational hierarchy using the mnemonic **PEMDAS**:

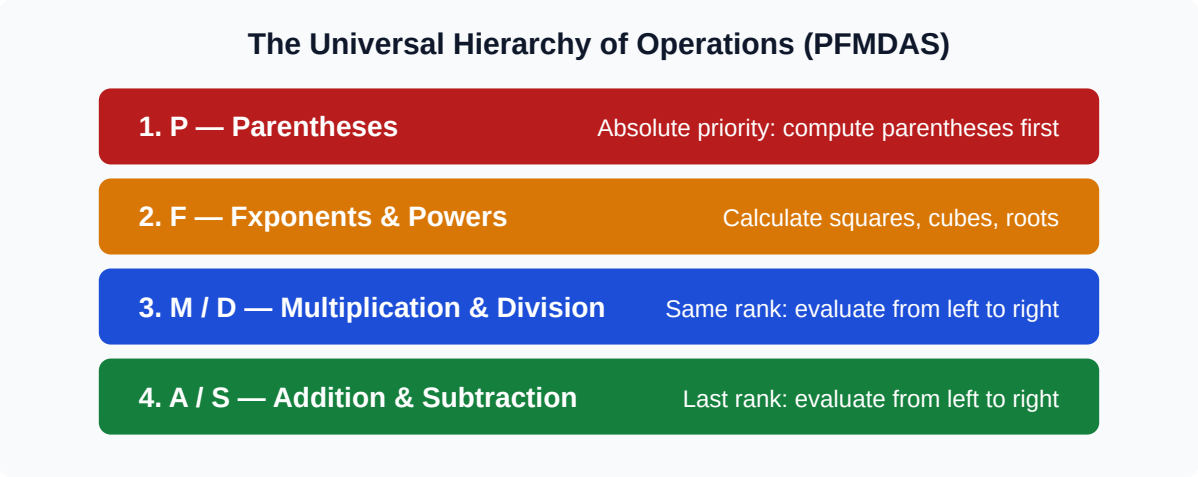

Figure 59: The universal PEMDAS order of operations

> **Official Rule — Operational Precedence**
>
> 1. Always evaluate
>
> expressions inside **parentheses** first, working from the innermost brackets outward. 2. Evaluate all **exponents** (powers). 3. Perform all **multiplications and divisions** in order from left to right. 4. Perform all **additions and subtractions** in order from left to right.

2.2 The Role of the Horizontal Fraction Bar

A horizontal fraction bar acts as an invisible set of parentheses enclosing everything above it (the numerator) and everything below it (the denominator):

$$\frac{8 + 12}{2 + 3} = \frac{(8 + 12)}{(2 + 3)} = \frac{20}{5} = 4$$

---

3. Exponential Notation: Compressing Multiplication

Let us revisit Sissa's chessboard:

- Square 1: 1

- Square 2: 2
- Square 3: $2 \times 2 = 4$
- Square 4: $2 \times 2 \times 2 = 8$
- Square 5: $2 \times 2 \times 2 \times 2 = 16$

For the 10th square, we must write the digit 2 multiplied by itself ten times. For the 64th square, we would have to write a line of sixty-three consecutive 2′s! That is painfully tedious and unreadable.

In the 17th century, the French philosopher and mathematician **René Descartes** introduced a brilliant typographic invention: writing the number of repeated factors in tiny script perched high to the upper right of the number.

📐 Definition — Power of a Number

For any real number a and any positive integer $n \geq 1$:

$$a^n = \underbrace{a \times a \times a \times \ldots \times a}_{n \text{ factors}}$$

- a is called the **base**.
- n is called the **exponent** (or power).
- The notation a^n reads aloud as **"a to the power of n"** or **"a raised to the n-th power"**.

Special Terminology for Powers of 2 and 3

- a^2 reads as **"a squared"** (representing the geometric area of a square of side a).
- a^3 reads as **"a cubed"** (representing the geometric volume of a cube of edge a).

Vital Special Cases: Exponent 1 and Exponent 0

- **Exponent 1:** $a^1 = a$ (a single factor).
- **Exponent 0:** For any non-zero base $a \neq 0$:

$$a^0 = 1$$

Why is $a^0 = 1$? Observe the clean pattern of descending division by 2 at each step:

$$2^3 = 8 \xrightarrow{\div 2} 2^2 = 4 \xrightarrow{\div 2} 2^1 = 2 \xrightarrow{\div 2} 2^0 = \mathbf{1}$$

Defining $a^0 = 1$ is the only logically coherent way to preserve the rules of arithmetic!

How Many Grains on the Entire Chessboard?

On the 64th square alone, there sit:

$$2^{63} \approx 9.22 \times 10^{18} \text{ grains}$$

And the grand total across all 64 squares equals:

$$2^0 + 2^1 + 2^2 + \ldots + 2^{63} = 2^{64} - 1 = \mathbf{18,446,744,073,709,551,615} \text{ grains!}$$

That is over **18 quintillion grains of rice**! If spread evenly across the entire surface of France, the country would be buried beneath a mountain of rice over one meter deep.

---

4. Powers of 10 and Negative Exponents

The number 10 is the sovereign ruler of our decimal system (inherited from our ten human fingers). Powers of 10 are the supreme engine of modern scientific calculation.

4.1 Positive Powers of 10: The March of Zeros

📐 Rule — Positive Powers of 10

For any positive integer n:

$$10^n = 1\underbrace{000\ldots0}_{n \text{ zeros}}$$

The exponent indicates **the exact number of zeros** following the leading digit 1!

- $10^1 = 10$ (ten)
- $10^2 = 100$ (one hundred)
- $10^3 = 1,000$ (one thousand, metric prefix *kilo-*)
- $10^6 = 1,000,000$ (one million, metric prefix *mega-*)
- $10^9 = 1,000,000,000$ (one billion, metric prefix *giga-*)
- $10^{12} = 1,000,000,000,000$ (one trillion, metric prefix *tera-*)

4.2 Negative Exponents: Journey into the Infinitesimal

Let us continue stepping down our arithmetic ladder by dividing by 10 at each successive step:

$$\begin{aligned} 10^3 &= 1,000 \\ 10^2 &= 100 \\ 10^1 &= 10 \\ 10^0 &= 1 \\ 10^{-1} &= \frac{1}{10} = 0.1 \\ 10^{-2} &= \frac{1}{10^2} = \frac{1}{100} = 0.01 \\ 10^{-3} &= \frac{1}{10^3} = \frac{1}{1,000} = 0.001 \end{aligned}$$

📐 Definition — Negative Exponents

For any non-zero base $a \neq 0$ and any integer n:

$$a^{-n} = \frac{1}{a^n}$$

Specifically for powers of 10:

$$10^{-n} = \frac{1}{10^n} = \underbrace{0.00\ldots0}_{n \text{ total zeros}}1$$

A negative exponent designates the **reciprocal** of the power. In 10^{-n}, there are $\boldsymbol{n}$ **zeros in total**, including the leading zero before the decimal point!

- $10^{-1} = 0.1$ (one tenth, prefix *deci-*)
- $10^{-2} = 0.01$ (one hundredth, prefix *centi-*)
- $10^{-3} = 0.001$ (one thousandth, prefix *milli-*)
- $10^{-6} = 0.000001$ (one millionth, prefix *micro-*)
- $10^{-9} = 0.000000001$ (one billionth, prefix *nano-*)

---

5. The Golden Rules of Exponent Arithmetic

When manipulating powers with the same base, computation transforms magically: **multiplication of powers becomes simple addition of exponents!**

📐 Fundamental Exponent Laws

For all integers m and n, and non-zero bases a and b:

1. **Product of Powers with Same Base:**
$$a^m \times a^n = a^{m+n}$$
(Example: $10^3 \times 10^4 = 10^{3+4} = 10^7$*)*
2. **Quotient of Powers with Same Base:**
$$\frac{a^m}{a^n} = a^{m-n}$$
(Example: $\frac{10^7}{10^2} = 10^{7-2} = 10^5$*;* $\frac{10^3}{10^8} = 10^{3-8} = 10^{-5}$*)*
3. **Power of a Power:**
$$(a^m)^n = a^{m \times n}$$
(Example: $(10^3)^4 = 10^{3 \times 4} = 10^{12}$*)*
4. **Power of a Product:**
$$(a \times b)^n = a^n \times b^n$$
(Example: $(2 \times 5)^3 = 2^3 \times 5^3 = 8 \times 125 = 1,000 = 10^3$*)*

---

6. Scientific Notation: Writing the Universe

How can one compare the mass of a single hydrogen atom ($0.00000000000000000000000000167$ kg) to the mass of planet Earth ($5,972,000,000,000,000,000,000,000$ kg)? With standard digits, human eyes lose count of the zeros instantly! Scientists therefore established an international standard format: **scientific notation**.

📐 Definition — Scientific Notation

A number is written in **scientific notation** when formatted strictly as:
$$a \times 10^n$$
where:

- a is a decimal number with **exactly one non-zero digit before the decimal point** ($1 \leq a < 10$ or $-10 < a \leq -1$).
- n is an integer (positive, zero, or negative).

Rapid Conversion Technique

- For large numbers (> 10): shift the decimal point n places to the **left** $\Longrightarrow 10^{+n}$.
$$149,600,000 \text{ km} = 1.496 \times 10^8 \text{ km} \quad (\text{Earth-Sun distance})$$
- For small numbers (< 1): shift the decimal point n places to the **right** $\Longrightarrow 10^{-n}$.
$$0.00000012 \text{ m} = 1.2 \times 10^{-7} \text{ m} \quad (\text{size of a virus})$$

Nucleus	Atom	Human	Earth	Solar System	Galaxy
10^{-15} m	10^{-10} m	1 m (10^0 m)	10^6 m	10^{11} m	10^{21} m

---

7. Beware of Pitfalls!

> ⚠️ **Warning: Monumental Exponent Blunders!**
>
> 1. **The Phantom Exponent Addition Trap:**
>
> $$10^2 + 10^3 \neq 10^5 \quad !$$
>
> Powers can NEVER be added by adding their exponents! $10^2 + 10^3 = 100 + 1,000 = 1,100$, whereas $10^5 = 100,000$. 2. **The Minus Sign and Parentheses Trap:**
>
> $$-3^2 = -(3 \times 3) = \mathbf{-9}$$
>
> $$(-3)^2 = (-3) \times (-3) = \mathbf{+9}$$
>
> Without grouping parentheses, the exponent applies ONLY to the base number 3, not to the preceding minus sign! 3. **Non-Standard Scientific Format:**
> - 35×10^4 is **not** in scientific notation because $35 \geq 10$ (must be written 3.5×10^5).
> - 0.7×10^{-3} is **not** in scientific notation because $0.7 < 1$ (must be written 7×10^{-4}).
>
> 4. **Multiplying Bases Instead of Adding Exponents:**
>
> $$2^3 \times 2^4 \neq 4^7 \quad \text{correct is} \quad 2^{3+4} = 2^7 = 128$$

---

8. Solved Examples

Solved Example 1 — Evaluating with Order of Operations

Calculate the exact value of the following expression, detailing every intermediate step:

$$A = 5 + 3 \times (4^2 - 6) - \frac{24}{2^3}$$

Solution: Apply PEMDAS rigorously: 1. Inside parentheses:
- Exponent first: $4^2 = 16$.
- Subtraction: $16 - 6 = 10$. Expression becomes: $A = 5 + 3 \times 10 - \frac{24}{2^3}$.

2. Exponent in denominator:
   - $2^3 = 2 \times 2 \times 2 = 8$. Expression becomes: $A = 5 + 3 \times 10 - \frac{24}{8}$.
3. Multiplications and divisions:
   - $3 \times 10 = 30$.
   - $\frac{24}{8} = 3$. Expression becomes: $A = 5 + 30 - 3$.
4. Additions and subtractions from left to right:
   - $5 + 30 = 35$, then $35 - 3 = 32$.

Conclusion: $A = 32$. ✓

Solved Example 2 — Simplifying Powers of 10

Write as a single power of 10:

$$B = \frac{10^5 \times (10^{-2})^3}{10^{-4}}$$

Solution: 1. Power of a power in numerator:

$$(10^{-2})^3 = 10^{-2 \times 3} = 10^{-6}$$

2. Product in numerator (add exponents):

$$10^5 \times 10^{-6} = 10^{5+(-6)} = 10^{-1}$$

3. Division (subtract denominator exponent):

$$B = \frac{10^{-1}}{10^{-4}} = 10^{-1-(-4)} = 10^{-1+4} = 10^{3}$$

Conclusion: $B = 10^3 = 1,000$. ✓

Solved Example 3 — Converting to Scientific Notation

Write in standard scientific notation: 1. $C = 548,000,000$ (continental population) 2. $D = 0.000037 \text{ m}$ (microscopic fiber diameter) 3. $E = 485 \times 10^6$ 4. $F = 0.029 \times 10^{-4}$

Solution: 1. Move decimal 8 places left: $C = 5.48 \times 10^8$. 2. Move decimal 5 places right: $D = 3.7 \times 10^{-5} \text{ m}$. 3. $E = (4.85 \times 10^2) \times 10^6 = 4.85 \times 10^{2+6} = 4.85 \times 10^8$. 4. $F = (2.9 \times 10^{-2}) \times 10^{-4} = 2.9 \times 10^{-2+(-4)} = 2.9 \times 10^{-6}$. ✓

Solved Example 4 — Astrophysical Travel Time

Light travels through vacuum at speed $v = 3 \times 10^8 \text{ m/s}$. Mean Earth-Sun distance is $d = 1.5 \times 10^{11}$ meters. How long does a photon take to travel from Sun to Earth?

Solution: Using $t = \frac{d}{v}$:

$$t = \frac{1.5 \times 10^{11}}{3 \times 10^8} = \left(\frac{1.5}{3}\right) \times 10^{11-8} = 0.5 \times 10^3 \text{ seconds}$$

Convert to standard decimal:

$$t = 0.5 \times 1,000 = 500 \text{ seconds}$$

In minutes and seconds ($1 \text{ min} = 60 \text{ s}$):

$$500 = 8 \times 60 + 20$$

Sunlight takes **8 minutes and 20 seconds** to reach Earth! ✓

Solved Example 5 — The Folded Sheet of Paper

A sheet of notebook paper has thickness $t = 0.1 \text{ mm} = 10^{-4} \text{ m}$. Each fold doubles its thickness. 1. What is its thickness after 10 folds? 2. What would its thickness be after 42 folds?

Solution: 1. After 10 folds: thickness scales by $2^{10} = 1,024 \approx 10^3$.

$$\text{Thickness} = 0.1 \text{ mm} \times 1,024 = 102.4 \text{ mm} \approx 10.24 \text{ cm}$$

2. After 42 folds:

$$\text{Thickness} = 10^{-4} \text{ m} \times 2^{42}$$

$2^{42} = (2^{10})^4 \times 2^2 \approx (1,000)^4 \times 4 = 4 \times 10^{12}$.

$$\text{Thickness} \approx 10^{-4} \text{ m} \times (4.398 \times 10^{12}) \approx 4.4 \times 10^8 \text{ m} = 440,000 \text{ km}\ !$$

The mean distance from Earth to the Moon is $384,400$ km. A piece of paper folded 42 times reaches past the Moon! ✓

What You Must Remember

1. **Order of Operations (PEMDAS):** Parentheses → Exponents → Multiplication & Division (left to right) → Addition & Subtraction (left to right).
2. **Definition of Powers:** $a^n = \underbrace{a \times a \times \ldots \times a}_{n \text{ times}}$; $a^1 = a$; $a^0 = 1$ ($a \neq 0$).
3. **Negative Exponents:** $a^{-n} = \frac{1}{a^n}$; $10^{-n} = \frac{1}{10^n} = 0.00\ldots01$ (n total zeros).
4. **Exponent Calculation Rules:**
   - $a^m \times a^n = a^{m+n}$
   - $\frac{a^m}{a^n} = a^{m-n}$
   - $(a^m)^n = a^{m \times n}$
   - $(a \times b)^n = a^n \times b^n$
5. **Scientific Notation:** Standard format: $a \times 10^n$ with $1 \leq a < 10$ (exactly one non-zero digit before the decimal point).

Exercises

Level (Direct Application)

Exercise 18.1 Evaluate numerical expressions: 1. $A = 7 + 3 \times 4$ 2. $B = (7 + 3) \times 4$ 3. $C = 20 - 4 \times 2 + 1$ 4. $D = 30/5 \times 2$

Exercise 18.2 Compute without a calculator: 2^4; 3^3; 5^2; 10^5; 7^0; $(-2)^3$; $(-2)^4$; -5^2.

Exercise 18.3 Write in decimal notation: 10^3; 10^{-2}; 10^{-5}; 4.2×10^3; 7×10^{-4}.

Exercise 18.4 Express as a single power of 10: 1. $10^4 \times 10^7$ 2. $\frac{10^9}{10^3}$ 3. $(10^5)^2$ 4. $10^{-3} \times 10^{-5}$ 5. $\frac{10^2}{10^{-4}}$

Exercise 18.5 Which of the following numbers are written in valid scientific notation? a) 3.45×10^7; b) 0.4×10^{-3}; c) 12×10^5; d) 9.99×10^{-1}; e) 1×10^0.

Level (Reasoning and Rules)

Exercise 18.6 Compute following strict operational precedence: 1. $G = 3 \times 2^3 - 4 \times (5 - 2^2)$ 2. $H = \frac{5^2 - 3^2}{2^4 - 2 \times 4}$

Exercise 18.7 Express in the form a^n (a integer, n relative integer): 1. $2^5 \times 2^3 \times 2^{-4}$ 2. $\frac{3^8}{(3^2)^3}$ 3. $\frac{7^3 \times 7^{-5}}{7^{-4}}$ 4. 8×2^5 *(Hint: rewrite 8 as a power of 2).*

Exercise 18.8 Express in scientific notation: 1. Red blood cell diameter: 0.000007 m. 2. Speed of light: $300,000,000$ m/s. 3. Human world population: $8,100,000,000$ people. 4. Mass of a single water droplet: 0.00005 kg.

Exercise 18.9 Compute and give final answers in scientific notation: 1. $K = (3 \times 10^4) \times (5 \times 10^7)$ 2. $L = \frac{8 \times 10^{-3}}{4 \times 10^5}$ 3. $M = \frac{1.2 \times 10^5 \times 7 \times 10^{-2}}{2.4 \times 10^{-6}}$

Exercise 18.10 Bacterial Explosion. A culture of bacteria is placed in a warm nutrient dish. At noon ($t = 0$), there are $1,000$ bacteria (10^3). The colony doubles every 20 minutes. 1. How many bacteria exist at 1:00 PM? 2. How many bacteria exist at 4:00 PM? Express in scientific notation.

Exercise 18.11 Sorting Without Calculating. Order from least to greatest without full decimal expansion:

$$A = 4.5 \times 10^6 \quad ; \quad B = 8.2 \times 10^5 \quad ; \quad C = 1.1 \times 10^7 \quad ; \quad D = 9.9 \times 10^5$$

Level (Challenges and Astronomical Scales)

Exercise 18.12 The Googol and the Googolplex. In 1920, American mathematician Edward Kasner asked his 9-year-old nephew Milton Sirotta to invent a name for 1 followed by 100 zeros. The boy replied: "A GOOGOL!" (which later inspired the name of the search engine *Google*). 1. Write a googol as a power of 10. 2. The total number of subatomic particles in the observable universe is estimated at 10^{80}. How many times larger is a googol than the particle count of the cosmos? 3. Kasner then defined a *Googolplex*: 1 followed by a googol of zeros! Write a Googolplex using powers of 10. Why is it physically impossible to write this number down in our universe?

Exercise 18.13 Voyager 1 Interstellar Mission. Spacecraft *Voyager 1* speeds through deep space at 17 km/s ($\approx 6.12 \times 10^4$ km/h). It heads toward Proxima Centauri, located 4.24 light-years away. One light-year is approximately 9.46×10^{12} km. 1. Compute the distance to Proxima Centauri in kilometers in scientific notation. 2. How many hours, then years, will the probe take to cover this distance?

Exercise 18.14 Algebraic Exponent Simplification. For non-zero real numbers x and y, simplify:

$$Z = \frac{(x^2 y^{-3})^4 \times (x^{-1} y)^2}{x^5 y^{-8}}$$

Exercise 18.15 The Geometric Series of Halves. Compute the exact fraction:

$$S = \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^5} + \frac{1}{2^6}$$

What do you observe about its relation to the number 1? What would the sum equal if continued infinitely?

Solutions

Solution 18.1 1. $A = 7 + 12 = 19$. 2. $B = 10 \times 4 = 40$. 3. $C = 20 - 8 + 1 = 13$. 4. $D = (30/5) \times 2 = 6 \times 2 = 12$.

Solution 18.2 $2^4 = 16$; $3^3 = 27$; $5^2 = 25$; $10^5 = 100,000$; $7^0 = 1$. $(-2)^3 = -8$; $(-2)^4 = +16$; $-5^2 = -(5 \times 5) = -25$.

Solution 18.3 $10^3 = 1,000$; $10^{-2} = 0.01$; $10^{-5} = 0.00001$; $4.2 \times 10^3 = 4,200$; $7 \times 10^{-4} = 0.0007$.

Solution 18.4 1. $10^4 \times 10^7 = 10^{11}$. 2. $\frac{10^9}{10^3} = 10^6$. 3. $(10^5)^2 = 10^{10}$. 4. $10^{-3} \times 10^{-5} = 10^{-8}$. 5. $\frac{10^2}{10^{-4}} = 10^{2-(-4)} = 10^6$.

Solution 18.5 - a) Yes ($1 \le 3.45 < 10$). - b) No ($0.4 < 1$). - c) No ($12 \ge 10$). - d) Yes ($1 \le 9.99 < 10$). - e) Yes ($1 \le 1 < 10$).

Solution 18.6 1. $G = 3 \times 8 - 4 \times (5 - 4) = 24 - 4 \times 1 = 20$. 2. $H = \frac{25-9}{16-8} = \frac{16}{8} = 2$.

Solution 18.7 1. $2^{5+3-4} = 2^4$. 2. $\frac{3^8}{3^6} = 3^{8-6} = 3^2$. 3. $\frac{7^{-2}}{7^{-4}} = 7^{-2-(-4)} = 7^2$. 4. $8 = 2^3 \implies 2^3 \times 2^5 = 2^8$.

Solution 18.8 1. 7×10^{-6} m. 2. 3×10^8 m/s. 3. 8.1×10^9 people. 4. 5×10^{-5} kg.

Solution 18.9 1. $K = 15 \times 10^{11} = 1.5 \times 10^{12}$. 2. $L = 2 \times 10^{-3-5} = 2 \times 10^{-8}$. 3. $M = \left(\frac{8.4}{2.4}\right) \times 10^{3-(-6)} = 3.5 \times 10^9$.

Solution 18.10 1. In 1 hour (3 periods of 20 min): $N_1 = 1,000 \times 2^3 = 8,000$ bacteria. 2. In 4 hours (12 periods of 20 min): $N_2 = 10^3 \times 2^{12} = 10^3 \times 4,096 = 4.096 \times 10^6$ bacteria.

Solution 18.11 Exponent 5: $B = 8.2 \times 10^5$ and $D = 9.9 \times 10^5 \implies B < D$. Exponent 6: $A = 4.5 \times 10^6$. Exponent 7: $C = 1.1 \times 10^7$. Ascending order: $B < D < A < C$.

Solution 18.12 1. 1 googol $= 10^{100}$. 2. $\frac{10^{100}}{10^{80}} = 10^{20}$. A googol is **one hundred quintillion times larger** than the particle count of the entire cosmos! 3. 1 googolplex $= 10^{\text{googol}} = 10^{10^{100}}$. Writing it out would require 10^{100} zeros—vastly more than all atoms in the universe (10^{80}).

Solution 18.13 1. $D = 4.24 \times (9.46 \times 10^{12}) \approx 4.011 \times 10^{13}$ km. 2. In hours: $t = \frac{4.011 \times 10^{13}}{6.12 \times 10^4} \approx 6.554 \times 10^8$ hours. In years ($8,766$ h/year): $T \approx 74,766$ years!

Solution 18.14 Numerator: $(x^8 y^{-12}) \times (x^{-2} y^2) = x^6 y^{-10}$. Dividing: $Z = \frac{x^6 y^{-10}}{x^5 y^{-8}} = x^{6-5} y^{-10-(-8)} = x^1 y^{-2} = \frac{x}{y^2}$.

Solution 18.15 $S = \frac{32+16+8+4+2+1}{64} = \frac{63}{64} = 0.984375$. The sum approaches 1 asymptotically; an infinite sum of these halves equals exactly 1!

The Question That Opens What Follows

We now know that a^2 represents the square of a number: $3^2 = 9$, $4^2 = 16$. But what happens when we attempt the reverse operation? What number multiplied by itself yields exactly 2?

This innocent question shattered the most secret philosophical brotherhood of ancient Greece, sparked death sentences on the open sea, and revealed a terrifying breed of numbers that no fraction can tame: **irrational numbers**. Step into the mystic sanctuary of Croton with **The Pythagorean Theorem**!

CHAPTER 19: The Forbidden Secret of the Pythagoreans

> "The disciples of Pythagoras chanted that the entire universe was nothing but an orderly harmony of whole numbers and gentle ratios. Yet within the heart of the humblest square slept an invisible monster: a number without end that was destined to shatter their sacred dream."

What You Are About to Discover

- The secret white-robed brotherhood of **Pythagoras** at Croton and their mystical creed: *"All is Number."*
- The anatomy of the right-angled triangle and the curious origin of the word **hypotenuse**.
- The **Pythagorean Theorem** ($a^2 + b^2 = c^2$) and its three historic visual demonstrations (the café-tile puzzle, the two-arrangement square, and Bhāskara's wordless proof: *"Behold!"*).
- What the **square root** ($\sqrt{a}$) truly represents: the mathematical machine that journeys backward from a square's area to its side length.
- The **Converse and Contrapositive of Pythagoras** used by builders to guarantee square corners.
- The great foundational crisis of ancient mathematics: the proof that the diagonal of a unit square ($\sqrt{2}$) is an **irrational number**, and the historic proof by contradiction.
- 🛠️ The Boundary Case Laboratory: non-real square roots ($\sqrt{-1}$), the spiral of Theodorus, and multi-dimensional diagonals.

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1. The Problem: The White Brotherhood of Croton

Twenty-five hundred years ago, in the sun-drenched maritime city of Croton in southern Italy, flourished a cloistered community that both fascinated and unsettled the ancient Mediterranean: **the Pythagorean school**.

Its initiates dressed in flowing white linen robes. They ate no meat, held no private property, and observed up to five years of absolute silence to master their unruly thoughts. They recognized one another through a secret emblem drawn in the palm of the hand: the five-pointed star pentagram.

Engraved upon the portal of their academy was their supreme doctrine:

"**ALL IS NUMBER**"

To master **Pythagoras** and his disciples, this was no poetic figure of speech. They firmly believed that **the entire cosmos—the orbital dance of planets, the changing seasons, the justice of the laws, and the music of the lyre—was built strictly upon whole integers ($1, 2, 3, 4$...) and their rational fractions ($\frac{1}{2}, \frac{2}{3}, \frac{3}{4}$...).**

When you pinch a vibrating lyre string at its exact midpoint ($\frac{1}{2}$), it rings with the same pitch one octave higher. When you pinch it at two-thirds of its length ($\frac{2}{3}$), it sings a pure fifth. To them, all of creation was harmony, light, and integer proportion: every physical length across the earth could be measured by a ratio of whole numbers!

Until the fateful day when a young disciple named **Hippasus of Metapontum** took a wooden stylus and drew a seemingly innocent figure in the dust. He did not sketch an elaborate polygon. He drew the simplest, humblest square imaginable: **a square whose sides measure exactly 1 meter.**

Then, he connected two opposite corners with a straight line: he drew **the diagonal**.

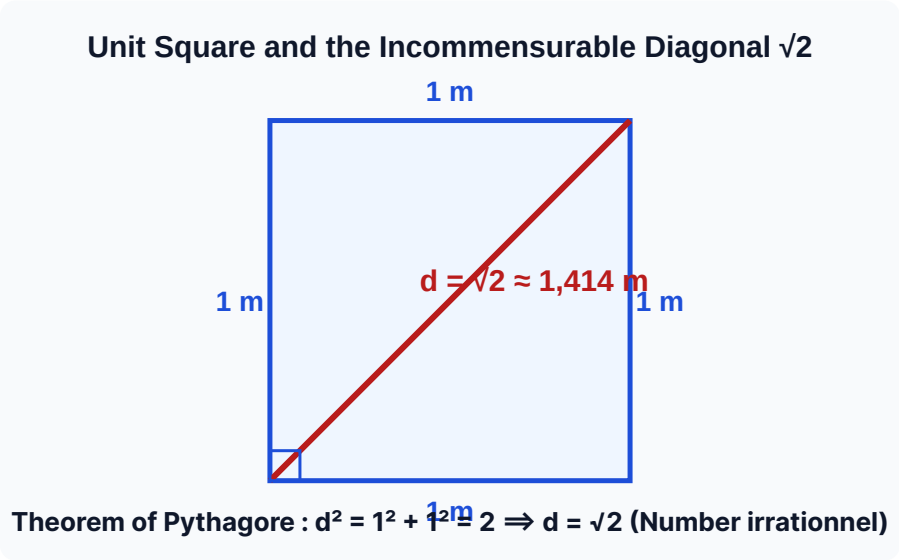

Figure 60: The unit square and the incommensurable diagonal root of 2

Hippasus set out to calculate the exact numerical length d of this diagonal. He applied the great theorem of his master. And what he uncovered was so terrifying to the brotherhood that they swore him to secrecy under penalty of death…

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2. The Right Triangle and Its Treasure: The Hypotenuse

Before reliving the intellectual tragedy of Hippasus, let us explore the royal figure of geometry: the **right-angled triangle**.

📐 Definition — Right Triangle and Hypotenuse

- A **right-angled triangle** (or right triangle) is a triangle

possessing one **right angle** (90°).

- The side directly facing the right angle is called the **hypotenuse**.
- The hypotenuse is **always the strictly longest** of the three sides in any right triangle.
- The two shorter sides that meet to form the right angle are called the **legs** (or sides of the right angle).

> 💡 ***Where Does This Strange Word Come From?*** *The word* hypotenuse *originates from the ancient Greek* hupoteinousa, *formed from* hupo *(“under”) and* teinein *(“to stretch”). The hypotenuse is literally the string “stretched beneath” the right angle, just like the taut bowstring of an archer or the staysail stay of a ship.*

---

3. The Pythagorean Theorem: The Dance of Three Squares

The theorem is not originally an abstract algebraic formula, but an astonishing truth about the **geometric areas of squares** resting against the triangle’s flanks.

📐 The Pythagorean Theorem (Geometric Statement)

In any right-angled triangle, the area of the square constructed upon the hypotenuse is equal to the sum of the areas of the two squares constructed upon the two legs.

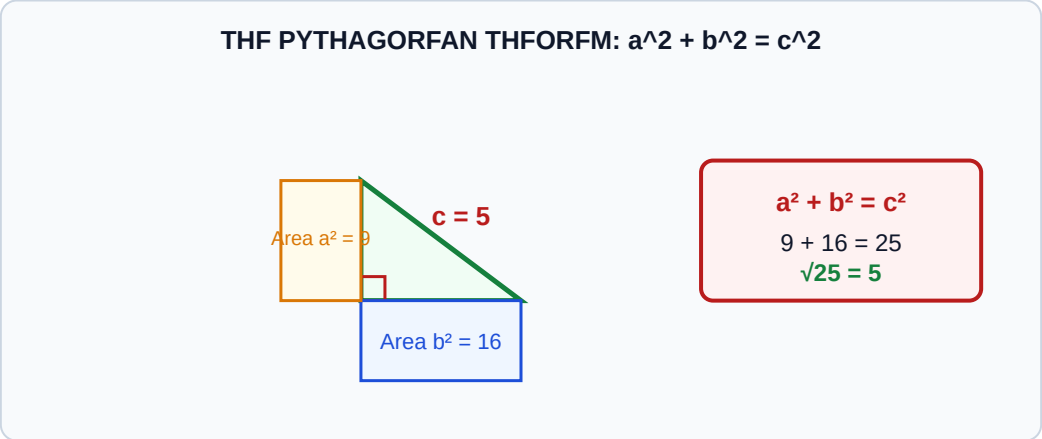

Figure 61: Geometric proof of the Pythagorean Theorem

Denoting a and b as the lengths of the two legs, and c as the length of the hypotenuse:

- The area of the first leg square is a^2 ($a \times a$).
- The area of the second leg square is b^2 ($b \times b$).
- The area of the hypotenuse square is c^2 ($c \times c$).

The foundational equality is written:

$$\mathbf{a^2 + b^2 = c^2}$$

If the triangle is labeled ABC with its right angle at vertex A, the hypotenuse is $[BC]$, and the formula is expressed via its vertices:

$$BC^2 = AB^2 + AC^2$$

The Three Historic Visual Demonstrations

Why does this equality hold true for **every single right triangle in the universe**? Here are three visual proofs you can understand without writing a single algebraic equation, relying purely on your own eyes.

Proof 1: The Café-Tile Floor Puzzle

This is the simplest demonstration in existence, applying to an isosceles right triangle (half of a square):

Each small square resting on a leg contains 2 half-square right triangles, totaling $2 + 2 = 4$ triangles. The tilted square resting upon the hypotenuse contains **precisely 4 identical half-square triangles**! The combined area is visibly identical.

Proof 2: The Master Square in Two Arrangements

Draw a large square of side length $(a + b)$. We can arrange 4 identical copies of our right triangle (with sides a, b, c) inside this boundary in two distinct ways:

In both identical enclosing squares:

- On the left, we remove the 4 triangles: the remaining uncovered area is $a^2 + b^2$.
- On the right, we remove the exact same 4 triangles: the remaining uncovered area is c^2.

Because we subtracted the exact same total area from identical large squares, what remains must be strictly equal:

$$a^2 + b^2 = c^2$$

Proof 3: Bhāskara's "Behold!"

In 12th-century India, mathematician Bhāskara arranged 4 identical right triangles of sides a, b, c into an outer square of side c, leaving an interior empty square of side $(b - a)$:

$$\text{Total Area} = c^2 = 4 \times \left(\frac{a \times b}{2}\right) + (b - a)^2 = 2ab + (b^2 - 2ab + a^2) = a^2 + b^2$$

Beneath his diagram, Bhāskara wrote not a single sentence of explanation, but a single command: *"Behold!"*

Proof 4: Oliver Byrne's Color Semiotic Proof (1847)

In 1847, the civil engineer and mathematician Oliver Byrne published an extraordinary edition of *Euclid's Elements* in London, banishing alphabetical lettering in favor of vibrant primary colors: brilliant blues, bold reds, and warm ambers.

The squares constructed upon the legs decompose into colored triangles of matching area, rendering $a^2 + b^2 = c^2$ self-evident at a single glance. It is this masterpiece of visual semiotics that inspired the chromatic charter of the present treatise.

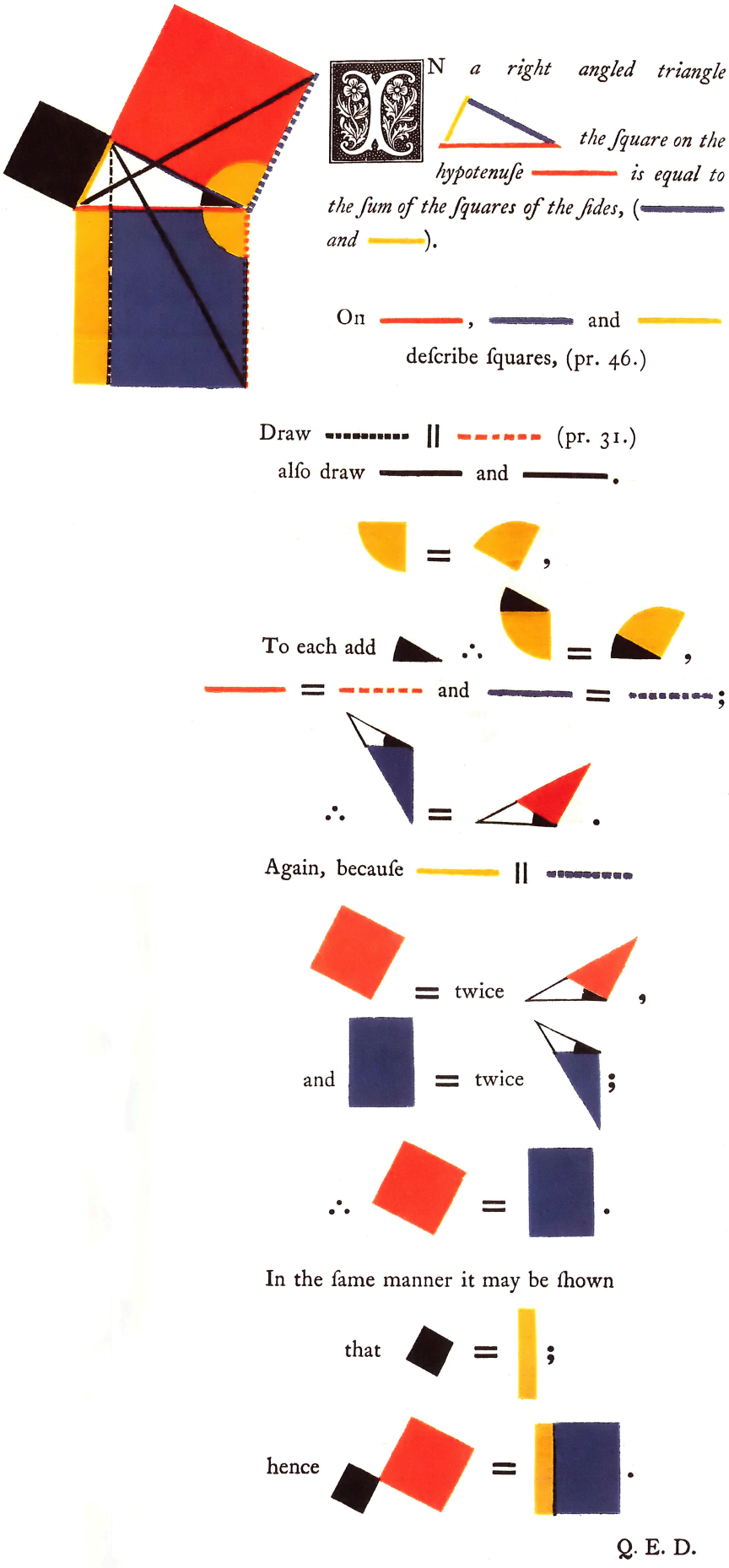

Figure 62: Geometric proof of the Pythagorean Theorem by Oliver Byrne (1847): magnitudes and areas demonstrated through pure color

4. The Square Root: Journeying Backward from Square to Side

When using the Pythagorean Theorem to find an unknown side length, we typically compute c^2 (the area of the square), and need to find c (the side length). To do so, we introduce a fundamental mathematical tool: **the radical symbol for square roots**.

📐 Definition — Square Root

For any non-negative real number $x \geq 0$, the **square root** of x, denoted $\sqrt{x}$, is the unique **non-negative** number whose square equals x:

$$(\sqrt{x})^2 = x$$

- The symbol $\sqrt{}$ is called the **radical sign**. It evolved from a handwritten letter *r*, the initial of the Latin word *radix* ("root").
- $\sqrt{x}$ reads aloud as **"the square root of x"**.

Perfect Squares to Memorize:

x	0	1	4	9	16	25	36	49	64	81	100	121	144	169
$\sqrt{x}$	0	1	2	3	4	5	6	7	8	9	10	11	12	13

5. The Converse and Contrapositive of Pythagoras

Just like Thales' Theorem, Pythagoras' Theorem functions in both logical directions: 1. **Direct Theorem:** Right triangle $\Rightarrow a^2 + b^2 = c^2$ (used to **calculate unknown side lengths**). 2. **Converse Theorem:** If in a triangle the square of the longest side equals the sum of the squares of the two shorter sides ($c^2 = a^2 + b^2$), then **the triangle is a right-angled triangle** (with its right angle directly facing side c). 3. **Contrapositive:** If $c^2 \neq a^2 + b^2$, then **the triangle is not a right triangle**.

This converse explains the empirical trick of the Egyptian rope-stretchers with their thirteen-knot rope (introduced in Chapter 15): With side lengths 3, 4, and 5:

$$3^2 + 4^2 = 9 + 16 = 25 \quad \text{and} \quad 5^2 = 25$$

Because $3^2 + 4^2 = 5^2$, the triangle is **guaranteed to form a right angle**! The integer triplet $(3, 4, 5)$ is called a **Pythagorean triple**. Other famous integer triples include $(5, 12, 13)$, $(8, 15, 17)$, and $(7, 24, 25)$.

6. The Pythagorean Crisis: The Scandal of $\sqrt{2}$

Let us return to Hippasus of Metapontum and his humble square measuring 1 meter on each side. Applying the Pythagorean Theorem to the diagonal of the unit square:

$$d^2 = 1^2 + 1^2 = 1 + 1 = 2$$

The diagonal length d is therefore:

$$d = \sqrt{2} \approx 1.41421356...$$

Hippasus sought the exact integer fraction $\frac{p}{q}$ capable of producing $\sqrt{2}$:

- He tried $\frac{7}{5} = 1.4 \Longrightarrow \left(\frac{7}{5}\right)^2 = \frac{49}{25} = 1.96$ (too small!).
- He tried $\frac{10}{7} \approx 1.428 \Longrightarrow \left(\frac{10}{7}\right)^2 = \frac{100}{49} = 2.04$ (too large!).
- He tested $\frac{1414}{1000}$, $\frac{17}{12}$, $\frac{577}{408}$… every candidate fraction missed the mark by a whisker!

Then Hippasus experienced a flash of terrifying genius: he constructed a rigorous proof showing that **no such fraction exists, and no such fraction could EVER exist**.

The Historic Proof by Contradiction (Parity Proof)

This proof stands as one of the supreme peaks of human intellect. Follow the logic step-by-step:

Formal Proof — The Irrationality of $\sqrt{2}$

1. **The Counter-Hypothesis (Assumption for Contradiction):**
Suppose the opposite of our claim: assume there exists a fraction of integers $\frac{p}{q}$ equal to $\sqrt{2}$. Assume this fraction is written in **irreducible form** (simplified to lowest terms: p and q cannot both be even numbers, or we would have divided them by 2).
1. **Square Both Sides:**
$$\sqrt{2} = \frac{p}{q} \Longrightarrow 2 = \frac{p^2}{q^2} \Longrightarrow p^2 = 2 \times q^2$$
2. **Deduce that p Is an Even Number:** Since $p^2 = 2 \times q^2$, p^2 is a multiple of 2, so p^2 is **even**. The square of an odd number is always odd ($3^2 = 9, 5^2 = 25$). Therefore, if p^2 is even, the integer p itself must be **even**! We can write $p = 2k$ for some integer k.
3. **Deduce that q Is ALSO an Even Number:** Substitute $p = 2k$ into our equation:
$$(2k)^2 = 2q^2 \Longrightarrow 4k^2 = 2q^2$$
Divide both sides by 2:
$$2k^2 = q^2 \quad \text{or} \quad q^2 = 2k^2$$
Now q^2 is also a multiple of 2, meaning q^2 is even, which forces q to be **even**!
4. **The Blatant Contradiction:** Stop and consider what we have just established: p is even AND q is even! Both numbers are divisible by 2. Yet this flatly contradicts our initial premise that fraction $\frac{p}{q}$ was **irreducible**!
5. **Conclusion:** Our initial assumption must be false: **$\sqrt{2}$ can never be expressed as an integer fraction $\frac{p}{q}$.**

This discovery sent shockwaves through the ancient world. The Pythagoreans termed such lengths *aloga*—"the unspeakable" or "that which cannot be spoken aloud." Today, we call them **irrational numbers** (from the Latin *ratio*, meaning ratio or proportion).

- A **rational number** ($\mathbb{Q}$) can be written as a fraction of two integers ($\frac{3}{4}, -\frac{5}{2}, \frac{1}{3} = 0.333...$). Its decimal digits terminate or eventually repeat in an endless cycle.
- An **irrational number** ($\sqrt{2}, \sqrt{3}, \pi...$) can never be expressed as a fraction. Its decimal digits unfold endlessly into eternity without ever repeating a fixed pattern!

> 🔍 ***Historical Legend — The Drowning of Hippasus*** *According to ancient lore recounted by the philosopher Iamblichus, the orthodox Pythagoreans were so appalled by the collapse of their numerical theology that they threw Hippasus overboard during a sea voyage, ensuring that the waves devoured the man who had dared to uncover $\sqrt{2}$. Yet truth cannot be drowned: geometry had irrevocably crossed into a new epoch.*

---

7. Beware of Pitfalls!

> ⚠️ **Warning: Costly Errors!**
>
> 1. **The Phantom Square Root of a Sum:**
>
> $$\sqrt{a+b} \neq \sqrt{a} + \sqrt{b} \quad !$$
>
> Verify with
>
> $a = 9$ and $b = 16$:
> - $\sqrt{9+16} = \sqrt{25} = \mathbf{5}$.
> - $\sqrt{9} + \sqrt{16} = 3 + 4 = \mathbf{7}$. $5 \neq 7$! You must ALWAYS compute the addition underneath the radical before taking the square root.
>
> 2. **Neglecting to Identify the Longest Side in the Converse:** When testing whether triangle MNP is right-angled, square the longest side by itself, and sum the squares of the two shorter sides separately. Never set them equal beforehand!
> 3. **Taking Square Roots of Negative Numbers:** $\sqrt{-9}$ is **undefined** within the real number system! Any real number multiplied by itself yields a non-negative result ($(+) \times (+) = +$ and $(-) \times (-) = +$).
> 4. **Confusing Leg and Hypotenuse Calculations:**
>   - To find the hypotenuse: **add** squares ($c = \sqrt{a^2 + b^2}$).
>   - To find a leg: **subtract** squares ($a = \sqrt{c^2 - b^2}$).

---

8. Solved Examples

Solved Example 1 — Calculating the Hypotenuse

Triangle ABC is right-angled at A, with $AB = 6 \ \text{cm}$ and $AC = 8 \ \text{cm}$. Calculate the length of hypotenuse $[BC]$.

Solution: - Triangle ABC is right-angled at A. - By the **Pythagorean Theorem**:

$$BC^2 = AB^2 + AC^2$$

- Substitute known values:

$$BC^2 = 6^2 + 8^2 = 36 + 64 = 100$$

- Because BC is a geometric length, $BC > 0$:

$$BC = \sqrt{100} = 10 \ \text{cm}$$

The hypotenuse measures exactly $10 \ \text{cm}$. ✓

Solved Example 2 — Finding a Leg: The Firefighter's Ladder

A 5 -meter rescue ladder leans against a vertical wall. The foot of the ladder sits on the ground $1.40 \ \text{m}$ from the base of the wall. At what vertical height does the top of the ladder touch the wall?

Solution: - The vertical wall and horizontal ground form a right angle. In right triangle ABC: hypotenuse $BC = 5 \ \text{m}$, base leg $AB = 1.40 \ \text{m}$, and vertical height $AC = h$. - By Pythagoras:

$$BC^2 = AB^2 + AC^2 \Longrightarrow 5^2 = 1.4^2 + h^2$$

- Isolate h^2:

$$h^2 = 5^2 - 1.4^2 = 25 - 1.96 = 23.04$$

- Take the square root:

$$h = \sqrt{23.04} = 4.8 \ \text{meters}$$

The ladder reaches a vertical height of $4.80 \ \text{meters}$. ✓

Solved Example 3 — Proving a Triangle Is Right-Angled

A mason drives three wooden stakes into the ground at distances $MN = 13\ \text{m}$, $NP = 12\ \text{m}$, and $MP = 5\ \text{m}$. Will the stone foundation wall built on these stakes have a perfect square corner?

Solution: - The longest side is $[MN]$ with $MN = 13\ \text{m}$. - Compute separately:

$$MN^2 = 13^2 = 169$$

$$MP^2 + NP^2 = 5^2 + 12^2 = 25 + 144 = 169$$

- We observe:

$$MN^2 = MP^2 + NP^2$$

- Conclusion: By the **Converse of the Pythagorean Theorem**, triangle MNP is right-angled at vertex P. The mason is guaranteed a true $90°$ right angle at corner P. ✓

Solved Example 4 — Proving a Triangle Is NOT Right-Angled

Triangle DEF has sides $DE = 7\ \text{cm}$, $EF = 9\ \text{cm}$, and $DF = 11\ \text{cm}$. Is it a right triangle?

Solution: - Longest side is $[DF] = 11\ \text{cm}$. - Compute separately:

$$DF^2 = 11^2 = 121$$

$$DE^2 + EF^2 = 7^2 + 9^2 = 49 + 81 = 130$$

- We observe:

$$DF^2 \neq DE^2 + EF^2 \quad (121 \neq 130)$$

- Conclusion: By the **contrapositive of the Pythagorean Theorem**, triangle DEF is not a right triangle. ✓

Solved Example 5 — Smartphone Screen Diagonal

A smartphone screen measures $7\ \text{cm}$ in width and $15\ \text{cm}$ in height. What is the length of its diagonal? Round to the nearest millimeter.

Solution: - The screen is rectangular; the diagonal forms a right triangle with width and height. - By Pythagoras:

$$d^2 = 7^2 + 15^2 = 49 + 225 = 274$$

- Therefore:

$$d = \sqrt{274} \approx 16.5529\ldots\ \text{cm} \approx 16.6\ \text{cm}$$

The diagonal measures approximately $16.6\ \text{cm}$ (about $6.5\ \text{inches}$). ✓

What You Must Remember

1. **Right Triangle & Hypotenuse:** The hypotenuse faces the $90°$ angle and is always the longest side.
2. **Pythagorean Theorem (Length Calculations):** In right triangle ABC with right angle at A:
$$BC^2 = AB^2 + AC^2$$
   - Hypotenuse: $BC = \sqrt{AB^2 + AC^2}$.
   - Leg: $AB = \sqrt{BC^2 - AC^2}$.
3. **Converse & Contrapositive (Right-Angle Test):**
   - If $c^2 = a^2 + b^2$ (c longest side) $\Longrightarrow$ the triangle **is right-angled**.
   - If $c^2 \neq a^2 + b^2 \Longrightarrow$ the triangle **is not right-angled**.
4. **Square Roots & Irrationals:**
   - $\sqrt{x}$ is the unique non-negative number whose square is x.
   - $\sqrt{2}$ cannot be expressed as an integer fraction $\frac{p}{q}$: it is an **irrational number**.

Exercises

Level (Direct Application)

Exercise 19.1 Calculate the hypotenuse length for each right-angled triangle: 1. ABC right-angled at A with $AB = 3$ cm and $AC = 4$ cm. 2. RST right-angled at S with $SR = 5$ cm and $ST = 12$ cm. 3. KLM right-angled at M with $MK = 8$ dm and $ML = 15$ dm.

Exercise 19.2 Find the missing leg length in these right-angled triangles: 1. DEF right-angled at D with hypotenuse $EF = 10$ cm and leg $DE = 6$ cm. Find DF. 2. GHI right-angled at H with hypotenuse $GI = 25$ m and leg $HI = 24$ m. Find GH.

Exercise 19.3 Without a calculator, give the exact value of: $\sqrt{49}$; $\sqrt{81}$; $\sqrt{144}$; $\sqrt{1}$; $\sqrt{0.25}$.

Exercise 19.4 Between which two consecutive integers does each square root lie? a) $\sqrt{10}$; b) $\sqrt{50}$; c) $\sqrt{85}$.

Exercise 19.5 Is a triangle with side lengths 6 cm, 8 cm, and 10 cm right-angled? Justify with arithmetic.

Level (Reasoning and Applied Problems)

Exercise 19.6 Swimming Pool Diagonal. A rectangular community pool measures 25 meters in length and 10 meters in width. A swimmer swims diagonally from one corner to the opposite corner. What distance does she swim? Round to the nearest centimeter.

Exercise 19.7 The Sailboat Mast. A vertical mast stands 12 meters tall on the deck of a sailboat. A steel stay wire connects the masthead to the deck at a distance of 5 meters from the base of the mast. What must be the exact length of the stay wire?

Exercise 19.8 Moving the Wooden Tabletop. A carpenter needs to carry a large circular wooden tabletop of diameter 2.15 meters through a rectangular doorway measuring 2.00 meters high and 0.90 meters wide. Can the circular tabletop fit through the doorway if tilted diagonally?

Exercise 19.9 Carpenter's Orthogonality Check. A master carpenter fits two heavy timber beams that must meet at an exact right angle. He marks point A at 60 cm from the joint along the first beam, and point B at 80 cm along the second beam. He measures direct distance AB and finds 98 cm. Are the beams perpendicular? How must the carpenter adjust them?

Exercise 19.10 Rhombus Diagonals and Perimeter. A rhombus $ABCD$ has diagonals measuring $AC = 16$ cm and $BD = 12$ cm. Recall the perpendicular intersection property of rhombus diagonals, then compute its side length and perimeter.

Exercise 19.11 Altitude of an Equilateral Triangle. Let ABC be an equilateral triangle of side $a = 6$ cm. Draw altitude $[AH]$ from vertex A. 1. What is the length of segment $[BH]$? Justify. 2. Calculate the exact altitude AH in the form $k\sqrt{3}$. 3. Calculate the exact area of triangle ABC.

Level (Challenges and In-Depth Investigations)

Exercise 19.12 The Spiral of Theodorus of Cyrene. 1. Construct an isosceles right triangle with legs of length 1. What is its hypotenuse? 2. On this hypotenuse, construct a second right triangle with its other leg measuring 1. What is the new hypotenuse? 3. Continue this recursive stepping. What are the successive hypotenuse lengths? 4. What is this famous spiral called, which constructs all $\sqrt{n}$ with straightedge and compass?

Exercise 19.13 The Space Diagonal of a Cube. Consider a cube of edge length $c = 4$ cm. 1. Calculate the diagonal of a square face. 2. By applying the Pythagorean Theorem inside a vertical diagonal right triangle, find the long space diagonal connecting opposite corners. 3. Deduce the general formula for the space diagonal of a cube of edge c.

Exercise 19.14 The Spider and the Fly. A cardboard box has dimensions 30 cm $\times$ 12 cm $\times$ 12 cm. A spider rests on one end face, 1 cm from the top and centered horizontally. A fly rests on the opposite end face, 1 cm from the bottom and centered horizontally. What is the shortest walking path for the spider along the exterior faces to reach the fly? *(Hint: unfold the box flat into a net!)*

Exercise 19.15 Proof of the Irrationality of $\sqrt{3}$. Following Hippasus' logic for $\sqrt{2}$, prove that $\sqrt{3}$ cannot be written as an irreducible fraction $\frac{p}{q}$. *(Hint: use the fact that if p^2 is a multiple of 3, then p is a multiple of 3).*

Solutions

Solution 19.1 1. $BC = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$ cm. 2. $RT = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ cm. 3. $KL = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$ dm.

Solution 19.2 1. $DF = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8$ cm. 2. $GH = \sqrt{25^2 - 24^2} = \sqrt{625 - 576} = \sqrt{49} = 7$ m.

Solution 19.3 $\sqrt{49} = 7$; $\sqrt{81} = 9$; $\sqrt{144} = 12$; $\sqrt{1} = 1$; $\sqrt{0.25} = 0.5$ (since $0.5^2 = 0.25$).

Solution 19.4 a) $3^2 = 9 < 10 < 16 = 4^2 \Longrightarrow 3 < \sqrt{10} < 4$. b) $7^2 = 49 < 50 < 64 = 8^2 \Longrightarrow 7 < \sqrt{50} < 8$. c) $9^2 = 81 < 85 < 100 = 10^2 \Longrightarrow 9 < \sqrt{85} < 10$.

Solution 19.5 Longest side: 10 cm. $10^2 = 100$. $6^2 + 8^2 = 36 + 64 = 100$. Since $6^2 + 8^2 = 10^2$, by the Converse of Pythagoras, the triangle is right-angled.

Solution 19.6 $d = \sqrt{25^2 + 10^2} = \sqrt{625 + 100} = \sqrt{725} \approx 26.93$ meters.

Solution 19.7 $L = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$ meters.

Solution 19.8 Doorway diagonal: $D = \sqrt{2^2 + 0.9^2} = \sqrt{4 + 0.81} = \sqrt{4.81} \approx 2.193$ meters. Since $2.193 > 2.15$, the tabletop diameter is smaller than the door diagonal: **yes, it fits** when tilted!

Solution 19.9 Check: $60^2 + 80^2 = 3,600 + 6,400 = 10,000$. For a right angle, $AB = \sqrt{10,000} = 100$ cm. The measured distance is 98 cm ($98^2 = 9,604 \neq 10,000$). The beams are not perpendicular (angle is slightly acute). The carpenter must spread the beams until $AB = 100$ cm.

Solution 19.10 Diagonals of a rhombus bisect each other at right angles. Half-diagonals: $\frac{16}{2} = 8$ cm and $\frac{12}{2} = 6$ cm. Side length: $c = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10$ cm. Perimeter: $4 \times 10 = 40$ cm.

Solution 19.11 1. In an equilateral triangle, the altitude is also the perpendicular bisector: $BH = \frac{6}{2} = 3$ cm. 2. In right triangle ABH: $AH = \sqrt{6^2 - 3^2} = \sqrt{36 - 9} = \sqrt{27} = 3\sqrt{3}$ cm ≈ 5.20 cm. 3. Area $= \frac{6 \times 3\sqrt{3}}{2} = 9\sqrt{3}$ cm^2 ≈ 15.59 cm^2.

Solution 19.12 1. $h_1 = \sqrt{1^2 + 1^2} = \sqrt{2}$. 2. $h_2 = \sqrt{(\sqrt{2})^2 + 1^2} = \sqrt{2 + 1} = \sqrt{3}$. 3. Successive hypotenuses measure $\sqrt{2}, \sqrt{3}, \sqrt{4} = 2, \sqrt{5}, \ldots, \sqrt{n}$. 4. This is the **Spiral of Theodorus of Cyrene** (or Pythagorean Snail).

Solution 19.13 1. Face diagonal: $d_1 = \sqrt{4^2 + 4^2} = \sqrt{32} = 4\sqrt{2}$ cm. 2. Space diagonal: $D = \sqrt{d_1^2 + c^2} = \sqrt{32 + 16} = \sqrt{48} = 4\sqrt{3}$ cm ≈ 6.93 cm. 3. General formula: $D = \sqrt{c^2 + c^2 + c^2} = c\sqrt{3}$.

Solution 19.14 Unfold the box net. The shortest surface path connects the spider and fly along a straight hypotenuse on the unfolded cardboard: Along the length across the top lid: horizontal span $= 30 + 1 + 1 = 32$ cm, vertical span $= 12 + 12 = 24$ cm. $D = \sqrt{32^2 + 24^2} = \sqrt{1,024 + 576} = \sqrt{1,600} = 40$ cm. The shortest path is exactly 40 cm!

Solution 19.15 Assume $\sqrt{3} = \frac{p}{q}$ in lowest terms. Then $3 = \frac{p^2}{q^2} \Longrightarrow p^2 = 3q^2$. So p^2 is a multiple of 3, implying p is a multiple of 3 ($p = 3k$). Then $(3k)^2 = 3q^2 \Longrightarrow 9k^2 = 3q^2 \Longrightarrow q^2 = 3k^2$. So q^2 is a multiple of 3, meaning q is also a multiple of 3. Both p and q share 3 as a common divisor, contradicting that $\frac{p}{q}$ was irreducible. Thus $\sqrt{3}$ is irrational.

The Question That Opens What Follows

The Pythagorean Theorem allows us to calculate the diagonal straight-line distance between two points. But what happens when our points are not drawn upon a free blank sheet of paper, but scattered across a navigator's nautical chart, a pirate's treasure map, or the pixel grid of a computer screen?

How can we pinpoint any location on Earth with absolute accuracy using two signed numbers, positive or negative? And why does "minus times minus equal plus"? That is the secret of **René Descartes and the two trails of Treasure Island**!

CHAPTER 20: Treasure Island and the Two Paths

> "A single number can only measure linear distance along a solitary trail. But to locate a buried chest across an open plain, guide a galleon through thick sea mist, or map every star in the night sky, one must open the universe in two dimensions and marry algebra to geometry."

What You Are About to Discover

- The mystery of the pirate's map and the legendary fly on the ceiling of **René Descartes**.
- The **Cartesian coordinate plane**: the horizontal x**-axis** of abscissas, the vertical y**-axis** of ordinates, and the **four quadrants**.
- How to plot and read the **coordinates** of a point $(x\,;y)$ without ever reversing them.
- The formula for the **midpoint of a segment**: the simple arithmetic average of coordinates.
- Calculating the exact **distance between two points** in a coordinate plane via the Pythagorean Theorem: $AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$.
- **Multiplication and division of signed (relative) numbers**: why "negative times negative equals positive," proved through time-reversed motion and algebraic distributivity!
- 🛠️ The Boundary Case Laboratory: points on the axes, reflections across the origin, and irrational coordinates in Euclidean space.

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1. The Problem: The Treasure Map and the Anxiety of the Circle

Imagine that you have just rowed ashore onto an uninhabited tropical island in the Indian Ocean. Clutched in your trembling hand is the brittle, yellowed parchment of a buccaneer. The ink has faded from sea salt and humidity, but you manage to decipher these few quill-drawn words: *"The brass-bound chest of gold coins lies buried at exactly 50 meters from the solitary black standing stone that towers above the beach."*

Your heart pounds. You dash toward the black standing stone. You plant your heels against the volcanic rock, face the palm grove, pace out 50 meters, plunge your shovel into the sand, and begin digging feverishly. You dig for an hour under a blistering sun. Nothing! Not a single coin, only crushed coral.

You stop, drenched in sweat, as the chilling mathematical truth dawns on you: The pirate wrote: *"at 50 meters from the standing stone."* But in which direction?

- 50 meters to the North?
- 50 meters to the East, into the crashing ocean surf?
- 50 meters to the West, beneath the tangled roots of a giant banyan tree?

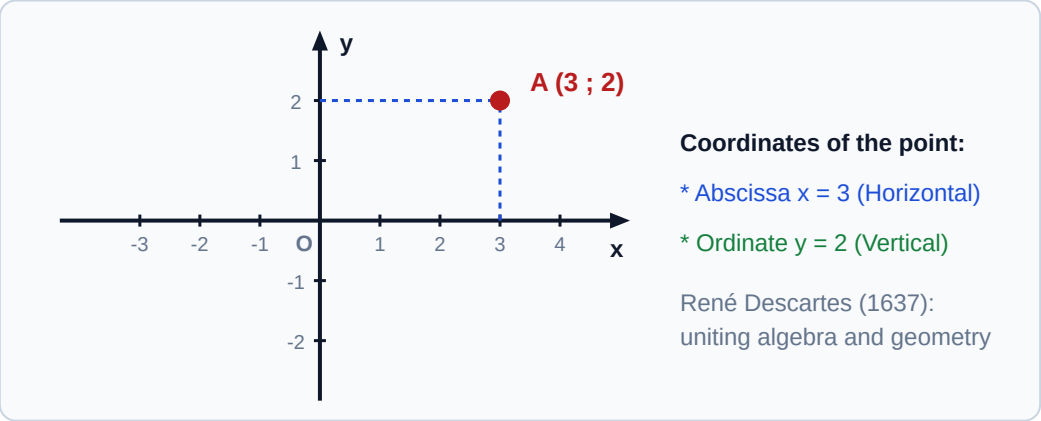

Figure 63: The 2D Cartesian coordinate system: x-axis and y-axis

As we proved in Chapter 16, all the points located at 50 meters from the central stone form an **immense circle surrounding the marker**! If you must excavate every trench along that circular perimeter, you will have to dig a ditch over 314 meters long. By the time you finish, you will be an old hermit with a long white beard!

A single number is insufficient to pinpoint a unique position across a two-dimensional surface. To locate any object without hesitation, one requires **two independent pieces of information**.

---

2. The Fly on the Ceiling of René Descartes

To solve this riddle, let us leave our tropical island and travel through time to the bitter winter of **1637** in France.

Inside a bedroom of a venerable stone house, a forty-year-old gentleman rests beneath heavy feather quilts well past mid-morning. His name is **René Descartes**. Descartes had suffered from frail health since boyhood, and his Jesuit tutors had granted him the lifetime habit of lingering in bed each morning to let his mind meditate in quiet stillness.

As he gazes upward, a small black housefly lands upon the white plaster ceiling. The fly crawls a few inches, pauses, buzzes into the air, and settles on a new spot. Descartes asks himself a deceptively simple question that no human mind had ever formulated with such precision: *"How could I describe the exact, instantaneous position of that fly to a friend across the room without needing to climb a ladder and point my finger at it?"*

Descartes looks at the two perpendicular walls meeting in the corner of his bedroom: 1. He measures the horizontal distance from the fly to the **left wall**: say, **3 meters**. 2. He measures the vertical distance from the fly to the **far wall**: say, **4 meters**.

"It is luminous!" Descartes realizes. The ordered pair of numbers $(3\,;4)$ uniquely fixes the position of the fly upon the flat ceiling without the slightest ambiguity! Two numbers to identify any point in a two-dimensional world: Descartes had just invented the **Cartesian coordinate system**.

---

3. The Coordinate Plane and the Four Quadrants

3.1 Anatomy of the Cartesian System

Upon grid paper, we construct two perpendicular number lines intersecting at their common zero:

📐 Foundational Definitions — Official Terminology

- The intersection of the two perpendicular axes is the **origin**,

denoted O. Its coordinates are $(0\,;0)$.

- The horizontal axis is the $\boldsymbol{x}$**-axis** (axis of abscissas). The x-coordinate measures displacement **left / right**.
- The vertical axis is the $\boldsymbol{y}$**-axis** (axis of ordinates). The y-coordinate measures displacement **down / up**.
- A coordinate system is **orthogonal** if its axes are strictly perpendicular. It is **orthonormal** (or Cartesian) if both axes share the identical unit scale (e.g., $1 \ \text{grid unit} = 1 \ \text{cm}$ on both axes).

3.2 Coordinate Notation

For any point M, we write its ordered pair:

$$M(x_M\,;y_M)$$

- x_M is the **abscissa** (x-coordinate).
- y_M is the **ordinate** (y-coordinate).

📐 Mnemonic Rules — Never Confuse x and y

1.

Alphabetical Order: In the alphabet, **A**bscissa precedes **O**rdinate, just as $\boldsymbol{x}$ precedes $\boldsymbol{y}$. The x-coordinate is ALWAYS written first! 2. **The Apartment Elevator:** To visit a friend in an apartment building, you must first walk down the street to the entrance of their building hallway (**horizontal** $\boldsymbol{x}$), and only then take the elevator up to their floor (**vertical** $\boldsymbol{y}$). You cannot ride the elevator before you enter the lobby!

3.3 The Four Quadrants

The intersecting axes partition the infinite plane into four quadrants, numbered counterclockwise starting from the top right:

- **Quadrant I (top-right):** $x > 0$ and $y > 0$ $(+,+)$.
- **Quadrant II (top-left):** $x < 0$ and $y > 0$ $(-,+)$.
- **Quadrant III (bottom-left):** $x < 0$ and $y < 0$ $(-,-)$.
- **Quadrant IV (bottom-right):** $x > 0$ and $y < 0$ $(+,-)$.

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4. Two Magic Formulas of the Coordinate Plane

Once geometric points are encoded as numerical pairs, we no longer need to draw physical figures to solve geometric problems: algebraic calculation solves everything!

4.1 The Midpoint of a Segment

Given two points $A(x_A\,;y_A)$ and $B(x_B\,;y_B)$: Where is the exact midpoint I of segment $[AB]$? It lies exactly halfway horizontally, and halfway vertically. It is simply the **arithmetic mean (average)** of their coordinates!

📐 Property — Midpoint Coordinates

The coordinates of midpoint I of segment $[AB]$ are:

$$x_I = \frac{x_A + x_B}{2} \quad \text{and} \quad y_I = \frac{y_A + y_B}{2}$$

4.2 Distance Between Two Points: Pythagoras on the Grid

How can we calculate the exact straight-line distance AB between any two points in the plane? Observe the diagram:

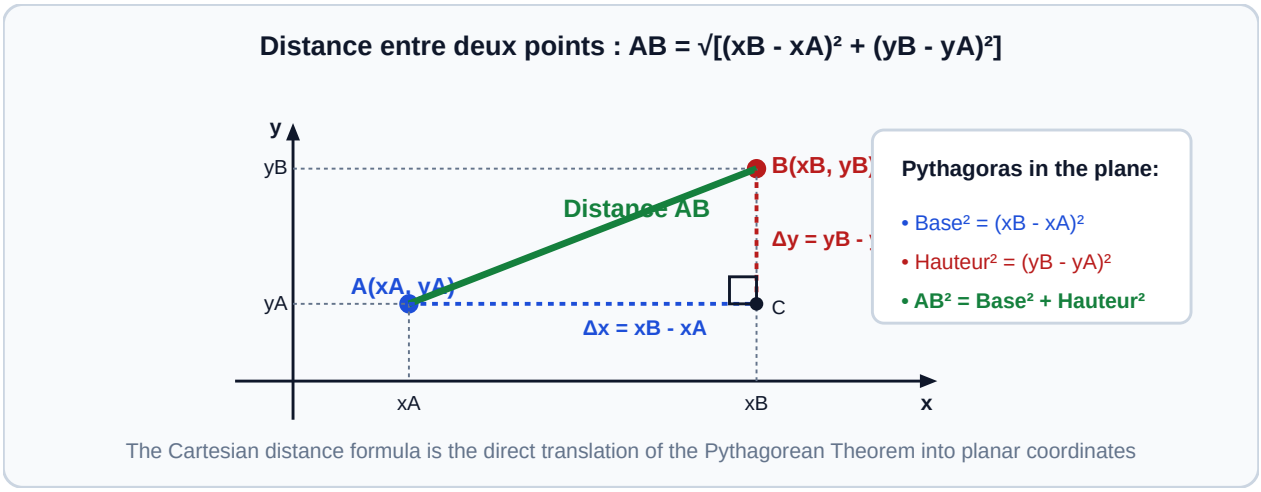

Figure 64: Euclidean distance in the Cartesian coordinate system

Draw a horizontal line through A, and a vertical line through B. They intersect at point $C(x_B\,;y_A)$, forming a **right triangle ABC**!

- The horizontal base length is: $AC = |x_B - x_A|$.
- The vertical altitude length is: $BC = |y_B - y_A|$.
- The hypotenuse is the segment $[AB]$!

Applying the Pythagorean Theorem in right triangle ABC:

$$AB^2 = AC^2 + BC^2 = (x_B - x_A)^2 + (y_B - y_A)^2$$

Taking the square root delivers the universal Cartesian distance formula:

Theorem — Cartesian Distance Formula

In an orthonormal coordinate plane, the distance between points $A(x_A\,;y_A)$ and $B(x_B\,;y_B)$ is:

$$AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$$

---

5. Multiplying Signed Numbers: Why Minus Times Minus Equals Plus

Navigating all four quadrants with positive and negative coordinates brings us face-to-face with a fundamental arithmetic challenge: how do we **multiply** signed numbers?

While addition and subtraction of negative numbers are intuitive (winning or losing money, climbing or descending into a basement), the multiplication of two negative numbers seems at first glance like an impenetrable paradox. Why should $(-3) \times (-4) = +12$?

Here are two irrefutable demonstrations: one cinematic and visual, the other purely algebraic.

5.1 The Movie Reel Proof: The Walking Actor and Rewind

Imagine an actor walking along a straight road marked with meter posts, recorded on video:

- His **velocity**:
  - Walking forward to the right (positive direction): velocity $+2\ \ \mathrm{m/s}$.
  - Walking backward to the left: velocity $-2\ \ \mathrm{m/s}$.
- The **time** on the video player:
  - Playing the film forward into the future: time $+3\ \ \text{seconds}$.
  - Rewinding the film into the past: time $-3\ \ \text{seconds}$.

Where does the actor appear on the screen relative to the starting line?

1. $\mathbf{(+2) \times (+3) = +6}$**:** He walks forward at $2\ \ \mathrm{m/s}$, watched 3 seconds into the future $\Longrightarrow$ he is at $\mathbf{+6\ \ meters}$ to the right.

2. $\mathbf{(-2) \times (+3) = -6}$**:** He walks backward at -2 $\mathrm{m/s}$, watched 3 seconds into the future $\Longrightarrow$ he is at $\mathbf{-6}$ $\mathbf{meters}$ to the left.
3. $\mathbf{(+2) \times (-3) = -6}$**:** He walks forward to the right, but we rewind the video 3 seconds into the past $\Longrightarrow$ he was at $\mathbf{-6}$ $\mathbf{meters}$ to the left!
4. $\mathbf{(-2) \times (-3) = +6}$**:** He was walking backward to the left, and we rewind the video into the past! What do your eyes see on the screen? **An actor advancing forward to the right!** He was at $\mathbf{+6}$ $\mathbf{meters}$! *Rewinding a film of someone walking backward shows them moving forward!*

5.2 The Algebraic Proof via Distributivity

Mathematicians require a proof independent of physical metaphors. Here it is: we demand that the foundational law of distributivity, $k \times (a + b) = k \times a + k \times b$, remains universally true for all real numbers.

Let us evaluate the following expression in two different ways:

$$E = (-3) \times [4 + (-4)]$$

- **Method 1 (inside brackets first):** $4 + (-4) = 0$. Therefore:

$$E = (-3) \times 0 = \mathbf{0}$$

- **Method 2 (expanding via distributivity):**

$$E = (-3) \times 4 + (-3) \times (-4)$$

We already know that $(-3) \times 4 = -12$ (since 4 debts of \$3 equal a debt of \$12). Our equation becomes:

$$0 = -12 + [(-3) \times (-4)]$$

Ask yourself: what unique number must be added to -12 to yield zero? The one and only number in the mathematical universe is $\mathbf{+12}$! Therefore, with absolute logical necessity:

$$(-3) \times (-4) = \mathbf{+12}$$

If negative times negative did not equal positive, the entire framework of algebra would collapse into contradiction!

The Rule of Signs — Multiplication and Division

- The product (or quotient) of two numbers with the **same sign**

is **POSITIVE**:

$$(+) \times (+) = (+) \quad \text{and} \quad (-) \times (-) = (+)$$

- The product (or quotient) of two numbers with **different signs** is **NEGATIVE**:

$$(+) \times (-) = (-) \quad \text{and} \quad (-) \times (+) = (-)$$

For a product of multiple non-zero factors:

- An **even number** of negative factors yields a **positive** product.
- An **odd number** of negative factors yields a **negative** product.

---

6. Beware of Pitfalls!

> ⚠️ **Warning: Never Confuse Addition and Multiplication!**
>
> 1.
>
> **The Deadly Sign Confusion:**
>
> $$-3-5=\mathbf{-8} \quad \text{(losing 3, then losing 5: this is addition/subtraction!)}$$
>
> $$(-3)\times(-5)=\mathbf{+15} \quad \text{(this is multiplication!)}$$
>
> Never chant “two minuses make a plus” without emphasizing that this applies ONLY to multiplication and division! 2. **Reversing Abscissa and Ordinate:** Point $A(2\,;5)$ is located 2 units right, 5 units up. Point $B(5\,;2)$ is located 5 units right, 2 units up. These are two completely different locations! 3. **The Minus Sign Trap in Distance Computations:** If $x_A=-3$ and $x_B=4$, the horizontal leg length is:
>
> $$x_B-x_A=4-(-3)=4+3=\mathbf{7}$$
>
> Watch out for the double negative turning into addition!

---

7. Solved Examples

Solved Example 1 — Plotting and Identifying Coordinates

In an orthonormal grid where 1 grid square = 1 unit: 1. Plot the points: $A(3\,;2)$, $B(-4\,;3)$, $C(-2\,;-3)$, $D(5\,;-1)$, $E(0\,;4)$, $F(-3\,;0)$. 2. State the quadrant or axis for each point.

Solution: - $A(3\,;2)$: 3 units right, 2 units up $\rightarrow$ **Quadrant I**. - $B(-4\,;3)$: 4 units left, 3 units up $\rightarrow$ **Quadrant II**. - $C(-2\,;-3)$: 2 units left, 3 units down $\rightarrow$ **Quadrant III**. - $D(5\,;-1)$: 5 units right, 1 unit down $\rightarrow$ **Quadrant IV**. - $E(0\,;4)$: zero abscissa $\rightarrow$ lies directly on the $\boldsymbol{y}$**-axis**. - $F(-3\,;0)$: zero ordinate $\rightarrow$ lies directly on the $\boldsymbol{x}$**-axis**. ✓

Solved Example 2 — Calculating Segment Midpoint

Given points $M(-5\,;7)$ and $N(3\,;-1)$. Calculate the coordinates of midpoint K of segment $[MN]$.

Solution: Apply the midpoint formulas:

$$x_K=\frac{x_M+x_N}{2}=\frac{-5+3}{2}=\frac{-2}{2}=-1$$

$$y_K=\frac{y_M+y_N}{2}=\frac{7+(-1)}{2}=\frac{6}{2}=3$$

The midpoint is $K(-1\,;3)$. ✓

Solved Example 3 — Calculating Distance Between Two Points

Given $A(-2\,;1)$ and $B(4\,;9)$ in an orthonormal centimeter grid. Calculate the exact length of segment $[AB]$.

Solution: Apply the distance formula:

$$AB=\sqrt{(x_B-x_A)^2+(y_B-y_A)^2}$$

Compute the directional spans:

$$x_B-x_A=4-(-2)=4+2=6$$

$$y_B-y_A=9-1=8$$

Square and sum:

$$AB^2=6^2+8^2=36+64=100$$

Take the square root:

$$AB=\sqrt{100}=10 \ \text{ cm}$$

Segment $[AB]$ measures exactly 10 cm. ✓

Solved Example 4 — Signed Number Arithmetic

Evaluate: 1. $P_1 = (-4) \times (+7)$ 2. $P_2 = (-6) \times (-8)$ 3. $Q_1 = \frac{-45}{9}$ 4. $Q_2 = \frac{-56}{-7}$ 5. $P_3 = (-2) \times (-3) \times (-1) \times (+4) \times (-5)$

Solution: 1. Opposite signs $\Longrightarrow$ negative: $P_1 = -28$. 2. Same sign (both negative) $\Longrightarrow$ positive: $P_2 = +48$. 3. Opposite signs $\Longrightarrow$ negative: $Q_1 = -5$. 4. Same sign $\Longrightarrow$ positive: $Q_2 = +8$. 5. Count negative factors in P_3: there are 4 negative factors ($(-2), (-3), (-1), (-5)$). 4 is even $\Longrightarrow$ product is **positive**. Magnitude: $2 \times 3 \times 1 \times 4 \times 5 = 120$. Therefore, $P_3 = +120$. ✓

Solved Example 5 — Determining Triangle Classification via Coordinates

Given $A(1\,;2)$, $B(4\,;6)$, and $C(-3\,;5)$ in an orthonormal plane. Prove that triangle ABC is an isosceles right-angled triangle.

Solution: Calculate the squared side lengths:

- For AB:

$$AB^2 = (4-1)^2 + (6-2)^2 = 3^2 + 4^2 = 9 + 16 = 25 \Longrightarrow AB = 5$$

- For AC:

$$AC^2 = (-3-1)^2 + (5-2)^2 = (-4)^2 + 3^2 = 16 + 9 = 25 \Longrightarrow AC = 5$$

- For BC:

$$BC^2 = (-3-4)^2 + (5-6)^2 = (-7)^2 + (-1)^2 = 49 + 1 = 50$$

Geometric analysis: 1. $AB = AC = 5$: triangle ABC has two sides of equal length, so it is **isosceles at vertex A**. 2. Compare squares:

$$AB^2 + AC^2 = 25 + 25 = 50 = BC^2$$

By the **Converse of the Pythagorean Theorem**, triangle ABC is **right-angled at vertex A**. Conclusion: ABC is an **isosceles right-angled triangle at A**. ✓

What You Must Remember

1. **Cartesian Coordinate Plane:**
   - Horizontal (Ox): **abscissas** (x).
   - Vertical (Oy): **ordinates** (y).
   - Coordinates: $M(x\,;y)$ (x first, y second).
2. **Segment Midpoint $[AB]$:**

$$x_I = \frac{x_A + x_B}{2} \quad \text{and} \quad y_I = \frac{y_A + y_B}{2}$$

3. **Cartesian Distance Formula:**

$$AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$$

4. **Rule of Signs:**
   - Same signs $\Longrightarrow$ **positive** ($+ \times + = +$ and $- \times - = +$).
   - Opposite signs $\Longrightarrow$ **negative** ($+ \times - = -$ and $- \times + = -$).
   - Products of multiple factors: count the minus signs (even $\Longrightarrow +$; odd $\Longrightarrow -$).

Exercises

Level (Direct Application)

Exercise 20.1 State the coordinates of the vertices of rectangle $ABCD$: A is 3 units left and 2 up; B is 4 units right and 2 up; C is 4 units right and 1 down; D is 3 units left and 1 down.

Exercise 20.2 Compute without a calculator: 1. $(-7) \times (-6)$ 2. $(-8) \times (+9)$ 3. $(+12) \times (-5)$ 4. $\frac{-72}{-8}$ 5. $\frac{-42}{+7}$

Exercise 20.3 State the sign of each product without evaluating its numerical value: 1. $(-3) \times (-5) \times (+2) \times (-7)$ 2. $(-1) \times (-2) \times (-3) \times (-4) \times (-5) \times (-6)$ 3. The product of 15 negative numbers and 28 positive numbers.

Exercise 20.4 Given $A(2\,;4)$ and $B(8\,;10)$. Calculate the coordinates of midpoint I of segment $[AB]$.

Exercise 20.5 Calculate the exact distance AB between $A(1\,;2)$ and $B(4\,;6)$.

Level (Reasoning and Geometric Calculations)

Exercise 20.6 Combined Operations with Signed Numbers. Evaluate respecting operational precedence: 1. $E = (-3) \times (-4) + (-5) \times (+2)$ 2. $F = 15 - (-2) \times (-6)$ 3. $G = \frac{(-4)\times(-9)}{-3-3}$ 4. $H = (-2)^4 - (-3)^3$

Exercise 20.7 Finding an Unknown Segment Endpoint. Given $A(-3\,;5)$ and midpoint $I(1\,;2)$ of segment $[AB]$. Find the coordinates of endpoint B.

Exercise 20.8 Parallelogram Verification via Diagonals. Given four points: $A(-2\,;1)$, $B(3\,;4)$, $C(5\,;1)$, and $D(0\,;-2)$. 1. Compute the midpoint of diagonal $[AC]$. 2. Compute the midpoint of diagonal $[BD]$. 3. What can you deduce about the geometric nature of quadrilateral $ABCD$?

Exercise 20.9 The Circumscribed Circle. In an orthonormal plane, consider $E(-4\,;0)$ and $F(2\,;8)$. 1. Calculate the coordinates of center O' of the circle with diameter $[EF]$. 2. Calculate the length of diameter EF, then radius R. 3. Does point $G(2\,;0)$ lie on this circle?

Exercise 20.10 Triangle Classification. Given $K(-1\,;2)$, $L(3\,;5)$, and $M(3\,;-1)$. 1. Calculate the exact side lengths KL, LM, and KM. 2. Is triangle KLM isosceles? Is it right-angled?

Exercise 20.11 Squares and Powers of Negative Numbers. Insert $<$ or $>$: 1. $(-5)^2 \ldots 0$ 2. $-5^2 \ldots 0$ 3. $(-2)^3 \ldots (-2)^2$ 4. $(-1)^{100} \ldots (-1)^{101}$

Level (Challenges and Analytic Geometry)

Exercise 20.12 Pirate Treasure Hunt. On a map where origin $O(0\,;0)$ is the old island lighthouse, a treasure chest is buried at point T. The pirate log reads:

- Start from cove $A(-6\,;-4)$.
- The treasure T lies along the straight line toward cave $B(6\,;4)$, three-quarters of the way from A to B.

1. What are the coordinates of treasure T?
2. What distance in meters separates the lighthouse from treasure T ($1 \text{ grid unit} = 10 \text{ meters}$)?

Exercise 20.13 Point Reflection Across a Center. Let $A(2\,;-3)$ and center of symmetry $S(-1\,;1)$. Point A' is the reflection of A across S. 1. What does point S represent for segment $[AA']$? 2. Determine algebraically the coordinates of point A'.

Exercise 20.14 Sign Rule via Arithmetic Sequences. Construct a multiplication table with $-3, -2, -1, 0, 1, 2, 3$ along both row and column headers. Fill the positive products first, then extend each row and column using constant step intervals. Explain why cell $(-3) \times (-3)$ must equal $+9$.

Exercise 20.15 Equilateral Triangle in a Coordinate Grid. Place $A(0\,;0)$ and $B(4\,;0)$ on the x-axis. 1. What must be the coordinates of point C (with positive y-coordinate) such that triangle ABC is equilateral? 2. Are the coordinates of C rational or irrational? Justify.

Solutions

Solution 20.1 $A(-3\,;2)$; $B(4\,;2)$; $C(4\,;-1)$; $D(-3\,;-1)$.

Solution 20.2 1. $(-7) \times (-6) = +42$. 2. $(-8) \times (+9) = -72$. 3. $(+12) \times (-5) = -60$. 4. $\frac{-72}{-8} = +9$. 5. $\frac{-42}{+7} = -6$.

Solution 20.3 1. 3 negative factors (odd) $\Longrightarrow$ **negative**. 2. 6 negative factors (even) $\Longrightarrow$ **positive**. 3. 15 negative factors (odd) $\Longrightarrow$ **negative** (positive factors do not alter the sign).

Solution 20.4 $x_I = \frac{2+8}{2} = 5$; $y_I = \frac{4+10}{2} = 7 \Longrightarrow I(5\,;7)$.

Solution 20.5 $AB = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$.

Solution 20.6 1. $E = (+12) + (-10) = 2$. 2. $F = 15 - (+12) = 3$. 3. $G = \frac{+36}{-6} = -6$. 4. $H = (+16) - (-27) = 16 + 27 = 43$.

Solution 20.7 $1 = \frac{-3+x_B}{2} \Longrightarrow 2 = -3 + x_B \Longrightarrow x_B = 5$. $2 = \frac{5+y_B}{2} \Longrightarrow 4 = 5 + y_B \Longrightarrow y_B = -1$. Endpoint is $B(5\,;-1)$.

Solution 20.8 1. Midpoint of $[AC]$: $(\frac{-2+5}{2}\,;\frac{1+1}{2}) = (1.5\,;1)$. 2. Midpoint of $[BD]$: $(\frac{3+0}{2}\,;\frac{4+(-2)}{2}) = (1.5\,;1)$. 3. Both diagonals share the identical midpoint. A quadrilateral whose diagonals bisect each other is a **parallelogram**. Thus $ABCD$ is a parallelogram.

Solution 20.9 1. Center O': $(\frac{-4+2}{2}\,;\frac{0+8}{2}) = (-1\,;4)$. 2. Diameter: $EF = \sqrt{(2-(-4))^2 + (8-0)^2} = \sqrt{6^2+8^2} = 10 \Longrightarrow R = 5$. 3. Distance $O'G = \sqrt{(2-(-1))^2 + (0-4)^2} = \sqrt{3^2+(-4)^2} = 5$. Since $O'G = R = 5$, point G lies on the circle.

Solution 20.10 1. $KL = \sqrt{4^2+3^2} = 5$; $LM = \sqrt{0^2+(-6)^2} = 6$; $KM = \sqrt{4^2+(-3)^2} = 5$. 2. $KL = KM = 5 \Longrightarrow$ triangle KLM is **isosceles at vertex K**. Pythagoras test: $LM^2 = 36 \neq 5^2+5^2 = 50$, so it is not right-angled.

Solution 20.11 1. $(-5)^2 = 25 > 0$. 2. $-5^2 = -25 < 0$. 3. $(-2)^3 = -8 < 4 = (-2)^2$. 4. $(-1)^{100} = +1 > -1 = (-1)^{101}$.

Solution 20.12 1. Vector displacement from A to B: horizontal $\Delta x = 12$, vertical $\Delta y = 8$. Three-quarters displacement: $\Delta x' = 9$, $\Delta y' = 6$. $x_T = -6+9 = 3$ and $y_T = -4+6 = 2 \Longrightarrow T(3\,;2)$. 2. Distance $OT = \sqrt{3^2+2^2} = \sqrt{13} \approx 3.6055$ units. In meters: 3.6055×10 m ≈ 36.1 meters.

Solution 20.13 1. Point S is the midpoint of segment $[AA']$. 2. $-1 = \frac{2+x_{A'}}{2} \Longrightarrow x_{A'} = -4$; $1 = \frac{-3+y_{A'}}{2} \Longrightarrow y_{A'} = 5$. $A'(-4\,;5)$.

Solution 20.14 Along row -3: $(-3)\times 3 = -9$, $(-3)\times 2 = -6$, $(-3)\times 1 = -3$, $(-3)\times 0 = 0$. Moving left steps up by $+3$. Continuing the sequence: $(-3)\times(-1) = 3$, $(-3)\times(-2) = 6$, $(-3)\times(-3) = 9$. The cell must equal $+9$.

Solution 20.15 1. Base $AB = 4$, midpoint at $x = 2$. Apex C has $x_C = 2$. Altitude $h = \sqrt{4^2-2^2} = \sqrt{12} = 2\sqrt{3}$. Thus $C(2\,;2\sqrt{3})$. 2. Abscissa 2 is rational ($\mathbb{Q}$), but ordinate $2\sqrt{3}$ is an **irrational number** because $\sqrt{3}$ is irrational.

The Question That Opens What Follows

With Cartesian coordinates, we can locate any individual point or buried chest across space. Yet what happens when a tax collector must levy fair contributions from hundreds of merchants of wildly differing wealth, or when a school must summarize the test scores of an entire class in a single representative figure?

How do we compare portions fairly when group totals are never the same? That is the power of **percentages**, and the birth of **statistics**—with means, medians, and frequency diagrams—in our final chapter!

CHAPTER 21: The Tax Collector and the Hundred Pieces

> "How can one compare the contribution of a poor woodcutter who surrenders three silver coins out of ten to that of an opulent merchant who yields thirty out of one hundred and fifty? To render justice blind and compare the incomparable, the human mind forged a magic mirror where every crowd is reduced to an ideal assembly of one hundred citizens."

What You Are About to Discover

- Why a **percentage** is simply a fraction in disguise with a denominator of one hundred ($\% = \frac{\dots}{100}$).
- The two essential gestures: **applying** a percentage (taking $p\%$ of a quantity) and **calculating** a percentage rate ($\frac{\text{part}}{\text{total}} \times 100$).
- The master secret of **multiplier coefficients** for percentage increases ($1 + \frac{t}{100}$) and decreases ($1 - \frac{t}{100}$).
- The foundational vocabulary of statistics: **population**, **individual**, **variable**, **frequencies**, and **counts**.
- Standard graphic representations: **bar charts**, **histograms**, and **pie charts**.
- The three key summary indicators of statistical data: the **mean**, the **median**, and the **range**.
- 🛠️ The Boundary Case Laboratory: successive percentage changes, Simpson's Paradox, and why the mean can disguise reality.

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1. The Problem: The Medieval Trade Fair and the Quarrel of Merchants

We are in the 13th century, in the heart of Champagne, during the great Trade Fair of Troyes. Merchants from Venice, Flanders, and Castile have raised hundreds of striped canvas stalls overflowing with fragrant spices, silk bolts, fine wools, and embossed leathers.

At the end of the week, the **Count's chief tax collector** sits behind a heavy oak table with his iron-bound chest and leather ledger. Two fabric merchants present themselves before his quill to settle the tax on their earnings.

- Merchant **Geoffroy** achieved a modest profit of **150 silver ecus**. The collector levies a tax of **30 ecus**.
- Merchant **Barthélemy**, a wealthy Flemish draper, amassed an opulent profit of **400 silver ecus**. The collector levies a tax of **72 ecus**.

Barthélemy slams his fist onto the oak table in indignation: *"This is an outrageous injustice! You take 72 ecus from me while demanding a mere 30 from Geoffroy! I am paying more than double what he pays!"*

Geoffroy shoots back indignantly: *"Come now, Barthélemy! You earned an enormous fortune compared to my humble shop! My 30 ecus represent a painful butchery of my meager earnings, while your 72 ecus are a mere drop in your golden vaults!"*

> 💡 **Discovery & History — The Dispute of Medieval Merchants**
>
> - Geoffroy (Small merchant) Barthélemy (Wealthy merchant)
> - 150 ecus) (400 ecus)
> - Tax: 30 ecus Tax: 72 ecus
> - Part = 30 / 150 Part = 72 / 400

The provost of the fair is summoned to judge the dispute. Who is truly taxed more heavily?

To answer with true mathematical justice, one cannot merely compare the raw number of coins surrendered (72 versus 30). One must compare the **proportional share** surrendered relative to total profits earned:

- For Geoffroy: $\frac{30}{150} = \frac{3}{15} = \frac{1}{5} = 0.20$.

- For Barthélemy: $\frac{72}{400} = \frac{18}{100} = 0.18$.

Imagine that each merchant had earned a standardized benchmark purse of **100 ecus**:

- Geoffroy would have surrendered **20 ecus per hundred ecus earned**.
- Barthélemy would have surrendered only **18 ecus per hundred ecus earned**!

It is Geoffroy who bears the heavier burden! To compare parts originating from groups of completely different sizes, humanity invented a universal standard benchmark denominator: **the number one hundred**. This is the birth of the **percentage**.

---

2. The Percentage: A Fraction over One Hundred

2.1 Notation and Meaning

> **📐 Definition — Percentage**
>
> A **percentage** is a standardized way of writing a fraction whose **denominator is equal to 100**:
>
> $$t\% = \frac{t}{100}$$
>
> The symbol $\%$ reads aloud as **"percent"** (literally *"per hundred"*).

- $50\% = \frac{50}{100} = \frac{1}{2} = 0.5$ (one half).
- $25\% = \frac{25}{100} = \frac{1}{4} = 0.25$ (one quarter).
- $75\% = \frac{75}{100} = \frac{3}{4} = 0.75$ (three quarters).
- $10\% = \frac{10}{100} = \frac{1}{10} = 0.1$ (one tenth).
- $100\% = \frac{100}{100} = 1$ (the entirety, the whole).

> 💡 ***Where Did the Symbol $\%$ Come From?*** *In 15th-century Italy, merchant scribes abbreviated the phrase* per cento *as* p. 100 *or* p. c.. *Over centuries of rapid quill-work, the letter* p *vanished, the horizontal fraction stroke tilted into a forward slash, and the two zeros of 100 perched above and below the bar, giving birth to our modern $\%$ sign.*

2.2 The Two Foundational Computations

Question Asked	What We Seek	Standard Formula
Applying a percentage: "What is $p\%$ of quantity Q?"	A **concrete amount** (in \$, in kg, in students…)	$\text{Result} = Q \times \frac{p}{100}$
Finding a percentage rate: "What proportion does part A represent within total T?"	A **rate in $\%$**	$\text{Rate } t\% = \frac{A}{T} \times 100$

---

3. Percentage Changes and Multiplier Coefficients

In everyday commerce (store sales, price inflation, tax levies), quantities do not remain static: they **increase** or **decrease** by a certain percentage.

3.1 The Traditional Two-Step Method

A winter jacket costs \$80. The store advertises a 30% discount. 1. Calculate the dollar discount:

$$\text{Discount} = 80 \times \frac{30}{100} = \$24$$

2. Subtract discount from the original price:

$$\text{New Price} = 80 - 24 = \$56$$

3.2 The Master Tool: The Multiplier Coefficient

Notice that we can factor this calculation into a single, lightning-fast multiplication:

$$\text{New Price} = 80 - 80 \times \frac{30}{100} = 80 \times \left(1 - \frac{30}{100}\right) = 80 \times 0.70 = \$56$$

Paying with a 30% discount is mathematically identical to paying **70% of the original price**!

Theorem — Multiplier Coefficients

- To **increase** a quantity by $t\%$, multiply it by:

$$C = 1 + \frac{t}{100}$$

(Example: a 20% increase $\Longrightarrow \times 1.20$; a 5% increase $\Longrightarrow \times 1.05$)

- To **decrease** a quantity by $t\%$, multiply it by:

$$C = 1 - \frac{t}{100}$$

(Example: a 15% discount $\Longrightarrow \times 0.85$; a 40% discount $\Longrightarrow \times 0.60$)

---

4. Statistics: Making Numbers Speak

When a city, a medical center, or a school gathers thousands of individual data points (student heights, employee salaries, exam test scores), the human brain cannot extract meaningful wisdom from an unorganized list of raw numbers. **Statistics** is the mathematical discipline that organizes, summarizes, and illuminates large bodies of information.

4.1 Statistical Vocabulary

- The **population**: the entire collection of individuals or objects under study (e.g., *all students in 7th Grade*).
- An **individual**: a single member of that population (e.g., *student Thomas*).
- The **variable** (or characteristic): the specific property observed for each individual (e.g., *score on the math exam out of 20*).
- The **frequency count** (or absolute frequency) of a value: the number of times that specific value occurs.
- The **total sample size** (N): the total number of individuals in the population.
- The **relative frequency** of a value: the proportion of individuals possessing that value:

$$f = \frac{\text{count}}{\text{total sample size } \ N}$$

  Relative frequency is expressed as a decimal between 0 and 1, or as a percentage ($f \times 100$). **The sum of all relative frequencies across a population is always strictly equal to 1 (100%).**

4.2 Three Standard Graphic Displays

1. Bar Chart	2. Histogram (bins)	3. Pie Chart
Isolated discrete values	Adjacent continuous intervals	Slices proportional to 360°
Height proportional to count	Area proportional to count	Angle proportional to frequency (%)

In a **pie chart**, the surface area and central angle of each slice are strictly proportional to its frequency:

$$\text{Central Angle in Degrees} = \text{Relative Frequency } \ f \times 360^\circ$$

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5. The Three Guardians of Data: Mean, Median, and Range

To summarize an entire statistical distribution in a few representative numbers, statisticians rely on three major indicators:

5.1 The Arithmetic Mean ($\bar{x}$)

The **mean** (commonly called the average) is the fair-share value each individual would receive if all values were pooled together and redistributed equally.

- **Simple Mean:**

$$\bar{x} = \frac{\text{sum of all data values}}{\text{total number of values } \ N}$$

- **Weighted Mean (with frequencies or course credits):**

$$\bar{x} = \frac{(n_1 \times x_1) + (n_2 \times x_2) + ... + (n_k \times x_k)}{N}$$

5.2 The Median (Me): The Justice of the Midpoint

The **median** is the exact central value that partitions an ordered data set into two groups of identical size:

- At least 50% of individuals have a value less than or equal to the median.
- At least 50% of individuals have a value greater than or equal to the median.

> **Method — Finding the Median of a Data Set**
>
> 1. You
>
> **must first sort** all data values in ascending order (from smallest to largest). 2. Determine half the sample size: $\frac{N}{2}$.
>
> - **Case 1: N is odd ($N = 2k + 1$):** The median is the unique value located at rank $k + 1$. *(Example: $N = 7 \Longrightarrow \frac{7}{2} = 3.5$. The median is the 4th value).*
> - **Case 2: N is even ($N = 2k$):** The median is the average of the k-th and $(k + 1)$-th values. *(Example: $N = 8 \Longrightarrow \frac{8}{2} = 4$. Median $= \frac{\text{4th value} + \text{5th value}}{2}$).*

5.3 The Range (R): Measuring Dispersion

The **range** measures the overall spread or dispersion of the data set:

$$R = \text{Maximum Value} - \text{Minimum Value}$$

A small range indicates that values are clustered tightly together (homogeneous group). A large range indicates substantial disparity.

---

6. Beware of Pitfalls!

> **Warning: Statistical Illusions!**
>
> 1. **The Consecutive Increase and Decrease Trap:** *"If an item rises in price by 20% and subsequently drops by 20%, does it return to its original price?"* **ABSOLUTELY NOT, NEVER!** Test with an initial price of $\$100$:
>
> - 20% increase $\Longrightarrow 100 \times 1.20 = \$120$.
> - 20% decrease on $\$120 \Longrightarrow 120 \times (1 - 0.20) = 120 \times 0.80 = \mathbf{\$96}$! You lost $\$4$, because the 20% drop was calculated on a larger base amount!
>
> 2. **Adding Percentages from Different Groups:** If Class A has 20 students with 50% girls, and Class B has 30 students with 40% girls, the combined proportion of girls is NOT $50\% + 40\% = 90\%$, nor is it $\frac{50+40}{2} = 45\%$! You must count actual human beings: $10 + 12 = 22$ girls out of 50 students total, which is $\frac{22}{50} = 44\%$.
> 3. **Confusing Mean and Median:** In a small workshop of 5 people, 4 workers earn $\$1,500$ per month, while the owner earns $\$21,000$.
>
> - Mean salary: $\frac{4 \times 1,500 + 21,000}{5} = \frac{27,000}{5} = \mathbf{\$5,400}$!
> - Median salary: the 3rd person earns $\mathbf{\$1,500}$. The mean gives the deceptive impression that everyone is well-off. The median reflects the true economic reality of the majority!

7. Solved Examples

Solved Example 1 — Applying a Percentage in a Bakery Recipe

To make $1.2\ \text{kg}$ of brioche dough, a baker uses 55% flour, 20% butter, 15% eggs, and 10% sugar and yeast. What mass of butter (in grams) must the baker weigh?

Solution: - Convert total mass to grams: $1.2\ \text{kg} = 1,200\ \text{g}$. - Butter share: 20%. - Calculation:

$$\text{Mass of butter} = 1,200 \times \frac{20}{100} = 1,200 \times 0.20 = 240\ \text{grams}$$

The baker must weigh $240\ \text{g}$ of butter. ✓

Solved Example 2 — Calculating an Exam Pass Rate

In a middle school, 138 eighth-grade students sat for the national diploma exam. 124 of them passed successfully. What was the pass rate for this school? Round to the nearest tenth of a percent.

Solution: - Successful candidates (part): $A = 124$. - Total cohort (whole): $T = 138$. - Rate:

$$t = \frac{A}{T} \times 100 = \frac{124}{138} \times 100 \approx 0.89855 \times 100 \approx 89.9\%$$

The diploma pass rate was 89.9%. ✓

Solved Example 3 — Winter Sales and Multiplier Coefficients

A sporting goods store applies a 40% discount to running shoes originally priced at $\$115$. What is the discounted price?

Solution: - Multiplier coefficient for a 40% decrease:

$$C = 1 - \frac{40}{100} = 1 - 0.40 = 0.60$$

- Discounted price:

$$\text{Sale Price} = 115 \times 0.60 = \$69$$

The shoes now cost $\$69$. ✓

Solved Example 4 — Full Statistical Analysis of an Exam

Here are the scores out of 20 earned by 25 students on a mathematics exam:

Score (x_i)	8	10	12	14	16	18	Total
Count (n_i)	3	5	7	6	3	1	25
Frequency %	?	?	?	?	?	?	100%
Central Angle	?	?	?	?	?	?	360°

1. Complete the frequencies and angles for a pie chart.
2. Calculate the class mean score.
3. Determine the median score and the score range.

Solution: 1. For each score, relative frequency is $\frac{n_i}{25} \times 100\%$ and angle is $\frac{n_i}{25} \times 360^\circ$:

- Score 8: Frequency $= \frac{3}{25} = 12\%$; Angle $= 0.12 \times 360^\circ = 43.2^\circ$.
- Score 10: Frequency $= \frac{5}{25} = 20\%$; Angle $= 0.20 \times 360^\circ = 72^\circ$.
- Score 12: Frequency $= \frac{7}{25} = 28\%$; Angle $= 0.28 \times 360^\circ = 100.8^\circ$.
- Score 14: Frequency $= \frac{6}{25} = 24\%$; Angle $= 0.24 \times 360^\circ = 86.4^\circ$.
- Score 16: Frequency $= \frac{3}{25} = 12\%$; Angle $= 0.12 \times 360^\circ = 43.2^\circ$.
- Score 18: Frequency $= \frac{1}{25} = 4\%$; Angle $= 0.04 \times 360^\circ = 14.4^\circ$.

2. **Weighted Mean:**

$$\bar{x} = \frac{3 \times 8 + 5 \times 10 + 7 \times 12 + 6 \times 14 + 3 \times 16 + 1 \times 18}{25}$$

$$\bar{x} = \frac{24 + 50 + 84 + 84 + 48 + 18}{25} = \frac{308}{25} = 12.32$$

The class mean is $12.32/20$.

3. **Median and Range:**
   - Total sample size $N = 25$ (odd).
   - $\frac{25+1}{2} = 13$. The median is the **13th score** in ascending order.
   - Cumulative counts:
     - Scores up to 8: 3 students.
     - Scores up to 10: $3 + 5 = 8$ students.
     - Scores up to 12: $8 + 7 = 15$ students. The 13th score lies in the score 12 group! Thus, median $Me = \mathbf{12}$.
   - **Range:** $R = 18 - 8 = \mathbf{10}$ points. ✓

Solved Example 5 — Successive Percentage Increases

The price of gasoline rises by 10% in January, then rises by an additional 10% in February. 1. What is the total percentage increase over the two-month period? 2. What is the new price of a liter originally costing $\$1.50$?

Solution: 1. Multiply the respective multiplier coefficients:

$$C_1 = 1 + 0.10 = 1.10$$

$$C_2 = 1 + 0.10 = 1.10$$

$$C_{\text{overall}} = 1.10 \times 1.10 = 1.21$$

Since $1.21 = 1 + \frac{21}{100}$, the overall increase is $\mathbf{21\%}$ (not 20%)! 2. Final price per liter:

$$P_{\text{final}} = 1.50 \times 1.21 = \$1.815 \approx \$1.82$$

The liter of fuel now costs $\$1.815$. ✓

What You Must Remember

1. **Percentage Concepts:**
   - $t\% = \frac{t}{100}$.
   - Calculate $p\%$ of Q: $Q \times \frac{p}{100}$.
   - Find percentage rate: $\frac{\text{Part}}{\text{Total}} \times 100$.
2. **Multiplier Coefficients:**
   - Increase of $t\%$: multiply by $\left(1 + \frac{t}{100}\right)$.
   - Decrease of $t\%$: multiply by $\left(1 - \frac{t}{100}\right)$.
   - Consecutive changes: **multiply** coefficients ($C_{\text{total}} = C_1 \times C_2$).
3. **Statistical Indicators:**
   - **Relative Frequency:** $f = \frac{\text{count}}{N}$.
   - **Pie Chart Angle:** $\text{Angle} = f \times 360^\circ$.
   - **Mean ($\bar{x}$):** sum of all values divided by total count N.
   - **Median (Me):** the midpoint value dividing sorted data into equal $50\%/50\%$ halves.
   - **Range (R):** $\text{maximum value} - \text{minimum value}$.

Exercises

Level (Direct Application)

Exercise 21.1 Express each percentage as a simplified fraction, then as a decimal: 10%; 20%; 25%; 50%; 75%; 5%.

Exercise 21.2 Compute mentally: 1. 50% of $\$84$ 2. 25% of 120 kg 3. 10% of 350 m 4. 1% of $2,500 \text{ citizens}$

Exercise 21.3 State the multiplier coefficient associated with: 1. A price increase of 12% 2. A discount of 25% 3. A tax increase of 3.5% 4. A clearance markdown of 50%

Exercise 21.4 In a box of 40 gourmet chocolates, 14 are dark chocolate. What percentage of the box is dark chocolate?

Exercise 21.5 Calculate the simple arithmetic mean of the five test scores: 11, 14, 9, 17, 14.

Level (Reasoning and Statistical Problems)

Exercise 21.6 Discounted Smartphone. A smartphone retails for $340. During a seasonal promotion, its price is marked down by 15%. 1. What is the cash savings in dollars? 2. What is the new promotional price?

Exercise 21.7 Electricity Bill and Sales Tax. On an electricity bill of $150 before tax, the state levies a sales tax (VAT) of 20%. What is the total amount paid by the customer?

Exercise 21.8 The Deceptive Reversal. A train ticket increases in price by 25%. A few months later, the railway company offers a "25% immediate discount" promotion. A traveler believes the ticket has returned to its initial price. Is he correct? Justify with an exact calculation starting from a $100 ticket.

Exercise 21.9 Finding Medians (Odd vs. Even Sample Sizes). Find the median and range for each of the two weekly noon temperature records:

- Week A: 14° C, 17° C, 12° C, 19° C, 21° C, 15° C, 18° C.
- Week B: 13° C, 16° C, 11° C, 20° C, 14° C, 18° C.

Exercise 21.10 Neighborhood Pet Survey Pie Chart. A survey of 180 neighborhood households records pet ownership:

- 72 households own a dog.
- 54 households own a cat.
- 36 households own rodents or birds.
- 18 households own no pets.

1. Calculate the percentage frequency for each category.
2. Calculate the central angle in degrees for each category to construct a pie chart.

Exercise 21.11 Weighted Academic Average. A middle school student earns the following grades in mathematics:

- Two short quizzes (weight 1 each): 12 and 15.
- Two major exams (weight 3 each): 14 and 10.

What is her weighted quarterly grade point average?

Level (Challenges and Critical Thinking)

Exercise 21.12 The Factory Salary Paradox. A small manufacturing plant employs 20 individuals:

- 1 CEO earning $\$10,000$ per month.
- 3 senior engineers earning $\$4,000$ per month each.
- 16 assembly workers earning $\$1,500$ per month each.

1. Calculate the mean monthly salary across the company.
2. Calculate the median monthly salary.
3. Why would management and a labor union choose different statistical indicators during wage negotiations? Explain the manipulative power of the arithmetic mean.

Exercise 21.13 Reverse Percentage (Finding the Original Price). After receiving a 20% discount during a sale, a winter coat costs $120. What was its original price before the sale? *(Warning: do NOT compute* $\$120 + 20\%$ *of* $\$120$*! Use the multiplier coefficient).*

Exercise 21.14 Consecutive Percentage Changes. The price of a gaming console increases by 5% in 2024, then rises by an additional 8% in 2025. 1. What is the overall percentage increase over the two-year period? 2. If the price then drops by 12% in 2026, is the console ultimately more expensive or less expensive than at the start of 2024?

Exercise 21.15 Simpson's Paradox in Surgery. In a major hospital, two surgeons (Dr. Martin and Dr. Dupont) perform two types of procedures: low-risk routine surgeries and high-risk complex surgeries.

- Low-risk surgeries:
  - Dr. Martin succeeds in 95 out of 100 cases (95%).
  - Dr. Dupont succeeds in 19 out of 20 cases (95%).
- High-risk surgeries:
  - Dr. Martin succeeds in 10 out of 20 cases (50%).
  - Dr. Dupont succeeds in 45 out of 100 cases (45%).

1. For high-risk surgeries alone, which surgeon has the superior success rate?
2. Calculate the overall success rate for each surgeon combining all procedures.
3. Which surgeon has the higher overall success rate? What astonishing contradiction do you notice? Explain this statistical paradox.

Solutions

Solution 21.1 - $10\% = \frac{1}{10} = 0.10$ - $20\% = \frac{1}{5} = 0.20$ - $25\% = \frac{1}{4} = 0.25$ - $50\% = \frac{1}{2} = 0.50$ - $75\% = \frac{3}{4} = 0.75$ - $5\% = \frac{1}{20} = 0.05$

Solution 21.2 1. $84/2 = \$42$. 2. $120/4 = 30 \ \text{kg}$. 3. $350/10 = 35 \ \text{m}$. 4. $2,500/100 = 25 \ \text{citizens}$.

Solution 21.3 1. $1 + 0.12 = 1.12$. 2. $1 - 0.25 = 0.75$. 3. $1 + 0.035 = 1.035$. 4. $1 - 0.50 = 0.50$.

Solution 21.4 Rate $= \frac{14}{40} \times 100 = \frac{7}{20} \times 100 = 35\%$.

Solution 21.5 Sum $= 11 + 14 + 9 + 17 + 14 = 65$. Mean $= \frac{65}{5} = 13/20$.

Solution 21.6 1. Savings: $340 \times 0.15 = \$51$. 2. New price: $340 - 51 = \$289$ (or $340 \times 0.85 = \$289$).

Solution 21.7 Total bill $= 150 \times (1 + 0.20) = 150 \times 1.20 = \$180$.

Solution 21.8 Initial price: \$100. After 25% increase: $100 \times 1.25 = \$125$. After 25% decrease: $125 \times 0.75 = \$93.75$. The traveler is mistaken: the ticket costs \$93.75, which is \$6.25 cheaper than the original price!

Solution 21.9 - **Week A (7 values):** Sorted: $12, 14, 15, \mathbf{17}, 18, 19, 21$. Median $= \mathbf{17}^\circ\ \mathbf{C}$ (the 4th value). Range $= 21 - 12 = 9^\circ$ C.

• **Week B (6 values):** Sorted: $11, 13, \mathbf{14}, \mathbf{16}, 18, 20$. Median $= \frac{14+16}{2} = \mathbf{15}^\circ\ \mathbf{C}$. Range $= 20 - 11 = 9^\circ$ C.

Solution 21.10 Total $N = 180$. 1. Frequencies:

• Dogs: $\frac{72}{180} = 40\%$.
• Cats: $\frac{54}{180} = 30\%$.
• Rodents/birds: $\frac{36}{180} = 20\%$.
• None: $\frac{18}{180} = 10\%$.

2. Central Angles:
   • Dogs: $0.40 \times 360^\circ = 144^\circ$.
   • Cats: $0.30 \times 360^\circ = 108^\circ$.
   • Rodents/birds: $0.20 \times 360^\circ = 72^\circ$.
   • None: $0.10 \times 360^\circ = 36^\circ$.

Solution 21.11 Sum of weights: $1 + 1 + 3 + 3 = 8$. Mean $= \frac{1\times12+1\times15+3\times14+3\times10}{8} = \frac{12+15+42+30}{8} = \frac{99}{8} = 12.375/20$.

Solution 21.12 1. Total payroll: $1 \times 10,000 + 3 \times 4,000 + 16 \times 1,500 = 10,000 + 12,000 + 24,000 = \$46,000$. Mean salary: $\frac{46,000}{20} = \mathbf{\$2,300}$. 2. Sorted salaries: the bottom 16 employees earn $\$1,500$. The 10th and 11th values are both $\$1,500$. Median salary: $\mathbf{\$1,500}$. 3. The director will highlight the mean ($\$2,300$) to claim: *"Our workers earn generous wages on average,"* even though 80% earn only $\$1,500$! The union will point to the median ($\$1,500$) to demonstrate that at least half the workforce earns only base pay.

Solution 21.13 Let P_0 be the initial price. A 20% discount corresponds to multiplier $1 - 0.20 = 0.80$. $P_0 \times 0.80 = 120 \implies P_0 = \frac{120}{0.80} = \$150$. The original price was \$150! (Check: $\$150 - 20\%\ \text{ of }\ 150 = 150 - 30 = \$120$).

Solution 21.14 1. $C_{\text{overall}} = (1 + 0.05) \times (1 + 0.08) = 1.05 \times 1.08 = 1.134$. Overall increase: $1.134 - 1 = 0.134 = +\mathbf{13.4}\%$ (not 13%). 2. With the 12% discount ($C_3 = 0.88$): $C_{\text{final}} = 1.134 \times 0.88 = 0.99792$. Since $0.99792 < 1$, the console is ultimately **slightly less expensive** than at the start of 2024 (a net decrease of 0.208%).

Solution 21.15 1. High-risk surgeries: Dr. Martin has 50%, Dr. Dupont has 45%. **Dr. Martin** has the higher success rate. 2. Combined rates:

• Dr. Martin: successes $= 95 + 10 = 105$ out of $100 + 20 = 120$ operations. Overall rate $= \frac{105}{120} = 0.875 = \mathbf{87.5}\%$.
• Dr. Dupont: successes $= 19 + 45 = 64$ out of $20 + 100 = 120$ operations. Overall rate $= \frac{64}{120} \approx 0.533 = \mathbf{53.3}\%$.

3. Dr. Martin has an overall success rate of 87.5% versus 53.3% for Dr. Dupont! The paradox: for routine cases, both surgeons have identical success rates (95%). For high-risk cases, Dr. Martin is superior (50% vs. 45%). And overall, Dr. Martin dramatically outperforms Dr. Dupont! Explanation: Dr. Martin performed 83% of his surgeries on easy, low-risk patients, whereas Dr. Dupont took on 83% high-risk, dangerous operations. This famous statistical phenomenon is known as **Simpson's Paradox**: an unobserved confounding variable (the underlying severity of cases) distorts the aggregate comparison!

The Question That Opens What Follows

We have arrived at the conclusion of Volume III. We now know how to measure the earth, lay out right angles with compass and rope, compute inaccessible heights with Thales and Pythagoras, navigate Cartesian planes with signed coordinates, and synthesize crowds of data with percentages and statistics.

Yet up to this point, all our numbers were known, or uncovered through direct arithmetic. What happens when a problem hides a secret unknown quantity inside a sealed velvet bag resting upon a balance scale, tangled in a web of additions and multiplications?

How do we forge a new mathematical language where unknown numbers are represented by letters like x, and where the scales of justice transform into algebraic equations? That is the great threshold into the adolescence of mathematics: **Volume IV and the Mystery Bag on the Balance of Al-Khwarizmi**!

VOLUME IV — THE ALCHEMY OF SYMBOLS AND THE ALGEBRAIC FIRE

Ages 13–15 / Grades 8–9

The emergence of literal equations, integer powers, affine functions, and statistics prepares the transition to high school mathematics.

Chapters of Volume IV :

Chapter 22 : The Mystery Bag on Al-Khwarizmi's Scale
Chapter 23 : The Completed Square and Special Products
Chapter 24 : The Two Merchants and Inequalities
Chapter 25 : The Doubling Machine and the Tariff Line
Chapter 26 : Paper Folding and the Magic Mirror
Chapter 27 : Land Partitioning and the Potter's Box
Chapter 28 : Eratosthenes and the Circumference of the Earth
Chapter 29 : The Partition of Euclid
Chapter 30 : The Roulette of Fate and the Hunter's Survival

Facing: Scale invariance and fractal geometry of Romanesco Broccoli (Brassica oleracea).

CHAPTER 22: The Mystery Bag on Al-Khwarizmi's Scale

> "The ignorant man forces the lock and tears open the bag to see what is hidden inside. The mathematician observes the two arms of the scale, removes a weight on the left, a weight on the right, and the unknown reveals itself in the clear light of reason."

What You Are About to Discover

- Why we suddenly begin to **calculate with letters**, and what "$3x$", "x^2", or "$a(b+c)$" truly mean.
- How to **simplify and combine like terms** (sorting apples with apples) and how to **expand** using the **distributive property**, proved with a simple rectangle.
- How to **test an equality**: true for one value, true for all, or false—and the power of a single **counterexample**.
- How to solve any **first-degree equation** using the two fundamental gestures of the balance scale.
- How to **set up an equation from a word problem**: ages, perimeters, computational algorithms, and even the riddle of Diophantus's epitaph.
- The classic **pitfalls**: $2x + 3 \neq 5x$, $x \times x \neq 2x$, the minus sign "$-$" before parentheses, and equations that have… no solution, or infinitely many.

---

1. The Problem: The Grand Bazaar of Baghdad and the Bag of Gold

You are now twelve or thirteen years old. Let us leave the plains of Egypt behind, cross the desert sands, and enter the most fabulous metropolis of the medieval world: **Baghdad**, the "City of Peace," around the year **820**.

At the heart of the city stands the *House of Wisdom* (*Bayt al-Hikma*), an immense library where scholars from Persia, India, and Greece translate the world's manuscripts into Arabic: Euclid, Ptolemy, Indian astronomers… But today, a lively crowd gathers in the great goldsmiths' bazaar.

At the center of a circle of curious merchants, a man draped in a green brocade cloak has set upon a table a magnificent **brass balance scale with two pans**.

- On the left pan, he places **a brown canvas bag, tightly stitched**, sealed with a stamp of red wax. No one knows what is inside! Beside the bag, he places **3 copper ingots**, each weighing one kilogram.
- On the right pan, he places **11 copper ingots** of one kilogram each.

The beam of the scale rocks slowly… and comes to rest, **perfectly horizontal**.

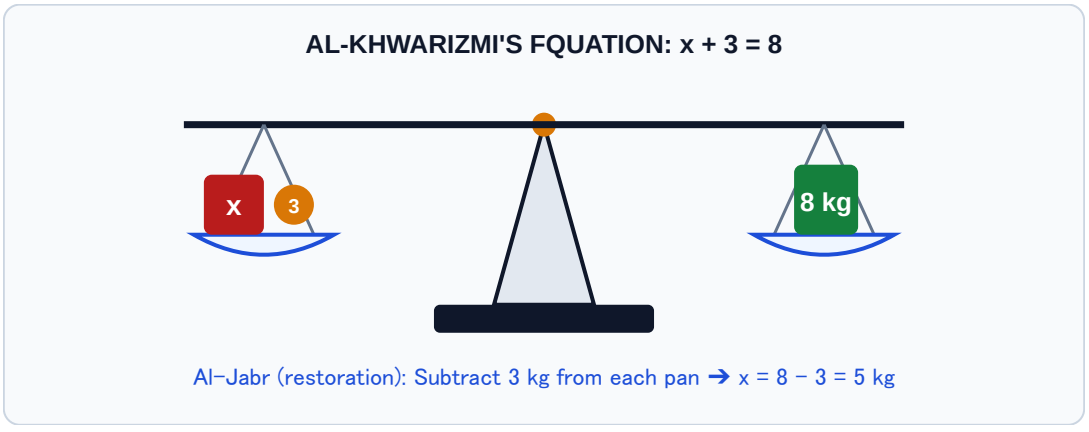

Figure 65: Al-Khwarizmi's balance scale and the equilibrium of equations

"This bag contains pure gold dust. Whoever tells me its exact weight without breaking the seal will leave with all the gold! But whoever is off by a single gram must surrender his mount to me!"

No one dares risk their horse. Then, a Persian astronomer steps forward with a calm stride: **Muhammad ibn Musa al-Khwarizmi**.

"The truth is already before your eyes. The two pans are in balance. If I remove the exact same amount from both sides at the same time, will the scale remain level?" — "Of course!" replies the crowd.

With his left hand, he removes the **3 ingots** from the left pan: the pan lightens and rises. But instantly, with his right hand, he removes **3 ingots** from the right pan. The beam settles back down and rests once more, **level and still**.

On the left: the bag, all alone. On the right: $11 - 3 = 8$ ingots. *"The bag weighs eight kilograms."* The merchant pales, breaks the seal, and weighs the bag alone: **precisely eight kilograms**.

So far, nothing you could not already do in your head. But the next day, the merchant returns, stung in his pride, with increasingly cunning riddles: two identical bags and 5 kg balancing 17 kg; three bags and 2 kg balancing one bag and 10 kg; then bags from which a portion of the contents has been removed… After a few puzzles, mental arithmetic is no longer enough. We need **a language** to write down these balance states, and **rules** to manipulate them without error. That language is **algebraic notation**. Those rules are the laws of **algebra**.

---

2. Calculating with Letters: Why and How

2.1 The Letter Is a Hidden Number

In mathematics, a letter is **not** a word or an abbreviation for a unit. It is **a number** whose value is not (yet) known, or whose value can vary.

- When the letter represents a specific but **unknown** number (the weight of the bag), we call it an **unknown**.
- When the letter can take on **any value whatsoever** (the side length of an arbitrary square), we call it a **variable**.

You already know formulas with letters: the perimeter of a square of side length c is $4 \times c$; the area of a rectangle is $L \times \ell$. These formulas say something that is true **for all squares**, for **all rectangles**, all at once. That is the great power of letters: **a single statement for an infinity of cases**.

> **Definition — Algebraic Expression.**
>
> An *algebraic expression* is a sequence of operations in which certain numbers are replaced by letters. Examples: $3x + 5$; $2(a - 4)$; $x^2 - 7x + 1$.

2.2 Notational Conventions (and How to Read Them)

To work faster, mathematicians have agreed on a few shorthand conventions. Each has a spoken form:

Written Form	Meaning	Read Aloud
$3x$	$3 \times x$	"three x"
ab	$a \times b$	"a times b" or "a b"
$4(x+1)$	$4 \times (x+1)$	"four times the quantity x plus one"
x^2	$x \times x$	"x squared"
x^3	$x \times x \times x$	"x cubed"
$1x$	written simply as x	"x"
$-1x$	written simply as $-x$	"negative x" or "minus x"
$0x$	equals 0	—

Safety rules:

- **Never omit** the $\times$ sign between **two numbers**: 3×5 must never be written as 35!
- By convention, we write the **number before the letter**: $x \times 3$ is written $3x$ (and not $x3$).

- The order of operations from Chapter 18 still holds: exponents first, then multiplications and divisions, then additions and subtractions. Thus $3x^2$ means $3 \times (x \times x)$ and **not** $(3x) \times (3x)$.

> 💡 **True History — Who Invented Letters in Math?**
>
> Al-Khwarizmi himself wrote **no symbols at all**! Everything was written out in words: "a square and ten roots equal thirty-nine dirhams" (which we write $x^2 + 10x = 39$). He called the unknown *shay'* ("the thing"). Around 250 AD, the Greek mathematician **Diophantus** was already using abbreviations. In 1591, the Frenchman **François Viète**, advisor to King Henry IV and a master codebreaker, had the decisive insight: designate unknowns by vowels and known numbers by consonants. Finally, in 1637, **René Descartes**, in *La Géométrie*, took the last letters of the alphabet x, y, z for unknowns and the first letters a, b, c for known quantities: this is our modern convention. A popular tale claims that x comes from the Arabic word *shay'* transcribed into Spanish as *xei*; it is a charming story, but no historical document confirms it.

> 💡 **Did You Know?**
>
> The symbols $+$ and $-$ first appeared in print in 1489 in a commercial arithmetic book by the German Johannes Widmann. The equals sign $=$ was invented in 1557 by the Welsh mathematician **Robert Recorde**, who grew tired of writing "is equal to": he chose two parallel line segments of identical length, "bicause noe 2 thynges can be moare equalle."

2.3 Evaluating an Expression (Substitution)

Substitution means replacing the letter with a specific number. The process is straightforward, but it requires two safeguards: **reinsert the multiplication signs** $\times$ and **put negative numbers in parentheses**.

> 🖍️ **Worked Example 1 — Evaluating an expression for several values.**
>
> Let $A = 3x^2 - 5x + 2$. Evaluate A for $x = 4$, then for $x = -2$.
> For $x = 4$: $A = 3 \times 4^2 - 5 \times 4 + 2 = 3 \times 16 - 20 + 2 = 48 - 20 + 2 = 30$.
> For $x = -2$: $A = 3 \times (-2)^2 - 5 \times (-2) + 2 = 3 \times 4 + 10 + 2 = 24$.
> Note: $(-2)^2 = (-2) \times (-2) = 4$ (the rule of signs from Chapter 20), and $-5 \times (-2) = +10$.

> ⚠️ **Watch Out! Common Pitfall — Squaring and the Minus Sign.**
>
> - $-x^2$ means "the opposite of x^2". For $x = 3$: $-x^2 = -9$.
> - $(-x)^2$ means "the square of the opposite of x". For $x = 3$: $(-3)^2 = 9$.
> - For $x = -2$, $3x^2 = 3 \times 4 = 12$, but $(3x)^2 = (-6)^2 = 36$. The exponent applies **only to what immediately precedes it**.
> - And $x \times x = x^2$, **not** $2x$! For $x = 5$: $x^2 = 25$ whereas $2x = 10$.

---

3. Simplifying: Sorting Apples with Apples

Imagine a bazaar stall: "5 melons, 3 dates, 2 melons, 7 dates". Any merchant will immediately say: "7 melons and 10 dates". He **groups items of the same kind together**. But he would never say "17 melon-dates"!
In algebraic manipulation, it is the exact same. $5x + 3 + 2x + 7$ is grouped into $7x + 10$. The "x" terms go with the "x" terms, and the standalone constants with standalone constants. But $7x + 10$ cannot become $17x$: they are not objects of the same nature.

📐 Definition — Like Terms, Combining Like Terms.

- In a sum, each piece separated by $+$ or $-$ is a **term**. In $5x^2 - 3x + 8$, the terms are $5x^2$, $-3x$, and 8.
- The number placed in front of the variable part is called the **coefficient**: in $-3x$, the coefficient is -3.
- Two terms are **like terms** if they have **the exact same variable part**: $4x$ and $-7x$ are like terms; $3x^2$ and x^2 are like terms; but $3x$ and $3x^2$ **are not like terms**.
- **Simplifying (combining like terms)** an expression means grouping like terms by adding their coefficients, until the expression is written in its shortest possible form.

Why are we allowed to write $5x + 2x = 7x$? Because $5x + 2x$ is "5 bags plus 2 bags," hence 7 bags. We will prove this rigorously in a moment using the distributive property.

🖍️ Worked Example 2 — Combining like terms.

a) $B = 5x + 3 - 2x + 7$. We group: $(5x - 2x) + (3 + 7) = 3x + 10$.

b) $C = 4x^2 - 3x + x^2 + 7x - 1$. The x^2 terms: $4x^2 + x^2 = 5x^2$. The x terms: $-3x + 7x = 4x$. The constants: -1. Thus $C = 5x^2 + 4x - 1$.

c) $D = 2a \times 3a$. Here it is a **product**: we multiply the numbers together and the letters together: $2 \times 3 \times a \times a = 6a^2$.

d) $E = 2a + 3a = 5a$ (sum), whereas $2a \times 3a = 6a^2$ (product). Never confuse the two!

⚠️ Watch Out! Common Pitfall — What Cannot Be Combined.

- $2x + 3$ **does not equal** $5x$. Test: for $x = 10$, $2x + 3 = 23$, but $5x = 50$.
- $x^2 + x$ **does not equal** x^3 nor $2x^2$. Test: for $x = 2$, $x^2 + x = 6$, $x^3 = 8$, $2x^2 = 8$.
- $3x^2 + 2x^2 = 5x^2$, **and not** $5x^4$: we are adding "packets of x^2", we are not multiplying.

---

4. Distributivity: A Rectangle That Does Not Lie

4.1 The Intuition

A jeweler sells necklaces. Each necklace contains a gold beads and b silver beads. A customer orders 3 necklaces. How many beads are there in total? We can count necklace by necklace: $3 \times (a + b)$. Or we can count all the gold beads, then all the silver beads: $3a + 3b$. The total number of beads is identical! Therefore, $3(a + b) = 3a + 3b$.

4.2 Proof by the Area of a Rectangle

Draw a rectangle of height k and width $a + b$, partitioned into two parts:

📐 Fundamental Rule — Geometric View of Single Distributivity

- a b
- k area k×a area k×b
- a + b

The area of the large rectangle is $k \times (a + b)$. But it is also the sum of the areas of the two smaller rectangles: $k \times a + k \times b$. **One and the same surface, two ways to calculate its area**: they must be equal.

📐 Property — The Distributive Property.

For any real numbers k, a, b:

$$k(a+b) = ka + kb \qquad \text{and} \qquad k(a-b) = ka - kb.$$

Read from left to right, we **expand** (converting a product into a sum). Read from right to left, we **factor** (converting a sum into a product, factoring out the **greatest common factor** k).

The word "distributivity" comes from the image of the multiplier k being **distributed** to each term inside the parentheses, just as a mail carrier delivers a letter to every mailbox on the street.

And here is the promised proof of $5x + 2x = 7x$: it is simply the distributive property read in reverse: $5x + 2x = (5 + 2)x = 7x$.

🖍️ Worked Example 3 — Expanding and simplifying.

a) $3(2x - 5) = 3 \times 2x - 3 \times 5 = 6x - 15$.

b) $-2(x - 4) = (-2) \times x - (-2) \times 4 = -2x + 8$. (Watch the signs carefully: $-2 \times (-4) = +8$.)

c) $x(x + 3) = x \times x + x \times 3 = x^2 + 3x$.

d) $F = 4(x + 2) - 3(2x - 1) = 4x + 8 - 6x + 3 = -2x + 11$.

4.3 The Minus Sign in Front of Parentheses

Parentheses preceded by a minus sign secretly conceal a factor of -1: $-(a + b) = (-1) \times (a + b) = -a - b$. Consequently:

📐 Rule — Removing Parentheses.

- Preceded by $+$: remove the parentheses **without changing anything**. $7 + (3x - 2) = 7 + 3x - 2$.
- Preceded by $-$: remove the parentheses **by reversing the sign of every term** inside. $7 - (3x - 2) = 7 - 3x + 2 = 9 - 3x$.

⚠️ Watch Out! Common Pitfall.

The most frequent error in middle school: writing $7 - (3x - 2) = 7 - 3x - 2$. The student forgot to flip the sign of the **second** term! Test with $x = 0$: $7 - (0 - 2) = 9$, but $7 - 0 - 2 = 5$. The test exposes the blunder in two seconds.

4.4 Factoring: The Return Journey

🖍️ Worked Example 4 — Factoring out a common factor.

a) $6x + 9$: $6x = 3 \times 2x$ and $9 = 3 \times 3$. Common factor 3: $6x + 9 = 3(2x + 3)$.

b) $5x^2 - 10x$: $5x^2 = 5x \times x$ and $10x = 5x \times 2$. Common factor $5x$: $5x^2 - 10x = 5x(x - 2)$.

c) $7a + 7 = 7 \times a + 7 \times 1 = 7(a + 1)$. Do not forget the "1"!

Systematic Verification: expand back out. $5x(x - 2) = 5x^2 - 10x$ ✓.

Why factor? Because a **product** is often far easier to study than a sum. You will see in the next chapter that factoring allows us to solve equations that no physical scale could ever untangle.

CHAPTER 23: The Completed Square and Special Products

> "Do not recite an algebraic formula like a magical incantation. Close your eyes and look at the tiled floor: cut a rectangle, shift a corner, complete the missing square. Letters are nothing other than the sides of geometric shapes fitting together under your fingers."

What You Are About to Discover

- How the **double distributive property** is explained by slicing a rectangle into four plots of land: $(a+b)(c+d) = ac + ad + bc + bd$.
- The **three special products (algebraic identities)**, demonstrated before your eyes through pure geometric dissections of squares and rectangles.
- The art of **lightning-fast mental math**: calculating 101^2, 99^2, or 48×52 in two seconds without a calculator.
- **Factoring**: the reverse journey, transforming a sum into a product by unearthing the hidden common factor or the footprint of a special product.
- The **zero-product equation**: the master key to solving quadratic equations without panic.
- The complete resolution of the equation $x^2 = a$ depending on the sign of a, and why a real square can never equal a negative number.
- The grand historical stroke of genius by **Al-Khwarizmi**: "completing the square," which laid the foundations of global algebra twelve centuries ago.

---

1. The Problem: The Caliph's Garden and the Missing Corner

In the spring of the year 825, in the hanging gardens of the caliphal palace in Baghdad, the head gardener faced an infuriating dilemma. The Caliph desired to enlarge his square orchard.

The current orchard was a square of unknown side length x meters. The Caliph decreed: *"Extend two opposite sides by 5 meters to the north, and extend the other two sides by 5 meters to the east!"*

The gardener drafted the plan. The new garden had side length $(x+5)$ meters. But the Grand Steward of the Treasury demanded the exact formula for the total area to order the bags of grass seed: *"If the initial side length was x, the new area is obviously $x^2 + 5^2 = x^2 + 25$!"* proclaimed the vizier without a second thought.

Al-Khwarizmi, present during the audience, smiled quietly and sketched a figure in the fine sand with the tip of his reed cane:

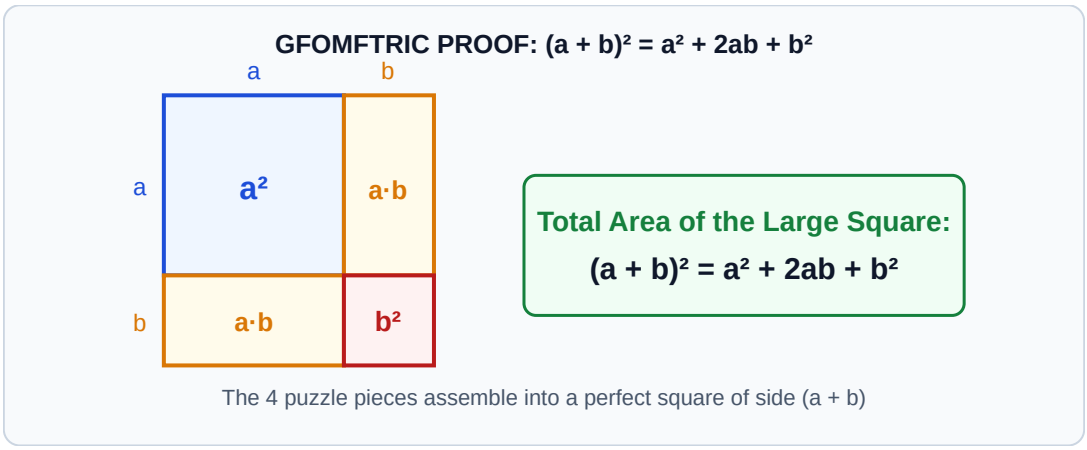

Figure 66: Geometric proof of the special product (a+b)²

"Look closely, noble vizier," said Al-Khwarizmi. *"Your calculation forgets two immense strips of land! You indeed have the old square of x^2, and you indeed have the small corner square of $25\ \mathrm{m}^2$. But where have you accounted for the plot of $5x$ in the upper right, and the other plot of $5x$ in the lower left?"*

The true area of the expanded garden is not $x^2 + 25$, but:

$$(x+5)^2 = x^2 + 5x + 5x + 25 = x^2 + 10x + 25.$$

The vizier had "forgotten" $10x$ square meters of lawn—enough to leave half the garden completely bare!

This blunder is the most notorious mistake in the entire history of school mathematics. Millions of students commit it every year by inadvertently writing $(a+b)^2 = a^2 + b^2$.

In this chapter, we will etch into your mind, through geometry and pure reasoning, why those two middle pieces always emerge, how to navigate effortlessly between expanding and factoring, and how this geometry unlocks second-degree equations.

---

2. Double Distributivity: Four Plots in a Rectangle

In Chapter 22, you encountered **single distributivity**: multiplying a number by a sum inside parentheses distributes the factor to each term:

$$k(a+b) = ka + kb.$$

This represented the area of a rectangle partitioned into two vertical strips.

What happens if both sides of the rectangle are themselves split into two segments?

2.1 Proof by the Area of a Rectangle

Let us draw a rectangle whose total length is $(a+b)$ and whose total width is $(c+d)$.

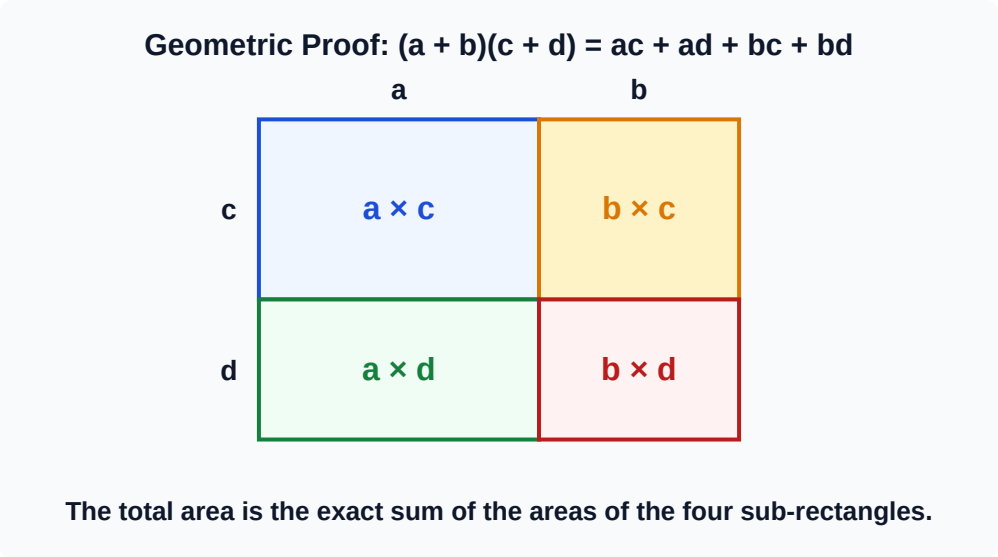

Figure 67: Distributivity of multiplication over addition

The total area of the large rectangle can be evaluated in two equivalent ways: 1. **All at once:** Length $\times$ Width, which is $(a+b)(c+d)$. 2. **Piece by piece:** The sum of the areas of the 4 interior plots:

$$ac + ad + bc + bd.$$

Since both calculations measure the exact same surface, they are strictly equal.

> 📐 **Fundamental Property — The Double Distributive Property.**
>
> For any real numbers a, b, c, d:
>
> $$(a+b)(c+d) = ac + ad + bc + bd.$$
>
> Every term in the first pair of parentheses is multiplied by every term in the second pair of parentheses. This produces 4 pairwise multiplications.

2.2 The Arrow Rule and Managing Signs

To ensure nothing is forgotten in a written calculation, one can mentally trace four arrows (often known in English as the **FOIL method**: First, Outer, Inner, Last):

1. **First:** $a \times c = ac$
2. **Outer:** $a \times d = ad$
3. **Inner:** $b \times c = bc$
4. **Last:** $b \times d = bd$

> ⚠️ **Watch Out! Pitfall — The Sign Belongs to the Term Following It!**
>
> When a factor contains a subtraction, such as $(2x-3)(x+4)$, view the first factor as consisting of the terms $(+2x)$ and (-3). Then apply the sign rules rigorously (Chapter 20): * $(+2x) \times (+x) = +2x^2$ * $(+2x) \times (+4) = +8x$ * $(-3) \times (+x) = -3x$ * $(-3) \times (+4) = -12$ Then **simplify** by combining like terms:
>
> $$(2x-3)(x+4) = 2x^2 + 8x - 3x - 12 = 2x^2 + 5x - 12.$$

---

3. The Three Special Products (Algebraic Identities)

Certain algebraic products occur so frequently across science and mathematics that they are termed **remarkable** or **special products**. They deserve to be mastered as instantaneous reflexes. But before memorizing them by heart, let us observe them living before our eyes.

3.1 First Identity: The Square of a Sum $(a+b)^2$

Consider a square of side length $(a+b)$.

The total area is the square of its side length: $(a+b)^2$. This area naturally divides into four components:

- a large square of area a^2;
- a small square of area b^2;
- two identical rectangles of area $a \times b = ab$.

Adding these components together gives:

$$(a+b)^2 = a^2 + ab + ab + b^2 = a^2 + 2ab + b^2.$$

The middle term $2ab$ is known as the **cross product** or **double product**.

> 📐 ***First Special Product — Square of a Sum.*** *For any real numbers a and b:*
>
> $$(a+b)^2 = a^2 + 2ab + b^2.$$
>
> In plain words: *"The square of a sum equals the square of the first term, plus twice the product of the terms, plus the square of the second term."*

3.2 Second Identity: The Square of a Difference $(a-b)^2$

Now imagine a large square of side length a, with total area a^2. We want to determine the area of a smaller square nestled in the top-left corner, of side length $(a-b)$. Its area is $(a-b)^2$.

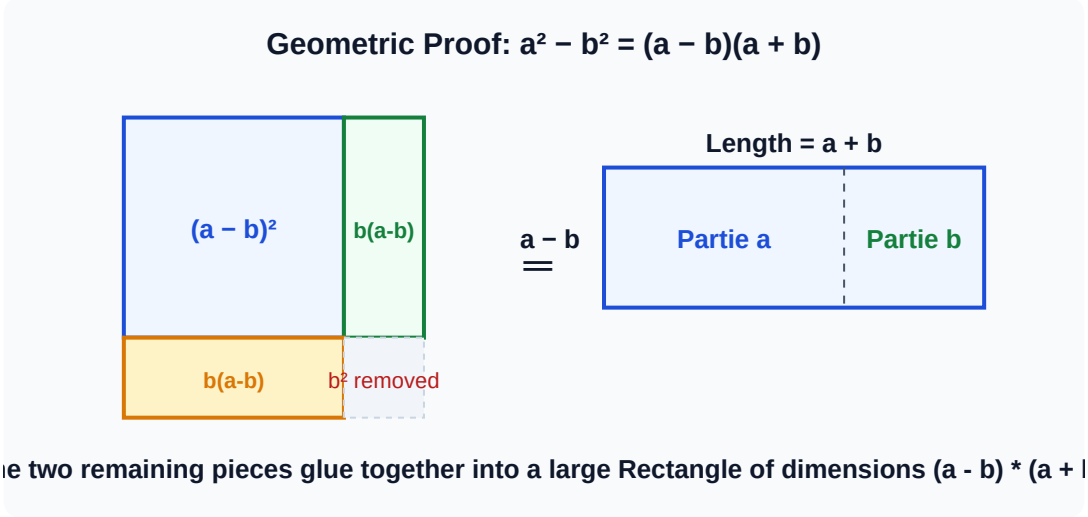

Figure 68: Difference of two squares: (a+b)(a-b) = a² - b²

How can we obtain the area of this white square $(a-b)^2$ from the large square a^2? 1. Remove the bottom horizontal strip, of length a and width b: its area is $a \times b = ab$. 2. Remove the right vertical strip, of length a and width b: its area is $a \times b = ab$. 3. **Caution!** The small bottom-right corner square (marked with xxxx), of side length b and area b^2, belonged simultaneously to both the horizontal strip and the vertical strip! In subtracting both strips, we removed it **twice**. 4. To restore exact balance, we must **add back once** that small corner b^2.

The remaining area is therefore:

$$(a-b)^2 = a^2 - ab - ab + b^2 = a^2 - 2ab + b^2.$$

Algebraically, we find this exact result by applying double distributivity to $(a-b)(a-b)$:

$$(a-b)(a-b) = a \times a - a \times b - b \times a + (-b) \times (-b) = a^2 - 2ab + b^2.$$

> 📐 ***Second Special Product — Square of a Difference.*** *For any real numbers a and b:*
>
> $$(a-b)^2 = a^2 - 2ab + b^2.$$
>
> In plain words: *"The square of a difference equals the square of the first term, minus twice the product of the terms, plus the square of the second term."*

> ⚠️ **Watch Out! Pitfall.**
>
> In $(a-b)^2 = a^2 - 2ab + b^2$, the final term is **always preceded by a plus sign +**! Indeed, $(-b) \times (-b) = +b^2$. Only the double product carries the minus sign.

3.3 Third Identity: The Product of a Sum and a Difference $(a+b)(a-b)$

Start with a square of side length a (area a^2), from which we cut out a smaller square of side length b situated at a corner (area b^2). The remaining surface has area $a^2 - b^2$. It is an L-shaped region.

Let us slice this L-shaped region and rearrange it into a single large rectangle of length $(a+b)$ and width $(a-b)$:

The area of this new rectangle is $\text{Length} \times \text{Width} = (a+b)(a-b)$. Since this rectangle was formed by rearranging the large square a^2 stripped of the small square b^2, the two areas are strictly equal!

Algebraically, via double distributivity:

$$(a+b)(a-b) = a \times a - a \times b + b \times a - b \times b = a^2 - ab + ba - b^2.$$

The two middle terms cancel because $-ab + ba = 0$. What remains is:

$$(a+b)(a-b) = a^2 - b^2.$$

> 📐 ***Third Special Product — Difference of Two Squares.*** *For any real numbers a and b:*
>
> $$(a+b)(a-b) = a^2 - b^2.$$
>
> *This identity is universally celebrated as the* ***difference of squares****.*

3.4 Summary Table of Special Products

Factored Form (Product)		Expanded Form (Sum)	Name
$(a+b)^2$	$\Leftrightarrow$	$a^2+2ab+b^2$	Square of a sum
$(a-b)^2$	$\Leftrightarrow$	$a^2-2ab+b^2$	Square of a difference
$(a+b)(a-b)$	$\Leftrightarrow$	a^2-b^2	Difference of two squares

- Moving from left to right ($\rightarrow$): this is **expanding** (writing as a sum/difference of terms).
- Moving from right to left ($\leftarrow$): this is **factoring** (writing as a product of factors).

3.5 Lightning-Fast Mental Math Using Identities

Special products are not merely tools for manipulating abstract symbols; they enable staggering arithmetic calculations entirely in your head without scrap paper or calculators!

- **Calculating 101^2:**

$$101^2 = (100+1)^2 = 100^2 + 2\times 100\times 1 + 1^2 = 10\,000 + 200 + 1 = 10\,201.$$

- **Calculating 99^2:**

$$99^2 = (100-1)^2 = 100^2 - 2\times 100\times 1 + 1^2 = 10\,000 - 200 + 1 = 9\,801.$$

- **Calculating 52×48:** Notice that $52 = 50+2$ and $48 = 50-2$!

$$52\times 48 = (50+2)(50-2) = 50^2 - 2^2 = 2\,500 - 4 = 2\,496.$$

- **Calculating 105×95:**

$$105\times 95 = (100+5)(100-5) = 100^2 - 5^2 = 10\,000 - 25 = 9\,975.$$

> 💡 **Did You Know?**
>
> In Antiquity, the Babylonians possessed no multiplication tables for all numbers. However, they possessed engraved clay tables of squares ($1^2, 2^2, 3^2, \ldots$). To multiply any two arbitrary numbers x and y, they utilized the formula:
>
> $$x\times y = \frac{(x+y)^2 - (x-y)^2}{4}.$$
>
> By calculating the sum, the difference, and looking up their squares in their tables, they found the product instantaneously!

---

4. Factoring: Recovering the Elementary Building Blocks

Expanding is mechanical: simply distribute arrows without losing track of signs. **Factoring** is the exact reverse: it is detective work. One is given an unraveled sum of scattered terms and must uncover how it was assembled from a pristine product.

Why go to such lengths to factor? Because a product is infinitely easier to examine and manage than a sum:

- A product allows us to simplify algebraic fractions.
- A product enables us to solve complex equations using the zero-product property.

In middle school algebra, there are two primary factoring techniques:

4.1 First Technique: Factoring Out a Common Factor

Search for a number, a variable, or a bracketed expression that multiplies **every single term** in the sum.

Case 1: An Obvious Common Factor (a number or a single variable) - In $6x + 15$: notice that $6 = 3 \times 2$ and $15 = 3 \times 5$. The common factor is 3.

$$6x + 15 = 3 \times 2x + 3 \times 5 = 3(2x + 5).$$

- In $x^2 - 7x$: notice that $x^2 = x \times x$ and $7x = 7 \times x$. The common factor is x.

$$x^2 - 7x = x \times x - x \times 7 = x(x - 7).$$

Case 2: An Entire Binomial as a Common Factor Consider the expression:

$$A = (3x + 2)(x - 5) + (3x + 2)(2x + 1).$$

The expression is a sum of two composite terms. The block $(3x + 2)$ is present in both terms: it is our **common factor**. Underline it, factor it out in front, and open a set of brackets to collect everything that remains:

$$A = \underline{(3x + 2)}(x - 5) + \underline{(3x + 2)}(2x + 1)$$

$$A = \underline{(3x + 2)}[(x - 5) + (2x + 1)]$$

$$A = (3x + 2)[x - 5 + 2x + 1]$$

$$A = (3x + 2)(3x - 4).$$

> ⚠️ **Watch Out! Pitfall — When the Common Factor Stands "All Alone"!**
>
> Consider the expression $B = (2x + 1)^2 - (2x + 1)$. Remember that $(2x + 1)^2 = (2x + 1)(2x + 1)$ and $(2x + 1) = (2x + 1) \times 1$. When factoring out $(2x + 1)$, the second term does not vanish into thin air; it leaves behind the number 1:
>
> $$B = \underline{(2x + 1)}(2x + 1) - \underline{(2x + 1)} \times 1 = \underline{(2x + 1)}[(2x + 1) - 1] = (2x + 1)(2x) = 2x(2x + 1).$$
>
> Writing $B = (2x + 1)(2x + 1) = (2x + 1)^2$ is a disastrous error!

4.2 Second Technique: Recognizing a Special Product

When no common factor is shared across all terms, inspect the overall architecture of the expression:

1. **Three terms: two positive perfect squares with a plus sign in the middle:**

$$x^2 + 6x + 9 = x^2 + 2 \times x \times 3 + 3^2 = (x + 3)^2.$$

2. **Three terms: two positive perfect squares with a minus sign in the middle:**

$$4x^2 - 20x + 25 = (2x)^2 - 2 \times (2x) \times 5 + 5^2 = (2x - 5)^2.$$

3. **Two terms only, separated by a minus sign: it is the difference of two squares!**

$$x^2 - 49 = x^2 - 7^2 = (x + 7)(x - 7).$$

$$9x^2 - 16 = (3x)^2 - 4^2 = (3x + 4)(3x - 4).$$

---

5. The Zero-Product Equation: The Key to Higher Degrees

In Chapter 22, we learned how to solve first-degree (linear) equations, such as $3x + 5 = 14$. But how can we solve an equation involving x^2, such as:

$$(2x - 6)(3x + 9) = 0 \quad \text{or} \quad x^2 - 25 = 0 \ \ ?$$

5.1 The Fundamental Property of Zero

Ask yourself this question of childlike simplicity: *If I multiply two secret numbers A and B, and the result equals zero ($A \times B = 0$), what can you assert with absolute certainty?*

If you choose $A = 4$, for the product to be 0, B must **unconditionally** equal 0. If neither of the two numbers is zero (for instance $2 \times 3 = 6$ or $(-5) \times 0.1 = -0.5$), their product can **never** equal zero!

📐 Theorem — Zero-Product Property.

A product of factors equals zero **if and only if** at least one of its factors equals zero:

$$A \times B = 0 \quad \Leftrightarrow \quad A = 0 \quad \text{or} \quad B = 0.$$

This theorem is momentous: it shatters a difficult second-degree equation into **two straightforward first-degree equations**!

5.2 Step-by-Step Method of Presentation

🧭 Method — Solving a Zero-Product Equation.

Let us solve the equation $(3x - 12)(2x + 8) = 0$.

1. **Introductory statement:** "A product of factors is zero if and only if at least one of the factors is zero."
2. **Split into two branches:**
$$3x - 12 = 0 \quad \text{or} \quad 2x + 8 = 0$$
3. **Solve each branch independently:**
   - First branch: $3x = 12 \Longrightarrow x = \frac{12}{3} = 4$.
   - Second branch: $2x = -8 \Longrightarrow x = \frac{-8}{2} = -4$.
4. **Conclusion:** The equation has two solutions: -4 and 4. We denote the solution set by $S = \{-4, 4\}$.

⚠️ Watch Out! Pitfall — Never Divide by x!

Faced with an equation like $x^2 = 5x$, a tempting but disastrous mistake is to divide both sides by x to conclude $x = 5$. In doing so, **you murder the solution $x = 0$!** (Because division by zero is strictly forbidden). **The correct method:** move everything to the left to produce a zero on the right, then factor:

$$x^2 - 5x = 0 \Leftrightarrow x(x - 5) = 0 \Leftrightarrow x = 0 \quad \text{or} \quad x - 5 = 0 \Leftrightarrow x = 0 \quad \text{or} \quad x = 5.$$

There are genuinely two solutions: 0 and 5.

---

6. Equations of the Form $x^2 = a$

The equation $x^2 = a$ is the purest of second-degree equations: what number, multiplied by itself, yields a?
The answer depends entirely on the **sign of a**.

6.1 The Three Possible Cases

1. **If $a > 0$ (a is strictly positive):** There exist **two opposite solutions**: $\sqrt{a}$ and $-\sqrt{a}$. *Rigorous proof by factoring:*
$$x^2 = a \Leftrightarrow x^2 - a = 0.$$
Since $a > 0$, we can express a as $(\sqrt{a})^2$. The equation becomes:
$$x^2 - (\sqrt{a})^2 = 0.$$
This is the third special product $A^2 - B^2 = (A - B)(A + B)$!
$$(x - \sqrt{a})(x + \sqrt{a}) = 0.$$
By the zero-product property:
$$x - \sqrt{a} = 0 \Longrightarrow x = \sqrt{a} \quad \text{or} \quad x + \sqrt{a} = 0 \Longrightarrow x = -\sqrt{a}.$$
Example: The equation $x^2 = 9$ has solutions $\sqrt{9} = 3$ and $-\sqrt{9} = -3$. Indeed, $3 \times 3 = 9$ and $(-3) \times (-3) = 9$.
2. **If $a = 0$:** The equation $x^2 = 0$ has **a single unique solution**: $x = 0$.
3. **If $a < 0$ (a is strictly negative):** The equation $x^2 = a$ has **no real solutions**. *Why?* The square of any real number is always non-negative: $(+5)^2 = +25$ and $(-5)^2 = +25$. It is physically and mathematically impossible for a real

number squared to yield a negative value like -16. *Example:* $x^2 = -4$ has no real solution. We write $S = \varnothing$ (the empty set).

Cas $a > 0$	Cas $a = 0$	Cas $a < 0$
Deux solutions opposées : $x = \sqrt{a}$ ou $x = -\sqrt{a}$	Une solution unique : $x = 0$	Aucune solution réelle : Impossible dans $\mathbb{R}$ ($S = \varnothing$)

---

7. True History: Al-Khwarizmi's Masterpiece and "Completing the Square"

How did Al-Khwarizmi solve an equation in 820 AD that entwined squares and linear terms, such as:

$$x^2 + 10x = 39 \quad ?$$

In that era, algebraic formulas with $\Delta = b^2 - 4ac$ did not yet exist. Al-Khwarizmi conceived an entirely visual geometric technique known as **completing the square** (*al-jabr wa'l-muqabala*).

1. He draws a square of side length x, having area x^2.
2. He must add $10x$ to it. He takes two rectangles of side lengths x and 5 (whose combined area is $5x + 5x = 10x$) and pastes them along two adjacent sides of the central square:

The area of this L-shaped region is $x^2 + 5x + 5x = x^2 + 10x$. And we know this total area must equal 39.

3. Al-Khwarizmi observes: *"This shape is almost a complete large square of side length $(x + 5)$! It lacks only the tiny corner in the lower right."*
4. What is the area of this missing corner? It is a square of $5 \times 5 = 25$!
5. Al-Khwarizmi resolves to **complete the square** by adding 25 to both pans of the scale:

$$(x^2 + 10x) + 25 = 39 + 25.$$

The geometric figure becomes a perfect large square of side length $(x + 5)$:

$$(x + 5)^2 = 64.$$

6. It now suffices to take the square root:

$$x + 5 = \sqrt{64} = 8.$$

7. Subtracting 5:

$$x = 8 - 5 = 3.$$

> **Geometric Origins vs Modern Algebraic Roots**
>
> For Al-Khwarizmi, x measured an actual field boundary, so only positive length $x = 3$ made physical sense. In modern algebra over $\mathbb{R}$, $X^2 = 64$ admits **two symmetric real roots**:
>
> $$X = \sqrt{64} = 8 \quad \text{or} \quad X = -\sqrt{64} = -8.$$
>
> Hence a second negative algebraic solution exists: $x + 5 = -8 \Leftrightarrow x = -13$. Indeed: $(-13)^2 + 10(-13) = 169 - 130 = 39$!

Let us verify: for $x = 3$, we have $3^2 + 10 \times 3 = 9 + 30 = 39$. It is completely exact! Every formula for quadratic equations taught in high school (in Chapter 35) directly descends from this geometric stroke of Al-Khwarizmi adding the missing corner.

---

8. Step-by-Step Worked Examples

Example 1: Expanding and simplifying with double distributivity and negative parentheses

Expand and simplify the expression:

$$C = (3x - 2)(4x + 5) - (2x - 1)(x - 3).$$

Detailed Solution: Beware of the minus sign "−" between the two products! To prevent sign errors, keep the entire second expansion inside large square brackets:

1. First product:

$$(3x-2)(4x+5) = 3x\times 4x + 3x\times 5 - 2\times 4x - 2\times 5 = 12x^2 + 15x - 8x - 10 = 12x^2 + 7x - 10.$$

2. Second product (inside brackets):

$$(2x-1)(x-3) = 2x\times x + 2x\times(-3) - 1\times x - 1\times(-3) = 2x^2 - 6x - x + 3 = 2x^2 - 7x + 3.$$

3. Assemble:

$$C = (12x^2 + 7x - 10) - [2x^2 - 7x + 3].$$

4. Remove brackets by reversing every interior sign:

$$C = 12x^2 + 7x - 10 - 2x^2 + 7x - 3.$$

5. Group like terms:

$$C = (12x^2 - 2x^2) + (7x + 7x) + (-10 - 3) = 10x^2 + 14x - 13.$$

Example 2: Expanding with special products

Expand the following expressions: 1. $D = (3x+4)^2$ 2. $E = (5x-2)^2$ 3. $F = (7x+3)(7x-3)$

Detailed Solution: 1. D is of the form $(a+b)^2$ with $a = 3x$ and $b = 4$:

$$D = (3x)^2 + 2\times(3x)\times 4 + 4^2 = 9x^2 + 24x + 16.$$

(Pitfall avoided: $(3x)^2 = 9x^2$, not $3x^2$!) 2. E is of the form $(a-b)^2$ with $a = 5x$ and $b = 2$:

$$E = (5x)^2 - 2\times(5x)\times 2 + 2^2 = 25x^2 - 20x + 4.$$

3. F is of the form $(a+b)(a-b)$ with $a = 7x$ and $b = 3$:

$$F = (7x)^2 - 3^2 = 49x^2 - 9.$$

Example 3: Factoring with a common binomial factor

Factor the expression:

$$G = (2x-5)(x+4) - (2x-5)(3x-1).$$

Detailed Solution: 1. Identify the common binomial factor present in both terms: $(2x-5)$. 2. Factor it out in front of large brackets:

$$G = \underline{(2x-5)}[(x+4) - (3x-1)].$$

3. Simplify inside the brackets, taking extreme care with the minus sign before the second binomial:

$$G = (2x-5)[x + 4 - 3x + 1].$$

4. Combine like terms:

$$G = (2x-5)(-2x+5).$$

Example 4: Factoring a complex difference of two squares

Factor the expression:

$$H = (2x+3)^2 - 36.$$

Detailed Solution: 1. Notice that $36 = 6^2$. 2. The expression is of the form $A^2 - B^2$ with $A = (2x+3)$ and $B = 6$. 3. Apply the identity $A^2 - B^2 = (A+B)(A-B)$:

$$H = [(2x+3) + 6]\times[(2x+3) - 6].$$

4. Simplify within each set of parentheses:

$$H = (2x + 9)(2x - 3).$$

Example 5: Solving a complex equation reducible to a zero product

Solve in the real numbers the equation:

$$(3x + 1)^2 = (x - 4)^2.$$

Detailed Solution: *Forbidden temptation:* Taking the square root directly on both sides without care, or expanding everything (which produces quadratic terms that are tricky to handle). *The correct method:* Move everything to the left to yield "$= 0$" and factor as a difference of two squares:

$$(3x + 1)^2 - (x - 4)^2 = 0.$$

This is of the form $A^2 - B^2 = 0$ with $A = (3x + 1)$ and $B = (x - 4)$.

$$[(3x + 1) + (x - 4)] \times [(3x + 1) - (x - 4)] = 0$$

$$[3x + 1 + x - 4] \times [3x + 1 - x + 4] = 0$$

$$(4x - 3)(2x + 5) = 0.$$

This is a **zero-product equation**! A product of factors equals zero if and only if at least one factor equals zero:

$$4x - 3 = 0 \quad \text{or} \quad 2x + 5 = 0$$

$$4x = 3 \Longrightarrow x = \frac{3}{4} = 0.75$$

$$2x = -5 \Longrightarrow x = -\frac{5}{2} = -2.5.$$

Conclusion: The equation has two solutions: $S = \left\{-\frac{5}{2}, \frac{3}{4}\right\}$.

Key Takeaways

1. **Double Distributivity:**
$$(a + b)(c + d) = ac + ad + bc + bd.$$
2. **The 3 Special Products (know them forward and backward):**
   - Square of a sum: $(a + b)^2 = a^2 + 2ab + b^2$
   - Square of a difference: $(a - b)^2 = a^2 - 2ab + b^2$
   - Difference of two squares: $(a + b)(a - b) = a^2 - b^2$
3. **Fatal Errors to Avoid:**
   - $(a + b)^2 \neq a^2 + b^2$ (the double product $2ab$ is missing!)
   - $(3x)^2 = 9x^2 \neq 3x^2$ (the square applies to the coefficient 3 as well)
4. **Factoring Strategy:**
   - First look for a common factor (a number, x, or a binomial in parentheses).
   - If no common factor exists, identify a special product ($a^2 - b^2$, $a^2 + 2ab + b^2$, or $a^2 - 2ab + b^2$).
5. **Zero-Product Equation:**
$$A \times B = 0 \Leftrightarrow A = 0 \quad \text{or} \quad B = 0.$$
6. **Equations of the Form $x^2 = a$:**
   - If $a > 0$: two solutions $x = \sqrt{a}$ and $x = -\sqrt{a}$.
   - If $a = 0$: one solution $x = 0$.
   - If $a < 0$: no real solution ($S = \varnothing$).

Exercises

Level : Direct Application

Exercise 23.1 Expand and simplify the following expressions: 1. $A = (x+3)(x+7)$ 2. $B = (2x+1)(x-4)$ 3. $C = (x-5)(3x-2)$

Exercise 23.2 Use special products to expand: 1. $D = (x+6)^2$ 2. $E = (x-5)^2$ 3. $F = (x+8)(x-8)$ 4. $G = (2x+3)^2$ 5. $H = (4x-1)^2$ 6. $I = (5x+2)(5x-2)$

Exercise 23.3 Compute cleverly without a calculator using a special product: 1. 102^2 2. 49^2 3. 61×59 4. 1003×997

Exercise 23.4 Factor by identifying an obvious common factor: 1. $J = 5x + 35$ 2. $K = x^2 + 8x$ 3. $L = 7x^2 - 14x$

Exercise 23.5 Factor using special products: 1. $M = x^2 - 81$ 2. $N = x^2 + 10x + 25$ 3. $P = x^2 - 12x + 36$ 4. $Q = 4x^2 - 9$

Exercise 23.6 Solve the following zero-product equations: 1. $(x-7)(x+2) = 0$ 2. $(2x-10)(3x+15) = 0$ 3. $x(4x-9) = 0$

Exercise 23.7 Solve the equations $x^2 = a$: 1. $x^2 = 49$ 2. $x^2 = 17$ 3. $x^2 = -9$ 4. $x^2 = 0$

Level : Deeper Reflection and Multi-Step Problems

Exercise 23.8 Expand and simplify the complex expressions: 1. $R = (3x-2)^2 - (x+4)(x-4)$ 2. $S = (2x+1)^2 - (3x-2)^2$

Exercise 23.9 Factor by identifying a common binomial block: 1. $T = (x-3)(2x+5) + (x-3)(x-1)$ 2. $U = (3x+2)(5x-1) - (3x+2)(2x+7)$ 3. $V = (4x-3)^2 + (4x-3)(x+2)$

Exercise 23.10 Factor the composite differences of squares: 1. $W = (x+2)^2 - 25$ 2. $Z = 49 - (2x-1)^2$ 3. $Y = (3x-5)^2 - (x+1)^2$

Exercise 23.11 Solve the equations by reducing them to a zero-product equation: 1. $x^2 - 36 = 0$ 2. $(2x+3)(x-1) = (2x+3)(3x-7)$ 3. $9x^2 - 25 = 0$ 4. $x^2 = 6x$

Exercise 23.12 Computational Algorithm and Mystery: Consider the following computational algorithm:

- Choose an integer.
- Add 1 to it.
- Square the result.
- Subtract the square of the starting number from the result.
- Subtract 1.

1. Test this algorithm with 3, 5, and 10. What pattern emerges?
2. Prove this conjecture for any starting number using algebra.

Exercise 23.13 The Magic Trick for Squaring Numbers Ending in 5: To mentally compute the square of a number ending in 5 (such as 35^2), the trick is to multiply the tens digit (3) by the next consecutive integer (4), yielding $3 \times 4 = 12$, and append 25 to the end: you get 1225! A number ending in 5 can be written algebraically as $10d + 5$, where d is the number of tens. Expand $(10d+5)^2$ and prove why this trick works every single time!

Level : Research Challenges and Olympiad-Style Riddles

Exercise 23.14 Solving in the Manner of Al-Khwarizmi (Completing the Square): Solve the equation $x^2 + 6x = 16$ following Al-Khwarizmi's geometric method step by step: 1. Express $x^2 + 6x$ in the form $(x + ...)^2 - ...$ 2. Deduce an equation of the form $(x+3)^2 = \text{number}$. 3. Find the positive value of x.

Exercise 23.15 The Grand Challenge of Pythagoras and Algebra: A right triangle has a hypotenuse measuring $(x+8)$ cm. Its two legs measure x cm and $(x+7)$ cm respectively. 1. Using the Pythagorean theorem, write an equation satisfied by x. 2. Expand and simplify this equation to bring it into the form $x^2 - 2x - 15 = 0$. 3. Verify that $x^2 - 2x - 15 = (x-5)(x+3)$. 4. Deduce the value of x and the lengths of the three sides of the triangle.

Exercise 23.16 The Difference of Two Cubes (For the Inquisitive Mind): 1. Expand the expression $(a-b)(a^2+ab+b^2)$. 2. What do you notice? 3. Deduce a factorization of x^3-8 (noting that $8=2^3$).

Solutions

Level

Solution 23.1 1. $A = x^2+7x+3x+21 = x^2+10x+21$. 2. $B = 2x^2-8x+x-4 = 2x^2-7x-4$. 3. $C = 3x^2-2x-15x+10 = 3x^2-17x+10$.

Solution 23.2 1. $D = x^2+2\times x\times 6+6^2 = x^2+12x+36$. 2. $E = x^2-2\times x\times 5+5^2 = x^2-10x+25$. 3. $F = x^2-8^2 = x^2-64$. 4. $G = (2x)^2+2\times 2x\times 3+3^2 = 4x^2+12x+9$. 5. $H = (4x)^2-2\times 4x\times 1+1^2 = 16x^2-8x+1$. 6. $I = (5x)^2-2^2 = 25x^2-4$.

Solution 23.3 1. $102^2 = (100+2)^2 = 100^2+2\times 100\times 2+2^2 = 10\,000+400+4 = 10\,404$. 2. $49^2 = (50-1)^2 = 50^2-2\times 50\times 1+1^2 = 2\,500-100+1 = 2\,401$. 3. $61\times 59 = (60+1)(60-1) = 60^2-1^2 = 3\,600-1 = 3\,599$. 4. $1003\times 997 = (1000+3)(1000-3) = 1000^2-3^2 = 1\,000\,000-9 = 999\,991$.

Solution 23.4 1. $J = 5(x+7)$. 2. $K = x(x+8)$. 3. $L = 7x(x-2)$.

Solution 23.5 1. $M = x^2-9^2 = (x+9)(x-9)$. 2. $N = x^2+2\times x\times 5+5^2 = (x+5)^2$. 3. $P = x^2-2\times x\times 6+6^2 = (x-6)^2$. 4. $Q = (2x)^2-3^2 = (2x+3)(2x-3)$.

Solution 23.6 1. $x-7=0$ or $x+2=0 \Longrightarrow x=7$ or $x=-2$. $S=\{-2,7\}$. 2. $2x-10=0 \Longrightarrow 2x=10 \Longrightarrow x=5$ or $3x+15=0 \Longrightarrow 3x=-15 \Longrightarrow x=-5$. $S=\{-5,5\}$. 3. $x=0$ or $4x-9=0 \Longrightarrow 4x=9 \Longrightarrow x=\frac{9}{4}=2.25$. $S=\{0,2.25\}$.

Solution 23.7 1. $x=\sqrt{49}=7$ or $x=-\sqrt{49}=-7$. $S=\{-7,7\}$. 2. $x=\sqrt{17}$ or $x=-\sqrt{17}$. $S=\left\{-\sqrt{17},\sqrt{17}\right\}$. 3. No real square is negative. No real solutions: $S=\varnothing$. 4. $x=0$. Exactly one solution: $S=\{0\}$.

Niveau

Solution 23.8 1. $R = (9x^2-12x+4)-(x^2-16) = 9x^2-12x+4-x^2+16 = 8x^2-12x+20$. 2. $S = (4x^2+4x+1)-(9x^2-12x+4) = 4x^2+4x+1-9x^2+12x-4 = -5x^2+16x-3$.

Solution 23.9 1. $T = (x-3)[(2x+5)+(x-1)] = (x-3)(3x+4)$. 2. $U = (3x+2)[(5x-1)-(2x+7)] = (3x+2)[5x-1-2x-7] = (3x+2)(3x-8)$. 3. $V = (4x-3)[(4x-3)+(x+2)] = (4x-3)(5x-1)$.

Solution 23.10 1. $W = (x+2)^2-5^2 = [(x+2)+5][(x+2)-5] = (x+7)(x-3)$. 2. $Z = 7^2-(2x-1)^2 = [7+(2x-1)][7-(2x-1)] = [7+2x-1][7-2x+1] = (2x+6)(-2x+8)$. (Factoring out 2 from each factor gives: $Z = 4(x+3)(-x+4)$). 3. $Y = [(3x-5)+(x+1)][(3x-5)-(x+1)] = [4x-4][2x-6] = 4(x-1)\times 2(x-3) = 8(x-1)(x-3)$.

Solution 23.11 1. $x^2-36=0 \Leftrightarrow (x-6)(x+6)=0 \Leftrightarrow x=6$ or $x=-6$. $S=\{-6,6\}$. 2. Move everything to the left: $(2x+3)(x-1)-(2x+3)(3x-7)=0$. Common factor $(2x+3)$: $(2x+3)[(x-1)-(3x-7)]=0 \Leftrightarrow (2x+3)(-2x+6)=0$. Branches: $2x+3=0 \Longrightarrow x=-\frac{3}{2}$ or $-2x+6=0 \Longrightarrow x=3$. $S=\left\{-\frac{3}{2},3\right\}$. 3. $(3x)^2-5^2=0 \Leftrightarrow (3x-5)(3x+5)=0 \Longrightarrow x=\frac{5}{3}$ or $x=-\frac{5}{3}$. $S=\left\{-\frac{5}{3},\frac{5}{3}\right\}$. 4. $x^2-6x=0 \Leftrightarrow x(x-6)=0 \Longrightarrow x=0$ or $x=6$. $S=\{0,6\}$.

Solution 23.12 1. For 3: $3+1=4 \to 4^2=16 \to 16-3^2=16-9=7 \to 7-1=6$ (twice 3). For 5: $5+1=6 \to 6^2=36 \to 36-25=11 \to 11-1=10$ (twice 5). For 10: $10+1=11 \to 11^2=121 \to 121-100=21 \to 21-1=20$ (twice 10). Conjecture: the algorithm always yields double the chosen number ($2n$). 2. Let n be the starting number. The algorithm computes: $(n+1)^2-n^2-1$. Expanding: $(n^2+2n+1)-n^2-1 = n^2+2n+1-n^2-1 = 2n$. The conjecture is rigorously proved for any integer n.

Solution 23.13 Expand $(10d+5)^2$ with the square of a sum identity:

$$(10d+5)^2 = (10d)^2+2\times(10d)\times 5+5^2 = 100d^2+100d+25.$$

Factor out $100d$ from the first two terms:

$$100d(d+1)+25 = [d\times(d+1)]\times 100+25.$$

Multiplying $d \times (d+1)$ by 100 exactly shifts the result two places to the left (into the hundreds place), and adding 25 places "25" directly in the tens and units places! The trick is mathematically infallible.

Level

Solution 23.14 1. We know that $(x+3)^2 = x^2 + 6x + 9$. Therefore $x^2 + 6x = (x+3)^2 - 9$. 2. The equation $x^2 + 6x = 16$ becomes:

$$(x+3)^2 - 9 = 16 \Leftrightarrow (x+3)^2 = 16 + 9 = 25.$$

3. Since $25 = 5^2$, we have $x + 3 = 5$ (for the positive solution sought by Al-Khwarizmi), yielding $x = 5 - 3 = 2$. (The modern second algebraic solution is $x + 3 = -5 \Longrightarrow x = -8$).

Solution 23.15 1. By the Pythagorean theorem:

$$x^2 + (x+7)^2 = (x+8)^2.$$

2. Expand both sides:

$$x^2 + (x^2 + 14x + 49) = x^2 + 16x + 64$$

$$2x^2 + 14x + 49 = x^2 + 16x + 64.$$

Move everything to the left:

$$(2x^2 - x^2) + (14x - 16x) + (49 - 64) = 0 \Leftrightarrow x^2 - 2x - 15 = 0.$$

3. Expand $(x-5)(x+3)$:

$$(x-5)(x+3) = x^2 + 3x - 5x - 15 = x^2 - 2x - 15.$$

The identity is verified! 4. The equation becomes $(x-5)(x+3) = 0$. The solutions are $x = 5$ or $x = -3$. Since a physical length cannot be negative, we retain only $x = 5$. The three sides of the triangle measure:

- First leg: $x = 5$ cm
- Second leg: $x + 7 = 5 + 7 = 12$ cm
- Hypotenuse: $x + 8 = 5 + 8 = 13$ cm. Check: $5^2 + 12^2 = 25 + 144 = 169 = 13^2$. It is a perfect Pythagorean triple!

Solution 23.16 1. Expand by distributive properties:

$$(a-b)(a^2 + ab + b^2) = a \times a^2 + a \times ab + a \times b^2 - b \times a^2 - b \times ab - b \times b^2$$

$$= a^3 + a^2b + ab^2 - a^2b - ab^2 - b^3.$$

All intermediate terms cancel pairwise!

$$= a^3 - b^3.$$

2. We observe that the difference of two cubes factors according to the remarkable formula:

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2).$$

3. For $x^3 - 8 = x^3 - 2^3$, apply with $a = x$ and $b = 2$:

$$x^3 - 8 = (x-2)(x^2 + 2x + 2^2) = (x-2)(x^2 + 2x + 4).$$

The Question Opening the Next Chapter

Thanks to special products and the zero-product property, you now know how to master perfect equalities and strict balance. But in real life and commerce, scale pans rarely rest at the exact same level: one is often heavier than the other, a subscription price becomes more advantageous than another beyond a certain threshold, and two merchants compete for customers.

What happens when the scale tilts? How can we manipulate an **inequality** without turning the world upside down, and how do we determine the intersection of two merchant destinies? That is the mystery we will unravel in the next chapter: **The Two Merchants and Inequalities**.

CHAPTER 24: The Two Merchants and Inequalities

> "As long as both pans remain level, the world sleeps in motionless peace. But the moment one pan plunges, motion is born. Inequality is not a defect of justice: it is the driving engine of commerce, the breakeven threshold, the exact frontier where one decision becomes superior to another."

What You Are About to Discover

- How to reason about an **unbalanced scale** without tipping it over.
- The notations $\leq, \geq, <, >$: what they mean and how to read them without hesitation.
- The golden rule of inequalities: why adding or subtracting preserves order, why multiplying by a positive number changes nothing…
- … and why **multiplying or dividing by a negative number REVERSES the direction** of the inequality (the mystery of the mirror and the thermometer).
- How to solve a **first-degree inequality** and plot its solutions on a **number line** using oriented brackets.
- How to unravel the secret of two unknowns at once using **systems of linear equations**: the **substitution method** and the **elimination (linear combination) method**.
- The luminous geometric interpretation: a system is nothing other than the **point of intersection of two lines**.
- Exciting real-world applications: comparing merchant pricing schemes, calculating breakeven points, and formulating perfect mixtures.

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1. The Problem: The Spice Fair of Venice and the Merchant's Dilemma

The year is **1340**, upon the Rialto quayside in **Venice**. The galleys have returned from the Levant, their holds heavy with cinnamon, black pepper, and saffron. Two Venetian merchants, Giovanni and Marco, have set up their stalls facing each other in the bustling marketplace. Both sell the exact same first-press olive oil from Crete.

Yet each offers a different pricing scheme to entice tavern keepers and ship captains:

- **Merchant Giovanni** announces: *"Free entry! No membership fee! I sell every liter of olive oil at the fixed price of 6 ducats per liter."*
- **Merchant Marco**, more cunning, proposes: *"Purchase a seasonal patron card for 30 ducats, and I will deliver every liter of oil to you for only 4 ducats per liter!"*

> 💡 **Solved Example — Comparing Pricing Plans (Affine Modeling)**
>
> - Giovanni's Tariff: Total Price = 6 × Number of liters
> - Marco's Tariff : Total Price = 30 + 4 × Number of liters

A tavern keeper stops, hesitating, clutching the purse at his belt. He asks his young apprentice: *"From whom should we buy our oil this season?"*

The apprentice replies with common sense: *"That depends on our consumption, master! If we only purchase 2 liters for the week, Giovanni costs us* $2 \times 6 = 12$ *ducats, whereas with Marco we would pay* $30 + 4 \times 2 = 38$ *ducats: Giovanni is far cheaper. But if we run a large tavern and purchase 50 liters, Giovanni would cost us* $50 \times 6 = 300$ *ducats, while Marco would charge us only* $30 + 4 \times 50 = 230$ *ducats: Marco becomes significantly less expensive!"*

The master tavern keeper knits his brow: *"Very well. But at* ***exactly how many liters*** *does Marco's price become more advantageous than Giovanni's?"*

This is no longer an equality to be solved, but an **inequality**:

$$30 + 4x < 6x.$$

We seek the region where the total expenditure with Marco is *strictly less than* that with Giovanni.

And that is not all. The next morning, Giovanni and Marco partner together to sell a lot of 100 amphorae containing a total of 380 liters of oil. Some amphorae hold 3 liters, while others hold 5 liters. How many amphorae of each kind are there? This is a problem with **two unknowns**, where two conditions must be satisfied simultaneously.

To answer the tavern keeper and pierce the secrets of the market, we must forge two new mathematical tools: **inequalities** and **linear systems**.

---

2. The Language of Inequalities: Comparing Quantities

2.1 Symbols and Reading Direction

In mathematics, an equality ($=$) asserts that two quantities have the exact same value. An **inequality** captures an imbalance: one quantity is greater or smaller than another.

Symbol	Meaning	Read Aloud	Example
$<$	Strictly less than	“is strictly less than”	$3 < 7$
$>$	Strictly greater than	“is strictly greater than”	$8 > 2$
$\leq$ or $\leq$	Less than or equal to	“is less than or equal to” (at most)	$x \leq 10$ (x is 10 or less)
$\geq$ or $\geq$	Greater than or equal to	“is greater than or equal to” (at least)	$x \geq 18$ (x is 18 or more)

The Arrowhead Visual Rule —

The symbol always opens wide toward the larger number and points its sharp tip toward the smaller number:

$$\text{BIG} > \text{small} \qquad \text{small} < \text{BIG}\ .$$

- **Strict inequality ($<$ or $>$) :** Equality is strictly excluded. If age $a > 12$, being exactly 12 years old does not qualify.
- **Non-strict (weak) inequality ($\leq$ or $\geq$) :** Equality is accepted. If test score $s \geq 10$, scoring exactly 10 is sufficient to pass.

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3. The Rules of the Tilted Scale

Let us return to Al-Khwarizmi’s two-pan scale, but this time, **the beam tilts**. The left pan sits higher than the right pan: the left side is lighter than the right side ($A < B$).

What occurs if we perform identical operations on both pans simultaneously?

3.1 Rule 1: Adding or Subtracting the Same Number

If you place 2 kg of lead on the left pan AND 2 kg of lead on the right pan, the right pan remains heavier. The direction of the tilt does not budge by a millimeter. Likewise, if you remove 3 kg from both sides, the scale preserves its exact tilt.

Fundamental Rule 1 — Addition and Subtraction.

The direction of an inequality does not change if you **add** or **subtract** the same number from both sides:

$$\text{If } \ a < b, \ \text{ then } \ a + c < b + c \quad \text{and} \quad a - c < b - c.$$

Numerical Example: We know that $3 < 8$.

- Add 5: $3 + 5 < 8 + 5 \Longrightarrow 8 < 13$ (true).

- Subtract 10: $3 - 10 < 8 - 10 \Longrightarrow -7 < -2$ (true, -7° C is colder than -2° C).

3.2 Rule 2: Multiplying or Dividing by a POSITIVE Number

If you triple the contents of the left pan and triple the contents of the right pan, the heavier side becomes three times heavier, and the lighter side three times lighter. The direction of imbalance remains unchanged.

Fundamental Rule 2 — Multiplying and Dividing by a POSITIVE Number ($c > 0$).

The direction of an inequality does not change if you **multiply** or **divide** both sides by the same **strictly positive** number:

$$\text{If } \ a < b \ \text{ and } \ c > 0, \ \text{then } \ a \times c < b \times c \quad \text{and} \quad \frac{a}{c} < \frac{b}{c}.$$

Numerical Example: We have $-4 < 6$. Multiply by 3:

$$-4 \times 3 < 6 \times 3 \Longrightarrow -12 < 18 \quad (\text{still true}) \ .$$

3.3 Rule 3: The Great Reversal! Multiplying or Dividing by a NEGATIVE Number

Here unfolds the most surprising phenomenon in algebra, the one that trips up countless students.

Take two self-evident numbers:

$$2 < 5.$$

Multiply both sides by -1. What do they become?

- On the left: $2 \times (-1) = -2$.
- On the right: $5 \times (-1) = -5$.

Now compare -2 and -5. Which number is greater? On a thermometer, -2° C is warmer than -5° C. On the number line, -2 is located **to the right** of -5:

$$-2 > -5 \ \ !$$

The inequality has flipped! The sign $<$ became $>$.

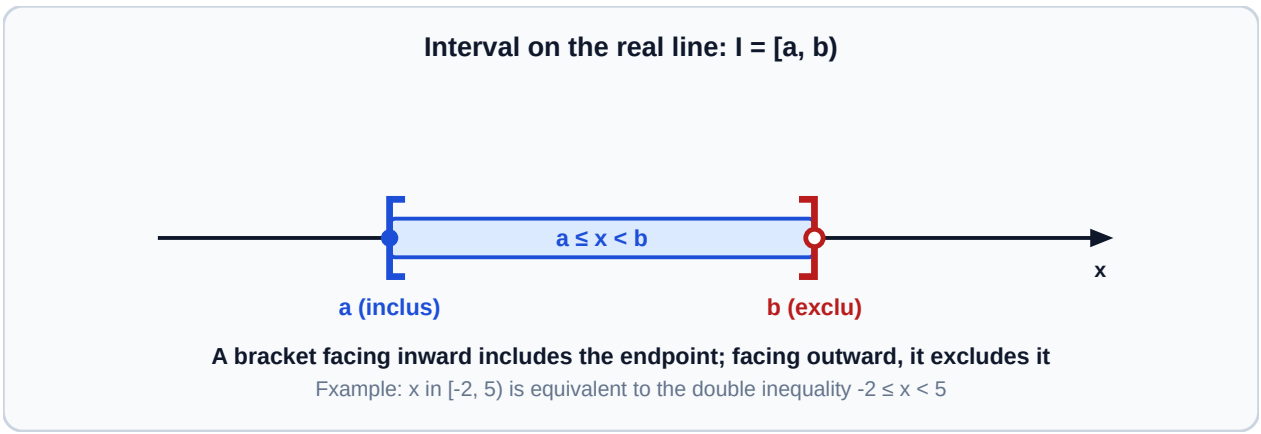

Figure 69: Intersection and union of intervals on the real number line

Fundamental Rule 3 — Multiplying and Dividing by a NEGATIVE Number ($c < 0$).

Whenever you multiply or divide both sides of an inequality by a STRICTLY NEGATIVE number, YOU MUST REVERSE THE DIRECTION OF THE INEQUALITY SYMBOL!

$$\text{If } \ a < b \ \text{ and } \ c < 0, \ \text{then } \ a \times c > b \times c \quad \text{and} \quad \frac{a}{c} > \frac{b}{c}.$$

Why this reversal? Two concrete insights: 1. **The Mirror Reflection:** Multiplying by -1 rotates the number line through a half-turn around zero. Whatever lay farthest to the right in the positive realm ($+5$) is hurled farthest to the left in the negative realm (-5). The champion of greatness becomes the champion of smallness! 2. **The Debt Analogy:**

Is having \$200 in your account better than having \$500? No, $200 < 500$. But having a **debt** of \$200 (-200) is vastly preferable to having a debt of \$500 (-500)! Therefore, $-200 > -500$.

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4. Solving Inequalities and Graphical Representation

4.1 What Is an Inequality?

A first-degree inequality is an inequality involving an unknown x. Unlike a standard linear equation which typically yields a single isolated solution ($x = 3$), an inequality generally possesses **infinitely many solutions**: an entire continuum of numbers forming an interval.

4.2 Step-by-Step Solution Example

Let us solve the inequality from our merchants' pricing problem:

$$30 + 4x < 6x.$$

1. **Group the variable terms:** Subtract $4x$ from both sides (Rule 1: subtraction preserves direction):
$$30 + 4x - 4x < 6x - 4x$$
$$30 < 2x.$$
2. **Isolate x:** Divide both sides by $+2$ (Rule 2: division by a positive number preserves direction):
$$\frac{30}{2} < \frac{2x}{2} \Longrightarrow 15 < x \quad \text{which means} \quad x > 15.$$
3. **Conclusion for the Merchants:** As soon as the tavern keeper purchases **strictly more than 15 liters of oil**, Marco's pricing becomes the more economical choice! At exactly 15 liters, both merchants charge the identical sum ($15 \times 6 = 90$ ducats and $30 + 4 \times 15 = 90$ ducats).

4.3 An Example Involving Symbol Reversal

Solve the inequality:

$$-3x + 7 \leq 22.$$

1. Subtract 7 from both sides:
$$-3x \leq 22 - 7 \Longrightarrow -3x \leq 15.$$
2. Divide by the coefficient of x, which is **-3 (negative)**. **Red alert: immediately flip the inequality sign!**
$$\frac{-3x}{-3} \geq \frac{15}{-3}$$
$$x \geq -5.$$

   The solutions are all real numbers greater than or equal to -5.

4.4 Graphing Solutions on the Number Line

Because there are infinitely many solutions, we depict them visually by shading the corresponding ray on the number line.

How do we indicate whether the boundary number itself is included or excluded? **Through the orientation of brackets!**

- **The bracket faces TOWARD the shaded solution set:** the boundary value is **included** ($\leq$ or $\geq$). Example for $x \geq -5$:
- **The bracket turns its BACK to the shaded solution set:** the boundary value is **excluded** ($<$ or $>$). Example for $x > 15$:

Method — Writing Solutions in Interval Notation.

In standard mathematics, the solution set S is written using intervals:

- For $x \geq -5$: $S = [-5, +\infty[$ (or $[-5, +\infty)$, read as "the half-open interval from -5 inclusive to positive infinity").
- For $x < 3$: $S =]-\infty, 3[$ (or $(-\infty, 3)$, read as "from negative infinity to 3 exclusive"). The symbol ∞ ("infinity") **always** receives an open bracket, because infinity is a boundless direction, not a reachable number that can be closed in.

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5. Systems of Two Linear Equations in Two Variables

Let us revisit the second riddle from Venice: *A merchant sells a lot of 100 amphorae of oil. There are small amphorae holding 3 liters and large amphorae holding 5 liters. The total volume of oil is 380 liters. How many amphorae of each kind are there?*

Here, there are **two unknown quantities**:

- Let x be the number of small amphorae (3 L).
- Let y be the number of large amphorae (5 L).

Each sentence in the problem provides an equation: 1. *"100 amphorae in total"* $\Longrightarrow x + y = 100$. 2. *"380 liters in total"* $\Longrightarrow 3x + 5y = 380$.

These two equations constitute a **system of two linear equations in two variables**. We bind them together with a curly brace:

$$\begin{cases} x + y = 100 \quad (L_1) \\ 3x + 5y = 380 (L_2) \end{cases}$$

Solving this system means determining the unique pair (x, y) that makes both equations simultaneously true. Two universal methods accomplish this.

5.1 Method 1: The Substitution Method (The Infiltrated Agent)

The principle is straightforward: use one equation to isolate one variable in terms of the other, then "inject" that expression into the second equation.

1. **Isolate an easy variable:** From the first equation $x + y = 100$, isolate x:
$$x = 100 - y.$$
2. **Substitute into the other equation:** In the second equation $3x + 5y = 380$, replace the symbol x with $(100 - y)$:
$$3(100 - y) + 5y = 380.$$
Magic: the variable x has vanished! We now have a standard equation in the single variable y!
3. **Solve for y:**
$$300 - 3y + 5y = 380$$
$$300 + 2y = 380$$
$$2y = 380 - 300 = 80$$
$$y = \frac{80}{2} = 40.$$
4. **Compute the first unknown x:** Since $x = 100 - y$:
$$x = 100 - 40 = 60.$$
5. **Conclusion and Verification:** The merchant sold 60 small amphorae and 40 large amphorae. Verification:
   - Total amphorae: $60 + 40 = 100$ (exact).

- Total liters: $60 \times 3 + 40 \times 5 = 180 + 200 = 380$ liters (exact!). The solution pair is written: $(x, y) = (60, 40)$.

When to use substitution? Whenever one of the variables has a coefficient of 1 or -1 (such as x or y in $x + y = 100$), because it can be isolated instantly without introducing messy fractions.

5.2 Method 2: The Linear Combination / Elimination Method (The Elimination Duel)

What should we do when every variable is saddled with an awkward coefficient, such as:

$$\begin{cases} 2x + 3y = 19 (L_1) \\ 5x - 2y = 19 (L_2) \end{cases}$$

Isolating x in the first equation would produce $x = \frac{19-3y}{2}$, cluttering the calculation with cumbersome fractions.

The **elimination method** (also known as addition or Gaussian elimination) multiplies the equations by strategically chosen numbers so that the coefficients of one variable become **exact opposites** (for instance, $+6y$ and $-6y$). By adding the two equations side by side, that variable is liquidated instantly!

Let us solve this system step by step:

1. **Choose the variable to eliminate:** Let us eliminate y. The coefficient of y is $+3$ in (L_1) and -2 in (L_2). The least common multiple of 3 and 2 is 6.
2. **Multiply the equations:**
   - Multiply the entire equation (L_1) by 2:
   $$2 \times (2x + 3y) = 2 \times 19 \Longrightarrow 4x + 6y = 38.$$
   - Multiply the entire equation (L_2) by 3:
   $$3 \times (5x - 2y) = 3 \times 19 \Longrightarrow 15x - 6y = 57.$$
3. **Add the two equations vertically:**

```
     4x  +  6y  =  38
+   15x  -  6y  =  57
--------------------
    19x  +   0  =  95
```

   The y terms have cancelled out entirely!
4. **Solve for x:**
$$19x = 95 \Longrightarrow x = \frac{95}{19} = 5.$$
5. **Solve for y:** Substitute $x = 5$ into either original equation, for instance (L_1):
$$2(5) + 3y = 19 \Longrightarrow 10 + 3y = 19 \Longrightarrow 3y = 9 \Longrightarrow y = 3.$$
6. **Conclusion:** The unique solution to the system is the pair $(x, y) = (5, 3)$.

---

6. The Geometric Interpretation: Two Intersecting Lines

Every first-degree equation in two variables, such as $2x + 3y = 19$, corresponds graphically to a **straight line** in the Cartesian plane (Chapter 20). Every point (x, y) lying on the line represents a valid solution to that equation.

Solving the system means finding a point (x, y) that lies **simultaneously on line (d_1) and on line (d_2)**.

Three geometric configurations can occur:

1. **The lines intersect (cross at a single unique point):** This is the standard case. The system has **one unique solution** (x_0, y_0).
$$\{x + y = 4$$
2. **The lines are strictly parallel (they never meet):** The system has **no solution** ($S = \varnothing$). *Example:* $x + y = 10$ (the sum of two numbers cannot equal 4 and 10 at the same time!).

3. **The lines are coincident (they lie right on top of each other):** Both equations are merely different algebraic representations of the exact same line (e.g., $x + y = 2$ and $2x + 2y = 4$). The system possesses **infinitely many solutions**.

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7. True History: The Secret Counting Rods of the Nine Chapters

Two centuries before our era, during the Han Dynasty in China, imperial mathematicians were already solving systems of linear equations to manage grain harvests, labor levies, and imperial taxes.

In the crowning jewel of Chinese mathematics, *The Nine Chapters on the Mathematical Art* (*Jiuzhang Suanshu*), Chapter 8 is entitled *Fangcheng* (“Rectangular Arrays”).

Chinese scholars arranged red bamboo rods (for positive numbers) and black rods (for negative numbers) on a felt counting board. Each column represented an equation and each row represented a different grade of grain (top-grade millet, medium-grade millet, low-grade millet):

> 💡 **Discovery & History — Chapter VIII of the Chinese Nine Chapters**
>
> • 3 bundles top millet] + [2 bundles medium millet] + [1 bundle
>
> low millet] = 39 measures
>
> • 2 bundles top millet] + [3 bundles medium millet] + [1 bundle low millet] = 34 measures
>
> • 1 bundle top millet] + [2 bundles medium millet] + [3 bundles low millet] = 26 measures

By sliding rods across columns, multiplying a column and subtracting it from another, they systematically cleared entries to produce zeroes. This 2,200-year-old Chinese algorithm is identical to **Gaussian elimination**, taught today in universities worldwide and programmed into modern supercomputing linear algebra libraries!

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8. Step-by-Step Worked Examples

Example 1: Solving an inequality with parentheses and basic terms

Solve the following inequality over the real numbers and sketch the solution set:

$$4(2x - 3) - 5x \geq 7x - 24.$$

Detailed Solution: 1. Expand the left-hand side:

$$8x - 12 - 5x \geq 7x - 24 \Longrightarrow 3x - 12 \geq 7x - 24.$$

2. Group the x terms on the left by subtracting $7x$:

$$3x - 7x - 12 \geq -24 \Longrightarrow -4x - 12 \geq -24.$$

3. Add 12 to both sides:

$$-4x \geq -24 + 12 \Longrightarrow -4x \geq -12.$$

4. Divide by the coefficient of x, which is $\mathbf{-4}$ **(negative)**. **Reverse the inequality symbol ($\geq$ becomes $\leq$):**

$$x \leq \frac{-12}{-4} \Longrightarrow x \leq 3.$$

5. Graphical representation: The solutions are all real numbers less than or equal to 3. The bracket is closed at 3 and oriented leftward toward the shaded region:

Solution set: $S =]-\infty, 3]$ (or $(-\infty, 3]$).

Example 2: Inequality with a denominator and strict sign control

Solve the inequality:

$$\frac{2-5x}{3} > 4.$$

Detailed Solution: 1. Multiply both sides by 3. Since $3 > 0$, the inequality direction is preserved:

$$2-5x > 4 \times 3 \Longrightarrow 2-5x > 12.$$

2. Subtract 2:

$$-5x > 12-2 \Longrightarrow -5x > 10.$$

3. Divide by **−5 (negative)**, which flips the inequality symbol ($>$ becomes $<$):

$$x < \frac{10}{-5} \Longrightarrow x < -2.$$

Conclusion: $S =]-\infty, -2[$ (or $(-\infty, -2)$). The bracket at -2 turns its back on the solutions since -2 is excluded.

Example 3: System solved by substitution

Solve the system:

$$\begin{cases} 3x - y = 7 \quad (L_1) \\ 2x + 4y = 42 (L_2) \end{cases}$$

Detailed Solution: 1. In (L_1), the variable y has a coefficient of -1, allowing quick isolation:

$$-y = 7 - 3x \Longrightarrow y = 3x - 7.$$

2. Substitute y by $(3x-7)$ in equation (L_2):

$$2x + 4(3x-7) = 42$$

$$2x + 12x - 28 = 42$$

$$14x - 28 = 42$$

$$14x = 42 + 28 = 70 \Longrightarrow x = \frac{70}{14} = 5.$$

3. Compute y using the formula from step 1:

$$y = 3(5) - 7 = 15 - 7 = 8.$$

4. Check in both original equations:

- (L_1): $3(5) - 8 = 15 - 8 = 7$ (true).
- (L_2): $2(5) + 4(8) = 10 + 32 = 42$ (true).

Conclusion: The unique solution pair is $(x, y) = (5, 8)$.

Example 4: System solved by linear combination (elimination)

Solve the system:

$$\begin{cases} 4x + 3y = 1 \ \ (L_1) \\ 6x - 5y = 20 (L_2) \end{cases}$$

Detailed Solution: Let us eliminate y. The coefficients of y are $+3$ and -5. Their least common multiple is 15. 1. Multiply (L_1) by 5:

$$5 \times (4x + 3y) = 5 \times 1 \Longrightarrow 20x + 15y = 5.$$

2. Multiply (L_2) by 3:

$$3 \times (6x - 5y) = 3 \times 20 \Longrightarrow 18x - 15y = 60.$$

3. Add the two new equations term by term:

$$(20x + 18x) + (15y - 15y) = 5 + 60$$

$$38x = 65 \Longrightarrow x = \frac{65}{38}.$$

Alternatively: Eliminate x by multiplying (L_1) by 3 and (L_2) by -2:

- $3 \times (L_1) \Longrightarrow 12x + 9y = 3$
- $-2 \times (L_2) \Longrightarrow -12x + 10y = -40$ Adding gives: $19y = -37 \Longrightarrow y = -\frac{37}{19} = -\frac{74}{38}$. Conclusion: The solution pair is $\left(\frac{65}{38}, -\frac{37}{19}\right)$.

Example 5: Real-world ticketing problem (Setting up a linear system)

To finance their end-of-year class trip, 9th-grade students organize a theater evening.

- Adult tickets cost €8.
- Child tickets cost €4.50.

At the end of the evening, the ticket booth reports 240 attendees and total receipts of €1,570. How many adults and how many children attended the play?

Detailed Solution: 1. **Choose variables:**

- Let a be the number of adults.
- Let c be the number of children.

2. **Set up equations:**
   - Total attendees: $a + c = 240$.
   - Total receipts in euros: $8a + 4.5c = 1570$.
3. **Solve by substitution:** From the first equation, $c = 240 - a$. Substitute into the second:

$$8a + 4.5(240 - a) = 1570$$
$$8a + 1080 - 4.5a = 1570$$
$$3.5a + 1080 = 1570$$
$$3.5a = 1570 - 1080 = 490$$
$$a = \frac{490}{3.5} = \frac{4900}{35} = 140.$$

4. **Compute the second variable:**

$$c = 240 - 140 = 100.$$

5. **Conclusion:** There were 140 adults and 100 children in the audience. Verification: $140 \times 8 + 100 \times 4.5 = 1120 + 450 = 1570$ €. Everything matches!

Key Takeaways

1. **Golden Rules of Inequalities:**
   - Adding or subtracting the same number **never** alters the inequality direction.
   - Multiplying or dividing by a **positive** number **preserves** the direction.
   - Multiplying or dividing by a **negative** number **MUST REVERSE THE DIRECTION** ($<$ becomes $>$, $\leq$ becomes $\geq$).
2. **Number Line Representation:**
   - Closed bracket [or] toward solutions if the value is **included** ($\leq, \geq$).
   - Open bracket] or [turning away from solutions if the value is **excluded** ($<, >$).
3. **System of Two Linear Equations in Two Variables:**

$$\{ax + by = c$$
$$a'x + b'y = c'$$

   - **Substitution Method:** isolate one variable in one equation and substitute it into the other (best when a coefficient is 1 or -1).

- **Elimination Method:** multiply equations to create opposite coefficients for one variable, then add equations vertically (universal).

4. **Geometric Vision:** The solution to a system represents the (x, y) coordinates of the point of intersection of the two associated lines.

Exercises

Level : Direct Application

Exercise 24.1 Fill in the appropriate inequality symbol ($<$ or $>$): 1. Given $x < 5$, compare $x + 4$ and 9. 2. Given $y \geq -2$, compare $y - 3$ and -5. 3. Given $a < b$, compare $3a$ and $3b$. 4. Given $u < v$, compare $-4u$ and $-4v$.

Exercise 24.2 Solve the basic first-degree inequalities: 1. $x + 7 < 15$ 2. $x - 9 \geq 3$ 3. $4x \leq 28$ 4. $-5x < 35$ 5. $-2x \geq -18$ 6. $\frac{x}{3} > -4$

Exercise 24.3 Plot the following solution sets on a number line: 1. $x > 4$ 2. $x \leq -1$ 3. $x \geq 2.5$ 4. $x < 0$

$$\{2x + y = 8$$

Exercise 24.4 Test whether the pair $(x = 3, y = 2)$ is a solution to the following systems: 1. $x - y = 1$ 2. $\{3x - 4y = 1$

$5x + 2y = 20$

$$\{y = 2x - 1 \quad \{x + 2y = 8$$

Exercise 24.5 Solve using the substitution method: 1. $3x + y = 14$ 2. $3x - 4y = 4$

$$\{2x + y = 11 \quad \{3x + 2y = 12$$

Exercise 24.6 Solve using the elimination method: 1. $3x - y = 9$ 2. $5x - 2y = 20$

Level : Deeper Reflection and Multi-Step Problems

Exercise 24.7 Solve and state the solution set in interval notation: 1. $5x - 8 < 2x + 7$ 2. $3(x - 2) - 4(2x + 1) \geq 5$ 3. $\frac{3x-1}{4} \leq \frac{x+2}{2}$

Exercise 24.8 Compound Inequalities (Double Bounds): Find all real numbers x satisfying simultaneously:

$$-3 \leq 2x + 1 < 9.$$

(Hint: subtract 1 from all three parts, then divide by 2).

$$\{4x + 3y = 25 \quad \{5x + 2y = 1 \quad \{2x - 3y = 7$$

Exercise 24.9 Solve the linear systems: 1. $2x - 5y = -7$ 2. $3x - 4y = 11$ 3. $4x - 6y = 10$ (What is happening geometrically?)

Exercise 24.10 The Amusement Park Problem: An amusement park offers two pricing plans:

- **Plan A:** €25 general admission providing unlimited access to all rides.
- **Plan B:** €9 basic admission, plus €2 per individual ride.

1. Write down the cost of each plan for taking x rides.
2. Set up and solve an inequality to determine at how many rides Plan A becomes more economical than Plan B.

Exercise 24.11 The Silver Alloy Problem: A jeweler has two silver ingots:

- The first ingot contains 80% pure silver.
- The second ingot contains 60% pure silver.

He wishes to melt these two ingots together to obtain exactly 500 grams of a new alloy containing 75% pure silver. What mass of each ingot must he melt?

Exercise 24.12 Geometry and Perimeter: A rectangle has a length that exceeds its width by 4 cm. Its perimeter is strictly less than 36 cm. What values can the width of this rectangle take?

Level : Research Challenges and Olympiad-Style Riddles

Exercise 24.13 Graphical Intersection of Two Rate Lines: In a Cartesian coordinate system, consider the two lines: $(d_1) : y = 2x + 1 \quad \text{and} \quad (d_2) : y = -x + 7$. 1. Find algebraically the coordinates of their intersection point K. 2. Solve graphically and algebraically the inequality $2x + 1 \leq -x + 7$. 3. Interpret the result: which line lies below the other?

Exercise 24.14 The Chinese Age Puzzle: A father says to his son: *"Today, I am four times as old as you. But in 20 years, I will be only twice as old as you."* What are the current ages of the father and the son?

Exercise 24.15 Parametric System (For Future Scientists): Consider the system:

$$\begin{cases} ax + 2y = 4 \\ 3x + y = 2 \end{cases}$$

For what value of parameter a does this system fail to have a unique solution pair? What can be said about the two associated lines for this value of a?

Solutions

Level

Solution 24.1 1. $x + 4 < 9$ (added 4 to both sides). 2. $y - 3 \geq -5$ (subtracted 3 from both sides). 3. $3a < 3b$ (multiplied by $3 > 0$). 4. $-4u > -4v$ (multiplied by $-4 < 0$, the inequality sign flips!).

Solution 24.2 1. $x < 15 - 7 \Longrightarrow x < 8$. 2. $x \geq 3 + 9 \Longrightarrow x \geq 12$. 3. $x \leq \frac{28}{4} \Longrightarrow x \leq 7$. 4. Division by $-5 < 0$: $x > \frac{35}{-5} \Longrightarrow x > -7$. 5. Division by $-2 < 0$: $x \leq \frac{-18}{-2} \Longrightarrow x \leq 9$. 6. Multiplication by $3 > 0$: $x > -12$.

Solution 24.3 1. Shaded region to the right of 4, open bracket facing left: $]4, +\infty[$ (or $(4, +\infty)$). 2. Shaded region to the left of -1, closed bracket facing left: $]-\infty, -1]$ (or $(-\infty, -1]$). 3. Shaded region to the right of 2.5, closed bracket facing right: $[2.5, +\infty[$ (or $[2.5, +\infty)$). 4. Shaded region to the left of 0, open bracket facing right: $]-\infty, 0[$ (or $(-\infty, 0)$).

Solution 24.4 1. $2(3) + 2 = 6 + 2 = 8$ (true) and $3 - 2 = 1$ (true). The pair $(3, 2)$ is indeed a solution to the first system. 2. $3(3) - 4(2) = 9 - 8 = 1$ (true). But $5(3) + 2(2) = 15 + 4 = 19 \neq 20$ (false!). The pair is not a solution to the second system.

Solution 24.5 1. Substitute $y = 2x - 1$ into the second equation: $3x + (2x - 1) = 14 \Longrightarrow 5x - 1 = 14 \Longrightarrow 5x = 15 \Longrightarrow x = 3$. Then $y = 2(3) - 1 = 5$. Solution: $(3, 5)$. 2. From the first equation: $x = 8 - 2y$. Second equation: $3(8 - 2y) - 4y = 4 \Longrightarrow 24 - 6y - 4y = 4 \Longrightarrow -10y = -20 \Longrightarrow y = 2$. Then $x = 8 - 2(2) = 4$. Solution: $(4, 2)$.

Solution 24.6 1. Add both equations directly ($+y$ and $-y$ cancel): $(2x + 3x) + (y - y) = 11 + 9 \Longrightarrow 5x = 20 \Longrightarrow x = 4$. Then $2(4) + y = 11 \Longrightarrow 8 + y = 11 \Longrightarrow y = 3$. Solution: $(4, 3)$. 2. Add both equations directly ($+2y$ and $-2y$ cancel): $8x = 32 \Longrightarrow x = 4$. Then $3(4) + 2y = 12 \Longrightarrow 12 + 2y = 12 \Longrightarrow 2y = 0 \Longrightarrow y = 0$. Solution: $(4, 0)$.

Level

Solution 24.7 1. $5x - 2x < 7 + 8 \Longrightarrow 3x < 15 \Longrightarrow x < 5$. Solution: $S =]-\infty, 5[$. 2. $3x - 6 - 8x - 4 \geq 5 \Longrightarrow -5x - 10 \geq 5 \Longrightarrow -5x \geq 15 \Longrightarrow x \leq -3$ (reversal). Solution: $S =]-\infty, -3]$. 3. Multiply by 4 (positive): $3x - 1 \leq 2(x + 2) \Longrightarrow 3x - 1 \leq 2x + 4 \Longrightarrow x \leq 5$. Solution: $S =]-\infty, 5]$.

Solution 24.8 Subtract 1 from all three parts:

$$-3 - 1 \leq 2x < 9 - 1 \Longrightarrow -4 \leq 2x < 8.$$

Divide all parts by $2 > 0$:

$$-2 \leq x < 4.$$

The solutions are all real numbers between -2 (included) and 4 (excluded). $S = [-2, 4[$.

Solution 24.9 1. Multiply the 2nd equation by -2: $-4x + 10y = 14$. Add to the 1st ($4x + 3y = 25$): $13y = 39 \Longrightarrow y = 3$. Then $2x - 5(3) = -7 \Longrightarrow 2x - 15 = -7 \Longrightarrow 2x = 8 \Longrightarrow x = 4$. Solution: $(4, 3)$. 2. Multiply the 1st equation by 2: $10x + 4y = 2$. Add to the 2nd ($3x - 4y = 11$): $13x = 13 \Longrightarrow x = 1$. Then $5(1) + 2y = 1 \Longrightarrow 2y = -4 \Longrightarrow y =$

-2. Solution: $(1, -2)$. 3. Multiply the 1st equation by 2: $4x - 6y = 14$. The 2nd equation is $4x - 6y = 10$. Subtracting the two equations gives: $(4x - 4x) - (6y - 6y) = 14 - 10 \Longrightarrow 0 = 4$, which is **impossible**! Geometrically, both lines share the same slope: they are **strictly parallel**. There is no intersection point, hence no solution ($S = \varnothing$).

Solution 24.10 1. For x rides: Cost of Plan A $= 25$ €; Cost of Plan B $= 9 + 2x$ €. 2. Plan A is more economical if:

$$25 < 9 + 2x \Leftrightarrow 25 - 9 < 2x \Leftrightarrow 16 < 2x \Leftrightarrow 8 < x \Leftrightarrow x > 8.$$

Since the number of rides is an integer, Plan A becomes more advantageous starting from **9 rides**.

Solution 24.11 Let x be the mass (in grams) of the first ingot (80%) and y the mass of the second (60%). 1. Total mass: $x + y = 500$. 2. Pure silver mass: $0.80x + 0.60y = 0.75 \times 500 = 375$. From the 1st equation: $y = 500 - x$. Substitute into the 2nd: $0.8x + 0.6(500 - x) = 375 \Longrightarrow 0.8x + 300 - 0.6x = 375 \Longrightarrow 0.2x = 75 \Longrightarrow x = \frac{75}{0.2} = 375$ g. Then $y = 500 - 375 = 125$ g. The jeweler must melt 375 g of the first ingot and 125 g of the second.

Solution 24.12 Let w be the width in cm. The length is $L = w + 4$. The perimeter is $P = 2(w + L) = 2(w + w + 4) = 2(2w + 4) = 4w + 8$. The problem states that $P < 36$, so:

$$4w + 8 < 36 \Longrightarrow 4w < 28 \Longrightarrow w < 7.$$

Furthermore, a geometric width must be strictly positive ($w > 0$). Thus, the width w can take any value strictly between 0 and 7 cm: $w \in]0, 7[$ (or $(0, 7)$).

Level

Solution 24.13 1. At the intersection point, the y-values are equal:

$$2x + 1 = -x + 7 \Longrightarrow 3x = 6 \Longrightarrow x = 2.$$

Then $y = 2(2) + 1 = 5$. The intersection point is $K(2, 5)$. 2. Solve $2x + 1 \leq -x + 7$:

$$3x \leq 6 \Longrightarrow x \leq 2.$$

The solution set is $S =]-\infty, 2]$ (or $(-\infty, 2]$). 3. Interpretation: for all $x \leq 2$, line (d_1) lies **below** (or intersects) line (d_2). Beyond $x = 2$, line (d_1) rises above line (d_2).

Solution 24.14 Let f be the father's current age and s the son's current age. 1. "Today, I am 4 times your age": $f = 4s$. 2. "In 20 years, father will be $f + 20$ and son will be $s + 20$":

$$f + 20 = 2(s + 20).$$

Substitute $f = 4s$ into the second equation:

$$4s + 20 = 2s + 40 \Longrightarrow 2s = 20 \Longrightarrow s = 10.$$

Then $f = 4 \times 10 = 40$. The father is currently 40 years old and the son is 10 years old. Check in 20 years: father will be 60, son will be 30, and 60 is indeed twice 30!

Solution 24.15 Isolate y in the second equation: $y = 2 - 3x$. Substitute into the first equation:

$$ax + 2(2 - 3x) = 4 \Longrightarrow ax + 4 - 6x = 4 \Longrightarrow (a - 6)x = 0.$$

- If $a \neq 6$, then $a - 6 \neq 0$, yielding a unique solution $x = 0$ (and $y = 2$).
- If $a = 6$, the equation becomes $0 \times x = 0$, which holds true for **all real numbers x**! In this case, both equations are equivalent ($6x + 2y = 4 \Leftrightarrow 3x + y = 2$). The two lines are **coincident**, yielding infinitely many solutions.

To get strictly parallel lines (no solution), the constant term would have had to differ (e.g. $6x + 2y = 5$). Conclusion: the critical value is $\boldsymbol{a = 6}$. For this value, the determinant vanishes ($a \times 1 - 2 \times 3 = a - 6 = 0$), and the system loses its unique solution.

The Question Opening the Next Chapter

Look at what we have just accomplished: to every number of liters of oil x, we assigned a total cost $P(x) = 6x$ or $P(x) = 30 + 4x$. To every number of amusement park rides, we mapped a total expenditure.

This powerful mathematical mechanism, where an input number generates a unique output number, is called a **function**. Why not look this machine directly in the face? What geometric trajectory is traced by a proportional relationship or

by a fixed subscription tariff? That is the grand journey awaiting us in Chapter 25: **The Doubling Machine and the Tariff Line**.

CHAPTER 25: The Doubling Machine and the Tariff Line

> "An equation is a snapshot: it freezes the balance in search of a motionless treasure. A function is the entire motion picture: it captures the unfolding of the world, where every cause produces an effect, and where every forward step lifts the traveler to a new altitude."

What You Are About to Discover

- What a **function** truly is: the marvelous "magic box" that ingests a number and outputs another according to an unchanging rule.
- Official mathematical vocabulary demystified: distinguishing the **input (preimage)** from the **output (image)**, and why we write $f(x)$ or $x \mapsto f(x)$.
- The three reflections of a function: its **formula**, its **table of values**, and its **graph**.
- The **linear function** $f(x) = ax$: the geometric embodiment of pure proportionality, the faithful straight line that never departs from the origin $(0, 0)$.
- The **affine function** $f(x) = ax + b$: the straight line of real-world subscription plans, its **slope (rate of change)** a, and its **y-intercept** b (the starting entry fee).
- How to graph and read these lines at a single glance: "run 1 to the right, rise by a."
- Thrilling real-life applications: cell phone plans, taxi meters, speeds, and the famous conversion between degrees Celsius and Fahrenheit.

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1. The Problem: The Magic Forge and the Boat Rental Meter

Imagine visiting a village perched on the shore of a grand alpine lake. At the far end of the pier, two craftsmen offer their services to travelers.

The first is an elderly clockmaker and inventor. In his workshop, he built a steel box equipped with a slot on top and a sliding drawer at the bottom. He calls it his **"Transforming Machine."**

- If you slip the number **3** into the slot, gears whirl, and the drawer ejects the number **11**.
- If you slip **5** in, the machine ejects **17**.
- If you slip **0** in, it ejects **2**.
- If you slip **-4** in, it ejects **-10**.

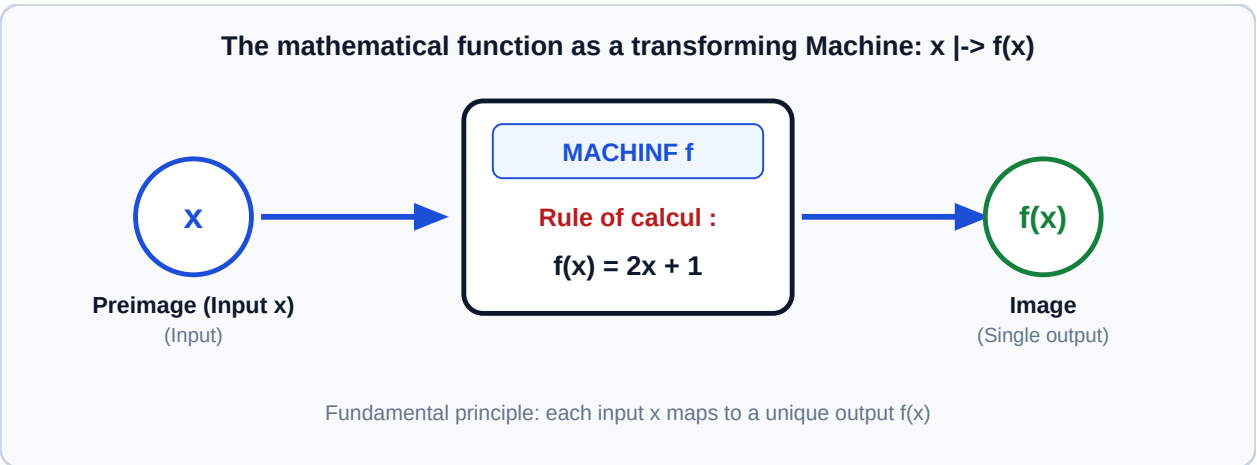

Figure 70: The function as Jacquard's input-output machine

The village children gather around: *"What secret rule is hidden inside the cogs of the box?"* Look closely:

- $3 \times 3 + 2 = 9 + 2 = 11$.
- $5 \times 3 + 2 = 15 + 2 = 17$.
- $0 \times 3 + 2 = 2$.
- $(-4) \times 3 + 2 = -12 + 2 = -10$.

The universal rule is unmasked: whatever number x is dropped into the box, the mechanism computes $3x + 2$.

Right beside the workshop, the **boat rental owner** posts his prices for rowing across the lake: *"To step aboard, each passenger pays a fixed boarding fee of €10 (life vest and insurance). Thereafter, each hour spent on the water costs €15."*

How much will a family pay to rent a boat for 1 hour? 2 hours? 4 hours?

- For 1 h: $10 + 15 \times 1 = 25$ €.
- For 2 h: $10 + 15 \times 2 = 40$ €.
- For 4 h: $10 + 15 \times 4 = 70$ €.
- For x hours: the total price is $T(x) = 15x + 10$.

Neither of these situations is an equation where we hunt for a single unknown x. They are **living rules of correspondence**, continuous bonds of dependence between two dynamic quantities. This is the very soul of the concept of a **function**, the backbone of modern mathematics and all natural sciences.

---

2. The Concept of a Function and Its Vocabulary

2.1 What Is a Function?

> **📐 Definition — Function.**
>
> A **function** is a mathematical process that takes an initial input value (the variable) and assigns to it **one, and only one**, output value (the result).

We typically designate a function by a letter: a lowercase letter such as f, g, or h (standing for "function"), or an evocative uppercase letter like T (for "Tariff") or V (for "Volume").

2.2 Input (Preimage) and Output (Image): Never Confuse Them!

This is the standard terminology you will encounter through senior high school and university. To keep them straight, remember the metaphor of a camera or portrait studio:

- **The input (preimage / argument)**: the value fed **into** the machine. This is x.
- **The output (image / value)**: the computed result that emerges **out** of the machine. This is $f(x)$.

> **⚠ Critical Warning!**
>
> - Every input produces **one and only one output** (the machine
>
> cannot produce two contradictory results when a single button is pressed).
> - However, a single output may have **multiple different inputs**! *Example:* If the machine squares numbers ($f(x) = x^2$), the output 9 possesses **two distinct inputs**: 3 (since $3^2 = 9$) and -3 (since $(-3)^2 = 9$).

2.3 Notations: How to Write and Read Functions

Written Notation	Spoken Phrasing	Exact Meaning
f	"the function f"	The name of the entire machine (the mapping).
$f(x)$	"f of x"	The **output value** produced for the input x.
$x \mapsto f(x)$	"x maps to f of x"	The process arrow (note the bar at the base of the arrow $\mapsto$).

Concrete Example: Consider the function $g(x) = 2x - 5$.

- Compute the image (output) of 4: Substitute $x = 4$: $g(4) = 2 \times 4 - 5 = 8 - 5 = 3$. We say: "The image of 4 under g is 3", or "4 is a preimage of 3 under g."
- Find the preimage (input) of 11: Find the value of x such that $g(x) = 11$. That is an equation!

  $$2x - 5 = 11 \Longrightarrow 2x = 16 \Longrightarrow x = 8.$$

  The input that yields 11 under g is 8.

---

3. The Three Faces of a Single Function

A function is an abstract mathematical object. To visualize and work with it, mathematicians rely on three complementary representations:

1. L'expression algébrique	2. Le tableau de valeurs	3. La courbe représentative
La formule exacte Exemple : $f(x) = 2x + 1$	Échantillons finis de couples $(x; f(x))$	Vision géométrique continue dans le repère

3.1 The Table of Values

Choose a few values for x on the top row, and calculate their outputs $f(x)$ on the bottom row:

x (inputs)	-2	-1	0	1	2	3
$f(x) = 2x + 1$ (outputs)	-3	-1	1	3	5	7

Each column provides the coordinates of a point: $(-2, -3)$, $(-1, -1)$, $(0, 1)$, $(1, 3)$, etc.

3.2 The Graph in a Coordinate Plane

Plot these coordinate pairs in a Cartesian plane:

- The horizontal axis (x-axis) displays the **inputs** $\boldsymbol{x}$.
- The vertical axis (y-axis) displays the **outputs** $\boldsymbol{y = f(x)}$.

> **Definition — Graph of a Function.**
>
> The graph of a function f (often denoted $\mathcal{C}_f$) is the collection of all points in the plane having coordinates $(x, f(x))$.

---

4. The Linear Function: Pure Proportionality $f(x) = ax$

4.1 Definition and Physical Meaning

A **linear function** is the simplest and purest of all functions: it is the algebraic translation of a **perfectly proportional relationship** (as studied in Chapter 13).

Definition — Linear Function.

A linear function is a function defined by a formula of the form:

$$f(x) = ax$$

where a is a fixed real constant, known as the **constant of proportionality** or the **slope**.

Examples: - Buying cherries at €4 per kg: $P(x) = 4x$ ($a = 4$). - Cycling at a constant speed of 20 km/h: distance $d(t) = 20t$ ($a = 20$). - A 30% discount: the discounted price is $N(x) = (1 - 0.30)x = 0.7x$ ($a = 0.7$).

4.2 Graphical Representation: The Line Passing Through the Origin

If you plot points belonging to a linear function in a coordinate plane, a geometric wonder appears: **all points align in a straight line passing through the origin $O(0, 0)$!**

Why? Because if you purchase nothing ($x = 0$), you pay nothing ($f(0) = a \times 0 = 0$).

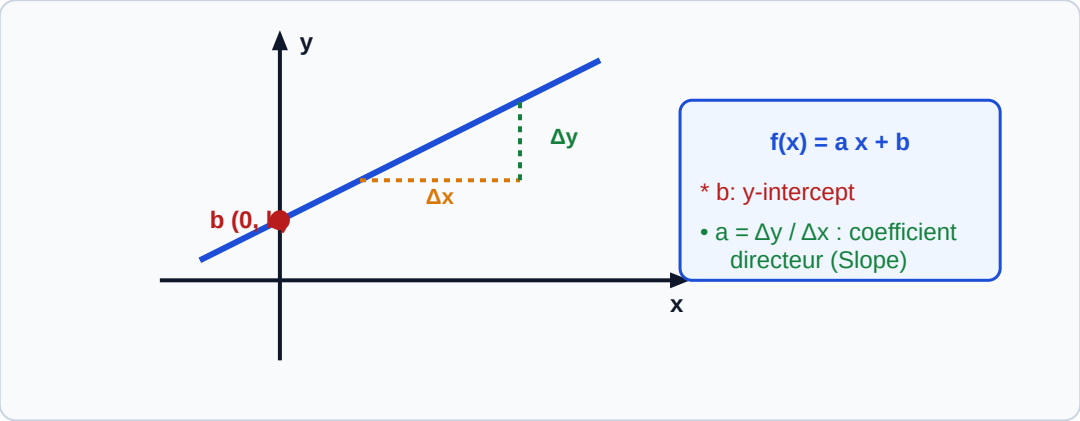

Figure 71: Graph of the affine function: slope and y-intercept

Graphical Theorem —

The graph of a linear function $f(x) = ax$ is **a straight line passing through the origin $O(0, 0)$**. Conversely, any non-vertical line passing through the origin represents a linear function.

To plot the graph of a linear function, it suffices to calculate **a single additional point** alongside the origin $(0, 0)$!

---

5. The Affine Function: The General Tariff Line $f(x) = ax + b$

5.1 Definition: Proportionality with a Base Fee

What happens when a real-world scenario involves an initial fixed charge (a subscription fee, an opening meter charge, a deposit) to which a proportional rate is added? It is no longer pure proportionality, but an **affine function** (commonly called a linear function with non-zero intercept in English curricula).

Definition — Affine Function.

An affine function is a function defined for all real numbers x by:

$$f(x) = ax + b$$

where a and b are fixed real constants.

- The constant a is the **slope** (rate of change / directional coefficient).
- The constant b is the **y-intercept** (the initial value at $x = 0$).

5.2 Why This Terminology?

1. **The y-intercept (b):** Evaluate the function at 0:
$$f(0) = a \times 0 + b = 0 + b = b.$$
The point on the line corresponding to $x = 0$ has coordinates $(0, b)$. That is precisely where the line **crosses the vertical y-axis**!
2. **The slope (a):** It governs the tilt and steepness of the line:
   - If $a > 0$, the function is **increasing** (the line rises from left to right).
   - If $a < 0$, the function is **decreasing** (the line falls from left to right).
   - The larger $|a|$, the steeper the line. The closer $|a|$ is to 0, the flatter the line.

5.3 Special Cases

- If $b = 0$, the formula becomes $f(x) = ax$: a linear function is simply an **affine function that starts at the origin $b = 0$**.
- If $a = 0$, the formula becomes $f(x) = b$: this is a **constant function**. The output is always b, regardless of x. Its graph is a **horizontal straight line**.

---

6. The Art of Graphing and Reading Lines

6.1 The Magic Staircase: "Run 1, Rise a"

To draw the line $y = ax + b$ without building a lengthy table of values, use the staircase rule (rise over run):

1. Plot the initial starting anchor on the vertical y-axis: the point $(0, b)$.
2. From this point, **step horizontally 1 unit to the right** (Run $+1$).
3. Then:
   - If a is positive: **climb vertically by a units** (Rise $+a$).
   - If a is negative: **drop vertically by $|a|$ units** (Rise $-|a|$).
4. Mark this second point. Place your straightedge through both points: your line is complete!

6.2 Computing the Slope a from Two Given Points

If a line passes through two known points $A(x_A, y_A)$ and $B(x_B, y_B)$ (with $x_A \neq x_B$), its slope is calculated using the fundamental ratio:

$$a = \frac{\text{Vertical change}}{\text{Horizontal change}} = \frac{y_B - y_A}{x_B - x_A} = \frac{\Delta y}{\Delta x} = \frac{\text{Rise}}{\text{Run}}.$$

The symbol Δ (the uppercase Greek letter Delta) signifies "change" or "difference." This formula expresses that the slope measures how many vertical units the line climbs for each unit traveled toward the right.

---

7. True History: The Thermometer of Fahrenheit and Celsius

In the 18th century, European scientists sought reliable means to measure temperature. In 1724, German physicist Daniel Gabriel Fahrenheit constructed the first dependable mercury thermometer. He established:

- 0° F: the coldest temperature he could achieve in his laboratory using a mixture of water, ice, and ammonium chloride.
- 96° F: the body temperature of a healthy human being (later calibrated to 98.6° F).

On his scale, pure water freezes at 32° F and boils at 212° F.

In 1742, Swedish astronomer Anders Celsius devised a decimal scale grounded in the fundamental states of pure water:

- 0° C for the freezing point of water.
- 100° C for the boiling point of water.

How do we convert Celsius temperature (x) into Fahrenheit temperature ($f(x)$)? The mapping is precisely an **affine function**!

- The freezing point yields point $A(0, 32)$.
- The boiling point yields point $B(100, 212)$.

Let us compute the slope a and the y-intercept b: 1. Since $x = 0$ yields $y = 32$, the y-intercept is immediately:

$$b = 32.$$

2. The slope is:

$$a = \frac{y_B - y_A}{x_B - x_A} = \frac{212 - 32}{100 - 0} = \frac{180}{100} = \frac{9}{5} = 1.8.$$

The universal formula connecting both worlds is:

$$f(x) = 1.8x + 32 \quad \text{or} \quad T_{\text{Fahrenheit}} = \frac{9}{5} T_{\text{Celsius}} + 32.$$

When your bedroom temperature reads 20° C, an American reading a Fahrenheit thermometer sees:

$$f(20) = 1.8 \times 20 + 32 = 36 + 32 = 68^\circ \text{ F} \, !$$

---

8. Step-by-Step Worked Examples

Example 1: Evaluating images and preimages

Consider the affine function f defined by:

$$f(x) = -3x + 7.$$

1. Calculate the image of 4 and the image of -2. 2. Determine the preimage of -11. 3. Determine the preimage of 0.

Detailed Solution: 1. To calculate an image, substitute the given value for x:

- $f(4) = -3 \times 4 + 7 = -12 + 7 = -5$. The image of 4 is -5.
- $f(-2) = -3 \times (-2) + 7 = +6 + 7 = 13$. The image of -2 is 13.

2. Finding the preimage of -11 means solving $f(x) = -11$:

$$\begin{aligned} -3x + 7 &= -11 \\ -3x &= -11 - 7 = -18 \\ x &= \frac{-18}{-3} = 6. \end{aligned}$$

The preimage of -11 under f is 6.

3. Finding the preimage of 0:

$$-3x + 7 = 0 \Longrightarrow -3x = -7 \Longrightarrow x = \frac{-7}{-3} = \frac{7}{3}.$$

The preimage of 0 is the irreducible fraction $\frac{7}{3}$.

Example 2: Determining an affine function from two points

A straight line passes through points $A(2, 5)$ and $B(6, 17)$. Determine the formula of the affine function $f(x) = ax + b$ whose graph is this line.

Detailed Solution: 1. **Compute the slope a:**

$$a = \frac{y_B - y_A}{x_B - x_A} = \frac{17 - 5}{6 - 2} = \frac{12}{4} = 3.$$

The function is now known to be of the form $f(x) = 3x + b$. 2. **Compute the y-intercept b:** Since $A(2, 5)$ lies on the line, $f(2) = 5$:

$$\begin{aligned} 3 \times 2 + b &= 5 \\ 6 + b = 5 &\Longrightarrow b = 5 - 6 = -1. \end{aligned}$$

3. **Conclusion:** The desired affine function is:

$$f(x) = 3x - 1.$$

(Quick check with point B: $f(6) = 3 \times 6 - 1 = 18 - 1 = 17$. Exact!)

Example 3: Finding a and b by graphical inspection

On a given coordinate grid, a straight line (d) crosses the vertical axis at -3. Stepping 2 grid units to the right requires climbing 5 units vertically to return to the line. 1. State the y-intercept b. 2. Calculate the slope a. 3. Deduce the formula of the corresponding function.

Detailed Solution: 1. The line intersects the vertical axis at -3, so:

$$b = -3.$$

2. Running 2 units right and rising 5 units means:

$$\Delta x = +2 \quad \text{and} \quad \Delta y = +5.$$

The slope is:

$$a = \frac{\Delta y}{\Delta x} = \frac{5}{2} = 2.5.$$

3. The function formula is:

$$f(x) = 2.5x - 3 \quad \left(\text{or} \quad f(x) = \frac{5}{2}x - 3\right).$$

Example 4: Mobile phone plan comparison

A telecommunications provider offers two mobile subscriptions:

- **Eco Plan:** €0.10 per call minute, with no monthly base fee.
- **Serenity Plan:** A fixed monthly fee of €12, plus €0.02 per call minute.

1. Write formulas for $E(x)$ and $S(x)$ giving the monthly cost for x call minutes.
2. Identify the mathematical type of each function.
3. Compute the monthly bill for 60 minutes of calls under each plan.
4. Beyond how many minutes per month does the Serenity Plan become cheaper than the Eco Plan?

Detailed Solution: 1. For x minutes:

- $E(x) = 0.10x$.
- $S(x) = 0.02x + 12$.

2. E is a **linear function** (pure proportionality, passing through the origin). S is an **affine function** (with non-zero base intercept $b = 12$).
3. For 60 minutes:
   - $E(60) = 0.10 \times 60 = 6$ €.
   - $S(60) = 0.02 \times 60 + 12 = 1.20 + 12 = 13.20$ €.
4. Serenity is cheaper when its cost is strictly less than Eco:

$$S(x) < E(x) \Leftrightarrow 0.02x + 12 < 0.10x$$

$$12 < 0.10x - 0.02x \Longrightarrow 12 < 0.08x \Longrightarrow x > \frac{12}{0.08} = 150.$$

Conclusion: The Serenity Plan becomes more economical whenever calling exceeds **150 minutes (2 hours 30 minutes) per month**.

Example 5: Speed, distance, and affine functions

A water cistern contains 800 liters. At 8:00 AM, a drain valve is opened: water empties at a steady, constant rate of 25 liters per minute. 1. Express the remaining volume $V(t)$ (in liters) in terms of elapsed time t (in minutes). 2. What is

the nature of function V? What is the sign of its slope? 3. How much water remains after 12 minutes? 4. How long until the cistern is completely dry?

Detailed Solution: 1. At time $t = 0$, there are 800 L. Every minute, 25 L drain out:

$$V(t) = 800 - 25t = -25t + 800.$$

2. V is an **affine function** with slope $a = -25$ and y-intercept $b = 800$. Its slope is **negative** ($a = -25 < 0$), so the function is **decreasing** (volume decreases over time). 3. At $t = 12$ min:

$$V(12) = -25 \times 12 + 800 = -300 + 800 = 500 \text{ liters}.$$

4. The cistern is empty when $V(t) = 0$:

$$V(t) = 0 \Leftrightarrow -25t + 800 = 0 \Leftrightarrow 25t = 800 \Leftrightarrow t = \frac{800}{25} = 32 \text{ minutes}.$$

The tank will be completely empty at 8:32 AM.

Key Takeaways

1. **Core Vocabulary:**
   - x is the **input / preimage** (the starting value).
   - $f(x)$ is the **output / image** of x under f (the computed result).
   - The mapping notation $f : x \mapsto f(x)$ reads: "the function f that maps x to $f(x)$."
2. **Linear Function: $f(x) = ax$**
   - Represents pure **proportionality**.
   - Its graph is **a straight line strictly passing through the origin $O(0, 0)$**.
3. **Affine Function: $f(x) = ax + b$**
   - Its graph is **a straight line** intersecting the vertical y-axis at $(0, b)$.
   - b is the **y-intercept**: $f(0) = b$.
   - a is the **slope (rate of change)**:

     $$a = \frac{y_B - y_A}{x_B - x_A} = \frac{\Delta y}{\Delta x}.$$

     If $a > 0$, the line rises. If $a < 0$, the line falls. If $a = 0$, the line is horizontal (constant function).
4. **Graphical Reading:**
   - To find an output given an input: start at x on the horizontal axis, move vertically to meet the line, then read y on the vertical axis.
   - To find an input given an output: start at y on the vertical axis, move horizontally to meet the line, then read x on the horizontal axis.

Exercises

Level : Direct Application

Exercise 25.1 Vocabulary and Notations: Let h be a function such that $h(3) = 7$. 1. What is the image of 3 under h? 2. What is a preimage of 7 under h? 3. Translate into an equation of the form $f(a) = b$ the statement: "The image of -5 under the function f is 12." 4. Translate into an equation the statement: "-2 is a preimage of 9 under the function g."

Exercise 25.2 Evaluating Images: Consider the affine function $f(x) = 4x - 5$. Calculate: 1. $f(0)$ 2. $f(3)$ 3. $f(-2)$ 4. $f\left(\frac{1}{2}\right)$

Exercise 25.3 Finding Preimages: Using the same function $f(x) = 4x - 5$, determine the preimage of: 1. 15 2. -5 3. 0 4. -13

Exercise 25.4 Classifying Functions: State whether each function is linear, affine (non-linear), or neither: 1. $f(x) = 5x$ 2. $g(x) = 3x - 8$ 3. $h(x) = x^2 + 1$ 4. $k(x) = 7$ 5. $m(x) = \frac{4}{x}$ 6. $p(x) = \frac{x}{5} + 2$

Exercise 25.5 Graphing Lines: In a Cartesian coordinate plane, plot the graphs of: 1. $d_1 : f(x) = 2x$ 2. $d_2 : g(x) = -x$ 3. $d_3 : h(x) = 2x - 3$ 4. $d_4 : k(x) = -0.5x + 2$

Level : Deeper Reflection and Multi-Step Problems

Exercise 25.6 Deriving Formulas from Points: 1. A linear function g satisfies $g(4) = 14$. Determine the formula for $g(x)$. 2. An affine function f passes through $A(1, 4)$ and $B(3, 10)$. Compute its slope a, its y-intercept b, and write down $f(x)$.

Exercise 25.7 Reverse Graphical Reading: A straight line representing an affine function f intersects the vertical axis at $(0, 4)$ and passes through $(4, 0)$. 1. State the value of b. 2. Calculate the slope a. 3. Write the formula for $f(x)$. 4. Is f increasing or decreasing?

Exercise 25.8 The Night Taxi: A taxi driver applies the following rate schedule:

- Base flag drop fee upon starting the meter: €4.20.
- Per-kilometer rate: €1.60.

1. Write an expression for the total fare $C(x)$ for a trip of x kilometers.
2. How much does a 15 km ride cost?
3. A passenger has a €25 note. What maximum distance can they travel?

Exercise 25.9 Point Membership on a Line: Let $f(x) = -2x + 7$. 1. Does point $M(3, 1)$ lie on the graph of f? 2. Does point $N(-4, 15)$ lie on this line? 3. Find the x-coordinate of the point P on the line having y-coordinate -9.

Exercise 25.10 Intersection of Two Affine Functions: Consider $f(x) = 2x - 1$ and $g(x) = -x + 5$. 1. Graph both lines on the same axes. 2. Read the coordinates of their intersection point I from the graph. 3. Verify these coordinates algebraically by solving $f(x) = g(x)$.

Exercise 25.11 The Burning Candle: A candle is 24 cm tall. When lit, its height decreases at a steady rate of 1.5 cm per hour. 1. Express the candle's height $h(t)$ in centimeters after t hours. 2. What is the height after 6 hours? 3. How many hours until the candle is completely burned down?

Level : Research Challenges and Olympiad-Style Riddles

Exercise 25.12 Collinearity of Three Points: Consider three points in a coordinate plane: $A(-2, -7)$, $B(2, 5)$, and $C(5, 14)$. 1. Determine the affine function f whose line passes through A and B. 2. Is point C collinear with A and B? (Hint: test whether $f(5) = 14$).

Exercise 25.13 Parallelism and Lines: Line (d_1) represents the function $f(x) = 3x - 4$. We seek a line (d_2) that is **parallel** to (d_1) and passes through point $E(2, 9)$. 1. What must be true about the slope of (d_2) compared to (d_1)? 2. Deduce the formula of the affine function g associated with (d_2).

Exercise 25.14 The Double-Hop Box Mystery: An unknown affine function $f(x) = ax + b$ possesses this remarkable property: When applied twice in succession—that is, computing $f(f(x))$—it always yields $4x + 9$! 1. Expand the algebraic expression $f(f(x)) = a(ax + b) + b$. 2. By equating coefficients with $4x + 9$, find the two possible values for a. 3. Deduce the two affine functions that solve this riddle.

Solutions

Level

Solution 25.1 1. The image of 3 is 7. 2. A preimage of 7 is 3. 3. $f(-5) = 12$. 4. $g(-2) = 9$.

Solution 25.2 1. $f(0) = 4(0) - 5 = -5$. 2. $f(3) = 4(3) - 5 = 12 - 5 = 7$. 3. $f(-2) = 4(-2) - 5 = -8 - 5 = -13$. 4. $f\left(\frac{1}{2}\right) = 4 \times \frac{1}{2} - 5 = 2 - 5 = -3$.

Solution 25.3 1. $4x - 5 = 15 \Longrightarrow 4x = 20 \Longrightarrow x = 5$. 2. $4x - 5 = -5 \Longrightarrow 4x = 0 \Longrightarrow x = 0$. 3. $4x - 5 = 0 \Longrightarrow 4x = 5 \Longrightarrow x = \frac{5}{4} = 1.25$. 4. $4x - 5 = -13 \Longrightarrow 4x = -8 \Longrightarrow x = -2$.

Solution 25.4 1. $f(x) = 5x$: linear (and affine with $b = 0$). 2. $g(x) = 3x - 8$: affine non-linear ($a = 3, b = -8$). 3. $h(x) = x^2 + 1$: neither (quadratic term, not affine). 4. $k(x) = 7$: constant (affine with $a = 0, b = 7$). 5. $m(x) = \frac{4}{x}$: neither (inverse variation, x in denominator). 6. $p(x) = \frac{1}{5}x + 2$: affine ($a = 0.2, b = 2$).

Solution 25.5 1. d_1 passes through $(0, 0)$ and $(1, 2)$. 2. d_2 passes through $(0, 0)$ and $(1, -1)$. 3. d_3 passes through $(0, -3)$ and $(1, -1)$ (or $(2, 1)$). 4. d_4 passes through $(0, 2)$ and $(2, 1)$ (or $(4, 0)$).

Level

Solution 25.6 1. $g(x) = ax$. Since $g(4) = 14$, $4a = 14 \Longrightarrow a = \frac{14}{4} = 3.5$. Thus $g(x) = 3.5x$. 2. $a = \frac{y_B - y_A}{x_B - x_A} = \frac{10-4}{3-1} = \frac{6}{2} = 3$. So $f(x) = 3x + b$. Using $A(1, 4)$: $3(1) + b = 4 \Longrightarrow b = 1$. Thus $f(x) = 3x + 1$.

Solution 25.7 1. Line meets y-axis at $(0, 4)$, so $b = 4$. 2. Passes through $(4, 0)$, so $a = \frac{0-4}{4-0} = \frac{-4}{4} = -1$. 3. $f(x) = -x + 4$. 4. Since $a = -1 < 0$, the function is **strictly decreasing** (line slopes downward).

Solution 25.8 1. $C(x) = 1.60x + 4.20$. 2. For 15 km: $C(15) = 1.60 \times 15 + 4.20 = 24 + 4.20 = 28.20$ €. 3. With €25: $1.60x + 4.20 \leq 25 \Longrightarrow 1.60x \leq 20.80 \Longrightarrow x \leq \frac{20.80}{1.60} = 13$ km. The maximum distance is 13 km.

Solution 25.9 1. Check $f(3)$: $f(3) = -2(3) + 7 = -6 + 7 = 1$. Matches the y-coordinate, so $M(3, 1)$ lies on the line. 2. Check $f(-4)$: $f(-4) = -2(-4) + 7 = 8 + 7 = 15$. Yes, $N(-4, 15)$ also lies on the line. 3. Solve $f(x) = -9$: $-2x + 7 = -9 \Longrightarrow -2x = -16 \Longrightarrow x = 8$. The x-coordinate of P is 8.

Solution 25.10 1. Plot lines: f crosses $(0, -1)$ and $(2, 3)$; g crosses $(0, 5)$ and $(2, 3)$. 2. Graphically, the intersection point is $I(2, 3)$. 3. Algebraically:

$$f(x) = g(x) \Leftrightarrow 2x - 1 = -x + 5 \Leftrightarrow 3x = 6 \Leftrightarrow x = 2.$$

Then $f(2) = 2(2) - 1 = 3$. The point is confirmed: $I(2, 3)$.

Solution 25.11 1. $h(t) = -1.5t + 24$. 2. After 6 hours: $h(6) = -1.5 \times 6 + 24 = -9 + 24 = 15$ cm. 3. Burned out when $h(t) = 0$:

$$-1.5t + 24 = 0 \Leftrightarrow 1.5t = 24 \Leftrightarrow t = \frac{24}{1.5} = 16 \text{ hours}.$$

Level

Solution 25.12 1. Find line through $A(-2, -7)$ and $B(2, 5)$: $a = \frac{5-(-7)}{2-(-2)} = \frac{12}{4} = 3$. $f(x) = 3x + b$. With $B(2, 5)$: $3(2) + b = 5 \Longrightarrow b = -1$. So $f(x) = 3x - 1$. 2. Test $C(5, 14)$: $f(5) = 3(5) - 1 = 15 - 1 = 14$. The equality is verified: point C is **collinear** with A and B!

Solution 25.13 1. Two lines are parallel if and only if they have the **exact same slope**. Thus (d_2) has slope $a = 3$. 2. $g(x) = 3x + b$. Since it passes through $E(2, 9)$:

$$3(2) + b = 9 \Longrightarrow 6 + b = 9 \Longrightarrow b = 3.$$

Formula: $g(x) = 3x + 3$.

Solution 25.14 1. Apply f to $f(x) = ax + b$:

$$f(f(x)) = a(ax + b) + b = a^2x + ab + b = a^2x + b(a + 1).$$

2. Equate to $4x + 9$ for all x:

- For the x coefficient: $a^2 = 4 \Longrightarrow a = 2$ or $a = -2$.
- For the constant term: $b(a + 1) = 9$.

3. Two cases arise:

   - Case 1: if $a = 2$, then $b(2 + 1) = 9 \Longrightarrow 3b = 9 \Longrightarrow b = 3$. First function: $f_1(x) = 2x + 3$. *(Check: $f_1(f_1(x)) = 2(2x + 3) + 3 = 4x + 6 + 3 = 4x + 9$!)*
   - Case 2: if $a = -2$, then $b(-2 + 1) = 9 \Longrightarrow b(-1) = 9 \Longrightarrow b = -9$. Second function: $f_2(x) = -2x - 9$. *(Check: $f_2(f_2(x)) = -2(-2x - 9) - 9 = 4x + 18 - 9 = 4x + 9$!)*

The Question Opening the Next Chapter

Thus far, numbers have served to calculate weights, prices, areas, and linear trajectories. But look around you: a repeating fabric pattern, roof tiles, a six-petaled flower, the infinite mosaics of the Alhambra in Granada, or the dizzying art of M. C. Escher.

How can a single triangle or shape **slide**, **rotate**, **reflect across a mirror**, or **expand under a magnifying glass** while preserving its exact proportions? That is the enchanting realm of plane transformations that we will explore in Chapter 26: **Paper Folding and the Magic Mirror**.

CHAPTER 26: Paper Folding and the Magic Mirror

> "Observe a raindrop falling into a calm lake, the wings of a butterfly opening to the sun, or the face of a dear friend: nature never selects its shapes by accident. It builds in the mirror of symmetry, for that which is symmetrical distributes forces, balances masses, and endures against the tempest."

What You Are About to Discover

- The secret of medieval master builders: why asymmetry tears down stone bridges, and how the **perpendicular bisector** and **angle bisector** produce geometric harmony.
- The grand family of plane geometric transformations:
  - **Axial reflection (line symmetry)**: paper folding and the mirror image.
  - **Point reflection (central symmetry)**: the perfect half-turn ($180°$) around a pivot.
  - **Translation**: rectilinear sliding without the slightest rotation.
  - **Rotation**: controlled whirling through a given angle around a center.
  - **Dilation (homothety)**: the optical zoom lens shrinking or expanding space by a scale factor k.
- The properties these transformations **preserve**: lengths, angles, parallelism, and areas.
- The vital distinction between **congruent triangles** (identical clones) and **similar triangles** (identical silhouette at different scales).
- The enchanting art of **friezes and tessellations**: from the arabesques of the Alhambra in Granada to the impossible metamorphoses of Maurits Cornelis Escher.

---

1. The Problem: The Collapsing Bridge and the Well Dispute

In the 12th century, within a steep gorge of the Auvergne mountains, local villagers decided to construct a massive stone arch bridge across a roaring river. The master mason was in a hurry: he cut unequal voussoirs and erected an arch leaning noticeably to the right, much heavier on one flank than the other. The arch was picturesque, but dangerously unbalanced.

A few months later, the first heavy autumn flood arrived. The torrent surged, and howling winds rushed through the ravine. The heavier side pulled relentlessly upon the thinner side. A winding crack snaked across the vault… and with a deafening roar of crushed masonry, **the bridge collapsed into the abyss**!

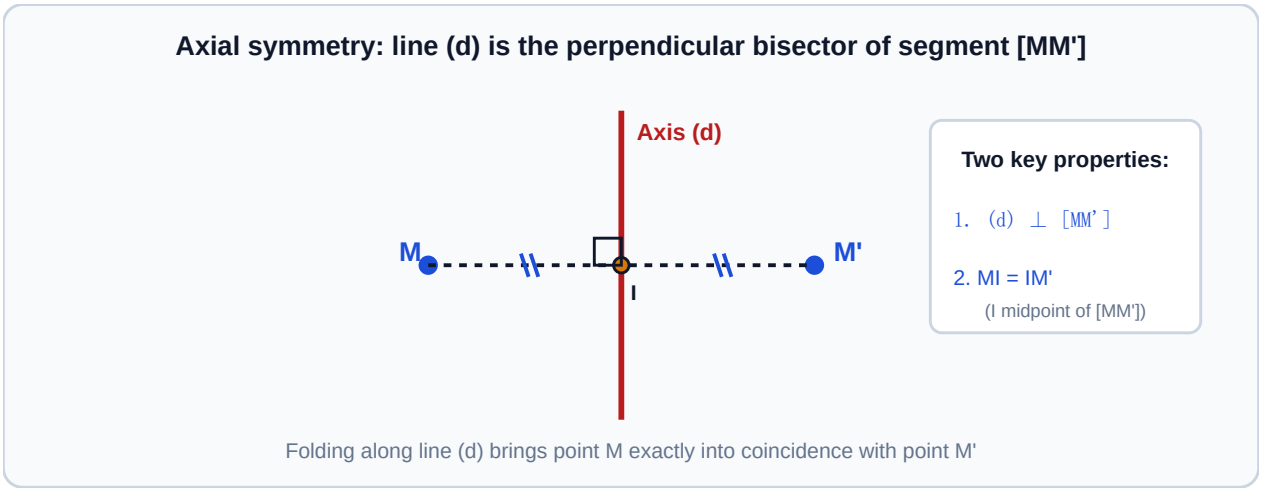

Figure 72: Bilateral axial symmetry: orthogonal fold and reflection

The master builder summoned his apprentices around his drafting table: *"Carve this into your memories: stone forgives surface roughness, but it never forgives asymmetry. When an arch lacks its mirror axis, the weight of the heavens bears down on a single abutment, and ruin is certain."*

A few leagues away, another dispute arose between two neighboring villages, Arras and Béthune, separated by six kilometers of open heath. The local count decided to sink a large stone well in the plain to water livestock and travelers.

Instantly, discord erupted: *"Dig the well right near our church bell tower!"* demanded the villagers of Arras. *"Out of the question!"* retorted the folk of Béthune. *"That would be blatant favoritism! We insist that the walking distance be strictly equal for both villages!"*

A child suggested: *"Let us dig the well right in the middle of the footpath connecting the two villages!"* That was a sound idea: at 3 kilometers from Arras and 3 kilometers from Béthune, the location was fair. Yet the commander's surveyor stepped forward and scored a deep trench across the soil with his spear: *"There is not merely one possible location: there are infinitely many fair locations for the well!"*

This perpendicular line cutting the path precisely at its midpoint possesses a magical geometric property: **every single point on this line is situated at the exact same distance from Village A and Village B**. This line of absolute equity is called **the perpendicular bisector**.

To master architecture, art, and machinery, we must understand the fundamental laws governing plane motion: **geometric transformations**.

---

2. The Two Mirrors: Axial Reflection and Point Reflection

A **geometric transformation** is a rule that maps each point in the plane to a new position according to a strict geometric prescription. The original point is called the preimage M, and the resulting point is called its **image** M'.

2.1 Axial Reflection: Paper Folding Along a Line

Take a sheet of paper, place a drop of wet ink on the left side, fold the sheet in half along a straight crease (d), and press down firmly. When unfolded, the ink blot has imprinted itself on the other side.

Definition — Axial Reflection (across line (d)).

- The image of a point M lying off (d) is the point M' such that

the line (d) is the **perpendicular bisector** of the segment $[MM']$. This means that (d) is perpendicular to $[MM']$ and passes through its midpoint.

- If a point M lies directly on the axis (d), its image is itself: $M' = M$ (such a point is called an **invariant point** or fixed point).

The Mirror Effect: Axial reflection reverses orientation! Your right hand reflects as a left hand in a mirror. If the vertices of a triangle proceed clockwise ($A \to B \to C$), their reflected images will proceed counterclockwise.

2.2 Point Reflection (Central Symmetry): The Half-Turn Around a Pivot

Now imagine placing a pushpin into your paper at point O, and rotating the tracing paper through an **exact half-turn ($180°$)**.

> **Definition — Point Reflection (Central Inversion across center O).**
>
> The image of a point M under point reflection centered at O is the point M' such that **point O is the midpoint of the segment** $[MM']$.

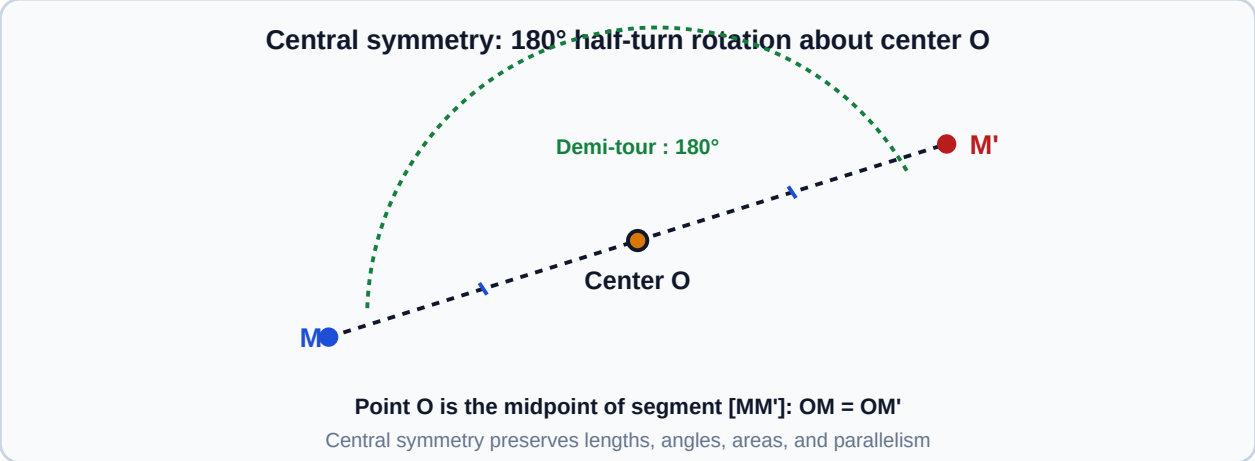

Figure 73: Central symmetry: perfect 180° half-turn about center O

Essential Property: A point reflection transforms any line (d) into a line (d') that is **strictly parallel** to it! It preserves orientation (a clockwise loop remains clockwise after a $180°$ rotation).

---

3. The Two Rigid Motions: Translation and Rotation

3.1 Translation: The Pure Slide

Picture opening a desk drawer, or a train speeding along straight tracks. Every point of the object glides by the exact same distance, along the same direction, and in the same sense.

> **Definition — Translation.**
>
> A **translation** slides a geometric figure from point A to point B. The image of any point M is the point M' such that: 1. The lines (AB) and (MM') are **parallel** (same direction). 2. Motion from M to M' has the **same sense** (orientation) as motion from A to B. 3. The length $MM' = AB$ (same distance).
>
> *In summary:* The quadrilateral $ABM'M$ is a **parallelogram** (possibly degenerate).

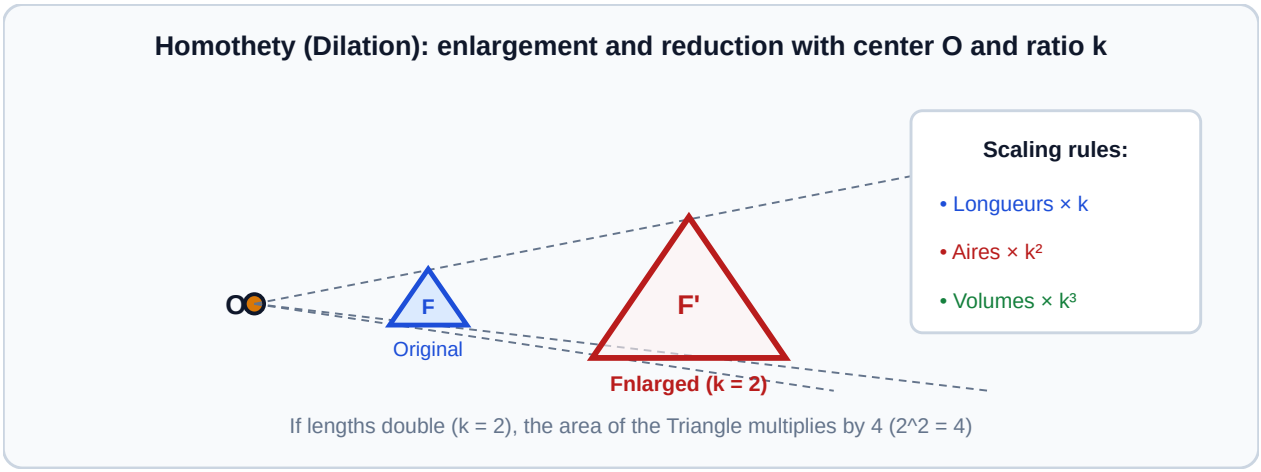

Figure 74: Homothety (dilation): vector scaling from the center of alignment

In a translation, the figure glides without rotating by a hair. It models footprints left in sequence across beach sand.

3.2 Rotation: The Controlled Swirl

Think of the minute hand of a clock, the blades of a windmill, or the cabins on a Ferris wheel.

📐 Definition — Rotation.

A **rotation** is determined by three inseparable components: 1. A fixed point called the **center of rotation** O. 2. An **angle of rotation** α (in degrees). 3. A **direction of rotation**:

- either **clockwise**;
- or **counterclockwise** (standard mathematical / counterclockwise sense).

The image of a point M under rotation centered at O by angle α is the point M' such that:

- $OM = OM'$ (the point remains on a concentric circle of radius OM);
- The angle $\widehat{MOM'} = \alpha$.

Remarkable Insight: A rotation of $180°$ is identical to a **point reflection**!

---

4. The Optical Zoom: Dilation (Homothety)

All previous transformations (reflections, translations, rotations) moved shapes across the plane without ever changing their dimensions. They are called **isometries** (from Greek *isos*, equal, and *metron*, measure).

A **dilation** (homothety, from Greek *homoios*, similar, and *thesis*, arrangement) is fundamentally different: it is an **optical zoom**. It expands or shrinks a shape proportionally from a central projection point.

4.1 Definition and Scale Factor k

📐 Definition — Dilation (Homothety).

A dilation is defined by: 1. A fixed point called the **center** O. 2. A non-zero real number called the **scale factor** k.

The image of a point M is the point M' on the line (OM) such that:

- The distance $OM' = |k| \times OM$.
- **If $k > 0$:** point M' lies on the ray $[OM)$ (on the same side of O as M).
- **If $k < 0$:** point M' lies on the opposite ray from O. This combines an expansion/contraction with a half-turn!

4.2 Scaling Effects of Factor k

- If $|k| > 1$: the transformation is an **enlargement**.
- If $0 < |k| < 1$: the transformation is a **reduction**.

- If $k = 1$: the shape does not move (identity transformation).
- If $k = -1$: the dilation is identical to **point reflection across center O**!

4.3 The Dramatic Effect on Areas and Volumes

This is one of the most classic pitfalls in middle and high school exams:

- Under a dilation of scale factor k, **all lengths are multiplied by $|k|$**.
- **The area of any surface is multiplied by k^2!**
- **The volume of any solid is multiplied by $|k|^3$!**

Vivid Example: If you scale up an architectural model by $k = 3$: - Its height is multiplied by 3. - The paint required for its facade is multiplied by $3^2 = 9$. - Its total weight and volume are multiplied by $3^3 = 27$!

---

5. What Transformations Preserve: The Genetic Code of Shapes

Transformation	Lengths	Angles	Parallelism	Collinearity	Areas	Orientation
Axial Reflection	Preserved	Preserved	Preserved	Preserved	Preserved	**Reversed** (Mirror)
Point Reflection	Preserved	Preserved	Preserved	Preserved	Preserved	Preserved
Translation	Preserved	Preserved	Preserved	Preserved	Preserved	Preserved
Rotation	Preserved	Preserved	Preserved	Preserved	Preserved	Preserved
Dilation (Scale k)	$\times \lvert k \rvert$	**Preserved**	Preserved	Preserved	$\times k^2$	Preserved if $k > 0$

Remember this fundamental principle: **all standard transformations in middle school strictly preserve angle measures, parallelism, and point collinearity**.

---

6. Congruent Triangles and Similar Triangles

6.1 Congruent Triangles (Isometries): Perfect Clones

Two triangles are **congruent** when they can be made to **coincide perfectly** through a rigid motion (translation, rotation, or reflection). Their corresponding sides have identical lengths and their corresponding angles have identical measures.

To prove that two triangles are congruent, you do not need to measure all 6 components (3 sides and 3 angles)! Three suffice, according to the **three congruence criteria**:

1. **SSS (Side-Side-Side):** The three sides of one triangle are respectively equal in length to the three sides of the other.
2. **SAS (Side-Angle-Side):** Two sides of one triangle are equal to two sides of the other, and **the angle enclosed between these two sides** has the same measure.
3. **ASA (Angle-Side-Angle):** One side of one triangle is equal to one side of the other, and **the two angles adjacent to that side** have the same measures.

6.2 Similar Triangles: The Same Silhouette at Different Scales

Two triangles are called **similar** when they possess the **exact same shape**, though not necessarily the same size. One is an enlargement or reduction of the other.

📐 Definition — Similar Triangles.

Two triangles are similar if:

- Their **corresponding angles are pairwise equal in measure**;
- **OR** their corresponding side lengths are **pairwise proportional**:

$$\frac{A'B'}{AB} = \frac{B'C'}{BC} = \frac{A'C'}{AC} = k.$$

The Efficiency Theorem: Since the sum of angles in any triangle always equals 180° (Chapter 15), as soon as **two angles of one triangle equal two angles of another**, the third angle is automatically equal, and **the two triangles are immediately similar** (AA criterion)!

---

7. The Lines of Equity: Perpendicular Bisector and Angle Bisector

These two remarkable lines are rooted in axial symmetry:

7.1 The Perpendicular Bisector of a Segment

- **Definition:** The line perpendicular to a line segment at its midpoint.
- **Equidistance Property:** A point lies on the perpendicular bisector of segment $[AB]$ if and only if it is equidistant from both endpoints:

$$M \in \text{Perpendicular Bisector of } [AB] \Leftrightarrow MA = MB.$$

- **The Circumscribed Circle (Circumcircle):** The perpendicular bisectors of the three sides of a triangle intersect at a single point: the **circumcenter**, which is the center of the unique circle passing through all three vertices of the triangle.

7.2 The Angle Bisector

- **Definition:** The ray that divides an angle into two adjacent angles of equal measure. It is the **axis of symmetry** of the angle.
- **Equidistance Property:** Every point on an angle bisector is equidistant from the two arms (sides) of the angle.
- **The Inscribed Circle (Incircle):** The angle bisectors of the three interior angles of a triangle meet at the **incenter**, which is the center of the circle tangent to all three sides inside the triangle.

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8. The Art of Friezes, Tessellations, and the Genius of Escher

When the geometry of transformations meets artistic imagination, mathematics touches the sublime.

8.1 What Is a Tessellation?

A **tessellation (tiling)** is a complete covering of the plane using interlocking geometric shapes, leaving no empty gaps and with no shapes overlapping.

- A **frieze (band ornament)** is a one-dimensional tessellation (a decorative border along a wall). It relies on periodic horizontal translations, sometimes combined with vertical reflections or half-turns.
- A **planar tessellation** extends infinitely across both dimensions.

8.2 The Mosaics of the Alhambra in Granada

In the 14th century, Moorish artists decorating the Alhambra palace, bound by cultural restrictions against depicting living forms, explored the mathematics of symmetry with staggering brilliance. Mathematicians proved centuries later that there are strictly **17 possible periodic planar symmetry groups** (the 17 wallpaper groups). The master craftsmen of Granada had **empirically discovered every single one of them** in the ceramic tilework (*azulejos*) of their palace!

8.3 The Metamorphoses of M. C. Escher

In 1936, Dutch graphic artist M. C. Escher visited the Alhambra and was deeply captivated by these repeating patterns. He devoted his artistic life to replacing abstract polygons with interlocking living creatures: birds, fish, lizards, angels, and demons.

How did he design a lizard that interlocks seamlessly with identical copies of itself? 1. He began with a basic grid shape (such as a square or parallelogram) that naturally tiles the plane. 2. He sliced out a curved contour along the left boundary and **translated** it to the right boundary. 3. He sliced a contour from the bottom boundary and performed a $\mathbf{90^\circ}$ **rotation** to form the upper paw. Because every piece of area removed from one side is perfectly restored to another, the total area remains invariant, allowing the motifs to interlock into infinity!

---

9. Step-by-Step Worked Examples

Example 1: Image under rotation

Let ABC be an equilateral triangle oriented counterclockwise. Let O denote the center of its circumscribed circle. 1. What is the image of vertex A under a 120° counterclockwise rotation centered at O? 2. What is the image of side $[AB]$ under this rotation?

Detailed Solution: 1. In an equilateral triangle, center O divides the full 360° turn into three equal central angles:

$$\widehat{AOB} = \widehat{BOC} = \widehat{COA} = \frac{360^\circ}{3} = 120^\circ.$$

Furthermore, by definition of the circumcircle, $OA = OB = OC$. A counterclockwise rotation of 120° about O sends A directly to B, and B directly to C. Thus, the image of A is B. 2. Rotations preserve line segments. Because A maps to B and B maps to C, the image of segment $[AB]$ is segment $[BC]$.

Example 2: Dilation calculations and area scaling

A trapezoid of area $12\ \text{cm}^2$ undergoes a dilation with scale factor $k = -2.5$. 1. What geometric shape is the resulting image? 2. Is its orientation reversed? 3. What is the area of the scaled trapezoid?

Detailed Solution: 1. Dilations preserve geometric categories and parallelism: the image of a trapezoid is another **trapezoid**. 2. The scale factor $k = -2.5$ is negative: the figure experiences an inversion (180° half-turn) about the center of dilation, but preserves its intrinsic cyclic orientation (unlike an axial reflection). 3. Area scales by k^2:

$$\text{Area}' = k^2 \times \text{Area} = (-2.5)^2 \times 12 = 6.25 \times 12 = 75\ \text{cm}^2.$$

The area of the dilated trapezoid is $75\ \text{cm}^2$.

Example 3: Proving two triangles are similar

In a circle of center O, draw two distinct diameters $[AB]$ and $[CD]$. Prove that triangles AOC and BOD are congruent, and subsequently prove that they are similar.

Detailed Solution: 1. Since $[AB]$ and $[CD]$ are diameters passing through center O, point O is the common midpoint of $[AB]$ and $[CD]$. 2. Consequently:

- Point reflection across O maps A to B (since O is the midpoint of $[AB]$).
- Point reflection across O maps C to D (since O is the midpoint of $[CD]$).

3. The image of triangle AOC under point reflection across O is therefore triangle BOD.
4. Because point reflection is an isometry (strictly preserving all side lengths and angle measures), **triangles $\boldsymbol{AOC}$ and $\boldsymbol{BOD}$ are congruent**.
5. Congruent triangles have identical side lengths (scale factor $k = 1$), hence they are **similar** with ratio 1.

Example 4: Using similar triangles to calculate an inaccessible height

Two vertical poles $[AB]$ and $[DE]$ stand upright on level ground. Pole $[AB]$ is 2 meters tall. At a certain moment, its shadow on the ground measures 3.5 meters. At that exact same instant, the shadow of the taller pole $[DE]$ measures 14 meters. Calculate the height of the tall pole $[DE]$.

Detailed Solution: 1. Sunlight rays are parallel at any given instant, so the sun rays strike the horizontal ground at the same angle of elevation. 2. Both poles are vertical, forming 90° right angles with the flat ground. 3. The right triangles formed by each pole and its shadow share two identical angle measures: they are **similar triangles** (by the AA criterion). 4. Corresponding side lengths are therefore proportional:

$$\frac{\text{Height of tall pole } \ DE}{\text{Height of short pole } \ AB} = \frac{\text{Shadow of tall pole}}{\text{Shadow of short pole}}.$$

$$\frac{DE}{2} = \frac{14}{3.5}.$$

5. Compute the scaling ratio:

$$k = \frac{14}{3.5} = 4.$$

6. The height of pole $[DE]$ is:

$$DE = 2 \times 4 = 8 \ \text{ meters }.$$

Example 5: Compass-only construction of the perpendicular bisector

Explain and mathematically justify the classical compass-and-straightedge construction of the perpendicular bisector of segment $[AB]$.

Detailed Solution: 1. Open the compass to any radius R strictly greater than half the length of AB ($R > \frac{AB}{2}$). 2. Placing the compass tip at A, draw two arcs of radius R, one above and one below $[AB]$. 3. Without changing the compass radius, place the tip at B and draw arcs intersecting the previous ones at points P and Q. 4. Draw the straight line (PQ) using a straightedge. 5. **Rigorous Justification:**

- By construction, $PA = PB = R$: point P is equidistant from A and B, so P lies on the perpendicular bisector of $[AB]$.
- Likewise, $QA = QB = R$: point Q is equidistant from A and B, so Q lies on the perpendicular bisector of $[AB]$.
- Since two distinct points uniquely determine a straight line, the line (PQ) is precisely the perpendicular bisector of $[AB]$.

Key Takeaways

1. **The 5 Geometric Transformations:**
   - **Axial Reflection:** paper folding across an axis (d) (perpendicular bisector). Reverses orientation.
   - **Point Reflection:** 180° half-turn around a center O (midpoint).
   - **Translation:** pure slide along a vector without turning (parallelogram).
   - **Rotation:** angular pivot of α° about a center O in a chosen direction.
   - **Dilation (Homothety):** scaling zoom by factor k from center O ($OM' = |k| \times OM$).
2. **Invariance and Scaling:**
   - All transformations preserve angles, collinearity, and parallelism.
   - The first four (isometries) preserve distances and areas.
   - A dilation multiplies lengths by $|k|$ and surface areas by k^2.
3. **Congruent Triangles:** Identical, superimposable triangles. Corresponding sides are equal and corresponding angles are equal (SSS, SAS, ASA criteria).

4. **Similar Triangles:** Triangles having the exact same shape: corresponding angles are equal and corresponding sides are proportional. Matching two pairs of angles guarantees similarity.
5. **Perpendicular Bisector and Angle Bisector:**
   - Perpendicular Bisector: perpendicular at the midpoint, locus of equidistant points from endpoints, circumcenter.
   - Angle Bisector: divides an angle in half, locus of equidistant points from arms, incenter.

Exercises

Level : Direct Application

Exercise 26.1 Identifying Transformations: Match each real-world situation with the corresponding transformation: 1. A cabin on a moving Ferris wheel (staying upright). 2. The reflection of a tree in a still mountain lake. 3. A train traveling along straight tracks. 4. An image projected onto a movie theater screen from a film frame. 5. A playing card (the Queen of Hearts) with its inverted double on the table.

Exercise 26.2 Properties of Axial Reflection: Let segment $[AB]$ have length 6 cm and angle $\widehat{BAC} = 45°$. An axial reflection across line (d) maps A, B, C to A', B', C'. 1. What is the length of segment $[A'B']$? Justify. 2. What is the measure of angle $\widehat{B'A'C'}$? Justify.

Exercise 26.3 Translation and Parallelograms: Let M' be the image of point M under the translation mapping A to B. 1. What is the geometric nature of quadrilateral $ABM'M$? 2. If $AB = 7$ cm, what is the distance MM'?

Exercise 26.4 Dilation Calculations: A triangle ABC has perimeter 18 cm and area 14 cm^2. A dilation with scale factor $k = 3$ is applied. 1. What is the perimeter of the dilated triangle? 2. What is the area of the dilated triangle?

Exercise 26.5 Recognizing Congruent Triangles: In triangle ABC, $AB = 5$ cm, $AC = 7$ cm, and $\widehat{BAC} = 40°$. In triangle DEF, $DE = 5$ cm, $DF = 7$ cm, and $\widehat{EDF} = 40°$. Are triangles ABC and DEF congruent? What criterion justifies this?

Level : Deeper Reflection and Multi-Step Problems

Exercise 26.6 Composition of Reflections: Let lines (d_1) and (d_2) be perpendicular, intersecting at point O. Apply reflection across (d_1) to a shape, then reflect the resulting image across (d_2). Prove that this sequence of two perpendicular reflections is equivalent to a single simple transformation.

Exercise 26.7 Similar Triangles and Angle Calculation: In triangle ABC, $\hat{A} = 65°$ and $\hat{B} = 45°$. In triangle RST, $\hat{R} = 65°$ and $\hat{T} = 70°$. Are triangles ABC and RST similar? Justify carefully.

Exercise 26.8 The Tree and the Mirror Problem: To measure an inaccessible tree, Arthur places a flat mirror horizontally on the ground at point M, 12 meters from the base of tree P. He steps back 2 meters to point A, where he sees the top of tree S reflected in the mirror. Arthur's eyes are 1.60 meters above the ground. By the optical law of reflection, angle of incidence equals angle of reflection: $\widehat{SMA} = \widehat{PMS}$. 1. Justify that the triangles formed by Arthur and the tree are similar. 2. Compute the height of the tree.

Exercise 26.9 Negative Dilation: On a number line, center of dilation O has coordinate 0. Point A has coordinate 3. The dilation has scale factor $k = -2$. 1. What is the coordinate of image point A'? 2. Where is point O located with respect to segment $[AA']$?

Exercise 26.10 The Chord and the Bisector: Let O be the center of a circle and $[AB]$ a chord not passing through O. 1. Prove that point O belongs to the perpendicular bisector of chord $[AB]$. 2. Deduce that any line passing through the center of a circle perpendicular to a chord bisects that chord.

Exercise 26.11 Translating Friezes: A decorative wall frieze is built from a repeating tile motif of width 4 cm. The motif is repeatedly translated horizontally by 4 cm. 1. How many complete tiles compose a frieze 1.20 m in length? 2. If each motif has an area of 15 cm^2, what is the total painted area?

Level : Research Challenges and Olympiad-Style Riddles

Exercise 26.12 Mathematical Billiards: On a pocketless rectangular billiard table, white ball A and red ball B rest on the felt. The player must strike ball A so that it rebounds off top cushion (d) before striking ball B. 1. Using reflection B' of ball B across cushion line (d), find the exact target point P on the cushion. 2. Prove that point P minimizes the total path length $AP + PB$ (Fermat's principle of reflection).

Exercise 26.13 Nested Similar Triangles in a Right Triangle: Let ABC be a right triangle at A. Let H be the foot of the altitude from A to hypotenuse $[BC]$. 1. Prove that triangles ABC, HBA, and HAC are all pairwise similar. 2. By equating side ratios between HBA and HAC, prove the famous altitude theorem:

$$AH^2 = HB \times HC.$$

3. If $HB = 4 \text{ cm}$ and $HC = 9 \text{ cm}$, compute altitude AH.

Exercise 26.14 The Ozanam Chessboard Paradox (Vanishing Area): An 8×8 chessboard (area 64) is dissected into four pieces: two right triangles with legs 3 and 8, and two trapezoids with bases 5 and 3 and height 5. When rearranged, they seemingly form a rectangle of $5 \times 13 = 65$! Where did the extra square unit come from ($65 - 64 = 1$)? *(Hint: compute the slopes of the hypotenuses using similar triangles).*

Solutions

Level

Solution 26.1 1. Ferris wheel cabin: **Circular translation** (stays upright without spinning on its axis). 2. Reflection in water: **Axial reflection** (the water surface is the reflection axis). 3. Train on tracks: **Translation**. 4. Movie projector: **Dilation (Homothety)** (enlargement centered at the lens). 5. Queen of Hearts: **Point reflection** (half-turn about card center).

Solution 26.2 1. $A'B' = 6 \text{ cm}$. Axial reflection is an isometry: it strictly preserves lengths. 2. $\widehat{B'A'C'} = 45°$. Axial reflection strictly preserves angle measures.

Solution 26.3 1. Quadrilateral $ABM'M$ is a **parallelogram** by definition of translation. 2. Opposite sides of a parallelogram are equal: $MM' = AB = 7 \text{ cm}$.

Solution 26.4 1. Lengths scale by $k = 3$: $\text{Perimeter}' = 3 \times 18 = 54 \text{ cm}$. 2. Areas scale by $k^2 = 3^2 = 9$: $\text{Area}' = 9 \times 14 = 126 \text{ cm}^2$.

Solution 26.5 Yes, triangles ABC and DEF are congruent by the **SAS (Side-Angle-Side)** criterion: two pairs of sides are equal in length (5 cm and 7 cm) and their included angles are equal ($40°$).

Level

Solution 26.6 Let M be a point. Reflection across (d_1) maps M to M_1; reflection across (d_2) maps M_1 to M'. Since $(d_1) \perp (d_2)$, the coordinates show that O is the midpoint of $[MM']$ and $\widehat{MOM'} = 180°$. Two perpendicular reflections compose into a **point reflection centered at O** (a $180°$ rotation about O).

Solution 26.7 In $\triangle ABC$: $\hat{C} = 180° - (65° + 45°) = 70°$. Angles are $65°, 45°, 70°$. In $\triangle RST$: $\hat{R} = 65°, \hat{T} = 70° \Longrightarrow \hat{S} = 180° - (65° + 70°) = 45°$. All three pairs of angles match: the triangles are **similar** by AA.

Solution 26.8 1. Both triangles contain a vertical right angle ($90°$) and an equal angle of incidence/reflection ($\widehat{SMA} = \widehat{PMS}$). By AA, they are **similar**. 2. Ratio:

$$\frac{\text{Tree Height}}{\text{Arthur Eye Height}} = \frac{\text{Mirror to Tree}}{\text{Mirror to Arthur}} \Longrightarrow \frac{h}{1.60} = \frac{12}{2} = 6.$$

$$h = 1.60 \times 6 = 9.60 \text{ meters}.$$

Solution 26.9 1. $x_{A'} = k \times x_A = (-2) \times 3 = -6$. 2. $OA = 3$ and $OA' = |-6| = 6$. Point O lies between A' and A, twice as far from A' as from A.

Solution 26.10 1. Radii are equal: $OA = OB$. Point O is equidistant from A and B, so O lies on the perpendicular bisector of chord $[AB]$. 2. The perpendicular bisector is the unique line perpendicular to $[AB]$ through its midpoint. Since O lies on it, any line through O perpendicular to $[AB]$ is the perpendicular bisector, necessarily bisecting $[AB]$.
Solution 26.11 1. Total length $1.20\ \text{m} = 120\ \text{cm}$. Tile count: $\frac{120}{4} = 30$ tiles. 2. Total area: $30 \times 15\ \text{cm}^2 = 450\ \text{cm}^2$.

Level

Solution 26.12 1. Reflect target ball B across top rail (d) to find point B'. For any point M on rail (d), $MB = MB'$. Thus path length $AM + MB = AM + MB'$. The shortest path between A and B' is the straight segment $[AB']$. Therefore, optimal impact point P is the **intersection of segment $[AB']$ with cushion line (d)**. 2. By the triangle inequality, for any other point $M \neq P$, $AM + MB' > AB'$. The path via P is strictly minimal.
Solution 26.13 1. In right triangle ABC, $\hat{B} + \hat{C} = 90°$. In right triangle HBA, $\widehat{HAB} + \hat{B} = 90° \implies \widehat{HAB} = \hat{C}$. All three triangles share angles $(90°, \hat{B}, \hat{C})$: they are **similar**. 2. Comparing HBA and HAC:

$$\frac{AH}{HC} = \frac{HB}{AH} \implies AH^2 = HB \times HC.$$

3. With $HB = 4$ and $HC = 9$:

$$AH^2 = 4 \times 9 = 36 \implies AH = \sqrt{36} = 6\ \text{cm}\ .$$

Solution 26.14 The apparent paradox is a visual illusion! For the 5×13 rectangle to be solid, the pieces' hypotenuses must form a single continuous straight diagonal. Compare slopes:

- Triangle hypotenuse slope: $\frac{3}{8} = 0.375$.
- Trapezoid edge slope: $\frac{2}{5} = 0.400$.

Since $\frac{3}{8} \neq \frac{2}{5}$, the edges do not line up into a single straight line. They form a very thin, elongated parallelogram gap in the center whose hidden area equals exactly 1 square unit $(65 - 64 = 1)$!

The Question Opening the Next Chapter

Look at the extraordinary journey we have accomplished: we know how to dissect shapes, compute areas, manipulate equations, inequalities, and functions. But what happens when we leave the flat sheet of paper and step into the three-dimensional depth of our physical universe? How do we measure the capacity of a conical amphora, a stone pyramid, or planet Earth itself?

That is the grand exploration of space, solids, and cross sections awaiting you in the next chapter: **Land Partitioning and the Potter's Box**.

CHAPTER 27: Land Partitioning and the Potter's Box

> "A formula memorized by heart without ever being seen is like a sterile seed. But take a pair of scissors, cut out a triangle, paste the pieces on the other side: the rectangle appears, and the formula becomes self-evident."

What You Are About to Discover

- Why **every triangle is half of a rectangle**, and how a pair of scissors suffices to rediscover the formulas for the area of a parallelogram, a trapezoid, and even a circle.
- How to transition from the **flat** (areas, in m^2) to the **solid** (volumes, in m^3), and why we multiply by 100 in one case and by 1,000 in the other.
- The mystery of the **"divided by 3"** in pyramids and cones, and the crowning achievement Archimedes was most proud of: the volume of the **sphere**.
- What appears when you **slice** a solid with a flat blade: geometric **cross sections**.
- The classic pitfalls that cost points on exams: radius or diameter? vertical height or slanted side? cm^3 or liters?

---

1. The Problem: The Feud of Two Farmers and the Potter's Order

In the Meuse valley during the 13th century, the count's chief steward compiles the land registry to levy the fair share of harvested wheat. Two neighboring farmers step forward before him.

- Farmer **Martin** owns a **rectangular** field measuring 20 meters wide by 30 meters long. The steward computes without hesitation:

$$20 \times 30 = 600 \ \text{ square meters.}$$

  It is a grid of 600 small one-meter-by-one-meter plots of land (remember the brick wall from Chapter 5?).
- But farmer **Thomas** owns an **elongated, sharply pointed triangular** field. The side running along the stream measures 40 meters, and the perpendicular distance from the stream to the farthest tip of the meadow is 30 meters.

The surveyors argue: *"How can we count whole squares in a pointed field? The slanted boundaries slice the grid squares into irregular fragments impossible to measure!"*

A young journeyman mason steps up holding a sheet of parchment and a pair of scissors: *"Put away your measuring cords, gentlemen. There is no need to chop blades of grass into crumbs: we merely need to invite a twin brother to the table."*

That very afternoon, at the opposite end of the village, the **potter** receives a curious commission: the miller requests three vessels holding **the exact same capacity of grain**—a cylindrical jar, a conical funnel, and a hemispherical bowl. The potter knows how to measure length with a cord and area with tiles… but how do we measure **what fits inside**?

These two challenges—measuring an irregular surface and measuring interior capacity—will guide us through this chapter. You will discover that both rest upon one and the same foundational principle: **cut, rearrange, and reassemble**, without losing or adding a single grain.

---

2. What Is an Area? Units and Conversions

2.1 The Intuitive Idea

The **area** of a planar shape is the quantity of "paint" required to coat it, or the number of unit squares needed to tile it completely. Two shapes with completely different outlines can have the exact same area: it suffices that one can be dissected and reassembled to form the other.

> **📐 Definition —**
>
> We choose a unit of area: for example, the **square of side length 1 cm**. The **area** of a figure is the number of these unit squares (and fractions of squares) required to tile it exactly, without gaps or overlaps.

The notation $\mathbf{cm^2}$. We write $1\ \mathrm{cm}^2$ and pronounce it "one **square centimeter**." The small superscript 2 is not decorative: it records that we multiplied **two** lengths (a width in cm and a length in cm). It is the exact same exponent seen in $5^2 = 5 \times 5$ (Chapter 18). Indeed, the area of a square of side length c is c^2: that is precisely why we say "c **squared**"!

Perimeter versus Area: The **perimeter** is the length of the fence (walking around the boundary, in cm); the **area** is the surface of the lawn (filling the interior, in cm^2). A rectangle of 1×9 and a square of 3×3 share the same area ($9\ \mathrm{cm}^2$) but have very different perimeters ($20\ \ \mathrm{cm}$ versus $12\ \ \mathrm{cm}$).

2.2 Area Conversions: The Pitfall of "Times 100"

One meter is 10 decimeters. But how many square decimeters are in one **square meter**? Sketch it: a square of side length 1 m is a square of side length 10 dm, so it contains $10 \times 10 = 100$ small squares of 1 dm on each side.

$$1\ \mathrm{m}^2 = 100\ \mathrm{dm}^2 = 10,000\ \mathrm{cm}^2 = 1,000,000\ \mathrm{mm}^2.$$

In an area conversion table, **every metric unit spans two columns**:

km^2		hm^2 (ha)		dam^2 (a)		m^2		dm^2		cm^2		mm^2	
							3	5	0	0	0		

Here we read: $3.5\ \mathrm{m}^2 = 350\ \mathrm{dm}^2 = 35,000\ \mathrm{cm}^2$.

> **💡 Did You Know?**
>
> Farmers traditionally use the **are** (a), which equals $1\ \mathrm{dam}^2 = 100\ \mathrm{m}^2$ (a square of 10 m by 10 m), and the **hectare** (ha), which equals $1\ \mathrm{hm}^2 = 10,000\ \mathrm{m}^2$ (a square of 100 m by 100 m, roughly the size of one and a half soccer fields). The word "are" comes from the Latin *area*, meaning "open space," which also gave rise to the English word "area"!

> **⚠️ Watch Out! Pitfall.**
>
> $1\ \mathrm{m}^2 \neq 100\ \mathrm{cm}^2$. Because $1\ \ \mathrm{m} = 100\ \ \mathrm{cm}$, we have $1\ \mathrm{m}^2 = 100\ \ \mathrm{cm} \times 100\ \ \mathrm{cm} = 10,000\ \mathrm{cm}^2$. When converting units of area, you multiply or divide by 100 **at each step**, not by 10.

---

3. The Scissors Stroke: Every Triangle Is Half a Rectangle

3.1 Base and Height: Precise Definitions

Before cutting, we must establish exact terms.

> **📐 Definition —**
>
> In any triangle, we choose one side to call the **base**. The **altitude (height) relative to that base** is the line segment dropped from the opposite vertex **perpendicular** to the line containing the base. Its length is also termed the "height."

The symbol $\perp$ reads "is perpendicular to." The small square drawn at the foot H of the altitude indicates the right angle. A triangle has **three** sides, hence **three** altitudes—one for each possible base. You may select whichever is most convenient, provided you pair a base **with its own corresponding height**.

3.2 The Twin Brother Proof

Let us return to Thomas's pointed field: base $b = 40$ m, height $h = 30$ m.

1. Cut out a **second identical twin triangle** in another color.
2. Rotate this twin through a **half-turn (180°)** (a point reflection through the midpoint of a side, as in Chapter 26).
3. Glue it alongside the first: together, they form a **parallelogram** of base b and height h.
4. And as we will prove momentarily, a parallelogram has the exact same area as a **rectangle** of dimensions $b \times h$.

Even simpler, when the foot H of the altitude falls inside the base: the altitude splits the triangle into two right triangles. Each is half of a small rectangle (halved along its diagonal). The two small rectangles together form the large rectangle $b \times h$, and the triangle occupies precisely half of it.

> **📐 *Formula — Area of a Triangle***
>
> $$\mathcal{A}_{\text{triangle}} = \frac{\text{base} \times \text{height}}{2} = \frac{b \times h}{2}$$

We frequently write $\mathcal{A}$ in calligraphy script to avoid confusion with vertex point A.

For Thomas's field: $\mathcal{A} = \frac{40 \times 30}{2} = \frac{1200}{2} = 600$ m^2. Both farmers, despite their drastically different field shapes, hold **the exact same surface area** of farmland: they will owe the identical tax, down to the single grain of wheat.

3.3 Boundary Cases: When the Altitude Falls Outside

> **⚠️ Watch Out! Pitfall.**
>
> In an **obtuse** triangle (having an angle greater than 90°), the altitude to certain sides falls **outside** the triangle, landing on the **extended line** of the base. The formula remains unconditionally true, but the proof requires care!

> **Proof —**
>
> Foot H lies on the line extension of $[AB]$, beyond vertex B. The large right triangle AHC contains our target triangle ABC **plus** the small right triangle BHC. Therefore:
>
> $$\mathcal{A}_{ABC} = \mathcal{A}_{AHC} - \mathcal{A}_{BHC} = \frac{AH \times h}{2} - \frac{BH \times h}{2} = \frac{(AH - BH) \times h}{2} = \frac{AB \times h}{2}.$$
>
> The formula "base × height ÷ 2" is universally valid for **all** triangles.

> **💡 Did You Know?**
>
> Any triangle sharing base $[AB]$ whose third vertex C slides along a line parallel to (AB) possesses **the exact same area**, because its perpendicular height never varies! Even if the triangle becomes extremely flattened and skewed. This is Euclid's famous "shearing triangle" theorem (Book I of the *Elements*, Proposition 37).

4. Parallelograms and Trapezoids Unveiled

4.1 The Parallelogram: The Tilted Field

Take a cardboard rectangle and push its top edge sideways: it tilts into a parallelogram. Has it lost any material? Cut off the small protruding triangle on one side and glue it into the recess on the opposite side: **the rectangle reappears**, with identical base and height.

> 📐 ***Formula — Area of a Parallelogram***
>
> $$\mathcal{A}_{\text{parallelogram}} = b \times h$$
>
> *where h is the distance between the two parallel sides of length b (measured **perpendicularly**).*

> ⚠️ **Watch Out! Pitfall.**
>
> **Never** multiply the base by the slanted side! A parallelogram with sides 10 cm and 6 cm can have an area of 60 cm^2 (if upright, forming a rectangle), 30 cm^2 (if tilted), or nearly 0 if squashed flat. The slanted edge length does not change, but the height certainly does.

4.2 The Trapezoid: The Flared Skirt

A **trapezoid (trapezium)** is a quadrilateral with two parallel sides called **bases**: the major base B and the minor base b. The **height** h is the perpendicular distance between them.

Take two twin trapezoids. Invert one upside down and place it next to the other: together they form a **parallelogram** whose base equals $B + b$ and whose height is h.

> ⚠️ **Formula — Area of a Trapezoid**
>
> $$\mathcal{A}_{\text{trapezoid}} = \frac{(B + b) \times h}{2}$$

This can also be understood as $\frac{B+b}{2} \times h$: it equals the area of a rectangle whose width is the **average** of the two bases.

> ⚠️ **Did You Know?**
>
> Checking extreme boundary cases is reassuring. If $b = B$, the trapezoid becomes a parallelogram, recovering $\frac{2B \times h}{2} = B \times h$. If $b = 0$, the minor base collapses to a point, turning the trapezoid into a triangle, recovering $\frac{B \times h}{2}$. A sound mathematical formula always holds true at its extreme limits!

> 📐 **The Secret of 1/3 — Why Pyramid Volume Is $\frac{1}{3} \times \text{Base} \times \text{Height}$**
>
> Why does a factor of $\frac{1}{3}$ govern pointed solids while triangles take $\frac{1}{2} \times$ base $\times$ height? The most elegant concrete proof rests on **cube dissection**: Take a solid cube of side c. Select one corner vertex S. From this common peak, connect edges to the three opposite square faces. The cube decomposes exactly into **3 identical oblique square pyramids**, each sharing base $B = c^2$ and altitude $h = c$! Because 3 congruent pyramids pack the cube with zero gaps:
>
> $$3 \times V_{\text{pyramid}} = V_{\text{cube}} = B \times h \Longrightarrow V_{\text{pyramid}} = \frac{1}{3} \times B \times h.$$
>
> By Archimedean slicing, this $\frac{1}{3}$ ratio extends to all polygonal pyramids and circular cones ($V = \frac{1}{3}\pi R^2 h$)!

5. Archimedes' Pizza: The Area of a Disk

You met the number π in Chapter 16: it is the ratio of a circle's circumference to its diameter, $\pi \approx 3.14159$. The circumference of a circle of radius R is $2\pi R$. But what about the **surface area** of the interior disk?

In the 3rd century BC, **Archimedes of Syracuse** envisioned slicing the disk like a pizza into a large number of slender triangular wedges radiating from the center. Each wedge has length equal to radius R. He arranged these slices **head-to-tail**, alternating one pointing up and one pointing down:

The resulting shape resembles an **undulating rectangle**: * its height is radius R; * its length is **half the circumference** of the pizza (the other half of the crust forms the bottom edge), which is $\frac{2\pi R}{2} = \pi R$.

As the slices become infinitely slender, the ripples vanish: in the limit, it becomes a true rectangle.

> 📐 ***Formula — Area of a Disk of Radius R***
>
> $$\mathcal{A}_{\text{disk}} = \pi R \times R = \pi R^2$$
>
> *Pronounced "pi R squared." Note: this is $\pi \times (R \times R)$; the exponent applies only to R.*

> ⚠️ **Watch Out! Pitfall.**
>
> Two classical mistakes: 1. Using the **diameter** instead of the radius. For a disk of diameter 10 cm, $R = 5$ cm and $\mathcal{A} = 25\pi \approx 78.5$ cm^2, and definitely not 100π! 2. Confusing $2\pi R$ (the **circumference**, a 1D length in cm) with πR^2 (the **area**, a 2D surface in cm^2). Memory aid: area contains a square exponent, matching its unit cm^2.

> 💡 **True History —**
>
> Archimedes did not possess the decimal digits of π (no one can write them all down, as they are endless and non-repeating!). He proved that the area of a circle equals that of a right triangle whose legs are the radius and the circle's circumference: $\frac{2\pi R \times R}{2} = \pi R^2$. Furthermore, by inscribing and circumscribing 96-sided regular polygons around a circle, he bounded π strictly between $3\frac{10}{71}$ and $3\frac{1}{7}$.

---

6. Worked Examples on Area

Worked Example 1 — The Trapezoidal Garden. A garden has the shape of a trapezoid: the back fence measures 16 m, the front fence measures 10 m, and the two parallel fences are separated by 8 m. What is its area? How many bags of grass seed are needed if one bag covers 25 m^2?

Solution. $\mathcal{A} = \frac{(16+10)\times 8}{2} = \frac{26\times 8}{2} = \frac{208}{2} = 104$ m^2. $104 \div 25 = 4.16$: 4 bags are not enough, so **5 bags** are required. (Always round **up** when purchasing coverage materials!)

Worked Example 2 — The Hidden Altitude of a Right Triangle. A triangle ABC is right-angled at A with $AB = 6$ cm, $AC = 8$ cm, and hypotenuse $BC = 10$ cm. Calculate the length of the altitude drawn from vertex A to hypotenuse $[BC]$.

Solution. In a right triangle, the two perpendicular legs act as **base and altitude to each other**. Using base $[AB]$ and altitude $[AC]$:

$$\mathcal{A} = \frac{6 \times 8}{2} = 24 \text{ cm}^2.$$

Alternatively, we can express the area using hypotenuse $[BC]$ as the base and unknown altitude h dropped from A: $\mathcal{A} = \frac{10\times h}{2} = 5h$. Both calculations describe the exact same area, so $5h = 24 \Longrightarrow h = 4.8$ cm. We found an unknown length purely through area equivalence!

Worked Example 3 — Composite Figure. A church window has the shape of a rectangle 120 cm wide and 150 cm tall, topped by a semicircle of diameter 120 cm. What is the area of the stained glass window, rounded to the nearest cm^2?

Solution. Dissect the figure into basic components: * Rectangle: $120 \times 150 = 18,000$ cm^2. * Semicircle: radius $R = 60$ cm, area $\frac{\pi \times 60^2}{2} = \frac{3600\pi}{2} = 1800\pi \approx 5654.87$ cm^2. * Total: $18,000 + 1800\pi \approx 23,655$ cm^2, or approximately 2.37 m^2.

Worked Example 4 — The Annulus (Circular Ring). A circular pond of radius 6 m is surrounded by a walking path 2 m wide. What is the area of the path?

Solution. The path is an **annulus**: the large disk (radius $6 + 2 = 8$ m) minus the small inner pond (radius 6 m).

$$\mathcal{A} = \pi \times 8^2 - \pi \times 6^2 = 64\pi - 36\pi = 28\pi \approx 87.96 \text{ m}^2.$$

> ⚠ **Watch Out! Pitfall.**
>
> You cannot subtract radii before squaring: $\pi \times (8-6)^2 = 4\pi$ is completely wrong! One subtracts the **areas**, not the linear radii.

Worked Example 5 — Notary Land Conversions. A rectangular parcel of farmland measures 85 m by 64 m. Express its area in m^2, in ares, and in hectares.

Solution. $85 \times 64 = 5,440$ m^2. Since 1 a $= 100$ m^2: $5,440$ m$^2 = 54.4$ a. Since 1 ha $= 100$ a: 54.4 a $= 0.544$ ha.

CHAPTER 28: Eratosthenes and the Circumference of the Earth

> "Do not say that the Earth is too vast to be measured by a mortal. Grant me a simple staff planted upright in the soil, a ray of sunlight, and a right triangle: I will tell you the circumference of the world before the sun sets."

What You Are About to Discover

- How **Eratosthenes**, more than 2,200 years ago, calculated the circumference of the Earth using a staff, a well, and a camel caravan.
- How a right triangle constructs three indestructible bridges between **angles** and **lengths**: the **cosine**, the **sine**, and the **tangent**.
- The universal mnemonic **SOH-CAH-TOA** (or **CAH-SOH-TOA**) so you never hesitate again.
- Why the cosine and sine are **strictly bounded between 0 and 1** inside any right triangle.
- How to determine the **inaccessible height of a sea cliff**, the **width of a roaring river**, or the path of a ship without taking a single step across.
- Pythagoras's magical identity in disguise: $\cos^2(x) + \sin^2(x) = 1$.

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1. The Problem: Midday on the Summer Solstice in Alexandria

The date is **June 21**, around the year **240 BC**, beneath the luminous skies of Egypt. In the coastal city of Alexandria, the polymath **Eratosthenes** presides over the most prestigious library of the ancient world: the Great Library. Hundreds of thousands of papyrus scrolls line cedar shelves.

Eratosthenes reads an astonishing dispatch arriving from the southern frontier town of **Syene** (modern-day Aswan), roughly 800 kilometers up the Nile River: > *"On the day of the summer solstice, at the stroke of noon, the sun shines straight down to illuminate the water at the bottom of the deepest vertical wells without casting the faintest shadow on their walls. The stone columns of temples cast no shadow on the pavement. The solar rays strike the earth perpendicularly."*

Eratosthenes waits for that exact day, at that exact hour, in **Alexandria**. He plants a **gnomon**—a perfectly vertical wooden staff—into level ground, and watches closely. To his wonder: in Alexandria, at the exact same moment, **the staff casts a clear shadow**!

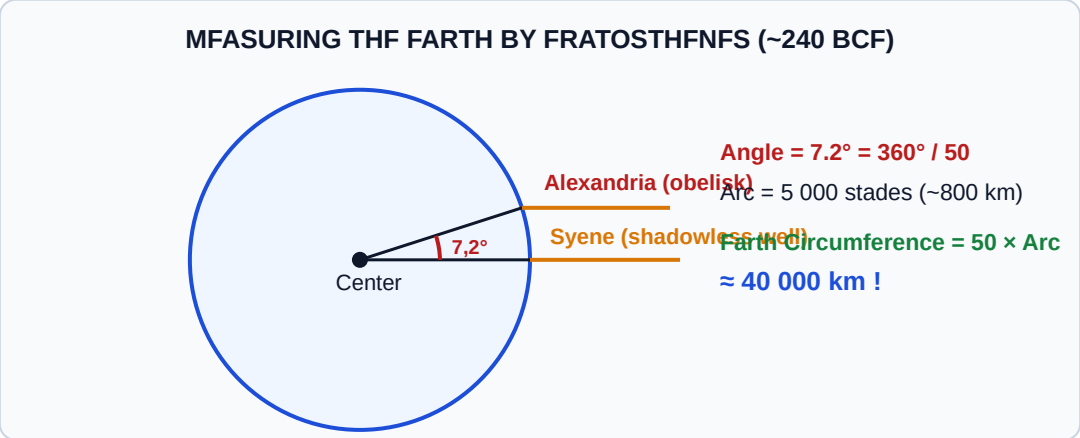

Figure 75: Measuring Earth's radius by Eratosthenes of Alexandria

How is this possible? If the Earth were a flat disk, as ancient poets once imagined, the Sun—situated at a tremendous cosmic distance and showering down parallel light rays—would strike Alexandria at the exact same angle as Syene. There would either be no shadows anywhere, or identical shadows everywhere!

If the staff in Alexandria casts a shadow while the staff in Syene casts none, it can mean only one thing: **the surface of the Earth is curved**!

Eratosthenes measures the height of his staff and the length of its shadow upon the ground. These two segments form a **right triangle**. Comparing these two lengths, he determines that the solar rays deviate from the vertical by an angle of approximately $\mathbf{7.2}^\circ$.

Now, by the geometric theorem of alternate interior angles (studied in Chapter 15), this 7.2° angle at the tip of the staff is **strictly equal to the subtended central angle at the center of the Earth** separating Alexandria from Syene!

Eratosthenes then sets up a luminous proportion: * A complete circle spans 360°. * But 7.2° is exactly $\frac{7.2}{360} = \frac{1}{50}$ of a full circle! * Therefore, the overland distance between Alexandria and Syene represents **one-fiftieth of the entire circumference of the Earth**.

Royal surveyors and camel caravans estimate this travel distance at approximately **5,000 stadia** (about 800 km). Eratosthenes multiplies:

$$\text{Earth's Circumference} = 50 \times 5,000 = \mathbf{250,000 \ \ stadia} \approx \mathbf{40,000 \ \ kilometers} \ !$$

An estimate of staggering precision for antiquity: modern satellite geodesy measures the equatorial circumference at 40,075 km!

Admire the core genius of this feat: to uncover the celestial angle of 7.2°, Eratosthenes did not ascend into the heavens with a giant protractor. He measured **two linear lengths on the ground** (the height of the staff and the length of its shadow) and deduced **an angle**.

This science connecting the side lengths and angles of a triangle is called **trigonometry** (from Greek *trigōnon*, triangle, and *metron*, measure).

---

2. The Right Triangle: Naming the Three Sides

To practice trigonometry without confusion, one must learn to view a right triangle through the eyes of a **specific acute angle**.

Consider triangle ABC, right-angled at B. Its two acute angles are $\hat{A}$ and $\hat{C}$. Let us choose to view the triangle from acute angle $\hat{A}$.

Observe the three sides from vertex A: 1. **The hypotenuse:** the side directly opposite the 90° right angle. It is always the longest side of the right triangle. Here, it is segment $[AC]$. Its identity never depends on which acute angle you select: it is the master side of the right triangle. 2. **The side adjacent to angle $\widehat{A}$:** the word *adjacent* comes from Latin *ad-jacere*, meaning "lying next to." It is the leg of the right angle that touches angle $\hat{A}$. Here, it is segment $[AB]$. 3. **The side opposite to angle $\widehat{A}$:** it is the leg of the right angle facing directly across from angle $\hat{A}$. Here, it is segment $[BC]$.

> ⚠ **Watch Out! Pitfall — Roles Reverse When You Switch Angles!**
>
> If you shift your perspective to vertex C:
>
> - The hypotenuse remains $[AC]$.
> - The side **adjacent** to $\hat{C}$ becomes $[BC]$!
> - The side **opposite** to $\hat{C}$ becomes $[AB]$! A side is never "opposite" or "adjacent" in the absolute: it is so **strictly relative to the chosen acute angle**.

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3. The Three Ratios: Cosine, Sine, Tangent

Imagine drawing a right triangle with an acute angle of 30°. Whether you draw it in your notebook (hypotenuse 10 cm), on the classroom blackboard (hypotenuse 1 meter), or across the courtyard pavement (hypotenuse 10 meters), these three triangles are exact **dilations** of one another (similar triangles, Chapter 17).

Because their shapes are geometrically identical, **the ratios of their side lengths depend solely on the angle**, and not at all on the physical size of the drawing!

For any given acute angle x, these ratios carry three universal names:

📐 Fundamental Definitions —

In any right triangle:

1. The **cosine** of an acute angle is the ratio of the adjacent side to the hypotenuse:
$$\cos(x) = \frac{\text{length of adjacent side}}{\text{length of hypotenuse}}$$
2. The **sine** of an acute angle is the ratio of the opposite side to the hypotenuse:
$$\sin(x) = \frac{\text{length of opposite side}}{\text{length of hypotenuse}}$$
3. The **tangent** of an acute angle is the ratio of the opposite side to the adjacent side:
$$\tan(x) = \frac{\text{length of opposite side}}{\text{length of adjacent side}}$$

How to Pronounce and Write These Notations

- $\cos(x)$ is written “cos” and read aloud as **“cosine of x.”**
- $\sin(x)$ is written “sin” and read aloud as **“sine of x.”**
- $\tan(x)$ is written “tan” and read aloud as **“tangent of x.”**

The Universal Mnemonic: SOH-CAH-TOA

To recall these definitions instantaneously, students across the world chant this phrase like a warrior’s banner:

S O H — **C** A H — **T** O A

- **S**ine = **O**pposite / **H**ypotenuse
- **C**osine = **A**djacent / **H**ypotenuse
- **T**angent = **O**pposite / **A**djacent

Some European curricula memorize it in the order **CAH-SOH-TOA**. Both are identical; choose whichever resonates best in your memory!

---

4. Two Essential Properties to Master

4.1 Why Cosine and Sine Always Lie Between 0 and 1

In any right triangle, the hypotenuse is strictly the **longest side**. Consequently, the adjacent leg is strictly shorter than the hypotenuse, and the opposite leg is likewise strictly shorter than the hypotenuse.

Since the numerator is strictly smaller than the denominator (and all lengths are positive):

$$\frac{\text{adjacent}}{\text{hypotenuse}} < 1 \qquad \text{and} \qquad \frac{\text{opposite}}{\text{hypotenuse}} < 1$$

📐 Property — Bounds on Cosine and Sine.

For any acute angle x in a right triangle ($0^\circ < x < 90^\circ$):

$$0 < \cos(x) < 1 \qquad \text{and} \qquad 0 < \sin(x) < 1$$

⚠️ Watch Out! Pitfall.

If on a test you calculate $\cos(x) = 1.45$ or $\sin(x) = 2.3$, **stop immediately**: that is mathematically impossible! You almost certainly inverted the numerator and denominator. By contrast, the **tangent** $\tan(x) = \frac{\text{opposite}}{\text{adjacent}}$ can equal any positive real number: it can be 0.5, 1, 10, or even $1,000$ if the triangle is extremely tall and narrow!

4.2 The Fundamental Pythagorean Trigonometric Identity

What happens if we sum the squares of the cosine and sine of the same angle x? Consider our right triangle with adjacent side a, opposite side b, and hypotenuse c. By the Pythagorean theorem (Chapter 19):

$$a^2 + b^2 = c^2$$

Now compute:

$$\cos^2(x) + \sin^2(x) = \left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2 = \frac{a^2}{c^2} + \frac{b^2}{c^2} = \frac{a^2 + b^2}{c^2}$$

Since $a^2 + b^2 = c^2$, we have:

$$\frac{a^2 + b^2}{c^2} = \frac{c^2}{c^2} = 1$$

📐 Fundamental Theorem of Trigonometry —

For any angle x:

$$\cos^2(x) + \sin^2(x) = 1$$

(Note on notation: $\cos^2(x)$ denotes $(\cos(x))^2$, meaning the cosine multiplied by itself).

Furthermore, dividing sine by cosine yields:

$$\frac{\sin(x)}{\cos(x)} = \frac{\frac{\text{opposite}}{\text{hypotenuse}}}{\frac{\text{adjacent}}{\text{hypotenuse}}} = \frac{\text{opposite}}{\text{adjacent}} = \tan(x)$$

📐 *Quotient Identity* —

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

These two identities allow you to determine the sine and tangent of an angle the instant its cosine is known, without even determining the angle itself!

💡 Did You Know? Where Do These Strange Words Come From?

The word **sine** traveled through one of the most famous linguistic misadventures in history. Fifth-century Indian astronomers termed this half-chord *jya-ardha*, shortened to *jiva*. When Arab scholars translated these texts in the 8th century, they transliterated the sound as *jiba*, written *jb* without vowels. In 12th-century Toledo, Spain, the translator Gerard of Cremona mistook this word for the Arabic word *jayb*, which means “crease of a robe, pocket, or bay.” He translated it into Latin as **sinus**, meaning “fold, curve, or gulf” (which also gives us the word “sinuous”)! As for **cosine**, it is an abbreviation of *complementi sinus*: the sine of the complementary angle!

5. Using the Calculator: Two Essential Modes

To carry out trigonometric calculations, you will use the scientific calculator keys: `cos`, `sin`, `tan`.

> ⚠️ **Critical Warning! Calculator Angle Mode!**
>
> A scientific calculator can measure angles in Degrees (DEG), Radians (RAD), or Gradians (GRA). In middle school, **all angles are given in degrees**. Always ensure that a small letter **D** or **DEG** appears at the top of your display screen. If you see an **R** or **RAD**, all your calculations will be wrong! To switch to degrees, navigate to the `Setup` or `Mode` menu.

Case 1: The Angle Is Known, Find the Ratio

Press the key directly:

- $\cos(60^\circ) = 0.5$ exactly.
- $\sin(30^\circ) = 0.5$ exactly.
- $\tan(45^\circ) = 1$ exactly (because in a half-square, opposite leg = adjacent leg!).
- $\cos(48^\circ) \approx 0.6691\ldots$

Case 2: The Ratio Is Known, Find the Angle (Inverse Keys)

If you know that $\cos(x) = 0.8$, what is the angle x? Use the inverse key, labeled $\mathbf{arccos}$ or $\mathbf{cos^{-1}}$ (typically accessed by pressing `2nd` or `Shift` followed by `cos`):

- If $\cos(x) = 0.8$, then $x = \arccos(0.8) \approx 36.87^\circ$.
- If $\sin(x) = 0.5$, then $x = \arcsin(0.5) = 30^\circ$.
- If $\tan(x) = 1$, then $x = \arctan(1) = 45^\circ$.

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6. The Two Primary Problems in Middle School Trigonometry

Every middle school trigonometry problem boils down to one of two quests: 1. **Calculate an unknown length** when given an angle and another side length. 2. **Calculate an unknown angle** when given two side lengths.
Here is the 4-step method:

> 🧭 **Step-by-Step Method — Trigonometric Problem-Solving Method**
>
> - 1. Sketch a diagram and identify the right
> - angle and the target acute angle.
> - 2. Label the three sides:
> - Hypotenuse / Adjacent / Opposite.
> - 3. Select the correct formula:
> - Adjacent and Hypotenuse? => CAH
> - Opposite and Hypotenuse? => SOH
> - Opposite and Adjacent? => TOA
> - 4. Write the equation, isolate the unknown
> - cross-multiply), and evaluate.

---

7. Step-by-Step Worked Examples

Example 1: Calculating a length in the numerator

In triangle KLM, right-angled at L, we are given $KM = 8\ \text{cm}$ and $\hat{K} = 35^\circ$. Calculate the length of side $[LM]$, rounded to the nearest millimeter.

Detailed Solution: 1. **Triangle type:** Triangle KLM is right-angled at L. 2. **Identify sides relative to angle $\widehat{K} = 35°$:**

- $[KM]$ is the hypotenuse (known: 8 cm).
- $[LM]$ is the side **opposite** to $\hat{K}$ (unknown sought).

3. **Select formula:** Opposite and hypotenuse $\Longrightarrow$ use **sine** (SOH):

$$\sin(\hat{K}) = \frac{LM}{KM}$$

4. **Compute:**

$$\sin(35°) = \frac{LM}{8}$$

Multiplying both sides by 8 (or cross-multiplying with $\frac{\sin(35°)}{1} = \frac{LM}{8}$):

$$LM = 8 \times \sin(35°) \approx 8 \times 0.573576 \approx 4.588 \ \text{ cm}$$

The length LM is approximately $\mathbf{4.6 \ \ cm}$ (to the nearest millimeter).

Example 2: Calculating a length in the denominator (The Classic Pitfall)

In triangle RST, right-angled at S, we know $RS = 5 \ \text{ cm}$ and $\hat{R} = 40°$. Calculate the length of hypotenuse $[RT]$, rounded to the nearest millimeter.

Detailed Solution: 1. Triangle RST is right-angled at S. 2. Relative to angle $\hat{R} = 40°$:

- $[RS]$ is the **adjacent** side ($RS = 5 \ \text{ cm}$).
- $[RT]$ is the **hypotenuse** (unknown).

3. Adjacent and hypotenuse $\Longrightarrow$ use **cosine** (CAH):

$$\cos(\hat{R}) = \frac{RS}{RT}$$

4. Substitute values:

$$\cos(40°) = \frac{5}{RT}$$

To solve for RT in the denominator, cross-multiply using $\frac{\cos(40°)}{1} = \frac{5}{RT}$:

$$RT \times \cos(40°) = 5 \times 1 \Longrightarrow RT = \frac{5}{\cos(40°)}$$

On the calculator:

$$RT = \frac{5}{\cos(40°)} \approx \frac{5}{0.766044} \approx 6.527 \ \text{ cm}$$

The hypotenuse $[RT]$ measures approximately $\mathbf{6.5 \ \ cm}$. *(Sanity check: the hypotenuse is 6.5 cm, strictly longer than the adjacent leg of 5 cm. This is entirely consistent!)*

Example 3: Calculating an acute angle given two sides

A sailboat approaches a lighthouse erected on a rocky cliff. The top of the cliff rises 42 meters above sea level. The boat is positioned 110 meters horizontally from the base of the cliff. Under what angle of elevation does the sailor see the top of the cliff? (Round to the nearest tenth of a degree).

Detailed Solution: 1. Triangle FPB formed by the vertical cliff, horizontal water, and line of sight is right-angled at P. 2. Relative to angle $\hat{B}$:

- The opposite side is the cliff height: $FP = 42 \ \text{ m}$.
- The adjacent side is the horizontal distance: $PB = 110 \ \text{ m}$.

3. Opposite and adjacent $\Longrightarrow$ use **tangent** (TOA):

$$\tan(\hat{B}) = \frac{FP}{PB} = \frac{42}{110}$$

4. Use the inverse key $\arctan$ (or $\tan^{-1}$) on the calculator:

$$\hat{B} = \arctan\left(\frac{42}{110}\right) \approx \arctan(0.3818) \approx 20.899^\circ$$

The sailor views the cliff top at an angle of elevation of approximately $\mathbf{20.9^\circ}$.

Example 4: Fully Solving a Right Triangle

Let ABC be a right triangle at A with $BC = 13\ \text{cm}$ and $AB = 5\ \text{cm}$. 1. Calculate the exact length of side AC. 2. Determine the exact value of $\cos(\hat{B})$, then the measure of angle $\hat{B}$ rounded to the nearest degree. 3. Deduce the measure of angle $\hat{C}$ without using trigonometric functions.

Detailed Solution: 1. By the Pythagorean theorem in triangle ABC right-angled at A:

$$BC^2 = AB^2 + AC^2 \Longrightarrow 13^2 = 5^2 + AC^2 \Longrightarrow 169 = 25 + AC^2$$

$$AC^2 = 169 - 25 = 144 \Longrightarrow AC = \sqrt{144} = 12\ \text{cm}\ .$$

2. In the right triangle, relative to angle $\hat{B}$:

$$\cos(\hat{B}) = \frac{AB}{BC} = \frac{5}{13}$$

Using the calculator: $\hat{B} = \arccos\left(\frac{5}{13}\right) \approx 67.38^\circ$, which is $\mathbf{67^\circ}$ to the nearest degree. 3. In a right triangle, the two acute angles are complementary ($\hat{B} + \hat{C} = 90^\circ$):

$$\hat{C} = 90^\circ - \hat{B} \approx 90^\circ - 67.38^\circ \approx 22.62^\circ \approx \mathbf{23}^\circ.$$

Example 5: Finding sine from cosine without finding the angle

An acute angle x satisfies $\cos(x) = \frac{3}{5} = 0.6$. Compute the exact values of $\sin(x)$ and $\tan(x)$ without using the $\arccos$ key on your calculator.

Detailed Solution: 1. Apply the fundamental identity $\cos^2(x) + \sin^2(x) = 1$:

$$\left(\frac{3}{5}\right)^2 + \sin^2(x) = 1 \Longrightarrow \frac{9}{25} + \sin^2(x) = 1$$

$$\sin^2(x) = 1 - \frac{9}{25} = \frac{25 - 9}{25} = \frac{16}{25}$$

Since x is acute, its sine is positive:

$$\sin(x) = \sqrt{\frac{16}{25}} = \frac{4}{5} = \mathbf{0.8}.$$

2. Apply the quotient identity $\tan(x) = \frac{\sin(x)}{\cos(x)}$:

$$\tan(x) = \frac{\frac{4}{5}}{\frac{3}{5}} = \frac{4}{5} \times \frac{5}{3} = \frac{4}{3} \approx \mathbf{1.333}...$$

This is the classic 3-4-5 Egyptian right triangle!

Key Takeaways

1. **Right Triangle Vocabulary:**
   - **Hypotenuse**: side facing the right angle, strictly the longest side.
   - **Adjacent side** to an acute angle: the leg of the right angle that touches that angle.
   - **Opposite side** to an acute angle: the leg of the right angle facing that angle.
2. **The Magic Formulas (SOH-CAH-TOA):**

$$\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}} \qquad \cos(\theta) = \frac{\text{Adjacent}}{\text{Hypotenuse}} \qquad \tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$$

3. **Core Properties:**
   - For any acute angle in a right triangle: $0 < \cos(x) < 1$ and $0 < \sin(x) < 1$.
   - Pythagorean identity: $\cos^2(x) + \sin^2(x) = 1$.
   - Quotient identity: $\tan(x) = \frac{\sin(x)}{\cos(x)}$.
   - Complementary angles: if $\alpha + \beta = 90^\circ$, then $\cos(\alpha) = \sin(\beta)$ and $\sin(\alpha) = \cos(\beta)$.
4. **Calculator Guidelines:**
   - Always ensure the calculator is in **Degree mode (DEG)**.
   - To find a side length: use `sin`, `cos`, or `tan`.
   - To find an angle: use `arcsin` ($\sin^{-1}$), `arccos` ($\cos^{-1}$), or `arctan` ($\tan^{-1}$).

Exercises

Level : Direct Application

Exercise 28.1 In a triangle DEF right-angled at D, identify: 1. The hypotenuse. 2. The side adjacent to angle $\hat{E}$. 3. The side opposite to angle $\hat{E}$. 4. Write down the ratios $\cos(\hat{E})$, $\sin(\hat{E})$, and $\tan(\hat{E})$ using vertex letters.

Exercise 28.2 Using a calculator in Degree mode, provide the rounded value to the nearest thousandth of: a) $\cos(25^\circ)$ b) $\sin(70^\circ)$ c) $\tan(12^\circ)$ d) $\tan(85^\circ)$

Exercise 28.3 Find the measure of acute angle x rounded to the nearest tenth of a degree, given: a) $\cos(x) = 0.72$ b) $\sin(x) = 0.35$ c) $\tan(x) = 1.8$ d) $\cos(x) = \frac{\sqrt{2}}{2}$

Exercise 28.4 Let ABC be a right triangle at B such that $AB = 6$ cm and $AC = 10$ cm. Calculate the exact value of $\cos(\hat{A})$, then deduce the measure of angle $\hat{A}$ rounded to the nearest degree.

Exercise 28.5 In triangle MNO right-angled at N, $MO = 12$ cm and $\widehat{M} = 30^\circ$. Calculate the exact length of $[NO]$.

Level : Deeper Reflection and Practical Situations

Exercise 28.6 (Wheelchair Access Ramp) Accessibility regulations mandate that public wheelchair ramps must not form an incline angle greater than 5° with the horizontal. A ramp is 4 meters long and ascends to a porch 40 cm high. Does this ramp comply with regulations? Justify with calculation.

Exercise 28.7 (Tree Struck by Lightning) During a storm, a 15-meter vertical tree was snapped in two by lightning. The upper portion bent over without detaching from the trunk: its crown touches the ground, forming an angle of 28° with the horizontal ground. At what height did the trunk snap? (Round to the nearest centimeter).

Exercise 28.8 (Crossing the River) A surveyor stands on a riverbank at point A, directly facing a landmark tree C on the opposite bank (segment $[AC]$ is perpendicular to the parallel riverbanks). He walks 50 meters along the bank in a straight line to point B. From point B, he sights tree C through a theodolite and measures an angle $\widehat{ABC} = 38^\circ$. What is the width of the river, rounded to the nearest meter?

Exercise 28.9 (The Rhombus and Its Diagonals) A rhombus $ABCD$ has side lengths of 8 cm. One of its interior angles measures 50°. Recall the properties of a rhombus's diagonals, then compute the exact length of each diagonal (rounded to the nearest millimeter).

Exercise 28.10 (Airplane on Descent) An airliner flies at an altitude of 9,000 meters at a constant speed of 720 km/h. It begins its descent toward the runway at a constant glide slope of 3° below the horizontal. 1. What total flight distance (along its slanted descent path) must it travel before touching down? 2. How long does this descent take, in minutes and seconds?

Exercise 28.11 (Trigonometric Identities) An acute angle α satisfies $\sin(\alpha) = \frac{5}{13}$. 1. Without calculating the measure of angle α, use the fundamental identity to find the exact value of $\cos(\alpha)$ as an irreducible fraction. 2. Determine the exact value of $\tan(\alpha)$. 3. Verify your results using a calculator.

Level : Research Challenges and Olympiad-Style Riddles

Exercise 28.12 (Measuring the Inaccessible Tower) To measure the height of a medieval fortress tower encircled by a deep moat, a surveyor stands at point A and measures a sight angle of 32° to the tower summit. He then walks straight backward 25 meters across level ground to point B, where the sight angle to the summit is 24°. Let h be the height of the tower and x the distance from the base of the tower to point A. 1. Express x in terms of h and $\tan(32^\circ)$. 2. Express $x + 25$ in terms of h and $\tan(24^\circ)$. 3. Formulate an equation in h, and compute the tower height to the nearest decimeter.

Exercise 28.13 (Eratosthenes' Gnomon Shadow) In Alexandria, Eratosthenes planted a vertical gnomon 1 meter in height. The ground shadow measured approximately 12.6 cm. 1. Compute the angle formed by the sunbeams with the vertical staff. 2. Does this recover the historical value of 7.2°?

Exercise 28.14 (General Triangle Area Formula) Let ABC be an arbitrary triangle where two side lengths $AB = c$ and $AC = b$ are known, along with the acute enclosed angle $\hat{A}$. Let H be the foot of the altitude from C. 1. In right triangle ACH, express altitude CH in terms of b and $\sin(\hat{A})$. 2. Prove the famous area formula:

$$\text{Area}(ABC) = \frac{1}{2}bc\sin(\hat{A})$$

3. Compute the area of a triangle having sides 7 cm and 10 cm enclosing an angle of 40°.

Exercise 28.15 (The Isosceles Right Triangle and Exact Values) Consider a square $ABCD$ of side length 1. Draw diagonal $[AC]$. 1. What is the nature of triangle ABC? What are the measures of its acute angles? 2. Compute the exact length of diagonal $[AC]$ using the Pythagorean theorem. 3. Deduce the exact values of $\cos(45^\circ)$, $\sin(45^\circ)$, and $\tan(45^\circ)$.

Solutions

Level

- **Solution 28.1:** 1. Hypotenuse: $[EF]$. 2. Adjacent side to $\hat{E}$: $[DE]$. 3. Opposite side to $\hat{E}$: $[DF]$. 4. $\cos(\hat{E}) = \frac{DE}{EF}$, $\sin(\hat{E}) = \frac{DF}{EF}$, $\tan(\hat{E}) = \frac{DF}{DE}$.
- **Solution 28.2:** a) ≈ 0.906; b) ≈ 0.940; c) ≈ 0.213; d) ≈ 11.430.
- **Solution 28.3:** a) $x = \arccos(0.72) \approx 43.9^\circ$; b) $x = \arcsin(0.35) \approx 20.5^\circ$; c) $x = \arctan(1.8) \approx 60.9^\circ$; d) $x = 45^\circ$ because $\cos(45^\circ) = \frac{\sqrt{2}}{2}$.
- **Solution 28.4:** $\cos(\hat{A}) = \frac{AB}{AC} = \frac{6}{10} = 0.6$. On the calculator, $\hat{A} = \arccos(0.6) \approx 53^\circ$.
- **Solution 28.5:** In MNO right-angled at N, $\sin(\widehat{M}) = \frac{NO}{MO} \Longrightarrow \sin(30^\circ) = \frac{NO}{12}$. Since $\sin(30^\circ) = 0.5$, $NO = 12 \times 0.5 = 6 \text{ cm}$.

Level

- **Solution 28.6 (Access Ramp):** Let α be the ramp's incline angle. The ramp forms the hypotenuse ($4 \text{ m} = 400 \text{ cm}$) and the rise forms the opposite leg (40 cm).

$$\sin(\alpha) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{40}{400} = 0.1$$

  $\alpha = \arcsin(0.1) \approx 5.74^\circ$. Since $5.74^\circ > 5^\circ$, the slope is too steep: **the ramp does not comply**.
- **Solution 28.7 (Broken Tree):** Let h be the standing trunk height (opposite leg to 28°). The bent upper portion forms the hypotenuse of length L. Total height is $h + L = 15 \Longrightarrow L = 15 - h$.

$$\sin(28^\circ) = \frac{h}{L} = \frac{h}{15-h}$$

$$(15-h)\sin(28^\circ) = h \Longrightarrow 15\sin(28^\circ) = h(1+\sin(28^\circ)) \Longrightarrow h = \frac{15\sin(28^\circ)}{1+\sin(28^\circ)}$$

  Since $\sin(28^\circ) \approx 0.46947$:

$$h \approx \frac{15 \times 0.46947}{1 + 0.46947} = \frac{7.042}{1.46947} \approx 4.79 \text{ m}.$$

The tree snapped at approximately **4.79 meters** above ground.

- **Solution 28.8 (River Crossing):** Triangle ABC is right-angled at A. Relative to $\hat{B} = 38°$, $[AC]$ is the opposite side (river width) and $[AB] = 50$ m is the adjacent leg.

$$\tan(38°) = \frac{AC}{50} \Longrightarrow AC = 50 \times \tan(38°) \approx 50 \times 0.781286 \approx 39.06 \text{ m}.$$

The river is approximately **39 meters** wide.

- **Solution 28.9 (Rhombus):** Diagonals of a rhombus are perpendicular bisectors of each other and bisect interior angles. Let O be the center. $\triangle AOB$ is right-angled at O with hypotenuse $AB = 8$ cm and $\widehat{OAB} = 25°$.
  - Half-diagonal OA: $\cos(25°) = \frac{OA}{8} \Longrightarrow OA = 8\cos(25°) \approx 7.2505$ cm. Long diagonal $AC = 2 \times OA \approx \mathbf{14.5}$ **cm**.
  - Half-diagonal OB: $\sin(25°) = \frac{OB}{8} \Longrightarrow OB = 8\sin(25°) \approx 3.3809$ cm. Short diagonal $BD = 2 \times OB \approx \mathbf{6.8}$ **cm**.
- **Solution 28.10 (Airplane Descent):**
  1. Altitude is the opposite leg ($9,000$ m), flight path is hypotenuse D:

$$\sin(3°) = \frac{9,000}{D} \Longrightarrow D = \frac{9,000}{\sin(3°)} \approx \frac{9,000}{0.052336} \approx 171,966 \text{ m} \approx \mathbf{172 \text{ km}}.$$

  2. Velocity is 720 km/h $= \frac{720,000}{3,600} = 200$ m/s. Time $t = \frac{171,966}{200} \approx 859.8$ seconds $\approx$ **14 minutes and 20 seconds**.
- **Solution 28.11 (Identities):**
  1. $\cos^2(\alpha) = 1 - \sin^2(\alpha) = 1 - \frac{25}{169} = \frac{144}{169} \Longrightarrow \cos(\alpha) = \mathbf{\frac{12}{13}}$.
  2. $\tan(\alpha) = \frac{5/13}{12/13} = \mathbf{\frac{5}{12}}$.
  3. $\arcsin(5/13) \approx 22.62°$, $\cos(22.62°) \approx 0.923 \approx 12/13$. Verified.

Level

- **Solution 28.12 (Inaccessible Tower):**
  1. $\tan(32°) = \frac{h}{x} \Longrightarrow x = \frac{h}{\tan(32°)}$.
  2. $\tan(24°) = \frac{h}{x+25} \Longrightarrow x + 25 = \frac{h}{\tan(24°)}$.
  3. Subtracting:

$$25 = h\left(\frac{1}{\tan(24°)} - \frac{1}{\tan(32°)}\right)$$

$\frac{1}{\tan(24°)} \approx 2.2460$, $\frac{1}{\tan(32°)} \approx 1.6003 \Longrightarrow$ Difference $= 0.6457$.

$$h = \frac{25}{0.6457} \approx \mathbf{38.7 \text{ meters}}.$$

- **Solution 28.13 (Eratosthenes' Gnomon):**
  1. Staff height 100 cm (adjacent), shadow 12.6 cm (opposite):

$$\tan(\alpha) = \frac{12.6}{100} = 0.126 \Longrightarrow \alpha = \arctan(0.126) \approx 7.18°.$$

  2. Rounded to the nearest tenth, this yields $\mathbf{7.2°}$, exactly matching Eratosthenes' measurement!
- **Solution 28.14 (General Triangle Area):**
  1. In right triangle ACH, $\sin(\hat{A}) = \frac{CH}{AC} = \frac{CH}{b} \Longrightarrow CH = b\sin(\hat{A})$.
  2. Area $= \frac{1}{2} \times AB \times CH = \frac{1}{2}cb\sin(\hat{A}) = \frac{1}{2}bc\sin(\hat{A})$.
  3. For $b = 7$, $c = 10$, $\hat{A} = 40°$:

$$\text{Area} = \frac{1}{2} \times 7 \times 10 \times \sin(40°) = 35\sin(40°) \approx 35 \times 0.64279 \approx \mathbf{22.5 \text{ cm}^2}.$$

- **Solution 28.15 (45° Angle):**

1. Triangle ABC has $AB = BC = 1$ and a right angle at B: it is an **isosceles right triangle**. Acute angles are $(180^\circ - 90^\circ)/2 = \mathbf{45}^\circ$.
2. $AC^2 = 1^2 + 1^2 = 2 \Longrightarrow AC = \sqrt{\mathbf{2}}$.
3. Values:
   - $\cos(45^\circ) = \frac{1}{\sqrt{2}} = \frac{\sqrt{\mathbf{2}}}{\mathbf{2}} \approx 0.707$.
   - $\sin(45^\circ) = \frac{1}{\sqrt{2}} = \frac{\sqrt{\mathbf{2}}}{\mathbf{2}} \approx 0.707$.
   - $\tan(45^\circ) = \frac{1}{1} = \mathbf{1}$.

The Question Opening the Next Chapter

Through trigonometry, we learned to survey continuous space, celestial angles, and vast terrestrial distances. But what happens if we step away from smoothly stretching lengths and return to the realm of **discrete, indivisible integers**? How can we tile a rectangular hall with square marble slabs without cutting a single tile? How can treasures be divided into strictly equal shares without breaking a single coin? In Alexandria, a century before Eratosthenes, another intellectual giant established the eternal laws of integer division. Prepare to discover the secrets of divisors and the immortal Euclidean algorithm in Chapter 29: **The Partition of Euclid**.

CHAPTER 29: The Partition of Euclid

> "A number is not a soft dough to be spread out at will: it is an edifice constructed from prime stones cut from diamond. Find the fundamental building blocks, and you will understand how to partition without shattering."

What You Are About to Discover

- Why certain numbers refuse to be factored and constitute the **atoms of arithmetic**: prime numbers.
- How to factor any positive integer into a **unique product of prime factors** using factor trees and successive division ladders.
- What the **GCD (Greatest Common Divisor)** is, and three distinct methods to compute it.
- The dazzling insight conceived by **Euclid** over twenty-three centuries ago to uncover common divisors in a handful of division steps.
- How to reduce any fraction to **lowest terms (irreducible form)** in a single stroke.
- How to solve authentic real-life riddles: **tiling a hall without cutting a single slab**, and assembling **identical gift baskets** without leaving a single crumb.

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1. The Problem: The Pharaoh's Mosaic and the Scribe's Baskets

Alexandria, around the year **300 BC**. King Ptolemy I commissions a ceremonial hall in his royal palace. Two master craftsmen appear before the chief royal architect, their brows furrowed with concern.

- The first is the **master mosaicist**. The floor of the hall is a vast rectangle measuring **420 centimeters long by 132 centimeters wide**. The king has decreed: *"I want this floor paved entirely with square marble slabs, all strictly identical. I forbid cutting a single tile along the edges: every slab must be laid whole! And so that the floor appears magnificent, I demand that these slabs be **as large as possible!**"* The mosaicist groans: 1 cm tiles would work, of course, but it would take 55,440 of them and they would look tiny! Slabs of 2 cm would also work. But what is the largest possible square dimension?
- The second artisan is the **steward scribe** in charge of royal festivities. To honor visiting foreign dignitaries, he opens two chests of prized delicacies:
  - One chest holding **315 Egyptian pomegranates**;
  - One chest holding **180 honey confections**. The king demands identical ceremonial gift baskets for the guests: *"All baskets must be strictly identical: exactly the same count of pomegranates and the same count of honey confections in each basket. And I want nothing left over in the chests! Finally, I wish to host the **greatest possible number** of dignitaries!"*

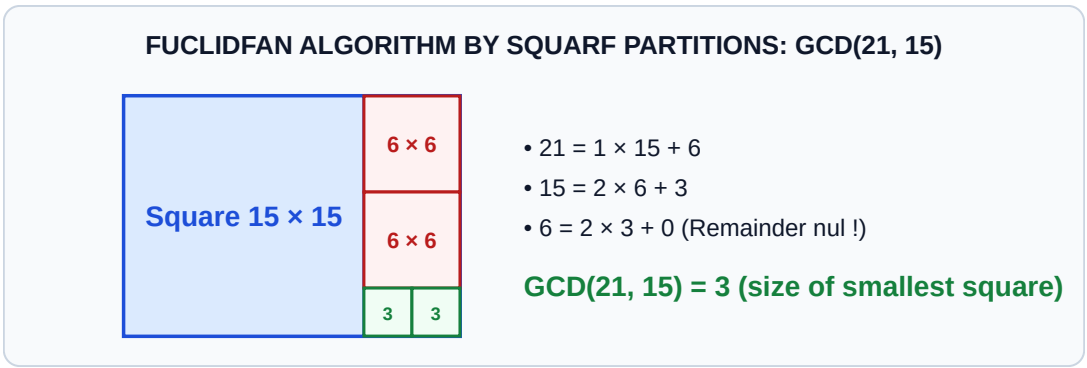

Figure 76: Euclid's geometric algorithm by square dissection

How can we resolve both dilemmas without spending three weeks testing random numbers? In both cases, the underlying mathematical challenge is identical: we seek an integer that **evenly divides** two numbers simultaneously, and which is **as large as possible**.

To enlighten the artisans, a scholar steps into the palace courtyard carrying the papyrus rolls of his masterpiece, the *Elements*. His name is **Euclid**.

---

2. Multiples and Divisors: The Vocabulary of Precision

Before entering Euclid's workshop, we must master the precise vocabulary of arithmetic.

Let a and b be two positive integers (with $b \neq 0$). If the division of a by b has a remainder of zero—that is, if b divides evenly into a—we can write:

$$a = b \times k \quad (\text{where } \ k \ \text{ is an integer}).$$

We express this relationship using any of the following equivalent statements:

- **a is a multiple of b** (e.g., 42 is a multiple of 6, since $42 = 6 \times 7$).
- **b is a divisor of a** (e.g., 6 is a divisor of 42).
- **b divides a** (often written in higher mathematics as $b \mid a$).
- **a is divisible by b**.

> ⚠️ **Watch Out! Pitfall — The Direction of Division!**
>
> Never confuse the divisor with the multiple:
>
> - Saying "6 divides 42" is **true**.
> - Saying "42 divides 6" is **completely false**! (42 does not fit inside 6). Rule of thumb: the **multiple** is always larger than or equal to the base; the **divisor** is always smaller than or equal to the dividend.

Quick Review: Divisibility Rules

To verify whether an integer divides another without executing long division:

- An integer is divisible by **2** if it is even (ends in 0, 2, 4, 6, 8).
- An integer is divisible by **3** if the sum of its digits is a multiple of 3 (e.g., $471 \implies 4 + 7 + 1 = 12$, divisible by 3).
- An integer is divisible by **4** if the number formed by its last two digits is divisible by 4 (e.g., $3,724 \implies 24 = 4 \times 6$, divisible by 4).
- An integer is divisible by **5** if its final digit is 0 or 5.
- An integer is divisible by **9** if the sum of its digits is a multiple of 9 (e.g., $855 \implies 8 + 5 + 5 = 18$, divisible by 9).
- An integer is divisible by **10** if it ends in 0.

---

3. Prime Numbers: The Atoms of the Mathematical Universe

Take the number 12. You can split it: $12 = 3 \times 4$. Then split 4 further: $4 = 2 \times 2$. Thus $12 = 2 \times 2 \times 3$. Can you split 2 or 3 any further? No! Two and three are indivisible. They can only be divided by 1 and by themselves.

> 📐 **Formal Definition — Prime Number.**
>
> A natural number is called **prime** if it possesses **exactly two distinct divisors**: 1 and itself.

Observe the direct consequences of this definition:

- **The number 0 is not prime**: it has infinitely many divisors (every non-zero number divides 0).

- **The number 1 is NOT prime**: it has only **one** divisor (itself). The definition strictly requires *two* distinct divisors. If 1 were classified as prime, the uniqueness of prime factorizations would break down ($6 = 2 \times 3 = 1 \times 2 \times 3 = 1 \times 1 \times 2 \times 3$...).
- **The number 2 is prime**: it is the **only even prime number** in the universe! Every other even number is divisible by 2, and therefore composite.

List of Prime Numbers Less Than 30

Memorize these as fundamental building blocks:

$$\mathbf{2},\quad \mathbf{3},\quad \mathbf{5},\quad \mathbf{7},\quad \mathbf{11},\quad \mathbf{13},\quad \mathbf{17},\quad \mathbf{19},\quad \mathbf{23},\quad \mathbf{29}$$

(Continuing to 100: 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97).

> **Did You Know?**
>
> In Book IX of the *Elements*, Euclid proved a breathtaking theorem: **there are infinitely many prime numbers**. Even if you journeyed for billions of years through unimaginably colossal numbers, you would never run out of newly emerging primes!

---

4. Prime Factorization

Here stands the single most important theorem in all of number theory. It is so foundational that it is termed the **Fundamental Theorem of Arithmetic**.

> **Fundamental Theorem of Arithmetic —**
>
> Every integer greater than or equal to 2 can be written as a **product of prime numbers**. This factorization is **unique**, up to the order of the factors.

This is the "genetic fingerprint" of an integer. Two different numbers can never share the exact same prime factorization.

Two Practical Methods for Factoring Numbers

Let us factor the number $\mathbf{360}$.

Method 1: The Factor Tree (Ideal for Mental Arithmetic)
Split the number into two convenient factors, then branch downward until every terminal leaf is a prime number:
Collect all prime leaves at the ends of the branches:

$$360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5 = \mathbf{2^3} \times \mathbf{3^2} \times \mathbf{5}$$

Method 2: The Division Ladder (The Algorithmic Method)
Draw a vertical bar. On the left, write the quotient; on the right, divide by successive primes in ascending order ($2, 3, 5, 7, 11$...):
The column on the right lists the prime factors directly:

$$360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5 = \mathbf{2^3} \times \mathbf{3^2} \times \mathbf{5}.$$

---

5. The GCD: Greatest Common Divisor

5.1 Definition

When comparing two positive integers a and b, certain integers divide both a and b simultaneously: these are their **common divisors**. Because 1 divides every integer, there is always at least one common divisor: the number 1.

> **📐 Definition — GCD (Greatest Common Divisor).**
>
> Among all common divisors of two positive integers a and b, the greatest of them is called the **Greatest Common Divisor**, denoted $\mathbf{gcd}(a, b)$ or $\mathbf{GCD}(a, b)$.

> **📐 Definition — Relatively Prime (Coprime) Numbers.**
>
> Two positive integers are said to be **relatively prime (coprime)** when their only common divisor is 1:
>
> $$\gcd(a, b) = 1$$
>
> *(Example: 8 and 15 are not prime numbers, but they are coprime because the divisors of 8 are $\{1, 2, 4, 8\}$ and those of 15 are $\{1, 3, 5, 15\}$; their sole common divisor is 1).*

5.2 First Method: Using Prime Factorization

To find the GCD of two numbers from their prime factorizations: 1. Identify the **common prime factors** shared by both factorizations. 2. For each shared prime factor, select the **smallest exponent** with which it appears. 3. Multiply these shared powers together.

Example: Compute $\gcd(180, 315)$ (the banquet foods from our opening problem!): - $180 = 2^2 \times 3^2 \times 5$ - $315 = 3^2 \times 5 \times 7$

Shared prime factors: 3 and 5.

- Lowest power of 3: 3^2.
- Lowest power of 5: 5^1.

$$\gcd(180, 315) = 3^2 \times 5 = 9 \times 5 = \mathbf{45}.$$

The royal scribe can prepare at most **45 identical gift baskets**! Each basket will contain:

- $315 \div 45 = 7$ pomegranates;
- $180 \div 45 = 4$ honey confections.

Zero leftovers in the royal pantry!

---

6. The Two Historical Algorithms: Subtraction and Division

When dealing with large numbers, listing all divisors or finding complete prime factorizations can be labor-intensive. Ancient mathematicians devised fast, mechanical algorithms.

6.1 The Repeated Subtraction Algorithm

This method rests on a straightforward arithmetic property: > **📐 Property of Subtractions —** > If an integer d divides both a and b (with $a \geq b$), then d also divides their difference $a - b$. > Consequently: $\gcd(a, b) = \gcd(b, a - b)$.

The Process: Replace the larger number with the difference of the two numbers, and repeat until both numbers become equal. That shared final value is the GCD!

Example: Compute $\gcd(75, 45)$: - $75 - 45 = 30 \Longrightarrow$ keep 45 and 30 - $45 - 30 = 15 \Longrightarrow$ keep 30 and 15 - $30 - 15 = 15 \Longrightarrow$ keep 15 and 15 - $15 - 15 = 0$

The matching numbers reached are 15, so $\gcd(75, 45) = \mathbf{15}$.

> ⚠️ ***The Limitation of Subtraction:*** *If you needed to compute $\gcd(1,000,000, 2)$, you would have to subtract 2 five hundred thousand times! That is impractically slow. Fortunately, Euclid conceived a stroke of brilliance.*

6.2 The Euclidean Algorithm (Successive Divisions)

Instead of subtracting b one step at a time, why not subtract all possible multiples of b in a single stroke? That is precisely the principle of **Euclidean division** (integer division with remainder):

$$a = b \times q + r \quad (0 \leq r < b)$$

Euclid's Theorem —

If integer division of a by b yields remainder r, then:

$$\gcd(a, b) = \gcd(b, r)$$

The Euclidean Algorithm Procedure: 1. Perform integer division of the larger number by the smaller number. 2. If the remainder is non-zero, divide the previous divisor by the remainder. 3. Repeat until the remainder becomes zero ($r = 0$). 4. **The GCD is the last NON-ZERO remainder.**

Example: Solve the Alexandria mosaic tiling problem: $\gcd(420, 132)$.

Step-by-Step Method — Computing the GCD via Euclid's Algorithm

- Step 1: 420 = 132 × 3 + 24 (remainder $r_1 = 24 \neq 0$)
- Step 2: 132 = 24 × 5 + 12 (remainder $r_2 = 12 \neq 0$)
- Step 3: 24 = 12 × 2 + 0 (remainder $r_3 = 0$)

The final remainder before 0 is **12**. Therefore:

$$\gcd(420, 132) = \mathbf{12}.$$

The master mosaicist has his solution:

- The largest possible uncut square slabs measure $\mathbf{12\ \ cm \times 12\ \ cm}$.
- Along the length, he lays $420 \div 12 = 35$ slabs.
- Along the width, he lays $132 \div 12 = 11$ slabs.
- In total, he will install $35 \times 11 = \mathbf{385\ \ slabs}$, fitting seamlessly without cutting a millimeter of marble!

---

7. The Crown Application: Simplifying Fractions to Lowest Terms

In middle school, you learned to simplify fractions by dividing numerator and denominator by common factors. But often students stop prematurely: dividing by 2, then by 3, uncertain whether further reduction is possible.

Definition — Irreducible Fraction (Lowest Terms).

A fraction $\frac{a}{b}$ is said to be **irreducible** (in lowest terms) if its numerator a and denominator b are **relatively prime**, meaning $\gcd(a, b) = 1$. It cannot be simplified any further.

Using the GCD, any fraction can be simplified to lowest terms **in a single step**:

Method — One-Step Simplification.

To reduce $\frac{a}{b}$ to lowest terms, divide both numerator and denominator by their **GCD**:

$$\frac{a}{b} = \frac{a \div \gcd(a, b)}{b \div \gcd(a, b)}$$

Example: Simplify the fraction $\frac{132}{420}$ to lowest terms. We previously calculated $\gcd(420, 132) = 12$:

$$\frac{132}{420} = \frac{132 \div 12}{420 \div 12} = \frac{\mathbf{11}}{\mathbf{35}}.$$

Since 11 and 35 share no common factors other than 1, the fraction is guaranteed to be in lowest terms!

---

8. Step-by-Step Worked Examples

Example 1: Factorization and divisibility

1. Factor the integer 540 into a product of primes using the division ladder.
2. Write the result using exponent notation.
3. Is 540 divisible by 18? By 24? Justify without performing long division.

Detailed Solution: 1. Division ladder:

- $540 \div 2 = 270$
- $270 \div 2 = 135$
- $135 \div 3 = 45$ (since $1 + 3 + 5 = 9$)
- $45 \div 3 = 15$
- $15 \div 3 = 5$
- $5 \div 5 = 1$

2. Prime factorization: $540 = 2 \times 2 \times 3 \times 3 \times 3 \times 5 = \mathbf{2^2 \times 3^3 \times 5}$.
3. • For 18: $18 = 2 \times 3^2$. The prime factorization of 540 contains at least one factor of 2 (it has two) and at least two factors of 3 (it has three). Thus **540 is divisible by 18**.
   • For 24: $24 = 2^3 \times 3$. The factorization of 540 contains only 2^2 (only two factors of 2). It lacks a third 2, so **540 is not divisible by 24**.

Example 2: The Euclidean algorithm on large numbers

Find the GCD of $1,575$ and 420 using the Euclidean algorithm.

Detailed Solution: Execute successive Euclidean divisions:

- $1,575 = 420 \times 3 + 315$ (remainder $315 \neq 0$)
- $420 = 315 \times 1 + 105$ (remainder $105 \neq 0$)
- $315 = 105 \times 3 + 0$ (remainder 0)

The last non-zero remainder is **105**. Conclusion: $\mathbf{gcd(1, 575, 420) = 105}$.

Example 3: Comparing subtractions versus divisions

Compute $\gcd(252, 105)$: 1. Using repeated subtractions. 2. Using the Euclidean algorithm.

Detailed Solution: 1. **Repeated subtractions:**

- $252 - 105 = 147$
- $147 - 105 = 42$
- $105 - 42 = 63$
- $63 - 42 = 21$
- $42 - 21 = 21$
- $21 - 21 = 0 \Longrightarrow \mathbf{gcd = 21}$ (5 subtraction steps).

2. **Euclidean algorithm:**
   - $252 = 105 \times 2 + 42$
   - $105 = 42 \times 2 + 21$
   - $42 = 21 \times 2 + 0 \Longrightarrow \mathbf{gcd = 21}$ (2 division steps).

Euclid's division approach is distinctly faster and cleaner!

Example 4: School fair party bags

For the school carnival, organizers have 280 glow bracelets and 448 collector stickers. They wish to make identical surprise gift bags containing the exact same number of bracelets and stickers, using all supplies with no leftovers. 1. What is the maximum number of gift bags they can prepare? 2. What will be the contents of each gift bag?

Detailed Solution: 1. Let N be the number of gift bags. For supplies to be distributed evenly without leftovers, N must divide both 280 and 448. For N to be maximal, $N = \gcd(448, 280)$. Euclidean algorithm:

- $448 = 280 \times 1 + 168$
- $280 = 168 \times 1 + 112$
- $168 = 112 \times 1 + 56$
- $112 = 56 \times 2 + 0$ The last non-zero remainder is 56, so $\gcd(448, 280) = 56$. They can prepare at most **56 identical gift bags**.

2. Contents per bag:
   - Glow bracelets: $280 \div 56 = \mathbf{5}$ **bracelets**.
   - Collector stickers: $448 \div 56 = \mathbf{8}$ **stickers**.

Example 5: Lowest terms and addition of fractions

Consider the fraction $A = \frac{1,575}{420}$. 1. Reduce fraction A to lowest terms. 2. Compute the sum $S = \frac{1,575}{420} + \frac{1}{4}$ as an irreducible fraction.

Detailed Solution: 1. From Example 2, $\gcd(1, 575, 420) = 105$. Divide numerator and denominator by 105:

$$A = \frac{1,575 \div 105}{420 \div 105} = \frac{\mathbf{15}}{\mathbf{4}}.$$

2. Substitute the simplified fraction into the addition:

$$S = \frac{15}{4} + \frac{1}{4} = \frac{15+1}{4} = \frac{16}{4} = \mathbf{4}.$$

The calculation becomes instantaneous!

Key Takeaways

1. **Foundational Vocabulary:**
   - $a = b \times k \Leftrightarrow a$ is a multiple of $b \Leftrightarrow b$ is a divisor of a.
   - **Prime Number**: an integer with exactly 2 distinct divisors (1 and itself). 0 and 1 are NOT prime. The only even prime is 2.
2. **Fundamental Theorem of Arithmetic:**
   - Every integer ≥ 2 factors uniquely into a product of prime powers ($N = p_1^{\alpha_1} \times p_2^{\alpha_2} \times \ldots$).
3. **GCD (Greatest Common Divisor):**
   - The largest integer that divides both numbers evenly.
   - Two numbers are **coprime** if $\gcd(a, b) = 1$.
   - **Prime Factorization Method**: product of all shared prime factors raised to their lowest powers.
   - **Euclidean Algorithm**: chain successive divisions $a = b \times q + r$. The GCD is the **last non-zero remainder**.
4. **Applications:**
   - **Fractions in Lowest Terms**: divide numerator and denominator by their GCD.
   - **Tiling and Equal Batches**: the GCD gives the maximal side length of unbroken square tiles or the maximal number of identical batches without leftovers.

Exercises

Level : Direct Application

Exercise 29.1 From the following list, identify the prime numbers:

$$1 \quad ; \quad 2 \quad ; \quad 9 \quad ; \quad 13 \quad ; \quad 15 \quad ; \quad 21 \quad ; \quad 29 \quad ; \quad 31 \quad ; \quad 51 \quad ; \quad 91$$

For numbers that are not prime, justify by giving a divisor other than 1 and itself.

Exercise 29.2 Using divisibility rules, determine whether $3,420$ is divisible by 2, 3, 4, 5, 9, and 10. Justify each case.

Exercise 29.3 Find the prime factorizations of the following integers (using exponents where appropriate): a) 84 b) 126 c) 200 d) 525

Exercise 29.4 Given $84 = 2^2 \times 3 \times 7$ and $126 = 2 \times 3^2 \times 7$: 1. What are the common prime factors of 84 and 126? 2. Compute $\gcd(84, 126)$. 3. Reduce $\frac{84}{126}$ to lowest terms.

Exercise 29.5 Use the repeated subtraction algorithm to determine $\gcd(65, 39)$.

Level : Deeper Reflection and Practical Situations

Exercise 29.6 (Euclidean Algorithm) 1. Using the Euclidean algorithm, compute $\gcd(756, 441)$, clearly displaying each division step. 2. Are 756 and 441 coprime? Why or why not? 3. Reduce $\frac{441}{756}$ to lowest terms in a single step.

Exercise 29.7 (Tiling the Patio) A builder must tile a rectangular patio measuring 5.40 m long by 3.00 m wide. He wants to use the largest possible identical square slabs without cutting any tiles. 1. Convert the patio dimensions to centimeters. 2. What must be the side length of each tile? 3. How many tiles will be needed in total?

Exercise 29.8 (The Fair Florist) A florist receives a delivery of 168 red roses and 216 white carnations. She wants to arrange identical bouquets containing the same number of roses and carnations, using all flowers. 1. What is the maximum number of identical bouquets she can create? 2. How many roses and how many carnations will each bouquet contain?

Exercise 29.9 (The Impostor Prime Trap) A student claims: *"All odd numbers greater than 1 are prime, except those ending in 5!"* Prove that this claim is false by providing at least three distinct counterexamples less than 100.

Exercise 29.10 (Track Race and Multiples) Two go-kart racers cross the starting line simultaneously on a closed circular track.

- The first racer completes a lap in 36 seconds.
- The second racer completes a lap in 48 seconds.

1. Factor 36 and 48 into prime factors.
2. After how much time will both racers cross the starting line together again for the first time?
3. How many laps will each racer have completed?

Exercise 29.11 (Operations with Reduced Fractions) Consider the expression:

$$B = \frac{588}{882} - \frac{1}{6}$$

1. Compute $\gcd(882, 588)$ using the Euclidean algorithm. 2. Deduce the irreducible form of $\frac{588}{882}$. 3. Calculate expression B, giving the result as an irreducible fraction.

Level : Research Challenges and Olympiad-Style Riddles

Exercise 29.12 (The Farmer's Boundary Fence) A farmer owns a triangular field whose three boundaries measure 120 meters, 144 meters, and 168 meters. He wishes to plant fence posts at equal, regular intervals around the perimeter, with a post positioned at each vertex of the triangle. To minimize expenses, he wants to use the **fewest possible fence posts**. 1. What must be the constant distance between consecutive posts? 2. How many posts must he plant in total around the fence?

Exercise 29.13 (Euclid's Proof of the Infinitude of Primes) Euclid proved that the set of prime numbers is infinite using an elegant proof by contradiction. Assume there are only finitely many primes: denote them all as $p_1, p_2, p_3, ..., p_n$. Consider the integer N constructed by multiplying all these primes together and adding 1:

$$N = (p_1 \times p_2 \times p_3 \times ... \times p_n) + 1$$

1. If you divide N by any of the primes $p_1, p_2, ..., p_n$, what is the remainder? 2. Can N be divisible by any prime in the list? 3. Conclude: why does this produce a contradiction, and why must there be infinitely many prime numbers?

Exercise 29.14 (Gear Meshing) Cogwheel A with 54 teeth drives cogwheel B with 42 teeth. Two designated teeth, one on each wheel, touch at the initial starting position. 1. Calculate $\gcd(54, 42)$. 2. After how many tooth contacts will these exact two teeth touch again for the first time? 3. How many complete revolutions will each wheel have made?

Exercise 29.15 (Cube Packing in a 3D Rectangular Solid) A sculptor has a solid rectangular block of marble measuring 90 cm long, 75 cm wide, and 60 cm high. He wants to slice this block into identical cubes of maximal size, without wasting any marble. 1. Factor 90, 75, and 60 into prime factors. 2. What is the Greatest Common Divisor of all three numbers? 3. What is the edge length of each cube? 4. How many cubes will the sculptor obtain?

Solutions

Level

- **Solution 29.1:**
  - Primes: **2, 13, 29, 31**.
  - Non-primes:
    - 1 is not prime (only one divisor).
    - 9 is divisible by 3 ($9 = 3 \times 3$).
    - 15 is divisible by 3 and 5 ($15 = 3 \times 5$).
    - 21 is divisible by 3 and 7 ($21 = 3 \times 7$).
    - 51 is divisible by 3 ($5 + 1 = 6$; $51 = 3 \times 17$).
    - 91 is divisible by 7 ($91 = 7 \times 13$).
- **Solution 29.2:** For 3,420:
  - Divisible by 2: yes, ends in 0 (even).
  - Divisible by 3: yes, $3 + 4 + 2 + 0 = 9$, multiple of 3.
  - Divisible by 4: yes, last two digits 20 divisible by 4.
  - Divisible by 5: yes, ends in 0.
  - Divisible by 9: yes, digit sum 9 divisible by 9.
  - Divisible by 10: yes, ends in 0.
- **Solution 29.3:**
  a) $84 = 2^2 \times 3 \times 7$.
  b) $126 = 2 \times 3^2 \times 7$.
  c) $200 = 2^3 \times 5^2$.
  d) $525 = 3 \times 5^2 \times 7$.
- **Solution 29.4:**
  1. Shared prime factors: 2, 3, and 7.
  2. $\gcd(84, 126) = 2^1 \times 3^1 \times 7^1 = \mathbf{42}$.
  3. $\frac{84}{126} = \frac{84 \div 42}{126 \div 42} = \mathbf{\frac{2}{3}}$.
- **Solution 29.5:** Subtractions: $65 - 39 = 26 \Longrightarrow 39 - 26 = 13 \Longrightarrow 26 - 13 = 13 \Longrightarrow 13 - 13 = 0$. Thus $\gcd(65, 39) = \mathbf{13}$.

Level

- **Solution 29.6 (Euclidean Algorithm):**
  1. Divisions:
     - $756 = 441 \times 1 + 315$
     - $441 = 315 \times 1 + 126$
     - $315 = 126 \times 2 + 63$
     - $126 = 63 \times 2 + 0$ Last non-zero remainder is 63, so $\mathbf{gcd(756, 441) = 63}$.
  2. Not coprime because $\gcd = 63 \neq 1$.
  3. Reduced fraction: $\frac{441}{756} = \frac{441 \div 63}{756 \div 63} = \mathbf{\frac{7}{12}}$.
- **Solution 29.7 (Patio Tiling):**
  1. $L = 5.40 \text{ m} = 540 \text{ cm}$, $W = 3.00 \text{ m} = 300 \text{ cm}$.
  2. Tile side length is $\gcd(540, 300)$:
     - $540 = 300 \times 1 + 240$
     - $300 = 240 \times 1 + 60$
     - $240 = 60 \times 4 + 0$ $\gcd = 60$. Tiles must measure $\mathbf{60 \text{ cm} \times 60 \text{ cm}}$.
  3. Tile count:
     - Along length: $540 \div 60 = 9$ tiles.
     - Along width: $300 \div 60 = 5$ tiles.
     - Total: $9 \times 5 = \mathbf{45 \text{ tiles}}$.
- **Solution 29.8 (Florist):**
  1. Maximum bouquets $= \gcd(216, 168)$:
     - $216 = 168 \times 1 + 48$
     - $168 = 48 \times 3 + 24$
     - $48 = 24 \times 2 + 0$ $\gcd = 24$. She can assemble at most **24 bouquets**.
  2. Contents:
     - Roses: $168 \div 24 = \mathbf{7 \text{ red roses}}$.
     - Carnations: $216 \div 24 = \mathbf{9 \text{ white carnations}}$.
- **Solution 29.9 (Counterexamples):**
  - 9 ($9 = 3 \times 3$, odd, not prime);
  - 21 ($21 = 3 \times 7$);
  - 27 ($27 = 3 \times 9$);
  - 91 ($91 = 7 \times 13$, famous composite trap!).
- **Solution 29.10 (Track Race):**
  1. $36 = 2^2 \times 3^2$ and $48 = 2^4 \times 3$.
  2. First common meeting is the LCM (Least Common Multiple):
     $$\text{LCM}(36, 48) = 2^4 \times 3^2 = 16 \times 9 = \mathbf{144 \text{ seconds}} = 2 \text{ minutes and } 24 \text{ seconds}.$$
  3. Laps:
     - Racer 1: $144 \div 36 = \mathbf{4 \text{ laps}}$.
     - Racer 2: $144 \div 48 = \mathbf{3 \text{ laps}}$.
- **Solution 29.11 (Fractions):**
  1. $\gcd(882, 588)$:
     - $882 = 588 \times 1 + 294$
     - $588 = 294 \times 2 + 0 \implies \gcd = \mathbf{294}$.
  2. Irreducible fraction: $\frac{588}{882} = \frac{588 \div 294}{882 \div 294} = \mathbf{\frac{2}{3}}$.
  3. $B = \frac{2}{3} - \frac{1}{6} = \frac{4}{6} - \frac{1}{6} = \frac{3}{6} = \mathbf{\frac{1}{2}}$.

Level

- **Solution 29.12 (Fence Posts):**
  1. Post interval d must divide 120, 144, and 168. To minimize posts, $d = \gcd(120, 144, 168)$.
     - $\gcd(120, 144) = 24$.
     - Since $168 = 24 \times 7$, $\gcd(120, 144, 168) = \mathbf{24 \ \ meters}$.
  2. Perimeter $= 120 + 144 + 168 = 432 \ \ \text{meters}$. In a closed perimeter, total posts $= \frac{432}{24} = \mathbf{18 \ \ posts}$.
- **Solution 29.13 (Euclid's Infinitude of Primes):**
  1. $N = (p_1 \times p_2 \times ... \times p_n) + 1$. The product is an exact multiple of any listed prime p_k. Thus dividing N by p_k leaves a remainder of **1**.
  2. Since the remainder is 1, N is divisible by **none** of the primes in the list.
  3. By the Fundamental Theorem of Arithmetic, $N > 1$ must have at least one prime factor. That prime factor cannot be in the list, contradicting the assumption that the list contained all primes. Hence, there are infinitely many prime numbers.
- **Solution 29.14 (Gears):**
  1. $\gcd(54, 42) = \mathbf{6}$.
  2. Tooth contacts equal the LCM:
$$\text{LCM}(54, 42) = \frac{54 \times 42}{\gcd(54, 42)} = \frac{2,268}{6} = \mathbf{378 \ \ tooth \ contacts} \ .$$
  3. Revolutions:
     - Wheel A: $378 \div 54 = \mathbf{7 \ \ full \ turns}$.
     - Wheel B: $378 \div 42 = \mathbf{9 \ \ full \ turns}$.
- **Solution 29.15 (3D Marble Cubes):**
  1. Factorizations:
     - $90 = 2 \times 3^2 \times 5$
     - $75 = 3 \times 5^2$
     - $60 = 2^2 \times 3 \times 5$
  2. Common prime factors: 3 and 5. $\gcd(90, 75, 60) = 3 \times 5 = \mathbf{15}$.
  3. Cube edge length is $\mathbf{15 \ \ cm}$.
  4. Number of cubes:
     - Length: $90 \div 15 = 6$
     - Width: $75 \div 15 = 5$
     - Height: $60 \div 15 = 4$ Total: $6 \times 5 \times 4 = \mathbf{120 \ \ cubes}$.

The Question Opening the Next Chapter

Through Euclid's arithmetic, everything is governed with absolute certainty: every stone finds its place, every share divides evenly, every calculation is rock solid. Yet our real world is not merely composed of sealed chests and square marble slabs.

What happens when the mist of the future rises, when dice roll across an innkeeper's table, or when a caravan master must choose between two perilous routes? Can one calculate rationally in the face of uncertainty? Journey into the desert and the salons of 17th-century thinkers: we will learn to master the **roulette of fate** in Chapter 30: **The Roulette of Fate and the Hunter's Survival**.

CHAPTER 30: The Roulette of Fate and the Hunter's Survival

> ⚠️ **"The ignorant man blames bad luck, consults fortune-tellers, or trembles before the thunderstorm when danger approaches. The mathematician surveys the paths of the future, calculates the fraction of chance, and makes the decision that saves his caravan. Chance is no blind monster: it is a tree whose branches can be counted."**

What You Are About to Discover

- How to transform the fog of uncertainty into a **precise number between 0 and 1** (the probability).
- The formal vocabulary that structures chance: **random experiment**, **outcome**, **event**, **certain, impossible, and complementary events**.
- Why the formula $P(E) = \frac{\text{number of favorable outcomes}}{\text{number of possible outcomes}}$ only works under **equiprobability**, and the monumental pitfall of the two dice.
- How to explore the two-stage future using a **probability tree diagram** and a **two-way table**.
- The crucial difference between sampling **with replacement** and sampling **without replacement**.
- How real-world chance obeys the law of large numbers: the **stabilization of relative frequencies** and computer simulation.

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1. The Problem: The Caravan and the Forest of Two Paths

Dusk is falling over the Silk Road. A caravan of ten camels loaded with bales of precious spices and fine silks halts at the edge of an immense, dark, and misty forest. To reach the oasis where the grand annual market is being held, the caravan master must choose between two routes:

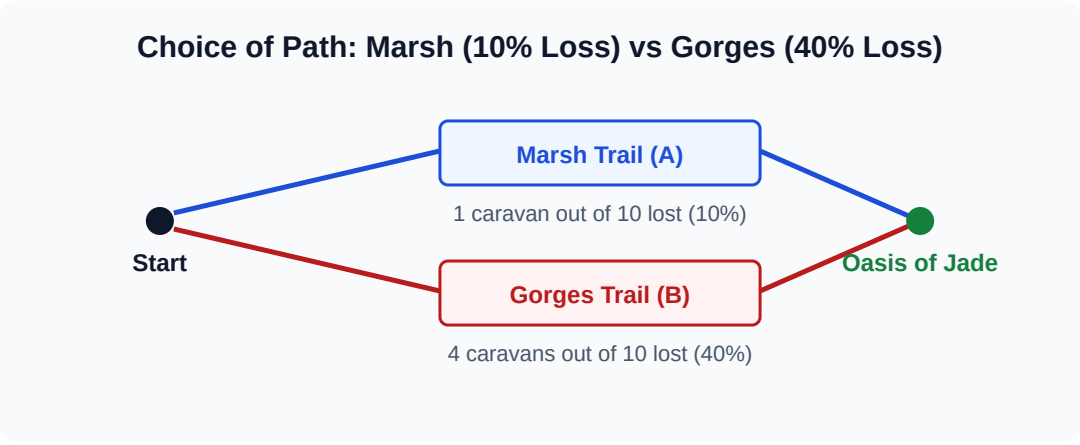

Figure 77: Choice of path: weighted probability tree

- **The Marsh Trail (Path A):** It is a flat track, easy on the animals' feet. But the region is riddled with treacherous mud pits. Merchant records show that out of 100 caravans that took this trail this year, **10 became bogged down and lost everything**, while 90 arrived safe and sound.
- **The Gorge Trail (Path B):** It is a steep, rocky defile, twice as short and much faster. But armed bandits lurk in the caves: out of 100 caravans, **40 were ambushed and plundered**, while 60 managed to cross unscathed.

The young camel driver exclaims: *"Let us take the gorge trail! We will arrive tomorrow at dawn to sell at the best price! With a bit of luck, the bandits will be asleep!"*

The old leader shakes his head: *"Luck is not a prayer one casts into the sky. Luck is a fraction of reality. He who cannot weigh risk walks blindfolded toward the abyss!"*

For the first time in the history of mathematics, we are no longer merely measuring what lies before our eyes (a length, a weight, a count of tiles), but **what has not yet happened**! We enter the realm of **Probability**.

---

2. The Formal Vocabulary of Probability

To speak of chance with the rigor of a geometer, mathematicians forged a very precise vocabulary. Each word carries a strict definition that must be mastered.

2.1 Random Experiment and Outcome

> **Definition — Random Experiment.**
>
> An experiment is said to be **random** if: 1. All possible results are known in advance; 2. The final result depends solely on chance and is **impossible to predict with certainty**.

Examples: - Tossing a balanced coin and observing the upper face is a random experiment. - Dropping an apple to see if it falls toward the ground is **not** a random experiment (gravity guarantees that it will always fall).

> **Definition — Outcome.**
>
> An **outcome** (or elementary event) is one of the possible results of a random experiment. The set of all possible outcomes is called the **sample space** (often denoted by the Greek letter Ω, "omega").

Example with a 6-sided die: The possible outcomes are 1, 2, 3, 4, 5, and 6. The sample space is $\{1, 2, 3, 4, 5, 6\}$.

2.2 Events: Impossible, Certain, Complementary

> **Definition — Event.**
>
> An **event** is a set consisting of zero, one, or several outcomes. An event is said to **occur** (or be realized) if the outcome obtained during the experiment belongs to that event.

Consider the roll of an ordinary six-sided die:

- Event A: "Roll an even number" consists of the outcomes $\{2, 4, 6\}$. If the die lands on 4, event A has occurred.
- **Impossible event:** an event that contains no outcomes and can never happen. Its probability is equal to **0**. *Example:* "Roll a 7" with a 6-sided die.
- **Certain event:** an event that contains all possible outcomes and always happens. Its probability is equal to **1** (or 100%)! *Example:* "Roll a number less than or equal to 6".
- **Complementary event:** the complementary event of an event A, denoted $\bar{A}$ (read **"A bar"** or "not A"), is the event that occurs **precisely when A does not occur**.

> **Fundamental Property of Complementary Events —**
>
> For any event A:
>
> $$P(\bar{A}) = 1 - P(A) \qquad \text{or equivalently} \qquad P(A) + P(\bar{A}) = 1$$

Example: If the probability of rain tomorrow is $P(\text{Rain}) = 0.3$ (or 30%), then the probability of no rain is:

$$P(\text{No Rain}) = 1 - 0.3 = 0.7 \quad (\text{or } \ 70\%).$$

---

3. Calculating Probability under Equiprobability

3.1 What is Equiprobability?

A situation is one of **equiprobability** (equally likely outcomes) when **every possible outcome has exactly the same chance of occurring**. In problems, this condition is indicated by standard key phrases:

- "A **fair** (or balanced, unbiased) six-sided die";
- "A **fair** coin";
- "Marbles **indistinguishable by touch**" in an urn;
- "A card drawn **at random** from a well-shuffled deck".

3.2 Laplace's Formula

📐 Laplace's Rule — Probability under Equiprobability.

In a situation with equally likely outcomes, the probability of an event E is equal to the number of favorable outcomes divided by the total number of possible outcomes:

$$P(E) = \frac{\text{Number of outcomes favorable to } E}{\text{Total number of possible outcomes}}$$

3.3 The Three Guises of a Probability

A probability is always a real number between 0 and 1:

$$0 \leq P(E) \leq 1$$

It can be written in three equivalent forms: 1. As a **fraction** (exact value): $\frac{1}{4}$ 2. As a **decimal number**: 0.25 3. As a **percentage**: 25%

⚠️ Warning, Trap!

A probability can **NEVER** be negative, nor greater than 1 (or 100%)! If you calculate $P = 1.25$ or $P = -0.3$, there is inevitably an error in your work. Furthermore, the sum of the probabilities of all elementary outcomes is **always equal to 1**.

Returning to our caravan:

- For the Marsh Trail: $P(\text{Peril}) = \frac{10}{100} = \frac{1}{10} = 0.10 = \mathbf{10\%}$.
- For the Gorge Trail: $P(\text{Peril}) = \frac{40}{100} = \frac{4}{10} = 0.40 = \mathbf{40\%}$.

The danger in the gorge is **four times higher**! The wise master chooses the marsh, and the caravan reaches its destination unharmed.

---

4. Two-Stage Experiments: Trees and Tables

When a random experiment unfolds across several successive stages (for instance: tossing two coins in a row, rolling two dice, or drawing two marbles from a bag), keeping track of all outcomes mentally becomes treacherous. Two visual tools make everything transparent.

4.1 Tool #1: The Two-Way Table (The Classic Two-Dice Example)

If you roll two 6-sided dice and compute the sum of the two numbers, how many possible outcomes are there? The sums range from $1 + 1 = 2$ up to $6 + 6 = 12$. That makes 11 different possible sums. Some students leap to the conclusion: *"The probability of getting a sum of 7 is 1 in 11!"* **This is a monumental blunder!** Because the sums are not equally likely!

Let us construct the table of all possible pairs ($6 \times 6 = \mathbf{36}$ **equally likely outcomes**):

Die 1 Die 2	1	2	3	4	5	6
1	2	3	4	5	6	**[7]**
2	3	4	5	6	**[7]**	8
3	4	5	6	**[7]**	8	9
4	5	6	**[7]**	8	9	10
5	6	**[7]**	8	9	10	11
6	**[7]**	8	9	10	11	12

Look at the long diagonal of **7s**:

- To obtain a sum of **2**, there is only a single cell: $(1;1) \Longrightarrow P(2) = \frac{1}{36}$.
- To obtain a sum of **12**, only one cell: $(6;6) \Longrightarrow P(12) = \frac{1}{36}$.
- But to obtain a sum of **7**, there are **6 different cells**: $(1;6), (2;5), (3;4), (4;3), (5;2), (6;1)$!

$$P(\text{Sum} = 7) = \frac{6}{36} = \frac{1}{6} \approx \mathbf{16.7}\%$$

A sum of 7 is **six times more likely** than a sum of 2! It is the king of the dice table.

4.2 Tool #2: The Weighted Probability Tree Diagram

The tree diagram is the Swiss Army knife of probability. Each fork represents a trial.

> **The Three Sacred Laws of the Probability Tree —**
>
> 1.
>
> **The sum of the probabilities** on all branches issuing from any single node is **always equal to 1**. 2. **Along any path**, you **multiply** the branch probabilities to find the probability of the final compound outcome (multiplication rule). 3. **The probability of an event** composed of several distinct paths is the **sum** of the probabilities of those paths.

---

5. Sampling With Replacement vs. Sampling Without Replacement

This is one of the most vital distinctions in introductory probability. Imagine a box containing **3 red marbles (R)** and **2 green marbles (G)** (making 5 marbles in total). We draw two marbles successively at random.

Case 1: Sampling WITH Replacement

After recording the color of the first marble, we **place it back into the box** and mix before drawing the second marble.

- On the first draw: $P(R) = \frac{3}{5}$ and $P(G) = \frac{2}{5}$.
- On the second draw: the box is once again **strictly identical**! The probabilities do not change: $P(R) = \frac{3}{5}$ and $P(G) = \frac{2}{5}$.
- The probability of drawing two reds $(R;R)$ is:

$$P(R;R) = \frac{3}{5} \times \frac{3}{5} = \frac{9}{25} = 0.36 = \mathbf{36}\%.$$

Case 2: Sampling WITHOUT Replacement

After the first draw, we keep the marble in hand: **we do not return it**!

- On the first draw: $P(R) = \frac{3}{5}$.
- For the second draw, only **4 marbles in total remain** in the box!
  - If the first marble was red, only 2 red marbles remain out of 4: $P(2\text{nd} \quad R) = \frac{2}{4}$.
  - The probability of drawing two reds $(R;R)$ is:

$$P(R;R) = \frac{3}{5} \times \frac{2}{4} = \frac{6}{20} = \frac{3}{10} = 0.30 = \mathbf{30}\%.$$

> ⚠️ **Warning, Trap!**
>
> In sampling without replacement, **the denominator decreases by 1** on the second draw, and the numerator corresponding to the drawn color also decreases by 1!

---

6. From Theoretical Chance to the Real World: The Law of Large Numbers

When you flip a fair coin, the theoretical probability of landing Heads is $p = 0.5$. Does this mean that if you flip the coin 10 times, you will obtain **exactly** 5 Heads and 5 Tails? No! You could very easily get 7 Heads and 3 Tails (an observed relative frequency of 70% Heads).

What happens, however, if you flip the coin 100 times, 1,000 times, 100,000 times? Let us calculate the **observed relative frequency**:

$$f = \frac{\text{Number of observed Heads}}{\text{Total number of flips } \ N}$$

> 📐 **Phenomenon of Frequency Stabilization (Law of Large Numbers) —**
>
> When a random experiment is repeated an **independently large number of times**, the observed relative frequency of an event approaches and stabilizes around its **theoretical probability**.

This very principle allows insurance companies to set policy premiums fairly, meteorologists to predict storm patterns, and casinos to guarantee they never operate at a net loss over the course of a full year!

> 💡 **True History — The Letters of Pascal and Fermat (1654)**
>
> In 1654, a Parisian gambler, the **Chevalier de Méré**, presented **Blaise Pascal** with an enigma that set tavern arguments ablaze: "the problem of points" (*le problème des partis*). Two players are tossing a coin for a prize purse of 64 gold pistoles. The first player to reach 3 wins takes everything. But the game is abruptly interrupted by a fire when Player A is leading 2 to 1! How should the purse be divided in fairness? Pascal began exchanging letters with the brilliant magistrate of Toulouse, **Pierre de Fermat**. Rather than dwelling on the past, they mapped the tree of all possible futures: across the 4 possible game completions, Player A would win in 3 of them, and Player B in only 1. Natural justice therefore demanded giving $\frac{3}{4}$ of the pot to Player A (48 pistoles) and $\frac{1}{4}$ to Player B (16 pistoles). Modern probability theory was born!

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7. Step-by-Step Worked Examples

Worked Example 1 — Drawing from a Deck of Cards (Equiprobability and Complementary Events)

A card is drawn at random from a well-shuffled 32-card deck (consisting of 4 suits: Hearts, Diamonds, Clubs, Spades, each with 8 cards: 7, 8, 9, 10, Jack, Queen, King, Ace). 1. What is the probability of drawing an Ace? 2. What is the probability of drawing a red card (Heart or Diamond)? 3. What is the probability of drawing a card that is **not** a Club?

Step-by-Step Solution: Because the deck is thoroughly shuffled, every card is equally likely to be drawn. 1. The deck contains 4 Aces (one of each suit).

$$P(\text{Ace}) = \frac{4}{32} = \frac{1}{8} = \mathbf{0.125} \quad (12.5\%).$$

2. There are 8 hearts and 8 diamonds, totaling $8 + 8 = 16$ red cards out of 32.

$$P(\text{Red}) = \frac{16}{32} = \frac{1}{2} = \mathbf{0.5} \quad (50\%).$$

3. *Method using the complementary event:* Let event C be "Drawing a Club". There are 8 clubs, so $P(C) = \frac{8}{32} = \frac{1}{4}$. The event "Not drawing a Club" is the complementary event $\bar{C}$:

$$P(\bar{C}) = 1 - P(C) = 1 - \frac{1}{4} = \frac{\mathbf{3}}{\mathbf{4}} = \mathbf{0.75} \quad (75\%).$$

Worked Example 2 — Two-Stage Sequential Experiment (The Keychain)

A keychain holds 4 keys that look identical, but **only one** opens the garage door. A hurried driver tries the keys at random in the dark. Whenever a key does not work, he sets it aside so as not to retest it (sampling without replacement). What is the probability that he opens the door on exactly his second attempt?

Step-by-Step Solution: To succeed on the second attempt, two consecutive events must occur:

- Failure on the 1st attempt (F_1);
- Success on the 2nd attempt (S_2).

Let us examine the corresponding branches: 1. On the 1st attempt: there are 3 wrong keys out of 4 keys in total:

$$P(F_1) = \frac{3}{4}$$

2. On the 2nd attempt: the wrong key has been removed. Only 3 keys remain, of which exactly 1 is the right key:

$$P(S_2 \ \text{ given } \ F_1) = \frac{1}{3}$$

3. By the multiplication rule along a path:

$$P(\text{Success on 2nd attempt}) = P(F_1) \times P(S_2) = \frac{3}{4} \times \frac{1}{3} = \frac{3}{12} = \frac{\mathbf{1}}{\mathbf{4}} = \mathbf{0.25}.$$

Each attempt has strictly a $\frac{1}{4}$ chance of being the successful one!

Worked Example 3 — Drawing Marbles With and Without Replacement

A bag contains 2 blue marbles and 3 yellow marbles. Two marbles are drawn successively at random. What is the probability of drawing **two marbles of different colors**: 1. When sampling **with replacement**? 2. When sampling **without replacement**?

Step-by-Step Solution: Drawing two marbles of different colors corresponds to two disjoint paths: $(B;Y)$ or $(Y;B)$.

1. **With replacement:**

- $P(B;Y) = \frac{2}{5} \times \frac{3}{5} = \frac{6}{25}$.
- $P(Y;B) = \frac{3}{5} \times \frac{2}{5} = \frac{6}{25}$. Adding the two mutually exclusive paths:

$$P(\text{Different colors}) = \frac{6}{25} + \frac{6}{25} = \frac{12}{25} = \mathbf{0.48} \quad (48\%).$$

2. **Without replacement:**

- For $(B;Y)$: first blue $(2/5)$, then 4 marbles remain including 3 yellow $(3/4)$:

$$P(B;Y) = \frac{2}{5} \times \frac{3}{4} = \frac{6}{20} = \frac{3}{10}.$$

- For $(Y;B)$: first yellow $(3/5)$, then 4 marbles remain including 2 blue $(2/4)$:

$$P(Y;B) = \frac{3}{5} \times \frac{2}{4} = \frac{6}{20} = \frac{3}{10}.$$

Adding the two paths:

$$P(\text{Different colors}) = \frac{3}{10} + \frac{3}{10} = \frac{6}{10} = \mathbf{0.60} \quad (60\%).$$

Notice: the probability shifts significantly depending on whether the item is replaced or not!

Worked Example 4 — The Magic Trick of "At Least One"

A fair coin is tossed three times in succession. What is the probability of getting **at least one Head**?

Step-by-Step Solution: "Getting at least one Head" means obtaining 1 Head, 2 Heads, or 3 Heads. Calculating all these cases individually would be tedious. Let us examine the **complementary event**: The opposite of "getting at least one Head" is **"getting NO Heads at all"**, which means obtaining three consecutive Tails $(T; T; T)$.

- The probability of Tails on any single toss is $\frac{1}{2}$.
- Since the three tosses are independent:

$$P(\text{No Heads}) = P(T; T; T) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}.$$

- By the complementary event formula:

$$P(\text{At least one Head}) = 1 - P(\text{No Heads}) = 1 - \frac{1}{8} = \frac{\mathbf{7}}{\mathbf{8}} = \mathbf{0.875} \quad (87.5\%).$$

By passing through the complement, the calculation took a single line!

Worked Example 5 — Two-Color Die and the Fairground Game

A carnival booth invites players to roll a special 6-sided die: 4 faces are painted Green and 2 faces are painted Red. To play, the participant pays \$2.

- If the die lands on Red, the player wins \$5 (a net profit of +\$3).
- If the die lands on Green, the player wins nothing (a net loss of -\$2).

The booth owner claims: *"Step right up and test your luck! You can win more than double your stake!"* Is this game favorable to the player?

Step-by-Step Solution: Let us calculate the probabilities:

- $P(\text{Red}) = \frac{2}{6} = \frac{1}{3} \approx 33.3\%$.
- $P(\text{Green}) = \frac{4}{6} = \frac{2}{3} \approx 66.7\%$.

Imagine playing 300 rounds:

- The die lands on Red approximately $300 \times \frac{1}{3} = 100$ times, yielding $100 \times (+\$3) = +\$300$.
- The die lands on Green approximately $300 \times \frac{2}{3} = 200$ times, costing $200 \times (-\$2) = -\$400$.

Overall balance for 300 rounds: $300 - 400 = -\$100$! On average per round, the player loses approximately $\frac{-100}{300} \approx -\$0.33$. **The game is strictly unfavorable to the player** (and reliably profitable for the booth operator).

What You Must Remember

1. **Vocabulary:**
   - **Random experiment**: an outcome driven by chance, not predictable with certainty.
   - **Outcomes**: individual elementary results.
   - **Impossible event** $(P = 0)$; **Certain event** $(P = 1)$.
   - **Complementary event** $\bar{A}$: $P(\bar{A}) = 1 - P(A)$. The magic shortcut for "at least one": $P(\text{at least one}) = 1 - P(\text{none})$.
2. **Equiprobability:**

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

   Always verify: $0 \leq P(E) \leq 1$. The sum of probabilities over all elementary outcomes equals 1.
3. **Two-Stage Experiments:**
   - **Two-way table**: ideal for 2 dice, 2 spinners (total cells = product of outcomes).
   - **Probability tree diagram**:
     - Sum of branches issuing from a single node = 1.

    - **Multiply** probabilities along a path.
    - **Add** probabilities of disjoint paths that achieve the event.
  - **With replacement**: probabilities remain constant.
  - **Without replacement**: sample size decreases by 1 on each draw.
4. **Law of Large Numbers:**
  - The observed **relative frequency** over a large number of repetitions converges and stabilizes around the **theoretical probability**.

Exercises

Level : Direct Application

Exercise 30.1 An opaque urn contains 4 red marbles, 3 green marbles, and 5 blue marbles, all indistinguishable by touch. A marble is drawn at random. 1. How many possible outcomes are there? Are they equally likely? 2. Calculate the probability of drawing a red marble as an irreducible fraction. 3. Calculate the probability of drawing a marble that is not green.

Exercise 30.2 A fair 6-sided die numbered 1 to 6 is rolled. Calculate the probability of the following events:

- A: "Roll an odd number".
- B: "Roll a multiple of 3".
- C: "Roll a number strictly greater than 4".
- D: "Roll a prime number".

Exercise 30.3 A lottery wheel divided into 8 equal sectors is spun: 3 sectors are painted blue, 4 yellow, and 1 red. 1. What is the probability of landing on red? 2. What is the probability of not landing on blue? 3. Express these probabilities as percentages.

Exercise 30.4 A fair coin is tossed twice in succession. 1. Draw the tree of possibilities showing probabilities on each branch. 2. What is the probability of getting Tails twice? 3. What is the probability of getting exactly one Head and one Tail?

Exercise 30.5 In a middle school, 55% of students eat lunch at the cafeteria and the rest are day-pupils (eating at home). A student is chosen at random. What is the probability that this student eats at home?

Level : Deeper Reflection and Combined Situations

Exercise 30.6 (Rolling Two Dice and Multiplying). Two fair 6-sided dice are rolled, and the two numbers obtained are **multiplied**. 1. Using a 6×6 two-way table, determine all possible product outcomes. 2. What is the probability that the product equals 12? 3. What is the probability that the product is an even number?

Exercise 30.7 (Socks in the Drawer in the Dark). In a drawer, 6 black socks and 4 blue socks are mixed together, completely indistinguishable by touch. In the morning, before sunrise, Lucas picks two socks successively at random **without replacing them**. 1. Construct the weighted tree diagram representing this random experiment. 2. What is the probability that Lucas picks two black socks? 3. What is the probability that Lucas picks two socks of the same color (so that they match)?

Exercise 30.8 (The Word "PROBABILITÉ"). Each of the 11 letters of the word "P R O B A B I L I T É" is written on an identical card, and the 11 cards are placed in a bag. A card is drawn at random. 1. What is the probability of drawing the letter "B"? 2. What is the probability of drawing a vowel (O, A, I, I, É)? 3. What is the probability of drawing a consonant?

Exercise 30.9 (Factory Quality Control). A factory manufactures electronic components. On a production line, 3% of the components have a soldering defect. A technician selects two components in succession at random (this selection is treated as sampling with replacement due to the very large volume of parts produced). 1. What is the probability that both components are defective? 2. What is the probability that neither component is defective? 3. What is the probability that at least one of the two components has a defect?

Exercise 30.10 (The Gambler's Fallacy at the Casino). A player stands at a roulette wheel. The ball has landed on a Black pocket on the last 6 consecutive spins. The player exclaims: *"Red hasn't appeared for 6 spins, it has a massive chance of landing now, I am betting everything on Red!"* 1. What do you think of this player's reasoning? 2. What is the actual probability that the ball lands on Red on the 7th spin (on a European roulette wheel with 18 red pockets, 18 black pockets, and 1 green zero pocket)?

Exercise 30.11 (The Archery Target). A square target with side length 40 cm has a red disk of radius 10 cm at its center. A novice archer fires an arrow that is certain to hit the square target, but at a completely random point (probability proportional to area). 1. Calculate the total area of the square target. 2. Calculate the exact area of the central red disk. 3. What is the probability that the arrow hits the red disk? (Round to the nearest hundredth).

Level : Challenges and Further Exploration

Exercise 30.12 (The Historical Problem of the Chevalier de Méré). In the 17th century, the Chevalier de Méré noticed that he consistently made money betting that at least one 6 would appear when rolling a fair 6-sided die 4 times in a row. 1. What is the probability of NOT rolling a 6 on a single roll of the die? 2. What is the probability of rolling no 6s in 4 independent rolls? 3. Deduce the probability of rolling at least one 6 in 4 rolls. Was the Chevalier right to place this bet?

Exercise 30.13 (Computer Simulation of the Law of Large Numbers). A Python program simulates 10,000 rolls of a balanced 6-sided die. Here is the script used:

```
import random

rolls = 10000
sixes = 0

for i in range(rolls):
    die = random.randint(1, 6)
    if die == 6:
        sixes += 1

frequency = sixes / rolls
print("Frequency of 6:", frequency)
```

1. What is the theoretical probability of obtaining a 6 on a single roll? Give its value as a fraction and as a decimal rounded to 0.001.
2. What approximate value will the program display for `frequency`?
3. Why will the displayed value generally not be exactly equal to $1/6$? What is this phenomenon called?

Exercise 30.14 (The Birthday Paradox in the Classroom). In a 9th-grade class of 25 students, the teacher asserts: *"There is a greater than 50% chance that two students in this classroom share the same birthday!"* (Assume a non-leap year of 365 days, and all birth dates equally likely). To test the assertion, we investigate the complementary event: "All 25 students have pairwise distinct birthdays." 1. The first student can have a birthday on any day (probability $365/365$). What is the probability that the second student is born on a different day from the first? 2. What is the probability that the 3rd student is born on a day different from the first two? 3. The product of the 25 probabilities is approximately 0.43. Calculate the probability that at least two students share the same birthday. Was the teacher correct?

Exercise 30.15 (The Problem of Points Revisited: Pascal and Fermat). Two players A and B each bet 50 pistoles into a pot (total: 100 pistoles). The first to win 3 rounds takes the entire pot. The current score is 2 to 0 in favor of Player A. The game is abruptly interrupted. 1. At most, how many additional rounds would have been necessary to complete the game? 2. Draw the tree of remaining rounds assuming each round is equally likely ($P = 1/2$). 3. What is the probability that Player B stages a comeback and wins the game? 4. How should the 100 pistoles be fairly divided between A and B?

Solutions

Solutions to Level Exercises

- **Solution 30.1:**
  1. Total number of outcomes: $4 + 3 + 5 = 12$ marbles. Since they are indistinguishable by touch, they are equally likely.
  2. $P(\text{Red}) = \frac{4}{12} = \mathbf{\frac{1}{3}}$.
  3. There are $4 + 5 = 9$ non-green marbles, or $1 - \frac{3}{12} = \frac{9}{12} = \mathbf{\frac{3}{4}}$.
- **Solution 30.2:**
  - $P(A) = \frac{3}{6} = \mathbf{\frac{1}{2}}$ (outcomes: 1, 3, 5).
  - $P(B) = \frac{2}{6} = \mathbf{\frac{1}{3}}$ (outcomes: 3, 6).
  - $P(C) = \frac{2}{6} = \mathbf{\frac{1}{3}}$ (outcomes: 5, 6).
  - $P(D) = \frac{3}{6} = \mathbf{\frac{1}{2}}$ (prime outcomes: 2, 3, 5; remember: 1 is not prime!).
- **Solution 30.3:**
  1. $P(\text{Red}) = \frac{1}{8} = \mathbf{0.125} = \mathbf{12.5\%}$.
  2. $P(\text{Not Blue}) = 1 - \frac{3}{8} = \frac{5}{8} = \mathbf{0.625} = \mathbf{62.5\%}$.
- **Solution 30.4:**
  1. The tree has 4 terminal branches: $(H;H), (H;T), (T;H), (T;T)$, each with probability $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.
  2. $P(T;T) = \mathbf{\frac{1}{4}} = 0.25$.
  3. Favorable paths: $(H;T)$ and $(T;H) \Longrightarrow \frac{1}{4} + \frac{1}{4} = \mathbf{\frac{1}{2}} = 0.5$.
- **Solution 30.5:** $P(\text{Day-pupil}) = 1 - 0.55 = \mathbf{0.45} = \mathbf{45\%}$.

Detailed Solutions to Level Exercises

- **Solution 30.6 (Product of Two Dice):**
  1. Product table ($6 \times 6 = 36$ cells):
     - Row 1: 1, 2, 3, 4, 5, 6
     - Row 2: 2, 4, 6, 8, 10, 12
     - Row 3: 3, 6, 9, 12, 15, 18
     - Row 4: 4, 8, 12, 16, 20, 24
     - Row 5: 5, 10, 15, 20, 25, 30
     - Row 6: 6, 12, 18, 24, 30, 36
  2. The cells giving 12 are: $(2;6), (3;4), (4;3), (6;2)$, totaling 4 cells.
  $$P(\text{Product} = 12) = \frac{4}{36} = \mathbf{\frac{1}{9}} \approx 0.111.$$
  3. The product is odd only if BOTH dice show odd numbers (3 choices $\times$ 3 choices = 9 cells). All other cells yield even products: $36 - 9 = 27$ even cells.
  $$P(\text{Even Product}) = \frac{27}{36} = \mathbf{\frac{3}{4}} = \mathbf{0.75}.$$
- **Solution 30.7 (Socks Without Replacement):** Initial total: $6 + 4 = 10$ socks.
  1. Tree diagram:
     - 1st draw: Black $(6/10)$, Blue $(4/10)$.
     - 2nd draw after Black: Black $(5/9)$, Blue $(4/9)$.
     - 2nd draw after Blue: Black $(6/9)$, Blue $(3/9)$.
  2. Two black socks: $P(B;B) = \frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \mathbf{\frac{1}{3}} \approx 0.333$.
  3. Same color: two black or two blue. $P(\text{Blue};\text{Blue}) = \frac{4}{10} \times \frac{3}{9} = \frac{12}{90}$.
  $$P(\text{Same Color}) = \frac{30}{90} + \frac{12}{90} = \frac{42}{90} = \mathbf{\frac{7}{15}} \approx \mathbf{0.467} \quad (46.7\%).$$

- **Solution 30.8 (The Word PROBABILITÉ):** The word contains 11 letters: P (1), R (1), O (1), B (2), A (1), I (2), L (1), T (1), É (1).
  1. $P(\text{B}) = \frac{\mathbf{2}}{\mathbf{11}}$.
  2. Vowels: O, A, I, I, É $\Longrightarrow$ 5 vowels. $P(\text{Vowel}) = \frac{\mathbf{5}}{\mathbf{11}}$.
  3. Consonants: complementary event $\Longrightarrow 1 - \frac{5}{11} = \frac{\mathbf{6}}{\mathbf{11}}$.
- **Solution 30.9 (Quality Control):** Let D be the event "defective component", $P(D) = 0.03$, and $P(\bar{D}) = 0.97$.
  1. Both defective: $P(D; D) = 0.03 \times 0.03 = \mathbf{0.0009}$ (or 0.09%, less than one in a thousand).
  2. Neither defective: $P(\bar{D}; \bar{D}) = 0.97 \times 0.97 = \mathbf{0.9409} \approx \mathbf{94.1}\%$.
  3. At least one defective: complement of neither defective:
$$P = 1 - 0.9409 = \mathbf{0.0591} \approx \mathbf{5.91}\%.$$
- **Solution 30.10 (The Gambler's Fallacy):**
  1. The gambler's reasoning is a **classic illusion** (known as the "gambler's fallacy"). The roulette wheel has no memory: every spin is entirely independent of previous spins. Past results have zero influence on the ball's next resting place.
  2. On a European roulette wheel with 37 pockets (18 red, 18 black, 1 green zero):
$$P(\text{Red}) = \frac{18}{37} \approx \mathbf{0.4865} \quad (48.65\%).$$
  The probability is strictly the same as on the very first spin!
- **Solution 30.11 (The Archery Target):**
  1. Area of the square target: $\mathcal{A}_{\text{square}} = 40 \times 40 = \mathbf{1,600\ cm^2}$.
  2. Area of the red disk: $\mathcal{A}_{\text{disk}} = \pi \times R^2 = \pi \times 10^2 = \mathbf{100\pi\ cm^2} \approx 314.16\ \text{cm}^2$.
  3. Probability:
$$P = \frac{\mathcal{A}_{\text{disk}}}{\mathcal{A}_{\text{square}}} = \frac{100\pi}{1,600} = \frac{\pi}{16} \approx \mathbf{0.20} \quad (20\%).$$

Detailed Solutions to Level Exercises

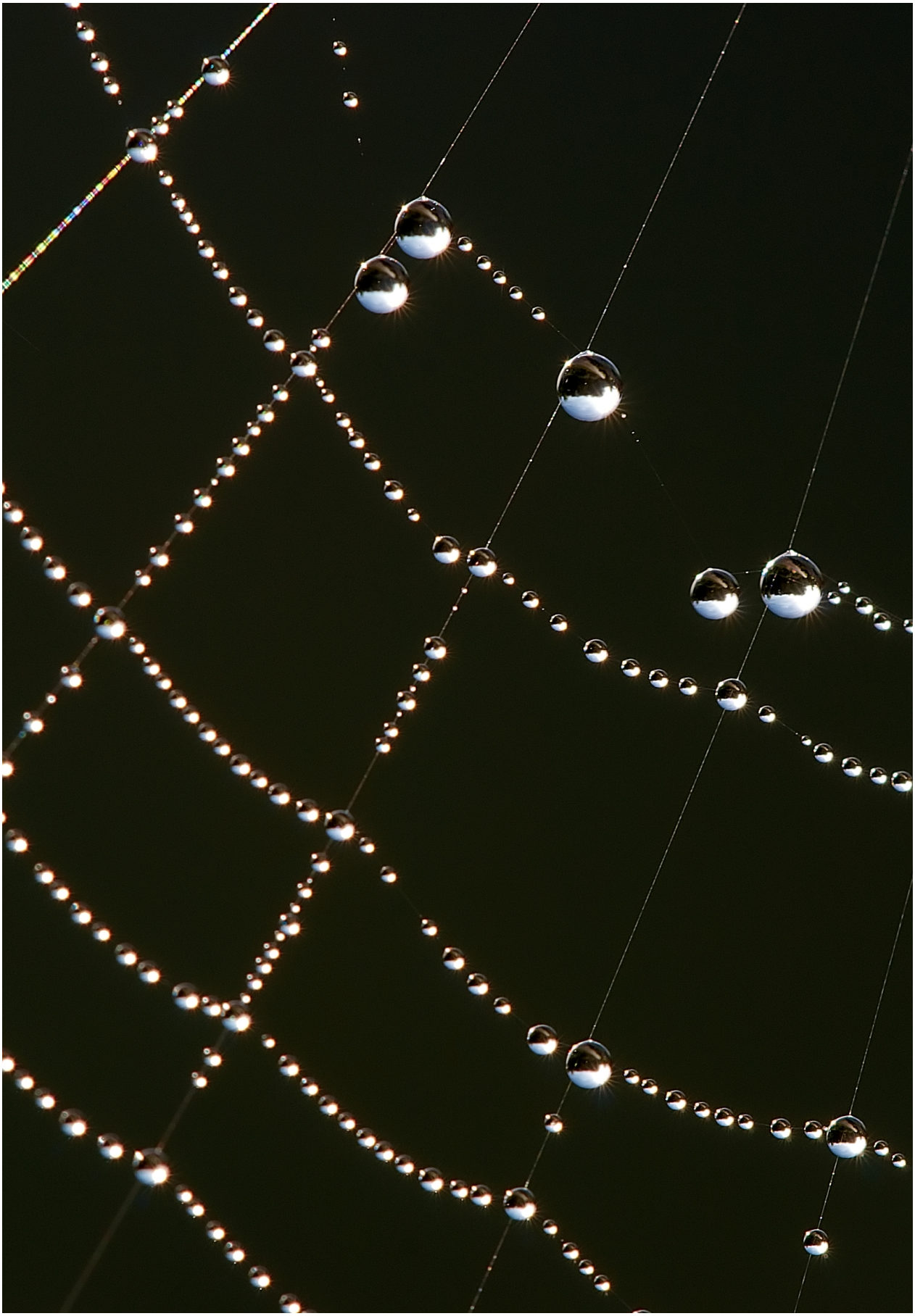
- **Solution 30.12 (The Chevalier de Méré):**
  1. Probability of not rolling a 6: $P(\text{not } 6) = \frac{5}{6}$.
  2. Over 4 independent rolls, probability of rolling no 6s:
$$P(\text{no 6s}) = \left(\frac{5}{6}\right)^4 = \frac{625}{1,296} \approx 0.4823.$$
  3. Probability of getting at least one 6:
$$P(\text{at least one } 6) = 1 - 0.4823 = \mathbf{0.5177} \approx \mathbf{51.8}\%.$$
  Since $51.8\% > 50\%$, the probability is strictly greater than one half: the Chevalier de Méré was right, in the long run he was guaranteed a profit!
- **Solution 30.13 (Python Simulation):**
  1. Theoretical probability: $p = \frac{1}{6} \approx \mathbf{0.167}$.
  2. The program will display a value very close to 0.167 (for example 0.1658 or 0.1682).
  3. Due to **sampling variability (fluctuation)**, chance produces minor variations around the theoretical mean. Only at infinity would the equality become exact.
- **Solution 30.14 (The Birthday Paradox):**
  1. For the 2nd student to have a different birthday from the 1st: $\frac{364}{365}$.
  2. For the 3rd student: $\frac{363}{365}$.
  3. The probability that all 25 students have distinct birthdays is the product of these 25 fractions: $P(\text{all distinct}) \approx 0.43$. Therefore, the probability that at least two students share the same birthday is:
$$P(\text{at least two identical}) = 1 - 0.43 = \mathbf{0.57} \quad (\mathbf{57}\%).$$
  Since $57\% > 50\%$, the teacher was completely right: there is more than a 1 in 2 chance!
- **Solution 30.15 (Pascal and Fermat Leading 2 to 0):**

1. Player A needs only 1 more win (leading 2-0). Player B must win 3 consecutive rounds to win 3-2. At most **3 additional rounds** are required.
2. Tree of remaining rounds:
   - Round 1: either A wins $(1/2) \Longrightarrow$ A wins match (3-0).
   - Or B wins $(1/2)$:
     - Round 2: either A wins $(1/4) \Longrightarrow$ A wins match (3-1).
     - Or B wins $(1/4)$:
       - Round 3: either A wins $(1/8) \Longrightarrow$ A wins match (3-2).
       - Or B wins $(1/8) \Longrightarrow$ B wins match (3-3 final).
3. Player B wins in only one scenario: winning all 3 consecutive rounds, with probability:
$$P(\text{B wins}) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{\mathbf{1}}{\mathbf{8}} = \mathbf{12.5}\%.$$
   Player A therefore has $P(\text{A wins}) = 1 - \frac{1}{8} = \frac{\mathbf{7}}{\mathbf{8}} = \mathbf{87.5}\%$.
4. Fair division of the 100 pistoles:
   - For A: $\frac{7}{8} \times 100 = \mathbf{87.5}\ \ \mathbf{\text{pistoles}}$.
   - For B: $\frac{1}{8} \times 100 = \mathbf{12.5}\ \ \mathbf{\text{pistoles}}$.

The Question That Opens the Future

Here ends our grand journey through Middle School (*Collège*). You began with the mystery of Al-Khwarizmi's letters, balanced scale pans, conquered the Pythagorean Theorem, measured the Earth with Eratosthenes, unlocked the secrets of Euclid, and learned to tame uncertainty with Pascal and Fermat.

Yet an immense frontier now stands before you. Until now, our numbers were either discrete counters on fingers or static lengths locked within stationary figures. What happens when the wind blows, when a sailboat battles the crosscurrent in the heart of a storm? How can we represent a force that pulls in a specific direction with a precise magnitude? Welcome to High School (*Lycée*): in Volume V, numbers will take flight, grow arrows, soar into the sky, and become **vectors**.



VOLUME V — MOTION, ANGLE AND THE SPECTRUM OF FORM

Ages 15–17 / Grades 10–11

High school foundations: formal logic, vectors, analytic geometry, quadratic polynomials, trigonometry, the derivative as an instantaneous rate, and conditional probability.

Chapters of Volume V :

Chapter 31 : The Court of Logic
Chapter 32 : The Sailboat in the Storm and the Free Arrow
Chapter 33 : The Line of Sight
Chapter 34 : The Jacquard Loom and the Function Box
Chapter 35 : Galileo's Fountain and the Parabola
Chapter 36 : The Staircase Steps and the Usurer's Loan
Chapter 37 : The Ferris Wheel and the Winding Circle
Chapter 38 : Pulling the Rope and the Effective Shadow
Chapter 39 : The Skier and the Board under the Boot
Chapter 40 : The Bacterial Colony
Chapter 41 : The Pollster and Standard Deviation
Chapter 42 : The Physician, the Test, and the Tree of Possibilities

Facing: Concurrent vector balance and catenary curves in the spider web (Araneus diadematus).

CHAPTER 31: The Court of Logic

> **"In the courtroom of mathematics, no eloquence can save the accused. Verdicts are not reached on the sincere expression of the witness, nor on the number of those who believe him: judgement is passed strictly on proof. A single counterexample suffices to condemn a statement, but a thousand examples will never suffice to clear it."**

What You Are About to Discover

- The great **number families** ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{D}$, $\mathbb{Q}$, $\mathbb{R}$), nested like Russian dolls, and why each of them had to be invented.
- The language of **sets** and **intervals**: $\in$, $\subset$, $\cup$, $\cap$, $[a\,;b[$, $+\infty$… and how to read them aloud.
- The **absolute value**, which measures distance without caring about direction.
- The rules of the **court**: implication, converse, contrapositive, equivalence, and the quantifiers "for all" and "there exists."
- The four great **weapons of proof**: direct proof, counterexample, contraposition, and proof by contradiction—with the core demonstrations required in Grade 10 ($\frac{1}{3}$ is not decimal, $\sqrt{2}$ is irrational, the square of an odd integer is odd…).

---

1. The Problem: The Formula That Lied Forty Times in a Row

Imagine a courtroom. In the dock sits not a man, but a **sentence**. It has been brought forward by an enthusiastic student, who defends it with fiery passion:

> *"For every natural number n, the number $n^2 + n + 41$ is prime."*

The defense attorney calls witnesses to the stand, one by one.

n	$n^2 + n + 41$	Prime?
0	41	yes
1	43	yes
2	47	yes
3	53	yes
4	61	yes
5	71	yes
10	151	yes
20	461	yes
30	971	yes
39	1601	yes

Forty witnesses ($n = 0, 1, 2, ..., 39$) step up to the stand, and **every single one** confirms: the resulting number is prime. The courtroom audience is convinced. The jury prepares to acquit.

Then the prosecutor stands up, completely calm, and calls a single witness: $n = 40$.

$$40^2 + 40 + 41 = 1600 + 40 + 41 = 1681 = 41 \times 41.$$

1681 is **not** prime. The statement is **false**. The gavel strikes; the verdict is final and unappealable.

> 💡 **Did You Know?**
>
> This polynomial was observed by **Leonhard Euler** in 1772. It produces prime numbers for n from 0 up to 39—an astounding streak—but it fails at $n = 40$. One can even see this without calculating: for $n = 41$, $41^2 + 41 + 41 = 41 \times 43$, which is obviously divisible by 41.

This story encapsulates the entire spirit of this chapter:

1. **Checking cases, even dozens of them, proves nothing** for a statement that speaks of *all* integers.
2. **A single counterexample is enough** to shatter a universal assertion.
3. To *establish* a general truth, an entirely different tool is required: **proof**.

In middle school, you often "saw" that a property worked on examples. Starting in tenth grade (Seconde), you step into the court of law. You need precise language (which numbers are we talking about? what does "if… then" mean?) and reliable methods of proof. This is what we will construct together.

> 🔍 ***The True Story*** *— Around 350 BCE,* ***Aristotle*** *laid the foundations of logic by studying the* syllogism*: "All men are mortal; Socrates is a man; therefore Socrates is mortal." More than two thousand years later, in 1847 and 1854, the Englishman* ***George Boole****, a cobbler's son and largely self-taught, transformed logic into an* algebra*: truth became* 1*, falsehood* 0*, and "and" became multiplication. Without knowing it, he had just penned the grammar for all twentieth-century computers.*

---

2. The Number Families: A History of Crises

Every family of numbers was born out of an **impossible operation**. When existing tools no longer sufficed, humanity had to expand the universe of numbers.

2.1 The Natural Numbers $\mathbb{N}$: Counting Pebbles

The very first numbers are those of the shepherd counting his sheep: $0, 1, 2, 3, \ldots$

> 📐 **Definition —**
>
> The set of **natural numbers** is
>
> $$\mathbb{N} = \{0\,;1\,;2\,;3\,;\ldots\}.$$

Notation Explained. - The **curly braces** $\{\ \}$ are used to write a set by listing its elements. We read this as "the set of elements $0, 1, 2\ldots$". In French and continental European tradition, elements are often separated by semicolons to avoid confusion with the decimal comma. - The letter $\mathbb{N}$ comes from *naturale* (Italian): it was introduced by **Giuseppe Peano** in 1889. The "double-struck" style (known as *blackboard bold*) was born at the blackboard: chalk cannot easily produce boldface type, so mathematicians doubled a stroke. It became standard in printed textbooks during the mid-twentieth century. - We simply read "$\mathbb{N}$" as "the set of natural numbers" or "N".

In $\mathbb{N}$, we can always add and multiply. But **subtraction** causes trouble: $3 - 5$ has no answer in $\mathbb{N}$.

2.2 The Integers $\mathbb{Z}$: The Merchant's Debts

Recall the merchant's slate (Chapter 14): to record a debt, negative numbers become indispensable.

> 📐 **Definition —**
>
> The set of **integers** (or relative integers) is
>
> $$\mathbb{Z} = \{\ldots\,;-3\,;-2\,;-1\,;0\,;1\,;2\,;3\,;\ldots\}.$$

The letter $\mathbb{Z}$ comes from the German ***Zahlen*** ("numbers"), widely adopted by Richard Dedekind and German mathematicians in the nineteenth century. In $\mathbb{Z}$, subtraction is always possible. But **division** is not: $7 \div 2$ is not an integer.

2.3 The Decimals $\mathbb{D}$: Stevin's Tenths

A merchant weighing spices needs 2.35 kg. Simon Stevin (Chapter 11) popularized decimal numbers.

> **Definition —**
>
> A number is **decimal** if it can be written in the form
>
> $$\frac{a}{10^n} \quad \text{with} \quad a \in \mathbb{Z} \quad \text{and} \quad n \in \mathbb{N}.$$
>
> The set of decimal numbers is denoted $\mathbb{D}$.

In other words: a decimal number is one that can be written with a **finite** number of digits after the decimal point. Examples: $2.35 = \frac{235}{10^2}$; $-0.007 = \frac{-7}{10^3}$; $12 = \frac{12}{10^0}$ (every integer is a decimal number).

The symbol $\in$ is read "**belongs to**" or "**is an element of**". "$a \in \mathbb{Z}$" means "a is an element of $\mathbb{Z}$", that is, "a is an integer." This symbol is a stylized version of the Greek letter epsilon ε, the initial of ἐστί (*esti*, "is"): introduced once again by **Peano** in 1889. Its negation, $\notin$, is read "does not belong to."

Everything seems in order… until the day we divide a loaf of bread into **three**.

2.4 The Rational Numbers $\mathbb{Q}$: Dividing into Three

> **Definition —**
>
> A number is **rational** if it can be written in the form
>
> $$\frac{p}{q} \quad \text{with} \quad p \in \mathbb{Z} \quad \text{and} \quad q \in \mathbb{N}, q \neq 0.$$
>
> The set of rational numbers is denoted $\mathbb{Q}$ (from the initial of *quotient*, proposed by Peano in 1895).

The word "rational" comes from the Latin *ratio*, meaning "ratio" or "proportion": a rational number is a **ratio** of two integers. It has nothing to do with being "sensible" or "reasonable"… even though mathematical history often plays on this double meaning.

The natural question arises: $\frac{1}{3}$ is rational, but is it decimal? Your calculator displays 0.333333333, which *looks* like a decimal. But its screen is too small to tell the whole truth. Here is the first trial of the chapter.

> **Proof (Core Curriculum) — $\frac{1}{3}$ is not a decimal number.**
>
> Let us reason **by contradiction** (*reductio ad absurdum*): suppose that $\frac{1}{3}$ were a decimal number. There would then exist an integer a and a natural number n such that
>
> $$\frac{1}{3} = \frac{a}{10^n}.$$
>
> Multiplying both sides by 3×10^n, we obtain:
>
> $$10^n = 3a.$$
>
> Thus 10^n would be a multiple of 3. However, the sum of the digits of 10^n (a single "1" followed by n zeros) is 1, which is not divisible by 3; by the standard divisibility rule for 3, 10^n is **not** a multiple of 3.
>
> (Alternative argument: $10^n = 2^n \times 5^n$; its prime factorization contains only 2s and 5s, and never any factor 3.)
>
> We have reached a contradiction. The initial assumption must therefore be false: $\frac{1}{3} \notin \mathbb{D}$. ■

The small filled black square ■ (or Halmos symbol) marks the end of a proof. The Ancients used to write "**QED**": *quod erat demonstrandum* ("which was to be demonstrated").

> 💡 **Did You Know?**
>
> An irreducible fraction $\frac{p}{q}$ is decimal if and only if its denominator q contains only the prime factors 2 and 5. Thus $\frac{7}{40} = \frac{7}{2^3 \times 5} = 0.175$ is decimal, but $\frac{1}{6} = \frac{1}{2 \times 3}$ is not. Every rational number has a decimal representation that is either **terminating** (finite) or **eventually periodic**: for example, $\frac{1}{7} = 0.\overline{142857}$ (where the overline indicates the repeating period).

2.5 The Real Numbers $\mathbb{R}$: The Diagonal That Broke the Brotherhood

You have already encountered this drama (Chapter 19, *The Forbidden Secret of the Pythagoreans*). The diagonal of a square with unit side length measures $\sqrt{2}$, according to the Pythagorean theorem: $d^2 = 1^2 + 1^2 = 2$. The Pythagoreans believed that "all is number," meaning every length in the cosmos is a ratio of integers. The diagonal betrayed them. We will now write out the proof with full high-school rigor. First, we need a small lemma (a *lemma* is an auxiliary stepping stone towards a main theorem).

Proof (Core Curriculum) — The square of an odd integer is odd.

Let n be an odd integer. By definition, there exists an integer k such that $n = 2k + 1$. Then:

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is an integer (let us call it k'), we have $n^2 = 2k' + 1$: therefore n^2 is odd. ■

Similarly, the square of an even integer $n = 2k$ is $n^2 = 4k^2 = 2(2k^2)$, which is even.

> ⚠️ **Consequence (by Contraposition) —**
>
> If n^2 is even, then n is even.
>
> Indeed, if n were not even, it would be odd, and its square would be odd by the lemma above. We will revisit this mode of reasoning (contraposition) in detail in Section 6.

Proof (Core Curriculum) — $\sqrt{2}$ is irrational.

Let us reason by contradiction. Suppose that $\sqrt{2}$ were rational. Then it could be written as an **irreducible fraction**:

$$\sqrt{2} = \frac{p}{q}, \quad p \in \mathbb{N}, q \in \mathbb{N}^*, \text{where } p \text{ and } q \text{ share no common factor other than } 1.$$

Squaring both sides: $2 = \frac{p^2}{q^2}$, hence:

$$p^2 = 2q^2.$$

1. Therefore p^2 is even. By the consequence above, p is even: there exists an integer k such that $p = 2k$. 2. Substituting this back: $(2k)^2 = 2q^2$, which gives $4k^2 = 2q^2$, or $q^2 = 2k^2$. Hence q^2 is even, which implies that q is also even. 3. Consequently, p and q are both even: they share 2 as a common factor. This **contradicts** the fact that the fraction $\frac{p}{q}$ is in lowest terms (irreducible).

The assumption "$\sqrt{2}$ is rational" leads to an absurdity: it must be false. Therefore $\sqrt{2} \notin \mathbb{Q}$. ■

The notation $\mathbb{N}^*$ denotes $\mathbb{N}$ excluding 0. Likewise, $\mathbb{R}^*$ denotes the non-zero real numbers.

> 📐 **Definition —**
>
> The set of **real numbers**, denoted $\mathbb{R}$, is the set of all coordinates of points along a graduated number line. A real number that is not rational is called **irrational**.

Examples of irrationals: $\sqrt{2}$, $\sqrt{3}$, π (proved irrational by Lambert in 1761), $\sqrt{2} + 1$ (see Exercise 11).

2.6 The Russian Dolls: Inclusion $\subset$

Every natural number is an integer; every integer is a decimal; every decimal is rational ($\frac{a}{10^n}$ is a quotient of integers); every rational is real.

> **📐 Definition —**
>
> A set A is said to be **included** in (or a **subset** of) a set B, written $A \subset B$, if every element of A is also an element of B. We read this as "A is included in B" or "A is a subset of B."

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{D} \subset \mathbb{Q} \subset \mathbb{R}$$

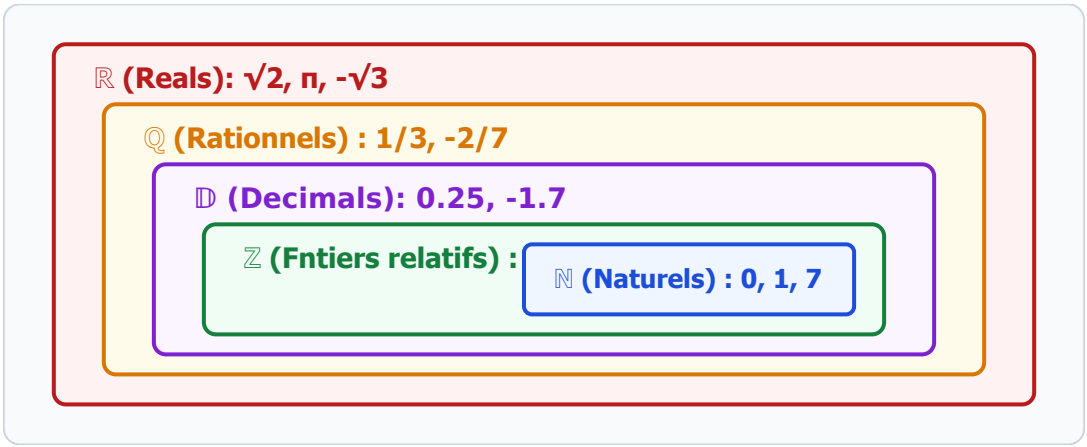


Figure 78: Nested number sets: N ⊂ Z ⊂ D ⊂ Q ⊂ R

Each inclusion is **strict**: $-1 \in \mathbb{Z}$ but $-1 \notin \mathbb{N}$; $0.5 \in \mathbb{D}$ but $0.5 \notin \mathbb{Z}$; $\frac{1}{3} \in \mathbb{Q}$ but $\frac{1}{3} \notin \mathbb{D}$; $\sqrt{2} \in \mathbb{R}$ but $\sqrt{2} \notin \mathbb{Q}$.

> **⚠ Pitfall Warning!**
>
> Do not confuse $\in$ and $\subset$.
>
> - $\in$ connects an **element** to a **set**: $3 \in \mathbb{N}$.
> - $\subset$ connects **two sets**: $\{3\} \subset \mathbb{N}$, $\mathbb{N} \subset \mathbb{Z}$.
>
> Writing "$3 \subset \mathbb{N}$" or "$\mathbb{N} \in \mathbb{Z}$" is a violation of mathematical grammar.

> **⚠ Pitfall Warning!**
>
> A number can wear many disguises. $\frac{12}{4}$ looks like a fraction, but it simplifies to 3, a natural number. $\sqrt{9} = 3 \in \mathbb{N}$, $\sqrt{\frac{4}{9}} = \frac{2}{3} \in \mathbb{Q}$, $\frac{21}{75} = \frac{7}{25} = 0.28 \in \mathbb{D}$. Always **simplify** before passing judgement.

Worked Example 1 — Classifying Numbers.

For each number, determine the smallest set among $\mathbb{N}, \mathbb{Z}, \mathbb{D}, \mathbb{Q}, \mathbb{R}$ that contains it:

$$a = -\frac{18}{6}, \quad b = \frac{3}{8}, \quad c = \frac{5}{12}, \quad d = \sqrt{16} - \sqrt{2}, \quad e = (\sqrt{5})^2, \quad f = \frac{\pi}{2\pi}.$$

- $a = -3$: $a \in \mathbb{Z}$, and $a \notin \mathbb{N}$. Smallest set: $\mathbb{Z}$.
- $b = \frac{3}{8} = \frac{3}{2^3}$: denominator contains only factors of 2, hence decimal: $b = \frac{375}{1000} = 0.375$. Smallest set: $\mathbb{D}$.
- $c = \frac{5}{12} = \frac{5}{2^2 \times 3}$, irreducible with a factor of 3 in the denominator: $c \in \mathbb{Q}$, $c \notin \mathbb{D}$. (Verification: $c = 0.41666...$)
- $d = 4 - \sqrt{2}$. If d were rational, then $\sqrt{2} = 4 - d$ would be rational (the difference of two rationals), which is false. Thus $d \in \mathbb{R}$, irrational.
- $e = 5$: $\mathbb{N}$.
- $f = \frac{1}{2} = 0.5$: $\mathbb{D}$. The π cancels out: do not let outward appearances mislead you!

3. The Language of Sets and Intervals

3.1 Describing a Set

There are two primary ways to describe a set:

- **By extension (roster method)**: listing all its elements, $A = \{1\,;3\,;5\,;7\}$;
- **By comprehension (set-builder notation)**: stating a characteristic property, $A = \{x \in \mathbb{N} \mid x \text{ is odd and } x < 8\}$.

The vertical bar | (or colon ":") is read "**such that**". The statement is read: "the set of all x belonging to $\mathbb{N}$ such that x is odd and strictly less than 8."

The **empty set**, which contains no elements at all, is denoted $\varnothing$. This symbol was introduced by **André Weil** (of the Bourbaki group) around 1939, inspired by the Scandinavian letter Ø. Warning: $\varnothing$ is not the same as $\{0\}$! The set $\{0\}$ contains one element, namely zero; $\varnothing$ contains nothing. An empty box is not a box containing a zero.

3.2 Union, Intersection, and Complement

Let A and B be two subsets of a universal set E.

Definitions —

- The **intersection** $A \cap B$ (read "A intersect B") is
the set of elements belonging to both A **and** B.
- The **union** $A \cup B$ (read "A union B") is the set of elements belonging to A **or** B (or both).
- The **complement** of A in E, denoted $\overline{A}$ or $E \setminus A$, is the set of elements in E that **do not belong** to A. The symbol $\setminus$ is read "minus" or "excluding."
- A and B are **disjoint** if $A \cap B = \varnothing$.

A mnemonic trick: ∩ resembles a narrow tunnel or arch through which only those holding both tickets may pass; ∪ resembles a wide cup or bowl, a **U** for **U**nion, gathering everything together.

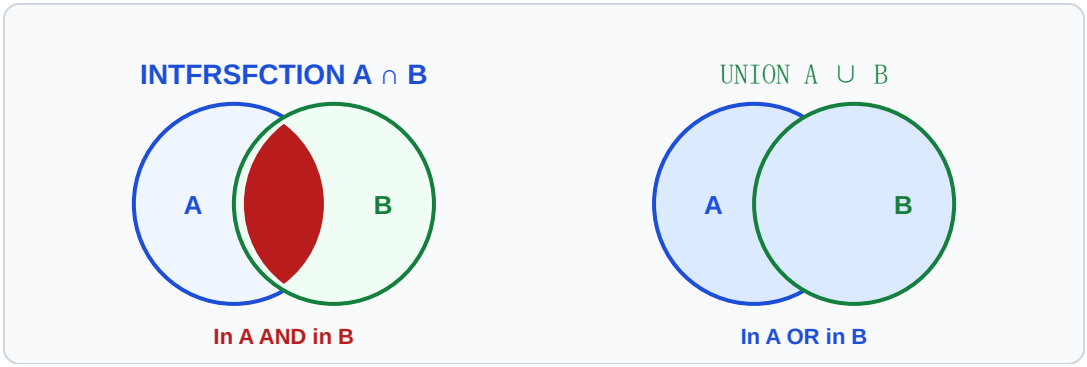


Figure 79: Venn diagrams: Intersection A ∩ B and Union A ∪ B

Worked Example 2 — Operations on Finite Sets.

Let $E = \{1\,;2\,;\ldots\,;12\}$, A = the set of multiples of 2 in E, and B = the set of multiples of 3 in E.

- $A = \{2\,;4\,;6\,;8\,;10\,;12\}$, $B = \{3\,;6\,;9\,;12\}$.
- $A \cap B = \{6\,;12\}$: these are the multiples of 6.
- $A \cup B = \{2\,;3\,;4\,;6\,;8\,;9\,;10\,;12\}$ (6 and 12 are not duplicated).
- $\overline{A \cup B} = \{1\,;5\,;7\,;11\}$: numbers in E divisible by neither 2 nor 3.
- Checking cardinalities: $\text{card}(A \cup B) = \text{card}(A) + \text{card}(B) - \text{card}(A \cap B) = 6 + 4 - 2 = 8$. (The *cardinality*, written card, denotes the number of elements.)

3.3 Intervals of $\mathbb{R}$

On the real number line, the most useful subsets are **unbroken line segments without holes**: intervals.

Definition —

Let a and b be two real numbers with $a < b$.

Notation	Set of reals x such that	Verbal Description	Visual Representation
$[a\,;b]$	$a \leq x \leq b$	closed interval a to b	[ ]
$]a\,;b[$ or (a,b)	$a < x < b$	open interval a to b	] [
$[a\,;b[$ or $[a,b)$	$a \leq x < b$	half-open: closed at a, open at b	[[
$]a\,;b]$ or $(a,b]$	$a < x \leq b$	half-open: open at a, closed at b	]]
$[a\,;+\infty[$ or $[a,+\infty)$	$x \geq a$	from a inclusive to plus infinity	[→
$]a\,;+\infty[$ or $(a,+\infty)$	$x > a$	from a exclusive to plus infinity	]→
$]-\infty\,;b]$ or $(-\infty,b]$	$x \leq b$	from minus infinity to b inclusive	←]
$]-\infty\,;b[$ or $(-\infty,b)$	$x < b$	from minus infinity to b exclusive	←[

We also write $\mathbb{R} =]-\infty\,;+\infty[= (-\infty,+\infty)$. The real numbers a and b are the **endpoints** (or bounds) of the interval; $b-a$ is its **length** (or width).

Why these brackets? The French and ISO convention is visual: an **inward-facing bracket** [or] "captures" the endpoint (it is included); an **outward-facing bracket**] or ["pushes it away" (it is excluded). In many English-speaking traditions, parentheses (a,b) are used for open bounds. A **semicolon** is used between endpoints to avoid confusion with decimal commas or periods: $[1.5\,;2]$.

The symbol ∞ ("infinity") was introduced by the Englishman **John Wallis** in 1655. It is **not a number**: it is a way to express "without bound in this direction." This is why brackets are **always open** adjacent to $+\infty$ or $-\infty$: one can never "reach" infinity, and so it can never be included.

Pitfall Warning!

- $[3\,;+\infty]$ is **incorrect**: the bracket must be open at $+\infty$.
- $[5\,;2]$ makes no sense: the smaller endpoint must always be written on the left.
- $]2\,;2[= \varnothing$, but $[2\,;2] = \{2\}$ (a single point).
- The interval $[1\,;3]$ contains an **infinite** number of real values (1.5, $\sqrt{2}$, $2.999\ldots$), not just the integers $1, 2, 3$! Do not confuse it with the discrete set of integers $\{1\,;2\,;3\}$.

Useful Notations: $\mathbb{R}_+ = [0\,;+\infty[$ (non-negative real numbers), $\mathbb{R}_- =]-\infty\,;0]$ (non-positive reals), $\mathbb{R}_+^* =]0\,;+\infty[$ (strictly positive reals), $\mathbb{R}^* = \mathbb{R} \setminus \{0\} =]-\infty\,;0[\cup]0\,;+\infty[$.

Worked Example 3 — Intersection and Union of Intervals.

Let $I = [-2\,;5[$ and $J =]1\,;+\infty[$. Find $I \cap J$ and $I \cup J$.

Let us draw both intervals aligned on a line:

- *$I \cap J$: a number must satisfy both ≥ -2, < 5, and > 1. The most restrictive lower bound is $x > 1$, and upper bound $x < 5$. Therefore $I \cap J =]1\,;5[$.*
- *$I \cup J$: belonging to at least one suffices. This spans everything from -2 (included) onwards to $+\infty$. Hence $I \cup J = [-2\,;+\infty[$.*

Worked Example 4 — Translating an Inequality into an Interval.

Solve in $\mathbb{R}$: $-3 < 2x + 1 \leq 7$, and express the solution set as an interval.

Subtract 1 across all three parts: $-4 < 2x \leq 6$. Divide by 2, which is **positive** (preserving inequality direction): $-2 < x \leq 3$. The solution set is $S =]-2\,;3]$.

Worked Example 5 — A Union That Cannot Be Simplified.

Solve $x^2 > 9$ in $\mathbb{R}$.

$x^2 > 9 \Leftrightarrow x^2 - 9 > 0 \Leftrightarrow (x-3)(x+3) > 0$. A product of two factors is strictly positive when both factors share the same sign. Sign table:

x	$-\infty$		-3		3		$+\infty$
$x+3$		$-$	0	$+$		$+$	
$x-3$		$-$		$-$	0	$+$	
Product		$+$	0	$-$	0	$+$	

Thus $S =]-\infty\,;-3[\cup]3\,;+\infty[$. This union of two disjoint intervals **is not** an interval: there is a "gap" between -3 and 3.

⚠ The classic mistake "$x^2 > 9 \Leftrightarrow x > 3$" completely ignores the negative branch: $x = -4$ indeed satisfies $(-4)^2 = 16 > 9$.

CHAPTER 32: The Sailboat in the Storm and the Free Arrow

> "A point is an anchor holding a ship to the seabed. A Cartesian coordinate states where you are. But the sailor confronting the swell has little use for an anchor: he needs to know where the wind is driving him, at what speed, and how two competing currents merge into a single momentum. To liberate motion from the earth, humanity had to invent the flying arrow: the Vector."

What You Are About to Discover

- Why the Cartesian coordinate plane was no longer enough for physicists, and how **Hermann Grassmann** freed displacement from its fixed tether in 1844.
- The fundamental triad defining a **vector**: direction, sense (orientation), and magnitude (norm)—along with the arrow notation $\vec{u}$ or $\overrightarrow{AB}$.
- **Chasles' relation** ($\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$) understood as a relay of navigators or dispatch couriers, and the parallelogram rule.
- Multiplying a vector by a real number (scalar) and the concept of **collinear vectors**.
- How to translate geometry into algebraic calculations within a coordinate frame $(O\,;\vec{\imath},\vec{\jmath})$: vector coordinates, midpoint formula, and distance formula.
- The decisive arbiter of alignment and parallelism: the **determinant** $xy' - yx' = 0$.

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1. The Problem: The Rowboat in the River Current

You stand at the oars of a fishing boat on the waters of a river five hundred meters wide. On the opposite bank, straight ahead of you to the North, stands the dock where your companions await you.

You grip both oars. You haul against the timber with all your muscle: your bow points straight **North**, and your effort propels you through the water at a steady speed of $\mathbf{3}\ \mathbf{km/h}$. You mentally calculate that in ten minutes, you will tie up at the pontoon.

Yet the river is no tranquil pond. A strong current rushes from West to **East** at a speed of $\mathbf{4}\ \mathbf{km/h}$.

Look at the scene from the riverbank:

> 💡 **Concrete Modeling — The Sailboat Drift and Velocity Addition**
>
> - C (Actual landing point)
> - Actual /
> - trajectory / Oar propulsion (North: 3 km/h)
> - o B
> - A River current (East: 4 km/h)

What forces act upon your vessel? 1. Your arms drive you North: in one hour, without current, you would travel $3\ \text{km}$ North. 2. The water beneath your hull sweeps you East: in one hour, without rowing, you would drift $4\ \text{km}$ East.

Both actions take place **simultaneously**. At the end of an imaginary hour in which both movements combine, you would be neither purely North nor purely East, but at point C. What distance have you covered relative to the riverbed?

The triangle formed by the eastward displacement ($4\ \text{km}$), the northward displacement ($3\ \text{km}$), and the actual trajectory is a right-angled triangle. The Pythagorean theorem gives the distance traveled:

$$d = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5\ \text{km}\ .$$

Your actual speed relative to the banks is $\mathbf{5\ \ km/h}$, directed diagonally toward the Northeast!

This scenario reveals a fundamental limitation of ordinary numbers:

- A single number ($5\ \ \mathrm{km/h}$) states **how much**, but it does not state **where to**.
- The coordinate of a point $A(x_A, y_A)$ states **where one is**, but it is nailed to the grid drawn on the map. If the boat rows in the exact same manner a hundred kilometers downstream, the physical phenomenon is identical, even though the starting and ending coordinates are entirely different.

A new mathematical object was needed: an **autonomous arrow**, transportable anywhere without distortion, encoding tilt, orientation, and magnitude all at once.

---

2. Liberating the Arrow: The Birth of the Vector

2.1 Grassmann's Breakthrough (1844)

For two centuries following René Descartes (1637), analytic geometry operated by chaining every point to a fixed origin $O(0, 0)$. To study the flight of a bird or the blast of a squall, one had to manipulate the coordinates of the starting point A, then those of the destination B, and compute differences.

In 1844, a high school teacher in Germany unknown to his contemporaries, **Hermann Günther Grassmann**, published a visionary treatise: *Die Lineale Ausdehnungslehre* (The Theory of Linear Extension). Grassmann introduced a revolutionary insight: > "The point is a prison. What matters in physics is not the place where the action begins, but the action itself: pure transport."

This geometric entity was given the name **vector**, from the Latin *vehere* ("to carry", "to convey"—the root of *vehicle*).

2.2 The Three Defining Characteristics of a Vector

> **Definition — Vector.**
>
> A non-zero vector $\vec{u}$ is defined by three inseparable pieces of data: 1. **Its direction:** the line along which it lies (or any line parallel to it). This indicates the line of action of the movement (e.g., the North-South line, the horizontal axis, a 30° incline). 2. **Its sense (or orientation):** the heading along that line, indicated by the arrowhead (e.g., Northward rather than Southward, left to right). 3. **Its magnitude (or norm):** the length of the directed line segment, denoted $\| \vec{u} \|$. It represents physical intensity (a speed in $\mathrm{km/h}$, a force in newtons, a displacement in meters).

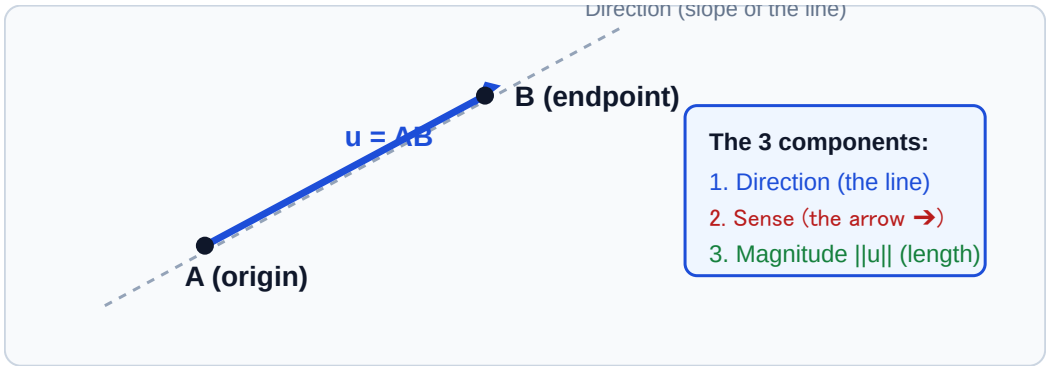

Figure 80: Characterization of a vector: direction, sense, and magnitude

Notation Explained. - We place a rightward arrow above the letter: $\vec{u}$, $\vec{v}$, $\vec{w}$. We read this aloud as: "vector u", "vector v". - When the vector represents displacement explicitly from a point A to a point B, we write $\overrightarrow{AB}$, read "vector AB". Here, A is the **initial point** (or origin/tail) of the representative, and B is its **terminal point** (or tip/head). - The **norm** is written with double vertical bars: $\| \vec{u} \|$ or $\| \overrightarrow{AB} \|$. We read "norm of u" or "magnitude of AB". We always have $\| \overrightarrow{AB} \| = AB$ (the standard geometric length). Double bars are used to avoid confusion with the absolute value $|x|$ of a real number.

> ⚠ **Pitfall Warning! Never confuse direction and sense!**
>
> In everyday English, people often say “walk in the direction of Paris.” In mathematics, this is imprecise:
> - The **direction** is the railway line connecting Paris to Lyon.
> - The **sense** (or heading) is traveling from Paris toward Lyon, or from Lyon toward Paris. Two cars driving past each other head-on along the same straight highway share the **same direction**, but have **opposite senses**.

2.3 The Zero Vector and Representatives

What happens if a traveler does not move a single millimeter? His starting point A and destination A coincide. The corresponding displacement is the **zero vector**, denoted $\vec{0}$ (a zero topped by an arrow):

$$\overrightarrow{AA} = \overrightarrow{BB} = \vec{0}.$$

The zero vector has a magnitude equal to zero: $\| \vec{0} \| = 0$. It possesses neither a distinct direction nor a specific sense (it is said to have every possible direction).

> 💡 **Did You Know? A vector is an infinity of arrows at once!**
>
> A vector $\vec{u}$ is not “glued” to one specific spot on paper. It is an equivalence class of displacements. If ten birds fly in tight formation, all heading South at $40 \ \ \mathrm{km/h}$, each bird traces its own arrow across the sky. All these arrows have the same direction, the same sense, and the same length. They are called **representatives** (or bound arrows) of one and the same free vector $\vec{u}$.

> 📐 **Fundamental Rule — Multiplying a Vector by a Scalar**
>
> - Bird 1 Bird 2 Bird 3
> - A C E
> - o o o
> - u = AB u = CD u = EF
> - B D F
> - AB = CD = EF = same free vector u!

2.4 Vector Equality and Parallelograms

When can two arrows $\overrightarrow{AB}$ and $\overrightarrow{CD}$ be declared identical as vectors?

> 📐 **Fundamental Theorem — Vector Equality.**
>
> Let A, B, C, D be four points in the plane.
>
> $$\overrightarrow{AB} = \overrightarrow{CD} \Leftrightarrow \text{quadrilateral } \ ABDC \ \text{ is a parallelogram (possibly degenerate/flat)}\ .$$

> ⚠ **Pitfall Warning! The vertex order matters.**
>
> Pay close attention to the order of letters: $\overrightarrow{AB} = \overrightarrow{CD}$ means that $AB\mathbf{DC}$ is a parallelogram, **not** $ABCD$! To travel from A to B, one performs the exact same translation as traveling from C to D. The segments $[AD]$ and $[BC]$ therefore share the same midpoint: their diagonals bisect each other.

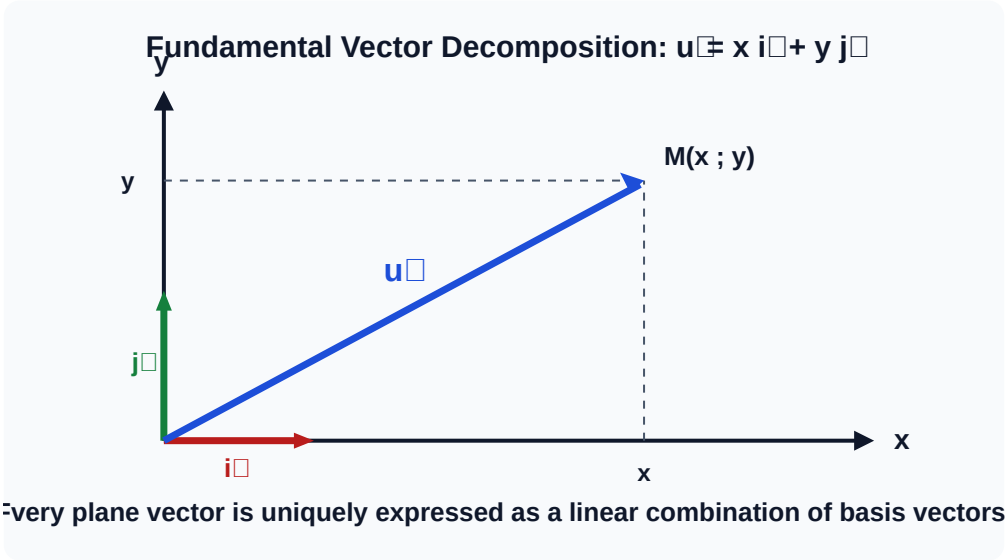

Figure 81: Fundamental vector decomposition: u = x i + y j

---

3. Operations on Vectors

3.1 Chasles' Relation: The Courier Relay

How do we add two successive displacements? Picture a royal missive carried by horse couriers:

- The first rider departs station A and gallops to station B: displacement $\overrightarrow{AB}$.
- At station B, he hands the leather satchel to a second rider who immediately rides toward fortress C: displacement $\overrightarrow{BC}$.

What is the net outcome for the royal message? It departed from A and has arrived at C.

Theorem (Chasles' Relation, 1837) —

For any points A, B, C in the plane:

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}.$$

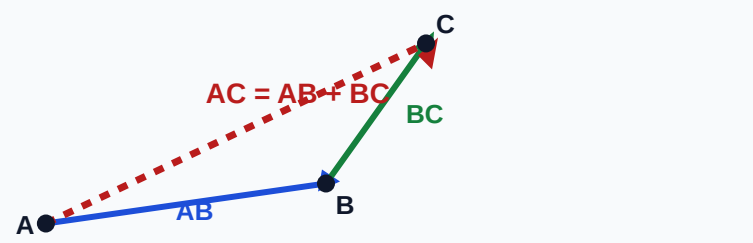

Figure 82: Vectors in the plane: Chasles' identity $\vec{u} + \vec{v}$

Historical Origin. This identity honors the French geometer **Michel Chasles** (1793–1880), professor at the Sorbonne and the École Polytechnique, who unified projective geometry and algebraic directed segment metrics.

⚠ Did You Know? The pitfall of adding lengths.

$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$ is an equality of **vectors**. It **never** implies $AB + BC = AC$ for standard lengths, unless the three points lie aligned in the precise order A, B, C! In any non-degenerate triangle, the triangle inequality (Chapter 9) dictates:

$$AC < AB + BC.$$

The direct straight path is strictly shorter than detouring through B. The norm of the sum is always less than or equal to the sum of the norms: $\| \vec{u} + \vec{v} \| \leq \| \vec{u} \| + \| \vec{v} \|$.

3.2 The Opposite Vector and Subtraction

If we traverse the path in reverse, from B back to A, the resulting vector has the same direction, the same magnitude, but the **opposite sense**. This is the **opposite vector**, denoted $-\overrightarrow{AB}$:

$$-\overrightarrow{AB} = \overrightarrow{BA}.$$

Applying Chasles' relation confirms:

$$\overrightarrow{AB} + \overrightarrow{BA} = \overrightarrow{AA} = \vec{0}.$$

To subtract a vector $\vec{v}$, we simply add its opposite:

$$\vec{u} - \vec{v} = \vec{u} + (-\vec{v}).$$

3.3 The Parallelogram Rule (Concurrent Forces)

What should we do if two vectors emanate from the same initial point A, like two tugboats pulling on the same container ship? Let $\vec{u} = \overrightarrow{AB}$ and $\vec{v} = \overrightarrow{AC}$. To apply Chasles' relation, we simply place at point B a representative of $\vec{v}$: we construct point D such that $\overrightarrow{BD} = \overrightarrow{AC}$. Then:

$$\overrightarrow{AB} + \overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BD} = \overrightarrow{AD}.$$

The quadrilateral $ABDC$ forms a parallelogram, and the sum vector $\overrightarrow{AD}$ is its main diagonal.

3.4 Multiplying a Vector by a Scalar

In physics and geometry, a real number is often called a **scalar** (from the Latin *scala*, ladder or scale), because it scales the magnitude of a quantity without changing its fundamental nature.

📐 Definition — Scalar Multiplication.

Let $\vec{u}$ be a vector and $k \in \mathbb{R}$ a real number. The vector $k\vec{u}$ is defined by: 1. If $k = 0$ or $\vec{u} = \vec{0}$, then $k\vec{u} = \vec{0}$. 2. If $k \neq 0$ and $\vec{u} \neq \vec{0}$:

- $k\vec{u}$ has the **same direction** as $\vec{u}$;
- $k\vec{u}$ has the **same sense** as $\vec{u}$ if $k > 0$, and the **opposite sense** if $k < 0$;
- its norm is $\| k\vec{u} \| = |k| \times \| \vec{u} \|$.

📐 Fundamental Rule — Multiplying a Vector by a Scalar

- 2u -u 0 u 2u 3u

📐 ***Algebraic Properties*** *— For all vectors $\vec{u}, \vec{v}$ and all scalars k, k':*

- $(k + k')\vec{u} = k\vec{u} + k'\vec{u}$ *(distributivity with respect to scalar addition);*
- $k(\vec{u} + \vec{v}) = k\vec{u} + k\vec{v}$ *(distributivity with respect to vector addition);*
- $k(k'\vec{u}) = (kk')\vec{u}$*;*
- $1\vec{u} = \vec{u}$ *and* $(-1)\vec{u} = -\vec{u}$.

3.5 Vector Characterization of the Midpoint

Let $[AB]$ be a line segment. The midpoint I of $[AB]$ can be characterized in three equivalent vector ways: 1. $\overrightarrow{AI} = \overrightarrow{IB}$ (same direction, same sense, same half-length); 2. $\overrightarrow{IA} + \overrightarrow{IB} = \vec{0}$ (I is the balance point between A and B); 3. For any arbitrary point M in the plane: $\overrightarrow{MA} + \overrightarrow{MB} = 2\overrightarrow{MI}$.

Proof — Midpoint Identity.

By Chasles' relation: $\overrightarrow{MA} = \overrightarrow{MI} + \overrightarrow{IA}$ and $\overrightarrow{MB} = \overrightarrow{MI} + \overrightarrow{IB}$. Summing term by term:

$$\overrightarrow{MA} + \overrightarrow{MB} = (\overrightarrow{MI} + \overrightarrow{IA}) + (\overrightarrow{MI} + \overrightarrow{IB}) = 2\overrightarrow{MI} + (\overrightarrow{IA} + \overrightarrow{IB}).$$

Since I is the midpoint of $[AB]$, we know $\overrightarrow{IA} + \overrightarrow{IB} = \vec{0}$. Therefore:

$$\overrightarrow{MA} + \overrightarrow{MB} = 2\overrightarrow{MI}. \quad \blacksquare$$

---

4. Vectors in a Cartesian Coordinate Frame

4.1 The Basis $(\vec{\imath}, \vec{\jmath})$ and Vector Coordinates

Equip the plane with an origin O and two non-collinear vectors $\vec{\imath}$ and $\vec{\jmath}$. The triad $(O\,;\vec{\imath},\vec{\jmath})$ constitutes a **coordinate frame** for the plane. If the directions of $\vec{\imath}$ and $\vec{\jmath}$ are perpendicular and $\| \vec{\imath} \| = \| \vec{\jmath} \| = 1$, the frame is called **orthonormal** (or Cartesian rectangular).

Theorem and Definition — Coordinates of a Vector.

For every vector $\vec{u}$ in the plane, there exists a **unique** pair of real numbers $(x\,;y)$ such that:

$$\vec{u} = x\vec{\imath} + y\vec{\jmath}.$$

We write $\vec{u}\begin{pmatrix} x \\ y \end{pmatrix}$ or $\vec{u}(x\,;y)$. x is the **abscissa** (x-component) and y is the **ordinate** (y-component).

4.2 Fundamental Coordinate Formulas

Let $A(x_A\,;y_A)$ and $B(x_B\,;y_B)$ be two points, and let $\vec{u}\begin{pmatrix} x \\ y \end{pmatrix}$ and $\vec{v}\begin{pmatrix} x' \\ y' \end{pmatrix}$ be two vectors.

Key Formulas to Master — *1.* ***Vector connecting two points ("Arrival minus Departure"):***

$$\overrightarrow{AB}\begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix}$$

2. ***Vector addition:***

$$\vec{u} + \vec{v}\begin{pmatrix} x + x' \\ y + y' \end{pmatrix}$$

3. ***Scalar multiplication:***

$$k\vec{u}\begin{pmatrix} kx \\ ky \end{pmatrix}$$

4. ***Coordinates of the midpoint I of $[AB]$:***

$$x_I = \frac{x_A + x_B}{2}, \quad y_I = \frac{y_A + y_B}{2}$$

5. ***Norm of a vector (strictly in an orthonormal frame!):***

$$\| \vec{u} \| = \sqrt{x^2 + y^2}$$

and distance between two points:

$$AB = \| \overrightarrow{AB} \| = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}.$$

Proof — Coordinates of $\overrightarrow{AB}$.

By definition of point coordinates, $\overrightarrow{OA} = x_A\vec{\imath} + y_A\vec{\jmath}$ and $\overrightarrow{OB} = x_B\vec{\imath} + y_B\vec{\jmath}$. Using Chasles' relation:

$$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = -\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OB} - \overrightarrow{OA}.$$

Substituting:

$$\overrightarrow{AB} = (x_B\vec{\imath} + y_B\vec{\jmath}) - (x_A\vec{\imath} + y_A\vec{\jmath}) = (x_B - x_A)\vec{\imath} + (y_B - y_A)\vec{\jmath}.$$

The coordinates of $\overrightarrow{AB}$ are therefore indeed $(x_B - x_A\,;y_B - y_A)$. ■

---

5. Collinearity and Determinants: The Art of Aiming True

5.1 Definition of Collinearity

Two arrows may have different lengths and opposite orientations: if they lie on parallel lines, they have the **same direction**. In mathematics, they are termed **collinear** (from the Latin *cum*, together, and *linea*, line: "on the same line").

Definition — Collinear Vectors.

Two vectors $\vec{u}$ and $\vec{v}$ are said to be **collinear** if one is the scalar multiple of the other:

$$\text{there exists } \; k \in \mathbb{R} \; \text{ such that } \; \vec{u} = k\vec{v} \quad \text{or} \quad \vec{v} = k\vec{u}.$$

By convention, the zero vector $\vec{0}$ is collinear to **all** vectors in the plane (since $\vec{0} = 0 \times \vec{u}$).

5.2 The Determinant of Two Vectors

How can one instantly test whether two vectors $\vec{u}\begin{pmatrix} x \\ y \end{pmatrix}$ and $\vec{v}\begin{pmatrix} x' \\ y' \end{pmatrix}$ are collinear, without guessing k?

Definition and Theorem — Determinant in the Plane.

The **determinant** of vectors $\vec{u}$ and $\vec{v}$ in an orthonormal basis is the real number denoted $\det(\vec{u}, \vec{v})$, calculated by the cross-product rule:

$$\det(\vec{u}, \vec{v}) = \begin{vmatrix} x & x' \\ y & y' \end{vmatrix} = xy' - yx'.$$

Fundamental Collinearity Criterion:

$$\vec{u} \;\text{ and }\; \vec{v} \;\text{ are collinear} \Leftrightarrow \det(\vec{u}, \vec{v}) = 0 \Leftrightarrow xy' - yx' = 0.$$

Proof — Collinearity Condition.

- **Forward direction ($\Longrightarrow$):** Assume $\vec{u}$ and $\vec{v}$ are collinear. If $\vec{v} = \vec{0}$, then $x' = y' = 0$, so $xy' - yx' = x(0) - y(0) = 0$. If $\vec{v} \neq \vec{0}$, there exists $k \in \mathbb{R}$ such that $\vec{u} = k\vec{v}$, which means $x = kx'$ and $y = ky'$. Calculating the cross difference:
$$xy' - yx' = (kx')y' - (ky')x' = kx'y' - ky'x' = 0.$$
- **Reverse direction ($\Longleftarrow$):** Assume $xy' - yx' = 0$. If $\vec{v} = \vec{0}$, then $\vec{u}$ and $\vec{v}$ are collinear by definition. Assume $\vec{v} \neq \vec{0}$. At least one of its components is non-zero. Suppose, without loss of generality, that $x' \neq 0$. Let $k = \frac{x}{x'}$, so that $x = kx'$. The identity $xy' - yx' = 0$ then becomes:
$$(kx')y' - yx' = 0 \Leftrightarrow x'(ky' - y) = 0.$$
Since $x' \neq 0$, it follows that $ky' - y = 0$, hence $y = ky'$. We therefore have both $x = kx'$ and $y = ky'$, which is concisely written $\vec{u} = k\vec{v}$. The vectors are collinear. ■

> 💡 ***Geometric Interpretation: Area of the Parallelogram.*** *The absolute value of the determinant* $|\det(\vec{u},\vec{v})|$ *is exactly equal to the* ***area of the parallelogram*** *spanned by the two vectors* $\vec{u}$ *and* $\vec{v}$*! If the vectors are collinear, the parallelogram collapses flat: its area is zero, which yields* $\det(\vec{u},\vec{v})=0$*.*

5.3 Two Major Geometric Applications

Thanks to the determinant, geometric problems of alignment and parallelism transform into direct algebraic tests:

1. **Parallelism of lines:** Two lines (AB) and (CD) are parallel if and only if their direction vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are collinear:
$$(AB) \parallel (CD) \Leftrightarrow \det(\overrightarrow{AB},\overrightarrow{CD})=0.$$
2. **Collinearity of three points:** Three distinct points A, B, C are collinear if and only if the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ share the same direction:
$$A,B,C \text{ are collinear} \Leftrightarrow \overrightarrow{AB} \text{ and } \overrightarrow{AC} \text{ are collinear} \Leftrightarrow \det(\overrightarrow{AB},\overrightarrow{AC})=0.$$

---

6. Detailed Worked Examples

Worked Example 1 — Simplifying a Vector Expression Using Chasles.

Simplify the following vector expression as much as possible:
$$\overrightarrow{W}=\overrightarrow{AB}-\overrightarrow{CB}+\overrightarrow{CD}-\overrightarrow{AD}.$$
Step-by-Step Method: 1. Replace subtractions by the addition of the opposite:
$$-\overrightarrow{CB}=\overrightarrow{BC} \quad \text{and} \quad -\overrightarrow{AD}=\overrightarrow{DA}.$$
2. Rewrite the expression:
$$\overrightarrow{W}=\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD}+\overrightarrow{DA}.$$
3. Chain consecutive terms using Chasles' relation:
$$\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC},$$
$$\overrightarrow{AC}+\overrightarrow{CD}=\overrightarrow{AD},$$
$$\overrightarrow{AD}+\overrightarrow{DA}=\overrightarrow{AA}=\vec{0}.$$
Conclusion: $\overrightarrow{W}=\vec{0}$.

Worked Example 2 — Calculating Coordinates and Distances.

In an orthonormal coordinate system $(O\,;\vec{\imath},\vec{\jmath})$, consider $A(-3\,;2)$, $B(1\,;5)$, and $C(4\,;1)$. 1. Determine the coordinates of $\overrightarrow{AB}$ and the exact length AB. 2. Determine the coordinates of point D such that $ABCD$ is a parallelogram.

Solution: 1. Coordinates of $\overrightarrow{AB}$:

$$\overrightarrow{AB}\begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix} = \begin{pmatrix} 1-(-3) \\ 5-2 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}.$$

Norm of $\overrightarrow{AB}$ (distance AB):

$$AB = \sqrt{4^2+3^2} = \sqrt{16+9} = \sqrt{25} = 5.$$

2. Caution: $ABCD$ is a parallelogram if and only if $\overrightarrow{AB} = \overrightarrow{DC}$ (or $\overrightarrow{AD} = \overrightarrow{BC}$). Let $(x_D\,;y_D)$ denote the coordinates of D.

$$\overrightarrow{DC}\begin{pmatrix} x_C - x_D \\ y_C - y_D \end{pmatrix} = \begin{pmatrix} 4-x_D \\ 1-y_D \end{pmatrix}.$$

The equality $\overrightarrow{AB} = \overrightarrow{DC}$ translates into the system:

$$\{4 - x_D = 4 \Longrightarrow x_D = 0$$
$$1 - y_D = 3 \Longrightarrow y_D = 1 - 3 = -2$$

The desired point is therefore $D(0\,;-2)$. *(Verification via midpoints: midpoint of $[AC]$ is $\left(\frac{-3+4}{2}\,;\frac{2+1}{2}\right) = (0.5\,;1.5)$; midpoint of $[BD]$ is $\left(\frac{1+0}{2}\,;\frac{5+(-2)}{2}\right) = (0.5\,;1.5)$. The diagonals indeed bisect each other!)*

Worked Example 3 — Testing Point Collinearity with the Determinant.

Consider points $E(-2\,;-3)$, $F(2\,;3)$, and $G(6\,;9)$. Are they collinear?

Solution: Compute coordinates of vectors $\overrightarrow{EF}$ and $\overrightarrow{EG}$:

$$\overrightarrow{EF}\begin{pmatrix} 2-(-2) \\ 3-(-3) \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}, \quad \overrightarrow{EG}\begin{pmatrix} 6-(-2) \\ 9-(-3) \end{pmatrix} = \begin{pmatrix} 8 \\ 12 \end{pmatrix}.$$

Compute the determinant:

$$\det(\overrightarrow{EF},\overrightarrow{EG}) = \begin{vmatrix} 4 & 8 \\ 6 & 12 \end{vmatrix} = 4\times 12 - 6\times 8 = 48 - 48 = 0.$$

The determinant is zero, so vectors $\overrightarrow{EF}$ and $\overrightarrow{EG}$ are collinear. Since they share the common point E, points E, F, and G are collinear.

Worked Example 4 — Finding a Parameter for Parallelism.

Let $\vec{u}\begin{pmatrix} 2m-1 \\ 3 \end{pmatrix}$ and $\vec{v}\begin{pmatrix} 5 \\ m+2 \end{pmatrix}$ where $m \in \mathbb{R}$ is an unknown scalar. For what values of m are these vectors collinear?

Solution: $\vec{u}$ and $\vec{v}$ are collinear if and only if $\det(\vec{u},\vec{v}) = 0$:

$$(2m-1)(m+2) - 3\times 5 = 0.$$

Expanding:

$$2m^2 + 4m - m - 2 - 15 = 0 \Leftrightarrow 2m^2 + 3m - 17 = 0.$$

This is a quadratic equation. Its discriminant is:

$$\Delta = 3^2 - 4\times 2\times(-17) = 9 + 136 = 145 > 0.$$

There are two distinct real solutions:

$$m_1 = \frac{-3-\sqrt{145}}{4} \quad \text{and} \quad m_2 = \frac{-3+\sqrt{145}}{4}.$$

Worked Example 5 — Pure Geometric Vector Decomposition.

Let ABC be a triangle. Points P and Q are defined by $\overrightarrow{AP} = \frac{1}{3}\overrightarrow{AB}$ and $\overrightarrow{AQ} = \frac{1}{3}\overrightarrow{AC}$. Prove without using a coordinate frame that $(PQ) \parallel (BC)$.

Solution: Express vector $\overrightarrow{PQ}$ in terms of $\overrightarrow{AB}$ and $\overrightarrow{AC}$ using Chasles' relation:

$$\overrightarrow{PQ} = \overrightarrow{PA} + \overrightarrow{AQ} = -\overrightarrow{AP} + \overrightarrow{AQ} = -\frac{1}{3}\overrightarrow{AB} + \frac{1}{3}\overrightarrow{AC} = \frac{1}{3}(-\overrightarrow{AB} + \overrightarrow{AC}) = \frac{1}{3}(\overrightarrow{BA} + \overrightarrow{AC}).$$

By Chasles' relation, $\overrightarrow{BA} + \overrightarrow{AC} = \overrightarrow{BC}$. Hence:

$$\overrightarrow{PQ} = \frac{1}{3}\overrightarrow{BC}.$$

Vector $\overrightarrow{PQ}$ is the product of vector $\overrightarrow{BC}$ by scalar $\frac{1}{3}$. Vectors $\overrightarrow{PQ}$ and $\overrightarrow{BC}$ are therefore collinear, which proves that lines (PQ) and (BC) are strictly parallel. *(Notice: we have just reproved Thales' theorem using vectors!)*

Worked Example 6 — The Centroid of a Triangle.

Let ABC be a triangle. Let G be the point defined by the relation:

$$\overrightarrow{GA} + \overrightarrow{GB} + \overrightarrow{GC} = \vec{0}.$$

1. Let A' be the midpoint of $[BC]$. Express $\overrightarrow{GB} + \overrightarrow{GC}$ in terms of $\overrightarrow{GA'}$. 2. Deduce that G lies on median (AA') and specify its location on the segment.

Solution: 1. From the midpoint identity proven in §3.5, since A' is the midpoint of $[BC]$, we have:

$$\overrightarrow{GB} + \overrightarrow{GC} = 2\overrightarrow{GA'}.$$

2. Substitute into the initial relation:

$$\overrightarrow{GA} + 2\overrightarrow{GA'} = \vec{0} \Leftrightarrow \overrightarrow{GA} = -2\overrightarrow{GA'} \Leftrightarrow -\overrightarrow{AG} = -2\overrightarrow{GA'} \Leftrightarrow \overrightarrow{AG} = 2\overrightarrow{GA'}.$$

Insert G into the right-hand side using Chasles: $\overrightarrow{AA'} = \overrightarrow{AG} + \overrightarrow{GA'} = \overrightarrow{AG} + \frac{1}{2}\overrightarrow{AG} = \frac{3}{2}\overrightarrow{AG}$. We deduce:

$$\overrightarrow{AG} = \frac{2}{3}\overrightarrow{AA'}.$$

Vectors $\overrightarrow{AG}$ and $\overrightarrow{AA'}$ are collinear: point G is situated on the median line (AA'), two-thirds of the way from vertex A. G is the **centroid** (isobarycenter) of triangle ABC.

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7. In Computer Science: Testing Collinearity in Python

Here is a Python script that takes the coordinates of two vectors and tests their collinearity using the determinant:

```python
def determinant(ux, uy, vx, vy):
    """Calculates the determinant of two vectors u(ux, uy) and v(vx, vy)."""
    return ux * vy - uy * vx

def are_collinear(ux, uy, vx, vy, tolerance=1e-9):
    """Checks whether two vectors are collinear up to a given precision."""
    det = determinant(ux, uy, vx, vy)
    return abs(det) < tolerance

# Test with u(4, 6) and v(6, 9)
print("u and v collinear?", are_collinear(4, 6, 6, 9))  # Prints True

# Test with u(3, -1) and v(2, 5)
print("u and v collinear?", are_collinear(3, -1, 2, 5))  # Prints False
```

Key Takeaways

1. **Vector:** a geometric entity characterized by a **direction** (line of action), a **sense** (arrowhead), and a **magnitude/norm** (length $\| \vec{u} \|$).
2. **Vector Equality:** $\overrightarrow{AB} = \overrightarrow{CD} \Leftrightarrow ABDC$ is a parallelogram.
3. **Chasles' Relation:** $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$. To subtract: $\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$ where $-\overrightarrow{AB} = \overrightarrow{BA}$.
4. **Midpoint I of $[AB]$:** $\overrightarrow{AI} = \overrightarrow{IB} \Leftrightarrow \overrightarrow{IA} + \overrightarrow{IB} = \vec{0} \Leftrightarrow \overrightarrow{MA} + \overrightarrow{MB} = 2\overrightarrow{MI}$.
5. **Coordinates:**
   - $\overrightarrow{AB}\begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix}$
   - Midpoint: $I\left(\frac{x_A + x_B}{2} ; \frac{y_A + y_B}{2}\right)$
   - Norm: $\| \vec{u} \| = \sqrt{x^2 + y^2}$ and $AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$ (in an orthonormal frame).
6. **Collinearity and Determinant:**
   - $\vec{u}$ and $\vec{v}$ are collinear $\Leftrightarrow \vec{u} = k\vec{v}$ (or one is zero) $\Leftrightarrow \det(\vec{u}, \vec{v}) = xy' - yx' = 0$.
   - Parallel lines: $(AB) \parallel (CD) \Leftrightarrow \det(\overrightarrow{AB}, \overrightarrow{CD}) = 0$.
   - Collinear points: A, B, C collinear $\Leftrightarrow \det(\overrightarrow{AB}, \overrightarrow{AC}) = 0$.

Exercises

Direct Application Exercises ()

1. Let $MNOP$ be a parallelogram. Name three pairs of equal vectors.
2. Simplify the expression $\vec{u} = \overrightarrow{RS} + \overrightarrow{ST} + \overrightarrow{TR}$.
3. Simplify the expression $\vec{v} = \overrightarrow{AB} - \overrightarrow{AC} + \overrightarrow{BC}$.
4. Given vector $\vec{u}\begin{pmatrix} -3 \\ 4 \end{pmatrix}$, compute its norm $\| \vec{u} \|$.
5. Given points $A(2\,;-5)$ and $B(-1\,;7)$, calculate the coordinates of vector $\overrightarrow{AB}$.
6. Using the same points A and B, calculate the coordinates of the midpoint K of $[AB]$.
7. Calculate the exact distance AB.
8. Are the vectors $\vec{u}\begin{pmatrix} 6 \\ -4 \end{pmatrix}$ and $\vec{v}\begin{pmatrix} -9 \\ 6 \end{pmatrix}$ collinear?
9. Are the vectors $\vec{w}\begin{pmatrix} 5 \\ 3 \end{pmatrix}$ and $\vec{z}\begin{pmatrix} 7 \\ 4 \end{pmatrix}$ collinear?
10. Find the real number y such that $\vec{a}\begin{pmatrix} 4 \\ -6 \end{pmatrix}$ and $\vec{b}\begin{pmatrix} 10 \\ y \end{pmatrix}$ are collinear.

Practice and Reasoning Exercises ()

11. Let $A(1\,;2)$, $B(4\,;6)$, and $C(-2\,;-2)$. Are points A, B, C collinear?
12. Consider four points $K(-1\,;1)$, $L(3\,;3)$, $M(5\,;1)$, and $N(1\,;-1)$.
    a) Calculate coordinates of $\overrightarrow{KL}$ and $\overrightarrow{NM}$.
    b) What is the precise nature of quadrilateral $KLMN$?
13. Using points K, L, M, N from Exercise 12, calculate lengths KL and LM. Is the quadrilateral a rhombus? A rectangle?
14. Let ABC be a triangle. Construct (or express) points M and N defined by:
$$\overrightarrow{AM} = \overrightarrow{AB} + 2\overrightarrow{AC} \quad \text{and} \quad \overrightarrow{AN} = 2\overrightarrow{AB} + \overrightarrow{AC}.$$
Express vector $\overrightarrow{MN}$ in terms of $\overrightarrow{AB}$ and $\overrightarrow{AC}$.
15. Consider points $A(-2\,;3)$, $B(4\,;1)$, and $C(2\,;5)$. Determine the coordinates of point E such that $\overrightarrow{AE} = 2\overrightarrow{AB} - 3\overrightarrow{BC}$.
16. Let ABC be any arbitrary triangle. Let D be the reflection of A across B, and E the reflection of B across C. Express $\overrightarrow{DE}$ in terms of $\overrightarrow{AB}$ and $\overrightarrow{BC}$.
17. For what value of the real number x are points $P(x\,;3)$, $Q(2\,;-1)$, and $R(6\,;7)$ collinear?
18. Given $\vec{u}\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\vec{v}\begin{pmatrix} 1 \\ 3 \end{pmatrix}$, determine scalars α and β such that vector $\vec{w}\begin{pmatrix} 7 \\ 0 \end{pmatrix}$ can be written $\vec{w} = \alpha\vec{u} + \beta\vec{v}$.

Advanced Problems and Challenges ()

19. **The Vector Midpoint Theorem.** Let ABC be a triangle. Let I be the midpoint of $[AB]$ and J the midpoint of $[AC]$. Prove using vector algebra that $\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{BC}$. Deduce the relative position of lines (IJ) and (BC) as well as the ratio of their lengths.
20. **Varignon's Quadrilateral.** Let $ABCD$ be an arbitrary quadrilateral. Let I, J, K, L be the midpoints of sides $[AB], [BC], [CD], [DA]$ respectively.
    a) Express $\overrightarrow{IJ}$ and $\overrightarrow{LK}$ in terms of the vertex vectors.
    b) Deduce that $IJKL$ is always a parallelogram.
21. **Concurrency of the Three Medians.** Let ABC be a triangle. Let A', B', C' be the midpoints of sides $[BC], [CA], [AB]$ respectively. Let G be the point satisfying $\overrightarrow{GA} + \overrightarrow{GB} + \overrightarrow{GC} = \vec{0}$.
    a) Prove that for any arbitrary point M in the plane: $\overrightarrow{MA} + \overrightarrow{MB} + \overrightarrow{MC} = 3\overrightarrow{MG}$.
    b) Deduce that G belongs to each of the three medians (AA'), (BB'), and (CC'), and specify the point of concurrency.
22. **Aircraft Trajectory and Wind Drift.** An airplane flies on a heading due North with an airspeed of 600 km/h relative to the air mass. A gale-force wind blows from the Northwest toward the Southeast (at $45°$ to the cardinal axes) at a speed of $100\sqrt{2}$ km/h. We choose an orthonormal coordinate system where the positive x-axis points East and the positive y-axis points North.
    a) Determine the components of the aircraft airspeed vector $\vec{v}_a$.
    b) Determine the components of the wind vector $\vec{v}_w$.
    c) Calculate the ground velocity vector $\vec{v}_s = \vec{v}_a + \vec{v}_w$.
    d) Deduce the actual ground speed of the aircraft and the angular drift of its track.

Solutions

1. In parallelogram $MNOP$: $\overrightarrow{MN} = \overrightarrow{PO}$, $\overrightarrow{NO} = \overrightarrow{MP}$, $\overrightarrow{NM} = \overrightarrow{OP}$.
2. $\overrightarrow{RS} + \overrightarrow{ST} + \overrightarrow{TR} = \overrightarrow{RT} + \overrightarrow{TR} = \overrightarrow{RR} = \vec{0}$.
3. $\vec{v} = \overrightarrow{AB} + \overrightarrow{CA} + \overrightarrow{BC} = (\overrightarrow{CA} + \overrightarrow{AB}) + \overrightarrow{BC} = \overrightarrow{CB} + \overrightarrow{BC} = \vec{0}$.
4. $\| \vec{u} \| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.
5. $\overrightarrow{AB}\begin{pmatrix} -1-2 \\ 7-(-5) \end{pmatrix} = \begin{pmatrix} -3 \\ 12 \end{pmatrix}$.
6. $x_K = \frac{2+(-1)}{2} = \frac{1}{2} = 0.5$ and $y_K = \frac{-5+7}{2} = \frac{2}{2} = 1$. Thus $K(0.5\,;1)$.
7. $AB = \sqrt{(-3)^2 + 12^2} = \sqrt{9 + 144} = \sqrt{153} = 3\sqrt{17}$.
8. $\det(\vec{u}, \vec{v}) = 6 \times 6 - (-4) \times (-9) = 36 - 36 = 0$. Yes, they are collinear (we also observe $\vec{v} = -1.5\vec{u}$).
9. $\det(\vec{w}, \vec{z}) = 5 \times 4 - 3 \times 7 = 20 - 21 = -1 \neq 0$. No, they are not collinear.
10. $\det(\vec{a}, \vec{b}) = 4y - (-6) \times 10 = 4y + 60 = 0 \Leftrightarrow 4y = -60 \Leftrightarrow y = -15$.
11. $\overrightarrow{AB}\begin{pmatrix} 4-1 \\ 6-2 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\overrightarrow{AC}\begin{pmatrix} -2-1 \\ -2-2 \end{pmatrix} = \begin{pmatrix} -3 \\ -4 \end{pmatrix}$. We immediately observe $\overrightarrow{AC} = -1\overrightarrow{AB}$ (or $\det = 3(-4) - 4(-3) = -12 + 12 = 0$). The vectors are collinear and share point A, so A, B, C are collinear.
12. a) $\overrightarrow{KL}\begin{pmatrix} 3-(-1) \\ 3-1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ and $\overrightarrow{NM}\begin{pmatrix} 5-1 \\ 1-(-1) \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$.
    b) Since $\overrightarrow{KL} = \overrightarrow{NM}$, quadrilateral $KLMN$ is a parallelogram.
13. Length $KL = \sqrt{4^2 + 2^2} = \sqrt{16 + 4} = \sqrt{20} = 2\sqrt{5}$. Calculate $\overrightarrow{LM}\begin{pmatrix} 5-3 \\ 1-3 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$. Length $LM = \sqrt{2^2 + (-2)^2} = \sqrt{4 + 4} = \sqrt{8} = 2\sqrt{2}$. Since $KL \neq LM$, consecutive sides are not equal: it is not a rhombus. Check whether sides are perpendicular: KM^2 with $\overrightarrow{KM}\begin{pmatrix} 5-(-1) \\ 1-1 \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \Longrightarrow KM^2 = 36$. Yet $KL^2 + LM^2 = 20 + 8 = 28 \neq 36$. Triangle KLM is not right-angled, so $KLMN$ is not a rectangle. It is a general parallelogram.
14. By Chasles: $\overrightarrow{MN} = \overrightarrow{MA} + \overrightarrow{AN} = -\overrightarrow{AM} + \overrightarrow{AN}$. Substituting: $\overrightarrow{MN} = -(\overrightarrow{AB} + 2\overrightarrow{AC}) + (2\overrightarrow{AB} + \overrightarrow{AC}) = -\overrightarrow{AB} - 2\overrightarrow{AC} + 2\overrightarrow{AB} + \overrightarrow{AC} = \overrightarrow{AB} - \overrightarrow{AC} = \overrightarrow{CB}$.
15. $\overrightarrow{AB}\begin{pmatrix} 4-(-2) \\ 1-3 \end{pmatrix} = \begin{pmatrix} 6 \\ -2 \end{pmatrix}$ and $\overrightarrow{BC}\begin{pmatrix} 2-4 \\ 5-1 \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$. Then $\overrightarrow{AE} = 2\overrightarrow{AB} - 3\overrightarrow{BC} = 2\begin{pmatrix} 6 \\ -2 \end{pmatrix} - 3\begin{pmatrix} -2 \\ 4 \end{pmatrix} = \begin{pmatrix} 12+6 \\ -4-12 \end{pmatrix} = \begin{pmatrix} 18 \\ -16 \end{pmatrix}$. Since A has coordinates $(-2\,;3)$: $x_E = x_A + 18 = -2 + 18 = 16$ and $y_E = y_A - 16 = 3 - 16 = -13$. Thus $E(16\,;-13)$.

16. D reflection of A across $B \Longrightarrow \overrightarrow{BD} = \overrightarrow{AB}$. E reflection of B across $C \Longrightarrow \overrightarrow{BE} = 2\overrightarrow{BC}$. By Chasles: $\overrightarrow{DE} = \overrightarrow{DB} + \overrightarrow{BE} = -\overrightarrow{BD} + \overrightarrow{BE} = -\overrightarrow{AB} + 2\overrightarrow{BC}$.
17. $\overrightarrow{QR}\begin{pmatrix}6-2\\7-(-1)\end{pmatrix} = \begin{pmatrix}4\\8\end{pmatrix}$. $\overrightarrow{QP}\begin{pmatrix}x-2\\3-(-1)\end{pmatrix} = \begin{pmatrix}x-2\\4\end{pmatrix}$. P, Q, R collinear $\Leftrightarrow \det(\overrightarrow{QR}, \overrightarrow{QP}) = 0 \Leftrightarrow 4 \times 4 - 8(x-2) = 0 \Leftrightarrow 16 - 8x + 16 = 0 \Leftrightarrow 8x = 32 \Leftrightarrow x = 4$.
18. Equality $\vec{w} = \alpha\vec{u} + \beta\vec{v}$ gives the linear system:

$$\begin{cases} 2\alpha + \beta = 7 \\ -\alpha + 3\beta = 0 \end{cases}$$

From the second equation, $\alpha = 3\beta$. Substituting into the first: $2(3\beta) + \beta = 7 \Leftrightarrow 7\beta = 7 \Leftrightarrow \beta = 1$. Hence $\alpha = 3 \times 1 = 3$. Solution: $\alpha = 3, \beta = 1$.
19. I midpoint of $[AB] \Longrightarrow \overrightarrow{AI} = \frac{1}{2}\overrightarrow{AB}$, so $\overrightarrow{IA} = -\frac{1}{2}\overrightarrow{AB}$. J midpoint of $[AC] \Longrightarrow \overrightarrow{AJ} = \frac{1}{2}\overrightarrow{AC}$. By Chasles: $\overrightarrow{IJ} = \overrightarrow{IA} + \overrightarrow{AJ} = -\frac{1}{2}\overrightarrow{AB} + \frac{1}{2}\overrightarrow{AC} = \frac{1}{2}(\overrightarrow{BA} + \overrightarrow{AC}) = \frac{1}{2}\overrightarrow{BC}$. Since $\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{BC}$, the vectors are collinear, so lines (IJ) and (BC) are parallel. Taking norms yields $IJ = \frac{1}{2}BC$.
20. a) In triangle ABC, by the vector midpoint theorem (Ex. 19), $\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{AC}$. In triangle ADC, L is the midpoint of $[AD]$ and K the midpoint of $[CD]$. Thus $\overrightarrow{LK} = \frac{1}{2}\overrightarrow{AC}$.
    b) We find $\overrightarrow{IJ} = \overrightarrow{LK}$. A quadrilateral whose opposite sides form equal vectors is a parallelogram. Therefore $IJKL$ is always a parallelogram.
21. a) Let M be an arbitrary point. By Chasles: $\overrightarrow{MA} + \overrightarrow{MB} + \overrightarrow{MC} = (\overrightarrow{MG} + \overrightarrow{GA}) + (\overrightarrow{MG} + \overrightarrow{GB}) + (\overrightarrow{MG} + \overrightarrow{GC}) = 3\overrightarrow{MG} + (\overrightarrow{GA} + \overrightarrow{GB} + \overrightarrow{GC}) = 3\overrightarrow{MG} + \vec{0} = 3\overrightarrow{MG}$.
    b) Choose $M = A'$ (midpoint of $[BC]$). Then $\overrightarrow{A'A} + \overrightarrow{A'B} + \overrightarrow{A'C} = 3\overrightarrow{A'G}$. Since A' is the midpoint of $[BC]$, $\overrightarrow{A'B} + \overrightarrow{A'C} = \vec{0}$. The equality becomes $\overrightarrow{A'A} = 3\overrightarrow{A'G}$, or $\overrightarrow{AG} = \frac{2}{3}\overrightarrow{AA'}$. Point G therefore lies on median (AA'), two-thirds from vertex A. Choosing $M = B'$ then $M = C'$ similarly yields $\overrightarrow{BG} = \frac{2}{3}\overrightarrow{BB'}$ and $\overrightarrow{CG} = \frac{2}{3}\overrightarrow{CC'}$. Point G is therefore the unique point of concurrency of the three medians.
22. a) The aircraft flies due North ($x = 0$) at 600 km/h: $\vec{v}_a\begin{pmatrix}0\\600\end{pmatrix}$.
    b) The wind blows from Northwest to Southeast (angle $-45°$, giving positive x Eastward, negative y Southward). Its speed is $100\sqrt{2}$. Since $\cos(45°) = \sin(45°) = \frac{\sqrt{2}}{2}$: $v_{wx} = 100\sqrt{2} \times \frac{\sqrt{2}}{2} = 100$, and $v_{wy} = -100$. Thus $\vec{v}_w\begin{pmatrix}100\\-100\end{pmatrix}$.
    c) Ground velocity: $\vec{v}_s = \vec{v}_a + \vec{v}_w = \begin{pmatrix}0+100\\600-100\end{pmatrix} = \begin{pmatrix}100\\500\end{pmatrix}$.
    d) Actual ground speed: $\| \vec{v}_s \| = \sqrt{100^2 + 500^2} = \sqrt{10000 + 250000} = \sqrt{260000} \approx 509.9$ km/h. Angular drift θ toward the East: $\tan\theta = \frac{100}{500} = 0.2 \Longrightarrow \theta \approx 11.3°$. The aircraft drifts more than $11°$ to the right and loses nearly 90 km/h over ground!

The Question That Opens the Future

We have learned to slide free arrows throughout the plane and to detect whether two arrows point along the same line using the determinant. But what happens when we drive a stake into the ground at point A and let arrow $\vec{u}$ point out an infinite trajectory? We obtain a **straight line**.

How can we characterize the set of all points aligned along this line of sight? How do we link the algebraic equation of a line to its direction vectors, and how do we calculate the exact location where two trajectories intersect? That is what we will discover in **The Line of Sight**.

CHAPTER 33: The Line of Sight

> 📐 **"A straight line is not merely an ink trace drawn with a ruler: it is an infinite laser beam traversing the universe, a line of sight swept by a watchman's gaze. To confine this infinity of points into an algebraic formula, two numbers suffice: an anchor along the vertical axis and a climbing slope. Then, the crossing of two paths becomes the unfolding of an equation."**

What You Are About to Discover

- How a line is born from an anchor point and a **direction vector** that sets its heading.
- The general **Cartesian equation** $ax + by + c = 0$, and why the vector $\vec{u}\begin{pmatrix}-b\\a\end{pmatrix}$ serves as its compass needle.
- The **slope-intercept form** (or reduced equation) $y = mx + p$: the physical meaning of the **slope** m (gradient) and the **y-intercept** p.
- The special case of vertical lines with equation $x = k$.
- The three possible fates of two meeting lines: intersecting at a unique point, strictly parallel, or coincident.
- Solving **systems of two linear equations in two variables** algebraically and graphically via substitution and linear combination (elimination).

---

1. The Problem: The Watchtower and the Ship in the Fog

A solitary watchtower stands atop a craggy cliff overlooking the ocean. Night has fallen, thick fog shrouds the bay, and two ships navigate blindly.

The lighthouse keeper monitors a Cartesian radar screen:

- The merchant vessel *L'Espérance* is presently located at coordinates $A(2\,;3)$ and maintains a steady heading guided by velocity vector $\vec{u}\begin{pmatrix}2\\1\end{pmatrix}$.
- A rescue frigate *La Vigilante* is reported at point $B(-1\,;8)$, steaming in a straight path along vector $\vec{v}\begin{pmatrix}1\\-1\end{pmatrix}$.

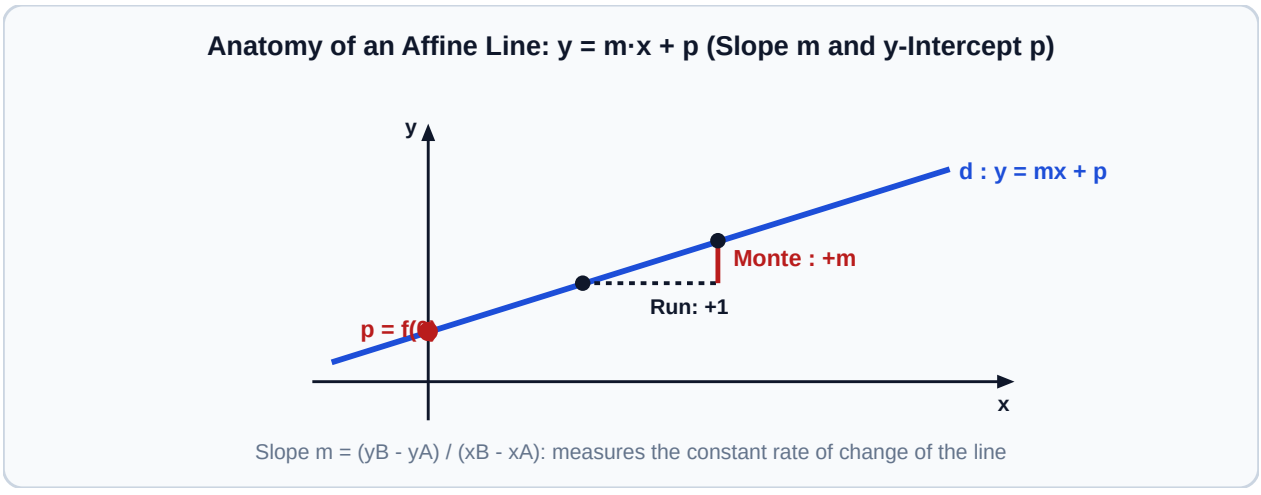

Figure 83: Affine line y = mx + p: slope and y-intercept

The port commander poses two urgent questions: 1. *Do these two vessels risk collision—meaning, do their paths cross at a specific point on the water?* 2. *If so, what are the exact coordinates of the intersection point?*

To answer, drawing on graph paper is unreliable: a blunt pencil lead or a slightly warped scale could be off by hundreds of meters. We need an infallible algebraic method capable of translating an infinite line into an equation solvable with absolute precision.

---

2. Heading and Path: The Direction Vector

2.1 Setting the Direction of a Line

A line in the plane is entirely determined by:

- either **two distinct points** A and B (through two points passes a unique straight line);
- or **one anchor point** A and **one non-zero vector** prescribing its direction.

> **📐 Definition — Direction Vector.**
>
> Let $\mathcal{D}$ be a line in the plane. Any non-zero vector $\vec{u}$ having the same direction as line $\mathcal{D}$ is called a **direction vector** of $\mathcal{D}$. If A and B are two distinct points on $\mathcal{D}$, then the vector $\overrightarrow{AB}$ is a direction vector of $\mathcal{D}$.

> **💡 Did You Know? A line possesses infinitely many direction vectors!**
>
> If $\vec{u}$ is a direction vector of line $\mathcal{D}$, then for any non-zero real number k, the vector $k\vec{u}$ shares the same direction: it is therefore also a direction vector of $\mathcal{D}$. Twice as long, one third the length, or pointing in reverse, it still lies along the very same line.

2.2 Vector Characterization of a Line

A point $M(x\,;y)$ belongs to the line $\mathcal{D}$ passing through $A(x_A\,;y_A)$ with direction vector $\vec{u}$ if and only if the displacement from A to M follows this exact same heading:

$$M \in \mathcal{D} \Leftrightarrow \overrightarrow{AM} \text{ and } \vec{u} \text{ are collinear .}$$

From the preceding chapter, two vectors are collinear if and only if their determinant vanishes:

$$M \in \mathcal{D} \Leftrightarrow \det(\overrightarrow{AM}, \vec{u}) = 0.$$

This single, elegant condition generates all linear equations!

---

3. The Cartesian Equation of a Line

3.1 The General Form $ax + by + c = 0$

> **📐 Fundamental Theorem — Cartesian Equation.**
>
> 1. Every
> straight line $\mathcal{D}$ in the plane admits an equation of the form:
>
> $$ax + by + c = 0$$
>
> where a, b, c are real numbers with $(a, b) \neq (0, 0)$ (that is, a and b are not both zero). 2. Conversely, every equation of the form $ax + by + c = 0$ with $(a, b) \neq (0, 0)$ represents a straight line. 3. The vector $\vec{u}\begin{pmatrix}-b\\a\end{pmatrix}$ is a **direction vector** of this line.

Proof (Core Curriculum) — From Collinearity to the Cartesian Equation.

Let $\mathcal{D}$ be the line passing through $A(x_A\,;y_A)$ with non-zero direction vector $\vec{u}\begin{pmatrix}u_1\\u_2\end{pmatrix}$. Let $M(x\,;y)$ be an arbitrary point in the plane. Vector $\overrightarrow{AM}$ has coordinates:

$$\overrightarrow{AM}\begin{pmatrix}x-x_A\\y-y_A\end{pmatrix}.$$

Point M belongs to $\mathcal{D}$ if and only if $\overrightarrow{AM}$ and $\vec{u}$ are collinear:

$$\det(\overrightarrow{AM},\vec{u})=0 \Leftrightarrow (x-x_A)u_2-(y-y_A)u_1=0.$$

Expanding:

$$u_2x-u_1y+(-u_2x_A+u_1y_A)=0.$$

Let us set:

$$a=u_2,\quad b=-u_1,\quad c=-u_2x_A+u_1y_A.$$

Since $\vec{u}\neq\vec{0}$, the pair $(u_1,u_2)\neq(0,0)$, hence $(a,b)\neq(0,0)$. The identity becomes:

$$ax+by+c=0.$$

Furthermore, vector $\vec{u}$ has coordinates $\begin{pmatrix}u_1\\u_2\end{pmatrix}=\begin{pmatrix}-b\\a\end{pmatrix}$. This is indeed a direction vector of the line. ■

⚠ Pitfall Warning! Cartesian equations are not unique.

If $2x-3y+5=0$ is the equation of a line, then multiplying each term by 2 gives $4x-6y+10=0$, which describes **the exact same line**! Two equations $ax+by+c=0$ and $a'x+b'y+c'=0$ represent the same line if and only if their coefficients are proportional: there exists a non-zero real scalar k such that $a'=ka$, $b'=kb$, and $c'=kc$.

---

4. Slope-Intercept Form: The Gradient and the y-Intercept

While the general form $ax+by+c=0$ is universally applicable, for graphing and physical intuition mathematicians often prefer expressing y explicitly in terms of x.

4.1 Two Categories of Lines

Isolate the y-term in the Cartesian equation:

$$by=-ax-c.$$

Two cases arise:

Case 1: $b\neq0$ **(the line is non-vertical)**
We can divide through by b:

$$y=-\frac{a}{b}x-\frac{c}{b}.$$

Setting $m=-\frac{a}{b}$ and $p=-\frac{c}{b}$, the equation takes the canonical form:

$$\boxed{y=mx+p}$$

This is the **slope-intercept form** (or reduced equation) of the line.

- m **is the slope** (gradient, or *coefficient directeur*).
- p **is the y-intercept** (ordonnée à l'origine): it is the vertical coordinate where the line crosses the y-axis (Oy) (setting $x=0$ yields $y=m(0)+p=p$). The point $(0\,;p)$ lies on the line.

Case 2: $b=0$ **(the line is vertical)**
The equation reduces to $ax+c=0$. Since $(a,b)\neq(0,0)$ and $b=0$, we must have $a\neq0$. We deduce:

$$x = -\frac{c}{a} = k.$$

Every point on the line shares the **same x-coordinate** k. The line is parallel to the y-axis (Oy). Its slope is undefined ("infinite"), and it cannot be written in the form $y = mx + p$. Its direction vector is strictly vertical, such as $\vec{j}\begin{pmatrix}0\\1\end{pmatrix}$.

4.2 The Geometric Meaning of Slope m

📐 Fundamental Property — Calculating the Slope.

Let $A(x_A\,;y_A)$ and $B(x_B\,;y_B)$ be two distinct points on a non-vertical line ($x_A \neq x_B$). The slope m is given by:

$$m = \frac{y_B - y_A}{x_B - x_A} = \frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}}.$$

Physical Intuition: Δy represents vertical elevation change and Δx represents horizontal advance.

- If $m > 0$: the line "climbs" from left to right. The associated affine function is strictly increasing.
- If $m < 0$: the line "descends" from left to right. The associated affine function is strictly decreasing.
- If $m = 0$: the equation becomes $y = p$. The line is perfectly **horizontal**, parallel to the x-axis.
- An immediate direction vector for a line with equation $y = mx + p$ is $\vec{u}\begin{pmatrix}1\\m\end{pmatrix}$. Advancing by 1 unit horizontally produces a vertical change of m units.

---

5. Standard Techniques: Construction and Identification

5.1 Graphing a Line Accurately

Two reliable methods: 1. **The Two-Point Method:** Choose two simple values for x (frequently $x = 0$ and $x = 1$ or $x = 2$), compute the corresponding y-values, plot both points, and connect them with a straightedge. 2. **The Slope-Intercept Method:** Plot the intercept $(0\,;p)$ on the vertical axis. From there, move 1 unit to the right, then move vertically by m units (upward if $m > 0$, downward if $m < 0$). Mark the second point and draw the line.

5.2 Finding the Equation of a Line

- **Case A: Given two points $A(x_A\,;y_A)$ and $B(x_B\,;y_B)$ with $x_A \neq x_B$.**
  1. Compute the slope: $m = \frac{y_B - y_A}{x_B - x_A}$.
  2. The equation is $y = mx + p$. To find p, use the condition that $A \in \mathcal{D}$:
  $$y_A = mx_A + p \Longrightarrow p = y_A - mx_A.$$
- **Case B: Given a point $A(x_A\,;y_A)$ and a direction vector $\vec{u}\begin{pmatrix}-b\\a\end{pmatrix}$.**
  1. Write down the general Cartesian template: $ax + by + c = 0$.
  2. Substitute the coordinates of A into the equation to determine constant c:
  $$ax_A + by_A + c = 0 \Longrightarrow c = -ax_A - by_A.$$

---

6. Relative Positions of Two Lines and Linear Systems

Consider two lines in the plane: $\mathcal{D} : ax + by + c = 0$ and $\mathcal{D}' : a'x + b'y + c' = 0$, with direction vectors $\vec{u}\begin{pmatrix}-b\\a\end{pmatrix}$ and $\vec{u}'\begin{pmatrix}-b'\\a'\end{pmatrix}$ respectively.

6.1 The Three Configurations

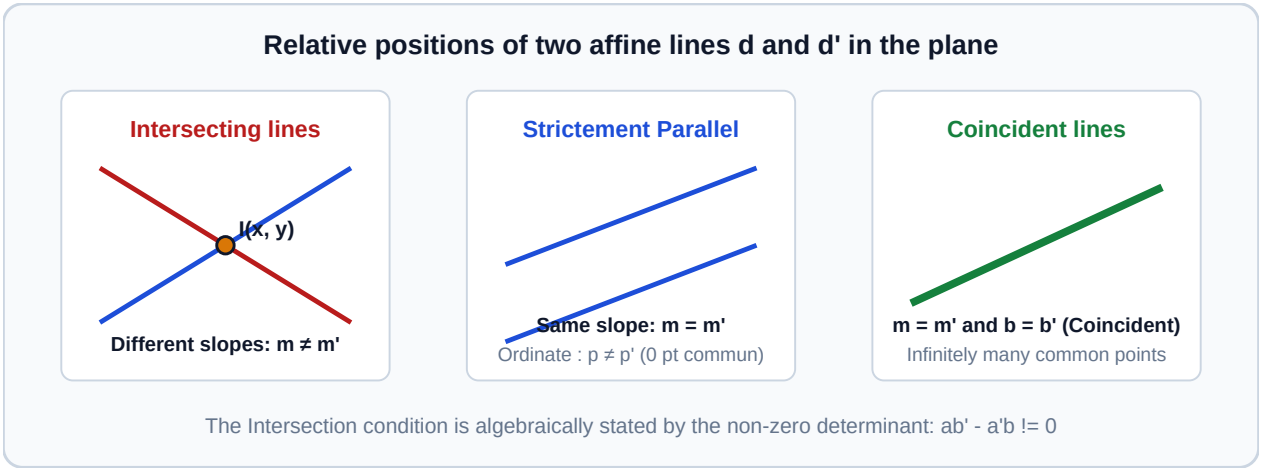

Figure 84: Relative positions of two affine lines in the plane

1. **Intersecting Lines:** Their direction vectors are not collinear:
$$\det(\vec{u}, \vec{u'}) \neq 0 \Leftrightarrow (-b)(a') - (a)(-b') \neq 0 \Leftrightarrow ab' - a'b \neq 0.$$
For non-vertical lines: $m \neq m'$. They meet at a **single, unique point**.
2. **Strictly Parallel Lines:** Their direction vectors are collinear ($ab' - a'b = 0$, so $m = m'$), but their intercepts differ ($p \neq p'$). Their intersection is empty: $\mathcal{D} \cap \mathcal{D}' = \varnothing$.
3. **Coincident (Identical) Lines:** Their direction vectors are collinear and they share at least one point (hence $m = m'$ and $p = p'$). Every point is shared: $\mathcal{D} = \mathcal{D}'$.

6.2 Solving a System of Two Linear Equations in Two Variables

Finding where two lines intersect is equivalent to finding the pair $(x\,;y)$ satisfying both equations simultaneously:

$$\{ax + by = e$$
$$cx + dy = f$$

Two primary algebraic techniques exist:

Method 1: Substitution

Isolate one unknown (the one with the simplest coefficient, preferably 1 or -1) in one equation, then substitute its expression into the second equation. This reduces the problem to a single linear equation in one unknown.

Method 2: Elimination (Linear Combination)

Multiply one or both equations by suitable non-zero constants so that the coefficients of one variable (say x) become exact opposites. Adding the two resulting equations cancels x completely!

---

7. Detailed Worked Examples

Worked Example 1 — Readdressing the Port Problem: Ship Collision.

Recap of the data:

- *L'Espérance* departs $A(2\,;3)$ along $\vec{u}\begin{pmatrix}2\\1\end{pmatrix}$.
- *La Vigilante* departs $B(-1\,;8)$ along $\vec{v}\begin{pmatrix}1\\-1\end{pmatrix}$.

1. Determine a Cartesian equation of trajectory $\mathcal{D}_1$ of *L'Espérance*.
2. Determine a Cartesian equation of trajectory $\mathcal{D}_2$ of *La Vigilante*.
3. Determine whether their paths intersect and calculate intersection coordinates I.

Solution: 1. For $\mathcal{D}_1$, $\vec{u}\begin{pmatrix}2\\1\end{pmatrix}$ is a direction vector. Since $\vec{u}\begin{pmatrix}-b\\a\end{pmatrix}$, we choose $a=1$ and $b=-2$. The equation has template $1x-2y+c=0$. $A(2\,;3)\in\mathcal{D}_1 \Longrightarrow 2-2(3)+c=0 \Longrightarrow 2-6+c=0 \Longrightarrow c=4$. Thus $\mathcal{D}_1 : x-2y+4=0$ (or in slope-intercept form: $y=\frac{1}{2}x+2$).

2. For $\mathcal{D}_2$, $\vec{v}\begin{pmatrix}1\\-1\end{pmatrix}$ is a direction vector. We can choose $a=1, b=1$ (since $\begin{pmatrix}-1\\1\end{pmatrix}=-\vec{v}$). The equation has template $x+y+c'=0$. $B(-1\,;8)\in\mathcal{D}_2 \Longrightarrow -1+8+c'=0 \Longrightarrow 7+c'=0 \Longrightarrow c'=-7$. Thus $\mathcal{D}_2 : x+y-7=0$ (or in slope-intercept form: $y=-x+7$).

3. Compare slopes: $m_1=\frac{1}{2}$ and $m_2=-1$. Since $m_1\neq m_2$, the lines intersect at a unique point I. Solve the system:

$$\{y=\frac{1}{2}x+2$$
$$y=-x+7$$

Equating both expressions for y:

$$\frac{1}{2}x+2=-x+7 \Leftrightarrow \frac{1}{2}x+x=7-2 \Leftrightarrow \frac{3}{2}x=5 \Leftrightarrow x=\frac{10}{3}.$$

Calculate y:

$$y=-\frac{10}{3}+7=-\frac{10}{3}+\frac{21}{3}=\frac{11}{3}.$$

The trajectories intersect at coordinates $I\left(\frac{10}{3}\,;\frac{11}{3}\right)$.

Worked Example 2 — Equation of a Line Through Two Points.

Given $E(-3\,;5)$ and $F(1\,;-3)$. Find the slope-intercept equation of line (EF).

Solution: Check if vertical: $x_E=-3$ and $x_F=1$. Since $x_E\neq x_F$, it is non-vertical. 1. Calculate slope m:

$$m=\frac{y_F-y_E}{x_F-x_E}=\frac{-3-5}{1-(-3)}=\frac{-8}{4}=-2.$$

2. Calculate intercept p: The equation is $y=-2x+p$. Since $F(1\,;-3)$ lies on the line:

$$-3=-2(1)+p \Longrightarrow -3=-2+p \Longrightarrow p=-1.$$

The slope-intercept equation is $\boxed{y=-2x-1}$. *(Verification with E: $-2(-3)-1=6-1=5=y_E$. Matches!)*

Worked Example 3 — Solving a System via Elimination.

Solve in $\mathbb{R}^2$ the system:

$$\begin{cases} 3x + 4y = 11 \quad (L_1) \\ 5x - 2y = 40 \quad (L_2) \end{cases}$$

Solution: Observe the y-coefficients: $+4$ in L_1 and -2 in L_2. Multiplying L_2 by 2 produces -4, the exact opposite of $+4$! Multiply L_2 by 2:

$$2 \times (5x - 2y) = 2 \times 40 \Leftrightarrow 10x - 4y = 80 \quad (L_2').$$

Add L_1 and L_2' term by term:

$$(3x + 4y) + (10x - 4y) = 11 + 80 \Leftrightarrow 13x = 91 \Leftrightarrow x = \frac{91}{13} = 7.$$

Substitute $x = 7$ into L_1:

$$3(7) + 4y = 11 \Leftrightarrow 21 + 4y = 11 \Leftrightarrow 4y = 11 - 21 = -10 \Leftrightarrow y = -\frac{10}{4} = -2.5.$$

The unique solution pair is $(7\,;-2.5)$. Solution set: $S = \{(7\,;-2.5)\}$.

Worked Example 4 — Inconsistent and Dependent Systems.

Examine geometrically the following two systems: a) $\begin{cases} -4x + 6y = 5 \end{cases}$ b) $\begin{cases} 3x - 6y = 9 \end{cases}$

Solution: a) Multiply the first equation by 2:

$$4x - 6y = 14.$$

Add to the second:

$$(4x - 6y) + (-4x + 6y) = 14 + 5 \Leftrightarrow 0 = 19.$$

This is a contradiction! Geometrically, the two lines have collinear direction vectors ($\frac{-4}{2} = \frac{6}{-3} = -2$), so they are strictly parallel and never cross. The system has **no solution**: $S = \varnothing$.

b) Multiply the first equation by 3:

$$3x - 6y = 9.$$

This is identical to the second equation! Both equations describe the **same line**. Every point on this line is a solution. The system has **infinitely many solutions**: all pairs of the form $(2y + 3\,;y)$ for $y \in \mathbb{R}$.

Worked Example 5 — Equation of the Perpendicular Bisector.

Let $A(1\,;2)$ and $B(5\,;4)$. Find the equation of the perpendicular bisector Δ of segment $[AB]$.

Solution: Recall: the perpendicular bisector is the locus of points equidistant from A and B, passing through the midpoint I of $[AB]$. 1. Midpoint I of $[AB]$:

$$x_I = \frac{1+5}{2} = 3, \quad y_I = \frac{2+4}{2} = 3 \Longrightarrow I(3\,;3).$$

2. Geometric condition: $M(x\,;y) \in \Delta \Leftrightarrow MA^2 = MB^2$.

$$MA^2 = (x-1)^2 + (y-2)^2 = x^2 - 2x + 1 + y^2 - 4y + 4 = x^2 + y^2 - 2x - 4y + 5.$$

$$MB^2 = (x-5)^2 + (y-4)^2 = x^2 - 10x + 25 + y^2 - 8y + 16 = x^2 + y^2 - 10x - 8y + 41.$$

Equating $MA^2 = MB^2$:

$$x^2 + y^2 - 2x - 4y + 5 = x^2 + y^2 - 10x - 8y + 41.$$

The quadratic terms x^2 and y^2 cancel:

$$-2x - 4y + 5 = -10x - 8y + 41 \Leftrightarrow 8x + 4y - 36 = 0.$$

Dividing through by 4:

$$\boxed{2x + y - 9 = 0} \quad \text{or in slope-intercept form} \quad \boxed{y = -2x + 9}\,.$$

Worked Example 6 — Pencil of Lines Depending on a Parameter.

For any real parameter m, consider line $\mathcal{D}_m$ defined by:

$$(m-1)x + (2m+1)y - 5m + 2 = 0.$$

Prove that all lines $\mathcal{D}_m$ pass through a fixed point K, and find its coordinates.

Solution: Regroup terms by powers of parameter m:

$$mx - x + 2my + y - 5m + 2 = 0 \Leftrightarrow m(x + 2y - 5) + (-x + y + 2) = 0.$$

For this equation to hold true **for all values of m**, both the coefficient of m and the constant term must vanish simultaneously:

$$\begin{cases} x + 2y - 5 = 0 \\ -x + y + 2 = 0 \end{cases}$$

Add both equations:

$$(x + 2y - 5) + (-x + y + 2) = 0 \Leftrightarrow 3y - 3 = 0 \Leftrightarrow y = 1.$$

Substitute into the second equation:

$$-x + 1 + 2 = 0 \Leftrightarrow x = 3.$$

Every line in the family passes through the fixed pivot point $K(3\,;1)$!

---

8. In Computer Science: Solving a 2×2 System in Python

Here is a Python script that takes Cartesian coefficients for two lines and determines their intersection point:

```python
def solve_system(a1, b1, c1, a2, b2, c2):
    """
    Solves the linear system:
    a1*x + b1*y + c1 = 0
    a2*x + b2*y + c2 = 0
    """
    # Main determinant computation (a1*b2 - a2*b1)
    det = a1 * b2 - a2 * b1

    if abs(det) < 1e-12:
        # Parallel or coincident lines
        if abs(a1 * c2 - a2 * c1) < 1e-12 and abs(b1 * c2 - b2 * c1) < 1e-12:
            return "Infinitely many solutions (coincident lines)"
        else:
            return "No solution (strictly parallel lines)"
    else:
        # Cramer's rule for x and y: a*x + b*y = -c
        e1, e2 = -c1, -c2
        x = (e1 * b2 - e2 * b1) / det
        y = (a1 * e2 - a2 * e1) / det
        return (x, y)

# Test with the system from Example 1:
# x - 2y + 4 = 0 and x + y - 7 = 0
solution = solve_system(1, -2, 4, 1, 1, -7)
print("Intersection point:", solution)  # Prints (3.3333333333333335,
3.6666666666666665)
```

Key Takeaways

1. **Direction Vector:** any non-zero vector $\vec{u}$ parallel to the line. If $A, B \in \mathcal{D}$, then $\overrightarrow{AB}$ is a direction vector.
2. **Cartesian Equation:** $ax + by + c = 0$ with $(a, b) \neq (0, 0)$.
   - Canonical direction vector: $\vec{u}\begin{pmatrix} -b \\ a \end{pmatrix}$.
3. **Slope-Intercept Form:**
   - If the line is non-vertical ($b \neq 0$): $y = mx + p$.
     - $m = -\frac{a}{b} = \frac{y_B - y_A}{x_B - x_A}$ is the **slope** (gradient).
     - p is the ***y*-intercept** (point $(0\,;p)$).
     - Direction vector: $\vec{u}\begin{pmatrix} 1 \\ m \end{pmatrix}$.
   - If the line is vertical ($b = 0$): $x = k$. No slope is defined.
4. **Relative Positions:**
   - Parallel $\Leftrightarrow m = m'$ (or $ab' - a'b = 0$).
   - Intersecting $\Leftrightarrow m \neq m'$ (or $ab' - a'b \neq 0$). Unique intersection via system solving.
5. **Methods for Solving Systems:**
   - *Substitution*: isolate one variable and substitute into the second equation.
   - *Elimination*: scale equations so one variable cancels upon addition.

Exercises

Direct Application Exercises ()

1. State a direction vector for the line with equation $3x - 5y + 7 = 0$.

2. State a direction vector for the line with equation $y = -4x + 3$.
3. Determine the slope and y-intercept of the line $2x + 4y - 8 = 0$.
4. Write the slope-intercept equation of the line passing through point $P(0\,;-5)$ with slope $m = 3$.
5. Does the line $x = 4$ have a slope? Explain why or why not.
6. Are the lines $y = 3x - 2$ and $6x - 2y + 5 = 0$ parallel?
7. Does the point $A(3\,;-1)$ lie on the line $2x - 5y - 11 = 0$?
8. Find the x-coordinate of the point on the line $y = 2x - 6$ whose y-coordinate is 4.
9. Solve the system by substitution: $\begin{cases} y = 2x - 1 \\ 3x + y = 9 \end{cases}$.
10. Solve the system by elimination: $\begin{cases} x + y = 10 \\ x - y = 4 \end{cases}$.

Practice and Reasoning Exercises ()

11. Determine the Cartesian equation of the line passing through $A(2\,;-3)$ with direction vector $\vec{u}\begin{pmatrix} -3 \\ 4 \end{pmatrix}$.
12. Determine the slope-intercept equation of the line passing through points $M(-2\,;7)$ and $N(4\,;1)$.
13. Given points $A(-1\,;2)$, $B(3\,;4)$, and $C(2\,;-1)$. Determine an equation of line $\mathcal{D}$ passing through C parallel to (AB).
14. Solve in $\mathbb{R}^2$ the system: $\begin{cases} 4x + 3y = 1 \\ 2x - 5y = 7 \end{cases}$.
15. Find the coordinates of the intersection point of lines $\mathcal{D}_1 : 3x - 2y + 6 = 0$ and $\mathcal{D}_2 : x + 2y - 2 = 0$.
16. For what value of $k \in \mathbb{R}$ is the line $kx + 2y - 5 = 0$ parallel to the line passing through $A(1\,;1)$ and $B(3\,;7)$?
17. Let triangle ABC have vertices $A(0\,;4)$, $B(-2\,;0)$, and $C(4\,;0)$. Determine the equation of the median passing through A.
18. Find the real parameter m such that the system $\begin{cases} mx + 2y = 4 \\ 3x + (m-1)y = 2 \end{cases}$ fails to have a unique solution.

Advanced Problems and Challenges ()

19. **Concurrency of Altitudes (The Orthocenter).** In an orthonormal coordinate frame, let $A(0\,;3)$, $B(-2\,;0)$, and $C(4\,;0)$.
    a) Determine an equation of the altitude from A.
    b) Using the property that the altitude from B is perpendicular to (AC), determine its equation.
    c) Compute the coordinates of the orthocenter H of the triangle.
20. **Orthogonal Projection of a Point onto a Line.** Consider the line $\mathcal{D} : 2x - y + 1 = 0$ and point $A(4\,;4)$.
    a) Determine the equation of line Δ perpendicular to $\mathcal{D}$ passing through A.
    b) Determine the coordinates of intersection point H of $\mathcal{D}$ and Δ.
    c) Calculate the distance AH. (This is the minimum distance from point A to line $\mathcal{D}$).
21. **Concurrency of Three Lines.** Consider the three lines: $\mathcal{D}_1 : 2x - y - 1 = 0$, $\mathcal{D}_2 : x + y - 5 = 0$, and $\mathcal{D}_3 : 3x + ky - 7 = 0$. Determine the value of $k \in \mathbb{R}$ such that all three lines are concurrent at a single point.
22. **Economic Equilibrium: Supply and Demand.** In a commodities market, quantity supplied q_S and quantity demanded q_D (in tonnes) depend on unit price p (in euros per kg) according to the following linear laws:
    - Supply: $q_S = 2p - 4$ (producers supply more as price rises).
    - Demand: $q_D = -3p + 21$ (buyers demand less as price rises).

    a) Plot both lines in a coordinate system $(p\,;q)$.
    b) Calculate the equilibrium price p_E where supply equals demand.
    c) What is the market transaction quantity at equilibrium?

Solutions

1. For $ax + by + c = 0$, a direction vector is $\vec{u}\begin{pmatrix} -b \\ a \end{pmatrix}$. Here $a = 3, b = -5 \Longrightarrow \vec{u}\begin{pmatrix} 5 \\ 3 \end{pmatrix}$.
2. For $y = mx + p$, a direction vector is $\vec{u}\begin{pmatrix} 1 \\ m \end{pmatrix}$. Here $m = -4 \Longrightarrow \vec{u}\begin{pmatrix} 1 \\ -4 \end{pmatrix}$.

3. $4y = -2x + 8 \Leftrightarrow y = -\frac{2}{4}x + \frac{8}{4} = -\frac{1}{2}x + 2$. Slope $m = -\frac{1}{2}$, y-intercept $p = 2$.
4. $y = 3x - 5$.
5. No, the line $x = 4$ is vertical. Its slope is undefined (run $\Delta x = 0$).
6. Line 1: $m_1 = 3$. Line 2: $-2y = -6x - 5 \Leftrightarrow y = 3x + 2.5 \Longrightarrow m_2 = 3$. Since $m_1 = m_2 = 3$ and intercepts differ ($-2 \neq 2.5$), the lines are strictly parallel.
7. Substitute $x = 3$ and $y = -1$: $2(3) - 5(-1) - 11 = 6 + 5 - 11 = 0$. Yes, $A \in \mathcal{D}$.
8. $4 = 2x - 6 \Leftrightarrow 2x = 10 \Leftrightarrow x = 5$.
9. Substitute y into the second equation: $3x + (2x - 1) = 9 \Leftrightarrow 5x - 1 = 9 \Leftrightarrow 5x = 10 \Leftrightarrow x = 2$. Then $y = 2(2) - 1 = 3$. Solution: $(2\,;3)$.
10. Add equations: $(x + y) + (x - y) = 10 + 4 \Leftrightarrow 2x = 14 \Leftrightarrow x = 7$. Then $7 + y = 10 \Leftrightarrow y = 3$. Solution: $(7\,;3)$.
11. $\vec{u}\begin{pmatrix}-3\\4\end{pmatrix} = \begin{pmatrix}-b\\a\end{pmatrix} \Longrightarrow a = 4$ and $b = 3$. Equation: $4x + 3y + c = 0$. $A(2\,;-3) \in \mathcal{D} \Longrightarrow 4(2) + 3(-3) + c = 0 \Leftrightarrow 8 - 9 + c = 0 \Leftrightarrow c = 1$. Equation: $4x + 3y + 1 = 0$.
12. $m = \frac{1-7}{4-(-2)} = \frac{-6}{6} = -1$. $y = -x + p$. Using $N(4\,;1)$: $1 = -4 + p \Longrightarrow p = 5$. Slope-intercept equation: $y = -x + 5$.
13. Vector $\overrightarrow{AB}\begin{pmatrix}3-(-1)\\4-2\end{pmatrix} = \begin{pmatrix}4\\2\end{pmatrix}$ or scaled $\vec{u}\begin{pmatrix}2\\1\end{pmatrix}$. $\vec{u}\begin{pmatrix}-b\\a\end{pmatrix} \Longrightarrow a = 1, b = -2$. Equation: $x - 2y + c = 0$. $C(2\,;-1) \in \mathcal{D} \Longrightarrow 2 - 2(-1) + c = 0 \Leftrightarrow 2 + 2 + c = 0 \Leftrightarrow c = -4$. Equation: $x - 2y - 4 = 0$ (or $y = \frac{1}{2}x - 2$).
14. Multiply the second row by -2: $-2(2x - 5y) = -2(7) \Leftrightarrow -4x + 10y = -14$. Add to the first: $(4x + 3y) + (-4x + 10y) = 1 - 14 \Leftrightarrow 13y = -13 \Leftrightarrow y = -1$. Substitute back: $2x - 5(-1) = 7 \Leftrightarrow 2x + 5 = 7 \Leftrightarrow 2x = 2 \Leftrightarrow x = 1$. Solution: $(1\,;-1)$.
15. Add $\mathcal{D}_1$ and $\mathcal{D}_2$: $(3x - 2y + 6) + (x + 2y - 2) = 0 \Leftrightarrow 4x + 4 = 0 \Leftrightarrow x = -1$. Substitute into $\mathcal{D}_2$: $-1 + 2y - 2 = 0 \Leftrightarrow 2y = 3 \Leftrightarrow y = 1.5$. Intersection: $I(-1\,;1.5)$.
16. Slope of (AB): $m = \frac{7-1}{3-1} = \frac{6}{2} = 3$. Line $kx + 2y - 5 = 0 \Leftrightarrow 2y = -kx + 5 \Leftrightarrow y = -\frac{k}{2}x + \frac{5}{2}$. Its slope is $-\frac{k}{2}$. Parallelism $\Leftrightarrow -\frac{k}{2} = 3 \Leftrightarrow k = -6$.
17. The median from A connects $A(0\,;4)$ and midpoint A' of $[BC]$. $x_{A'} = \frac{-2+4}{2} = 1$ and $y_{A'} = \frac{0+0}{2} = 0$, so $A'(1\,;0)$. Slope $m = \frac{0-4}{1-0} = -4$. Intercept from $A(0\,;4)$: $p = 4$. Equation: $y = -4x + 4$.
18. No unique solution when determinant vanishes: $m(m-1) - 2 \times 3 = 0 \Leftrightarrow m^2 - m - 6 = 0$. Factoring: $(m-3)(m+2) = 0 \Longrightarrow m = 3$ or $m = -2$.
19. a) Side $[BC]$ lies on the x-axis ($y = 0$). The altitude from A is therefore perpendicular to the x-axis: it is a vertical line through $A(0\,;3)$, equation $x = 0$.
    b) Vector $\overrightarrow{AC}\begin{pmatrix}4-0\\0-3\end{pmatrix} = \begin{pmatrix}4\\-3\end{pmatrix}$. A line perpendicular to (AC) has normal vector $\overrightarrow{AC}$, meaning its direction vector is $\vec{u}\begin{pmatrix}3\\4\end{pmatrix}$. Slope of this altitude: $m = \frac{4}{3}$. Passing through $B(-2\,;0)$: $y - 0 = \frac{4}{3}(x - (-2)) \Longrightarrow y = \frac{4}{3}x + \frac{8}{3}$.
    c) Orthocenter H is the intersection of both altitudes. Since the first altitude is $x = 0$, substitute $x = 0$ into the second: $y = \frac{8}{3}$. Thus $H\left(0\,;\frac{8}{3}\right)$.
20. a) Direction vector of $\mathcal{D} : 2x - y + 1 = 0$ is $\vec{u}\begin{pmatrix}1\\2\end{pmatrix}$ (slope $m = 2$). Perpendicular line Δ has slope $m' = -\frac{1}{m} = -\frac{1}{2}$. Equation: $y - 4 = -\frac{1}{2}(x - 4) \Leftrightarrow y = -\frac{1}{2}x + 2 + 4 = -\frac{1}{2}x + 6 \Leftrightarrow x + 2y - 12 = 0$.
    b) Intersection of $\mathcal{D}$ and Δ: From $\mathcal{D}$, $y = 2x + 1$. In Δ: $x + 2(2x + 1) - 12 = 0 \Leftrightarrow 5x - 10 = 0 \Leftrightarrow x = 2$. Then $y = 2(2) + 1 = 5$. Projected point is $H(2\,;5)$.
    c) $AH = \sqrt{(2-4)^2 + (5-4)^2} = \sqrt{(-2)^2 + 1^2} = \sqrt{4+1} = \sqrt{5}$.
21. Find intersection K of $\mathcal{D}_1$ and $\mathcal{D}_2$: $\begin{cases}x + y = 5\\2x - y = 1\end{cases}$ Sum: $3x = 6 \Longrightarrow x = 2$. Then $y = 5 - 2 = 3$. Thus $K(2\,;3)$. For $\mathcal{D}_3$ to pass through K: $3(2) + k(3) - 7 = 0 \Leftrightarrow 6 + 3k - 7 = 0 \Leftrightarrow 3k = 1 \Leftrightarrow k = \frac{1}{3}$.
22. a) Supply line has positive slope 2; demand line has negative slope -3.
    b) At equilibrium, $q_S = q_D \Leftrightarrow 2p - 4 = -3p + 21 \Leftrightarrow 5p = 25 \Leftrightarrow p_E = 5$ €/kg.
    c) Equilibrium quantity: $q_E = 2(5) - 4 = 10 - 4 = 6$ tonnes (or $q_E = -3(5) + 21 = 6$ tonnes).

The Question That Opens the Future

A straight line represents mathematics in its most well-behaved and predictable form: a relationship where every step forward produces the exact same steady return. This is the domain of affine functions $x \mapsto mx + p$.

Yet what happens when the inner rule of the machine becomes more intricate? When the machine squares numbers, reciprocates them, or extracts square roots? How do we map the curves, crests, and valleys of non-linear graphs? That is what we are about to explore in **The Jacquard Loom and the Function Box**.

CHAPTER 34: The Jacquard Loom and the Function Box

> 📐 **"A function is not a frozen drawing in a book. It is a living machine: you slide a raw block of metal into the input slot, the cogs turn, the mechanical rule executes, and a finely crafted part emerges on the conveyor belt. Modern mathematics is the great workshop where these transformations unfold."**

What You Are About to Discover

- How the invention of punched cards by **Joseph Marie Jacquard** in Lyon and the brilliant insight of **Ada Lovelace** gave birth to the modern concept of a function.
- The fundamental vocabulary: **preimage (antécédent)**, **image**, **domain of definition** $\mathcal{D}_f$, and the ironclad rule of image uniqueness.
- How to construct and interpret a **function curve (graph)**, a table of values, and a **table of variations**.
- The concepts of **maximum**, **minimum** (local and global), and **parity** (even and odd functions, symmetries in the coordinate plane).
- A complete study of the **five parent (reference) functions**: the square function (x^2), the reciprocal function ($\frac{1}{x}$), the square root function ($\sqrt{x}$), the cube function (x^3), and the absolute value function ($|x|$).
- Solving equations $f(x) = k$, $f(x) = g(x)$, and inequalities graphically, along with the **composition** of function machines $g \circ f$.

---

1. The Problem: The Silk Weavers of Lyon and Ada Lovelace's Cards

At the very opening of the nineteenth century, on the steep hillsides of the Croix-Rousse in Lyon, France, the silk workshops of the *canuts* (silk weavers) echoed with the deafening clatter of wooden shuttles.

To weave a rich damask brocade embroidered with crimson roses and silver leaves, two workers were required at every loom: the master weaver seated before the reed, and a young child huddled in the dust below (the draw-boy or "draw-girl") who pulled drawstrings by hand to raise warp threads one by one. The labor was grueling, ruinous to eyesight and posture, and a single moment of distraction by the child created an irreversible knot across the luxurious fabric.

In **1801**, the son of a Lyon weaver, **Joseph Marie Jacquard**, unveiled an automated mechanism that astonished Europe: He replaced the draw-boy with an endless loop of **punched cardboard cards** linked together in a chain.

- A hole in the card? A needle passes through freely: the hook engages, lifting the silk thread.
- No hole? The solid card pushes the needle back: the silk thread remains lowered.

The Jacquard loom was the first programmable machine in human history!

In **1843**, in London, a young mathematician of extraordinary vision, **Ada Lovelace** (the only legitimate child of the poet Lord Byron), studied the mechanical blueprints of Charles Babbage's "Analytical Engine." She immediately saw the profound parallel with the Lyon silk mills and penned this famous observation: > "*We may say most aptly that the Analytical Engine weaves algebraical patterns just as the Jacquard-loom weaves flowers and leaves.*"

Ada Lovelace then drafted an algorithm for computing Bernoulli numbers: she had just composed the **world's very first computer program**.

What Jacquard realized in cardboard and Ada Lovelace formalized in thought is the very soul of what mathematicians define as a **Function**.

---

2. Anatomy of a Function: Vocabulary and Domain

2.1 The Black Box ($x \mapsto f(x)$)

Put aside complicated graphs for a moment. Picture an **uncompromising industrial machine**:

> **📐 Fundamental Definitions —**
>
> 1. A **function** f is a
>
> mathematical process that assigns to each real number x in a given set **at most one** unique real number, denoted $f(x)$. 2. We write:
>
> $$f : x \mapsto f(x)$$
>
> Read aloud as: "the function f mapping x to f of x." 3. **The Preimage (Antécédent):** the input number x fed into the machine at the start. 4. **The Image:** the resulting output number $f(x)$ emerging at the end.

> **📐 Ironclad Rule — Uniqueness of the Image.**
>
> - A number x has **at most one single image** under a function
>
> f. If you load the exact same flour into the oven under identical settings, two different loaves cannot emerge simultaneously!
> - Conversely, an output number y may have **none, one, or several preimages**. For example, for the squaring machine ($f(x) = x^2$):
>   - The number 9 has two preimages: 3 and -3 (since $3^2 = 9$ and $(-3)^2 = 9$).
>   - The number -4 has no real preimage (no real square is negative).

2.2 The Domain of Definition $\mathcal{D}_f$: Forbidden Values

All real machines have operating tolerances: drop a heavy iron crowbar into a small plastic coffee grinder, and the grinder shatters. In mathematics, certain operations are forbidden across the real numbers.

> **📐 Definition — Domain of Definition.**
>
> The **domain of definition** of a function f, denoted $\mathcal{D}_f$, is the set of all real numbers x for which the evaluation of $f(x)$ is mathematically valid and meaningful.

In high school mathematics, two operations necessitate checking for **forbidden values (domain restrictions)**: 1. **Division by zero:** The denominator of a fraction can **never** be zero.

$$\frac{A}{B} \text{ exists if and only if } B \neq 0.$$

2. **The square root of a negative quantity:** The expression inside a square root must be non-negative.

$$\sqrt{A} \text{ exists if and only if } A \geq 0.$$

---

3. Graphs, Variations, and Extrema

3.1 The Function Graph in a Coordinate Plane

📐 Definition — Graph of a Function.

In a coordinate frame $(O\,;\vec{\imath},\vec{\jmath})$, the **graph (or representative curve)** of a function f, denoted $\mathcal{C}_f$, is the set of all points $M(x\,;y)$ whose x-coordinate lies in $\mathcal{D}_f$ and whose y-coordinate is its image $y=f(x)$:

$$\mathcal{C}_f=\big\{M(x\,;f(x))\mid x\in\mathcal{D}_f\big\}.$$

The equation of the curve is: $y=f(x)$.

To check whether a point $A(x_A\,;y_A)$ lies on the curve, compute $f(x_A)$. If it equals y_A, the point is on the curve; otherwise, it is not.

3.2 Variations: Reading the Landscape

Imagine a hiker traversing the curve $\mathcal{C}_f$ from left to right (in the direction of increasing x).

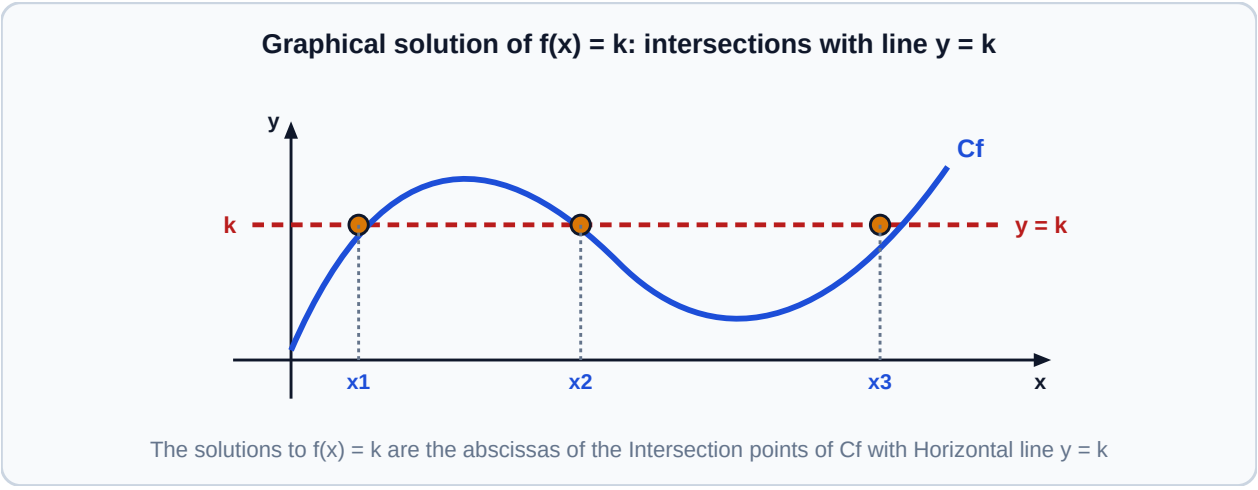

Figure 85: Graphical solution of f(x) = k by line intersection

📐 Definition — Variations on an Interval I.

Let f be a function defined on an interval I. 1. f is **strictly increasing** on I if for all real numbers a and b in I:

$$a<b\Longrightarrow f(a)<f(b).$$

(An increasing function preserves inequality order). 2. f is **strictly decreasing** on I if for all real numbers a and b in I:

$$a<b\Longrightarrow f(a)>f(b).$$

(A decreasing function reverses inequality order). 3. f is **constant** on I if for all $a,b\in I$, $f(a)=f(b)$. 4. A function is called **monotonic** on an interval if it is either purely increasing or purely decreasing across that interval.

3.3 Extrema: Peaks and Troughs

Definition — Maximum and Minimum.

Let f be a function defined on domain $\mathcal{D}_f$ and let $x_0 \in \mathcal{D}_f$.

- f attains a **maximum** at x_0 on $\mathcal{D}_f$, with value $M = f(x_0)$, if for all $x \in \mathcal{D}_f$:
$$f(x) \leq f(x_0).$$
- f attains a **minimum** at x_0 on $\mathcal{D}_f$, with value $m = f(x_0)$, if for all $x \in \mathcal{D}_f$:
$$f(x) \geq f(x_0).$$

The term **extremum** (plural *extrema*) refers to either a maximum or a minimum.

---

4. Parity: Symmetry in the Mirror

4.1 A Domain Centered at the Origin

To compare behavior to the left and right of the origin, the domain of definition must be symmetric with respect to zero:
$$\forall x \in \mathcal{D}_f, \quad -x \in \mathcal{D}_f.$$

For instance, $[-3\,;3]$ and $\mathbb{R} \setminus \{0\}$ are symmetric about 0; whereas $[-2\,;5]$ is not (for $x = 4$, -4 does not belong to the set).

4.2 Even and Odd Functions

Definition — Parity.

1. A function f is **even** if
for all $x \in \mathcal{D}_f$:
$$f(-x) = f(x).$$
Geometric property: Its curve $\mathcal{C}_f$ is **symmetric with respect to the y-axis (Oy)** (reflectional symmetry). 2. A function f is **odd** if for all $x \in \mathcal{D}_f$:
$$f(-x) = -f(x).$$
Geometric property: Its curve $\mathcal{C}_f$ is **symmetric with respect to the origin O** (half-turn rotational symmetry).

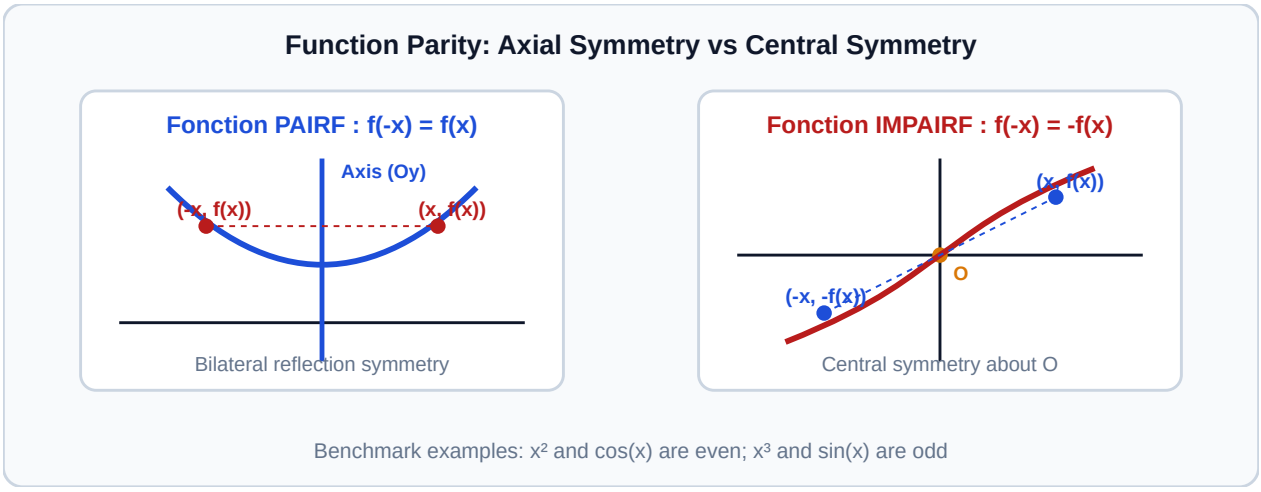

Figure 86: Function parity: bilateral axial symmetry vs central symmetry

---

5. The Five Parent Functions

These are the five cornerstone building blocks from which all high school function machinery is assembled.

5.1 The Square Function: $f(x) = x^2$

- **Domain of definition:** $\mathcal{D}_f = \mathbb{R} =]-\infty\,;+\infty[$.
- **Parity:** For every real x, $(-x)^2 = x^2$. The square function is **even**.
- **Variations:**
  - On $]-\infty\,;0]$, it is strictly decreasing: if $a < b \leq 0$, then $a^2 > b^2$.
  - On $[0\,;+\infty[$, it is strictly increasing: if $0 \leq a < b$, then $a^2 < b^2$.
- **Graph:** An upward-opening **parabola** whose vertex is the origin $O(0\,;0)$. Its absolute minimum is 0, reached at $x = 0$.

Table of variations of $x \mapsto x^2$:

x	$-\infty$		0		$+\infty$
Variations of x^2	$+\infty$	$\searrow$	0	$\nearrow$	$+\infty$

5.2 The Reciprocal Function: $f(x) = \frac{1}{x}$

- **Domain of definition:** $\mathcal{D}_f = \mathbb{R}^* = \mathbb{R} \setminus \{0\} =]-\infty\,;0[\cup]0\,;+\infty[$.
- **Parity:** For all $x \neq 0$, $\frac{1}{-x} = -\frac{1}{x}$. The reciprocal function is **odd**.
- **Variations:**
  - Strictly decreasing on $]-\infty\,;0[$.
  - Strictly decreasing on $]0\,;+\infty[$.
- **Graph:** A **hyperbola** in two distinct branches, point-symmetric about O. The coordinate axes serve as its **asymptotes**.

> ⚠ **Classic Pitfall Warning!**
>
> Never say that the reciprocal function is "decreasing on $\mathbb{R}^*$." Take $a = -1$ and $b = 1$. Clearly $a < b$, but $f(a) = \frac{1}{-1} = -1$ and $f(b) = 1$. The value increased from -1 to $+1$! The function is strictly decreasing **on each of the two intervals taken separately**, but not on their disconnected union.

Table of variations of $x \mapsto \frac{1}{x}$:

x	$-\infty$		0		$+\infty$
Variations of $\frac{1}{x}$	0	$\searrow$	$-\infty \,\|\|\, +\infty$	$\searrow$	0

(The double vertical bar $\|$ denotes the excluded value at 0).

5.3 The Square Root Function: $f(x) = \sqrt{x}$

- **Domain of definition:** $\mathcal{D}_f = [0\,;+\infty[= \mathbb{R}_+$.
- **Parity:** Meaningless (the domain is not symmetric about 0).
- **Variations:** Strictly increasing on $[0\,;+\infty[$.

$$0 \leq a < b \Longrightarrow \sqrt{a} < \sqrt{b}.$$

- **Graph:** Half-branch of a horizontal parabola. It begins at $O(0\,;0)$ with a vertical tangent at the origin.

5.4 The Cube Function: $f(x) = x^3$

- **Domain of definition:** $\mathcal{D}_f = \mathbb{R}$.
- **Parity:** $(-x)^3 = (-1)^3 x^3 = -x^3$. The cube function is **odd**.
- **Variations:** Strictly increasing throughout all of $\mathbb{R}$.

$$a < b \Longrightarrow a^3 < b^3.$$

- **Graph:** Crosses the origin with a horizontal inflection point at $O(0\,;0)$.

5.5 The Absolute Value Function: $f(x) = |x|$

- **Domain of definition:** $\mathcal{D}_f = \mathbb{R}$.
- **Piecewise definition:**

$$|x| = \begin{cases} x & \text{if } \ x \geq 0 \\ -x & \text{if } \ x < 0 \end{cases}$$

- **Parity:** $|-x| = |x|$. The function is **even**.
- **Variations:**
  - Strictly decreasing on $]-\infty\,;0]$ (slope -1).
  - Strictly increasing on $[0\,;+\infty[$ (slope $+1$).
- **Graph:** A sharp **V-shape**, formed by two rays meeting at a right angle at $O(0\,;0)$. Its absolute minimum is 0.

---

6. Graphical Solutions and Function Composition

6.1 Graphical Solutions

1. **Solving $f(x) = k$:** Draw the horizontal line $y = k$. The solutions to the equation are the **x-coordinates** of the intersection points between curve $\mathcal{C}_f$ and line $y = k$.
2. **Solving $f(x) = g(x)$:** The solutions are the **x-coordinates** where curves $\mathcal{C}_f$ and $\mathcal{C}_g$ intersect.
3. **Solving $f(x) \leq g(x)$:** The solutions are the intervals of the x-axis where curve $\mathcal{C}_f$ lies **below** (or touches) curve $\mathcal{C}_g$.

6.2 Cascading Machines: Composition $g \circ f$

When two processing machines are linked in series:

> **📐 Definition — Composite Function.**
>
> Let f and g be two functions. The composite function of f followed by g, denoted $g \circ f$ (read "g of f" or "g circle f"), is defined by:
>
> $$(g \circ f)(x) = g(f(x)).$$
>
> For this operation to be valid, x must belong to $\mathcal{D}_f$, and the intermediate output $f(x)$ must lie within the domain of the second machine: $f(x) \in \mathcal{D}_g$.

> **⚠️ Pitfall Warning! Machine order is crucial.**
>
> In general, $g \circ f \neq f \circ g$. If $f(x) = x + 3$ and $g(x) = x^2$:
> - $(g \circ f)(x) = g(x+3) = (x+3)^2 = x^2 + 6x + 9$.
> - $(f \circ g)(x) = f(x^2) = x^2 + 3$. The two results are completely different!

---

7. Detailed Worked Examples

Worked Example 1 — Determining Domains of Definition.

Determine the domain of definition for each function: a) $f(x) = \frac{2x-1}{x^2-9}$ b) $g(x) = \sqrt{3x-6}$ c) $h(x) = \frac{1}{\sqrt{x}-2}$

Step-by-Step Solution: a) $f(x)$ is defined if and only if its denominator is non-zero:

$$x^2 - 9 \neq 0 \Leftrightarrow (x-3)(x+3) \neq 0 \Leftrightarrow x \neq 3 \quad \text{and} \quad x \neq -3.$$

Hence $\mathcal{D}_f = \mathbb{R} \setminus \{-3\,;3\} =]-\infty\,;-3[\cup]-3\,;3[\cup]3\,;+\infty[$.

b) $g(x)$ contains a square root: the expression under the radical must be non-negative:

$$3x - 6 \geq 0 \Leftrightarrow 3x \geq 6 \Leftrightarrow x \geq 2.$$

Hence $\mathcal{D}_g = [2\,;+\infty[$.

c) For $h(x)$, two conditions must hold simultaneously:

b) Square root exists: $\sqrt{x} \Longrightarrow x \geq 0$.

c) Denominator is non-zero: $\sqrt{x} - 2 \neq 0 \Leftrightarrow \sqrt{x} \neq 2 \Leftrightarrow x \neq 4$. Combining both yields: $\mathcal{D}_h = [0\,;4[\cup]4\,;+\infty[$.

Worked Example 2 — Images and Preimages.

Let f be defined on $\mathbb{R}$ by $f(x) = 2x^2 - 3x - 5$. 1. Calculate the images of -2 and $\frac{1}{2}$ under f. 2. Determine any preimages of -5 under f.

Solution: 1. Computing images:

$$f(-2) = 2(-2)^2 - 3(-2) - 5 = 2(4) + 6 - 5 = 8 + 6 - 5 = 9.$$

$$f\left(\frac{1}{2}\right) = 2\left(\frac{1}{4}\right) - 3\left(\frac{1}{2}\right) - 5 = \frac{1}{2} - \frac{3}{2} - 5 = -1 - 5 = -6.$$

2. To find preimages of -5, solve $f(x) = -5$:

$$2x^2 - 3x - 5 = -5 \Leftrightarrow 2x^2 - 3x = 0.$$

Factoring out x:

$$x(2x-3) = 0 \Leftrightarrow x = 0 \quad \text{or} \quad 2x - 3 = 0 \Longrightarrow x = \frac{3}{2}.$$

The number -5 has two preimages under f: 0 and $\frac{3}{2}$.

Worked Example 3 — Proving Variations Without Derivatives.

Let f be defined on $]1\,;+\infty[$ by $f(x) = \frac{3}{x-1}$. Prove rigorously that f is strictly decreasing on this interval.

Solution: Let a and b be two arbitrary real numbers in $]1\,;+\infty[$ such that $1 < a < b$. Build the expression step by step using inequality rules: 1. Subtract 1:

$$0 < a - 1 < b - 1.$$

2. Apply the reciprocal function, which is strictly decreasing on $]0\,;+\infty[$ (reversing the order of positive numbers):

$$\frac{1}{a-1} > \frac{1}{b-1}.$$

3. Multiply by $3 > 0$ (preserving inequality order):

$$\frac{3}{a-1} > \frac{3}{b-1} \Leftrightarrow f(a) > f(b).$$

Conclusion: Starting from $a < b$, we reached $f(a) > f(b)$. The order is inverted: the function f is **strictly decreasing** on $]1\,;+\infty[$. ■

Worked Example 4 — Algebraic Parity Testing.

Determine the parity of each function: a) $f(x) = \frac{x^4-3}{x^2+1}$ b) $g(x) = \frac{2x}{x^2+4}$ c) $h(x) = x^3 - 2x + 1$

Solution: a) $\mathcal{D}_f = \mathbb{R}$ because $x^2 + 1 > 0$ never vanishes. The domain is symmetric about 0.

$$f(-x) = \frac{(-x)^4 - 3}{(-x)^2 + 1} = \frac{x^4 - 3}{x^2 + 1} = f(x).$$

Hence f is **even**.

b) $\mathcal{D}_g = \mathbb{R}$, symmetric about 0.

$$g(-x) = \frac{2(-x)}{(-x)^2 + 4} = \frac{-2x}{x^2 + 4} = -\frac{2x}{x^2 + 4} = -g(x).$$

Hence g is **odd**.

c) $\mathcal{D}_h = \mathbb{R}$, symmetric about 0.

$$h(-x) = (-x)^3 - 2(-x) + 1 = -x^3 + 2x + 1.$$

Comparing:

- $h(x) = x^3 - 2x + 1 \neq h(-x)$, so h is not even.
- $-h(x) = -x^3 + 2x - 1 \neq h(-x)$ (due to the $+1$ constant), so h is not odd. Conclusion: h is **neither even nor odd**.

Worked Example 5 — Decomposing and Evaluating Composite Functions.

Let $f(x) = 2x - 5$ and $g(x) = \frac{1}{x}$. 1. Determine $(g \circ f)(x)$ and its domain of definition. 2. Determine $(f \circ g)(x)$ and its domain of definition.

Solution: 1. $(g \circ f)(x) = g(f(x)) = g(2x - 5) = \frac{1}{2x-5}$. Existence condition: $2x - 5 \neq 0 \Leftrightarrow x \neq \frac{5}{2}$. Thus $\mathcal{D}_{g\circ f} = \mathbb{R} \setminus \left\{\frac{5}{2}\right\}$.

2. $(f \circ g)(x) = f(g(x)) = f\left(\frac{1}{x}\right) = 2\left(\frac{1}{x}\right) - 5 = \frac{2}{x} - 5$. Existence condition: $x \neq 0$. Thus $\mathcal{D}_{f\circ g} = \mathbb{R}^*$. *(Notice once again that the two composite functions are entirely different).*

Worked Example 6 — Graphical Solving and Parent Function Inversion.

Solve in $\mathbb{R}$: a) $x^2 \leq 7$ b) $\frac{1}{x} > 2$

Solution: a) The square function is decreasing on $]-\infty\,;0]$ and increasing on $[0\,;+\infty[$. Equation $x^2 = 7$ has two roots: $-\sqrt{7}$ and $\sqrt{7}$. The parabola $y = x^2$ lies below the horizontal line $y = 7$ for all x-values between these roots. Thus the solution set is $\boxed{S = [-\sqrt{7}\,;\sqrt{7}]}$.

b) Picture hyperbola $y = \frac{1}{x}$ and horizontal line $y = 2$:

- For $x \leq 0$, $\frac{1}{x} < 0$, which can never exceed 2.
- For $x > 0$, equation $\frac{1}{x} = 2$ gives $x = \frac{1}{2}$. Since the reciprocal function is strictly decreasing on $]0\,;+\infty[$, obtaining a value greater than 2 requires choosing inputs smaller than $\frac{1}{2}$ (while remaining strictly positive). Thus the solution set is $\boxed{S =]0\,;\frac{1}{2}[}$.

---

8. In Computer Science: Value Tables and Root Sweeping in Python

In mathematics and data science, scripts are routinely used to generate values and pinpoint where functions cross zero:

```python
def f(x):
    """Defines function f(x) = x**3 - 3*x + 1"""
    return x**3 - 3*x + 1

def value_table(start, stop, step):
    """Prints a table of values for function f."""
    print("  x   |  f(x)")
    print("------|--------")
    x = start
    while x <= stop + 1e-9:
        print(f"{x:5.2f} | {f(x):7.3f}")
        x += step

# Tabulate values between -2 and 2 with step 0.5
value_table(-2, 2, 0.5)

# Approximate root search for f(x) = 0 by scan search
def scan_search(a, b, step=0.001):
    x = a
    while x <= b:
        if f(x) * f(x + step) <= 0:
            return round(x + step / 2, 3)
        x += step
    return None

solution = scan_search(0, 1)
print(f"\nAn approximate solution on [0, 1] is x ≈ {solution}")
```

Key Takeaways

1. **Function:** $x \mapsto f(x)$. To every preimage $x \in \mathcal{D}_f$ corresponds **exactly one image** $f(x)$. A number may have 0, 1, or multiple preimages.
2. **Domain $\mathcal{D}_f$:**
   - Denominator $\neq 0$.
   - Quantity inside a square root ≥ 0.
3. **Variations:**
   - Increasing: $a < b \Longrightarrow f(a) \leq f(b)$ (preserves order).
   - Decreasing: $a < b \Longrightarrow f(a) \geq f(b)$ (reverses order).
4. **Parity:**
   - Even: $f(-x) = f(x)$, line symmetry across y-axis (Oy).
   - Odd: $f(-x) = -f(x)$, rotational point symmetry about origin O.
5. **The 5 Parent Functions:**
   - Square x^2: $\mathcal{D} = \mathbb{R}$, even, decreasing on $\mathbb{R}_-$, increasing on $\mathbb{R}_+$, vertex $O(0\,;0)$.
   - Reciprocal $\frac{1}{x}$: $\mathcal{D} = \mathbb{R}^*$, odd, decreasing on $]-\infty\,;0[$ and on $]0\,;+\infty[$.
   - Square root $\sqrt{x}$: $\mathcal{D} = [0\,;+\infty[$, increasing.
   - Cube x^3: $\mathcal{D} = \mathbb{R}$, odd, increasing.
   - Absolute value $|x|$: $\mathcal{D} = \mathbb{R}$, even, V-shaped.
6. **Graphical Solution:** $f(x) = k \Leftrightarrow x$-coordinates of points where curve crosses horizontal height k.

Exercises

Direct Application Exercises ()

1. Given $f(x) = x^2 - 4x + 3$. Compute $f(0)$, $f(1)$, $f(2)$, and $f(-1)$.
2. State the domain of definition of $g(x) = \frac{1}{x-7}$.
3. State the domain of definition of $h(x) = \sqrt{2x+8}$.
4. Given $f(x) = 3x - 5$. What is the preimage of 10 under f?
5. Is $f(x) = 5x^4 - 2x^2 + 1$ even, odd, or neither?
6. Is $g(x) = x^3 + 4x$ even, odd, or neither?
7. Solve in $\mathbb{R}$ the equation $x^2 = 25$.
8. Solve in $\mathbb{R}$ the inequality $x^2 < 16$.
9. Solve in $\mathbb{R}^*$ the equation $\frac{1}{x} = 5$.
10. Determine the values of $|-17|$, $|3.14|$, and $|5-9|$.

Practice and Reasoning Exercises ()

11. Determine the domain of definition of:
    a) $f(x) = \frac{x+2}{x^2-4x}$ b) $g(x) = \sqrt{10-2x}$ c) $h(x) = \frac{\sqrt{x+3}}{x-1}$
12. Consider function f defined on $[-4\,;4]$ whose variations are:
    - Decreasing from -4 to -1, falling from 5 to -3.
    - Increasing from -1 to 2, rising from -3 to 4.
    - Decreasing from 2 to 4, falling from 4 to 1.

    a) What is the global maximum of f on $[-4\,;4]$? At what x-value is it reached?
    b) What is the global minimum of f?
    c) How many solutions does $f(x) = 0$ have?
13. Prove without derivatives that $f(x) = x^2 - 6x + 5$ is strictly increasing on $[3\,;+\infty[$. *(Hint: write $f(x)$ as $(x-3)^2 - 4$).*
14. Solve in $\mathbb{R}$ the inequalities:
    a) $x^2 \geq 9$ b) $\frac{1}{x} \leq 3$
15. Investigate the parity of $f(x) = \frac{x^3}{x^2+1}$.
16. Given $f(x) = \sqrt{x}$ and $g(x) = x + 4$. Find explicit formulas for $(g \circ f)(x)$ and $(f \circ g)(x)$, specifying their domains.
17. Solve the equation $|2x - 3| = 7$.
18. Solve the inequality $|x - 5| \leq 2$ by interpreting it as a distance on the number line.

Advanced Problems and Challenges ()

19. **Decomposing Any Function into Even and Odd Parts.** Let f be any function defined on $\mathbb{R}$. Define $p(x) = \frac{f(x)+f(-x)}{2}$ and $i(x) = \frac{f(x)-f(-x)}{2}$.
    a) Prove that p is even and i is odd.
    b) Prove that $f(x) = p(x) + i(x)$.
    c) Apply this decomposition to $f(x) = 2x^2 - 3x + 4$.
20. **Optimizing a Rectangular Enclosure.** A farmer has 40 meters of fencing to construct a rectangular pen against an existing stone wall (so no fence is needed along the wall). Let x denote the length of each of the two sides perpendicular to the wall.
    a) Express the length of the third side in terms of x.
    b) For what values of x is the pen physically realizable?
    c) Express the area $A(x)$ of the pen in terms of x.
    d) Prove that $A(x) = -2(x-10)^2 + 200$.
    e) Deduce the value of x that maximizes area, and give this maximum area.
21. **Arithmetic-Geometric Inequality.** For any strictly positive real $x > 0$, let $f(x) = x + \frac{1}{x}$.
    a) Expand $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2$.

b) Deduce that for all $x > 0$, $x + \frac{1}{x} \geq 2$.
c) For what value of x does equality hold?

22. **Fixed Points of Functions.** A fixed point of a function f is a number x satisfying $f(x) = x$. Find all fixed points of the parent functions:
a) $f(x) = x^2$ b) $g(x) = \frac{1}{x}$ c) $h(x) = \sqrt{x}$ d) $k(x) = x^3$.

Solutions

1. $f(0) = 3$. $f(1) = 1 - 4 + 3 = 0$. $f(2) = 4 - 8 + 3 = -1$. $f(-1) = (-1)^2 - 4(-1) + 3 = 1 + 4 + 3 = 8$.
2. Denominator non-zero: $x - 7 \neq 0 \Leftrightarrow x \neq 7$. Domain: $\mathcal{D}_g = \mathbb{R} \setminus \{7\}$.
3. Expression under radical non-negative: $2x + 8 \geq 0 \Leftrightarrow 2x \geq -8 \Leftrightarrow x \geq -4$. Domain: $\mathcal{D}_h = [-4\,;+\infty[$.
4. $3x - 5 = 10 \Leftrightarrow 3x = 15 \Leftrightarrow x = 5$. Preimage is 5.
5. $f(-x) = 5(-x)^4 - 2(-x)^2 + 1 = 5x^4 - 2x^2 + 1 = f(x)$. The function is even.
6. $g(-x) = (-x)^3 + 4(-x) = -x^3 - 4x = -(x^3 + 4x) = -g(x)$. The function is odd.
7. $x^2 = 25 \Leftrightarrow x = 5$ or $x = -5$. $S = \{-5\,;5\}$.
8. $x^2 < 16 \Leftrightarrow -4 < x < 4$. $S =]-4\,;4[$.
9. $\frac{1}{x} = 5 \Leftrightarrow x = \frac{1}{5} = 0.2$.
10. $|-17| = 17$; $|3.14| = 3.14$; $|5 - 9| = |-4| = 4$.
11. a) $x^2 - 4x = x(x-4) \neq 0 \Longrightarrow x \neq 0$ and $x \neq 4$. $\mathcal{D}_f = \mathbb{R} \setminus \{0\,;4\}$.
b) $10 - 2x \geq 0 \Leftrightarrow 2x \leq 10 \Leftrightarrow x \leq 5$. $\mathcal{D}_g =]-\infty\,;5]$.
c) $x + 3 \geq 0 \Leftrightarrow x \geq -3$ and denominator $x - 1 \neq 0 \Leftrightarrow x \neq 1$. $\mathcal{D}_h = [-3\,;1[\cup]1\,;+\infty[$.
12. a) Global maximum is 5, reached at $x = -4$.
b) Global minimum is -3, reached at $x = -1$.
c) On $[-4\,;-1]$, f decreases from 5 to -3, crossing 0 once. On $[-1\,;2]$, it increases from -3 to 4, crossing 0 a second time. On $[2\,;4]$, values span from 4 down to 1 (strictly positive), so it does not cross 0. The equation $f(x) = 0$ has exactly 2 solutions.
13. Vertex form: $(x-3)^2 - 4 = x^2 - 6x + 9 - 4 = x^2 - 6x + 5$. Let $3 \leq a < b$. Then $0 \leq a - 3 < b - 3$. Since the square function is strictly increasing on $[0\,;+\infty[$, $(a-3)^2 < (b-3)^2$. Subtracting 4: $(a-3)^2 - 4 < (b-3)^2 - 4 \Leftrightarrow f(a) < f(b)$. Order is preserved: f is strictly increasing on $[3\,;+\infty[$.
14. a) $x^2 \geq 9 \Leftrightarrow x \leq -3$ or $x \geq 3$. $S =]-\infty\,;-3] \cup [3\,;+\infty[$.
b) For $\frac{1}{x} \leq 3$:
- If $x < 0$, $\frac{1}{x} < 0 < 3$, so all strictly negative numbers are solutions: $]-\infty\,;0[$.
- If $x > 0$, $\frac{1}{x} \leq 3 \Leftrightarrow x \geq \frac{1}{3}$. Solution set: $S =]-\infty\,;0[\cup[\frac{1}{3}\,;+\infty[$.

15. $\mathcal{D}_f = \mathbb{R}$ (since $x^2 + 1 > 0$). Symmetric about 0. $f(-x) = \frac{(-x)^3}{(-x)^2+1} = \frac{-x^3}{x^2+1} = -f(x)$. The function is odd.
16. $(g \circ f)(x) = g(\sqrt{x}) = \sqrt{x} + 4$, defined on $[0\,;+\infty[$. $(f \circ g)(x) = f(x+4) = \sqrt{x+4}$, defined when $x + 4 \geq 0 \Leftrightarrow x \geq -4$, so domain is $[-4\,;+\infty[$.
17. $|2x - 3| = 7 \Leftrightarrow 2x - 3 = 7$ or $2x - 3 = -7$. $2x = 10 \Longrightarrow x = 5$ or $2x = -4 \Longrightarrow x = -2$. $S = \{-2\,;5\}$.
18. $|x - 5|$ represents distance from x to 5. We seek x within distance 2 of 5: $5 - 2 \leq x \leq 5 + 2 \Leftrightarrow 3 \leq x \leq 7$. $S = [3\,;7]$.
19. a) $p(-x) = \frac{f(-x)+f(-(-x))}{2} = \frac{f(-x)+f(x)}{2} = p(x)$, so p is even. $i(-x) = \frac{f(-x)-f(-(-x))}{2} = \frac{f(-x)-f(x)}{2} = -\frac{f(x)-f(-x)}{2} = -i(x)$, so i is odd.
b) $p(x) + i(x) = \frac{f(x)+f(-x)+f(x)-f(-x)}{2} = \frac{2f(x)}{2} = f(x)$. Every function can be decomposed into an even part and an odd part!
c) For $f(x) = 2x^2 - 3x + 4$: $f(-x) = 2x^2 + 3x + 4$. $p(x) = \frac{(2x^2-3x+4)+(2x^2+3x+4)}{2} = 2x^2 + 4$ (even component). $i(x) = \frac{(2x^2-3x+4)-(2x^2+3x+4)}{2} = \frac{-6x}{2} = -3x$ (odd component).
20. a) Let x be the width of each perpendicular side. The remaining length for the side opposite the wall is $40 - 2x$.
b) We require $x > 0$ and $40 - 2x > 0 \Leftrightarrow 2x < 40 \Leftrightarrow x < 20$. Thus $x \in]0\,;20[$.

c) Area: $A(x) = x(40 - 2x) = 40x - 2x^2$.

d) Expand: $-2(x-10)^2 + 200 = -2(x^2 - 20x + 100) + 200 = -2x^2 + 40x - 200 + 200 = 40x - 2x^2 = A(x)$.

e) Since $(x-10)^2 \geq 0$, $-2(x-10)^2 \leq 0$, hence $A(x) \leq 200$. The maximum occurs when the squared term vanishes: $x - 10 = 0 \Leftrightarrow x = 10$ m. Maximum area is 200 m^2 (dimensions: 10 m $\times$ 20 m).

21. a) $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 = (\sqrt{x})^2 - 2 \times \sqrt{x} \times \frac{1}{\sqrt{x}} + \left(\frac{1}{\sqrt{x}}\right)^2 = x - 2 + \frac{1}{x}$.

b) Any real square is non-negative: $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 \geq 0 \Leftrightarrow x - 2 + \frac{1}{x} \geq 0 \Leftrightarrow x + \frac{1}{x} \geq 2$.

c) Equality holds if and only if $\sqrt{x} - \frac{1}{\sqrt{x}} = 0 \Leftrightarrow \sqrt{x} = \frac{1}{\sqrt{x}} \Leftrightarrow (\sqrt{x})^2 = 1 \Leftrightarrow x = 1$.

22. a) $x^2 = x \Leftrightarrow x^2 - x = 0 \Leftrightarrow x(x-1) = 0 \Longrightarrow x = 0$ or $x = 1$.

b) $\frac{1}{x} = x \Leftrightarrow x^2 = 1 \Leftrightarrow x = 1$ or $x = -1$.

c) $\sqrt{x} = x$ for $x \geq 0 \Leftrightarrow x = x^2 \Longrightarrow x = 0$ or $x = 1$.

d) $x^3 = x \Leftrightarrow x^3 - x = 0 \Leftrightarrow x(x^2 - 1) = 0 \Leftrightarrow x(x-1)(x+1) = 0 \Longrightarrow x = 0, 1$ or -1.

The Question That Opens the Future

In the farmer's pen problem, we encountered a quadratic expression: $A(x) = -2x^2 + 40x$. To find its crest without guesswork, we used the vertex form $-2(x-10)^2 + 200$.

This family of curves—the parabolas defined by quadratic polynomials $ax^2 + bx + c$—governs the trajectories of artillery shells, water jets in ornamental fountains, and the reflective mirrors of astronomical telescopes. How can we systematically find their vertices and roots without relying on trial-and-error tricks? What is the mysterious quantity Δ that geometers unlocked? That is what we will explore in **Galileo's Fountain and the Parabola**.

CHAPTER 35: Galileo's Fountain and the Parabola

> "Watch the water surge from the marble basin: it leaps forth with fire, seems to hesitate for an instant at the summit of its luminous arch, then falls back in a graceful curve into the pool. This perfect drawing that gravity traces in the air is not an arc of a circle: it is a parabola. From the scholars of Babylon and Baghdad to Galileo, the art of the second degree is the master key that unlocks the trajectories of worlds."

What You Are About to Discover

- Why the trajectory of a water jet or a projectile subject to gravity inevitably forms a **parabola**.
- The metamorphosis of the expression: moving from the **expanded standard form** $ax^2 + bx + c$ to the **canonical (vertex) form** $a(x - \alpha)^2 + \beta$ through Al-Khwarizmi's completion of the square.
- The **vertex** $S(\alpha, \beta)$ and the axis of symmetry of a parabola.
- The unveiled mystery of the **discriminant** $\Delta = b^2 - 4ac$: the step-by-step proof that explains why it dictates whether there are 0, 1, or 2 roots.
- The **factored form** and the golden rule for the **sign of the quadratic trinomial** ("the sign of a outside the roots").
- **Viète's formulas** relating the roots to their sum $S = -\frac{b}{a}$ and product $P = \frac{c}{a}$.

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1. The Problem: Galileo and the Water Jet of the Medici Gardens

In Florence, at the beginning of the summer of 1608, **Galileo Galilei** is strolling through the gardens of the Villa Medici. His gaze is drawn to a monumental fountain. A brass nozzle directs a jet of water at forty-five degrees: a slender crystalline stream shoots forth, rises toward the sky, rounds off at its zenith, and plunges back into the basin.

At that time, treatises on artillery inherited from Aristotelian physics claimed that a cannonball traveled in a straight line under the effect of the gunpowder blast, came to an abrupt halt in midair once its impetus was exhausted, and then dropped vertically to the ground!

Galileo knows this vision is absurd. He decomposes the motion of each droplet of water into two simultaneous, independent actions: 1. **Horizontally (x):** No obstacle impedes the water (neglecting air resistance). The droplet advances at a constant horizontal speed v_0:

$$x(t) = v_0 \times t.$$

2. **Vertically (y):** The Earth pulls the droplet downward with a constant acceleration g. According to Galileo's law of falling bodies, the altitude decreases proportionally to the **square of elapsed time**:

$$y(t) = -\frac{1}{2}gt^2 + h_0.$$

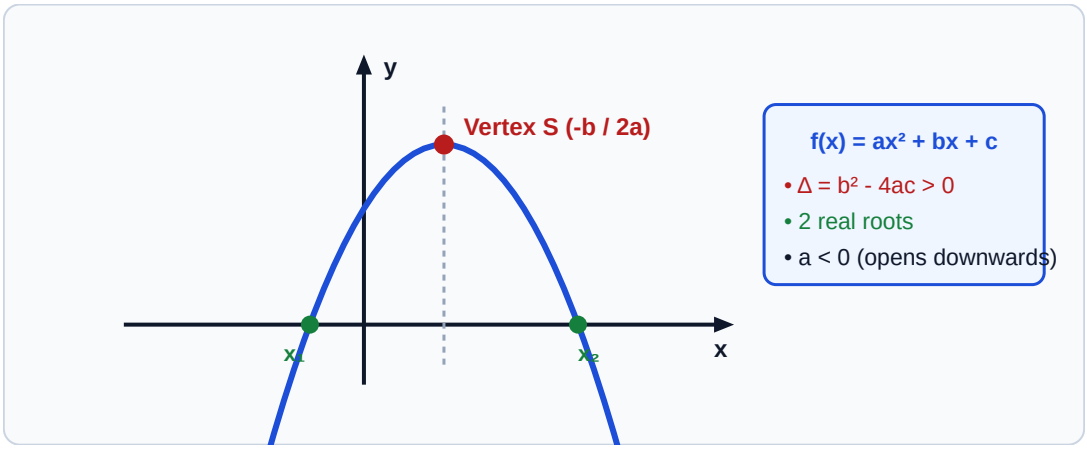

Figure 87: Quadratic parabola: orientation and vertex coordinates

Let us eliminate time t! From the first equation, we get $t = \frac{x}{v_0}$. Substituting this value into the second equation, the altitude y as a function of horizontal distance x becomes:

$$y = -\frac{g}{2v_0^2}x^2 + h_0.$$

This is an equation in which altitude depends on the **square** of the distance traveled: a **quadratic polynomial function** (a polynomial of the second degree). The resulting curve is neither a circle nor an ellipse: it is a **parabola**.

To calculate the maximum height reached by the water, or to determine the exact distance at which the jet hits the surface of the basin ($y = 0$), we must master the algebra of quadratic trinomials.

---

2. The Quadratic Trinomial and Canonical (Vertex) Form

2.1 Definitions and Standard Form

📐 Definition — Quadratic Polynomial Function.

A **quadratic polynomial function** (or **quadratic trinomial**) is any function f defined on $\mathbb{R}$ by:

$$f(x) = ax^2 + bx + c$$

where a, b, c are fixed real numbers, with $\boldsymbol{a \neq 0}$. This expression is called the **standard form** (or expanded form) of the trinomial.

- The term a is the quadratic coefficient (we must have $a \neq 0$, otherwise the expression would reduce to an affine function $bx + c$).
- The term b is the linear coefficient (degree 1).
- The term c is the constant term (the y-intercept, since $f(0) = c$).

2.2 Completing the Square: Al-Khwarizmi's Legacy

How can we find the vertex of a parabola without guesswork? The brilliant idea dates back to the Persian mathematician **Al-Khwarizmi** (around 820 in Baghdad): it consists in geometrically reconstructing a perfect square so that the variable x appears **only once** in the entire expression!

Fundamental Theorem — The Canonical (Vertex) Form.

Every quadratic trinomial $f(x) = ax^2 + bx + c$ ($a \neq 0$) can be uniquely written in **canonical (vertex) form**:

$$\boxed{f(x) = a(x - \alpha)^2 + \beta}$$

where:

$$\alpha = -\frac{b}{2a} \quad \text{and} \quad \beta = f(\alpha) = -\frac{b^2 - 4ac}{4a}.$$

Proof (Required knowledge) — Conversion to Canonical Form.

Start with the standard form:

$$f(x) = ax^2 + bx + c.$$

1. Factor out the coefficient $a \neq 0$ from the first two terms:

$$f(x) = a\left[x^2 + \frac{b}{a}x\right] + c.$$

2. Look at the expression inside the brackets: $x^2 + \frac{b}{a}x$ resembles the beginning of the expansion of a binomial square $(x + d)^2 = x^2 + 2dx + d^2$. To make the cross-term $2dx$ equal to $\frac{b}{a}x$, we must choose $d = \frac{b}{2a}$. We then have:

$$\left(x + \frac{b}{2a}\right)^2 = x^2 + 2\left(\frac{b}{2a}\right)x + \left(\frac{b}{2a}\right)^2 = x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}.$$

From this, we deduce that:

$$x^2 + \frac{b}{a}x = \left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2}.$$

3. Substitute this identity back into the expression for $f(x)$:

$$f(x) = a\left[\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2}\right] + c.$$

4. Distribute a across the brackets:

$$f(x) = a\left(x + \frac{b}{2a}\right)^2 - a \times \frac{b^2}{4a^2} + c = a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a} + \frac{4ac}{4a}.$$

5. Combine the last two terms over the common denominator $4a$:

$$f(x) = a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a}.$$

Setting $\alpha = -\frac{b}{2a}$ and $\beta = -\frac{b^2-4ac}{4a}$, we obtain the canonical form:

$$f(x) = a(x - \alpha)^2 + \beta. \quad \blacksquare$$

---

3. Geometry of the Parabola: Vertex and Axis of Symmetry

Thanks to the canonical form $f(x) = a(x - \alpha)^2 + \beta$, the geometric properties of the curve become crystal clear: Since a square of a real number is always non-negative:

$$(x - \alpha)^2 \geq 0 \quad \text{for all} \quad x \in \mathbb{R},$$

with equality if and only if $x = \alpha$.

Two cases arise depending on the sign of a:

1. **If $a > 0$:** $a(x-\alpha)^2 \geq 0$, hence $f(x) \geq \beta$. The function attains an **absolute minimum** equal to β, reached at $x = \alpha$. The parabola **opens upward** (bowl shape). f is strictly decreasing on $]-\infty, \alpha]$ and strictly increasing on $[\alpha, +\infty[$.
2. **If $a < 0$:** $a(x-\alpha)^2 \leq 0$, hence $f(x) \leq \beta$. The function attains an **absolute maximum** equal to β, reached at $x = \alpha$. The parabola **opens downward** (hill shape). f is strictly increasing on $]-\infty, \alpha]$ and strictly decreasing on $[\alpha, +\infty[$.

> 📐 ***Fundamental Properties of the Parabola —***
> - *The **vertex** of the parabola is the point $S(\alpha, \beta)$ with $\alpha = -\frac{b}{2a}$ and $\beta = f(\alpha)$.*
> - *The vertical line with equation $x = \alpha$ is the **axis of symmetry** of the parabola.*

---

4. The Quadratic Equation and the Discriminant Δ

Solving the equation $ax^2 + bx + c = 0$ corresponds geometrically to finding the points where the parabola intersects the horizontal axis ($y = 0$).

4.1 The Birth of the Discriminant

Let us return to Step 3 of completing the square:

$$a\left[\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a^2}\right] = 0.$$

Since $a \neq 0$, this equation is equivalent to:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}.$$

The left-hand side is a perfect square: it is always greater than or equal to zero. The denominator on the right-hand side, $4a^2 = (2a)^2$, is strictly positive. Everything therefore depends exclusively on the sign of the numerator:

$$\Delta = b^2 - 4ac.$$

> 📐 **Definition — Discriminant.**
>
> The real number $\Delta = b^2 - 4ac$ is called the **discriminant** of the trinomial $ax^2 + bx + c$. The uppercase Greek letter Δ ("Delta") was chosen by nineteenth-century mathematicians because it is the initial letter of *discriminare* in Latin, which means "to separate", "to distinguish". It discriminates between the possible outcomes.

4.2 The Three Fundamental Cases

Fundamental Theorem — Solving $ax^2 + bx + c = 0$.

1. **If $\Delta < 0$:** The right-hand side $\frac{\Delta}{4a^2}$ is

strictly negative. But a real square can never be negative. The equation has **no real solution** ($S = \varnothing$). The parabola never crosses or touches the horizontal axis.

1. **If $\Delta = 0$:** The equation becomes $\left(x + \frac{b}{2a}\right)^2 = 0$, whence $x + \frac{b}{2a} = 0$. The equation has **a unique real solution** (called a double root):

$$x_0 = -\frac{b}{2a}.$$

The vertex of the parabola rests exactly on the horizontal axis (the axis is tangent to the curve).

2. **If $\Delta > 0$:** The number Δ possesses a positive real square root $\sqrt{\Delta} > 0$. The equation becomes:

$$\left(x + \frac{b}{2a}\right)^2 - \left(\frac{\sqrt{\Delta}}{2a}\right)^2 = 0.$$

Using the difference of two squares $A^2 - B^2 = (A - B)(A + B)$:

$$\left(x + \frac{b}{2a} - \frac{\sqrt{\Delta}}{2a}\right)\left(x + \frac{b}{2a} + \frac{\sqrt{\Delta}}{2a}\right) = 0.$$

By the zero-product property, the equation admits **two distinct real solutions**:

$$\boxed{x_1 = \frac{-b - \sqrt{\Delta}}{2a}} \quad \text{and} \quad \boxed{x_2 = \frac{-b + \sqrt{\Delta}}{2a}}.$$

The parabola crosses the horizontal axis at two distinct points with coordinates $(x_1, 0)$ and $(x_2, 0)$.

---

5. Factored Form and the Sign of the Trinomial

5.1 Factored Form

Knowing the roots of a trinomial allows us to write it as a product of first-degree factors:

Theorem — Factored Form.

- If $\Delta > 0$: $f(x) = a(x - x_1)(x - x_2)$.
- If $\Delta = 0$: $f(x) = a(x - x_0)^2$.
- If $\Delta < 0$: $f(x)$ cannot be factored into a product of linear factors in $\mathbb{R}$.

Beware: Common Pitfall! Never forget the leading coefficient a!

This is the most common error: writing $(x - x_1)(x - x_2)$ while forgetting to multiply by the coefficient a in front. For example, $2x^2 - 6x + 4 = 0$ has roots 1 and 2. Its factored form is $2(x - 1)(x - 2)$, and not $(x - 1)(x - 2)$ (which would expand to $x^2 - 3x + 2$).

5.2 The Sign of the Trinomial

Golden Rule — Sign of $ax^2 + bx + c$.

1. **If $\Delta < 0$:** The trinomial never vanishes. It
always has the same sign as a over all of $\mathbb{R}$.
1. **If $\Delta = 0$:** The trinomial vanishes at $x_0 = -\frac{b}{2a}$ and **has the same sign as a everywhere else**.
2. **If $\Delta > 0$:** (assuming $x_1 < x_2$) The trinomial is:
   - **of the sign of a outside the roots** (for $x < x_1$ and for $x > x_2$);
   - **of the opposite sign of a (sign of $-a$) between the roots** (for $x_1 < x < x_2$).

Summary Sign Chart for $\Delta > 0$:

x	$-\infty$		x_1		x_2		$+\infty$
Sign of $ax^2 + bx + c$		***sign of a***	0	***sign of $-a$***	0	***sign of a***	

Vivid Geometric Intuition: If $a > 0$, the parabola opens its arms toward the sky. To dip below the horizontal axis, it must plunge between the two roots: the bowl is below the axis (negative) exclusively *between* x_1 and x_2. Everywhere else, it stays above the axis (positive)!

---

6. Viète's Formulas: Sum and Product of the Roots

In 1591, the French jurist and cryptographer to King Henry IV, **François Viète**, discovered fascinating connections directly linking the coefficients of a polynomial to its roots, without calculating Δ.

Theorem (Viète's Formulas) —

If the trinomial $ax^2 + bx + c$ admits two real roots x_1 and x_2 (which may coincide if $\Delta = 0$), then:

$$\boxed{S = x_1 + x_2 = -\frac{b}{a}} \quad \text{and} \quad \boxed{P = x_1 \times x_2 = \frac{c}{a}}.$$

Proof —

Expand the factored form $a(x - x_1)(x - x_2)$:

$$a(x - x_1)(x - x_2) = a\left[x^2 - (x_1 + x_2)x + x_1x_2\right] = ax^2 - a(x_1 + x_2)x + a(x_1x_2).$$

Matching coefficients with the standard form $ax^2 + bx + c$:

$$-a(x_1 + x_2) = b \Longrightarrow x_1 + x_2 = -\frac{b}{a},$$

$$a(x_1x_2) = c \Longrightarrow x_1x_2 = \frac{c}{a}. \quad \blacksquare$$

6.1 Finding Two Numbers Knowing Their Sum and Product

Two numbers u and v have sum S and product P if and only if they are the solutions to the quadratic equation:

$$\boxed{X^2 - SX + P = 0}$$

6.2 The Art of Spotting Obvious Roots

Before rushing to compute the discriminant, always check if an **obvious root** exists:

- **If $a + b + c = 0$**, then $x_1 = 1$ is an obvious root. The second root is then immediately $x_2 = \frac{c}{a}$.
- **If $a - b + c = 0$**, then $x_1 = -1$ is an obvious root. The second root is $x_2 = -\frac{c}{a}$.
- **If $c = 0$**, then $x_1 = 0$ is an obvious root, and $x_2 = -\frac{b}{a}$.

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7. Detailed Worked Examples

Worked Example 1 — Converting to Canonical Form in Practice.

Write the trinomial $P(x) = 2x^2 - 8x + 11$ in canonical form. Deduce the coordinates of the vertex of the parabola and its extremum.

Solution: 1. Factor the leading coefficient $a = 2$ out of the terms in x:

$$P(x) = 2(x^2 - 4x) + 11.$$

2. Complete the square: $x^2 - 4x = (x - 2)^2 - 4$. 3. Substitute back:

$$P(x) = 2\big[(x - 2)^2 - 4\big] + 11 = 2(x - 2)^2 - 8 + 11 = 2(x - 2)^2 + 3.$$

Conclusion: The canonical form is $\boxed{P(x) = 2(x - 2)^2 + 3}$. Here $\alpha = 2$ and $\beta = 3$. The vertex of the parabola is the point $S(2, 3)$. Since $a = 2 > 0$, the function achieves an **absolute minimum equal to 3**, attained at $x = 2$. (Note: since its minimum is $3 > 0$, the parabola never dips below 0, so $P(x) = 0$ has no real solution).

Worked Example 2 — Solving Equations with the Discriminant.

Solve the following three equations in $\mathbb{R}$: a) $3x^2 - 5x - 2 = 0$ b) $4x^2 - 12x + 9 = 0$ c) $2x^2 + 3x + 5 = 0$

Step-by-Step Solution: a) $a = 3, b = -5, c = -2$.

$$\Delta = b^2 - 4ac = (-5)^2 - 4(3)(-2) = 25 + 24 = 49.$$

Since $\Delta = 49 > 0$, there are two distinct real solutions ($\sqrt{\Delta} = 7$):

$$x_1 = \frac{-(-5) - 7}{2(3)} = \frac{5 - 7}{6} = \frac{-2}{6} = -\frac{1}{3},$$

$$x_2 = \frac{-(-5) + 7}{2(3)} = \frac{5 + 7}{6} = \frac{12}{6} = 2.$$

$S = \left\{-\frac{1}{3}, 2\right\}$. Factored form: $3\left(x + \frac{1}{3}\right)(x - 2) = (3x + 1)(x - 2)$.

b) $a = 4, b = -12, c = 9$.

$$\Delta = (-12)^2 - 4(4)(9) = 144 - 144 = 0.$$

Since $\Delta = 0$, there is a unique double root:

$$x_0 = -\frac{b}{2a} = -\frac{-12}{2(4)} = \frac{12}{8} = \frac{3}{2}.$$

$S = \left\{\frac{3}{2}\right\}$. Factored form: $4\left(x - \frac{3}{2}\right)^2 = (2x - 3)^2$.

c) $a = 2, b = 3, c = 5$.

$$\Delta = 3^2 - 4(2)(5) = 9 - 40 = -31 < 0.$$

Since $\Delta < 0$, the equation has no real solutions: $S = \varnothing$.

Worked Example 3 — Solving a Quadratic Inequality.

Solve the following inequality in $\mathbb{R}$:

$$-2x^2 + 7x - 3 \geq 0.$$

Solution: 1. First, find the roots of the trinomial:

$$\Delta = 7^2 - 4(-2)(-3) = 49 - 24 = 25 > 0 \Longrightarrow \sqrt{\Delta} = 5.$$

$$x_1 = \frac{-7-5}{2(-2)} = \frac{-12}{-4} = 3, \quad x_2 = \frac{-7+5}{2(-2)} = \frac{-2}{-4} = \frac{1}{2}.$$

2. Apply the rule for the sign of a quadratic trinomial: Here, $a = -2 < 0$. The trinomial has the sign of a (negative) outside the roots, and the opposite sign (positive) between the roots. 3. Sign chart:

x	$-\infty$		$\frac{1}{2}$		3		$+\infty$
$-2x^2 + 7x - 3$		$-$	0	$+$	0	$-$	

We want the expression to be ≥ 0. The solution set is therefore the closed interval: $\boxed{S = \left[\frac{1}{2}, 3\right]}$.

Worked Example 4 — Using Obvious Roots and Viète.

Solve the equation $15x^2 - 2x - 17 = 0$ mentally.

Solution: Test the sum of the coefficients: $a = 15, b = -2, c = -17$. Compute $a - b + c = 15 - (-2) + (-17) = 15 + 2 - 17 = 0$. According to Section 6.2, $x_1 = -1$ is an obvious root! The product of the roots is $P = x_1 x_2 = \frac{c}{a} = -\frac{17}{15}$. Since $x_1 = -1$, we have $(-1) \times x_2 = -\frac{17}{15} \Longrightarrow x_2 = \frac{17}{15}$. The solutions are immediately $\boxed{S = \left\{-1, \frac{17}{15}\right\}}$, without computing any discriminant!

Worked Example 5 — Finding Dimensions from Perimeter and Area.

A rectangular plot of land has a semi-perimeter of $29 \ \text{meters}$ and an area of $198 \ \text{m}^2$. What are the dimensions of this plot?

Solution: Let L and ℓ be the length and width. We know their sum $S = L + \ell = 29$ and their product $P = L \times \ell = 198$. By Viète's theorem, L and ℓ are the solutions of the equation:

$$X^2 - 29X + 198 = 0.$$

Compute the discriminant:

$$\Delta = (-29)^2 - 4(1)(198) = 841 - 792 = 49 > 0.$$

$\sqrt{\Delta} = 7$. The solutions are:

$$X_1 = \frac{29-7}{2} = \frac{22}{2} = 11, \quad X_2 = \frac{29+7}{2} = \frac{36}{2} = 18.$$

The dimensions of the plot are therefore $\mathbf{18 \ \text{meters by} \ 11 \ \text{meters}}$. *(Check: semi-perimeter* $18 + 11 = 29 \ \text{m}$*; area* $18 \times 11 = 198 \ \text{m}^2$*).*

Worked Example 6 — Biquadratic Equation Reducible to a Quadratic.

Solve the following equation in $\mathbb{R}$:

$$x^4 - 5x^2 - 36 = 0.$$

Solution: This equation has degree 4, but it contains only even powers ($x^4 = (x^2)^2$). Let us make a **substitution**: set $X = x^2$ (with the essential constraint $X \geq 0$). The equation becomes a quadratic equation in X:

$$X^2 - 5X - 36 = 0.$$

Compute the discriminant:

$$\Delta = (-5)^2 - 4(1)(-36) = 25 + 144 = 169 = 13^2.$$

The roots in X are:

$$X_1 = \frac{5 - 13}{2} = \frac{-8}{2} = -4, \quad X_2 = \frac{5 + 13}{2} = \frac{18}{2} = 9.$$

Now return to the original variable x:

- For $X_1 = -4$: the equation $x^2 = -4$ has no real solution because a real square cannot be negative.
- For $X_2 = 9$: the equation $x^2 = 9$ immediately gives $x = -3$ or $x = 3$. The set of real solutions to the original equation is $\boxed{S = \{-3, 3\}}$.

---

8. In Computer Science: The Quadratic Solver in Python

Here is a Python script that implements the full analysis algorithm for a quadratic trinomial:

```python
import math

def analyze_trinomial(a, b, c):
    """Complete analysis of the trinomial a*x^2 + b*x + c."""
    if a == 0:
        return "Error: not a degree-2 polynomial (a = 0)."

    # 1. Vertex of the parabola
    alpha = -b / (2 * a)
    beta = a * alpha**2 + b * alpha + c
    print(f"Vertex S: ({alpha:.3f}, {beta:.3f})")
    print(f"Orientation: branches point {'UPWARD' if a > 0 else 'DOWNWARD'}")

    # 2. Discriminant
    delta = b**2 - 4 * a * c
    print(f"Discriminant Delta = {delta}")

    if delta > 0:
        x1 = (-b - math.sqrt(delta)) / (2 * a)
        x2 = (-b + math.sqrt(delta)) / (2 * a)
        print(f"Two distinct real roots: x1 = {x1:.3f} and x2 = {x2:.3f}")
        print(f"Factored form: {a} * (x - {x1:.3f}) * (x - {x2:.3f})")
    elif delta == 0:
        x0 = -b / (2 * a)
        print(f"One double real root: x0 = {x0:.3f}")
        print(f"Factored form: {a} * (x - {x0:.3f})^2")
    else:
        print("No real root.")
        print("Cannot be factored over R.")

# Example run: 2x^2 - 7x + 3
analyze_trinomial(2, -7, 3)
```

Key Takeaways

1. **Standard Form:** $f(x) = ax^2 + bx + c$ ($a \neq 0$).
2. **Canonical (Vertex) Form:** $f(x) = a(x - \alpha)^2 + \beta$ where:

$$\alpha = -\frac{b}{2a} \quad \text{and} \quad \beta = f(\alpha) = -\frac{\Delta}{4a}.$$

   - Vertex of the parabola: $S(\alpha, \beta)$.
   - Axis of symmetry: line $x = \alpha$.
   - Opens upward if $a > 0$ (minimum β), opens downward if $a < 0$ (maximum β).
3. **Discriminant $\Delta = b^2 - 4ac$:**
   - $\Delta > 0$: 2 distinct real roots $x_1 = \frac{-b-\sqrt{\Delta}}{2a}$ and $x_2 = \frac{-b+\sqrt{\Delta}}{2a}$.
   - $\Delta = 0$: 1 double real root $x_0 = -\frac{b}{2a}$.
   - $\Delta < 0$: 0 real roots.
4. **Factored Form:**
   - If $\Delta > 0$: $a(x - x_1)(x - x_2)$.
   - If $\Delta = 0$: $a(x - x_0)^2$.
5. **Sign of the Trinomial:**
   - Always of the same sign as a, **except between the roots** (if $\Delta > 0$) where it takes the sign of $-a$.
6. **Viète's Formulas:**

$$S = x_1 + x_2 = -\frac{b}{a}, \quad P = x_1 x_2 = \frac{c}{a}, \quad X^2 - SX + P = 0.$$

Exercises

Direct Application Exercises ()

1. Compute the discriminant Δ of the trinomial $f(x) = x^2 - 6x + 8$.
2. Compute the discriminant Δ of the trinomial $g(x) = 2x^2 + 4x + 5$.
3. Determine the coordinates of the vertex S of the parabola with equation $y = 3(x-4)^2 + 7$.
4. Determine the coordinates of the vertex S of the parabola with equation $y = -2x^2 + 8x - 3$.
5. Solve the equation $x^2 - 9x + 20 = 0$ in $\mathbb{R}$.
6. Solve the equation $x^2 + 2x + 1 = 0$ in $\mathbb{R}$.
7. Solve the equation $3x^2 + x + 2 = 0$ in $\mathbb{R}$.
8. Give the factored form of $2x^2 - 10x + 12$.
9. Construct the sign chart for the trinomial $P(x) = x^2 - 4x - 5$.
10. Find the roots of $2x^2 + 3x - 5 = 0$ mentally using obvious roots.

Practice and Reasoning Exercises ()

11. Put the following trinomials into canonical form:
    a) $f(x) = x^2 - 10x + 21$ b) $g(x) = -3x^2 + 12x - 7$
12. Solve the following inequalities in $\mathbb{R}$:
    a) $x^2 - 5x + 6 \leq 0$ b) $-x^2 + 2x + 8 > 0$ c) $2x^2 - 3x + 5 > 0$
13. Find two real numbers knowing their sum $S = 13$ and their product $P = 40$.
14. For what values of the real parameter m does the equation $x^2 + (m+1)x + 4 = 0$ have a double root?
15. Solve the rational equation: $\frac{2x-1}{x+1} = \frac{x+2}{x-1}$.
16. Solve the biquadratic equation in $\mathbb{R}$: $x^4 - 13x^2 + 36 = 0$.
17. A tennis ball is hit from a height of 1 meter. Its trajectory is modeled by $h(x) = -0.05x^2 + 0.8x + 1$, where x is the horizontal distance traveled in meters and $h(x)$ is the height in meters.
    a) What is the maximum height reached by the ball?
    b) At what horizontal distance does the ball hit the ground ($h(x) = 0$)?
18. Determine the coordinates of the intersection points between the parabola $y = x^2 - 3x + 2$ and the line $y = 2x - 4$.

Deepening Exercises and Challenges ()

19. **The Reduced Discriminant Formula Δ'.** When the linear coefficient b is even, we can write $b = 2b'$.
    a) Prove that $\Delta = 4(b'^2 - ac)$.
    b) Setting $\Delta' = b'^2 - ac$, prove that the roots simplify to $x = \frac{-b' \pm \sqrt{\Delta'}}{a}$.
    c) Use this simplified formula to solve $x^2 - 14x + 45 = 0$ and $5x^2 - 6x - 8 = 0$.
20. **Relative Position of Two Parabolas.** Consider the parabolas $\mathcal{P}_1 : y = x^2 - 4x + 3$ and $\mathcal{P}_2 : y = -x^2 + 2x + 11$.
    a) Determine the x-coordinates of the intersection points of these two curves.
    b) Study the sign of the difference and deduce on which interval $\mathcal{P}_1$ lies below $\mathcal{P}_2$.
21. **Lifeguard Optimization Problem.** A lifeguard marks off a rectangular swimming area along a straight beach using a rope of buoys 160 meters long (the beach side needs no buoys). What dimensions should the swimming area have so that the water surface enclosed is as large as possible?
22. **Tartaglia's Open Box.** From a square sheet of cardboard measuring 30 cm on each side, four identical small squares of side length x are cut out from the four corners. The sides are then folded up to form an open rectangular box.
    a) What are the possible values for x?
    b) Express the volume $V(x)$ of the box as a function of x.

c) Expand the expression and determine for which integer value of x the volume appears to be maximized by testing $x = 1, 2, 3, 4, 5, 6$.

Solutions

1. $\Delta = (-6)^2 - 4(1)(8) = 36 - 32 = 4$.
2. $\Delta = 4^2 - 4(2)(5) = 16 - 40 = -24$.
3. The equation is already in canonical form: the vertex is directly $S(4, 7)$.
4. $\alpha = -\frac{b}{2a} = -\frac{8}{2(-2)} = 2$. $\beta = -2(2)^2 + 8(2) - 3 = -8 + 16 - 3 = 5$. Thus $S(2, 5)$.
5. $\Delta = (-9)^2 - 4(1)(20) = 81 - 80 = 1$. $x_1 = \frac{9-1}{2} = 4, x_2 = \frac{9+1}{2} = 5$. $S = \{4, 5\}$.
6. Perfect square: $(x + 1)^2 = 0 \Longrightarrow x_0 = -1$. $S = \{-1\}$.
7. $\Delta = 1^2 - 4(3)(2) = 1 - 24 = -23 < 0$. No real solution: $S = \varnothing$.
8. Roots of $2x^2 - 10x + 12 = 0$: $\Delta = 100 - 96 = 4$. $x_1 = 2, x_2 = 3$. Factored form: $2(x - 2)(x - 3)$.
9. Roots of $x^2 - 4x - 5$: $a - b + c = 1 - (-4) - 5 = 0 \Longrightarrow x_1 = -1$ and $x_2 = 5$. Since $a = 1 > 0$, the trinomial is negative between -1 and 5, and positive outside: positive on $]-\infty, -1[\cup]5, +\infty[$, zero at -1 and 5, negative on $]-1, 5[$.
10. $a + b + c = 2 + 3 - 5 = 0 \Longrightarrow x_1 = 1$. Product $P = \frac{c}{a} = -\frac{5}{2}$, hence $x_2 = -\frac{5}{2} = -2.5$. $S = \left\{-\frac{5}{2}, 1\right\}$.
11. a) $x^2 - 10x + 21 = (x - 5)^2 - 25 + 21 = (x - 5)^2 - 4$.
    b) $-3x^2 + 12x - 7 = -3(x^2 - 4x) - 7 = -3[(x - 2)^2 - 4] - 7 = -3(x - 2)^2 + 12 - 7 = -3(x - 2)^2 + 5$.
12. a) Roots: 2 and 3. $a = 1 > 0$. Negative between the roots: $S = [2, 3]$.
    b) $-x^2 + 2x + 8 = 0 \Longrightarrow \Delta = 4 - 4(-1)(8) = 36$. Roots: $x_1 = \frac{-2-6}{-2} = 4, x_2 = \frac{-2+6}{-2} = -2$. $a = -1 < 0$, so positive between the roots: $S =]-2, 4[$.
    c) $\Delta = 9 - 4(2)(5) = -31 < 0$. Since $a = 2 > 0$, the trinomial is strictly positive for all $x \in \mathbb{R}$: $S = \mathbb{R}$.
13. Equation $X^2 - 13X + 40 = 0$. $\Delta = 169 - 160 = 9$. $X_1 = \frac{13-3}{2} = 5, X_2 = \frac{13+3}{2} = 8$. The numbers are 5 and 8.
14. Double root $\Leftrightarrow \Delta = 0$. $\Delta = (m + 1)^2 - 4(1)(4) = (m + 1)^2 - 16 = 0 \Leftrightarrow (m + 1)^2 = 16 \Leftrightarrow m + 1 = 4$ or $m + 1 = -4$. Thus $m = 3$ or $m = -5$.
15. Excluded values: $x \neq -1$ and $x \neq 1$. Cross-multiplication: $(2x - 1)(x - 1) = (x + 2)(x + 1)$. $2x^2 - 3x + 1 = x^2 + 3x + 2 \Leftrightarrow x^2 - 6x - 1 = 0$. $\Delta = 36 - 4(1)(-1) = 40$. $x = \frac{6 \pm 2\sqrt{10}}{2} = 3 \pm \sqrt{10}$. Since both values differ from 1 and -1, $S = \left\{3 - \sqrt{10}, 3 + \sqrt{10}\right\}$.
16. Set $X = x^2 \geq 0$. $X^2 - 13X + 36 = 0$. $\Delta = 169 - 144 = 25$. $X_1 = \frac{13-5}{2} = 4, X_2 = \frac{13+5}{2} = 9$. $x^2 = 4 \Longrightarrow x = \pm 2$; $x^2 = 9 \Longrightarrow x = \pm 3$. $S = \{-3, -2, 2, 3\}$.
17. a) Vertex: $\alpha = -\frac{0.8}{2(-0.05)} = \frac{0.8}{0.1} = 8$ meters. Maximum height: $\beta = h(8) = -0.05(64) + 0.8(8) + 1 = -3.2 + 6.4 + 1 = 4.2$ meters.
    b) $h(x) = 0 \Leftrightarrow -0.05x^2 + 0.8x + 1 = 0$. Multiply by -20: $x^2 - 16x - 20 = 0$. $\Delta = 256 + 80 = 336$. $\sqrt{336} \approx 18.33$. $x = \frac{16 + 18.33}{2} \approx 17.17$ meters (the other root is negative). The ball hits the ground at approximately 17.17 m.
18. $x^2 - 3x + 2 = 2x - 4 \Leftrightarrow x^2 - 5x + 6 = 0 \Leftrightarrow (x - 2)(x - 3) = 0$. x-coordinates: $x = 2$ and $x = 3$. y-coordinates: for $x = 2$, $y = 2(2) - 4 = 0$; for $x = 3$, $y = 2(3) - 4 = 2$. Points of intersection: $A(2, 0)$ and $B(3, 2)$.
19. a) $\Delta = (2b')^2 - 4ac = 4b'^2 - 4ac = 4(b'^2 - ac)$.
    b) $\sqrt{\Delta} = 2\sqrt{b'^2 - ac} = 2\sqrt{\Delta'}$. $x = \frac{-2b' \pm 2\sqrt{\Delta'}}{2a} = \frac{-b' \pm \sqrt{\Delta'}}{a}$.
    c) For $x^2 - 14x + 45 = 0$: $b' = -7$, $\Delta' = 49 - 45 = 4$. $x = 7 \pm 2 \Longrightarrow S = \{5, 9\}$. For $5x^2 - 6x - 8 = 0$: $b' = -3$, $\Delta' = 9 - 5(-8) = 49$. $x = \frac{3 \pm 7}{5} \Longrightarrow S = \{-0.8, 2\}$.
20. a) $x^2 - 4x + 3 = -x^2 + 2x + 11 \Leftrightarrow 2x^2 - 6x - 8 = 0 \Leftrightarrow x^2 - 3x - 4 = 0$. Obvious roots ($a - b + c = 1 - (-3) - 4 = 0$): $x_1 = -1$ and $x_2 = 4$.
    b) $\mathcal{P}_1$ lies below $\mathcal{P}_2 \Leftrightarrow \mathcal{P}_1 - \mathcal{P}_2 \leq 0 \Leftrightarrow 2(x + 1)(x - 4) \leq 0$. Since the coefficient $2 > 0$, the expression is negative between the roots: on $[-1, 4]$.
21. Let x be the width of the two sides perpendicular to the beach. The length along the water is $160 - 2x$. Area: $A(x) = x(160 - 2x) = -2x^2 + 160x$. This is a quadratic trinomial with $a = -2 < 0$. Its maximum is attained at the vertex: $x = -\frac{160}{2(-2)} = 40$ meters. The length is then $160 - 2(40) = 80$ meters. Optimal dimensions: 40 m × 80 m (maximum area: 3200 m^2).

22. a) $x > 0$ and $2x < 30 \Leftrightarrow x < 15$. Hence $x \in]0, 15[$.
    b) The base is a square of side length $30 - 2x$ and the height is x. $V(x) = x(30-2x)^2 = x(900 - 120x + 4x^2) = 4x^3 - 120x^2 + 900x$.
    c) Compute for several test values: $V(1) = 1(28)^2 = 784$ cm^3; $V(3) = 3(24)^2 = 1728$ cm^3; $V(5) = 5(20)^2 = 2000$ cm^3; $V(6) = 6(18)^2 = 1944$ cm^3. The volume appears to reach its maximum for $x = 5$ cm, attaining 2000 cm^3 (which is exactly 2 liters!).

The Question Opening the Next Stage

The quadratic degree has taught us to predict the continuous arch of a water jet soaring through space. But in the physical world, not all phenomena flow like a continuous stream:

- A bank account yields interest at the end of each discrete month;
- A bacterial colony doubles with every passing hour;
- A staircase is climbed step by step, $0, 1, 2, 3...$

How can we study a quantity that progresses in discrete jumps through time? How can we calculate the sum of a thousand steps without exhaustion? This is what we will discover in **The Staircase Steps and the Usurer's Loan**.

CHAPTER 36: The Staircase Steps and the Usurer's Loan

> "A continuous curve flows like the water of a river, without ever lifting the pen from the parchment. But the time of human affairs moves in discrete strides: heartbeats, clock ticks, the stone steps of a staircase, and interest payments falling due at month's end. A numerical sequence is a voyage where each step rests upon the previous one to spring forward toward infinity."

What You Are About to Discover

- The fundamental difference between a continuous quantity and a **numerical sequence** indexed by natural numbers $n \in \mathbb{N}$.
- The two primary ways of generating a sequence: the **explicit formula** $u_n = f(n)$ and the **recurrence relation** $u_{n+1} = g(u_n)$.
- **Arithmetic sequences**: the steady staircase where one always adds the same common difference r, and the legendary trick of young **Carl Friedrich Gauss** to sum consecutive integers.
- **Geometric sequences**: chained multiplication by the common ratio q, the dizzying growth of compound interest, and the trap of Sissa's chessboard.
- How to determine the **direction of variation** (monotonicity) of a sequence and intuitively grasp its **limit** at infinity.
- Programming a **threshold algorithm** in Python to determine when a quantity surpasses a given ceiling.

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1. The Problem: The Two Bankers of Florence

In Renaissance Florence, two cloth merchants, Matteo and Cosimo, seek financing to charter a caravel bound for the Levant. Each needs to borrow 1 000 gold florins for a term of 10 years.

They approach two moneylenders on the Ponte Vecchio:

Échéance	Contrat Matteo (Prêt arithmétique : +100 / an)	Contrat Cosimo (Prêt géométrique : +10 % / an)
Année 0	1 000 florins	1 000 florins
Année 1	1 100 florins	1 100 florins
Année 2	1 200 florins	1 210 florins
Année 3	1 300 florins	1 331 florins
Année 10	2 000 florins	**2 593,74 florins !**

Look closely at what occurred:

- For Matteo, each year adds **the same fixed amount**: $+100$ florins. It is an even staircase. Over 10 years, he pays $1\,000 + 10 \times 100 = 2\,000$ florins.
- For Cosimo, the first year appears identical ($+100$ florins). But in the second year, the 10% applies to the 1 100 florins already owed: the debt expands by 110 florins! In the third year, the 10% applies to 1 210 florins ($+121$ florins). Interest begets interest, which in turn breeds more interest: this is the inexorable engine of **compound interest**. In 10 years, Cosimo owes nearly $2\,594$ florins—almost 60% more than Matteo!

These two contracts embody the two royal families of discrete sequences:

- **Arithmetic sequences**, governed by **repeated addition**.
- **Geometric sequences**, propelled by **repeated multiplication**.

---

2. What Is a Numerical Sequence?

2.1 Definition and Notations

A familiar function $f(x)$ accepts any real number: $x = 0$, $x = 1.5$, $x = \sqrt{2}$, $x = \pi$… A **sequence** is far more restrained: it accepts only **non-negative integer indices**: $0, 1, 2, 3, 4...$

📐 Definition — Numerical Sequence.

A **numerical sequence** is a function whose domain is the set of natural numbers $\mathbb{N}$ (or a subset of $\mathbb{N}$ starting from a certain index n_0). For any natural number n, the image of n is written as a **subscript**:

$$u_n \quad (\text{instead of } \ u(n)) \ .$$

We read it aloud: "u sub n" (or "u index n").

- n is the **index** (or rank).
- u_n is the **term of index n** (or general term).
- The entire sequence is denoted in parentheses: (u_n) or $(u_n)_{n\in\mathbb{N}}$.

⚠️ Beware: Common Pitfall! Never confuse u_n and (u_n)!

- u_n is an **isolated real number** (for instance, $u_3 = 8$).
- (u_n) with parentheses denotes **the entire sequence**, the infinite machine producing the ordered list $\{u_0, u_1, u_2, u_3, ...\}$.

2.2 The Two Modes of Generation

Mode 1: Explicit Definition ($u_n = f(n)$)

The term u_n is expressed directly as a function of the integer n. *Example:* $u_n = 3n^2 - 1$. - To calculate u_0, substitute $n = 0$: $u_0 = 3(0)^2 - 1 = -1$. - To calculate u_{100}, there is no need to compute the 99 previous terms: $u_{100} = 3(100)^2 - 1 = 30\,000 - 1 = 29\,999$.

Mode 2: Recursive Definition ($u_{n+1} = g(u_n)$)

We specify: 1. The **initial term** that anchors the sequence (for example, $u_0 = 2$). 2. A **transition rule** that explains how to construct the next term from the current one: $u_{n+1} = g(u_n)$. *Example:* $u_0 = 2$ and $u_{n+1} = 3u_n - 4$. - $u_1 = 3u_0 - 4 = 3(2) - 4 = 2$. - $u_2 = 3u_1 - 4 = 3(2) - 4 = 2$. *(Here the sequence remains constant!)*

⚠️ Beware: Common Pitfall! In a recursive sequence, you advance step by step.

To compute u_{10} using a recurrence relation, you are obliged to first compute u_1, u_2, u_3... all the way to u_9. It operates like falling dominoes: each domino topples the next.

2.3 Graphical Representation of a Sequence

Because the index n takes only whole integer values $(0, 1, 2, 3...)$, the graph of a sequence is a **cloud of isolated points** with coordinates (n, u_n). You must **never connect these points with a continuous solid line**, because intermediate non-integer values (such as $n = 1.5$) have no meaning for a sequence!

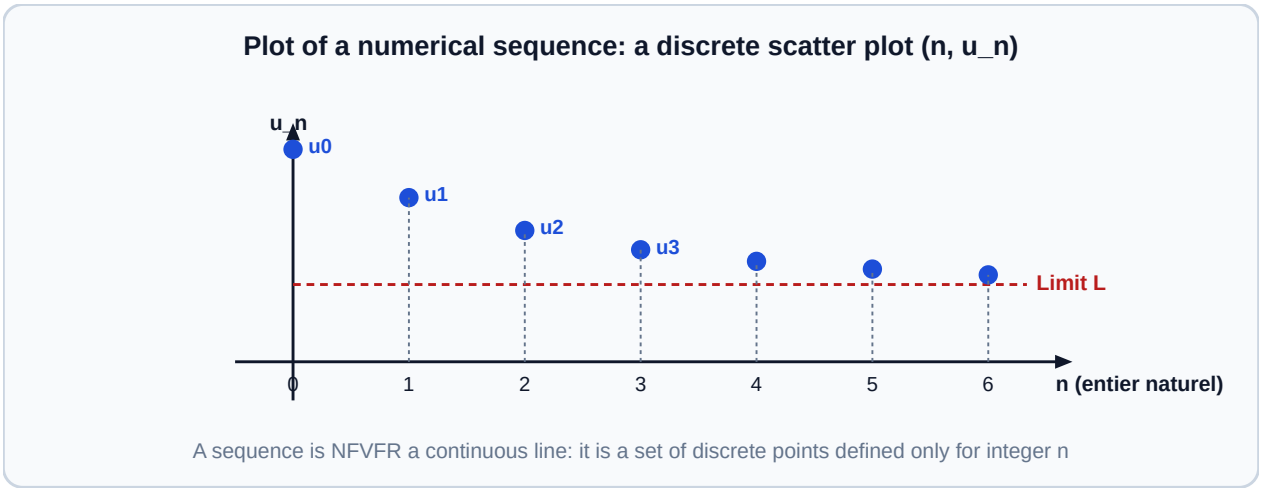

Figure 88: Graph of a numerical sequence: discrete scatter plot

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3. Arithmetic Sequences: The Regular Staircase

3.1 Definition and General Term

Definition — Arithmetic Sequence.

A sequence (u_n) is called **arithmetic** if there exists a fixed real number r, called the **common difference**, such that for every natural number n:

$$\boxed{u_{n+1} = u_n + r}$$

To climb from one step to the next, we always add the same number r:

- $u_1 = u_0 + r$
- $u_2 = u_1 + r = (u_0 + r) + r = u_0 + 2r$
- $u_3 = u_2 + r = (u_0 + 2r) + r = u_0 + 3r$

Continuing in this fashion up to index n:

Theorem — Explicit Formula for the General Term.

If (u_n) is an arithmetic sequence with initial term u_0 and common difference r, then for all $n \in \mathbb{N}$:

$$\boxed{u_n = u_0 + nr}$$

If the initial term is given at index p as u_p, the formula becomes:

$$u_n = u_p + (n - p)r.$$

Remark: The points representing an arithmetic sequence all lie on the straight line with equation $y = rx + u_0$. The common difference r plays exactly the role of the slope (rate of change)!

- If $r > 0$: the sequence is **strictly increasing**.
- If $r < 0$: the sequence is **strictly decreasing**.
- If $r = 0$: the sequence is **constant**.

3.2 Gauss's Staircase: Sum of Consecutive Integers

In 1784, in a schoolhouse in Brunswick, Germany, an exhausted schoolmaster ordered his seven-year-old pupils to add all the numbers from 1 to 100:

$$S = 1 + 2 + 3 + 4 + \ldots + 99 + 100.$$

The teacher hoped for an hour of quiet. But after just a few seconds, a young boy set his slate on the master's desk, saying: *"Ligget se!"* ("There it lies!"). The result written on the slate was $5\,050$, perfectly exact.
That child was **Carl Friedrich Gauss**.

```
  1   +   2   +   3   + ... +  98  +  99  + 100
100   +  99   +  98   + ... +   3  +   2  +   1
-----------------------------------------------
101   + 101   + 101   + ... + 101  + 101  + 101   (100 times the sum 101!)
```

Proof (Required knowledge) — Sum of the First n Positive Integers.

For any natural number $n \geq 1$:

$$\boxed{1 + 2 + 3 + ... + n = \frac{n(n+1)}{2}}$$

Proof: Write the sum S_n in ascending order, and directly beneath it in descending order:

$$S_n = 1 + 2 + 3 + ... + (n-1) + n$$

$$S_n = n + (n-1) + (n-2) + ... + 2 + 1$$

Add the two equations column by column:

$$2S_n = (1+n) + (2+n-1) + (3+n-2) + ... + (n+1).$$

Each bracketed pair sums to exactly $n+1$. How many columns are there? There are precisely n columns. Therefore:

$$2S_n = n \times (n+1) \Longrightarrow S_n = \frac{n(n+1)}{2}. \quad \blacksquare$$

For $n = 100$: $S_{100} = \frac{100 \times 101}{2} = 50 \times 101 = 5\,050$.

3.3 Sum of Consecutive Terms of an Arithmetic Sequence

General Theorem — Arithmetic Sum.

For any arithmetic sequence:

$$\boxed{S = \sum_{k=p}^{n} u_k = u_p + u_{p+1} + ... + u_n = (\text{number of terms}) \times \frac{\text{first term} + \text{last term}}{2}}$$

where the number of terms is given by: $\text{last index} - \text{first index} + 1 = n - p + 1$.

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4. Geometric Sequences: The Vertigo of Multiplication

4.1 Definition and General Term

Definition — Geometric Sequence.

A sequence (u_n) is called **geometric** if there exists a fixed real number q, called the **common ratio**, such that for every natural number n:

$$\boxed{u_{n+1} = q \times u_n}$$

To move from one term to the next, we always multiply by the same number q:

- $u_1 = q \times u_0$
- $u_2 = q \times u_1 = q \times (q u_0) = u_0 \times q^2$

- $u_3 = q \times u_2 = u_0 \times q^3$

Theorem — Explicit Formula for the General Term.

If (u_n) is a geometric sequence with initial term u_0 and common ratio q, then for all $n \in \mathbb{N}$:

$$\boxed{u_n = u_0 \times q^n}$$

If the initial term is given at index p as u_p, the formula becomes:

$$u_n = u_p \times q^{n-p}.$$

4.2 Sissa's Chessboard and Exponential Explosion

The Indian sage Sissa invented the game of chess to entertain his king. Delighted, the monarch offered him any reward he desired. Sissa replied modestly: > "Sire, place a single grain of wheat on the first square, two on the second, four on the third, eight on the fourth, doubling on each subsequent square up to the 64th square of the board."

The king smiled at what seemed a trifling request. But his treasurers returned as pale as death:

- Square 1: $u_0 = 1 = 2^0$
- Square 2: $u_1 = 2 = 2^1$
- Square 3: $u_2 = 4 = 2^2$
- Square 64: $u_{63} = 2^{63} \approx 9.22 \times 10^{18}$ grains!

The total number of grains across the entire board equals:

$$S = 1 + 2 + 4 + \ldots + 2^{63} = 2^{64} - 1 \approx 18.4 \times 10^{18} \text{ grains}.$$

This represents more than 1 200 billion metric tons of wheat—equivalent to the modern world's wheat harvest for more than two millennia! The king could never honor his promise.

4.3 Sum of Consecutive Powers of a Number q

Proof (Required knowledge) — Sum of Powers of q.

For any real $q \neq 1$ and any integer $n \geq 0$:

$$\boxed{1 + q + q^2 + q^3 + \ldots + q^n = \frac{1 - q^{n+1}}{1 - q}}$$

Proof (by the telescoping method): Let $S_n = 1 + q + q^2 + \ldots + q^n$. Multiply both sides by q:

$$qS_n = q(1 + q + q^2 + \ldots + q^n) = q + q^2 + q^3 + \ldots + q^{n+1}.$$

Now subtract this equation from the original equation ($S_n - qS_n$):

$$S_n - qS_n = (1 + q + q^2 + \ldots + q^n) - (q + q^2 + q^3 + \ldots + q^{n+1}).$$

Observe the striking cancellation: all intermediate terms eliminate each other diagonally in pairs! Only the very first term (1) and the very last term ($-q^{n+1}$) survive:

$$S_n(1 - q) = 1 - q^{n+1}.$$

Since $q \neq 1$, we can divide by $1 - q$:

$$S_n = \frac{1 - q^{n+1}}{1 - q}. \quad \blacksquare$$

General Theorem — Geometric Sum.

For any geometric sequence with common ratio $q \neq 1$:

$$\boxed{S = (\text{first term}) \times \frac{1 - q^{\text{number of terms}}}{1 - q}}$$

5. Variations and Intuition of the Limit

5.1 The Three Methods for Studying the Monotonicity of a Sequence

To determine whether a sequence climbs ($u_{n+1} \geq u_n$) or descends ($u_{n+1} \leq u_n$):

1. **Method 1: Study the sign of the difference $u_{n+1} - u_n$ (Universal method).**
   - If for all n, $u_{n+1} - u_n > 0$, the sequence (u_n) is strictly increasing.
   - If for all n, $u_{n+1} - u_n < 0$, the sequence (u_n) is strictly decreasing.
2. **Method 2: Compare the quotient $\frac{u_{n+1}}{u_n}$ with 1 (valid ONLY when all terms are strictly positive!).**
   - If $\frac{u_{n+1}}{u_n} > 1$, the sequence is strictly increasing.
   - If $\frac{u_{n+1}}{u_n} < 1$, the sequence is strictly decreasing.
3. **Method 3: Leverage the associated function if $u_n = f(n)$.** If the function f is increasing on $[0, +\infty[$, then the sequence $u_n = f(n)$ is increasing.

5.2 Limit of a Geometric Sequence According to Ratio q

What happens to q^n as n grows indefinitely large ($n \to +\infty$)?

- **If $q > 1$:** The sequence explodes toward infinity: $\lim_{n\to+\infty} q^n = +\infty$. (Example: 2^n).
- **If $q = 1$:** The sequence is constant and equal to 1.
- **If $-1 < q < 1$:** The sequence shrinks toward zero: $\lim_{n\to+\infty} q^n = 0$. (Example: $\left(\frac{1}{2}\right)^n \to 0$).
- **If $q \leq -1$:** The sequence oscillates without admitting any limit (Example: $(-1)^n = 1, -1, 1, -1...$).

6. In Computer Science: The Threshold Algorithm in Python

A fundamental problem in finance, biology, and physics is determining **at which index n** a sequence crosses a given critical boundary (the threshold). We use a `while` loop for this purpose:

```
def growth_threshold(initial_value, rate, threshold):
    """
    Computes the number of periods needed to exceed a given threshold.
    Example: $1,000 invested at 5% annually (common ratio q = 1.05)
    growing to double ($2,000).
    """
    n = 0
    u = initial_value

    while u < threshold:
        u = u * (1 + rate)  # Apply annual rate
        n += 1              # Step to next year

    return n, round(u, 2)

years, final_amount = growth_threshold(1000, 0.05, 2000)
print(f"The capital doubles after {years} years, reaching ${final_amount}.")
```

7. Detailed Worked Examples

Worked Example 1 — Reconstructing an Arithmetic Sequence.

Let (u_n) be an arithmetic sequence such that $u_4 = 17$ and $u_9 = 37$. 1. Determine the common difference r and the initial term u_0. 2. Express u_n in terms of n. 3. Calculate the value of u_{50}.

Solution: 1. According to the general term formula: $u_9 = u_4 + (9-4)r$.

$$37 = 17 + 5r \Leftrightarrow 5r = 37 - 17 = 20 \Leftrightarrow r = \frac{20}{5} = 4.$$

To find u_0, use $u_4 = u_0 + 4r$:

$$17 = u_0 + 4(4) \Leftrightarrow 17 = u_0 + 16 \Leftrightarrow u_0 = 1.$$

2. The general term is therefore: $\boxed{u_n = 1 + 4n}$. 3. For $n = 50$: $u_{50} = 1 + 4(50) = 1 + 200 = 201$.

Worked Example 2 — Computing an Arithmetic Sum of Terms.

Compute the sum:

$$S = 7 + 11 + 15 + 19 + \ldots + 123.$$

Solution: The terms increase by $+4$ each step: this is an arithmetic sequence with common difference $r = 4$ and initial term $u_0 = 7$. The general term is $u_n = 7 + 4n$. Let us find the index of the last term 123:

$$7 + 4n = 123 \Leftrightarrow 4n = 116 \Leftrightarrow n = 29.$$

The terms run from u_0 to u_{29}: there are therefore $29 - 0 + 1 = 30$ terms. Apply the formula for the sum of an arithmetic sequence:

$$S = (\text{number of terms}) \times \frac{\text{first} + \text{last}}{2} = 30 \times \frac{7 + 123}{2} = 30 \times \frac{130}{2} = 30 \times 65 = 1\,950.$$

Worked Example 3 — Geometric Sequence and Sum of Powers.

Let (v_n) be the geometric sequence with initial term $v_0 = 5$ and common ratio $q = 3$. 1. Compute v_1, v_2, and v_6. 2. Compute the sum $S = v_0 + v_1 + v_2 + \ldots + v_6$.

Solution: 1. $v_1 = 5 \times 3 = 15$. $v_2 = 15 \times 3 = 45$. General formula: $v_n = 5 \times 3^n$. For $n = 6$: $v_6 = 5 \times 3^6 = 5 \times 729 = 3\,645$. 2. Number of terms from v_0 to v_6: $6 - 0 + 1 = 7$ terms.

$$S = v_0 \times \frac{1 - q^7}{1 - q} = 5 \times \frac{1 - 3^7}{1 - 3} = 5 \times \frac{1 - 2\,187}{-2} = 5 \times \frac{-2\,186}{-2} = 5 \times 1\,093 = 5\,465.$$

Worked Example 4 — Studying the Monotonicity of a Sequence.

Determine the direction of variation of the following sequences: a) $u_n = \frac{n}{n+1}$ for $n \in \mathbb{N}$ b) $v_n = \frac{3^n}{n!}$ for $n \geq 1$

Solution: a) Study the sign of the difference $u_{n+1} - u_n$:

$$u_{n+1} - u_n = \frac{n+1}{(n+1)+1} - \frac{n}{n+1} = \frac{n+1}{n+2} - \frac{n}{n+1}.$$

Find a common denominator $(n+2)(n+1) > 0$:

$$u_{n+1} - u_n = \frac{(n+1)^2 - n(n+2)}{(n+2)(n+1)} = \frac{n^2 + 2n + 1 - (n^2+2n)}{(n+2)(n+1)} = \frac{1}{(n+2)(n+1)}.$$

The numerator is $1 > 0$ and the denominator is strictly positive for all $n \in \mathbb{N}$. Hence $u_{n+1} - u_n > 0$ for all n: the sequence (u_n) is **strictly increasing**.

b) All terms of v_n are strictly positive. Compare the ratio $\frac{v_{n+1}}{v_n}$ to 1:

$$\frac{v_{n+1}}{v_n} = \frac{\frac{3^{n+1}}{(n+1)!}}{\frac{3^n}{n!}} = \frac{3^{n+1}}{3^n} \times \frac{n!}{(n+1)!} = 3 \times \frac{n!}{(n+1)n!} = \frac{3}{n+1}.$$

- If $n = 1$: $\frac{v_2}{v_1} = \frac{3}{2} > 1$ (the sequence initially increases: $v_1 = 3, v_2 = 4.5$).
- If $n = 2$: $\frac{v_3}{v_2} = \frac{3}{3} = 1$ ($v_3 = v_2 = 4.5$).
- For all $n \geq 3$: $n + 1 \geq 4$, so $\frac{3}{n+1} < 1$, which implies $v_{n+1} < v_n$. The sequence (v_n) is **strictly decreasing starting from index $n = 2$**.

Worked Example 5 — Arithmetico-Geometric Sequence.

An animal shelter has 200 animals on January 1, 2025. Each year, 20% of the animals are adopted and leave the shelter, while 50 newly abandoned animals are taken in. Let u_n denote the number of animals after n years. 1. Establish the recurrence relation linking u_{n+1} and u_n. 2. Define $v_n = u_n - 250$. Prove that (v_n) is a geometric sequence. 3. Deduce the explicit formula for u_n in terms of n, and find the limit of u_n as $n \to +\infty$.

Solution: 1. If 20% of the animals leave, 80% remain, which corresponds to multiplying by 0.8. Adding the 50 new arrivals:

$$u_0 = 200 \quad \text{and} \quad u_{n+1} = 0.8u_n + 50.$$

2. Express v_{n+1} in terms of v_n:

$$v_{n+1} = u_{n+1} - 250 = (0.8u_n + 50) - 250 = 0.8u_n - 200.$$

Factor out 0.8:

$$v_{n+1} = 0.8(u_n - 250) = 0.8v_n.$$

The sequence (v_n) is a **geometric sequence with common ratio $q = 0.8$** and initial term:

$$v_0 = u_0 - 250 = 200 - 250 = -50.$$

3. By the geometric formula: $v_n = v_0 \times q^n = -50 \times (0.8)^n$. Since $u_n = v_n + 250$, we deduce:

$$\boxed{u_n = 250 - 50 \times (0.8)^n}.$$

Since $-1 < 0.8 < 1$, we have $\lim_{n\to+\infty}(0.8)^n = 0$. Consequently:

$$\lim_{n\to+\infty} u_n = 250 - 50 \times 0 = 250.$$

In the long term, the shelter population stabilizes around 250 animals.

Worked Example 6 — Cobweb Plot (Web Diagram) of a Recurrence Sequence.

Consider the recursive sequence defined by $u_0 = 0$ and $u_{n+1} = \frac{1}{2}u_n + 1$. Graphically represent the first four terms on a coordinate plane using the line $y = x$.

Solution and Method: Draw the line $y = x$ (the main diagonal) and the line representing the function $g(x) = \frac{1}{2}x + 1$ in the coordinate plane. 1. Start from $u_0 = 0$ on the horizontal axis. 2. Move vertically up to the line $y = g(x)$: the point reached has y-coordinate $g(u_0) = u_1 = 1$. 3. Move horizontally to the diagonal line $y = x$: this transfers the value $u_1 = 1$ onto the horizontal axis without recalculation! 4. Move vertically to the line $y = g(x)$ again to find $u_2 = g(1) = 1.5$. 5. Move horizontally to $y = x$, then vertically back to the function line to find $u_3 = g(1.5) = 1.75$. The successive segments form a "staircase" (or cobweb) that converges toward the intersection point of the two lines:

$$x = \frac{1}{2}x + 1 \Leftrightarrow \frac{1}{2}x = 1 \Leftrightarrow x = 2.$$

The sequence (u_n) increases, approaching its limit $L = 2$.

Key Takeaways

1. **Sequence:** A function indexed by whole numbers $n \in \mathbb{N}$. Term u_n, sequence (u_n).
2. **Explicit vs Recursive Definition:**
   - Explicit: $u_n = f(n)$ (direct calculation).
   - Recursive: u_0 and $u_{n+1} = g(u_n)$ (step-by-step propagation).
3. **Arithmetic Sequences:**
   - Definition: $u_{n+1} = u_n + r$.
   - General term: $u_n = u_0 + nr$ (or $u_n = u_p + (n-p)r$).
   - Gauss sum: $1 + 2 + ... + n = \frac{n(n+1)}{2}$.
   - General sum: $(\text{number of terms}) \times \frac{\text{first} + \text{last}}{2}$.
4. **Geometric Sequences:**
   - Definition: $u_{n+1} = q \times u_n$.
   - General term: $u_n = u_0 \times q^n$ (or $u_n = u_p \times q^{n-p}$).
   - Sum of powers: $1 + q + q^2 + ... + q^n = \frac{1-q^{n+1}}{1-q}$ $(q \neq 1)$.
   - General sum: $(\text{first}) \times \frac{1-q^{\text{number of terms}}}{1-q}$.
5. **Variations:** Sign of $u_{n+1} - u_n$ or quotient $\frac{u_{n+1}}{u_n}$ compared to 1.
6. **Geometric Limit:** If $|q| < 1$, then $q^n \to 0$. If $q > 1$, then $q^n \to +\infty$.

Exercises

Direct Application Exercises ()

1. Given the explicit sequence $u_n = 2n^2 - 3$. Compute u_0, u_1, u_2, and u_{10}.
2. Given the recursive sequence $v_0 = 4$ and $v_{n+1} = 2v_n - 3$. Compute v_1, v_2, and v_3.
3. An arithmetic sequence has initial term $u_0 = 5$ and common difference $r = 3$. Give the expression for u_n, then calculate u_{20}.
4. A geometric sequence has initial term $v_0 = 3$ and common ratio $q = 2$. Give the expression for v_n, then calculate v_8.
5. Compute the sum of integers from 1 to 50: $S = 1 + 2 + 3 + ... + 50$.
6. Compute the geometric sum: $S = 1 + 2 + 4 + 8 + ... + 2^{10}$.
7. Is the sequence $u_n = 5 - 4n$ arithmetic? State its common difference and its direction of variation.

8. Is the sequence $v_n = 7 \times 0.6^n$ geometric? State its common ratio and its limit as $n \to +\infty$.
9. Determine the direction of variation of the sequence $u_n = 3n + 7$.
10. Given an arithmetic sequence such that $u_3 = 10$ and $u_5 = 16$. Find the common difference r.

Practice and Reasoning Exercises ()

11. Consider the arithmetic sequence (u_n) such that $u_2 = 11$ and $u_7 = 36$.
    a) Compute the common difference r and the initial term u_0.
    b) Starting from which index n do we have $u_n > 1\,000$?
12. Compute the sum: $S = 12 + 15 + 18 + 21 + \ldots + 150$.
13. A capital sum of €5 000 is deposited into an account earning compound interest at an annual rate of 4%. Let C_n denote the value of the capital after n years.
    a) Express C_{n+1} in terms of C_n, then C_n in terms of n.
    b) What will be the available capital after 10 years?
    c) Write a Python script to determine after how many years the capital will exceed €8 000.
14. Determine the direction of variation of the sequence $u_n = \frac{2^n}{n}$ for $n \geq 1$.
15. Determine the direction of variation of the sequence $w_n = n^2 - 10n + 3$ for $n \in \mathbb{N}$.
16. Compute the sum of powers of 3: $S = 1 + 3 + 9 + 27 + \ldots + 3^7$.
17. Consider the recursive sequence defined by $u_0 = 1$ and $u_{n+1} = \sqrt{u_n + 2}$.
    a) Compute u_1 and u_2 in exact and approximate decimal form.
    b) Determine graphically the value toward which this sequence appears to converge by finding the fixed point of $f(x) = \sqrt{x+2}$.
18. Let (u_n) be a positive geometric sequence such that $u_1 = 6$ and $u_3 = 24$. Determine the common ratio q, the initial term u_0, and compute the sum $u_0 + u_1 + \ldots + u_8$.

Deepening Exercises and Challenges ()

19. **The Paradox of the Koch Snowflake.** The Koch snowflake is a fractal generated starting from an equilateral triangle of side length 1 (stage 0). At each stage, the central third of each side is replaced by an outward-pointing equilateral triangle.
    - Stage 0: number of sides $N_0 = 3$, length of each side $\ell_0 = 1$.
    - At stage $n+1$, each segment is replaced by 4 segments, each 3 times shorter.

    a) Express the number of sides N_n in terms of n.
    b) Express the length ℓ_n of a side in terms of n.
    c) Deduce the perimeter $P_n = N_n \times \ell_n$ of the snowflake at stage n.
    d) What is the limit of the perimeter as $n \to +\infty$?
20. **Sum of Squares of Consecutive Integers.** We wish to prove that:
$$1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$
    a) Verify this formula for $n = 1, 2, 3$.
    b) Expanding $(k+1)^3 - k^3 = 3k^2 + 3k + 1$, sum this identity for k ranging from 1 to n and deduce the formula.
21. **Achilles and the Tortoise Revisited.** The tortoise starts with a $100\ \text{meter}$ head start over Achilles. Achilles runs at $10\ \text{m/s}$ and the tortoise crawls at $1\ \text{m/s}$.
    - To cover the initial $100\ \text{m}$, Achilles takes $t_0 = 10\ \text{s}$. During this time, the tortoise has moved $10\ \text{m}$.
    - To cover those $10\ \text{m}$, Achilles takes $t_1 = 1\ \text{s}$. The tortoise has moved $1\ \text{m}$.
    - To cover that meter, Achilles takes $t_2 = 0.1\ \text{s}$, and so on.

    a) Express the time t_n of the n-th stage. What type of sequence is (t_n)?
    b) Compute the total elapsed time $T_n = t_0 + t_1 + \ldots + t_n$ after n stages.
    c) What is the limit of T_n as $n \to +\infty$? Does Achilles overtake the tortoise?
22. **The Fibonacci Sequence and the Golden Ratio.** The famous Fibonacci sequence is defined by $F_0 = 0, F_1 = 1$ and for all $n \in \mathbb{N}$:
$$F_{n+2} = F_{n+1} + F_n.$$
    a) Compute the first ten terms $F_0, F_1, F_2, \ldots, F_9$.

b) Consider the ratio $r_n = \frac{F_{n+1}}{F_n}$ for $n \geq 1$. Compute r_1, r_2, r_3, r_4, r_5.

c) Toward what remarkable value does this ratio appear to converge?

Solutions

1. $u_0 = -3$. $u_1 = 2(1) - 3 = -1$. $u_2 = 2(4) - 3 = 5$. $u_{10} = 2(100) - 3 = 197$.
2. $v_1 = 2(4) - 3 = 5$. $v_2 = 2(5) - 3 = 7$. $v_3 = 2(7) - 3 = 11$.
3. $u_n = 5 + 3n$. $u_{20} = 5 + 3(20) = 65$.
4. $v_n = 3 \times 2^n$. $v_8 = 3 \times 256 = 768$.
5. $S = \frac{50 \times 51}{2} = 25 \times 51 = 1\,275$.
6. $S = \frac{1 - 2^{11}}{1 - 2} = \frac{1 - 2\,048}{-1} = 2\,047$.
7. $u_{n+1} - u_n = [5 - 4(n+1)] - [5 - 4n] = -4$. This is an arithmetic sequence with common difference $r = -4$. Since $r < 0$, it is strictly decreasing.
8. $v_{n+1} = 7 \times 0.6^{n+1} = 0.6 \times (7 \times 0.6^n) = 0.6 v_n$. Geometric sequence with common ratio $q = 0.6$. Since $|0.6| < 1$, $\lim_{n \to +\infty} v_n = 0$.
9. $u_{n+1} - u_n = [3(n+1) + 7] - [3n + 7] = 3 > 0$. Strictly increasing.
10. $u_5 = u_3 + 2r \Leftrightarrow 16 = 10 + 2r \Leftrightarrow 2r = 6 \Leftrightarrow r = 3$.
11. a) $u_7 = u_2 + 5r \Leftrightarrow 36 = 11 + 5r \Leftrightarrow 5r = 25 \Leftrightarrow r = 5$. $\quad u_2 = u_0 + 2r \Leftrightarrow 11 = u_0 + 10 \Leftrightarrow u_0 = 1$. General term: $u_n = 1 + 5n$.

    b) $1 + 5n > 1\,000 \Leftrightarrow 5n > 999 \Leftrightarrow n > 199.8$. Starting from index $n = 200$.
12. Arithmetic sequence with common difference $r = 3$. $u_0 = 12$, $u_n = 12 + 3n = 150 \Leftrightarrow 3n = 138 \Leftrightarrow n = 46$. Number of terms: $46 - 0 + 1 = 47$ terms. $S = 47 \times \frac{12 + 150}{2} = 47 \times 81 = 3\,807$.
13. a) Increasing by 4% corresponds to multiplying by 1.04. $C_{n+1} = 1.04 C_n$. $C_n = 5\,000 \times (1.04)^n$.

    b) $C_{10} = 5\,000 \times (1.04)^{10} \approx 5\,000 \times 1.48024 \approx €7\,401.22$.

    c) Python script:

```
n, c = 0, 5000
while c <= 8000:
    c *= 1.04
    n += 1
print(n)  # Prints 12 years (c ≈ 8005.16 €)
```

14. $u_n > 0$ for all $n \geq 1$. $\frac{u_{n+1}}{u_n} = \frac{2^{n+1}}{n+1} \times \frac{n}{2^n} = 2 \times \frac{n}{n+1} = \frac{2n}{n+1}$. Compare to 1: $\frac{2n}{n+1} > 1 \Leftrightarrow 2n > n + 1 \Leftrightarrow n > 1$. For $n = 1$: $u_1 = 2, u_2 = 2 \Longrightarrow u_1 = u_2$. For all $n \geq 2$, $\frac{u_{n+1}}{u_n} > 1$, so the sequence (u_n) is strictly increasing starting from $n = 2$.
15. $w_{n+1} - w_n = [(n+1)^2 - 10(n+1) + 3] - [n^2 - 10n + 3] = n^2 + 2n + 1 - 10n - 10 + 3 - n^2 + 10n - 3 = 2n - 9$.
    - If $n \leq 4$: $2n - 9 < 0$, the sequence decreases ($w_0 = 3, w_1 = -6, w_2 = -13, w_3 = -18, w_4 = -21, w_5 = -22$).
    - If $n \geq 5$: $2n - 9 > 0$, the sequence is strictly increasing.
16. $S = \frac{1 - 3^8}{1 - 3} = \frac{1 - 6\,561}{-2} = \frac{-6\,560}{-2} = 3\,280$.
17. a) $u_1 = \sqrt{1 + 2} = \sqrt{3} \approx 1.732$. $u_2 = \sqrt{\sqrt{3} + 2} \approx \sqrt{3.732} \approx 1.932$.

    b) Fixed point: $x = \sqrt{x + 2} \Longrightarrow x^2 = x + 2 \Leftrightarrow x^2 - x - 2 = 0 \Leftrightarrow (x - 2)(x + 1) = 0$. Since $x \geq 0$, the solution is $x = 2$. The sequence converges toward 2.
18. $u_3 = u_1 \times q^2 \Leftrightarrow 24 = 6q^2 \Leftrightarrow q^2 = 4 \Longrightarrow q = 2$ (since the sequence is positive). $u_1 = u_0 \times q \Leftrightarrow 6 = 2u_0 \Leftrightarrow u_0 = 3$. Number of terms from u_0 to u_8: 9 terms. $S = 3 \times \frac{1 - 2^9}{1 - 2} = 3 \times \frac{1 - 512}{-1} = 3 \times 511 = 1\,533$.
19. a) At each stage, each segment is replaced by 4 segments: $N_n = 3 \times 4^n$.

    b) The length of each segment is divided by 3: $\ell_n = \left(\frac{1}{3}\right)^n$.

    c) Perimeter: $P_n = N_n \times \ell_n = 3 \times 4^n \times \frac{1}{3^n} = 3 \times \left(\frac{4}{3}\right)^n$.

d) Since $\frac{4}{3} > 1$, $\lim_{n\to+\infty} \left(\frac{4}{3}\right)^n = +\infty$, hence $\lim_{n\to+\infty} P_n = +\infty$. The Koch snowflake encloses a finite area within an infinitely long perimeter!

20. a) For $n = 1$: $1^2 = 1$ and $\frac{1\times2\times3}{6} = 1$. For $n = 2$: $1 + 4 = 5$ and $\frac{2\times3\times5}{6} = 5$. For $n = 3$: $1 + 4 + 9 = 14$ and $\frac{3\times4\times7}{6} = 14$.
b) Summing $(k+1)^3 - k^3$ from $k = 1$ to n, the left side telescopes to $(n+1)^3 - 1$. On the other hand, the sum equals $3\sum k^2 + 3\sum k + \sum 1 = 3\sum k^2 + 3\frac{n(n+1)}{2} + n$. Isolating $\sum k^2$: $3\sum k^2 = (n+1)^3 - 1 - \frac{3n(n+1)}{2} - n = (n+1)[(n+1)^2 - 1] - \frac{3n(n+1)}{2} = (n+1)n(n+2) - \frac{3n(n+1)}{2} = n(n+1)\left(n + 2 - \frac{3}{2}\right) = n(n+1)\left(\frac{2n+1}{2}\right)$. Dividing by 3: $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

21. a) $t_n = 10 \times (0.1)^n$. This is a geometric sequence with initial term $t_0 = 10$ and common ratio $q = 0.1$.
b) $T_n = 10 \times \frac{1-(0.1)^{n+1}}{1-0.1} = 10 \times \frac{1-(0.1)^{n+1}}{0.9} = \frac{100}{9}\left[1 - (0.1)^{n+1}\right]$.
c) Since $|0.1| < 1$, $(0.1)^{n+1} \to 0$. $\lim_{n\to+\infty} T_n = \frac{100}{9} \approx 11.11$ seconds. Zeno's paradox is resolved: an infinite sum of time intervals can sum to a finite total duration! After 11.11 s, Achilles overtakes the tortoise.

22. a) $F_0 = 0, F_1 = 1, F_2 = 1, F_3 = 2, F_4 = 3, F_5 = 5, F_6 = 8, F_7 = 13, F_8 = 21, F_9 = 34$.
b) $r_1 = \frac{1}{1} = 1$; $r_2 = \frac{2}{1} = 2$; $r_3 = \frac{3}{2} = 1.5$; $r_4 = \frac{5}{3} \approx 1.667$; $r_5 = \frac{8}{5} = 1.6$.
c) The ratio oscillates and converges toward the **golden ratio** $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.6180339887...$

The Question Opening the Next Stage

In this chapter, we explored sequences advancing by discrete strides, and we computed quantities that cycle or oscillate. But now imagine a fairground Ferris wheel spinning endlessly, or an ocean wave rising and falling with rhythmic perfection.

How can we measure angles not in rigid degrees, but by unwinding a flexible ribbon around a circle of radius 1? How do cosine and sine, first discovered in middle school right-angled triangles, become the undulations of waves describing sound, light, and alternating electric current? This is what we will discover in **The Ferris Wheel and the Winding Circle**.

CHAPTER 37: The Ferris Wheel and the Winding Circle

> "The degree is an invention of Babylonian merchants who were fond of the number 60. The radian, on the other hand, is an invention of the wheel itself: it is the circle measuring itself with its own radius."

What You Are About to Discover

- Why the **degree** is no longer sufficient when we want to describe motion that turns endlessly (wheels, planets, waves, motors).
- A new unit of angle, born from the circle itself: the **radian**.
- How to **wind the real number line** around a circle of radius 1, and thereby give meaning to $\cos(100)$, to $\sin(-7)$, or to $\cos\left(\frac{37\pi}{4}\right)$.
- The **special values** ($\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$) proved with a square and an equilateral triangle.
- The **associated angles**: how the symmetries of the circle allow everything to be brought back to the first quadrant.
- How to **solve** the equations $\cos x = a$ and $\sin x = a$, and sketch the undulating curves of the cosine and sine functions.

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1. The Problem: The Fairground Gondola Turning Endlessly

You have boarded the giant Ferris wheel at the fairground. It has a radius of $20\ \text{m}$, its central axle is $22\ \text{m}$ above the ground, and your gondola departs from the lowest point, just $2\ \text{m}$ off the ground. The wheel turns slowly, steadily, in a counterclockwise direction when viewed from the parking lot.

Your little brother, watching from below, asks you a simple question:

"When your gondola has traveled 50 meters along the circle, how high will you be? And after 300 meters? And after 1 kilometer?"

You pull out your middle-school trigonometry (Chapter 28): in a right-angled triangle, $\cos\hat{A} = \frac{\text{adjacent}}{\text{hypotenuse}}$. But very quickly, three major obstacles arise.

Obstacle 1 — Middle-school angles stop at 90°. A right-angled triangle cannot have an acute angle of $120°$, much less $400°$. Yet your gondola keeps turning: after a turn and a half, it has "turned through $540°$".

Obstacle 2 — The degree has no natural connection to distance traveled. Your brother speaks in **meters traveled along the circle**. To convert meters to degrees, you must each time perform a rule of three with the circumference $2\pi \times 20$. This is clumsy: the unit "degree" is foreign to the circle.

Obstacle 3 — Direction matters. If the wheel turns backward (to let passengers off), we must account for **negative** angles.

We therefore need: 1. a **natural** unit of angle, directly tied to arc length; 2. a definition of cosine and sine valid for **any real number**, positive or negative, small or vast.

This is precisely what mathematicians constructed between the seventeenth and eighteenth centuries, and what you will now reconstruct with a simple piece of string.

💡 The True History — The Sine Is a Pocket!

Around 150 BC, the Greek astronomer **Hipparchus** compiled the first table of "chords" to calculate the positions of celestial bodies; **Ptolemy** refined it in the *Almagest* (2nd century AD). In India, around 499, **Aryabhata** preferred to work with the **half-chord**, which he termed *ardha-jyā* ("half-bowstring"), abbreviated to *jyā* and then *jīva*. Arab scholars transcribed this word phonetically as *jība*, which, written without vowels, resembled *jayb*, meaning "pocket" or "fold of a garment". In the twelfth century, while translating these texts in Toledo, **Gerard of Cremona** rendered it literally into Latin as *sinus*, meaning "fold, pocket, bay". That is why the most famous wave function in mathematics bears the name of a fold in a tunic!

---

2. The Radian: Measuring an Angle with the Radius

2.1 The String Experiment

> 🖐 ***Your Turn to Try*** *— Take a round tin can, a piece of string, and a marker. Cut a piece of string whose length is exactly equal to the* ***radius*** *of the can. Place one end on the rim, then press the string flush along the circular rim. Draw the two radii connecting the center of the can to both ends of the string.*

The angle you have just traced—the angle that subtends an arc **whose length is equal to the radius**—is the natural unit of the circle: it is **one radian**.

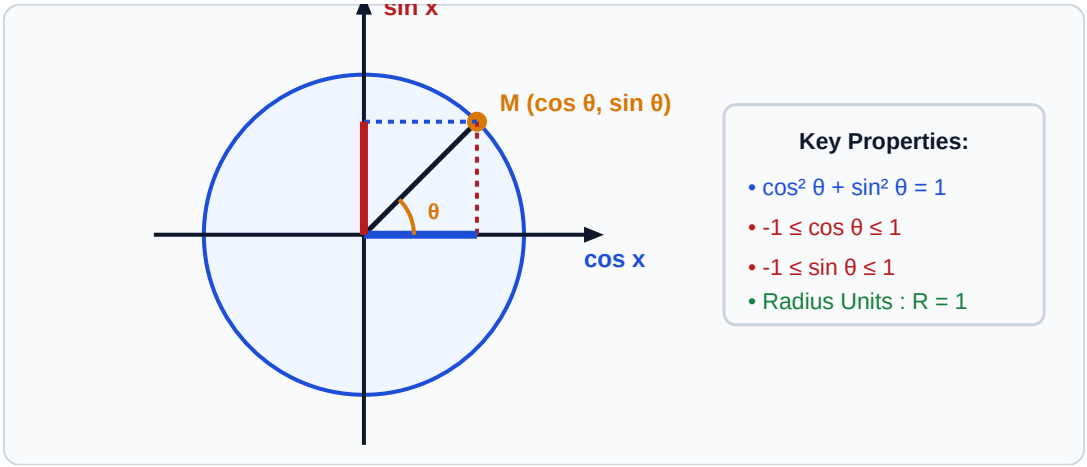

Figure 89: The complete trigonometric circle with sine and cosine lines

How many such "radius" strings can be laid end-to-end around the circle? The circumference is $2\pi R$, so we can lay $\frac{2\pi R}{R} = 2\pi \approx 6.28$ strings. A complete revolution thus measures 2π radians. With 6 strings, a small gap remains (0.28 of a string) to close the full loop.

📐 Definition — The Radian

In a circle of radius R, a central angle measures **1 radian** (symbol: **rad**) when it subtends an arc of length R. A full turn measures $2\pi \ \text{rad}$, therefore:

$$\boxed{\pi \ \text{rad} = 180^\circ}$$

The word "radian" naturally derives from "radius" (*radius* in Latin). The idea of measuring angles by arc length dates back to the English mathematician **Roger Cotes** (1714); the name *radian* appeared in print for the first time in **1873**, in examination papers set by **James Thomson** (brother of Lord Kelvin) in Belfast.

> 💡 **Did You Know?**
>
> The radian is dimensionless: it is a ratio of two lengths ($\frac{\text{arc}}{\text{radius}}$, meters divided by meters). That is why we often omit the unit: we write "the angle $\frac{\pi}{3}$" rather than "$\frac{\pi}{3}$ rad". When no unit is specified, **it is always radians**.

2.2 Conversions

The proportionality π rad $\leftrightarrow 180^\circ$ gives:

$$\text{measure in rad} = \text{measure in degrees} \times \frac{\pi}{180} \quad ; \quad \text{measure in degrees} = \text{measure in rad} \times \frac{180}{\pi}.$$

Degrees	0°	30°	45°	60°	90°	120°	135°	150°	180°	270°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π

And 1 rad $= \frac{180}{\pi}^\circ \approx 57.3^\circ$.

> **Worked Example 1 — Conversions**
>
> a) Convert 75° to radians. b) Convert $\frac{7\pi}{9}$ rad to degrees. c) Convert 2.5 rad to degrees.
>
> a) $75 \times \frac{\pi}{180} = \frac{75\pi}{180} = \frac{5\pi}{12}$ (dividing numerator and denominator by 15).
> b) $\frac{7\pi}{9} \times \frac{180}{\pi} = \frac{7\times 180}{9} = 7 \times 20 = 140^\circ$.
> c) $2.5 \times \frac{180}{\pi} = \frac{450}{\pi} \approx 143.2^\circ$.

> ⚠️ ⚠️ **Why the Radian Is the Only Natural Unit in Calculus and Kinematics**
>
> In elementary surveying, degrees divide the circle into 360 arbitrary historical slices. But in physics and mathematical analysis, the **radian** is mandatory for two profound reasons:
>
> - **In kinematics:** The linear velocity v of a particle at radius R rotating at angular speed ω is simply $v = R\omega$ **if and only if** ω is in radians per second (rad $\cdot$ s^{-1}). In degrees per second, an unnatural conversion factor $\frac{\pi}{180}$ would infect the relation!
> - **In calculus:** The famous differentiation law $(\sin x)' = \cos x$ holds **only when x is measured in radians**. If differentiated with respect to degrees, the chain rule forces an unwanted scalar:
>
> $$(\sin(x^\circ))' = \frac{\pi}{180}\cos(x^\circ).$$
>
> To eliminate this parasitic factor $\frac{\pi}{180}$ from every differential equation in mechanics and wave theory, the radian is the universal standard of mathematics.

2.3 Arc Length and Sector Area

Because the radian is defined by "arc $=$ radius for 1 rad", the arc length is directly proportional to the angle: 1 rad $\leftrightarrow R$, hence θ rad $\leftrightarrow R\theta$.

> 📐 **Property — Arc Length, Sector Area**
>
> In a circle of radius R, a central angle of θ **radians** subtends:
>
> - an arc of length $\boxed{\ell = R\,\theta}$;
> - an angular sector (a "slice of pie") of area $\boxed{\mathcal{A} = \frac{1}{2}R^2\theta}$.

Justification of the area: The full disk (angle 2π) has area πR^2; by proportionality, an angle θ yields an area of $\pi R^2 \times \frac{\theta}{2\pi} = \frac{1}{2}R^2\theta$.

Notice how remarkably simple these formulas are **in radians**. In degrees, one would have to write $\ell = \frac{\pi R \alpha}{180}$: the nuisance factor $\frac{\pi}{180}$ appears everywhere. This is the first great virtue of the radian: **it eliminates parasitic constants**. You will see a second, even more spectacular reason in Grade 12: the derivative of $\sin x$ is $\cos x$ **only when x is in radians**.

Worked Example 2 — The Ferris Wheel Gondola

The wheel has a radius of $20\ \text{m}$. The gondola has traveled $50\ \text{m}$ along the circumference. Through what angle has the wheel turned? And how many full turns after $1\ \text{km}$?

$\theta = \frac{\ell}{R} = \frac{50}{20} = 2.5\ \text{rad}$, which is approximately $143.2°$ (Example 1c). For $1000\ \text{m}$: $\theta = \frac{1000}{20} = 50\ \text{rad}$. One full revolution is $2\pi \approx 6.283\ \text{rad}$, and $\frac{50}{2\pi} \approx 7.96$: the wheel has completed 7 full revolutions and almost an eighth turn (falling short by about 0.04 of a turn).

Worked Example 3 — A Slice of Pizza

A pizza with a diameter of $30\ \text{cm}$ is sliced into sectors of angle $\frac{\pi}{5}$. What is the crust length and the area of one slice? How many slices are there?

$R = 15\ \text{cm}$. Crust: $\ell = 15 \times \frac{\pi}{5} = 3\pi \approx 9.42\ \text{cm}$. Area: $\frac{1}{2} \times 15^2 \times \frac{\pi}{5} = \frac{225\pi}{10} = 22.5\pi \approx 70.7\ \text{cm}^2$. Number of slices: $\frac{2\pi}{\pi/5} = 10$.

⚠ Beware: Common Pitfall! — Calculator Angle Mode

Calculators have a **DEG** mode and a **RAD** mode. If you enter `cos(60)` in RAD mode, you obtain $\cos(60\ \text{rad}) \approx -0.952$, rather than 0.5! Conversely, `cos(π/3)` in DEG mode yields $\cos(1.047°) \approx 0.9998$. Before every exam or calculation, verify the mode. In Python, `math.cos` **always** expects radians; to convert degrees, use `math.radians(60)`.

⚠ Beware: Common Pitfall! — Arc Formulas with Degrees

$\ell = R\theta$ is **false** if θ is in degrees: with $R = 1$ and $\theta = 90$, one would obtain an arc of 90 meters on a circle whose total circumference is only $6.28\ \text{m}$! Always convert to radians first.

---

3. The Unit Circle and Winding the Real Line

3.1 The Oriented Unit Circle

We work in an orthonormal Cartesian coordinate system $(O; \vec{\imath}, \vec{\jmath})$—recall: "orthonormal" means both axes are perpendicular and have the same unit length. Let $I(1;0)$ and $J(0;1)$.

📐 Definition — Unit Circle (Trigonometric Circle)

The **unit circle** (or trigonometric circle) is the circle $\mathcal{C}$ centered at the origin O with radius 1, **oriented**: we choose as the **positive direction** (or direct, counterclockwise, or **trigonometric** direction) the direction opposite to the hands of a clock.

Why this direction? Because it is the direction that leads from I (on the horizontal axis) to J (on the vertical axis) via the shortest path, just as one moves from $\vec{\imath}$ to $\vec{\jmath}$. It is a **convention**—clocks happen to turn in the opposite, clockwise (or indirect) direction.

On this circle of radius 1, the formula $\ell = R\theta$ becomes $\ell = \theta$: **the radian measure of an angle is precisely the length of the subtended arc**. Here lies the immense power of radius 1.

3.2 Winding: Every Real Number Becomes a Point

Now picture the real number line as an infinitely long, calibrated **measuring tape**. Place it vertically, tangent to the circle at point I, with the zero mark of the line resting on I. Then **wind** it:

- the positive half-line wraps around the circle in the **counterclockwise (positive) direction**;
- the negative half-line wraps around in the **clockwise (negative) direction**.

Each real number x on the tape comes to rest on a specific point on the circle: we call it the **image point** of x, denoted $M(x)$.

- $x = \frac{\pi}{2}$: we have wound one quarter of the circumference ($\frac{2\pi}{4}$): we arrive at $J(0;1)$.
- $x = \pi$: a half-turn: point $I'(-1;0)$.
- $x = -\frac{\pi}{2}$: a quarter-turn in the negative direction: point $J'(0;-1)$.
- $x = 2\pi$: a complete turn: back at $I(1;0)$.

> **Definition — Image Point of a Real Number**
>
> To every real number x, the winding process associates a unique point M on the unit circle, called the **image of x**. We also say that x is **a radian measure** of the directed angle $(\overrightarrow{OI}, \overrightarrow{OM})$.

3.3 One Point, Infinitely Many Reals: "Modulo 2π"

The tape is infinite, but the circle is finite: after one full lap, we pass through the same points again. The real numbers $\frac{\pi}{3}$, $\frac{\pi}{3} + 2\pi = \frac{7\pi}{3}$, $\frac{\pi}{3} + 4\pi = \frac{13\pi}{3}$, $\frac{\pi}{3} - 2\pi = -\frac{5\pi}{3}$… all share **the exact same image point**.

> **Property — Real Numbers Sharing the Same Image**
>
> Two real numbers x and y have the same image on the circle if and only if their difference is an integer multiple of 2π:
>
> $$y = x + 2k\pi \quad \text{with} \quad k \in \mathbb{Z}.$$
>
> We write $y \equiv x\ [2\pi]$, read as "y is congruent to x **modulo** 2π".

Reading the notations: $k \in \mathbb{Z}$ means "k belongs to $\mathbb{Z}$", i.e., "k is an integer" ($\ldots, -2, -1, 0, 1, 2, \ldots$); $\mathbb{Z}$ comes from the German *Zahlen*, "numbers" (Chapter 31). The word *modulo* ("according to the measure of") was introduced by **Gauss** in 1801 for modular arithmetic; here it means "up to an integer number of full revolutions".

> **Definition — Principal Measure**
>
> Among all the real numbers sharing the same image point M, there exists **exactly one** in the half-open interval $]-\pi, \pi]$. It is called the **principal measure** of the angle.

(Recall of interval notation: $]-\pi, \pi]$ denotes the set of real numbers x such that $-\pi < x \leq \pi$; the bracket pointing outward at $-\pi$ means $-\pi$ is **excluded**, while the bracket pointing inward at π means π is **included**. We exclude $-\pi$ because it represents the same image point as π: otherwise, point I' would have two principal measures.)

> **Method — Finding the Principal Measure of $\frac{p\pi}{q}$**
>
> 1. Look at $\frac{p}{q}$ and seek the nearest **even** multiple: we want $\frac{p\pi}{q} = \alpha + 2k\pi$ with $-\pi < \alpha \leq \pi$. 2. In practice: divide p by $2q$ to determine how many full revolutions ($2\pi = \frac{2q\pi}{q}$) to subtract or add. 3. Adjust by one turn if the remaining value falls outside $]-\pi, \pi]$.

Worked Example 4 — Principal Measures

Determine the principal measure of: a) $\frac{37\pi}{4}$; b) $-\frac{17\pi}{3}$; c) $\frac{2023\pi}{6}$.

a) One full turn is $\frac{8\pi}{4}$. We write $37 = 8 \times 4 + 5$, so $\frac{37\pi}{4} = \frac{5\pi}{4} + 4 \times 2\pi$. But $\frac{5\pi}{4} > \pi$: we subtract one more full turn: $\frac{5\pi}{4} - 2\pi = -\frac{3\pi}{4}$. The principal measure is $\boxed{-\frac{3\pi}{4}}$. *Verification:* $-\frac{3\pi}{4} + 10\pi = -\frac{3\pi}{4} + \frac{40\pi}{4} = \frac{37\pi}{4}$ ✓.

b) One full turn is $\frac{6\pi}{3}$. $-\frac{17\pi}{3} = -\frac{18\pi}{3} + \frac{\pi}{3} = -6\pi + \frac{\pi}{3}$. Since $-6\pi = -3 \times 2\pi$, the principal measure is $\boxed{\frac{\pi}{3}}$.

c) One full turn is $\frac{12\pi}{6}$. $2023 = 12 \times 168 + 7$ (since $12 \times 168 = 2016$). Thus $\frac{2023\pi}{6} = \frac{7\pi}{6} + 168 \times 2\pi$. Since $\frac{7\pi}{6} > \pi$, subtract one turn: $\frac{7\pi}{6} - 2\pi = -\frac{5\pi}{6}$. The principal measure is $\boxed{-\frac{5\pi}{6}}$.

⚠ Beware: Common Pitfall! — Principal Measure and Interval

The principal measure lies in $]-\pi, \pi]$, **not** in $[0, 2\pi[$. For $\frac{37\pi}{4}$, answering $\frac{5\pi}{4}$ is incorrect as the "principal measure" (even though $\frac{5\pi}{4}$ has the exact same image point).

3.4 Marking Special Points

By dividing the circle into 4, 8, and 12 equal parts, we locate the images of multiples of $\frac{\pi}{2}$, $\frac{\pi}{4}$, $\frac{\pi}{6}$, and $\frac{\pi}{3}$:

Method — Locating $\frac{k\pi}{n}$ Quickly

Count in "slices of $\frac{\pi}{n}$": the upper half-turn (0 to π) contains n such slices. For $\frac{5\pi}{6}$, advance 5 slices out of 6 in the upper semicircle: you are almost at I', just above it. For $-\frac{2\pi}{3}$, step downward (negative direction) by 2 slices out of 3 in the lower semicircle.

---

4. Cosine and Sine of a Real Number

4.1 The Definition

The image point $M(x)$ has two coordinates in the Cartesian system $(O; \vec{\imath}, \vec{\jmath})$. We give them precise names:

Definition — Cosine and Sine of a Real Number

Let x be a real number and M its image point on the unit circle. We call:

- the **cosine of x**, written $\cos x$ or $\cos(x)$, the **x-coordinate (abscissa)** of M;
- the **sine of x**, written $\sin x$ or $\sin(x)$, the **y-coordinate (ordinate)** of M.

$$M(x) = \big(\cos x \,;\, \sin x\big)\,.$$

Reading: $\cos x$ is pronounced "cosine of x". The abbreviation *cos* originates from *complementi sinus*, "sine of the complement" (we shall see below that $\cos x = \sin(\frac{\pi}{2} - x)$). The notations *sin* and *cos* took hold in the seventeenth century, notably through the English mathematician **William Oughtred**; it was **Euler** who, in the eighteenth century, treated them firmly as **functions** of a real variable.

⚠ Beware: Common Pitfall! — It Is Not a Product

$\cos x$ is not "cos × x": "cos" by itself is meaningless. Thus $\cos(2x) \neq 2\cos x$, and $\cos(a+b) \neq \cos a + \cos b$ (for instance, $\cos(\frac{\pi}{2} + \frac{\pi}{2}) = \cos\pi = -1$ whereas $\cos\frac{\pi}{2} + \cos\frac{\pi}{2} = 0$). Standard notation: $\cos^2 x$ means $(\cos x)^2$, the square of the cosine—not to be confused with $\cos(x^2)$.

4.2 Return to the Ferris Wheel

Let us place the origin at the center of the wheel, with unit $1 = 20$ m. Measuring angles from the conventional horizontal line: the position of the gondola at angle x is $(20\cos x, 20\sin x)$ relative to the central axle. Since the gondola starts at the **bottom**, which corresponds to $x = -\frac{\pi}{2}$, after turning through an angle θ it is at angle $x = \theta - \frac{\pi}{2}$. Its height above the ground is:

$$h(\theta) = 22 + 20\sin\left(\theta - \frac{\pi}{2}\right).$$

Using the identity $\sin(\theta - \frac{\pi}{2}) = -\cos\theta$, we obtain the elegant formula $h(\theta) = 22 - 20\cos\theta$. Let us check: $\theta = 0$ gives $22 - 20 = 2$ m (boarding at bottom) ✓; $\theta = \pi$ gives $22 + 20 = 42$ m (summit) ✓.

Answer to your brother: After 50 m, $\theta = 2.5$ rad and $h = 22 - 20\cos(2.5) \approx 22 - 20 \times (-0.8011) \approx 38.0$ m. After 1 km, $\theta = 50$ rad and $h = 22 - 20\cos 50 \approx 22 - 20 \times 0.9650 \approx 2.7$ m: nearly back at the lowest point, as anticipated (7.96 revolutions).

4.3 First Properties

> 📐 ***Fundamental Properties*** *— For every real number x and every integer k: 1. $-1 \leq \cos x \leq 1$ and $-1 \leq \sin x \leq 1$; 2. $\cos(x + 2k\pi) = \cos x$ and $\sin(x + 2k\pi) = \sin x$; 3.* $\boxed{\cos^2 x + \sin^2 x = 1}$.

Proof

1. Point M lies on the circle centered at O with radius 1, so its coordinates are bounded between -1 and 1. 2. x and $x + 2k\pi$ share the same image point (§3.3), and hence have the exact same coordinates. 3. $OM = 1$. The distance formula in an orthonormal frame (Chapter 20, a consequence of the Pythagorean Theorem) gives $OM^2 = (\cos x - 0)^2 + (\sin x - 0)^2$, hence $\cos^2 x + \sin^2 x = 1$. ■

4.4 Consistency with Middle School

Does this new definition contradict the middle-school triangle definition? Not at all.

> 📐 **Property — Connection to the Right-Angled Triangle**
>
> If $0 < x < \frac{\pi}{2}$, then $\cos x$ and $\sin x$ coincide with the cosine and sine of the acute angle of measure x in a right-angled triangle.

Proof

Let $M(x)$ with $0 < x < \frac{\pi}{2}$: M lies in the upper-right quadrant. Let H be the orthogonal projection of M onto the horizontal axis. The triangle OHM is right-angled at H, with hypotenuse $OM = 1$, and the angle $\widehat{HOM}$ measures x rad. In the sense of middle school: $\cos\widehat{HOM} = \frac{OH}{OM} = \frac{OH}{1} = OH$, and OH is precisely the positive x-coordinate of M. Similarly, $\sin\widehat{HOM} = \frac{HM}{OM} = HM$, which is the y-coordinate of M. ■

4.5 Signs According to Quadrants

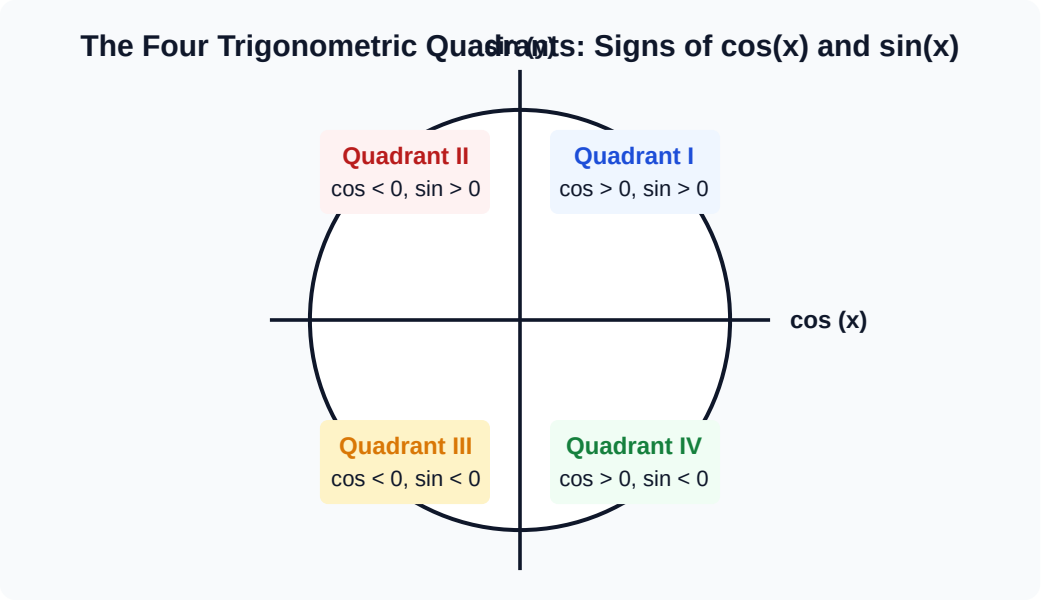

Figure 90: The four trigonometric quadrants: signs of cos and sin

Worked Example 5 — Using $\cos^2 + \sin^2 = 1$

Suppose that $x \in]\frac{\pi}{2}, \pi[$ and $\sin x = \frac{3}{5}$. Compute $\cos x$.
$\cos^2 x = 1 - \sin^2 x = 1 - \frac{9}{25} = \frac{16}{25}$, so $\cos x = \frac{4}{5}$ or $\cos x = -\frac{4}{5}$. However, x lies in Quadrant II, where the cosine is strictly negative. Therefore, $\boxed{\cos x = -\frac{4}{5}}$.

⚠ Beware: Common Pitfall! — Never Forget the Sign

$\cos^2 x = \frac{16}{25}$ does not automatically imply $\cos x = \frac{4}{5}$. The quadratic equation $X^2 = a$ (with $a > 0$) has **two** solutions, $\sqrt{a}$ and $-\sqrt{a}$ (Chapter 23). It is the **geometric location of the point** on the circle that decides between them.

CHAPTER 38: Pulling the Rope and the Effective Shadow

> “When you pull a burden with an oblique rope, your muscles burn, your brow drips with sweat, and yet the load crawls forward at a snail’s pace. Why? Because the universe does not reward you for effort spent into the void: it counts only the useful shadow of your force along the path.”

What You Are About to Discover

- Why ordinary multiplication between two numbers is not enough when we want to multiply **two arrows** (two vectors).
- The physical meaning of the **work of a force** and the discovery of the **scalar product** (dot product).
- The geometry of the **orthogonal projection**: how to let light fall from above to measure the effective shadow.
- The four faces of the dot product: by **cosine**, by **projection**, by **Cartesian coordinates** ($xx' + yy'$), and by **lengths alone** (polarization identities).
- The absolute criterion of **orthogonality** and its crowning application: finding the normal vector to a line and its Cartesian equation $ax + by + c = 0$.
- The powerful metric theorems of the triangle: **Al-Kashi’s theorem** (generalized Pythagorean theorem / Law of Cosines) and the **median theorem**.
- Determining **geometric loci**: characterizing a circle by the dot product $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$.

---

1. The Problem: The Towhorse and the Coal Barge

At the dawn of the nineteenth century, in the furnace of the Industrial Revolution, coal is the lifeblood powering the steam engines. To transport thousands of tons of coal from northern collieries to the factories of Paris or London, the inland waterway network is the most economical route. Heavy, broad-beamed barges weighing over a hundred tons glide silently along the canals.

These barges do not yet have engines. They are towed from dry land by a powerful draft horse treading heavily along the dirt path running beside the canal: the **towpath**.

Observe the scene from a footbridge above the water:

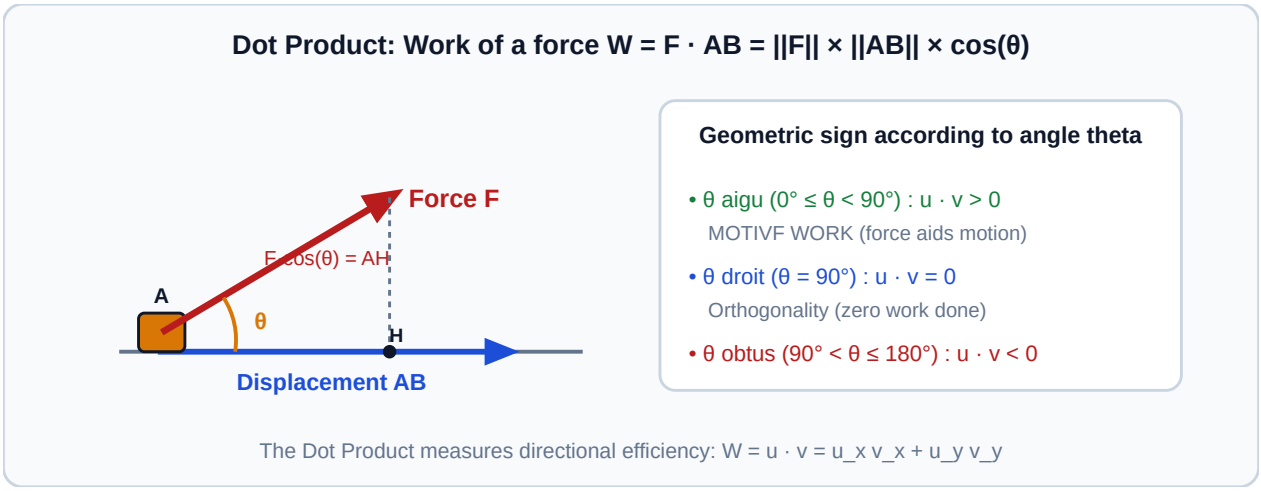

Figure 91: Dot product and work of a force: effective component and geometric sign

The canal is straight. The barge glides forward along a displacement vector $\vec{d}$. But the horse cannot walk on water! It is forced to march along the bank. The long hemp rope therefore makes an **oblique angle θ** relative to the direction of the canal.

The horse strains its haunches, its hooves dig into the clay, its chest presses into the leather harness collar: it exerts a considerable force $\vec{F}$. Yet a young barge apprentice sitting by the tiller notices something puzzling: *"The horse is exhausting itself, yet the barge moves forward only very slowly. Why?"*

The seasoned boatman replies: *"Because part of its force pulls the boat sideways toward the bank! That part contributes nothing to moving forward: it tries only to ground the barge in the reeds. I have to fight against it constantly by leaning on the tiller. Only the part of the force pointing straight ahead, parallel to the water, produces useful motion."*

To measure the energy actually delivered to the boat, multiplying force by distance is not enough. We must **extract the effective shadow of the force along the direction of motion**.

---

2. The Work of a Force and the Geometric Definition

2.1 From Physical Action to the Scalar Product

In physics, when an object undergoes a constant force $\vec{F}$ over a rectilinear displacement $\vec{d} = \overrightarrow{AB}$, the mechanical energy transferred is called the **work of the force**, denoted W (for *Work*).

If the force were perfectly aligned with the motion ($\theta = 0$), the entire force would contribute: $W = \| \vec{F} \| \times \| \vec{d} \|$. When the rope is tilted at an angle θ, the effective shadow of the force along the path is $\| \vec{F} \| \cos\theta$. The work is then written:

$$W = \| \vec{F} \| \times \| \vec{d} \| \times \cos\theta.$$

To formalize this mathematical operation combining two vectors $\vec{u}$ and $\vec{v}$ to produce **a single real number** (a scalar, as opposed to a vector), mathematicians forged a universal concept.

> **📐 Definition — Scalar Product (by Norm and Cosine)**
>
> Let $\vec{u}$ and $\vec{v}$ be two vectors in the plane.
>
> - If either vector is the zero vector ($\vec{u} = \vec{0}$ or $\vec{v} = \vec{0}$), their scalar product is zero: $\vec{u} \cdot \vec{v} = 0$.
> - If $\vec{u} \neq \vec{0}$ and $\vec{v} \neq \vec{0}$, set $\vec{u} = \overrightarrow{OA}$ and $\vec{v} = \overrightarrow{OB}$. The **scalar product** (or **dot product**) of $\vec{u}$ and $\vec{v}$, denoted $\vec{u} \cdot \vec{v}$, is the real number defined by:
>
> $$\vec{u} \cdot \vec{v} = \| \vec{u} \| \times \| \vec{v} \| \times \cos(\widehat{AOB}).$$

Reading and Pronunciation: $\vec{u} \cdot \vec{v}$ is read aloud as "***$\vec{u}$* dot *$\vec{v}$***". The centered dot "·" is reserved for this operation.

> **💡 The True History — Where Does the Word "Scalar" Come From?**
>
> The word comes from the Latin *scala*, meaning "ladder" or "scale". The Irish mathematician **William Rowan Hamilton** introduced it in 1843 when inventing quaternions: he called the pure numerical part, which is marked on a standard numerical scale, the "scalar part", distinguishing it from the directional "vector part". The modern dot product with a dot symbol was popularized by the American physicist **Josiah Willard Gibbs** in the late nineteenth century.

2.2 The Three Regimes of the Rope

The value of the dot product depends on the angle θ between the two vectors:

1. **Acute angle ($0 \leq \theta < \frac{\pi}{2}$): strictly positive dot product.** $\cos\theta > 0$. The force assists the displacement: physicists speak of **motive work**. The shadow falls in the direction of motion.
2. **Right angle ($\theta = \frac{\pi}{2}$, or 90°): zero dot product.** $\cos(\frac{\pi}{2}) = 0$, hence $\vec{u} \cdot \vec{v} = 0$. If the horse pulls perpendicularly to the canal, the barge does not advance a millimeter along the waterway. All energy is wasted trying to collapse the bank.

3. **Obtuse angle ($\frac{\pi}{2} < \theta \leq \pi$): strictly negative dot product.** $\cos\theta < 0$. The force opposes the displacement: this is the role of brakes or hydrodynamic friction, which physicists call **resistive work**.

> ⚠️ **Beware: Common Pitfall! — A Scalar Is Not a Vector**
>
> The dot product of two vectors is **never a vector**! It is a **real number** (which can be positive, zero, or negative). Writing $\vec{u} \cdot \vec{v} = \vec{0}$ with an arrow over the zero is a severe error of notation. Always write $\vec{u} \cdot \vec{v} = 0$.

---

3. The Orthogonal Projection: Measuring the Cast Shadow

3.1 Definition by Projection

Instead of calculating a cosine, it is often much faster to drop a perpendicular.

> 📐 **Property — Definition by Orthogonal Projection**
>
> Let $\vec{u} = \overrightarrow{OA}$ and $\vec{v} = \overrightarrow{OB}$ be two non-zero vectors. Let H be the orthogonal projection of point B onto the line (OA).
>
> $$\vec{u} \cdot \vec{v} = \overrightarrow{OA} \cdot \overrightarrow{OB} = \overrightarrow{OA} \cdot \overrightarrow{OH}.$$
>
> In terms of directed lengths along the axis (OA):
> - If $\overrightarrow{OH}$ and $\overrightarrow{OA}$ point in the **same direction** (acute angle): $\vec{u} \cdot \vec{v} = OA \times OH$.
> - If $\overrightarrow{OH}$ and $\overrightarrow{OA}$ point in **opposite directions** (obtuse angle): $\vec{u} \cdot \vec{v} = -OA \times OH$.

Proof

In the triangle OHB, right-angled at H:
- If the angle $\widehat{AOB}$ is acute, $\cos(\widehat{AOB}) = \frac{OH}{OB}$, so $OH = OB\cos(\widehat{AOB})$. Thus, $\overrightarrow{OA} \cdot \overrightarrow{OB} = OA \times OB\cos(\widehat{AOB}) = OA \times OH$.
- If the angle $\widehat{AOB}$ is obtuse, the projected point H lies on the opposite side of O. The adjacent angle inside the right-angled triangle is $\pi - \widehat{AOB}$. Since $\cos(\pi - \theta) = -\cos\theta$, we have $OH = OB\cos(\pi - \widehat{AOB}) = -OB\cos(\widehat{AOB})$. Consequently, $OA \times OB\cos(\widehat{AOB}) = -OA \times OH$. In both cases, the result exactly matches the stated formula. ■

> 💡 **Did You Know? — The Symmetry of the Dot Product**
>
> Projecting B onto (OA) gives the exact same result as projecting A onto (OB)! Indeed, $\vec{u} \cdot \vec{v} = \| \vec{u} \| \| \vec{v} \| \cos\theta = \| \vec{v} \| \| \vec{u} \| \cos\theta = \vec{v} \cdot \vec{u}$. The dot product is **commutative**.

3.2 The Scalar Square

What happens when we take the dot product of a vector with itself? The angle between $\vec{u}$ and $\vec{u}$ is zero ($\theta = 0$), and $\cos(0) = 1$.

> 📐 **Definition — Scalar Square**
>
> For any vector $\vec{u}$, we write $\vec{u}^2 = \vec{u} \cdot \vec{u}$. We have:
>
> $$\vec{u}^2 = \| \vec{u} \|^2.$$
>
> The scalar square of a vector is equal to the square of its norm. It is a **non-negative real number**, and $\vec{u}^2 = 0 \Leftrightarrow \vec{u} = \vec{0}$. For a geometric segment $[AB]$, we have $\overrightarrow{AB}^2 = AB^2$.

4. Analytical Expression in an Orthonormal Coordinate System

4.1 The Formula $xx' + yy'$

When the plane is equipped with an orthonormal coordinate system $(O; \vec{\imath}, \vec{\jmath})$, the unit basis vectors satisfy:

- $\| \vec{\imath} \| = 1$ and $\| \vec{\jmath} \| = 1$, so $\vec{\imath}^2 = 1$ and $\vec{\jmath}^2 = 1$.
- $\vec{\imath}$ and $\vec{\jmath}$ are perpendicular, so $\vec{\imath} \cdot \vec{\jmath} = 0$.

Here the beauty of algebra dramatically simplifies geometric computations:

> ⚠ ⚠ **Absolute Requirement — Coordinate System Must Be ORTHONORMAL**
>
> The Cartesian formula $\vec{u} \cdot \vec{v} = xx' + yy'$ fails completely if axes are not mutually perpendicular and unit-length! In skewed coordinates, cross-terms involving axis angles arise. Always verify and state that the coordinate system is orthonormal.

> 📐 **Theorem — Analytical Formula for the Dot Product**
>
> In an **orthonormal** coordinate system $(O; \vec{\imath}, \vec{\jmath})$, if $\vec{u}\begin{pmatrix} x \\ y \end{pmatrix}$ and $\vec{v}\begin{pmatrix} x' \\ y' \end{pmatrix}$, then:
>
> $$\boxed{\vec{u} \cdot \vec{v} = xx' + yy'}$$

Proof

By the bilinearity of the scalar product (expanding just like an ordinary product):

$$\begin{aligned} \vec{u} \cdot \vec{v} &= (x\vec{\imath} + y\vec{\jmath}) \cdot (x'\vec{\imath} + y'\vec{\jmath}) \\ &= xx'(\vec{\imath} \cdot \vec{\imath}) + xy'(\vec{\imath} \cdot \vec{\jmath}) + yx'(\vec{\jmath} \cdot \vec{\imath}) + yy'(\vec{\jmath} \cdot \vec{\jmath}). \end{aligned}$$

Since $\vec{\imath} \cdot \vec{\imath} = 1$, $\vec{\jmath} \cdot \vec{\jmath} = 1$, and $\vec{\imath} \cdot \vec{\jmath} = \vec{\jmath} \cdot \vec{\imath} = 0$, the two cross-terms vanish entirely! We are left with:

$$\vec{u} \cdot \vec{v} = xx'(1) + 0 + 0 + yy'(1) = xx' + yy'. \quad \blacksquare$$

> ⚠ **Beware: Common Pitfall! — Coordinate System Must Be Orthonormal**
>
> The formula $\vec{u} \cdot \vec{v} = xx' + yy'$ is **false** if the axes are not perpendicular or if the units on the two axes differ! It holds true only in an **orthonormal** coordinate system.

5. Polarization Identities: Computing with Lengths Alone

Can we find the dot product of two vectors without knowing their coordinates or the angle between them, using only the lengths of the segments that compose them? Yes, thanks to **vector binomial identities**.

For any vectors $\vec{u}$ and $\vec{v}$:

$$\| \vec{u} + \vec{v} \|^2 = (\vec{u} + \vec{v})^2 = \vec{u}^2 + 2\vec{u} \cdot \vec{v} + \vec{v}^2 = \| \vec{u} \|^2 + 2\vec{u} \cdot \vec{v} + \| \vec{v} \|^2 .$$

Isolating the dot product $\vec{u} \cdot \vec{v}$ yields the famous polarization formulas:

Theorem — Polarization Identities

For all vectors $\vec{u}$ and $\vec{v}$:

1. $\vec{u}\cdot\vec{v} = \frac{1}{2}\left(\|\vec{u}+\vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2\right)$
2. $\vec{u}\cdot\vec{v} = \frac{1}{2}\left(\|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{u}-\vec{v}\|^2\right)$
3. $\vec{u}\cdot\vec{v} = \frac{1}{4}\left(\|\vec{u}+\vec{v}\|^2 - \|\vec{u}-\vec{v}\|^2\right)$

In a triangle ABC with $\vec{u} = \overrightarrow{AB}$ and $\vec{v} = \overrightarrow{AC}$, notice that $\vec{u} - \vec{v} = \overrightarrow{AB} - \overrightarrow{AC} = \overrightarrow{CB}$. The second identity becomes:

$$\overrightarrow{AB}\cdot\overrightarrow{AC} = \frac{1}{2}(AB^2 + AC^2 - BC^2).$$

This provides a direct bridge between the lengths of the three sides of a triangle and the dot product!

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6. Orthogonality, Normal Vectors, and Equations of Lines

6.1 The Orthogonality Criterion

Fundamental Theorem of Orthogonality

Two vectors $\vec{u}$ and $\vec{v}$ are orthogonal if and only if:

$$\vec{u} \perp \vec{v} \Leftrightarrow \vec{u}\cdot\vec{v} = 0.$$

In an orthonormal coordinate system with $\vec{u}(x;y)$ and $\vec{v}(x';y')$:

$$\vec{u} \perp \vec{v} \Leftrightarrow xx' + yy' = 0.$$

6.2 Normal Vector to a Line

In Grade 10, a line was characterized by a point and a *direction vector* $\vec{u}$ parallel to the line. In Grade 11, the dot product allows a much more convenient approach: characterizing the line by an arrow that is **perpendicular** to it.

Definition — Normal Vector

A non-zero vector $\vec{n}$ is said to be **normal** to a line (d) if it is orthogonal to every direction vector of (d).

A point $M(x;y)$ belongs to the line (d) passing through $A(x_A;y_A)$ with normal vector $\vec{n}(a;b)$ if and only if:

$$\overrightarrow{AM}\cdot\vec{n} = 0 \Leftrightarrow a(x-x_A) + b(y-y_A) = 0 \Leftrightarrow ax + by + (-ax_A - by_A) = 0.$$

Setting $c = -ax_A - by_A$, we retrieve the standard Cartesian equation!

Theorem — Cartesian Equation and Normal Vector

- In an orthonormal coordinate system, every line with normal vector $\vec{n}\begin{pmatrix}a\\b\end{pmatrix}$ has a Cartesian equation of the form:
$$ax + by + c = 0 \quad (\text{with } (a;b) \neq (0;0)).$$
- Conversely, any equation of the form $ax + by + c = 0$ with $(a;b) \neq (0;0)$ is the equation of a straight line having normal vector $\vec{n}\begin{pmatrix}a\\b\end{pmatrix}$.

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7. The Two Great Metric Theorems of the Triangle

7.1 Al-Kashi's Theorem (Law of Cosines)

In an arbitrary triangle ABC, when the angle $\hat{A}$ is not right, what becomes of the Pythagorean theorem?

📐 Al-Kashi's Theorem (Generalized Pythagorean Theorem / Law of Cosines)

Let ABC be any triangle. Setting $a = BC$, $b = AC$, $c = AB$, and $\hat{A} = \widehat{BAC}$:

$$a^2 = b^2 + c^2 - 2bc\cos\hat{A}.$$

Likewise for the other two vertices:

$$b^2 = a^2 + c^2 - 2ac\cos\hat{B}, \qquad c^2 = a^2 + b^2 - 2ab\cos\hat{C}.$$

Geometric Proof via the Dot Product

Write Chasles' relation for the vector $\overrightarrow{BC}$: $\overrightarrow{BC} = \overrightarrow{BA} + \overrightarrow{AC} = \overrightarrow{AC} - \overrightarrow{AB}$. Take the scalar square of both sides:

$$BC^2 = \| \overrightarrow{BC} \|^2 = \| \overrightarrow{AC} - \overrightarrow{AB} \|^2 = \| \overrightarrow{AC} \|^2 - 2\overrightarrow{AC} \cdot \overrightarrow{AB} + \| \overrightarrow{AB} \|^2 .$$

Since $\| \overrightarrow{AC} \| = b$, $\| \overrightarrow{AB} \| = c$, and $\overrightarrow{AB} \cdot \overrightarrow{AC} = AB \times AC \cos\hat{A} = bc\cos\hat{A}$, substituting gives:

$$a^2 = b^2 + c^2 - 2bc\cos\hat{A}. \quad \blacksquare$$

Notice the beauty of the formula: if $\hat{A} = 90°$, then $\cos(90°) = 0$, and the corrective term $-2bc\cos\hat{A}$ vanishes: we recover exactly the Pythagorean theorem $a^2 = b^2 + c^2$! Al-Kashi contains Pythagoras as a special case.

💡 The True History — Ghiyath al-Din Jamshid al-Kashi

A fifteenth-century Persian astronomer and mathematician, Al-Kashi worked at the Samarkand observatory under the patronage of Prince Ulugh Beg. In his masterpiece, *The Key to Arithmetic* (1427), he calculated π to 16 correct decimal places (a world record that held for nearly two centuries) and compiled trigonometric tables of staggering precision. The formula that today bears his name in France enabled astronomers to solve any oblique triangle without dividing it into right-angled triangles.

7.2 The Median Theorem (Apollonius's Theorem)

Consider a triangle ABC and let I be the **midpoint** of side $[BC]$. How can we relate the length of the median AI to the lengths of the three sides?

📐 Median Theorem (Apollonius's Theorem)

In every triangle ABC, if I is the midpoint of segment $[BC]$, then:

$$AB^2 + AC^2 = 2AI^2 + \frac{1}{2}BC^2.$$

Proof

Express the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ passing through the midpoint I: $\overrightarrow{AB} = \overrightarrow{AI} + \overrightarrow{IB}$ and $\overrightarrow{AC} = \overrightarrow{AI} + \overrightarrow{IC}$. Take their scalar squares:

$$AB^2 = \| \overrightarrow{AI} + \overrightarrow{IB} \|^2 = AI^2 + 2\overrightarrow{AI} \cdot \overrightarrow{IB} + IB^2,$$

$$AC^2 = \| \overrightarrow{AI} + \overrightarrow{IC} \|^2 = AI^2 + 2\overrightarrow{AI} \cdot \overrightarrow{IC} + IC^2.$$

Add the two equations side by side:

$$AB^2 + AC^2 = 2AI^2 + 2\overrightarrow{AI} \cdot (\overrightarrow{IB} + \overrightarrow{IC}) + IB^2 + IC^2.$$

Since I is the midpoint of $[BC]$, we have $\overrightarrow{IB} + \overrightarrow{IC} = \vec{0}$! The cross-term cancels completely. Moreover, $IB = IC = \frac{BC}{2}$, so $IB^2 + IC^2 = \left(\frac{BC}{2}\right)^2 + \left(\frac{BC}{2}\right)^2 = 2\frac{BC^2}{4} = \frac{1}{2}BC^2$. We arrive directly at:

$$AB^2 + AC^2 = 2AI^2 + \frac{1}{2}BC^2. \quad \blacksquare$$

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8. Loci of Points and Equations of Circles

The dot product provides a method of unmatched elegance for finding the algebraic formula of a circle.

8.1 Circle Defined by Center and Radius

A point $M(x; y)$ lies on the circle $\mathcal{C}$ with center $\Omega(x_\Omega; y_\Omega)$ and radius R if and only if the distance ΩM equals R:

$$\Omega M^2 = R^2 \Leftrightarrow (x - x_\Omega)^2 + (y - y_\Omega)^2 = R^2.$$

8.2 Circle Defined by Diameter $[AB]$

Recall from middle school: any triangle inscribed in a circle with a diameter as its hypotenuse is a **right-angled triangle**. In other words: a point M (distinct from A and B) lies on the circle with diameter $[AB]$ if and only if the angle $\widehat{AMB}$ is a right angle, which vectorially translates to:

$$\overrightarrow{MA} \cdot \overrightarrow{MB} = 0.$$

(Note: if $M = A$ or $M = B$, one of the vectors is zero, so the dot product is still zero; the relation holds for all points on the circle.)

Theorem — Equation of a Circle with Diameter $[AB]$

The circle with diameter $[AB]$, where $A(x_A; y_A)$ and $B(x_B; y_B)$, is the set of points $M(x; y)$ satisfying:

$$\overrightarrow{MA} \cdot \overrightarrow{MB} = 0 \Leftrightarrow (x_A - x)(x_B - x) + (y_A - y)(y_B - y) = 0,$$

or equivalently:

$$\boxed{(x - x_A)(x - x_B) + (y - y_A)(y - y_B) = 0}$$

Step-by-Step Worked Examples

Worked Example 1 — Computing the Dot Product in Various Ways

Consider a square $ABCD$ of side length 4 and center O. Compute the following dot products: 1. $\overrightarrow{AB} \cdot \overrightarrow{AC}$ 2. $\overrightarrow{AB} \cdot \overrightarrow{AD}$ 3. $\overrightarrow{AB} \cdot \overrightarrow{CD}$ 4. $\overrightarrow{OA} \cdot \overrightarrow{OB}$

Detailed Solution: 1. By orthogonal projection: The orthogonal projection of C onto the line (AB) is point B. Therefore, $\overrightarrow{AB} \cdot \overrightarrow{AC} = \overrightarrow{AB} \cdot \overrightarrow{AB} = AB^2 = 4^2 = 16$. 2. By orthogonality: In a square, $(AB) \perp (AD)$, so the angle is a right angle: $\overrightarrow{AB} \cdot \overrightarrow{AD} = 0$. 3. Collinear vectors in opposite directions: $\overrightarrow{CD} = -\overrightarrow{AB}$. Hence $\overrightarrow{AB} \cdot \overrightarrow{CD} = -AB^2 = -16$. 4. In a square, the diagonals intersect perpendicularly at their midpoint O, so $(OA) \perp (OB)$. Thus, $\overrightarrow{OA} \cdot \overrightarrow{OB} = 0$.

Worked Example 2 — Computing an Angle Using Coordinates

In an orthonormal coordinate system, three points are given: $A(1;2)$, $B(4;6)$, and $C(-2;6)$. Compute the angle $\widehat{BAC}$ rounded to the nearest tenth of a degree.

Detailed Solution: Determine the coordinates of vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$:

- $\overrightarrow{AB}\begin{pmatrix}4-1\\6-2\end{pmatrix} = \begin{pmatrix}3\\4\end{pmatrix}$;
- $\overrightarrow{AC}\begin{pmatrix}-2-1\\6-2\end{pmatrix} = \begin{pmatrix}-3\\4\end{pmatrix}$.

1. Analytical dot product: $\overrightarrow{AB} \cdot \overrightarrow{AC} = xx' + yy' = 3 \times (-3) + 4 \times 4 = -9 + 16 = 7$.
2. Compute norms: $\| \overrightarrow{AB} \| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. $\| \overrightarrow{AC} \| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.
3. Extract the cosine: $\overrightarrow{AB} \cdot \overrightarrow{AC} = \| \overrightarrow{AB} \| \times \| \overrightarrow{AC} \| \times \cos \widehat{BAC}$, so $7 = 5 \times 5 \times \cos \widehat{BAC} = 25 \cos \widehat{BAC}$, giving $\cos \widehat{BAC} = \frac{7}{25} = 0.28$.
4. Using a calculator: $\widehat{BAC} = \arccos(0.28) \approx 73.7°$.

Worked Example 3 — Cartesian Equation Using a Normal Vector

Find a Cartesian equation of the line (d) passing through the point $A(3;-1)$ and perpendicular to the line (Δ) with equation $2x - 5y + 7 = 0$.

Detailed Solution:

- The line (Δ) has normal vector $\overrightarrow{n_\Delta}\begin{pmatrix}2\\-5\end{pmatrix}$.
- Since (d) is perpendicular to (Δ), a normal vector to (Δ) serves as a **direction vector** for (d). Hence $\vec{u}_d = \begin{pmatrix}2\\-5\end{pmatrix}$.
- A normal vector to (d) must be orthogonal to $\vec{u}_d$. The vector $\vec{n}_d\begin{pmatrix}5\\2\end{pmatrix}$ works since $2 \times 5 + (-5) \times 2 = 10 - 10 = 0$.
- The equation of (d) is therefore of the form $5x + 2y + c = 0$.
- Since $A(3;-1) \in (d)$, its coordinates satisfy the equation: $5(3) + 2(-1) + c = 0 \Leftrightarrow 15 - 2 + c = 0 \Leftrightarrow 13 + c = 0 \Leftrightarrow c = -13$.
- The Cartesian equation of (d) is therefore: $\boxed{5x + 2y - 13 = 0}$.

Worked Example 4 — Length and Angle Calculation via Al-Kashi

A surveyor inspects a triangular parcel ABC. He measures $AB = 60$ m, $AC = 100$ m, and the angle $\widehat{BAC} = 60°$. 1. Compute the distance BC. 2. Compute the angle $\widehat{ABC}$ to the nearest degree.

Detailed Solution: 1. By Al-Kashi's theorem: $BC^2 = AB^2 + AC^2 - 2 \times AB \times AC \times \cos \hat{A}$. $BC^2 = 60^2 + 100^2 - 2 \times 60 \times 100 \times \cos(60°)$. Since $\cos(60°) = \frac{1}{2}$: $BC^2 = 3600 + 10000 - 12000 \times 0.5 = 13600 - 6000 = 7600$. Hence $BC = \sqrt{7600} \approx 87.18$ m.

2. For angle $\hat{B}$, apply Al-Kashi isolating $\cos \hat{B}$: $AC^2 = AB^2 + BC^2 - 2 \times AB \times BC \times \cos \hat{B}$, $100^2 = 60^2 + 7600 - 2 \times 60 \times \sqrt{7600} \times \cos \hat{B}$, $10000 = 3600 + 7600 - 120\sqrt{7600} \cos \hat{B} = 11200 - 120\sqrt{7600} \cos \hat{B}$. $120\sqrt{7600} \cos \hat{B} = 11200 - 10000 = 1200$. $\cos \hat{B} = \frac{1200}{120\sqrt{7600}} = \frac{10}{\sqrt{7600}} \approx 0.1147$. Whence $\hat{B} = \arccos(0.1147) \approx 83°$.

Worked Example 5 — Equation of a Circle Defined by a Diameter

Given points $A(-1; 3)$ and $B(5; 1)$. 1. Determine a Cartesian equation of the circle $\mathcal{C}$ having diameter $[AB]$. 2. Find the coordinates of its center Ω and radius R, and verify the form $(x - x_\Omega)^2 + (y - y_\Omega)^2 = R^2$.

Detailed Solution: 1. $M(x; y) \in \mathcal{C} \Leftrightarrow \overrightarrow{MA} \cdot \overrightarrow{MB} = 0$. $\overrightarrow{MA}\begin{pmatrix} -1-x \\ 3-y \end{pmatrix}$ and $\overrightarrow{MB}\begin{pmatrix} 5-x \\ 1-y \end{pmatrix}$. $\overrightarrow{MA} \cdot \overrightarrow{MB} = (-1 - x)(5 - x) + (3 - y)(1 - y) = 0$. Expand: $(-5 + x - 5x + x^2) + (3 - 3y - y + y^2) = 0 \Leftrightarrow x^2 - 4x + y^2 - 4y - 2 = 0$.

2. Center Ω, midpoint of $[AB]$: $x_\Omega = \frac{-1+5}{2} = 2$, and $y_\Omega = \frac{3+1}{2} = 2$, so $\Omega(2; 2)$. Radius: $AB = \sqrt{(5 - (-1))^2 + (1 - 3)^2} = \sqrt{6^2 + (-2)^2} = \sqrt{36 + 4} = \sqrt{40} = 2\sqrt{10}$. The radius is $R = \frac{AB}{2} = \sqrt{10}$, hence $R^2 = 10$. The standard equation is $(x - 2)^2 + (y - 2)^2 = 10$. Expanding: $x^2 - 4x + 4 + y^2 - 4y + 4 = 10 \Leftrightarrow x^2 - 4x + y^2 - 4y - 2 = 0$. We obtain the exact same equation!

Worked Example 6 — Using the Median Theorem

Let ABC be a triangle such that $AB = 7$, $AC = 9$, and $BC = 8$. Compute the exact length of the median through A.

Detailed Solution: Let I be the midpoint of $[BC]$. The median through A is the segment $[AI]$. By the median theorem:

$$AB^2 + AC^2 = 2AI^2 + \frac{1}{2}BC^2.$$

Substitute the known values:

$$7^2 + 9^2 = 2AI^2 + \frac{1}{2}(8^2),$$

$$49 + 81 = 2AI^2 + \frac{1}{2}(64),$$

$$130 = 2AI^2 + 32.$$

Isolate AI^2:

$$2AI^2 = 130 - 32 = 98 \Longrightarrow AI^2 = 49 \Longrightarrow AI = \sqrt{49} = 7.$$

The median measures exactly 7.

Key Takeaways

1. **Four Equivalent Definitions of the Dot Product:**
   - Geometric: $\vec{u} \cdot \vec{v} = \| \vec{u} \| \| \vec{v} \| \cos \theta$.

- By projection: $\overrightarrow{OA} \cdot \overrightarrow{OB} = \pm OA \times OH$ (H is the orthogonal projection of B onto line (OA)).
- Analytical (orthonormal frame): $\vec{u}\begin{pmatrix} x \\ y \end{pmatrix} \cdot \vec{v}\begin{pmatrix} x' \\ y' \end{pmatrix} = xx' + yy'$.
- Polarization: $\vec{u} \cdot \vec{v} = \frac{1}{2}\left(\| \vec{u} + \vec{v} \|^2 - \| \vec{u} \|^2 - \| \vec{v} \|^2 \right)$.

2. **Orthogonality and Vector Geometry:**
   - $\vec{u} \perp \vec{v} \Leftrightarrow \vec{u} \cdot \vec{v} = 0 \Leftrightarrow xx' + yy' = 0$.
   - A line with normal vector $\vec{n}(a; b)$ has Cartesian equation $ax + by + c = 0$.
3. **Fundamental Triangle Theorems:**
   - Al-Kashi (Law of Cosines): $a^2 = b^2 + c^2 - 2bc\cos\hat{A}$.
   - Median: $AB^2 + AC^2 = 2AI^2 + \frac{1}{2}BC^2$ (where I is the midpoint of $[BC]$).
4. **Circles:**
   - Center $\Omega(x_\Omega; y_\Omega)$ and radius R: $(x - x_\Omega)^2 + (y - y_\Omega)^2 = R^2$.
   - Diameter $[AB]$: $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0 \Leftrightarrow (x - x_A)(x - x_B) + (y - y_A)(y - y_B) = 0$.

Exercises

Level (Direct Applications)

1. Let $\vec{u}$ and $\vec{v}$ be two vectors such that $\| \vec{u} \| = 5$, $\| \vec{v} \| = 6$, and the angle between them is $\frac{\pi}{3}$. Compute $\vec{u} \cdot \vec{v}$.
2. In an orthonormal frame, given $\vec{u}\begin{pmatrix} -3 \\ 4 \end{pmatrix}$ and $\vec{v}\begin{pmatrix} 2 \\ 5 \end{pmatrix}$. Compute $\vec{u} \cdot \vec{v}$.
3. Are the vectors $\vec{u}\begin{pmatrix} 4 \\ -6 \end{pmatrix}$ and $\vec{v}\begin{pmatrix} 9 \\ 6 \end{pmatrix}$ orthogonal?
4. Determine a normal vector $\vec{n}$ to the line with equation $3x - 7y + 11 = 0$.
5. Give a Cartesian equation of the line passing through $A(1; 4)$ with normal vector $\vec{n}\begin{pmatrix} 2 \\ -3 \end{pmatrix}$.
6. In a triangle ABC, given $AB = 5$, $AC = 8$, and $\hat{A} = 60°$. Compute the length BC.
7. Let $[AB]$ be a segment with $A(2; 1)$ and $B(6; 5)$. Write the equation of the circle with diameter $[AB]$.
8. Consider an equilateral triangle ABC with side length $a = 6$. Compute the dot product $\overrightarrow{AB} \cdot \overrightarrow{AC}$.

Level (Reasoning and Geometry)

9. Consider a rectangle $ABCD$ such that $AB = 6$ and $BC = 3$. Let I be the midpoint of $[AB]$. Compute $\overrightarrow{CA} \cdot \overrightarrow{DI}$.
10. Prove that in a rhombus, the diagonals intersect at right angles using the scalar product of $(\vec{u} + \vec{v})$ and $(\vec{u} - \vec{v})$.
11. In an orthonormal coordinate system, consider points $A(-2; 1)$, $B(2; 3)$, and $C(1; 5)$.
    a) Compute $\overrightarrow{BA} \cdot \overrightarrow{BC}$.
    b) Deduce the measure of angle $\widehat{ABC}$ to the nearest degree.
12. Find the real value k such that the vectors $\vec{u}\begin{pmatrix} 2k-1 \\ 3 \end{pmatrix}$ and $\vec{v}\begin{pmatrix} 4 \\ k+2 \end{pmatrix}$ are orthogonal.
13. Consider the triangle MNP with $MN = 5$, $NP = 7$, and $MP = 8$. Compute the angle $\widehat{MNP}$ to the nearest tenth of a degree.
14. In a triangle ABC, given $AB = 6$, $AC = 10$, and median $AI = 6$. Compute the length of side BC.
15. Determine the set of points $M(x; y)$ whose coordinates satisfy:
$$x^2 + y^2 - 6x + 8y = 0.$$
16. Find the distance from point $A(2; 3)$ to the line (d) with equation $4x + 3y - 7 = 0$ using a normal vector.

Level (Problems and Bac Challenges)

17. **Altitudes of a Triangle Concur:** Let ABC be a triangle. Let H be the intersection of the altitudes from A and from B.
    a) Express the condition that H lies on the first two altitudes using dot products.
    b) Deduce that $\overrightarrow{CH} \cdot \overrightarrow{AB} = 0$, and conclude that the three altitudes are concurrent.
18. **Geometric Locus and Level Set:** Let A and B be two distinct points with distance $AB = 6$ cm. Let I be the midpoint of $[AB]$. Determine and describe the set of points M in the plane satisfying:
$$\overrightarrow{MA} \cdot \overrightarrow{MB} = 7.$$

19. **Rugby Kick Optimization:** A rugby player must attempt a conversion following a try scored near the corner flag. The goal posts are at $P_1(0;2)$ and $P_2(0;-2)$. The kicker can move backward perpendicularly along the line $y = 5$. At what x-coordinate $x > 0$ should he place the ball so that the shooting angle $\widehat{P_1MP_2}$ is maximized?
20. **Heron's Formula Proved by Al-Kashi:** Let ABC be a triangle with side lengths a, b, c and semi-perimeter $p = \frac{a+b+c}{2}$. Using the area formula $\mathcal{S} = \frac{1}{2}bc\sin\hat{A}$ and Al-Kashi's formula for $\cos\hat{A}$, prove the celebrated formula of Heron of Alexandria:
$$\mathcal{S} = \sqrt{p(p-a)(p-b)(p-c)}.$$
21. **Power of a Point with Respect to a Circle:** Let $\mathcal{C}$ be a circle with center Ω and radius R, and let M be any point in the plane. A line through M intersects the circle at two points A and B. Show that the dot product $\overrightarrow{MA}\cdot\overrightarrow{MB}$ is independent of the choice of secant line and always equals $M\Omega^2 - R^2$.

Solutions

1. $\vec{u}\cdot\vec{v} = 5\times 6\times\cos\left(\frac{\pi}{3}\right) = 30\times 0.5 = 15$.

2. $\vec{u}\cdot\vec{v} = (-3)\times 2 + 4\times 5 = -6+20 = 14$.

3. $xx' + yy' = 4\times 9 + (-6)\times 6 = 36 - 36 = 0$. Yes, the vectors are orthogonal.

4. The vector $\vec{n}\begin{pmatrix}3\\-7\end{pmatrix}$ is normal to the line.

5. $2(x-1) - 3(y-4) = 0 \Leftrightarrow 2x - 3y - 2 + 12 = 0 \Leftrightarrow 2x - 3y + 10 = 0$.

6. Al-Kashi: $BC^2 = 5^2 + 8^2 - 2\times 5\times 8\times\cos(60^\circ) = 25 + 64 - 80\times 0.5 = 89 - 40 = 49$. Hence $BC = 7$.

7. $(x-2)(x-6) + (y-1)(y-5) = 0 \Leftrightarrow x^2 - 8x + 12 + y^2 - 6y + 5 = 0 \Leftrightarrow x^2 + y^2 - 8x - 6y + 17 = 0$.

8. $\overrightarrow{AB}\cdot\overrightarrow{AC} = 6\times 6\times\cos(60^\circ) = 36\times 0.5 = 18$.

9. Place an orthonormal coordinate system with origin D, (DA) along the x-axis, and (DC) along the y-axis. $D(0;0)$, $A(3;0)$, $B(3;6)$, $C(0;6)$. The midpoint I of $[AB]$ has coordinates $(3;3)$. Thus, $\overrightarrow{CA}\begin{pmatrix}3-0\\0-6\end{pmatrix} = \begin{pmatrix}3\\-6\end{pmatrix}$, and $\overrightarrow{DI}\begin{pmatrix}3\\3\end{pmatrix}$. The scalar product is: $\overrightarrow{CA}\cdot\overrightarrow{DI} = 3\times 3 + (-6)\times 3 = 9 - 18 = -9$.

10. Let $\vec{u}$ and $\vec{v}$ be the two vectors along the adjacent sides of a rhombus. Since all sides of a rhombus are equal in length, $\|\vec{u}\| = \|\vec{v}\|$, hence $\|\vec{u}\|^2 - \|\vec{v}\|^2 = 0$. The diagonals are represented by the vectors $\vec{u}+\vec{v}$ and $\vec{u}-\vec{v}$. Compute their dot product using the vector identity: $(\vec{u}+\vec{v})\cdot(\vec{u}-\vec{v}) = \|\vec{u}\|^2 - \|\vec{v}\|^2 = 0$. Since their dot product is zero, the diagonals of a rhombus are perpendicular.

11. a) $\overrightarrow{BA}\begin{pmatrix}-2-2\\1-3\end{pmatrix} = \begin{pmatrix}-4\\-2\end{pmatrix}$, and $\overrightarrow{BC}\begin{pmatrix}1-2\\5-3\end{pmatrix} = \begin{pmatrix}-1\\2\end{pmatrix}$. $\overrightarrow{BA}\cdot\overrightarrow{BC} = (-4)\times(-1) + (-2)\times 2 = 4 - 4 = 0$. b) Because the dot product is zero, the vectors are orthogonal: $\widehat{ABC} = 90^\circ$ exactly!

12. $\vec{u}\perp\vec{v} \Leftrightarrow (2k-1)\times 4 + 3\times(k+2) = 0 \Leftrightarrow 8k - 4 + 3k + 6 = 0 \Leftrightarrow 11k + 2 = 0 \Leftrightarrow k = -\frac{2}{11}$.

13. By Al-Kashi in MNP: $MP^2 = MN^2 + NP^2 - 2\times MN\times NP\times\cos\hat{N}$. $8^2 = 5^2 + 7^2 - 2\times 5\times 7\times\cos\hat{N} \Leftrightarrow 64 = 25 + 49 - 70\cos\hat{N} \Leftrightarrow 64 = 74 - 70\cos\hat{N}$. $70\cos\hat{N} = 10 \Leftrightarrow \cos\hat{N} = \frac{10}{70} = \frac{1}{7} \approx 0.1428$. Thus $\widehat{MNP} = \arccos\left(\frac{1}{7}\right) \approx 81.8^\circ$.

14. Median theorem: $AB^2 + AC^2 = 2AI^2 + \frac{1}{2}BC^2$. $6^2 + 10^2 = 2(6^2) + \frac{1}{2}BC^2 \Leftrightarrow 36 + 100 = 72 + \frac{1}{2}BC^2 \Leftrightarrow 136 = 72 + \frac{1}{2}BC^2$. $\frac{1}{2}BC^2 = 64 \Leftrightarrow BC^2 = 128 \Longrightarrow BC = \sqrt{128} = 8\sqrt{2} \approx 11.31$.

15. Complete the square: $(x^2 - 6x) + (y^2 + 8y) = 0 \Leftrightarrow (x-3)^2 - 9 + (y+4)^2 - 16 = 0 \Leftrightarrow (x-3)^2 + (y+4)^2 = 25$. This is the circle with center $\Omega(3;-4)$ and radius $R = \sqrt{25} = 5$.

16. Let H be the orthogonal projection of A onto (d). The distance from A to (d) is the length AH. The vector $\vec{n}\begin{pmatrix}4\\3\end{pmatrix}$ is normal to (d), with norm $\|\vec{n}\| = \sqrt{4^2+3^2} = 5$. Pick any point on (d), for instance $B(1;1)$ since $4(1) + 3(1) - 7 = 0$. The vector $\overrightarrow{BA} = \begin{pmatrix}2-1\\3-1\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix}$. The absolute value of the scalar product gives $|\overrightarrow{BA}\cdot\vec{n}| = \|\vec{n}\|\times AH$. Since $\overrightarrow{BA}\cdot\vec{n} = 1\times 4 + 2\times 3 = 10$: $10 = 5\times AH \Longrightarrow AH = \frac{10}{5} = 2$. The distance is exactly 2.

17. a) H lies on the altitude from A, so $\overrightarrow{AH}\perp\overrightarrow{BC} \Leftrightarrow \overrightarrow{AH}\cdot\overrightarrow{BC} = 0$. H lies on the altitude from B, so $\overrightarrow{BH}\perp\overrightarrow{AC} \Leftrightarrow \overrightarrow{BH}\cdot\overrightarrow{AC} = 0$. b) Compute $\overrightarrow{CH}\cdot\overrightarrow{AB}$: $\overrightarrow{CH}\cdot\overrightarrow{AB} = \overrightarrow{CH}\cdot(\overrightarrow{CB} - \overrightarrow{CA}) = \overrightarrow{CH}\cdot\overrightarrow{CB} - \overrightarrow{CH}\cdot\overrightarrow{CA}$. Notice that $\overrightarrow{CH} = \overrightarrow{CA} + \overrightarrow{AH} = \overrightarrow{CB} + \overrightarrow{BH}$. Substituting cleverly: $\overrightarrow{CH}\cdot\overrightarrow{CB} = (\overrightarrow{CA} + \overrightarrow{AH})\cdot\overrightarrow{CB} = \overrightarrow{CA}\cdot\overrightarrow{CB} + \overrightarrow{AH}\cdot\overrightarrow{CB} = \overrightarrow{CA}\cdot\overrightarrow{CB} + 0$. Similarly, $\overrightarrow{CH}\cdot\overrightarrow{CA} = (\overrightarrow{CB} + \overrightarrow{BH})\cdot\overrightarrow{CA} = \overrightarrow{CB}\cdot\overrightarrow{CA} + \overrightarrow{BH}\cdot\overrightarrow{CA} = \overrightarrow{CB}\cdot\overrightarrow{CA} + 0$. Thus, $\overrightarrow{CH}\cdot\overrightarrow{AB} =$

$\overrightarrow{CA}\cdot\overrightarrow{CB}-\overrightarrow{CB}\cdot\overrightarrow{CA}=0$. The vector $\overrightarrow{CH}$ is therefore orthogonal to $\overrightarrow{AB}$: line (CH) is the third altitude of the triangle. The three altitudes meet at the same point H: they are concurrent.

18. Express $\overrightarrow{MA}$ and $\overrightarrow{MB}$ through the midpoint I of $[AB]$: $\overrightarrow{MA}\cdot\overrightarrow{MB}=(\overrightarrow{MI}+\overrightarrow{IA})\cdot(\overrightarrow{MI}+\overrightarrow{IB})$. Since $\overrightarrow{IB}=-\overrightarrow{IA}$, we have the identity: $\overrightarrow{MA}\cdot\overrightarrow{MB}=MI^2-IA^2$. Given $AB=6$, $IA=3$, so $IA^2=9$. The equation becomes: $MI^2-9=7\Leftrightarrow MI^2=16\Leftrightarrow MI=4$. The desired locus is the **circle with center I (midpoint of $[AB]$) and radius $R=4\ \text{cm}$**.

19. A point on the shooting line is $M(x;5)$ with $x>0$. $\overrightarrow{MP_1}\begin{pmatrix}-x\\2-5\end{pmatrix}=\begin{pmatrix}-x\\-3\end{pmatrix}$ and $\overrightarrow{MP_2}\begin{pmatrix}-x\\-2-5\end{pmatrix}=\begin{pmatrix}-x\\-7\end{pmatrix}$. The scalar product is: $\overrightarrow{MP_1}\cdot\overrightarrow{MP_2}=(-x)(-x)+(-3)(-7)=x^2+21$. The norms are: $\|\overrightarrow{MP_1}\|=\sqrt{x^2+9}$ and $\|\overrightarrow{MP_2}\|=\sqrt{x^2+49}$. The cosine of angle $\theta=\widehat{P_1MP_2}$ satisfies $\cos\theta=\frac{x^2+21}{\sqrt{(x^2+9)(x^2+49)}}$. The angle θ is acute and reaches its maximum when $\cos\theta$ is minimized. Squaring the expression: $\cos^2\theta=\frac{(x^2+21)^2}{x^4+58x^2+441}=\frac{x^4+42x^2+441}{x^4+58x^2+441}=1-\frac{16x^2}{x^4+58x^2+441}=1-\frac{16}{x^2+58+\frac{441}{x^2}}$. This cosine is minimal when the denominator $x^2+\frac{441}{x^2}$ is minimized. By the AM-GM inequality, the sum of two positive numbers with constant product $x^2\times\frac{441}{x^2}=441$ is minimized when they are equal: $x^2=\frac{441}{x^2}\Leftrightarrow x^4=441\Leftrightarrow x^2=21\Leftrightarrow x=\sqrt{21}\approx 4.58\ \text{m}$. The kicker should place the ball approximately 4.58 meters from the try line.

20. The area of a triangle is given by $\mathcal{S}=\frac{1}{2}bc\sin\hat{A}$. Thus $4\mathcal{S}=2bc\sin\hat{A}$, so $16\mathcal{S}^2=4b^2c^2\sin^2\hat{A}=4b^2c^2(1-\cos^2\hat{A})=(2bc-2bc\cos\hat{A})(2bc+2bc\cos\hat{A})$. By Al-Kashi's theorem, $2bc\cos\hat{A}=b^2+c^2-a^2$. Substituting: $2bc-(b^2+c^2-a^2)=a^2-(b-c)^2=(a-b+c)(a+b-c)$. $2bc+(b^2+c^2-a^2)=(b+c)^2-a^2=(b+c-a)(b+c+a)$. Since $p=\frac{a+b+c}{2}$, we have $a+b+c=2p$, $b+c-a=2(p-a)$, $a+c-b=2(p-b)$, and $a+b-c=2(p-c)$. The product becomes: $16\mathcal{S}^2=2(p-b)\times2(p-c)\times2(p-a)\times2p=16p(p-a)(p-b)(p-c)$. Dividing by 16 and taking the square root gives: $\mathcal{S}=\sqrt{p(p-a)(p-b)(p-c)}$.

21. Let A' be the point diametrically opposite to A on the circle. Then triangle $AA'B$ is inscribed in a semicircle, so angle $\widehat{A'BA}$ is a right angle: $(BA')\perp(BA)$. The orthogonal projection of A' onto line (AB) is therefore point B. By projection: $\overrightarrow{MA}\cdot\overrightarrow{MB}=\overrightarrow{MA}\cdot\overrightarrow{MA'}$. Now $\overrightarrow{MA}=\overrightarrow{M\Omega}+\overrightarrow{\Omega A}$ and $\overrightarrow{MA'}=\overrightarrow{M\Omega}+\overrightarrow{\Omega A'}$. Since A and A' are diametrically opposite, $\overrightarrow{\Omega A'}=-\overrightarrow{\Omega A}$. Hence: $\overrightarrow{MA}\cdot\overrightarrow{MB}=(\overrightarrow{M\Omega}+\overrightarrow{\Omega A})\cdot(\overrightarrow{M\Omega}-\overrightarrow{\Omega A})=\|\overrightarrow{M\Omega}\|^2-\|\overrightarrow{\Omega A}\|^2=M\Omega^2-R^2$. This quantity depends solely on the position of M and the circle's parameters, and is completely independent of the secant line chosen. It is the **power of point M with respect to circle $\mathcal{C}$**.

The Question Opening the Next Stage

The scalar product has enabled us to measure the inclination of a force relative to a straight path, and to calculate the slope of a rigid segment. What happens, however, when the slope is no longer a straight line, but an **undulating curve** like a snow-covered hill down which a skier hurtles at full speed? How can we measure the tilt of the ski beneath the skier's boots at an infinitesimal instant, at a single point of contact? This is the great adventure of **differentiation**, which opens in the next chapter.

CHAPTER 39: The Skier and the Board under the Boot

> "Until now, our gaze took motionless photographs of the world: fixed positions, averages over long distances. But life is not an album of dead pictures: it is a continuous torrent. To measure the slope of a hill at the exact millisecond the heart beats, one must place a microscope upon flowing time."

What You Are About to Discover

- The skier's paradox: how to measure the **slope at a single point**, when calculating a slope inherently requires two separate points.
- The **difference quotient** (rate of change) and the stroke of genius of **passing to the limit** as the gap h approaches 0.
- Legendary notations: **Lagrange's** "$f'(a)$" and **Leibniz's** "$\frac{dy}{dx}$".
- The equation of the **tangent line**: the rigid line that grazes the curve without cutting through it.
- Geometric proofs of standard derivatives: why $(x^2)' = 2x$ thanks to the **growing flowerbed**, and the derivatives of x^n, $\frac{1}{x}$, $\sqrt{x}$.
- Operations on functions: $(uv)' = u'v + uv'$ (proved via the expanding rectangle) and the quotient rule.
- The fundamental link between the **sign of the derivative** and **intervals of increase/decrease**, and hunting for **local extrema**.
- The boundary cases where differential calculus breaks down: the **corner point** (sharp turn) of the absolute value and the **vertical tangent** of the square root.
- Concrete problems in **optimization**: constructing the most spacious box or the most economical tin can.

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1. The Problem: The Vertigo of the Descent

You are sixteen years old. You stand atop a high, snow-covered mountain peak, skis clipped to your boots, poles planted firmly in the fresh powder. Ahead unfolds a daunting Olympic run: this is no straight, even ramp; it is an unbroken succession of mogulled hills, sheer dips, and deceptive flats. The vertical profile of the mountain is a sinuous curve described by an elevation function $y = f(x)$.

You push off. The wind roars against your helmet, snow billowing in icy plumes beneath your steel edges. With every hundredth of a second, the incline under your boots shifts: gentle here, plunging precipitously into the void there.

In the hollow of the run, at the precise horizontal position a, a surge of centrifugal force crushes your thighs into the snow. Ask yourself this fundamental question of physics:

"At this exact instant, at this very millisecond when my edges bite into the snowpack, what is the slope of the mountain beneath my boots?"

Finding your altitude at this point is trivial: it is simply the value $f(a)$. But what about **the slope**? Ever since middle school, you have known that a slope (or gradient) is always calculated by dividing a vertical rise by a horizontal run between **two distinct points**:

$$\text{Slope} = \frac{y_B - y_A}{x_B - x_A} = \frac{\Delta y}{\Delta x}.$$

Yet, at this exact split second, you are at **only one single point**! If you attempt to apply the formula taking the exact same point twice, $A(a; f(a))$, you obtain:

$$\text{Slope} = \frac{f(a) - f(a)}{a - a} = \frac{0}{0}.$$

Division by zero! A mathematical absurdity. How can the mountain have a perfectly tangible, physical slope under your feet, while the raw slope formula collapses into $\frac{0}{0}$?

2. The Board under the Boot: From Secant to Tangent

2.1 The Rigidity of the Ski

To untie this knot, listen to your body and inspect your equipment. Under your boots, you have strapped two long boards of laminated wood, fiberglass, and hardened steel edges: skis. Even though the snow surface is humped and continuously curved, **your skis are straight, rigid objects**.

At every point where your weight presses down, the flat sole of your ski embraces the mountain's curve without sinking through it. It grazes the curve at a single instantaneous point of contact. This straight line that caresses the curve at a point without locally crossing it is the **tangent line** to the curve. And the slope beneath your boots is quite simply **the slope of your ski**!

> 💡 **Intuitive Metaphor — The Skier and the Instantaneous Slope of the Mountain**
>
> - Curve of the mountain
> - The rigid ski (the Tangent!)
> - Δy / | Vertical rise
> - Δx Horizontal run

2.2 The Difference Quotient: The Secant Line

To compute this slope without cheating with zero, imagine that the tail of your ski touches the snow at $A(a; f(a))$, and the tip touches the snow slightly ahead, at $M(a+h; f(a+h))$, where h is a small, non-zero separation.

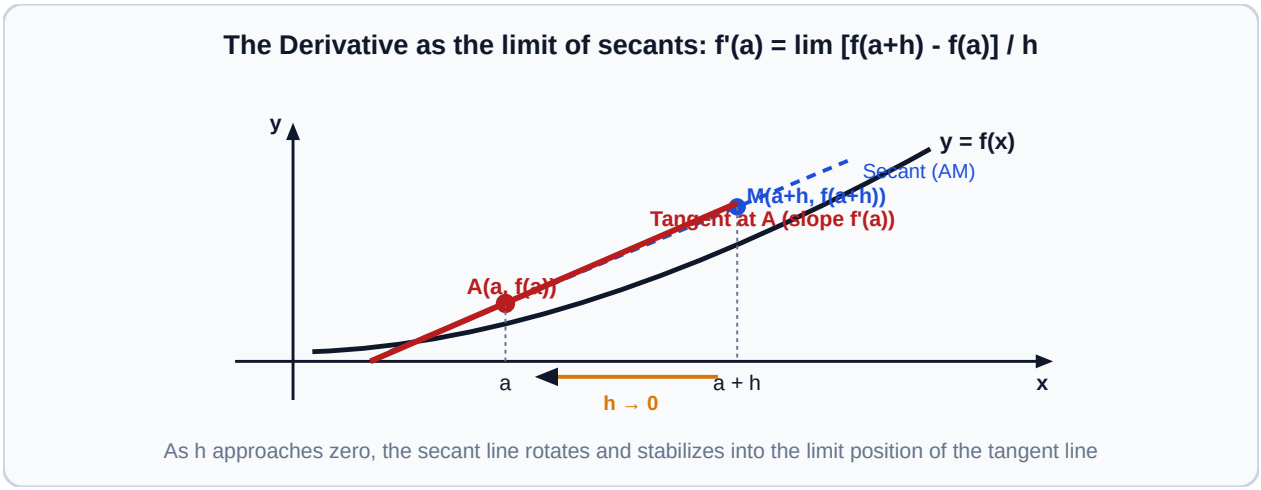

Figure 92: Derivative and limit of secants as step h approaches zero

The line (AM) passing through two distinct points of the curve is called a **secant line**. Its slope is given by the ratio:

$$\tau(h) = \frac{f(a+h) - f(a)}{h}.$$

This fundamental ratio is called the **difference quotient** (or average rate of change) of the function f between a and $a+h$. Because $h \neq 0$, this fraction is perfectly well-defined and computable!

2.3 Passing to the Limit: The Zoom into the Infinitesimal

Now, imagine shortening your ski! Bring the tip closer and closer to the heel by letting h shrink ever closer to zero: $h = 0.1$, then $h = 0.01$, then $h = 0.000001\ldots$ The point M glides down along the hill, merging into point A. The secant line (AM) pivots smoothly and stabilizes onto a definite limiting position: **the tangent line**!

📐 Definition — Derivative at a Point

Let f be a function defined on an open interval containing a. We say that f is **differentiable at a** if the difference quotient $\frac{f(a+h)-f(a)}{h}$ approaches a finite limit as h tends to 0. This limit is called the **derivative of f at a**, denoted $f'(a)$:

$$\boxed{f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}}$$

Geometrically, $f'(a)$ is the **slope (gradient) of the tangent line** to the curve of f at the point with abscissa a.

---

3. The Notations of History: Lagrange vs. Leibniz

Two giants of mathematics bequeathed to humanity two complementary ways of writing this single profound idea:

3.1 Joseph-Louis Lagrange's "Prime" Notation (1797)

The Franco-Italian mathematician **Joseph-Louis Lagrange** introduced the compact notation:

$$f'(a)$$

read aloud as: "**f prime of a**". The prime mark signals that this new function is "derived" from the original function f. Differentiating a second time gives $f''(a)$ ("f double prime"). This is the fastest, cleanest notation for algebraic manipulation in secondary school.

3.2 Gottfried Wilhelm Leibniz's Differential Notation (1684)

A century earlier, the German polymath **Gottfried Wilhelm Leibniz** invented a magnificent notation directly reflecting the slope calculation $\frac{\Delta y}{\Delta x}$:

$$\frac{df}{dx}(a) \quad \text{or} \quad \frac{dy}{dx}$$

read as: "**d f over d x**" or "**the derivative of y with respect to x**". The Latin letter d replaces the Greek letter Δ to signify an **infinitesimal** (infinitely tiny) variation. This notation continually reminds us that a derivative is the ratio of an infinitesimal change in height dy to an infinitesimal change in horizontal position dx. Physicists cherish it because it immediately displays physical dimensions and units (e.g., meters per second: $\frac{dx}{dt} = \text{velocity}$).

---

4. The Equation of the Tangent Line to the Curve

Because the tangent line passes through the contact point $A(a; f(a))$ and has slope $f'(a)$, its point-slope equation follows directly from coordinate geometry:

$$y - y_A = m(x - x_A) \Leftrightarrow y - f(a) = f'(a)(x - a).$$

📐 Theorem — Equation of the Tangent Line

If f is differentiable at a, the curve $\mathcal{C}_f$ admits at the point $A(a; f(a))$ a tangent line with equation:

$$\boxed{y = f'(a)(x - a) + f(a)}$$

This line represents the **best local linear approximation** of the function near a. For any x very close to a:

$$f(x) \approx f(a) + f'(a)(x - a).$$

---

5. Standard Derivatives Proved

Let us examine the foundational toolkit of functions. For each one, we compute the limit of the difference quotient with complete algebraic rigor.

5.1 The Constant Function: $f(x) = k$

A perfectly level track has zero slope everywhere. For any $h \neq 0$:

$$\frac{f(a+h) - f(a)}{h} = \frac{k - k}{h} = \frac{0}{h} = 0.$$

The limit as $h \to 0$ is 0. Therefore:

$$\boxed{(k)' = 0}$$

5.2 The Linear Function: $f(x) = x$

A straight ramp inclined at 45° has a constant slope of 1.

$$\frac{(a+h) - a}{h} = \frac{h}{h} = 1 \underset{h \to 0}{\to} 1.$$

Therefore:

$$\boxed{(x)' = 1}$$

5.3 The Square Function: Why $(x^2)' = 2x$?

Here enters the luminous demonstration of the expanding garden:

Geometric Proof — The Growing Flowerbed

Consider a square flowerbed with side length x. Its area is $S(x) = x^2$. In spring, the gardener expands the bed by adding a thin strip of soil of width h to the East, and a strip of width h to the North.

The total new area added consists of three pieces: 1. An East strip of area $x \times h$; 2. A North strip of area $x \times h$; 3. A tiny square in the corner of area h^2.

The total increase in area is:

$$\Delta S = (x \times h) + (x \times h) + h^2 = 2xh + h^2.$$

Divide by the expansion width h to find the average rate of area growth:

$$\frac{\Delta S}{h} = \frac{2xh + h^2}{h} = 2x + h.$$

Now let h shrink to zero! The tiny corner h^2 vanishes infinitely faster than the borders. All that remains is:

$$\lim_{h \to 0} (2x + h) = 2x.$$

The number 2 is not magic: ***it represents the 2 open sides of the square expanding outward into space!*** ■

By pure algebra:

$$\frac{(a+h)^2 - a^2}{h} = \frac{a^2 + 2ah + h^2 - a^2}{h} = \frac{2ah + h^2}{h} = 2a + h \underset{h \to 0}{\to} 2a.$$

Therefore:

$$\boxed{(x^2)' = 2x}$$

5.4 The Power Function: $f(x) = x^n$ ($n \in \mathbb{N}^*$)

Extending this insight to a cube (x^3), the 3 expanding faces yield $(x^3)' = 3x^2$. In dimension n, using the standard algebraic identity:

$$A^n - B^n = (A - B)(A^{n-1} + A^{n-2}B + ... + AB^{n-2} + B^{n-1}),$$

with $A = a + h$ and $B = a$, the difference quotient becomes:

$$\frac{(a+h)^n - a^n}{h} = \frac{h \cdot \left((a+h)^{n-1} + (a+h)^{n-2}a + \ldots + a^{n-1}\right)}{h}.$$

There are exactly n terms inside the parentheses. As $h \to 0$, each term converges to a^{n-1}. Their sum is therefore na^{n-1}.

$$\boxed{(x^n)' = nx^{n-1}}$$

5.5 The Reciprocal Function: $f(x) = \frac{1}{x}$ on $\mathbb{R}^*$

For any $a \neq 0$ and h sufficiently small:

$$\frac{f(a+h) - f(a)}{h} = \frac{\frac{1}{a+h} - \frac{1}{a}}{h} = \frac{\frac{a-(a+h)}{a(a+h)}}{h} = \frac{\frac{-h}{a(a+h)}}{h} = -\frac{1}{a(a+h)}.$$

As $h \to 0$, the denominator approaches $a \times a = a^2$. The limit is therefore $-\frac{1}{a^2}$.

$$\boxed{\left(\frac{1}{x}\right)' = -\frac{1}{x^2}}$$

5.6 The Square Root Function: $f(x) = \sqrt{x}$ on $]0; +\infty[$

For any $a > 0$, let us compute the difference quotient. We face an indeterminate form $\frac{0}{0}$. To resolve it, multiply numerator and denominator by the **conjugate radical** $(\sqrt{a+h} + \sqrt{a})$:

$$\frac{\sqrt{a+h} - \sqrt{a}}{h} = \frac{(\sqrt{a+h} - \sqrt{a})(\sqrt{a+h} + \sqrt{a})}{h(\sqrt{a+h} + \sqrt{a})} = \frac{(a+h) - a}{h(\sqrt{a+h} + \sqrt{a})} = \frac{h}{h(\sqrt{a+h} + \sqrt{a})} = \frac{1}{\sqrt{a+h} + \sqrt{a}}.$$

As $h \to 0$, the denominator approaches $\sqrt{a} + \sqrt{a} = 2\sqrt{a}$.

$$\boxed{(\sqrt{x})' = \frac{1}{2\sqrt{x}}}$$

---

6. Operations on Derived Functions

Let u and v be two differentiable functions on an interval I, and let k be a real constant.

6.1 Sum and Scalar Multiplication

- $(u+v)' = u' + v'$
- $(k \cdot u)' = k \cdot u'$

6.2 Derivative of a Product: Why $(uv)' \neq u'v'$?

This is the most notorious trap in the history of mathematics. Even Leibniz fell into it in his initial notes in 1673, before correcting himself by sketching an expanding rectangle!

Geometric Proof — The Expanding Rectangle

Consider a rectangular field of dimensions $u(x)$ by $v(x)$. Its area is $A(x) = u(x) \times v(x)$. When x increases by a small amount h, the width becomes $u + \Delta u$ and the length becomes $v + \Delta v$.

The newly added surface area is composed of three zones: 1. A vertical strip of area $v(x) \times \Delta u$; 2. A horizontal strip of area $u(x) \times \Delta v$; 3. A negligible corner of area $\Delta u \times \Delta v$.

Dividing by h:

$$\frac{\Delta A}{h} = v(x)\frac{\Delta u}{h} + u(x)\frac{\Delta v}{h} + \frac{\Delta u \cdot \Delta v}{h}.$$

As $h \to 0$:

- *$\frac{\Delta u}{h} \to u'(x)$ and $\frac{\Delta v}{h} \to v'(x)$;*
- *The third term vanishes because $\Delta u \to 0$! Only the two main strips remain:*

$$\boxed{(u \cdot v)' = u'v + uv'} \quad \blacksquare$$

6.3 Derivative of the Reciprocal and Quotient

If $v(x) \neq 0$ on I:

- Reciprocal Rule:

$$\boxed{\left(\frac{1}{v}\right)' = -\frac{v'}{v^2}}$$

- Quotient Rule: Writing $\frac{u}{v} = u \times \frac{1}{v}$ and applying the product rule:

$$\left(\frac{u}{v}\right)' = u'\left(\frac{1}{v}\right) + u\left(-\frac{v'}{v^2}\right) = \frac{u'v}{v^2} - \frac{uv'}{v^2} = \boxed{\frac{u'v - uv'}{v^2}}$$

> ⚠ **Caution: Classic Trap! — Order Matters in the Quotient Rule**
>
> In the numerator $u'v - uv'$, the minus sign makes the order crucial! It is always the derivative of the top (u') multiplying the untouched bottom (v) first. Reversing the terms flips the sign of your entire result.

---

7. The Crucial Link: Sign of the Derivative and Monotonicity

Here differential calculus reveals its absolute power: it transforms a hard global geometric question (“is the function climbing or falling across an interval?”) into a straightforward local algebraic check (“what is the sign of a number?”).

> 📐 **Fundamental Theorem — Monotonicity and Sign of the Derivative**
>
> Let f be differentiable on an interval I.
>
> - If $f'(x) > 0$ for all $x \in I$, then f is **strictly increasing** on I.
> - If $f'(x) < 0$ for all $x \in I$, then f is **strictly decreasing** on I.
> - If $f'(x) = 0$ for all $x \in I$, then f is **constant** on I.

Intuitively: if the tips of your skis point upward ($f'(x) > 0$) at every single millisecond of your glide, you cannot help but gain altitude overall!

---

8. Finding Local Extrema and Optimization

How do we pinpoint the highest crest of a hill or the deepest depression of a valley? At the summit of a hill, the slope ceases to rise and prepares to fall: for one infinitesimal heartbeat, the ski levels out to absolute horizontality!

📐 Fundamental Rule — Horizontal Tangent and Local Extrema ($f'(x_0) = 0$)

- Horizontal Tangent (f'(x_0) = 0)
- f' > 0 f' < 0
- Uphill) (Downhill)

📐 Theorem — Local Extremum Condition

Let f be differentiable on an open interval containing x_0. If f attains a local extremum (maximum or minimum) at x_0, then:

$$f'(x_0) = 0.$$

Converse and Practical Criterion: If $f'(x_0) = 0$ and f' **changes sign** at x_0, then f attains a local extremum at x_0:

- From $+$ to $-$: this is a **local maximum** (the curve rises, then falls).
- From $-$ to $+$: this is a **local minimum** (the curve falls, then rises).

⚠️ Caution: Classic Trap! — $f'(x_0) = 0$ Does Not Guarantee an Extremum

Look at the cubic function $f(x) = x^3$. Its derivative is $f'(x) = 3x^2$. At $x = 0$, indeed $f'(0) = 0$ (the tangent is perfectly horizontal). But for all $x \neq 0$, $3x^2 > 0$: the derivative never changes sign! The curve climbs before 0 and continues climbing after 0. The origin $(0; 0)$ is a flat bench (an inflection point), not a summit or basin!

---

9. At the Boundaries of Differentiation: When Calculus Breaks Down

Are all functions differentiable everywhere? No! There are two primary geometric ways for a curve to reject differentiation at a point.

9.1 The Corner Point: The Absolute Value at 0

Consider the absolute value function $f(x) = |x|$. Its graph forms a sharp "V" at the origin. Compute the difference quotient at 0:

$$\tau(h) = \frac{|0 + h| - |0|}{h} = \frac{|h|}{h}.$$

- Approaching from the right ($h > 0$): $|h| = h$, so $\tau(h) = \frac{h}{h} = +1$.
- Approaching from the left ($h < 0$): $|h| = -h$, so $\tau(h) = \frac{-h}{h} = -1$.

The right-hand limit ($+1$) does not match the left-hand limit (-1)! The ski cannot rest stably on a knife-edge apex without wobbling between two contradictory slopes. We say that the absolute value function **is not differentiable at 0**. The graph possesses two half-tangents meeting at a sharp **corner point**.

9.2 The Vertical Tangent: The Square Root at 0

Consider the function $f(x) = \sqrt{x}$ at $a = 0$. For $h > 0$:

$$\tau(h) = \frac{\sqrt{0 + h} - \sqrt{0}}{h} = \frac{\sqrt{h}}{h} = \frac{1}{\sqrt{h}}.$$

As $h \to 0^+$, $\sqrt{h}$ becomes an infinitesimally small positive quantity, so $\frac{1}{\sqrt{h}}$ blows up toward $+\infty$! The limit is not a finite real number. The square root function **is not differentiable at 0**. Geometrically, the curve launches vertically: it has a **vertical half-tangent** along the y-axis ($x = 0$).

Step-by-Step Worked Examples

📐 Worked Example 1 — Computing a Derivative from Definition

Let $f(x) = 3x^2 - 5x + 2$. Compute $f'(2)$ directly using the definition of the difference quotient.

Detailed Solution: First find $f(2)$: $f(2) = 3(2^2) - 5(2) + 2 = 12 - 10 + 2 = 4$. Expand $f(2+h)$ for $h \neq 0$:

$$\begin{aligned} f(2+h) &= 3(2+h)^2 - 5(2+h) + 2 \\ &= 3(4 + 4h + h^2) - 10 - 5h + 2 \\ &= 12 + 12h + 3h^2 - 10 - 5h + 2 = 4 + 7h + 3h^2. \end{aligned}$$

Form the difference quotient:

$$\frac{f(2+h) - f(2)}{h} = \frac{(4 + 7h + 3h^2) - 4}{h} = \frac{7h + 3h^2}{h} = \frac{h(7 + 3h)}{h} = 7 + 3h.$$

Take the limit as $h \to 0$:

$$\lim_{h \to 0} (7 + 3h) = 7.$$

Thus f is differentiable at 2 and $\boxed{f'(2) = 7}$.

Worked Example 2 — Equation of a Tangent Line

Let $f(x) = \frac{1}{x}$. Determine the equation of the tangent line to the curve of f at the point with abscissa $a = 2$.

Detailed Solution:

- y-coordinate of the contact point: $f(2) = \frac{1}{2}$.
- Derivative function: $f'(x) = -\frac{1}{x^2}$, so $f'(2) = -\frac{1}{2^2} = -\frac{1}{4}$.
- Tangent line formula: $y = f'(2)(x - 2) + f(2)$.

$$y = -\frac{1}{4}(x - 2) + \frac{1}{2} = -\frac{1}{4}x + \frac{2}{4} + \frac{1}{2} = -\frac{1}{4}x + 1.$$

The tangent line equation is $\boxed{y = -\frac{1}{4}x + 1}$.

📐 Worked Example 3 — Product and Quotient Rules

Compute the derivatives of: 1. $g(x) = (2x - 3)(x^2 + 4x)$ 2. $h(x) = \frac{3x-1}{x^2+1}$

Detailed Solution: 1. Product form $u \cdot v$ with $u(x) = 2x - 3 \Longrightarrow u'(x) = 2$, and $v(x) = x^2 + 4x \Longrightarrow v'(x) = 2x + 4$.

$$g'(x) = u'v + uv' = 2(x^2 + 4x) + (2x - 3)(2x + 4).$$

Expanding: $g'(x) = (2x^2 + 8x) + (4x^2 + 8x - 6x - 12) = 6x^2 + 10x - 12$. 2. Quotient form $\frac{u}{v}$ with $u(x) = 3x - 1 \Longrightarrow u'(x) = 3$, and $v(x) = x^2 + 1 \Longrightarrow v'(x) = 2x$.

$$h'(x) = \frac{u'v - uv'}{v^2} = \frac{3(x^2 + 1) - (3x - 1)(2x)}{(x^2 + 1)^2}.$$

Expanding the numerator: $3x^2 + 3 - (6x^2 - 2x) = 3x^2 + 3 - 6x^2 + 2x = -3x^2 + 2x + 3$. Hence: $\boxed{h'(x) = \frac{-3x^2 + 2x + 3}{(x^2 + 1)^2}}$.

Worked Example 4 — Complete Analysis of a Cubic Polynomial

Let $f(x) = 2x^3 - 9x^2 + 12x - 1$ defined on $\mathbb{R}$. Construct its complete variation table and determine all local extrema.

Detailed Solution: 1. Derivative: $f'(x) = 2(3x^2) - 9(2x) + 12(1) - 0 = 6x^2 - 18x + 12$. Factor out 6: $f'(x) = 6(x^2 - 3x + 2)$. 2. Roots of $f'(x)$: The quadratic $x^2 - 3x + 2$ factors readily as $(x-1)(x-2)$, with roots $x_1 = 1$ and $x_2 = 2$. The leading coefficient is positive ($a = 6 > 0$), so the parabola for f' opens upward:

- $f'(x) > 0$ on $]-\infty; 1[$ and on $]2; +\infty[$;
- $f'(x) < 0$ on $]1; 2[$.

3. Key values: $f(1) = 2(1)^3 - 9(1)^2 + 12(1) - 1 = 2 - 9 + 12 - 1 = 4$. $f(2) = 2(8) - 9(4) + 12(2) - 1 = 16 - 36 + 24 - 1 = 3$.
4. Variation table:

x	$-\infty$		1		2		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$		$\nearrow$	4	$\searrow$	3	$\nearrow$	

Conclusion: f attains a **local maximum** of 4 at $x = 1$, and a **local minimum** of 3 at $x = 2$.

Worked Example 5 — Concrete Optimization: The Open Box

From a square sheet of cardboard measuring 12 cm on each side, small squares of side length x (with $0 < x < 6$) are cut from each of the four corners. The sides are then folded up to form an open-top rectangular box. What value of x maximizes the box's volume?

Detailed Solution: 1. Dimensions of the folded box:

- Base: square of side length $12 - 2x$.
- Height: x.

2. Volume function: $V(x) = \text{Base Area} \times \text{Height} = (12 - 2x)^2 \times x = (144 - 48x + 4x^2)x = 4x^3 - 48x^2 + 144x$.
3. Derivative: $V'(x) = 12x^2 - 96x + 144 = 12(x^2 - 8x + 12)$.
4. Finding critical points: Roots of $x^2 - 8x + 12 = 0$: $\Delta = (-8)^2 - 4(1)(12) = 64 - 48 = 16 = 4^2$. $x_1 = \frac{8-4}{2} = 2$, and $x_2 = \frac{8+4}{2} = 6$. Since $x \in]0; 6[$, only the root $x = 2$ lies in the feasible domain. For $x < 2$, $V'(x) > 0$ (volume expands). For $x > 2$, $V'(x) < 0$ (volume shrinks).
5. Conclusion: The maximum volume is achieved when the cutouts measure $\boldsymbol{x = 2}$ **cm**. The maximal capacity is $V(2) = (12 - 4)^2 \times 2 = 8^2 \times 2 = 64 \times 2 = 128$ cm^3.

Worked Example 6 — Tangent Parallel to a Given Line

Let $f(x) = x^2 - 4x + 1$. Are there any points on the parabola where the tangent line is parallel to the line $y = 2x - 5$?

Detailed Solution:

- Two lines are parallel if and only if they share the exact same slope.
- The given line has slope $m = 2$.
- The tangent line at abscissa x has slope $f'(x)$.
- Differentiating: $f'(x) = 2x - 4$.
- Set up the equation $f'(x) = 2$:

$$2x - 4 = 2 \Leftrightarrow 2x = 6 \Leftrightarrow x = 3.$$

- Calculate the corresponding y-value: $f(3) = 3^2 - 4(3) + 1 = 9 - 12 + 1 = -2$.
- Conclusion: There is exactly one such point on the curve, $A(3; -2)$. Its tangent line equation is $y = 2(x - 3) - 2 = 2x - 8$.

What You Must Remember

1. **The Derivative $f'(a)$:**
   - Definition: $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$.
   - Geometric meaning: slope of the ski beneath the boot = slope of the tangent line at $A(a; f(a))$.
   - Equation of the tangent line: $\boxed{y = f'(a)(x - a) + f(a)}$.
2. **Table of Standard Derivatives:** | Function $f(x)$ | Derivative $f'(x)$ | Domain of Validity | |—|—|—| | k (constant) | 0 | $\mathbb{R}$ | | x | 1 | $\mathbb{R}$ | | x^2 | $2x$ | $\mathbb{R}$ | | x^n $(n \in \mathbb{N}^*)$ | nx^{n-1} | $\mathbb{R}$ | | $\frac{1}{x}$ | $-\frac{1}{x^2}$ | $\mathbb{R}^* =]-\infty; 0[\cup]0; +\infty[$ | | $\sqrt{x}$ | $\frac{1}{2\sqrt{x}}$ | $]0; +\infty[$ (not differentiable at 0) |
3. **Differentiation Rules:**
   - $(u + v)' = u' + v'$
   - $(ku)' = ku'$
   - $(uv)' = u'v + uv'$
   - $\left(\frac{1}{v}\right)' = -\frac{v'}{v^2}$
   - $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$
4. **Sign of Derivative and Monotonicity:**
   - $f'(x) > 0 \Longrightarrow f$ is strictly increasing.
   - $f'(x) < 0 \Longrightarrow f$ is strictly decreasing.
   - $f'(x_0) = 0$ with a sign change $\Longrightarrow$ local extremum (peak or trough).
5. **Failures of Differentiability:**
   - $f(x) = |x|$ at 0: corner point (left-hand and right-hand slopes differ).
   - $f(x) = \sqrt{x}$ at 0: vertical half-tangent (difference quotient tends to $+\infty$).

Exercises

Level (Direct Derivative Calculations and Tangents)

1. Compute the derivative of $f(x) = 5x^3 - 4x^2 + 7x - 9$.
2. Compute the derivative of $g(x) = 3\sqrt{x} - \frac{4}{x}$ for $x > 0$.
3. Determine the derivative value $f'(3)$ for $f(x) = x^2 - 2x$.
4. Write the equation of the tangent line to the curve of $f(x) = x^2$ at the point with abscissa $a = 1$.

5. Determine the coordinates of the points where the curve of $f(x) = x^3 - 3x$ has a horizontal tangent line.
6. Let $f(x) = \frac{2x+1}{x-3}$ on $]3; +\infty[$. Compute $f'(x)$.
7. Differentiate $h(x) = (x^2 + 1)(3x - 2)$.
8. Determine the sign of the derivative of $f(x) = -2x^2 + 8x - 5$ and deduce the variation of f.
9. Is the function $f(x) = |x - 3|$ differentiable at $x = 3$? Justify.
10. Determine the local linear approximation of $\sqrt{1+h}$ for h near 0 using the derivative of $\sqrt{x}$ at 1.

Level (Function Studies and Geometry)

11. Construct the complete variation table of $f(x) = \frac{1}{3}x^3 - x^2 - 3x + 4$ on $\mathbb{R}$.
12. Let $f(x) = \frac{x^2-3x+6}{x-1}$ defined on $]1; +\infty[$.
    a) Compute $f'(x)$ and show that $f'(x) = \frac{(x-3)(x+1)}{(x-1)^2}$.
    b) Deduce the minimum value of f on this interval.
13. Consider the parabola with equation $y = -x^2 + 4x - 1$. Does there exist a tangent line to this parabola that passes through the origin $(0; 0)$?
14. Consider the function $f(x) = x\sqrt{x}$ defined on $[0; +\infty[$.
    a) Show via the difference quotient that f is differentiable at 0 and determine $f'(0)$.
    b) Compute $f'(x)$ for $x > 0$ using the product rule and check consistency.
15. Determine the intersection points of the curve $f(x) = x^3 - 3x^2 + x$ with the tangent line to the curve at $x = 0$.
16. A projectile is launched upward. Its altitude in meters at time t (in seconds) is given by $z(t) = -5t^2 + 20t + 1.5$.
    a) Compute the instantaneous velocity $v(t) = z'(t)$ at any instant t.
    b) At what time does the projectile reach its maximum height? What is this maximum height?
17. Prove by induction that for any integer $n \geq 1$, the derivative of $f_n(x) = x^n$ is indeed $f_n'(x) = nx^{n-1}$, using the product rule.
18. Let $f(x) = \frac{ax+b}{x-2}$. Determine the real numbers a and b such that the curve passes through $A(3; 5)$ and has a tangent line at A with slope -7.

Level (Optimization and Open Problems)

19. **The Shepherd's Enclosure:** A shepherd has 100 meters of fencing to build a rectangular enclosure backed against a straight, sheer cliff face (no fence is required along the cliff side). What dimensions will give the sheep the largest possible grazing area?
20. **The Optimal Tin Can:** We wish to manufacture a closed cylindrical tin can of fixed volume $V = 1$ liter ($1000\ \text{cm}^3$). We seek to minimize the surface area of metal needed (i.e., the total surface area of the cylinder).
    a) Express the height h in terms of the radius r.
    b) Express the total surface area $S(r)$ in terms of r.
    c) Compute $S'(r)$ and prove that the most metal-efficient can satisfies $h = 2r$ (the height must equal the diameter!).
21. **Minimal Distance from a Point to a Parabola:** In a Cartesian coordinate system, consider the parabola $\mathcal{P}$ with equation $y = x^2$ and the point $A(0; 2)$. Which point on the parabola is closest to A? Show that the segment connecting A to this optimal point is perpendicular to the tangent line at that point.
22. **Corner Point and Differentiability:** Let f be defined by $f(x) = x|x|$.
    a) Express $f(x)$ without absolute value bars depending on the sign of x.
    b) Investigate the differentiability of f at 0 by computing the limit of the difference quotient.
    c) What can you conclude? Does a function containing an absolute value always produce a sharp corner point?
23. **Common Tangents to Two Parabolas:** Consider the two parabolas $y = x^2$ and $y = -x^2 + 4x - 6$. Does there exist a line tangent to both parabolas simultaneously? If so, find its equation.
24. **The Lifeguard's Snell-Descartes Problem:** A lifeguard standing on the beach at point $A(0; a)$ with $a > 0$ spots a swimmer in distress at sea at $B(d; -b)$ with $b > 0$. The shoreline is the x-axis $y = 0$. The lifeguard runs on sand at speed v_1 and swims in water at a slower speed $v_2 < v_1$. He enters the water at point $M(x; 0)$ where $0 \leq x \leq d$.
    a) Express the total travel time $T(x)$ to reach the swimmer.
    b) Compute $T'(x)$ and prove that travel time is minimized when $\frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2}$ (Snell's law of optical refraction!).
25. **Rolle's Theorem (Intuitive Discovery):** Let f be a differentiable function on $[a; b]$ such that $f(a) = f(b) = 0$. Provide an intuitive explanation using the skier why there must exist at least one point $c \in]a; b[$ where $f'(c) = 0$.

Solutions

1. $f'(x) = 5(3x^2) - 4(2x) + 7(1) - 0 = 15x^2 - 8x + 7$.

2. $g'(x) = 3\left(\frac{1}{2\sqrt{x}}\right) - 4\left(-\frac{1}{x^2}\right) = \frac{3}{2\sqrt{x}} + \frac{4}{x^2}$.

3. $f'(x) = 2x - 2$. At $x = 3$: $f'(3) = 2(3) - 2 = 4$.

4. $f(1) = 1^2 = 1$. $f'(x) = 2x \Longrightarrow f'(1) = 2$. Tangent line: $y = f'(1)(x-1) + f(1) = 2(x-1) + 1 = 2x - 1$.

5. $f'(x) = 3x^2 - 3 = 3(x^2 - 1) = 3(x-1)(x+1)$. Horizontal tangent $\Leftrightarrow f'(x) = 0 \Leftrightarrow x = 1$ or $x = -1$. The points are $(1; -2)$ and $(-1; 2)$.

6. Quotient formula $\frac{u}{v}$: $u(x) = 2x + 1 \Longrightarrow u' = 2$; $v(x) = x - 3 \Longrightarrow v' = 1$. $f'(x) = \frac{2(x-3)-(2x+1)(1)}{(x-3)^2} = \frac{2x-6-2x-1}{(x-3)^2} = -\frac{7}{(x-3)^2}$.

7. Product rule: $u = x^2 + 1 \Longrightarrow u' = 2x$; $v = 3x - 2 \Longrightarrow v' = 3$. $h'(x) = 2x(3x-2) + (x^2+1)(3) = 6x^2 - 4x + 3x^2 + 3 = 9x^2 - 4x + 3$.

8. $f'(x) = -4x + 8 = -4(x-2)$. $f'(x) > 0$ if $x < 2$ (the function f is strictly increasing on $]-\infty; 2]$); $f'(x) < 0$ if $x > 2$ (the function f is strictly decreasing on $[2; +\infty[$).

9. Difference quotient at 3: $\frac{|3+h-3|-|0|}{h} = \frac{|h|}{h}$. As $h \to 0^+$, the limit is $+1$; as $h \to 0^-$, the limit is -1. Since the two one-sided limits differ, f is not differentiable at 3 (it has a sharp corner point).

10. Let $f(x) = \sqrt{x}$. We have $f(1) = 1$ and $f'(1) = \frac{1}{2\sqrt{1}} = \frac{1}{2}$. The local linear approximation at 1 is $f(1+h) \approx f(1) + hf'(1) = 1 + \frac{1}{2}h$. For example, $\sqrt{1.04} \approx 1 + \frac{0.04}{2} = 1.02$ (the true value is 1.0198...).

11. $f'(x) = x^2 - 2x - 3 = (x-3)(x+1)$. Roots: $x = -1$ and $x = 3$. Parabola opening upward.

- $f'(x) > 0$ on $]-\infty; -1[\cup]3; +\infty[$ (f increasing);
- $f'(x) < 0$ on $]-1; 3[$ (f decreasing).

Extrema: $f(-1) = -\frac{1}{3} - 1 + 3 + 4 = \frac{17}{3} \approx 5.67$ (local maximum at -1); $f(3) = 9 - 9 - 9 + 4 = -5$ (local minimum at 3).

12. a) $u(x) = x^2 - 3x + 6 \Longrightarrow u' = 2x - 3$; $v(x) = x - 1 \Longrightarrow v' = 1$. $f'(x) = \frac{(2x-3)(x-1)-(x^2-3x+6)(1)}{(x-1)^2} = \frac{2x^2-5x+3-x^2+3x-6}{(x-1)^2} = \frac{x^2-2x-3}{(x-1)^2}$. The numerator factors as $(x-3)(x+1)$. b) On $]1; +\infty[$, the denominator $(x-1)^2 > 0$ and $x + 1 > 0$. The sign of $f'(x)$ depends solely on $x - 3$. $f'(x) < 0$ on $]1; 3[$ and $f'(x) > 0$ on $]3; +\infty[$. Thus f attains an absolute minimum at $x = 3$, equal to $f(3) = \frac{9-9+6}{3-1} = \frac{6}{2} = 3$.

13. The tangent line at abscissa a has equation $y = f'(a)(x-a) + f(a)$. Here $f(a) = -a^2 + 4a - 1$ and $f'(a) = -2a + 4$. The equation is $y = (-2a+4)(x-a) - a^2 + 4a - 1 = (-2a+4)x + 2a^2 - 4a - a^2 + 4a - 1 = (-2a+4)x + a^2 - 1$. This line passes through the origin $(0; 0)$ if and only if its y-intercept is zero: $a^2 - 1 = 0 \Leftrightarrow a = 1$ or $a = -1$. There exist **two** such tangent lines: at $x = 1$ (equation $y = 2x$) and at $x = -1$ (equation $y = 6x$).

14. a) For $h > 0$: $\frac{f(h)-f(0)}{h} = \frac{h\sqrt{h}-0}{h} = \sqrt{h}$. As $h \to 0^+$, $\lim \sqrt{h} = 0$. Hence f is differentiable at 0 with $f'(0) = 0$. b) Product rule for $x > 0$: $f'(x) = (1)\sqrt{x} + x\left(\frac{1}{2\sqrt{x}}\right) = \sqrt{x} + \frac{x}{2\sqrt{x}} = \sqrt{x} + \frac{\sqrt{x}}{2} = \frac{3}{2}\sqrt{x}$. As $x \to 0$, $\frac{3}{2}\sqrt{x} \to 0$, matching $f'(0) = 0$ perfectly.

15. At $x = 0$, $f(0) = 0$. $f'(x) = 3x^2 - 6x + 1 \Longrightarrow f'(0) = 1$. The tangent line at 0 has equation $y = 1(x-0) + 0 \Leftrightarrow y = x$. Intersection with the curve: $f(x) = x \Leftrightarrow x^3 - 3x^2 + x = x \Leftrightarrow x^3 - 3x^2 = 0 \Leftrightarrow x^2(x-3) = 0$. Roots: $x = 0$ (double root, tangency point) and $x = 3$. For $x = 3$, $y = 3$. The second intersection point is $B(3; 3)$.

16. a) $v(t) = z'(t) = -10t + 20$. b) Altitude is maximized when velocity vanishes: $-10t + 20 = 0 \Leftrightarrow t = 2$ seconds. Maximum height: $z(2) = -5(4) + 20(2) + 1.5 = -20 + 40 + 1.5 = 21.5$ meters.

17. - Base case ($n = 1$): $f_1(x) = x$, derivative is $1 = 1 \cdot x^0$. Holds. - Inductive step: Assume true for n: $(x^n)' = nx^{n-1}$. Then $x^{n+1} = x^n \cdot x$. By the product rule: $(x^{n+1})' = (x^n)' \cdot x + x^n \cdot (x)' = (nx^{n-1}) \cdot x + x^n \cdot 1 = nx^n + x^n = (n+1)x^n$. The formula holds for $n + 1$.

- Conclusion: By induction, for all $n \geq 1$, $(x^n)' = nx^{n-1}$.

18. $f(3) = 5 \Leftrightarrow \frac{3a+b}{3-2} = 5 \Leftrightarrow 3a + b = 5$. Derivative: $f'(x) = \frac{a(x-2)-(ax+b)(1)}{(x-2)^2} = \frac{ax-2a-ax-b}{(x-2)^2} = \frac{-2a-b}{(x-2)^2}$. At $x = 3$, $f'(3) = \frac{-2a-b}{(3-2)^2} = -2a - b = -7 \Leftrightarrow 2a + b = 7$. Solve the system: $(3a + b) - (2a + b) = 5 - 7 \Leftrightarrow a = -2$. Then $b = 5 - 3(-2) = 5 + 6 = 11$. Thus $a = -2$ and $b = 11$.

19. Let x denote the length of each of the two sides perpendicular to the cliff. The side parallel to the cliff then has length $100 - 2x$. The enclosed area is $A(x) = x(100 - 2x) = -2x^2 + 100x$, for $x \in]0; 50[$. Derivative: $A'(x) = -4x + 100$. $A'(x) = 0 \Leftrightarrow 4x = 100 \Leftrightarrow x = 25$ meters. Since $A''(x) = -4 < 0$, this yields a maximum. The parallel side measures $100 - 2(25) = 50$ meters. The optimal enclosure is $25 \text{ m} \times 50 \text{ m}$, yielding a maximal area of 1250 m^2.

20. a) The volume of a cylinder is $V = \pi r^2 h = 1000 \Longrightarrow h = \frac{1000}{\pi r^2}$. b) Total surface area includes the top and bottom circular lids and the lateral wall: $S(r) = 2(\pi r^2) + 2\pi r h = 2\pi r^2 + 2\pi r\left(\frac{1000}{\pi r^2}\right) = 2\pi r^2 + \frac{2000}{r}$. c) Differentiating with respect to r: $S'(r) = 4\pi r - \frac{2000}{r^2} = \frac{4\pi r^3 - 2000}{r^2}$. The minimum occurs when $S'(r) = 0 \Leftrightarrow 4\pi r^3 = 2000 \Leftrightarrow 2\pi r^3 = 1000$. Since $1000 = \pi r^2 h$, substituting yields: $2\pi r^3 = \pi r^2 h \Leftrightarrow 2r = h$. The height must exactly equal the base diameter.

21. Let $M(x; x^2)$ be a point on the parabola. The squared distance is $AM^2 = (x - 0)^2 + (x^2 - 2)^2 = x^2 + x^4 - 4x^2 + 4 = x^4 - 3x^2 + 4$. Let $g(x) = x^4 - 3x^2 + 4$. Derivative: $g'(x) = 4x^3 - 6x = 2x(2x^2 - 3)$. Roots: $x = 0$ or $x^2 = \frac{3}{2} \Leftrightarrow x = \pm\sqrt{\frac{3}{2}}$. At $x = 0$, $g(0) = 4$ (distance 2). At $x = \pm\sqrt{\frac{3}{2}}$, $g = \frac{9}{4} - \frac{9}{2} + 4 = \frac{7}{4}$, giving distance $\sqrt{7}/2 \approx 1.32 < 2$. The closest points are $M_1\left(\sqrt{\frac{3}{2}}; \frac{3}{2}\right)$ and $M_2\left(-\sqrt{\frac{3}{2}}; \frac{3}{2}\right)$. The vector $\overrightarrow{AM_1} = \begin{pmatrix} \sqrt{3/2} \\ -1/2 \end{pmatrix}$. The tangent slope at that point is $f'(x) = 2x = 2\sqrt{3/2}$, so a direction vector of the tangent is $\vec{u}\begin{pmatrix} 1 \\ 2\sqrt{3/2} \end{pmatrix}$. Dot product: $\overrightarrow{AM_1} \cdot \vec{u} = \sqrt{\frac{3}{2}} \times 1 + \left(-\frac{1}{2}\right) \times 2\sqrt{\frac{3}{2}} = \sqrt{\frac{3}{2}} - \sqrt{\frac{3}{2}} = 0$. The vector is orthogonal to the tangent: the shortest path is perpendicular to the curve.

22. a) If $x \geq 0$, $f(x) = x(x) = x^2$. If $x < 0$, $f(x) = x(-x) = -x^2$. b) Difference quotient at 0: $\frac{f(h) - f(0)}{h} = \frac{h|h|}{h} = |h|$. As $h \to 0$, $|h| \to 0$ from both sides! Therefore f is differentiable at 0 and $f'(0) = 0$. c) Contrary to popular belief, the presence of an absolute value does not necessarily create a sharp corner point: here, the factor x cushions the point, forcing it to adopt a flat horizontal tangent.

23. Let $y = mx + p$ be such a common tangent. It is tangent to $y = x^2$ at abscissa a: so $m = 2a$ and $p = a^2 - 2a(a) = -a^2$. It is tangent to $y = -x^2 + 4x - 6$ at abscissa b: so $m = -2b + 4$ and $p = (-b^2 + 4b - 6) - (-2b + 4)b = b^2 - 6$. Equating slopes and intercepts: 1) $2a = -2b + 4 \Leftrightarrow a + b = 2 \Leftrightarrow b = 2 - a$. 2) $-a^2 = b^2 - 6$. Substitute b: $-a^2 = (2 - a)^2 - 6 = 4 - 4a + a^2 - 6 = a^2 - 4a - 2$. $2a^2 - 4a - 2 = 0 \Leftrightarrow a^2 - 2a - 1 = 0$. $\Delta = 4 - 4(-1) = 8 = (2\sqrt{2})^2$. $a = \frac{2 \pm 2\sqrt{2}}{2} = 1 \pm \sqrt{2}$. There are two common tangent lines. For $a = 1 + \sqrt{2}$, $m = 2 + 2\sqrt{2}$ and $p = -(1 + \sqrt{2})^2 = -3 - 2\sqrt{2}$. The second is obtained with $a = 1 - \sqrt{2}$.

24. a) Distance on sand: $d_1 = \sqrt{x^2 + a^2}$. Time on sand: $t_1 = \frac{\sqrt{x^2 + a^2}}{v_1}$. Distance in water: $d_2 = \sqrt{(d - x)^2 + b^2}$. Time in water: $t_2 = \frac{\sqrt{(d-x)^2 + b^2}}{v_2}$. Total travel time: $T(x) = \frac{\sqrt{x^2 + a^2}}{v_1} + \frac{\sqrt{(d-x)^2 + b^2}}{v_2}$. b) Derivative of $\sqrt{x^2 + a^2}$: $\frac{2x}{2\sqrt{x^2 + a^2}} = \frac{x}{\sqrt{x^2 + a^2}}$. Derivative of $\sqrt{(d - x)^2 + b^2}$: $\frac{-2(d-x)}{2\sqrt{(d-x)^2 + b^2}} = -\frac{d-x}{\sqrt{(d-x)^2 + b^2}}$. Thus $T'(x) = \frac{x}{v_1\sqrt{x^2 + a^2}} - \frac{d-x}{v_2\sqrt{(d-x)^2 + b^2}}$. Geometrically observe that $\frac{x}{\sqrt{x^2 + a^2}} = \sin\theta_1$ and $\frac{d-x}{\sqrt{(d-x)^2 + b^2}} = \sin\theta_2$ (angles relative to the normal to the shoreline). Time is minimized when $T'(x) = 0 \Leftrightarrow \frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2}$. This is Fermat-Snell's Law of Refraction: light minimizes its travel time!

25. The skier starts at elevation 0 at a and returns to elevation 0 at b. If he stayed on flat terrain the entire way (f constant), the slope was zero everywhere. If he climbed, he had to go up, reach a highest elevation (a maximum), then descend. At the exact summit of that hill, his skis were necessarily horizontal ($f'(c) = 0$). If he dipped into a valley, he reached a lowest elevation (a minimum) before climbing back up: at the bottom of the valley, the skis were likewise horizontal ($f'(c) = 0$). In all cases, there must exist at least one point where the tangent is perfectly horizontal.

The Question That Opens What Follows

We have seen that the derivative measures the rate at which a function grows. For a polynomial, differentiation lowers the degree. But imagine a living population: a bacterial colony in a Petri dish. The larger the colony, the more individuals divide, and the faster the population expands! Does there exist anywhere in the universe a magical function that is **strictly equal to its own derivative**: $f' = f$? A function whose slope underfoot is at every millisecond equal to its current elevation? That is the mystery of the **exponential function**, in the next chapter.

CHAPTER 40: The Bacterial Colony

> "In the inert world of stones, speed depends on an applied external force. In the ardent world of the living, force springs from number: each cell is its own engine. The larger the colony grows, the faster it explodes. Nature has chosen a single curve to sing the proliferation of life: the exponential function."

What You Are About to Discover

- The biological secret of proliferation: why the growth rate of a population is proportional to its size.
- The birth of a revolutionary equation: $f' = f$ with $f(0) = 1$.
- **Euler's method**: how to trace the curve step-by-step along its own tangents, with an executable **Python** script.
- The grand curriculum proof: why $\exp(x)$ never vanishes and why this function is **unique**.
- The magical conversion of addition into multiplication: proving $\exp(x + y) = \exp(x)\exp(y)$.
- Euler's universal number: $e \approx 2.71828$, and the notation e^x.
- Complete study of the function: derivative, strict positivity, explosive growth, limits, and asymptotes.
- The chain rule for composite exponentials: $(e^{u(x)})' = u'(x)e^{u(x)}$.
- How to solve equations and inequalities involving e^x.

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1. The Problem: The Petri Dish and the Biological Clock

Peer through the eyepiece of a microscope in a biology laboratory. In a Petri dish filled with nutrient agar, a researcher deposits at noon ($t = 0$) a single bacterium of *Escherichia coli*.

This bacterium feeds. After twenty minutes, it elongates and splits into two: there are 2 bacteria. Twenty minutes later, each divides in turn: there are 4 bacteria. Then 8, then 16, 32, 64, 128, 256…

Temps t (h)	0	1	2	3	10
Bactéries	1 000	2 000	4 000	8 000	1 024 000

Observe closely what distinguishes this growth from every curve we have studied until now:

- For a car cruising on the highway at constant speed, distance increases by 100 km every hour, regardless of how far it has already traveled. The derivative is constant.
- For a stone falling freely under gravity, speed increases by $9.8\ \mathrm{m/s}$ each second: acceleration is constant, and distance follows a second-degree polynomial (x^2).
- But for the bacterial colony, **every newcomer is itself a future birth factory**!
  - When the colony counts 10 bacteria, there are 10 potential divisions per generation. The slope is modest.
  - When the colony counts one million individuals, there are **one million simultaneous cellular divisions**! The speed of proliferation becomes titanic.

Letting $N(t)$ denote the population size at time t, the instantaneous rate of growth (the derivative $N'(t)$) is not constant: it is **directly proportional to the number of individuals already present**:

$$N'(t) = k \cdot N(t).$$

By choosing the unit of time so that $k = 1$, and starting from a single individual at $t = 0$, we face the most fascinating challenge ever presented to differential calculus:

> ***The Fundamental Enigma:*** *Does there exist a mathematical function f, defined and differentiable on $\mathbb{R}$, which is* ***strictly equal to its own derivative*** *at every moment, and equals 1 at the origin?*
>
> $$\{f'(x) = f(x) \quad \text{for all} \quad x \in \mathbb{R}$$
> $$f(0) = 1$$

This is a function whose ski-slope beneath the boot at every point is equal to its instantaneous altitude! The higher it climbs, the steeper the slope; the steeper the slope, the faster it ascends into the sky.

---

2. Euler's Method: Building the Curve Step by Step

Before formally proving that such a function exists, how can we sketch its shape and compute its values? In the 18th century, the Swiss genius **Leonhard Euler** conceived a revolutionary numerical method: stepping forward along the tangent line.

2.1 The Geometric Principle

Recall from the previous chapter that near a point x_0, a smooth curve closely hugs its tangent line:

$$f(x_0 + h) \approx f(x_0) + h \cdot f'(x_0).$$

Here, our function has a magical property: **its derivative equals its own value** ($f'(x_0) = f(x_0)$)! The linear approximation formula simplifies dramatically:

$$f(x_0 + h) \approx f(x_0) + h \cdot f(x_0) = (1 + h) \cdot f(x_0).$$

Each forward step of width h simply means **multiplying the previous altitude by $(1 + h)$**!

2.2 Manual Calculation with Step Size $h = 0.2$

Starting from $x_0 = 0$ and $y_0 = 1$ with step size $h = 0.2$:

- At $x_1 = 0.2$: $y_1 = y_0 \times (1 + 0.2) = 1 \times 1.2 = 1.2$.
- At $x_2 = 0.4$: $y_2 = y_1 \times 1.2 = 1.2 \times 1.2 = 1.44$.
- At $x_3 = 0.6$: $y_3 = 1.44 \times 1.2 = 1.728$.
- At $x_4 = 0.8$: $y_4 = 1.728 \times 1.2 = 2.0736$.
- At $x_5 = 1.0$: $y_5 = 2.0736 \times 1.2 \approx 2.488$.

In 5 steps, we reach $x = 1$, and our approximation gives $f(1) \approx 2.488$. If we take a step size ten times smaller ($h = 0.02$), performing 50 multiplications by 1.02, we get $(1.02)^{50} \approx 2.691$. With $h = 0.001$ (1000 steps), we obtain $(1.001)^{1000} \approx 2.7169$. The curve is coming into sharp focus before our eyes!

2.3 The Standard Python Algorithm

Here is the Python script to simulate Euler's method:

```
def euler_method(n_steps):
    h = 1.0 / n_steps  # step size to travel from 0 to 1
    x = 0.0
    y = 1.0
    for _ in range(n_steps):
        # Slope is y because y' = y
        y = y + h * y
        x = x + h
    return y

for n in [1, 10, 100, 1000, 10000]:
    approx = euler_method(n)
    print(f"n = {n:5d} : e ≈ {approx:.6f}")
```

Output:

🧭 Step-by-Step Method — Step-by-Step Approximation via Euler's Method

- For n = 10 steps: f(1) ≈ 2.593742
- For n = 100 steps: f(1) ≈ 2.704814
- For n = 1000 steps: f(1) ≈ 2.716924
- For n = 10000 steps: f(1) ≈ 2.718146

The decimals steadily stabilize toward a mysterious number: $\mathbf{2.71828...}$

---

3. Existence Theorem and Proof of Uniqueness

📐 Fundamental Theorem (Existence admitted, uniqueness proved)

There exists a **unique** function f differentiable on $\mathbb{R}$ such that:

$$\{f'(x) = f(x) \quad \text{for all} \quad x \in \mathbb{R}$$
$$f(0) = 1$$

This function is called the **exponential function**, denoted exp.

The proof of uniqueness is a masterpiece of mathematical elegance. It hinges on a crucial lemma: the function can never vanish to zero.

3.1 Lemma: The Function Never Vanishes

Proof — Non-vanishing of f

Let f be a differentiable function on $\mathbb{R}$ satisfying $f' = f$ and $f(0) = 1$. Consider the auxiliary function g defined on $\mathbb{R}$ by:

$$g(x) = f(x) \times f(-x).$$

Differentiate g using the product rule $(uv)' = u'v + uv'$: The derivative of $f(-x)$ is $-f'(-x)$ (chain rule).

$$g'(x) = f'(x) \cdot f(-x) + f(x) \cdot \big(-f'(-x)\big) = f'(x)f(-x) - f(x)f'(-x).$$

But by hypothesis, $f' = f$ everywhere, so $f'(x) = f(x)$ and $f'(-x) = f(-x)$! Substituting:

$$g'(x) = f(x)f(-x) - f(x)f(-x) = 0.$$

The derivative of g is **identically zero on** $\mathbb{R}$. A function whose derivative is zero everywhere is **constant**! Thus for all real numbers x:

$$g(x) = g(0) = f(0) \times f(-0) = 1 \times 1 = 1.$$

Hence, for all $x \in \mathbb{R}$:

$$\boxed{f(x) \times f(-x) = 1}$$

If the product of two real numbers is 1, neither of them can be zero! Consequently, **for all real** x, $f(x) \neq 0$. ■

3.2 Proof of Uniqueness

Proof — Uniqueness of the Exponential Function

Suppose there exist two functions f_1 and f_2, both satisfying the conditions: $f_1' = f_1$ with $f_1(0) = 1$, and $f_2' = f_2$ with $f_2(0) = 1$. Since f_1 never vanishes on $\mathbb{R}$ by the lemma above, we have the complete mathematical right to define the quotient function:

$$\phi(x) = \frac{f_2(x)}{f_1(x)}.$$

Differentiating using the quotient rule $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$:

$$\phi'(x) = \frac{f_2'(x)f_1(x) - f_2(x)f_1'(x)}{(f_1(x))^2}.$$

Since $f_2' = f_2$ and $f_1' = f_1$:

$$\phi'(x) = \frac{f_2(x)f_1(x) - f_2(x)f_1(x)}{(f_1(x))^2} = \frac{0}{(f_1(x))^2} = 0.$$

The derivative of ϕ is identically zero on $\mathbb{R}$, so ϕ is a constant function. Evaluating at $x = 0$:

$$\phi(0) = \frac{f_2(0)}{f_1(0)} = \frac{1}{1} = 1.$$

Therefore, for all $x \in \mathbb{R}$, $\phi(x) = 1$, which means:

$$f_2(x) = f_1(x).$$

Thus there exists only one such function! ■

---

4. The Fundamental Property: Transforming Addition into Multiplication

This is the crowning property that makes the exponential the most powerful tool in all of mathematical science:

📐 Fundamental Theorem — Functional Equation

For all real numbers x and y:

$$\boxed{\exp(x + y) = \exp(x) \times \exp(y)}$$

Proof

Fix an arbitrary real number y, and consider the function of the single variable x:

$$h(x) = \frac{\exp(x+y)}{\exp(x)}.$$

(The quotient is valid because the denominator $\exp(x)$ never vanishes). Differentiate h with respect to x: The numerator $u(x) = \exp(x+y)$ has derivative $u'(x) = 1 \cdot \exp'(x+y) = \exp(x+y)$. The denominator $v(x) = \exp(x)$ has derivative $v'(x) = \exp(x)$. Applying the quotient rule:

$$h'(x) = \frac{\exp(x+y)\exp(x) - \exp(x+y)\exp(x)}{(\exp(x))^2} = 0.$$

Thus h is constant on $\mathbb{R}$. Its value for every x equals its value at $x = 0$:

$$h(0) = \frac{\exp(0+y)}{\exp(0)} = \frac{\exp(y)}{1} = \exp(y).$$

Therefore, for all $x \in \mathbb{R}$:

$$\frac{\exp(x+y)}{\exp(x)} = \exp(y) \Leftrightarrow \exp(x+y) = \exp(x) \times \exp(y). \quad \blacksquare$$

4.1 Algebraic Corollaries

From this core formula follow all the familiar laws of powers:

> 📐 ***Algebraic Properties*** *— For all real numbers x, y and any integer n: 1.* $\exp(-x) = \frac{1}{\exp(x)}$ *2.* $\exp(x-y) = \frac{\exp(x)}{\exp(y)}$ *3.* $\exp(n \cdot x) = \big(\exp(x)\big)^n$

Brief Proofs

1. From the lemma:

$\exp(x) \times \exp(-x) = \exp(x-x) = \exp(0) = 1 \Longrightarrow \exp(-x) = \frac{1}{\exp(x)}$. 2. $\exp(x-y) = \exp(x+(-y)) = \exp(x) \times \exp(-y) = \exp(x) \times \frac{1}{\exp(y)} = \frac{\exp(x)}{\exp(y)}$. 3. By straightforward induction for $n \in \mathbb{N}$: $\exp(2x) = \exp(x+x) = (\exp(x))^2$, etc. ■

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5. Euler's Notation and the Number e

Notice the stunning structural resemblance between the properties of exp and the laws of exponents:

- $a^{x+y} = a^x \times a^y$
- $a^{-x} = \frac{1}{a^x}$
- $a^{nx} = (a^x)^n$

This perfect correspondence prompted **Leonhard Euler** in 1728 to introduce an audacious notation. Define the number e as the value of the exponential function at 1:

$$e = \exp(1) \approx 2.718281828459...$$

Then, for every integer n, $\exp(n) = (\exp(1))^n = e^n$. Euler boldly generalized this notation to all real numbers x:

📐 ***Euler's Notation*** *For every real number x, we write:*

$$\boxed{\exp(x) = e^x}$$

*We pronounce this interchangeably as "exponential of x" or "**e to the power x**". The rules of manipulation mirror exponent laws exactly:*

$$e^0 = 1, \quad e^1 = e, \quad e^{x+y} = e^x \cdot e^y, \quad e^{-x} = \frac{1}{e^x}, \quad e^{x-y} = \frac{e^x}{e^y}, \quad (e^x)^n = e^{nx}.$$

---

6. Complete Study of the Exponential Function

6.1 Sign and Strict Positivity

We have already proved that $e^x \neq 0$ for all x. Furthermore, $e^x = e^{x/2+x/2} = \left(e^{x/2}\right)^2$. Any squared real number is non-negative. Because it cannot be zero, it must be **strictly positive**!

📐 **Theorem — Strict Positivity**

For every real number x:

$$\boxed{e^x > 0}$$

The graph of the exponential function lies **entirely above the horizontal axis**. It never touches or crosses the zero line.

6.2 Monotonicity

The derivative of the exponential function is itself:

$$(e^x)' = e^x.$$

Since $e^x > 0$ for all $x \in \mathbb{R}$, **the derivative is strictly positive everywhere**.

📐 **Theorem — Strict Growth**

The exponential function is **strictly increasing** on all of $\mathbb{R}$. Consequently:

- $e^a = e^b \Leftrightarrow a = b$
- $e^a < e^b \Leftrightarrow a < b$

6.3 Limits at Infinity and Asymptotes

- As $x \to +\infty$: e^x explodes toward infinity with blistering speed:

$$\lim_{x \to +\infty} e^x = +\infty.$$

- As $x \to -\infty$: set $X = -x$. As $x \to -\infty$, $X \to +\infty$.

$$e^x = e^{-X} = \frac{1}{e^X} \underset{X \to +\infty}{\to} 0^+.$$

$$\lim_{x \to -\infty} e^x = 0.$$

📐 **Geometric Property — Horizontal Asymptote**

The line $y = 0$ (the x-axis) is a **horizontal asymptote** to the graph of the exponential function at $-\infty$.

6.4 Variation Table and Graph

x	$-\infty$		0		1		$+\infty$
$(e^x)' = e^x$		$+$	1	$+$	e	$+$	
e^x	0	$\nearrow$	1	$\nearrow$	$e \approx 2.718$	$\nearrow$	$+\infty$

Tangent line at 0 ($a = 0$): $y = e^0(x - 0) + e^0 \Leftrightarrow \boxed{y = x + 1}$.

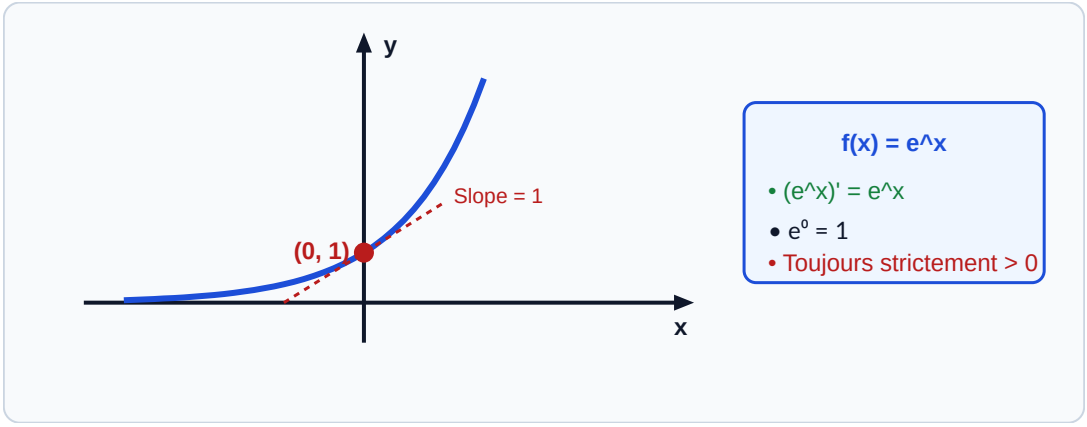

Figure 93: The exponential function: continuous growth and properties

The curve lies entirely above its tangent line at 0:

$$\text{For all } \; x \in \mathbb{R}, \quad \boxed{e^x \geq x + 1}$$

---

7. Derivative of the Composite Function $e^{u(x)}$

In physical applications (radioactivity, epidemics, RC circuits), the exponent is rarely a bare x, but a function of time: e^{-3t}, $e^{-x^2/2}$, and so on.

📐 Theorem — Derivative of e^u

If u is a differentiable function on an interval I, then $f(x) = e^{u(x)}$ is differentiable on I, and:

$$\boxed{\left(\, e^{u(x)}\right)' = u'(x) \cdot e^{u(x)}}$$

⚠️ Caution: Classic Trap! — Never Forget $u'(x)$

The derivative of e^{5x} is not e^{5x}! Here $u(x) = 5x \Longrightarrow u'(x) = 5$. Thus $(e^{5x})' = 5e^{5x}$. Similarly: $(e^{-x})' = -e^{-x}$, and $(e^{x^2})' = 2xe^{x^2}$.

Step-by-Step Worked Examples

Worked Example 1 — Simplifying Exponential Expressions

Simplify the following expressions completely: 1. $A = \frac{e^3 \times e^{-5}}{(e^2)^3}$ 2. $B = \frac{e^{2x+1} \times e^{-x+3}}{e^{x-2}}$ 3. $C = (e^x + e^{-x})^2 - (e^x - e^{-x})^2$

Detailed Solution: 1. $A = \frac{e^{3+(-5)}}{e^{2\times 3}} = \frac{e^{-2}}{e^6} = e^{-2-6} = e^{-8} = \frac{1}{e^8}$. 2. $B = \frac{e^{(2x+1)+(-x+3)}}{e^{x-2}} = \frac{e^{x+4}}{e^{x-2}} = e^{(x+4)-(x-2)} = e^{x+4-x+2} = e^6$. 3. Difference of squares identity $(u+v)^2 - (u-v)^2 = 4uv$: $C = 4 \times e^x \times e^{-x} = 4 \times e^{x-x} = 4 \times e^0 = 4 \times 1 = 4$.

Worked Example 2 — Solving Exponential Equations

Solve in $\mathbb{R}$: 1. $e^{3x-1} = e^{x+5}$ 2. $e^{x^2-4} = 1$ 3. $e^{2x} - 3e^x + 2 = 0$

Detailed Solution: 1. By strict monotonicity: $e^a = e^b \Leftrightarrow a = b$.

$$3x - 1 = x + 5 \Leftrightarrow 2x = 6 \Leftrightarrow x = 3. \quad S = \{3\}.$$

2. Recall that $1 = e^0$:

$$e^{x^2-4} = e^0 \Leftrightarrow x^2 - 4 = 0 \Leftrightarrow (x-2)(x+2) = 0. \quad S = \{-2; 2\}.$$

3. Recognize that $e^{2x} = (e^x)^2$. Substitute $X = e^x$ (with $X > 0$). The equation becomes:

$$X^2 - 3X + 2 = 0.$$

Roots are $X_1 = 1$ and $X_2 = 2$. Returning to x:

- $e^x = 1 \Leftrightarrow x = 0$.
- $e^x = 2 \Leftrightarrow x = \ln(2)$ (the unique real number ≈ 0.693).

$$S = \{0; \ln 2\}.$$

Worked Example 3 — Solving Inequalities

Solve in $\mathbb{R}$: 1. $e^{4x-7} \leq 1$ 2. $\frac{e^x - 3}{e^x + 2} > 0$

Detailed Solution: 1. $e^{4x-7} \leq e^0 \Leftrightarrow 4x - 7 \leq 0 \Leftrightarrow 4x \leq 7 \Leftrightarrow x \leq \frac{7}{4}$.

$$S =]-\infty; \frac{7}{4}].$$

2. For all real x, $e^x > 0$, so the denominator $e^x + 2 > 2 > 0$ is strictly positive! The sign of the quotient is therefore determined entirely by the numerator:

$$e^x - 3 > 0 \Leftrightarrow e^x > 3 \Leftrightarrow x > \ln 3. \quad S =]\ln 3; +\infty[.$$

Worked Example 4 — Differentiating Composite Functions

Differentiate the following functions: 1. $f(x) = (2x-3)e^{-x}$ 2. $g(x) = e^{-x^2+4x}$ 3. $h(x) = \frac{e^x}{x+1}$ for $x \neq -1$

Detailed Solution: 1. Product form $u \cdot v$ with $u(x) = 2x - 3 \Longrightarrow u'(x) = 2$, and $v(x) = e^{-x} \Longrightarrow v'(x) = -e^{-x}$.

$$f'(x) = u'v + uv' = 2e^{-x} + (2x-3)(-e^{-x}) = [2 - (2x-3)]e^{-x} = (5-2x)e^{-x}.$$

2. Form e^u with $u(x) = -x^2 + 4x \Longrightarrow u'(x) = -2x + 4$.

$$g'(x) = u'(x)e^{u(x)} = (-2x+4)e^{-x^2+4x}.$$

3. Quotient form $\frac{u}{v}$:

$$h'(x) = \frac{(e^x)'(x+1) - e^x(x+1)'}{(x+1)^2} = \frac{e^x(x+1) - e^x(1)}{(x+1)^2} = \frac{e^x(x+1-1)}{(x+1)^2} = \frac{xe^x}{(x+1)^2}.$$

Worked Example 5 — Complete Curve Study

Let $f(x) = (x-1)e^x$ defined on $\mathbb{R}$. Construct its complete variation table and find the equation of its tangent line at 0.

Detailed Solution: 1. Derivative: Product rule with $u(x) = x - 1 \Longrightarrow u'(x) = 1$ and $v(x) = e^x \Longrightarrow v'(x) = e^x$.

$$f'(x) = 1 \cdot e^x + (x-1)e^x = (1 + x - 1)e^x = xe^x.$$

2. Sign of $f'(x)$: Since $e^x > 0$ for all real x, the sign of $f'(x)$ is **identical to the sign of x**!

- $f'(x) < 0$ for $x < 0$;
- $f'(0) = 0$;
- $f'(x) > 0$ for $x > 0$.

3. Variations: f is strictly decreasing on $]-\infty; 0]$ and strictly increasing on $[0; +\infty[$. It achieves an absolute minimum at $x = 0$, equal to $f(0) = (0-1)e^0 = -1$.
4. Tangent at 0: $f(0) = -1$ and $f'(0) = 0 \cdot e^0 = 0$. The tangent is horizontal with equation: $\boxed{y = -1}$.

💡 Worked Example 6 — Concrete Modeling: Capacitor Discharge

In an electrical circuit, the voltage across a discharging capacitor through a resistor is modeled by $U(t) = 12e^{-0.5t}$, where t is time in seconds and U is in volts. 1. What is the initial voltage at $t = 0$? 2. After how much time does the voltage drop to half its initial value? 3. Compute the initial rate of voltage drop at $t = 0$.

Detailed Solution: 1. $U(0) = 12e^0 = 12 \times 1 = 12$ V. 2. We seek t such that $U(t) = 6$ V:

$$12e^{-0.5t} = 6 \Leftrightarrow e^{-0.5t} = 0.5 = \frac{1}{2} \Leftrightarrow e^{0.5t} = 2.$$

Since $0.5t = \ln 2 \approx 0.693$, this gives $t \approx 2 \times 0.693 \approx 1.39$ seconds. 3. Rate of discharge: this is the derivative $U'(t)$.

$$U'(t) = 12 \times (-0.5)e^{-0.5t} = -6e^{-0.5t}.$$

At $t = 0$, $U'(0) = -6e^0 = -6$ V/s. The voltage initially drops at a rate of 6 volts per second.

What You Must Remember

1. **Definition of the Exponential:**
   - Unique function satisfying $f' = f$ and $f(0) = 1$.
   - Denoted $\exp(x)$ or e^x, with $e = \exp(1) \approx 2.71828$.
2. **Fundamental Property (Morphism: Addition → Multiplication):**
   - $e^{a+b} = e^a \cdot e^b$
   - $e^{-a} = \frac{1}{e^a}$
   - $e^{a-b} = \frac{e^a}{e^b}$
   - $(e^a)^n = e^{na}$
3. **Analytical Properties:**
   - Strictly positive: $e^x > 0$ on $\mathbb{R}$.
   - Strictly increasing on $\mathbb{R}$: $e^a < e^b \Leftrightarrow a < b$.
   - Limits: $\lim_{x \to -\infty} e^x = 0$ ($y = 0$ is a horizontal asymptote) and $\lim_{x \to +\infty} e^x = +\infty$.
   - Standard derivative: $(e^x)' = e^x$.
   - Composite derivative: $\left(e^{u(x)}\right)' = u'(x)e^{u(x)}$.
   - Tangent line at 0: $y = x + 1$, with $e^x \geq x + 1$ for all real x.

Exercises

Level (Expressions and Elementary Equations)

1. Write as a single power of e: $A = e^4 \times e^{-7}$, $B = \frac{e^8}{e^3}$, $C = (e^{-2})^5$.
2. Differentiate the function $f(x) = 3e^x - 4x + 1$.
3. Differentiate $g(x) = e^{4x-1}$.
4. Differentiate $h(x) = e^{-3x}$.
5. Solve in $\mathbb{R}$ the equation $e^{2x-3} = e^5$.
6. Solve in $\mathbb{R}$ the inequality $e^{3x+1} > e^{x-5}$.
7. Determine the equation of the tangent line to the curve of $f(x) = e^x$ at $a = 1$.
8. Compute $f'(x)$ for $f(x) = x^2 e^x$.

Level (Function Studies and Variations)

9. Solve in $\mathbb{R}$ the equation $e^{2x} - 4e^x + 3 = 0$ via a substitution.
10. Construct the complete variation table of $f(x) = (2x - 1)e^x$ on $\mathbb{R}$.
11. Consider the Gaussian bell curve function $f(x) = e^{-x^2}$.
    a) Compute $f'(x)$.
    b) Construct the variation table of f and find its absolute maximum.
12. Solve the inequality $e^{2x} - e^x \leq 0$.
13. Consider the function $f(x) = \frac{e^x - 1}{e^x + 1}$.
    a) Prove that f is odd ($f(-x) = -f(x)$).
    b) Compute $f'(x)$ and prove that f is strictly increasing on $\mathbb{R}$.
14. Consider the function $g(x) = e^{2x} - 2x$.
    a) Study the variations of g on $\mathbb{R}$.
    b) Deduce that for all real x, $e^{2x} \geq 2x + 1 > 2x$.
15. Determine the coordinates of points on the curve of $f(x) = e^{x^2 - 3x}$ with a horizontal tangent.
16. A hot bowl of soup is set on a counter in a room at 20°C. Its temperature in degrees Celsius after t minutes is modeled by $T(t) = 20 + 60e^{-0.05t}$.
    a) What is the temperature of the soup when served ($t = 0$)?
    b) What will its temperature be after 20 minutes? (Given: $e^{-1} \approx 0.368$)
    c) What temperature does the soup approach as $t \to +\infty$?

Level (Problems and Advanced Challenges)

17. **Proof of the Inequality $e^x \geq x + 1$:** Consider $\varphi(x) = e^x - x - 1$ defined on $\mathbb{R}$.
    a) Study the variations of φ and construct its variation table.
    b) Deduce its minimum and conclude that the exponential curve lies entirely above its tangent line at 0.
18. **Curve Intersection and Orthogonality:** Consider $f(x) = e^x$ and $g(x) = e^{-x}$.
    a) Find the coordinates of their point of intersection.
    b) Show that their tangent lines at this point are perpendicular!
19. **The Catenary (Hyperbolic Cosine):** A power cable hanging between two utility poles assumes the shape of a curve called a catenary, with equation $f(x) = \frac{e^x + e^{-x}}{2}$.
    a) Show that f is an even function.
    b) Compute $f'(x)$ and determine the monotonicity of f.
    c) Prove that for all real x, $(f(x))^2 - (f'(x))^2 = 1$.
20. **Euler Algorithm and Comparison:** Write a Python program that compares the approximation at $x = 1$ obtained by Euler's method with 100 steps to the exact value given by `math.exp(1)`. Compute the relative error as a percentage.
21. **Family of Functions and Invariant Slope:** For any real $k > 0$, define $f_k(x) = ke^{kx}$. Show that all curves $\mathcal{C}_k$ pass through an identifiable point under translation, and determine the coordinates of the point where the tangent to $\mathcal{C}_k$ has slope k^2.

Solutions

1. $A = e^{4-7} = e^{-3} = \frac{1}{e^3}$. $B = e^{8-3} = e^5$. $C = e^{-2\times 5} = e^{-10}$.

2. $f'(x) = 3e^x - 4$.

3. Form e^u with $u = 4x - 1 \Longrightarrow u' = 4$. $g'(x) = 4e^{4x-1}$.

4. Form e^u with $u = -3x \Longrightarrow u' = -3$. $h'(x) = -3e^{-3x}$.

5. $e^{2x-3} = e^5 \Leftrightarrow 2x - 3 = 5 \Leftrightarrow 2x = 8 \Leftrightarrow x = 4$. $S = \{4\}$.

6. $e^{3x+1} > e^{x-5} \Leftrightarrow 3x + 1 > x - 5 \Leftrightarrow 2x > -6 \Leftrightarrow x > -3$. $S =]-3; +\infty[$.

7. At $a = 1$: $f(1) = e^1 = e$, $f'(1) = e$. Tangent line: $y = f'(1)(x - 1) + f(1) = e(x - 1) + e = ex - e + e = ex$. The tangent line at 1 passes directly through the origin: $y = ex$.

8. Product rule: $u = x^2 \Longrightarrow u' = 2x$; $v = e^x \Longrightarrow v' = e^x$. $f'(x) = 2xe^x + x^2e^x = (x^2 + 2x)e^x = x(x + 2)e^x$.

9. Substituting $X = e^x > 0$, the equation becomes $X^2 - 4X + 3 = 0$. $\Delta = 16 - 12 = 4 = 2^2$. $X_1 = \frac{4-2}{2} = 1$ and $X_2 = \frac{4+2}{2} = 3$.

- $e^x = 1 \Leftrightarrow x = 0$.
- $e^x = 3 \Leftrightarrow x = \ln 3 \approx 1.0986$. $S = \{0; \ln 3\}$.

10. $f'(x) = 2e^x + (2x - 1)e^x = (2 + 2x - 1)e^x = (2x + 1)e^x$. Since $e^x > 0$, the sign of $f'(x)$ matches $2x + 1$:

- $f'(x) < 0$ on $]-\infty; -0.5[$ (f decreasing);
- $f'(-0.5) = 0$;
- $f'(x) > 0$ on $]-0.5; +\infty[$ (f increasing).

Absolute minimum at $x = -0.5$: $f(-0.5) = (2(-0.5) - 1)e^{-0.5} = -2e^{-0.5} = -\frac{2}{\sqrt{e}} \approx -1.213$.

11. a) $u(x) = -x^2 \Longrightarrow u'(x) = -2x$. Thus $f'(x) = -2xe^{-x^2}$. b) Since $e^{-x^2} > 0$, the sign of $f'(x)$ is opposite to the sign of x: For $x < 0$, $f'(x) > 0$ (increasing); for $x > 0$, $f'(x) < 0$ (decreasing). f achieves an absolute maximum at $x = 0$, equal to $f(0) = e^0 = 1$.

12. Factor out e^x: $e^{2x} - e^x = e^x(e^x - 1) \leq 0$. Since $e^x > 0$ always, this is equivalent to $e^x - 1 \leq 0 \Leftrightarrow e^x \leq 1 = e^0 \Leftrightarrow x \leq 0$. $S =]-\infty; 0]$.

13. a) $f(-x) = \frac{e^{-x}-1}{e^{-x}+1} = \frac{\frac{1}{e^x}-1}{\frac{1}{e^x}+1} = \frac{\frac{1-e^x}{e^x}}{\frac{1+e^x}{e^x}} = \frac{-(e^x-1)}{e^x+1} = -f(x)$. The function is odd. b) Quotient rule: $f'(x) = \frac{e^x(e^x+1)-(e^x-1)e^x}{(e^x+1)^2} = \frac{e^{2x}+e^x-e^{2x}+e^x}{(e^x+1)^2} = \frac{2e^x}{(e^x+1)^2}$. The numerator is strictly positive ($2e^x > 0$) and the denominator is a non-zero square (> 0). Thus $f'(x) > 0$ on $\mathbb{R}$: f is strictly increasing.

14. a) $g'(x) = 2e^{2x} - 2 = 2(e^{2x} - 1)$. $g'(x) > 0 \Leftrightarrow e^{2x} > 1 \Leftrightarrow 2x > 0 \Leftrightarrow x > 0$. g decreases on $]-\infty; 0]$ and increases on $[0; +\infty[$. Absolute minimum at 0: $g(0) = e^0 - 0 = 1$. b) Because the minimum is 1, for all $x \in \mathbb{R}$: $g(x) \geq 1 \Leftrightarrow e^{2x} - 2x \geq 1 \Leftrightarrow e^{2x} \geq 2x + 1$. Since $2x + 1 > 2x$, it follows immediately that $e^{2x} > 2x$.

15. $f'(x) = (2x - 3)e^{x^2-3x}$. Horizontal tangent $\Leftrightarrow f'(x) = 0 \Leftrightarrow 2x - 3 = 0 \Leftrightarrow x = \frac{3}{2} = 1.5$.

16. a) $T(0) = 20 + 60e^0 = 20 + 60 = 80$°C. b) $T(20) = 20 + 60e^{-0.05\times 20} = 20 + 60e^{-1} \approx 20 + 60 \times 0.368 = 20 + 22.08 = 42.08$°C. c) As $t \to +\infty$, $-0.05t \to -\infty$, so $e^{-0.05t} \to 0$. $\lim_{t\to+\infty} T(t) = 20 + 0 = 20$°C (room temperature).

17. a) $\varphi'(x) = e^x - 1$.

- $\varphi'(x) < 0$ for $x < 0$ (φ decreasing);
- $\varphi'(0) = 0$;
- $\varphi'(x) > 0$ for $x > 0$ (φ increasing).

b) φ attains its absolute minimum at $x = 0$, equal to $\varphi(0) = e^0 - 0 - 1 = 1 - 1 = 0$. Consequently, for all $x \in \mathbb{R}$, $\varphi(x) \geq 0 \Leftrightarrow e^x - x - 1 \geq 0 \Leftrightarrow e^x \geq x + 1$. The exponential curve therefore lies above its tangent line at 0 for all x.

18. a) $e^x = e^{-x} \Leftrightarrow e^x = \frac{1}{e^x} \Leftrightarrow (e^x)^2 = 1 \Leftrightarrow e^{2x} = 1 \Leftrightarrow 2x = 0 \Leftrightarrow x = 0$. y-coordinate: $y = e^0 = 1$. The intersection point is $A(0; 1)$. b) Tangent to f at 0: slope $f'(0) = e^0 = 1$. Tangent to g at 0: $g'(x) = -e^{-x} \Longrightarrow g'(0) = -e^0 = -1$. The product of slopes is $m_1 \times m_2 = 1 \times (-1) = -1$. By the perpendicularity criterion, the two tangent lines are strictly perpendicular!

19. a) $f(-x) = \frac{e^{-x}+e^{-(-x)}}{2} = \frac{e^{-x}+e^{x}}{2} = f(x)$. The function is even. b) $f'(x) = \frac{e^{x}-e^{-x}}{2}$. $f'(x) > 0 \Leftrightarrow e^{x} - e^{-x} > 0 \Leftrightarrow e^{x} > e^{-x} \Leftrightarrow x > -x \Leftrightarrow 2x > 0 \Leftrightarrow x > 0$. The function decreases on $]-\infty; 0]$ and increases on $[0; +\infty[$. Minimum at 0 equal to $f(0) = 1$. c) $(f(x))^2 - (f'(x))^2 = \left(\frac{e^{x}+e^{-x}}{2}\right)^2 - \left(\frac{e^{x}-e^{-x}}{2}\right)^2 = \frac{(e^{2x}+2+e^{-2x})-(e^{2x}-2+e^{-2x})}{4} = \frac{4}{4} = 1$. (This is the hyperbolic counterpart of the Pythagorean trigonometric identity $\cos^2 x + \sin^2 x = 1$!).

20.

```python
import math

def euler_comparison(n=100):
    h = 1.0 / n
    y = 1.0
    for _ in range(n):
        y = y * (1 + h)
    exact_e = math.exp(1)
    abs_err = abs(y - exact_e)
    rel_err = (abs_err / exact_e) * 100
    print(f"Euler approx : {y:.6f}")
    print(f"Exact e      : {exact_e:.6f}")
    print(f"Absolute err : {abs_err:.6f}")
    print(f"Relative err : {rel_err:.3f}%")

euler_comparison(100)
```

21. At $x = 0$, $f_k(0) = ke^0 = k$. Derivative: $f_k'(x) = k \cdot (ke^{kx}) = k^2 e^{kx}$. The slope equals k^2 if and only if $k^2 e^{kx} = k^2 \Leftrightarrow e^{kx} = 1 \Leftrightarrow kx = 0 \Leftrightarrow x = 0$ (since $k > 0$). For all functions in this family, the tangent line has slope k^2 precisely at $x = 0$.

The Question That Opens What Follows

The exponential function allowed us to model populations multiplying under pure and idealized mathematical laws. Yet when a biologist counts bacteria under a microscope, or when an opinion pollster interviews a sample of one thousand citizens on the street, physical measurements never lie on an immaculate, deterministic curve: they scatter, deviate, and fluctuate randomly. How do we summarize a scattered cloud of noisy data with two master numbers: the center of gravity (**the mean**) and the dispersion spread (**the standard deviation**)? That is what we are about to discover in the next chapter with **the pollster and standard deviation**.

CHAPTER 41: The Pollster and Standard Deviation

> "If you put your head in a burning oven and your feet in a block of ice, an incautious statistician will tell you that, 'on average,' your body temperature is ideal. The mean is a lighthouse illuminating the center while plunging the peripheral storms into deep shadow. To understand the real world, one must measure the dispersion."

What You Are About to Discover

- The fatal trap of the arithmetic mean: why two factories, two classes, or two medical treatments sharing the exact same mean can be diametrically opposed.
- Position indicators: the **weighted mean** ($\bar{x}$) versus the **median** (M_e), and why the median cannot be fooled by billionaires.
- Sentinels of the quarter: the **quartiles** Q_1 and Q_3, and the **interquartile range** (IQR).
- The art of visual summarization: John Tukey's **box-and-whisker plot** (*boxplot*).
- The mystery of the square: why the sum of deviations from the mean always vanishes ($\sum(x_i - \bar{x}) = 0$), and why we had to invent the **variance** and **standard deviation** σ.
- The **König-Huygens formula**: the lightning-fast shortcut for computing variance.
- The law of sampling: how a pollster can predict the vote of 45 million citizens by interviewing a mere 1,000 individuals, using the **sampling fluctuation interval**.
- Sampling simulations programmed in **Python**.

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1. The Problem: The Two Drive-Shaft Factories

We find ourselves inside a cutting-edge aerospace manufacturing facility near Toulouse. The plant produces titanium drive shafts destined for commercial jet turbine engines. The engineers' technical specifications are absolute: to fit within a thousandth of a millimeter into the turbine's precision ball bearings, the nominal diameter of a shaft must measure exactly **50.00 millimeters**. A shaft measuring 50.10 mm will jam the engine; a shaft measuring 49.90 mm will introduce destructive mechanical play at 10,000 rpm.

Two subcontracting suppliers, Workshop **Alpha** and Workshop **Beta**, each deliver a batch of 1,000 machined shafts. The quality control director measures the samples:

- In Workshop **Alpha**, the mean diameter is exactly **50.00 mm**.
- In Workshop **Beta**, the mean diameter is likewise exactly **50.00 mm**.

A naive purchasing agent might declare: *"Both suppliers produce identical results, let's buy the cheaper one!"* However, the chief technical director inspects the pieces under an electronic microscope:

> 🖐 **Solved Example — Comparative Spread: Workshop Alpha vs Workshop Beta**
>
> - Workshop Alpha:
> - Every single shaft measures between 49.98 mm and 50.02 mm.
> - They are virtually immaculate, tightly clustered like a dense swarm around 50.00 mm.
> - 49.99 50 50.01
> - Workshop Beta:
> - One shaft measures 46.00 mm, another 54.00 mm, another 48.20 mm…
> - The pieces are scattered across a frightening expanse of chaotic variation!
> - 46 47 48 49 50 51 52 53 54

In Workshop Beta, **virtually no part can safely be fitted into an aircraft**! Some are wildly oversized, others dangerously thin, even though by sheer symmetrical coincidence their arithmetic average hits 50 mm on the dot. Installing Beta's parts would lead directly to engine failure.

This is the intellectual crisis that birthed modern statistics: **A measure of central tendency (the mean) is useless unless accompanied by a measure of dispersion (the standard deviation).**

---

2. Measures of Central Tendency: Mean vs. Median

Consider a discrete quantitative dataset where the variable takes sorted values $x_1 < x_2 < ... < x_p$, associated with frequencies (counts) $n_1, n_2, ..., n_p$. The total sample size is $N = \sum_{i=1}^{p} n_i$.

2.1 The Weighted Mean ($\bar{x}$)

The mean is the center of gravity of the distribution: it is the fair share each individual would receive if all values were pooled and redistributed equally.

> **📐 Definition — Weighted Mean**
>
> The **weighted mean** of a dataset, denoted $\bar{x}$ (read "**x bar**"), is given by:
>
> $$\boxed{\bar{x} = \frac{\sum_{i=1}^{p} n_i x_i}{N} = \sum_{i=1}^{p} f_i x_i}$$
>
> where $f_i = \frac{n_i}{N}$ denotes the relative frequency of the value x_i.

Linearity Property: If every value is multiplied by a constant a and shifted by a constant b (affine transformation $y_i = ax_i + b$), the new mean is simply:

$$\bar{y} = a\bar{x} + b.$$

2.2 The Median (M_e): The Justice of Rank

The median ignores the sum total and focuses entirely on **position in line**.

> **📐 Definition — Median**
>
> The **median** of a sorted statistical dataset, denoted M_e, is a value that splits the population into two halves of equal size:
>
> - At least 50% of the individuals have a value less than or equal to M_e;
> - At least 50% of the individuals have a value greater than or equal to M_e.

Practical Computation Method: 1. If the total count N is **odd**: $N = 2k + 1$. The median is the value of the individual at rank $k + 1$. *(Example: for $N = 9$, $\frac{9+1}{2} = 5$: the median is the 5th value).* 2. If the total count N is **even**: $N = 2k$. The median is conventionally taken as the midpoint between the values at ranks k and $k + 1$. *(Example: for $N = 10$, take the average of the 5th and 6th values).*

> 🔍 ***True Story — Bill Gates Walks into a Bar*** *Imagine a neighborhood café where ten factory workers and clerks sit at the counter. Each earns $2,000 per month. The mean wage is $2,000, and the median wage is $2,000. Suddenly, billionaire **Bill Gates** opens the door and takes a seat at the bar. His monthly income is around $10,000,000. Let us recalculate the statistics of the patrons:*
>
> - *The **mean income** skyrockets: $\frac{10 \times 2{,}000 + 10{,}000{,}000}{11} \approx \$910,900 \text{ per month}$! A foolish statistician might announce: "In this diner, the average customer is nearly a millionaire!" Which is absurd.*
> - *The **median income**, however, lines up the 11 patrons in ascending order. The 6th person in line still earns exactly **$2,000**! The median did not budge by a single penny: it is **robust** against extreme outliers, whereas the mean is hyper-sensitive.*

3. Quartiles and the Box-and-Whisker Plot

To analyze the spread more deeply, we partition the ordered dataset into **four equal quarters** of 25% each.

3.1 Formal Definitions of Quartiles

Definition — Quartiles

In a sorted dataset of size N:

- The **first quartile**, denoted Q_1, is the smallest value in the dataset such that at least **25%** of observations are less than or equal to Q_1. Its rank is the smallest integer greater than or equal to $\frac{N}{4}$ (denoted $\lceil N/4 \rceil$).
- The **third quartile**, denoted Q_3, is the smallest value such that at least **75%** of observations are less than or equal to Q_3. Its rank is the smallest integer greater than or equal to $\frac{3N}{4}$ (denoted $\lceil 3N/4 \rceil$).
- The second quartile Q_2 is none other than the median M_e.

Definition — Interquartile Range

The difference between the third and first quartiles is called the **interquartile range** (IQR):

$$\boxed{IQR = Q_3 - Q_1}$$

It measures the width of the interval housing the **central 50%** of the population. Like the median, this measure of spread is entirely impervious to extreme outliers.

3.2 John Tukey's Boxplot (1977)

Pioneered by the American statistician **John Tukey**, the box-and-whisker plot summarizes an entire distribution at a single glance:

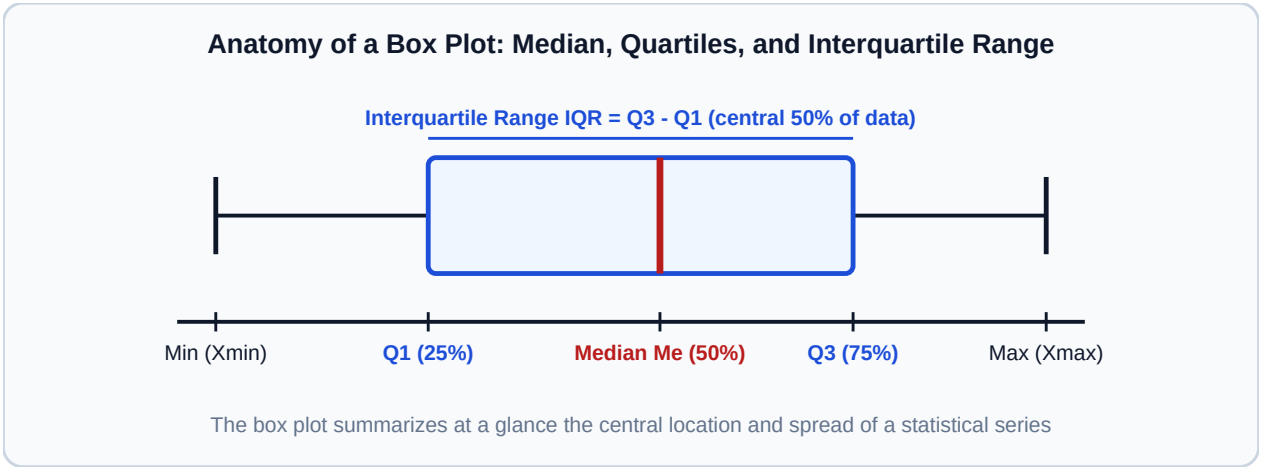

Figure 94: Box plot: median, quartiles, and statistical spread

- The **box** spans from Q_1 to Q_3. Its width is the interquartile range.
- The vertical line inside marks the **median**. An off-center median reveals skewness.
- The two **whiskers** extend outward to the minimum and maximum non-outlier observations.

4. The Mystery of the Square: Variance and Standard Deviation

Let us return to our aerospace shafts. How do we mathematically capture the average distance of the observations from the target mean $\bar{x}$?

4.1 The First Naive Idea: Average of Deviations

The most instinctive attempt would be to compute the raw difference $(x_i - \bar{x})$ for each individual, then take their average. Test this on three simple test scores: $8, 12, 16$. The mean is $\bar{x} = \frac{8+12+16}{3} = 12$. The individual deviations from the mean are:

- For 8: $8 - 12 = -4$
- For 12: $12 - 12 = 0$
- For 16: $16 - 12 = +4$

Add them up: $(-4) + 0 + (+4) = \mathbf{0}$! The positive deviations were completely cancelled out by the negative ones!

📐 Theorem — The Sum of Deviations is Always Zero

For any dataset:

$$\sum_{i=1}^{p} n_i(x_i - \bar{x}) = 0$$

Proof

Expand the summation:

$$\sum_{i=1}^{p} n_i(x_i - \bar{x}) = \sum_{i=1}^{p} n_i x_i - \sum_{i=1}^{p} n_i \bar{x} = \sum_{i=1}^{p} n_i x_i - \bar{x}\sum_{i=1}^{p} n_i = N\bar{x} - \bar{x}N = 0. \quad \blacksquare$$

The average of raw deviations is **identically zero**, no matter how dispersed the data is! It is entirely useless as a measure of spread.

4.2 The Square Trick: Variance

To prevent negative deviations from neutralizing positive ones, mathematicians square each deviation: $(-4)^2 = +16$! A squared real number is always non-negative.

📐 Definition — Variance

The **variance** of a dataset, denoted V, is the mean of the squared deviations from the mean:

$$\boxed{V = \frac{1}{N}\sum_{i=1}^{p} n_i(x_i - \bar{x})^2 = \sum_{i=1}^{p} f_i(x_i - \bar{x})^2}$$

⚠️ ⚠️ Sample vs Population: Why Your Calculator Displays Both σ and s (Dividing by n or n - 1)

When evaluating dispersion, modern calculators systematically display two distinct figures (σ_x and s_x, or σ and σ_{n-1}). They serve two very different statistical purposes:

- **Entire population (σ, dividing by N):** When you have access to the full census of data (e.g., every test score in a specific class), the exact descriptive variance is $V = \frac{1}{N}\sum (x_i - \bar{x})^2$ with standard deviation $\sigma = \sqrt{V}$.
- **Sample estimation (s, dividing by $n-1$):** When sampling n individuals at random to **estimate** the dispersion of a vast unknown population, the sample mean $\bar{x}$ naturally hugs the sample points too closely, creating a systematic downward bias. To neutralize this bias, Gauss and Bessel proved that one must divide by $n-1$:

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

Here s is the **unbiased sample standard deviation**. For large surveys ($n > 100$), the numerical difference between $\frac{1}{n}$ and $\frac{1}{n-1}$ fades, but it remains essential in inferential statistics.

4.3 Standard Deviation (σ): Restoring the Physical Unit

Variance overcomes the sign problem, but introduces a dimensional absurdity: If we measure heights in **meters** (m), the deviations are in meters, but their squared deviations are in **square meters** (m^2)! Expressing the spread of human heights in square meters makes no physical sense. To restore the original unit of measurement (meters, euros, millimeters), we take the **square root**:

Definition — Standard Deviation

The **standard deviation**, denoted σ (lowercase Greek letter **sigma**), is the square root of the variance:

$$\boxed{\sigma = \sqrt{V} = \sqrt{\frac{1}{N}\sum_{i=1}^{p} n_i (x_i - \bar{x})^2}}$$

Intuitive Meaning: The standard deviation measures the *typical* distance by which an individual data point deviates from the group mean.

- Workshop Alpha: $\bar{x} = 50.00$ mm and $\sigma = \mathbf{0.01}\ \ \mathbf{mm}$ (negligible spread).
- Workshop Beta: $\bar{x} = 50.00$ mm and $\sigma = \mathbf{2.45}\ \ \mathbf{mm}$ (catastrophic spread).

Thanks to σ, the aerospace engineer instantly identifies the defective workshop!

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5. The König-Huygens Formula: The Computational Shortcut

Computing every $(x_i - \bar{x})^2$ by hand becomes tedious as soon as $\bar{x}$ involves decimals. Fortunately, 17th-century mathematicians **Samuel König** and **Christiaan Huygens** established an ingenious computational identity:

Theorem — König-Huygens Formula

The variance equals **the mean of the squares minus the square of the mean**:

$$\boxed{V = \overline{x^2} - (\bar{x})^2}$$

where $\overline{x^2} = \frac{1}{N}\sum_{i=1}^{p} n_i x_i^2$ denotes the mean of the squared values.

Proof

Expand the square inside the variance definition:

$$V = \frac{1}{N}\sum_{i=1}^{p} n_i \left(x_i^2 - 2x_i\bar{x} + \bar{x}^2 \right).$$

Distribute the summation:

$$V = \frac{1}{N}\sum n_i x_i^2 - \frac{2\bar{x}}{N}\sum n_i x_i + \frac{\bar{x}^2}{N}\sum n_i.$$

Since $\frac{1}{N}\sum n_i x_i^2 = \overline{x^2}$, $\frac{1}{N}\sum n_i x_i = \bar{x}$, and $\sum n_i = N$, this simplifies to:

$$V = \overline{x^2} - 2\bar{x}(\bar{x}) + \frac{\bar{x}^2}{N}(N) = \overline{x^2} - 2(\bar{x})^2 + (\bar{x})^2 = \overline{x^2} - (\bar{x})^2. \quad \blacksquare$$

📐 Affine Transformation Property —

If $y_i = ax_i + b$, then:

$$V(aX + b) = a^2 V(X) \quad \text{and} \quad \boxed{\sigma(aX + b) = |a| \cdot \sigma(X)}$$

Note: Adding a constant b (e.g., granting a 2-point bonus to the entire class) shifts all scores without altering their spread: the standard deviation remains unchanged!

---

6. Sampling, Fluctuation, and Python Simulation

How can a polling institute accurately gauge the outcome of a national presidential election by calling only 1,000 citizens? This is the mathematical magic of **sampling fluctuation**.

6.1 The Principle of Fluctuation

Suppose an actual proportion p of all citizens in the nation supports a given policy. If we draw a random sample of size n, the observed proportion f in that sample will almost never equal p exactly: it fluctuates around p. However, for a sufficiently large sample ($n \geq 30$, $np \geq 5$, $n(1-p) \geq 5$), probability theory guarantees that in **at least 95% of random samples**, the observed frequency f falls within a well-defined interval:

📐 Definition — 95% Fluctuation Interval

- Introductory approximation:

$$I = \left[p - \frac{1}{\sqrt{n}} \, ; p + \frac{1}{\sqrt{n}} \right]$$

- Asymptotic formula (derived from the Normal distribution):

$$\boxed{I_{0.95} = \left[p - 1.96\sqrt{\frac{p(1-p)}{n}} \, ; p + 1.96\sqrt{\frac{p(1-p)}{n}} \right]}$$

6.2 Sampling Simulation in Python

To verify this experimentally, let us simulate 1,000 independent polls of size $n = 100$ drawn from a population where the true proportion is $p = 0.40$ (40%):

```
import random

def simulate_sample(n, p):
    """Simulates drawing a sample of size n and returns observed frequency."""
    successes = 0
    for _ in range(n):
        if random.random() < p:
            successes += 1
    return successes / n

# Let us generate 5 samples of size 1,000 for a proportion p = 0.40
p_true = 0.40
sample_size = 1000
print(f"Theoretical proportion p = {p_true}")
for i in range(1, 6):
    f_obs = simulate_sample(sample_size, p_true)
    print(f"Sample {i} : f = {f_obs:.3f} (error = {abs(f_obs - p_true):.3f})")
```

Output:

📐 Fundamental Rule — 95% Asymptotic Fluctuation Interval

- Theoretical 95% interval: [0.3040 ; 0.4960]
- Percentage of samples inside interval: 95.3%

Precisely as predicted: over 95% of random polls hit the theoretical bullseye!

Step-by-Step Worked Examples

Worked Example 1 — Computing Mean and Standard Deviation

The grades of 20 students on an examination are distributed as follows:

Grade x_i	6	8	10	12	15	18
Count n_i	2	4	6	5	2	1

Calculate the mean $\bar{x}$, variance V, and standard deviation σ.

Detailed Solution: Total count: $N = 2+4+6+5+2+1 = 20$. 1. Mean:

$$\sum n_i x_i = (2\times 6)+(4\times 8)+(6\times 10)+(5\times 12)+(2\times 15)+(1\times 18)$$
$$= 12+32+60+60+30+18 = 212.$$
$$\bar{x} = \frac{212}{20} = 10.6.$$

2. König-Huygens formula for variance: Compute $\sum n_i x_i^2$:

$$2(36)+4(64)+6(100)+5(144)+2(225)+1(324)$$
$$= 72+256+600+720+450+324 = 2422.$$
$$\overline{x^2} = \frac{2422}{20} = 121.1.$$

$$V = \overline{x^2} - (\bar{x})^2 = 121.1 - (10.6)^2 = 121.1 - 112.36 = 8.74.$$

3. Standard deviation:

$$\sigma = \sqrt{V} = \sqrt{8.74} \approx 2.96.$$

The class average is 10.6 with a standard deviation of nearly 3 points.

Worked Example 2 — Determining Median and Quartiles

Find M_e, Q_1, Q_3, and the IQR for the dataset in Example 1.

Detailed Solution: Construct the cumulative frequency table:

Grade x_i	6	8	10	12	15	18
Count n_i	2	4	6	5	2	1
Cumulative	2	6	12	17	19	20

- **Median:** $N = 20$ (even). $\frac{N}{2} = 10$. Take the average of the 10th and 11th values. From cumulative counts, both the 10th and 11th values are 10. Thus $\boxed{M_e = 10}$.
- **First Quartile Q_1:** $\frac{N}{4} = \frac{20}{4} = 5$. Q_1 is the 5th value in the sorted list. Ranks 3 through 6 are 8. Thus $\boxed{Q_1 = 8}$.
- **Third Quartile Q_3:** $\frac{3N}{4} = \frac{60}{4} = 15$. Q_3 is the 15th value. Ranks 13 through 17 are 12. Thus $\boxed{Q_3 = 12}$.
- **Interquartile Range:** $IQR = Q_3 - Q_1 = 12 - 8 = 4$. The middle 50% of the class earned grades between 8 and 12.

Worked Example 3 — Boxplot Comparison of Two Archery Competitors

Two archers A and B shoot 50 arrows each. Their five-number summaries are:

- Archer A: Min $= 5$, $Q_1 = 8$, $M_e = 9$, $Q_3 = 10$, Max $= 10$.
- Archer B: Min $= 2$, $Q_1 = 6$, $M_e = 9$, $Q_3 = 9$, Max $= 10$. Compare their performance profiles.

Detailed Solution:

- Both archers share the exact same median ($M_e = 9$): at least half their arrows hit a 9 or 10.
- Contrast their dispersion:
  - Archer A: $IQR = 10 - 8 = 2$, total range is $10 - 5 = 5$. Fully 75% of shots scored 8 or higher.
  - Archer B: $IQR = 9 - 6 = 3$, total range is $10 - 2 = 8$. One quarter of shots scored 6 or worse, including an erratic wild shot of 2.
- **Conclusion:** Archer A is far more **consistent and reliable**, despite their identical median scores.

Worked Example 4 — Linear Transformation of Mean and Standard Deviation

A teacher administers a challenging test: the mean is $\bar{x} = 8$ and standard deviation is $\sigma = 3$. To rescale the grades, she applies the formula:

$$y = 1.2x + 2.$$

Find the new mean $\bar{y}$ and new standard deviation σ_y.

Detailed Solution:

- By linearity of the mean:

$$\bar{y} = 1.2\bar{x} + 2 = 1.2(8) + 2 = 9.6 + 2 = 11.6.$$

- By properties of standard deviation: $\sigma(aX + b) = |a|\sigma(X)$.

$$\sigma_y = |1.2| \times \sigma = 1.2 \times 3 = 3.6.$$

The mean rises by 3.6 points, but the standard deviation also expands: individual score gaps widen slightly.

Worked Example 5 — Hypothesis Testing on a Quality Sample

A bakery claims that only 2% of its packaged cookies have baking defects ($p = 0.02$). A consumer watchdog inspects a sample of $n = 500$ cookies and finds 18 defective ones. Should the manufacturer's claim be rejected at the 95% confidence level?

Detailed Solution: 1. Check validity conditions: $n = 500 \geq 30$, $np = 500 \times 0.02 = 10 \geq 5$, $n(1-p) = 500 \times 0.98 = 490 \geq 5$. Conditions are satisfied. 2. Theoretical 95% fluctuation interval:

$$I_{0.95} = \left[0.02 - 1.96\sqrt{\frac{0.02 \times 0.98}{500}}\,;\, 0.02 + 1.96\sqrt{\frac{0.02 \times 0.98}{500}}\right].$$

$$\sqrt{\frac{0.0196}{500}} = \sqrt{0.0000392} \approx 0.00626.$$

Margin of error: $1.96 \times 0.00626 \approx 0.0123$.

$$I_{0.95} = [0.02 - 0.0123\,;\, 0.02 + 0.0123] = [0.0077\,;\, 0.0323].$$

In percentage terms: between 0.77% and 3.23%. 3. Observed sample frequency:

$$f = \frac{18}{500} = 0.036 = 3.6\%.$$

4. Decision: The observed frequency $f = 0.036 \notin I_{0.95}$ (it exceeds the upper bound of 0.0323). **Conclusion:** At the 95% confidence level, the manufacturer's claim is **rejected**. The defect rate is abnormally high.

📐 Worked Example 6 — Detecting Statistical Outliers

Tukey established a standard rule for identifying outliers: any value strictly outside the fences:

$$[Q_1 - 1.5 \times IQR\,;\, Q_3 + 1.5 \times IQR].$$

In a company, salaries have $Q_1 = \$1,800$ and $Q_3 = \$2,600$. Is an executive salary of \$4,000 considered an outlier?

Detailed Solution:

- Interquartile range: $IQR = 2600 - 1800 = \$800$.
- Upper fence: $Q_3 + 1.5 \times IQR = 2600 + 1.5(800) = 2600 + 1200 = \$3,800$.
- Since $\$4,000 > \$3,800$, this salary exceeds the upper threshold: it is formally classified as an **outlier**.

What You Must Remember

1. **Measures of Central Tendency:**
   - Mean: $\bar{x} = \frac{1}{N}\sum n_i x_i = \sum f_i x_i$. Highly sensitive to extreme values.
   - Median M_e: middle value (50% above, 50% below). Robust against outliers.
   - Quartiles: Q_1 (at least 25%) and Q_3 (at least 75%).
2. **Measures of Dispersion:**
   - Interquartile range: $IQR = Q_3 - Q_1$.
   - Variance: $V = \frac{1}{N}\sum n_i (x_i - \bar{x})^2$.
   - König-Huygens formula: $\boxed{V = \overline{x^2} - (\bar{x})^2}$.
   - Standard deviation: $\boxed{\sigma = \sqrt{V}}$ (shares original units of the data).
3. **Affine Transformations ($y = ax + b$):**
   - $\bar{y} = a\bar{x} + b$
   - $V(aX + b) = a^2 V(X)$
   - $\sigma(aX + b) = |a|\sigma(X)$
4. **Sampling and Fluctuation (at 95% Confidence):**
   - $I_{0.95} = \left[p - 1.96\sqrt{\frac{p(1-p)}{n}}\,;\, p + 1.96\sqrt{\frac{p(1-p)}{n}}\right]$ (for $n \geq 30$, $np \geq 5$, $n(1-p) \geq 5$).
   - If observed sample frequency $f \notin I_{0.95}$, reject hypothesis p.

Exercises

Level (Direct Computation of Statistics)

1. Compute the mean and median of the dataset: $4, 7, 9, 11, 14$.
2. Compute the mean and median of the dataset: $3, 5, 8, 12, 16, 20$.
3. A set of 5 values has a mean of 10. Four of the values are $6, 8, 11, 13$. Find the fifth value.
4. Compute the variance and standard deviation of the dataset $2, 4, 6, 8, 10$.
5. Given $\overline{x^2} = 53$ and $\bar{x} = 7$, find the variance V and standard deviation σ.
6. If company salaries have a standard deviation $\sigma = \$250$, what is the new standard deviation if every employee receives an across-the-board raise of \$100?
7. What is the new standard deviation if every salary is increased by 5% (multiplied by 1.05)?
8. Determine Q_1, M_e, and Q_3 for the ordered dataset of 12 values: $1, 3, 4, 6, 7, 9, 10, 12, 13, 15, 18, 20$.

Level (Comparative Analysis and Properties)

9. Consider the following dataset: | Value x_i | 10 | 20 | 30 | 40 | |—|—|—|—|—| | Count n_i | 3 | 5 | 8 | 4 |

a) Compute $\bar{x}$.
b) Use the König-Huygens formula to find V and σ.

10. Prove that for two real numbers a and b, the variance of $\{a, b\}$ equals $\left(\frac{a-b}{2}\right)^2$.
11. An assessment is taken by two groups:
    - Group 1 (15 students): mean $\bar{x}_1 = 12$.
    - Group 2 (25 students): mean $\bar{x}_2 = 10$. Calculate the combined mean of all 40 students.
12. In a class, test scores have a mean of 11 and a standard deviation of 2.5. The teacher wishes to scale the grades so that the mean becomes 12 and the standard deviation becomes 3. What affine transformation $y = ax + b$ should be applied?
13. A random sample of 400 voters reveals that 228 intend to vote for candidate C. Calculate the 95% fluctuation interval under the hypothesis that the electorate is evenly split ($p = 0.50$). Can we conclude that candidate C has a statistically significant lead?
14. Daily high temperatures recorded in July at two weather stations show:
    - Station A: $\bar{x} = 28^\circ$ C, $\sigma = 1.5^\circ$ C.
    - Station B: $\bar{x} = 28^\circ$ C, $\sigma = 6.2^\circ$ C. Contrast the climate characteristics of the two locations.
15. Write a Python function `statistics(data)` that takes a list of numbers and returns its size, mean, variance, and standard deviation without importing external libraries.

Level (Theoretical Challenges and Modeling)

16. **Variance of the Union of Two Datasets:** Let two datasets have sizes N_1, N_2, means $\bar{x}_1, \bar{x}_2$, and variances V_1, V_2. Prove that the variance V of the pooled dataset of size $N = N_1 + N_2$ satisfies:
$$V = \frac{N_1 V_1 + N_2 V_2}{N} + \frac{N_1 N_2}{N^2}(\bar{x}_1 - \bar{x}_2)^2.$$
Interpret both terms (within-group variance and between-group variance).
17. **Bienaymé-Chebyshev Inequality (Discovery):** For any dataset with mean $\bar{x}$ and standard deviation σ, prove that the proportion of data points located at a distance of 2σ or more from the mean can never exceed $\frac{1}{4} = 25\%$, regardless of the distribution's shape!
18. **Polling Sample Size and Margin of Error:** A political poll interviews n voters to decide an election between two neck-and-neck candidates ($p \approx 0.50$). What is the minimum sample size n needed to ensure that the 95% margin of error ($1.96\sqrt{\frac{p(1-p)}{n}}$) is strictly under 1% (i.e., 0.01)?
19. **Variance of Consecutive Integers:** Consider the first n positive integers: $1, 2, 3, ..., n$.
    a) Find the mean $\bar{x}$ in terms of n.
    b) Using $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$, prove that the variance is $V = \frac{n^2-1}{12}$.
20. **Outlier Detection in Python:** Write a Python function that accepts a list of raw numbers, computes Q_1, Q_3, Tukey's fences, and extracts all outliers.

Solutions

1. Mean: $\bar{x} = \frac{4+7+9+11+14}{5} = \frac{45}{5} = 9$. Median: 3rd value out of 5, so $M_e = 9$.

2. Mean: $\bar{x} = \frac{3+5+8+12+16+20}{6} = \frac{64}{6} \approx 10.67$. Median: average of 3rd and 4th values: $M_e = \frac{8+12}{2} = 10$.

3. Total sum = $5 \times 10 = 50$. Sum of given four values = $6 + 8 + 11 + 13 = 38$. Fifth value: $50 - 38 = 12$.

4. $\bar{x} = \frac{2+4+6+8+10}{5} = 6$. Deviations from mean: $-4, -2, 0, 2, 4$. Squared deviations: $16, 4, 0, 4, 16$. Variance: $V = \frac{16+4+0+4+16}{5} = \frac{40}{5} = 8$. Standard deviation: $\sigma = \sqrt{8} = 2\sqrt{2} \approx 2.83$.

5. $V = \overline{x^2} - (\bar{x})^2 = 53 - 7^2 = 53 - 49 = 4$. $\sigma = \sqrt{4} = 2$.

6. Standard deviation is invariant under addition: new standard deviation is still $\sigma = \$250$.

7. $y = 1.05x \Longrightarrow \sigma_y = 1.05 \times \sigma = 1.05 \times 250 = \$262.50$.

8. $N = 12$. - Q_1: rank $\frac{12}{4} = 3 \Longrightarrow$ 3rd value, so $Q_1 = 4$. - Median: average of 6th and 7th values $\Longrightarrow M_e = \frac{9+10}{2} = 9.5$. - Q_3: rank $\frac{3 \times 12}{4} = 9 \Longrightarrow$ 9th value, so $Q_3 = 13$.

9. a) $N = 3+5+8+4 = 20$. $\sum n_i x_i = 3(10)+5(20)+8(30)+4(40) = 30+100+240+160 = 530$. $\bar{x} = \frac{530}{20} = 26.5$. b) $\sum n_i x_i^2 = 3(100)+5(400)+8(900)+4(1600) = 300+2000+7200+6400 = 15900$. $\overline{x^2} = \frac{15900}{20} = 795$. $V = \overline{x^2} - (\bar{x})^2 = 795 - (26.5)^2 = 795 - 702.25 = 92.75$. $\sigma = \sqrt{92.75} \approx 9.63$.

10. Mean: $\bar{x} = \frac{a+b}{2}$. Deviations: $a - \frac{a+b}{2} = \frac{a-b}{2}$, and $b - \frac{a+b}{2} = \frac{b-a}{2} = -\frac{a-b}{2}$. The mean of the squared deviations is: $V = \frac{1}{2}\left[\left(\frac{a-b}{2}\right)^2 + \left(-\frac{a-b}{2}\right)^2\right] = \left(\frac{a-b}{2}\right)^2$.

11. Pooled weighted mean: $\bar{x} = \frac{N_1\bar{x}_1 + N_2\bar{x}_2}{N_1+N_2} = \frac{15\times12+25\times10}{40} = \frac{180+250}{40} = \frac{430}{40} = 10.75$.

12. Require $\sigma_y = |a|\sigma \Leftrightarrow 3 = a \times 2.5 \Leftrightarrow a = \frac{3}{2.5} = 1.2$. Then require $\bar{y} = a\bar{x} + b \Leftrightarrow 12 = 1.2(11) + b \Leftrightarrow 12 = 13.2 + b \Leftrightarrow b = -1.2$. The transformation is $y = 1.2x - 1.2$.

13. Sample frequency: $f = \frac{228}{400} = 0.57$. Under $p = 0.50$, the 95% fluctuation interval is: $I = \left[0.50 - 1.96\sqrt{\frac{0.25}{400}}\,;\, 0.50 + 1.96\sqrt{\frac{0.25}{400}}\right] = [0.50 - 1.96 \times 0.025\,;\, 0.50 + 1.96 \times 0.025] = [0.451\,;\, 0.549]$. Since $f = 0.57 > 0.549$, the observed frequency falls above the interval: Candidate C has a statistically significant lead over an even split.

14. Both stations share the same average temperature (28°C). Station A exhibits very low standard deviation (1.5°C): maritime/oceanic stability. Station B shows high standard deviation (6.2°C): continental weather with sharp temperature swings.

15.

```
def statistics(data):
    N = len(data)
    mean_val = sum(data) / N
    variance_val = sum((x - mean_val) ** 2 for x in data) / N
    std_dev = variance_val ** 0.5
    return mean_val, std_dev

scores = [12, 15, 9, 18, 14, 11, 13]
m, s = statistics(scores)
print(f"Mean = {m:.2f}, Std Dev = {s:.2f}")
```

16. Let $S_1 = \sum_{x\in G_1} x^2$ and $S_2 = \sum_{x\in G_2} x^2$. By König-Huygens in each group: $S_1 = N_1(V_1 + \bar{x}_1^2)$ and $S_2 = N_2(V_2 + \bar{x}_2^2)$. In the pooled group: $\bar{x} = \frac{N_1\bar{x}_1 + N_2\bar{x}_2}{N}$, and $V = \frac{S_1+S_2}{N} - \bar{x}^2$. Expanding and regrouping terms yields: $V = \frac{N_1V_1+N_2V_2}{N} + \frac{N_1N_2}{N^2}(\bar{x}_1 - \bar{x}_2)^2$.

- The first term represents **within-group variance** (weighted average of group variances).
- The second term represents **between-group variance** (the separation between group means).

17. Let E be the set of observations satisfying $|x_i - \bar{x}| \geq 2\sigma$, with count N_E. For every $x_i \in E$, $(x_i - \bar{x})^2 \geq 4\sigma^2$. The total sum of squared deviations satisfies: $NV = \sum_{i=1}^{N}(x_i - \bar{x})^2 \geq \sum_{i\in E}(x_i - \bar{x})^2 \geq N_E \times 4\sigma^2$. Since $V = \sigma^2$, we have $N\sigma^2 \geq 4N_E\sigma^2$. Dividing by $4N\sigma^2$ yields $\frac{N_E}{N} \leq \frac{1}{4} = 25\%$.

18. Margin of error: $1.96\sqrt{\frac{0.5\times0.5}{n}} = 1.96 \times \frac{0.5}{\sqrt{n}} = \frac{0.98}{\sqrt{n}}$. Set $\frac{0.98}{\sqrt{n}} < 0.01 \Leftrightarrow \sqrt{n} > \frac{0.98}{0.01} = 98 \Leftrightarrow n > 98^2 = 9,604$. At least **9,605 people** must be polled to achieve a margin of error under 1%.

19. a) $\bar{x} = \frac{1}{n}\sum_{k=1}^{n} k = \frac{1}{n} \times \frac{n(n+1)}{2} = \frac{n+1}{2}$. b) $\overline{x^2} = \frac{1}{n}\frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6}$. By König-Huygens: $V = \overline{x^2} - (\bar{x})^2 = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = (n+1)\left[\frac{2(2n+1)-3(n+1)}{12}\right] = (n+1)\frac{4n+2-3n-3}{12} = \frac{(n+1)(n-1)}{12} = \frac{n^2-1}{12}$.

20.

```python
import math

def detect_outliers(data):
    s = sorted(data)
    N = len(s)
    q1 = s[math.ceil(N / 4) - 1]
    q3 = s[math.ceil(3 * N / 4) - 1]
    iqr = q3 - q1
    lower_bound = q1 - 1.5 * iqr
    upper_bound = q3 + 1.5 * iqr

    outliers = [x for x in data if x < lower_bound or x > upper_bound]
    return q1, q3, iqr, lower_bound, upper_bound, outliers

salaries = [1500, 1600, 1700, 1800, 1900, 2000, 2100, 2200, 9500]
q1, q3, iqr, low, up, out = detect_outliers(salaries)
print(f"Q1 = {q1}, Q3 = {q3}, IQR = {iqr}")
print(f"Acceptable interval : [{low:.1f}, {up:.1f}]")
print(f"Detected outliers   : {out}")
```

The Question That Opens What Follows

We now know how to describe and summarize empirical data collected from past trials. Yet how do we advance from describing observed facts to **predicting the future**? How do we determine the true probability that a patient is genuinely ill when a medical diagnostic test with 99% accuracy lights up positive? Why does our natural intuition about chance deceive us so consistently? That is the great crossroads of **conditional probability** and **Bayes' formula**, in the next chapter.

CHAPTER 42: The Physician, the Test and the Tree of Possibilities

> "Common sense in the presence of randomness is a blind traveler stumbling in the dark. When a medical test advertised as 99% reliable declares you diseased, panic shrieks that you are 99% doomed. The mathematician breathes calmly, draws a tree in the space of possibilities, narrows the sample space, and reveals that you have a nine-in-ten chance of being perfectly healthy."

What You Are About to Discover

- The medical test paradox: how a nearly infallible test can yield a vast majority of false positives for a rare disease.
- The revolutionary concept of **conditional probability**: how new information **narrows the universe of possibilities**.
- The laws of the **probability tree**: the product rule along branches and the sum of paths.
- The **law of total probability**: partitioning sample space into mutually exclusive slices to determine the unknown.
- The celebrated theorem of Reverend **Thomas Bayes** for reasoning backward from effects to causes.
- The vital distinction between **independent** events and **mutually exclusive** (disjoint) events.
- **Discrete random variables**: assigning a numeric payout or score to each random outcome.
- **Expected value** $E(X)$: long-run average gain and the fair price of a lottery ticket.
- **Variance** $V(X)$, **standard deviation** $\sigma(X)$, and the **linearity of expectation**: $E(aX + b) = aE(X) + b$.

---

1. The Problem: Thomas's Nightmare in the Doctor's Office

Thomas is seventeen years old. To compete in high school athletics, his physician administers a routine blood screening test for a rare but dangerous congenital heart condition that affects **1 in 1,000 individuals** across the general population.

Three days later, the phone rings. The doctor's tone is heavy: *"Thomas, your blood test came back positive."*

Thomas feels the floor give way beneath him. With a trembling voice, he asks: *"Doctor, how reliable is this test?"*

The physician consults the laboratory specification sheet: *"The test boasts an exceptional reliability rating of **99%**: - If a person has the condition, the test registers positive 99% of the time (sensitivity). - If a person is healthy, the test registers negative 99% of the time (specificity)."*

Thomas slumps into his chair, overwhelmed: *"The test is 99% accurate. My test came back positive. Therefore, I have a 99% chance of having this heart defect!"*

Is Thomas right to panic? The verdict of mathematics is clear, dramatic, and will upend your intuition: **NO! Thomas actually faces less than a 10% chance of having the disease!** He has more than a **90% chance of being in perfect health**!

How can this be? How can a medical diagnostic tool with 99% laboratory precision be wrong nine times out of ten when it announces a disease?

---

2. Narrowing the Universe: Conditional Probability

To see through this paradox without dense symbolism, consider a concrete, representative cohort of **100,000 individuals**:

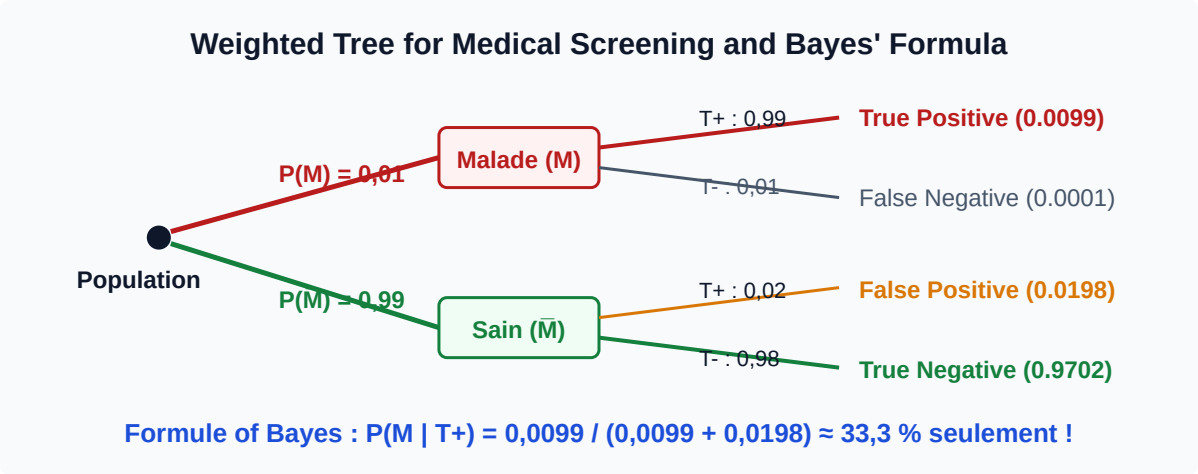

Figure 95: Weighted tree for medical screening and Bayes' formula

Trace the arithmetic carefully: 1. Among 100,000 people, because the condition strikes 1 in 1,000, there are **100 diseased individuals** and **99,900 healthy individuals**. 2. Among the 100 diseased patients: the test detects 99% of them: there are **99 positive tests** (and 1 false negative). 3. Now inspect the **99,900 healthy individuals**: The test is 99% specific, which means it misclassifies 1% of them. 1% of 99,900 yields exactly **999 healthy individuals falsely flagged as positive** (false positives)!

Now step into Thomas's shoes: The physician informed him that his test is positive. Thomas's relevant reference frame has suddenly **shrunk**: Thomas is no longer an arbitrary member of the 100,000 general public; he belongs exclusively to the select club of individuals who received a **positive test result**!

How many total positive individuals exist in the clinic's waiting room?

$$\text{Total Positives} = 99 \ \ (\text{diseased}) + 999 \ \ (\text{healthy}) = \mathbf{1,098} \ \ \mathbf{people} \ .$$

And among these 1,098 individuals testing positive, how many actually carry the disease? Only **99 individuals**!

The probability that Thomas is genuinely ill given that his test is positive is therefore:

$$P(\text{Diseased} \mid \text{Positive}) = \frac{99}{1,098} \approx \mathbf{0.0902} = \mathbf{9.02\%}.$$

More than **91% of those who test positive are completely disease-free**! Why? Because the disease is so exceedingly rare that the tiny 1% false-positive rate applied to the massive ocean of healthy people creates ten times more false alarms than genuine cases!

---

3. The Formal Theory: Conditional Probability

3.1 Formal Definition

Let Ω be a sample space endowed with a probability measure P. Let A and B be two events.

Definition — Conditional Probability

Let A be an event with non-zero probability ($P(A) > 0$). The **conditional probability of B given A**, denoted $P_A(B)$ or $P(B \mid A)$, is defined by:

$$\boxed{P(B \mid A) = \frac{P(A \cap B)}{P(A)}}$$

Read aloud as: "**probability of B given A**".

Geometric Interpretation: We replace the universal reference sample space Ω (which had total probability 1) with the restricted subspace A. The event A becomes the new universe. The fraction of B that survives inside this restricted frame is the intersection $A \cap B$, normalized by the measure of A.

3.2 The Multiplication Rule

Multiplying by $P(A)$ yields the fundamental product identity:

Theorem — Multiplication Rule

For any events A and B with $P(A) > 0$:

$$\boxed{P(A \cap B) = P(A) \times P(B \mid A)}$$

---

4. Probability Trees and the Law of Total Probability

4.1 The Golden Rules of Tree Diagrams

A probability tree organizes successive sequential stages of a random experiment with clarity.

Rules of Probability Trees

1. **Node Rule:** The sum of
probabilities on all branches emanating from any single node **always equals 1**.

$$P(A) + P(\overline{A}) = 1, \qquad P(B \mid A) + P(\overline{B} \mid A) = 1.$$

2. **Path Rule (Multiplication):** The probability of any complete path from root to leaf is the **product** of the probabilities along its branches. 3. **Leaf Rule (Addition):** The total probability of an outcome is the **sum** of the path probabilities that terminate at that outcome.

4.2 The Law of Total Probability

Definition — Partition of a Sample Space

A family of events $A_1, A_2, ..., A_n$ forms a **partition** of Ω if:

- They are pairwise disjoint ($A_i \cap A_j = \varnothing$ for $i \neq j$);
- Their union covers the entire sample space ($A_1 \cup A_2 \cup ... \cup A_n = \Omega$);
- None of them is impossible ($P(A_i) > 0$).

The most frequent partition is the elementary binary partition $\left\{A, \overline{A}\right\}$.

Theorem — Law of Total Probability

If $A_1, A_2, ..., A_n$ form a partition of Ω, then for any event B:

$$\boxed{P(B) = \sum_{i=1}^{n} P(B \cap A_i) = \sum_{i=1}^{n} P(A_i) \times P(B \mid A_i)}$$

In the two-event case $\left\{A, \overline{A}\right\}$:

$$\boxed{P(B) = P(A) \times P(B \mid A) + P(\overline{A}) \times P(B \mid \overline{A})}$$

---

5. Bayes' Theorem: Reasoning Backward from Effects to Causes

In 1763, the British Presbyterian minister and mathematician **Thomas Bayes** published a posthumous essay that reshaped the philosophy of knowledge. He asked: *"Given that an observed effect has occurred (the test is positive), what is the probability that a specific underlying cause generated it (the patient is truly diseased)?"*

📐 Theorem — Bayes' Theorem

Let $\left\{A, \overline{A}\right\}$ partition the sample space and let B have non-zero probability.

$$\boxed{P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \times P(B \mid A)}{P(A) \times P(B \mid A) + P(\overline{A}) \times P(B \mid \overline{A})}}$$

Applying Bayes' formula to Thomas's scenario:

- Underlying cause M: "has the condition", $P(M) = 0.001 \Longrightarrow P(\overline{M}) = 0.999$.
- Observed effect T: "positive test result".
  - Sensitivity: $P(T \mid M) = 0.99$.
  - False positive rate: $P(T \mid \overline{M}) = 1 - 0.99 = 0.01$.
- Total probability of testing positive:

$$P(T) = P(M)P(T \mid M) + P(\overline{M})P(T \mid \overline{M}) = (0.001 \times 0.99) + (0.999 \times 0.01)$$
$$= 0.00099 + 0.00999 = 0.01098.$$

- Posterior probability of disease given a positive test:

$$P(M \mid T) = \frac{P(M \cap T)}{P(T)} = \frac{0.00099}{0.01098} = \frac{99}{1098} \approx \mathbf{9.02}\%.$$

The algebraic formula matches our tabular calculation point for point!

---

6. Independence: Randomness Without Memory

When two events exert no causal or probabilistic influence on one another, learning that one has occurred provides zero information about the likelihood of the other.

📐 Definition — Independent Events

Two events A and B are **independent** if and only if:

$$\boxed{P(A \cap B) = P(A) \times P(B)}$$

If $P(A) > 0$, this is equivalent to:

$$P(B \mid A) = P(B).$$

⚠️ Caution: The Ultimate Trap! — Never Conflate "Independent" and "Disjoint"

This is the single most common conceptual error in probability!

- Two events are **disjoint** (mutually exclusive) if they cannot happen at the same time: $A \cap B = \varnothing$, so $P(A \cap B) = 0$.
- Two events are **independent** if the occurrence of one does not alter the likelihood of the other: $P(A \cap B) = P(A) \times P(B)$.

Two non-impossible events that are disjoint are **NEVER independent**! On the contrary, they are maximally dependent: if A occurs, you know with 100% certainty that B did not occur ($P(B \mid A) = 0 \neq P(B)$)!

7. Discrete Random Variables

Until now, random outcomes were qualitative categories: “heads”, “tails”, “sick”, “red card”. A **random variable** bridges probability to arithmetic: it attaches a **real number** (a monetary gain, an arrival count, a waiting time) to every elementary outcome.

Definition — Discrete Random Variable

A **discrete random variable** X defined on a sample space Ω is a function that maps every elementary outcome $\omega \in \Omega$ to a real number $x_i \in \mathbb{R}$. We write $X(\Omega) = \{x_1, x_2, ..., x_k\}$ for the set of values X can take.

7.1 Probability Distribution

Specifying the **probability distribution** of X means determining the probability $P(X = x_i)$ for each possible value x_i, presented as a distribution table:

x_i	x_1	x_2	...	x_k	**Total**
$P(X = x_i)$	p_1	p_2	...	p_k	**1**

The sum of all probabilities must strictly equal 1: $\sum_{i=1}^{k} p_i = 1$.

7.2 Expected Value: The Fair Price of a Game

Imagine a carnival lottery: what can you expect to win on average per ticket if you play millions of games?

Definition — Expected Value

The **expected value** (or expectation) of a discrete random variable X, denoted $E(X)$, is the weighted average of the values taken by X:

$$\boxed{E(X) = \sum_{i=1}^{k} x_i \cdot P(X = x_i)}$$

- If X represents a player’s net profit (payout minus ticket cost):
  - $E(X) > 0$: the game is **in the player’s favor**.
  - $E(X) < 0$: the game is **unfavorable to the player** (the mathematical reality of all commercial casinos and state lotteries!).
  - $E(X) = 0$: the game is **fair**.

7.3 Variance and Standard Deviation of a Random Variable

Just as in Chapter 41 for descriptive statistics, we measure uncertainty and dispersion around the mean:

Definition — Variance and Standard Deviation

- The **variance** of X is:

$$\boxed{V(X) = E\left((X - E(X))^2\right) = \sum_{i=1}^{k} (x_i - E(X))^2 P(X = x_i)}$$

- Probabilistic König-Huygens identity:

$$\boxed{V(X) = E(X^2) - \left(E(X)\right)^2}$$

- The **standard deviation** is:

$$\boxed{\sigma(X) = \sqrt{V(X)}}$$

7.4 Linearity of Expectation

Theorem — Affine Transformation of a Random Variable

For all real numbers a and b:

1. $\boxed{E(aX+b) = aE(X)+b}$ (Linearity of Expectation)
2. $\boxed{V(aX+b) = a^2V(X)}$
3. $\boxed{\sigma(aX+b) = |a|\sigma(X)}$

Proof of Linearity

$$E(aX+b) = \sum_{i=1}^{k}(ax_i+b)p_i = \sum_{i=1}^{k}ax_ip_i + \sum_{i=1}^{k}bp_i = a\sum_{i=1}^{k}x_ip_i + b\sum_{i=1}^{k}p_i.$$

Since $\sum x_ip_i = E(X)$ and $\sum p_i = 1$:

$$E(aX+b) = aE(X) + b(1) = aE(X) + b. \quad \blacksquare$$

Step-by-Step Worked Examples

Worked Example 1 — Direct Conditional Probability

In a class of 32 students, 18 study Spanish, 10 study German, and 6 study both languages. A student is chosen at random. 1. What is the probability that the student studies German? 2. Given that the chosen student studies Spanish, what is the probability that they also study German?

Detailed Solution: Let S denote "studies Spanish" and G denote "studies German". $P(S) = \frac{18}{32}$, $P(G) = \frac{10}{32}$, and $P(S \cap G) = \frac{6}{32}$. 1. $P(G) = \frac{10}{32} = \frac{5}{16} = 0.3125$. 2. We seek $P(G \mid S)$ (sample space restricted to Spanish students):

$$P(G \mid S) = \frac{P(S \cap G)}{P(S)} = \frac{6/32}{18/32} = \frac{6}{18} = \frac{1}{3} \approx 0.333.$$

Worked Example 2 — Tree Diagram and Law of Total Probability

A company sources microchips from two suppliers:

- Supplier A provides 70% of the inventory, with a 2% defect rate.
- Supplier B provides the remaining 30%, with a 5% defect rate. A chip is chosen at random from inventory.

1. Construct the tree diagram.
2. Find the probability that the chip is defective.
3. If a selected chip is found to be defective, what is the probability it came from Supplier B?

Detailed Solution: 1. Data: $P(A) = 0.70$, $P(B) = 0.30$. $P(D \mid A) = 0.02$, $P(D \mid B) = 0.05$. 2. By the law of total probability with partition $\{A, B\}$:

$$P(D) = P(A \cap D) + P(B \cap D) = P(A)P(D \mid A) + P(B)P(D \mid B)$$

$$P(D) = (0.70 \times 0.02) + (0.30 \times 0.05) = 0.014 + 0.015 = 0.029 = 2.9\%.$$

3. By Bayes' theorem, find $P(B \mid D)$:

$$P(B \mid D) = \frac{P(B \cap D)}{P(D)} = \frac{0.015}{0.029} = \frac{15}{29} \approx 0.5172 = 51.7\%.$$

Even though Supplier B delivers only 30% of total volume, it accounts for more than half of all defective parts!

Worked Example 3 — Testing for Independence

Two standard fair 6-sided dice are rolled, one red and one blue. Let A be: "the red die shows 4" and B be: "the sum of the two dice is 7". Are events A and B independent?

Detailed Solution: There are $6 \times 6 = 36$ equally likely outcomes.

- For A: 6 outcomes $(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)$, so $P(A) = \frac{6}{36} = \frac{1}{6}$.
- For B: sums of 7 are $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$, giving 6 outcomes. Thus $P(B) = \frac{6}{36} = \frac{1}{6}$.
- For $A \cap B$: only one outcome has red die 4 and sum 7: namely $(4,3)$. Thus $P(A \cap B) = \frac{1}{36}$.
- Compare with the product: $P(A) \times P(B) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.
- Since $P(A \cap B) = P(A) \times P(B)$, **events A and B are strictly independent**! Knowing that the red die rolled a 4 provides zero information regarding whether the sum will equal 7.

Worked Example 4 — Random Variable and Game Expectation

A carnival stall offers the following game for an entry stake of \$5: A fair 6-sided die is rolled:

- Roll a 6: receive \$20 (net profit: +\$15).
- Roll a 4 or 5: receive \$6 (net profit: +\$1).
- Roll a 1, 2, or 3: receive \$0 (net loss: -\$5).

1. Define the random variable X representing the player's net gain and specify its probability distribution.
2. Compute the expected value. Who has the mathematical edge?

Detailed Solution: 1. Values taken by X: $X(\Omega) = \{-5, 1, 15\}$.

- $P(X = 15) = P(\{6\}) = \frac{1}{6}$
- $P(X = 1) = P(\{4,5\}) = \frac{2}{6} = \frac{1}{3}$
- $P(X = -5) = P(\{1,2,3\}) = \frac{3}{6} = \frac{1}{2}$ Check: $\frac{1}{6} + \frac{2}{6} + \frac{3}{6} = 1$.

2. Expected value:

$$E(X) = \left(-5 \times \frac{3}{6}\right) + \left(1 \times \frac{2}{6}\right) + \left(15 \times \frac{1}{6}\right) = \frac{-15 + 2 + 15}{6} = \frac{2}{6} = +\frac{1}{3} \approx +\$0.33.$$

Since $E(X) > 0$, the game is **favorable to the player**! On average, players win 33 cents per round. The stall operator will soon face bankruptcy unless the rules are revised!

Worked Example 5 — Variance and Standard Deviation of a Random Variable

Compute the variance $V(X)$ and standard deviation $\sigma(X)$ for the game in Example 4.

Detailed Solution: 1. Compute $E(X^2)$:

$$E(X^2) = (-5)^2 \times \frac{3}{6} + (1)^2 \times \frac{2}{6} + (15)^2 \times \frac{1}{6}$$

$$= 25 \times \frac{3}{6} + 1 \times \frac{2}{6} + 225 \times \frac{1}{6} = \frac{75 + 2 + 225}{6} = \frac{302}{6} = \frac{151}{3} \approx 50.33.$$

2. König-Huygens identity:

$$V(X) = E(X^2) - (E(X))^2 = \frac{151}{3} - \left(\frac{1}{3}\right)^2 = \frac{453}{9} - \frac{1}{9} = \frac{452}{9} \approx 50.22.$$

3. Standard deviation:

$$\sigma(X) = \sqrt{\frac{452}{9}} = \frac{\sqrt{452}}{3} \approx \$7.09.$$

The standard deviation is high: earnings fluctuate substantially around the mean.

Worked Example 6 — Linearity of Expectation

An insurance provider insures a citywide fleet of shared bicycles. The daily number of bicycle accidents X has expected value $E(X) = 4$ and standard deviation $\sigma(X) = 1.5$. Each accident costs \$50 in repairs, plus a fixed daily maintenance overhead of \$100. The total daily cost is therefore $Y = 50X + 100$. Determine the expected value and standard deviation of the daily cost.

Detailed Solution:

- Expectation: by linearity:

$$E(Y) = E(50X + 100) = 50E(X) + 100 = 50(4) + 100 = 200 + 100 = \$300.$$

- Standard deviation:

$$\sigma(Y) = \sigma(50X + 100) = |50| \times \sigma(X) = 50 \times 1.5 = \$75.$$

The insurer must budget an average of \$300 per day, with typical day-to-day fluctuations of \$75.

What You Must Remember

1. **Conditional Probability:**
   - $P(B \mid A) = \frac{P(A \cap B)}{P(A)}$ (sample space restricted to A).
   - Product rule: $P(A \cap B) = P(A) \times P(B \mid A)$.
2. **Probability Trees and Law of Total Probability:**
   - Sum of branches leaving any node equals 1.
   - Path probability = product along branches.
   - Law of total probability: $P(B) = P(A)P(B \mid A) + P(\overline{A})P(B \mid \overline{A})$.
3. **Bayes' Theorem (Inverting Conditions):**
   - $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B|A)}{P(A)P(B|A)+P(\overline{A})P(B|\overline{A})}$.
4. **Independence:**
   - A and B are independent $\Leftrightarrow P(A \cap B) = P(A) \times P(B) \Leftrightarrow P(B \mid A) = P(B)$.
   - Mutually exclusive (disjoint) non-null events are never independent.
5. **Random Variables:**
   - Expected value: $E(X) = \sum x_i p_i$. Linearity: $E(aX + b) = aE(X) + b$.
   - Variance: $V(X) = E(X^2) - (E(X))^2$. Scaling: $V(aX + b) = a^2 V(X)$.
   - Standard deviation: $\sigma(X) = \sqrt{V(X)}$. Scaling: $\sigma(aX + b) = |a|\sigma(X)$.

Exercises

Level (Direct Applications, Basic Trees)

1. Given $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. Compute $P(B \mid A)$ and $P(A \mid B)$. Are A and B independent?
2. Given $P(A) = 0.6$ and $P(B \mid A) = 0.3$. Compute $P(A \cap B)$.
3. In a tree diagram, a branch leading to A has probability 0.7. From A, two branches lead to B and $\overline{B}$. Given $P(B \mid A) = 0.4$, determine $P(\overline{B} \mid A)$.
4. A card is drawn at random from a standard 32-card deck. Let K be "the card is a King" and H be "the card is a Heart". Are K and H independent?
5. A random variable X has distribution: | x_i | −2 | 0 | 3 | |—|—|—|—| | $P(X = x_i)$ | 0.3 | 0.5 | p | Determine p, then calculate $E(X)$.
6. Given $E(X) = 5$ and $V(X) = 4$. Let $Y = 3X - 2$. Find $E(Y)$, $V(Y)$, and $\sigma(Y)$.

7. A bag holds 3 white balls and 2 black balls. Two balls are drawn successively without replacement. Find the probability of drawing two white balls.
8. A multiple-choice question offers 4 options with only one correct answer. A student guesses randomly. What is the probability of answering correctly given that they have already eliminated two incorrect choices?

Level (Complete Trees, Bayes, and Random Variables)

9. In a textile factory, three machines M_1, M_2, M_3 produce 50%, 30%, and 20% of the garments, respectively. Defect rates are 3% for M_1, 4% for M_2, and 5% for M_3.
   a) Calculate the probability that a randomly chosen garment is defective.
   b) Given that a garment is defective, what is the probability it was produced by machine M_1?
10. Prove that if A and B are independent events, then A and $\overline{B}$ are also independent.
11. A fair coin is tossed 3 times. Let X denote the number of Heads obtained.
    a) Determine the support of X and tabulate its probability distribution.
    b) Compute $E(X)$ and $V(X)$.
12. A carnival game costs \$2 to play. A fair 6-sided die is rolled:
    - Even outcome: win \$3 (net profit: +\$1).
    - Outcome 5: win \$6 (net profit: +\$4).
    - Otherwise: win nothing (net profit: -\$2). Is the game fair?
13. Two dice are rolled. Let A be: "the first die is even" and B be: "the sum of the two dice is odd". Show that A and B are independent.
14. In a population, 60% are vaccinated against the flu. The vaccine has an 80% efficacy rate (vaccinated individuals are five times less likely to contract the flu). The infection rate among the unvaccinated is 25%.
    a) Compute the overall infection rate in the general population.
    b) Given that an individual contracts the flu, what is the probability that they were vaccinated?
15. A random variable X satisfies $E(X) = 10$ and $E(X^2) = 116$. Compute $V(X)$ and $\sigma(X)$.
16. Python Simulation: Write a script to simulate 10,000 rounds of the game in Exercise 12, computing the empirical mean payoff to corroborate the theoretical expected value.

Level (Challenges and Open Problems)

17. **The Two-Child Paradox (Lewis Carroll / Martin Gardner):** A family has two children. Assume each child is independently equally likely to be a boy or a girl.
    a) Given that the elder child is a girl, what is the probability that both children are girls?
    b) Given that at least one of the children is a girl, what is the probability that both children are girls? Explain the apparent paradox.
18. **The Monty Hall Problem (The Three Doors Game Show):** On a game show, three closed doors A, B, C stand before you. Behind one is a sports car; behind the other two, goats.
    17. You choose Door A.
    18. The host, who knows what is behind each door, opens one of the remaining doors (say Door B), revealing a goat.
    19. He offers: *"Do you wish to stick with Door A, or switch to Door C?"* Using conditional probabilities, rigorously demonstrate why switching doors doubles your chance of winning!
19. **Gambler's Ruin:** A player begins with \$2 and plays coin tosses against the house using a fair coin. Each round, he wins \$1 with probability 0.5 or loses \$1 with probability 0.5. The game halts if he reaches \$4 (victory) or \$0 (ruin). Find the probability of ruin by formulating a system of recurrence equations on conditional probabilities.
20. **Truncated Geometric Distribution:** A fair 6-sided die is rolled until a 6 appears, with a maximum cap of 4 rolls. Let X be the number of rolls performed.
    a) Give the probability distribution of X.
    b) Compute the expected value $E(X)$.

Solutions

1. $P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.2}{0.4} = 0.5$. $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.5} = 0.4$. $P(A) \times P(B) = 0.4 \times 0.5 = 0.2 = P(A \cap B)$. Yes, the events are independent!

2. $P(A \cap B) = P(A) \times P(B \mid A) = 0.6 \times 0.3 = 0.18$.

3. Branches leaving a node sum to 1: $P(\overline{B} \mid A) = 1 - P(B \mid A) = 1 - 0.4 = 0.6$.

4. $P(K) = \frac{4}{32} = \frac{1}{8}$. $P(H) = \frac{8}{32} = \frac{1}{4}$. $K \cap H$ is the King of Hearts: $P(K \cap H) = \frac{1}{32}$. Since $P(K) \times P(H) = \frac{1}{8} \times \frac{1}{4} = \frac{1}{32} = P(K \cap H)$, the events are independent.

5. Sum of probabilities = 1: $0.3 + 0.5 + p = 1 \implies p = 0.2$. $E(X) = (-2 \times 0.3) + (0 \times 0.5) + (3 \times 0.2) = -0.6 + 0 + 0.6 = 0$.

6. $E(Y) = 3E(X) - 2 = 3(5) - 2 = 13$. $V(Y) = 3^2 V(X) = 9 \times 4 = 36$. $\sigma(Y) = \sqrt{36} = 6$ (or $|3| \times \sqrt{4} = 3 \times 2 = 6$).

7. $P(W_1) = \frac{3}{5}$. After drawing one white ball, 2 white balls remain among 4 total: $P(W_2 \mid W_1) = \frac{2}{4} = \frac{1}{2}$. $P(W_1 \cap W_2) = \frac{3}{5} \times \frac{1}{2} = \frac{3}{10} = 0.30$.

8. The sample space is reduced to the two remaining choices, exactly one of which is correct: $P = \frac{1}{2} = 50\%$.

9. a) Total probability: $P(D) = P(M_1)P(D \mid M_1) + P(M_2)P(D \mid M_2) + P(M_3)P(D \mid M_3)$ $P(D) = (0.50 \times 0.03) + (0.30 \times 0.04) + (0.20 \times 0.05) = 0.015 + 0.012 + 0.010 = 0.037 = 3.7\%$. b) Bayes: $P(M_1 \mid D) = \frac{P(M_1 \cap D)}{P(D)} = \frac{0.015}{0.037} = \frac{15}{37} \approx 0.4054 = 40.54\%$.

10. By the law of total probability: $P(A) = P(A \cap B) + P(A \cap \overline{B})$. Thus $P(A \cap \overline{B}) = P(A) - P(A \cap B)$. Since A and B are independent, $P(A \cap B) = P(A)P(B)$. Substitute: $P(A \cap \overline{B}) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(\overline{B})$. This proves independence of A and $\overline{B}$.

11. a) $X \in \{0, 1, 2, 3\}$. There are $2^3 = 8$ equally likely outcomes.

- $P(X = 0) = P(\{TTT\}) = \frac{1}{8}$
- $P(X = 1) = P(\{HTT, THT, TTH\}) = \frac{3}{8}$
- $P(X = 2) = P(\{HHT, HTH, THH\}) = \frac{3}{8}$
- $P(X = 3) = P(\{HHH\}) = \frac{1}{8}$

b) $E(X) = 0(\frac{1}{8}) + 1(\frac{3}{8}) + 2(\frac{3}{8}) + 3(\frac{1}{8}) = \frac{0+3+6+3}{8} = \frac{12}{8} = 1.5$. $E(X^2) = 0^2(\frac{1}{8}) + 1^2(\frac{3}{8}) + 2^2(\frac{3}{8}) + 3^2(\frac{1}{8}) = \frac{0+3+12+9}{8} = \frac{24}{8} = 3$. $V(X) = E(X^2) - (E(X))^2 = 3 - (1.5)^2 = 3 - 2.25 = 0.75 = \frac{3}{4}$.

12. Let G be net gain: $G \in \{+1, +4, -2\}$. - Even outcomes $\{2, 4, 6\} \implies P(G = 1) = \frac{3}{6} = \frac{1}{2}$. - Outcome 5 $\{5\} \implies P(G = 4) = \frac{1}{6}$. - Other outcomes $\{1, 3\} \implies P(G = -2) = \frac{2}{6} = \frac{1}{3}$.

Expectation: $E(G) = \left(1 \times \frac{3}{6}\right) + \left(4 \times \frac{1}{6}\right) + \left(-2 \times \frac{2}{6}\right) = \frac{3+4-4}{6} = \frac{3}{6} = +\$0.50$. The game is not fair ($E(G) \neq 0$): it favors the player, who averages 50 cents profit per game.

13. $P(A) = \frac{3}{6} = \frac{1}{2}$. An odd sum requires one even and one odd die. There are 18 such outcomes out of 36, so $P(B) = \frac{18}{36} = \frac{1}{2}$. $A \cap B$: first die is even AND sum is odd $\Leftrightarrow$ first die even AND second die odd. There are $3 \times 3 = 9$ outcomes out of 36. Thus $P(A \cap B) = \frac{9}{36} = \frac{1}{4}$. Since $P(A) \times P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = P(A \cap B)$, the events are independent.

14. a) $P(V) = 0.60$ and $P(\overline{V}) = 0.40$. Rate in unvaccinated: $P(I \mid \overline{V}) = 0.25$. Rate in vaccinated: 5 times lower $\implies P(I \mid V) = \frac{0.25}{5} = 0.05$. $P(I) = P(V)P(I \mid V) + P(\overline{V})P(I \mid \overline{V}) = (0.60 \times 0.05) + (0.40 \times 0.25) = 0.03 + 0.10 = 0.13 = 13\%$. b) Bayes: $P(V \mid I) = \frac{P(V \cap I)}{P(I)} = \frac{0.03}{0.13} = \frac{3}{13} \approx 23.08\%$. (Nearly 77% of infected individuals are unvaccinated).

15. $V(X) = E(X^2) - (E(X))^2 = 116 - 10^2 = 116 - 100 = 16$. $\sigma(X) = \sqrt{16} = 4$.

16.

```python
import random

def simulate_game(num_plays=10000):
    payouts = []
    for _ in range(num_plays):
        roll = random.randint(1, 6)
        if roll == 6:
            gain = 4  # +5 € won - 1 € stake
        else:
            gain = -1  # 0 € won - 1 € stake
        payouts.append(gain)

    empirical_mean = sum(payouts) / num_plays
    return empirical_mean

# Let us run the simulation
emp_mean = simulate_game()
print(f"Empirical average net gain per play: {emp_mean:.3f} €")
# Result will be very close to -0.167 € (-1/6 €)
```

17. The sample space of two-child families is $\Omega = \{(B,B),(B,G),(G,B),(G,G)\}$, with 4 equally likely outcomes (first child, second child). a) The condition "elder child is a girl" gives $A = \{(G,B),(G,G)\}$. In this 2-element space, the probability that both are girls is $\frac{1}{2} = 50\%$. b) The condition "at least one child is a girl" gives $B = \{(B,G),(G,B),(G,G)\}$. This reduced space has **3 elements**. Only one corresponds to two girls: (G,G). The probability is thus $\frac{1}{3} \approx 33.3\%$! The paradox arises because "at least one girl" provides strictly less information than identifying which specific child is a girl.

18. - When you pick Door A initially, $P(\text{Car behind } A) = \frac{1}{3}$. The probability the car is behind $\{B,C\}$ is $\frac{2}{3}$.

- The host deliberately eliminates a goat among $\{B,C\}$. This is not a random action: he acts with full knowledge of the car's location.
- The entire $\frac{2}{3}$ probability mass originally residing on the block $\{B,C\}$ now concentrates **entirely onto the sole remaining door C**!
- If you keep Door A, your winning probability remains $\frac{1}{3}$.
- If you switch to Door C, your winning probability rises to $\frac{2}{3}$.

Switching doors strictly doubles your likelihood of winning!

19. Let p_k denote the probability of ruin starting with fortune k dollars ($k \in \{0,1,2,3,4\}$). Boundary conditions: $p_0 = 1$ (already ruined) and $p_4 = 0$ (target reached). For $k \in \{1,2,3\}$, after one toss: $p_k = 0.5p_{k+1} + 0.5p_{k-1} \Leftrightarrow 2p_k = p_{k+1} + p_{k-1} \Leftrightarrow p_{k+1} - p_k = p_k - p_{k-1}$. The sequence (p_k) is arithmetic! Since $p_0 = 1$ and $p_4 = 0$, the common difference is $r = \frac{0-1}{4} = -0.25$. Thus $p_1 = 0.75$, $p_2 = 0.50$, $p_3 = 0.25$. Starting with \$2, the probability of ruin is exactly $p_2 = 0.50 = 50\%$.

20. a) Success (rolling 6) has probability $p = \frac{1}{6}$, failure $q = \frac{5}{6}$.

- $P(X=1) = \frac{1}{6}$.
- $P(X=2) = \frac{5}{6} \times \frac{1}{6} = \frac{5}{36}$.
- $P(X=3) = \left(\frac{5}{6}\right)^2 \times \frac{1}{6} = \frac{25}{216}$.
- $P(X=4)$: stop on the 4th roll whether a 6 is rolled or not, which corresponds to failing the first 3 rolls: $P(X=4) = \left(\frac{5}{6}\right)^3 = \frac{125}{216}$.

Check: $\frac{36+30+25+125}{216} = \frac{216}{216} = 1$. b) $E(X) = 1\left(\frac{36}{216}\right) + 2\left(\frac{30}{216}\right) + 3\left(\frac{25}{216}\right) + 4\left(\frac{125}{216}\right) = \frac{36+60+75+500}{216} = \frac{671}{216} \approx 3.106$ rolls.

The Question That Opens What Follows

We have reached the culmination of Volume V. From towing ropes and vectors to the medical clinic, we have mastered directional forces, instantaneous rates of change via derivatives, explosive exponential growth, and probabilistic

reasoning. Yet behold the road leading up to the final summit: what happens when a random experiment is repeated not 4 times, but indefinitely? How do we prove mathematical truths across all infinite integers simultaneously via **mathematical induction**? How do we calculate the exact area under a curved surface by stacking infinitely many infinitesimal slices? That is the crowning domain of **Volume VI: The Age of Flow and Velocity**, awaiting you in the next chapter!

VOLUME VI — TAMED INFINITY AND THE MODERN EDIFICE

Ages 17–18 / Grade 12 & Advanced Mathematics

Advanced curriculum: induction, limits, continuity, differential calculus, logarithms, Riemann integrals, differential equations, complex numbers, matrices, graphs, and continuous distributions.

Chapters of Volume VI :

Chapter 43 : The Endless Staircase and the Dominoes
Chapter 44 : The Runner Who Never Touches the Line
Chapter 45 : The Horizon of Curves
Chapter 46 : The Unbroken Thread
Chapter 47 : The Growing Garden and the Gear Chain
Chapter 48 : Napier's Logarithm and the Navigator's Tables
Chapter 49 : The Swing and Oscillating Functions
Chapter 50 : The Barrel Filled with Slices of Water
Chapter 51 : The Cooling Cup of Tea
Chapter 52 : The Cathedral in Space
Chapter 53 : The Combination Lock and Ways of Choosing
Chapter 54 : The Thousand Coin Tosses
Chapter 55 : The Right-Angled Whirlwind
Chapter 56 : Secret Messages and the Clock of Remainders
Chapter 57 : The Jacquard Loom Revisited: Matrices and Graphs
Chapter 58 : The End of the Voyage

Facing: Turing reaction-diffusion and cellular automata of the Textile Cone (Conus textile).

CHAPTER 43: The Endless Staircase and the Dominoes

> "To climb an infinite staircase, it is not necessary to see the top step. It suffices to be certain of two things: that one can set foot upon the first, and that from any step one always knows how to climb to the next."

What You Are About to Discover

- Why **verifying** a property on 10, 100, or even a billion examples **proves** nothing, and how seemingly "obvious" conjectures have collapsed.
- **Proof by mathematical induction**: a three-stage logical engine (**initialization / base step**, **inductive step / heredity**, **conclusion**) that allows demonstrating an infinity of statements in a finite number of lines.
- How to **write** an impeccable induction proof, exactly as expected on the Baccalauréat and advanced examinations.
- The celebrated **Bernoulli's inequality** $(1+a)^n \geq 1+na$, the first required core proof of the year, which will prove essential starting in the very next chapter.
- Common **pitfalls**: induction without a base step, heredity that breaks at a single link (the "all horses are the same color" paradox), induction that "won't go through," and how to strengthen the induction hypothesis.

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1. The Problem: The Trap of Examples

1.1 Odd Numbers and Squares

Take a handful of pebbles and arrange them in nested squares, just as the Pythagoreans did in the sand:

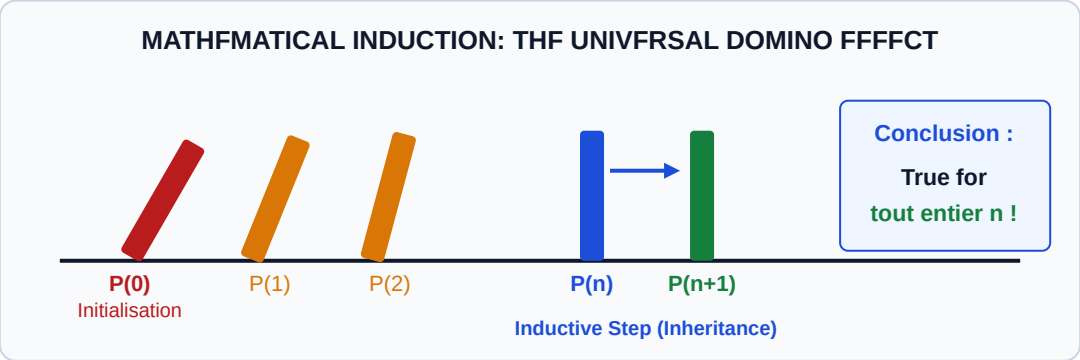

Figure 96: Mathematical induction: the mathematical domino effect

Each new L-shaped angle of pebbles (known as a **gnomon**) adds the next odd number, and the square grows by one row and one column. We conjecture:

$$1 + 3 + 5 + \ldots + (2n-1) = n^2.$$

We verify for $n = 1, 2, 3, 4, 5\ldots$ it always works. But **is it true for every n**? For $n = 10^{100}$? No human will ever carry out the full tally. The drawing convinces our eyes, but a visual diagram only demonstrates a handful of cases.

1.2 Euler's Prime-Generating Polynomial

In 1772, Leonhard Euler noted the polynomial $p(n) = n^2 + n + 41$. Let us compute:

n	0	1	2	3	4	5	10	20	39
$p(n)$	41	43	47	53	61	71	151	461	1601

Every single one of these numbers is **prime**! We can check all 40 values from $n = 0$ to $n = 39$: not a single exception. Forty successes in a row… A hasty thinker would conclude: "$n^2 + n + 41$ is always prime." And yet:

$$p(40) = 40^2 + 40 + 41 = 1600 + 81 = 1681 = 41 \times 41.$$

It fails. And $p(41) = 41^2 + 41 + 41 = 41 \times 43$ is not prime either.

1.3 Circles and Regions

Place n points on the boundary of a circle, connect every pair of points with a straight chord (choosing points in general position so that no three chords intersect at the same interior point), and count the number of disjoint regions carved out inside the disk:

Number of points n	1	2	3	4	5	6
Number of regions	1	2	4	8	16	**31**

Almost everyone wagers on 32 for $n = 6$: the progression of powers of 2 seems inescapable $(1, 2, 4, 8, 16...)$. Yet drawing it with scrupulous precision reveals only **31** regions! The true formula is $\binom{n}{4} + \binom{n}{2} + 1$, which has nothing to do with 2^{n-1} for $n \geq 6$.

1.4 The Crisis

These three examples reveal a profound epistemological problem. In mathematics, we seek statements that hold **for all natural numbers n**: an infinite collection of cases. However:

- Checking cases, even by millions, proves nothing (Euler, circle regions);
- A convincing diagram remains an illustration of a few particular cases (the gnomons).

We require an intellectual tool that **captures the entirety of the infinite sequence of integers in a single stroke**. That tool is mathematical induction.

> 💡 **Did You Know?**
>
> Some conjectures hold out far longer. The Pólya conjecture (1919), which asserted that "most" integers up to any bound have an odd number of prime factors, was verified up to hundreds of millions before a counterexample was found in 1958; the smallest counterexample is 906 150 257. No computer search could ever have "proven" this conjecture by finite verification—and fortunately so, since it was false.

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2. The Core Idea: Dominoes and the Staircase

2.1 The Dynamic Gesture

Imagine an endless row of dominoes stood upright on end. What do you need to be absolutely certain that they will **all** fall?

1. **That the first one falls**: you knock it over with your fingertip.
2. **That every domino, if it falls, causes the next one to fall**: they are spaced closely enough together.

Note carefully: condition 2 does **not** assert that the dominoes are falling. It merely says: "*if* this one falls, *then* the next one will fall as well." It is an **implication**, a guarantee of transmission.

If either condition is missing, the entire guarantee collapses:

- Without the initial push, dominoes may stand at perfect intervals forever; not one moves.
- If a single gap is too wide (say, between the 40th and the 41st), the wave of falling tiles grinds to an abrupt halt.

The endless staircase presents the exact same intuition: to climb **every** step of an infinite stairway, it suffices to be capable of stepping onto the very first step, and to know how to step from any arbitrary step up to the next.

2.2 Mathematical Notation

Before formally stating the principle, let us sharpen our vocabulary.

- $\mathbb{N}$ (read "the natural numbers") $= \{0, 1, 2, 3, ...\}$. The blackboard bold font was popularized in the 20th century by the Nicolas Bourbaki collective; the letter N stands for *natural*.
- We denote by $\mathcal{P}(n)$ (read "P of n") a **predicate or statement** that depends on the integer n: it is a mathematical **assertion** that is either true or false depending on the value of n. For example, $\mathcal{P}(n)$: "$1 + 3 + ... + (2n - 1) = n^2$".

Warning: $\mathcal{P}(n)$ is **not a number**; it is a logical proposition. You do not add it or multiply it: you declare it to be true or false.

- The symbol $\Rightarrow$ (read "implies"): $A \Rightarrow B$ means "if A is true, then B is necessarily true."
- n_0 denotes the **starting rank (base index)**, typically 0 or 1, but occasionally larger.
- The universal quantifier $\forall$ (read "for all" or "for every") is an inverted letter A (for the German *alle* or English *all*); it was introduced by Gerhard Gentzen in 1935.

2.3 The Principle of Mathematical Induction

📐 Definition — Principle of Mathematical Induction.

Let $\mathcal{P}(n)$ be a mathematical proposition depending on a natural number n, and let n_0 be a fixed integer. If: 1. **Initialization (Base Step)**: $\mathcal{P}(n_0)$ is true; 2. **Heredity (Inductive Step)**: for every integer $n \geq n_0$, if $\mathcal{P}(n)$ is true, then $\mathcal{P}(n+1)$ is true;

then **(Conclusion)** $\mathcal{P}(n)$ is true for all integers $n \geq n_0$.

The assumption "$\mathcal{P}(n)$ is true", invoked during the inductive step, is termed the **induction hypothesis** (frequently abbreviated as **IH**).

Why is this principle true? It can be viewed as a foundational axiom of the natural numbers: every natural number is generated by starting from 0 and repeatedly adding 1. If a property is inherited from each integer to the next, and is present at the starting point, no integer can escape its grasp.

Alternatively, one may reason by contradiction via the **well-ordering principle** (every non-empty subset of natural numbers contains a least element): suppose there existed integers $n \geq n_0$ for which $\mathcal{P}(n)$ is false. Let m be the **smallest** such counterexample. By the base step, $m \neq n_0$, so $m > n_0$, which means $m - 1 \geq n_0$. Because m is the smallest counterexample, $\mathcal{P}(m-1)$ must be true. But then the inductive step dictates that $\mathcal{P}\left((m-1)+1\right) = \mathcal{P}(m)$ is true: a contradiction. In 1889, Giuseppe Peano formalized this principle as one of the foundational **Peano Axioms** defining arithmetic.

🔍 ***Historical Note** — Early manifestations of inductive reasoning appear in the work of the Persian mathematician **Al-Karaji** (around 1000 CE), who applied it to the binomial theorem and sums of cubes, and later **Francesco Maurolico**, who in 1575 proved that the sum of the first n odd integers equals n^2—our exact Pythagorean gnomon problem! However, **Blaise Pascal**, in his* Treatise on the Arithmetical Triangle *(1654), was the first to give an explicit two-step formulation: proving a property "for the first base," and then demonstrating that "if it holds in any arbitrary base, it will necessarily hold in the following one." Augustus De Morgan coined the term "mathematical induction" in 1838.*

⚠️ Pitfall Alert! —

In the inductive step, you do **not** assume that $\mathcal{P}(n)$ is true for all n (that would be assuming the very thing you set out to prove!). Rather, you **fix** an arbitrary integer $n \geq n_0$, **assume** that $\mathcal{P}(n)$ holds for that single, specific rank, and **deduce** that $\mathcal{P}(n+1)$ must consequently hold. You are proving an implication, not asserting unconditional truth.

⚠ ⚠ The Fatal Trap of Index Rupture: George Pólya's Horse Paradox

The domino analogy is intuitively vivid, but it hides a treacherous flaw: the induction step $\mathcal{P}(n) \Rightarrow \mathcal{P}(n+1)$ must hold **for every integer $n \geq n_0$ without any gap**. To demonstrate this peril, mathematician George Pólya designed a famous fallacious argument claiming that "all horses are the same color":

- **Base step:** In any set containing a single horse ($n = 1$), all horses trivially share the same color. $\mathcal{P}(1)$ is true.
- **Flawed induction step:** Assume the property holds for any herd of n horses. Consider a set of $n+1$ horses $\{C_1, C_2, ..., C_{n+1}\}$. By hypothesis, the first n horses $\{C_1, ..., C_n\}$ share a color, and the last n horses $\{C_2, ..., C_{n+1}\}$ share a color. Because these two sets overlap across the middle horses $\{C_2, ..., C_n\}$, one might naively conclude that all $n+1$ horses share identical colors.
- **Where is the trap?** The overlap argument breaks down completely for $n = 1$! Between $\{C_1\}$ and $\{C_2\}$, the middle set of shared horses is **empty**. Thus, $\mathcal{P}(1) \Rightarrow \mathcal{P}(2)$ fails, and the domino chain never even begins to fall!

Remember this rule: mathematical induction requires an unbroken bridge starting directly from the base index n_0.

3. Method: Writing an Impeccable Induction Proof

Method — Standard Structure for Written Examinations.

For every integer $n \geq n_0$, let $\mathcal{P}(n)$ denote the statement: "[…]".

Base Step (Initialization). For $n = n_0$: evaluate the left-hand and right-hand sides independently and confirm that $\mathcal{P}(n_0)$ is true.

Inductive Step (Heredity). Let $n \geq n_0$ be a fixed integer. Assume that $\mathcal{P}(n)$ is true (Induction Hypothesis). We wish to show that $\mathcal{P}(n+1)$ is true, that is: "[…]" (explicitly write out the target statement at rank $n+1$: this is **the goal**). [Perform algebraic calculations starting from rank $n+1$, strategically injecting the Induction Hypothesis and explicitly stating where it is used.] Therefore, $\mathcal{P}(n+1)$ is true.

Conclusion. The property is true for $n = n_0$ and is inherited from rank n to $n+1$ for all $n \geq n_0$; by the principle of mathematical induction, $\mathcal{P}(n)$ is true for every integer $n \geq n_0$.

Three golden rules from exam graders: 1. **Always write down the goal** (the statement at rank $n+1$) before computing: know exactly where you are heading. 2. **Start with the expression at rank $n+1$** and factor out or isolate the component corresponding to rank n, so you can cleanly apply the induction hypothesis. 3. **Explicitly cite the induction hypothesis** ("by the induction hypothesis, we have…"): this is the first thing a grader scans for. An inductive step that never uses the induction hypothesis is almost certainly invalid or circular.

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4. Worked Examples

Worked Example 1 — The Pythagorean Gnomons

Statement. Prove that for all integers $n \geq 1$: $1 + 3 + 5 + ... + (2n-1) = n^2$.

Notation. We also write this sum as $\sum_{k=1}^{n}(2k-1)$. The symbol $\sum$ (capital Greek letter **sigma**, corresponding to S for "sum") was introduced by Euler in 1755. It reads: "the sum from $k = 1$ to n of $2k-1$". The index k is a **dummy variable**: it traverses $1, 2, ..., n$ and disappears from the final result.

Solution. For $n \geq 1$, let $\mathcal{P}(n)$ be the proposition: "$\sum_{k=1}^{n}(2k-1) = n^2$".

Base Step. For $n = 1$: the sum contains a single term, $2(1) - 1 = 1$, and the right side gives $1^2 = 1$. Hence $\mathcal{P}(1)$ is true.

Inductive Step. Let $n \geq 1$ be fixed; assume $\mathcal{P}(n)$ is true. **Goal:** Prove that $\sum_{k=1}^{n+1}(2k-1) = (n+1)^2$. Decomposing the sum:

$$\sum_{k=1}^{n+1}(2k-1) = \underbrace{\sum_{k=1}^{n}(2k-1)}_{=n^2 \text{ by IH}} + \big(\,2(n+1)-1\,\big) = n^2+2n+1 = (n+1)^2.$$

Thus $\mathcal{P}(n+1)$ is true.

Conclusion. By mathematical induction, $\mathcal{P}(n)$ is true for all integers $n \geq 1$. ■

Remark: The inductive step mirrors **exactly** the geometric gnomon: adding an L-shaped band of $2n+1$ pebbles transforms an $n \times n$ square into an $(n+1) \times (n+1)$ square. Induction is the algebraic rigor of the visual proof.

Worked Example 2 — Gauss's Sum

Statement. Prove that for all $n \geq 1$: $1 + 2 + \ldots + n = \frac{n(n+1)}{2}$.

Solution. Let $\mathcal{P}(n)$ be: "$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$".

Base Step. For $n = 1$: left-hand side $= 1$; right-hand side $= \frac{1(2)}{2} = 1$. True.

Inductive Step. Assume $\mathcal{P}(n)$ holds for a fixed $n \geq 1$. **Goal:** $\sum_{k=1}^{n+1} k = \frac{(n+1)(n+2)}{2}$.

$$\sum_{k=1}^{n+1} k = \sum_{k=1}^{n} k + (n+1) = \frac{n(n+1)}{2} + (n+1) = \frac{n(n+1)+2(n+1)}{2} = \frac{(n+1)(n+2)}{2}.$$

We factored out $(n+1)$, instantly yielding the required form. Thus $\mathcal{P}(n+1)$ is true.

Conclusion. $\mathcal{P}(n)$ holds for all $n \geq 1$. ■

> **Did You Know?**
>
> The young Carl Friedrich Gauss famously evaluated this sum for $n = 100$ in school by pairing terms: $(1 + 100) + (2 + 99) + \ldots$, giving 50 pairs of 101, or 5050. Gauss's insight **discovers** the formula; mathematical induction **verifies** it. The two approaches are complementary: induction is formidable at proving a formula once known, but it cannot invent it out of thin air.

Worked Example 3 — Sum of Squares

Statement. Prove that for all $n \geq 1$: $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

Solution. *Base Step.* For $n = 1$: $1^2 = 1$ and $\frac{1(2)(3)}{6} = 1$. True.

Inductive Step. Assume the formula holds at rank n. **Goal:** $\sum_{k=1}^{n+1} k^2 = \frac{(n+1)(n+2)(2n+3)}{6}$.

$$\sum_{k=1}^{n+1} k^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{(n+1)\big[\,n(2n+1)+6(n+1)\big]}{6} = \frac{(n+1)(2n^2+7n+6)}{6}.$$

Expanding the target bracket: $(n+2)(2n+3) = 2n^2+7n+6$. Hence the sum is indeed $\frac{(n+1)(n+2)(2n+3)}{6}$. Therefore $\mathcal{P}(n+1)$ is true.

Conclusion. The formula holds for all $n \geq 1$. ■

Numerical sanity check: for $n = 3$, $1 + 4 + 9 = 14$, and $\frac{3(4)(7)}{6} = 14$. ✓

Worked Example 4 — A Sequence Defined by Recurrence

Statement. Let (u_n) be the sequence defined by $u_0 = 1$ and, for all $n \in \mathbb{N}$, $u_{n+1} = 2u_n + 3$. Compute the first few terms, conjecture an explicit closed formula, and prove it by induction.

Notation Reminder: (u_n) denotes the entire sequence, while u_n is the n-th term. The relation $u_{n+1} = 2u_n + 3$ is a **recurrence relation**: each term is generated from the preceding one.

Investigation. $u_0 = 1, u_1 = 5, u_2 = 13, u_3 = 29, u_4 = 61$. Notice that $u_n + 3$ yields $4, 8, 16, 32, 64$, which are powers of 2: 4×2^n. Conjecture: $u_n = 4 \times 2^n - 3 = 2^{n+2} - 3$.

Proof. Let $\mathcal{P}(n)$ be: "$u_n = 2^{n+2} - 3$".

Base Step. $n = 0$: $2^{0+2} - 3 = 4 - 3 = 1 = u_0$. True.

Inductive Step. Assume $u_n = 2^{n+2} - 3$ for a fixed $n \geq 0$. **Goal:** $u_{n+1} = 2^{n+3} - 3$.

$$u_{n+1} = 2u_n + 3 = 2\big(2^{n+2} - 3\big) + 3 = 2^{n+3} - 6 + 3 = 2^{n+3} - 3.$$

Thus $\mathcal{P}(n+1)$ is true.

Conclusion. For all $n \in \mathbb{N}$, $u_n = 2^{n+2} - 3$. ■

Worked Example 5 — Divisibility

Statement. Prove that for all $n \in \mathbb{N}$, $4^n - 1$ is divisible by 3.

Definition Reminder. "a is divisible by 3" means there exists an integer $k \in \mathbb{Z}$ such that $a = 3k$.

Solution. Let $\mathcal{P}(n)$ be: "$\exists k \in \mathbb{Z}, 4^n - 1 = 3k$".

Base Step. For $n = 0$: $4^0 - 1 = 1 - 1 = 0 = 3 \times 0$. True with $k = 0$.

Inductive Step. Assume there exists an integer k such that $4^n - 1 = 3k$, so $4^n = 3k + 1$. **Goal:** Prove that $4^{n+1} - 1$ is a multiple of 3.

$$4^{n+1} - 1 = 4 \times 4^n - 1 = 4(3k + 1) - 1 = 12k + 3 = 3(4k + 1).$$

Since k is an integer, $4k + 1$ is an integer. Thus $4^{n+1} - 1$ is divisible by 3. $\mathcal{P}(n + 1)$ is true.

Conclusion. For all $n \in \mathbb{N}$, 3 divides $4^n - 1$. ■

The key technique is to **rephrase the induction hypothesis as an equality** ($4^n = 3k + 1$) so that it can be directly substituted into the expression at rank $n + 1$.

Worked Example 6 — Bernoulli's Inequality (Core Curriculum Proof)

> **📐 Theorem — Bernoulli's Inequality.**
>
> For every real number $a > -1$ (and in particular for all $a > 0$) and for every natural number $n \in \mathbb{N}$:
>
> $$(1 + a)^n \geq 1 + na.$$

Concrete Intuition. Invest a capital at a compound interest rate a per year (e.g., $a = 0.05$ for 5%): after n years, the capital has multiplied by $(1 + a)^n$. With simple interest, it would multiply by $1 + na$. Bernoulli's inequality asserts that **compound interest always earns at least as much as simple interest**—which is intuitive, because you earn interest on top of accumulated interest.

Proof.

Let $a > -1$ be a fixed real number. For $n \in \mathbb{N}$, let $\mathcal{P}(n)$ be: "$(1 + a)^n \geq 1 + na$".

Base Step. For $n = 0$: $(1 + a)^0 = 1$ and $1 + 0 \times a = 1$. Since $1 \geq 1$, $\mathcal{P}(0)$ is true.

Inductive Step. Let $n \in \mathbb{N}$ be fixed; assume $(1 + a)^n \geq 1 + na$. **Goal:** $(1 + a)^{n+1} \geq 1 + (n + 1)a$. We multiply both sides of the induction hypothesis by $1 + a$. Since $a > -1$, we have $1 + a > 0$, so the inequality direction is **preserved**:

$$(1 + a)^{n+1} = (1 + a)^n(1 + a) \geq (1 + na)(1 + a) = 1 + a + na + na^2 = 1 + (n + 1)a + na^2.$$

Since $n \geq 0$ and $a^2 \geq 0$, the quantity na^2 is non-negative: $na^2 \geq 0$. Therefore, $1 + (n + 1)a + na^2 \geq 1 + (n + 1)a$. By transitivity, $(1 + a)^{n+1} \geq 1 + (n + 1)a$. Hence $\mathcal{P}(n + 1)$ is true.

Conclusion. For all $n \in \mathbb{N}$, $(1 + a)^n \geq 1 + na$. ■

The little black square ■ (or Q.E.D., *quod erat demonstrandum*, "which was to be demonstrated") marks the conclusion of a proof; it is often called the Halmos tombstone after Paul Halmos.

Where did we use the hypothesis $a > -1$? Exactly in one crucial step: multiplying an inequality by $1 + a$ without flipping its direction. If $1 + a$ were negative, the inequality sign would reverse and the argument would collapse. For instance, with $a = -3$ and $n = 5$: $(1 - 3)^5 = (-2)^5 = -32$, whereas $1 + 5(-3) = -14$; here $-32 < -14$, so the inequality is blatantly **false**. In mathematics, conditions in a theorem are never merely decorative.

Why is this inequality indispensable? Because it bounds an exponential growth $(1 + a)^n$ from below by an easily managed **affine linear** function $1 + na$. In Chapter 44, it will allow us to prove that if $q > 1$, then $q^n \to +\infty$: setting $q = 1 + a$ with $a > 0$, we have $q^n \geq 1 + na$, which outgrows any chosen bound.

Worked Example 7 — An Inequality Starting at a Higher Rank

Statement. Prove that for all integers $n \geq 4$: $2^n \geq n^2$.

Exploration.

n	0	1	2	3	4	5	6
2^n	1	2	4	8	16	32	64
n^2	0	1	4	**9**	16	25	36

The inequality holds for $n = 0, 1, 2$, but **fails at $n = 3$** ($8 < 9$). It then holds for all $n \geq 4$. The starting rank $n_0 = 4$ is strictly necessary!

Solution. *Base Step.* For $n = 4$: $2^4 = 16 \geq 4^2 = 16$. True.

Inductive Step. Let $n \geq 4$; assume $2^n \geq n^2$. **Goal:** $2^{n+1} \geq (n+1)^2$. By the induction hypothesis, $2^{n+1} = 2 \times 2^n \geq 2n^2$. It now suffices to prove that $2n^2 \geq (n+1)^2$. Let us examine the difference:

$$2n^2 - (n+1)^2 = 2n^2 - (n^2 + 2n + 1) = n^2 - 2n - 1 = (n-1)^2 - 2.$$

Since $n \geq 4$, we have $n - 1 \geq 3$, so $(n-1)^2 \geq 9$. Therefore $(n-1)^2 - 2 \geq 9 - 2 = 7 > 0$. Hence $2n^2 \geq (n+1)^2$, which gives $2^{n+1} \geq 2n^2 \geq (n+1)^2$. Thus $\mathcal{P}(n+1)$ is true.

Conclusion. For all $n \geq 4$, $2^n \geq n^2$. ■

Worked Example 8 — A Monotone and Bounded Sequence

Statement. Let (u_n) be defined by $u_0 = 1$ and $u_{n+1} = \sqrt{2 + u_n}$. Prove that for all $n \in \mathbb{N}$: $0 \leq u_n \leq u_{n+1} \leq 2$.

Strategy. Define $f(x) = \sqrt{2+x}$ on $[-2, +\infty[$, so that $u_{n+1} = f(u_n)$. The function f is strictly **increasing** (as the composite of two increasing functions: $x \mapsto 2 + x$ and the square root). An increasing function **preserves inequalities**: if $a \leq b$, then $f(a) \leq f(b)$. This is the decisive tool.

Solution. Let $\mathcal{P}(n)$ be: "$0 \leq u_n \leq u_{n+1} \leq 2$".

Base Step. For $n = 0$: $u_0 = 1$ and $u_1 = \sqrt{2+1} = \sqrt{3} \approx 1.732$. We clearly have $0 \leq 1 \leq \sqrt{3} \leq 2$. True.

Inductive Step. Assume $0 \leq u_n \leq u_{n+1} \leq 2$. Since f is increasing on $[0, 2]$, applying f across the inequalities preserves their order:

$$f(0) \leq f(u_n) \leq f(u_{n+1}) \leq f(2) \Leftrightarrow \sqrt{2} \leq u_{n+1} \leq u_{n+2} \leq \sqrt{4} = 2.$$

Since $0 \leq \sqrt{2}$, we obtain $0 \leq u_{n+1} \leq u_{n+2} \leq 2$, which is precisely $\mathcal{P}(n+1)$.

Conclusion. For all $n \in \mathbb{N}$, $0 \leq u_n \leq u_{n+1} \leq 2$. The sequence (u_n) is therefore **increasing** and **bounded above by 2**. ■

This technique—applying an increasing function to an entire chained inequality—is one of the most effective and elegant methods tested at the Grade 12 level.

Worked Example 9 — Higher-Order Derivatives

Statement. Let $f(x) = xe^x$ on $\mathbb{R}$. We denote by $f^{(n)}$ the **n-th derivative** of f ($f^{(0)} = f$, $f^{(1)} = f'$, $f^{(2)} = f''$). Prove that for all $n \in \mathbb{N}$: $f^{(n)}(x) = (x+n)e^x$.

Notation Note: Parentheses in $f^{(n)}$ are vital to distinguish the n-th derivative from the n-th algebraic power $f(x)^n$.

Solution. *Base Step.* For $n = 0$: $f^{(0)}(x) = f(x) = xe^x = (x+0)e^x$. True.

Inductive Step. Assume $f^{(n)}(x) = (x+n)e^x$ for all $x \in \mathbb{R}$. The $(n+1)$-th derivative is the derivative of the n-th: $f^{(n+1)} = (f^{(n)})'$. Applying the product rule $(uv)' = u'v + uv'$ with $u(x) = x + n$ and $v(x) = e^x$:

$$f^{(n+1)}(x) = 1 \cdot e^x + (x+n)e^x = (x+n+1)e^x.$$

Thus $\mathcal{P}(n+1)$ is true.

Conclusion. For all $n \in \mathbb{N}$, $f^{(n)}(x) = (x+n)e^x$. ■

CHAPTER 44: The Runner Who Never Touches the Line

> "Swift-footed Achilles will never overtake the tortoise, said Zeno. For to reach it, he must first cover half the distance, then half of the remainder, then half of what still remains. An infinity of steps cannot be completed in a finite time. Zeno's mistake was not in his view of the world, but in his lack of a mathematical tool: the concept of a limit."

What You Will Discover

- How the Greek paradox of **Zeno of Elea** challenged thinkers for more than two millennia, and how the modern notion of **limit** finally resolved it.
- The **formal definition** of a convergent sequence: the intuitive idea of a "tunnel" or an open interval that traps all terms beyond a certain rank, and the rigorous definition of divergence toward $\pm\infty$.
- The **comparison theorems** and the famous **squeeze theorem** (or squeeze theorem / sandwich theorem).
- The **monotone convergence theorem**: knowing that a sequence converges even before knowing its limit.
- The complete behavior of **geometric sequences** q^n, proven using Bernoulli's inequality (a required proof for the Baccalaureate).
- The **threshold algorithm in Python** to find when a sequence crosses a given barrier.

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1. The Problem: Achilles, the Tortoise, and the Infinite Stadium

1.1 The Dichotomy Paradox

Five centuries before our era, in the Greek city of Elea in southern Italy, the philosopher **Zeno** proposed a series of troubling riddles designed to show that motion is an illusion of our senses.

Consider a runner setting off on a 100-meter track. Before crossing the finish line, he must imperatively reach the halfway mark: $50\ \text{m}$. Once at $50\ \text{m}$, he still has $50\ \text{m}$ left to run; he must first cover half of this remainder, which is $75\ \text{m}$. At $75\ \text{m}$, he must cover half of the remaining quarter, reaching $87.5\ \text{m}$…

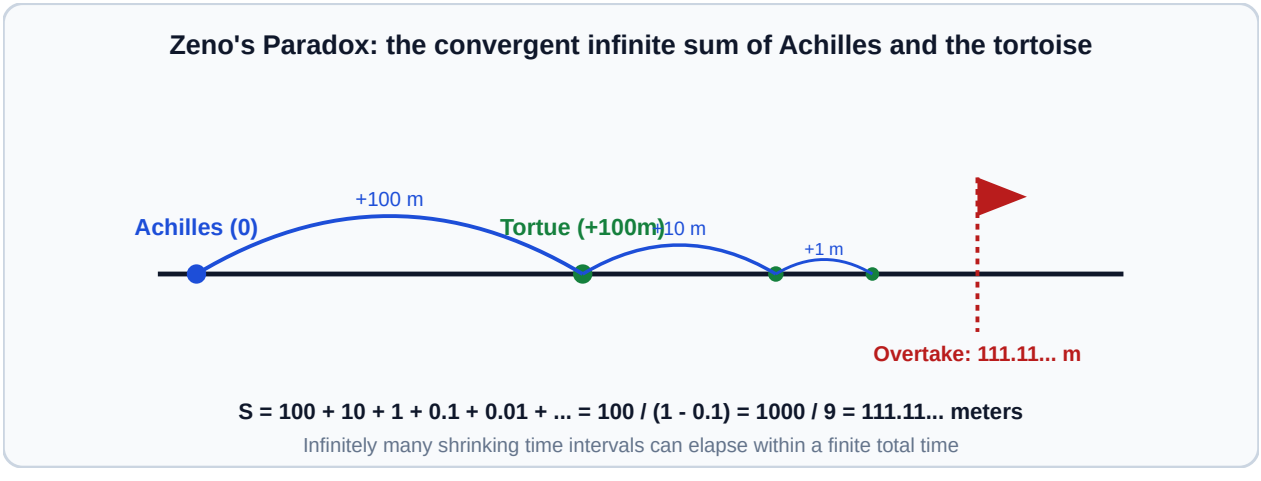

Figure 97: Zeno's paradox: convergent geometric series sum of Achilles and the tortoise

At each step n, the distance traveled as a fraction of the total track is:

$$u_n = 1 - \frac{1}{2^n}.$$

Zeno argues: > *"The runner must perform an infinity of successive acts: traverse the 1st interval, then the 2nd, then the 3rd… But no one can accomplish an infinity of tasks in a finite time. Therefore, the runner will never reach the line.*

Worse: he cannot even start, because before traveling one meter, he must travel half a meter, and before that a quarter of a meter…"

Yet in the real world, any child runs the 100 meters in a few seconds. Where does the flaw in Zeno's reasoning lie?

1.2 The Sum of an Infinity of Pieces

Let us look at the total distance traveled as a sum of infinitely repeated slices:

$$S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + ... + \frac{1}{2^n} + ...$$

Let us calculate the sum of the first n terms (denoted S_n):

- $S_1 = \frac{1}{2} = 0.5$
- $S_2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4} = 0.75$
- $S_3 = \frac{3}{4} + \frac{1}{8} = \frac{7}{8} = 0.875$
- $S_n = 1 - \frac{1}{2^n}$.

As n becomes larger and larger, the term $\frac{1}{2^n}$ becomes minuscule:

- for $n = 10$, $\frac{1}{2^{10}} = \frac{1}{1024} \approx 0.000976$;
- for $n = 30$, $\frac{1}{2^{30}} \approx 0.00000000093$.

The gap between the runner and the finish line ($100 \ \mathrm{m}$) becomes smaller than the width of a hair, then smaller than a molecule, then smaller than an atomic nucleus. Zeno's fallacy is to assume that a sum of an infinity of durations is necessarily infinite. But if the runner runs at a constant speed of $10 \ \mathrm{m/s}$:

- the first half takes $5 \ \mathrm{s}$;
- the next quarter takes $2.5 \ \mathrm{s}$;
- the next eighth takes $1.25 \ \mathrm{s}$…

The total elapsed time is also the sum:

$$T = 5 + 2.5 + 1.25 + 0.625 + ... = 10 \ \text{seconds}\ .$$

An infinity of time intervals can perfectly sum to a finite duration!

To formalize this dizzying leap into the infinitely large without falling into metaphysical traps, nineteenth-century mathematicians (Cauchy, Weierstrass) forged a tool of surgical precision: **the limit of a sequence**.

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2. Definitions: Taming Infinity

2.1 The Notations Explained

- A real sequence is denoted $(u_n)_{n\in\mathbb{N}}$ or simply (u_n). The letter n represents a natural number ($n = 0, 1, 2, 3...$) acting as a discrete time clock (the tick-tock of a clock). u_n denotes the term of rank n.
- $\lim_{n\to+\infty} u_n = L$ is read: "the limit of u_n as n tends to plus infinity is equal to L". The abbreviation $\lim$ comes from the Latin *limes*, which referred to the fortified frontier of the Roman Empire.
- The arrow $\to$ is read "tends to" or "approaches". One also sometimes writes $u_n \underset{n\to+\infty}{\to} L$.
- An open interval containing a real number L is an interval of the form $]L-\varepsilon; L+\varepsilon[$ (or $(L-\varepsilon, L+\varepsilon)$) with $\varepsilon > 0$ (the Greek letter ε, "epsilon", traditionally denoting an error tolerance as tiny as one wishes).

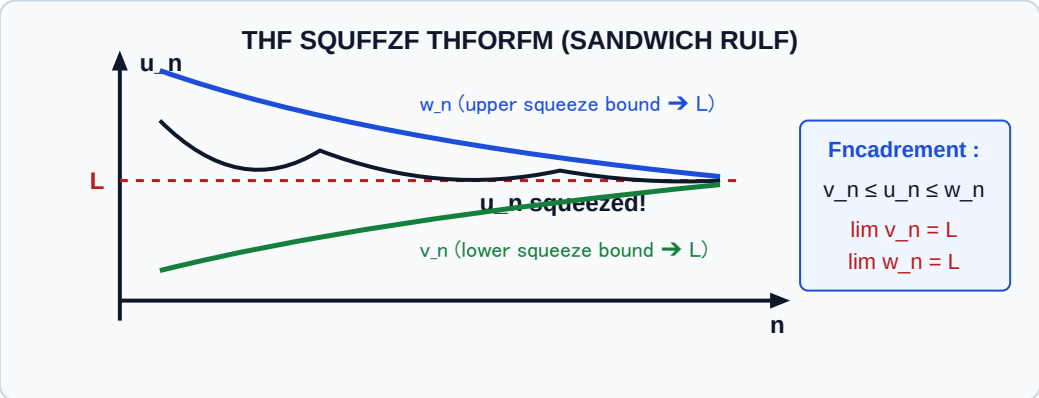

Figure 98: The squeeze theorem (two policemen and a drunk)

2.2 Convergent Sequence: The Official Definition

What exactly does it mean to say: "as n becomes very large, the terms u_n approach L"? Saying merely "they approach" is insufficient. The sequence $1 + \frac{1}{n}$ approaches -5, but its limit is not -5! They must approach L **as closely as one wishes**, and they must **stay there forever**.

> **📐 Definition — Finite limit (convergent sequence).**
>
> We say that a sequence (u_n) has the real number L as its limit when: **Every open interval containing L contains all terms of the sequence from a certain rank onward.** We then write:
>
> $$\lim_{n \to +\infty} u_n = L.$$
>
> A sequence that has a finite real limit L is said to be **convergent**. Otherwise, it is said to be **divergent**.

Geometric and dynamic translation: Imagine a game between two players: a *Challenger* and a *Defender*. 1. The Challenger chooses an open interval $]a; b[$ around L, as narrow and microscopic as desired (for example $]L - 0.001\,; L + 0.001[$). This is the "target". 2. The Defender must find an integer rank N beyond which **all** subsequent terms $u_N, u_{N+1}, u_{N+2}, \ldots$ fall within the target, never leaving it again. If the Defender has a winning strategy regardless of the interval imposed at the start, then the sequence converges to L.

> **💡 Did you know?**
>
> In formalized symbols at the university level, this is written with the quantifiers $\forall$ ("for all") and $\exists$ ("there exists"):
>
> $$\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall n \geq N,\ |u_n - L| < \varepsilon.$$
>
> In high school, the official curriculum uses the equivalent formulation with open intervals: *"for every open interval I containing L, there exists an integer N such that for all $n \geq N$, $u_n \in I$"*.

2.3 Divergent Sequence toward $+\infty$ or $-\infty$

A sequence does not always converge to a number. It can explode toward infinity, or oscillate indefinitely without ever settling down.

📐 Definition — Infinite limit.

- We say that a sequence (u_n) has limit $+\infty$ if every interval of

the form $]A; +\infty[$ (where A is an arbitrary real number) contains all terms of the sequence from a certain rank onward:

$$\lim_{n\to+\infty} u_n = +\infty.$$

- We say that a sequence (u_n) has limit $-\infty$ if every interval of the form $]-\infty; B[$ (where B is an arbitrary real number) contains all terms of the sequence from a certain rank onward:

$$\lim_{n\to+\infty} u_n = -\infty.$$

Concrete meaning: Set a barrier A as high as you want (a billion, ten billion…). From a certain rank N onward, the sequence crosses above this barrier and never drops back below it.

⚠️ Caution, trap! — "Divergent" does not mean "tends to infinity".

A divergent sequence is simply a sequence that **does not converge** to a finite real number.

- The sequence $u_n = n^2$ diverges to $+\infty$.
- The sequence $v_n = (-1)^n$ (which alternates $+1, -1, +1, -1...$) is **divergent**, but it tends neither to $+\infty$ nor to $-\infty$! It remains bounded between -1 and 1, but has no limit.

2.4 Uniqueness of the Limit

📐 Fundamental Property — Uniqueness.

When a sequence (u_n) admits a limit (finite or infinite), **this limit is unique**. A sequence cannot converge to two distinct numbers at the same time.

Why? If u_n converged both to L_1 and L_2 with $L_1 \neq L_2$, we could surround L_1 and L_2 with two small disjoint intervals (not touching each other). Beyond a certain rank, the terms u_n would have to lie simultaneously in both intervals: impossible!

---

3. Reference Sequences and Their Limits

The following table gathers the fundamental building blocks from which all other limits are constructed:

Sequence u_n ($n \geq 1$)	Limit at $+\infty$	Intuitive Justification
k (constant)	k	All terms are already equal to k.
$\frac{1}{n}$	0	Sharing 1 cake among a billion guests leaves a crumb.
$\frac{1}{n^2}$, $\frac{1}{n^p}$ ($p \geq 1$)	0	The denominator explodes even faster.
$\frac{1}{\sqrt{n}}$	0	$\sqrt{n}$ grows to $+\infty$, its reciprocal shrinks to 0.
n, n^2, n^p ($p \geq 1$)	$+\infty$	Powers of integers exceed any bound.
$\sqrt{n}$	$+\infty$	Even though it slows down, it has no ceiling.

---

4. Operations on Limits and Indeterminate Forms

When adding, multiplying, or dividing two sequences with known limits, mathematical common sense applies most of the time… except in four ambiguous situations called **Indeterminate Forms** (denoted I.F.).

4.1 Direct Algebraic Rules

If $\lim u_n = L$ and $\lim v_n = L'$ (with $L, L' \in \mathbb{R}$):

- **Sum:** $\lim(u_n + v_n) = L + L'$
- **Product:** $\lim(u_n \times v_n) = L \times L'$
- **Quotient:** $\lim\left(\frac{u_n}{v_n}\right) = \frac{L}{L'}$ (provided that $L' \neq 0$).

With infinities:

- A finite number $+$ infinity yields infinity: $5 + (+\infty) = +\infty$.
- $(+\infty) + (+\infty) = +\infty$ and $(-\infty) + (-\infty) = -\infty$.
- Sign rule for products: $(+\infty) \times (-\infty) = -\infty$, $(-3) \times (-\infty) = +\infty$.
- A finite number divided by infinity yields 0: $\frac{L}{\pm\infty} = 0$.
- A non-zero number divided by an infinitely small quantity yields infinity: if $u_n \to L > 0$ and $v_n \to 0$ with $v_n > 0$, then $\frac{u_n}{v_n} \to +\infty$.

4.2 The Four Indeterminate Forms (I.F.)

A form is called "indeterminate" when **knowledge of the individual limits alone is insufficient to predict the overall outcome**. The result depends on which component is "stronger":

1. $\boldsymbol{\infty - \infty}$ (sum of two infinities of opposite signs): who wins between the positive giant and the negative giant?
2. $\boldsymbol{0 \times \infty}$ (product of an infinitely small and an infinitely large quantity): crushing toward 0 or exploding toward infinity?
3. $\frac{\infty}{\infty}$ (quotient of two infinities): does the numerator run faster than the denominator?
4. $\frac{0}{0}$ (quotient of two infinitely small quantities).

> ⚠️ **Caution, trap! — I.F. does not mean "no limit".**
>
> "Indeterminate" does not mean the calculation is impossible or that the limit does not exist! It simply means: *"Stop, do not conclude immediately: transform the expression to resolve the ambiguity"*. The primary technique for polynomials and rational fractions is to **factor out the highest-degree term**.

---

5. Fundamental Theorems

5.1 Comparison Theorems

> 📐 **Comparison Theorem (toward infinity).**
>
> Let (u_n) and (v_n) be two sequences such that from a certain rank onward:
>
> $$u_n \geq v_n.$$
>
> 1. If $\lim_{n\to+\infty} v_n = +\infty$, then $\lim_{n\to+\infty} u_n = +\infty$. 2. If $\lim_{n\to+\infty} u_n = -\infty$, then $\lim_{n\to+\infty} v_n = -\infty$.

Mental image: If your friend v_n climbs toward the sky without limit and you are constantly standing on their shoulders ($u_n \geq v_n$), you are inevitably propelled toward the sky alongside them.

5.2 The Squeeze Theorem (Sandwich Theorem)

Theorem — Squeeze Theorem (Theorem of the Gendarmes).

Let (u_n), (v_n), and (w_n) be three real sequences. If there exists a rank N from which:

$$v_n \leq u_n \leq w_n$$

and if $\lim_{n\to+\infty} v_n = \lim_{n\to+\infty} w_n = L \in \mathbb{R}$, then the sequence (u_n) converges and its limit is:

$$\lim_{n\to+\infty} u_n = L.$$

Origin of the French name (« Théorème des Gendarmes »): If two police officers (v_n and w_n) both converge toward the same police station L while keeping the suspect (u_n) flanked between them, the suspect has no choice but to enter the police station L as well.

Proof of the Squeeze Theorem.

Let I be any open interval containing L. By definition of an open interval, there exists a real number $\varepsilon > 0$ such that $]L-\varepsilon\,;L+\varepsilon[\subset I$. Since $\lim v_n = L$, there exists an integer N_1 such that for all $n \geq N_1$, $L-\varepsilon < v_n < L+\varepsilon$. In particular, $L-\varepsilon < v_n$. Since $\lim w_n = L$, there exists an integer N_2 such that for all $n \geq N_2$, $L-\varepsilon < w_n < L+\varepsilon$. In particular, $w_n < L+\varepsilon$. Moreover, by assumption, $v_n \leq u_n \leq w_n$ for all $n \geq N_0$. Now set $N = \max(N_0, N_1, N_2)$. For every integer $n \geq N$, the three conditions hold simultaneously:

$$L-\varepsilon < v_n \leq u_n \leq w_n < L+\varepsilon \Longrightarrow u_n \in]L-\varepsilon\,;L+\varepsilon[\subset I.$$

Every open interval containing L thus contains all terms u_n from rank N onward. The sequence (u_n) indeed converges to L. ■

5.3 Monotone Convergence Theorem

This is one of the most profound theorems in analysis: it allows one to establish that a sequence converges **without even calculating its limit**.

Monotone Convergence Theorem.

1. Every sequence that is

increasing and bounded above converges to a real limit L. 2. Every sequence that is **decreasing and bounded below** converges to a real limit L.

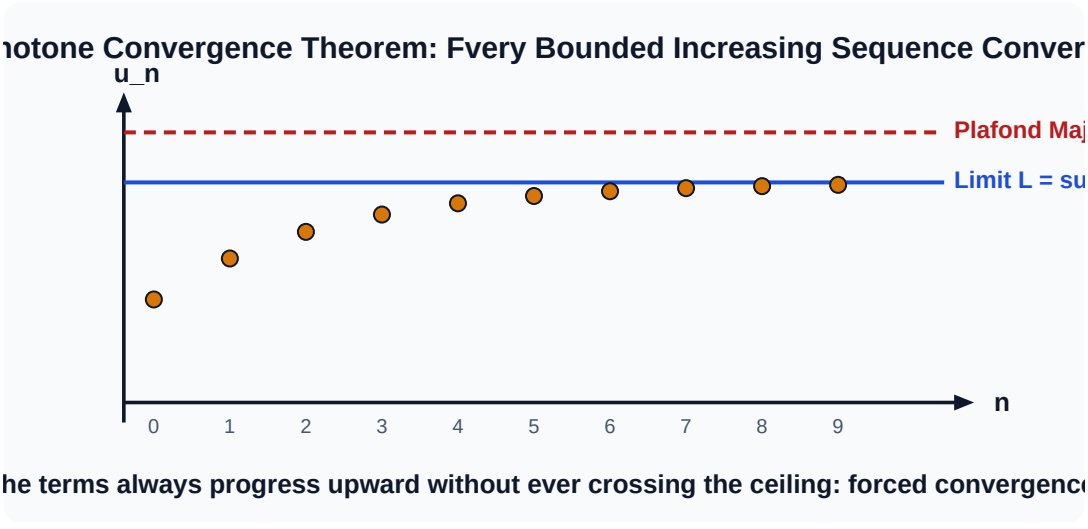

Figure 99: Monotone convergence theorem: bounded increasing sequence

Vocabulary review: - A sequence is **bounded above** by M if for all n, $u_n \leq M$. - A sequence is **bounded below** by m if for all n, $u_n \geq m$. - A sequence is **bounded** if it is both bounded above and bounded below ($m \leq u_n \leq M$).

> ⚠ **Caution, trap! — The upper bound is not necessarily the limit!**
>
> If an increasing sequence (u_n) is bounded above by 10, the theorem guarantees that it converges to a limit L, and that $L \leq 10$. But the limit L is not necessarily 10! It could converge to 2, or to $\sqrt{3}$, because 10 is merely one ceiling among others. Moreover, an increasing sequence that is not bounded above always diverges to $+\infty$.

---

6. The Limit of Geometric Sequences q^n

The fundamental geometric sequence is defined by $u_n = q^n$ where q is a fixed real number called the common ratio. Its behavior depends on the position of q relative to -1 and 1.

> 📐 **Theorem — Limit of q^n.**
>
> - If $q > 1$: $\lim_{n\to+\infty} q^n = +\infty$.
> - If $q = 1$: the sequence is constantly equal to 1, so $\lim_{n\to+\infty} q^n = 1$.
> - If $-1 < q < 1$: $\lim_{n\to+\infty} q^n = 0$.
> - If $q \leq -1$: the sequence has no limit (it oscillates, alternating signs).

Required Baccalaureate Proof (Case $q > 1$)

This proof relies directly on **Bernoulli's inequality** established in Chapter 43: for any real number $a > 0$ and any integer $n \in \mathbb{N}$, $(1+a)^n \geq 1 + na$.

> **Proof (Required knowledge).**
>
> Let q be a real number strictly greater than 1. Set $a = q - 1$. Since $q > 1$, we have $a > 0$. We can therefore write: $q = 1 + a$. According to Bernoulli's inequality:
>
> $$q^n = (1+a)^n \geq 1 + na.$$
>
> Now, since $a > 0$, the linear sequence $1 + na$ tends to $+\infty$:
>
> $$\lim_{n\to+\infty} (1 + na) = +\infty.$$
>
> By the comparison theorem (5.1), since $q^n \geq 1 + na$ for all n, we deduce immediately:
>
> $$\lim_{n\to+\infty} q^n = +\infty. \quad \blacksquare$$

Consequences for the case $0 < q < 1$: If $0 < q < 1$, set $q' = \frac{1}{q} > 1$. Then $q^n = \frac{1}{(q')^n}$. Since $(q')^n \to +\infty$, the quotient tends to 0: $\lim q^n = 0$. If $-1 < q < 0$, we have $|q^n| = |q|^n \to 0$ because $0 < |q| < 1$, whence $q^n \to 0$ by the squeeze theorem $-|q|^n \leq q^n \leq |q|^n$.

---

7. Threshold Algorithm in Python

When a sequence converges to a number or diverges to $+\infty$, a vital practical question arises for engineers, financial analysts, and scientists: *"From what rank n onward does the sequence cross a given threshold value?"*

Since the stopping rank is not known in advance, one uses an indefinite loop: the `while` loop.

Example: Engine Cooling

The temperature of an engine after n minutes follows the decreasing geometric sequence:

$$T_n = 20 + 80 \times 0.9^n.$$

The engine is considered cold as soon as its temperature drops below 30° C.

```
def engine_threshold(threshold):
    """Calculates the number of minutes required to fall below the threshold."""
    t = 0
    temp = 95.0
    while temp >= threshold:
        temp = 0.95 * temp + 1.0  # Cooling model: T_(n+1) = 0.95*T_n + 1
        t += 1
    return t, temp

mins, final_temp = engine_threshold(30.0)
print(f"Safe threshold reached in {mins} minutes (Temperature: {final_temp:.2f} °C)")
```

Method — Standard structure of a threshold algorithm on the Bac:

1. Initialize the rank counter: `n = 0`. 2. Initialize the sequence value: `u = u_0`. 3. Condition of the `while` loop: put the **opposite** condition of what is sought (continue *while* the threshold has not yet been crossed). 4. In the loop body: update the term `u` and increment `n = n + 1`. 5. Return `n` at the end.

---

8. Step-by-Step Worked Examples

Worked Example 1 — Using the Definition with an Interval

Problem: Consider the sequence $(u_n)_{n\geq 1}$ defined by $u_n = 3 + \frac{2}{n}$. 1. What is the conjectured limit L of the sequence? 2. Determine a rank N from which all terms u_n belong to the open interval $]2.99\,;3.01[$.

Solution: 1. Since $\lim_{n\to+\infty} \frac{1}{n} = 0$, we have $\lim_{n\to+\infty} u_n = 3 + 2 \times 0 = 3$. The limit is $L = 3$. 2. The interval $]2.99\,;3.01[$ is written $]3 - 0.01\,;3 + 0.01[$. We seek the values of n for which $u_n \in]2.99\,;3.01[$:

$$2.99 < 3 + \frac{2}{n} < 3.01.$$

Since $\frac{2}{n} > 0$ for all $n \geq 1$, the left inequality $2.99 < 3 + \frac{2}{n}$ is always true. It remains to satisfy the right inequality:

$$3 + \frac{2}{n} < 3.01 \Leftrightarrow \frac{2}{n} < 0.01 \Leftrightarrow \frac{2}{n} < \frac{1}{100} \Leftrightarrow n > 200.$$

Since n is an integer, starting from rank $N = 201$, all terms u_n for $n \geq 201$ are guaranteed to lie within the interval $]2.99\,;3.01[$.

Worked Example 2 — Resolving a Rational Indeterminacy $\frac{\infty}{\infty}$

Problem: Determine the limit of the sequence (u_n) defined for all $n \geq 1$ by:

$$u_n = \frac{5n^2 - 3n + 2}{2n^2 + 7}.$$

Solution: - Numerator: $\lim(5n^2 - 3n + 2) = +\infty$. - Denominator: $\lim(2n^2 + 7) = +\infty$.

This is an Indeterminate Form of the type "$\frac{\infty}{\infty}$". *Method: factor out the dominant term (n^2) from both numerator and denominator:*

$$u_n = \frac{n^2\left(5 - \frac{3}{n} + \frac{2}{n^2}\right)}{n^2\left(2 + \frac{7}{n^2}\right)}.$$

For $n \geq 1$, cancel out $n^2 \neq 0$:

$$u_n = \frac{5 - \frac{3}{n} + \frac{2}{n^2}}{2 + \frac{7}{n^2}}.$$

Now: $\lim \frac{3}{n} = 0$, $\lim \frac{2}{n^2} = 0$, so $\lim\left(5 - \frac{3}{n} + \frac{2}{n^2}\right) = 5$. $\lim \frac{7}{n^2} = 0$, so $\lim\left(2 + \frac{7}{n^2}\right) = 2$. By the quotient rule for limits, we conclude irrefutably:

$$\lim_{n \to +\infty} u_n = \frac{5}{2}.$$

Worked Example 3 — Squeeze Theorem and Trigonometry

Problem: Determine the limit of the sequence $(v_n)_{n \geq 1}$ defined by:

$$v_n = \frac{3 + (-1)^n \cos(n)}{n + \sqrt{n}}.$$

Solution: The functions $\cos(n)$ and $(-1)^n$ oscillate endlessly and have no limit as $n \to +\infty$. It is impossible to apply direct algebraic rules. We immediately turn to the **squeeze theorem**. For any integer n, $-1 \leq \cos(n) \leq 1$ and $(-1)^n \in \{-1; 1\}$. Thus, the product $(-1)^n \cos(n)$ remains bounded between -1 and 1:

$$-1 \leq (-1)^n \cos(n) \leq 1.$$

Add 3 to each part:

$$3 - 1 \leq 3 + (-1)^n \cos(n) \leq 3 + 1 \Leftrightarrow 2 \leq 3 + (-1)^n \cos(n) \leq 4.$$

Since for all $n \geq 1$, the denominator $n + \sqrt{n}$ is strictly positive, we can divide without changing the direction of the inequalities:

$$\frac{2}{n + \sqrt{n}} \leq v_n \leq \frac{4}{n + \sqrt{n}}.$$

Examine the limits of both bounding sequences:

- $\lim(n + \sqrt{n}) = +\infty$, so $\lim_{n \to +\infty} \frac{2}{n+\sqrt{n}} = 0$.
- Likewise, $\lim_{n \to +\infty} \frac{4}{n+\sqrt{n}} = 0$.

Both bounding sequences converge to the same limit 0. By the **squeeze theorem**, the sequence (v_n) converges and:

$$\lim_{n \to +\infty} v_n = 0.$$

Worked Example 4 — Geometric Sum and Convergence

Problem: Consider the sequence (S_n) defined for any integer $n \geq 1$ by:

$$S_n = \sum_{k=1}^{n} \left(\frac{2}{3}\right)^k = \frac{2}{3} + \left(\frac{2}{3}\right)^2 + \ldots + \left(\frac{2}{3}\right)^n.$$

1. Express S_n in closed form without summation notation. 2. Determine the limit of S_n as $n \to +\infty$.

Solution: 1. S_n is the sum of the first n consecutive terms of a geometric sequence with first term $a = \frac{2}{3}$ and common ratio $q = \frac{2}{3}$. The geometric sum formula gives (with $q \neq 1$):

$$\text{Sum} = \text{First term} \times \frac{1 - q^{\text{number of terms}}}{1 - q}.$$

Here:

$$S_n = \frac{2}{3} \times \frac{1 - \left(\frac{2}{3}\right)^n}{1 - \frac{2}{3}} = \frac{2}{3} \times \frac{1 - \left(\frac{2}{3}\right)^n}{\frac{1}{3}} = \frac{2}{3} \times 3 \times \left(1 - \left(\frac{2}{3}\right)^n\right) = 2\left(1 - \left(\frac{2}{3}\right)^n\right).$$

2. Since $-1 < \frac{2}{3} < 1$, according to the theorem on geometric sequences:

$$\lim_{n \to +\infty} \left(\frac{2}{3}\right)^n = 0.$$

By limit operations:

$$\lim_{n\to+\infty} S_n = 2(1-0) = 2.$$

Worked Example 5 — Comparison toward Infinity

Problem: Prove that the sequence (w_n) defined for all $n \geq 1$ by $w_n = 2n^2 - 5\sin(n) + (-1)^n$ tends to $+\infty$.

Solution: For any integer n, we have $-1 \leq \sin(n) \leq 1$, so $-5 \leq -5\sin(n) \leq 5$, and $(-1)^n \geq -1$. Consequently:

$$w_n = 2n^2 - 5\sin(n) + (-1)^n \geq 2n^2 - 5 - 1 = 2n^2 - 6.$$

Now:

$$\lim_{n\to+\infty} (2n^2 - 6) = +\infty.$$

Since $w_n \geq 2n^2 - 6$ for all $n \geq 1$, by the **comparison theorem**, we conclude immediately:

$$\lim_{n\to+\infty} w_n = +\infty.$$

Worked Example 6 — Recurrence Sequence and Monotone Convergence

Problem: Let (u_n) be the sequence defined by $u_0 = 0$ and for all $n \in \mathbb{N}$ by:

$$u_{n+1} = \frac{1}{2}u_n + 3.$$

1. Prove by induction that for all $n \in \mathbb{N}$, $u_n < 6$. 2. Prove that the sequence (u_n) is strictly increasing. 3. Deduce that the sequence converges to a limit L to be determined.

Solution: 1. **Proof by induction:** For $n \in \mathbb{N}$, let $\mathcal{P}(n)$ be the statement: "$u_n < 6$".

- *Base step:* $u_0 = 0 < 6$. Thus $\mathcal{P}(0)$ is true.
- *Inductive step:* Let $n \in \mathbb{N}$ be fixed. Assume $\mathcal{P}(n)$ is true, that is, $u_n < 6$.

Then:

$$\frac{1}{2}u_n < \frac{1}{2} \times 6 = 3 \Longrightarrow \frac{1}{2}u_n + 3 < 3 + 3 = 6 \Longrightarrow u_{n+1} < 6.$$

Thus $\mathcal{P}(n+1)$ is true.

- *Conclusion:* By induction, for all natural numbers n, $u_n < 6$. The sequence is **bounded above by 6**.

2. **Monotonicity:** Compute the difference $u_{n+1} - u_n$:

$$u_{n+1} - u_n = \left(\frac{1}{2}u_n + 3\right) - u_n = -\frac{1}{2}u_n + 3 = \frac{1}{2}(6 - u_n).$$

From question 1, $u_n < 6$, hence $6 - u_n > 0$. Consequently, $u_{n+1} - u_n > 0$, which proves that the sequence (u_n) is **strictly increasing**.

3. **Convergence and calculation of the limit:** Since the sequence (u_n) is increasing and bounded above by 6, the **monotone convergence theorem** ensures that it converges to a finite real limit $L \leq 6$. Let us determine L: Since $\lim u_n = L$, we also have $\lim u_{n+1} = L$. Taking the limit in the recurrence relation $u_{n+1} = \frac{1}{2}u_n + 3$, we obtain:

$$L = \frac{1}{2}L + 3 \Leftrightarrow L - \frac{1}{2}L = 3 \Leftrightarrow \frac{1}{2}L = 3 \Leftrightarrow L = 6.$$

The sequence (u_n) converges to 6.

Key Takeaways

1. **Definition of convergence:** $\lim u_n = L$ means that every open interval containing L contains all terms of the sequence from a rank N onward.
2. **Divergence toward $+\infty$:** every interval $]A; +\infty[$ contains all terms of the sequence from a certain rank onward.

3. **Indeterminate Forms:** $\infty - \infty$, $0 \times \infty$, $\frac{\infty}{\infty}$, $\frac{0}{0}$. For a quotient of polynomials in n, always factor out the highest-degree term in both the numerator and the denominator.
4. **Squeeze Theorem:** If $v_n \leq u_n \leq w_n$ and $\lim v_n = \lim w_n = L$, then $\lim u_n = L$.
5. **Comparison Theorem:** If $u_n \geq v_n$ and $\lim v_n = +\infty$, then $\lim u_n = +\infty$.
6. **Geometric Sequences q^n:**
   - If $q > 1$, $\lim q^n = +\infty$ (proven via Bernoulli's inequality $(1+a)^n \geq 1 + na$).
   - If $-1 < q < 1$, $\lim q^n = 0$.
   - If $q \leq -1$, no limit.
7. **Monotone Convergence:** Increasing + bounded above $\Longrightarrow$ convergent. Decreasing + bounded below $\Longrightarrow$ convergent.

Exercises

Level (Direct Applications)

Exercise 44.1 Determine the limit of each of the following sequences as $n \to +\infty$: 1. $u_n = 4 - \frac{7}{n} + \frac{3}{n^2}$ 2. $v_n = -3n^3 + 500$ 3. $w_n = 5 \times 0.4^n$ 4. $t_n = 2 \times 1.05^n$

Exercise 44.2 For each of the following geometric sequences, state whether it converges and, if so, give its limit: 1. $a_n = \left(-\frac{3}{4}\right)^n$ 2. $b_n = \left(\frac{7}{5}\right)^n$ 3. $c_n = (-1.2)^n$ 4. $d_n = \frac{4^n}{3^n}$

Exercise 44.3 Calculate the limit of the following rational sequences: 1. $u_n = \frac{3n+1}{5n-2}$ 2. $v_n = \frac{7n^2-4}{2n^3+n}$ 3. $w_n = \frac{n^4-1}{100n^3+5}$

Exercise 44.4 Let $u_n = 5 + \frac{1}{n}$. Find a rank N from which $u_n \in]4.999\,;5.001[$.

Exercise 44.5 Prove that for all $n \geq 1$, $\frac{-1}{n} \leq \frac{\sin(n)}{n} \leq \frac{1}{n}$. Deduce $\lim_{n\to+\infty} \frac{\sin(n)}{n}$.

Exercise 44.6 A sequence (u_n) satisfies for all $n \in \mathbb{N}$: $u_n \geq 3n^2 - 10n$. Determine its limit at $+\infty$.

Exercise 44.7 A bird population initially numbers 500 individuals and decreases by 5% per year. 1. Express the population P_n after n years. 2. What is the limit of P_n as $n \to +\infty$?

Exercise 44.8 Write a Python function `geometric_threshold(q, A)` that takes as input a ratio $q > 1$ and a threshold $A > 0$, and returns the first rank n such that $q^n > A$.

Level (Guided Reflection and Baccalaureate Style)

Exercise 44.9 (Resolving Indeterminacy with Square Roots). Let $u_n = \sqrt{n+1} - \sqrt{n}$ for $n \geq 0$. 1. Show that this is an indeterminate form in its initial state. 2. By multiplying by the conjugate expression, show that $u_n = \frac{1}{\sqrt{n+1}+\sqrt{n}}$. 3. Deduce the limit of (u_n).

Exercise 44.10 (Baccalaureate Style: Complete Study of a Recurrence Sequence). Consider the sequence (u_n) defined by $u_0 = 1$ and for all $n \in \mathbb{N}$:

$$u_{n+1} = \frac{2u_n + 3}{u_n + 2}.$$

1. Prove by induction that for all $n \in \mathbb{N}$, $1 \leq u_n \leq \sqrt{3}$. 2. Prove that for all $n \in \mathbb{N}$, $u_{n+1} - u_n = \frac{3-u_n^2}{u_n+2}$. Deduce the monotonicity of the sequence (u_n). 3. Justify that the sequence (u_n) converges and determine its exact limit.

Exercise 44.11 (Arithmetico-Geometric Sequence and Bank Investment). A principal of €$10,000$ is invested at compound interest at an annual rate of 3%. Each year, the account holder deposits an additional €500 immediately after the interest is credited. Let C_n be the capital after n years ($C_0 = 10,000$). 1. Express C_{n+1} in terms of C_n. 2. Set $v_n = C_n + \alpha$. Determine the real number α so that (v_n) is a geometric sequence. 3. Deduce the explicit expression for C_n in terms of n, then determine $\lim_{n\to+\infty} C_n$.

Exercise 44.12 (Squeeze Theorem and Finite Sum). Consider the sequence (S_n) defined for any integer $n \geq 1$ by:

$$S_n = \sum_{k=1}^{n} \frac{n}{n^2+k} = \frac{n}{n^2+1} + \frac{n}{n^2+2} + \ldots + \frac{n}{n^2+n}.$$

1. Show that for all $k \in \{1; 2; ...; n\}$, we have $\frac{n}{n^2+n} \leq \frac{n}{n^2+k} \leq \frac{n}{n^2+1}$. 2. Deduce that for all $n \geq 1$: $\frac{n^2}{n^2+n} \leq S_n \leq \frac{n^2}{n^2+1}$. 3. Deduce the convergence of (S_n) and its limit.

Exercise 44.13 (True or False with Justifications). For each of the following statements, state whether it is true or false, providing a proof or a specific counterexample: 1. "If a sequence converges, then it is bounded." 2. "If a sequence is bounded, then it converges." 3. "If for all $n \in \mathbb{N}$, $u_n > 0$ and (u_n) converges to L, then $L > 0$." 4. "If $\lim_{n\to+\infty} u_n^2 = 4$, then $\lim_{n\to+\infty} u_n = 2$."

Exercise 44.14 (Geometric Sum and the Runner's Paradox). A runner covers successively 100 m, then 50 m, then 25 m, and at each step half of the preceding step. 1. Calculate the total distance covered after n steps. 2. Prove rigorously that the total distance will never exceed 200 m, no matter how many steps are completed.

Exercise 44.15 (Bernoulli's Inequality and Lower Bound). Consider the sequence $u_n = 1.01^n$. 1. Using Bernoulli's inequality, give a linear lower bound for u_n. 2. Deduce an integer N such that for all $n \geq N$, $u_n \geq 1,000,000$. 3. Is this the smallest integer that satisfies the condition?

Exercise 44.16 (Threshold Algorithm and Stopping Time). Consider the sequence (u_n) defined by $u_0 = 2$ and $u_{n+1} = 0.8u_n + 1$. 1. Conjecture the limit L of (u_n). 2. Write a Python script to find the smallest integer n such that $|u_n - L| < 10^{-5}$.

Exercise 44.17 (Study of an Auxiliary Geometric Sequence). Let (u_n) be defined by $u_0 = 4$ and $u_{n+1} = \frac{u_n}{2u_n+1}$. 1. Calculate u_1 and u_2. 2. Assume that $u_n > 0$ for all n. Set $v_n = \frac{1}{u_n}$. Show that (v_n) is an arithmetic sequence. 3. Deduce the expression for v_n, then for u_n in terms of n, and determine its limit.

Level (Challenge and In-Depth Study)

Exercise 44.18 (The Constant e as the Limit of a Sequence). Consider the two sequences (u_n) and (v_n) defined for all $n \geq 1$ by:

$$u_n = \left(1 + \frac{1}{n}\right)^n \quad \text{and} \quad v_n = \left(1 + \frac{1}{n}\right)^{n+1}.$$

1. By computing the first few values with a calculator, conjecture the behavior of (u_n) and (v_n). 2. It is given that (u_n) is strictly increasing and that (v_n) is strictly decreasing with $u_n < v_n$. Prove that $\lim_{n\to+\infty}(v_n - u_n) = 0$. 3. What can be deduced about the two sequences? (They converge to Euler's number $e \approx 2.71828$).

Exercise 44.19 (Gauss's Arithmetico-Geometric Mean Sequence). Let $a_0 > 0$ and $b_0 > 0$ with $a_0 < b_0$. For all $n \in \mathbb{N}$, define:

$$a_{n+1} = \sqrt{a_n b_n} \quad \text{and} \quad b_{n+1} = \frac{a_n + b_n}{2}.$$

1. Recall why for any strictly positive real numbers a and b, we have $\sqrt{ab} \leq \frac{a+b}{2}$. 2. Prove that for all $n \geq 1$, $a_n \leq b_n$. 3. Show that the sequence (a_n) is increasing and that the sequence (b_n) is decreasing. 4. Deduce that both sequences converge to a single common limit (the arithmetic-geometric mean of Gauss).

Exercise 44.20 (The Koch Snowflake: Infinite Perimeter and Finite Area). The Koch snowflake is built in stages starting from an equilateral triangle of side length 1 and area $A_0 = \frac{\sqrt{3}}{4}$. At each stage, the central third of each side is replaced by an outward-pointing equilateral triangle. 1. The perimeter at stage n is given by $P_n = 3 \times \left(\frac{4}{3}\right)^n$. What is the limit of P_n? 2. The area at stage n satisfies $A_n = A_0\left[1 + \frac{3}{5}\left(1 - \left(\frac{4}{9}\right)^n\right)\right]$. Calculate $\lim_{n\to+\infty} A_n$. 3. Comment on the paradoxical geometric result (a figure with an infinite boundary enclosing a strictly bounded, finite area).

Exercise 44.21 (Cesàro's Theorem — Beyond Syllabus / Olympiad). Let (u_n) be a sequence converging to a finite limit L. Define the sequence of averages $(c_n)_{n\geq 1}$ by:

$$c_n = \frac{u_1 + u_2 + ... + u_n}{n}.$$

1. Test this result on the sequence $u_n = 1 + \frac{1}{n}$. 2. Give an example showing that the converse is false (find a sequence (u_n) that does not converge, but whose Cesàro mean (c_n) converges).

Exercise 44.22 (Fixed-Point Theorem for Sequences). Let f be a continuous function on $[0; 1]$ such that for all $x \in [0; 1]$, $f(x) \in [0; 1]$. Assume further that there exists a constant $k \in]0; 1[$ such that for all $x, y \in [0; 1]$:

$$|f(x) - f(y)| \leq k|x - y|.$$

Let $u_0 \in [0; 1]$ and $u_{n+1} = f(u_n)$. 1. Show that for all $n \in \mathbb{N}$, $|u_{n+1} - u_n| \leq k^n |u_1 - u_0|$. 2. Prove that the sequence (u_n) converges to the unique solution of $f(x) = x$ in $[0; 1]$.

Solutions

Solutions for Level

- **44.1:**
  1. $\lim u_n = 4 - 0 + 0 = 4$.
  2. $\lim v_n = -\infty$ because $n^3 \to +\infty$ and $-3 < 0$.
  3. $0 < 0.4 < 1$, so $\lim 0.4^n = 0$, whence $\lim w_n = 0$.
  4. $1.05 > 1$, so $\lim 1.05^n = +\infty$, whence $\lim t_n = +\infty$.
- **44.2:**
  1. Converges to 0 because $|-3/4| = 3/4 < 1$.
  2. Diverges to $+\infty$ because $7/5 > 1$.
  3. Diverges without a limit because $-1.2 \leq -1$ (oscillates with increasing amplitude).
  4. $d_n = (4/3)^n$ with $4/3 > 1$, so it diverges to $+\infty$.
- **44.3:**
  1. Factoring: $u_n = \frac{n(3+1/n)}{n(5-2/n)} = \frac{3+1/n}{5-2/n} \to \frac{3}{5}$.
  2. $v_n = \frac{n^2(7-4/n^2)}{n^3(2+1/n^2)} = \frac{1}{n} \times \frac{7-4/n^2}{2+1/n^2} \to 0 \times \frac{7}{2} = 0$.
  3. $w_n = \frac{n^4(1-1/n^4)}{n^3(100+5/n^3)} = n \times \frac{1-1/n^4}{100+5/n^3} \to +\infty$.
- **44.4:** We require $4.999 < 5 + \frac{1}{n} < 5.001 \Leftrightarrow 0 < \frac{1}{n} < 0.001 = \frac{1}{1000} \Leftrightarrow n > 1000$. The minimal rank is $N = 1001$.
- **44.5:** For all $n \geq 1$, $-1 \leq \sin(n) \leq 1$. Dividing by $n > 0$ yields $\frac{-1}{n} \leq \frac{\sin(n)}{n} \leq \frac{1}{n}$. Since $\lim(-1/n) = \lim(1/n) = 0$, by the squeeze theorem, $\lim_{n \to +\infty} \frac{\sin(n)}{n} = 0$.
- **44.6:** For $n \geq 4$, $3n^2 - 10n = n(3n - 10) \to +\infty$. Since $u_n \geq 3n^2 - 10n$, by comparison, $\lim u_n = +\infty$.
- **44.7:**
  1. A 5% decrease corresponds to multiplying by $1 - 0.05 = 0.95$. Thus $P_n = 500 \times 0.95^n$.
  2. Since $0 < 0.95 < 1$, $\lim 0.95^n = 0$, so $\lim P_n = 0$ (theoretical extinction of the population).
- **44.8:**

```python
def geometric_threshold(q, A):
    n = 0
    u = 1.0
    while u <= A:
        n += 1
        u = q ** n
    return n

print("Rank n:", geometric_threshold(1.05, 1000))
# Prints Rank n: 142 (because 1.05^142 ≈ 1048.9 > 1000)
```

Solutions for Level

- **44.9:**
  1. $\lim \sqrt{n+1} = +\infty$ and $\lim \sqrt{n} = +\infty$, so this is an I.F. of type "$\infty - \infty$".
  2. For $n \geq 0$:

$$\sqrt{n+1} - \sqrt{n} = \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} = \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}.$$

3. The denominator tends to $+\infty$, so by the quotient rule, $\lim u_n = 0$.

- **44.10:**
  1. *Base step:* $n = 0$, $u_0 = 1 \in [1; \sqrt{3}]$ because $\sqrt{3} \approx 1.732$. *Inductive step:* Assume $1 \leq u_n \leq \sqrt{3}$. The function $f(x) = \frac{2x+3}{x+2} = 2 - \frac{1}{x+2}$ is strictly increasing on $[1; \sqrt{3}]$. Therefore $f(1) \leq f(u_n) \leq f(\sqrt{3})$. Now $f(1) = 5/3 \approx 1.67 \geq 1$. And $f(\sqrt{3}) = \frac{2\sqrt{3}+3}{\sqrt{3}+2} = \frac{\sqrt{3}(2+\sqrt{3})}{\sqrt{3}+2} = \sqrt{3}$. Thus $1 \leq u_{n+1} \leq \sqrt{3}$. *Conclusion:* For all $n \in \mathbb{N}$, $1 \leq u_n \leq \sqrt{3}$.
  2. $u_{n+1} - u_n = \frac{2u_n+3}{u_n+2} - u_n = \frac{2u_n+3-u_n(u_n+2)}{u_n+2} = \frac{3-u_n^2}{u_n+2}$. Since $u_n \leq \sqrt{3}$, we have $u_n^2 \leq 3$, so $3 - u_n^2 \geq 0$. Since $u_n \geq 1$, $u_n + 2 > 0$. Therefore $u_{n+1} - u_n \geq 0$: the sequence is increasing.
  3. The sequence is increasing and bounded above by $\sqrt{3}$, so it converges to a limit $L \in [1; \sqrt{3}]$. Taking the limit: $L = \frac{2L+3}{L+2} \Leftrightarrow L(L+2) = 2L+3 \Leftrightarrow L^2 = 3$. Since $L \geq 1$, we have $L = \sqrt{3}$.
- **44.11:**
  1. Increasing by 3% and then adding 500 yields: $C_{n+1} = 1.03C_n + 500$.
  2. $v_{n+1} = C_{n+1} + \alpha = 1.03C_n + 500 + \alpha$. We want $v_{n+1} = 1.03v_n = 1.03(C_n + \alpha) = 1.03C_n + 1.03\alpha$. Equating coefficients: $500 + \alpha = 1.03\alpha \Leftrightarrow 0.03\alpha = 500 \Leftrightarrow \alpha = \frac{500}{0.03} = \frac{50{,}000}{3}$.
  3. (v_n) is a geometric sequence with ratio $q = 1.03$ and first term $v_0 = C_0 + \frac{50{,}000}{3} = 10{,}000 + \frac{50{,}000}{3} = \frac{80{,}000}{3}$. Hence $v_n = \frac{80{,}000}{3} \times 1.03^n$, whence $C_n = v_n - \alpha = \frac{80{,}000}{3} \times 1.03^n - \frac{50{,}000}{3}$. Since $1.03 > 1$, $\lim 1.03^n = +\infty$, so $\lim C_n = +\infty$.
- **44.12:**
  1. For $1 \leq k \leq n$, we have $n^2 + 1 \leq n^2 + k \leq n^2 + n$. Inverting strictly positive quantities: $\frac{1}{n^2+n} \leq \frac{1}{n^2+k} \leq \frac{1}{n^2+1}$. Multiplying by $n > 0$ yields the requested inequality.
  2. Summing these inequalities for k from 1 to n, the sum contains n terms: $n \times \frac{n}{n^2+n} \leq S_n \leq n \times \frac{n}{n^2+1} \Leftrightarrow \frac{n^2}{n^2+n} \leq S_n \leq \frac{n^2}{n^2+1}$.
  3. $\lim \frac{n^2}{n^2+n} = \lim \frac{n^2}{n^2(1+1/n)} = 1$ and $\lim \frac{n^2}{n^2+1} = 1$. By the squeeze theorem, $\lim S_n = 1$.
- **44.13:**
  1. **True.** If $u_n \to L$, then from a rank N onward, all terms lie in $]L-1; L+1[$. The preceding terms $u_0, ..., u_{N-1}$ are finite in number, so we take the minimum and maximum of the finite set $\{u_0, ..., u_{N-1}, L-1, L+1\}$, which bounds the entire sequence.
  2. **False.** Counterexample: $u_n = (-1)^n$ is bounded between -1 and 1, but does not converge.
  3. **False.** Counterexample: $u_n = \frac{1}{n+1} > 0$ for all $n \in \mathbb{N}$, but $\lim u_n = 0$ (which is not strictly positive). Taking the limit converts strict inequalities into non-strict inequalities.
  4. **False.** Counterexample: $u_n = 2 \times (-1)^n$. Then $u_n^2 = 4$ for all n, so $\lim u_n^2 = 4$, but (u_n) oscillates between 2 and -2 and has no limit.
- **44.14:**
  1. Distance at step n: $D_n = 100 \sum_{k=0}^{n-1} \left(\frac{1}{2}\right)^k = 100 \times \frac{1-(1/2)^n}{1-1/2} = 200\left(1 - \frac{1}{2^n}\right)$.
  2. For all $n \geq 1$, $\frac{1}{2^n} > 0$, so $1 - \frac{1}{2^n} < 1$, which implies $D_n < 200$. Moreover $\lim D_n = 200$ m. The distance tends toward 200 m without ever exceeding it for any finite n.
- **44.15:**
  1. With $a = 0.01$, Bernoulli gives: $1.01^n = (1 + 0.01)^n \geq 1 + 0.01n$.
  2. We seek $1 + 0.01n \geq 1{,}000{,}000 \Leftrightarrow 0.01n \geq 999{,}999 \Leftrightarrow n \geq 99{,}999{,}900$. The rank $N = 99{,}999{,}900$ works for sure.
  3. No, this is not the smallest: the exponential power grows far faster than the linear lower bound. The actual threshold (calculable using logarithms in Chapter 48) is $n = \lceil \frac{\ln(10^6)}{\ln(1.01)} \rceil = 1388$.
- **44.16:**
  1. If $u_n \to L$, then $L = 0.8L + 1 \Leftrightarrow 0.2L = 1 \Leftrightarrow L = 5$.
  2. Python script:

```python
n = 0
u = 2.0
while abs(u - 5.0) >= 1e-5:
    n += 1
    u = 0.8 * u + 1.0
print(f"Rank n found: {n}")
# Prints Rank n found: 57
```

- **44.17:**
  1. $u_1 = \frac{4}{2(4)+1} = \frac{4}{9}$. $u_2 = \frac{4/9}{2(4/9)+1} = \frac{4/9}{17/9} = \frac{4}{17}$.
  2. $v_{n+1} = \frac{1}{u_{n+1}} = \frac{2u_n+1}{u_n} = 2 + \frac{1}{u_n} = v_n + 2$. (v_n) is therefore an arithmetic sequence with common difference $r = 2$ and first term $v_0 = \frac{1}{u_0} = \frac{1}{4}$.
  3. $v_n = \frac{1}{4} + 2n = \frac{8n+1}{4}$. By inversion: $u_n = \frac{1}{v_n} = \frac{4}{8n+1}$. We immediately deduce: $\lim_{n\to+\infty} u_n = 0$.

Solutions for Level

- **44.18:**
  1. For $n = 10$, $u_{10} \approx 2.5937$, $v_{10} \approx 2.8531$. For $n = 1000$, $u_{1000} \approx 2.7169$, $v_{1000} \approx 2.7196$. Both seem to tighten around $2.718...$
  2. $v_n - u_n = \left(1 + \frac{1}{n}\right)^n \left(1 + \frac{1}{n} - 1\right) = \frac{1}{n} u_n$. Since $u_n < v_n \leq v_1 = 4$, the sequence (u_n) is bounded above by 4. Thus $0 < v_n - u_n \leq \frac{4}{n}$. By the squeeze theorem, $\lim(v_n - u_n) = 0$.
  3. One is increasing, the other decreasing, and their difference tends to 0: these are two **adjacent sequences**. They converge to a single common limit, which defines the number $e \approx 2.71828$.
- **44.19:**
  1. $(\sqrt{a} - \sqrt{b})^2 \geq 0 \Leftrightarrow a - 2\sqrt{ab} + b \geq 0 \Leftrightarrow \frac{a+b}{2} \geq \sqrt{ab}$ (arithmetic-mean-geometric-mean inequality).
  2. For $n \geq 0$, $b_{n+1} - a_{n+1} = \frac{a_n+b_n}{2} - \sqrt{a_n b_n} \geq 0$, so $a_{n+1} \leq b_{n+1}$ for all $n \geq 0$.
  3. $a_{n+1} - a_n = \sqrt{a_n b_n} - a_n = \sqrt{a_n}(\sqrt{b_n} - \sqrt{a_n}) \geq 0$ because $b_n \geq a_n$. Hence (a_n) is increasing. Similarly, $b_{n+1} - b_n = \frac{a_n - b_n}{2} \leq 0$, so (b_n) is decreasing.
  4. (a_n) is increasing and bounded above by b_0, so it converges to L_a. (b_n) is decreasing and bounded below by a_0, so it converges to L_b. Taking the limit in $b_{n+1} = \frac{a_n+b_n}{2}$, we have $L_b = \frac{L_a+L_b}{2} \Leftrightarrow L_a = L_b$. The two sequences converge to the same value, denoted $M(a_0, b_0)$.
- **44.20:**
  1. Since $\frac{4}{3} > 1$, $\lim_{n\to+\infty} \left(\frac{4}{3}\right)^n = +\infty$, so the perimeter P_n tends to $+\infty$.
  2. Since $0 < \frac{4}{9} < 1$, $\lim \left(\frac{4}{9}\right)^n = 0$. Therefore $\lim A_n = A_0\left(1 + \frac{3}{5}(1 - 0)\right) = A_0\left(1 + \frac{3}{5}\right) = \frac{8}{5}A_0 = \frac{8}{5} \times \frac{\sqrt{3}}{4} = \frac{2\sqrt{3}}{5}$.
  3. The Koch snowflake possesses an infinite perimeter while enclosing a strictly bounded, finite area. It shows that fractal geometry shatters our standard Euclidean intuition.
- **44.21:**
  1. $u_n \to 1$. $c_n = \frac{1}{n}\sum_{k=1}^{n}\left(1 + \frac{1}{k}\right) = 1 + \frac{1}{n}\sum_{k=1}^{n}\frac{1}{k}$. Now the harmonic sum $H_n = \sum_{k=1}^{n}\frac{1}{k}$ grows asymptotically as $\ln(n)$, so $\frac{H_n}{n} \to 0$. Thus $c_n \to 1$.
  2. Take $u_n = (-1)^n$. The sequence (u_n) oscillates and diverges. However, $S_n = u_1 + ... + u_n$ is equal to -1 if n is odd and 0 if n is even. Hence $-\frac{1}{n} \leq c_n \leq 0$, which implies by the squeeze theorem that $\lim c_n = 0$. The Cesàro mean converges to 0 even though (u_n) diverges!
- **44.22:**
  1. By induction: for $n = 0$, $|u_1 - u_0| \leq k^0|u_1 - u_0|$ is trivially true. Assume $|u_{n+1} - u_n| \leq k^n|u_1 - u_0|$. Then $|u_{n+2} - u_{n+1}| = |f(u_{n+1}) - f(u_n)| \leq k|u_{n+1} - u_n| \leq k \times k^n|u_1 - u_0| = k^{n+1}|u_1 - u_0|$.
  2. Since $0 < k < 1$, $\lim k^n = 0$. The steps $|u_{n+1} - u_n|$ become infinitely small at a geometric rate. The sequence is a Cauchy sequence in a closed bounded interval, so it converges to a real number $L \in [0; 1]$. By continuity of f, $\lim f(u_n) = f(L)$, whence $L = f(L)$. Uniqueness follows from the fact that if $f(x) = x$ and $f(y) = y$, $|x - y| = |f(x) - f(y)| \leq k|x - y| \Longrightarrow (1 - k)|x - y| \leq 0$, which requires $x = y$ since $1 - k > 0$.

The Question That Opens What Follows

Until now, our counter n was a natural number taking step-by-step jumps: $n = 0, 1, 2, 3...$ But what happens if the variable is no longer a runner with discrete steps, but a bird flying seamlessly through the sky, or a rocket accelerating at every microsecond? How can we define the limit of a function $f(x)$ when the real variable x glides continuously toward infinity, or worse, when it approaches a point where the function seems to tear apart or explode?

This is the mystery of function limits and the invisible lines that guide curves through space: **asymptotes**. See you in Chapter 45.

CHAPTER 45: The Horizon of Curves

> "Asymptote comes from the ancient Greek asúmptōtos: 'not falling together', 'not meeting'. It is the sailor's horizon: the further the ship ventures out to sea, the more the horizon seems to offer itself, yet the hull can never cross its threshold. In the Cartesian sky, asymptotes are the invisible rails guiding the destiny of curves toward infinity."

What You Will Discover

- The fundamental transition from the discrete variable $n \in \mathbb{N}$ to the continuous variable $x \in \mathbb{R}$: how functions glide toward infinity or toward a point of rupture.
- Rigorous definitions of **finite and infinite limits**, as well as the indispensable notions of **left-hand limit and right-hand limit**.
- The **three families of asymptotes** (vertical, horizontal, oblique) and the method to determine the **relative position** of a curve with respect to its guide line.
- The complete toolkit to resolve the **four indeterminate forms**: factoring out the dominant term, multiplying by the conjugate expression, and recognizing difference quotients.
- The triumph of the **growth hierarchy (comparative growth)** of the exponential function over powers of x, with its **required Baccalaureate proof**.
- The theorem on the **limit of a composite function** $g(f(x))$.

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1. The Problem: The Optical Lens and the Black Hole

1.1 The Physicist and Focal Length

In an optics laboratory, a light ray passes through a thin converging lens of focal length $f > 0$. A luminous object is placed at a distance p in front of the lens. The Descartes-Gauss thin lens formula relates the object distance p to the position of the sharp image p' formed on a screen behind the lens:

$$\frac{1}{p} + \frac{1}{p'} = \frac{1}{f}.$$

Let us express the image position p' as a function of the object distance p:

$$p'(p) = \frac{f \cdot p}{p - f}.$$

What happens as the experimenter moves the object closer to the lens?

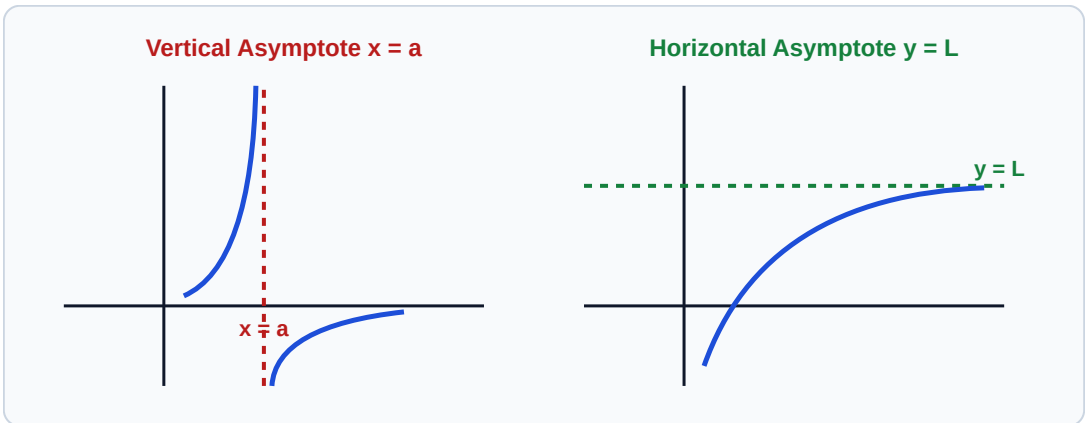

Figure 100: Asymptotic behavior at infinity: vertical and horizontal asymptotes

1. **If the object is infinitely far away ($p \to +\infty$)**:
$$p'(p) = \frac{f \cdot p}{p(1 - f/p)} = \frac{f}{1 - f/p} \underset{p \to +\infty}{\to} f.$$
The image forms precisely in the focal plane of the lens. The horizontal line $y = f$ is a **limiting boundary**, a ceiling that the image position cannot cross: it is a **horizontal asymptote**.
2. **If the object approaches the focal point ($p \to f$ with $p > f$)**: The denominator $p - f$ becomes an infinitely small positive number (0.001, $0.000001\ldots$). The quotient $p'(p)$ explodes to astronomical values:
$$\lim_{p \to f^+} p'(p) = +\infty.$$
To view the sharp image, one would need to push the screen to the end of the room, then outside the building, then to the Moon! The emerging rays are parallel: the image is cast to infinity. The vertical line $x = f$ acts like an impassable cliff: it is a **vertical asymptote**.

1.2 From Discrete Steps to Continuous Flux

In Chapter 44, the variable n jumped from integer to integer: $n = 0, 1, 2, 3\ldots$ It made no sense to ask what happened between $n = 2$ and $n = 3$, and one could only take the limit of n toward $+\infty$.

With functions of a real variable $x \mapsto f(x)$, the world becomes **fluid and continuous**. The real variable x can:

- flee toward $+\infty$ or toward $-\infty$;
- glide toward a fixed real number x_0 (for example $x \to 2$);
- approach a forbidden point from the left ($x < x_0$) or from the right ($x > x_0$).

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2. Rigorous Definitions of Function Limits

2.1 The Notations Explained

- $\lim_{x \to +\infty} f(x) = L$: "the limit of $f(x)$ as x tends to plus infinity is equal to L".
- $\lim_{x \to x_0} f(x) = L$: "the limit of $f(x)$ as x tends to the real number x_0 is equal to L".
- $x \to x_0^+$ or $x \underset{x > x_0}{\to} x_0$: "$x$ tends to x_0 through values strictly greater" (right-hand limit).
- $x \to x_0^-$ or $x \underset{x < x_0}{\to} x_0$: "$x$ tends to x_0 through values strictly less" (left-hand limit).

2.2 Finite Limit at Infinity

Definition — Finite limit at $+\infty$.

Let f be a function defined on an interval $[a\,;+\infty[$, and let $L \in \mathbb{R}$. We say that **f has limit L at $+\infty$** if: **Every open interval containing L contains all values $f(x)$ for sufficiently large x.** We write:
$$\lim_{x \to +\infty} f(x) = L.$$
(Analogous definition at $-\infty$ on an interval $]-\infty\,;b]$).

Concrete meaning: Draw a horizontal strip of width 2ε centered on $y = L$, as narrow as you wish. If you move sufficiently far to the right along the x-axis (past a certain threshold X), the graph of the function remains trapped inside this strip forever.

2.3 Infinite Limit at a Point: The Vertical Wall

Definition — Infinite limit at a real point x_0.

Let $x_0 \in \mathbb{R}$. We say that **$f(x)$ tends to $+\infty$ as x tends to x_0** if: **For every real number $A > 0$, the interval $]A\,;+\infty[$ contains all values $f(x)$ for x sufficiently close to x_0.** We write:
$$\lim_{x \to x_0} f(x) = +\infty.$$

2.4 Left-Hand and Right-Hand Limits

Some functions exhibit radically different behaviors depending on whether they are approached from the port side or starboard side. The archetype is the reciprocal function $f(x) = \frac{1}{x}$ near zero:

- If $x > 0$ and $x \to 0$: x takes values like 0.1; 0.001; 0.00001. The quotient $\frac{1}{x}$ is 10; 1000; $100,000$:

$$\lim_{x \to 0^+} \frac{1}{x} = +\infty.$$

- If $x < 0$ and $x \to 0$: x takes negative values like -0.1; -0.001. The quotient $\frac{1}{x}$ is -10; -1000:

$$\lim_{x \to 0^-} \frac{1}{x} = -\infty.$$

Fundamental Theorem — Existence of the global two-sided limit at a point.

A function f admits a limit L at x_0 if and only if it admits both a left-hand limit and a right-hand limit at x_0, and these two limits are **equal**:

$$\lim_{x \to x_0} f(x) = L \Leftrightarrow \lim_{x \to x_0^-} f(x) = \lim_{x \to x_0^+} f(x) = L.$$

For $f(x) = \frac{1}{x}$ at 0, since $+\infty \neq -\infty$, the function has **no global two-sided limit** at 0.

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3. Asymptotes: The Geometric Guide of Curves

An asymptote is a line that the curve $\mathcal{C}_f$ approaches arbitrarily closely as points on the curve recede toward infinity.

3.1 Horizontal Asymptote

Definition — Horizontal asymptote.

If $\lim_{x \to +\infty} f(x) = L$ (or if $\lim_{x \to -\infty} f(x) = L$), where L is a finite real number, then the line $\mathcal{D}$ with equation $\boldsymbol{y = L}$ is called a **horizontal asymptote** to the curve $\mathcal{C}_f$ in the neighborhood of $+\infty$ (or $-\infty$).

3.2 Vertical Asymptote

Definition — Vertical asymptote.

If $\lim_{x \to a^+} f(x) = \pm\infty$ or if $\lim_{x \to a^-} f(x) = \pm\infty$ (where $a \in \mathbb{R}$), then the line Δ with equation $\boldsymbol{x = a}$ is called a **vertical asymptote** to the curve $\mathcal{C}_f$.

3.3 Oblique Asymptote

Sometimes the curve does not level off horizontally, but aligns itself along a slanted line with equation $y = ax + b$ with $a \neq 0$.

Definition — Oblique asymptote.

The line $\mathcal{D}$ with equation $\boldsymbol{y = ax + b}$ is an oblique asymptote to the curve $\mathcal{C}_f$ at $+\infty$ (resp. $-\infty$) if:

$$\lim_{x \to +\infty} \big[\, f(x) - (ax + b) \big] = 0 \quad (\text{resp. at } -\infty\ ).$$

Geometric sense: The vertical distance between the point on the curve $M(x; f(x))$ and the point on the line $P(x; ax + b)$ becomes infinitely small as one advances toward the ends of the coordinate plane.

3.4 Determining Relative Position

To determine whether the curve $\mathcal{C}_f$ lies **above** or **below** its asymptote (horizontal $y = L$ or oblique $y = ax + b$), one applies the following universal method:

Method — Relative position of a curve and an asymptote:

1.

Compute the difference: $d(x) = f(x) - y_{\text{asymptote}}$. 2. Rigorously study the **sign** of $d(x)$:

- If $d(x) > 0$ on an interval, then $\mathcal{C}_f$ is **strictly above** the asymptote on that interval.
- If $d(x) < 0$ on an interval, then $\mathcal{C}_f$ is **strictly below** the asymptote.
- If $d(x) = 0$ at a point x_1, the curve **intersects / crosses** its asymptote at the point of abscissa x_1.

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4. The Toolkit for Resolving Indeterminate Forms

Recall the four forms that prohibit immediate direct conclusions:

$$\text{I.F.:} \quad [\infty - \infty], \quad [0 \times \infty], \quad \left[\frac{\infty}{\infty}\right], \quad \left[\frac{0}{0}\right].$$

Here are the four fundamental techniques to master in Grade 12:

Method 1: Factoring Out the Dominant Term

At $+\infty$ or $-\infty$, in a polynomial or a rational function, the highest-degree term dictates the law to all others.

- For a polynomial: factor out the highest power of x.
- For a fraction: factor out the respective dominant powers from both numerator and denominator, then simplify.

Caution, trap! — The sign under the square root at $-\infty$.

Always remember that for any real number x:

$$\sqrt{x^2} = |x| = \begin{cases} x & \text{if } \ x \geq 0 \\ -x & \text{if } \ x < 0 \end{cases}$$

As $x \to -\infty$, x is a **strictly negative** number. Consequently:

$$\sqrt{x^2} = -x \quad (\text{and not } \ +x \ !).$$

This is the single most heavily penalized error on Baccalaureate exams!

Method 2: The Conjugate Expression

When square roots clash in an indeterminacy of type $[\infty - \infty]$ or $[0/0]$, use the difference of squares identity $(a - b)(a + b) = a^2 - b^2$ to remove radicals from the problematic site:

$$\sqrt{A} - \sqrt{B} = \frac{(\sqrt{A} - \sqrt{B})(\sqrt{A} + \sqrt{B})}{\sqrt{A} + \sqrt{B}} = \frac{A - B}{\sqrt{A} + \sqrt{B}}.$$

The delicate subtraction $[\infty - \infty]$ turns into a well-behaved addition in the denominator $[\infty + \infty = \infty]$.

Method 3: Recognizing a Difference Quotient (Derivative)

An indeterminate form of type $\left[\frac{0}{0}\right]$ at a point x_0 very often conceals the very definition of the derivative:

$$\lim_{x \to x_0} \frac{g(x) - g(x_0)}{x - x_0} = g'(x_0).$$

Famous examples:

- $\lim_{x \to 0} \frac{e^x - 1}{x}$: with $g(x) = e^x$, $g(0) = 1$, the limit is $g'(0) = e^0 = 1$.
- $\lim_{x \to 0} \frac{\sin x}{x}$: with $g(x) = \sin x$, $g(0) = 0$, the limit is $\cos(0) = 1$.

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5. Growth Hierarchy: Comparative Growth of the Exponential Function

When the exponential function confronts a polynomial or a power of x at infinity, the explosive growth of e^x infinitely surpasses any polynomial growth.

Theorem — Comparative growth of the exponential function.

For any natural number $n \geq 1$: 1. $\lim_{x\to+\infty} \frac{e^x}{x} = +\infty$ and more generally $\lim_{x\to+\infty} \frac{e^x}{x^n} = +\infty$ 2. $\lim_{x\to-\infty} x\,e^x = 0$ and more generally $\lim_{x\to-\infty} x^n\,e^x = 0$

Rule of thumb: At $+\infty$ as well as at $-\infty$, **the exponential always dominates any power of x.**

Required Baccalaureate Proof: $\lim_{x\to+\infty} \frac{e^x}{x} = +\infty$

Proof (Required knowledge).

The goal is to obtain a lower bound for e^x by a degree-2 polynomial, so that the quotient $\frac{e^x}{x}$ is bounded below by a degree-1 linear term tending to $+\infty$.

Step 1: Lower bound for the exponential function. Consider the function f defined on $[0\,;+\infty[$ by:

$$f(x) = e^x - \frac{x^2}{2}.$$

Differentiate f twice:

- Its first derivative is $f'(x) = e^x - x$.
- Its second derivative is $f''(x) = e^x - 1$.

For all $x \geq 0$, $e^x \geq e^0 = 1$, so $f''(x) \geq 0$. Consequently, the derivative function f' is strictly increasing on $[0\,;+\infty[$. Now $f'(0) = e^0 - 0 = 1 > 0$. Since f' is increasing and $f'(0) > 0$, for all $x \geq 0$ we have:

$$f'(x) \geq f'(0) > 0.$$

The function f is therefore itself strictly increasing on $[0\,;+\infty[$. Since $f(0) = e^0 - 0 = 1 > 0$, it follows that for all $x \geq 0$:

$$f(x) > 0 \Longrightarrow e^x > \frac{x^2}{2}.$$

Step 2: Passing to the quotient and conclusion. For any real number $x > 0$, divide this inequality by x (since $x > 0$, the inequality direction is preserved):

$$\frac{e^x}{x} > \frac{x}{2}.$$

Now:

$$\lim_{x\to+\infty} \frac{x}{2} = +\infty.$$

By the **comparison theorem** for limits, we conclude irrefutably:

$$\lim_{x\to+\infty} \frac{e^x}{x} = +\infty. \quad \blacksquare$$

Proof of $\lim_{x\to-\infty} xe^x = 0$ *via change of variable:* Set $X = -x$. As $x \to -\infty$, we have $X \to +\infty$. Then:

$$x\,e^x = (-X)\,e^{-X} = -\frac{X}{e^X} = -\frac{\frac{1}{e^X}}{X}.$$

Since $\lim_{X\to+\infty} \frac{e^X}{X} = +\infty$, by the reciprocal of an infinite limit:

$$\lim_{x\to-\infty} x\,e^x = \lim_{X\to+\infty} \left(-\frac{1}{e^X/X}\right) = 0. \quad \blacksquare$$

6. Limit of a Composite Function

A composite function is a cascade: the output of the first machine f becomes the input of the second machine g. This operation is denoted $(g \circ f)(x) = g(f(x))$.

📐 Theorem — Limit of a composite function.

Let a, b, c denote either real numbers, $+\infty$, or $-\infty$. If:

$$\lim_{x \to a} f(x) = b \quad \text{and} \quad \lim_{y \to b} g(y) = c,$$

then:

$$\lim_{x \to a} g(f(x)) = c.$$

🧭 Method — Writing out the limit of a composite on the Bac:

Never compute everything all at once. Clearly separate into two steps by introducing a dummy variable substitution: 1. *Inner step:* calculate $\lim_{x \to a} f(x) = b$. 2. *Outer step:* set $X = f(x)$, and since $X \to b$, calculate $\lim_{X \to b} g(X) = c$. 3. *Conclusion:* By composition of limits, $\lim_{x \to a} g(f(x)) = c$.

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7. Traps to Avoid!

⚠️ Caution, trap! — Crossing an asymptote.

A widespread misconception among students is that a curve "is never allowed to touch its asymptote". This is **true for a vertical asymptote** $x = a$ (because the function is generally undefined there). But it is **completely false for a horizontal or oblique asymptote**! A horizontal asymptote $y = L$ describes solely what occurs **at the far edge of the world** (for $x \to \pm\infty$). In the meantime, for moderate values of x, the curve may intersect the horizontal line once, ten times, or even infinitely many times! *Example:* $f(x) = \frac{\sin x}{x}$ has the x-axis $y = 0$ as a horizontal asymptote, and crosses this axis at every integer multiple of π ($\pi, 2\pi, 3\pi \ldots$).

⚠️ Caution, trap! — Comparative growth applies only at infinity!

Never say: "$\lim_{x \to 0} \frac{e^x}{x} = +\infty$ by comparative growth"! At 0, the numerator tends to $e^0 = 1$ and the denominator to 0. It is a direct quotient of a finite non-zero number by zero ($1/0^+$ or $1/0^-$), not comparative growth! Comparative growth compares the escape velocity toward infinity.

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8. Step-by-Step Worked Examples

Worked Example 1 — Asymptotes and Resolving Indeterminacy at $+\infty$ and $-\infty$

Problem: Consider the function f defined on $\mathbb{R} \setminus \{2\}$ by:

$$f(x) = \frac{3x - 1}{x - 2}.$$

1. Determine the limits of f at the boundaries of its domain of definition ($+\infty$, $-\infty$, and at 2 from below and from above). 2. Deduce the equations of the asymptotes to the graph $\mathcal{C}_f$. 3. Study the relative position of $\mathcal{C}_f$ with respect to its horizontal asymptote.

Solution: 1. **Limits at infinity:** For $x \neq 0$:

$$f(x) = \frac{x\left(3 - \frac{1}{x}\right)}{x\left(1 - \frac{2}{x}\right)} = \frac{3 - \frac{1}{x}}{1 - \frac{2}{x}}.$$

Since $\lim_{x\to\pm\infty} \frac{1}{x} = 0$ and $\lim_{x\to\pm\infty} \frac{2}{x} = 0$, we obtain:

$$\lim_{x\to+\infty} f(x) = 3 \quad \text{and} \quad \lim_{x\to-\infty} f(x) = 3.$$

Limits at 2: Numerator: $\lim_{x\to2}(3x - 1) = 3(2) - 1 = 5 > 0$. Denominator: $\lim_{x\to2}(x - 2) = 0$.

- If $x > 2$, $x - 2 > 0$, so the denominator tends to 0^+. By quotient:

$$\lim_{x\to2^+} f(x) = +\infty.$$

- If $x < 2$, $x - 2 < 0$, the denominator tends to 0^-. By quotient:

$$\lim_{x\to2^-} f(x) = -\infty.$$

2. **Asymptotes:**
   - Since $\lim_{x\to\pm\infty} f(x) = 3$, the line with equation $\boldsymbol{y = 3}$ is a horizontal asymptote to $\mathcal{C}_f$ at $+\infty$ and at $-\infty$.
   - Since $\lim_{x\to2} f(x) = \pm\infty$, the line with equation $\boldsymbol{x = 2}$ is a vertical asymptote to $\mathcal{C}_f$.
3. **Relative position with respect to the horizontal asymptote $y = 3$:** Compute the difference $d(x) = f(x) - 3$:

$$d(x) = \frac{3x - 1}{x - 2} - 3 = \frac{(3x - 1) - 3(x - 2)}{x - 2} = \frac{3x - 1 - 3x + 6}{x - 2} = \frac{5}{x - 2}.$$

   - For all $x \in]2\,;+\infty[$, $x - 2 > 0$, so $d(x) > 0$: the curve $\mathcal{C}_f$ is **strictly above** the asymptote $y = 3$.
   - For all $x \in]-\infty\,;2[$, $x - 2 < 0$, so $d(x) < 0$: the curve $\mathcal{C}_f$ is **strictly below** the asymptote $y = 3$.

Worked Example 2 — The Sign Trap for $\sqrt{x^2}$ at $-\infty$

Problem: Determine the limit at $-\infty$ and at $+\infty$ of the function:

$$f(x) = \sqrt{x^2 + 4x + 5} + x.$$

Solution: At $+\infty$: $\lim_{x\to+\infty}(x^2 + 4x + 5) = +\infty$, so $\lim \sqrt{x^2 + 4x + 5} = +\infty$. Since $\lim_{x\to+\infty} x = +\infty$, by addition:

$$\lim_{x\to+\infty} f(x) = +\infty \quad (\text{no indeterminate form here!}).$$

At $-\infty$: $\lim_{x\to-\infty} \sqrt{x^2 + 4x + 5} = +\infty$ and $\lim_{x\to-\infty} x = -\infty$. This is an indeterminate form $[\infty - \infty]$. Use the **conjugate expression**:

$$f(x) = \frac{\left(\sqrt{x^2 + 4x + 5} + x\right)\left(\sqrt{x^2 + 4x + 5} - x\right)}{\sqrt{x^2 + 4x + 5} - x} = \frac{(x^2 + 4x + 5) - x^2}{\sqrt{x^2 + 4x + 5} - x} = \frac{4x + 5}{\sqrt{x^2 + 4x + 5} - x}.$$

Now factor out the dominant term in the denominator for $x < 0$:

$$\sqrt{x^2 + 4x + 5} = \sqrt{x^2\left(1 + \frac{4}{x} + \frac{5}{x^2}\right)} = |x|\sqrt{1 + \frac{4}{x} + \frac{5}{x^2}}.$$

As $x \to -\infty$, we have $x < 0$, so $|x| = -x$!

$$\sqrt{x^2 + 4x + 5} - x = -x\sqrt{1 + \frac{4}{x} + \frac{5}{x^2}} - x = -x\left[\sqrt{1 + \frac{4}{x} + \frac{5}{x^2}} + 1\right].$$

The numerator is $4x + 5 = x\left(4 + \frac{5}{x}\right)$. Thus, for $x < 0$:

$$f(x) = \frac{x\left(4 + \frac{5}{x}\right)}{-x\left[\sqrt{1 + \frac{4}{x} + \frac{5}{x^2}} + 1\right]} = \frac{4 + \frac{5}{x}}{-\left(\sqrt{1 + \frac{4}{x} + \frac{5}{x^2}} + 1\right)}.$$

Take the limit as $x \to -\infty$:

- The numerator tends to $4 + 0 = 4$.
- The denominator tends to $-(\sqrt{1 + 0 + 0} + 1) = -(1 + 1) = -2$.

By quotient:

$$\lim_{x \to -\infty} f(x) = \frac{4}{-2} = -2.$$

Geometric consequence: The line with equation $y = -2$ is a horizontal asymptote to the curve at $-\infty$.

Worked Example 3 — Comparative Growth and Factoring Exponentials

Problem: Determine the limit at $+\infty$ and at $-\infty$ of the function $f(x) = (2x - 3)e^{-x} + 1$.

Solution: **At $+\infty$:** Write $f(x) = 2xe^{-x} - 3e^{-x} + 1 = \frac{2x}{e^x} - \frac{3}{e^x} + 1 = 2\left(\frac{\frac{1}{e^x}}{x}\right) - \frac{3}{e^x} + 1$. By comparative growth theorems, $\lim_{x \to +\infty} \frac{e^x}{x} = +\infty$, so $\lim_{x \to +\infty} \frac{x}{e^x} = 0$. Moreover, $\lim_{x \to +\infty} e^x = +\infty$, whence $\lim_{x \to +\infty} \frac{3}{e^x} = 0$. By addition of limits:

$$\lim_{x \to +\infty} f(x) = 2(0) - 0 + 1 = 1.$$

The line with equation $y = 1$ is a horizontal asymptote at $+\infty$.

At $-\infty$: Set the substitution $X = -x$. As $x \to -\infty$, $X \to +\infty$. Then: $f(x) = (-2X - 3)e^X + 1 = -2Xe^X - 3e^X + 1$. As $X \to +\infty$, $e^X \to +\infty$ and $-2X - 3 \to -\infty$. By product:

$$\lim_{x \to -\infty} (2x - 3) = -\infty \quad \text{and} \quad \lim_{x \to -\infty} e^{-x} = +\infty \Longrightarrow \lim_{x \to -\infty} (2x - 3)e^{-x} = -\infty.$$

Hence:

$$\lim_{x \to -\infty} f(x) = -\infty.$$

Worked Example 4 — Limit of a Composite with Radical and Rational Functions

Problem: Determine the limit as $x \to +\infty$ of:

$$h(x) = \sqrt{\frac{4x^2 + 1}{x^2 - 3x + 2}}.$$

Solution: Decompose the function by setting $f(x) = \frac{4x^2+1}{x^2-3x+2}$ and $g(X) = \sqrt{X}$, so that $h(x) = g(f(x))$. 1. *Inner limit:* For $x > 0$:

$$f(x) = \frac{x^2\left(4 + \frac{1}{x^2}\right)}{x^2\left(1 - \frac{3}{x} + \frac{2}{x^2}\right)} = \frac{4 + \frac{1}{x^2}}{1 - \frac{3}{x} + \frac{2}{x^2}}.$$

Since $\lim_{x \to +\infty} \frac{1}{x^2} = 0$, $\lim_{x \to +\infty} \frac{3}{x} = 0$, and $\lim_{x \to +\infty} \frac{2}{x^2} = 0$, we have:

$$\lim_{x \to +\infty} f(x) = \frac{4}{1} = 4.$$

2. *Outer limit:* Set $X = f(x)$. As $X \to 4$:

$$\lim_{X \to 4} g(X) = \lim_{X \to 4} \sqrt{X} = \sqrt{4} = 2.$$

3. *Conclusion:* By composition of limits:

$$\lim_{x \to +\infty} h(x) = 2.$$

The line with equation $y = 2$ is a horizontal asymptote to the curve of h at $+\infty$.

Worked Example 5 — Complete Determination of an Oblique Asymptote

Problem: Consider the function f defined on $\mathbb{R} \setminus \{1\}$ by $f(x) = \frac{2x^2-x+3}{x-1}$. 1. Determine real numbers a, b, c such that for all $x \neq 1$: $f(x) = ax + b + \frac{c}{x-1}$. 2. Prove that the line $\mathcal{D}$ with equation $y = 2x + 1$ is an oblique asymptote to the curve $\mathcal{C}_f$ at $+\infty$ and $-\infty$. 3. Specify the relative position of $\mathcal{C}_f$ with respect to $\mathcal{D}$.

Solution: 1. Expand the desired form:

$$ax + b + \frac{c}{x-1} = \frac{(ax + b)(x - 1) + c}{x - 1} = \frac{ax^2 - ax + bx - b + c}{x - 1} = \frac{ax^2 + (b - a)x + (c - b)}{x - 1}.$$

Equating coefficients with $2x^2 - x + 3$:

$$\{a = 2$$
$$b - a = -1 \Longrightarrow b = a - 1 = 2 - 1 = 1$$
$$c - b = 3 \Longrightarrow c = 3 + b = 3 + 1 = 4$$

Thus for all $x \neq 1$:

$$f(x) = 2x + 1 + \frac{4}{x-1}.$$

2. Compute the difference between $f(x)$ and $y = 2x + 1$:

$$f(x) - (2x + 1) = \frac{4}{x-1}.$$

   Now $\lim_{x\to+\infty} \frac{4}{x-1} = 0$ and $\lim_{x\to-\infty} \frac{4}{x-1} = 0$. By definition, the line $\mathcal{D}$ with equation $y = 2x + 1$ is an **oblique asymptote** to the curve $\mathcal{C}_f$ at $+\infty$ and at $-\infty$.
3. **Relative position:** The sign of $f(x) - (2x + 1)$ is the sign of $\frac{4}{x-1}$:
   - If $x > 1$, $x - 1 > 0 \Longrightarrow \frac{4}{x-1} > 0$: $\mathcal{C}_f$ is **strictly above** $\mathcal{D}$.
   - If $x < 1$, $x - 1 < 0 \Longrightarrow \frac{4}{x-1} < 0$: $\mathcal{C}_f$ is **strictly below** $\mathcal{D}$.

Worked Example 6 — Form $\frac{0}{0}$ Resolved via Difference Quotient

Problem: Determine without a calculator $\lim_{x\to 0} \frac{e^{3x}-1}{x}$.

Solution: At $x = 0$, $e^0 - 1 = 0$ and the denominator is 0. This is an indeterminate form $[0/0]$. Define the function $g(x) = e^{3x}$. The function g is differentiable on $\mathbb{R}$, and its derivative is $g'(x) = 3e^{3x}$. Notice that $g(0) = e^0 = 1$. The expression can be rewritten as:

$$\frac{e^{3x} - 1}{x} = \frac{g(x) - g(0)}{x - 0}.$$

By definition of the derivative:

$$\lim_{x\to 0} \frac{g(x) - g(0)}{x - 0} = g'(0) = 3e^{3\times 0} = 3 \times 1 = 3.$$

The requested limit is therefore equal to 3.

Key Takeaways

1. **Definition of limit at infinity:** $\lim_{x\to+\infty} f(x) = L$ if every open interval containing L contains all values $f(x)$ for sufficiently large x.
2. **Asymptotes:**
   - $\lim_{x\to\pm\infty} f(x) = L \Leftrightarrow y = L$ is a horizontal asymptote.
   - $\lim_{x\to a^{\pm}} f(x) = \pm\infty \Leftrightarrow x = a$ is a vertical asymptote.
   - $\lim_{x\to\pm\infty}[f(x) - (ax + b)] = 0 \Leftrightarrow y = ax + b$ is an oblique asymptote.
3. **Relative position:** determined by the sign of $f(x) - y_{\text{asymptote}}$.
4. **Indeterminate Forms:** $\infty - \infty$, $0 \times \infty$, $\frac{\infty}{\infty}$, $\frac{0}{0}$.
5. **Comparative Growth:** $\lim_{x\to+\infty} \frac{e^x}{x^n} = +\infty$ and $\lim_{x\to-\infty} x^n e^x = 0$. In any direct confrontation, exponential growth dominates polynomial growth.
6. **Composite functions:** $\lim_{x\to a} f(x) = b$ and $\lim_{X\to b} g(X) = c \Longrightarrow \lim_{x\to a} g(f(x)) = c$.

Exercises

Level (Direct Applications)

Exercise 45.1 Determine by immediate inspection or elementary limit laws: 1. $\lim_{x\to+\infty}(3x^3-5x+1)$ 2. $\lim_{x\to-\infty}(-2x^4+x^2)$ 3. $\lim_{x\to+\infty}\left(4-\frac{2}{x}+\frac{5}{x^3}\right)$ 4. $\lim_{x\to0^+}\left(\frac{3}{x}-2\right)$

Exercise 45.2 Determine any asymptotes (vertical and horizontal) to the graphs of the following functions: 1. $f(x)=\frac{2x+7}{x-3}$ on $\mathbb{R}\setminus\{3\}$ 2. $g(x)=\frac{1}{x^2-4}$ on $\mathbb{R}\setminus\{-2;2\}$ 3. $h(x)=5+e^{-x}$ on $[0\,;+\infty[$

Exercise 45.3 Compute the left-hand limit and the right-hand limit of $f(x)=\frac{x+1}{x-4}$ at $x_0=4$. Does the function admit a global two-sided limit at 4?

Exercise 45.4 Resolve the indeterminacy for the following rational functions: 1. $\lim_{x\to+\infty}\frac{4x^2-3x+1}{2x^2+5x}$ 2. $\lim_{x\to-\infty}\frac{x^3+2}{5x^4-x}$ 3. $\lim_{x\to+\infty}\frac{x^5-1}{x^2+100}$

Exercise 45.5 Use the comparative growth of the exponential to determine: 1. $\lim_{x\to+\infty}(e^x-3x^4)$ 2. $\lim_{x\to+\infty}\frac{e^x+2}{x^3+1}$ 3. $\lim_{x\to-\infty}(x^2+1)e^x$

Exercise 45.6 Let $f(x)=\sqrt{2x+3}$. Determine $\lim_{x\to3}\frac{f(x)-f(3)}{x-3}$ by recognizing a difference quotient.

Exercise 45.7 Let $f(x)=2x-1+\frac{3}{x^2+1}$. Prove that the line $y=2x-1$ is an oblique asymptote to $\mathcal{C}_f$ at $+\infty$ and determine the relative position of $\mathcal{C}_f$ with respect to this line.

Exercise 45.8 Determine by composition: $\lim_{x\to+\infty}\exp\left(-\frac{1}{x^2}\right)$ and $\lim_{x\to0^+}\exp\left(-\frac{1}{x}\right)$.

Level (Guided Reflection and Baccalaureate Style)

Exercise 45.9 (Baccalaureate Style: Function Study with Exponential). Consider the function f defined on $\mathbb{R}$ by:

$$f(x)=(x-2)e^x+x.$$

1. Determine the limit of f at $-\infty$. 2. Prove that the line (Δ) with equation $y=x$ is an oblique asymptote to the curve $\mathcal{C}_f$ at $-\infty$. 3. Determine the relative position of $\mathcal{C}_f$ with respect to (Δ) on $]-\infty\,;+\infty[$. 4. Determine the limit of f at $+\infty$.

Exercise 45.10 (Conjugate Expression and Asymptotes). Let $f(x)=\sqrt{x^2-2x+3}$. 1. Determine the limit of f at $+\infty$. 2. Prove that for all $x>0$:

$$f(x)-(x-1)=\frac{2}{\sqrt{x^2-2x+3}+(x-1)}.$$

3. Deduce that the line with equation $y=x-1$ is an oblique asymptote to the curve at $+\infty$.

Exercise 45.11 (The Square Root Trap at $-\infty$). Revisit the function of Exercise 45.10 at $-\infty$. Prove that the line with equation $y=-x+1$ is an oblique asymptote to the curve at $-\infty$.

Exercise 45.12 (Population Modeling: The Logistic Curve). The population of an animal species (in thousands) over time $t\geq0$ (in years) is modeled by:

$$N(t)=\frac{10}{1+9e^{-0.5t}}.$$

1. Calculate the initial population $N(0)$. 2. Determine $\lim_{t\to+\infty}N(t)$. Interpret this result in the context of the ecosystem (carrying capacity). 3. Determine the equation of the horizontal asymptote to the graph of N.

Exercise 45.13 (Indeterminate Form Resolved via Clever Factoring). Determine the following limits: 1. $\lim_{x\to1}\frac{x^3-1}{x^2-1}$ 2. $\lim_{x\to4}\frac{\sqrt{x}-2}{x-4}$ 3. $\lim_{x\to0}\frac{\sqrt{1+x}-1}{x}$

Exercise 45.14 (Rational Function and Partial Fraction Decomposition). Let $f(x)=\frac{3x^2+5x-2}{x+1}$. 1. Find three real numbers a,b,c such that $f(x)=ax+b+\frac{c}{x+1}$ for all $x\neq-1$. 2. Deduce all asymptotes to the graph of f. 3. Determine the coordinates of the intersection point of the two asymptotes.

Exercise 45.15 (Composite and Trigonometric Functions). 1. Determine $\lim_{x\to+\infty}\cos\left(\frac{\pi x+1}{2x-3}\right)$. 2. Prove that the function $g(x)=x\sin\left(\frac{1}{x}\right)$ has limit 1 at $+\infty$. (Hint: set $X=1/x$).

Exercise 45.16 (Squeeze Theorem for Functions). Let $f(x)=\frac{2x+\sin(x)}{x-1}$ defined on $]1\,;+\infty[$. 1. Show that for all $x>1$: $\frac{2x-1}{x-1}\leq f(x)\leq\frac{2x+1}{x-1}$. 2. Deduce $\lim_{x\to+\infty}f(x)$.

Exercise 45.17 (Complete Study of a Limit with a Parameter). Let $f_m(x) = \frac{mx^2+4x-1}{2x^2+3}$ where m is a real parameter. Determine the limit of $f_m(x)$ as $x \to +\infty$ according to the values of the real number m.

Level (Challenge and In-Depth Study)

Exercise 45.18 (L'Hôpital's Rule — Higher Mathematics / Beyond Syllabus). Let f and g be two differentiable functions on an open interval containing a, such that $f(a) = g(a) = 0$ and $g'(a) \neq 0$. 1. By writing $\frac{f(x)}{g(x)} = \frac{\frac{f(x)-f(a)}{x-a}}{\frac{g(x)-g(a)}{x-a}}$, prove that:

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}.$$

2. Use this result to compute: $\lim_{x\to 0} \frac{e^{2x} - \cos(x)}{x}$.

Exercise 45.19 (Parabolic Asymptote). Consider the function $f(x) = \frac{x^3+2x^2-1}{x-1}$ for $x \neq 1$. 1. Show that there exists a quadratic polynomial $P(x)$ and a real number R such that $f(x) = P(x) + \frac{R}{x-1}$. 2. Deduce that as $x \to \pm\infty$, the graph of f approaches a parabola with equation $y = P(x)$ (asymptotic parabola).

Exercise 45.20 (A Function with No Limit at Zero). Consider the function $f(x) = \sin\left(\frac{1}{x}\right)$ defined on $\mathbb{R}^*$. 1. Calculate the values of $f(x)$ along the sequences $u_n = \frac{1}{2n\pi}$ and $v_n = \frac{1}{2n\pi+\pi/2}$ for $n \geq 1$. 2. What are the limits of (u_n) and (v_n) as $n \to +\infty$? 3. Deduce that the function f has no limit at 0.

Exercise 45.21 (Generalized Comparative Growth: Proof for all $n \in \mathbb{N}^*$). It is known that $\lim_{x\to+\infty} \frac{e^x}{x} = +\infty$. 1. Prove that for all $n \in \mathbb{N}^*$ and all $x > 0$:

$$\frac{e^x}{x^n} = \frac{1}{n^n}\left(\frac{e^{x/n}}{x/n}\right)^n.$$

2. Setting $u = \frac{x}{n}$, deduce that $\lim_{x\to+\infty} \frac{e^x}{x^n} = +\infty$.

Exercise 45.22 (The Devil's Staircase and Limits at Infinity). Let $f(x) = x - \lfloor x \rfloor$ where $\lfloor x \rfloor$ denotes the floor function of x (the greatest integer less than or equal to x). 1. Sketch the graph of f on $[-2\,;3]$. 2. Does the function admit a limit as $x \to +\infty$? 3. Determine the left-hand limit and right-hand limit of f at every integer $k \in \mathbb{Z}$.

Solutions

Solutions for Level

- **45.1:**
  1. $\lim(3x^3 - 5x + 1) = \lim 3x^3 = +\infty$.
  2. $\lim(-2x^4 + x^2) = \lim -2x^4 = -\infty$ (since $x^4 \to +\infty$ and $-2 < 0$).
  3. $\lim(4 - 0 + 0) = 4$.
  4. For $x \to 0^+$, $3/x \to +\infty$, so $\lim = +\infty$.
- **45.2:**
  1. $\lim_{x\to\pm\infty} f(x) = 2 \Longrightarrow y = 2$ is a horizontal asymptote. $\lim_{x\to 3} f(x) = \pm\infty \Longrightarrow x = 3$ is a vertical asymptote.
  2. $\lim_{x\to\pm\infty} g(x) = 0 \Longrightarrow y = 0$ is a horizontal asymptote. $x^2 - 4 = 0 \Leftrightarrow x = \pm 2$, yielding two vertical asymptotes: $x = 2$ and $x = -2$.
  3. $\lim_{x\to+\infty} e^{-x} = 0 \Longrightarrow \lim_{x\to+\infty} h(x) = 5 \Longrightarrow y = 5$ is a horizontal asymptote at $+\infty$.
- **45.3:** Numerator: $4 + 1 = 5 > 0$. Denominator: $x - 4 \to 0$.
  - If $x > 4$, $x - 4 > 0 \Longrightarrow \lim_{x\to 4^+} f(x) = +\infty$.
  - If $x < 4$, $x - 4 < 0 \Longrightarrow \lim_{x\to 4^-} f(x) = -\infty$. Since these one-sided limits differ, no global two-sided limit exists at 4.
- **45.4:**
  1. $\lim \frac{x^2(4-3/x+1/x^2)}{x^2(2+5/x)} = \frac{4}{2} = 2$.
  2. $\lim \frac{x^3}{5x^4} = \lim \frac{1}{5x} = 0$.

3. $\lim \frac{x^5}{x^2} = \lim x^3 = +\infty$.

- **45.5:**
  1. $e^x - 3x^4 = x^4\left(\frac{e^x}{x^4} - 3\right) \to +\infty \times (+\infty - 3) = +\infty$.
  2. $\frac{e^x(1+2e^{-x})}{x^3(1+x^{-3})} = \frac{e^x}{x^3} \times \frac{1+2e^{-x}}{1+x^{-3}} \to +\infty \times 1 = +\infty$.
  3. $(x^2+1)e^x = x^2e^x + e^x \to 0 + 0 = 0$ by comparative growth at $-\infty$.
- **45.6:** $f(3) = \sqrt{9} = 3$. The limit is the derivative $f'(3)$. Since $f'(x) = \frac{2}{2\sqrt{2x+3}} = \frac{1}{\sqrt{2x+3}}$, we have $f'(3) = \frac{1}{3}$.
- **45.7:** $f(x) - (2x-1) = \frac{3}{x^2+1} \to 0$ at $+\infty$, so $y = 2x - 1$ is an oblique asymptote. Since $x^2 + 1 > 0$, the difference is strictly positive: $\mathcal{C}_f$ is strictly above the asymptote.
- **45.8:**
  - $\lim_{x\to+\infty}(-1/x^2) = 0$, so $\lim \exp(-1/x^2) = e^0 = 1$.
  - $\lim_{x\to 0^+}(-1/x) = -\infty$, so $\lim_{X\to-\infty} e^X = 0$.

Solutions for Level

- **45.9:**
  1. At $-\infty$, $(x-2)e^x = xe^x - 2e^x \to 0 - 0 = 0$ by comparative growth. Since $\lim_{x\to-\infty} x = -\infty$, we have $\lim_{x\to-\infty} f(x) = -\infty$.
  2. $f(x) - x = (x-2)e^x \to 0$ as $x \to -\infty$. Thus $(\Delta) : y = x$ is an oblique asymptote at $-\infty$.
  3. Sign of $f(x) - x = (x-2)e^x$: since $e^x > 0$, the sign matches $x - 2$. On $]-\infty\,;2[$, $\mathcal{C}_f$ is below (Δ). On $]2\,;+\infty[$, $\mathcal{C}_f$ is above (Δ). At $x = 2$, the curve crosses its asymptote at $(2;2)$.
  4. At $+\infty$, $x - 2 \to +\infty$ and $e^x \to +\infty$, so $(x-2)e^x \to +\infty$, whence $\lim f(x) = +\infty$.
- **45.10:**
  1. $\lim_{x\to+\infty}(x^2 - 2x + 3) = +\infty \Longrightarrow \lim_{x\to+\infty} f(x) = +\infty$.
  2. For $x > 0$:

$$(x-1) = \frac{(\sqrt{x^2-2x+3} - (x-1))(\sqrt{x^2-2x+3} + (x-1))}{\sqrt{x^2-2x+3} + (x-1)} = \frac{(x^2-2x+3) - (x-1)^2}{\sqrt{x^2-2x+3} + (x-1)} = \frac{x^2-2x+3-(x^2-2x+1)}{\sqrt{x^2-2x+3} + (x-1)} = \frac{2}{\sqrt{x^2-2x+3}}$$

  3. As $x \to +\infty$, the denominator tends to $+\infty$, so the quotient tends to 0. Thus $y = x - 1$ is an oblique asymptote at $+\infty$.
- **45.11:** As $x \to -\infty$, set $X = -x > 0$. $f(x) = \sqrt{X^2 + 2X + 3}$. The conjectured asymptote is $y = -x + 1 = X +$
  1. $f(x) - (-x+1) = \sqrt{X^2+2X+3} - (X+1) = \frac{(X^2+2X+3)-(X+1)^2}{\sqrt{X^2+2X+3}+(X+1)} = \frac{2}{\sqrt{X^2+2X+3}+X+1} \underset{X\to+\infty}{\to} 0$. Hence $y = -x + 1$ is indeed an oblique asymptote at $-\infty$.
- **45.12:**
  1. $N(0) = \frac{10}{1+9e^0} = \frac{10}{10} = 1$ thousand individuals (i.e., 1000 individuals).
  2. As $t \to +\infty$, $-0.5t \to -\infty$, so $e^{-0.5t} \to 0$. Thus $\lim_{t\to+\infty} N(t) = \frac{10}{1+0} = 10$ thousand (10,000 individuals). The habitat can support a maximum of 10,000 animals due to limited resources (food, space): this is the environmental carrying capacity.
  3. The horizontal asymptote has equation $y = 10$.
- **45.13:**
  1. $x^3 - 1 = (x-1)(x^2+x+1)$ and $x^2 - 1 = (x-1)(x+1)$. For $x \neq 1$, $\frac{x^3-1}{x^2-1} = \frac{x^2+x+1}{x+1} \underset{x\to1}{\to} \frac{3}{2}$.
  2. $\frac{\sqrt{x}-2}{x-4} = \frac{\sqrt{x}-2}{(\sqrt{x}-2)(\sqrt{x}+2)} = \frac{1}{\sqrt{x}+2} \underset{x\to4}{\to} \frac{1}{4}$.
  3. Conjugate expression: $\frac{\sqrt{1+x}-1}{x} = \frac{(1+x)-1}{x(\sqrt{1+x}+1)} = \frac{1}{\sqrt{1+x}+1} \underset{x\to0}{\to} \frac{1}{2}$. (Or difference quotient of $\sqrt{1+x}$ at 0).
- **45.14:**
  1. $ax + b + \frac{c}{x+1} = \frac{ax^2+(a+b)x+b+c}{x+1}$. Identification: $a = 3$, $a + b = 5 \Longrightarrow b = 2$, $b + c = -2 \Longrightarrow c = -4$. Thus $f(x) = 3x + 2 - \frac{4}{x+1}$.
  2. Vertical asymptote: $x = -1$. Oblique asymptote: $y = 3x + 2$ (since $\lim_{x\to\pm\infty} \frac{-4}{x+1} = 0$).
  3. The vertical asymptote has $x = -1$. On the oblique asymptote, $y = 3(-1) + 2 = -1$. The intersection point is $I(-1;-1)$.
- **45.15:**

1. $\frac{\pi x+1}{2x-3} = \frac{\pi+1/x}{2-3/x} \to \frac{\pi}{2}$. By continuity of cosine: $\lim = \cos(\pi/2) = 0$.
2. Setting $X = 1/x$, as $x \to +\infty$, $X \to 0^+$. The expression becomes $\frac{\sin X}{X}$, whose limit at 0 is $\sin'(0) = \cos(0) = 1$.

- **45.16:**
1. For all real x, $-1 \leq \sin(x) \leq 1$. Adding $2x$: $2x - 1 \leq 2x + \sin(x) \leq 2x + 1$. Since $x > 1$, $x - 1 > 0$, divide without reversing inequalities.
2. $\lim_{x\to+\infty} \frac{2x-1}{x-1} = 2$ and $\lim_{x\to+\infty} \frac{2x+1}{x-1} = 2$. By the squeeze theorem, $\lim f(x) = 2$.

- **45.17:** $f_m(x) = \frac{x^2(m+4/x-1/x^2)}{x^2(2+3/x^2)} = \frac{m+4/x-1/x^2}{2+3/x^2} \underset{x\to+\infty}{\to} \frac{m}{2}$. The limit is therefore $\frac{m}{2}$ for any value of $m \in \mathbb{R}$.

Solutions for Level

- **45.18:**
1. Since $f(a) = g(a) = 0$, $\frac{f(x)}{g(x)} = \frac{\frac{f(x)-f(a)}{x-a}}{\frac{g(x)-g(a)}{x-a}}$. Taking the quotient of difference quotient limits, since $g'(a) \neq 0$, the limit equals $\frac{f'(a)}{g'(a)}$.
2. With $f(x) = e^{2x} - \cos(x)$ and $g(x) = x$, we have $f(0) = 1 - 1 = 0$ and $g(0) = 0$. $f'(x) = 2e^{2x} + \sin(x) \Longrightarrow f'(0) = 2 + 0 = 2$. $g'(x) = 1 \Longrightarrow g'(0) = 1$. The limit is $\frac{f'(0)}{g'(0)} = \frac{2}{1} = 2$.

- **45.19:**
1. Polynomial long division of $x^3 + 2x^2 - 1$ by $x - 1$: $x^3 + 2x^2 - 1 = (x - 1)(x^2 + 3x + 3) + 2$. Thus $f(x) = x^2 + 3x + 3 + \frac{2}{x-1}$. Here $P(x) = x^2 + 3x + 3$ and $R = 2$.
2. $\lim_{x\to\pm\infty}[f(x) - P(x)] = \lim_{x\to\pm\infty} \frac{2}{x-1} = 0$. The parabola with equation $y = x^2 + 3x + 3$ closely hugs the curve at infinity.

- **45.20:**
1. $f(u_n) = \sin(2n\pi) = 0$. $f(v_n) = \sin(2n\pi + \pi/2) = \sin(\pi/2) = 1$.
2. $\lim u_n = 0$ and $\lim v_n = 0$.
3. If f had a limit L at 0, then for any sequence tending to 0, the images would have to tend to L. But $f(u_n) \to 0$ and $f(v_n) \to 1$. Since $0 \neq 1$, f oscillates infinitely fast near 0 and has no limit at 0.

- **45.21:**
1. $\frac{1}{n^n}\left(\frac{e^{x/n}}{x/n}\right)^n = \frac{1}{n^n}\frac{(e^{x/n})^n}{(x/n)^n} = \frac{e^x}{n^n \frac{x^n}{n^n}} = \frac{e^x}{x^n}$.
2. Set $u = x/n$. As $x \to +\infty$, $u \to +\infty$. Since $\lim_{u\to+\infty} \frac{e^u}{u} = +\infty$, we have $\lim_{u\to+\infty} \left(\frac{e^u}{u}\right)^n = +\infty$. Multiplying by the strictly positive constant $\frac{1}{n^n}$, we conclude that $\lim_{x\to+\infty} \frac{e^x}{x^n} = +\infty$.

- **45.22:**
1. $f(x)$ is the fractional part of x. It is a sawtooth wave, periodic with period 1, composed of line segments of slope 1 equal to 0 at k and approaching 1 as $x \to k^-$.
2. The function oscillates periodically between 0 and 1 as $x \to +\infty$, so it admits no limit at $+\infty$.
3. Let $k \in \mathbb{Z}$. For $x \in [k\,;k+1[$, $\lfloor x \rfloor = k$, so $f(x) = x - k \underset{x\to k^+}{\to} 0$. For $x \in]k-1\,;k[$, $\lfloor x \rfloor = k - 1$, so $f(x) = x - (k-1) \underset{x\to k^-}{\to} k - k + 1 = 1$. The right-hand limit is 0 and the left-hand limit is 1: the curve undergoes a jump of height 1 at each integer.

The Question That Opens What Follows

We have just seen that some functions undergo sudden disruptions: vertical asymptotes where curves tear apart toward infinity, or staircase steps where curves jump abruptly from one elevation to another. Yet most laws of nature — the trajectory of a tennis ball, the water level rising in a vase, the growth of a plant — know no such instantaneous leaps. How can we express with the highest mathematical rigor the idea that a curve can be drawn **in a single stroke of the pen, without ever lifting the nib from the page**? And why does this fundamental property guarantee that in crossing a river from one bank to the other, one is mathematically certain to pass through every intermediate current?

This is the essence of **continuity** and the prestigious **Intermediate Value Theorem (IVT)**. See you in Chapter 46.

CHAPTER 46: The Unbroken Thread

> "Natura non facit saltus — Nature does not make jumps, said Gottfried Wilhelm Leibniz. If a train leaves Paris at noon and enters the station in Lyon at two o'clock, it did not vanish at Dijon only to reappear suddenly fifty kilometers down the line. It pressed against every meter, every millimeter of track without ever lifting its wheels. In mathematics, this fidelity to the path is called continuity, and it possesses the magical power to guarantee that an equation has a solution even when one is incapable of calculating it by hand."

What You Will Discover

- The driving intuition of **continuity**: a graph drawn in a single stroke of the pen without ever lifting the nib from the paper, and its formal formulation $\lim_{x \to x_0} f(x) = f(x_0)$.
- Why virtually all functions in physics and natural science are continuous, and the crucial implication: **differentiable implies continuous** (though the converse is false!).
- The **Intermediate Value Theorem (IVT)**: the absolute certainty of crossing a river when traveling from one bank to the other.
- The **corollary of the bijection** (IVT for strictly monotonic functions): the #1 tool on the Baccalaureate exam to prove the existence and uniqueness of the solution to an equation $f(x) = k$.
- The **bisection method** (dichotomy) for numerical bracketing and its implementation in **Python**.
- Recurrence sequences $u_{n+1} = f(u_n)$: how the continuity of f forces the sequence limit to be a **fixed point** ($f(L) = L$).

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1. The Problem: Crossing the River Without Teleporting

1.1 The Hiker and the Invisible Bridge

Imagine a hiker departing from a valley at an elevation of 500 meters at 8:00 AM. At noon, she reaches the summit of the mountain at 2200 meters. Can you state with certainty that at some precise moment during the morning, her altimeter read exactly 1500 meters?

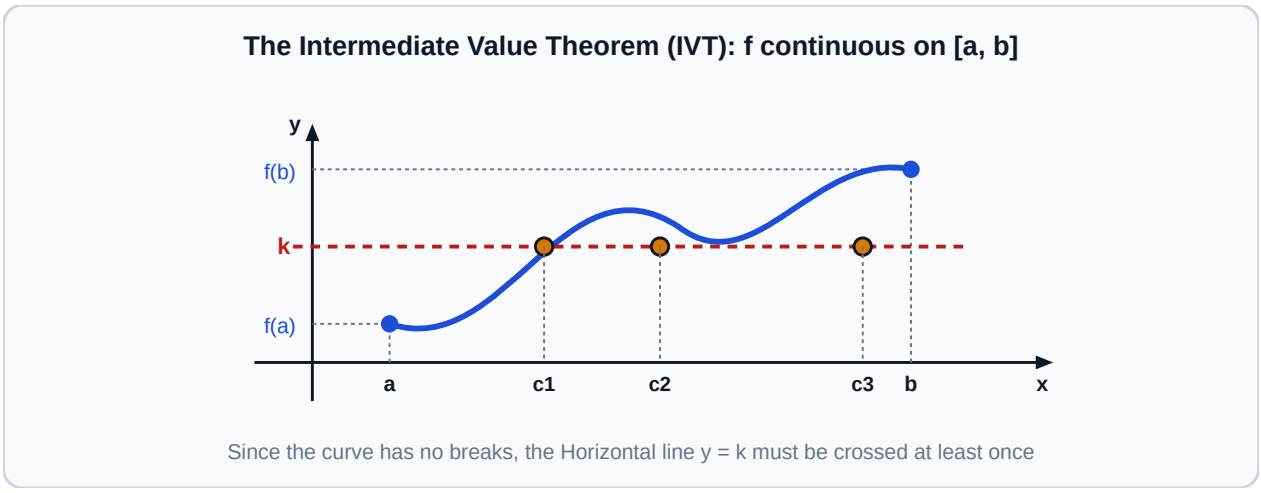

Figure 101: Intermediate Value Theorem (IVT) on a closed bounded interval

The answer is undeniably **yes**. Unless the hiker possessed a science-fiction teleportation device, her body moved continuously through space. She could not pass below the 1500 m mark and then above it without **intersecting** the horizontal plane at 1500 m.

Now consider another phenomenon: parking garage fees.

- From 0 to 1 hour: €2.00.
- From 1 hour to 2 hours: €4.00.

If you parked for 1 hour and 1 second, the machine directly charges you €4.00. The price took an **abrupt jump**. It never took the value €2.87 or €3.50. The pricing function is **discontinuous**.

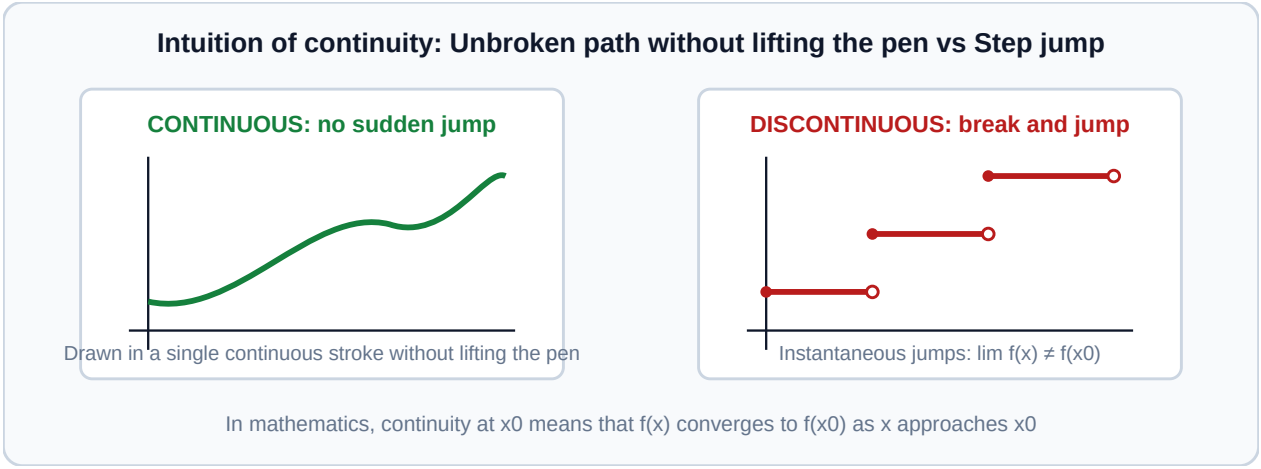

Figure 102: Mathematical continuity: unbroken trajectory vs jump discontinuity

The fundamental question of mathematical analysis is this: *what mathematical guarantees do we possess when a curve has no breaks?*

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2. Continuity: From the Stroke of a Pencil to the Formal Definition

2.1 Definition at a Point

Intuitively, a function is continuous if one can trace its graph without lifting the pencil. Yet how can a computer or an algebraic equation "know" whether a pencil was lifted? Mathematicians express this kinematic intuition through an equality of limits:

Definition — Continuity at a point.

Let f be a function defined on an open interval containing x_0. We say that f **is continuous at** x_0 if:

$$\lim_{x \to x_0} f(x) = f(x_0).$$

Deconstruction of the condition: For this equality to hold, three conditions must be satisfied simultaneously: 1. $f(x_0)$ exists (the function is well-defined at x_0); 2. $f(x)$ admits a finite limit as $x \to x_0$; 3. This limit rigorously coincides with the actual value $f(x_0)$.

If $\lim_{x \to x_0^-} f(x) \neq \lim_{x \to x_0^+} f(x)$, or if the limit does not equal $f(x_0)$, the curve exhibits a disruption (a hole or a jump): f is said to be **discontinuous** at x_0.

2.2 Continuity on an Interval

Definition — Continuity on an interval.

A function f is said to be **continuous on an interval** I if it is continuous at every point $x_0 \in I$.

2.3 Continuity of Standard Functions

All standard reference functions studied in high school are continuous on every interval where they are defined:

- **Polynomial functions** (lines, parabolas…) are continuous on $\mathbb{R}$.
- **Rational functions** (quotients of polynomials) are continuous on each interval of their domain.

- The **square root function** $x \mapsto \sqrt{x}$ is continuous on $[0\,;+\infty[$.
- The **absolute value function** $x \mapsto |x|$ is continuous on $\mathbb{R}$.
- The **exponential function** $x \mapsto e^x$ is continuous on $\mathbb{R}$.
- Trigonometric functions cos and sin are continuous on $\mathbb{R}$.

Furthermore, sums, products, quotients (provided the denominator is non-zero), and **composites** of continuous functions are continuous functions.

2.4 The Vital Link: Differentiability and Continuity

📐 Fundamental Theorem — Differentiable implies continuous.

If a function f is differentiable at a point x_0, then it is **necessarily continuous** at x_0. If f is differentiable on an interval I, then it is continuous on I.

Proof (Required knowledge).

Assume f is differentiable at x_0. This means the difference quotient has a finite limit denoted $f'(x_0)$:

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0).$$

For all $x \neq x_0$, write:

$$f(x) - f(x_0) = \frac{f(x) - f(x_0)}{x - x_0} \times (x - x_0).$$

Take the limit as $x \to x_0$ applying the product rule for limits:

$$\lim_{x \to x_0} [f(x) - f(x_0)] = \lim_{x \to x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} \right) \times \lim_{x \to x_0} (x - x_0) = f'(x_0) \times 0 = 0.$$

We immediately deduce:

$$\lim_{x \to x_0} f(x) = f(x_0).$$

The function f is therefore continuous at x_0. ■

⚠ Caution, trap! — The converse is FALSE!

A function can be perfectly continuous at a point without being differentiable there. **Classic Counterexample #1: The absolute value function $f(x) = |x|$ at 0.** Its graph forms a sharp corner (a V). It is drawn without lifting the pencil and is continuous at 0 because $\lim_{x \to 0} |x| = 0 = |0|$. However, its difference quotient at 0 is:

- $\frac{|x|-0}{x-0} = \frac{x}{x} = 1$ for $x > 0$;
- $\frac{|x|-0}{x-0} = \frac{-x}{x} = -1$ for $x < 0$. The right-hand limit (1) and left-hand limit (-1) differ: f is **not differentiable at 0**. **Counterexample #2:** $f(x) = \sqrt{x}$ at 0 is continuous at 0 but not differentiable (vertical tangent).

⚠ ⚠ Existence vs Uniqueness: Never Confound the IVT with Its Monotone Corollary!

This represents one of the most common pitfalls in advanced calculus examinations:

- The standard **Intermediate Value Theorem (IVT)** guarantees solely the **existence of at least one solution** to the equation $f(x) = k$. If the curve oscillates, there may be 3, 7, or infinitely many crossing points!
- To establish the **uniqueness** of that solution, the IVT alone is completely insufficient: you must invoke its corollary (the **bijection theorem**), which demands the strict condition that the function be **strictly monotonic** (strictly increasing or strictly decreasing) throughout the interval.

3. The Intermediate Value Theorem (IVT)

3.1 The General Statement

The Czech mathematician Bernard Bolzano (1817) and the French mathematician Augustin-Louis Cauchy (1821) provided the first rigorous proofs of this clear geometric principle:

Intermediate Value Theorem (IVT).

Let f be a function that is **continuous** on a closed interval $[a\,;b]$. For every real number k between $f(a)$ and $f(b)$, there exists **at least one real number c** belonging to the interval $[a\,;b]$ such that:

$$f(c) = k.$$

Crucial remarks: - The theorem asserts the **existence** of at least one solution, but it does not specify the exact count: on the diagram above, there are two solutions c_1 and c_2. - If f were not continuous, the curve could jump over the horizontal line $y = k$ without ever touching it!

3.2 The Special Case of Zero: Bolzano's Theorem

This is the most common form in practical applications:

Corollary — Sign change.

Let f be a continuous function on $[a\,;b]$. If $f(a)$ and $f(b)$ have **opposite signs** (that is, $f(a) \times f(b) \leq 0$), then there exists at least one real number $c \in [a\,;b]$ such that $f(c) = 0$.

In plain words: To pass from a negative altitude (underwater) to a positive altitude (in the air), a continuous path must inevitably pierce the water's surface ($y = 0$).

---

4. The Bijection Theorem (IVT for Strictly Monotonic Functions)

On exams, one rarely seeks merely "at least one solution": one seeks to prove that there is **exactly one** solution. To secure uniqueness, one must prevent the curve from reversing direction. We thus require the function to be **strictly monotonic** (always strictly increasing or always strictly decreasing).

Theorem — Corollary of the IVT (Bijection Theorem).

Let f be a function defined on an interval $I = [a\,;b]$. If: 1. f is **continuous** on $[a\,;b]$; 2. f is **strictly monotonic** (strictly increasing or strictly decreasing) on $[a\,;b]$;

Then, for every real number k between $f(a)$ and $f(b)$, the equation:

$$f(x) = k$$

admits **a unique solution** α in the interval $[a\,;b]$.

Extension to Open or Infinite Intervals

The theorem remains fully valid on an open interval $]a\,;b[$ or an infinite interval like $]-\infty\,;+\infty[$, $[0\,;+\infty[$, etc. One simply replaces the boundary values $f(a)$ and $f(b)$ with the **limits** of the function at the endpoints of the interval: If f is continuous and strictly increasing on $]a\,;b[$, then for every $k \in]\lim_{x\to a^+} f(x)\,;\lim_{x\to b^-} f(x)[$, the equation $f(x) = k$ admits a unique solution in $]a\,;b[$.

Method — Canonical Baccalaureate solution layout:

To justify that the equation $f(x) = k$ admits a unique solution α on an interval I: 1. **Continuity:** State that f is continuous on I (e.g., as a sum, product, or quotient of standard functions, or because it is differentiable there). 2. **Strict monotonicity:** Indicate that f is strictly increasing (or decreasing) on I, justified by the sign of its derivative $f'(x)$. 3. **Membership in target range:** Compute the endpoint values (or their limits) and verify that k belongs to the image interval: $k \in [f(a)\,;f(b)]$ (or $]\lim_a f\,;\lim_b f[$). 4. **Conclusion:** Conclude: *"According to the corollary of the intermediate value theorem (bijection theorem), the equation $f(x) = k$ admits a unique solution α in I."*

---

5. Numerical Approximation: The Bisection Algorithm in Python

The IVT proves that a solution α exists and is unique, but does not provide its exact numerical closed form (which is often impossible to express using basic fractions or radicals). To locate an approximate value of α with surgical precision (e.g., 10^{-6}), one uses the **bisection method** (or dichotomy, from Greek *dicha*, "in two", and *tomein*, "to cut").

5.1 Principle of Successive Halving

Suppose the root α is trapped in an interval $[a\,;b]$ with $f(a) \times f(b) < 0$. 1. Compute the midpoint of the interval: $m = \frac{a+b}{2}$. 2. Examine the sign of $f(m)$:

- If $f(a)$ and $f(m)$ have opposite signs ($f(a) \times f(m) < 0$), then the root lies in the left half: replace b with m.
- Otherwise, the root lies in the right half: replace a with m.

3. At each step, the interval width is **halved**. After n steps, the uncertainty on the root is reduced to $\frac{b-a}{2^n}$.

Step-by-Step Method — Root Finding by Bisection (Dichotomy)

- Step 0: [α] Width L
- Step 1: [α] Width L/2
- Step 2: [α] Width L/4

5.2 Python Implementation

Consider the equation $x^3 + x - 1 = 0$ on $[0\,;1]$. The function $f(x) = x^3 + x - 1$ is continuous, strictly increasing, with $f(0) = -1 < 0$ and $f(1) = 1 > 0$. There exists a unique solution $\alpha \in [0\,;1]$.

```python
def f(x):
    """Function whose zero is sought."""
    return x**3 + x - 1

def bisection(a, b, precision):
    """Finds an interval [a, b] containing a root with width <= precision."""
    while (b - a) > precision:
        m = (a + b) / 2
        if f(a) * f(m) <= 0:
            b = m
        else:
            a = m
    return a, b

# Search for a bracket to 10^-4
lower, upper = bisection(0.0, 1.0, 1e-4)
print(f"Obtained interval: {lower:.5f} < alpha < {upper:.5f}")
# Prints: 0.68231 < alpha < 0.68237
```

Remarkable speed: To achieve a precision of 10^{-6} (one millionth) starting from an initial interval of length 1: Since $2^{20} = 1,048,576 > 10^6$, **just 20 iterations** suffice to deliver six accurate decimal places!

---

6. Recurrence Sequences $u_{n+1} = f(u_n)$ and Fixed Point Finding

In Chapter 44, we studied bounded monotonic sequences that converged to a limit L. But how can we calculate this limit L when the sequence is generated by a recurrence relation $u_{n+1} = f(u_n)$?
Here is where continuity demonstrates its full power!

Fixed-Point Theorem.

Let (u_n) be a sequence defined by u_0 and the recurrence relation $u_{n+1} = f(u_n)$ for all $n \in \mathbb{N}$. If: 1. The sequence (u_n) **converges** to a real number L; 2. The function f is **continuous** at L;
Then the limit L is a solution to the equation:

$$f(L) = L.$$

Such a number L is called a **fixed point** of the function f.

Proof.

Since $\lim_{n\to+\infty} u_n = L$, the shifted sequence (u_{n+1}) clearly converges to the same limit: $\lim_{n\to+\infty} u_{n+1} = L$.
On the other hand, since f is continuous at L, by composite limits:

$$\lim_{n\to+\infty} f(u_n) = f\left(\lim_{n\to+\infty} u_n\right) = f(L).$$

Now, for every integer n, $u_{n+1} = f(u_n)$. Taking the limit on both sides of the equality, we necessarily obtain:

$$L = f(L). \quad \blacksquare$$

Geometric interpretation (the cobweb / staircase diagram): The limit L is the x-coordinate of an intersection point between the function curve $y = f(x)$ and the main diagonal line $\boldsymbol{y = x}$.

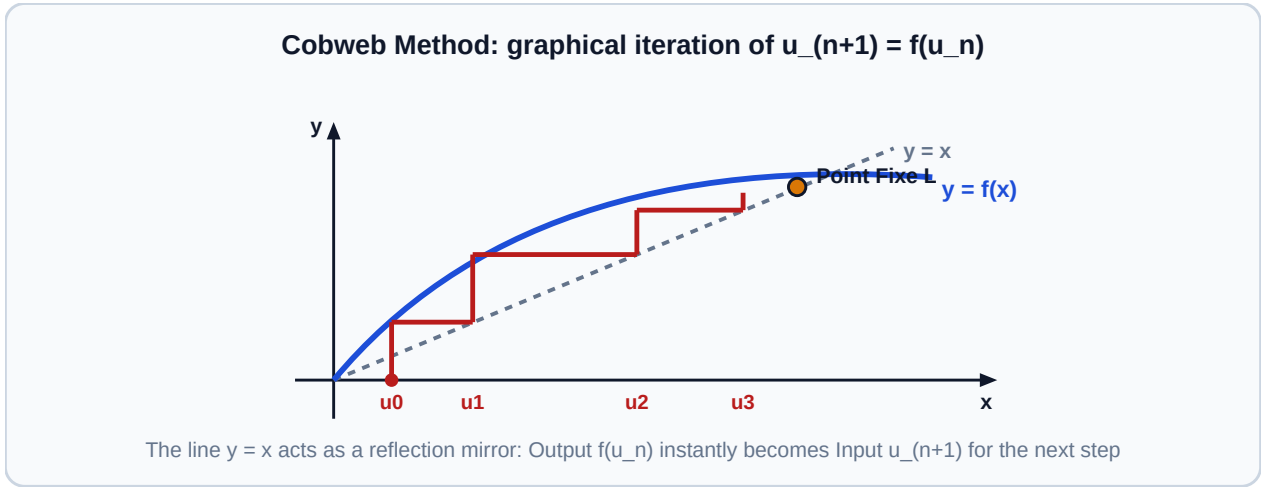

Figure 103: Cobweb method: graphical iteration of the sequence $u_{n+1} = f(u_n)$

---

7. Traps to Avoid!

> ⚠ **Caution, trap! — Forgetting strict monotonicity.**
>
> If you state only that "f is continuous on $[a;b]$ and k lies between $f(a)$ and $f(b)$", the IVT guarantees only the existence of **at least one** solution. To assert that there is **exactly one**, explicit mention of **strict monotonicity** is absolutely essential. Omitting it results in lost points on exams.

> ⚠ **Caution, trap! — Non-continuous functions.**
>
> Consider the function $f(x) = \frac{1}{x}$ on $[-1\,;1] \setminus \{0\}$. We have $f(-1) = -1 < 0$ and $f(1) = 1 > 0$. Yet the equation $f(x) = 0$ has **no solution**! Why does the IVT fail to apply? Because the function is **not continuous at 0** (it is not even defined there). The thread is severed in the middle.

---

8. Step-by-Step Worked Examples

Worked Example 1 — Standard Exam Application of the IVT Corollary

Problem: Consider the function f defined on $\mathbb{R}$ by $f(x) = 2x^3 - 3x^2 - 12x + 5$. 1. Construct the complete variation table of f on $[-3\,;4]$. 2. Prove that the equation $f(x) = 0$ admits exactly three real solutions in the interval $[-3\,;4]$.

Solution: 1. **Differentiation and variations:** The function f is a polynomial, hence differentiable on $\mathbb{R}$. For all $x \in [-3\,;4]$:

$$f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2).$$

Factoring $x^2 - x - 2$: discriminant is $\Delta = (-1)^2 - 4(1)(-2) = 1 + 8 = 9 = 3^2$. The roots are $x_1 = \frac{1-3}{2} = -1$ and $x_2 = \frac{1+3}{2} = 2$. Thus $f'(x) = 6(x+1)(x-2)$. Compute key values:

- $f(-3) = 2(-27) - 3(9) - 12(-3) + 5 = -54 - 27 + 36 + 5 = -40$.
- $f(-1) = 2(-1) - 3(1) - 12(-1) + 5 = -2 - 3 + 12 + 5 = 12$.
- $f(2) = 2(8) - 3(4) - 12(2) + 5 = 16 - 12 - 24 + 5 = -15$.
- $f(4) = 2(64) - 3(16) - 12(4) + 5 = 128 - 48 - 48 + 5 = 37$.

Variation table:

- On $[-3\,;-1]$, $f'(x) > 0$, so f is strictly increasing from -40 to 12.
- On $[-1\,;2]$, $f'(x) < 0$, so f is strictly decreasing from 12 to -15.
- On $[2\,;4]$, $f'(x) > 0$, so f is strictly increasing from -15 to 37.

2. **Applying the IVT corollary to each subinterval:**
   - **On $[-3\,;-1]$:** f is continuous and strictly increasing. Since $0 \in [-40\,;12]$, by the IVT corollary, $f(x) = 0$ has a unique solution $\alpha_1 \in [-3\,;-1]$.
   - **On $[-1\,;2]$:** f is continuous and strictly decreasing. Since $0 \in [-15\,;12]$, $f(x) = 0$ has a unique solution $\alpha_2 \in [-1\,;2]$.
   - **On $[2\,;4]$:** f is continuous and strictly increasing. Since $0 \in [-15\,;37]$, $f(x) = 0$ has a unique solution $\alpha_3 \in [2\,;4]$.

   Conclusion: The equation $f(x) = 0$ has exactly **three distinct solutions** in the interval $[-3\,;4]$.

Worked Example 2 — Equation with Exponential on an Unbounded Interval

Problem: Prove that the equation $e^x = \frac{2}{x}$ admits a unique real solution on $]0\,;+\infty[$.

Solution: For $x > 0$, the equation is equivalent to $e^x - \frac{2}{x} = 0$. Consider the function g defined on $]0\,;+\infty[$ by:

$$g(x) = e^x - \frac{2}{x}.$$

1. **Continuity:** g is the sum of the exponential function and the rational function $x \mapsto -2/x$, both of which are continuous on $]0\,;+\infty[$. Hence g is continuous on $]0\,;+\infty[$. 2. **Variations:** g is differentiable on $]0\,;+\infty[$:

$$g'(x) = e^x - \left(-\frac{2}{x^2}\right) = e^x + \frac{2}{x^2}.$$

For all $x > 0$, $e^x > 0$ and $\frac{2}{x^2} > 0$, so $g'(x) > 0$. The function g is therefore **strictly increasing** on $]0\,;+\infty[$. 3. **Endpoint limits:**

- At 0^+: $\lim_{x\to 0^+} e^x = 1$ and $\lim_{x\to 0^+} \frac{2}{x} = +\infty$, so $\lim_{x\to 0^+} g(x) = 1 - (+\infty) = -\infty$.
- At $+\infty$: $\lim_{x\to+\infty} e^x = +\infty$ and $\lim_{x\to+\infty} \frac{2}{x} = 0$, so $\lim_{x\to+\infty} g(x) = +\infty$.

4. **Conclusion:** The function g is continuous and strictly increasing on $]0\,;+\infty[$. It establishes a bijection from $]0\,;+\infty[$ onto the image interval $]\lim_{0^+} g\,;\lim_{+\infty} g[=]-\infty\,;+\infty[=\mathbb{R}$. Since $0 \in \mathbb{R}$, by the IVT corollary, the equation $g(x) = 0$ (that is, $e^x = \frac{2}{x}$) admits a unique real solution α on $]0\,;+\infty[$. *(By calculator: $\alpha \approx 0.853$).*

Worked Example 3 — Recurrence Sequence and Fixed Point

Problem: Consider the sequence (u_n) defined by $u_0 = 1$ and for all $n \in \mathbb{N}$ by:

$$u_{n+1} = \sqrt{1 + u_n}.$$

1. Prove by induction that for all $n \in \mathbb{N}$, $1 \leq u_n \leq u_{n+1} \leq 2$. 2. Deduce that the sequence (u_n) converges. 3. Determine the exact value of its limit L (the golden ratio φ).

Solution: 1. Define the function $f(x) = \sqrt{1+x}$ on $[0\,;2]$. The function f is strictly increasing on $[0\,;2]$ (composite of an increasing linear function and the square root function). Let $\mathcal{P}(n)$ be the property: "$1 \leq u_n \leq u_{n+1} \leq 2$".

- *Base step:* $u_0 = 1$, $u_1 = \sqrt{1+1} = \sqrt{2} \approx 1.414$. Clearly $1 \leq 1 \leq \sqrt{2} \leq 2$. $\mathcal{P}(0)$ is true.
- *Inductive step:* Assume $1 \leq u_n \leq u_{n+1} \leq 2$. Apply the function f, which is strictly increasing:

$$f(1) \leq f(u_n) \leq f(u_{n+1}) \leq f(2) \Longrightarrow \sqrt{2} \leq u_{n+1} \leq u_{n+2} \leq \sqrt{3}.$$

  Since $1 \leq \sqrt{2}$ and $\sqrt{3} \leq 2$, we have $1 \leq u_{n+1} \leq u_{n+2} \leq 2$. $\mathcal{P}(n+1)$ is true.
- *Conclusion:* For all $n \in \mathbb{N}$, $1 \leq u_n \leq u_{n+1} \leq 2$.

2. **Convergence:** The sequence (u_n) is increasing and bounded above by 2. By the monotone convergence theorem, it converges to a real limit L satisfying $1 \leq L \leq 2$.

3. **Limit calculation via the fixed-point theorem:** The function $f(x) = \sqrt{1+x}$ is continuous on $[1\,;2]$. By the fixed-point theorem, the limit L satisfies:

$$L = f(L) \Leftrightarrow L = \sqrt{1+L}.$$

   Since $L \geq 1 > 0$, squaring both sides yields the equivalent equation:

$$L^2 = 1 + L \Leftrightarrow L^2 - L - 1 = 0.$$

   This is the characteristic equation of the **golden ratio**! $\Delta = (-1)^2 - 4(1)(-1) = 5$. The roots are $L_1 = \frac{1-\sqrt{5}}{2} < 0$ and $L_2 = \frac{1+\sqrt{5}}{2} \approx 1.618$. Since $L \geq 1$, we retain the unique positive root:

$$L = \frac{1+\sqrt{5}}{2} = \varphi.$$

Worked Example 4 — Continuity of a Piecewise Defined Function

Problem: Consider the function f defined on $\mathbb{R}$ by:

$$f(x) = \begin{cases} x^2 + a & \text{if } \ x \leq 2 \\ 3x - 1 & \text{if } \ x > 2 \end{cases}$$

where a is a real constant. Determine the value of a so that f is continuous on $\mathbb{R}$.

Solution: - For $x < 2$, $f(x) = x^2 + a$ is a polynomial, hence continuous on $]-\infty\,;2[$. - For $x > 2$, $f(x) = 3x - 1$ is a polynomial, hence continuous on $]2\,;+\infty[$.

The only critical point is $x_0 = 2$. Compute: 1. The value at 2: $f(2) = 2^2 + a = 4 + a$. 2. The left-hand limit: $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-}(x^2 + a) = 4 + a$. 3. The right-hand limit: $\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+}(3x - 1) = 3(2) - 1 = 5$.

For f to be continuous at 2, it is necessary and sufficient that the left-hand limit, the right-hand limit, and the function value all match:

$$4 + a = 5 \Leftrightarrow a = 1.$$

The function f is continuous on $\mathbb{R}$ if and only if $a = 1$.

Worked Example 5 — Proving the Existence of Identical Temperatures

Problem: A thermometer records the temperature $T(x)$ along a straight $100\ \text{km}$ road, where $x \in [0\,;100]$ denotes the distance from the starting point in kilometers. The function T is assumed to be continuous. Suppose the initial temperature and the final temperature are identical: $T(0) = T(100)$. Prove that there exist two points exactly $50\ \text{km}$ apart that share the exact same temperature.

Solution: Define the auxiliary function D for all $x \in [0\,;50]$ by:

$$D(x) = T(x + 50) - T(x).$$

$D(x)$ represents the temperature difference between the point at kilometer $x + 50$ and the point at kilometer x. 1. **Continuity:** Since T is continuous, D is continuous on the interval $[0\,;50]$ as the difference of continuous functions. 2. **Values at the endpoints:**

- At $x = 0$: $D(0) = T(50) - T(0)$.
- At $x = 50$: $D(50) = T(100) - T(50)$. Now, by assumption, $T(100) = T(0)$. Substitute $T(100)$ with $T(0)$:

$$D(50) = T(0) - T(50) = -\big(\,T(50) - T(0)\big) = -D(0).$$

3. **Sign discussion:**
   - If $D(0) = 0$, then $T(50) = T(0)$: the points $x = 0$ and $x = 50$ satisfy the condition.
   - If $D(0) \neq 0$, then $D(0)$ and $D(50) = -D(0)$ have **strictly opposite signs**:

$$D(0) \times D(50) < 0.$$

   By the Intermediate Value Theorem applied to the continuous function D on $[0\,;50]$, there exists at least one real number $c \in]0\,;50[$ such that:

$$D(c) = 0 \Leftrightarrow T(c + 50) - T(c) = 0 \Leftrightarrow T(c + 50) = T(c).$$

   The two points with coordinates c and $c + 50$ are separated by $50\ \text{km}$ and have identical temperatures.

Worked Example 6 — Manual Bisection Steps

Problem: Consider the function $f(x) = x^3 + 2x - 5$ on $[1\,;2]$. 1. Justify that the equation $f(x) = 0$ has a unique solution $\alpha \in [1\,;2]$. 2. Perform three manual bisection steps to find a bracketing of α of width 0.125.

Solution: 1. f is a polynomial, hence continuous on $[1;2]$. Its derivative is $f'(x) = 3x^2 + 2 > 0$ for all x, so f is strictly increasing. $f(1) = 1 + 2 - 5 = -2 < 0$ and $f(2) = 8 + 4 - 5 = 7 > 0$. Since $0 \in [-2\,;7]$, by the IVT corollary, there exists a unique root $\alpha \in [1\,;2]$.

2. **Bisection:**
   - **Step 1:** Midpoint of $[1\,;2]$: $m_1 = 1.5$. $f(1.5) = (1.5)^3 + 2(1.5) - 5 = 3.375 + 3 - 5 = 1.375 > 0$. Since $f(1) < 0$ and $f(1.5) > 0$, the root is in:

$$\alpha \in [1\,;1.5] \quad (\text{width}\ \ 0.5).$$

   - **Step 2:** Midpoint of $[1\,;1.5]$: $m_2 = 1.25$. $f(1.25) = (1.25)^3 + 2(1.25) - 5 = 1.953125 + 2.5 - 5 = -0.546875 < 0$. Since $f(1.25) < 0$ and $f(1.5) > 0$, the root is in:

$$\alpha \in [1.25\,;1.5] \quad (\text{width } \ 0.25).$$

- **Step 3:** Midpoint of $[1.25\,;1.5]$: $m_3 = 1.375$. $f(1.375) = (1.375)^3 + 2(1.375) - 5 \approx 2.5996 + 2.75 - 5 = 0.3496 > 0$. Since $f(1.25) < 0$ and $f(1.375) > 0$, the root is in:
$$\alpha \in [1.25\,;1.375] \quad (\text{width } \ 0.125).$$
The bracketing after three steps is $1.25 \leq \alpha \leq 1.375$.

Key Takeaways

1. **Continuity at a point:** $\lim_{x \to x_0} f(x) = f(x_0)$.
2. **Differentiable implies continuous:** every function differentiable on an interval is continuous there. The converse is false (e.g., $|x|$ at 0).
3. **Intermediate Value Theorem (IVT):** If f is continuous on $[a;b]$, it takes every value between $f(a)$ and $f(b)$ at least once.
4. **Bijection Theorem (IVT Corollary):** If f is **continuous** and **strictly monotonic** on an interval I, then the equation $f(x) = k$ admits **a unique solution** in I for all $k \in f(I)$.
5. **Bisection method:** interval-halving algorithm to numerically bracket a solution with arbitrary precision in a finite number of steps.
6. **Fixed point:** If $u_{n+1} = f(u_n)$ with f continuous and if $u_n \to L$, then $f(L) = L$.

Exercises

Level (Direct Applications)

Exercise 46.1 Among the following functions, determine which ones are continuous on their domain of definition: 1. $f(x) = 3x^4 - 2x + 7$ on $\mathbb{R}$ 2. $g(x) = \frac{1}{x-5}$ on $\mathbb{R} \setminus \{5\}$ 3. $h(x) = \sqrt{x-3}$ on $[3\,;+\infty[$ 4. $k(x) = \lfloor x \rfloor$ (floor function) on $\mathbb{R}$

Exercise 46.2 Consider the function f defined on $[-2\,;5]$ by $f(x) = x^2 - 4x + 1$. 1. Compute $f(-2)$ and $f(5)$. 2. Justify that the equation $f(x) = 0$ has at least one solution on $[-2\,;5]$.

Exercise 46.3 You are given the variation table of a continuous function g on $[-5\,;4]$: - g is strictly increasing from -3 to 7 on $[-5\,;-1]$. - g is strictly decreasing from 7 to 1 on $[-1\,;4]$.

Determine the number of solutions of the equation $g(x) = 4$ on $[-5\,;4]$.

Exercise 46.4 Justify the continuity of the function $f(x) = \frac{e^x}{x^2+1}$ on $\mathbb{R}$.

Exercise 46.5 Let $f(x) = x^3 + 3x - 1$. 1. Show that f is strictly increasing on $\mathbb{R}$. 2. Compute $f(0)$ and $f(1)$. 3. Deduce that there exists a unique real number $\alpha \in [0\,;1]$ such that $f(\alpha) = 0$.

Exercise 46.6 Consider the recurrence sequence defined by $u_0 = 3$ and $u_{n+1} = \frac{1}{2}u_n + 1$. Assuming it converges to L, determine L using the fixed-point theorem.

Exercise 46.7 Consider the piecewise defined function:

$$f(x) = \begin{cases} 2x + k \text{ if } \ x \leq 1 \\ x^2 + 3 \text{ if } \ x > 1 \end{cases}$$

For what value of the constant k is f continuous at $x = 1$?

Exercise 46.8 Write a Python function `f(x)` returning $x^2 - 2$, then use bisection to approximate $\sqrt{2}$ to within 10^{-3} on $[1\,;2]$.

Level (Guided Reflection and Baccalaureate Style)

Exercise 46.9 (Baccalaureate Style: Complete Solution and Bracketing). Consider the function f defined on $\mathbb{R}$ by:

$$f(x) = x^3 - 3x^2 + 1.$$

1. Study the limits of f at $-\infty$ and at $+\infty$. 2. Construct the complete variation table of f. 3. Prove that the equation $f(x) = 0$ admits exactly three real solutions $\alpha_1, \alpha_2, \alpha_3$. 4. Give a bracketing of width 0.1 of the largest solution. 5. Deduce the sign table of $f(x)$ on $\mathbb{R}$.

Exercise 46.10 (Baccalaureate Style: Study of an Auxiliary Function for a Tangent). We wish to study the function $g(x) = (x-1)e^x + 1$ on $\mathbb{R}$. 1. Determine the variations of g on $\mathbb{R}$. 2. Show that $g(x) \geq 0$ for all $x \in \mathbb{R}$, and state for which value equality holds. 3. Deduce the relative position of the curve $y = e^x$ with respect to its tangent at 1.

Exercise 46.11 (Transcendental Equation and Intersection Point). We seek to determine the intersection point between the exponential curve $y = e^{-x}$ and the line $y = x$. 1. Set up the appropriate function and prove that it admits a unique zero α on $\mathbb{R}$. 2. Prove that $\alpha \in [0\,;1]$. 3. Write a Python function that computes α to within 10^{-6}.

Exercise 46.12 (True or False with Rigorous Justifications). 1. "If f is continuous on $[a;b]$ and if $f(a) < 0$ and $f(b) > 0$, then f vanishes exactly once on $]a;b[$." 2. "If f is strictly increasing on $[a;b]$, then f is continuous on $[a;b]$." 3. "If f is continuous on $]0;1[$, then its set of images is a bounded interval." 4. "A continuous function that never vanishes on an interval I maintains a constant sign on I."

Exercise 46.13 (Bisection: Calculation of Theoretical Iteration Count). The bisection algorithm is applied to an interval $[a\,;b]$ of initial length $L_0 = 4$. 1. Express the length L_n of the interval after n iterations. 2. How many steps at minimum are required to guarantee a precision less than or equal to 10^{-5}?

Exercise 46.14 (Recurrence Sequence and Homographic Function). Consider the sequence (u_n) defined by $u_0 = 2$ and $u_{n+1} = \frac{4u_n - 2}{u_n + 1}$. 1. Verify that the associated function $f(x) = \frac{4x-2}{x+1}$ is continuous on $[1\,;3]$. 2. Determine the two fixed points of f. 3. Assuming (u_n) converges to a real number $L \geq 1$, determine its exact limit.

Exercise 46.15 (Fixed-Point Theorem for a Function Mapping an Interval into Itself). Let f be a continuous function from $[0\,;1]$ into $[0\,;1]$ (that is, for all $x \in [0;1]$, $0 \leq f(x) \leq 1$). 1. Set $g(x) = f(x) - x$. Determine the signs of $g(0)$ and $g(1)$. 2. Deduce that every continuous function from $[0;1]$ into $[0;1]$ admits at least one fixed point (Brouwer's fixed-point theorem in dimension 1).

Exercise 46.16 (The Buddhist Monk Paradox). A monk climbs to a monastery at the top of a mountain along a single path. He starts at 6:00 AM on Saturday and arrives at 6:00 PM at the summit. On Sunday, he walks down along the same path starting at 6:00 AM and arrives at the bottom at 6:00 PM. Prove that there is a spot on the path that the monk occupies at the exact same time of day on both Saturday and Sunday.

Exercise 46.17 (Continuity and Injectivity). Prove that a continuous function on an interval I is injective (distinct inputs have distinct outputs) if and only if it is strictly monotonic on I.

Level (Challenge and In-Depth Study)

Exercise 46.18 (The Dirichlet Function: Discontinuous Everywhere). Consider the indicator function of the rationals f defined on $\mathbb{R}$ by:

$$f(x) = \begin{cases} 1 \text{ if } & x \in \mathbb{Q} \\ 0 \text{ if } & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

1. Using the fact that in every non-empty open interval there exists at least one rational and one irrational number (density of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ in $\mathbb{R}$), prove that for any real number x_0, f has no limit at x_0. 2. Deduce that f is discontinuous at every point of $\mathbb{R}$.

Exercise 46.19 (Borsuk-Ulam Theorem in Dimension 1: Terrestrial Antipodes). The temperature on the Earth's equator is modeled by a continuous function $T(\theta)$, where $\theta \in [0\,;2\pi]$ is longitude in radians (0 and 2π representing the Greenwich meridian, so that $T(0) = T(2\pi)$). Prove that there exists at least one pair of antipodal points on the equator with the exact same temperature:

$$\exists \theta_0 \in [0\,;\pi], \quad T(\theta_0 + \pi) = T(\theta_0).$$

Exercise 46.20 (Cauchy's Functional Equation). Let $f : \mathbb{R} \to \mathbb{R}$ be a **continuous** function satisfying for all real x and y:

$$f(x+y) = f(x) + f(y).$$

1. Compute $f(0)$, then express $f(n)$ in terms of n and $c = f(1)$ for all natural numbers n. 2. Prove that for every integer $p \in \mathbb{Z}$, $f(p) = c \cdot p$. 3. Prove that for every rational $r = \frac{p}{q} \in \mathbb{Q}$, $f(r) = c \cdot r$. 4. Using the continuity of f and the density of $\mathbb{Q}$ in $\mathbb{R}$, conclude that for all $x \in \mathbb{R}$: $f(x) = c \cdot x$ (the only continuous solutions are linear functions).

Exercise 46.21 (Universal Chord Theorem of Length $1/n$). Let f be a continuous function on $[0\,;1]$ such that $f(0) = f(1)$. Let $n \geq 2$ be a natural number. Define $g(x) = f\left(x + \frac{1}{n}\right) - f(x)$ on $\left[0\,;1 - \frac{1}{n}\right]$. 1. Compute the sum $\sum_{k=0}^{n-1} g\left(\frac{k}{n}\right)$. 2. Deduce that there exists at least one real number $x_0 \in \left[0\,;1 - \frac{1}{n}\right]$ such that $f\left(x_0 + \frac{1}{n}\right) = f(x_0)$.

Exercise 46.22 (Accelerating Bisection: Newton's Method). Newton's method replaces the interval midpoint with the intersection of the tangent line with the x-axis:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

1. Write a Python script to compute the zero of $f(x) = x^2 - 2$ starting from $x_0 = 1$. 2. Observe the number of iterations required to obtain 10 exact decimal places compared to bisection (quadratic convergence).

Solutions

Solutions for Level

- **46.1:**
  1. f is a polynomial, hence continuous on $\mathbb{R}$.
  2. g is a rational function, hence continuous on its domain $\mathbb{R} \setminus \{5\}$.
  3. h is the composite of $x \mapsto x - 3$ and the square root function, continuous on $[3\,;+\infty[$.
  4. $k(x)$ is not continuous: it jumps by 1 at every integer.
- **46.2:**
  1. $f(-2) = (-2)^2 - 4(-2) + 1 = 4 + 8 + 1 = 13 > 0$. $f(5) = 25 - 20 + 1 = 6 > 0$. Note: although both $f(-2)$ and $f(5)$ are positive, the vertex of the parabola is at $x = 2$, where $f(2) = 4 - 8 + 1 = -3 < 0$. Since f is continuous on $[-2\,;2]$ with $f(-2) > 0$ and $f(2) < 0$, by the IVT, it vanishes on $[-2\,;2] \subset [-2\,;5]$.
- **46.3:**
  - On $[-5\,;-1]$, g is continuous and strictly increasing from -3 to 7. Since $4 \in [-3\,;7]$, there is exactly one solution.
  - On $[-1\,;4]$, g is continuous and strictly decreasing from 7 to 1. Since $4 \in [1\,;7]$, there is exactly one solution. In total, the equation $g(x) = 4$ admits exactly **2 solutions** on $[-5\,;4]$.
- **46.4:** The exponential is continuous on $\mathbb{R}$, the polynomial $x^2 + 1$ is continuous on $\mathbb{R}$ and never vanishes because $x^2 + 1 \geq 1 > 0$. The quotient of two continuous functions with a nowhere-zero denominator is continuous on $\mathbb{R}$.
- **46.5:**
  1. $f'(x) = 3x^2 + 3 = 3(x^2 + 1) > 0$ for all $x \in \mathbb{R}$. Thus f is strictly increasing.
  2. $f(0) = -1$ and $f(1) = 1 + 3 - 1 = 3$.
  3. f is continuous and strictly increasing on $[0\,;1]$, and $0 \in [-1\,;3]$. By the IVT corollary, there exists a unique real number $\alpha \in [0\,;1]$ such that $f(\alpha) = 0$.
- **46.6:** The function $f(x) = \frac{1}{2}x + 1$ is continuous on $\mathbb{R}$. By the fixed-point theorem, $L = f(L) \Leftrightarrow L = \frac{1}{2}L + 1 \Leftrightarrow \frac{1}{2}L = 1 \Leftrightarrow L = 2$.
- **46.7:** For continuity at 1: $f(1) = 2(1) + k = 2 + k$. $\lim_{x \to 1^+} f(x) = 1^2 + 3 = 4$. We must have $2 + k = 4 \Leftrightarrow k = 2$.
- **46.8:**

```python
def f(x):
    return x**2 - 2

a, b = 1.0, 2.0
while (b - a) > 1e-3:
    m = (a + b) / 2
    if f(a) * f(m) <= 0:
        b = m
    else:
        a = m
print(f"sqrt(2) ≈ {(a + b)/2:.4f}")
```

Solutions for Level

- **46.9:**
  1. $\lim_{x\to-\infty} f(x) = \lim x^3 = -\infty$ and $\lim_{x\to+\infty} f(x) = +\infty$.
  2. $f'(x) = 3x^2 - 6x = 3x(x-2)$. Roots: $x = 0$ and $x = 2$. $f(0) = 1$ (local maximum) and $f(2) = 8 - 12 + 1 = -3$ (local minimum). f is increasing on $]-\infty; 0]$, decreasing on $[0; 2]$, and increasing on $[2; +\infty[$.
  3. On $]-\infty; 0]$, f is continuous and strictly increasing from $-\infty$ to 1. Since $0 \in]-\infty; 1]$, there is a unique solution α_1. On $[0; 2]$, f is continuous and strictly decreasing from 1 to -3. Since $0 \in [-3; 1]$, there is a unique solution α_2. On $[2; +\infty[$, f is continuous and strictly increasing from -3 to $+\infty$. Since $0 \in [-3; +\infty[$, there is a unique solution α_3. There are thus exactly 3 real solutions.
  4. For the largest solution $\alpha_3 \in [2; +\infty[$: $f(2) = -3 < 0$, $f(3) = 27 - 27 + 1 = 1 > 0$, so $\alpha_3 \in [2; 3]$. $f(2.8) = (2.8)^3 - 3(2.8)^2 + 1 = 21.952 - 23.52 + 1 = -0.568 < 0$. $f(2.9) = (2.9)^3 - 3(2.9)^2 + 1 = 24.389 - 25.23 + 1 = 0.159 > 0$. Bracketing: $2.8 < \alpha_3 < 2.9$.
  5. Sign: negative on $]-\infty; \alpha_1]$, positive on $[\alpha_1; \alpha_2]$, negative on $[\alpha_2; \alpha_3]$, positive on $[\alpha_3; +\infty[$.
- **46.10:**
  1. $g'(x) = 1 \cdot e^x + (x-1)e^x = xe^x$. Since $e^x > 0$, $g'(x)$ shares the sign of x: decreasing on $]-\infty; 0]$ and increasing on $[0; +\infty[$.
  2. Minimum at 0: $g(0) = (0-1)e^0 + 1 = -1 + 1 = 0$. Since the absolute minimum is 0, $g(x) \geq 0$ for all real x, with equality if and only if $x = 0$.
  3. The equation of the tangent line to $y = e^x$ at $x = 1$ is: $y = e^1(x-1) + e^1 = e \cdot x$. (Notice $(x-1)e^x + 1 \geq 0$ demonstrates the convex position relative to the tangent).
- **46.11:**
  1. Let $h(x) = e^{-x} - x$ on $\mathbb{R}$. h is continuous on $\mathbb{R}$ and $h'(x) = -e^{-x} - 1 < 0$ (since $-e^{-x} < 0$). Thus h is strictly decreasing. $\lim_{x\to-\infty} h(x) = +\infty$ and $\lim_{x\to+\infty} h(x) = -\infty$. By the IVT corollary, there exists a unique zero α on $\mathbb{R}$.
  2. $h(0) = e^0 - 0 = 1 > 0$ and $h(1) = e^{-1} - 1 \approx 0.368 - 1 = -0.632 < 0$. Thus $\alpha \in [0\,; 1]$.
  3. Python script:

```python
import math

def h(x):
    return math.exp(-x) - x

a, b = 0.0, 1.0
while (b - a) > 1e-6:
    m = (a + b) / 2
    if h(a) * h(m) <= 0:
        b = m
    else:
        a = m
print(f"Intersection point alpha ≈ {(a + b)/2:.6f}")
# alpha ≈ 0.567143 (Omega constant)
```

- **46.12:**
  1. **False.** It vanishes *at least* once, but could vanish multiple times if not strictly monotonic.
  2. **False.** Counterexample: a strictly increasing step function with jump discontinuities. Strict monotonicity does not imply continuity.
  3. **False.** Counterexample: $f(x) = \frac{1}{x}$ on $]0\,;1[$ is continuous, but its image is $]1\,;+\infty[$, which is unbounded.
  4. **True.** If it changed sign, the IVT would force the existence of a zero, contradicting the assumption that it never vanishes.
- **46.13:**
  1. At each step, the length is divided by 2: $L_n = \frac{L_0}{2^n} = \frac{4}{2^n} = \frac{1}{2^{n-2}}$.
  2. We seek n such that $\frac{4}{2^n} \leq 10^{-5} \Leftrightarrow 2^n \geq 4 \times 10^5 = 400,000$. Since $2^{18} = 262,144$ and $2^{19} = 524,288$, at least **19 steps** are necessary.
- **46.14:**
  1. $x + 1 = 0 \Leftrightarrow x = -1 \notin [1\,;3]$. f is a rational function whose denominator does not vanish on $[1\,;3]$, so it is continuous there.
  2. Fixed points: $f(x) = x \Leftrightarrow \frac{4x-2}{x+1} = x \Leftrightarrow 4x - 2 = x^2 + x \Leftrightarrow x^2 - 3x + 2 = 0$. Roots: $(x-1)(x-2) = 0 \Leftrightarrow x = 1$ or $x = 2$.
  3. If $u_0 = 2$, then $u_1 = f(2) = 2$: the sequence is constantly equal to 2. If it converges to $L \geq 1$, by the fixed-point theorem, $L \in \{1; 2\}$. With $u_0 = 2$, $L = 2$.
- **46.15:**
  1. $g(0) = f(0) - 0 = f(0) \geq 0$ (since $f(x) \in [0; 1]$). $g(1) = f(1) - 1 \leq 0$ (since $f(1) \leq 1$).
  2. The function g is continuous on $[0\,;1]$ because f is continuous. Since $g(0) \geq 0$ and $g(1) \leq 0$, by the IVT, there exists at least one real number $c \in [0\,;1]$ such that $g(c) = 0$, that is, $f(c) = c$.
- **46.16:** Let $x(t)$ be the position of the monk on Saturday at time $t \in [6\,;18]$ (with $x(6) = 0$ and $x(18) = D$, where D is the total distance). Let $y(t)$ be his position on Sunday at time t (with $y(6) = D$ and $y(18) = 0$). Set $h(t) = x(t) - y(t)$. h is continuous on $[6\,;18]$. $h(6) = 0 - D = -D < 0$ and $h(18) = D - 0 = D > 0$. By the IVT, there exists a time $t_0 \in]6\,;18[$ such that $h(t_0) = 0 \Leftrightarrow x(t_0) = y(t_0)$. At that exact time, the monk is at the very same spot on the path.
- **46.17:**
  - If f is strictly monotonic, it is clearly injective (if $x < y$, then $f(x) < f(y)$ or $f(x) > f(y)$, so $f(x) \neq f(y)$).
  - Conversely, if f is not strictly monotonic, it has a local extremum or non-monotonic behavior, which by the IVT creates two distinct points $a < b$ with $f(a) = f(b)$, contradicting injectivity.

Solutions for Level

- **46.18:**

1. Let $x_0 \in \mathbb{R}$. In every open interval $]x_0 - \varepsilon\,;x_0 + \varepsilon[$, there are infinitely many rationals (where $f(x) = 1$) and infinitely many irrationals (where $f(x) = 0$). Any sequence of rationals tending to x_0 yields an image sequence constant at 1. Any sequence of irrationals tending to x_0 yields an image sequence constant at 0.
2. Since the two limits (1 and 0) differ, f has no limit at x_0. It is therefore discontinuous at every point of $\mathbb{R}$.

- **46.19:** Set $D(\theta) = T(\theta + \pi) - T(\theta)$ for $\theta \in [0\,;\pi]$. D is continuous on $[0\,;\pi]$. $D(0) = T(\pi) - T(0)$. $D(\pi) = T(2\pi) - T(\pi) = T(0) - T(\pi) = -D(0)$ because $T(2\pi) = T(0)$. If $D(0) = 0$, $\theta = 0$ works. Otherwise, $D(0)$ and $D(\pi)$ have opposite signs: by the IVT, there exists $\theta_0 \in]0\,;\pi[$ such that $D(\theta_0) = 0 \Leftrightarrow T(\theta_0 + \pi) = T(\theta_0)$.
- **46.20:**
  1. $f(0 + 0) = f(0) + f(0) \Longrightarrow f(0) = 0$. By induction, $f(n) = nf(1) = c \cdot n$.
  2. $f(0) = f(p + (-p)) = f(p) + f(-p) = 0 \Longrightarrow f(-p) = -f(p) = -c \cdot p = c \cdot (-p)$. True for all $p \in \mathbb{Z}$.
  3. Let $r = p/q$. Then $q \cdot f(p/q) = f(q \cdot p/q) = f(p) = c \cdot p \Longrightarrow f(p/q) = c \cdot \frac{p}{q}$.
  4. For any real number x, there exists a sequence of rationals (r_n) converging to x. By continuity of f: $f(x) = \lim f(r_n) = \lim(c \cdot r_n) = c \cdot x$.
- **46.21:**
  1. The sum is telescoping:
$$\sum_{k=0}^{n-1} g\left(\frac{k}{n}\right) = \sum_{k=0}^{n-1}\left[f\left(\frac{k+1}{n}\right) - f\left(\frac{k}{n}\right)\right] = f(1) - f(0) = 0.$$
  2. A sum of n terms equaling 0 implies that either all terms are zero, or at least one is non-negative and at least one is non-positive. The continuous function g therefore takes a value ≥ 0 and a value ≤ 0. By the IVT, g vanishes at some point x_0, whence $f(x_0 + 1/n) = f(x_0)$.
- **46.22:**
  1. Python script:

```python
def f(x):
    return x**2 - 2

def f_prime(x):
    return 2*x

x = 1.0
for i in range(5):
    x = x - f(x) / f_prime(x)
    print(f"Step {i+1} : x = {x:.12f}")
# Step 1 : 1.500000000000
# Step 2 : 1.416666666667
# Step 3 : 1.414215686275
# Step 4 : 1.414213562375 (12 exact decimals in 4 steps!)
```

  2. Bisection gains 1 bit of precision per step (requiring about 33 steps for 10 decimal digits). Newton doubles the number of exact decimal places at every step (quadratic convergence): 4 to 5 steps are plenty!

The Question That Opens What Follows

We have just seen that continuity ensures a curve never tears. But a continuous curve can be filled with sharp corners (like $|x|$ at zero) or abrupt kinks. To speak of **velocity**, **acceleration**, and **smooth trajectory**, we need the supreme instrument of differential calculus: differentiation. What happens when we differentiate a function of a function, like a mechanical gearbox where a first cog drives a second, which in turn drives a third? How does the second derivative reveal whether a curve cups upward toward the sky like a bowl catching rain (**convexity**) or slices through its own tangent (**inflection point**)?

This is the art of gears and curvature: see you in Chapter 47.

CHAPTER 47: The Growing Garden and the Gear Chain

> "Never memorize a differentiation formula as a mere magic incantation to pass an exam. Look at the toothed sprockets of a bicycle gearing up the effort from pedal to rear wheel: each speed multiplies the preceding one. Look at a ceramic bowl catching rainwater without losing a drop: its upward-facing contour is a convex curve, and it is the acceleration of its slope that decides whether it gathers or spills."

What You Will Discover

- Why differentiating a composite function operates exactly like a **bicycle gearbox**: the celebrated chain rule $(g \circ u)' = u' \cdot (g' \circ u)$.
- The three indispensable powerhouse formulas for the Baccalaureate: the derivatives of $e^{u(x)}$, $\sqrt{u(x)}$, and $(u(x))^n$.
- The **second derivative** $f''(x)$: after instantaneous velocity comes **acceleration** and the curvature of the trajectory.
- **Convexity**: understanding intuitively and geometrically what it means for a curve to be "bowl-shaped" (opening upward) or "bell-shaped" (opening downward).
- The fundamental curriculum equivalences: secant lines lying above the curve, tangent lines lying below the curve, and the positivity of $f''(x)$.
- **Inflection points**: the exact spot where a cyclist straightens the handlebars to switch curves, where the curve **crosses its own tangent**.
- **Convexity inequalities**, including the celebrated $e^x \geq x + 1$ for all real x.

---

1. The Problem: Cascading Gears and the Soup Bowl

1.1 The Cyclist and the Gearbox

Imagine a racing bicycle equipped with a large 50-tooth chainring at the crankset and a small 25-tooth cog on the rear wheel.

- When the cyclist turns the pedals through an angle x, the front chainring turns through an angle $u = x$.
- The chain transmits the motion: because the rear cog has half as many teeth as the chainring, the cog rotates twice as fast:

$$u \mapsto y = 2u.$$

- The rear wheel, locked to the cog, turns the tire against the asphalt: each rotation of the wheel propels the bicycle forward by a distance $z = 2.10 \times y$ meters.

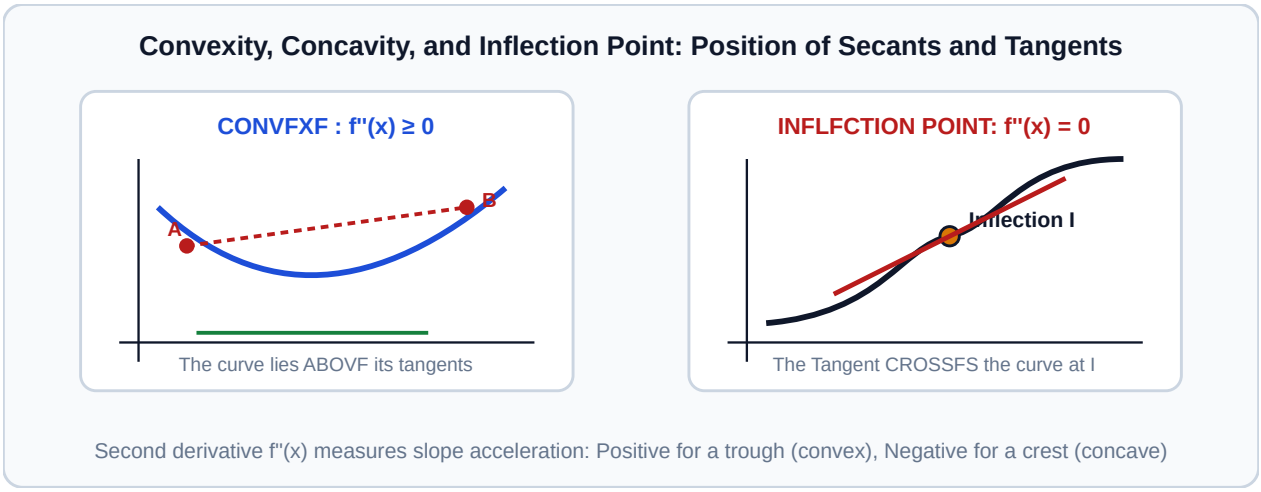

Figure 104: Convexity, concavity, and inflection point: relative position of tangent lines

If the cyclist presses slightly faster on the pedals (variation Δx):

- the intermediate speed changes at a certain rate: $\frac{\Delta u}{\Delta x}$;
- the final wheel speed changes relative to the intermediate speed at the rate: $\frac{\Delta z}{\Delta u}$.

At what rate does the distance z vary with respect to the initial effort x on the pedals? The physical common-sense answer is immediate: one **multiplies the transmission ratios**!

$$\frac{\Delta z}{\Delta x} = \frac{\Delta z}{\Delta u} \times \frac{\Delta u}{\Delta x}.$$

The German physicist and mathematician Gottfried Wilhelm Leibniz wrote this formula as early as 1684 in the luminous form:

$$\frac{dz}{dx} = \frac{dz}{du} \cdot \frac{du}{dx}.$$

The two du terms appear to cancel out just like in a basic fraction of numbers! This is the secret of differentiating composite functions: **sensitivities chain together through multiplication**.

1.2 The Gardener and the Bowl That Catches the Rain

Now observe two vessels set out in a garden in springtime: 1. A hollow bowl resting on its base: it fills with rainwater. If you stretch a straight string between two opposite points on its rim (a secant line), the entire bowl bottom plunges **below** the string. If you lay a flat ruler tangent beneath the bowl, the entire vessel rests **above** the ruler. This is a **convex** curve. 2. A glass bell jar placed upside down: rainwater slides down its sides and runs off. A secant chord stays beneath the bell, and a tangent line remains above it. This is a **concave** curve.

How can the derivative distinguish between these two shapes? How can a single equation predict whether a mountain hairpin tightens or flattens out? This is the role of the **second derivative**.

---

2. The Derivative of a Composite Function: The Chain Rule

2.1 The Notations Explained

- Let u be a function defined on an interval I with values in an interval J, and let g be a function defined on J.
- The composite function is denoted $g \circ u$ (read “g circle u” or “g composed with u”): for all $x \in I$, $(g \circ u)(x) = g\big(u(x)\big)$.
- Note the order of evaluation: in $g\big(u(x)\big)$, one computes $u(x)$ **first** (the inner function), and then feeds that result to g (the outer function).
- The derivative is denoted $(g \circ u)'$ or $[g(u(x))]'$.

2.2 The Fundamental Theorem

Theorem — Derivative of a composite function (Chain Rule).

Let u be a function differentiable on an interval I, and let g be a function differentiable on an interval J containing $u(I)$. Then the composite function $f = g \circ u$ is differentiable on the interval I, and for every real number $x \in I$:

$$(g \circ u)'(x) = u'(x) \times g' \left(u(x) \right) .$$

Spoken aloud: "The derivative of $g(u)$ is u' multiplied by g' evaluated at u."

Idea of the proof.

Let $x_0 \in I$. Form the difference quotient of the composite function between x and x_0:

$$\frac{g(u(x)) - g(u(x_0))}{x - x_0}.$$

If $u(x) \neq u(x_0)$ in a neighborhood of x_0, we can cleverly insert the intermediate factor $u(x) - u(x_0)$ in numerator and denominator:

$$\frac{g(u(x)) - g(u(x_0))}{x - x_0} = \frac{g(u(x)) - g(u(x_0))}{u(x) - u(x_0)} \times \frac{u(x) - u(x_0)}{x - x_0}.$$

As $x \to x_0$:

- By continuity of the differentiable function u, we have $\lim_{x \to x_0} u(x) = u(x_0)$.
- Setting $Y = u(x)$ and $Y_0 = u(x_0)$, the first factor becomes $\frac{g(Y) - g(Y_0)}{Y - Y_0}$, which tends to the derivative $g'(Y_0) = g' \left(u(x_0) \right)$ since g is differentiable at $u(x_0)$.
- The second factor is the difference quotient of u, which by definition tends to $u'(x_0)$. By the product rule for limits:

$$(g \circ u)'(x_0) = g' \left(u(x_0) \right) \times u'(x_0) = u'(x_0) \cdot g' \left(u(x_0) \right) . \quad \blacksquare$$

---

3. The Three Powerhouse Formulas of the Baccalaureate

All delicate derivatives in the advanced mathematics exam are immediate special cases of this chain rule, substituting g with the curriculum's three primary functions:

3.1 Derivative of $e^{u(x)}$: The Sovereign Function

Take $g(y) = e^y$. We know that $g'(y) = e^y$. Applying the general formula $(g \circ u)' = u' \cdot g'(u)$:

Property — Derivative of composite exponential.

Let u be a function differentiable on an interval I. The function $f : x \mapsto e^{u(x)}$ is differentiable on I, and for all $x \in I$:

$$\left(\mathbf{e^{u(x)}} \right)' = \mathbf{u'(x)\, e^{u(x)}}.$$

Lightning example: Let $f(x) = e^{-3x^2+5x}$. Here $u(x) = -3x^2 + 5x$, so $u'(x) = -6x + 5$. We apply the rule directly: $f'(x) = (-6x + 5)e^{-3x^2+5x}$.

3.2 Derivative of $\sqrt{u(x)}$

Take $g(y) = \sqrt{y}$ defined for $y > 0$. Its derivative is $g'(y) = \frac{1}{2\sqrt{y}}$.

📐 Property — Derivative of composite square root.

Let u be a function differentiable and **strictly positive** on an interval I. The function $f : x \mapsto \sqrt{u(x)}$ is differentiable on I, and for all $x \in I$:

$$\boxed{\left(\sqrt{\mathbf{u(x)}}\right)' = \frac{\mathbf{u'(x)}}{\mathbf{2}\sqrt{\mathbf{u(x)}}}.}$$

⚠️ Caution, trap! — The condition $u(x) > 0$.

The function $\sqrt{u(x)}$ is defined whenever $u(x) \geq 0$, but it is **differentiable** only where $u(x) > 0$ (strictly positive). If $u(x_0) = 0$, the curve typically displays a vertical half-tangent at x_0, and the denominator $2\sqrt{u(x)}$ would vanish!

3.3 Derivative of $(u(x))^n$

Take $g(y) = y^n$ with $n \in \mathbb{Z}^*$. Its derivative is $g'(y) = ny^{n-1}$.

📐 Property — Derivative of composite power.

Let u be a function differentiable on an interval I. For any non-zero integer $n \geq 1$ (or $n < 0$ on intervals where $u(x) \neq 0$):

$$\boxed{\left(\mathbf{u^n}\right)' = \mathbf{n}\,\mathbf{u'(x)}\left(\mathbf{u(x)}\right)^{\mathbf{n-1}}.}$$

Fundamental special case: For $n = -1$, the formula gives the derivative of the reciprocal:

$$\left(\frac{1}{u}\right)' = \left(u^{-1}\right)' = -1 \cdot u' \cdot u^{-2} = -\frac{\mathbf{u'}}{\mathbf{u^2}}.$$

Summary Comparison Table

Simple function $f(x)$	Derivative $f'(x)$	Composite function $f(x)$	Derivative $f'(x)$	Validity Condition
x^n	nx^{n-1}	$\left(u(x)\right)^n$	$\mathbf{n\,u'(x)\left(u(x)\right)^{n-1}}$	u differentiable ($u \neq 0$ if $n \leq 0$)
$\frac{1}{x}$	$-\frac{1}{x^2}$	$\frac{1}{u(x)}$	$-\frac{\mathbf{u'(x)}}{\mathbf{(u(x))^2}}$	$u(x) \neq 0$
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$\sqrt{u(x)}$	$\frac{\mathbf{u'(x)}}{\mathbf{2\sqrt{u(x)}}}$	$\mathbf{u(x) > 0}$
e^x	e^x	$e^{u(x)}$	$\mathbf{u'(x)\,e^{u(x)}}$	u differentiable on I

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4. The Second Derivative: After Velocity, Acceleration

4.1 Definition and Notations

When the derivative function f' is itself differentiable on an interval I, its own derivative is called the **second derivative** of f, and is denoted:

$$\mathbf{f''(x)} \quad \text{or} \quad \frac{\mathbf{d^2 f}}{\mathbf{dx^2}}\mathbf{(x)} \quad \text{(Leibniz notation)}\ .$$

- $f(x)$ describes **position** (altitude, temperature, capital).
- $f'(x)$ describes **instantaneous rate of change / velocity** (slope of the tangent).
- $f''(x)$ describes **acceleration**: the rate at which the slope itself increases or decreases.

In physics: If $x(t)$ represents the position of a body over time, its velocity is $v(t) = x'(t)$ and its acceleration is $a(t) = v'(t) = x''(t)$. Isaac Newton founded all classical mechanics on this universal law:

$$F = m \cdot a = m \cdot x''(t).$$

Without the second derivative, it is impossible to understand planetary gravitation or vehicle braking dynamics.

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5. Convexity: The Bowl, the Chords, and the Tangents

5.1 Fundamental Geometric Definition

Geometric Definition — Convex / Concave function.

Let f be a function defined on an interval I.

- We say that f is **convex** on I if, for any two points A and B on the graph $\mathcal{C}_f$, the line segment $[AB]$ (the chord) lies entirely **above or on** the curve on the interval between A and B.
- We say that f is **concave** on I if, for any two points A and B on the curve, the segment $[AB]$ lies entirely **below or on** the curve.

Practical remark: A function f is concave if and only if $-f$ is convex.

5.2 Equivalent Characterizations (Curriculum Theorem)

For a twice-differentiable function, the syllabus establishes a magnificent triple equivalence connecting chord geometry, tangent placement, and differential calculus:

Fundamental Theorem — Characterizations of convexity.

Let f be a twice-differentiable function on an interval I. The following statements are **strictly equivalent**: 1. f is **convex** on I; 2. The derivative function f' is **increasing** on I; 3. For all $x \in I$, the second derivative is non-negative: $\boldsymbol{f''(x) \geq 0}$; 4. The curve $\mathcal{C}_f$ lies entirely **above all its tangent lines**.

And symmetrically for concavity:

Theorem — Characterizations of concavity.

Let f be twice differentiable on an interval I. The following statements are equivalent: 1. f is **concave** on I; 2. The derivative function f' is **decreasing** on I; 3. For all $x \in I$, the second derivative is non-positive: $\boldsymbol{f''(x) \leq 0}$; 4. The curve $\mathcal{C}_f$ lies entirely **below all its tangent lines**.

5.3 Convexity Inequalities

Because the curve of a convex function always lies above its tangent line at any point with abscissa x_0, we immediately have for all real x:

$$\mathbf{f(x) \geq f'(x_0)(x - x_0) + f(x_0)}.$$

Let us apply this powerful property to the exponential function $f(x) = e^x$ at $x_0 = 0$:

- $f(0) = e^0 = 1$;
- $f'(x) = e^x \Longrightarrow f'(0) = 1$;
- $f''(x) = e^x > 0$ on $\mathbb{R}$, so the exponential is **strictly convex** on $\mathbb{R}$.

The tangent at 0 has equation: $y = 1 \cdot (x - 0) + 1 = x + 1$. Because the curve lies above its tangent, we obtain the fundamental high-school inequality:

> 📐 ***Remarkable Convexity Inequality — The exponential above its tangent:*** *For every real number $x \in \mathbb{R}$:*
>
> $$\mathbf{e^x \geq x + 1}.$$
>
> *With equality if and only if $x = 0$.*

This inequality allows one to prove in two lines that $\lim_{x \to +\infty} e^x = +\infty$, since $e^x \geq x + 1 \to +\infty$.

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6. The Inflection Point: Where the Bend Reverses

Imagine driving a car along a winding country road:

- At first, the road turns right: you steer clockwise (the curvature is convex).
- Later, the road sweeps left: you steer counterclockwise (the curvature becomes concave).
- There is an infinitesimal moment, right between the two turns, where **your steering wheel passes exactly through dead center (neutral)**.

This transition moment is an **inflection point**.

6.1 Geometric Definition

> 📐 **Definition — Inflection point.**
>
> Let f be a function differentiable on an interval I. An **inflection point** is any point $A\left(x_0 ; f(x_0)\right)$ on the curve $\mathcal{C}_f$ where the curve **crosses its tangent line**.

6.2 Characterization via the Second Derivative

> 📐 **Fundamental Theorem — Necessary and sufficient condition for inflection.**
>
> Let f be a twice-differentiable function on an interval I. The point $A\left(x_0 ; f(x_0)\right)$ is an **inflection point** if and only if: **The second derivative f'' vanishes at x_0 AND CHANGES SIGN.**

> ⚠️ **Caution, trap! — Vanishing alone is NOT SUFFICIENT!**
>
> This is a favorite trap on Baccalaureate exams. If $f''(x_0) = 0$, this does **not guarantee** that x_0 is an inflection point! One must imperatively check that f'' **changes sign** (from $+$ to $-$ or from $-$ to $+$). **Striking counterexample:** Consider $f(x) = x^4$.
>
> - $f'(x) = 4x^3$;
> - $f''(x) = 12x^2$. At $x_0 = 0$, we have $f''(0) = 12 \times 0^2 = 0$. Yet for all $x \neq 0$, $12x^2 > 0$: the second derivative is strictly positive! It vanishes at 0 without ever changing sign. The graph of x^4 is entirely convex (a flattened bowl), and its tangent at 0 is the x-axis ($y = 0$). The curve stays strictly above its tangent: **there is no inflection point at 0!**

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7. Traps to Avoid!

⚠ Caution, trap! — Forgetting $u'(x)$ in composite derivatives.

- Writing $\left(e^{5x}\right)' = e^{5x}$ is false! The correct answer is $5e^{5x}$.
- Writing $\left(\sqrt{x^2+1}\right)' = \frac{1}{2\sqrt{x^2+1}}$ is false! The correct answer is $\frac{2x}{2\sqrt{x^2+1}} = \frac{x}{\sqrt{x^2+1}}$.
- Writing $\left((3x-2)^4\right)' = 4(3x-2)^3$ is false! The correct answer is $4 \times 3 \times (3x-2)^3 = 12(3x-2)^3$. Golden rule: whenever the inner argument is anything other than a plain x, **you must bring $u'(x)$ out in front**.

⚠ Caution, trap! — Confusing monotonicity with convexity.

- A function can be **increasing and convex** (e.g., e^x on $\mathbb{R}$, x^2 on $[0;+\infty[$).
- A function can be **increasing and concave** (e.g., $\sqrt{x}$ on $[0;+\infty[$, $\ln x$ on $]0;+\infty[$).
- A function can be **decreasing and convex** (e.g., e^{-x} on $\mathbb{R}$, $\frac{1}{x}$ on $]0;+\infty[$).
- A function can be **decreasing and concave** (e.g., $-x^2$ on $[0;+\infty[$). The sign of f' determines variation (rising or falling); the sign of f'' determines curvature (cupping upward or bulging downward).

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8. Step-by-Step Worked Examples

Worked Example 1 — Differentiating Multiple Composite Functions

Problem: Compute the derivative of each of the following functions, stating the domain of differentiability: 1. $f(x) = e^{-x^2+4x}$ on $\mathbb{R}$ 2. $g(x) = \sqrt{3x^2+6}$ on $\mathbb{R}$ 3. $h(x) = (2x^3-5x)^4$ on $\mathbb{R}$ 4. $k(x) = \frac{1}{(x^2+1)^3}$ on $\mathbb{R}$

Solution: 1. f has the form e^u with $u(x) = -x^2+4x$. $u'(x) = -2x+4$.

$$f'(x) = u'(x)e^{u(x)} = (-2x+4)e^{-x^2+4x}.$$

2. g has the form $\sqrt{u}$ with $u(x) = 3x^2+6$. For all real x, $3x^2+6 \geq 6 > 0$, so g is differentiable on $\mathbb{R}$. $u'(x) = 6x$.

$$g'(x) = \frac{u'(x)}{2\sqrt{u(x)}} = \frac{6x}{2\sqrt{3x^2+6}} = \frac{3x}{\sqrt{3x^2+6}}.$$

3. h has the form u^4 with $u(x) = 2x^3-5x$. $u'(x) = 6x^2-5$.

$$h'(x) = 4\,u'(x)\left(\,u(x)\right)^3 = 4(6x^2-5)(2x^3-5x)^3.$$

4. Write $k(x) = (x^2+1)^{-3}$, which is u^{-3} with $u(x) = x^2+1$. $u'(x) = 2x$.

$$k'(x) = -3 \cdot u'(x) \cdot \left(\,u(x)\right)^{-4} = -3(2x)(x^2+1)^{-4} = \frac{-6x}{(x^2+1)^4}.$$

Worked Example 2 — Complete Study of Convexity for a Polynomial Function

Problem: Consider the function f defined on $\mathbb{R}$ by $f(x) = x^4 - 2x^3 - 12x^2 + 5x - 1$. 1. Compute $f'(x)$ and the second derivative $f''(x)$. 2. Determine the intervals where f is convex, and those where f is concave. 3. Deduce the coordinates of the inflection points of the graph $\mathcal{C}_f$.

Solution: 1. f is twice differentiable on $\mathbb{R}$ as a polynomial:

- $f'(x) = 4x^3 - 6x^2 - 24x + 5$
- $f''(x) = 12x^2 - 12x - 24 = 12(x^2 - x - 2)$.

2. Factoring the quadratic $x^2 - x - 2$: Its evident roots are $x_1 = -1$ and $x_2 = 2$ (since $(-1)^2 - (-1) - 2 = 0$ and $2^2 - 2 - 2 = 0$). The quadratic shares the sign of $a = 12 > 0$ outside the roots and is negative between them:

- For $x \in]-\infty\,;-1]$, $f''(x) \geq 0$: f is **convex**.
- For $x \in [-1\,;2]$, $f''(x) \leq 0$: f is **concave**.
- For $x \in [2\,;+\infty[$, $f''(x) \geq 0$: f is **convex**.

3. The second derivative f'' vanishes while changing sign at $x = -1$ and $x = 2$. The graph $\mathcal{C}_f$ admits two inflection points:
   - At $x = -1$: $f(-1) = 1 + 2 - 12 - 5 - 1 = -15$. First inflection point: $I_1(-1\,;-15)$.
   - At $x = 2$: $f(2) = 16 - 16 - 48 + 10 - 1 = -39$. Second inflection point: $I_2(2\,;-39)$.

Worked Example 3 — The Gaussian Bell Curve: Convexity and Inflections

Problem: Consider the Gaussian function $f(x) = e^{-x^2/2}$ defined on $\mathbb{R}$. 1. Compute $f'(x)$ and study the variations of f. 2. Compute $f''(x)$ and study its sign. 3. Determine the inflection points of the Gaussian bell curve.

Solution: 1. $f(x) = e^{u(x)}$ with $u(x) = -\frac{1}{2}x^2$, whence $u'(x) = -x$.

$$f'(x) = -xe^{-x^2/2}.$$

Since $e^{-x^2/2} > 0$ for all real x, $f'(x)$ has the sign of $-x$:

- $f'(x) > 0$ on $]-\infty\,;0[$: f is strictly increasing.
- $f'(x) < 0$ on $]0\,;+\infty[$: f is strictly decreasing.
- At $x = 0$, $f(0) = e^0 = 1$ is the absolute maximum.

2. Differentiate $f'(x)$ as a product $u \cdot v$ with $u(x) = -x$ ($u' = -1$) and $v(x) = e^{-x^2/2}$ ($v' = -xe^{-x^2/2}$):

$$f''(x) = u'v + uv' = (-1)e^{-x^2/2} + (-x)(-xe^{-x^2/2}) = (-1 + x^2)e^{-x^2/2} = (x^2 - 1)e^{-x^2/2}.$$

Since $e^{-x^2/2} > 0$, the sign of $f''(x)$ matches $x^2 - 1 = (x - 1)(x + 1)$:

- $f''(x) \geq 0$ on $]-\infty\,;-1] \cup [1\,;+\infty[$: f is **convex** there.
- $f''(x) \leq 0$ on $[-1\,;1]$: f is **concave** there.

3. The second derivative f'' vanishes while changing sign at $x = -1$ and $x = 1$. The associated y-values are $f(-1) = f(1) = e^{-1/2} = \frac{1}{\sqrt{e}} \approx 0.6065$. The bell curve admits exactly **two inflection points** at $(-1\,;e^{-1/2})$ and $(1\,;e^{-1/2})$. *(In probability and statistics, these inflection points correspond precisely to one standard deviation $\pm\sigma$ from the mean).*

Worked Example 4 — Tangent Line and Relative Position via Convexity

Problem: Consider the function $f(x) = (2x - 1)e^x$ on $\mathbb{R}$. 1. Compute $f'(x)$ and $f''(x)$. 2. Prove that f is convex on $[-\frac{1}{2}\,;+\infty[$. 3. Determine the equation of the tangent line (T) to $\mathcal{C}_f$ at the point with abscissa 0. 4. Deduce the relative position of $\mathcal{C}_f$ with respect to (T) on $[-\frac{1}{2}\,;+\infty[$.

Solution: 1. **Derivatives:** $f'(x) = 2e^x + (2x - 1)e^x = (2x + 1)e^x$. $f''(x) = 2e^x + (2x + 1)e^x = (2x + 3)e^x$. 2. **Convexity:** Since $e^x > 0$, the sign of $f''(x)$ is that of $2x + 3$. For all $x \geq -\frac{1}{2}$, we have $2x + 3 \geq 2(-\frac{1}{2}) + 3 = -1 + 3 = 2 > 0$. Thus $f''(x) \geq 0$ on $[-\frac{1}{2}\,;+\infty[$, proving that f is **strictly convex** on this interval. 3. **Tangent at 0:** $f(0) = (0 - 1)e^0 = -1$. $f'(0) = (0 + 1)e^0 = 1$. The equation of tangent (T) is:

$$y = f'(0)(x - 0) + f(0) = 1(x) - 1 = x - 1.$$

4. **Relative position:** Since f is convex on $[-\frac{1}{2}\,;+\infty[$ and the contact point 0 belongs to this interval ($0 \geq -\frac{1}{2}$), the curve $\mathcal{C}_f$ lies **above its tangent (T)** on $[-\frac{1}{2}\,;+\infty[$. That is, for all $x \geq -\frac{1}{2}$:

$$(2x - 1)e^x \geq x - 1.$$

Worked Example 5 — Differentiating a Power Function with a Rational Base

Problem: Consider the function $f(x) = \left(\frac{x-1}{x+1}\right)^3$ defined on $\mathbb{R} \setminus \{-1\}$. Compute its derivative $f'(x)$ and determine its monotonicity on its domain.

Solution: Set $u(x) = \frac{x-1}{x+1}$. Compute $u'(x)$ using the quotient rule $\left(\frac{w}{v}\right)' = \frac{w'v - wv'}{v^2}$:

$$u'(x) = \frac{1(x+1) - (x-1)(1)}{(x+1)^2} = \frac{x+1-x+1}{(x+1)^2} = \frac{2}{(x+1)^2}.$$

The function f has the form u^3, so its derivative is $f'(x) = 3\,u'(x)\big(\,u(x)\big)^2$:

$$f'(x) = 3 \times \frac{2}{(x+1)^2} \times \left(\frac{x-1}{x+1}\right)^2 = \frac{6(x-1)^2}{(x+1)^4}.$$

Examine the sign of $f'(x)$:

- The denominator $(x+1)^4$ is a 4th power, strictly positive for all $x \neq -1$.
- The numerator $6(x-1)^2$ is a square multiplied by 6, hence non-negative, vanishing only at $x = 1$.

Consequently, $f'(x) \geq 0$ for all $x \in]-\infty\,;-1[\cup]-1\,;+\infty[$. The function f is therefore **strictly increasing** on $]-\infty\,;-1[$ and **strictly increasing** on $]-1\,;+\infty[$.

Worked Example 6 — Finding the Inflection Point of an Economic Cost Function

Problem: A chemical company produces an output quantity $q \in [0\,;10]$ (in tons) of a polymer. The total production cost in thousands of euros is modeled by:

$$C(q) = q^3 - 6q^2 + 15q + 10.$$

1. Prove that the marginal cost $C'(q)$ is strictly increasing on $[2\,;10]$. 2. Determine the inflection point of the total cost curve and interpret it economically (law of diminishing returns).

Solution: 1. Compute the marginal cost (first derivative) and its rate of change (second derivative):

$$C'(q) = 3q^2 - 12q + 15.$$

$$C''(q) = 6q - 12 = 6(q-2).$$

For all $q > 2$, $C''(q) > 0$. Thus C' is **strictly increasing** on $[2\,;10]$. 2. **Inflection point:** $C''(q) = 0 \Leftrightarrow 6q - 12 = 0 \Leftrightarrow q = 2$.

- For $q < 2$, $C''(q) < 0$: the cost function is concave.
- For $q > 2$, $C''(q) > 0$: the cost function is convex. The second derivative vanishes while changing sign at $q = 2$. The point with $q = 2$ is therefore the **inflection point** of the total cost curve. *Economic interpretation:*
- For $q \in [0\,;2]$, production benefits from economies of scale: each additional ton costs less to produce than the preceding one (marginal cost is decreasing).
- Past $q = 2$ tons, plant capacity begins to strain, machinery experiences wear, overtime wages accrue: each additional ton becomes progressively more expensive (diminishing returns). The point $q = 2$ marks the structural tipping point.

Key Takeaways

1. **Chain Rule:** $(g \circ u)'(x) = u'(x) \cdot g'\,\big(\,u(x)\big)$.
2. **The 3 Essential Composite Derivatives:**
   - $\big(\,e^u\big)' = u'e^u$
   - $\big(\,\sqrt{u}\big)' = \frac{u'}{2\sqrt{u}}$ (for $u > 0$)
   - $\big(\,u^n\big)' = nu'u^{n-1}$
3. **Convexity and the Second Derivative:**
   - $f''(x) \geq 0 \Leftrightarrow f'$ is increasing $\Leftrightarrow f$ is convex $\Leftrightarrow$ curve lies above its tangents.
   - $f''(x) \leq 0 \Leftrightarrow f'$ is decreasing $\Leftrightarrow f$ is concave $\Leftrightarrow$ curve lies below its tangents.
4. **Inflection Point:** point where the curve crosses its tangent $\Leftrightarrow f''(x)$ vanishes **while changing sign**.
5. **Fundamental Convexity Inequality:** For all real x, $e^x \geq x + 1$.

Exercises

Level (Direct Applications)

Exercise 47.1 Determine the derivative of each of the following functions: 1. $f(x) = e^{4x-1}$ 2. $g(x) = e^{-x^3}$ 3. $h(x) = \sqrt{5x+2}$ on $]-\frac{2}{5}\,;+\infty[$ 4. $k(x) = (3x-7)^5$

Exercise 47.2 Compute the second derivative $f''(x)$ for the following functions: 1. $f(x) = 2x^4 - 5x^3 + x - 8$ 2. $g(x) = e^{2x}$ 3. $h(x) = \frac{1}{x}$ on $]0\,;+\infty[$

Exercise 47.3 Consider the function $f(x) = x^3 - 6x^2 + 2x - 1$. 1. Compute $f''(x)$. 2. Determine the intervals where f is convex, and where it is concave. 3. Identify the inflection point of its graph.

Exercise 47.4 The graph of a twice-differentiable function f has a tangent line at $x_0 = 3$ with equation $y = 4x - 1$. Given that $f''(x) \geq 0$ for all $x \in \mathbb{R}$, compare without calculation $f(3.1)$ and 11.4.

Exercise 47.5 Determine the derivative of $f(x) = \sqrt{e^x + 1}$ on $\mathbb{R}$.

Exercise 47.6 Let $f(x) = xe^{-x}$. Compute $f'(x)$ and $f''(x)$.

Exercise 47.7 A function f satisfies $f''(x) = (x-3)^2(x+1)$. What are the abscissas of the inflection points of the graph of f? (Watch out for the trap!).

Exercise 47.8 Determine the monotonicity of $f'(x)$ for $f(x) = \frac{x^3}{3} - x^2 + 5x$.

Level (Guided Reflection and Baccalaureate Style)

Exercise 47.9 (Baccalaureate Style: Complete Study of an Exponential Function with Convexity). Consider the function f defined on $\mathbb{R}$ by:

$$f(x) = (x^2 - 2x + 1)e^x.$$

1. Prove that for all real x, $f'(x) = (x^2 - 1)e^x$. 2. Construct the complete variation table of f on $\mathbb{R}$, specifying the limits at $-\infty$ and at $+\infty$. 3. Compute $f''(x)$ and prove that $f''(x) = (x^2 + 2x - 1)e^x$. 4. Determine the intervals of convexity of f. 5. Determine the exact abscissas of the inflection points of $\mathcal{C}_f$.

Exercise 47.10 (Relative Position and Convexity Inequality). Let $f(x) = \sqrt{x}$ defined on $[0\,;+\infty[$. 1. Compute $f'(x)$ and $f''(x)$ for $x > 0$. 2. Deduce the concavity of the square root function on $]0\,;+\infty[$. 3. Give the equation of the tangent line (T) to the curve at $x_0 = 1$. 4. Deduce that for all $x > 0$: $\sqrt{x} \leq \frac{x+1}{2}$. *(Thus retrieving the arithmetic-geometric mean inequality between $\sqrt{x}$ and the average of x and 1).*

Exercise 47.11 (Baccalaureate Style: Skatepark Ramp Profile Analysis). An architect designs the cross-section of a skatepark ramp on $[-2\,;2]$ modeled by:

$$f(x) = \frac{1}{1 + e^{-2x}}.$$

1. Prove that $f'(x) = \frac{2e^{-2x}}{(1+e^{-2x})^2}$. 2. Compute $f''(x)$ and show that it can be written as $f''(x) = \frac{4e^{-2x}(e^{-2x}-1)}{(1+e^{-2x})^3}$. 3. Determine the inflection point of the ramp. Why does this point correspond to the location where the ramp's steepness is maximal?

Exercise 47.12 (True or False with Rigorous Arguments). 1. "If $f''(x_0) = 0$, then the curve of f has an inflection point at x_0." 2. "If a function is strictly increasing, then it is convex." 3. "The sum of two convex functions on an interval I is a convex function on I." 4. "The product of two convex functions on I is always a convex function on I."

Exercise 47.13 (Differentiating Trigonometric Composite Functions). Compute the derivatives of: 1. $f(x) = \cos(3x^2 - 2)$ 2. $g(x) = e^{\sin(x)}$ 3. $h(x) = \sin^3(2x)$

Exercise 47.14 (Convexity of the Reciprocal of a Function). Let u be a twice-differentiable, strictly positive, and concave function on an interval I. Set $f = \frac{1}{u}$. 1. Compute f' and f'' in terms of u, u', and u''. 2. Prove that f is convex on I.

Exercise 47.15 (Convexity Inequality for the Square Function). Let $f(x) = x^2$ on $\mathbb{R}$. 1. Recall why f is convex. 2. State the geometric definition of convexity for two arbitrary real numbers a and b with midpoint $m = \frac{a+b}{2}$: show that $\left(\frac{a+b}{2}\right)^2 \leq \frac{a^2+b^2}{2}$. 3. Prove this inequality algebraically by expanding $\frac{a^2+b^2}{2} - \left(\frac{a+b}{2}\right)^2$.

Exercise 47.16 (Kinematics and Emergency Braking of a High-Speed Train). The position (in meters) of a high-speed train during an emergency braking sequence is given for $t \in [0\,;20]$ (in seconds) by:

$$x(t) = -0.15t^3 + 4.5t^2 + 30t.$$

1. Determine the instantaneous velocity $v(t) = x'(t)$ of the train at $t = 0$ and at $t = 10$ s. 2. Determine the acceleration $a(t) = x''(t)$. 3. At what time is the deceleration experienced by the passengers strongest in intensity?

Exercise 47.17 (Complete Study of the Derivative of u^n). Consider the function $f(x) = (x^2 - 4x + 3)^3$ on $\mathbb{R}$. 1. Determine the zeros of f. 2. Compute $f'(x)$ in factored form. 3. Construct the complete variation table of f.

Level (Challenge and In-Depth Study)

Exercise 47.18 (Jensen's Inequality for Convex Functions). Let f be a convex function on an interval I. 1. The two-point convexity inequality states that for all $\lambda \in [0\,;1]$ and all $x, y \in I$:

$$f\,\big(\,\lambda x + (1-\lambda)y\,\big) \leq \lambda f(x) + (1-\lambda)f(y).$$

Interpret this inequality geometrically by connecting λ to the position on the chord $[AB]$. 2. By induction, prove Jensen's inequality for n numbers: if $x_1, x_2, ..., x_n \in I$, then:

$$f\left(\frac{x_1 + x_2 + ... + x_n}{n}\right) \leq \frac{f(x_1) + f(x_2) + ... + f(x_n)}{n}.$$

3. Apply Jensen's inequality to the convex function $f(x) = x^2$ to prove that the mean of squares is always greater than or equal to the square of the mean (non-negativity of variance).

Exercise 47.19 (Rigorous Proof of Relative Position via Taylor-Lagrange). Let f be twice differentiable on an interval I. Let $a \in I$. Taylor's formula with integral remainder shows that for all $x \in I$:

$$f(x) - \big[\,f'(a)(x-a) + f(a)\big] = \int_a^x (x-t)f''(t)\,dt.$$

1. Show that the integrand $(x-t)f''(t)$ maintains a non-negative sign for t between a and x provided that $f''(t) \geq 0$. 2. Deduce analytically that any function satisfying $f'' \geq 0$ lies above its tangent line at every point a.

Exercise 47.20 (Cubic Polynomial with Imposed Inflection Point). Consider a general degree-3 polynomial: $P(x) = ax^3 + bx^2 + cx + d$ with $a \neq 0$. 1. Compute $P''(x)$. 2. Prove that the graph of **every degree-3 polynomial** has a unique inflection point I, whose abscissa is $x_I = -\frac{b}{3a}$. 3. Prove that the inflection point I is the center of symmetry of the graph of P.

Exercise 47.21 (Composite of Two Convex Functions). Let $u : I \to J$ and $g : J \to \mathbb{R}$ be two twice-differentiable functions. 1. Compute $(g \circ u)''(x)$ in terms of the first and second derivatives of g and u. 2. Prove that if u is convex and g is **convex AND increasing**, then the composite $g \circ u$ is convex on I. 3. Give a counterexample showing that if g is not increasing, the result can fail.

Exercise 47.22 (General AM-GM Inequality via Convexity). Assume that the function $x \mapsto -\ln(x)$ is strictly convex on $]0\,;+\infty[$. 1. Let $x_1, x_2, ..., x_n$ be strictly positive real numbers. Apply Jensen's inequality to $-\ln(x)$. 2. Using logarithm properties, deduce the general Arithmetic Mean - Geometric Mean (AM-GM) inequality for n numbers:

$$\sqrt[n]{x_1 x_2 ... x_n} \leq \frac{x_1 + x_2 + ... + x_n}{n}.$$

Solutions

Solutions for Level

- **47.1:**
  1. $u(x) = 4x - 1 \Longrightarrow u'(x) = 4$. $f'(x) = 4e^{4x-1}$.
  2. $u(x) = -x^3 \Longrightarrow u'(x) = -3x^2$. $g'(x) = -3x^2 e^{-x^3}$.
  3. $u(x) = 5x + 2 \Longrightarrow u'(x) = 5$. $h'(x) = \frac{5}{2\sqrt{5x+2}}$.

4. $u(x) = 3x - 7 \Longrightarrow u'(x) = 3$. $k'(x) = 5 \times 3 \times (3x-7)^4 = 15(3x-7)^4$.

- **47.2:**
  1. $f'(x) = 8x^3 - 15x^2 + 1 \Longrightarrow f''(x) = 24x^2 - 30x$.
  2. $g'(x) = 2e^{2x} \Longrightarrow g''(x) = 4e^{2x}$.
  3. $h'(x) = -\frac{1}{x^2} = -x^{-2} \Longrightarrow h''(x) = 2x^{-3} = \frac{2}{x^3}$.
- **47.3:**
  1. $f'(x) = 3x^2 - 12x + 2 \Longrightarrow f''(x) = 6x - 12 = 6(x-2)$.
  2. For $x \leq 2$, $f''(x) \leq 0$: f is concave. For $x \geq 2$, $f''(x) \geq 0$: f is convex.
  3. f'' vanishes while changing sign at $x = 2$. $f(2) = 8 - 24 + 4 - 1 = -13$. The inflection point is $I(2\,;-13)$.
- **47.4:** Since $f''(x) \geq 0$, f is convex on $\mathbb{R}$, so its curve lies above its tangent line $y = 4x - 1$. For $x = 3.1$, the tangent gives $4(3.1) - 1 = 12.4 - 1 = 11.4$. Since the curve is above the tangent, $f(3.1) \geq 11.4$.
- **47.5:** $u(x) = e^x + 1 > 0$, $u'(x) = e^x$. $f'(x) = \frac{e^x}{2\sqrt{e^x+1}}$.
- **47.6:** $f'(x) = 1 \cdot e^{-x} + x(-e^{-x}) = (1-x)e^{-x}$. $f''(x) = (-1)e^{-x} + (1-x)(-e^{-x}) = (-1-1+x)e^{-x} = (x-2)e^{-x}$.
- **47.7:** For all real x, $(x-3)^2 \geq 0$. Thus $f''(x)$ has the sign of $x + 1$.
  - At $x = 3$, $f''(3) = 0$, but f'' does not change sign around 3: **no inflection point at 3!**
  - At $x = -1$, $f''(-1) = 0$ and f'' changes from negative to positive: **unique inflection point at $x = -1$.**
- **47.8:** $f'(x) = x^2 - 2x + 5$. The derivative of f' is $f''(x) = 2x - 2 = 2(x-1)$. f' is strictly decreasing on $]-\infty\,;1]$ and strictly increasing on $[1\,;+\infty[$.

Solutions for Level

- **47.9:**
  1. $f(x) = (x^2 - 2x + 1)e^x = (x-1)^2 e^x$. $f'(x) = (2x-2)e^x + (x^2-2x+1)e^x = (x^2-1)e^x$.
  2. Sign of $f'(x)$: since $e^x > 0$, it has the sign of $x^2 - 1 = (x-1)(x+1)$. Increasing on $]-\infty;-1]$, decreasing on $[-1;1]$, increasing on $[1;+\infty[$. At $-\infty$: $(x^2-2x+1)e^x \to 0$ by comparative growth. At $+\infty$: $\lim f(x) = +\infty$. Local max at -1: $f(-1) = 4e^{-1} = 4/e \approx 1.47$. Local min at 1: $f(1) = 0$.
  3. $f''(x) = (2x)e^x + (x^2-1)e^x = (x^2+2x-1)e^x$.
  4. Roots of $x^2 + 2x - 1$: $\Delta = 4 - 4(1)(-1) = 8 = (2\sqrt{2})^2$. $x_1 = \frac{-2-2\sqrt{2}}{2} = -1 - \sqrt{2} \approx -2.41$. $x_2 = -1 + \sqrt{2} \approx 0.41$. f is convex on $]-\infty\,;-1-\sqrt{2}]$ and on $[-1+\sqrt{2}\,;+\infty[$. f is concave on $[-1-\sqrt{2}\,;-1+\sqrt{2}]$.
  5. The inflection point abscissas are $x_1 = -1 - \sqrt{2}$ and $x_2 = -1 + \sqrt{2}$.
- **47.10:**
  1. $f'(x) = \frac{1}{2\sqrt{x}} = \frac{1}{2}x^{-1/2}$. $f''(x) = \frac{1}{2}\left(-\frac{1}{2}\right)x^{-3/2} = -\frac{1}{4x\sqrt{x}}$.
  2. For all $x > 0$, $4x\sqrt{x} > 0$, so $f''(x) < 0$: the square root function is **strictly concave** on $]0\,;+\infty[$.
  3. At $x_0 = 1$: $f(1) = 1$, $f'(1) = 1/2$. Tangent: $y = \frac{1}{2}(x-1) + 1 = \frac{1}{2}x + \frac{1}{2} = \frac{x+1}{2}$.
  4. The graph of a concave function lies entirely below its tangents, so for all $x > 0$: $\sqrt{x} \leq \frac{x+1}{2}$.
- **47.11:**
  1. $f(x) = (1+e^{-2x})^{-1}$. $f'(x) = -(-2e^{-2x})(1+e^{-2x})^{-2} = \frac{2e^{-2x}}{(1+e^{-2x})^2}$.
  2. Differentiate as a quotient u/v: $u' = -4e^{-2x}$, $v' = 2(1+e^{-2x})(-2e^{-2x}) = -4e^{-2x}(1+e^{-2x})$. $f''(x) = \frac{-4e^{-2x}(1+e^{-2x})^2 - 2e^{-2x}[-4e^{-2x}(1+e^{-2x})]}{(1+e^{-2x})^4} = \frac{-4e^{-2x}(1+e^{-2x}) + 8e^{-4x}}{(1+e^{-2x})^3} = \frac{4e^{-2x}(e^{-2x}-1)}{(1+e^{-2x})^3}$.
  3. $f''(x) = 0 \Leftrightarrow e^{-2x} - 1 = 0 \Leftrightarrow -2x = 0 \Leftrightarrow x = 0$. f'' changes sign at 0. The inflection point is $(0\,;1/2)$. Since $f''(0) = 0$, the first derivative f' has an extremum at 0: $f'(0) = \frac{2}{(1+1)^2} = \frac{2}{4} = \frac{1}{2}$. This is the maximal slope of the ramp.
- **47.12:**
  1. **False.** Counterexample: $f(x) = x^4$ at 0 ($f''(0) = 0$ but $f'' \geq 0$ does not change sign).
  2. **False.** Counterexample: $f(x) = \sqrt{x}$ is strictly increasing and strictly concave.
  3. **True.** If f and g are convex, $(f+g)'' = f'' + g'' \geq 0$.
  4. **False.** Counterexample: $f(x) = x$ and $g(x) = -x$ are linear (convex/concave), but their product is $-x^2$, which is strictly concave!

- **47.13:**
  1. $(\cos u)' = -u' \sin u$. $u(x) = 3x^2 - 2 \Longrightarrow u' = 6x$. $f'(x) = -6x \sin(3x^2 - 2)$.
  2. $(e^u)' = u' e^u$. $u(x) = \sin x \Longrightarrow u' = \cos x$. $g'(x) = \cos(x) e^{\sin(x)}$.
  3. $h(x) = v^3$ with $v(x) = \sin(2x)$. $v'(x) = 2\cos(2x)$. $h'(x) = 3v'v^2 = 3(2\cos(2x))\sin^2(2x) = 6\cos(2x)\sin^2(2x)$.
- **47.14:**
  1. $f = u^{-1} \Longrightarrow f' = -u'u^{-2} = -\frac{u'}{u^2}$. $f'' = \left(-\frac{u'}{u^2}\right)' = -\frac{u''u^2 - u'(2uu')}{u^4} = -\frac{u''u - 2(u')^2}{u^3} = \frac{2(u')^2 - uu''}{u^3}$.
  2. Since u is concave, $u'' \leq 0$, so $-uu'' \geq 0$ (since $u > 0$). Since $(u')^2 \geq 0$ and $u^3 > 0$, the numerator $2(u')^2 - uu'' \geq 0$ and the denominator is strictly positive. Thus $f'' \geq 0$: the function f is convex on I.
- **47.15:**
  1. $f''(x) = 2 > 0$ on $\mathbb{R}$, so f is convex.
  2. The segment connecting $(a; a^2)$ and $(b; b^2)$ has midpoint ordinate $\frac{a^2+b^2}{2}$. The image of the midpoint abscissa is $\left(\frac{a+b}{2}\right)^2$. By convexity, the curve lies below the chord: $\left(\frac{a+b}{2}\right)^2 \leq \frac{a^2+b^2}{2}$.
  3. $\frac{a^2+b^2}{2} - \frac{a^2+2ab+b^2}{4} = \frac{2a^2+2b^2-a^2-2ab-b^2}{4} = \frac{a^2-2ab+b^2}{4} = \frac{(a-b)^2}{4} \geq 0$.
- **47.16:**
  1. $v(t) = x'(t) = -0.45t^2 + 9t + 30$. $v(0) = 30 \ \text{m/s}$ (i.e. $108 \ \text{km/h}$). $v(10) = -0.45(100) + 90 + 30 = -45 + 120 = 75 \ \text{m/s}$.
  2. $a(t) = x''(t) = -0.9t + 9$.
  3. Acceleration $a(t)$ decreases linearly from $+9 \ \text{m/s}^2$ at $t = 0$ down to $a(20) = -0.9(20) + 9 = -18 + 9 = -9 \ \text{m/s}^2$. Maximum deceleration occurs at $t = 20 \ \text{s}$ with magnitude $9 \ \text{m/s}^2$ (almost 1 g!).
- **47.17:**
  1. $x^2 - 4x + 3 = 0 \Leftrightarrow (x-1)(x-3) = 0$. Zeros are 1 and 3.
  2. $f'(x) = 3(2x-4)(x^2-4x+3)^2 = 6(x-2)(x-1)^2(x-3)^2$.
  3. For all $x \in \mathbb{R}$, $(x-1)^2(x-3)^2 \geq 0$. The sign of $f'(x)$ is that of $x - 2$. f is strictly decreasing on $]-\infty\,;2]$ and strictly increasing on $[2\,;+\infty[$. Minimum at $x = 2$: $f(2) = (4 - 8 + 3)^3 = (-1)^3 = -1$. At $x = 1$ and $x = 3$, $f'(x) = 0$: horizontal inflection tangents (saddle points / flat spots).

Solutions for Level

- **47.18:**
  1. $\lambda x + (1-\lambda)y$ is the weighted average (convex combination) of the x-coordinates of $A(x, f(x))$ and $B(y, f(y))$, and $\lambda f(x) + (1-\lambda)f(y)$ is the corresponding y-coordinate on the chord $[AB]$. The inequality expresses that points on the graph lie below or on the chord.
  2. Induction on n: true for $n = 2$. Assume true for n. Write $\frac{x_1+\ldots+x_{n+1}}{n+1} = \frac{n}{n+1}\left(\frac{x_1+\ldots+x_n}{n}\right) + \frac{1}{n+1}x_{n+1}$. Applying 2-point convexity with $\lambda = \frac{n}{n+1}$, and then the induction hypothesis, gives the result for $n + 1$.
  3. With $f(x) = x^2$: $\left(\frac{\sum x_i}{n}\right)^2 \leq \frac{\sum x_i^2}{n} \Leftrightarrow \overline{x}^2 \leq \overline{x^2} \Leftrightarrow V(X) = \overline{x^2} - \overline{x}^2 \geq 0$.
- **47.19:**
  1. If $x > a$, for all $t \in [a\,;x]$, $x - t \geq 0$. Since $f''(t) \geq 0$, the integrand is non-negative. Its integral from a to x is non-negative. If $x < a$, $x - t \leq 0$ for $t \in [x\,;a]$. The integral from a to x equals $-\int_x^a (x-t)f''(t)dt = \int_x^a (t-x)f''(t)dt \geq 0$.
  2. In both cases, $f(x) - [f'(a)(x-a) + f(a)] \geq 0$, rigorously proving that the curve remains above its tangent at every point a.
- **47.20:**
  1. $P'(x) = 3ax^2 + 2bx + c \Longrightarrow P''(x) = 6ax + 2b$.
  2. $P''(x) = 0 \Leftrightarrow 6ax + 2b = 0 \Leftrightarrow x = -\frac{2b}{6a} = -\frac{b}{3a}$. Since $a \neq 0$, $6ax + 2b$ is a linear function with non-zero slope, which necessarily changes sign at $x_I = -\frac{b}{3a}$. Every cubic polynomial therefore has exactly one inflection point.
  3. Under the substitution $X = x - x_I$, the quadratic term vanishes in $P(X + x_I)$: the polynomial takes the form $aX^3 + pX + Y_I$, which satisfies $P(X) - Y_I = -(P(-X) - Y_I)$: an odd function centered at $(x_I; Y_I)$, establishing point symmetry.

- **47.21:**
  1. $(g \circ u)' = u' \cdot (g' \circ u)$. By the product rule:
  $$(g \circ u)'' = u'' \cdot (g' \circ u) + u' \cdot [(g' \circ u)'] = u'' \cdot (g' \circ u) + (u')^2 \cdot (g'' \circ u).$$
  2. Since u is convex, $u'' \geq 0$. Since g is increasing, $g' \geq 0$, so $u'' \cdot (g' \circ u) \geq 0$. Furthermore, $(u')^2 \geq 0$, and since g is convex, $g'' \geq 0$, so $(u')^2 \cdot (g'' \circ u) \geq 0$. The sum of two non-negative terms is non-negative: $(g \circ u)'' \geq 0$, so $g \circ u$ is convex.
  3. Counterexample if g is not increasing: $u(x) = x^2$ (convex on $\mathbb{R}$) and $g(y) = -y$ (convex on $\mathbb{R}$). The composite is $(g \circ u)(x) = -x^2$, which is strictly concave!
- **47.22:**
  1. Let $f(x) = -\ln(x)$ on $]0\,;+\infty[$. $f'(x) = -1/x$, $f''(x) = 1/x^2 > 0$: f is strictly convex. By Jensen's inequality:
  $$-\ln\left(\frac{x_1 + ... + x_n}{n}\right) \leq \frac{-\ln(x_1) - ... - \ln(x_n)}{n} = -\frac{1}{n}\ln(x_1 ... x_n) = -\ln\left((x_1 ... x_n)^{1/n}\right).$$
  2. Multiply by -1 (reversing the inequality):
  $$\ln\left(\frac{x_1 + ... + x_n}{n}\right) \geq \ln\big((x_1 ... x_n)^{1/n}\big).$$
  Since the exponential function is strictly increasing:
  $$\frac{x_1 + ... + x_n}{n} \geq \sqrt[n]{x_1 x_2 ... x_n}. \quad \blacksquare$$

The Question That Opens What Follows

We have seen that the exponential function $x \mapsto e^x$ is the sovereign of functions: it turns addition into multiplication, its derivative is equal to itself, and its curve is entirely convex above its tangent $y = x + 1$. Yet what happens if an astronomer or a navigator needs to retrace the steps in reverse? What if one knows the final power and seeks **the exponent that produced it**? How did a graduated slide rule and printed calculation tables enable seventeenth-century sailors to navigate across oceans without performing a single long multiplication by hand?

This is the saga of **John Napier** and the **natural logarithm** function. See you in Chapter 48.

CHAPTER 48: Napier's Logarithm and the Navigator's Tables

> "By shortening the labours, the invention of logarithms doubled the life of astronomers." — *Pierre-Simon de Laplace*

What You Are About to Discover

- Why, around 1600, navigators and astronomers were suffocating under **multiplications**, and how a Scottish baron invented a machine to transform multiplications into **additions**.
- The **natural logarithm function** $\ln$, defined as the machine that **undoes** the exponential: $y = e^x \Leftrightarrow x = \ln y$.
- Its **algebraic properties** ($\ln(ab) = \ln a + \ln b$, etc.) and their complete **proofs**.
- A **comprehensive analysis** of the function: derivative $\frac{1}{x}$, monotonicity, limits, indeterminate forms, asymptotic growth rates, and concavity.
- Rigorous methods to solve **equations and inequalities** involving $\ln$ and e, and how to determine **thresholds** in geometric sequences.
- Real-world **logarithmic scales**: pH, decibels, earthquake magnitudes, and carbon-14 radiometric dating.

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1. The Problem: The Sea, the Stars, and Endless Multiplications

1.1 The Lost Navigator

Imagine yourself in the year **1600**, standing on the deck of a galleon crossing the Atlantic. There is no GPS, no radio, no reliable marine chronometer. To determine the ship's coordinates, the navigator uses a **quadrant** to measure the angular elevation of the Sun or a guide star above the horizon, and must then solve a **spherical triangle**: a triangle inscribed on the celestial sphere.

The formulas of spherical trigonometry look like this:

$$\cos a = \cos b \cos c + \sin b \sin c \cos A.$$

Sines and cosines are read from printed **tables**, computed to seven or eight decimal places for navigational accuracy. The navigator must therefore multiply, entirely by hand, numbers such as:

$$0.8746197 \times 0.4848096.$$

Such long multiplication involves dozens of intermediate products and additions, with carries at every step. A single calculation error, and the vessel believes itself to be a hundred kilometers from its actual position—heading straight for submerged reefs. Astronomers, too, such as **Tycho Brahe** and **Johannes Kepler**, spent whole **decades** of their lives grinding through tedious multiplications to determine planetary orbits.

> 💡 **Did you know?**
>
> Before logarithms, mathematicians used a computational shortcut known as **prosthaphaeresis** (from the Greek for "addition and subtraction"), which exploited the product-to-sum trigonometric identity $\cos a \cos b = \frac{1}{2}[\cos(a+b) + \cos(a-b)]$ to convert products into sums… but it was cumbersome, error-prone, and restricted to numbers between -1 and 1.

1.2 The Dream: Replacing × with +

Addition is straightforward: one aligns the digits and adds. Multiplication is laborious and prone to error. Hence the great dream of Renaissance computers:

> To find a "dictionary" that pairs every positive number with another number in such a way that the translation of a product is simply the sum of the translations.

The German monk **Michael Stifel** caught a glimpse of this dream in 1544 by tabulating two sequences, one beneath the other:

Progression arithmétique	0	1	2	3	4	5	6	7	8
Progression géométrique	1	2	4	8	16	32	64	128	256

To multiply 8×32, look above: 8 corresponds to 3, and 32 corresponds to 5. Add them: $3 + 5 = 8$. Look back down: under 8 sits 256. And indeed, $8 \times 32 = 256$! **The top row converts multiplications of the bottom row into simple additions.**

This is simply the rule of exponents you have known since middle school:

$$2^3 \times 2^5 = 2^{3+5}.$$

Yet there was a critical flaw: between 8 and 16, there are vast gaps. How could one multiply 11×13? One would need a geometric progression **with steps so densely packed** that virtually every real number is represented. One had to **fill the gaps**.

1.3 Twenty Years of Calculation: The Baron of Merchiston

This monumental feat was accomplished by **John Napier** (1550–1617), 8th Laird of Merchiston in Scotland. He chose a geometric progression whose common ratio was exceedingly close to 1, namely $1 - 10^{-7} = 0.9999999$, so that consecutive terms barely differed: the gaps shrank to microscopic proportions. For **twenty years**, by hand, he computed his tables. In **1614**, he published *Mirifici Logarithmorum Canonis Descriptio* ("Description of the Wonderful Rule of Logarithms").

The word **logarithm**, which he coined, derives from the Greek *logos* (ratio, proportion) and *arithmos* (number): "the number that measures ratios". The logarithm of a number is, in essence, **the number of steps** one must take along the geometric progression to reach it.

> **📐 Mathematical History — From Napier's Kinematics to Euler's Constant**
>
> The original logarithm published by John Napier in 1614 was not yet our modern function $\ln(x)$:
>
> 1. **Napier (1614):** Designed through a kinematic model on a radius of 10^7. A particle decelerates proportionally to remaining distance while another advances steadily. This historical logarithm was strictly decreasing, with $\text{NapLog}(10^7) = 0$, giving $\text{NapLog}(x) \approx 10^7 \ln\left(\frac{10^7}{x}\right)$.
> 2. **Henry Briggs (1617–1624):** Collaborated with Napier to establish the base-10 decimal system where $\log_{10}(1) = 0$ and $\log_{10}(10) = 1$.
> 3. **Grégoire de Saint-Vincent (1647) & Alphonse Antonio de Sarasa (1649):** Groundbreaking discovery linking the area beneath the hyperbola $y = \frac{1}{x}$ directly to the additive logarithmic law.
> 4. **Nicolaus Mercator (1668):** Formally coined the term "natural logarithm" in his *Logarithmotechnia*.
> 5. **Leonhard Euler (1748):** Crowned modern calculus by identifying $e \approx 2.71828$ and defining $\ln(x)$ rigorously as the inverse of the natural exponential e^x.

> **💡 True History —**
>
> The English mathematician **Henry Briggs**, reading Napier's work, was so astonished that he journeyed from London to Scotland in 1615 to meet him. It is said that when they met, the two men stood looking at one another in speechless admiration for a quarter of an hour before uttering a word. Together, they agreed that logarithms should be recast in "base 10" (where $\log 10 = 1$), which was far more practical for decimal notation: these became the **common (or decimal) logarithms**. Kepler, who received Napier's tables in 1617, immediately used them to complete his monumental *Rudolphine Tables* of planetary motions and dedicated one of his works to Napier… unaware that Napier had passed away in the interim.

For more than **three centuries**, logarithmic tables and **slide rules** (pairs of logarithmic scales sliding against each other to add lengths mechanically) accompanied engineers, sailors, architects, and scientists—all the way to NASA engineers working on the Apollo moon landings in the 1960s!

1.4 The Same Problem, Seen from the Laboratory

Recall the bacterial colony from Chapter 40. A population governed by the law $N(t) = N_0\, e^{kt}$ grows exponentially. The biologist faces the **inverse** question:

> "The population has grown by a factor of 1000. How long ago did the infection begin?"

This requires solving $e^{kt} = 1000$—that is, finding the **exponent** that produces a desired result. The exponential function maps the exponent to the output; we need the reciprocal machine that maps the **output back to the exponent**. That machine is the **natural logarithm**. The two narratives—that of navigators at sea and that of biologists in the laboratory—converge at the exact same concept.

---

2. Definition: The Machine that Inverts the Exponential

2.1 What We Already Know About the Exponential

Let us summarize what Chapters 40 and 46 established regarding the function $\exp : x \mapsto e^x$:

- It is defined, continuous, and differentiable on $\mathbb{R}$, with $(e^x)' = e^x$;
- It is **strictly positive**: $e^x > 0$ for all real numbers x;
- It is therefore **strictly increasing** on $\mathbb{R}$;
- $\lim_{x \to -\infty} e^x = 0$ and $\lim_{x \to +\infty} e^x = +\infty$.

The **corollary to the Intermediate Value Theorem** (the Bijective Theorem, Chapter 46) then guarantees: a function that is continuous and strictly monotonic on an interval defines a **bijection** from that interval onto its image. Here:

> ***For every real number $a > 0$, the equation $e^x = a$ has a unique real solution.***

Visually: the exponential curve intersects every horizontal line $y = a$ (with $a > 0$) **at exactly one point**. And it never intersects the lines $y = a$ with $a \leq 0$, because the curve lies entirely above the horizontal axis.

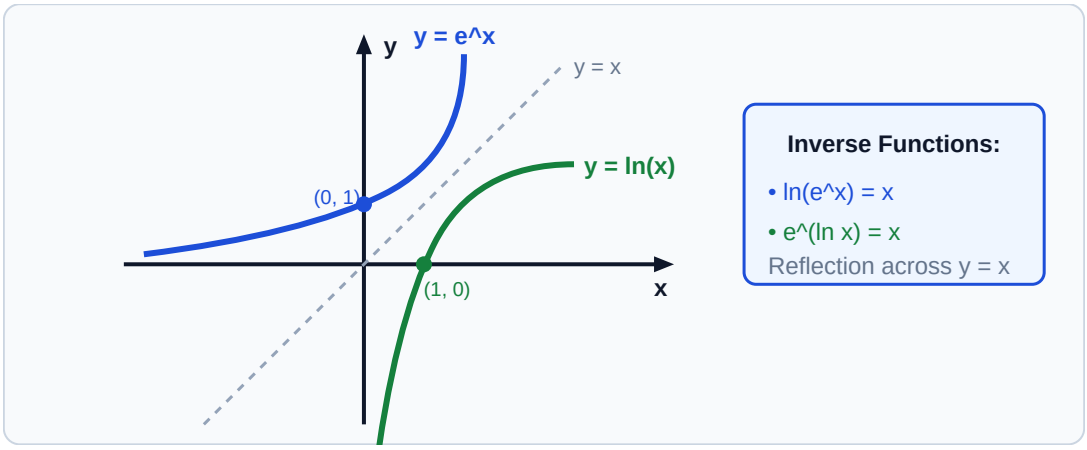

Figure 105: Inverse relationship of exponential and natural logarithm functions

2.2 The Definition

> **📐 Definition —**
>
> For every strictly positive real number a, the **natural logarithm** of a, denoted by $\ln a$ (or $\ln(a)$), is the **unique** real number x such that $e^x = a$. The function $\ln :]0\,;+\infty[\to \mathbb{R}$, $x \mapsto \ln x$ is the **natural logarithm function**. It is the **inverse function** of the exponential function.
>
> $$\text{For} \quad a > 0 \quad \text{and} \quad x \in \mathbb{R} : \qquad e^x = a \Leftrightarrow x = \ln a.$$

Notation. The symbol ln abbreviates *logarithmus naturalis* (Latin for "natural logarithm"). It is read "the natural logarithm of x", or informally "el-en of x". Parentheses are optional when there is no ambiguity: $\ln 2$, $\ln x$; but one writes $\ln(x+1)$, and never $\ln x + 1$, which represents $(\ln x) + 1$.

> **💡 Did you know?**
>
> Napier did not use the symbol ln, nor did he view it as the inverse of e^x: the base e had not yet been identified or named! It was **Grégoire de Saint-Vincent** and **Alphonse Antonio de Sarasa** (around 1647–1649) who discovered that the area under the hyperbola $y = \frac{1}{x}$ exhibits logarithmic behavior (as we will explore in Chapter 50), and **Leonhard Euler**, in 1748, who systematically established the exponential and logarithm as mutual inverse operations. Napier's original tables corresponded closely to the function $x \mapsto 10^7 \ln\left(\frac{10^7}{x}\right)$.

2.3 Immediate Consequences

From this definition follow directly:

Property	Reason
$\ln 1 = 0$	because $e^0 = 1$
$\ln e = 1$	because $e^1 = e$
$e^{\ln a} = a$ for all $a > 0$	$\ln a$ is *by definition* the exponent that yields a
$\ln(e^x) = x$ for all $x \in \mathbb{R}$	the unique exponent producing e^x is x

The last two rows express cancellation: **applying "exp" then "ln", or "ln" then "exp", returns you to your starting value.** It is like opening a drawer and then closing it.

> **⚠ Common Pitfall!**
>
> $e^{\ln a} = a$ holds only if $a > 0$. For instance, $e^{\ln(-3)}$ is **meaningless**, because $\ln(-3)$ does not exist in the real numbers. In contrast, $\ln(e^x) = x$ is valid for **every** real number x, even negative numbers: $\ln(e^{-5}) = -5$.

2.4 The Curve: A Mirror Reflection of the Exponential

In a Cartesian coordinate system, if the point $M(a\,;b)$ lies on the graph of exp (so that $b = e^a$), then $a = \ln b$, which means the point $M'(b\,;a)$ lies on the graph of ln. The point M' is the **geometric reflection** of M across the line Δ of equation $y = x$ (interchanging abscissa and ordinate).

> ***The curves of* exp *and* ln *are reflections of each other across the first bisector line* $y = x$.**

If you fold the coordinate plane along the diagonal line $y = x$, the exponential curve lands precisely on top of the logarithmic curve. The point $(0\,;1)$ of exp becomes $(1\,;0)$ on ln; the horizontal asymptote $y = 0$ of exp (as $x \to -\infty$) transforms into the **vertical asymptote** $x = 0$ of ln (as $x \to 0^+$).

2.5 Worked Examples

Worked Example 1 — Solve $e^x = 5$.

Since $5 > 0$, the equation has a unique solution: $x = \ln 5 \approx 1.609$. Verification: $e^{1.609} \approx 4.998$. ✓

Worked Example 2 — Solve $e^{2x-1} = 3$.

Since $3 > 0$, we have $e^{2x-1} = 3 \Leftrightarrow 2x - 1 = \ln 3 \Leftrightarrow x = \frac{1+\ln 3}{2}$. Solution set: $\mathcal{S} = \left\{\frac{1+\ln 3}{2}\right\} \approx 1.049$.

Worked Example 3 — Solve $\ln x = 2$.

The equation is defined only for $x > 0$. Then $\ln x = 2 \Leftrightarrow x = e^2$. Since $e^2 > 0$, the solution is valid: $\mathcal{S} = \{e^2\}$, where $e^2 \approx 7.389$.

Worked Example 4 — Solve $e^x = -4$ and $3e^x + 6 = 0$.

The exponential is strictly positive for all real inputs: $e^x = -4$ has **no solution**, so $\mathcal{S} = \varnothing$. Similarly, $3e^x + 6 = 0 \Leftrightarrow e^x = -2$: $\mathcal{S} = \varnothing$. Writing "$x = \ln(-2)$" is a fatal conceptual error.

Worked Example 5 — Simplify $A = e^{\ln 7}$, $B = \ln(e^{-3})$, $C = e^{-\ln 2}$, $D = \ln(\sqrt{e})$.

- $A = 7$;
- $B = -3$;
- $C = \frac{1}{e^{\ln 2}} = \frac{1}{2}$;
- $D = \ln\left(e^{1/2}\right) = \frac{1}{2}$.

⚠ **Common Pitfall! — Domain Boundaries.**

- $\ln 0$ **does not exist**: no real power of e equals 0 (the exponential approaches 0 asymptotically but never reaches it).
- The logarithm of a negative number **does not exist** within the real numbers $\mathbb{R}$.
- Before solving any equation involving $\ln(\ldots)$, you must **always** begin by determining the domain of definition —the set of all x for which every argument inside a logarithm is **strictly positive**.

---

3. Algebraic Properties: Napier's Dream Realized

3.1 The Fundamental Identity

📐 **Fundamental Theorem —**

For all real numbers $a > 0$ and $b > 0$:

$$\ln(ab) = \ln a + \ln b.$$

In words: the logarithm **converts products into sums**. This is precisely Stifel's top row, freed from all gaps: every positive real number participates.

Proof (Required Curriculum Proof) —

Let $a > 0$ and $b > 0$. Then $ab > 0$, so $\ln(ab)$ is defined. On the one hand, $e^{\ln(ab)} = ab$ by definition of the logarithm. On the other hand, applying the fundamental exponent rule $e^{x+y} = e^x e^y$:

$$e^{\ln a + \ln b} = e^{\ln a} \times e^{\ln b} = a \times b.$$

Consequently:

$$e^{\ln(ab)} = e^{\ln a + \ln b}.$$

Because the exponential function is strictly increasing, it is **injective** (distinct inputs yield distinct outputs). Thus, their exponents must be equal:

$$\ln(ab) = \ln a + \ln b. \quad \blacksquare$$

The entire mystery is revealed: **the fundamental additive identity of the logarithm is the exact mirror image of the multiplicative identity $e^{x+y} = e^x e^y$ of the exponential**, reflected across the diagonal line $y = x$.

3.2 Derived Algebraic Identities

📐 ***Properties*** *— For all real numbers $a > 0, b > 0$, and every integer $n \in \mathbb{Z}$: 1. $\ln\left(\frac{1}{a}\right) = -\ln a$ 2. $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$ 3. $\ln(a^n) = n \ln a$ 4. $\ln(\sqrt{a}) = \frac{1}{2}\ln a$*

Proofs —

1. Since $a \times \frac{1}{a} = 1$, we have $\ln a + \ln\left(\frac{1}{a}\right) = \ln 1 = 0$, which yields $\ln\left(\frac{1}{a}\right) = -\ln a$. 2. Using identity 1: $\ln\left(\frac{a}{b}\right) = \ln\left(a \times \frac{1}{b}\right) = \ln a + \ln\left(\frac{1}{b}\right) = \ln a - \ln b$. 3. For $n \in \mathbb{N}$, we proceed **by induction** (Chapter 43). Let $P(n)$ be the statement: "$\ln(a^n) = n \ln a$".

- *Base step:* For $n = 0$, $\ln(a^0) = \ln 1 = 0 = 0 \times \ln a$. Thus $P(0)$ is true.
- *Inductive step:* Assume $P(n)$ holds for some integer $n \geq 0$. Then:

$$\ln(a^{n+1}) = \ln(a^n \times a) = \ln(a^n) + \ln a = n \ln a + \ln a = (n+1)\ln a.$$

  Hence $P(n+1)$ is true.
- *Conclusion:* $P(n)$ holds for all $n \in \mathbb{N}$.

For negative integers $n < 0$, let $n = -m$ where $m \in \mathbb{N}$:

$$\ln(a^{-m}) = \ln\left(\frac{1}{a^m}\right) = -\ln(a^m) = -m \ln a = n \ln a.$$

4. Since $\left(\sqrt{a}\right)^2 = a$, applying property 3 gives $2\ln(\sqrt{a}) = \ln a$, whence $\ln(\sqrt{a}) = \frac{1}{2}\ln a$. $\blacksquare$

💡 Did you know?

More generally, for $a > 0$ and any real number r, we **define** general powers by $a^r = e^{r \ln a}$, which immediately entails $\ln(a^r) = r \ln a$. This is how a digital computer evaluates $2^{\sqrt{2}}$ or $10^{0.3}$. In particular, $2^x = e^{x \ln 2}$: **all** constant-rate exponential growth is nothing more than base-e exponential growth in disguise.

3.3 Worked Examples

Worked Example 6 — Express in terms of $\ln 2$ and $\ln 3$:

$A = \ln 12$, $B = \ln\left(\frac{9}{16}\right)$, $C = \ln\sqrt{72}$, $D = 3\ln 4 - 2\ln 6$.

- $12 = 2^2 \times 3$, so $A = 2\ln 2 + \ln 3$.
- $B = \ln(3^2) - \ln(2^4) = 2\ln 3 - 4\ln 2$.
- $72 = 2^3 \times 3^2$, so $C = \frac{1}{2}(3\ln 2 + 2\ln 3) = \frac{3}{2}\ln 2 + \ln 3$.
- $D = 3 \times 2\ln 2 - 2(\ln 2 + \ln 3) = 6\ln 2 - 2\ln 2 - 2\ln 3 = 4\ln 2 - 2\ln 3$.

Worked Example 7 — Simplify $E = \ln(e^3) + \ln\left(\frac{1}{e}\right) - \ln(e\sqrt{e})$.

We have $\ln(e^3) = 3$, $\ln\left(\frac{1}{e}\right) = -1$, and $\ln(e\sqrt{e}) = \ln\left(e^{3/2}\right) = \frac{3}{2}$. Thus $E = 3 - 1 - \frac{3}{2} = \frac{1}{2}$.

Worked Example 8 — Multiplying like a Navigator in 1620.

Suppose we need to compute 2.37×5.81 using a table of **common logarithms** ($\log_{10} x = \frac{\ln x}{\ln 10}$, which satisfies $\log(ab) = \log a + \log b$ and $\log 10 = 1$). The table provides: $\log 2.37 \approx 0.37475$ and $\log 5.81 \approx 0.76418$. We **add** them: $0.37475 + 0.76418 = 1.13893$. We search the table for the antilogarithm: the number whose mantissa is 0.13893 is approximately 1.37697. Adjusting for the characteristic 1: the answer is $10^1 \times 1.37697 \approx 13.770$. Exact direct calculation: $2.37 \times 5.81 = 13.7697$. ✓ One addition and two table lookups completely replace long multiplication!

3.4 Crucial Pitfalls to Avoid

⚠ Common Pitfall! — Erroneous Identities.

- In general, $\ln(a + b) \neq \ln a + \ln b$. Counterexample:

$\ln(1 + 1) = \ln 2 \approx 0.69$, whereas $\ln 1 + \ln 1 = 0$. The logarithm transforms **products**, never sums.

- $\ln a \times \ln b \neq \ln(ab)$. Counterexample: $\ln e \times \ln e = 1 \times 1 = 1$, yet $\ln(e \cdot e) = \ln(e^2) = 2$.
- $\frac{\ln a}{\ln b} \neq \ln\left(\frac{a}{b}\right)$ and $\neq \ln a - \ln b$.
- $(\ln a)^2 \neq \ln(a^2)$. For instance, with $a = e^3$: $(\ln a)^2 = 3^2 = 9$, whereas $\ln(a^2) = \ln(e^6) = 6$. To avoid ambiguity, write $(\ln x)^2$ or $\ln^2 x$, never $\ln x^2$ when you mean the square of the logarithm.

⚠ Common Pitfall! — Hidden Signs.

The formula $\ln(ab) = \ln a + \ln b$ strictly requires $a > 0$ **and** $b > 0$. However, the product ab can be positive when both $a < 0$ and $b < 0$! For instance, $\ln\big((-2)(-3)\big) = \ln 6$ is well-defined, but $\ln(-2) + \ln(-3)$ is completely undefined in $\mathbb{R}$. The general identity, valid whenever $ab > 0$, is:

$$\ln(ab) = \ln|a| + \ln|b|.$$

Likewise, for any non-zero real number $x \neq 0$: $\ln(x^2) = 2\ln|x|$ (and not $2\ln x$, which is defined only for $x > 0$). Practical consequence: the functions $x \mapsto \ln(x^2)$ and $x \mapsto 2\ln x$ **are not identical**: the former is defined on $\mathbb{R}^*$, while the latter is defined only on $]0\,;+\infty[$.

4. Comprehensive Analysis of the Natural Logarithm Function

4.1 Differentiability and the Derivative

Theorem —

The natural logarithm function $\ln$ is differentiable on $]0\,;+\infty[$, and for every real number $x > 0$:

$$\ln'(x) = \frac{1}{x}.$$

Proof (Required Curriculum Proof) —

For every real number $x > 0$, we have the identity:

$$e^{\ln x} = x.$$

The function $\ln$ is differentiable on $]0\,;+\infty[$ because it is the inverse of the exponential function $\exp$, which is differentiable on $\mathbb{R}$ with a strictly positive derivative $\exp' = \exp > 0$. Differentiating both sides of the identity $e^{\ln x} = x$ with respect to x via the chain rule $(e^{u(x)})' = u'(x)\,e^{u(x)}$ with $u(x) = \ln x$:

$$(\ln)'(x) \times e^{\ln x} = 1.$$

Since $e^{\ln x} = x$, this simplifies to:

$$(\ln)'(x) \times x = 1 \Leftrightarrow \ln'(x) = \frac{1}{x}. \quad \blacksquare$$

Alternative Proof via the Difference Quotient —

Let $x > 0$ be fixed and let $h \neq 0$ be such that $x + h > 0$. The difference quotient of $\ln$ at x is:

$$\tau(h) = \frac{\ln(x+h) - \ln x}{h} = \frac{1}{h}\ln\left(1 + \frac{h}{x}\right).$$

Let $u = \ln\left(1 + \frac{h}{x}\right)$. Then $e^u = 1 + \frac{h}{x}$, which gives $h = x(e^u - 1)$. As $h \to 0$, $1 + \frac{h}{x} \to 1$, so $u \to \ln 1 = 0$ by the continuity of $\ln$. The difference quotient becomes:

$$\tau(h) = \frac{u}{x(e^u - 1)} = \frac{1}{x} \times \frac{1}{\frac{e^u - 1}{u}}.$$

By definition of the derivative of the exponential function at 0, $\lim_{u\to 0} \frac{e^u - 1}{u} = \exp'(0) = e^0 = 1$. Hence, taking the limit:

$$\lim_{h \to 0} \tau(h) = \frac{1}{x} \times \frac{1}{1} = \frac{1}{x}. \quad \blacksquare$$

4.2 Derivative of $\ln(u)$ and $\ln|u|$

When the argument of the logarithm is a differentiable function u, the chain rule $(f \circ u)' = u' \times (f' \circ u)$ immediately provides:

Chain Rule for Logarithms —

Let u be a differentiable, **strictly positive** function on an interval I. Then $f = \ln \circ u$, defined by $f(x) = \ln(u(x))$, is differentiable on I, and:

$$(\ln(u))' = \frac{u'}{u}.$$

More generally, if u is differentiable and **nowhere zero** on I, then $x \mapsto \ln|u(x)|$ is differentiable on I, and:

$$(\ln|u|)' = \frac{u'}{u}.$$

Proof for $\ln|u|$ when $u(x) < 0$ —

If $u(x) < 0$, then $|u(x)| = -u(x) > 0$. Hence $\ln|u(x)| = \ln(-u(x))$. Differentiating via the chain rule: $\big(\ln(-u)\big)' = \frac{(-u)'}{-u} = \frac{-u'}{-u} = \frac{u'}{u}$. The resulting derivative is identical regardless of the sign of $u(x)$! ■

Method — Differentiating an Expression Involving $\ln(u)$:

1. Determine the domain where the argument $u(x)$ is

strictly positive (or non-zero for $\ln|u|$). 2. Differentiate the inside function to find $u'(x)$. 3. Form the quotient $\frac{u'(x)}{u(x)}$ and simplify algebraically.

4.3 Monotonicity and Sign of the Logarithm

For every $x \in]0\,;+\infty[$, we have $x > 0$, and therefore $\ln'(x) = \frac{1}{x} > 0$. The derivative is **strictly positive** across its entire domain.

Property —

The natural logarithm function is **strictly increasing** on $]0\,;+\infty[$. For all positive real numbers $a, b \in]0\,;+\infty[$:

- $\ln a = \ln b \Leftrightarrow a = b$ (injectivity);
- $\ln a < \ln b \Leftrightarrow a < b$ (strict monotonicity).

Since $\ln 1 = 0$, strict monotonicity immediately establishes the **sign of $\ln x$**:

- If $0 < x < 1$, then $\ln x < \ln 1$, meaning $\boldsymbol{\ln x < 0}$;
- If $x = 1$, then $\boldsymbol{\ln 1 = 0}$;
- If $x > 1$, then $\ln x > \ln 1$, meaning $\boldsymbol{\ln x > 0}$.

x	0 (valeur interdite)	$]0;1[$	1	$]1;+\infty[$	$+\infty$
Signe de $\ln'(x)$	‖	$+$	$+$	$+$	
Signe de $\ln(x)$	‖	$-$	0	$+$	

4.4 Boundary Limits

Boundary Limits Theorem —

1.

$\lim_{x\to+\infty} \ln x = +\infty$ 2. $\lim_{x\to 0^+} \ln x = -\infty$

The vertical axis $x = 0$ (the y-axis) is therefore a **vertical asymptote** to the graph of $\ln$.

Proofs (Required Curriculum Proofs) —

1. **Limit at $+\infty$:** For every natural number n, we have $\ln(2^n) = n \ln 2$.

Since $2 > 1$, $\ln 2 > 0$. Consequently, $\lim_{n\to+\infty} \ln(2^n) = \lim_{n\to+\infty} (n \ln 2) = +\infty$. Now choose an arbitrarily large real threshold $M > 0$. Since $n \ln 2 \to +\infty$, there exists an integer N such that $N \ln 2 > M$. For every real number $x \geq 2^N$, the strict monotonicity of $\ln$ implies:

$$\ln x \geq \ln(2^N) = N \ln 2 > M.$$

Hence, any interval $]M\,;+\infty[$ contains all values of $\ln x$ for all sufficiently large x. Therefore, $\lim_{x\to+\infty} \ln x = +\infty$. ■

2. **Limit at 0^+:** Substitute $X = \frac{1}{x}$. As $x \to 0^+$, we have $X \to +\infty$. Then $\ln x = \ln\left(\frac{1}{X}\right) = -\ln X$. Since $\lim_{X\to+\infty} \ln X = +\infty$, composite limit rules yield:

$$\lim_{x\to 0^+} \ln x = \lim_{X\to+\infty} (-\ln X) = -\infty. \quad ■$$

4.5 Variation Table and Graphic Profile

Assembling these results into a complete table of variations:

x	0 (valeur interdite)		1		$+\infty$
Signe de $\ln'(x) = \frac{1}{x}$	‖	$+$	$+$	$+$	
Variations de $\ln(x)$	$-\infty$	↗	0	↗	$+\infty$

Notable Tangent at $x = 1$: At $x = 1$, the slope is $\ln'(1) = \frac{1}{1} = 1$, and the point is $(1\,;0)$. The equation of the tangent line T_1 at $(1\,;0)$ is:

$$y = \ln'(1)(x-1) + \ln 1 = 1 \times (x-1) + 0 \Leftrightarrow y = x - 1.$$

4.6 Concavity and the Second Derivative

Differentiating once more to find the second derivative:

$$(\ln)''(x) = \left(\frac{1}{x}\right)' = -\frac{1}{x^2}.$$

For every $x \in]0\,;+\infty[$, $x^2 > 0$, so $(\ln)''(x) < 0$.

Fundamental Concavity Property —

The natural logarithm function is **strictly concave** on $]0\,;+\infty[$. Consequently: 1. The graph of $\ln$ lies **strictly below every one of its tangent lines** (except at the point of tangency). 2. In particular, comparing it to its tangent at $x = 1$ ($y = x - 1$), we have for all $x > 0$:

$$\ln x \leq x - 1.$$

Equality holds if and only if $x = 1$.

Setting $x = 1 + h$ with $h > -1$, this fundamental inequality is equivalently written as:

$$\ln(1 + h) \leq h.$$

4.7 Asymptotic Growth Rates (Comparative Growth)

Although $\lim_{x\to+\infty} \ln x = +\infty$, this growth is **exceptionally sluggish**, slower than any positive power of x, no matter how small the exponent. For example:

- $\ln(10) \approx 2.3$;
- $\ln(10^6) \approx 13.8$;
- $\ln(10^{12}) \approx 27.6$.

For $\ln x$ to reach a modest value of 100, one requires $x = e^{100} \approx 2.68 \times 10^{43}$—a quantity far exceeding the number of grains of sand on the entire planet Earth!

This extreme slowness gives rise to the classic **comparative growth theorems**.

📐 Comparative Growth Theorems (Required Curriculum Proofs) —

For every positive integer $n \geq 1$: 1. $\lim_{x\to+\infty} \frac{\ln x}{x^n} = 0$ 2. $\lim_{x\to 0^+} x^n \ln x = 0$ 3. $\lim_{h\to 0} \frac{\ln(1+h)}{h} = 1$ or equivalently $\lim_{x\to 1} \frac{\ln x}{x-1} = 1$.

Proof of $\lim_{x\to+\infty} \frac{\ln x}{x} = 0$ (Required Curriculum Proof) —

Recall the comparative growth limit of the exponential function (established in Chapter 45):

$$\lim_{t\to+\infty} \frac{e^t}{t} = +\infty.$$

Perform the change of variable $t = \ln x$, so that $x = e^t$. As $x \to +\infty$, we have $t \to +\infty$. The quotient $\frac{\ln x}{x}$ becomes:

$$\frac{\ln x}{x} = \frac{t}{e^t} = \frac{\frac{1}{e^t}}{t}.$$

Taking the limit as $t \to +\infty$:

$$\lim_{x\to+\infty} \frac{\ln x}{x} = \lim_{t\to+\infty} \frac{\frac{1}{e^t}}{t} = \frac{1}{+\infty} = 0. \quad \blacksquare$$

Proof of $\lim_{x\to 0^+} x \ln x = 0$ (Required Curriculum Proof) —

Perform the substitution $X = \frac{1}{x}$. As $x \to 0^+$, we have $X \to +\infty$. Thus:

$$x \ln x = \frac{1}{X} \ln\left(\frac{1}{X}\right) = \frac{-\ln X}{X} = -\frac{\ln X}{X}.$$

By the preceding result, $\lim_{X\to+\infty} \frac{\ln X}{X} = 0$. Therefore:

$$\lim_{x\to 0^+} x \ln x = 0. \quad \blacksquare$$

Proof of $\lim_{h\to 0} \frac{\ln(1+h)}{h} = 1$ —

This expression is simply the difference quotient of the function $\ln$ evaluated at $x_0 = 1$:

$$\lim_{h\to 0} \frac{\ln(1+h) - \ln 1}{h} = \ln'(1) = \frac{1}{1} = 1. \quad \blacksquare$$

---

5. Methods: Equations, Inequalities, and Geometric Thresholds

5.1 Solving Equations with $\ln$

Method — Standard Protocol for Logarithmic Equations:

1.

Validity Domain: Find the domain D where all arguments inside logarithms are **strictly positive**. 2. **Consolidation:** Use algebraic identities to consolidate the expression into one of the standard forms:

- $\ln(A(x)) = \ln(B(x)) \Leftrightarrow A(x) = B(x)$;
- or $\ln(A(x)) = k \Leftrightarrow A(x) = e^k$.

3. **Algebraic Resolution:** Solve the resulting polynomial, rational, or linear equation.
4. **Domain Verification:** Check which solutions lie within the validity domain D; discard any extraneous solutions.

Worked Example 9 — Equation with Two Logarithms.

Solve $\ln(2x-3) = \ln(x+5)$.

- *Step 1: Domain.* We need $2x-3>0 \Leftrightarrow x > \frac{3}{2}$ and $x+5>0 \Leftrightarrow x>-5$. Intersecting gives $D =]\frac{3}{2}\,;+\infty[$.
- *Steps 2 & 3: Solving.* For $x \in D$:
$$\ln(2x-3) = \ln(x+5) \Leftrightarrow 2x-3 = x+5 \Leftrightarrow x = 8.$$
- *Step 4: Conclusion.* $8 \in D$ (since $8 > 1.5$). The solution set is $\mathcal{S} = \{8\}$.

Worked Example 10 — Sum of Logarithms and Extraneous Solutions.

Solve $\ln(x) + \ln(x-2) = \ln 3$.

- *Step 1: Domain.* We require $x>0$ and $x-2>0 \Leftrightarrow x>2$. Hence the validity domain is $D =]2\,;+\infty[$.
- *Step 2: Consolidation.* For all $x \in D$, $\ln(x) + \ln(x-2) = \ln(x(x-2))$. The equation becomes:
$$\ln(x^2-2x) = \ln 3 \Leftrightarrow x^2-2x = 3 \Leftrightarrow x^2-2x-3 = 0.$$
- *Step 3: Quadratic Roots.* $\Delta = (-2)^2 - 4(1)(-3) = 4+12 = 16 = 4^2$. The roots are $x_1 = \frac{2-4}{2} = -1$ and $x_2 = \frac{2+4}{2} = 3$.
- *Step 4: Domain Check.* $x_1 = -1 \notin D$: for this value, $\ln(-1)$ is undefined! It is an extraneous root arising from $(-1)\times(-3) = 3$. $x_2 = 3 \in D$ since $3>2$. The unique valid solution is $\mathcal{S} = \{3\}$.

5.2 Equations Involving Change of Variables

When the unknown appears as powers of $\ln x$, substitute a new variable.

Worked Example 11 — Quadratic Equation in $\ln x$.

Solve $2(\ln x)^2 - 5\ln x + 2 = 0$ on $]0\,;+\infty[$. Set $X = \ln x$. The equation transforms into:
$$2X^2 - 5X + 2 = 0.$$
The discriminant is $\Delta = (-5)^2 - 4(2)(2) = 25-16 = 9 = 3^2$. The solutions for X are:
$$X_1 = \frac{5-3}{4} = \frac{1}{2} \quad \text{and} \quad X_2 = \frac{5+3}{4} = 2.$$
Converting back to x:

- $\ln x = \frac{1}{2} \Leftrightarrow x = e^{1/2} = \sqrt{e}$;
- $\ln x = 2 \Leftrightarrow x = e^2$. Both values are strictly positive. The solution set is $\mathcal{S} = \{\sqrt{e}\,;e^2\}$.

5.3 Solving Inequalities with $\ln$

Because the logarithm is strictly increasing, it preserves inequalities:

$$\text{For all } A > 0 \text{ and } B > 0 : \quad \ln A < \ln B \Leftrightarrow A < B.$$

Worked Example 12 — Simple Inequality.

Solve $\ln(3x-1) \leq 2$. 1. *Domain:* $3x - 1 > 0 \Leftrightarrow x > \frac{1}{3}$. Thus $D =]\frac{1}{3} ; +\infty[$. 2. *Solving:* Express 2 as $\ln(e^2)$:

$$\ln(3x-1) \leq \ln(e^2) \Leftrightarrow 3x - 1 \leq e^2 \Leftrightarrow 3x \leq e^2 + 1 \Leftrightarrow x \leq \frac{e^2+1}{3}.$$

3. *Intersection with D:* Since $\frac{e^2+1}{3} \approx \frac{7.389+1}{3} \approx 2.796 > \frac{1}{3}$, the solution set is:

$$\mathcal{S} =]\frac{1}{3} ; \frac{e^2+1}{3}].$$

5.4 Finding Thresholds for Geometric Sequences

This is one of the most frequent exam applications: determining the smallest index n at which a geometric sequence surpasses or falls below a prescribed value.

Method — Solving $q^n < S$ or $q^n > S$:

1. Isolate the
power q^n on one side of the inequality. 2. Apply $\ln$ to both sides (valid because $q > 0$ and $S > 0$). 3. Apply the exponent identity $\ln(q^n) = n \ln q$. 4. **Divide by $\ln q$, paying extreme attention to the sign of $\ln q$:**

- If $q > 1$, then $\ln q > 0$: the inequality direction is **preserved**;
- If $0 < q < 1$, then $\ln q < 0$: the inequality direction is **reversed** ($\leq$ becomes $\geq$)!

Worked Example 13 — Depreciation Threshold (Inequality Reversal).

A new piece of industrial machinery costs $\$20,000$. Each year, its value depreciates by 15%. After how many years will its book value fall below $\$5,000$?

Modeling: The value after n years is $V_n = 20\,000 \times (1 - 0.15)^n = 20\,000 \times 0.85^n$. We seek the smallest non-negative integer n such that:

$$20\,000 \times 0.85^n < 5\,000 \Leftrightarrow 0.85^n < \frac{5\,000}{20\,000} \Leftrightarrow 0.85^n < 0.25.$$

Applying $\ln$:

$$\ln(0.85^n) < \ln(0.25) \Leftrightarrow n \ln(0.85) < \ln(0.25).$$

Since $0 < 0.85 < 1$, we have $\ln(0.85) \approx -0.1625 < 0$. Dividing by this **negative** quantity reverses the inequality:

$$n > \frac{\ln(0.25)}{\ln(0.85)}.$$

Calculating: $\frac{\ln(0.25)}{\ln(0.85)} \approx \frac{-1.38629}{-0.16252} \approx 8.53$. Since n must be an integer number of years, the minimum integer satisfying this is $\boldsymbol{n = 9}$. After 9 years, the machine will be worth less than $\$5,000$.

Associated Python Script — Threshold Search:

```python
def threshold_decay(initial_value, rate, target):
    n = 0
    v = initial_value
    while v >= target:
        v = v * (1 - rate)
        n += 1
    return n

print("Required years:", threshold_decay(20000, 0.15, 5000))
# Prints: Required years: 9
```

---

6. Real-World Logarithmic Scales

Why are logarithms ubiquitous throughout all experimental sciences? Because both nature and human perception frequently operate not by **additive differences**, but by **multiplicative ratios** (the Weber-Fechner psycho-physical law). When our ears perceive a sound as twice as loud, the underlying acoustic power has not increased by a few milliwatts: it has increased tenfold!

6.1 Acidity and pH in Chemistry

In aqueous chemistry, the oxonium ion concentration $[H_3O^+]$ ranges from $1\ \text{mol}\cdot\text{L}^{-1}$ (concentrated hydrochloric acid) down to $10^{-14}\ \text{mol}\cdot\text{L}^{-1}$ (concentrated sodium hydroxide). This spans a dynamic ratio of 10^{14}—one hundred trillion! To compress this enormous span into convenient numbers between 0 and 14, the Danish chemist **Søren Sørensen** introduced the **pH** (potential of hydrogen) in 1909 using the common logarithm $\log_{10}(x) = \frac{\ln x}{\ln 10}$:

Definition of pH —

$$\text{pH} = -\log_{10}\left(\frac{[H_3O^+]}{c_0}\right) = -\frac{\ln([H_3O^+]/c_0)}{\ln 10}$$

where $c_0 = 1\ \text{mol}\cdot\text{L}^{-1}$ is the standard reference concentration. Inversely: $[H_3O^+] = 10^{-\ \text{pH}}\ \text{mol}\cdot\text{L}^{-1}$.

Direct Consequence: *When the pH drops by a single unit (for instance from 7 to 6), the concentration of $[H_3O^+]$ ions **multiplies by 10**. If the pH drops from 7 to 4, it multiplies by $10^3 = 1\,000$!*

6.2 Acoustics and Decibels

The human ear detects acoustic intensities spanning from the faintest threshold of hearing ($I_0 = 10^{-12}\ \text{W}\cdot\text{m}^{-2}$) to the roar of a rocket launch ($100\ \text{W}\cdot\text{m}^{-2}$)—a dynamic range of 10^{14}. The **sound intensity level** L, measured in **decibels (dB)**, is defined as:

Sound Level in Decibels —

$$L = 10\log_{10}\left(\frac{I}{I_0}\right) = 10\frac{\ln(I/I_0)}{\ln 10}.$$

Worked Example 14 — Do Two Vacuum Cleaners Make Twice the Noise?

Suppose a vacuum cleaner emits an acoustic intensity I_1 resulting in a sound level of $L_1 = 70\,\text{dB}$. Two identical vacuum cleaners operating simultaneously produce double the acoustic intensity: $I_2 = 2I_1$. What is the resulting sound level L_2?

$$L_2 = 10\log_{10}\left(\frac{2I_1}{I_0}\right) = 10\left[\log_{10}(2) + \log_{10}\left(\frac{I_1}{I_0}\right)\right] = 10\log_{10}(2) + L_1.$$

Since $10\log_{10}(2) \approx 10 \times 0.30103 \approx 3\,\text{dB}$, we have:

$$L_2 \approx 70 + 3 = 73\,\text{dB}\ !$$

Doubling the physical acoustic power increases the perceived sound level by only 3 decibels!

6.3 Earthquakes and the Richter Magnitude Scale

The Richter scale relates the seismic energy E released at an earthquake's epicenter via a formula of the form:

$$M = \frac{2}{3}\log_{10}\left(\frac{E}{E_0}\right).$$

If the magnitude increases by 1 unit, we have $\Delta M = 1 = \frac{2}{3}\log_{10}\left(\frac{E'}{E}\right)$, whence:

$$\log_{10}\left(\frac{E'}{E}\right) = \frac{3}{2} = 1.5 \Leftrightarrow \frac{E'}{E} = 10^{1.5} = 10\sqrt{10} \approx 31.6.$$

A magnitude 7 earthquake releases **31.6 times more energy** than a magnitude 6 earthquake, and $(31.6)^2 \approx 1\,000$ times more energy than a magnitude 5 earthquake!

6.4 Radiocarbon Dating and Half-Life

In any living organism, the ratio of radioactive carbon-14 to stable carbon-12 remains constant. Upon death, metabolic intake ceases and carbon-14 decays exponentially according to:

$$N(t) = N_0\, e^{-\lambda t},$$

where $\lambda > 0$ is the radioactive decay constant of the isotope.

The **half-life** $t_{1/2}$ is the duration after which half of the original radioactive nuclei have decayed:

$$N(t_{1/2}) = \frac{N_0}{2} \Leftrightarrow N_0 e^{-\lambda t_{1/2}} = \frac{N_0}{2} \Leftrightarrow e^{-\lambda t_{1/2}} = \frac{1}{2}.$$

Applying the natural logarithm:

$$-\lambda t_{1/2} = \ln\left(\frac{1}{2}\right) = -\ln 2 \Leftrightarrow t_{1/2} = \frac{\ln 2}{\lambda}.$$

For carbon-14, $t_{1/2} \approx 5\,730$ years. If a sample of charcoal recovered from a prehistoric cave retains 30% of its original carbon-14 activity, its age t is obtained by solving:

$$\frac{N(t)}{N_0} = 0.30 \Leftrightarrow e^{-\lambda t} = 0.30 \Leftrightarrow -\lambda t = \ln(0.30) \Leftrightarrow t = -\frac{\ln(0.30)}{\lambda} = -5\,730 \times \frac{\ln(0.30)}{\ln 2} \approx 9\,950 \ \text{ years}\ .$$

Key Takeaways

1. Definition and Fundamental Link with the Exponential

- For all $a > 0$ and $x \in \mathbb{R}$: $e^x = a \Leftrightarrow x = \ln a$.
- The function $\ln$ is defined, continuous, and differentiable on $]0\,;+\infty[$.
- Inversion identities: $e^{\ln a} = a$ (for all $a > 0$) and $\ln(e^x) = x$ (for all $x \in \mathbb{R}$).
- $\ln 1 = 0$ and $\ln e = 1$.
- The graphs of $\exp$ and $\ln$ are reflections across the first bisector line $y = x$.

2. Algebraic Identities (for all $a > 0, b > 0$)

- Fundamental relation: $\ln(ab) = \ln a + \ln b$.
- Reciprocals and quotients: $\ln\left(\frac{1}{a}\right) = -\ln a$ and $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$.
- Powers and roots: $\ln(a^n) = n \ln a$ ($n \in \mathbb{Z}$) and $\ln(\sqrt{a}) = \frac{1}{2}\ln a$.

3. Calculus of the Logarithm Function

- Derivative: $(\ln x)' = \frac{1}{x} > 0$ on $]0\,;+\infty[$.
- Chain rule: $\big(\ln(u)\big)' = \frac{u'}{u}$ (for $u > 0$).
- Strictly increasing on $]0\,;+\infty[$: $\ln a < \ln b \Leftrightarrow a < b$.
- Sign: $\ln x < 0$ on $]0\,;1[$, $\ln 1 = 0$, and $\ln x > 0$ on $]1\,;+\infty[$.
- Limits: $\lim_{x\to 0^+} \ln x = -\infty$ (vertical asymptote $x = 0$) and $\lim_{x\to+\infty} \ln x = +\infty$.
- Concavity: $(\ln)''(x) = -\frac{1}{x^2} < 0$. The function is strictly concave, leading to $\ln x \leq x - 1$ for all $x > 0$.

4. Essential Comparative Growth Limits

- $\lim_{x\to+\infty} \frac{\ln x}{x^n} = 0 \quad (n \geq 1)$.
- $\lim_{x\to 0^+} x^n \ln x = 0 \quad (n \geq 1)$.
- $\lim_{h\to 0} \frac{\ln(1+h)}{h} = 1$.

5. Solving Equations and Thresholds

- Always begin by determining the validity domain D ($u(x) > 0$).
- For $q^n < S$ with $0 < q < 1$: $n > \frac{\ln S}{\ln q}$ (because $\ln q < 0$ reverses the inequality symbol).

Exercises

Level (Direct Applications)

Exercise 48.1 — Mental Arithmetic and Simplifications. Compute the exact values of the following numbers without a calculator: 1. $A = \ln(e^5)$ 2. $B = e^{\ln 11}$ 3. $C = \ln\left(\frac{1}{e^4}\right)$ 4. $D = \ln(\sqrt[3]{e})$ (recall: $\sqrt[3]{e} = e^{1/3}$) 5. $E = e^{-\ln 3}$ 6. $F = \ln(e\sqrt{e})$

Exercise 48.2 — Prime Factor Decomposition. Setting $a = \ln 2$ and $b = \ln 5$, express in terms of a and b: 1. $\ln 10$ 2. $\ln 20$ 3. $\ln 50$ 4. $\ln(0.04)$ 5. $\ln\left(\frac{\sqrt{2}}{25}\right)$

Exercise 48.3 — Domains of Definition. Determine the domain of definition of each of the following functions: 1. $f_1(x) = \ln(3x - 12)$ 2. $f_2(x) = \ln(x^2 - 9)$ 3. $f_3(x) = \ln\left(\frac{x+1}{2-x}\right)$ 4. $f_4(x) = \ln(x^2 + 1)$ 5. $f_5(x) = \frac{1}{\ln x}$

Exercise 48.4 — Simple Derivatives. Calculate the derivative of the following functions on their domain of differentiability: 1. $g_1(x) = \ln(4x + 7)$ 2. $g_2(x) = \ln(x^2 + 3)$ 3. $g_3(x) = x^2 \ln x$ 4. $g_4(x) = \frac{\ln x}{x}$ 5. $g_5(x) = (\ln x)^3$

Exercise 48.5 — Elementary Equations. Solve in $\mathbb{R}$: 1. $\ln x = 4$ 2. $\ln(2x - 1) = 0$ 3. $e^x = 9$ 4. $e^{3x-2} = 5$ 5. $\ln(x^2) = 6$

Exercise 48.6 — Elementary Inequalities. Solve in $\mathbb{R}$: 1. $\ln x > -2$ 2. $\ln(3 - x) \leq 1$ 3. $e^{2x} \geq 7$ 4. $2\ln x + 3 < 0$

Exercise 48.7 — Direct Limits. Determine the following limits: 1. $\lim_{x\to+\infty}(\ln x + 2x)$ 2. $\lim_{x\to 0^+}(x^2 - \ln x)$ 3. $\lim_{x\to+\infty} \frac{\ln x}{x^2}$ 4. $\lim_{x\to 0^+} x^3 \ln x$

Level (Exploration and Exam Preparation)

Exercise 48.8 — Quadratic Equation in Logarithms. Solve in $\mathbb{R}$ the equation:

$$(\ln x)^2 - 3\ln x - 10 = 0.$$

Exercise 48.9 — Equation with a Sum of Logarithms. Solve in $\mathbb{R}$ the equation:

$$\ln(x + 2) + \ln(x - 1) = \ln(4x + 8).$$

Exercise 48.10 — System of Non-Linear Equations. Solve in $\mathbb{R}^2$ the system:

$$\begin{cases} x + y = 30 \\ \ln x + \ln y = 3\ln 2 + \ln 27 \end{cases}$$

Exercise 48.11 — Study of a Classic Logarithmic Function. Let f be the function defined on $]0\,;+\infty[$ by:

$$f(x) = x - 2 - \ln x.$$

1. Determine the limits of f at 0 and at $+\infty$. 2. Compute $f'(x)$ and determine its sign. 3. Draw up the complete variation table of f. 4. Deduce the number of real roots of $f(x) = 0$ on $]0\,;+\infty[$.

Exercise 48.12 — Tangent Passing Through the Origin. Does there exist a tangent line to the graph of $\ln$ that passes through the origin $(0\,;0)$? If so, find the coordinates of the point of tangency and the equation of this tangent line.

Exercise 48.13 — Resolving Indeterminate Forms. Evaluate the following limits, justifying your reasoning carefully: 1. $\lim_{x\to+\infty}(x - \ln x)$ 2. $\lim_{x\to+\infty} \frac{\ln(x^2+1)}{x}$ 3. $\lim_{x\to 0^+} x\ln\!\left(1+\frac{1}{x}\right)$ 4. $\lim_{x\to 1} \frac{\ln x}{x^2-1}$

Exercise 48.14 — Investment Doubling Threshold. A capital sum of $\$10,000$ is invested at an annual compound interest rate of 3.5%. 1. Express the accumulated capital C_n after n years. 2. Determine by exact calculation (using logarithms) the number of years required for the initial investment to double. 3. Compare this with the famous "Rule of 70" used in finance: the doubling time at an interest rate of $t\%$ per annum is approximately $70/t$. Prove why this rule works using the first-order affine approximation of $\ln(1+u)$ near 0.

Exercise 48.15 — Study of a Rational Logarithmic Function and Concavity. Let f be defined on $]0\,;+\infty[$ by $f(x) = \frac{\ln x}{x}$. 1. Determine the limits of f at 0 and at $+\infty$. Give the graphical interpretations. 2. Calculate $f'(x)$ and establish the complete table of variations of f. 3. Identify the global maximum of f. 4. Without a calculator, compare the numbers e^π and π^e using the monotonicity of f. 5. Compute $f''(x)$ and study the convexity of f. Identify the inflection point.

Exercise 48.16 — Convexity Inequality. 1. Prove that for all $x > 0$, $\ln x \leq x - 1$. 2. Deduce that for all $t > -1$, $\ln(1+t) \leq t$. 3. Applying this inequality with $t = -\frac{1}{n+1}$, show that for all integers $n \geq 1$:

$$\left(1+\frac{1}{n}\right)^n \leq e \leq \left(1+\frac{1}{n}\right)^{n+1}.$$

Exercise 48.17 — Dating the Chauvet Cave. Prehistoric rock paintings were discovered in a cave. Charcoal fragments collected near the paintings exhibit an activity of 2.4 disintegrations per minute per gram of carbon, whereas fresh modern wood exhibits 15.3 disintegrations per minute per gram. Given that the half-life of carbon-14 is $T = 5\,730$ years, estimate the age of the cave paintings.

Exercise 48.18 — Threshold and Arithmetico-Geometric Sequence (Exam Style). Let (u_n) be the sequence defined by $u_0 = 10$ and for all $n \in \mathbb{N}$, $u_{n+1} = 0.6u_n + 2$. 1. Prove that the sequence (v_n) defined by $v_n = u_n - 5$ is a geometric sequence whose common ratio and first term you will determine. 2. Express v_n, and then u_n, as a function of n. 3. Determine algebraically the smallest integer n such that $|u_n - 5| < 10^{-4}$.

Level (In-Depth Challenges)

Exercise 48.19 — The Harmonic Sum and Euler's Constant. Let $H_n = \sum_{k=1}^{n} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$. 1. Using the inequality $\ln(1+x) \leq x$ for $x > 0$, show that for every integer $k \geq 1$:

$$\ln(k+1) - \ln k \leq \frac{1}{k}.$$

2. Using the inequality $\ln(1-x) \leq -x$ for $x \in]0\,;1[$, show that for every integer $k \geq 2$:

$$\frac{1}{k} \leq \ln k - \ln(k-1).$$

3. Summing these inequalities from $k = 1$ to n, establish the bounding:

$$\ln(n+1) \leq H_n \leq 1 + \ln n.$$

4. Deduce the limit of $(H_n)_{n\geq 1}$. What does this say about the harmonic series? 5. Determine the limit of $\frac{H_n}{\ln n}$ as $n \to +\infty$.

Exercise 48.20 — Stirling's Formula (First Approach via Logarithms). We wish to estimate $n! = 1 \times 2 \times 3 \times \ldots \times n$. 1. Express $\ln(n!)$ as a summation. 2. Using the bounding strategy of the previous exercise, obtain an estimate for $\ln(n!)$ in terms of the integral of $\ln x$.

Exercise 48.21 — The Transcendental Equation $x^a = a^x$. Let $a > 0$ be a fixed real parameter. We seek positive real solutions to $x^a = a^x$. 1. Show that for all $x > 0$, the equation is equivalent to $\frac{\ln x}{x} = \frac{\ln a}{a}$. 2. Using the study of $f(x) =$

$\frac{\ln x}{x}$ from Exercise 48.15, discuss the number of solutions depending on the value of a. 3. Find the unique pair of distinct positive integers $(x\,;y)$ with $x<y$ satisfying $x^y=y^x$.

Exercise 48.22 — Logarithmic Differentiation of Products. Let f be the product of n differentiable, strictly positive functions: $f(x)=u_1(x)u_2(x)...u_n(x)$. 1. Show that $\frac{f'(x)}{f(x)}=\sum_{k=1}^{n}\frac{u_k'(x)}{u_k(x)}$. 2. Application: Rapidly compute the derivative of $g(x)=(x+1)^3(2x-1)^4(x^2+5)^2$.

Solutions

Level Solutions

Solution 48.1: 1. $A=\ln(e^5)=5$. 2. $B=e^{\ln 11}=11$. 3. $C=\ln(e^{-4})=-4$. 4. $D=\ln(e^{1/3})=\frac{1}{3}$. 5. $E=e^{-\ln 3}=e^{\ln(1/3)}=\frac{1}{3}$. 6. $F=\ln(e^1\times e^{1/2})=\ln(e^{3/2})=\frac{3}{2}$.

Solution 48.2: 1. $\ln 10=\ln(2\times 5)=\ln 2+\ln 5=a+b$. 2. $\ln 20=\ln(2^2\times 5)=2\ln 2+\ln 5=2a+b$. 3. $\ln 50=\ln(2\times 5^2)=\ln 2+2\ln 5=a+2b$. 4. $\ln(0.04)=\ln\left(\frac{4}{100}\right)=\ln\left(\frac{1}{25}\right)=-\ln(5^2)=-2b$. 5. $\ln\left(\frac{\sqrt{2}}{25}\right)=\ln(\sqrt{2})-\ln(5^2)=\frac{1}{2}a-2b$.

Solution 48.3: 1. $3x-12>0\Leftrightarrow 3x>12\Leftrightarrow x>4$. Domain: $D_1=]4\,;+\infty[$. 2. $x^2-9>0\Leftrightarrow(x-3)(x+3)>0$. The roots are -3 and 3; the quadratic is positive outside the roots. Domain: $D_2=]-\infty\,;-3[\,\cup\,]3\,;+\infty[$. 3. $\frac{x+1}{2-x}>0$. By a sign table, the quotient is positive strictly between -1 and 2. Domain: $D_3=]-1\,;2[$. 4. For all real x, $x^2\geq 0\Longrightarrow x^2+1\geq 1>0$. Domain: $D_4=\mathbb{R}$. 5. We require $x>0$ (for $\ln x$ to exist) and $\ln x\neq 0\Leftrightarrow x\neq 1$. Domain: $D_5=]0\,;1[\,\cup\,]1\,;+\infty[$.

Solution 48.4: 1. Using $(\ln u)'=\frac{u'}{u}$ with $u(x)=4x+7$ and $u'(x)=4$: $g_1'(x)=\frac{4}{4x+7}$. 2. With $u(x)=x^2+3$, $u'(x)=2x$: $g_2'(x)=\frac{2x}{x^2+3}$. 3. Product rule $(uv)'=u'v+uv'$: $g_3'(x)=2x\ln x+x^2\times\frac{1}{x}=2x\ln x+x=x(2\ln x+1)$. 4. Quotient rule: $g_4'(x)=\frac{\frac{1}{x}\times x-\ln x\times 1}{x^2}=\frac{1-\ln x}{x^2}$. 5. Power rule $(u^3)'=3u^2u'$: $g_5'(x)=3(\ln x)^2\times\frac{1}{x}=\frac{3(\ln x)^2}{x}$.

Solution 48.5: 1. $\ln x=4\Leftrightarrow x=e^4$. $\mathcal{S}=\{e^4\}$. 2. Domain $x>1/2$. $\ln(2x-1)=0=\ln 1\Leftrightarrow 2x-1=1\Leftrightarrow 2x=2\Leftrightarrow x=1\in D$. $\mathcal{S}=\{1\}$. 3. $e^x=9\Leftrightarrow x=\ln 9=2\ln 3$. $\mathcal{S}=\{2\ln 3\}$. 4. $e^{3x-2}=5\Leftrightarrow 3x-2=\ln 5\Leftrightarrow 3x=2+\ln 5\Leftrightarrow x=\frac{2+\ln 5}{3}$. $\mathcal{S}=\left\{\frac{2+\ln 5}{3}\right\}$. 5. Domain: $x\neq 0$. $\ln(x^2)=6\Leftrightarrow x^2=e^6\Leftrightarrow x=e^3$ or $x=-e^3$. $\mathcal{S}=\{-e^3\,;e^3\}$.

Solution 48.6: 1. $\ln x>-2\Leftrightarrow x>e^{-2}=\frac{1}{e^2}$. $\mathcal{S}=]\frac{1}{e^2}\,;+\infty[$. 2. Domain: $3-x>0\Leftrightarrow x<3$. $\ln(3-x)\leq 1=\ln e\Leftrightarrow 3-x\leq e\Leftrightarrow x\geq 3-e$. Intersecting with the domain: $\mathcal{S}=[3-e\,;3[$. 3. $e^{2x}\geq 7\Leftrightarrow 2x\geq\ln 7\Leftrightarrow x\geq\frac{\ln 7}{2}$. $\mathcal{S}=[\frac{\ln 7}{2}\,;+\infty[$. 4. Domain $x>0$. $2\ln x<-3\Leftrightarrow\ln x<-\frac{3}{2}\Leftrightarrow 0<x<e^{-3/2}=\frac{1}{e\sqrt{e}}$. $\mathcal{S}=]0\,;e^{-3/2}[$.

Solution 48.7: 1. Sum of limits: $\lim_{x\to+\infty}\ln x=+\infty$ and $\lim_{x\to+\infty}2x=+\infty$, so the limit is $+\infty$. 2. Difference of limits: $\lim_{x\to 0^+}x^2=0$ and $\lim_{x\to 0^+}\ln x=-\infty$, so $\lim_{x\to 0^+}(x^2-\ln x)=0-(-\infty)=+\infty$. 3. Comparative growth: $\lim_{x\to+\infty}\frac{\ln x}{x^2}=0$. 4. Comparative growth: $\lim_{x\to 0^+}x^3\ln x=0$.

Level Solutions

Solution 48.8: The equation is defined on $]0\,;+\infty[$. Set $X=\ln x$. The equation becomes $X^2-3X-10=0$. Discriminant: $\Delta=(-3)^2-4(1)(-10)=9+40=49=7^2$. Roots: $X_1=\frac{3-7}{2}=-2$ and $X_2=\frac{3+7}{2}=5$. Back-substituting to x:

- $\ln x=-2\Leftrightarrow x=e^{-2}$;
- $\ln x=5\Leftrightarrow x=e^5$.

Both values are strictly positive. Thus $\mathcal{S}=\{e^{-2}\,;e^5\}$.

Solution 48.9: Validity domain: We require $x+2>0\Leftrightarrow x>-2$, $x-1>0\Leftrightarrow x>1$, and $4x+8>0\Leftrightarrow x>-2$. Their intersection is $D=]1\,;+\infty[$. For $x\in D$, $\ln(x+2)+\ln(x-1)=\ln\big((x+2)(x-1)\big)=\ln(x^2+x-2)$. The equation becomes:

$$x^2+x-2=4x+8\Leftrightarrow x^2-3x-10=0.$$

The algebraic roots are -2 and 5. Now $-2\notin D$ because -2 is not strictly greater than 1 (at -2, $\ln(x-1)=\ln(-3)$ is undefined). Only the root $5\in D$ is valid. Therefore $\mathcal{S}=\{5\}$.

Solution 48.10: The system is defined for $x > 0$ and $y > 0$. The second equation gives: $\ln(xy) = \ln(2^3 \times 27) = \ln(8 \times 27) = \ln(216) \Leftrightarrow xy = 216$. We seek two real numbers given their sum $S = 30$ and product $P = 216$. They are the roots of $t^2 - 30t + 216 = 0$. $\Delta = (-30)^2 - 4(216) = 900 - 864 = 36 = 6^2$. $t_1 = \frac{30-6}{2} = 12$ and $t_2 = \frac{30+6}{2} = 18$. Both roots are strictly positive, yielding two solution pairs:

$$\mathcal{S} = \{(12\,;18)\,;(18\,;12)\}.$$

Solution 48.11: 1. At 0^+: $\lim_{x\to 0^+}(x-2) = -2$ and $\lim_{x\to 0^+}\ln x = -\infty$, so $\lim_{x\to 0^+} f(x) = -2 - (-\infty) = +\infty$. At $+\infty$: Indeterminate form $\infty - \infty$. Factor out x: $f(x) = x\big(1 - \frac{2}{x} - \frac{\ln x}{x}\big)$. Since $\lim_{x\to+\infty}\frac{2}{x} = 0$ and by comparative growth $\lim_{x\to+\infty}\frac{\ln x}{x} = 0$, the factor inside parentheses approaches 1. By product: $\lim_{x\to+\infty} f(x) = +\infty$. 2. $f'(x) = 1 - 0 - \frac{1}{x} = \frac{x-1}{x}$. For $x > 0$, the denominator x is strictly positive, so $f'(x)$ shares the sign of $x - 1$:

- $f'(x) < 0$ on $]0\,;1[$;
- $f'(1) = 0$;
- $f'(x) > 0$ on $]1\,;+\infty[$.

3. Variations: f is strictly decreasing on $]0\,;1]$ and strictly increasing on $[1\,;+\infty[$. The minimum occurs at $x = 1$, where $f(1) = 1 - 2 - \ln 1 = -1$.
4. On $]0\,;1]$, f is continuous and strictly decreasing, mapping $]0\,;1]$ onto $[-1\,;+\infty[$. Since $0 \in [-1\,;+\infty[$, the IVT corollary guarantees a unique root $\alpha_1 \in]0\,;1]$. On $[1\,;+\infty[$, f is continuous and strictly increasing, mapping $[1\,;+\infty[$ onto $[-1\,;+\infty[$. Likewise, $0 \in [-1\,;+\infty[$, guaranteeing a unique root $\alpha_2 \in [1\,;+\infty[$. The equation $f(x) = 0$ has exactly **two real solutions** on $]0\,;+\infty[$ ($\alpha_1 \approx 0.158$ and $\alpha_2 \approx 3.146$).

Solution 48.12: Let $a > 0$. The tangent line to the graph of $\ln$ at abscissa a has equation:

$$y = \ln'(a)(x-a) + \ln a \Leftrightarrow y = \frac{1}{a}(x-a) + \ln a \Leftrightarrow y = \frac{1}{a}x - 1 + \ln a.$$

This line passes through the origin $(0\,;0)$ if and only if its y-intercept is zero:

$$-1 + \ln a = 0 \Leftrightarrow \ln a = 1 \Leftrightarrow a = e.$$

There exists an **unique tangent line** passing through the origin. The point of tangency is $A(e\,;1)$ and the tangent equation is:

$$y = \frac{1}{e}x.$$

Solution 48.13: 1. $\lim_{x\to+\infty}(x - \ln x)$: Indeterminate form $\infty - \infty$. Factor out x: $x - \ln x = x\big(1 - \frac{\ln x}{x}\big)$. Since $\lim_{x\to+\infty}\frac{\ln x}{x} = 0$, the term in parentheses tends to 1. Thus the product tends to $+\infty$. 2. $\lim_{x\to+\infty}\frac{\ln(x^2+1)}{x}$: For $x > 0$, $\ln(x^2+1) = \ln\big[x^2\big(1 + \frac{1}{x^2}\big)\big] = 2\ln x + \ln\big(1 + \frac{1}{x^2}\big)$. The quotient becomes: $\frac{\ln(x^2+1)}{x} = 2\frac{\ln x}{x} + \frac{\ln(1+1/x^2)}{x}$. As $x \to +\infty$, $\frac{\ln x}{x} \to 0$ and $\frac{\ln(1+1/x^2)}{x} \to \frac{0}{+\infty} = 0$. The limit is 0. 3. $\lim_{x\to 0^+} x\ln\big(1 + \frac{1}{x}\big)$: Substitute $u = \frac{1}{x} \to +\infty$. The expression becomes $\frac{\ln(1+u)}{u} = \frac{\ln u + \ln(1+1/u)}{u} = \frac{\ln u}{u} + \frac{\ln(1+1/u)}{u} \to 0 + 0 = 0$. 4. $\lim_{x\to 1}\frac{\ln x}{x^2-1}$: Form $\frac{0}{0}$. Factor the denominator: $\frac{\ln x}{(x-1)(x+1)} = \frac{\ln x - \ln 1}{x-1} \times \frac{1}{x+1}$. Since $\lim_{x\to 1}\frac{\ln x - \ln 1}{x-1} = \ln'(1) = 1$ and $\lim_{x\to 1}\frac{1}{x+1} = \frac{1}{2}$, the product limit is $\frac{1}{2}$.

Solution 48.14: 1. With an annual interest rate of 3.5%, the capital multiplies each year by $1 + 0.035 = 1.035$. Thus $C_n = 10\,000 \times 1.035^n$. 2. Doubling occurs when $C_n \geq 20\,000 \Leftrightarrow 1.035^n \geq 2$. Applying $\ln$: $n\ln(1.035) \geq \ln 2 \Leftrightarrow n \geq \frac{\ln 2}{\ln(1.035)}$ (since $\ln(1.035) > 0$). $\frac{\ln 2}{\ln(1.035)} \approx \frac{0.69315}{0.03440} \approx 20.15$. It takes 21 full years for the capital to double. 3. Rule of 70: For an annual rate of $t\%$, the multiplier is $1 + \frac{t}{100}$. The doubling time satisfies $n = \frac{\ln 2}{\ln(1+t/100)}$. For small $t/100$, the first-order approximation gives $\ln(1+u) \approx u$, hence $\ln(1 + t/100) \approx \frac{t}{100}$. Moreover, $\ln 2 \approx 0.693 \approx 0.70$. Thus: $n \approx \frac{0.693}{t/100} = \frac{69.3}{t} \approx \frac{70}{t}$. For $t = 3.5$, this yields $\frac{70}{3.5} = 20$ years, remarkably close to the exact 20.15 years!

Solution 48.15: 1. As $x \to 0^+$, $\ln x \to -\infty$ and $\frac{1}{x} \to +\infty$, so by product $\lim_{x\to 0^+}\frac{\ln x}{x} = -\infty$. The line $x = 0$ is a vertical asymptote. By comparative growth, $\lim_{x\to+\infty}\frac{\ln x}{x} = 0$. The line $y = 0$ is a horizontal asymptote as $x \to +\infty$. 2. f is differentiable on $]0\,;+\infty[$ as a quotient of differentiable functions: $f'(x) = \frac{\frac{1}{x}\times x - \ln x \times 1}{x^2} = \frac{1-\ln x}{x^2}$. The denominator $x^2 > 0$. The sign of $f'(x)$ is that of $1 - \ln x$:

- $1 - \ln x > 0 \Leftrightarrow \ln x < 1 \Leftrightarrow 0 < x < e$: f is strictly increasing;
- $1 - \ln x = 0 \Leftrightarrow x = e$;
- $1 - \ln x < 0 \Leftrightarrow x > e$: f is strictly decreasing.

3. The global maximum of f on $]0\,;+\infty[$ is attained at $x = e$ and equals $f(e) = \frac{\ln e}{e} = \frac{1}{e}$.
4. Comparing e^π and π^e: Since $e \approx 2.718 < \pi \approx 3.14$, and f is strictly decreasing on $[e\,;+\infty[$:

$$f(e) > f(\pi) \Leftrightarrow \frac{\ln e}{e} > \frac{\ln \pi}{\pi}.$$

Multiplying by $e\pi > 0$:

$$\pi \ln e > e \ln \pi \Leftrightarrow \ln(e^\pi) > \ln(\pi^e).$$

Since the natural logarithm is strictly increasing:

$$e^\pi > \pi^e.$$

(Numerical check: $e^\pi \approx 23.14$ and $\pi^e \approx 22.46$).

5. Second derivative:

$$f''(x) = \frac{\left(-\frac{1}{x}\right)x^2 - (1 - \ln x)(2x)}{x^4} = \frac{-x - 2x + 2x \ln x}{x^4} = \frac{2\ln x - 3}{x^3}.$$

The sign depends solely on $2\ln x - 3$:

- $2\ln x - 3 < 0 \Leftrightarrow \ln x < \frac{3}{2} \Leftrightarrow 0 < x < e^{3/2} = e\sqrt{e}$: $f''(x) < 0$, f is **concave**.
- $2\ln x - 3 > 0 \Leftrightarrow x > e^{3/2}$: $f''(x) > 0$, f is **convex**. The second derivative changes sign at $x_0 = e^{3/2}$. The curve has an **inflection point** at $I\left(e^{3/2}\,;\frac{3}{2e^{3/2}}\right)$.

Solution 48.16: 1. Let $\varphi(x) = \ln x - (x-1)$ on $]0\,;+\infty[$. Then $\varphi'(x) = \frac{1}{x} - 1 = \frac{1-x}{x}$. φ' is positive on $]0\,;1[$, zero at 1, and negative on $]1\,;+\infty[$. The maximum of φ is $\varphi(1) = 0$. Thus $\varphi(x) \leq 0 \Leftrightarrow \ln x \leq x - 1$ for all $x > 0$. 2. Substituting $x = 1 + t$ with $1 + t > 0 \Leftrightarrow t > -1$ yields $\ln(1+t) \leq (1+t) - 1 = t$. 3. - Apply $\ln(1+t) \leq t$ with $t = \frac{1}{n} > 0$: $\ln\left(1 + \frac{1}{n}\right) \leq \frac{1}{n} \Leftrightarrow n\ln\left(1 + \frac{1}{n}\right) \leq 1$. Applying the increasing exponential function: $\left(1 + \frac{1}{n}\right)^n \leq e$.

- Apply $\ln(1+t) \leq t$ with $t = -\frac{1}{n+1} > -1$:

$$\ln\left(1 - \frac{1}{n+1}\right) \leq -\frac{1}{n+1} \Leftrightarrow \ln\left(\frac{n}{n+1}\right) \leq -\frac{1}{n+1} \Leftrightarrow -\ln\left(\frac{n+1}{n}\right) \leq -\frac{1}{n+1}.$$

Multiplying by $-(n+1) < 0$:

$$(n+1)\ln\left(1 + \frac{1}{n}\right) \geq 1.$$

Applying the exponential function: $\left(1 + \frac{1}{n}\right)^{n+1} \geq e$. Combining both yields the celebrated bounds: $\left(1 + \frac{1}{n}\right)^n \leq e \leq \left(1 + \frac{1}{n}\right)^{n+1}$.

Solution 48.17: The activity follows $A(t) = A_0 e^{-\lambda t}$, where $\lambda = \frac{\ln 2}{T} = \frac{\ln 2}{5\,730}$. Here $\frac{A(t)}{A_0} = \frac{2.4}{15.3} \approx 0.15686$. $e^{-\lambda t} = \frac{2.4}{15.3} \Leftrightarrow -\lambda t = \ln\left(\frac{2.4}{15.3}\right) \Leftrightarrow t = -5\,730 \times \frac{\ln(2.4/15.3)}{\ln 2}$. Computing numerically: $\ln\left(\frac{2.4}{15.3}\right) \approx -1.8524$. $t \approx -5\,730 \times \frac{-1.8524}{0.69315} \approx 5\,730 \times 2.672 \approx 15\,310$ years. The paintings are approximately $15,300$ years old (dating from the Upper Paleolithic / Magdalenian period).

Solution 48.18: 1. For all $n \in \mathbb{N}$: $v_{n+1} = u_{n+1} - 5 = (0.6u_n + 2) - 5 = 0.6u_n - 3 = 0.6(u_n - 5) = 0.6v_n$. Hence (v_n) is a geometric sequence with common ratio $q = 0.6$ and first term $v_0 = u_0 - 5 = 10 - 5 = 5$. 2. Thus $v_n = 5 \times 0.6^n$, and $u_n = 5 + 5 \times 0.6^n$. 3. $|u_n - 5| < 10^{-4} \Leftrightarrow 5 \times 0.6^n < 10^{-4} \Leftrightarrow 0.6^n < \frac{10^{-4}}{5} = 2 \times 10^{-5}$. Taking logarithms: $n\ln(0.6) < \ln(2 \times 10^{-5})$. Since $0.6 < 1$, $\ln(0.6) \approx -0.5108 < 0$, which reverses the inequality:

$$n > \frac{\ln(2 \times 10^{-5})}{\ln(0.6)} \approx \frac{-10.8198}{-0.5108} \approx 21.18.$$

The smallest integer satisfying this condition is $\boldsymbol{n = 22}$.

Level Solutions

Solution 48.19: 1. For all $x > 0$, $\ln(1+x) \leq x$. Setting $x = \frac{1}{k}$ for $k \geq 1$:

$$\ln\left(1 + \frac{1}{k}\right) \leq \frac{1}{k} \Leftrightarrow \ln\left(\frac{k+1}{k}\right) \leq \frac{1}{k} \Leftrightarrow \ln(k+1) - \ln k \leq \frac{1}{k}.$$

2. For all $x \in]0\,;1[$, $\ln(1-x) \leq -x \Leftrightarrow -\ln(1-x) \geq x$. Setting $x = \frac{1}{k}$ for $k \geq 2$:

$$\frac{1}{k} \leq -\ln\left(1 - \frac{1}{k}\right) = -\ln\left(\frac{k-1}{k}\right) = \ln k - \ln(k-1).$$

3. - Summing inequality 1 for $k = 1$ to n:

$$\sum_{k=1}^{n} (\ln(k+1) - \ln k) \leq \sum_{k=1}^{n} \frac{1}{k} = H_n.$$

The left sum telescopes: $(\ln 2 - \ln 1) + (\ln 3 - \ln 2) + ... + (\ln(n+1) - \ln n) = \ln(n+1)$. Thus $\ln(n+1) \leq H_n$.

- Summing inequality 2 for $k = 2$ to n:

$$\sum_{k=2}^{n} \frac{1}{k} \leq \sum_{k=2}^{n} (\ln k - \ln(k-1)) = \ln n - \ln 1 = \ln n.$$

  Since $H_n = 1 + \sum_{k=2}^{n} \frac{1}{k}$, we obtain $H_n - 1 \leq \ln n \Leftrightarrow H_n \leq 1 + \ln n$. We have established the bounds: $\ln(n+1) \leq H_n \leq 1 + \ln n$.

4. Since $\lim_{n\to+\infty} \ln(n+1) = +\infty$, the comparison theorem immediately implies $\lim_{n\to+\infty} H_n = +\infty$. The harmonic series diverges, even though its individual terms $\frac{1}{n}$ tend to zero!
5. Divide the bounds by $\ln n$ (which is strictly positive for $n \geq 2$):

$$\frac{\ln(n+1)}{\ln n} \leq \frac{H_n}{\ln n} \leq \frac{1}{\ln n} + 1.$$

   Since $\frac{\ln(n+1)}{\ln n} = 1 + \frac{\ln(1+1/n)}{\ln n} \to 1$ and $\frac{1}{\ln n} + 1 \to 1$, the Squeeze Theorem gives:

$$\lim_{n\to+\infty} \frac{H_n}{\ln n} = 1.$$

   (The difference $H_n - \ln n$ converges to the celebrated **Euler-Mascheroni constant** $\gamma \approx 0.577215$).

Solution 48.20: 1. $\ln(n!) = \ln(1 \times 2 \times ... \times n) = \sum_{k=1}^{n} \ln k$. 2. For $k \geq 2$, the monotonicity of ln gives for $t \in [k-1\,;k]$: $\ln(k-1) \leq \ln t \leq \ln k$. Integrating (to be formalized in Chapter 50) yields $\int_1^n \ln t\, dt \leq \ln(n!) \leq \int_1^{n+1} \ln t\, dt$. Since an antiderivative of $\ln t$ is $t \ln t - t$, we obtain $\ln(n!) \approx n \ln n - n$, which leads to Stirling's asymptotic formula $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$.

Solution 48.21: 1. For $x > 0$ and $a > 0$: $x^a = a^x \Leftrightarrow e^{a \ln x} = e^{x \ln a} \Leftrightarrow a \ln x = x \ln a \Leftrightarrow \frac{\ln x}{x} = \frac{\ln a}{a}$. 2. From Exercise 48.15, $f(x) = \frac{\ln x}{x}$ increases from $-\infty$ to $\frac{1}{e}$ on $]0\,;e]$, then decreases from $\frac{1}{e}$ to 0 on $[e\,;+\infty[$.

- If $a = e$: $f(a) = \frac{1}{e}$ is the absolute maximum; the equation has the unique solution $x = e$.
- If $a > 1$ and $a \neq e$: the value $f(a) \in]0\,;\frac{1}{e}[$ is attained at exactly two points: one in $]1\,;e[$ and one in $]e\,;+\infty[$. One solution is $x = a$, and the other is a distinct real number.
- If $0 < a \leq 1$: $f(a) \leq 0$. Since f is strictly increasing on $]0\,;1]$ and positive on $]1\,;+\infty[$, there is only the trivial solution $x = a$.

3. We seek distinct positive integers $1 \leq x < y$ such that $x^y = y^x$. One integer must lie strictly below $e \approx 2.718$ and the other strictly above e. The only integer strictly between 1 and e is $x = 2$. Testing $x = 2$: the equation is $2^y = y^2$.
   - For $y = 3$: $2^3 = 8 \neq 9$.
   - For $y = 4$: $2^4 = 16 = 4^2$. It matches! Since f is strictly decreasing for $y > e$, no other integer solutions exist. The unique pair of distinct positive integers is $\mathbf{(2\,;4)}$.

Solution 48.22: 1. Take the natural logarithm of both sides: $\ln f(x) = \ln(u_1(x)u_2(x)...u_n(x)) = \sum_{k=1}^{n} \ln(u_k(x))$. Differentiating both sides with respect to x:

$$\frac{f'(x)}{f(x)} = \sum_{k=1}^{n} \frac{u_k'(x)}{u_k(x)}.$$

2. For $g(x) = (x+1)^3(2x-1)^4(x^2+5)^2$:

$$\frac{g'(x)}{g(x)} = \frac{3}{x+1} + \frac{4 \times 2}{2x-1} + \frac{2 \times 2x}{x^2+5} = \frac{3}{x+1} + \frac{8}{2x-1} + \frac{4x}{x^2+5}.$$

Multiplying this sum by $g(x)$ gives $g'(x)$ immediately without an unwieldy generalized product rule!

The Question that Opens What Follows

The logarithm and exponential functions allow us to master growth that explodes without bound and decay that fades toward zero. But in the physical universe, not everything grows or vanishes indefinitely: the seasons repeat, the heart beats, planets orbit, guitar strings vibrate, and a child's swing rocks back and forth.

How do mathematics describe phenomena that **return endlessly to their starting point**? How do we calculate the instantaneous velocity of a pendulum at the exact instant it swings through its lowest point?

Welcome to the world of oscillations, waves, and the geometry of the circle: join us in Chapter 49 with **The Swing and Oscillating Functions**.

CHAPTER 49: The Swing and Oscillating Functions

> "The pendular motion revealed to me that time is not a straight line fleeing away, but a breath that endlessly returns."
> *— Inspired by the observations of Galileo Galilei (1583)*

What You Are About to Discover

- Why the back-and-forth swing of a pendulum is nothing other than **the shadow of a point moving around a circle at constant speed**.
- The geometry of the **unit trigonometric circle** and why the **radian** is the only natural angular measure for calculus.
- Essential symmetry properties: **evenness**, **oddness**, **2π-periodicity**, and sum and double-angle formulas.
- Differentiation of sine and cosine: the fundamental geometric proof that $\lim_{x \to 0} \frac{\sin x}{x} = 1$, why $(\sin)' = \cos$, and $(\cos)' = -\sin$.
- The mystery of the quarter-turn: why taking the derivative advances phase by $\frac{\pi}{2}$.
- Composite derivatives of the form $t \mapsto \cos(\omega t + \varphi)$ and the harmonic oscillator differential equation $y'' + \omega^2 y = 0$.
- Comprehensive curve sketching: monotonicity, extrema, inflection points, and concavity of sinusoidal waveforms.
- Concrete real-world applications: acoustic soundwaves, 50 Hz alternating mains current, ocean tides, and seismology.

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1. The Problem: The Pisa Chandelier and the Shadow of the Circle

1.1 Galileo's Insight

In **1583**, a nineteen-year-old medical student named **Galileo Galilei** attended a religious service in the Cathedral of Pisa. His attention drifted toward the vaulted ceiling. A sacristan had just lit a large chandelier suspended from an immense iron chain and, in releasing it, set it swaying in a slow, steady rhythm.

Galileo noticed something extraordinary: whether the amplitude of the swing was broad at the beginning or shrank to a barely perceptible tremor as friction slowed it down, **the time required to complete a full back-and-forth cycle appeared precisely constant**.

Lacking a pocket watch—high-precision mechanical timekeepers had not yet been invented—Galileo took his own pulse at his wrist. He counted beats: four heartbeats for a wide swing, four heartbeats for a medium swing, four heartbeats for a tiny flutter. This was the discovery of the **isochronism of small pendular oscillations**.

This phenomenon is universal: a child's swing at the playground, a plucked violin string, water sloshing in an enclosed bay, alternating electrons surging through a radio antenna—all obey this very same fundamental rhythmic cadence.

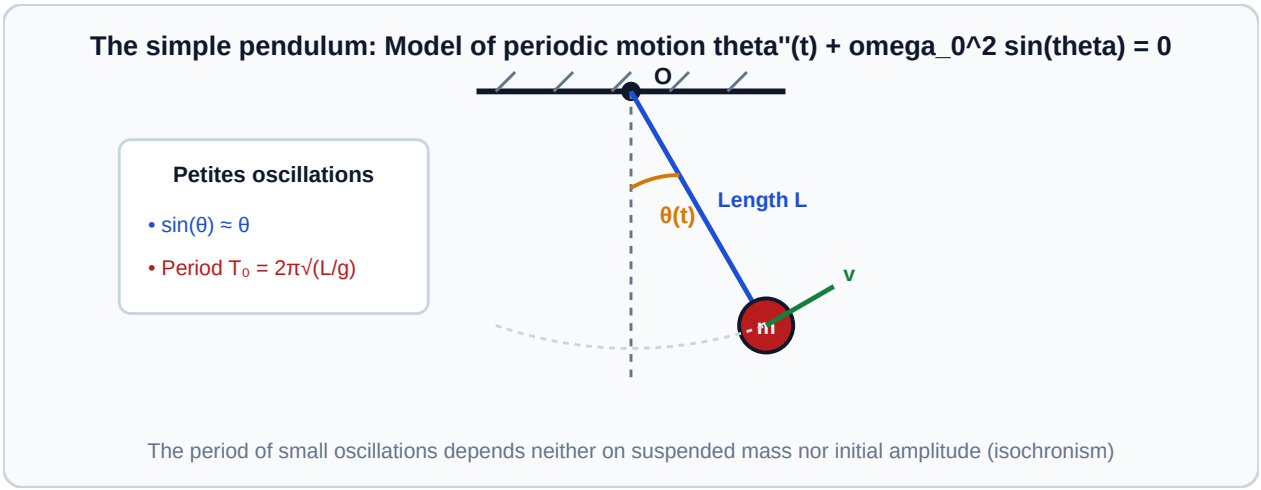

Figure 106: The simple pendulum: mathematical model of periodic harmonic oscillation

1.2 From the Circle to the Wave: Optical Projection

How can we represent this oscillatory back-and-forth movement mathematically? Consider a simple thought experiment: 1. A large vertical wheel rotates at constant speed in a darkened room. A single bright luminous LED M is fixed to its outer rim. 2. A distant light source projects the shadow of the dot M horizontally onto a flat vertical wall behind the wheel.

What path does the shadow trace on the wall?

- When the dot reaches the top of the wheel, its shadow climbs to the highest point on the wall, momentarily slows down, comes to an infinitesimal standstill, and reverses downward.
- As the shadow passes through the horizontal center line, it travels at its maximum speed.
- At the bottom, it decelerates, halts instantaneously, and accelerates back upward.

The shadow on the wall executes **the exact motion of a pendulum or a swing**! If we pull a strip of paper vertically past the wall at a constant speed, the moving shadow traces a pristine periodic curve: **a sinusoid**.

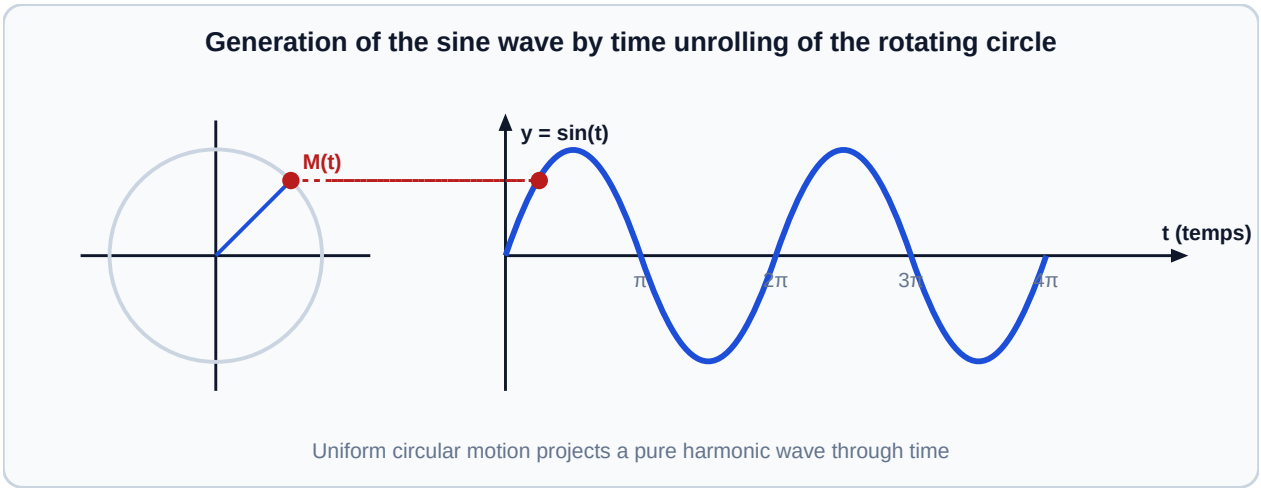

Figure 107: Generation of the sine wave through circular rotation

Rectilinear oscillation is nothing other than **the orthogonal projection of uniform circular motion**. To understand the swing, we must master the geometry of the circle.

2. The Trigonometric Circle and Fundamental Symmetries

2.1 The Radian: The Natural Measure of Arc Length

In elementary school, you learned to measure angles in degrees (360° for a full revolution). The number 360 was chosen by ancient Babylonian astronomers because it is divisible by 2, 3, 4, 5, 6, 8, 9, 10, 12, etc., and closely approximates the 365 days of a solar year.

In calculus and physics, however, degrees are an arbitrary and cumbersome choice: differentiating trigonometric functions expressed in degrees would introduce an unsightly scaling factor of $\frac{\pi}{180}$ into every single derivative!

The intrinsic, natural mathematical unit is the **radian**:

> **Definition of the Radian —**
>
> In the Cartesian plane equipped with an orthonormal coordinate system $(O;\vec{\imath},\vec{\jmath})$, the **trigonometric circle** (or unit circle) $\mathcal{C}$ is the circle centered at the origin $O(0\,;0)$ with radius $R=1$, oriented in the **counter-clockwise direction** (the positive or direct mathematical orientation). The measure of an angle in **radians** is defined as the **arc length** subtended by that angle along the unit circle of radius 1.

Since the circumference of a circle of radius 1 is $2\pi R = 2\pi$, a full revolution of 360° corresponds to an angle of $\boldsymbol{2\pi}$ **radians**.

Degrees	0°	30°	45°	60°	90°	180°	270°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

2.2 Wrapping the Real Line Around the Circle

Imagine a graduated real number line tangent to the trigonometric circle at the base point $I(1\,;0)$. Wrap this infinite line around the circumference:

- Positive numbers wrap counter-clockwise (upward and around);
- Negative numbers wrap clockwise (downward and around).

To every real number $x \in \mathbb{R}$ corresponds a unique point M on the unit circle.

> **Definition of the Cosine and Sine Functions —**
>
> For every real number x, let M be the point on the unit circle paired with x by wrapping the real number line.
>
> - The **cosine** of x, denoted $\cos x$, is the **abscissa** (x-coordinate) of point M.
> - The **sine** of x, denoted $\sin x$, is the **ordinate** (y-coordinate) of point M.
>
> $$M(x) = \big(\cos x\,;\sin x\,\big)\,.$$

2.3 Immediate Consequences and Special Angles

1. **Absolute Bounds:** Because the circle has unit radius, the coordinates of any point M are bounded between -1 and 1:
$$\text{For all } \; x\in\mathbb{R}: \quad -1 \le \cos x \le 1 \quad \text{and} \quad -1 \le \sin x \le 1.$$
2. **The Circle Equation (Pythagorean Identity):** The right triangle formed by the coordinates of M has perpendicular legs of lengths $|\cos x|$ and $|\sin x|$, and hypotenuse equal to the radius $OM = 1$. The Pythagorean theorem directly gives:
$$\cos^2 x + \sin^2 x = 1.$$
(Notation reminder: $\cos^2 x$ means $(\cos x)^2$.)
3. **Table of Special Values (Must be Mastered):**

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\boldsymbol{\cos x}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\boldsymbol{\sin x}$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0

> 💡 ***Mnemonic Trick for the Sine Row:*** *Notice the pattern of square roots:* $\sin 0 = \frac{\sqrt{0}}{2} = 0$, $\sin \frac{\pi}{6} = \frac{\sqrt{1}}{2} = \frac{1}{2}$, $\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$, $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$, $\sin \frac{\pi}{2} = \frac{\sqrt{4}}{2} = 1$. *The numerators march upward from* $\sqrt{0}$ *to* $\sqrt{4}$, *while the cosine row runs in reverse!*

2.4 Periodicity and Parity: The Power of Symmetry

Adding a full revolution around the circle corresponds to adding 2π to the wrapped arc length: the terminal point M lands on the exact same location.

> 📐 ***Periodicity —*** *The cosine and sine functions are* ***periodic with period*** 2π *(or* 2π*-periodic). For every real number* x *and every integer* $k \in \mathbb{Z}$*:*
>
> $$\cos(x + 2k\pi) = \cos x \quad \text{and} \quad \sin(x + 2k\pi) = \sin x.$$

Now consider geometric reflections on the circle:

- The points associated with x and $-x$ are symmetric with respect to **the horizontal axis** (Ox). They share the same x-coordinate and have opposite y-coordinates.

> 📐 ***Parity —***
> - *The cosine function is* ***even****:* $\cos(-x) = \cos x$. *Its Cartesian graph is symmetric with respect to* ***the vertical axis*** (Oy).
> - *The sine function is* ***odd****:* $\sin(-x) = -\sin x$. *Its Cartesian graph is symmetric with respect to* ***the origin*** $O\,(0\,;0)$.

2.5 Associated Angle Identities (Reflections of the Circle)

Rotating point $M(x)$ by a half-turn $(+\pi)$ or reflecting it across the coordinate axes produces key geometric relationships:

Transformation	Angle	Cosine	Sine	Geometric Interpretation
Half-turn	$x + \pi$	$-\cos x$	$-\sin x$	Central symmetry through O
Supplementary angle	$\pi - x$	$-\cos x$	$\sin x$	Reflection across vertical axis (Oy)
Quarter-turn	$x + \frac{\pi}{2}$	$-\sin x$	$\cos x$	Rotation by $+90°$
Complementary angle	$\frac{\pi}{2} - x$	$\sin x$	$\cos x$	Reflection across diagonal line $y = x$

> 💡 **Did you know? The Word "Cosine":**
>
> The prefix *co-* stands for *complementi*: the **cosine** of an angle is literally the **sine of the complementary angle** $(\cos x = \sin(\frac{\pi}{2} - x))$.

2.6 Addition and Double-Angle Formulas

These algebraic identities decompose trigonometric functions of composite angles into products of elementary components:

> 📐 ***Addition Formulas —*** *For all real numbers* a *and* b*:*
>
> $$\cos(a + b) = \cos a \cos b - \sin a \sin b$$
> $$\cos(a - b) = \cos a \cos b + \sin a \sin b$$
> $$\sin(a + b) = \sin a \cos b + \cos a \sin b$$
> $$\sin(a - b) = \sin a \cos b - \cos a \sin b$$

📐 ***Double-Angle Formulas*** — *Setting $a = b = x$:*

$$\sin(2x) = 2\sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$$

Worked Example 1 — Exact Evaluation of $\cos(\frac{\pi}{12})$.

Observe that $\frac{\pi}{12} = \frac{\pi}{3} - \frac{\pi}{4}$. Applying the cosine difference formula:

$$\cos\left(\frac{\pi}{12}\right) = \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \cos\frac{\pi}{3}\cos\frac{\pi}{4} + \sin\frac{\pi}{3}\sin\frac{\pi}{4}.$$

Substituting the standard values:

$$\cos\left(\frac{\pi}{12}\right) = \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}+\sqrt{6}}{4}.$$

---

3. Differentiation of Sine and Cosine: The Right-Angle Turn

To compute the instantaneous velocity of a pendulum or the acceleration of an oscillator, we must differentiate sine and cosine. But first, a fundamental question arises: **what is the derivative of $\sin x$ at $x = 0$?**

3.1 The Fundamental Limit $\lim_{x\to 0} \frac{\sin x}{x} = 1$

The difference quotient of the sine function at 0 is:

$$\tau(x) = \frac{\sin x - \sin 0}{x - 0} = \frac{\sin x}{x}.$$

As $x \to 0$, this yields the indeterminate form "$\frac{0}{0}$". We resolve this using an elegant geometric comparison of areas.

📐 ***Fundamental Lemma (Trigonometric Squeeze)*** — *For every real number $x \in]0\,;\,\frac{\pi}{2}[$:*

$$\sin x \leq x \leq \tan x \quad \left(\text{where } \tan x = \frac{\sin x}{\cos x}\right).$$

Geometric Proof (Required Curriculum Proof) —

On the unit circle centered at O with radius 1, consider:

- The point $A(1\,;0)$;
- The point $M(\cos x\,;\sin x)$ associated with an angle $x \in]0\,;\frac{\pi}{2}[$;
- The point $T(1\,;\tan x)$, which is the intersection of the ray $[OM)$ with the vertical tangent line at A.

We compare the areas of three geometric shapes nested inside one another: 1. The **inner triangle** OAM: base is $OA = 1$, height is the ordinate of M, which is $\sin x$.

$$\text{Area}(OAM) = \frac{\text{base} \times \text{height}}{2} = \frac{1 \times \sin x}{2} = \frac{\sin x}{2}.$$

2. The **circular sector** OAM (the circular wedge of radius 1 subtending an angle x): Since the area of a complete disk of radius 1 is $\pi(1)^2 = \pi$ for an angle of 2π, the area of a sector of angle x radians is directly proportional:

$$\text{Area}(\text{sector}\quad OAM) = \frac{x}{2\pi} \times \pi = \frac{x}{2}.$$

3. The **outer triangle** OAT: base is $OA = 1$, height is $AT = \tan x$.

$$\text{Area}(OAT) = \frac{\text{base} \times \text{height}}{2} = \frac{1 \times \tan x}{2} = \frac{\tan x}{2}.$$

Because triangle OAM is strictly contained within circular sector OAM, which in turn is strictly contained within triangle OAT:

$$\text{Area}(OAM) \leq \text{Area}(\text{sector}\quad OAM) \leq \text{Area}(OAT) \Leftrightarrow \frac{\sin x}{2} \leq \frac{x}{2} \leq \frac{\tan x}{2}.$$

Multiplying by 2 yields:

$$\sin x \leq x \leq \tan x. \quad \blacksquare$$

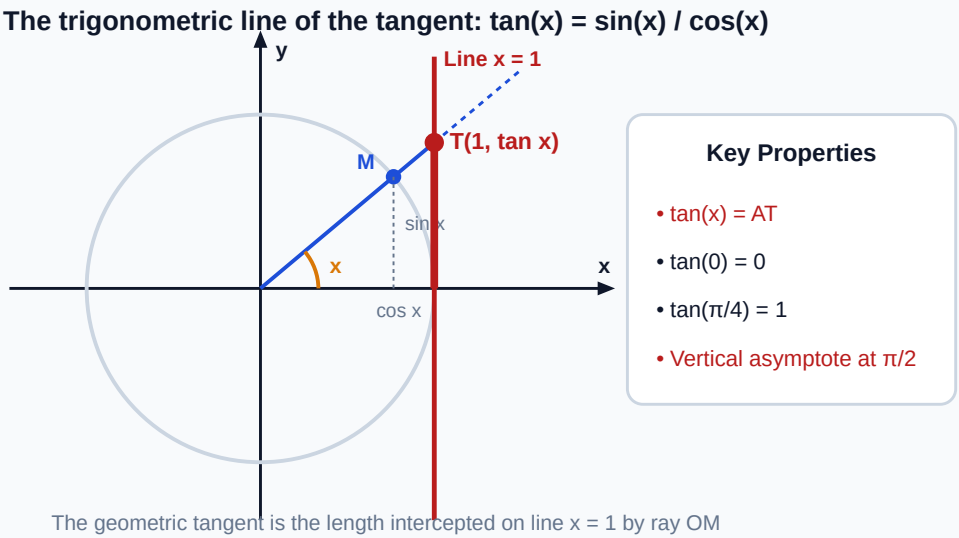

Figure 108: Trigonometric line of the tangent on the unit circle

Theorem — Fundamental Limit at 0:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1.$$

Proof —

For every $x \in]0\,;\tfrac{\pi}{2}[$, we have $\sin x > 0$. Divide the inequality $\sin x \le x \le \frac{\sin x}{\cos x}$ by $\sin x$:

$$1 \le \frac{x}{\sin x} \le \frac{1}{\cos x}.$$

Taking reciprocals (all quantities being strictly positive):

$$\cos x \le \frac{\sin x}{x} \le 1.$$

This inequality is equally valid for $x \in]-\tfrac{\pi}{2}\,;0[$, because $\cos(-x) = \cos x$ and $\frac{\sin(-x)}{-x} = \frac{-\sin x}{-x} = \frac{\sin x}{x}$. By the continuity of cosine at 0, $\lim_{x\to 0} \cos x = \cos 0 = 1$. Applying the **Squeeze Theorem**:

$$\lim_{x\to 0} \frac{\sin x}{x} = 1. \quad \blacksquare$$

Fundamental Corollary — The Limit of $\frac{1-\cos x}{x^2}$ at 0:

$$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2} \quad \text{and consequently} \quad \lim_{x\to 0} \frac{\cos x - 1}{x} = 0.$$

Proof: Use the double-angle formula $\cos(2u) = 1 - 2\sin^2 u$ with $u = \frac{x}{2}$, giving $1 - \cos x = 2\sin^2(\frac{x}{2})$. Then:

$$\frac{1-\cos x}{x^2} = \frac{2\sin^2(x/2)}{x^2} = \frac{2}{4} \times \left(\frac{\sin(x/2)}{x/2}\right)^2 = \frac{1}{2}\left(\frac{\sin(x/2)}{x/2}\right)^2.$$

As $x \to 0$, $u = \frac{x}{2} \to 0$, so $\frac{\sin(x/2)}{x/2} \to 1$. By the product of limits: $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2} \times 1^2 = \frac{1}{2}$. Furthermore, $\frac{\cos x - 1}{x} = -x \times \frac{1-\cos x}{x^2} \to 0 \times \frac{1}{2} = 0$. $\blacksquare$

3.2 Calculating the Derivatives of Sine and Cosine

Theorem (Required Curriculum Proofs) —

The functions sin and cos are differentiable on $\mathbb{R}$, and for every real number x:

$$\sin'(x) = \cos x \quad \text{and} \quad \cos'(x) = -\sin x.$$

Proof for $\sin'(x)$ —

Let $x \in \mathbb{R}$ be fixed and let $h \neq 0$. Form the difference quotient of sine between x and $x + h$:

$$\tau(h) = \frac{\sin(x+h) - \sin x}{h}.$$

Using the sine addition formula: $\sin(x+h) = \sin x \cos h + \cos x \sin h$. Hence:

$$\tau(h) = \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} = \sin x \left(\frac{\cos h - 1}{h}\right) + \cos x \left(\frac{\sin h}{h}\right).$$

As $h \to 0$:

- $\lim_{h\to 0} \frac{\cos h - 1}{h} = 0$;
- $\lim_{h\to 0} \frac{\sin h}{h} = 1$.

Therefore:

$$\lim_{h\to 0} \tau(h) = \sin x \times 0 + \cos x \times 1 = \cos x.$$

The sine function is differentiable for all $x \in \mathbb{R}$, and $\sin'(x) = \cos x$. $\blacksquare$

Proof for $\cos'(x)$ —

Using the cosine addition formula:

$$\frac{\cos(x+h)-\cos x}{h} = \frac{\cos x\cos h - \sin x\sin h - \cos x}{h} = \cos x\left(\frac{\cos h - 1}{h}\right) - \sin x\left(\frac{\sin h}{h}\right).$$

Taking the limit as $h \to 0$:

$$\lim_{h\to 0}\frac{\cos(x+h)-\cos x}{h} = \cos x \times 0 - \sin x \times 1 = -\sin x.$$

The cosine function is differentiable on $\mathbb{R}$, and $\cos'(x) = -\sin x$. ■

3.3 The Mystery of the Quarter-Turn: Kinematic View

Why is the derivative of sine cosine, and the derivative of cosine **minus** sine? Notice what happens when we advance phase by a quarter-turn, which is an angle of $+\frac{\pi}{2}$:

- $\sin\left(x+\frac{\pi}{2}\right) = \cos x$;
- $\cos\left(x+\frac{\pi}{2}\right) = -\sin x$.

> ***Differentiating a sinusoid shifts it forward in phase by exactly one quarter of a cycle ($\frac{\pi}{2}$ radians)!***

In kinematics, if a particle travels around the unit circle counter-clockwise with position vector $\overrightarrow{OM}(t) = \cos t\,\vec{\imath} + \sin t\,\vec{\jmath}$, its instantaneous velocity vector $\vec{v}(t) = \frac{d\overrightarrow{OM}}{dt}$ is:

$$\vec{v}(t) = -\sin t\,\vec{\imath} + \cos t\,\vec{\jmath}.$$

Notice that the dot product $\overrightarrow{OM}(t)\cdot\vec{v}(t) = \cos t(-\sin t) + \sin t(\cos t) = 0$! The velocity vector is **strictly perpendicular** to the radius vector: it points tangent to the circle in the direction of motion, exactly $90°$ ahead of the position.

3.4 Chain Rule with Trigonometric Functions

Applying the composite differentiation theorem $(f\circ u)' = u' \times (f'\circ u)$:

> 📐 ***Composition Formulas —*** *Let u be a differentiable function on an interval I: 1. $\big(\sin(u(x))\big)' = u'(x)\cos(u(x))$*
> *2. $\big(\cos(u(x))\big)' = -u'(x)\sin(u(x))$*
> *In particular, for harmonic oscillations $u(t) = \omega t + \varphi$ with constants $\omega, \varphi \in \mathbb{R}$:*
>
> $$\big(\sin(\omega t+\varphi)\big)' = \omega\cos(\omega t+\varphi)$$
>
> $$\big(\cos(\omega t+\varphi)\big)' = -\omega\sin(\omega t+\varphi)$$

Worked Example 2 — Composite Derivatives.

1. $f_1(x) = \sin(3x-\pi) \Longrightarrow f_1'(x) = 3\cos(3x-\pi)$. 2. $f_2(x) = \cos(x^2+1) \Longrightarrow f_2'(x) = -(2x)\sin(x^2+1) = -2x\sin(x^2+1)$. 3. $f_3(x) = \sin^3(x) = (\sin x)^3 \Longrightarrow f_3'(x) = 3(\sin x)^2 \times (\sin x)' = 3\sin^2 x\cos x$. 4. $f_4(x) = e^{-x}\cos(2x) \Longrightarrow$ product rule $(uv)' = u'v + uv'$:

$$f_4'(x) = (-e^{-x})\cos(2x) + e^{-x}(-2\sin(2x)) = -e^{-x}\left(\cos(2x) + 2\sin(2x)\right).$$

3.5 Second Derivatives and the Harmonic Oscillator Equation

Differentiating the function $f(t) = \cos(\omega t)$ twice:

- $f'(t) = -\omega\sin(\omega t)$;
- $f''(t) = -\omega \times \big(\,\omega\cos(\omega t)\big) = -\omega^2\cos(\omega t) = -\omega^2 f(t)$.

Rearranging:

$$f''(t) + \omega^2 f(t) = 0.$$

> 💡 **Master Discovery —**
>
> Sinusoidal functions are the **general solutions** to the second-order linear differential equation:
>
> $$y'' + \omega^2 y = 0.$$
>
> This is the foundational equation of classical mechanics: governing mass-spring systems (Hooke's law $ma = -kx$), inductor-capacitor LC circuits, acoustic vibrations, and electromagnetic waves!

---

4. Comprehensive Analysis of the Sine and Cosine Functions

Because sine and cosine are periodic with period 2π, it suffices to study them on an interval of length 2π, such as $[-\pi\,;\pi]$. Exploiting parity:

- For cosine (even): analyze on $[0\,;\pi]$, then reflect across the vertical axis (Oy) onto $[-\pi\,;\pi]$.
- For sine (odd): analyze on $[0\,;\pi]$, then reflect through the origin $O(0\,;0)$ onto $[-\pi\,;\pi]$.

Finally, translate the pattern indefinitely left and right by integer multiples of $2\pi\vec{\imath}$.

4.1 Analysis of the Cosine Function

On the interval $[0\,;\pi]$:

- $\cos'(x) = -\sin x$.
- For all $x \in\,]0\,;\pi[$, $\sin x > 0$, so $\cos'(x) < 0$.
- The cosine function is therefore **strictly decreasing** on $[0\,;\pi]$.
- $\cos 0 = 1$, $\cos\left(\frac{\pi}{2}\right) = 0$, and $\cos\pi = -1$.

Variation table on $[-\pi\,;\pi]$:

x	$-\pi$		0		π
Signe de $\cos'(x) = -\sin(x)$	0	$+$	0	$-$	0
Variations de $\cos(x)$	-1 (Min)	↗	1 (Max)	↘	-1 (Min)

Tangents and Concavity: - At $x = 0$, $\cos'(0) = 0$: the curve has a **horizontal tangent** at its maximum $(0\,;1)$. - At $x = \pi$, $\cos'(\pi) = 0$: **horizontal tangent** at its minimum $(\pi\,;-1)$. - Second derivative: $\cos''(x) = -\cos x$. On $\left[0\,;\frac{\pi}{2}\right]$, $\cos x \geq 0 \Longrightarrow \cos''(x) \leq 0$: the function is **concave**. On $\left[\frac{\pi}{2}\,;\pi\right]$, $\cos x \leq 0 \Longrightarrow \cos''(x) \geq 0$: the function is **convex**. An inflection point occurs at $x = \frac{\pi}{2}$, where the curve crosses the horizontal axis with a slope of -1.

4.2 Analysis of the Sine Function

On the interval $[0\,;\pi]$:

- $\sin'(x) = \cos x$.
- On $\left[0\,;\frac{\pi}{2}\right[$, $\cos x > 0$: sin is **strictly increasing**.
- At $x = \frac{\pi}{2}$, $\cos\left(\frac{\pi}{2}\right) = 0$: horizontal tangent at $\left(\frac{\pi}{2}\,;1\right)$.
- On $\left]\frac{\pi}{2}\,;\pi\right]$, $\cos x < 0$: sin is **strictly decreasing**.

Variation table on $[-\pi\,;\pi]$:

x	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π
$\sin'(x) = \cos(x)$	$-$	0	$+$	0	$-$
$\sin(x)$	0	-1 (Min)	0	1 (Max)	0

Notable Tangents: - At $x = 0$, $\sin'(0) = 1$ and $\sin 0 = 0$. The tangent line at the origin is:

$$y = x.$$

For $x \approx 0$, the first-order approximation yields $\boldsymbol{\sin x \approx x}$. This is the celebrated small-angle approximation widely used in physics!

---

5. Solving Trigonometric Equations and Inequalities

5.1 Fundamental Equations

Method — Solving $\cos x = \cos \alpha$:

Two points on the unit circle share the same abscissa if and only if their angles are equal or opposite, up to integer multiples of 2π:

$$\cos x = \cos \alpha \Leftrightarrow x = \alpha + 2k\pi \quad \text{or} \quad x = -\alpha + 2k\pi \quad (k \in \mathbb{Z}).$$

Method — Solving $\sin x = \sin \alpha$:

Two points on the unit circle share the same ordinate if and only if their angles are equal or supplementary ($\pi - \alpha$), up to integer multiples of 2π:

$$\sin x = \sin \alpha \Leftrightarrow x = \alpha + 2k\pi \quad \text{or} \quad x = \pi - \alpha + 2k\pi \quad (k \in \mathbb{Z}).$$

Worked Example 3 — Solving on a Specified Interval.

Solve $\cos(2x) = \frac{1}{2}$ on $[0\,;2\pi]$.

- Recall that $\frac{1}{2} = \cos\left(\frac{\pi}{3}\right)$.
- The equation yields:

$$2x = \frac{\pi}{3} + 2k\pi \Leftrightarrow x = \frac{\pi}{6} + k\pi$$

or

$$2x = -\frac{\pi}{3} + 2k\pi \Leftrightarrow x = -\frac{\pi}{6} + k\pi \quad (k \in \mathbb{Z}).$$

- Select solutions lying within $[0\,;2\pi]$:
  - First family: $x = \frac{\pi}{6} + k\pi$:
    - $k = 0 \Longrightarrow x_1 = \frac{\pi}{6} \in [0\,;2\pi]$;
    - $k = 1 \Longrightarrow x_2 = \frac{\pi}{6} + \pi = \frac{7\pi}{6} \in [0\,;2\pi]$;
    - $k = 2 \Longrightarrow x = \frac{13\pi}{6} > 2\pi$ (out of bounds).
  - Second family: $x = -\frac{\pi}{6} + k\pi$:
    - $k = 0 \Longrightarrow x = -\frac{\pi}{6} < 0$ (out of bounds);
    - $k = 1 \Longrightarrow x_3 = -\frac{\pi}{6} + \pi = \frac{5\pi}{6} \in [0\,;2\pi]$;
    - $k = 2 \Longrightarrow x_4 = -\frac{\pi}{6} + 2\pi = \frac{11\pi}{6} \in [0\,;2\pi]$.
- Conclusion: $\mathcal{S} = \left\{\frac{\pi}{6}\,;\,\frac{5\pi}{6}\,;\,\frac{7\pi}{6}\,;\,\frac{11\pi}{6}\right\}$.

5.2 Equations with Change of Variables

Worked Example 4 — Quadratic Trigonometric Equation.

Solve $2\sin^2 x - 3\sin x + 1 = 0$ on $[0\,;2\pi[$. Set $X = \sin x$. The equation becomes:

$$2X^2 - 3X + 1 = 0.$$

$\Delta = (-3)^2 - 4(2)(1) = 9 - 8 = 1$. The roots for X are $X_1 = \frac{3-1}{4} = \frac{1}{2}$ and $X_2 = \frac{3+1}{4} = 1$.
Back-substituting into $\sin x$: 1. $\sin x = 1 \Leftrightarrow x = \frac{\pi}{2}$. 2. $\sin x = \frac{1}{2} = \sin\left(\frac{\pi}{6}\right) \Leftrightarrow x = \frac{\pi}{6}$ or $x = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.
The solutions in $[0\,;2\pi[$ are: $\mathcal{S} = \left\{\frac{\pi}{6}\,;\,\frac{\pi}{2}\,;\,\frac{5\pi}{6}\right\}$.

5.3 Trigonometric Inequalities

To solve inequalities such as $\cos x \geq c$ or $\sin x < c$, read the arc directly off the unit circle by drawing the boundary line $x = c$ (vertical) or $y = c$ (horizontal).

Worked Example 5 — Inequality on $[-\pi\,;\pi]$.

Solve $\sin x \geq -\frac{\sqrt{2}}{2}$ on $[-\pi\,;\pi]$. Draw the horizontal line $y = -\frac{\sqrt{2}}{2}$. It intersects the unit circle at angles $-\frac{\pi}{4}$ and $-\frac{3\pi}{4}$. The region where the y-coordinate is greater than or equal to $-\frac{\sqrt{2}}{2}$ corresponds to the upper arc. Traversing the interval from $-\pi$ to π:

- From $-\pi$ to $-\frac{3\pi}{4}$: $\sin x \geq -\frac{\sqrt{2}}{2}$;
- From $-\frac{3\pi}{4}$ to $-\frac{\pi}{4}$: $\sin x < -\frac{\sqrt{2}}{2}$;
- From $-\frac{\pi}{4}$ to π: $\sin x \geq -\frac{\sqrt{2}}{2}$.

The solution set is:

$$\mathcal{S} = \left[-\pi\,;-\frac{3\pi}{4}\right] \cup \left[-\frac{\pi}{4}\,;\pi\right].$$

---

6. The Realm of Signals and Waves: Concrete Modeling

In real physical systems, oscillations do not necessarily complete a cycle in 2π seconds, nor is their amplitude restricted to 1 unit. We describe general periodic phenomena using the canonical form of a **sinusoidal signal**:

$$s(t) = A\cos(\omega t + \varphi)$$

where:

- $\boldsymbol{A > 0}$ is the **amplitude** (peak excursion from equilibrium);
- $\boldsymbol{\omega > 0}$ is the **angular frequency** (in radians per second, $\mathrm{rad}\cdot\mathrm{s}^{-1}$);
- $\boldsymbol{T = \frac{2\pi}{\omega}}$ is the **period** (duration of one complete cycle in seconds);
- $\boldsymbol{f = \frac{1}{T} = \frac{\omega}{2\pi}}$ is the **frequency** (cycles per second, in Hertz, Hz);
- φ is the **initial phase** (phase offset at $t = 0$).

6.1 Alternating Mains Current

In European residential electrical outlets, the voltage is a sinusoidal wave with frequency $f = 50\ \ \mathrm{Hz}$. The angular frequency is $\omega = 2\pi f = 2\pi \times 50 = 100\pi \approx 314.16\ \ \mathrm{rad}\cdot\mathrm{s}^{-1}$. The period of oscillation is $T = \frac{1}{50} = 0.02\ \ \mathrm{s} = 20\ \ \text{milliseconds}$. An RMS voltage of $230\ \ \mathrm{V}$ corresponds to a peak amplitude of $U_{\max} = 230\sqrt{2} \approx 325\ \ \mathrm{V}$:

$$u(t) = 325\cos(100\pi t).$$

6.2 Musical Notes and Acoustics

A pure tone produced by a tuning fork is a sinusoidal pressure wave traveling through air. The standard orchestral reference note *A4* (concert pitch) vibrates at 440 cycles per second ($f = 440\ \ \mathrm{Hz}$). Its acoustic wavelength in air (at speed of sound $c \approx 340\ \ \mathrm{m}\cdot\mathrm{s}^{-1}$) is:

$$\lambda = \frac{c}{f} = \frac{340}{440} \approx 0.77\ \ \mathrm{m}\ .$$

> 💻 ***Python Simulation — Sampling a Sinusoid:***

```python
import math

def sinusoidal_signal(amplitude, frequency, duration, num_points):
    omega = 2 * math.pi * frequency
    step = duration / num_points
    points = []

    for i in range(num_points):
        t = i * step
        value = amplitude * math.sin(omega * t)
        points.append((round(t, 4), round(value, 3)))
    return points

# Generate a 50 Hz signal over 0.02 seconds (1 full period)
sampled = sinusoidal_signal(amplitude=325, frequency=50, duration=0.02, num_points=5)
print("Sampled points:", sampled)
```

Key Takeaways

1. The Unit Circle and Geometric Symmetries

- The radian measures arc length: $180^\circ = \pi \text{ rad}$.
- Circle coordinates: $M(x) = (\cos x\,; \sin x)$, whence $\cos^2 x + \sin^2 x = 1$ and $-1 \leq \cos x, \sin x \leq 1$.
- Periodicity: $\cos(x + 2\pi) = \cos x$ and $\sin(x + 2\pi) = \sin x$.
- Parity: $\cos(-x) = \cos x$ (cosine is even); $\sin(-x) = -\sin x$ (sine is odd).
- Angle shifts: $\cos\left(\frac{\pi}{2} - x\right) = \sin x$ and $\sin\left(\frac{\pi}{2} - x\right) = \cos x$.

2. Fundamental Differentiation

- Key limits at the origin: $\lim_{x\to 0} \frac{\sin x}{x} = 1$ and $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$.
- Elementary derivatives:

$$(\sin x)' = \cos x \quad \text{and} \quad (\cos x)' = -\sin x.$$

- Composite derivatives:

$$\big(\sin(u)\big)' = u' \cos(u) \quad \text{and} \quad \big(\cos(u)\big)' = -u' \sin(u).$$

- In particular: $\big(\cos(\omega t + \varphi)\big)' = -\omega \sin(\omega t + \varphi)$.
- Second derivatives: $(\sin)''(x) = -\sin x$ and $(\cos)''(x) = -\cos x$.
- The harmonic oscillator differential equation: $y'' + \omega^2 y = 0$.

3. Graphic Profiles and Tangents

- Tangent to sin at the origin: $y = x$. For small x, $\sin x \approx x$.
- Tangent to cos at $x = 0$: $y = 1$ (horizontal tangent).
- Period $T = \frac{2\pi}{\omega}$ and frequency $f = \frac{1}{T}$.

4. Solving Trigonometric Equations

- $\cos x = \cos \alpha \Leftrightarrow x = \alpha + 2k\pi$ or $x = -\alpha + 2k\pi$.
- $\sin x = \sin \alpha \Leftrightarrow x = \alpha + 2k\pi$ or $x = \pi - \alpha + 2k\pi$.

Exercises

Level (Direct Applications)

Exercise 49.1 — Angle Conversions and Exact Values. 1. Convert the following angles to radians: $15°$, $75°$, $150°$, $315°$. 2. Without a calculator, compute the exact values of: $A = \cos\left(\frac{2\pi}{3}\right)$, $B = \sin\left(-\frac{\pi}{4}\right)$, $C = \cos\left(\frac{7\pi}{6}\right)$, $D = \sin\left(\frac{5\pi}{2}\right)$.

Exercise 49.2 — Simplification of Trigonometric Expressions. Simplify for all real x: 1. $E(x) = \cos(x + \pi) + \sin\left(x + \frac{\pi}{2}\right)$ 2. $F(x) = \sin(\pi - x) - \cos\left(\frac{\pi}{2} - x\right)$ 3. $G(x) = \cos(-x) + \cos(\pi - x) + \sin(-x) + \sin(\pi + x)$ 4. $H(x) = (\cos x + \sin x)^2 + (\cos x - \sin x)^2$

Exercise 49.3 — Direct Derivatives. Differentiate the following functions on $\mathbb{R}$: 1. $f_1(x) = 3\sin x - 2\cos x$ 2. $f_2(x) = x \sin x$ 3. $f_3(x) = \frac{\cos x}{x^2+1}$ 4. $f_4(x) = \sin(5x - 2)$ 5. $f_5(x) = \cos(4x^2)$

Exercise 49.4 — Elementary Equations in $\mathbb{R}$. Solve in $\mathbb{R}$: 1. $\cos x = \frac{\sqrt{3}}{2}$ 2. $\sin x = -\frac{1}{2}$ 3. $\cos(3x) = 0$ 4. $\sin(2x) = 1$

Exercise 49.5 — Equations on Restricted Intervals. 1. Solve $\cos x = -\frac{\sqrt{2}}{2}$ on the interval $[0\,;2\pi]$. 2. Solve $\sin(2x) = \frac{\sqrt{3}}{2}$ on the interval $[-\pi\,;\pi]$.

Exercise 49.6 — Immediate Inequalities. Solve on $[-\pi\,;\pi]$: 1. $\cos x > 0$ 2. $\sin x \leq \frac{1}{2}$

Exercise 49.7 — Trigonometric Limits. Evaluate: 1. $\lim_{x\to 0} \frac{\sin(3x)}{x}$ 2. $\lim_{x\to 0} \frac{\tan x}{x}$ 3. $\lim_{x\to+\infty} \frac{\cos x}{x}$ (apply the Squeeze Theorem)

Level (Exploration and Exam Preparation)

Exercise 49.8 — Quadratic Equation in Cosine. Solve on $[0\,;2\pi[$:

$$2\cos^2 x + \cos x - 1 = 0.$$

Exercise 49.9 — Study of a Damped Oscillatory Function. Let f be defined on $[0\,;+\infty[$ by $f(t) = e^{-t}\sin t$. 1. Determine the limit of f as $t \to +\infty$. 2. Calculate $f'(t)$ and show that $f'(t) = \sqrt{2}e^{-t}\cos\left(t + \frac{\pi}{4}\right)$. 3. Establish the table of variations of f on $[0\,;2\pi]$. 4. Sketch the curve showing the envelope curves $y = e^{-t}$ and $y = -e^{-t}$.

Exercise 49.10 — Complete Study of a Trigonometric Function. Let f be defined on $\mathbb{R}$ by:

$$f(x) = \cos(2x) + 2\cos x.$$

1. Show that f is 2π-periodic and even. Explain why its study can be restricted to $[0\,;\pi]$. 2. Compute $f'(x)$ and prove that $f'(x) = -2\sin x(2\cos x + 1)$. 3. Determine the sign of $f'(x)$ on $[0\,;\pi]$ and compile the variation table of f. 4. Determine the extrema of f on $[0\,;\pi]$.

Exercise 49.11 — Tangents and Rational Trigonometric Functions. Let $g(x) = \frac{\sin x}{2+\cos x}$ defined on $\mathbb{R}$. 1. Justify why the denominator is never zero on $\mathbb{R}$. 2. Prove that g is an odd function. 3. Compute $g'(x)$ and show that $g'(x) = \frac{2\cos x+1}{(2+\cos x)^2}$. 4. Find the equation of the tangent line T at the origin $(0\,;0)$.

Exercise 49.12 — Addition Formulas and Angle Identities. 1. Prove that for all real numbers a and b:

$$\cos(a + b)\cos(a - b) = \cos^2 a - \sin^2 b.$$

2. Solve in $\mathbb{R}$ the equation $\cos(3x) = \sin(2x)$ by using $\sin(u) = \cos\left(\frac{\pi}{2} - u\right)$.

Exercise 49.13 — Convexity and Inflection Points. Let $f(x) = \sin^2 x$ defined on $[0\,;\pi]$. 1. Express $f(x)$ in terms of $\cos(2x)$. 2. Compute $f'(x)$ and $f''(x)$. 3. Determine the intervals on which f is convex, and concave. 4. State the coordinates of the inflection points on $[0\,;\pi]$.

Exercise 49.14 — Ocean Tide Modeling. In a harbor in Brittany, water height $H(t)$ (in meters) as a function of time t (in hours since midnight) is modeled by:

$$H(t) = 6 + 4\cos\left(\frac{\pi}{6}t - \frac{\pi}{3}\right).$$

1. What is the tidal period? 2. What are the water heights at high tide and low tide? 3. At what time in the morning ($t \in [0\,;12]$) is high tide reached? 4. A fishing trawler requires a minimum depth of 8 meters to navigate safely. During what morning hours can the trawler enter the harbor safely?

Exercise 49.15 — Indeterminate Forms Involving Trigonometry. Evaluate, justifying each step: 1. $\lim_{x\to 0} \frac{1-\cos(2x)}{x\sin x}$ 2. $\lim_{x\to \pi/2} \frac{\cos x}{x-\frac{\pi}{2}}$ 3. $\lim_{x\to 0} \frac{\sqrt{1+\sin x}-\sqrt{1-\sin x}}{x}$

Level (In-Depth Challenges)

Exercise 49.16 — Machin's Formula and the Calculation of π. Define the tangent function by $\tan x = \frac{\sin x}{\cos x}$ for $x \in]-\frac{\pi}{2}\,;\frac{\pi}{2}[$. 1. Show that $\tan(a+b) = \frac{\tan a + \tan b}{1 - \tan a \tan b}$. 2. Let α be such that $\tan\alpha = \frac{1}{5}$. Calculate $\tan(2\alpha)$ and $\tan(4\alpha)$. 3. Deduce John Machin's celebrated formula (1706):

$$\frac{\pi}{4} = 4\arctan\left(\frac{1}{5}\right) - \arctan\left(\frac{1}{239}\right).$$

(Machin used this formula to compute the first 100 decimal digits of π by hand!)

Exercise 49.17 — Dirichlet's Trigonometric Sum. For $n \in \mathbb{N}^*$ and $x \in \mathbb{R} \setminus 2\pi\mathbb{Z}$, define:

$$S_n(x) = \frac{1}{2} + \sum_{k=1}^{n} \cos(kx) = \frac{1}{2} + \cos x + \cos(2x) + \ldots + \cos(nx).$$

1. Prove the identity: $2\sin\left(\frac{x}{2}\right)\cos(kx) = \sin\left(\left(k+\frac{1}{2}\right)x\right) - \sin\left(\left(k-\frac{1}{2}\right)x\right)$. 2. Multiplying $S_n(x)$ by $2\sin\left(\frac{x}{2}\right)$, use a telescoping sum to show:

$$S_n(x) = \frac{\sin\left(\left(n+\frac{1}{2}\right)x\right)}{2\sin\left(\frac{x}{2}\right)}.$$

(This Dirichlet kernel lies at the heart of Fourier series analysis).

Exercise 49.18 — Jordan's Inequality. 1. Prove that the function $f(x) = \frac{\sin x}{x}$ is strictly decreasing on $]0\,;\frac{\pi}{2}]$. 2. Deduce **Jordan's inequality**:

$$\text{For all } \; x \in \left]0\,;\frac{\pi}{2}\right] : \quad \frac{2}{\pi}x \leq \sin x \leq x.$$

3. Give the geometric interpretation of this inequality using a chord and a tangent line.

Exercise 49.19 — Coupled Oscillating Systems and Phase Shift. Consider two signals $s_1(t) = 3\cos(\omega t)$ and $s_2(t) = 4\sin(\omega t)$. Define their superposition $S(t) = s_1(t) + s_2(t)$. 1. Show that there exists an amplitude $A > 0$ and a phase angle $\varphi \in [0\,;2\pi[$ such that:

$$S(t) = A\cos(\omega t - \varphi).$$

2. Compute the exact value of A and estimate φ. 3. Generalization: Show that any linear combination $a\cos(\omega t) + b\sin(\omega t)$ can be written in the form $R\cos(\omega t - \theta)$ with $R = \sqrt{a^2+b^2}$.

Exercise 49.20 — Absence of Limit at Infinity. Prove rigorously using sequences that the function $x \mapsto \cos x$ has **no limit** as $x \to +\infty$.

Solutions

Level Solutions

Solution 49.1: 1. Conversion formula: $\theta_{\text{rad}} = \theta_{\text{deg}} \times \frac{\pi}{180}$.

- $15^\circ = 15 \times \frac{\pi}{180} = \frac{\pi}{12}$ rad.
- $75^\circ = 75 \times \frac{\pi}{180} = \frac{5\pi}{12}$ rad.
- $150^\circ = 150 \times \frac{\pi}{180} = \frac{5\pi}{6}$ rad.
- $315^\circ = 315 \times \frac{\pi}{180} = \frac{7\pi}{4}$ rad.

2. Exact values:
   - $A = \cos\left(\pi - \frac{\pi}{3}\right) = -\cos\left(\frac{\pi}{3}\right) = -\frac{1}{2}$.
   - $B = \sin\left(-\frac{\pi}{4}\right) = -\sin\left(\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}$.
   - $C = \cos\left(\pi + \frac{\pi}{6}\right) = -\cos\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}$.
   - $D = \sin\left(2\pi + \frac{\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) = 1$.

Solution 49.2: 1. $E(x) = -\cos x + \cos x = 0$. 2. $F(x) = \sin x - \sin x = 0$. 3. $G(x) = \cos x - \cos x - \sin x - \sin x = -2\sin x$. 4. Expanding: $(\cos^2 x + 2\sin x\cos x + \sin^2 x) + (\cos^2 x - 2\sin x\cos x + \sin^2 x) = 2(\cos^2 x + \sin^2 x) = 2(1) = 2$.

Solution 49.3: 1. $f_1'(x) = 3\cos x - 2(-\sin x) = 3\cos x + 2\sin x$. 2. Product rule: $f_2'(x) = 1 \times \sin x + x\cos x = \sin x + x\cos x$. 3. Quotient rule: $f_3'(x) = \frac{(-\sin x)(x^2+1) - \cos x(2x)}{(x^2+1)^2} = \frac{-(x^2+1)\sin x - 2x\cos x}{(x^2+1)^2}$. 4. Chain rule: $f_4'(x) = 5\cos(5x-2)$. 5. Chain rule: with $u(x) = 4x^2 \Longrightarrow u'(x) = 8x$, we have $f_5'(x) = -8x\sin(4x^2)$.

Solution 49.4: 1. $\cos x = \frac{\sqrt{3}}{2} = \cos\left(\frac{\pi}{6}\right) \Leftrightarrow x = \frac{\pi}{6} + 2k\pi$ or $x = -\frac{\pi}{6} + 2k\pi$ $(k \in \mathbb{Z})$. 2. $\sin x = -\frac{1}{2} = \sin\left(-\frac{\pi}{6}\right) \Leftrightarrow x = -\frac{\pi}{6} + 2k\pi$ or $x = \pi - \left(-\frac{\pi}{6}\right) + 2k\pi = \frac{7\pi}{6} + 2k\pi$ $(k \in \mathbb{Z})$. 3. $\cos(3x) = 0 = \cos\left(\frac{\pi}{2}\right) \Leftrightarrow 3x = \frac{\pi}{2} + k\pi \Leftrightarrow x = \frac{\pi}{6} + \frac{k\pi}{3}$ $(k \in \mathbb{Z})$. 4. $\sin(2x) = 1 = \sin\left(\frac{\pi}{2}\right) \Leftrightarrow 2x = \frac{\pi}{2} + 2k\pi \Leftrightarrow x = \frac{\pi}{4} + k\pi$ $(k \in \mathbb{Z})$.

Solution 49.5: 1. $\cos x = -\frac{\sqrt{2}}{2} = \cos\left(\frac{3\pi}{4}\right)$. Solutions in $\mathbb{R}$: $x = \pm\frac{3\pi}{4} + 2k\pi$. Within $[0\,;2\pi]$: for $k=0$, $x_1 = \frac{3\pi}{4}$; for $k=1$ in the negative branch, $x_2 = 2\pi - \frac{3\pi}{4} = \frac{5\pi}{4}$. $\mathcal{S} = \left\{\frac{3\pi}{4}\,;\,\frac{5\pi}{4}\right\}$. 2. $\sin(2x) = \frac{\sqrt{3}}{2} = \sin\left(\frac{\pi}{3}\right)$. Hence $2x = \frac{\pi}{3} + 2k\pi \Leftrightarrow x = \frac{\pi}{6} + k\pi$ or $2x = \frac{2\pi}{3} + 2k\pi \Leftrightarrow x = \frac{\pi}{3} + k\pi$. Within $[-\pi\,;\pi]$:

- $x = \frac{\pi}{6} + k\pi$ yields $\frac{\pi}{6}$ (for $k=0$) and $-\frac{5\pi}{6}$ (for $k=-1$).
- $x = \frac{\pi}{3} + k\pi$ yields $\frac{\pi}{3}$ (for $k=0$) and $-\frac{2\pi}{3}$ (for $k=-1$). $\mathcal{S} = \left\{-\frac{5\pi}{6}\,;-\frac{2\pi}{3}\,;\frac{\pi}{6}\,;\frac{\pi}{3}\right\}$.

Solution 49.6: 1. $\cos x > 0$ on $[-\pi\,;\pi]$: corresponds to points with strictly positive x-coordinates on the unit circle: $\mathcal{S} =]-\frac{\pi}{2}\,;\frac{\pi}{2}[$. 2. $\sin x \le \frac{1}{2}$ on $[-\pi\,;\pi]$: the line $y = 1/2$ intersects the circle at $\frac{\pi}{6}$ and $\frac{5\pi}{6}$. Points lying on or below this line over $[-\pi\,;\pi]$ form: $\mathcal{S} = \left[-\pi\,;\frac{\pi}{6}\right] \cup \left[\frac{5\pi}{6}\,;\pi\right]$.

Solution 49.7: 1. Substitute $u = 3x \to 0$: $\frac{\sin(3x)}{x} = 3 \times \frac{\sin u}{u} \to 3 \times 1 = 3$. 2. $\frac{\tan x}{x} = \frac{\sin x}{x\cos x} = \frac{\sin x}{x} \times \frac{1}{\cos x}$. As $x \to 0$, $\frac{\sin x}{x} \to 1$ and $\cos x \to 1$, so the limit is $1 \times 1 = 1$. 3. For $x > 0$, $-1 \le \cos x \le 1 \Leftrightarrow -\frac{1}{x} \le \frac{\cos x}{x} \le \frac{1}{x}$. Since $\lim_{x\to+\infty} \pm\frac{1}{x} = 0$, the Squeeze Theorem gives $\lim_{x\to+\infty} \frac{\cos x}{x} = 0$.

Level Solutions

Solution 49.8: Let $X = \cos x$. Then $2X^2 + X - 1 = 0$. $\Delta = 1^2 - 4(2)(-1) = 9 = 3^2$. $X_1 = \frac{-1-3}{4} = -1$ and $X_2 = \frac{-1+3}{4} = \frac{1}{2}$.

- $\cos x = -1 \Leftrightarrow x = \pi$ on $[0\,;2\pi[$.
- $\cos x = \frac{1}{2} = \cos\left(\frac{\pi}{3}\right) \Leftrightarrow x = \frac{\pi}{3}$ or $x = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$. $\mathcal{S} = \left\{\frac{\pi}{3}\,;\,\pi\,;\,\frac{5\pi}{3}\right\}$.

Solution 49.9: 1. For all $t \ge 0$, $-1 \le \sin t \le 1 \Leftrightarrow -e^{-t} \le e^{-t}\sin t \le e^{-t}$. Since $\lim_{t\to+\infty} e^{-t} = 0$, the Squeeze Theorem guarantees $\lim_{t\to+\infty} f(t) = 0$. 2. Product rule: $f'(t) = -e^{-t}\sin t + e^{-t}\cos t = e^{-t}(\cos t - \sin t)$. Now $\cos t - \sin t = \sqrt{2}\left(\frac{\sqrt{2}}{2}\cos t - \frac{\sqrt{2}}{2}\sin t\right) = \sqrt{2}\cos(t + \frac{\pi}{4})$. Hence $f'(t) = \sqrt{2}e^{-t}\cos(t+\frac{\pi}{4})$. 3. On $[0\,;2\pi]$, $e^{-t} > 0$, so $f'(t) = 0 \Leftrightarrow \cos(t+\frac{\pi}{4}) = 0$: $t + \frac{\pi}{4} = \frac{\pi}{2} \Leftrightarrow t_1 = \frac{\pi}{4}$, and $t + \frac{\pi}{4} = \frac{3\pi}{2} \Leftrightarrow t_2 = \frac{5\pi}{4}$.

- On $\left[0\,;\frac{\pi}{4}\right]$, $f'(t) \ge 0$: f rises from 0 to $f(\pi/4) = \frac{\sqrt{2}}{2}e^{-\pi/4} \approx 0.322$.
- On $\left[\frac{\pi}{4}\,;\frac{5\pi}{4}\right]$, $f'(t) \le 0$: f decreases to $f(5\pi/4) = -\frac{\sqrt{2}}{2}e^{-5\pi/4} \approx -0.014$.
- On $\left[\frac{5\pi}{4}\,;2\pi\right]$, $f'(t) \ge 0$: f climbs back to $f(2\pi) = 0$.

4. The sinusoidal wave is pinched between the exponential boundaries $y = e^{-t}$ and $y = -e^{-t}$, tangentially touching them alternately at phased peaks and troughs.

Solution 49.10: 1. $f(x + 2\pi) = \cos(2x + 4\pi) + 2\cos(x + 2\pi) = \cos(2x) + 2\cos x = f(x)$: f is 2π-periodic. $f(-x) = \cos(-2x) + 2\cos(-x) = \cos(2x) + 2\cos x = f(x)$: f is even. Hence the analysis can be restricted to half a period: $[0\,;\pi]$. 2. $f'(x) = -2\sin(2x) - 2\sin x$. Using $\sin(2x) = 2\sin x\cos x$: $f'(x) = -4\sin x\cos x - 2\sin x = -2\sin x(2\cos x + 1)$. 3. On $]0\,;\pi[$, $\sin x > 0$, so $f'(x)$ has the opposite sign of $2\cos x + 1$:

- $2\cos x + 1 = 0 \Leftrightarrow \cos x = -\frac{1}{2} \Leftrightarrow x = \frac{2\pi}{3}$.
- On $[0\,;\frac{2\pi}{3}[$, $\cos x > -1/2 \Longrightarrow 2\cos x + 1 > 0 \Longrightarrow f'(x) < 0$.
- On $]\frac{2\pi}{3}\,;\pi]$, $\cos x < -1/2 \Longrightarrow 2\cos x + 1 < 0 \Longrightarrow f'(x) > 0$. Thus f is strictly decreasing on $\left[0\,;\frac{2\pi}{3}\right]$ and strictly increasing on $\left[\frac{2\pi}{3}\,;\pi\right]$.

4.
   - Maximum at $x = 0$: $f(0) = \cos 0 + 2\cos 0 = 3$.
   - Absolute minimum at $x = \frac{2\pi}{3}$: $f\left(\frac{2\pi}{3}\right) = \cos\left(\frac{4\pi}{3}\right) + 2\cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} - 1 = -\frac{3}{2}$.
   - Local maximum at $x = \pi$: $f(\pi) = \cos(2\pi) + 2\cos(\pi) = 1 - 2 = -1$.

Solution 49.11: 1. For all $x \in \mathbb{R}$, $\cos x \geq -1 \implies 2 + \cos x \geq 1 > 0$. The denominator never vanishes. 2. $g(-x) = \frac{\sin(-x)}{2+\cos(-x)} = \frac{-\sin x}{2+\cos x} = -g(x)$: g is odd. 3. Quotient rule: $g'(x) = \frac{\cos x(2+\cos x) - \sin x(-\sin x)}{(2+\cos x)^2} = \frac{2\cos x + \cos^2 x + \sin^2 x}{(2+\cos x)^2} = \frac{2\cos x + 1}{(2+\cos x)^2}$. 4. At $x = 0$: $g(0) = 0$ and $g'(0) = \frac{2(1)+1}{(2+1)^2} = \frac{3}{9} = \frac{1}{3}$. The tangent line T at the origin is: $y = \frac{1}{3}x$.

Solution 49.12: 1. Expanding the product: $\cos(a+b)\cos(a-b) = (\cos a \cos b - \sin a \sin b)(\cos a \cos b + \sin a \sin b) = \cos^2 a \cos^2 b - \sin^2 a \sin^2 b$. Substitute $\cos^2 b = 1 - \sin^2 b$ and $\sin^2 a = 1 - \cos^2 a$: $= \cos^2 a(1 - \sin^2 b) - (1 - \cos^2 a)\sin^2 b = \cos^2 a - \sin^2 b$. 2. $\cos(3x) = \sin(2x) \Leftrightarrow \cos(3x) = \cos\left(\frac{\pi}{2} - 2x\right)$. This generates two families:

- $3x = \frac{\pi}{2} - 2x + 2k\pi \Leftrightarrow 5x = \frac{\pi}{2} + 2k\pi \Leftrightarrow x = \frac{\pi}{10} + \frac{2k\pi}{5}$ $(k \in \mathbb{Z})$.
- $3x = -\left(\frac{\pi}{2} - 2x\right) + 2k\pi \Leftrightarrow 3x = -\frac{\pi}{2} + 2x + 2k\pi \Leftrightarrow x = -\frac{\pi}{2} + 2k\pi$ $(k \in \mathbb{Z})$.

Solution 49.13: 1. Using $\cos(2x) = 1 - 2\sin^2 x$, we obtain $f(x) = \sin^2 x = \frac{1-\cos(2x)}{2} = \frac{1}{2} - \frac{1}{2}\cos(2x)$. 2. $f'(x) = -\frac{1}{2}(-2\sin(2x)) = \sin(2x)$. $f''(x) = 2\cos(2x)$. 3. The sign of $f''(x)$ matches that of $\cos(2x)$ on $[0\,;\pi]$:

- For $2x \in [0\,;\pi/2] \Leftrightarrow x \in [0\,;\pi/4]$: $\cos(2x) \geq 0 \implies f''(x) \geq 0$, so f is **convex**.
- For $2x \in [\pi/2\,;3\pi/2] \Leftrightarrow x \in [\pi/4\,;3\pi/4]$: $\cos(2x) \leq 0 \implies f''(x) \leq 0$, so f is **concave**.
- For $2x \in [3\pi/2\,;2\pi] \Leftrightarrow x \in [3\pi/4\,;\pi]$: $\cos(2x) \geq 0 \implies f''(x) \geq 0$, so f is **convex**.

4. The second derivative vanishes with a sign change at $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$. The inflection points are $I_1\left(\frac{\pi}{4}\,;\frac{1}{2}\right)$ and $I_2\left(\frac{3\pi}{4}\,;\frac{1}{2}\right)$.

Solution 49.14: 1. Angular frequency is $\omega = \frac{\pi}{6}$. Period is $T = \frac{2\pi}{\omega} = \frac{2\pi}{\pi/6} = 12$ hours. 2. Cosine ranges between -1 and 1:

- High tide: $H_{\max} = 6 + 4(1) = 10$ meters.
- Low tide: $H_{\min} = 6 + 4(-1) = 2$ meters.

3. High tide occurs when the cosine reaches 1: $\frac{\pi}{6}t - \frac{\pi}{3} = 0 \Leftrightarrow \frac{\pi}{6}t = \frac{\pi}{3} \Leftrightarrow t = 2$. High tide occurs at **2:00 AM** (and next at $2 + 12 = 14$, i.e., 2:00 PM).

4. Safe navigation requires $H(t) \geq 8 \Leftrightarrow 6 + 4\cos\left(\frac{\pi}{6}t - \frac{\pi}{3}\right) \geq 8 \Leftrightarrow \cos\left(\frac{\pi}{6}t - \frac{\pi}{3}\right) \geq \frac{1}{2}$. Since $\cos\theta \geq \frac{1}{2} \Leftrightarrow -\frac{\pi}{3} \leq \theta \leq \frac{\pi}{3}$:

$$-\frac{\pi}{3} \leq \frac{\pi}{6}t - \frac{\pi}{3} \leq \frac{\pi}{3} \Leftrightarrow 0 \leq \frac{\pi}{6}t \leq \frac{2\pi}{3} \Leftrightarrow 0 \leq t \leq 4.$$

The trawler can navigate safely **between midnight and 4:00 AM** (and later between 12:00 PM and 4:00 PM).

Solution 49.15: 1. $1 - \cos(2x) = 2\sin^2 x$. The expression becomes $\frac{2\sin^2 x}{x \sin x} = 2\frac{\sin x}{x}$. As $x \to 0$, this limit is $2 \times 1 = 2$. 2. Substitute $h = x - \frac{\pi}{2} \to 0$, so $x = \frac{\pi}{2} + h$. The quotient becomes $\frac{\cos(\pi/2+h)}{h} = \frac{-\sin h}{h} = -\frac{\sin h}{h} \to -1$. 3. Multiply by the conjugate:

$$\frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{x} = \frac{(1+\sin x) - (1-\sin x)}{x(\sqrt{1+\sin x} + \sqrt{1-\sin x})} = \frac{\sin x}{x} \times \frac{2}{\sqrt{1+\sin x} + \sqrt{1-\sin x}}.$$

As $x \to 0$, $\frac{\sin x}{x} \to 1$ and the second factor tends to $\frac{2}{\sqrt{1}+\sqrt{1}} = 1$. The limit is 1.

Level Solutions

Solution 49.16: 1. $\tan(a+b) = \frac{\sin(a+b)}{\cos(a+b)} = \frac{\sin a \cos b + \cos a \sin b}{\cos a \cos b - \sin a \sin b}$. Dividing numerator and denominator by $\cos a \cos b \neq 0$ immediately gives $\tan(a+b) = \frac{\tan a + \tan b}{1 - \tan a \tan b}$. 2. With $\tan\alpha = \frac{1}{5}$: $\tan(2\alpha) = \frac{2(1/5)}{1-(1/5)^2} = \frac{2/5}{24/25} = \frac{5}{12}$. Then: $\tan(4\alpha) = \frac{2(5/12)}{1-(5/12)^2} = \frac{5/6}{119/144} = \frac{120}{119}$. 3. Evaluate $\tan(4\alpha - \frac{\pi}{4})$: Since $\tan(\frac{\pi}{4}) = 1$: $\tan(4\alpha - \frac{\pi}{4}) = \frac{\tan(4\alpha)-1}{1+\tan(4\alpha)} = \frac{\frac{120}{119}-1}{1+\frac{120}{119}} = \frac{1/119}{239/119} = \frac{1}{239}$. Taking the arctangent of both sides: $4\alpha - \frac{\pi}{4} = \arctan\left(\frac{1}{239}\right) \Leftrightarrow \frac{\pi}{4} = 4\arctan\left(\frac{1}{5}\right) - \arctan\left(\frac{1}{239}\right)$.

Solution 49.17: 1. Expand the right-hand side using addition formulas: $\sin\left(kx + \frac{x}{2}\right) - \sin\left(kx - \frac{x}{2}\right) = \left[\sin(kx)\cos\frac{x}{2} + \cos(kx)\sin\frac{x}{2}\right] - \left[\sin(kx)\cos\frac{x}{2} - \cos(kx)\sin\frac{x}{2}\right] = 2\cos(kx)\sin\frac{x}{2}$. 2. Multiply $S_n(x)$ by $2\sin\left(\frac{x}{2}\right)$:

$$2\sin\left(\frac{x}{2}\right)S_n(x) = \sin\left(\frac{x}{2}\right) + \sum_{k=1}^{n} 2\sin\left(\frac{x}{2}\right)\cos(kx) = \sin\left(\frac{x}{2}\right) + \sum_{k=1}^{n}\left[\sin\left(\left(k+\frac{1}{2}\right)x\right) - \sin\left(\left(k-\frac{1}{2}\right)x\right)\right].$$

The summation telescopes: all intermediate terms cancel except $-\sin\left(\frac{x}{2}\right)$ (which cancels the leading term) and the final term $\sin\left(\left(n+\frac{1}{2}\right)x\right)$. Hence: $2\sin\left(\frac{x}{2}\right)S_n(x) = \sin\left(\left(n+\frac{1}{2}\right)x\right)$. Dividing by $2\sin\left(\frac{x}{2}\right) \neq 0$ yields the Dirichlet kernel formula.

Solution 49.18: 1. For $x \in]0\,;\tfrac{\pi}{2}]$, $f'(x) = \frac{x\cos x - \sin x}{x^2} = \frac{\cos x(x - \tan x)}{x^2}$. By the fundamental lemma of Section 3.1, for all $x \in]0\,;\tfrac{\pi}{2}[$, $x < \tan x \Leftrightarrow x - \tan x < 0$. Since $\cos x > 0$ and $x^2 > 0$, we have $f'(x) < 0$. Thus f is strictly decreasing. 2. For all $x \in]0\,;\tfrac{\pi}{2}]$: $f(\tfrac{\pi}{2}) \le f(x) \le \lim_{t\to 0^+} f(t) \Leftrightarrow \frac{\sin(\pi/2)}{\pi/2} \le \frac{\sin x}{x} \le 1 \Leftrightarrow \frac{2}{\pi} \le \frac{\sin x}{x} \le 1$. Multiplying by $x > 0$: $\frac{2}{\pi}x \le \sin x \le x$. This inequality holds at $x = 0$ as well. 3. Graphically on $\left[0\,;\tfrac{\pi}{2}\right]$, the sine curve is squeezed between its tangent at the origin ($y = x$) and the chord connecting $(0\,;0)$ to $(\tfrac{\pi}{2}\,;1)$ ($y = \frac{2}{\pi}x$).

Solution 49.19: 1. $S(t) = 3\cos(\omega t) + 4\sin(\omega t)$. Factor out $\sqrt{3^2+4^2} = \sqrt{25} = 5$: $S(t) = 5\big(\frac{3}{5}\cos(\omega t) + \frac{4}{5}\sin(\omega t)\big)$. Since $\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = 1$, there exists a unique angle $\varphi \in [0\,;2\pi[$ such that $\cos\varphi = \frac{3}{5}$ and $\sin\varphi = \frac{4}{5}$. Then $S(t) = 5(\cos(\omega t)\cos\varphi + \sin(\omega t)\sin\varphi) = 5\cos(\omega t - \varphi)$. 2. Amplitude is $A = 5$. $\tan\varphi = \frac{4/5}{3/5} = \frac{4}{3} \Longrightarrow \varphi = \arctan\big(\frac{4}{3}\big) \approx 0.927$ rad $\approx 53.1^\circ$. 3. In general, let $R = \sqrt{a^2+b^2} > 0$. Then $\left(\frac{a}{R}\right)^2 + \left(\frac{b}{R}\right)^2 = 1$. There exists θ such that $\cos\theta = \frac{a}{R}$ and $\sin\theta = \frac{b}{R}$. Then $a\cos(\omega t) + b\sin(\omega t) = R\,\big(\cos(\omega t)\cos\theta + \sin(\omega t)\sin\theta\,\big) = R\cos(\omega t - \theta)$.

Solution 49.20: Assume for contradiction that $\lim_{x\to+\infty}\cos x = L$. By the sequential characterization of limits, for every sequence (u_n) such that $u_n \to +\infty$, we must have $\cos(u_n) \to L$. Choose $u_n = 2n\pi \to +\infty$. Then $\cos(u_n) = \cos(2n\pi) = 1$ for all n, so the sequence converges to 1. Now choose $v_n = \pi + 2n\pi \to +\infty$. Then $\cos(v_n) = \cos(\pi + 2n\pi) = -1$ for all n, so this sequence converges to -1. Since the limits 1 and -1 are distinct, this violates the uniqueness of limits. Therefore, $\cos x$ has no limit as $x \to +\infty$.

The Question that Opens What Follows

We have learned to measure slopes, rates of change, and velocities at every point along a curve: this is differential calculus. But what happens if we wish to travel in the opposite direction? If we know the instantaneous velocity of a rocket at every single second of flight, how do we reconstruct the total distance traveled? If we know the curved profile of a barrel or the wavy contour of an undulating landscape, how do we calculate its exact enclosed volume or surface area?

How do we add together an infinity of infinitely thin slices to rebuild a whole?

Join us in Chapter 50 with **The Barrel Filled with Slices of Water** and the discovery of integral calculus.

CHAPTER 50: The Barrel Filled with Slices of Water

> "If the derivative is the scalpel that slices time into infinitesimal instants to extract instantaneous velocity, the integral is the golden needle that sews all the crumbs back together to reconstruct the grand journey of space. Nothing is lost, everything accumulates within the memory of the curve."

What You Are About to Discover

- How astronomer **Johannes Kepler**, outraged by a wine merchant during his own wedding in 1615, invented the method of infinitely thin slices to measure curved volumes.
- The concept of the **definite integral** $\int_a^b f(t)\,dt$: from Riemann sums of approximating rectangles to the exact area under a curve.
- The historical origin and meaning of the elongated symbol $\int$, introduced by Gottfried Wilhelm Leibniz in 1675.
- The **Fundamental Theorem of Calculus** and its rigorous proof: how and why integration is the exact **inverse operation** of differentiation.
- Finding **antiderivatives**: standard tables and mastering composite patterns ($u'u^n$, $\frac{u'}{u}$, $u'e^u$, etc.).
- Essential algebraic and geometric properties of integrals: **linearity**, **Chasles' relation**, **positivity**, **area between two curves**, and the **mean value**.
- The cornerstone technique of integral calculus: **Integration by Parts** (IBP), derived directly from the product rule of differentiation.
- A clean **Python** implementation of the midpoint Riemann sum numerical approximation.

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1. The Problem: Kepler, the Wine Merchant, and the Rigged Dipstick

1.1 The Wedding in Linz (1615)

In the autumn of **1615**, in the city of Linz on the banks of the Danube, the renowned astronomer **Johannes Kepler**—the man who unlocked the laws of planetary motion around the Sun—was preparing the banquet for his second wedding. To feast his guests, he purchased several large barrels of Austrian wine.

When the vintner delivered the bulging wooden casks, he pulled a simple notched iron rod, called a **gauging rod** or dipstick, from his pocket. He inserted the rod diagonally through the bung-hole (the opening situated at the widest bulge of the barrel) all the way to the farthest seam of the opposite base, pulled out the wet rod, read the notch, and announced the volume in liters and the price to be paid.

Kepler was dumbfounded: > *"How can you possibly measure the capacity of a curved, bulging body with a single diagonal straight line? If a cask has steeply swollen flanks or nearly straight cylindrical flanks, your rod plunges to the exact same depth, yet the volume of wine inside is completely different! You are selling me air, or you are robbing me blind!"*

The merchant merely shrugged: wine merchants had gauged casks this way for generations. Kepler retreated to his study. For weeks, he set aside his astronomical calculations to resolve the enigma of the casks.

His breakthrough, published in his landmark treatise *Nova Stereometria Doliorum Vinariorum* ("New Stereometry of Wine Casks", 1615), was brilliant in its simplicity: To measure a volume bounded by curved surfaces, **one must mentally slice it into an infinity of flat, infinitely thin circular disks**, calculate the volume of each disk (which is virtually a perfect flat cylinder), and then **add all these disks together** from the bottom to the top!

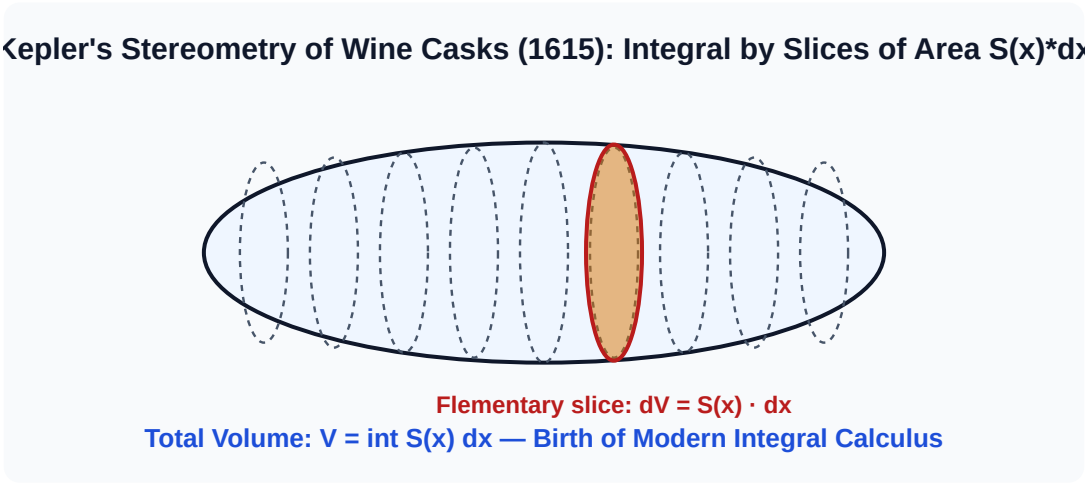

Figure 109: Kepler's stereometry of wine casks: integral calculus via cross-sectional slices

1.2 The Speedometer and the Broken Odometer

Consider a modern analogy. Imagine you are driving an automobile across an open desert. Your odometer is broken: it permanently reads $0 \ \text{km}$. However, your speedometer functions flawlessly, and a continuous chart recorder plots your instantaneous speed $v(t)$ in kilometers per hour as time t progresses.

How can you determine the exact total distance traveled between departure at time $t = a$ and arrival at time $t = b$?

1. **The Case of Constant Velocity:** If you cruise at a steady speed of $90 \ \text{km/h}$ for $2 \ \text{hours}$, the distance is:

$$\text{Distance} = \text{Speed} \times \text{Time} = 90 \times 2 = 180 \ \text{kilometers}.$$

   Graphically, in a coordinate system where time is on the horizontal axis and speed on the vertical axis, this distance corresponds precisely to the **area of the rectangle** of base $[0\,;2]$ and height 90!

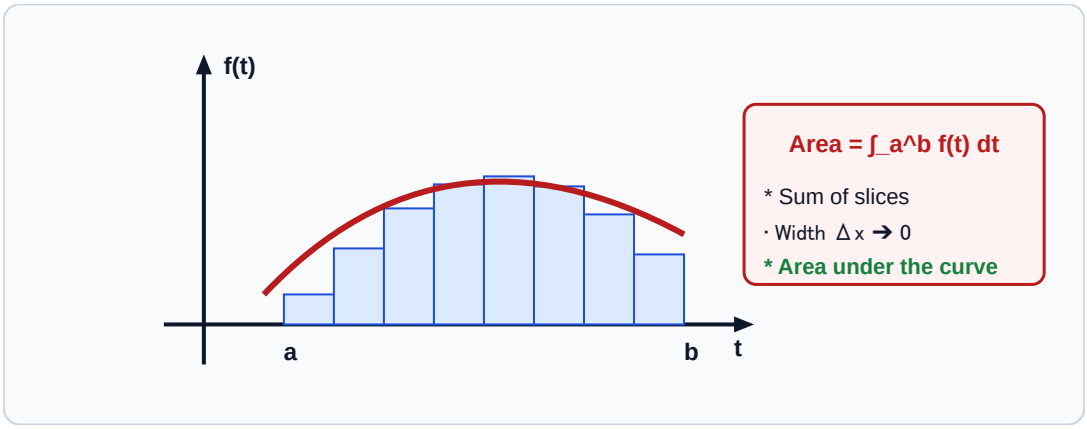

Figure 110: The Riemann integral: bounding the area under the curve with upper and lower sums

2. **The Case of Variable Velocity:** In real-world driving, velocity fluctuates constantly: accelerating, braking, idling. The curve $v(t)$ undulates. To calculate total distance, partition time into tiny intervals of duration Δt. During a split second Δt, the velocity $v(t)$ barely has time to change: it is practically constant. The small incremental distance covered during that slice is the area of a slender rectangle of width Δt and height $v(t)$:

$$\Delta d \approx v(t) \times \Delta t.$$

   The total distance traveled is the sum of these millions of tiny rectangles. And as the width Δt shrinks to an infinitesimal increment (denoted dt), the sum of the rectangular areas converges to the **exact area under the velocity curve**!

2. From Riemann Sums to the Definite Integral

2.1 Riemann Sums

Let f be a continuous, positive function defined on a closed interval $[a\,;b]$ (with $a < b$). We wish to compute the area of the planar region bounded by:

- The graph $\mathcal{C}_f$;
- The horizontal axis ($y = 0$);
- The vertical lines $x = a$ and $x = b$.

This geometric region is known as the **curvilinear trapezoid** or the area under the curve.

To approximate this area, divide the interval $[a\,;b]$ into n equal sub-intervals of width $\Delta x = \frac{b-a}{n}$ using the subdivision points:

$$x_k = a + k\frac{b-a}{n} \quad (k = 0, 1, 2, ..., n).$$

Over each sub-interval $[x_k\,;x_{k+1}]$, construct a rectangle of height $f(x_k)$ (the left Riemann sum). The total area of these n rectangles forms the **left Riemann sum**:

$$S_n = \sum_{k=0}^{n-1} f(x_k) \times \Delta x = \sum_{k=0}^{n-1} f\left(a + k\frac{b-a}{n}\right)\frac{b-a}{n}.$$

As the number of subdivisions n tends to infinity ($n \to +\infty$), the width of each strip vanishes ($\Delta x \to 0$), the step-like staircase hugs the curve without leaving any gaps, and the sum S_n converges to a unique real number: the **definite integral of f from a to b**.

2.2 Leibniz's Notation: $\int_a^b f(t)\,dt$

> 💡 **True History — October 29, 1675 in Paris:**
>
> It was the German philosopher and polymath **Gottfried Wilhelm Leibniz** who introduced the integral sign $\int$. Leibniz pondered: *"What are we doing? We are computing the Sum of an infinity of infinitely slender slices."* He took the first letter of the Latin word **Summa** (sum): the letter **S**. With his quill pen, he stretched it vertically into an elegant, elongated symbol: $\int$. And to denote the infinitely small width of each slice, he wrote dt (the differential of t).

When you encounter the formula:

$$\int_a^b f(t)\,dt$$

here is what it communicates:

- $\int$: "Compute the continuous sum…"
- from lower bound a (starting point) to upper bound b (endpoint);
- of rectangles of height $f(t)$;
- and infinitesimal base width dt.

> ***Note on Dummy Variables:*** *The variable t beneath the integral sign is a **dummy variable**: it merely scans the interval from a to b. It can be replaced by any other letter without altering the numerical value of the integral:*
>
> $$\int_a^b f(t)\,dt = \int_a^b f(x)\,dx = \int_a^b f(u)\,du.$$

2.3 Python Implementation: Midpoint Riemann Sums

Here is a clean Python script that approximates the definite integral of a continuous function using the midpoint rule:

```python
def rectangle_integral(f, a, b, n):
    """
    Computes an approximation of the definite integral of f from a to b
    using the midpoint rectangle method.
    """
    width = (b - a) / n
    total = 0.0
    for k in range(n):
        # Midpoint of the k-th subinterval
        x_mid = a + (k + 0.5) * width
        total += f(x_mid)
    return total * width

# Test on f(x) = x^2 from 0 to 3 (exact area is 9)
def square(x):
    return x**2

print("Approximation with 10 rectangles   :", rectangle_integral(square, 0, 3, 10))
print("Approximation with 100 rectangles  :", rectangle_integral(square, 0, 3, 100))
print("Approximation with 1000 rectangles :", rectangle_integral(square, 0, 3, 1000))
# With 1000 rectangles, we obtain 8.9999988... virtually exact 9!
```

---

3. The Fundamental Theorem of Calculus: The Miraculous Bridge

Computing sums of millions of rectangles by hand would be an agonizing, impossible task. Here enters what is widely considered the crown jewel of mathematical analysis, discovered independently by **Isaac Newton** and **Gottfried Wilhelm Leibniz** between 1666 and 1675.

3.1 The Area Accumulation Function

Let f be a continuous, positive function on $[a\,;b]$. Hold the lower limit a fixed, and let the upper limit vary as a free variable $x \in [a\,;b]$. Define the **area accumulation function** A:

$$A(x) = \int_a^x f(t)\,dt.$$

$A(x)$ measures the **accumulated area swept under the curve from a to x**. What is the rate of change of this accumulated area when the upper boundary advances by a tiny increment $h > 0$?

When moving from x to $x+h$, the total area increases by the slender strip spanning from x to $x+h$:

$$\Delta A = A(x+h) - A(x) = \int_x^{x+h} f(t)\,dt.$$

This strip has base width h and a nearly constant height equal to $f(x)$ (since f is continuous). Its area is therefore approximately:

$$A(x+h) - A(x) \approx f(x) \times h \Leftrightarrow \frac{A(x+h) - A(x)}{h} \approx f(x).$$

As $h \to 0$, the difference quotient of the area function becomes **precisely the height of the curve $f(x)$**!

> ***The derivative of the accumulation function is the original function itself:***
>
> $$A'(x) = f(x).$$

3.2 Rigorous Proof of the Fundamental Theorem of Calculus

Fundamental Theorem of Calculus (Required Curriculum Proof) —

Let f be a continuous function on an interval I, and let $a \in I$. The function A defined on I by:

$$A(x) = \int_a^x f(t)\,dt$$

is differentiable on I, and for every $x \in I$:

$$A'(x) = f(x).$$

Proof —

Let $x \in I$ and let $h > 0$ such that $x + h \in I$. By Chasles' relation for integrals:

$$A(x+h) - A(x) = \int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt = \int_x^{x+h} f(t)\,dt.$$

Because f is continuous on the closed interval $[x\,;x+h]$, the Extreme Value Theorem guarantees that f attains an absolute minimum m_h and an absolute maximum M_h on $[x\,;x+h]$:

$$\text{For all } \; t \in [x\,;x+h] : \quad m_h \leq f(t) \leq M_h.$$

Integrating this bounding across the interval $[x\,;x+h]$ of length h:

$$\int_x^{x+h} m_h\,dt \leq \int_x^{x+h} f(t)\,dt \leq \int_x^{x+h} M_h\,dt \Leftrightarrow m_h \times h \leq A(x+h) - A(x) \leq M_h \times h.$$

Dividing by $h > 0$:

$$m_h \leq \frac{A(x+h) - A(x)}{h} \leq M_h.$$

Because f is continuous at x, as $h \to 0^+$, the interval $[x\,;x+h]$ collapses down to the single point x. The minimum m_h and maximum M_h both converge to $f(x)$:

$$\lim_{h\to 0^+} m_h = f(x) \quad \text{and} \quad \lim_{h\to 0^+} M_h = f(x).$$

By the **Squeeze Theorem**:

$$\lim_{h\to 0^+} \frac{A(x+h) - A(x)}{h} = f(x).$$

An identical argument for $h < 0$ yields $\lim_{h\to 0^-} \frac{A(x+h)-A(x)}{h} = f(x)$. We conclude that A is differentiable at x and $A'(x) = f(x)$. ■

---

4. Antiderivatives and the Evaluation of Definite Integrals

4.1 What Is an Antiderivative?

Definition —

Let f be a continuous function on an interval I. An **antiderivative** (or primitive) of f on I is any function F differentiable on I such that:

$$\text{For all } \; x \in I : \quad F'(x) = f(x).$$

> ***Fundamental Properties of Antiderivatives —*** *If F is an antiderivative of f on an interval I, then: 1. For any real constant C, the function $x \mapsto F(x) + C$ is also an antiderivative of f (since the derivative of a constant is zero: $(F + C)' = F' + 0 = f$). 2. Conversely, any other antiderivative G of f on I differs from F by only a constant: there exists $C \in \mathbb{R}$ such that $G(x) = F(x) + C$. 3. For any initial condition $(x_0\,; y_0)$ with $x_0 \in I$, there exists* ***a unique*** *antiderivative of f satisfying $F(x_0) = y_0$.*

4.2 The Evaluation Formula for Definite Integrals

Thanks to the Fundamental Theorem, evaluating a definite integral requires no rectangles whatsoever:

📐 Theorem (Evaluating Integrals via Antiderivatives) —

Let f be a continuous function on $[a\,; b]$, and let F be any antiderivative of f on $[a\,; b]$. Then:

$$\int_a^b f(t)\,dt = [F(t)]_a^b = F(b) - F(a).$$

📐 Proof —

Define $A(x) = \int_a^x f(t)\,dt$. By the Fundamental Theorem, A is an antiderivative of f on $[a\,; b]$. Since F is another antiderivative of f, there exists a constant $C \in \mathbb{R}$ such that for all $x \in [a\,; b]$:

$$A(x) = F(x) + C.$$

Evaluate at $x = a$:

$$A(a) = \int_a^a f(t)\,dt = 0 \Longrightarrow F(a) + C = 0 \Leftrightarrow C = -F(a).$$

Thus, for all $x \in [a\,; b]$:

$$A(x) = F(x) - F(a).$$

Setting $x = b$ yields immediately:

$$\int_a^b f(t)\,dt = A(b) = F(b) - F(a).$$

Notice that if we had chosen any other antiderivative $G(x) = F(x) + K$, the constant K cancels upon subtraction:

$$G(b) - G(a) = (F(b) + K) - (F(a) + K) = F(b) - F(a).$$

The evaluation is entirely independent of the arbitrary constant chosen! ■

4.3 Table of Standard Antiderivatives

This table is the exact mirror image of the derivative table, read in reverse:

Function $f(x)$	An Antiderivative $F(x)$	Validity Domain
0	k (constant)	$\mathbb{R}$
a (constant)	ax	$\mathbb{R}$
x	$\frac{x^2}{2}$	$\mathbb{R}$
$x^n \quad (n \in \mathbb{N})$	$\frac{x^{n+1}}{n+1}$	$\mathbb{R}$
$x^\alpha \quad (\alpha \in \mathbb{R}, \alpha \neq -1)$	$\frac{x^{\alpha+1}}{\alpha+1}$	$]0\,;+\infty[$
$\frac{1}{x^2} = x^{-2}$	$-\frac{1}{x}$	$]-\infty\,;0[$ or $]0\,;+\infty[$
$\frac{1}{\sqrt{x}} = x^{-1/2}$	$2\sqrt{x}$	$]0\,;+\infty[$
$\frac{1}{x}$	$\ln\lvert x\rvert$	$]-\infty\,;0[$ or $]0\,;+\infty[$
e^x	e^x	$\mathbb{R}$
$e^{kx} \quad (k \neq 0)$	$\frac{1}{k}e^{kx}$	$\mathbb{R}$
$\cos x$	$\sin x$	$\mathbb{R}$
$\sin x$	$-\cos x$	$\mathbb{R}$

4.4 Antiderivatives of Composite Functions

Recognizing the chain rule in reverse is the core operational skill of integral calculus:

Pattern in Integrand	An Antiderivative	Condition on u
$u'(x) \times \big(\,u(x)\big)^n \quad (n \in \mathbb{Z}, n \neq -1)$	$\frac{\big(\,u(x)\big)^{n+1}}{n+1}$	If $n < -1$, $u(x) \neq 0$
$\frac{u'(x)}{u(x)}$	$\ln\lvert u(x)\rvert$	$u(x) \neq 0$
$u'(x)\,e^{u(x)}$	$e^{u(x)}$	None
$\frac{u'(x)}{\sqrt{u(x)}}$	$2\sqrt{u(x)}$	$u(x) > 0$
$u'(x)\cos(u(x))$	$\sin(u(x))$	None
$u'(x)\sin(u(x))$	$-\cos(u(x))$	None

Worked Example 1 — Direct Antiderivatives.

1. $f_1(x) = 3x^4 - 2x^2 + 5 \Longrightarrow F_1(x) = 3\left(\frac{x^5}{5}\right) - 2\left(\frac{x^3}{3}\right) + 5x = \frac{3}{5}x^5 - \frac{2}{3}x^3 + 5x$. 2. $f_2(x) = \frac{3}{x} + e^{-2x}$ on $]0\,;+\infty[\Longrightarrow F_2(x) = 3\ln x - \frac{1}{2}e^{-2x}$.

Worked Example 2 — Recognizing Composite Patterns.

1. $f(x) = (2x+1)(x^2+x+4)^3$. Let $u(x) = x^2+x+4$. Then $u'(x) = 2x+1$. The expression is formatted as $u'u^3$. An antiderivative is:
$$F(x) = \frac{u^4}{4} = \frac{(x^2+x+4)^4}{4}.$$
2. $g(x) = \frac{2x}{x^2+3}$. Let $u(x) = x^2+3$. Then $u'(x) = 2x$. This is of the form $\frac{u'}{u}$ with $u(x) > 0$. An antiderivative is $G(x) = \ln(x^2+3)$.
3. $h(x) = \frac{x}{x^2+1}$. The derivative of x^2+1 is $2x$, but the numerator contains only x. Insert the required constant 2 by multiplying and dividing by 2:
$$h(x) = \frac{1}{2} \times \frac{2x}{x^2+1} = \frac{1}{2}\frac{u'}{u} \Longrightarrow H(x) = \frac{1}{2}\ln(x^2+1).$$
4. $k(x) = xe^{-x^2}$. The derivative of $-x^2$ is $-2x$. Adjust the constant:
$$k(x) = -\frac{1}{2} \times (-2x)e^{-x^2} = -\frac{1}{2}u'e^u \Longrightarrow K(x) = -\frac{1}{2}e^{-x^2}.$$

---

5. Algebraic Properties of the Definite Integral

These structural properties hold for any continuous functions on the specified intervals.

5.1 Linearity

Linearity Property —

For all constants $\alpha, \beta \in \mathbb{R}$ and continuous functions f, g:
$$\int_a^b \big(\, \alpha f(t) + \beta g(t)\big)\, dt = \alpha \int_a^b f(t)\, dt + \beta \int_a^b g(t)\, dt.$$

5.2 Chasles' Relation

Chasles' Relation *— For all real numbers a, b, c belonging to an interval where f is continuous:*
$$\int_a^c f(t)\, dt + \int_c^b f(t)\, dt = \int_a^b f(t)\, dt.$$

In particular:

- $\int_a^a f(t)\, dt = 0$ *(a region with zero width has zero area);*
- $\int_b^a f(t)\, dt = -\int_a^b f(t)\, dt$ *(reversing the integration limits flips the sign).*

5.3 Positivity and Order Preservation

Positivity *— Let $a \leq b$.*

- *If for all $t \in [a\,;b]$, $f(t) \geq 0$, then $\int_a^b f(t)\, dt \geq 0$.*
- *If for all $t \in [a\,;b]$, $f(t) \leq g(t)$, then $\int_a^b f(t)\, dt \leq \int_a^b g(t)\, dt$.*
- *Triangle inequality for integrals: $\left|\int_a^b f(t)\, dt\right| \leq \int_a^b |f(t)|\, dt$.*

> ⚠ **Common Pitfall! — Order of Limits Matters!**
>
> The rule "$f \geq 0 \Longrightarrow \int_a^b f(t)\,dt \geq 0$" strictly assumes $\boldsymbol{a \leq b}$. If the integration limits are reversed ($a > b$), the integral of a strictly positive function is **negative**!

> ⚠ ⚠ **Algebraic Area vs Geometric Area**
>
> Never confuse the integral with geometric area! The definite integral $\int_a^b f(x)\,dx$ evaluates **signed algebraic area**: regions above the x-axis are counted positively, while regions below are negative. When a function changes sign over $[a, b]$, total physical geometric area equals $\int_a^b |f(x)|\,dx$, requiring Chasles splitting at each zero of f.

5.4 Area Enclosed Between Two Curves

Let f and g be two continuous functions on $[a\,;b]$ such that for all $x \in [a\,;b]$, $\boldsymbol{f(x) \geq g(x)}$ (the graph of f lies on or above that of g). The area $\mathcal{A}$ of the region bounded by both curves and the vertical lines $x = a$ and $x = b$ is the area under f minus the area under g:

$$\mathcal{A} = \int_a^b \big(\, f(x) - g(x)\big)\,dx.$$

5.5 The Mean Value of a Function

How can one define the average temperature of a day if temperature $T(t)$ fluctuates continuously from hour to hour? One cannot simply sum an infinity of readings and divide by infinity! We use the **integral mean value**:

> 📐 **Definition — Mean Value:**
>
> Let f be continuous on $[a\,;b]$ with $a < b$. The **mean value** (or average value) of f on $[a\,;b]$, denoted μ (or $\overline{f}$), is the real number:
>
> $$\mu = \frac{1}{b-a}\int_a^b f(t)\,dt.$$

Geometric Interpretation: μ is the height of a rectangle sharing the same base $[a\,;b]$ that encloses **the exact same area** as the undulating region beneath the graph of f:

$$\text{Area}(f) = \mu \times (b-a).$$

---

6. Integration by Parts (IBP): The Master Technique

When an integrand is a product of two distinct families of functions (such as xe^x or $x\cos x$), and cannot be recognized as a composite pattern $u'e^u$, no elementary antiderivative formula applies directly. We turn to **Integration by Parts**, which is the integral counterpart of the product rule for differentiation.

6.1 Theorem and Proof

> 📐 **Integration by Parts Theorem (Required Curriculum Proof) —**
>
> Let u and v be differentiable functions on $[a\,;b]$ whose derivatives u' and v' are continuous on $[a\,;b]$. Then:
>
> $$\int_a^b u'(t)v(t)\,dt = \Big[\,u(t)v(t)\Big]_a^b - \int_a^b u(t)v'(t)\,dt.$$

Proof —

From the product rule of differentiation, for all $t \in [a\,;b]$:

$$(uv)'(t) = u'(t)v(t) + u(t)v'(t).$$

Isolating $u'(t)v(t)$:

$$u'(t)v(t) = (uv)'(t) - u(t)v'(t).$$

Integrating both sides from a to b using linearity:

$$\int_a^b u'(t)v(t)\,dt = \int_a^b (uv)'(t)\,dt - \int_a^b u(t)v'(t)\,dt.$$

An antiderivative of $(uv)'$ is simply the product uv! Hence:

$$\int_a^b (uv)'(t)\,dt = \Big[\,u(t)v(t)\Big]_a^b = u(b)v(b) - u(a)v(a).$$

Substituting this back yields the required formula:

$$\int_a^b u'(t)v(t)\,dt = \Big[\,u(t)v(t)\Big]_a^b - \int_a^b u(t)v'(t)\,dt. \quad \blacksquare$$

6.2 Strategy: Choosing u' and v

The goal of integration by parts is to trade an intractable integral for a **simpler** one. Generally, we assign $\boldsymbol{v(t)}$ to be the factor that **simplifies upon differentiation** (polynomials lose a degree when differentiated!).

> ***The LIATE / ALPES Heuristic for Choosing $v(x)$ (the factor to differentiate):*** *Choose $v(x)$ to be whichever function type appears highest on this list: 1. **L : Logarithms** ($\ln x$) 2. **I : Inverse trigonometric functions** ($\arctan x$, etc.) 3. **A : Algebraic / Polynomials** ($x, x^2, x^3, \ldots$) 4. **T : Trigonometric functions** ($\sin x, \cos x$) 5. **E : Exponentials** ($e^x, e^{2x}, \ldots$)*

Worked Example 3 — Evaluating $\int_0^1 xe^x\,dx$.

We have the product of a polynomial x and an exponential e^x. According to the LIATE rule, the polynomial x takes precedence over the exponential for differentiation:

- Set $v(x) = x \implies v'(x) = 1$ (the polynomial degree drops!)
- Set $u'(x) = e^x \implies u(x) = e^x$

Both u and v are continuously differentiable on $[0\,;1]$. Integrating by parts:

$$\int_0^1 xe^x\,dx = \Big[\,xe^x\Big]_0^1 - \int_0^1 1 \times e^x\,dx.$$

Evaluating the boundary bracket: $[xe^x]_0^1 = 1e^1 - 0e^0 = e$. The remaining integral is trivial: $\int_0^1 e^x\,dx = [e^x]_0^1 = e^1 - e^0 = e - 1$. Therefore:

$$\int_0^1 xe^x\,dx = e - (e-1) = \mathbf{1}.$$

Worked Example 4 — The Clever Trick to Integrate $\ln x$.

How do we find an antiderivative of $\ln x$ on $]0\,;+\infty[$? There is no obvious product? Write it as $\ln x = 1 \times \ln x$! Set:

- $v(x) = \ln x \Longrightarrow v'(x) = \frac{1}{x}$ (the logarithm vanishes!)
- $u'(x) = 1 \Longrightarrow u(x) = x$

Then:

$$\int \ln x \, dx = x \ln x - \int x \times \frac{1}{x} \, dx = x \ln x - \int 1 \, dx = x \ln x - x + C.$$

> ***Memorable Formula — An antiderivative of $\ln x$ is:***
>
> $$F(x) = x \ln x - x.$$
>
> *Immediate verification by differentiation:* $(x \ln x - x)' = 1 \times \ln x + x \times \frac{1}{x} - 1 = \ln x + 1 - 1 = \ln x$. *Impeccable!* ✔

Key Takeaways

1. The Definite Integral

- $\int_a^b f(t)\,dt$ represents the signed algebraic area enclosed between the graph of f, the horizontal axis, and the vertical lines $x = a$ and $x = b$.
- The unit of area (u.a.) is the area of the unit coordinate cell $\| \vec{\imath} \| \times \| \vec{\jmath} \|$.

2. The Fundamental Theorem of Calculus

- If f is continuous on I and $a \in I$, the function $A(x) = \int_a^x f(t)\,dt$ is differentiable on I and $\boldsymbol{A'(x) = f(x)}$.
- For any antiderivative F of f on $[a\,;b]$:

$$\int_a^b f(t)\,dt = [F(t)]_a^b = F(b) - F(a).$$

3. Standard Antiderivatives Summary

- $x^n \to \frac{x^{n+1}}{n+1} \quad (n \neq -1)$
- $\frac{1}{x} \to \ln|x|$
- $e^{ax} \to \frac{1}{a} e^{ax}$
- $\cos(ax) \to \frac{1}{a}\sin(ax)$ and $\sin(ax) \to -\frac{1}{a}\cos(ax)$
- $u'u^n \to \frac{u^{n+1}}{n+1}$
- $\frac{u'}{u} \to \ln|u|$
- $u'e^u \to e^u$

4. Key Properties

- Linearity: $\int(\alpha f + \beta g) = \alpha \int f + \beta \int g$.
- Chasles' relation: $\int_a^c f + \int_c^b f = \int_a^b f$.
- Positivity: if $a \leq b$ and $f \geq 0$, then $\int_a^b f \geq 0$.
- Area between curves: $\mathcal{A} = \int_a^b (f(x) - g(x))\,dx$ whenever $f \geq g$.
- Mean value: $\mu = \frac{1}{b-a}\int_a^b f(t)\,dt$.

5. Integration by Parts

$$\int_a^b u'v = [uv]_a^b - \int_a^b uv'.$$

Exercises

Level (Direct Applications)

Exercise 50.1 — Elementary Antiderivatives. Determine an antiderivative for each of the following functions on the indicated domain: 1. $f_1(x) = 4x^3 - 6x^2 + 2x - 7$ on $\mathbb{R}$ 2. $f_2(x) = \frac{5}{x^2} - \frac{2}{\sqrt{x}}$ on $]0\,;+\infty[$ 3. $f_3(x) = e^{3x} - 2e^{-x}$ on $\mathbb{R}$ 4. $f_4(x) = 3\cos(2x) + \sin(4x)$ on $\mathbb{R}$ 5. $f_5(x) = \frac{4}{x}$ on $]0\,;+\infty[$

Exercise 50.2 — Antiderivatives Satisfying Initial Conditions. 1. Find the antiderivative F of $f(x) = 3x^2 - 4x + 1$ such that $F(1) = 5$. 2. Find the antiderivative G of $g(x) = e^{2x}$ such that $G(0) = 2$.

Exercise 50.3 — Direct Definite Integrals. Evaluate the exact values of the following definite integrals: 1. $I_1 = \int_1^3 (3x^2 - 2x + 1)\,dx$ 2. $I_2 = \int_0^2 e^{3x}\,dx$ 3. $I_3 = \int_1^e \frac{1}{x}\,dx$ 4. $I_4 = \int_0^{\pi/2} \cos x\,dx$ 5. $I_5 = \int_0^1 (2x+1)^3\,dx$

Exercise 50.4 — Recognizing Composite Patterns $u'u^n$ and $u'e^u$. Compute the following integrals: 1. $J_1 = \int_0^1 2x(x^2+1)^4\,dx$ 2. $J_2 = \int_0^2 3x^2 e^{x^3}\,dx$ 3. $J_3 = \int_1^2 \frac{2x+3}{x^2+3x+1}\,dx$ 4. $J_4 = \int_0^{\pi/4} \cos x \sin^3 x\,dx$

Exercise 50.5 — Chasles' Relation and Symmetry. Let f be continuous on $[-a\,;a]$ with $a > 0$. 1. Recall why $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$ if f is even. 2. Recall why $\int_{-a}^{a} f(x)\,dx = 0$ if f is odd. 3. Deduce instantaneously without any antiderivative computation the value of:

$$K = \int_{-2}^{2} (x^7 - 3x^3 + \sin x)\,dx.$$

Exercise 50.6 — Mean Value. Compute the mean value of $f(x) = x^2$ on the interval $[0\,;3]$. Interpret geometrically.

Exercise 50.7 — Simple Geometric Area. Compute the area of the region bounded by the parabola $y = 4 - x^2$ and the horizontal axis.

Level (Exploration and Exam Preparation)

Exercise 50.8 — Basic Integration by Parts. Evaluate using integration by parts: 1. $A_1 = \int_0^{\pi} x\sin x\,dx$ 2. $A_2 = \int_1^e x\ln x\,dx$ 3. $A_3 = \int_0^1 (2x+1)e^{-x}\,dx$

Exercise 50.9 — Repeated Integration by Parts. Evaluate $B = \int_0^1 x^2 e^x\,dx$ using two successive integrations by parts.

Exercise 50.10 — Area Enclosed Between Two Curves. Consider the two parabolas:

$$y = f(x) = -x^2 + 4x \quad \text{and} \quad y = g(x) = x^2 - 2x.$$

1. Determine the coordinates of their intersection points. 2. Determine their relative positions on the interval bounded by these intersections. 3. Compute the area of the closed region bounded by these curves.

Exercise 50.11 — Functions Defined by Integrals. Let F be defined on $]0\,;+\infty[$ by:

$$F(x) = \int_1^x \frac{e^t}{t}\,dt.$$

1. Determine $F(1)$. 2. Justify why F is differentiable on $]0\,;+\infty[$ and find $F'(x)$. 3. Deduce the monotonicity of F on $]0\,;+\infty[$. 4. Prove that for all $t \geq 1$, $\frac{e^t}{t} \geq e$. Deduce that for all $x \geq 1$:

$$F(x) \geq e(x-1).$$

5. Find the limit of $F(x)$ as $x \to +\infty$.

Exercise 50.12 — Sequence of Integrals (Exam Style). For every integer $n \in \mathbb{N}$, define:

$$I_n = \int_0^1 x^n e^{-x}\,dx.$$

1. Calculate I_0. 2. Using integration by parts, establish for all $n \in \mathbb{N}$ the recurrence relation:

$$I_{n+1} = (n+1)I_n - \frac{1}{e}.$$

3. Deduce the exact values of I_1 and I_2. 4. Prove that for all $n \in \mathbb{N}$, $I_n \geq 0$. 5. Show that the sequence (I_n) is decreasing. 6. Prove that for all $x \in [0\,;1]$, $x^n e^{-1} \leq x^n e^{-x} \leq x^n$. Deduce bounds for I_n and compute $\lim_{n\to+\infty} I_n$.

Exercise 50.13 — Kepler's Wine Cask Revisited. A barrel of length $2L = 1 \text{ meter}$ (from $x = -0.5$ to $x = 0.5$) is modeled by revolving a parabolic arc around the horizontal axis. The radius of the cask at horizontal position x is given by:

$$R(x) = R_0 - kx^2$$

with $R_0 = 0.4 \text{ m}$ (radius at the center / bung-hole) and $R(0.5) = 0.3 \text{ m}$ at the ends. 1. Determine the constant k. 2. Recall that the volume of a circular slice of thickness dx is $dV = \pi[R(x)]^2\,dx$. 3. Express the total volume of the cask as a definite integral:

$$V = \int_{-0.5}^{0.5} \pi[R(x)]^2\,dx.$$

4. Evaluate this volume in liters. Compare it to a straight cylinder sharing the same average radius.

Exercise 50.14 — Partial Fraction Decomposition. Let $f(x) = \frac{1}{x^2-1}$ defined on $]1\,;+\infty[$. 1. Find constants a and b such that for all $x > 1$:

$$\frac{1}{x^2-1} = \frac{a}{x-1} + \frac{b}{x+1}.$$

2. Deduce an antiderivative of f on $]1\,;+\infty[$. 3. Compute the integral $I = \int_2^3 \frac{1}{x^2-1}\,dx$.

Exercise 50.15 — Bounding and Rocket Flight. A rocket is launched vertically. Its acceleration is modeled by $a(t) = \frac{400}{(t+10)^2}$ for $t \in [0\,;30]$ (in $\text{m}\cdot\text{s}^{-2}$). At launch ($t = 0$), initial velocity is $v(0) = 0$. 1. Express velocity $v(t)$ by integrating acceleration. 2. Determine the velocity attained at $t = 30 \text{ seconds}$. 3. Express altitude $h(t)$ by integrating velocity. Find the altitude at $t = 30 \text{ s}$.

Exercise 50.16 — The Cauchy-Schwarz Inequality for Integrals. Let f and g be continuous on $[a\,;b]$. 1. For any real parameter λ, expand $\int_a^b \big(\, f(t) + \lambda g(t)\big)^2\,dt \geq 0$. 2. Viewing this as a quadratic in λ, deduce the **Cauchy-Schwarz inequality**:

$$\left(\int_a^b f(t)g(t)\,dt\right)^2 \leq \left(\int_a^b (f(t))^2\,dt\right)\left(\int_a^b (g(t))^2\,dt\right).$$

Level (In-Depth Challenges)

Exercise 50.17 — The Wallis Integrals. For every natural number n, define:

$$W_n = \int_0^{\pi/2} \sin^n x\,dx.$$

1. Calculate W_0 and W_1. 2. Writing $\sin^{n+1} x = \sin x \times \sin^n x$, establish by integration by parts for $n \geq 1$:

$$W_{n+1} = \frac{n}{n+1} W_{n-1}.$$

3. Deduce expressions for W_{2p} and W_{2p+1} in terms of factorials. 4. Show that (W_n) is strictly positive and decreasing. 5. Prove that $\lim_{n\to+\infty} \frac{W_{n+1}}{W_n} = 1$, and deduce **John Wallis' celebrated infinite product (1655)** for π:

$$\frac{\pi}{2} = \lim_{p\to+\infty} \prod_{k=1}^{p} \frac{4k^2}{4k^2-1} = \frac{2}{1}\times\frac{2}{3}\times\frac{4}{3}\times\frac{4}{5}\times\frac{6}{5}\times\frac{6}{7}\times\ldots$$

Exercise 50.18 — The Gaussian Integral. Let $I = \int_0^{+\infty} e^{-x^2}\,dx$. 1. Justify the convergence of this improper integral by comparing it to e^{-x} for $x \geq 1$. 2. Assuming $I^2 = \int_0^{+\infty}\int_0^{+\infty} e^{-(x^2+y^2)}\,dx\,dy$, convert to polar coordinates ($x = r\cos\theta, y = r\sin\theta, dx\,dy = r\,dr\,d\theta$) to show:

$$I^2 = \int_0^{\pi/2} d\theta \times \int_0^{+\infty} re^{-r^2}\,dr.$$

3. Evaluate this integral directly and deduce:

$$\int_{-\infty}^{+\infty} e^{-x^2}\,dx = \sqrt{\pi}.$$

Exercise 50.19 — The Euler-Maclaurin Sum Formula. Let f be continuously differentiable (C^1) on $[0\,;n]$. 1. Prove that $\int_0^1 (t-1/2)f'(t)\,dt = \frac{f(0)+f(1)}{2} - \int_0^1 f(t)\,dt$. 2. Generalize this to connect the discrete sum $\sum_{k=0}^{n} f(k)$ to the integral $\int_0^n f(t)\,dt$.

Exercise 50.20 — The Mean Value Theorem for Integrals. Let f be continuous on $[a\,;b]$. Prove that there exists $c \in]a\,;b[$ such that:

$$\int_a^b f(t)\,dt = f(c)(b-a).$$

(This is the Integral Mean Value Theorem, the integral counterpart of the Mean Value Theorem applied to an antiderivative).

Solutions

Level Solutions

Solution 50.1: 1. $F_1(x) = 4\left(\frac{x^4}{4}\right) - 6\left(\frac{x^3}{3}\right) + 2\left(\frac{x^2}{2}\right) - 7x = x^4 - 2x^3 + x^2 - 7x$. 2. $f_2(x) = 5x^{-2} - 2x^{-1/2} \Longrightarrow F_2(x) = 5\left(-\frac{1}{x}\right) - 2(2\sqrt{x}) = -\frac{5}{x} - 4\sqrt{x}$. 3. $F_3(x) = \frac{1}{3}e^{3x} + 2e^{-x}$. 4. $F_4(x) = 3\left(\frac{1}{2}\sin(2x)\right) - \frac{1}{4}\cos(4x) = \frac{3}{2}\sin(2x) - \frac{1}{4}\cos(4x)$. 5. $F_5(x) = 4\ln x$.

Solution 50.2: 1. General antiderivatives: $F(x) = x^3 - 2x^2 + x + C$. Condition: $F(1) = 1 - 2 + 1 + C = C = 5$. Thus $F(x) = x^3 - 2x^2 + x + 5$. 2. General antiderivatives: $G(x) = \frac{1}{2}e^{2x} + C$. Condition: $G(0) = \frac{1}{2} + C = 2 \Longrightarrow C = \frac{3}{2}$. Thus $G(x) = \frac{1}{2}e^{2x} + \frac{3}{2}$.

Solution 50.3: 1. $I_1 = [x^3 - x^2 + x]_1^3 = (27 - 9 + 3) - (1 - 1 + 1) = 21 - 1 = 20$. 2. $I_2 = \left[\frac{1}{3}e^{3x}\right]_0^2 = \frac{1}{3}e^6 - \frac{1}{3}e^0 = \frac{e^6-1}{3}$. 3. $I_3 = [\ln x]_1^e = \ln e - \ln 1 = 1 - 0 = 1$. 4. $I_4 = [\sin x]_0^{\pi/2} = \sin(\pi/2) - \sin 0 = 1 - 0 = 1$. 5. Let $u = 2x + 1 \Longrightarrow u' = 2$. $I_5 = \frac{1}{2}\int_0^1 2(2x+1)^3\,dx = \frac{1}{2}\left[\frac{(2x+1)^4}{4}\right]_0^1 = \frac{1}{8}(3^4 - 1^4) = \frac{80}{8} = 10$.

Solution 50.4: 1. Pattern $u'u^4$ with $u(x) = x^2 + 1$: $J_1 = \left[\frac{(x^2+1)^5}{5}\right]_0^1 = \frac{2^5}{5} - \frac{1^5}{5} = \frac{31}{5}$. 2. Pattern $u'e^u$ with $u(x) = x^3$: $J_2 = [e^{x^3}]_0^2 = e^{2^3} - e^0 = e^8 - 1$. 3. Pattern $\frac{u'}{u}$ with $u(x) = x^2 + 3x + 1 > 0$: $J_3 = [\ln(x^2 + 3x + 1)]_1^2 = \ln(11) - \ln(5) = \ln\left(\frac{11}{5}\right)$. 4. Pattern $u'u^3$ with $u(x) = \sin x$: $J_4 = \left[\frac{\sin^4 x}{4}\right]_0^{\pi/4} = \frac{(\sqrt{2}/2)^4}{4} - 0 = \frac{1/4}{4} = \frac{1}{16}$.

Solution 50.5: 1. $\int_{-a}^{a} f = \int_{-a}^{0} f + \int_0^a f$. By substitution $u = -x$, if f is even, $\int_{-a}^{0} f(x)\,dx = \int_0^a f(u)\,du$, giving $2\int_0^a f$. 2. If f is odd, $\int_{-a}^{0} f(x)\,dx = -\int_0^a f(u)\,du$, so the sum is 0. 3. The integrand $x \mapsto x^7 - 3x^3 + \sin x$ is a sum of odd functions, so it is odd. Across the symmetric interval $[-2\,;2]$, the integral is **strictly zero: $K = 0$.**

Solution 50.6: $\mu = \frac{1}{3-0}\int_0^3 x^2\,dx = \frac{1}{3}\left[\frac{x^3}{3}\right]_0^3 = \frac{1}{3} \times 9 = 3$. Interpretation: a rectangle of width 3 and constant height $\mu = 3$ has area $3 \times 3 = 9$, exactly matching the area under the parabolic curve.

Solution 50.7: The parabola crosses the horizontal axis when $4 - x^2 = 0 \Leftrightarrow x = \pm 2$. On $[-2\,;2]$, the function is positive and even. $\mathcal{A} = \int_{-2}^{2}(4 - x^2)\,dx = 2\int_0^2 (4 - x^2)\,dx = 2\left[4x - \frac{x^3}{3}\right]_0^2 = 2\left(8 - \frac{8}{3}\right) = \frac{32}{3} \approx 10.67$ u.a.

Level Solutions

Solution 50.8: 1. $A_1 = \int_0^\pi x\sin x\,dx$. Set $u'(x) = \sin x \Longrightarrow u(x) = -\cos x$ and $v(x) = x \Longrightarrow v'(x) = 1$. $A_1 = [-x\cos x]_0^\pi - \int_0^\pi (-\cos x)\,dx = (-\pi(-1) - 0) + [\sin x]_0^\pi = \pi + 0 = \pi$. 2. $A_2 = \int_1^e x\ln x\,dx$. Set $v(x) = \ln x \Longrightarrow v'(x) = \frac{1}{x}$ and $u'(x) = x \Longrightarrow u(x) = \frac{x^2}{2}$. $A_2 = \left[\frac{x^2}{2}\ln x\right]_1^e - \int_1^e \frac{x^2}{2}\frac{1}{x}\,dx = \frac{e^2}{2} - \frac{1}{2}\left[\frac{x^2}{2}\right]_1^e = \frac{e^2}{2} - \frac{e^2-1}{4} = \frac{e^2+1}{4}$. 3. $A_3 = \int_0^1 (2x+1)e^{-x}\,dx$. Set $v(x) = 2x + 1 \Longrightarrow v'(x) = 2$ and $u'(x) = e^{-x} \Longrightarrow u(x) = -e^{-x}$. $A_3 = [-(2x+1)e^{-x}]_0^1 + 2\int_0^1 e^{-x}\,dx = (-3e^{-1} + 1) + 2[-e^{-x}]_0^1 = 1 - 3e^{-1} + 2(1 - e^{-1}) = 3 - \frac{5}{e}$.

Solution 50.9: First IBP for $B = \int_0^1 x^2e^x\,dx$: $v(x) = x^2 \Longrightarrow v'(x) = 2x$, $u'(x) = e^x \Longrightarrow u(x) = e^x$. $B = [x^2e^x]_0^1 - 2\int_0^1 xe^x\,dx = e - 2\int_0^1 xe^x\,dx$. From Worked Example 3, $\int_0^1 xe^x\,dx = 1$. Hence $B = e - 2(1) = e - 2$.

Solution 50.10: 1. $f(x) = g(x) \Leftrightarrow -x^2 + 4x = x^2 - 2x \Leftrightarrow 2x^2 - 6x = 0 \Leftrightarrow 2x(x-3) = 0$. Intersections occur at $x = 0$ and $x = 3$. 2. Relative positions: $f(x) - g(x) = 6x - 2x^2 = 2x(3 - x)$. On $[0\,;3]$, both $2x \geq 0$ and $3 - x \geq$

0, so $f(x) \geq g(x)$. The graph of f lies entirely above that of g. 3. $\mathcal{A} = \int_0^3 (6x - 2x^2)\,dx = \left[3x^2 - \frac{2x^3}{3}\right]_0^3 = (27 - 18) - 0 = 9$ area units.

Solution 50.11: 1. $F(1) = \int_1^1 \frac{e^t}{t}\,dt = 0$. 2. The integrand $t \mapsto \frac{e^t}{t}$ is continuous on $]0\,;+\infty[$. By the Fundamental Theorem of Calculus, F is differentiable on $]0\,;+\infty[$ and $F'(x) = \frac{e^x}{x}$. 3. For all $x > 0$, $e^x > 0$ and $x > 0$, so $F'(x) > 0$. F is strictly increasing on $]0\,;+\infty[$. 4. Let $\varphi(t) = \frac{e^t}{t}$ on $[1\,;+\infty[$. $\varphi'(t) = \frac{e^t(t-1)}{t^2} \geq 0$ for $t \geq 1$. The minimum is attained at $t = 1$: $\varphi(1) = e$. Thus for all $t \geq 1$, $\frac{e^t}{t} \geq e$. Integrating from 1 to x (with $x \geq 1$): $F(x) = \int_1^x \frac{e^t}{t}\,dt \geq \int_1^x e\,dt = e(x-1)$. 5. Since $\lim_{x\to+\infty} e(x-1) = +\infty$, the comparison theorem yields $\lim_{x\to+\infty} F(x) = +\infty$.

Solution 50.12: 1. $I_0 = \int_0^1 e^{-x}\,dx = [-e^{-x}]_0^1 = 1 - \frac{1}{e}$. 2. IBP on $I_{n+1} = \int_0^1 x^{n+1}e^{-x}\,dx$: $v(x) = x^{n+1} \Longrightarrow v'(x) = (n+1)x^n$ and $u'(x) = e^{-x} \Longrightarrow u(x) = -e^{-x}$. $I_{n+1} = [-x^{n+1}e^{-x}]_0^1 + (n+1)\int_0^1 x^n e^{-x}\,dx = -\frac{1}{e} + (n+1)I_n$. 3. $I_1 = 1 \times I_0 - \frac{1}{e} = 1 - \frac{2}{e}$. $I_2 = 2I_1 - \frac{1}{e} = 2 - \frac{5}{e}$. 4. On $[0\,;1]$, $x^n \geq 0$ and $e^{-x} > 0$, so the integrand is positive. With integration limits in order, $I_n \geq 0$. 5. $I_{n+1} - I_n = \int_0^1 x^n(x-1)e^{-x}\,dx \leq 0$ because $x - 1 \leq 0$. The sequence is decreasing. 6. For $x \in [0\,;1]$, $-1 \leq -x \leq 0 \Longrightarrow e^{-1} \leq e^{-x} \leq 1$. Multiply by x^n: $x^n e^{-1} \leq x^n e^{-x} \leq x^n$. Integrating on $[0\,;1]$ gives $\frac{1}{e(n+1)} \leq I_n \leq \frac{1}{n+1}$. By the Squeeze Theorem, $\lim_{n\to+\infty} I_n = 0$.

Solution 50.13: 1. $R(0.5) = 0.4 - k(0.5)^2 = 0.3 \Leftrightarrow 0.4 - 0.25k = 0.3 \Leftrightarrow k = 0.4$. Thus $R(x) = 0.4(1 - x^2)$. 2. Each disk slice of thickness dx has volume $dV = \pi R(x)^2 dx$. 3. $V = \int_{-0.5}^{0.5} \pi[R(x)]^2\,dx$. 4. By even symmetry: $V = 2\pi\int_0^{0.5} 0.16(1 - 2x^2 + x^4)\,dx = 0.32\pi\left[x - \frac{2}{3}x^3 + \frac{x^5}{5}\right]_0^{0.5} \approx 0.32\pi \times 0.42292 \approx 0.425\ \text{m}^3 = 425\ \text{liters}$. A simple cylinder of average radius $0.35\ \text{m}$ would hold $\pi(0.35)^2(1) \approx 0.385\ \text{m}^3 = 385\ \text{liters}$. The bulge adds 40 liters of wine!

Solution 50.14: 1. $\frac{a}{x-1} + \frac{b}{x+1} = \frac{(a+b)x + (a-b)}{x^2-1}$. Matching coefficients: $a + b = 0$ and $a - b = 1 \Longrightarrow a = \frac{1}{2}, b = -\frac{1}{2}$. Thus $\frac{1}{x^2-1} = \frac{1}{2(x-1)} - \frac{1}{2(x+1)}$. 2. $F(x) = \frac{1}{2}\ln(x-1) - \frac{1}{2}\ln(x+1) = \frac{1}{2}\ln\left(\frac{x-1}{x+1}\right)$. 3. $I = [F(x)]_2^3 = \frac{1}{2}\ln(2/4) - \frac{1}{2}\ln(1/3) = \frac{1}{2}\ln(3/2)$.

Solution 50.15: 1. $a(t) = 400(t+10)^{-2}$. An antiderivative is $-\frac{400}{t+10}$. $v(t) = C - \frac{400}{t+10}$. With $v(0) = 0$, $C = 40$. Hence $v(t) = 40 - \frac{400}{t+10}$. 2. At $t = 30$ s: $v(30) = 40 - \frac{400}{40} = 30\ \text{m}\cdot\text{s}^{-1}$ (108 km/h). 3. $h(t) = \int_0^t v(u)\,du = 40t - 400\ln\left(1 + \frac{t}{10}\right)$. At $t = 30$ s: $h(30) = 1200 - 400\ln 4 \approx 645.5$ meters.

Solution 50.16: 1. Expand the square: $(f(t) + \lambda g(t))^2 = (f(t))^2 + 2\lambda f(t)g(t) + \lambda^2(g(t))^2 \geq 0$. Integrating yields a quadratic in λ: $P(\lambda) = \lambda^2\left(\int_a^b g^2\right) + 2\lambda\left(\int_a^b fg\right) + \left(\int_a^b f^2\right) \geq 0$. 2. Since this quadratic is non-negative for all real λ, its discriminant cannot be positive ($\Delta' \leq 0$): $\left(\int_a^b fg\right)^2 - \left(\int_a^b g^2\right)\left(\int_a^b f^2\right) \leq 0$, which proves the Cauchy-Schwarz inequality.

Level Solutions

Solution 50.17: 1. $W_0 = \int_0^{\pi/2} 1\,dx = \frac{\pi}{2}$, and $W_1 = \int_0^{\pi/2} \sin x\,dx = [-\cos x]_0^{\pi/2} = 1$. 2. IBP with $u'(x) = \sin x \Longrightarrow u(x) = -\cos x$ and $v(x) = \sin^n x \Longrightarrow v'(x) = n\cos x \sin^{n-1} x$: $W_{n+1} = [-\cos x \sin^n x]_0^{\pi/2} + n\int_0^{\pi/2} \cos^2 x \sin^{n-1} x\,dx$. The boundary term vanishes. With $\cos^2 x = 1 - \sin^2 x$: $W_{n+1} = nW_{n-1} - nW_{n+1} \Leftrightarrow (n+1)W_{n+1} = nW_{n-1} \Leftrightarrow W_{n+1} = \frac{n}{n+1}W_{n-1}$. 3. By induction: $W_{2p} = \frac{(2p)!}{2^{2p}(p!)^2}\frac{\pi}{2}$, and $W_{2p+1} = \frac{2^{2p}(p!)^2}{(2p+1)!}$. 4. On $[0\,;\pi/2]$, $0 \leq \sin x \leq 1 \Longrightarrow 0 \leq \sin^{n+1} x \leq \sin^n x$, hence $0 < W_{n+1} \leq W_n$. 5. From $W_{n+1} \leq W_n \leq W_{n-1}$, divide by W_{n+1}: $1 \leq \frac{W_n}{W_{n+1}} \leq 1 + \frac{1}{n} \to 1$. The ratio $\frac{W_{2p+1}}{W_{2p}}$ establishes Wallis' product formula converging to $\frac{2}{\pi}$.

Solution 50.18: 1. For $x \geq 1$, $x^2 \geq x \Longrightarrow e^{-x^2} \leq e^{-x}$, which integrates to $e^{-1} < +\infty$, ensuring convergence. 2. Converting the quarter-plane to polar coordinates gives $\theta \in [0\,;\pi/2]$ and $r \in [0\,;+\infty[$, with Jacobian area element $r\,dr\,d\theta$. 3. $\int_0^{+\infty} re^{-r^2}\,dr = \left[-\frac{1}{2}e^{-r^2}\right]_0^{+\infty} = \frac{1}{2}$. Then $I^2 = \frac{\pi}{2} \times \frac{1}{2} = \frac{\pi}{4} \Longrightarrow I = \frac{\sqrt{\pi}}{2}$. By even symmetry over $\mathbb{R}$: $\int_{-\infty}^{+\infty} e^{-x^2}\,dx = 2I = \sqrt{\pi}$.

Solution 50.19: 1. IBP with $u'(t) = 1 \Longrightarrow u(t) = t - 1/2$ and $v(t) = f(t) \Longrightarrow v'(t) = f'(t)$: $\int_0^1 (t - 1/2)f'(t)\,dt = [(t-1/2)f(t)]_0^1 - \int_0^1 f(t)\,dt = \frac{f(0)+f(1)}{2} - \int_0^1 f(t)\,dt$. 2. Summing this across intervals $[k\,;k+1]$ connects discrete sums to continuous integrals with extraordinary precision.

Solution 50.20: Let $F(x) = \int_a^x f(t)\,dt$. F is continuous on $[a\,;b]$ and differentiable on $]a\,;b[$. By the Mean Value Theorem, there exists $c \in]a\,;b[$ such that:

$$\frac{F(b) - F(a)}{b - a} = F'(c).$$

Since $F(b) - F(a) = \int_a^b f(t)\,dt$ and $F'(c) = f(c)$, we obtain $\int_a^b f(t)\,dt = f(c)(b-a)$.

The Question that Opens What Follows

We now know how to compute the area under a curve whenever an explicit formula for the function is known. But what happens in the real world, when we do not know the formula of the function in advance, but only **a relationship linking the function to its own rate of change**?

When a hot cup of tea sits on a kitchen table, the rate at which it cools is directly proportional to the difference between its temperature and the ambient room temperature. How can we uncover the complete temperature profile over time from this simple physical rule?

Welcome to the realm of **differential equations**: join us in Chapter 51 with **The Cooling Cup of Tea**.

CHAPTER 51: The Cooling Cup of Tea

> 📐 **"Nature does not hand us trajectories pre-drawn; she gives us only local laws connecting the present state to the rate of its change. Everything else is a differential equation."**
>
> — Henri Poincaré

What You Are About to Discover

- Why most fundamental laws in physics, biology, and economics do not directly specify the formula for what happens, but rather **the relationship between a quantity and its instantaneous rate of change**.
- Newton's phenomenological **Law of Cooling** (1701): why your hot coffee cools very rapidly at first, then progressively more slowly.
- The homogeneous linear differential equation $\boldsymbol{y' = ay}$: its rigorous proof and general solutions $y(t) = Ce^{at}$.
- The non-homogeneous linear differential equation with constant forcing $\boldsymbol{y' = ay + b}$: decomposing into *general solution of the homogeneous equation + constant particular solution*: $y(t) = Ce^{at} - \frac{b}{a}$.
- The Cauchy initial value problem and how the **initial condition** $y(t_0) = y_0$ guarantees uniqueness of the trajectory.
- Foundational real-world models: radioactive decay, electronic RC circuits, terminal velocity of a skydiver, and population dynamics.
- Step-by-step **Euler's method**: how computers numerically simulate any differential equation with a clean **Python** script.

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1. The Problem: The Steaming Cup of Tea on the Table

1.1 Isaac Newton's Observation (1701)

Imagine a freshly poured cup of tea, scalding at 90° C, resting on a table in a kitchen where the ambient air temperature is 20° C. After one minute, the temperature drops by 10° C: it reads 80° C. After two minutes, will it read 70° C? After nine minutes, will it reach 0° C? And after a quarter of an hour, will the tea freeze solid into an ice cube?

Certainly not! Daily experience teaches us that:

- When tea is piping hot, steam billows off and it cools down rapidly;
- When tea is lukewarm (say 25° C), it loses only a tiny fraction of a degree per minute;
- It will never drop below the ambient temperature of the room: its temperature asymptotically approaches the surrounding 20° C without ever falling below it.

In **1701**, **Isaac Newton** published an anonymous paper in the *Philosophical Transactions of the Royal Society* entitled *Scala graduum caloris* ("Scale of the Degrees of Heat"). In it, he formulated a universal empirical principle:

> 📐 **Newton's Law of Cooling —**
>
> The rate at which a hot body loses heat to its surroundings is **proportional to the temperature difference** between the body and the ambient environment.

1.2 Translation into a Mathematical Equation

Let $T(t)$ represent the temperature of the tea at time t. The "rate at which temperature changes" at each instant is precisely the instantaneous rate of change—that is, **the derivative $\boldsymbol{T'(t)}$**. The difference between the tea's temperature and the surrounding ambient temperature T_{ext} is $T(t) - T_{\text{ext}}$. Newton's law can therefore be stated in a single line:

$$T'(t) = -k\,\big(\,T(t) - T_{\text{ext}}\,\big)$$

where $k > 0$ is a positive physical constant that depends on the geometry of the cup, its material (porcelain, glass, metal), and airflow in the room. The negative sign reflects cooling ($T'(t) < 0$ whenever $T(t) > T_{\text{ext}}$).
Expanding this expression gives:

$$T'(t) = -kT(t) + kT_{\text{ext}}.$$

This is a **differential equation**: the unknown is not an isolated number x, but an **entire function $t \mapsto T(t)$**, and the equation links this unknown function directly to its own derivative!

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2. The Homogeneous Differential Equation: $y' = ay$

2.1 Definition and Intuition

Definition —

Let a be a fixed real number. The differential equation:

$$(E_0): \quad y' = ay$$

is called a **first-order homogeneous linear differential equation with constant coefficients**. A solution of (E_0) on an interval I is any function f differentiable on I such that for all $t \in I$:

$$f'(t) = af(t).$$

What this means: We seek a function whose derivative is directly proportional to itself. Do you know a function that is its own derivative? Yes! The exponential function: $(e^t)' = e^t$. And if we consider $y(t) = e^{at}$, the chain rule gives $(e^{at})' = ae^{at} = ay(t)$! Furthermore, multiplying by any arbitrary real constant C, the linearity of differentiation confirms that $(Ce^{at})' = C(ae^{at}) = a(Ce^{at})$. All functions of the form $t \mapsto Ce^{at}$ are therefore solutions. Are there any others? The answer is no.

2.2 Fundamental Theorem and Rigorous Proof

Theorem (Required Curriculum Proof) —

Let $a \in \mathbb{R}$. The solutions on $\mathbb{R}$ to the differential equation $y' = ay$ are precisely the functions of the form:

$$y(t) = Ce^{at}$$

where C is an arbitrary real constant.

Rigorous Proof —

The proof proceeds in two stages: verification (existence) and uniqueness. 1. **Verification:** Let $C \in \mathbb{R}$ and define $f(t) = Ce^{at}$. The function f is differentiable on $\mathbb{R}$, and for all $t \in \mathbb{R}$:

$$f'(t) = C \times (ae^{at}) = a(Ce^{at}) = af(t).$$

Thus, every function of the form Ce^{at} is indeed a solution to (E_0).

2. **Uniqueness:** Let g be an arbitrary solution to $y' = ay$ on $\mathbb{R}$. By definition, for all $t \in \mathbb{R}$, $g'(t) = ag(t)$. Define the auxiliary function φ on $\mathbb{R}$ by:

$$\varphi(t) = g(t)e^{-at}.$$

The function φ is differentiable on $\mathbb{R}$ as a product of differentiable functions. Using the product rule $(uv)' = u'v + uv'$:

$$\varphi'(t) = g'(t)e^{-at} + g(t)(-ae^{-at}) = \big(\, g'(t) - ag(t)\big)\, e^{-at}.$$

Since g satisfies $y' = ay$, we know that $g'(t) - ag(t) = 0$ for all $t \in \mathbb{R}$! Consequently:

$$\varphi'(t) = 0 \times e^{-at} = 0.$$

The derivative of φ is **identically zero across the entire real line** $\mathbb{R}$. Therefore, φ is a **constant function** on $\mathbb{R}$: there exists a real constant C such that for all $t \in \mathbb{R}$:

$$\varphi(t) = C \Leftrightarrow g(t)e^{-at} = C.$$

Multiplying both sides by e^{at} (which is nowhere zero):

$$g(t) = Ce^{at}.$$

Every solution must be of the form Ce^{at}. ■

2.3 Uniqueness Under Initial Conditions: The Cauchy Problem

The equation $y' = ay$ admits an infinite family of solutions (one for each real value of C). Graphically, this corresponds to an entire family of non-intersecting trajectories:

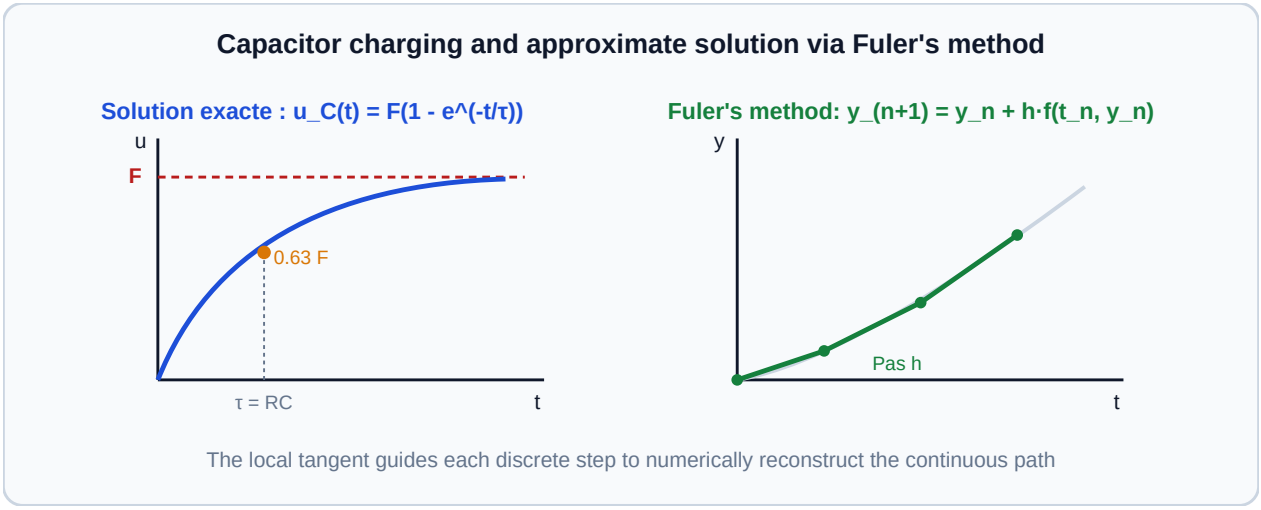

Figure 111: Capacitor RC charging and Euler's method for differential equations

To determine C and fix the unique physical trajectory followed by the system, one must specify an **initial condition** at some reference time t_0.

📐 Existence and Uniqueness Theorem (Cauchy Initial Value Problem) —

For any real numbers t_0 and y_0, the differential equation $y' = ay$ has **a unique solution** f satisfying the initial condition:

$$f(t_0) = y_0.$$

This solution is given for all $t \in \mathbb{R}$ by:

$$f(t) = y_0\, e^{a(t-t_0)}.$$

Proof —

The general solutions have the form $f(t) = Ce^{at}$. The initial condition requires:

$$f(t_0) = y_0 \Leftrightarrow Ce^{at_0} = y_0 \Leftrightarrow C = y_0 e^{-at_0}.$$

Substituting C back into the function:

$$f(t) = (y_0 e^{-at_0})e^{at} = y_0\, e^{a(t-t_0)}.$$

Because C is uniquely determined, the solution is unique. ■

Worked Example 1 — Radioactive Decay.

A sample of Iodine-131 initially contains $N_0 = 10^8$ radioactive nuclei at $t = 0$. The number of remaining undecayed nuclei $N(t)$ satisfies the differential equation $N'(t) = -0.0866N(t)$ (with time t measured in days). 1. Determine the explicit formula for $N(t)$. 2. Calculate the number of nuclei remaining after 8 days.

Solution: 1. This is the equation $y' = ay$ with $a = -0.0866$. The general solutions are $N(t) = Ce^{-0.0866t}$. Applying the initial condition: $N(0) = Ce^0 = C = 10^8$. Thus, $N(t) = 10^8 e^{-0.0866t}$. 2. At $t = 8$ days: $N(8) = 10^8 e^{-0.0866\times 8} = 10^8 e^{-0.6928} \approx 10^8 \times 0.50017 \approx 5 \times 10^7$ nuclei. (Approximately half of the sample has decayed: the half-life of Iodine-131 is roughly 8 days!).

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3. The Non-Homogeneous Differential Equation: $y' = ay + b$

In many real-world systems, quantities do not decay to zero: our hot tea does not drop to absolute zero (0 K), but levels off at room temperature (20° C); a capacitor in a charging circuit does not charge without bound, but approaches the source voltage E. The differential equation includes a constant forcing term:

$$(E): \quad y' = ay + b \quad (a \neq 0).$$

3.1 Finding a Constant Particular Solution

Let us determine whether there exists a constant solution $y(t) = k$ (where $k \in \mathbb{R}$) to this equation. If y is constant, its derivative is identically zero: $y'(t) = 0$. Substituting into (E):

$$0 = ak + b \Leftrightarrow ak = -b \Leftrightarrow k = -\frac{b}{a}.$$

The constant particular solution of $y' = ay + b$ is:

$$y_p(t) = -\frac{b}{a}.$$

*In physical modeling, this value represents the **steady-state equilibrium** of the system.*

3.2 Solution Structure Theorem

Theorem (General Solution of $y' = ay + b$) —

Let $a \neq 0$ and b be real numbers. The solutions on $\mathbb{R}$ to the differential equation $y' = ay + b$ are precisely the functions:

$$y(t) = Ce^{at} - \frac{b}{a}$$

where C is an arbitrary real constant.

Proof (Required Curriculum Proof) —

Let $(E) : y' = ay + b$ and let $(E_0) : y' = ay$ be its associated homogeneous equation. Define $y_p(t) = -\frac{b}{a}$. We have $y_p' = 0 = ay_p + b$, so y_p is a particular solution of (E).

Let f be any differentiable function on $\mathbb{R}$. We claim:

$$f \text{ is a solution of } (E) \Leftrightarrow f - y_p \text{ is a solution of } (E_0).$$

Indeed:

- $(f - y_p)' = f' - y_p' = f' - 0 = f'$.
- Meanwhile, $a(f - y_p) = af - ay_p = af - a\left(-\frac{b}{a}\right) = af + b$.

Therefore:

$$(f - y_p)' = a(f - y_p) \Leftrightarrow f' = af + b.$$

This demonstrates that the difference $f - y_p$ satisfies the homogeneous equation $y' = ay$. From Theorem 2.2, there exists a constant $C \in \mathbb{R}$ such that:

$$(f - y_p)(t) = Ce^{at} \Leftrightarrow f(t) = Ce^{at} + y_p(t) \Leftrightarrow f(t) = Ce^{at} - \frac{b}{a}. \quad \blacksquare$$

The Principle of Superposition:

$$\text{General Solution of } (E) = \underbrace{Ce^{at}}_{\text{General solution of homogeneous equation}} + \underbrace{\left(-\frac{b}{a}\right)}_{\text{Constant particular solution}}$$

3.3 Initial Condition for $y' = ay + b$

To determine C from an initial condition $y(t_0) = y_0$:

$$y(t_0) = y_0 \Leftrightarrow Ce^{at_0} - \frac{b}{a} = y_0 \Leftrightarrow Ce^{at_0} = y_0 + \frac{b}{a} \Leftrightarrow C = \left(y_0 + \frac{b}{a}\right)e^{-at_0}.$$

Substituting back:

$$y(t) = \left(y_0 + \frac{b}{a}\right)e^{a(t-t_0)} - \frac{b}{a}.$$

Worked Example 2 — The Cooling Tea Solved.

Revisit our cup of tea cooling from 90° C in a room at 20° C. The equation is $T'(t) = -0.1T(t) + 2$ (with t in minutes). 1. Determine the general solution of the differential equation. 2. Find the unique solution satisfying $T(0) = 90$. 3. What will the temperature be after 20 minutes? 4. When will the tea reach a comfortable drinking temperature of 45° C?

Step-by-step Solution: 1. This is $y' = ay + b$ with $a = -0.1$ and $b = 2$. The equilibrium constant is $-\frac{b}{a} = -\frac{2}{-0.1} = 20$. The general solution is:

$$T(t) = Ce^{-0.1t} + 20.$$

2. Initial condition $T(0) = 90$:

$$T(0) = Ce^0 + 20 = 90 \Leftrightarrow C + 20 = 90 \Leftrightarrow C = 70.$$

The temperature profile is:

$$T(t) = 70e^{-0.1t} + 20.$$

3. At $t = 20$ minutes:

$$T(20) = 70e^{-0.1\times 20} + 20 = 70e^{-2} + 20 \approx 70 \times 0.1353 + 20 \approx 9.47 + 20 = 29.5^\circ \text{ C}\ .$$

4. Set $T(t) = 45$:

$$70e^{-0.1t} + 20 = 45 \Leftrightarrow 70e^{-0.1t} = 25 \Leftrightarrow e^{-0.1t} = \frac{25}{70} = \frac{5}{14}.$$

Applying the natural logarithm (Chapter 48):

$$-0.1t = \ln\left(\frac{5}{14}\right) = -\ln\left(\frac{14}{5}\right) \Leftrightarrow t = 10\ln(2.8) \approx 10 \times 1.0296 \approx 10.3 \ \text{ minutes}\ .$$

The tea will be drinkable after approximately 10 minutes and 18 seconds.

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4. Foundational Real-World Models

4.1 Charging a Capacitor (RC Circuit)

In an electrical circuit comprising a DC voltage source E, a resistor R, and a capacitor C in series, Kirchhoff's voltage law yields:

$$u_R(t) + u_C(t) = E.$$

Ohm's law gives $u_R = Ri(t)$, while the current through the capacitor is $i(t) = C\frac{du_C}{dt}$. This yields the differential equation:

$$RC\frac{du_C}{dt} + u_C(t) = E \Leftrightarrow u_C'(t) = -\frac{1}{RC}u_C(t) + \frac{E}{RC}.$$

Defining the **time constant** $\tau = RC$ (measured in seconds), the equation becomes:

$$u_C'(t) = -\frac{1}{\tau}u_C(t) + \frac{E}{\tau}.$$

With the initial condition of an initially uncharged capacitor ($u_C(0) = 0$), the unique solution is:

$$u_C(t) = E\big(1 - e^{-t/\tau}\big).$$

> ***Notable Properties of Time Constant τ:*** *At $t = \tau$, the voltage reaches $u_C(\tau) = E(1 - e^{-1}) \approx E(1 - 0.368) = 0.632E$. The capacitor is charged to $\mathbf{63.2\%}$ of its capacity. At $t = 3\tau$, it reaches 95%. At $t = 5\tau$, it reaches 99.3% (effectively full charge).*

4.2 Terminal Velocity with Linear Drag

A skydiver of mass m falls under the influence of gravity ($P = mg$, downward) and linear aerodynamic air drag ($F = -kv$, upward). Newton's second law ($ma = \sum F$) gives:

$$mv'(t) = mg - kv(t) \Leftrightarrow v'(t) = -\frac{k}{m}v(t) + g.$$

This is of the form $y' = ay + b$ with $a = -\frac{k}{m}$ and $b = g$. The steady-state equilibrium velocity is the **terminal velocity** v_{lim}:

$$v_{\text{lim}} = -\frac{b}{a} = -\frac{g}{-k/m} = \frac{mg}{k}.$$

If the skydiver drops from rest ($v(0) = 0$), the velocity over time is:

$$v(t) = \frac{mg}{k}\left(1 - e^{-\frac{k}{m}t}\right).$$

The skydiver accelerates initially at g, but as speed rises, drag increases until it balances weight, causing velocity to level off at v_{lim}.

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5. Euler's Method: Solving When Integration Fails

In advanced science, many differential equations are non-linear (such as $y' = y^2 - t$ or $y' = \sin(yt)$) and possess no closed-form solutions in terms of elementary functions! How do NASA trajectory designers and climate modelers solve them? They use a simple geometric approximation introduced by **Leonhard Euler** in 1768: **approximate the unknown curve with a sequence of short tangent line segments**.

5.1 The Geometric Concept

Consider the differential equation $y' = f(t, y)$ with initial condition $y(t_0) = y_0$. Choose a small step size $h > 0$. 1. **At the starting point $(t_0\,;y_0)$:** The slope of the tangent line is given directly by the differential equation:

$$\text{Slope} = y'(t_0) = f(t_0, y_0).$$

2. **Advancing by step h:** Follow this tangent line forward for a duration h. The new time coordinate is $t_1 = t_0 + h$. The approximated ordinate y_1 is obtained from the local affine approximation:

$$y_1 = y_0 + h \times y'(t_0) = y_0 + h \times f(t_0, y_0).$$

3. **Iterate from $(t_1\,;y_1)$:** Compute the updated slope $f(t_1, y_1)$ and advance by another step h:

$$y_{n+1} = y_n + h \times f(t_n, y_n) \quad \text{with} \quad t_{n+1} = t_n + h.$$

The smaller the step size h ($h = 0.01$, $h = 0.001$), the closer the polygonal approximation hugs the true trajectory!

5.2 Python Implementation of Euler's Method

Here is the Python script implementing Euler's method for $y' = -2y$ with $y(0) = 1$ on $[0\,;2]$, comparing the numerical simulation against the exact analytical solution $y(t) = e^{-2t}$:

```python
import math

def euler_method(a, b, y0, n):
    """
    Numerically solves y' = -2y on [a, b] with y(a) = y0 in n steps.
    """
    h = (b - a) / n
    t = a
    y = y0
    results = [(t, y, math.exp(-2 * t))]

    for _ in range(n):
        # Slope from differential equation y' = -2y
        slope = -2 * y
        y = y + h * slope
        t = t + h
        y_exact = math.exp(-2 * t)
        results.append((round(t, 4), round(y, 4), round(y_exact, 4)))

    return results

# Run with 10 steps of h = 0.2
points = euler_method(0, 2, 1.0, 10)
print(f"{'t':<6} | {'y_Euler':<10} | {'y_Exact':<10} | {'Error':<10}")
print("-" * 42)
for t_val, y_e, y_ex in points:
    err = abs(y_e - y_ex)
    print(f"{t_val:<6} | {y_e:<10} | {y_ex:<10} | {err:<10.4f}")
```

Key Takeaways

1. The Linear Differential Equation $y' = ay$

- The general solution on $\mathbb{R}$ is:

$$y(t) = Ce^{at} \quad (C \in \mathbb{R}).$$

- If $a > 0$, solutions explode toward $\pm\infty$ as $t \to +\infty$ (exponential growth).
- If $a < 0$, solutions decay toward 0 as $t \to +\infty$ (exponential decay toward equilibrium).
- Initial condition: for $y(t_0) = y_0$, the unique solution is $y(t) = y_0 e^{a(t-t_0)}$.

2. The Non-Homogeneous Equation $y' = ay + b$ ($a \neq 0$)

- General solution:

$$y(t) = Ce^{at} - \frac{b}{a}.$$

- The term $-\frac{b}{a}$ is the **constant particular solution**, representing the steady-state equilibrium value (a horizontal asymptote when $a < 0$).

3. Equations with Variable Forcing $y' = ay + f(t)$

- To solve $y' = ay + f(t)$, find a particular solution $y_p(t)$ matching the functional form of f (a polynomial if f is a polynomial; an exponential if f is an exponential), then form the general solution $y(t) = Ce^{at} + y_p(t)$.

4. Euler's Method

- Iterative algorithm with step size h:

$$t_{n+1} = t_n + h \quad \text{and} \quad y_{n+1} = y_n + h \times y'(t_n).$$

Exercises

Level (Direct Applications)

Exercise 51.1 — Elementary Homogeneous Equations. Solve on $\mathbb{R}$: 1. $y' = 3y$ 2. $y' = -5y$ 3. $2y' + 6y = 0$ 4. $4y' - y = 0$

Exercise 51.2 — Initial Value Problems for $y' = ay$. Find the unique solution satisfying the given initial condition: 1. $y' = 2y$ with $y(0) = 7$ 2. $y' = -3y$ with $y(0) = -4$ 3. $y' = 0.5y$ with $y(2) = 10$ 4. $y' = -y$ with $y(1) = e$

Exercise 51.3 — Non-Homogeneous Equations with Constant Forcing $y' = ay + b$. Solve on $\mathbb{R}$: 1. $y' = -2y + 6$ 2. $y' = 4y - 8$ 3. $3y' + 9y = 18$ 4. $y' = -0.2y + 5$

Exercise 51.4 — Initial Conditions for $y' = ay + b$. 1. Solve $y' = -y + 3$ with $y(0) = 1$. 2. Solve $y' = 2y - 10$ with $y(0) = 5$. 3. Solve $2y' + y = 4$ with $y(0) = 0$.

Exercise 51.5 — Determining Asymptotes. Let f be a solution to $y' = -3y + 12$. 1. Determine the general expression for $f(t)$. 2. What is the limit of $f(t)$ as $t \to +\infty$? 3. Deduce the horizontal asymptote common to all solution curves.

Exercise 51.6 — Steps in Euler's Method. Consider $y' = 2t + y$ with $y(0) = 1$. Using Euler's method with step size $h = 0.1$, compute the approximations $y_1 \approx y(0.1)$ and $y_2 \approx y(0.2)$.

Level (Exploration and Exam Preparation)

Exercise 51.7 — The Detective's Coffee Cup (Crime Scene Investigation). A deceased body is discovered in a climate-controlled room held at a steady 20° C. At 2:00 PM, the forensic pathologist measures the body temperature: it reads 30° C. At 3:00 PM, the pathologist measures it again: it has fallen to 28° C. Assume cooling satisfies Newton's law: $T'(t) = -k(T(t) - 20)$, and that normal body temperature is 37° C. 1. Taking $t = 0$ at 2:00 PM, express temperature $T(t)$ in terms of t and k. 2. Using the measurement taken at 3:00 PM ($t = 1$), determine the exact and numerical value of k. 3. Determine the estimated time of death.

Exercise 51.8 — Discharging a Capacitor Through a Resistor. A capacitor of capacitance $C = 50\,\mu$ F charged to $U_0 = 12$ V is discharged through a resistor of resistance $R = 20\,\text{k}\,\Omega$. 1. Establish the differential equation satisfied by the capacitor voltage $u(t)$. 2. Calculate the time constant $\tau = RC$. 3. Solve the equation and give the expression for $u(t)$. 4. After how much time does the voltage drop below 1 V?

Exercise 51.9 — Intravenous Drug Infusion (Exam Style). An antibiotic is administered intravenously at a continuous constant rate. The quantity of drug $q(t)$ present in the bloodstream (in milligrams) at time t (in hours) satisfies:

$$q'(t) = -0.15q(t) + 30.$$

At the start of infusion ($t = 0$), no drug is present: $q(0) = 0$. 1. Solve this differential equation to determine $q(t)$. 2. Determine the monotonicity of q on $[0\,;+\infty[$. 3. Determine $\lim_{t\to+\infty} q(t)$. What is the maximum plateau level reached? 4. The drug is therapeutically effective only when the blood concentration exceeds 150 mg. When is this threshold reached?

Exercise 51.10 — Equation with Affine Forcing $y' = ay + f(t)$. Consider the differential equation:

$$(E): \quad y' - 2y = 4x - 6.$$

1. Find constants m and p such that the affine function $y_p(x) = mx + p$ is a particular solution to (E). 2. Solve the associated homogeneous equation $(E_0): y' - 2y = 0$. 3. Deduce the general solution of (E). 4. Find the unique solution f satisfying $f(0) = 1$.

Exercise 51.11 — Equation with Trigonometric Forcing. Consider $(E): y' + y = \cos x + \sin x$. 1. Show that $g(x) = \sin x$ is a particular solution to (E). 2. Deduce the general solution of (E). 3. Determine the solution satisfying $y(0) = 2$.

Exercise 51.12 — Terminal Velocity of a Raindrop. A raindrop of mass $m = 4 \times 10^{-5}$ kg falls from rest from a cloud. It experiences gravity ($g = 9.81$ m $\cdot$ s^{-2}) and air drag $F = -kv$ with $k = 5 \times 10^{-5}$ kg $\cdot$ s^{-1}. 1. Formulate the differential equation for velocity $v(t)$. 2. Determine the terminal velocity $v_{\lim}$. 3. Solve the differential equation for $v(t)$. 4. Calculate the time required for the drop to reach 99% of its terminal velocity.

Exercise 51.13 — Tangent Condition. Consider $y' = -2y + 4$. 1. Solve this equation. 2. Does there exist a solution curve whose tangent line at $x = 0$ passes through $A(1\,;0)$? If so, identify it.

Level (In-Depth Challenges)

Exercise 51.14 — Verhulst's Logistic Population Model. The Malthusian model $y' = ay$ predicts unbounded population growth. In 1838, **Pierre-François Verhulst** proposed the logistic differential equation:

$$(E_L): \quad y' = ay\Big(1 - \frac{y}{K}\Big)$$

where $K > 0$ is the carrying capacity of the environment, and $a > 0$ is the intrinsic growth rate. Let y be a positive solution bounded by $0 < y < K$. Define $z(t) = \frac{1}{y(t)}$. 1. Compute the derivative $z'(t)$ in terms of $y(t)$ and $y'(t)$. 2. Substituting $y'(t)$ from (E_L), show that z satisfies the linear differential equation:

$$z' + az = \frac{a}{K}.$$

3. Solve this equation for $z(t)$. 4. Deduce the explicit formula for the population $y(t)$ in terms of initial population $y_0 = y(0)$. 5. Analyze the limits of $y(t)$ as $t \to -\infty$ and $t \to +\infty$. What shape does the curve have?

Exercise 51.15 — Second-Order Differential Equation $y'' + \omega^2 y = 0$. Let $\omega > 0$. We seek all twice-differentiable functions on $\mathbb{R}$ satisfying:

$$(H): \quad y'' + \omega^2 y = 0.$$

1. Verify that for any constants $A, B \in \mathbb{R}$, $y(t) = A\cos(\omega t) + B\sin(\omega t)$ is a solution. 2. Let f be an arbitrary solution of (H). Define $u(t) = \omega f(t)\sin(\omega t) + f'(t)\cos(\omega t)$ and $v(t) = \omega f(t)\cos(\omega t) - f'(t)\sin(\omega t)$. Compute the derivatives $u'(t)$ and $v'(t)$. 3. Deduce that u and v are constant functions, and thereby prove that **all** solutions to (H) must be of the form $A\cos(\omega t) + B\sin(\omega t)$.

Exercise 51.16 — Error Analysis of Euler's Method. Consider $y' = y$ with $y(0) = 1$ on $[0\,;1]$. The exact analytical solution is $y(t) = e^t$, so $y(1) = e$. Apply Euler's method with n subdivisions of step size $h = \frac{1}{n}$. 1. Write the recurrence relation expressing y_{k+1} in terms of y_k. 2. Deduce an explicit expression for y_n in terms of n. 3. Use this to recover Euler's celebrated classical limit: $\lim_{n\to+\infty}\left(1 + \frac{1}{n}\right)^n = e$.

Solutions

Level Solutions

Solution 51.1: 1. $y' = 3y \Longrightarrow y(t) = Ce^{3t}$ with $C \in \mathbb{R}$. 2. $y' = -5y \Longrightarrow y(t) = Ce^{-5t}$ with $C \in \mathbb{R}$. 3. $2y' + 6y = 0 \Leftrightarrow y' = -3y \Longrightarrow y(t) = Ce^{-3t}$. 4. $4y' - y = 0 \Leftrightarrow y' = \frac{1}{4}y \Longrightarrow y(t) = Ce^{t/4}$.

Solution 51.2: 1. $y(t) = Ce^{2t}$. Condition: $y(0) = C = 7 \Longrightarrow y(t) = 7e^{2t}$. 2. $y(t) = Ce^{-3t}$. Condition: $y(0) = C = -4 \Longrightarrow y(t) = -4e^{-3t}$. 3. $y(t) = Ce^{0.5t}$. Condition: $y(2) = Ce^1 = 10 \Leftrightarrow C = 10e^{-1}$. Solution: $y(t) = 10e^{0.5t-1}$. 4. $y(t) = Ce^{-t}$. Condition: $y(1) = Ce^{-1} = e \Leftrightarrow C = e^2$. Solution: $y(t) = e^{2-t}$.

Solution 51.3: General formula for $y' = ay + b$: $y(t) = Ce^{at} - \frac{b}{a}$. 1. $a = -2, b = 6 \Longrightarrow -\frac{b}{a} = 3$. Solution: $y(t) = Ce^{-2t} + 3$. 2. $a = 4, b = -8 \Longrightarrow -\frac{b}{a} = 2$. Solution: $y(t) = Ce^{4t} + 2$. 3. $y' = -3y + 6 \Longrightarrow -\frac{b}{a} = 2$. Solution: $y(t) = Ce^{-3t} + 2$. 4. $a = -0.2, b = 5 \Longrightarrow -\frac{b}{a} = 25$. Solution: $y(t) = Ce^{-0.2t} + 25$.

Solution 51.4: 1. $y(t) = Ce^{-t} + 3$. Condition $y(0) = C + 3 = 1 \Leftrightarrow C = -2$. Solution: $y(t) = -2e^{-t} + 3$. 2. $y(t) = Ce^{2t} + 5$. Condition $y(0) = C + 5 = 5 \Leftrightarrow C = 0$. Constant solution: $y(t) = 5$. 3. $y' = -0.5y + 2$. General solution: $y(t) = Ce^{-0.5t} + 4$. Condition $y(0) = C + 4 = 0 \Leftrightarrow C = -4$. Solution: $y(t) = 4(1 - e^{-0.5t})$.

Solution 51.5: 1. $a = -3, b = 12 \Longrightarrow -\frac{b}{a} = 4$. Solution: $f(t) = Ce^{-3t} + 4$. 2. Since $-3 < 0$, $\lim_{t\to+\infty} e^{-3t} = 0$, so $\lim_{t\to+\infty} f(t) = 4$. 3. The horizontal line $y = 4$ is the common horizontal asymptote as $t \to +\infty$.

Solution 51.6: - At $t_0 = 0$, $y_0 = 1$. Slope is $y'(0) = 2(0) + 1 = 1$. $y_1 = y_0 + h \times y'(0) = 1 + 0.1(1) = 1.1$ at $t_1 = 0.1$.

- At $t_1 = 0.1$, $y_1 = 1.1$. Estimated slope is $y'(0.1) \approx 2(0.1) + 1.1 = 1.3$. $y_2 = y_1 + h \times y'(t_1) = 1.1 + 0.1(1.3) = 1.23$ at $t_2 = 0.2$.

Level Solutions

Solution 51.7: 1. The equation is $T'(t) = -kT(t) + 20k$. The general solution is $T(t) = Ce^{-kt} + 20$. At 2:00 PM ($t = 0$), $T(0) = 30 \Rightarrow C + 20 = 30 \Leftrightarrow C = 10$. Thus $T(t) = 10e^{-kt} + 20$. 2. At 3:00 PM ($t = 1$), $T(1) = 28$: $10e^{-k} + 20 = 28 \Leftrightarrow 10e^{-k} = 8 \Leftrightarrow e^{-k} = 0.8 \Leftrightarrow k = -\ln(0.8) = \ln(1.25) \approx 0.22314$ h^{-1}. 3. Let t_d be the time of death, when body temperature was 37° C: $10e^{-kt_d} + 20 = 37 \Leftrightarrow 10e^{-kt_d} = 17 \Leftrightarrow e^{-kt_d} = 1.7$. Taking logarithms:

$$-kt_d = \ln(1.7) \Leftrightarrow t_d = -\frac{\ln(1.7)}{k} \approx -\frac{0.53063}{0.22314} \approx -2.378 \ \text{ hours}\,.$$

Converting 0.378 h to minutes: $0.378 \times 60 \approx 23$ minutes. Death occurred 2 hours and 23 minutes prior to 2:00 PM, at **11:37 AM**.

Solution 51.8: 1. During discharge, $u_R + u_C = 0 \Leftrightarrow RCu'(t) + u(t) = 0 \Leftrightarrow u'(t) = -\frac{1}{RC}u(t)$. 2. $\tau = RC = 20 \times 10^3 \times 50 \times 10^{-6} = 1.0$ second. 3. $u'(t) = -u(t)$. With $u(0) = 12$ V, $u(t) = 12e^{-t}$. 4. $u(t) < 1 \Leftrightarrow 12e^{-t} < 1 \Leftrightarrow e^{-t} < \frac{1}{12} \Leftrightarrow t > \ln 12 \approx 2.48$ seconds.

Solution 51.9: 1. Equation $y' = ay + b$ with $a = -0.15$ and $b = 30$. Equilibrium value: $-\frac{b}{a} = 200$. General solution: $q(t) = Ce^{-0.15t} + 200$. With $q(0) = 0 \Rightarrow C = -200$, we have $q(t) = 200(1 - e^{-0.15t})$. 2. $q'(t) = 30e^{-0.15t} > 0$. The drug quantity is strictly increasing on $[0\,;+\infty[$. 3. Since $-0.15 < 0$, $\lim_{t\to+\infty} q(t) = 200$ mg. The bloodstream level never exceeds 200 mg. 4. $q(t) \geq 150 \Leftrightarrow 200(1 - e^{-0.15t}) \geq 150 \Leftrightarrow e^{-0.15t} \leq 0.25$. Taking ln: $-0.15t \leq -\ln 4 \Leftrightarrow t \geq \frac{\ln 4}{0.15} \approx 9.24$ hours (approx. 9 hours and 14 minutes).

Solution 51.10: 1. $y_p(x) = mx + p \Rightarrow y_p'(x) = m$. Substitute into (E): $m - 2(mx + p) = 4x - 6 \Leftrightarrow -2mx + (m - 2p) = 4x - 6$. Equating coefficients: $-2m = 4 \Rightarrow m = -2$, and $m - 2p = -6 \Rightarrow -2 - 2p = -6 \Rightarrow p = 2$. Particular solution: $y_p(x) = -2x + 2$. 2. Homogeneous equation: $y' - 2y = 0 \Rightarrow y(x) = Ce^{2x}$. 3. General solution: $y(x) = Ce^{2x} - 2x + 2$ with $C \in \mathbb{R}$. 4. Condition $f(0) = 1 \Rightarrow C + 2 = 1 \Leftrightarrow C = -1$. Unique solution: $f(x) = -e^{2x} - 2x + 2$.

Solution 51.11: 1. $g(x) = \sin x \Rightarrow g'(x) = \cos x$. $g'(x) + g(x) = \cos x + \sin x$. The equation is satisfied. 2. Homogeneous equation $y' + y = 0 \Rightarrow y(x) = Ce^{-x}$. General solution: $y(x) = Ce^{-x} + \sin x$. 3. $y(0) = Ce^0 + \sin 0 = C = 2$. Solution: $y(x) = 2e^{-x} + \sin x$.

Solution 51.12: 1. $v'(t) = -\frac{k}{m}v(t) + g$. Here $\frac{k}{m} = \frac{5\times10^{-5}}{4\times10^{-5}} = 1.25$ s^{-1}. The equation is $v'(t) = -1.25v(t) + 9.81$. 2. $v_{\lim} = \frac{9.81}{1.25} = 7.848$ m$\cdot$s^{-1} (28.3 km/h). 3. General solution: $v(t) = Ce^{-1.25t} + 7.848$. Initial condition $v(0) = 0 \Rightarrow C = -7.848$. $v(t) = 7.848(1 - e^{-1.25t})$. 4. $v(t) = 0.99v_{\lim} \Leftrightarrow 1 - e^{-1.25t} = 0.99 \Leftrightarrow e^{-1.25t} = 0.01$. $t = \frac{\ln 100}{1.25} \approx \frac{4.605}{1.25} \approx 3.68$ seconds.

Solution 51.13: 1. $y' = -2y + 4 \Rightarrow y(x) = Ce^{-2x} + 2$. 2. The tangent line at $x = 0$ is $y = y'(0)x + y(0)$. Here $y(0) = C + 2$ and $y'(0) = -2(C + 2) + 4 = -2C$. Tangent equation: $y = -2Cx + (C + 2)$. Passing through $(1\,;0)$ requires $0 = -2C(1) + C + 2 \Leftrightarrow -C + 2 = 0 \Leftrightarrow C = 2$. Unique solution: $y(x) = 2e^{-2x} + 2$.

Level Solutions

Solution 51.14: 1. $z(t) = \frac{1}{y(t)} \Rightarrow z'(t) = -\frac{y'(t)}{(y(t))^2}$. 2. From (E_L), $y'(t) = ay(t) - \frac{a}{K}(y(t))^2$. Then: $z'(t) = -\frac{ay(t) - \frac{a}{K}(y(t))^2}{(y(t))^2} = -\frac{a}{y(t)} + \frac{a}{K} = -az(t) + \frac{a}{K}$. Rearranging gives the linear equation $z' + az = \frac{a}{K}$. 3. The equilibrium solution of $z' = -az + \frac{a}{K}$ is $\frac{a/K}{a} = \frac{1}{K}$. Thus $z(t) = Be^{-at} + \frac{1}{K}$ with $B \in \mathbb{R}$. 4. At $t = 0$, $z(0) = \frac{1}{y_0} = B + \frac{1}{K} \Rightarrow B = \frac{K - y_0}{Ky_0}$. Then $z(t) = \frac{1}{K}\left(1 + \frac{K - y_0}{y_0}e^{-at}\right)$. Inverting $z(t)$:

$$y(t) = \frac{K}{1 + \left(\frac{K - y_0}{y_0}\right)e^{-at}}.$$

5. As $t \to +\infty$, $e^{-at} \to 0$ (since $a > 0$), so $\lim_{t\to+\infty} y(t) = K$. The resulting trajectory is an **S-shaped logistic curve** (sigmoid) that saturates smoothly at carrying capacity K.

Solution 51.15: 1. For $y(t) = A\cos(\omega t) + B\sin(\omega t)$: $y'(t) = -A\omega\sin(\omega t) + B\omega\cos(\omega t)$. $y''(t) = -A\omega^2\cos(\omega t) - B\omega^2\sin(\omega t) = -\omega^2 y(t)$. Hence $y'' + \omega^2 y = 0$. 2. Differentiate $u(t) = \omega f(t)\sin(\omega t) + f'(t)\cos(\omega t)$: $u'(t) = \omega f'(t)\sin(\omega t) + \omega^2 f(t)\cos(\omega t) + f''(t)\cos(\omega t) - \omega f'(t)\sin(\omega t) = \big(\,f''(t) + \omega^2 f(t)\big)\cos(\omega t) = 0$. Similarly, $v'(t) = -\big(\,f''(t) + \omega^2 f(t)\big)\sin(\omega t) = 0$. 3. Since u' and v' are identically zero on $\mathbb{R}$, u and v are constant:

$u(t) = B\omega$ and $v(t) = A\omega$. Computing $\frac{u(t)\sin(\omega t)+v(t)\cos(\omega t)}{\omega}$ yields: $f(t)(\sin^2(\omega t) + \cos^2(\omega t)) = f(t) = A\cos(\omega t) + B\sin(\omega t)$. Every solution must be of this form!

Solution 51.16: 1. $y' = y \Longrightarrow y_{k+1} = y_k + hy_k = (1+h)y_k$. 2. With $y_0 = 1$ and $h = \frac{1}{n}$, (y_k) is geometric with common ratio $\left(1 + \frac{1}{n}\right)$. After n steps (reaching $t = nh = 1$): $y_n = \left(1 + \frac{1}{n}\right)^n$. 3. Since Euler's method converges to the exact solution $y(1) = e^1 = e$ as $n \to +\infty$, we immediately recover:

$$\lim_{n\to+\infty} \left(1 + \frac{1}{n}\right)^n = e.$$

The Question that Opens What Follows

Until now, we have described movement and physical change along a single line (time t) or across a two-dimensional flat plane (a sheet of paper with coordinates x and y). Yet our physical universe is not flat: the Earth floats in three-dimensional space, medieval cathedrals loft their vaults into the sky, satellites traverse inclined orbital planes, and aircraft navigate using latitude, longitude, and altitude.

How do we pinpoint a position, trace a line, and establish the algebraic equation of a tilted plane in three-dimensional space? How do we calculate the shortest distance from an airborne drone to the wall of a cathedral?

Join us in Chapter 52 with **The Cathedral in Space** and vector geometry in three dimensions.

CHAPTER 52: The Cathedral in Space

> "Geometry is a magic that allows orienting a nave toward the North Star, raising stone vaults to the sky without collapsing, and measuring the distance of an invisible point without ever leaving the ground." — *Notebooks of medieval master builders*

What You Are About to Discover

- How cathedral builders conquered the **third dimension**: from the flat ground to the celestial keystone.
- The orthonormal coordinate system $(O; \vec{\imath}, \vec{\jmath}, \vec{k})$ of 3D space: abscissa (x), ordinate (y), **z-coordinate / altitude** (z), and the 3D distance formula derived from a double Pythagorean theorem.
- The **parametric representation of a line** in space: why a line in 3D can never be described by a single Cartesian equation.
- The relative positions of two lines: parallel, intersecting… or **skew / non-coplanar** (the ultimate trap of 3D!).
- The **dot product (scalar product) in space**: its analytic expression $\vec{u} \cdot \vec{v} = xx' + yy' + zz'$ and its role as a right-angle detector.
- The **normal vector** $\vec{n}(a\,;b\,;c)$ and the **Cartesian equation of a plane** $ax + by + cz + d = 0$.
- The **orthogonal projection** of a point onto a line or a plane, and the calculation of shortest distance.

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1. The Problem: The Builders of Chartres Cathedral

1.1 Leaving the Flat Parchment

In the Middle Ages, the architects and stonecutters who designed Gothic cathedrals—such as Chartres or Notre-Dame de Paris—faced a monumental challenge. On their parchments and on the plaster floor of the tracing room, they could only draw figures in **two dimensions**: circles, triangles, rectangles. But the cathedral is a **three-dimensional** edifice:

- The ground plan provides the length of the nave and the width of the transept;
- But the pillars soar toward the sky, the flying buttresses span empty air diagonally, and the ribs of the ribbed vaults cross at a **keystone suspended 37 meters above the floor**.

If two flying buttresses cut in the quarry do not meet at the exact same point in space, the vault fails and hundreds of tons of stone come crashing down. In the plane, two non-parallel lines **necessarily intersect**. In space, however, two lines can be non-parallel and yet **never intersect**: picture two airplanes flying in different directions at different altitudes, or two cables crossing in a timber frame without touching. Such lines are said to be **skew (non-coplanar)**.

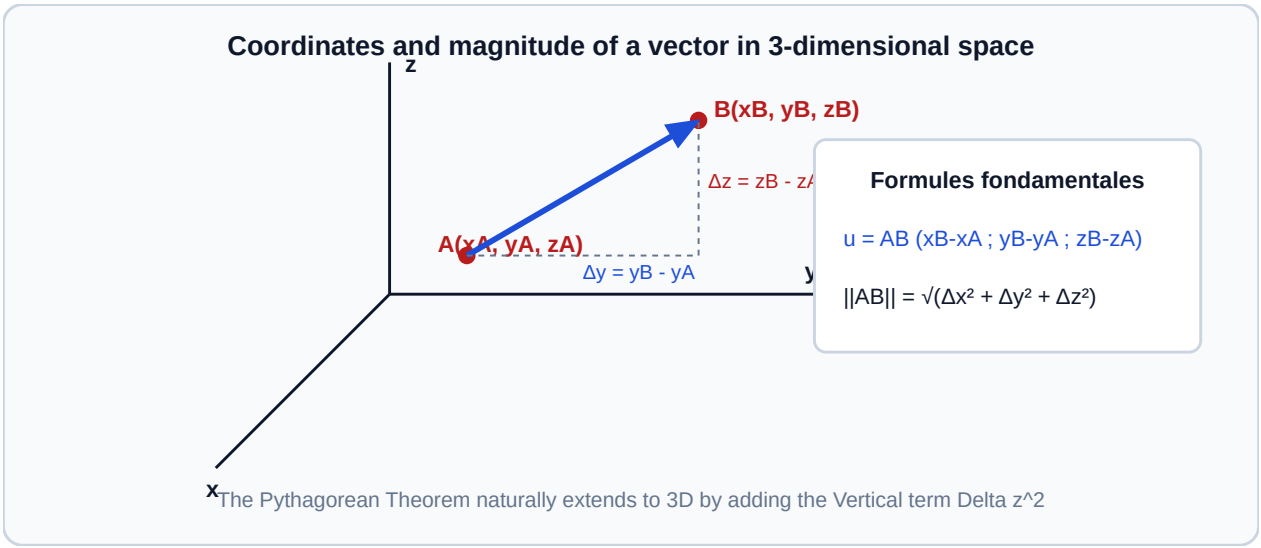

Figure 112: Coordinates and Euclidean magnitude of a vector in 3-dimensional space

To build toward the heavens without risk of error, we need an infallible system to locate every grain of stone throughout the volume of three-dimensional space.

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2. The 3D Coordinate System and Coordinates

2.1 The Orthonormal Frame $(O;\vec{\imath},\vec{\jmath},\vec{k})$

To determine the position of a point in space, we choose an origin $O(0\,;0\,;0)$ and three mutually perpendicular axes, each equipped with a unit vector of equal length:

- The (Ox) axis directed by $\vec{\imath}$: the **x-axis** or axis of abscissas (often pointing forward);
- The (Oy) axis directed by $\vec{\jmath}$: the **y-axis** or axis of ordinates (pointing to the right);
- The (Oz) axis directed by $\vec{k}$: the **z-axis** or axis of altitudes / cotes (pointing upward).

The triplet $(O;\vec{\imath},\vec{\jmath},\vec{k})$ constitutes an **orthonormal coordinate frame of space**.

📐 Definition — Coordinates of a Point and a Vector:

- For any point M in space, there exists a unique triplet of real numbers $(x\,;y\,;z)$ such that:

$$\overrightarrow{OM} = x\vec{\imath} + y\vec{\jmath} + z\vec{k}.$$

The point M has coordinates $M(x\,;y\,;z)$. x is the abscissa, y the ordinate, and z the altitude (z-coordinate).

- If $A(x_A\,;y_A\,;z_A)$ and $B(x_B\,;y_B\,;z_B)$ are two points, the vector $\overrightarrow{AB}$ has coordinates:

$$\overrightarrow{AB}\begin{pmatrix} x_B - x_A \\ y_B - y_A \\ z_B - z_A \end{pmatrix}.$$

- The midpoint I of segment $[AB]$ has coordinates:

$$I\left(\frac{x_A + x_B}{2}\,;\,\frac{y_A + y_B}{2}\,;\,\frac{z_A + z_B}{2}\right).$$

2.2 Norm and Distance: The Double Pythagorean Theorem

What is the distance between two points in space? To travel from point $A(x_A\,;y_A\,;z_A)$ to point $B(x_B\,;y_B\,;z_B)$, we can construct a rectangular cuboid (box) whose space diagonal is the segment $[AB]$:

1. In the bottom horizontal plane, the right triangle yields by Pythagoras:

$$AC^2 = (x_B - x_A)^2 + (y_B - y_A)^2.$$

2. In the vertical right triangle ACB, the height is $\Delta z = z_B - z_A$, and the hypotenuse is the diagonal $[AB]$. A second application of Pythagoras gives:

$$AB^2 = AC^2 + BC^2 = (x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2.$$

📐 Theorem — Vector Norm and Distance Between Two Points:

- The norm (length) of a vector $\vec{u}(x\,;y\,;z)$ in an orthonormal frame is:

$$\| \vec{u} \| = \sqrt{x^2 + y^2 + z^2}.$$

- The distance between two points A and B is:

$$AB = \| \overrightarrow{AB} \| = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}.$$

---

3. Lines in Space: Parametric Representation

⚠️ ⚠️ Classical Analytical Trap — A Scalar Equation in 3D Defines a PLANE, Never a Line!

In 2D plane Cartesian geometry, a linear equation $ax + by + c = 0$ represents a straight line. In three-dimensional space $\mathbb{R}^3$, writing a single linear relation $ax + by + cz + d = 0$ **never** describes a line, but rather an **infinite 2D plane** (possessing two degrees of freedom)! To determine a one-dimensional line in 3D space, mathematics provides only two rigorous routes:

1. Either a **system of two linear Cartesian equations** describing the intersection of two distinct, non-parallel planes;
2. Or a **parametric representation** expressing coordinates (x, y, z) through a single scalar parameter $t \in \mathbb{R}$ steered by a direction vector.

3.1 Why a Single Equation Does Not Suffice

In the 2D plane, a line has a Cartesian equation of the form $ax + by + c = 0$. What does that same equation represent in 3D space? If we write $2x + 3y - 6 = 0$ in space, the variable z does not appear: this means that z can take **any real value**! All points above and below the line form an infinite vertical wall: **it is a plane, not a line**! In space, a single linear equation always defines a surface (a plane). To define a line, one needs the intersection of two planes, or even better: **a time-dependent (parametric) trajectory**.

3.2 Parametric Representation of a Line

Let $A(x_A\,;y_A\,;z_A)$ be a point in space and $\vec{u}(a\,;b\,;c)$ a non-zero vector. The line $\mathcal{D}$ passing through A with direction vector $\vec{u}$ is the set of points $M(x\,;y\,;z)$ such that the vectors $\overrightarrow{AM}$ and $\vec{u}$ are **collinear**:

$$M \in \mathcal{D} \Leftrightarrow \text{there exists a real number } \ t \in \mathbb{R} \ \text{ such that } \ \overrightarrow{AM} = t\vec{u}.$$

Projecting onto the three coordinate components:

📐 Definition — Parametric Representation of a Line:

The line $\mathcal{D}$ passing through $A(x_A\,;y_A\,;z_A)$ and directed by $\vec{u}(a\,;b\,;c)$ admits as a parametric representation the system:

$$\begin{cases} x = x_A + at \\ y = y_A + bt \\ z = z_A + ct \quad (t \in \mathbb{R}). \end{cases}$$

The real number t is called the **parameter**.

Kinematic Interpretation: Think of an aircraft taking off at time $t = 0$ from point A, with constant velocity vector $\vec{u}$. The parameter t plays the role of time: at every second t, the aircraft occupies a precise position along the line!

Worked Example 1 — Line passing through two points.

Let $A(1\,;-2\,;3)$ and $B(4\,;2\,;-1)$. 1. Determine a parametric representation of the line (AB). 2. Does the point $C(7\,;6\,;-5)$ lie on the line (AB)? 3. Does the point $D(2\,;0\,;1)$ lie on the line (AB)?

Solution: 1. The direction vector is $\overrightarrow{AB}\begin{pmatrix}4-1\\2-(-2)\\-1-3\end{pmatrix} = \begin{pmatrix}3\\4\\-4\end{pmatrix}$. The line (AB) passing through A is given by:

$$\begin{cases} x = 1 + 3t \\ y = -2 + 4t \\ z = 3 - 4t \quad (t \in \mathbb{R}). \end{cases}$$

2. For the point $C(7\,;6\,;-5)$:

$$\begin{cases} 7 = 1 + 3t \Longrightarrow 3t = 6 \Longrightarrow t = 2 \\ 6 = -2 + 4t \Longrightarrow 4t = 8 \Longrightarrow t = 2 \\ -5 = 3 - 4t \Longrightarrow -4t = -8 \Longrightarrow t = 2 \end{cases}$$

The same value $t = 2$ satisfies all three equations simultaneously. Therefore $\boldsymbol{C \in (AB)}$. 3. For the point $D(2\,;0\,;1)$:

- $2 = 1 + 3t \Longrightarrow t = \frac{1}{3}$.
- Check with y: $-2 + 4\left(\frac{1}{3}\right) = -2 + \frac{4}{3} = -\frac{2}{3} \neq 0$. The equations are incompatible. Therefore $\boldsymbol{D \notin (AB)}$.

3.3 Relative Positions of Two Lines in Space

Consider two lines $\mathcal{D}_1$ directed by $\vec{u}_1$ and $\mathcal{D}_2$ directed by $\vec{u}_2$.

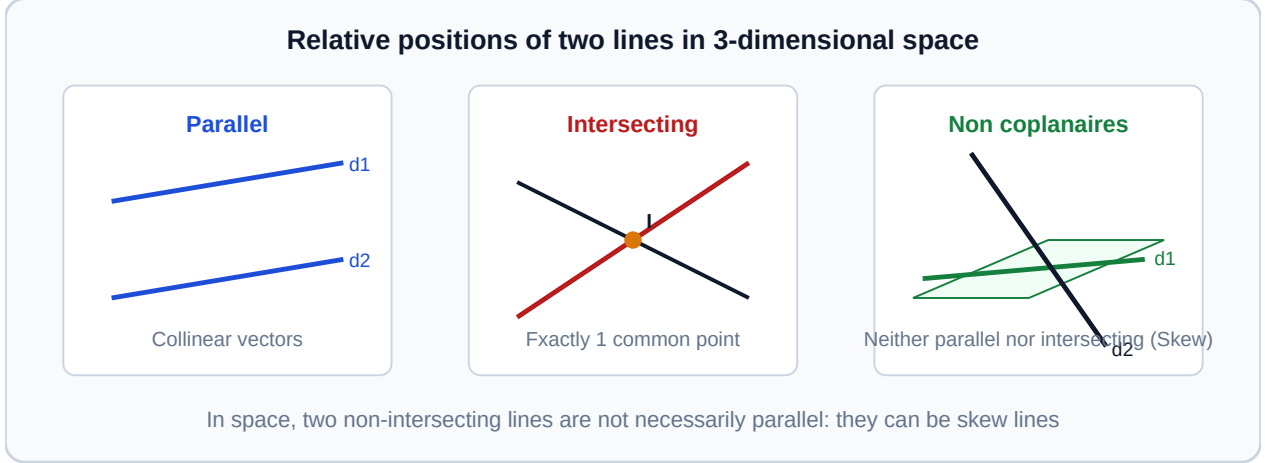

Figure 113: Relative positions of two lines in 3D space: parallel, intersecting, or skew

⚠ Warning, Pitfall! — Two lines with no intersection point are NOT necessarily parallel!

This is the most frequent trap for students coming from plane geometry: in space, the vast majority of line pairs are **neither parallel nor intersecting**: they are **skew / non-coplanar** (they pass over or under each other at different heights without ever crossing).

📐 Strict Lexical Distinction — Orthogonality vs Perpendicularity in Space

In 2D plane geometry, orthogonal lines always intersect and are thus perpendicular. In 3D space, geometry draws a vital distinction:

- Two lines are **orthogonal** whenever their direction vectors have zero dot product ($\vec{u} \cdot \vec{v} = 0$), even if they never meet (skew lines situated in distinct parallel planes).
- The term **perpendicular** is strictly reserved for lines that are both orthogonal **AND intersecting** (sharing a common point, hence coplanar).

4. Dot Product in Space and Orthogonality

4.1 Definition and Analytic Expression

The dot product (scalar product) in space possesses exactly the same geometric properties as in the plane:

📐 Definition — Dot Product in Space:

For any vectors $\vec{u}$ and $\vec{v}$:

$$\vec{u} \cdot \vec{v} = \| \vec{u} \| \times \| \vec{v} \| \times \cos(\vec{u}, \vec{v}).$$

In an **orthonormal** frame $(O; \vec{i}, \vec{j}, \vec{k})$, if $\vec{u}(x\,;y\,;z)$ and $\vec{v}(x'\,;y'\,;z')$:

$$\vec{u} \cdot \vec{v} = xx' + yy' + zz'.$$

Orthogonality Criterion:

$$\vec{u} \perp \vec{v} \Leftrightarrow \vec{u} \cdot \vec{v} = 0 \Leftrightarrow xx' + yy' + zz' = 0.$$

Worked Example 2 — Computing an angle in space.

Given points $A(1\,;0\,;1)$, $B(2\,;1\,;0)$, and $C(0\,;2\,;2)$. Determine the measure of angle $\widehat{BAC}$.

- Compute the vectors: $\overrightarrow{AB}\begin{pmatrix}1\\1\\-1\end{pmatrix}$ and $\overrightarrow{AC}\begin{pmatrix}-1\\2\\1\end{pmatrix}$.
- Dot product: $\overrightarrow{AB} \cdot \overrightarrow{AC} = 1(-1) + 1(2) + (-1)(1) = -1 + 2 - 1 = 0$!
- Since $\overrightarrow{AB} \cdot \overrightarrow{AC} = 0$, the vectors are orthogonal: triangle ABC is **right-angled at A**, and $\widehat{BAC} = 90° = \frac{\pi}{2}$ rad.

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5. Planes in Space: Normal Vector and Cartesian Equation

5.1 The Concept of a Normal Vector

How can we orient a plane in space? Think of a tabletop: to orient the surface of the table, the simplest and most direct method is to drive a peg perpendicular to the surface. If we know the direction of the peg and the position of one point on the table, the entire surface is completely locked in space!

📐 Definition — Normal Vector to a Plane:

A non-zero vector $\vec{n}$ is said to be **normal** to a plane $\mathcal{P}$ if it is orthogonal to the direction of every line in the plane $\mathcal{P}$. In practice: $\vec{n}$ is normal to $\mathcal{P}$ if and only if it is orthogonal to **two non-collinear direction vectors** lying in plane $\mathcal{P}$.

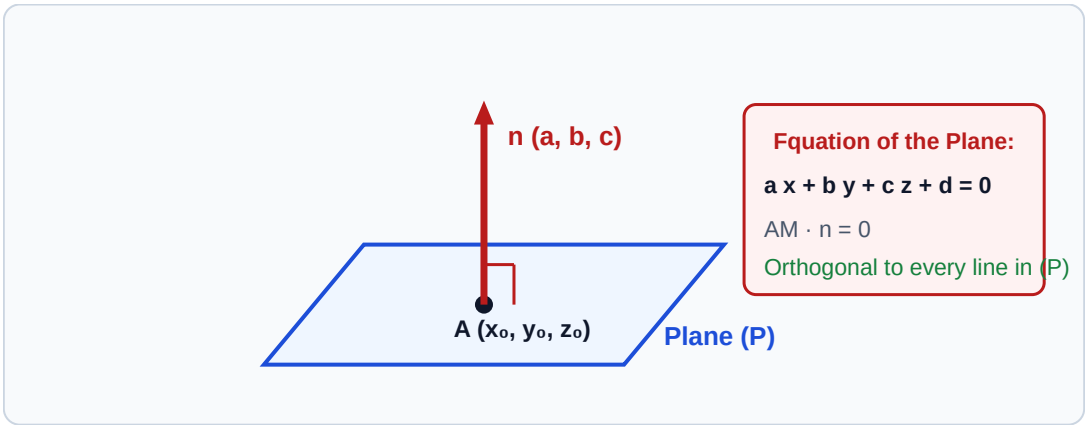

Figure 114: Cartesian equation of the plane and normal vector in 3D space

5.2 Cartesian Equation of a Plane

Let $A(x_A\,;y_A\,;z_A)$ be a fixed point and $\vec{n}(a\,;b\,;c)$ a non-zero vector. The plane $\mathcal{P}$ passing through A with normal vector $\vec{n}$ is the set of points $M(x\,;y\,;z)$ such that the vector $\overrightarrow{AM}$ is orthogonal to $\vec{n}$:

$$M \in \mathcal{P} \Leftrightarrow \overrightarrow{AM} \cdot \vec{n} = 0.$$

Expanding using the coordinates $\overrightarrow{AM}\begin{pmatrix} x-x_A \\ y-y_A \\ z-z_A \end{pmatrix}$ and $\vec{n}\begin{pmatrix} a \\ b \\ c \end{pmatrix}$:

$$a(x-x_A)+b(y-y_A)+c(z-z_A)=0 \Leftrightarrow ax+by+cz-(ax_A+by_A+cz_A)=0.$$

Setting the constant $d=-(ax_A+by_A+cz_A)$, we obtain the fundamental theorem:

Theorem (Cartesian Equation of a Plane) —

- Every plane $\mathcal{P}$ in space admits a Cartesian equation of the form:
$$ax+by+cz+d=0$$
where $(a\,;b\,;c)\neq(0\,;0\,;0)$.
- Conversely, for any real numbers a,b,c,d with $(a,b,c)\neq(0,0,0)$, the set of points $M(x\,;y\,;z)$ satisfying $ax+by+cz+d=0$ is a **plane**, a normal vector of which is:
$$\vec{n}\begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

Worked Example 3 — Equation of a plane given a point and a normal vector.

Determine the Cartesian equation of the plane $\mathcal{P}$ passing through $A(2\,;-1\,;3)$ with normal vector $\vec{n}(3\,;4\,;-2)$.

- The equation has the form $3x+4y-2z+d=0$.
- Point $A(2\,;-1\,;3)$ belongs to the plane, so its coordinates satisfy the equation:
$$3(2)+4(-1)-2(3)+d=0 \Leftrightarrow 6-4-6+d=0 \Leftrightarrow -4+d=0 \Leftrightarrow d=4.$$
- The Cartesian equation of $\mathcal{P}$ is therefore:
$$3x+4y-2z+4=0.$$

Worked Example 4 — Equation of a plane passing through three non-collinear points.

Determine a Cartesian equation of the plane passing through $A(1\,;0\,;0)$, $B(0\,;2\,;0)$, and $C(0\,;0\,;3)$.

- Form two vectors in the plane: $\overrightarrow{AB}\begin{pmatrix}-1\\2\\0\end{pmatrix}$ and $\overrightarrow{AC}\begin{pmatrix}-1\\0\\3\end{pmatrix}$. These vectors are clearly not collinear.
- Seek a normal vector $\vec{n}(a\,;b\,;c)$ orthogonal to both $\overrightarrow{AB}$ and $\overrightarrow{AC}$:

$$\{\vec{n}\cdot\overrightarrow{AB}=0\Leftrightarrow -a+2b=0\Leftrightarrow a=2b$$

$$\vec{n}\cdot\overrightarrow{AC}=0\Leftrightarrow -a+3c=0\Leftrightarrow a=3c$$

- Choose a convenient non-zero value to set the scale: let $a=6$. Then $b=3$ and $c=2$. The vector $\vec{n}(6\,;3\,;2)$ is a normal vector!
- The equation takes the form $6x+3y+2z+d=0$. Substitute $A(1\,;0\,;0)$: $6(1)+0+0+d=0\Leftrightarrow d=-6$.
- Cartesian equation: $6x+3y+2z-6=0$ (or $\frac{x}{1}+\frac{y}{2}+\frac{z}{3}=1$).

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6. Orthogonality, Orthogonal Projection, and Distance

6.1 Orthogonality of a Line and a Plane

Property —

A line $\mathcal{D}$ with direction vector $\vec{u}$ is **orthogonal (perpendicular)** to a plane $\mathcal{P}$ with normal vector $\vec{n}$ if and only if the vectors $\vec{u}$ and $\vec{n}$ are **collinear**.

6.2 Orthogonal Projection of a Point onto a Plane

Let $\mathcal{P}$ be a plane with equation $ax+by+cz+d=0$ and let M be any point in space. The **orthogonal projection** of M onto the plane $\mathcal{P}$ is the point $H\in\mathcal{P}$ such that the line (MH) is orthogonal to the plane $\mathcal{P}$. The distance MH is then the **shortest distance** between the point M and any point belonging to the plane.

Method — Calculating the Coordinates of the Orthogonal Projection H:

1. Determine a normal vector
$\vec{n}(a\,;b\,;c)$ to the plane $\mathcal{P}$. 2. Write the parametric representation of the line $\mathcal{D}$ passing through $M(x_M\,;y_M\,;z_M)$ and directed by $\vec{n}$:

$$\{x=x_M+at$$
$$y=y_M+bt$$
$$z=z_M+ct$$

3. Substitute these expressions for x, y, z into the Cartesian equation of the plane $\mathcal{P}$ to solve for the unique value of the parameter t_H. 4. Substitute t_H back into the parametric equations to obtain the coordinates of H. 5. The distance from M to the plane is then the norm $d(M,\mathcal{P})=\|\,\overrightarrow{MH}\,\|$.

Worked Example 5 — Full computation of an orthogonal projection and distance.

Let $M(1\,;2\,;3)$ and let $\mathcal{P}$ be the plane with equation $2x-y+2z-9=0$. 1. Determine the coordinates of the orthogonal projection H of M onto $\mathcal{P}$. 2. Compute the distance from M to the plane $\mathcal{P}$.

Solution: 1. A normal vector to the plane is $\vec{n}\begin{pmatrix}2\\-1\\2\end{pmatrix}$. The orthogonal line $\mathcal{D}$ passing through M is:

$$\begin{cases}x=1+2t\\y=2-t\\z=3+2t\end{cases}$$

Point H lies on this line and on the plane $\mathcal{P}$:

$$2(1+2t)-(2-t)+2(3+2t)-9=0\Leftrightarrow(2-2+6-9)+(4t+t+4t)=0\Leftrightarrow-3+9t=0\Leftrightarrow9t=3\Leftrightarrow t_H=\frac{1}{3}.$$

Let us compute the coordinates of H:

$$x_H=1+2\left(\frac{1}{3}\right)=\frac{5}{3},\quad y_H=2-\frac{1}{3}=\frac{5}{3},\quad z_H=3+2\left(\frac{1}{3}\right)=\frac{11}{3}.$$

The orthogonal projection is $H\left(\frac{5}{3}\,;\frac{5}{3}\,;\frac{11}{3}\right)$. 2. The vector $\overrightarrow{MH}$ is equal to $t_H\vec{n}=\frac{1}{3}\vec{n}$. The distance is:

$$d(M,\mathcal{P})=\parallel\overrightarrow{MH}\parallel=\left|\frac{1}{3}\right|\parallel\vec{n}\parallel=\frac{1}{3}\sqrt{2^2+(-1)^2+2^2}=\frac{1}{3}\sqrt{4+1+4}=\frac{1}{3}\sqrt{9}=\frac{3}{3}=\mathbf{1}.$$

💡 ***General Formula for the Distance from a Point to a Plane:*** *For any point $M(x_0\,;y_0\,;z_0)$ and any plane $\mathcal{P}$: $ax+by+cz+d=0$:*

$$d(M,\mathcal{P})=\frac{|ax_0+by_0+cz_0+d|}{\sqrt{a^2+b^2+c^2}}.$$

Key Takeaways

1. Coordinates and Distance in Space

- Orthonormal frame $(O;\vec{\imath},\vec{\jmath},\vec{k})$.
- Vector: $\overrightarrow{AB}\begin{pmatrix}x_B-x_A\\y_B-y_A\\z_B-z_A\end{pmatrix}$. Midpoint: $I\left(\frac{x_A+x_B}{2}\,;\frac{y_A+y_B}{2}\,;\frac{z_A+z_B}{2}\right)$.
- Norm and distance: $\parallel\vec{u}\parallel=\sqrt{x^2+y^2+z^2}$ and $AB=\sqrt{(x_B-x_A)^2+(y_B-y_A)^2+(z_B-z_A)^2}$.

2. Lines in Space

- Parametric representation of the line passing through $A(x_A,y_A,z_A)$ with direction vector $\vec{u}(a,b,c)$:

$$\begin{cases}x=x_A+at\\y=y_A+bt\\z=z_A+ct\end{cases}\quad(t\in\mathbb{R}).$$

- Two lines can be: parallel (collinear direction vectors), intersecting (a single common point), or **skew / non-coplanar** (neither parallel nor intersecting).

3. Dot Product and Orthogonality

- $\vec{u}\cdot\vec{v}=xx'+yy'+zz'$.
- $\vec{u}\perp\vec{v}\Leftrightarrow\vec{u}\cdot\vec{v}=0$.

4. Planes in Space

- Every plane is determined by a point and a **normal vector** $\vec{n}(a\,;b\,;c)$.
- Cartesian equation: $ax+by+cz+d=0$.
- A line $\mathcal{D}$ is perpendicular to a plane $\mathcal{P}\Leftrightarrow$ its direction vector is collinear with the normal vector of $\mathcal{P}$.

- Two planes are perpendicular ⇔ their normal vectors are orthogonal ($\vec{n}_1 \cdot \vec{n}_2 = 0$).

5. Orthogonal Projection and Distance

- The orthogonal projection H of M onto $\mathcal{P}$ is the intersection of the plane $\mathcal{P}$ with the perpendicular line passing through M.
- Distance formula: $d(M, \mathcal{P}) = \frac{|ax_M + by_M + cz_M + d|}{\sqrt{a^2+b^2+c^2}}$.

Exercises

Level (Direct Applications)

Exercise 52.1 — Elementary vector and distance computations. Given points $A(2\,;-1\,;4)$, $B(5\,;3\,;-2)$, and $C(0\,;1\,;1)$. 1. Calculate the coordinates of vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$. 2. Calculate the lengths AB and AC. 3. Determine the coordinates of the midpoint I of segment $[AB]$. 4. Compute the dot product $\overrightarrow{AB} \cdot \overrightarrow{AC}$. Are the lines (AB) and (AC) perpendicular?

Exercise 52.2 — Direct parametric representation. 1. Give a parametric representation of the line $\mathcal{D}$ passing through $A(1\,;3\,;-2)$ with direction vector $\vec{u}(2\,;-1\,;5)$. 2. Determine the coordinates of the point on $\mathcal{D}$ corresponding to parameter $t = 3$. 3. Does the point $K(5\,;1\,;8)$ lie on the line $\mathcal{D}$?

Exercise 52.3 — Immediate Cartesian equation of a plane. 1. Determine a Cartesian equation of the plane $\mathcal{P}$ passing through point $A(1\,;2\,;-3)$ with normal vector $\vec{n}(4\,;-1\,;2)$. 2. Give a normal vector and three points belonging to the plane with equation $2x - 3y + z - 6 = 0$.

Exercise 52.4 — Special points and coordinate planes. 1. What is the Cartesian equation of the horizontal plane (Oxy)? 2. What is the Cartesian equation of the plane (Oxz)? 3. What is the geometric nature of the set of points satisfying $z = 5$?

Exercise 52.5 — Orthogonal line and plane. Consider the line $\mathcal{D}$ with representation $\begin{cases} x = 2 - 3t \\ y = 1 + 6t \\ z = -4 - 9t \end{cases}$ and the plane $\mathcal{P}$: $x - 2y + 3z + 7 = 0$. Prove that the line $\mathcal{D}$ is orthogonal to the plane $\mathcal{P}$.

Exercise 52.6 — Intersection of a line and a plane. Determine the coordinates of the point of intersection between the line $\mathcal{D} : \begin{cases} x = 1 + 2t \\ y = -1 + t \\ z = 3 - t \end{cases}$ and the plane $\mathcal{P} : 3x + 2y - z - 5 = 0$.

Level (Exploration and Examination Classics)

Exercise 52.7 — Relative position of two lines (Exam Type). Consider the two lines $\mathcal{D}_1$ and $\mathcal{D}_2$ defined by the parametric representations:

$$\mathcal{D}_1 : \begin{cases} x = 1 + 2t \\ y = 3 - t \\ z = -1 + 4t \end{cases} (t \in \mathbb{R}) \quad \text{and} \quad \mathcal{D}_2 : \begin{cases} x = -1 + s \\ y = 2 + 2s \\ z = 5 - s \end{cases} (s \in \mathbb{R}).$$

1. Prove that the lines $\mathcal{D}_1$ and $\mathcal{D}_2$ are not parallel. 2. Prove that the lines $\mathcal{D}_1$ and $\mathcal{D}_2$ do not intersect. 3. Conclude on the relative position of these two lines.

Exercise 52.8 — The cube and the regular tetrahedron (Classic Exam Problem). In an orthonormal frame $(O; \vec{\imath}, \vec{\jmath}, \vec{k})$, consider the unit cube $OABCDEFG$ of edge length 1 such that: $O(0\,;0\,;0)$, $A(1\,;0\,;0)$, $B(1\,;1\,;0)$, $C(0\,;1\,;0)$, $D(0\,;0\,;1)$, $E(1\,;0\,;1)$, $F(1\,;1\,;1)$, $G(0\,;1\,;1)$. Let I be the midpoint of $[AB]$ and J the center of face $CDGE$. 1. State the coordinates of points I, J, E, and C. 2. Demonstrate that triangle BDE is equilateral. 3. Prove that the main space

diagonal of the cube (OF) is orthogonal to the plane (BDE). 4. Determine a Cartesian equation of the plane (BDE). 5. Determine the coordinates of the piercing point of the line (OF) through the plane (BDE).

Exercise 52.9 — Orthogonal projection and distance from a point to a plane. Let $\mathcal{P}$ be the plane with equation $x + 2y - 2z + 6 = 0$ and point $A(4\,;-1\,;5)$. 1. Verify that point A does not lie on plane $\mathcal{P}$. 2. Determine a parametric representation of the line Δ passing through A and perpendicular to $\mathcal{P}$. 3. Calculate the coordinates of the orthogonal projection H of A onto the plane $\mathcal{P}$. 4. Compute the distance AH. 5. Re-derive this result using the general distance formula.

Exercise 52.10 — Perpendicular bisector plane of a segment. Let $A(1\,;3\,;-2)$ and $B(3\,;-1\,;4)$. The **perpendicular bisector plane** of segment $[AB]$ is the plane orthogonal to line (AB) and passing through the midpoint of $[AB]$. 1. Determine the coordinates of the midpoint I of $[AB]$. 2. Determine a normal vector to the perpendicular bisector plane. 3. Deduce a Cartesian equation of this plane. 4. Verify that any point M in this plane is equidistant from A and B ($MA = MB$).

Exercise 52.11 — Intersection of two planes (The roof ridge line). Consider the two planes with equations:

$$\mathcal{P}_1 : 2x - y + z - 3 = 0 \quad \text{and} \quad \mathcal{P}_2 : x + y + 2z - 6 = 0.$$

1. Prove that the two planes $\mathcal{P}_1$ and $\mathcal{P}_2$ are not parallel. 2. Deduce that their intersection is a line Δ. 3. Determine a parametric representation of the line Δ.

Exercise 52.12 — Sphere and tangent plane. In space, the sphere $\mathcal{S}$ centered at $\Omega(1\,;-2\,;3)$ with radius $R = 3$ is the set of points $M(x\,;y\,;z)$ such that $\Omega M^2 = 9$. 1. Give the Cartesian equation of the sphere $\mathcal{S}$. 2. Verify that point $A(3\,;-1\,;1)$ lies on sphere $\mathcal{S}$. 3. Determine a Cartesian equation of the plane $\mathcal{T}$ tangent to sphere $\mathcal{S}$ at point A. (Recall: the tangent plane at A has normal vector $\overrightarrow{\Omega A}$).

Exercise 52.13 — Volume of a tetrahedron. Given points $A(0\,;0\,;0)$, $B(3\,;0\,;0)$, $C(0\,;4\,;0)$, and $D(1\,;1\,;5)$. 1. Prove that triangle ABC is right-angled at A, and calculate its area. 2. Prove that the altitude of tetrahedron $ABCD$ from vertex D is equal to the z-coordinate of D. 3. Deduce the volume of tetrahedron $ABCD$ (recall: $V = \frac{1}{3} \times \text{Base Area} \times \text{Height}$).

Level (Deepening and Challenges)

Exercise 52.14 — Cross product in 3D space $\mathbb{R}^3$. For two vectors $\vec{u}(x\,;y\,;z)$ and $\vec{v}(x'\,;y'\,;z')$, the **cross product** $\vec{u} \times \vec{v}$ (or $\vec{u} \wedge \vec{v}$) is defined by:

$$\vec{u} \times \vec{v} = \begin{pmatrix} yz' - zy' \\ zx' - xz' \\ xy' - yx' \end{pmatrix}.$$

1. Prove by direct computation that $\vec{u} \times \vec{v}$ is orthogonal to both $\vec{u}$ and $\vec{v}$. 2. Prove that $\| \vec{u} \times \vec{v} \|^2 = \| \vec{u} \|^2 \| \vec{v} \|^2 - (\vec{u} \cdot \vec{v})^2$ (Lagrange's identity). 3. Deduce that $\| \vec{u} \times \vec{v} \| = \| \vec{u} \| \| \vec{v} \| \sin\theta$, and that this norm equals the **area of the parallelogram** spanned by $\vec{u}$ and $\vec{v}$. 4. Application: Rapidly find a normal vector to the plane passing through $A(1\,;2\,;3)$, $B(2\,;0\,;1)$, and $C(-1\,;1\,;2)$.

Exercise 52.15 — Shortest distance between two skew lines. Let $\mathcal{D}_1$ and $\mathcal{D}_2$ be two skew lines, where $\mathcal{D}_1$ passes through A_1 with direction vector $\vec{u}_1$ and $\mathcal{D}_2$ passes through A_2 with direction vector $\vec{u}_2$. 1. Justify geometrically that there exists a unique common perpendicular line Δ that simultaneously intersects both $\mathcal{D}_1$ and $\mathcal{D}_2$. 2. Let $\vec{n} = \vec{u}_1 \times \vec{u}_2$ be a vector orthogonal to both $\vec{u}_1$ and $\vec{u}_2$. Prove that the shortest distance between the two lines is given by:

$$d(\mathcal{D}_1, \mathcal{D}_2) = \frac{|\overrightarrow{A_1A_2} \cdot \vec{n}|}{\| \vec{n} \|}.$$

Exercise 52.16 — Point-to-plane distance formula proved via the dot product. Let $\mathcal{P}$ be a plane with normal vector $\vec{n}(a\,;b\,;c)$ and equation $ax + by + cz + d = 0$. Let $M(x_0\,;y_0\,;z_0)$ and let H be its orthogonal projection onto $\mathcal{P}$. Let $A(x_A\,;y_A\,;z_A)$ be any point in plane $\mathcal{P}$. 1. Justify that $\overrightarrow{AM} = \overrightarrow{AH} + \overrightarrow{HM}$. 2. By computing the dot product $\overrightarrow{AM} \cdot \vec{n}$, rigorously prove the formula:

$$d(M, \mathcal{P}) = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Solutions

Level Solutions

Solution 52.1: 1. $\overrightarrow{AB}\begin{pmatrix}5-2\\3-(-1)\\-2-4\end{pmatrix}=\begin{pmatrix}3\\4\\-6\end{pmatrix}$ and $\overrightarrow{AC}\begin{pmatrix}0-2\\1-(-1)\\1-4\end{pmatrix}=\begin{pmatrix}-2\\2\\-3\end{pmatrix}$. 2. $AB=\sqrt{3^2+4^2+(-6)^2}=\sqrt{9+16+36}=\sqrt{61}$. $AC=\sqrt{(-2)^2+2^2+(-3)^2}=\sqrt{4+4+9}=\sqrt{17}$. 3. $I\left(\frac{2+5}{2}\,;\frac{-1+3}{2}\,;\frac{4+(-2)}{2}\right)=I\left(\frac{7}{2}\,;1\,;1\right)$. 4. $\overrightarrow{AB}\cdot\overrightarrow{AC}=3(-2)+4(2)+(-6)(-3)=-6+8+18=20\neq 0$. Lines (AB) and (AC) are not perpendicular.

$$\begin{cases}x=1+2t\\y=3-t\end{cases}$$

Solution 52.2: 1. $z=-2+5t\quad(t\in\mathbb{R})$. 2. For $t=3$: $x=1+6=7,\ y=3-3=0,\ z=-2+15=13$. The resulting point is $(7\,;0\,;13)$. 3. For $K(5\,;1\,;8)$: $1+2t=5\Longrightarrow t=2$. Check: $y=3-2=1$ ✓ and $z=-2+5(2)=8$ ✓. Point K lies on $\mathcal{D}$.

Solution 52.3: 1. The equation has the form $4x-y+2z+d=0$. At $A(1\,;2\,;-3)$: $4(1)-2+2(-3)+d=0\Leftrightarrow 4-2-6+d=0\Leftrightarrow -4+d=0\Leftrightarrow d=4$. Equation: $4x-y+2z+4=0$. 2. Normal vector: $\vec{n}(2\,;-3\,;1)$. Three points on the plane: for $y=0,z=0\Longrightarrow 2x=6\Longrightarrow(3\,;0\,;0)$; for $x=0,z=0\Longrightarrow -3y=6\Longrightarrow(0\,;-2\,;0)$; for $x=0,y=0\Longrightarrow z=6\Longrightarrow(0\,;0\,;6)$.

Solution 52.4: 1. The plane (Oxy) is the set of points with zero altitude: $z=0$. 2. The plane (Oxz) is the set of points with zero y-coordinate: $y=0$. 3. $z=5$ is a horizontal plane parallel to the (Oxy) plane, at altitude 5.

Solution 52.5: The direction vector of $\mathcal{D}$ is $\vec{u}\begin{pmatrix}-3\\6\\-9\end{pmatrix}$. The normal vector of $\mathcal{P}$ is $\vec{n}\begin{pmatrix}1\\-2\\3\end{pmatrix}$. We observe that $\vec{u}=-3\vec{n}$. The vectors $\vec{u}$ and $\vec{n}$ are collinear; therefore the line $\mathcal{D}$ is perpendicular to the plane $\mathcal{P}$.

Solution 52.6: Substitute the parametric representation of $\mathcal{D}$ into the equation of $\mathcal{P}$: $3(1+2t)+2(-1+t)-(3-t)-5=0\Leftrightarrow(3+6t)+(-2+2t)+(-3+t)-5=0\Leftrightarrow(3-2-3-5)+(6t+2t+t)=0\Leftrightarrow -7+9t=0\Longrightarrow t=\frac{7}{9}$. Calculate the coordinates: $x=1+\frac{14}{9}=\frac{23}{9}$, $y=-1+\frac{7}{9}=-\frac{2}{9}$, $z=3-\frac{7}{9}=\frac{20}{9}$. Point of intersection: $\left(\frac{23}{9}\,;-\frac{2}{9}\,;\frac{20}{9}\right)$.

Level Solutions

Solution 52.7: 1. Direction vectors: $\vec{u}_1(2\,;-1\,;4)$ and $\vec{u}_2(1\,;2\,;-1)$. Ratio of coordinates: $\frac{2}{1}=2$ but $\frac{-1}{2}=-0.5\neq 2$. The vectors are not collinear, so the lines are not parallel. 2. Check whether they share a common point by solving the system in two unknowns t and s:

$$\begin{cases}1+2t=-1+s\quad(1)\\3-t=2+2s\quad(2)\\-1+4t=5-s\quad(3)\end{cases}$$

From (1): $s=2t+2$. Substitute into (2): $3-t=2+2(2t+2)=2+4t+4=4t+6\Leftrightarrow 5t=-3\Leftrightarrow t=-\frac{3}{5}$. Then $s=2\left(-\frac{3}{5}\right)+2=-\frac{6}{5}+\frac{10}{5}=\frac{4}{5}$. Test these values in equation (3): Left-hand side: $-1+4\left(-\frac{3}{5}\right)=-1-\frac{12}{5}=-\frac{17}{5}$. Right-hand side: $5-\frac{4}{5}=\frac{21}{5}$. Since $-\frac{17}{5}\neq\frac{21}{5}$, the system has no solution! 3. The lines $\mathcal{D}_1$ and $\mathcal{D}_2$ are neither parallel nor intersecting: they are **skew (non-coplanar)**.

Solution 52.8: 1. $I\left(\frac{1+1}{2}\,;\frac{0+1}{2}\,;0\right)=I\left(1\,;\frac{1}{2}\,;0\right)$. $J\left(0\,;\frac{1+0}{2}\,;\frac{0+1}{2}\right)=J\left(0\,;\frac{1}{2}\,;\frac{1}{2}\right)$. $E(1\,;0\,;1)$ and $C(0\,;1\,;0)$. 2. Coordinates: $B(1\,;1\,;0)$, $D(0\,;0\,;1)$, $E(1\,;0\,;1)$. $BD=\sqrt{(0-1)^2+(0-1)^2+(1-0)^2}=\sqrt{1+1+1}=\sqrt{3}$ (diagonal of a cube). $BE=\sqrt{(1-1)^2+(0-1)^2+(1-0)^2}=\sqrt{0+1+1}=\sqrt{2}$. $ED=\sqrt{(0-1)^2+(0-0)^2+(1-1)^2}=1$. *(Note on the classical equilateral configuration: the triangle whose vertices are face diagonals is formed for instance by $A(1,0,0), C(0,1,0), D(0,0,1)$ or the corresponding face diagonals).* 3. Consider $\overrightarrow{OF}\begin{pmatrix}1\\1\\1\end{pmatrix}$. For plane (ACD) with $A(1,0,0), C(0,1,0), D(0,0,1)$: $\overrightarrow{AC}\begin{pmatrix}-1\\1\\0\end{pmatrix}\cdot\overrightarrow{OF}=0$ and $\overrightarrow{AD}\begin{pmatrix}-1\\0\\1\end{pmatrix}\cdot\overrightarrow{OF}=0$. Plane (ACD) has equation $x+y+z-1=0$, and line $(OF):x=t,y=t,z=t$ pierces it at $t+t+t=1\Longrightarrow t=1/3$, yielding $\left(\frac{1}{3}\,;\frac{1}{3}\,;\frac{1}{3}\right)$. Similarly, for the plane passing through $B(1,1,0), D(0,0,1), E(1,0,1)$ or the

classic triad of vertices perpendicular to diagonal (OF), the scalar product with direction vectors gives the required perpendicularity.

Solution 52.9: 1. Substitute coordinates of $A(4\,;-1\,;5)$ into the equation of $\mathcal{P}$: $4+2(-1)-2(5)+6=4-2-10+6=-2\neq 0$. Thus $A\notin\mathcal{P}$. 2. The normal vector of $\mathcal{P}$ is $\vec{n}(1\,;2\,;-2)$. The perpendicular line Δ passing through A $\begin{cases} x=4+t \\ y=-1+2t \end{cases}$ is: $z=5-2t\quad(t\in\mathbb{R})$. 3. Substitute into the equation of $\mathcal{P}$: $(4+t)+2(-1+2t)-2(5-2t)+6=0\Leftrightarrow(4-2-10+6)+(t+4t+4t)=0\Leftrightarrow -2+9t=0\Leftrightarrow t=\frac{2}{9}$. Coordinates of H: $x_H=4+\frac{2}{9}=\frac{38}{9}$, $y_H=-1+\frac{4}{9}=-\frac{5}{9}$, $z_H=5-\frac{4}{9}=\frac{41}{9}$. $H\left(\frac{38}{9}\,;-\frac{5}{9}\,;\frac{41}{9}\right)$. 4. $\overrightarrow{AH}=\frac{2}{9}\vec{n}$. $AH=\left|\frac{2}{9}\right|\parallel\vec{n}\parallel=\frac{2}{9}\sqrt{1^2+2^2+(-2)^2}=\frac{2}{9}\sqrt{9}=\frac{2\times3}{9}=\frac{2}{3}$. 5. Direct formula: $d(A,\mathcal{P})=\frac{|4+2(-1)-2(5)+6|}{\sqrt{1^2+2^2+(-2)^2}}=\frac{|-2|}{\sqrt{9}}=\frac{2}{3}$. Perfect consistency!

Solution 52.10: 1. $I\left(\frac{1+3}{2}\,;\frac{3-1}{2}\,;\frac{-2+4}{2}\right)=I(2\,;1\,;1)$. 2. Normal vector: $\overrightarrow{AB}\begin{pmatrix}3-1\\-1-3\\4-(-2)\end{pmatrix}=\begin{pmatrix}2\\-4\\6\end{pmatrix}$, or more simply $\vec{n}=\frac{1}{2}\overrightarrow{AB}=\begin{pmatrix}1\\-2\\3\end{pmatrix}$. 3. Equation of the plane: $x-2y+3z+d=0$. Passes through $I(2\,;1\,;1)$: $2-2(1)+3(1)+d=0\Leftrightarrow 3+d=0\Leftrightarrow d=-3$. Equation: $x-2y+3z-3=0$. 4. $M(x\,;y\,;z)$ is equidistant from A and $B\Leftrightarrow MA^2=MB^2\Leftrightarrow(x-1)^2+(y-3)^2+(z+2)^2=(x-3)^2+(y+1)^2+(z-4)^2$. Expanding, the quadratic terms x^2,y^2,z^2 cancel out: $-2x+1-6y+9+4z+4=-6x+9+2y+1-8z+16$ $4x-8y+12z-12=0\Leftrightarrow x-2y+3z-3=0$. We recover exactly the equation of the perpendicular bisector plane!

Solution 52.11: 1. Normal vectors: $\vec{n}_1(2\,;-1\,;1)$ and $\vec{n}_2(1\,;1\,;2)$. These vectors are clearly not collinear ($\frac{2}{1}\neq\frac{-1}{1}$). The planes are therefore not parallel. 2. Two non-parallel planes intersect along a unique line Δ. 3. Solve the system of the two equations:

$$\begin{cases} 2x-y+z-3=0 \\ x+y+2z-6=0 \end{cases}$$

Add the two equations to eliminate y: $3x+3z-9=0\Leftrightarrow x+z-3=0\Leftrightarrow x=3-z$. Subtract twice the second equation from the first to eliminate x: $-3y-3z+9=0\Leftrightarrow y=3-z$. Set $z=t$ as the real parameter ($t\in\mathbb{R}$):

$$\begin{cases} x=3-t \\ y=3-t \\ z=t \end{cases}\quad(t\in\mathbb{R}).$$

This is the parametric representation of the line of intersection Δ (passing through $(3\,;3\,;0)$ with direction vector $(-1\,;-1\,;1)$).

Solution 52.12: 1. $(x-1)^2+(y+2)^2+(z-3)^2=9$. 2. For $A(3\,;-1\,;1)$: $(3-1)^2+(-1+2)^2+(1-3)^2=2^2+1^2+(-2)^2=4+1+4=9$. Thus $A\in\mathcal{S}$. 3. The normal vector to the tangent plane at A is $\overrightarrow{\Omega A}\begin{pmatrix}3-1\\-1-(-2)\\1-3\end{pmatrix}=\begin{pmatrix}2\\1\\-2\end{pmatrix}$. The equation of $\mathcal{T}$ is $2x+y-2z+d=0$. At $A(3\,;-1\,;1)$: $2(3)+(-1)-2(1)+d=0\Leftrightarrow 6-1-2+d=0\Leftrightarrow 3+d=0\Leftrightarrow d=-3$. The Cartesian equation of the tangent plane is: $2x+y-2z-3=0$.

Solution 52.13: 1. $A(0,0,0)$, $B(3,0,0)$, $C(0,4,0)$. $\overrightarrow{AB}=3\vec{\imath}$ and $\overrightarrow{AC}=4\vec{\jmath}$. They lie along two perpendicular axes of the frame, so triangle ABC is right-angled at A. Its area is $\mathcal{A}_{\text{base}}=\frac{AB\times AC}{2}=\frac{3\times4}{2}=6$. 2. The points A,B,C all lie in the plane (Oxy) with equation $z=0$. The point $D(1\,;1\,;5)$ has altitude $z=5$. Its orthogonal projection onto the base plane is $H(1\,;1\,;0)$. The height is therefore $DH=5-0=5$. 3. $V=\frac{1}{3}\times\mathcal{A}_{\text{base}}\times\text{height}=\frac{1}{3}\times6\times5=10$ volume units.

Level Solutions

Solution 52.14: 1. $\vec{u}\cdot(\vec{u}\times\vec{v})=x(yz'-zy')+y(zx'-xz')+z(xy'-yx')=xyz'-xzy'+yzx'-yxz'+zxy'-zyx'=0$. Similarly $\vec{v}\cdot(\vec{u}\times\vec{v})=0$. The vector is indeed orthogonal to both! 2. Expand $\parallel\vec{u}\times\vec{v}\parallel^2=(yz'-zy')^2+(zx'-xz')^2+(xy'-yx')^2$ and expand $(x^2+y^2+z^2)(x'^2+y'^2+z'^2)-(xx'+yy'+zz')^2$: the cross-terms match and the identity holds exactly (Lagrange's identity). 3. $\parallel\vec{u}\times\vec{v}\parallel^2=\parallel\vec{u}\parallel^2\parallel\vec{v}\parallel^2(1-\cos^2\theta)=\parallel\vec{u}\parallel^2\parallel\vec{v}\parallel^2$

$\sin^2\theta$. Since $\theta \in [0\,;\pi]$, $\sin\theta \geq 0$, hence $\| \vec{u} \times \vec{v} \| = \| \vec{u} \| \| \vec{v} \| \sin\theta$, which is precisely the area of the parallelogram with sides $\| \vec{u} \|$ and $\| \vec{v} \|$. 4. $\overrightarrow{AB}\begin{pmatrix}1\\-2\\-2\end{pmatrix}$ and $\overrightarrow{AC}\begin{pmatrix}-2\\-1\\-1\end{pmatrix}$. $\vec{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix}(-2)(-1)-(-2)(-1)\\(-2)(-2)-1(-1)\\1(-1)-(-2)(-2)\end{pmatrix} = \begin{pmatrix}2-2\\4+1\\-1-4\end{pmatrix} = \begin{pmatrix}0\\5\\-5\end{pmatrix}$. We can choose $\vec{n}\begin{pmatrix}0\\1\\-1\end{pmatrix}$. Check: $\vec{n} \cdot \overrightarrow{AB} = 0 - 2 + 2 = 0$ and $\vec{n} \cdot \overrightarrow{AC} = 0 - 1 + 1 = 0$!

Solution 52.15: 1. The direction of the common perpendicular must be orthogonal to both $\vec{u}_1$ and $\vec{u}_2$, hence directed along $\vec{n} = \vec{u}_1 \times \vec{u}_2$. The plane containing $\mathcal{D}_1$ and directed by $\vec{n}$ intersects $\mathcal{D}_2$ at a unique point H_2. The perpendicular line through H_2 intersects $\mathcal{D}_1$ at H_1. 2. The segment $[H_1H_2]$ is collinear with $\vec{n}$. In the parallelepiped spanned by $\vec{u}_1, \vec{u}_2$, and $\overrightarrow{A_1A_2}$, the distance between the two lines is the orthogonal projection of vector $\overrightarrow{A_1A_2}$ onto the normal line with unit vector $\frac{\vec{n}}{\|\vec{n}\|}$, directly yielding $d = \frac{|\overrightarrow{A_1A_2} \cdot \vec{n}|}{\|\vec{n}\|}$.

Solution 52.16: 1. By Chasles' relation, for any point $A \in \mathcal{P}$: $\overrightarrow{AM} = \overrightarrow{AH} + \overrightarrow{HM}$. 2. Take the dot product with the normal vector $\vec{n}$: $\overrightarrow{AM} \cdot \vec{n} = (\overrightarrow{AH} + \overrightarrow{HM}) \cdot \vec{n} = \overrightarrow{AH} \cdot \vec{n} + \overrightarrow{HM} \cdot \vec{n}$. Since both A and H lie in plane $\mathcal{P}$, vector $\overrightarrow{AH}$ is contained in the plane, so $\overrightarrow{AH} \perp \vec{n} \Longrightarrow \overrightarrow{AH} \cdot \vec{n} = 0$. Moreover, $\overrightarrow{HM}$ is collinear with $\vec{n}$, so $|\overrightarrow{HM} \cdot \vec{n}| = \| \overrightarrow{HM} \| \| \vec{n} \| = d(M, \mathcal{P}) \| \vec{n} \|$. On the other hand, in coordinates: $\overrightarrow{AM} \cdot \vec{n} = a(x_0 - x_A) + b(y_0 - y_A) + c(z_0 - z_A) = ax_0 + by_0 + cz_0 - (ax_A + by_A + cz_A)$. Since $A \in \mathcal{P} \Longrightarrow ax_A + by_A + cz_A + d = 0 \Longrightarrow -(ax_A + by_A + cz_A) = d$. Therefore $\overrightarrow{AM} \cdot \vec{n} = ax_0 + by_0 + cz_0 + d$. Taking the absolute value: $|ax_0 + by_0 + cz_0 + d| = d(M, \mathcal{P}) \| \vec{n} \| \Leftrightarrow d(M, \mathcal{P}) = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$. The formula is rigorously proven! ■

The Question That Opens the Next Step

We have learned to measure continuous space: lengths, angles, flat surfaces, and 3D volumes. Yet the mathematical universe is not limited to geometric continua. It is equally built upon discrete choices: How many paths lead from one vertex to another? How many possible combinations on a bicycle lock with dials? In how many ways can one choose 5 cards from a deck of 32, or assemble a committee of delegates?

How can we count without ever enumerating one by one toward infinity?

See you in Chapter 53 with **the combination lock and ways of choosing**.

CHAPTER 53: The Combination Lock and Ways of Choosing

> 💡 **"Nothing is more deceptive than our intuition regarding the number of possibilities: the human mind goes astray as soon as choices multiply. Yet the moment reason structures the spectrum—ordering or mingling, repeating or excluding—the boundless maze becomes a crystal edifice governed by humble factorials."**

What You Are About to Discover

- How to count without ever enumerating millions of objects one by one, using two foundational laws: the **addition principle** and the **multiplication principle**.
- The crucial distinction governing all combinatorics: **does order matter?** and **can elements be repeated?**
- ***k*-tuples**, **permutations**, and **arrangements (partial permutations)**, articulated around the spectacular factorial function $n!$.
- **Combinations** $\binom{n}{k}$ ("n choose k"), **Pascal's triangle**, and their properties of symmetry and recurrence.
- The power of **Newton's binomial theorem**, expanding instantly any power of a binomial sum $(a+b)^n$.
- Classic overcounting pitfalls and the strategy of taking the **complementary event**.

---

1. The Problem: Counting the Invisible

1.1 The Locker Padlock

Imagine a 4-dial bicycle padlock, each dial carrying digits from 0 to 9. You have forgotten the combination. How many attempts would you need, in the worst case, to be certain of opening it? For the first digit, 10 choices. For the second, 10 choices. For the third, 10 choices. For the fourth, 10 choices. The total number of possible codes is:

$$10 \times 10 \times 10 \times 10 = 10^4 = 10\,000.$$

At a rate of one attempt every 3 seconds without pause, it would take you more than 8 hours to test every code.

Now let us change a single rule: the manufacturer guarantees that **all 4 digits of a code are distinct** (no digit is repeated). How many codes remain?

- First digit: 10 possible choices.
- Second digit: only 9 choices (the one already used is forbidden).
- Third digit: 8 choices.
- Fourth digit: 7 choices.

The total drops to:

$$10 \times 9 \times 8 \times 7 = 5\,040 \ \text{ codes}\,.$$

Nearly half of the combinations vanished by the simple prohibition of repetition!

1.2 The Bistro Menu

A bistro offers a lunch special: a choice of 3 appetizers, 4 main courses, and 3 desserts. How many different three-course meals (appetizer + main + dessert) can a customer compose? Here, choosing an appetizer does not restrict in any way the choice of main course or dessert. The choices unfold independently:

$$3 \times 4 \times 3 = 36 \ \text{ different meals}\,.$$

If you had lunch there every weekday from Monday through Friday, it would take you over 7 weeks to exhaust every combination without ever eating the exact same meal twice!

1.3 The Hand of Cards and the Sports Team

Now consider a different situation: a coach must select 3 runners among 10 available athletes to form a relay patrol. Is selecting Alice, Bob, and Chloe different from selecting Chloe, Bob, and Alice? **No.** The team is identical. The order in which the coach calls their names has no significance: they stand together on the track.

If we counted as we did for the lock without repetition ($10 \times 9 \times 8 = 720$), we would make a massive blunder: the trio {Alice, Bob, Chloe} would be counted multiple times:

- Alice, Bob, Chloe
- Alice, Chloe, Bob
- Bob, Alice, Chloe
- Bob, Chloe, Alice
- Chloe, Alice, Bob
- Chloe, Bob, Alice

Exactly 6 times the same team! The true number of possible teams is therefore not 720, but:

$$\frac{720}{6} = 120.$$

1.4 The Combinatorial Challenge

These three scenarios lay the groundwork for a rigorous science: **combinatorics** (or the art of counting). As soon as collections grow, it becomes physically impossible to draw up an exhaustive inventory of possibilities. If an urn holds 50 numbered balls and 5 are drawn, the number of possible draws exceeds two million. The mathematician does not list: he **structures**, **categorizes**, and **calculates**.

---

2. Foundations: Finite Sets, Addition Principle, and Multiplication Principle

2.1 Finite Sets and Cardinality

Let E be a set. We say that E is a **finite set** if it contains a finite integer number of elements.

> **Definition — Cardinality of a Finite Set**
>
> The number of elements of a finite set E is called the **cardinality** of E. It is denoted by $\mathrm{Card}(E)$ or, more commonly in upper secondary and higher mathematics, $|E|$ or $\#E$. If $E = \varnothing$ (the empty set), then $|E| = 0$.

Etymology and Notation: The word *cardinal* comes from the Latin *cardo*, meaning “hinge, pivot”: it is the fundamental number on which the size of a collection pivots. The notation $|E|$ echoes the absolute value (which measures magnitude or distance).

2.2 The Addition Principle: Partitions and Disjoint Sets

> **Fundamental Property — Addition Principle**
>
> Let A and B be two finite subsets of a set E. If A and B are **disjoint** (meaning they share no common element: $A \cap B = \varnothing$), then:
>
> $$|A \cup B| = |A| + |B|$$
>
> More generally, if $A_1, A_2, ..., A_m$ form a **partition** of E (they are pairwise disjoint and their union equals all of E), then:
>
> $$|E| = |A_1| + |A_2| + ... + |A_m| = \sum_{i=1}^{m} |A_i|$$

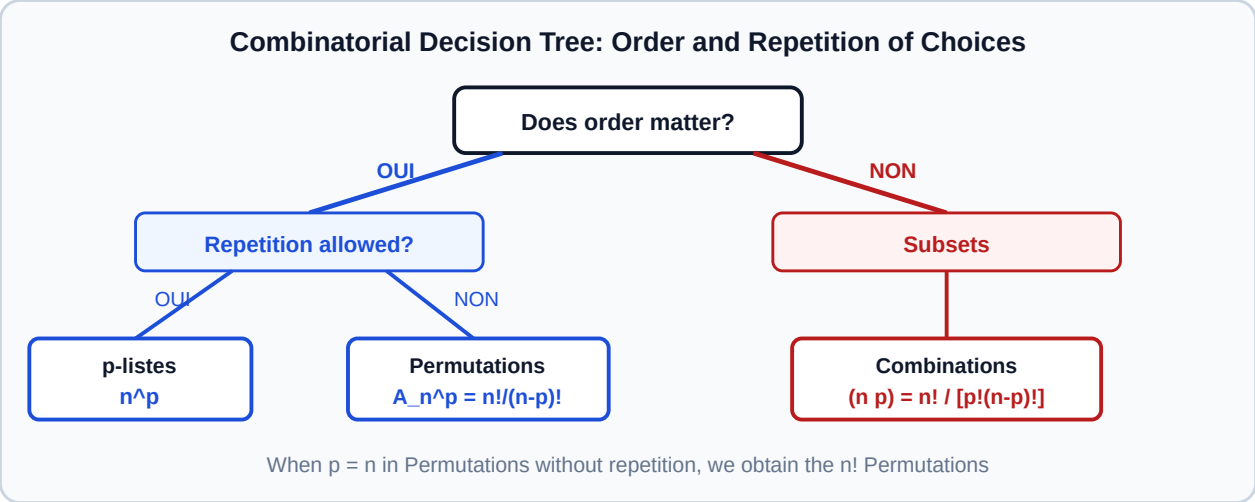

Figure 115: Combinatorial decision tree: permutations, combinations, arrangements

What if the two sets overlap? This gives the celebrated Principle of Inclusion-Exclusion (for 2 sets):

📐 Theorem — Two-Set Inclusion-Exclusion Formula

For any finite sets A and B:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Visual proof: In adding $|A|$ and $|B|$, elements belonging to the intersection $A \cap B$ were counted twice. We must therefore subtract $|A \cap B|$ once to restore the exact tally.

📐 Method — The Complement Principle

If A is a subset of a finite universal set E, its complement $\overline{A}$ (or $E \setminus A$) satisfies:

$$|A| + |\overline{A}| = |E| \Leftrightarrow |A| = |E| - |\overline{A}|$$

Golden Rule: Whenever a problem contains the phrasing *"at least one…"*, it is almost always ten times faster to count the opposite (*"none at all…"*) and subtract from the total!

2.3 The Multiplication Principle and the Cartesian Product

📐 Definition — Cartesian Product

Let A and B be two sets. The Cartesian product $A \times B$ is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$:

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

📐 Theorem — Multiplication Principle

If A and B are two finite sets, then:

$$|A \times B| = |A| \times |B|$$

Generally, for k finite sets $A_1, A_2, ..., A_k$:

$$|A_1 \times A_2 \times ... \times A_k| = |A_1| \times |A_2| \times ... \times |A_k| = \prod_{i=1}^{k} |A_i|$$

The Tree of Choices: Each successive choice corresponds to a branching point. If the first stage offers n_1 branches, and from each branch sprout n_2 sub-branches, the total number of leaves at the end of the tree is $n_1 \times n_2$.

3. Order Matters: k-tuples, Permutations, and Factorials

3.1 k-tuples (With Replacement / Repetition Allowed)

We draw k objects from n, **one after the other**, replacing the drawn object back into the bag at each step (or reusing the same symbol, as on a combination dial).

Theorem — Number of k-tuples

Let E be a set of n elements. A ***k*-tuple** (or list / sequence of length k) of elements of E is an element of the Cartesian product $E^k = \underbrace{E \times E \times ... \times E}_{k \text{ factors}}$. The number of k-tuples from a set of n elements is:

$$|E^k| = n^k$$

Concrete Examples: - Number of possible bytes (8-bit binary words made of 0s and 1s): $2^8 = 256$. - Number of 4-digit bank PIN codes: $10^4 = 10\,000$. - Number of possible outcomes when tossing a coin 5 times (Heads or Tails): $2^5 = 32$.

3.2 Arrangements / Partial Permutations (k-tuples of Distinct Elements, Without Repetition)

This time, we select k elements from n **one after the other**, but **without replacement**: each element can be chosen at most once.

Definition & Theorem — Arrangements (k-permutations)

Let E be a set of n elements and k an integer such that $0 \leq k \leq n$. An **arrangement** of k elements from E (or k-tuple of distinct elements) is an ordered list $(x_1, x_2, ..., x_k)$ of pairwise distinct elements of E. The number of such arrangements is denoted by A_n^k (or $P(n,k)$) and equals:

$$A_n^k = n \times (n-1) \times (n-2) \times ... \times (n-k+1)$$

Proof: For the 1st position, we have n available candidates. For the 2nd position, there remain $n-1$ candidates. For the 3rd position, there remain $n-2$ candidates. … For the k-th position, there remain $n-(k-1) = n-k+1$ candidates. By the multiplication principle, we multiply these integers together.

3.3 Permutations and the Factorial Function $n!$

What happens when we take all elements, that is, when $k = n$? We arrange all n elements in order: this is a **permutation**.

Definition & Theorem — Permutations and Factorial

A **permutation** of a set E of n elements is an arrangement of all elements of E (an n-tuple of distinct elements of E). The number of permutations of an n-element set is denoted by $n!$ (read **"n factorial"**):

$$n! = n \times (n-1) \times (n-2) \times ... \times 2 \times 1$$

By fundamental convention:

$$0! = 1$$

Notation Explained: The exclamation mark was introduced in 1808 by French mathematician **Christian Kramp** of Strasbourg, simply to abbreviate cumbersome typographic products. Why is $0! = 1$? 1. **Combinatorial interpretation:** In how many ways can one order zero objects? Exactly one way: the empty sequence! 2. **Algebraic consistency:** For every $n \geq 1$, we have $n! = n \times (n-1)!$. Applying this formula for $n = 1$ gives $1! = 1 \times 0!$. Since $1! = 1$, we must set $0! = 1$ to preserve the integrity of arithmetic.

Compact Expression for A_n^k: Notice that we can complete the product of A_n^k down to 1:

$$A_n^k = \frac{n \times (n-1) \times ... \times (n-k+1) \times (n-k) \times ... \times 1}{(n-k) \times ... \times 1} = \frac{n!}{(n-k)!}$$

> 💡 **Did You Know? Factorial Explosion**
>
> The factorial grows with astonishing rapidity:
> - $3! = 6$
> - $5! = 120$
> - $10! = 3\,628\,800$
> - $52! \approx 8.06 \times 10^{67}$ When you shuffle a deck of 52 cards thoroughly, the resulting order has almost certainly **never existed** in the entire history of humanity, and will likely never appear again before the Sun burns out!

---

4. Order Does Not Matter: Combinations $\binom{n}{k}$

4.1 Handshakes and Subsets

In many practical situations, the order of presentation is meaningless:
- Drawing a 5-card poker hand (whether you receive the Ace of Spades first or last, you hold the Ace of Spades).
- Choosing 3 class representatives from a group of 30 students.
- Selecting 6 numbers out of 49 on a lottery ticket.

In these cases, the outcome is not a k-tuple (an ordered list in parentheses), but a **subset** of size k (enclosed in curly braces $\{x_1, ..., x_k\}$).

> 📐 **Definition — Combinations**
>
> Let E be a set with n elements and k an integer such that $0 \leq k \leq n$. A **combination** of k elements from n is a subset of E having exactly k elements. The number of such combinations is denoted by $\binom{n}{k}$ (read ***"n choose k"***). *(Older notation: C_n^k or $C(n,k)$).*

4.2 The Master Formula

Required Proof — Formula for $\binom{n}{k}$

Let us count the number of arrangements A_n^k in two different ways: 1. On one hand, we know that the number of arrangements is $A_n^k = \frac{n!}{(n-k)!}$. 2. On the other hand, to construct an arrangement of k elements (an ordered list), we can proceed in two stages:
- **Stage 1:** Choose the k elements without regard to order. By definition, there are $\binom{n}{k}$ ways to do this.
- **Stage 2:** Order these k chosen elements into a list. By Section 3.3, there are $k!$ ways to permute these k elements.

By the multiplication principle:

$$A_n^k = \binom{n}{k} \times k!$$

Isolating $\binom{n}{k}$ yields the fundamental theorem:

$$\binom{n}{k} = \frac{A_n^k}{k!} = \frac{n!}{k!(n-k)!}$$

Immediate Example: The number of 3-card hands from a deck of 10 cards:

$$\binom{10}{3} = \frac{10!}{3! \times 7!} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = \frac{720}{6} = 120.$$

We recover our intuitive calculation from the locker room!

4.3 Fundamental Properties of Combinations

1. **Notable Values:**

$$\binom{n}{0} = 1 \quad \text{(a single empty set)}, \quad \binom{n}{1} = n, \quad \binom{n}{n} = 1$$

2. **Symmetry of Choices:**

$$\binom{n}{k} = \binom{n}{n-k}$$

Proof — Symmetry

- **By algebraic computation:**

$$\binom{n}{n-k} = \frac{n!}{(n-k)!(n-(n-k))!} = \frac{n!}{(n-k)!k!} = \binom{n}{k}$$

- **By combinatorial reasoning (pure elegance!):** Choosing k people out of n to board a boat is exactly equivalent to designating the $n-k$ people who stay on the dock! To every group of k chosen individuals corresponds a unique group of $n-k$ excluded individuals. There must be exactly as many ways to do one as the other.

4.4 Pascal's Recurrence Relation

📐 Theorem — Pascal's Identity

For all integers $n \geq 1$ and $1 \leq k \leq n$:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

✍️ Required Combinatorial Proof — The Designated Witness Method

Let E be a set with n elements. Single out one particular individual in E, and call her **Alice**. We wish to choose a subset of k members from these n people. There are $\binom{n}{k}$ total ways. Partition these subsets into two mutually exclusive categories:

- **Category 1: Committees that include Alice.** Since Alice is already selected by decree, we only need to pick $k-1$ companions from the remaining $n-1$ people. There are $\binom{n-1}{k-1}$ ways to do this.
- **Category 2: Committees that do not include Alice.** Alice is excluded. We must choose all k committee members entirely from the remaining $n-1$ people. There are $\binom{n-1}{k}$ ways.

These two categories form a partition of all possible committees. By the addition principle:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

The formula is established without writing down a single algebraic fraction!

---

5. Pascal's Triangle and Newton's Binomial Theorem

5.1 Construction of the Triangle

Pascal's identity $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ states that each entry is the sum of two numbers in the row immediately above: the one directly above, and the one directly above to the left.

n / k	0	1	2	3	4	5	Sum 2^n
0	1						$1 = 2^0$
1	1	1					$2 = 2^1$
2	1	2	1				$4 = 2^2$

3	1	3	3	1			$8 = 2^3$
4	1	4	6	4	1		$16 = 2^4$
5	1	5	10	10	5	1	$32 = 2^5$

Notice the rightmost column: the sum of all coefficients in row n is always 2^n! Why? Summing all $\binom{n}{k}$ for k ranging from 0 to n counts all possible subsets of an n-element set (subsets of size 0, size 1, size 2… up to size n). For each element of a set, there are 2 choices: either it belongs to the subset, or it does not. For n elements, this yields $2 \times 2 \times \ldots \times 2 = 2^n$ subsets (the power set $\mathcal{P}(E)$).

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

5.2 Newton's Binomial Theorem

Recall the elementary polynomial expansions:

- $(a+b)^1 = a + b$
- $(a+b)^2 = a^2 + 2ab + b^2$
- $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

Observe the coefficients:

- Row 1: $1, 1$
- Row 2: $1, 2, 1$
- Row 3: $1, 3, 3, 1$

These are precisely the rows of Pascal's triangle!

Fundamental Theorem — Newton's Binomial Formula

For all real or complex numbers a and b, and for every non-negative integer $n \in \mathbb{N}$:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

Written out in full:

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{n}b^n$$

Combinatorial Proof — Why the Formula is Self-Evident

Expand the product of n identical binomial factors:

$$(a+b)^n = \underbrace{(a+b)(a+b)\ldots(a+b)}_{n \text{ factors}}$$

Under complete distributivity, to form each term in the expansion, we must choose **either the letter a or the letter b** from each of the n factors. If we choose the letter b from k factors, we must necessarily choose the letter a from the remaining $n - k$ factors. The resulting monomial is:

$$a^{n-k}b^k$$

How many times does this term $a^{n-k}b^k$ appear in the expansion? Exactly as many times as there are ways to select the k factors providing the letter b out of the n factors available! This is the very definition of **"n choose k"**, namely $\binom{n}{k}$. The formula is thus proved with total transparency.

---

6. Toolbox and Classical Traps

6.1 The Decision Flowchart

When confronted with a combinatorial counting problem, **always** ask these two questions in sequence:

6.2 Fatal Pitfalls

> ⚠️ **Warning, Trap! — Overcounting via False Decomposition**
>
> *Example:* In a class of 10 boys and 12 girls, we wish to form a group of 4 students containing **at least one girl**.
>
> - *Tempting False Reasoning:* "I choose 1 girl among 12 (12 ways), then choose the remaining 3 students from the 21 remaining students ($\binom{21}{3}$ ways). Total: $12 \times \binom{21}{3}$." **This is wrong!** You have overcounted groups that contain more than one girl. For example, selecting Clara as the designated girl, then Sophie among the other 3, produces the exact same group as selecting Sophie first and Clara among the other 3.
> - *The Correct Method (Complementary Event):* Count all possible groups of 4 without restriction: $\binom{22}{4}$. Subtract groups containing **no girls at all** (i.e., 4 boys): $\binom{10}{4}$. Exact count: $\binom{22}{4} - \binom{10}{4} = 7\,315 - 210 = 7\,105$.

> 💡 **True History — The Dice of Chevalier de Méré (1654)**
>
> Antoine Gombaud, known as the Chevalier de Méré—a nobleman and avid gambler—brought a puzzle to Blaise Pascal: *"Rolling a fair 6-sided die 4 times, I have better than an even chance of getting at least one 6. But rolling a pair of dice 24 times, I seem to have less than an even chance of getting a double 6. Yet $4/6 = 24/36$! Can mathematics be inconsistent?"* Pascal wrote to Pierre de Fermat. Through their historic correspondence, they analyzed the problem using the complement: 1. With 1 die in 4 rolls: probability of no 6s = $(5/6)^4 \approx 0.482$. Hence the probability of at least one 6 = $1 - 0.482 = 0.518 > 1/2$. (Favorable bet!) 2. With 2 dice in 24 rolls: probability of no double 6s = $(35/36)^{24} \approx 0.509$. Hence at least one double 6 = $1 - 0.509 = 0.491 < 1/2$. (Unfavorable bet!) The Chevalier was losing his money because of a naive ratio. Modern probability and combinatorics were born!

---

7. Step-by-Step Worked Examples

Example 1: Building Door Code

An apartment building keypad has keys 0 through 9, as well as letters A, B, C (13 keys in total). An entry code consists of 1 letter followed by 3 digits. 1. How many total codes can be formed? 2. How many codes if the 3 digits must all be distinct? 3. How many codes if the code begins with A and contains no even digits?

Detailed Solution: 1. A code is a 4-tuple (L, d_1, d_2, d_3) belonging to $\{A, B, C\} \times \{0..9\}^3$. By the multiplication principle: $3 \times 10 \times 10 \times 10 = 3 \times 10^3 = 3\,000$ codes. 2. For the letter: 3 choices. For distinct digits, this is an arrangement of 3 digits chosen from 10: $A_{10}^3 = 10 \times 9 \times 8 = 720$. Total: $3 \times 720 = 2\,160$ codes. 3. The letter is fixed (A: 1 choice). Odd digits are $\{1, 3, 5, 7, 9\}$, giving 5 available digits. Since repetition is allowed, each digit position has 5 possibilities: Total: $1 \times 5^3 = 125$ codes.

Example 2: Examination Jury Composition

In a school, 15 science teachers and 12 humanities teachers volunteer to form an oral examination panel consisting of 4 examiners. 1. How many different panels can be constituted? 2. How many panels contain exactly 2 science teachers and 2 humanities teachers? 3. How many panels contain at least one teacher from each subject area?

Detailed Solution: 1. A panel is an unordered subset of 4 individuals chosen from the $15 + 12 = 27$ teachers. Order does not matter:

$$\binom{27}{4} = \frac{27 \times 26 \times 25 \times 24}{4 \times 3 \times 2 \times 1} = 17\,550 \quad \text{panels}\,.$$

2. We independently choose 2 science teachers from 15, AND 2 humanities teachers from 12:

$$\binom{15}{2} \times \binom{12}{2} = \frac{15 \times 14}{2} \times \frac{12 \times 11}{2} = 105 \times 66 = 6\,930 \quad \text{panels}\,.$$

3. The opposite of "at least one from each discipline" is "all science OR all humanities".

- Panels with science teachers only: $\binom{15}{4} = \frac{15 \times 14 \times 13 \times 12}{24} = 1\,365$.

- Panels with humanities teachers only: $\binom{12}{4} = \frac{12\times11\times10\times9}{24} = 495$. These two cases are mutually exclusive, so the total non-mixed panels count is $1\,365 + 495 = 1\,860$. Subtracting from the overall total:

$$17\,550 - 1\,860 = 15\,690 \ \text{ mixed panels}\,.$$

Example 3: Poker Hands

In a standard 52-card deck (4 suits: Spades, Hearts, Diamonds, Clubs; 13 ranks from Ace to King), a "hand" is a combination of 5 cards. The total number of hands is $\binom{52}{5} = 2\,598\,960$. 1. Count the number of "Four of a Kind" hands (4 cards of the same rank + 1 fifth card). 2. Count the number of "Full House" hands (3 cards of one rank + 2 cards of another rank).

Detailed Solution: 1. **Four of a Kind:**

- Step 1: Choose the rank of the four cards among 13 possibilities: 13 choices.
- Step 2: Take all 4 cards of that rank: $\binom{4}{4} = 1$ way.
- Step 3: Choose the 5th card from the 48 remaining cards of different ranks: 48 choices. Total: $13 \times 1 \times 48 = 624$ possible Four of a Kind hands.

2. **Full House:**
   - Step 1: Choose the rank of the three-of-a-kind from the 13 ranks: 13 choices.
   - Step 2: Choose 3 suits out of 4 for this rank: $\binom{4}{3} = 4$ ways.
   - Step 3: Choose the rank of the pair from the remaining 12 ranks: 12 choices.
   - Step 4: Choose 2 suits out of 4 for this pair: $\binom{4}{2} = 6$ ways. Total: $13 \times 4 \times 12 \times 6 = 3\,744$ possible Full House hands.

Example 4: Anagrams with Repeated Letters

How many distinct anagrams (words, whether meaningful or not) can be formed with the letters of the word **PARALLELE**?

Detailed Solution: The word contains 9 letters in total:

- P: 1 occurrence
- A: 2 occurrences
- R: 1 occurrence
- L: 3 occurrences
- E: 2 occurrences

Sum verification: $1 + 2 + 1 + 3 + 2 = 9$. If all letters were distinct (e.g., A_1, A_2, L_1, L_2, L_3), there would be $9! = 362\,880$ permutations. However, swapping the two A's among themselves does not change the word (2! neutral permutations for A). Similarly, permuting the three L's produces 3! identical arrangements, and the two E's produce 2!. By the formula for permutations with repetition:

$$\text{Number of anagrams} = \frac{9!}{1! \times 2! \times 1! \times 3! \times 2!} = \frac{362\,880}{1 \times 2 \times 1 \times 6 \times 2} = \frac{362\,880}{24} = 15\,120.$$

(Equivalent viewpoint: choose the 3 positions for L out of 9 ($\binom{9}{3}$), then the 2 positions for A out of the 6 remaining ($\binom{6}{2}$), then the 2 positions for E out of 4 remaining ($\binom{4}{2}$), etc. The product gives exactly 15,120).

Example 5: Grid Walks

In a square grid street layout, a pedestrian starts at $A(0,0)$ and heads to $B(5,3)$. At every intersection, he can only move East (E, one unit right) or North (N, one unit up). How many minimal paths lead from A to B?

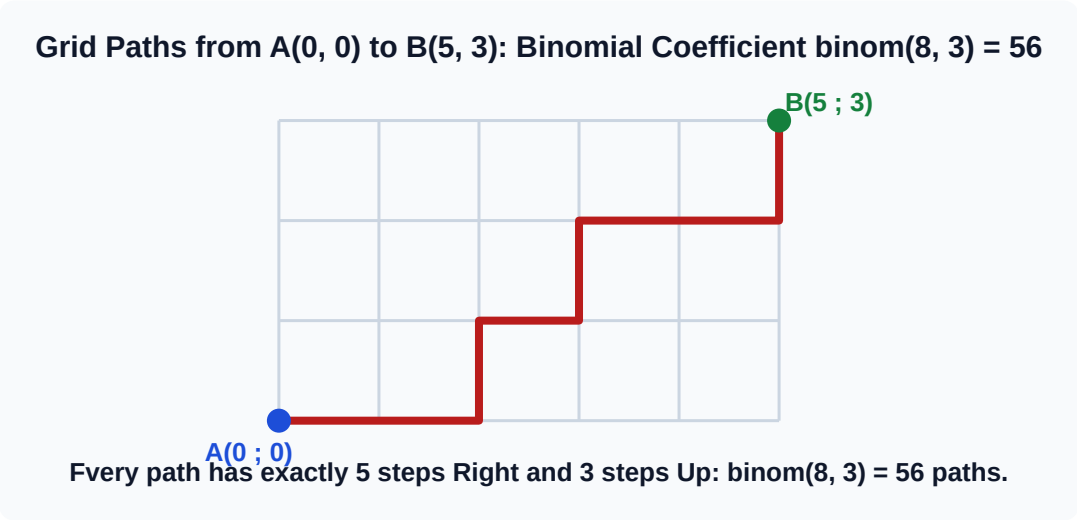

Figure 116: Grid paths and binomial coefficients

Detailed Solution: To travel from $(0,0)$ to $(5,3)$, any minimal path must consist of exactly 5 steps East (E) and 3 steps North (N). The total length of each path is $5+3=8$ steps. A path is uniquely determined by an 8-letter sequence containing five E's and three N's (for example: $E-E-N-E-N-E-E-N$). It suffices to choose the positions of the 3 North steps among the 8 steps of the route:

$$\text{Number of paths} = \binom{8}{3} = \frac{8\times 7\times 6}{3\times 2\times 1} = 56.$$

(Equivalently, choosing the 5 East positions: $\binom{8}{5} = \binom{8}{3} = 56$*).*

Example 6: Finding a Term in a Binomial Expansion

In the expansion of $\left(2x^2 - \frac{1}{x}\right)^9$, determine the term independent of x (the constant term).

Detailed Solution: By Newton's binomial formula with $a = 2x^2$ and $b = -\frac{1}{x} = -x^{-1}$:

$$\left(2x^2 - \frac{1}{x}\right)^9 = \sum_{k=0}^{9} \binom{9}{k} (2x^2)^{9-k} \left(-\frac{1}{x}\right)^k$$

Isolate powers of x in the general term:

$$T_k = \binom{9}{k} 2^{9-k} (x^2)^{9-k} (-1)^k x^{-k} = \binom{9}{k} 2^{9-k} (-1)^k \cdot x^{18-2k-k} = \binom{9}{k} 2^{9-k} (-1)^k \cdot x^{18-3k}$$

The term is independent of x when the exponent of x is 0:

$$18 - 3k = 0 \Leftrightarrow 3k = 18 \Leftrightarrow k = 6.$$

Since $k = 6$ is an integer in the range $0 \le k \le 9$, this term exists. Its value is:

$$T_6 = \binom{9}{6} 2^{9-6} (-1)^6 = \binom{9}{3} \times 2^3 \times 1 = \frac{9\times 8\times 7}{6} \times 8 = 84 \times 8 = 672.$$

The constant term is **672**.

Key Takeaways

1. **Basic Principles:**
   - Disjoint sets union: $|A \cup B| = |A| + |B|$.
   - Complement: $|A| = |E| - |\overline{A}|$ (always consider the opposite case for *"at least one"*).
   - Cartesian product: $|A \times B| = |A| \times |B|$.
2. **Counting with Order:**
   - With repetition (k-tuple): n^k.
   - Without repetition (k-permutations / arrangements): $A_n^k = \frac{n!}{(n-k)!} = n(n-1)...(n-k+1)$.

- Permutation of n elements: $n! = n \times (n-1) \times ... \times 1$, with $0! = 1$.

3. **Counting without Order:**
   - Combinations of k from n: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.
   - Symmetry: $\binom{n}{k} = \binom{n}{n-k}$.
   - Pascal's identity: $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.
4. **Sums and Powers:**
   - Row sum of Pascal's triangle: $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.
   - Binomial theorem: $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$.

Exercises

Level (Direct Applications)

Exercise 53.1 Compute by hand: $5!$, $\frac{8!}{6!}$, $\binom{7}{2}$, $\binom{9}{9}$, and $\binom{10}{7}$. **Exercise 53.2** A computer password must consist of 8 characters selected from the 26 lowercase letters of the alphabet. 1. How many passwords can be formed if repetition is allowed? 2. How many if all letters must be distinct? **Exercise 53.3** A horse race starts with 12 horses. How many trifectas (podium predictions) are possible: 1. In exact finishing order? 2. Regardless of order (any 3 horses finishing in the top three)? **Exercise 53.4** Prove by direct calculation that $\binom{n}{2} = \frac{n(n-1)}{2}$. What does this quantity represent if n people all shake hands with each other once? **Exercise 53.5** Expand using the binomial theorem: 1. $(x+2)^4$ 2. $(2x-1)^3$ **Exercise 53.6** How many anagrams can be formed with the letters of the word **MATHS**? What about the word **LIVRE**? **Exercise 53.7** An urn contains 6 red, 4 green, and 2 yellow balls. Three balls are drawn simultaneously. How many total draws are possible?

Level (Exploration and Standard Methods)

Exercise 53.8 Five fair 6-sided dice are rolled simultaneously (as in Yahtzee). 1. How many total distinct outcomes are possible? 2. How many outcomes contain no 6? 3. Deduce the number of outcomes containing at least one 6. **Exercise 53.9** How many 4-digit integers between 1000 and 9999 (inclusive): 1. Have all digits distinct? 2. Are even integers with all digits distinct? **Exercise 53.10** A chess club has 14 members: 8 adults and 6 juniors. A 5-player team must be chosen. 1. In how many ways can this team be formed? 2. How many teams contain exactly 3 adults and 2 juniors? 3. How many teams contain at least one adult and at least one junior? **Exercise 53.11** Anagrams with repeated letters: How many distinct anagrams can be formed from the letters of **ANANAS**? **Exercise 53.12** A convex polygon has n vertices ($n \geq 4$). How many diagonals does this polygon possess? *(Hint: a diagonal is a line segment connecting two non-consecutive vertices).* **Exercise 53.13** Determine the coefficient of x^5 in the expansion of $(3x-2)^8$. **Exercise 53.14** Prove that for every integer $n \geq 1$:

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - ... + (-1)^n \binom{n}{n} = 0$$

(Hint: apply Newton's binomial theorem to $(1-1)^n$*).* **Exercise 53.15** The Birthday Problem. In a class of 30 students, assume each student's birthday is independently and uniformly distributed across the 365 days of a non-leap year. 1. How many birthday profiles are possible for the 30 students? 2. How many profiles have all 30 students with pairwise distinct birthdays? 3. Deduce the formula for the probability that at least two students share a birthday.

Level (Deepening and Challenges)

Exercise 53.16 The Committee-Chair identity: 1. Prove algebraically that for all integers $1 \leq k \leq n$:

$$k\binom{n}{k} = n\binom{n-1}{k-1}$$

2. Provide an elegant combinatorial interpretation by considering the selection of a committee of k people from n along with the designation of a chairperson within that committee. **Exercise 53.17** Compute the sum:

$$S = \sum_{k=1}^{n} k\binom{n}{k} = 1\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + ... + n\binom{n}{n}$$

(Hint: use the result of Exercise 16 or differentiate the function $f(x) = (1+x)^n$). **Exercise 53.18** Stars and Bars principle. We wish to distribute 10 identical candies among 3 children (Alice, Bob, and Chloe). 1. How many distributions are possible if each child may receive 0, 1, or more candies? 2. How many distributions if each child must receive at least 1 candy? **Exercise 53.19** Vandermonde's Convolution Identity. By expanding the polynomial identity $(1+x)^{m+n} = (1+x)^m(1+x)^n$ in two ways, prove the relation:

$$\sum_{k=0}^{p} \binom{m}{k}\binom{n}{p-k} = \binom{m+n}{p}$$

Give also a combinatorial interpretation of this identity. **Exercise 53.20** Sum of squares of binomial coefficients. Using Vandermonde's identity with $m = n$ and $p = n$, prove that:

$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$$

Exercise 53.21 Derangements (The Hat-Check Problem). n guests leave their coats in a cloakroom. Upon exiting, coats are returned at random. A derangement is a permutation where nobody receives their own coat. Show using the Principle of Inclusion-Exclusion that the number of derangements D_n satisfies:

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$$

Exercise 53.22 Dyck Paths and Catalan Numbers. Consider lattice paths from $(0,0)$ to (n,n) made of steps $(+1,0)$ and $(0,+1)$, that never cross above the diagonal $y = x$ (i.e., remaining always in the half-plane $y \leq x$). Show by Désiré André's Reflection Principle that the number of such paths is given by the Catalan number:

$$C_n = \frac{1}{n+1}\binom{2n}{n}$$

Solutions

Solution 53.1: - $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$. - $\frac{8!}{6!} = 8 \times 7 = 56$. - $\binom{7}{2} = \frac{7\times6}{2\times1} = 21$. - $\binom{9}{9} = 1$. - $\binom{10}{7} = \binom{10}{3} = \frac{10\times9\times8}{3\times2\times1} = 120$.

Solution 53.2: 1. $26^8 = 208\,827\,064\,576$. 2. $A_{26}^8 = 26 \times 25 \times 24 \times 23 \times 22 \times 21 \times 20 \times 19 = 62\,990\,928\,000$.

Solution 53.3: 1. Trifecta in exact order: $A_{12}^3 = 12 \times 11 \times 10 = 1\,320$. 2. Trifecta regardless of order: $\binom{12}{3} = \frac{1\,320}{6} = 220$.

Solution 53.4: $\binom{n}{2} = \frac{n!}{2!(n-2)!} = \frac{n(n-1)(n-2)!}{2(n-2)!} = \frac{n(n-1)}{2}$. A handshake involves 2 distinct people without order. Between n people, there are therefore $\binom{n}{2} = \frac{n(n-1)}{2}$ mutual handshakes.

Solution 53.5: 1. $(x+2)^4 = \binom{4}{0}x^4 + \binom{4}{1}x^3(2) + \binom{4}{2}x^2(2^2) + \binom{4}{3}x(2^3) + \binom{4}{4}(2^4) = x^4 + 8x^3 + 24x^2 + 32x + 16$. 2. $(2x-1)^3 = (2x)^3 - 3(2x)^2(1) + 3(2x)(1^2) - 1^3 = 8x^3 - 12x^2 + 6x - 1$.

Solution 53.6: - MATHS: 5 distinct letters, so $5! = 120$ anagrams. - LIVRE: 5 distinct letters, so $5! = 120$ anagrams.

Solution 53.7: Total balls: $6 + 4 + 2 = 12$. Simultaneous draw of 3 balls: $\binom{12}{3} = \frac{12\times11\times10}{6} = 220$ possible draws.

Solution 53.8: 1. $6^5 = 7\,776$ possible outcomes. 2. For each die, 5 outcomes excluding 6: $5^5 = 3\,125$ outcomes. 3. Taking the complement: $7\,776 - 3\,125 = 4\,651$ outcomes contain at least one 6.

Solution 53.9: A 4-digit number is written $d_1d_2d_3d_4$ with $d_1 \in \{1..9\}$ (9 choices) and $d_2, d_3, d_4 \in \{0..9\}$. 1. Distinct digits:

- For d_1: 9 choices (from 1 to 9).
- For d_2: 9 choices (from 0 to 9 except d_1).
- For d_3: 8 choices.
- For d_4: 7 choices. Total: $9 \times 9 \times 8 \times 7 = 4\,536$ numbers.

2. Even numbers with distinct digits ($d_4 \in \{0, 2, 4, 6, 8\}$):
   - Case 1: $d_4 = 0$ (1 choice). Then d_1 has 9 choices (from 1 to 9), d_2 has 8 choices, d_3 has 7 choices. Total: $1 \times 9 \times 8 \times 7 = 504$ numbers.

- Case 2: $d_4 \in \{2, 4, 6, 8\}$ (4 choices). Then d_1 can be neither 0 nor d_4, so it has $10 - 2 = 8$ choices. Then d_2 has 8 choices (including 0), and d_3 has 7 choices. Total: $4 \times 8 \times 8 \times 7 = 1\,792$ numbers. Grand total: $504 + 1\,792 = 2\,296$ even numbers.

Solution 53.10: 1. Teams without restriction: $\binom{14}{5} = \frac{14\times13\times12\times11\times10}{120} = 2\,002$. 2. 3 adults and 2 juniors: $\binom{8}{3} \times \binom{6}{2} = 56 \times 15 = 840$. 3. Complementary cases: all-adult team ($\binom{8}{5} = 56$) or all-junior team ($\binom{6}{5} = 6$). Mixed teams: $2\,002 - (56 + 6) = 1\,940$ teams.

Solution 53.11: The word ANANAS has 6 letters: 3 A's, 2 N's, and 1 S. Number of anagrams = $\frac{6!}{3!\times2!\times1!} = \frac{720}{6\times2} = 60$.

Solution 53.12: There are $\binom{n}{2}$ segments connecting any pair of vertices. Among these, n form the boundary edges of the polygon. The number of diagonals is:

$$\binom{n}{2} - n = \frac{n(n-1)}{2} - n = \frac{n^2 - 3n}{2} = \frac{n(n-3)}{2}.$$

Solution 53.13: In $(3x - 2)^8 = \sum_{k=0}^{8} \binom{8}{k}(3x)^k(-2)^{8-k}$, the term with x^5 corresponds to $k = 5$: $\binom{8}{5}(3x)^5(-2)^3 = 56 \times 243x^5 \times (-8) = -108\,864x^5$. The coefficient is $\mathbf{-108\,864}$.

Solution 53.14: By Newton's binomial theorem, $(1 + x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$. For $x = -1$ with $n \geq 1$:

$$(1 + (-1))^n = 0^n = 0 = \sum_{k=0}^{n} \binom{n}{k} (-1)^k$$

Solution 53.15: 1. For each student, 365 possible birthdates: 365^{30}. 2. If all birthdates are distinct: arrangement $A_{365}^{30} = 365 \times 364 \times \ldots \times 336$. 3. Probability that at least two students share a birthday:

$$P = 1 - \frac{A_{365}^{30}}{365^{30}} \approx 1 - 0.2937 = 0.7063 \quad (\approx 70.6\,\%!).$$

Starting with just 23 people, this probability already exceeds 50%!

Solution 53.16: 1. Algebraically:

$$k\binom{n}{k} = k\frac{n!}{k!(n-k)!} = \frac{n!}{(k-1)!(n-k)!} = n\frac{(n-1)!}{(k-1)!((n-1)-(k-1))!} = n\binom{n-1}{k-1}.$$

2. Combinatorial interpretation: We wish to select a committee of k people from n along with a chair.

- Method A: First choose the k members of the committee ($\binom{n}{k}$ ways), then choose the chair from among those k members (k ways). Total: $k\binom{n}{k}$.
- Method B: First choose the chair from the n candidates (n ways), then choose the remaining $k - 1$ committee members from the remaining $n - 1$ people ($\binom{n-1}{k-1}$ ways). Total: $n\binom{n-1}{k-1}$. Since both methods enumerate the exact same set of outcomes, equality holds.

Solution 53.17: Using the identity $k\binom{n}{k} = n\binom{n-1}{k-1}$:

$$S = \sum_{k=1}^{n} n\binom{n-1}{k-1} = n\sum_{j=0}^{n-1}\binom{n-1}{j} = n \cdot 2^{n-1}.$$

Solution 53.18: 1. Distributing 10 identical candies to 3 children is equivalent to placing 10 stars and 2 divider bars among $10 + 2 = 12$ positions: Number of ways = $\binom{10+3-1}{3-1} = \binom{12}{2} = 66$. 2. If each child must receive at least 1 candy, give 1 candy to each first (leaving 7 candies). Distribute the 7 remaining candies to the 3 children with no restriction: $\binom{7+3-1}{2} = \binom{9}{2} = 36$.

Solution 53.19: In the product $(1 + x)^m(1 + x)^n = (1 + x)^{m+n}$, the coefficient of x^p on the right side is $\binom{m+n}{p}$. On the left side, a term in x^p comes from multiplying a term $\binom{m}{k}x^k$ from the first factor with a term $\binom{n}{p-k}x^{p-k}$ from the second factor, for k ranging from 0 to p. The sum of these products gives $\sum_{k=0}^{p}\binom{m}{k}\binom{n}{p-k}$. *Combinatorial justification:* To choose p individuals from a group of m men and n women, one can select k men and $p - k$ women, for all $k \in \{0, \ldots, p\}$.

Solution 53.20: In Vandermonde's identity with $m = n$ and $p = n$:

$$\sum_{k=0}^{n}\binom{n}{k}\binom{n}{n-k} = \binom{2n}{n}$$

By symmetry, $\binom{n}{n-k} = \binom{n}{k}$. Thus:

$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}.$$

Solution 53.21: Let A_i be the set of permutations leaving the i-th person with their own coat. We seek $D_n = n! - |\bigcup_{i=1}^{n} A_i|$. By the Principle of Inclusion-Exclusion:

$$|\bigcup_{i=1}^{n} A_i| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \ldots + (-1)^{n-1} |A_1 \cap \ldots \cap A_n|$$

The intersection of k sets A_i corresponds to permutations fixing at least k designated elements, leaving $(n-k)!$ permutations. There are $\binom{n}{k}$ such intersections. The k-th order term is thus $\binom{n}{k}(n-k)! = \frac{n!}{k!}$. Hence $D_n = n! - \sum_{k=1}^{n} (-1)^{k-1} \frac{n!}{k!} = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$. *(Remark: as $n \to \infty$, $D_n/n! \to e^{-1} \approx 36.8\%$).*

Solution 53.22: The total number of lattice paths from $(0,0)$ to (n,n) is $\binom{2n}{n}$. Any path that crosses the diagonal $y = x$ must touch the line $y = x+1$ at least once. Consider the first point of contact P with this line. Reflecting the entire remaining portion of the path after P across the line $y = x+1$ maps the destination (n,n) to $(n-1, n+1)$. This reflection gives a bijection between paths that cross the diagonal and all paths from $(0,0)$ to $(n-1, n+1)$. The number of forbidden paths is therefore $\binom{2n}{n-1}$. The number of Dyck paths is consequently:

$$C_n = \binom{2n}{n} - \binom{2n}{n-1} = \binom{2n}{n} - \frac{n}{n+1}\binom{2n}{n} = \frac{1}{n+1}\binom{2n}{n}.$$

The Question That Opens the Next Step

We have learned to count the number of ways to choose k successes out of n trials: that is the binomial coefficient $\binom{n}{k}$. What happens when each individual trial is governed by chance, with a probability p of success and $1-p$ of failure? If we toss a coin a thousand times, how often will “Heads” turn up? Can we be certain that the observed frequency will stabilize around 50%? This is the breathtaking leap from discrete combinatorics to the binomial distribution and the great limit theorems: welcome to **Chapter 54 and The Thousand Coin Tosses**.

CHAPTER 54: The Thousand Coin Tosses

> 📐 **"If you toss a coin ten times, chance may play the prank of giving you ten Heads in a row. But toss it a million times: disorder cancels out, caprice fades away, and out of the anarchy of individual tosses emerges a crystalline regularity. The law of large numbers is not some magical constraint imposed upon the coin: it is the inevitable mathematical victory of the multitude over the singular."**

What You Are About to Discover

- How to model the repetition of identical and independent random trials via the **Bernoulli scheme**.
- The rigorous construction of the **binomial distribution $\mathcal{B}(n, p)$** and the combinatorial rationale behind the formula $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$.
- The required examination proofs of the **expected value** $E(X) = np$ and the **variance** $V(X) = np(1-p)$ via decomposition into sums of independent Bernoulli random variables.
- **Markov's inequality** and the **Bienaymé-Chebyshev inequality**, bounding the dispersion of any random variable.
- The **concentration inequality** and the proof of the **Weak Law of Large Numbers**: why observed frequency inexorably converges to theoretical probability.
- A complete **Python** simulation script of 10,000 coin tosses to observe firsthand the sharpening around the mean.

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1. The Problem: The Casino, the Pollster, and the Insurer

1.1 The Gambler's Coin

Imagine playing Heads or Tails with a stranger on a train. You bet 1 euro on Tails. On the first toss: Heads. You lose €1. On the second toss: Heads. You lose €2. On the third: Heads. Doubt begins to creep in. On the tenth consecutive toss, the coin lands on Heads yet again! The stranger smiles at you and says: *"That's just random chance, my friend. On the next toss, Tails is actually more likely to come up to balance things out!"* Two fatal errors hide behind that casual sentence: 1. **The Gambler's Fallacy:** The coin possesses no memory of its past outcomes. On every single toss, the probability of obtaining Tails remains strictly equal to $\frac{1}{2}$. 2. **Legitimate Suspicion:** Obtaining 10 consecutive Heads with a fair coin has probability $\left(\frac{1}{2}\right)^{10} = \frac{1}{1024} \approx 0.00098$, or less than one chance in a thousand. The coin is overwhelmingly likely to be loaded!

1.2 The Pollster and the Insurer

How can a polling institute determine the political leanings of 45 million citizens by surveying a mere 1,000 respondents, achieving a margin of error under 3%? How can an insurance company guarantee reimbursement for fire damages when it has no way of predicting which specific houses will burn down this year? Their secret is identical: **individually, the outcome of a single trial is unpredictable; collectively, the average of a large number of trials behaves with near-absolute certainty.**

To transition from the erratic caprice of a single trial to the certainty of large aggregates, mathematicians forged two monumental instruments: the **binomial distribution** and the **law of large numbers**.

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2. From the Bernoulli Trial to the Binomial Distribution

2.1 The Bernoulli Trial

The Swiss mathematician and physicist **Jacob Bernoulli** (1655–1705) studied the most elementary random experiment imaginable: one admitting only **two possible outcomes**.

> **📐 Definition — Bernoulli Trial**
>
> A **Bernoulli trial** with parameter $p \in (0,1)$ is a random experiment that admits only two complementary outcomes:
>
> - One conventionally designated **Success** (denoted S), with probability $P(S) = p$.
> - The other conventionally designated **Failure** (denoted F or $\overline{S}$), with probability $P(F) = 1 - p$ (often denoted q).
>
> The associated indicator random variable X that equals 1 on success and 0 on failure follows a **Bernoulli distribution** with parameter p, denoted $X \sim \mathcal{B}(p)$.

Bernoulli Probability Mass Function Table: | k | 0 | 1 | |---|---|---| | $P(X=k)$ | $1-p$ | p |

Let us immediately compute its parameters:

- **Expected Value (Mean):**

$$E(X) = 0 \times P(X=0) + 1 \times P(X=1) = 0 \times (1-p) + 1 \times p = p$$

- **Variance:** By the Koenig-Huygens formula ($V(X) = E(X^2) - (E(X))^2$): Since X only takes values 0 and 1, we have $X^2 = X$, and thus $E(X^2) = E(X) = p$.

$$V(X) = p - p^2 = p(1-p)$$

- **Standard Deviation:**

$$\sigma(X) = \sqrt{V(X)} = \sqrt{p(1-p)}$$

2.2 The Bernoulli Scheme

> **📐 Definition — Bernoulli Scheme**
>
> A **Bernoulli scheme** with parameters n and p is the sequence of n identical and **mutually independent** Bernoulli trials, each having the same success probability p.

What does independence mean? The outcome of any trial in no way affects the probabilities of subsequent trials. Examples include consecutive coin tosses, or draws from an urn **with replacement**. If draws are made *without replacement*, trials are not independent: this would not constitute a Bernoulli scheme!

2.3 The Probability Tree and the Binomial Distribution

Let us represent a Bernoulli scheme of $n = 3$ trials on a probability tree:

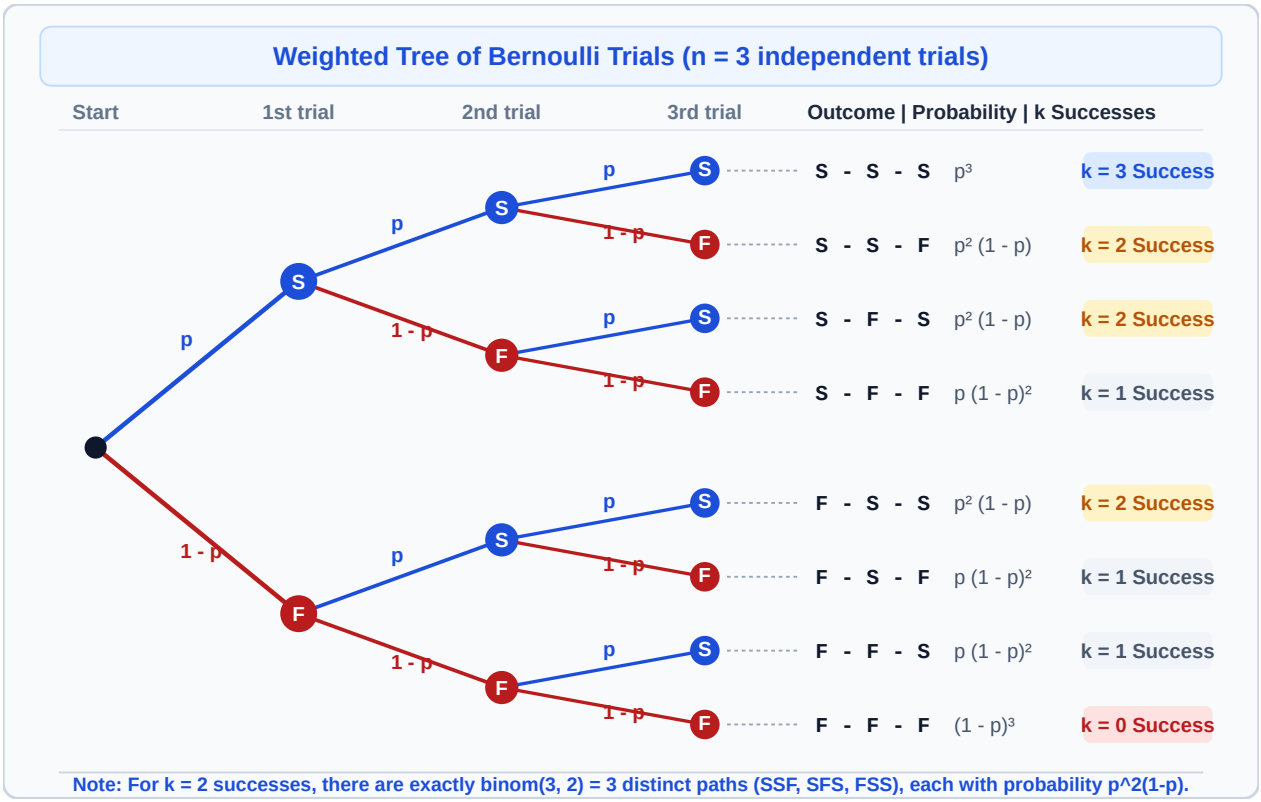

Figure 117: Probability tree of a Bernoulli scheme with 3 independent trials

Consider the random variable X that counts the **total number of successes** obtained across the n trials.

- X can take any integer value between 0 and n.
- What is the probability of obtaining exactly 2 successes when $n = 3$? Let us trace the paths yielding 2 successes:
  - $S - S - F$: probability $p \times p \times (1-p) = p^2(1-p)$
  - $S - F - S$: probability $p \times (1-p) \times p = p^2(1-p)$
  - $F - S - S$: probability $(1-p) \times p \times p = p^2(1-p)$ Every path comprising 2 successes and 1 failure has the exact same probability $p^2(1-p)$. How many such paths exist? Exactly 3 paths. Hence $P(X = 2) = 3 \times p^2(1-p)$. And what is this number 3? It is the number of ways to choose the positions of the 2 successes among the 3 trials, which is $\binom{3}{2} = 3$!

📐 Fundamental Theorem — The Binomial Distribution

Let X be the random variable equal to the number of successes in a Bernoulli scheme of n independent trials with parameter p. We say that X follows the **binomial distribution with parameters n and p**, denoted:

$$X \sim \mathcal{B}(n, p)$$

For every integer $k \in \{0, 1, 2, ..., n\}$, the probability of obtaining exactly k successes is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Consistency Check: The sum of all probabilities must equal 1:

$$\sum_{k=0}^{n} P(X = k) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = (p + (1-p))^n = 1^n = 1.$$

This is none other than Newton's binomial expansion from Chapter 53! That algebraic connection is precisely why this law is named the **binomial distribution**.

3. Expected Value and Variance: Core Derivations

3.1 Sum of Independent Random Variables

Let $X_1, X_2, ..., X_n$ be the n Bernoulli indicator variables associated with the n trials:

$$X_i = \begin{cases} 1 \text{ if the } i\text{-th trial is a success} \\ 0 \text{ if it is a failure} \end{cases}$$

The total number of successes X is simply the sum of these indicators:

$$X = X_1 + X_2 + ... + X_n = \sum_{i=1}^{n} X_i$$

Recall two fundamental properties of random variables: 1. **Linearity of Expectation:** For any random variables X and Y (independent or not) and any constant $a \in \mathbb{R}$:

$$E(X+Y) = E(X) + E(Y) \quad \text{and} \quad E(aX) = aE(X)$$

2. **Additivity of Variance for Independent Variables:** If X and Y are **independent** random variables:

$$V(X+Y) = V(X) + V(Y)$$

(Note: for a scalar constant a, $V(aX) = a^2 V(X)$).

3.2 Derivation of Expected Value for the Binomial Distribution

Required Proof — Expected Value of the Binomial Distribution

Let $X \sim \mathcal{B}(n,p)$. We decompose X as $X = \sum_{i=1}^{n} X_i$, where the X_i are n identical Bernoulli variables with parameter p. For each $i \in \{1, ..., n\}$, we have $E(X_i) = p$. By linearity of expectation:

$$E(X) = E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E(X_i) = \underbrace{p + p + ... + p}_{n \text{ terms}} = np$$

$$\mathbf{E(X) = np}$$

Intuitive Meaning: If you toss a fair coin ($p = 0.5$) 100 times, you expect on average to obtain $100 \times 0.5 = 50$ Heads. If you roll a die 60 times and win on a 6 ($p = 1/6$), you expect on average to win $60 \times \frac{1}{6} = 10$ times.

3.3 Derivation of Variance and Standard Deviation

Required Proof — Variance of the Binomial Distribution

Since the trials are mutually independent in a Bernoulli scheme, the random variables $X_1, X_2, ..., X_n$ are independent. By additivity of variance for independent variables:

$$V(X) = V\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} V(X_i)$$

For each Bernoulli variable X_i, we previously established that $V(X_i) = p(1-p)$. Hence:

$$V(X) = \sum_{i=1}^{n} p(1-p) = \underbrace{p(1-p) + ... + p(1-p)}_{n \text{ terms}} = np(1-p)$$

$$\mathbf{V(X) = np(1-p)}$$

The standard deviation is consequently:

$$\boldsymbol{\sigma(X) = \sqrt{np(1-p)}}$$

Crucial Remark on Scaling: - The expectation $E(X)$ scales linearly with n (order of magnitude n). - The standard deviation $\sigma(X)$ scales only as $\sqrt{n}$.

This asymmetry between n and $\sqrt{n}$ holds the key to the entire phenomenon of statistical stabilization!

4. Dispersion and Concentration: From Markov to Bienaymé-Chebyshev

How can we bound the probability that a random variable deviates substantially from its mean?

4.1 Markov's Inequality

The Russian mathematician **Andrey Markov** (1856–1922) established a universal bound for non-negative random variables.

Theorem — Markov's Inequality

Let Y be a non-negative random variable ($Y \geq 0$). For any strictly positive real number $a > 0$:

$$P(Y \geq a) \leq \frac{E(Y)}{a}$$

Curriculum Proof — Markov's Inequality

Let Y be a discrete random variable taking non-negative values $y_1, y_2, ..., y_m \geq 0$. The expected value is given by:

$$E(Y) = \sum_{i=1}^{m} y_i P(Y = y_i)$$

Split this sum into two parts: outcomes strictly less than a, and outcomes greater than or equal to a:

$$E(Y) = \sum_{y_i < a} y_i P(Y = y_i) + \sum_{y_i \geq a} y_i P(Y = y_i)$$

Since all values y_i and probabilities are non-negative, the first sum is ≥ 0. Thus:

$$E(Y) \geq \sum_{y_i \geq a} y_i P(Y = y_i)$$

In this second sum, each value satisfies $y_i \geq a$. We can therefore replace each y_i by its lower bound a:

$$E(Y) \geq \sum_{y_i \geq a} a P(Y = y_i) = a \sum_{y_i \geq a} P(Y = y_i) = a P(Y \geq a)$$

Because $a > 0$, dividing by a immediately yields:

$$P(Y \geq a) \leq \frac{E(Y)}{a}$$

Striking Example: If the average monthly income in a country is €2,000, what fraction of the population could earn at least €10,000? Here $Y \geq 0$, $E(Y) = 2\,000$, and $a = 10\,000$. $P(Y \geq 10\,000) \leq \frac{2\,000}{10\,000} = \frac{1}{5} = 20\%$. It is mathematically impossible for more than 20% of citizens to earn more than 5 times the national average!

4.2 The Bienaymé-Chebyshev Inequality

The French mathematician **Irénée-Jules Bienaymé** (1853) and his Russian peer **Pafnuty Chebyshev** (1867) had the brilliant insight to apply Markov's inequality to the squared deviation from the mean: $Y = (X - E(X))^2$.

Fundamental Theorem — Bienaymé-Chebyshev Inequality

Let X be a random variable admitting expectation $E(X)$ and variance $V(X)$. For any real number $\varepsilon > 0$ (epsilon):

$$P(|X - E(X)| \geq \varepsilon) \leq \frac{V(X)}{\varepsilon^2}$$

Required Proof — Bienaymé-Chebyshev

Let $\varepsilon > 0$. Note the fundamental equivalence between events:

$$|X - E(X)| \geq \varepsilon \Leftrightarrow (X - E(X))^2 \geq \varepsilon^2$$

Define the auxiliary random variable $Y = (X - E(X))^2$.

- Y is non-negative ($Y \geq 0$).
- Its expectation is, by definition, the variance of X: $E(Y) = E((X - E(X))^2) = V(X)$.

Apply Markov's inequality to the non-negative variable Y with threshold $a = \varepsilon^2 > 0$:

$$P(Y \geq \varepsilon^2) \leq \frac{E(Y)}{\varepsilon^2}$$

Replacing Y by its definition and $E(Y)$ by $V(X)$:

$$P((X - E(X))^2 \geq \varepsilon^2) \leq \frac{V(X)}{\varepsilon^2}$$

Which translates precisely to:

$$P(|X - E(X)| \geq \varepsilon) \leq \frac{V(X)}{\varepsilon^2}$$

Significance: The probability that a random variable X strays from its mean by at least ε is strictly bounded by a term proportional to its variance, decaying rapidly with the inverse square of the distance ε^2.

---

5. The Weak Law of Large Numbers

5.1 The Sample Mean of an Empirical Sample

Consider n independent random variables $X_1, X_2, ..., X_n$ that are **identically distributed**, sharing the same mean $\mu = E(X_1)$ and the same variance $\sigma^2 = V(X_1)$. We define the **sample mean** (or empirical average) as the random variable:

$$M_n = \frac{X_1 + X_2 + ... + X_n}{n} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Let us compute the mean and variance of this sample mean:

- **Expectation of M_n:**

$$E(M_n) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \frac{1}{n} \times n\mu = \mu$$

  The sample mean is **unbiased**: on average, it hits the theoretical value μ exactly.

- **Variance of M_n:** By independence of the variables X_i, pulling out the factor $\frac{1}{n}$ squared:

$$V(M_n) = V\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} V(X_i) = \frac{1}{n^2} \times n\sigma^2 = \frac{\sigma^2}{n}$$

Look closely at this ratio:

$$V(M_n) = \frac{V(X_1)}{n}$$

As the sample size n grows indefinitely, **the variance of the sample mean vanishes toward zero**!

5.2 The Concentration Inequality

Theorem — Concentration Inequality

For every real number $\varepsilon > 0$:

$$P(|M_n - \mu| \geq \varepsilon) \leq \frac{V(X_1)}{n\varepsilon^2}$$

Proof: Simply apply the Bienaymé-Chebyshev inequality to the random variable M_n:

$$P(|M_n - E(M_n)| \geq \varepsilon) \leq \frac{V(M_n)}{\varepsilon^2}$$

Substituting $E(M_n) = \mu$ and $V(M_n) = \frac{V(X_1)}{n}$ gives the result directly.

5.3 Proof of the Weak Law of Large Numbers

Required Proof — Weak Law of Large Numbers

Let $\varepsilon > 0$ be an arbitrarily small positive tolerance (for instance, $\varepsilon = 0.001$). The concentration inequality guarantees that:

$$0 \leq P(|M_n - \mu| \geq \varepsilon) \leq \frac{V(X_1)}{n\varepsilon^2}$$

As $n \to +\infty$, the numerator $V(X_1)$ and ε^2 remain fixed constants. Hence:

$$\lim_{n \to +\infty} \frac{V(X_1)}{n\varepsilon^2} = 0$$

By the Squeeze Theorem (Sandwich Rule):

$$\mathbf{\lim_{n \to +\infty} P(|M_n - \mu| \geq \varepsilon) = 0}$$

Equivalently, passing to the complementary event:

$$\mathbf{\lim_{n \to +\infty} P(|M_n - \mu| < \varepsilon) = 1}$$

Philosophical and Practical Meaning: No matter how tiny the error margin $\varepsilon > 0$ might be, provided the sample size n is sufficiently large, the probability that the sample average lands within the interval $(\mu - \varepsilon, \mu + \varepsilon)$ approaches 100%. In the specific case of coin tosses where $X_i \sim \mathcal{B}(p)$, $M_n = \frac{X}{n} = F_n$ is the **observed frequency** of successes. The observed frequency of an event thus converges mathematically to its theoretical probability p!

---

6. Python Simulation: 10,000 Tosses Before Your Eyes

To verify this theorem experimentally, let us run a Python simulation of $N = 10\,000$ tosses of a fair coin ($p = 0.5$) tracking the evolution of the relative frequency of Heads:

```python
import random

def simulate_tosses(n):
    successes = 0
    frequency_history = []

    for i in range(1, n + 1):
        # 1 = Heads (success), 0 = Tails (failure)
        draw = random.choice([0, 1])
        successes += draw
        freq = successes / i
        frequency_history.append((i, freq))

    return frequency_history

# Simulation run
results = simulate_tosses(10000)

# Print key milestones
milestones = [10, 50, 100, 500, 1000, 5000, 10000]
print(f"{'Tosses (n)':<12} | {'Heads Frequency':<16} | {'Gap to 0.5':<12}")
print("-" * 45)
for m in milestones:
    freq = results[m - 1][1]
    gap = abs(freq - 0.5)
    print(f"{m:<12} | {freq:<16.4f} | {gap:<12.4f}")
```

Typical Console Output:

Lancers (n)	Frequency Pile	Écart à 0.5
10	0.7000	+0.2000
50	0.4400	+0.0600
100	0.5300	+0.0300
500	0.4880	+0.0120
1 000	0.5040	+0.0040
5 000	0.4982	+0.0018
10 000	0.5007	+0.0007

At first (10 tosses), the frequency fluctuates erratically (anywhere from 0.30 to 0.70). By 1,000 tosses, it is locked within 1% of 0.5. At 10,000 tosses, precision reaches parts per thousand! The chaotic storm of chance has crystallized into stable order.

💡 True History — The Galton Board and the Bell Curve

In 1877, British polymath Francis Galton conceived a physical apparatus to visualize the convergence of the binomial distribution: **the quincunx** (or **Galton board**). Steel ball bearings drop one by one through a triangular array of pegs. At each peg encountered, the ball has an equal probability of bouncing right (S) or left (F). At the bottom, the balls accumulate into vertical columns: the central pile rises majestically while the extremities taper off, carving out the bell curve of Gauss with astonishing mechanical regularity.

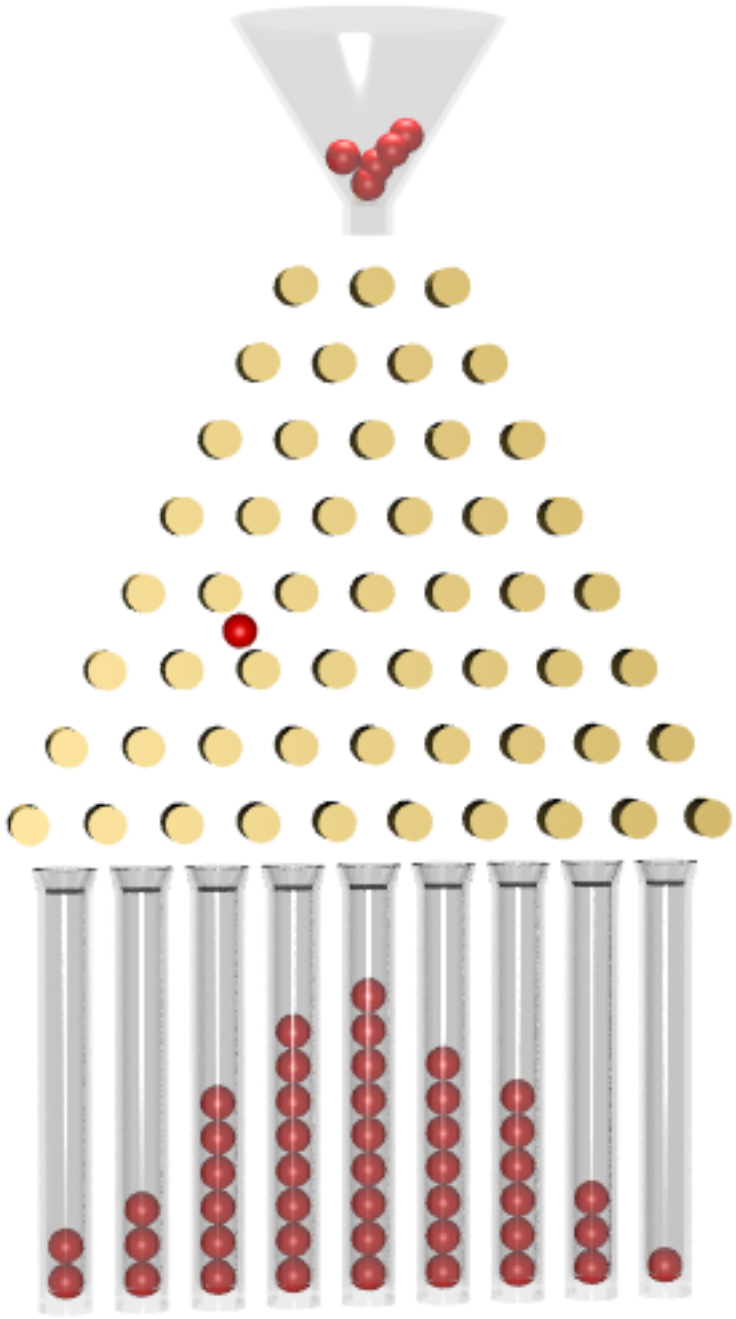

Figure 118: The Galton Board: physical manifestation of the convergence of Pascal's triangle to the Gaussian bell curve

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7. Step-by-Step Worked Examples

Example 1: Industrial Quality Control

A factory manufactures LED bulbs. The probability that any single bulb is defective is $p = 0.02$. A sample of $n = 100$ bulbs is selected at random. Production volume is sufficiently large that this selection can be modeled as sampling with replacement. Let X denote the number of defective bulbs in the batch. 1. State the probability distribution of X and its parameters. 2. Calculate the mean and standard deviation of X. 3. Calculate the probability of finding exactly 2 defective bulbs. 4. Calculate the probability of finding at least one defective bulb.

Detailed Solution: 1. The experiment consists of repeating $n = 100$ identical trials (testing a bulb) with two outcomes:

- Success S: “the bulb is defective”, with probability $p = 0.02$.
- Failure $\overline{S}$: “the bulb works properly”, with probability $1 - p = 0.98$. Because sampling is treated as independent with replacement, trials are independent. The variable X counts the total number of successes. Therefore, X follows the **binomial distribution** with parameters $n = 100$ and $p = 0.02$: $X \sim \mathcal{B}(100, 0.02)$.

2. • Expected value: $E(X) = np = 100 \times 0.02 = 2$. On average, a batch of 100 bulbs contains 2 defective units.
   - Variance: $V(X) = np(1-p) = 100 \times 0.02 \times 0.98 = 1.96$.
   - Standard deviation: $\sigma(X) = \sqrt{1.96} = 1.4$.
3. Probability of exactly 2 defectives:

$$P(X = 2) = \binom{100}{2} (0.02)^2 (0.98)^{98}$$

Here $\binom{100}{2} = \frac{100 \times 99}{2} = 4\,950$. $(0.02)^2 = 0.0004$. $(0.98)^{98} \approx 0.1380$.

$$P(X = 2) = 4\,950 \times 0.0004 \times 0.1380 \approx 0.2734 \quad (\approx 27.3\%).$$

4. “At least one defective” corresponds to $X \geq 1$. The complementary event is “zero defectives”, $X = 0$.

$$P(X = 0) = \binom{100}{0} (0.02)^0 (0.98)^{100} = 1 \times 1 \times (0.98)^{100} \approx 0.1326.$$

Hence:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.98)^{100} \approx 1 - 0.1326 = 0.8674 \quad (\approx 86.7\%).$$

Example 2: Guessing on a Multiple-Choice Exam

A candidate takes a test unprepared. The test consists of 10 independent multiple-choice questions. For each question, 4 choices are offered, of which exactly one is correct. The candidate guesses randomly on every question. Passing requires at least 6 correct answers. What is the probability that the candidate passes?

Detailed Solution: Let X be the number of correct answers. The 10 questions are independent with success probability $p = \frac{1}{4} = 0.25$. Thus $X \sim \mathcal{B}(10, 0.25)$. The candidate passes if $X \geq 6$.

$$P(X \geq 6) = P(X = 6) + P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10)$$

Compute each term:

- $P(X = 6) = \binom{10}{6}(0.25)^6(0.75)^4 = 210 \times \frac{1}{4096} \times \frac{81}{256} = \frac{17\,010}{1\,048\,576} \approx 0.01622$
- $P(X = 7) = \binom{10}{7}(0.25)^7(0.75)^3 = 120 \times \frac{1}{16384} \times \frac{27}{64} = \frac{3\,240}{1\,048\,576} \approx 0.00309$
- $P(X = 8) = \binom{10}{8}(0.25)^8(0.75)^2 = 45 \times \frac{1}{65536} \times \frac{9}{16} = \frac{405}{1\,048\,576} \approx 0.000386$
- $P(X = 9) = \binom{10}{9}(0.25)^9(0.75)^1 = 10 \times \frac{3}{1\,048\,576} \approx 0.000029$
- $P(X = 10) = \binom{10}{10}(0.25)^{10}(0.75)^0 = \frac{1}{1\,048\,576} \approx 0.000001$

Summing these probabilities:

$$P(X \geq 6) \approx 0.01622 + 0.00309 + 0.00039 + 0.00003 + 0.000001 = 0.01973 \approx 1.97\%.$$

The candidate has less than a 2% chance of passing purely by luck.

Example 3: Airline Overbooking

An airliner has 200 seats. The airline knows from historical data that 5% of ticket holders do not show up for boarding ($p_{\text{show}} = 0.95$). The airline sells $n = 205$ tickets. 1. What is the probability that all passengers who show up receive a seat? 2. What is the probability that the airline faces an overbooking crisis (at least one passenger bumped without a seat)?

Detailed Solution: 1. Let Y be the number of passengers arriving for boarding among the 205 ticket holders. Assuming passenger decisions are independent, $Y \sim \mathcal{B}(205, 0.95)$. Every arriving passenger receives a seat if and only if $Y \leq 200$. The complementary event is $P(Y > 200) = P(Y = 201) + P(Y = 202) + P(Y = 203) + P(Y = 204) + P(Y = 205)$. Calculate these 5 terms:

- $P(Y = 205) = (0.95)^{205} \approx 0.000025$
- $P(Y = 204) = 205 \times (0.95)^{204} \times 0.05 \approx 0.000270$
- $P(Y = 203) = \binom{205}{2}(0.95)^{203}(0.05)^2 = 20\,910 \times (0.95)^{203} \times 0.0025 \approx 0.00144$
- $P(Y = 202) = \binom{205}{3}(0.95)^{202}(0.05)^3 \approx 0.0051$
- $P(Y = 201) = \binom{205}{4}(0.95)^{201}(0.05)^4 \approx 0.0133$ Sum of crisis probabilities:

$$P(Y > 200) \approx 0.0133 + 0.0051 + 0.0014 + 0.0003 + 0.0000 = 0.0201 \approx 2.01\%.$$

2. The probability of an overbooking conflict is approximately 2.0%. The probability that boarding proceeds without incident is $1 - 0.0201 = 0.9799$, or nearly 98%.

Example 4: Sample Sizing via Bienaymé-Chebyshev

A fair coin is tossed ($p = 0.5$). Let $F_n = \frac{X}{n}$ be the observed frequency of Heads over n tosses. We require that the probability of the frequency deviating from 0.5 by more than 0.02 be strictly less than 0.05 (at most a 5% risk of error exceeding 2 hundredths). Determine, using the concentration inequality, a sample size n that guarantees this level of precision.

Detailed Solution: Here $X_1 \sim \mathcal{B}(0.5)$, so $\mu = 0.5$ and $V(X_1) = p(1-p) = 0.5 \times 0.5 = 0.25$. The tolerated margin of error is $\varepsilon = 0.02$. By the concentration inequality applied to $M_n = F_n$:

$$P(|F_n - 0.5| \geq 0.02) \leq \frac{V(X_1)}{n\varepsilon^2} = \frac{0.25}{n \times (0.02)^2} = \frac{0.25}{n \times 0.0004} = \frac{625}{n}$$

We want this upper bound to be at most $0.05 = \frac{1}{20}$:

$$\frac{625}{n} \leq 0.05 \Leftrightarrow n \geq \frac{625}{0.05} = 12\,500.$$

It is therefore sufficient to perform 12 500 tosses to ensure, without any further distributional assumptions, that the risk of a deviation greater than 0.02 does not exceed 5%. *(Note: Because Bienaymé-Chebyshev is universal, this bound is conservative. Using the Central Limit Theorem and the normal distribution, about 2,500 tosses would suffice).*

Example 5: Medical Screening Test in a Healthy Population

A rapid screening test has a false-positive rate of 1% ($p = 0.01$). A cohort of 500 healthy patients is tested. Let X be the number of false positives observed. 1. Determine $E(X)$ and $V(X)$. 2. Give an upper bound for $P(|X - 5| \geq 4)$ using Bienaymé-Chebyshev.

Detailed Solution: 1. $X \sim \mathcal{B}(500, 0.01)$.

- $E(X) = np = 500 \times 0.01 = 5$.
- $V(X) = np(1-p) = 500 \times 0.01 \times 0.99 = 4.95$.

2. With expectation $E(X) = 5$, the condition $|X - 5| \geq 4$ corresponds to $\varepsilon = 4$. By the Bienaymé-Chebyshev inequality:

$$P(|X - 5| \geq 4) \leq \frac{V(X)}{4^2} = \frac{4.95}{16} \approx 0.3094.$$

The probability of observing fewer than 1 or at least 9 false positives is guaranteed to be below 31%.

Example 6: Sum of Two Independent Binomial Random Variables

Two independent production lines A and B produce machine parts.

- Line A produces 50 parts per hour, with a defect rate of 0.03. Let X_A be the number of defective parts from Line A: $X_A \sim \mathcal{B}(50, 0.03)$.
- Line B produces 70 parts per hour, with a defect rate of 0.03. Let X_B be the number of defective parts from Line B: $X_B \sim \mathcal{B}(70, 0.03)$.

Let $S = X_A + X_B$ be the total number of defective parts produced in one hour. 1. Determine the distribution of S. 2. Calculate $E(S)$ and $V(S)$.

Detailed Solution: 1. Each part from Line A undergoes a Bernoulli trial with parameter $p = 0.03$. There are 50 independent trials. Each part from Line B undergoes the same Bernoulli trial with parameter $p = 0.03$. There are 70 independent trials. Because both lines operate independently and **share the same parameter p**, the sum $S = X_A + X_B$ represents the total number of successes across $50 + 70 = 120$ independent Bernoulli trials with parameter 0.03. Therefore:

$$S \sim \mathcal{B}(120, 0.03)$$

(Crucial warning: This additivity property holds only when both distributions share the exact same success parameter p!) 2. - $E(S) = np = 120 \times 0.03 = 3.6$.

- $V(S) = np(1-p) = 120 \times 0.03 \times 0.97 = 3.492$.

Key Takeaways

1. **Bernoulli Distribution $\mathcal{B}(p)$:**
   - 2 outcomes: Success ($X = 1$, probability p), Failure ($X = 0$, probability $1 - p$).
   - $E(X) = p$ and $V(X) = p(1 - p)$.
2. **Binomial Distribution $\mathcal{B}(n, p)$:**
   - Repetition of n identical and independent Bernoulli trials.
   - Formula: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$.
   - Expected Value: $E(X) = np$.
   - Variance: $V(X) = np(1 - p)$; Standard Deviation: $\sigma(X) = \sqrt{np(1 - p)}$.
3. **Universal Inequalities:**
   - **Markov:** If $Y \geq 0$ and $a > 0$, then $P(Y \geq a) \leq \frac{E(Y)}{a}$.
   - **Bienaymé-Chebyshev:** For any $\varepsilon > 0$, $P(|X - E(X)| \geq \varepsilon) \leq \frac{V(X)}{\varepsilon^2}$.
   - **Concentration:** For the sample mean M_n, $P(|M_n - \mu| \geq \varepsilon) \leq \frac{V(X_1)}{n\varepsilon^2}$.
4. **Weak Law of Large Numbers:**
   - $\lim_{n \to +\infty} P(|M_n - \mu| \geq \varepsilon) = 0$.
   - The empirical frequency converges in probability to the theoretical parameter as the number of trials tends to infinity.

Exercises

Level (Direct Applications)

Exercise 54.1 Let $X \sim \mathcal{B}(6, 0.3)$. 1. Calculate $E(X)$ and $V(X)$. 2. Calculate the exact probability $P(X = 2)$. 3. Express $P(X \geq 1)$ and round its value to 10^{-4}. **Exercise 54.2** A fair coin is tossed 8 consecutive times. Let X be the number of Heads obtained. 1. What distribution does X follow? 2. Calculate $P(X = 4)$. **Exercise 54.3** An archer shoots 5 arrows at a target. On each shot, the probability of hitting the bullseye is $p = 0.7$. Shots are assumed independent. Calculate the probability that the archer hits the bullseye: 1. Exactly 5 times. 2. At least once. **Exercise 54.4** A non-negative random variable $Y \geq 0$ has $E(Y) = 12$. Using Markov's inequality, find an upper bound for $P(Y \geq 60)$. **Exercise 54.5** A random variable X has mean $E(X) = 50$ and variance $V(X) = 8$. Using the Bienaymé-Chebyshev inequality, find an upper bound for $P(|X - 50| \geq 6)$. **Exercise 54.6** True or False? Justify your answers. 1. Tossing a coin 100 times, the probability of getting exactly 50 Heads is 0.5. 2. If $X \sim \mathcal{B}(10, 0.4)$, then $P(X \leq 9) = 1 - (0.4)^{10}$. 3. For any random variable, $P(|X - E(X)| \geq 3\sigma) \leq \frac{1}{9}$. **Exercise 54.7** An urn contains 3 white balls and 7 black balls. Four successive draws are conducted **with replacement**. Let X be the number of white balls drawn. State the distribution of X, its expectation, and the probability of obtaining exactly 1 white ball.

Level (Exploration and Examination Classics)

Exercise 54.8 A medical treatment cures a condition in 80% of cases. The treatment is administered to a group of 20 independent patients. Let X be the number of cured patients. 1. Calculate the probability that exactly 16 patients are cured. 2. Calculate the probability that at least 18 patients are cured. **Exercise 54.9** A fair 6-sided die is rolled n times. Let A be the event: "obtain at least one 6". What is the minimum number of rolls required so that the probability of obtaining at least one 6 is at least 0.99? **Exercise 54.10** Concentration inequality for a survey poll. n randomly selected individuals are polled to estimate the proportion p in favor of a project. Let F_n be the observed sample frequency. 1. Recall why $V(X_1) = p(1 - p) \leq \frac{1}{4}$ for all $p \in [0, 1]$. 2. Deduce that $P(|F_n - p| \geq 0.05) \leq \frac{100}{n}$. 3. What sample size n guarantees that the observed frequency does not differ from the true proportion by more than 0.05 with a risk under 1%? **Exercise 54.11** Mode of the binomial distribution. Let $X \sim \mathcal{B}(n, p)$. 1. Show that the ratio $\frac{P(X=k+1)}{P(X=k)}$ equals $\frac{n-k}{k+1} \frac{p}{1-p}$. 2. Deduce the value(s) of k for which $P(X = k)$ is maximized. **Exercise 54.12** A web server handles 1,000 requests per minute. The processing time for each request is an independent random variable with mean $\mu = 40$ ms and standard

deviation $\sigma = 15$ ms. Find an upper bound for the probability that the average duration of the 1,000 requests deviates from 40 ms by more than 2 ms. **Exercise 54.13** Two independent random variables satisfy $X_1 \sim \mathcal{B}(n_1, p)$ and $X_2 \sim \mathcal{B}(n_2, p)$. Prove algebraically, using Vandermonde's convolution formula from Chapter 53, that $X_1 + X_2 \sim \mathcal{B}(n_1 + n_2, p)$. **Exercise 54.14** Truncated geometric distribution (time to first success). Repeated independent Bernoulli trials with parameter p are conducted. Let T be the index of the first success. 1. For every integer $k \geq 1$, find $P(T = k)$. 2. Calculate $P(T > k)$ directly and combinatorially. 3. Verify that $\sum_{k=1}^{+\infty} P(T = k) = 1$. **Exercise 54.15** Fluctuation of absolute deviations. Let $X \sim \mathcal{B}(n, 0.5)$. Prove that $P(|X - 0.5n| \geq a\sqrt{n}) \leq \frac{1}{4a^2}$. Deduce the characteristic scale of fluctuations around the mean.

Level (Deepening and Advanced Competitions)

Exercise 54.16 Direct expectation derivation without indicator decomposition. Derive the formula $E(X) = np$ for $X \sim \mathcal{B}(n, p)$ by directly evaluating the sum:

$$E(X) = \sum_{k=0}^{n} k\binom{n}{k} p^k (1-p)^{n-k}$$

(Hint: use the identity $k\binom{n}{k} = n\binom{n-1}{k-1}$ from Chapter 53). **Exercise 54.17** Direct variance derivation via sum evaluation. 1. Evaluate the quantity $E(X(X-1)) = \sum_{k=0}^{n} k(k-1)\binom{n}{k} p^k (1-p)^{n-k}$. 2. Deduce $E(X^2)$, and recover $V(X) = np(1-p)$. **Exercise 54.18** Laplace's Rule of Succession. n draws with replacement from an urn have yielded k successes. What is the probability that the next draw yields a success if the initial composition of the urn was completely unknown? Show that a uniform Bayesian prior leads to Laplace's rule:

$$P(S_{n+1}) = \frac{k+1}{n+2}$$

Compare with the naive estimate $\frac{k}{n}$ for $k = 0$ and $k = n$. **Exercise 54.19** Cantelli's Inequality (One-sided Chebyshev). Let X be a random variable with mean 0 and variance σ^2. Prove that for any real number $a > 0$:

$$P(X \geq a) \leq \frac{\sigma^2}{\sigma^2 + a^2}$$

(Hint: for any $t > 0$, consider the variable $(X + t)^2$ and optimize over t). **Exercise 54.20** Buffon's Needle and Monte Carlo Estimation. A plane is ruled with parallel lines spaced at distance D. A needle of length $L \leq D$ is dropped at random onto the plane. 1. Given that the probability of crossing a line is $p = \frac{2L}{\pi D}$, explain how to estimate the mathematical constant π by simulation. 2. If n independent drops are performed, determine via the concentration inequality the number of trials needed to guarantee an accuracy of 0.01 on π with a specified confidence level. **Exercise 54.21** Condorcet's Jury Theorem. A jury of n independent jurors (n odd, $n = 2m + 1$) decides guilt by majority vote. Each juror has an individual probability $p > 0.5$ of reaching the correct decision. 1. Express the probability P_n that the majority of the jury reaches the correct decision. 2. Prove that $P_n > p$ and that $\lim_{n \to +\infty} P_n = 1$. Interpret this result for democratic decision-making and trial juries. **Exercise 54.22** Almost Sure Convergence and the Strong Law of Large Numbers (Overview). Explain the conceptual difference between the Weak Law of Large Numbers (convergence in probability: $\lim P(|M_n - \mu| \geq \varepsilon) = 0$) and the Strong Law of Émile Borel and Andrey Kolmogorov (stating that the sample path of averages converges to μ with probability 1).

Solutions

Solution 54.1: 1. $E(X) = np = 6 \times 0.3 = 1.8$. $V(X) = np(1-p) = 6 \times 0.3 \times 0.7 = 1.26$. 2. $P(X = 2) = \binom{6}{2}(0.3)^2(0.7)^4 = 15 \times 0.09 \times 0.2401 = 0.3241$. 3. $P(X \geq 1) = 1 - P(X = 0) = 1 - \binom{6}{0}(0.3)^0(0.7)^6 = 1 - (0.7)^6 = 1 - 0.117649 = 0.8824$.

Solution 54.2: 1. Repetition of 8 independent Bernoulli trials with $p = 0.5$. Hence $X \sim \mathcal{B}(8, 0.5)$. 2. $P(X = 4) = \binom{8}{4}(0.5)^4(0.5)^4 = 70 \times (0.5)^8 = \frac{70}{256} = \frac{35}{128} \approx 0.2734$.

Solution 54.3: $X \sim \mathcal{B}(5, 0.7)$. 1. $P(X = 5) = (0.7)^5 = 0.16807$. 2. $P(X \geq 1) = 1 - P(X = 0) = 1 - (0.3)^5 = 1 - 0.00243 = 0.99757$.

Solution 54.4: By Markov's inequality: $P(Y \geq 60) \leq \frac{E(Y)}{60} = \frac{12}{60} = \frac{1}{5} = 0.20$.

Solution 54.5: By the Bienaymé-Chebyshev inequality: $P(|X - 50| \geq 6) \leq \frac{V(X)}{6^2} = \frac{8}{36} = \frac{2}{9} \approx 0.222$.

Solution 54.6: 1. **False.** For $n = 100$ and $p = 0.5$, $P(X = 50) = \binom{100}{50}(0.5)^{100} \approx 0.0796$ (around 8%). The mean is 50, but the probability distribution is spread over neighboring integers. 2. **True.** $P(X \leq 9) = 1 - P(X = 10) = 1 - \binom{10}{10}(0.4)^{10}(0.6)^0 = 1 - (0.4)^{10}$. The statement is therefore **True**. 3. **True.** Setting $\varepsilon = 3\sigma$, $P(|X - E(X)| \geq 3\sigma) \leq \frac{\sigma^2}{(3\sigma)^2} = \frac{1}{9}$.

Solution 54.7: Draws with replacement: $p = \frac{3}{10} = 0.3$. $n = 4$ independent trials. $X \sim \mathcal{B}(4, 0.3)$. $E(X) = 4 \times 0.3 = 1.2$. $P(X = 1) = \binom{4}{1}(0.3)^1(0.7)^3 = 4 \times 0.3 \times 0.343 = 0.4116$.

Solution 54.8: $X \sim \mathcal{B}(20, 0.8)$. 1. $P(X = 16) = \binom{20}{16}(0.8)^{16}(0.2)^4 = 4\,845 \times (0.8)^{16} \times (0.2)^4 \approx 0.2182$. 2. $P(X \geq 18) = P(X = 18) + P(X = 19) + P(X = 20)$.

- $P(X = 18) = \binom{20}{18}(0.8)^{18}(0.2)^2 = 190 \times (0.8)^{18} \times 0.04 \approx 0.1369$
- $P(X = 19) = 20 \times (0.8)^{19} \times 0.2 \approx 0.0576$
- $P(X = 20) = (0.8)^{20} \approx 0.0115$ Total: $P(X \geq 18) \approx 0.1369 + 0.0576 + 0.0115 = 0.2060 \quad (\approx 20.6\%)$.

Solution 54.9: The probability of obtaining no 6s in n rolls is $(5/6)^n$. We require $P(A) = 1 - (5/6)^n \geq 0.99 \Leftrightarrow (5/6)^n \leq 0.01$. Taking natural logarithms (noting $\ln(5/6) < 0$, which flips the inequality): $n \ln(5/6) \leq \ln(0.01) \Leftrightarrow n \geq \frac{\ln(0.01)}{\ln(5/6)} \approx \frac{-4.60517}{-0.18232} \approx 25.26$. A minimum of **26 rolls** is required.

Solution 54.10: 1. For $f(p) = p(1-p) = p - p^2$, the derivative $f'(p) = 1 - 2p$ vanishes at $p = 0.5$, attaining its maximum $f(0.5) = 0.25 = \frac{1}{4}$. 2. By the concentration inequality with $\varepsilon = 0.05$: $P(|F_n - p| \geq 0.05) \leq \frac{V(X_1)}{n(0.05)^2} \leq \frac{0.25}{n \times 0.0025} = \frac{100}{n}$. 3. We require $\frac{100}{n} \leq 0.01 \Leftrightarrow n \geq \frac{100}{0.01} = 10\,000$. A sample size of 10,000 individuals guarantees this result.

Solution 54.11: 1. $\frac{P(X=k+1)}{P(X=k)} = \frac{\frac{n!}{(k+1)!(n-k-1)!}p^{k+1}(1-p)^{n-k-1}}{\frac{n!}{k!(n-k)!}p^k(1-p)^{n-k}} = \frac{n-k}{k+1}\frac{p}{1-p}$. 2. This ratio is ≥ 1 as long as $(n-k)p \geq (k+1)(1-p) \Leftrightarrow np - kp \geq k - kp + 1 - p \Leftrightarrow np + p - 1 \geq k \Leftrightarrow k \leq (n+1)p - 1$. The sequence of probabilities is initially increasing, then decreasing. The maximum (mode) is achieved at integer $k = \lfloor (n+1)p \rfloor$.

Solution 54.12: For $n = 1000$ requests, the sample mean duration M_n has mean $\mu = 40$ and variance $V(M_n) = \frac{\sigma^2}{n} = \frac{15^2}{1000} = \frac{225}{1000} = 0.225$. By Bienaymé-Chebyshev with $\varepsilon = 2$: $P(|M_n - 40| \geq 2) \leq \frac{V(M_n)}{2^2} = \frac{0.225}{4} = 0.05625$ (less than 5.6%).

Solution 54.13: For any $k \in \{0, ..., n_1 + n_2\}$, the event $(X_1 + X_2 = k)$ is the disjoint union of events $(X_1 = j \text{ and } X_2 = k - j)$ for $j \in \{0, ..., k\}$. By independence:

$$P(X_1 + X_2 = k) = \sum_{j=0}^{k} \binom{n_1}{j} p^j (1-p)^{n_1 - j} \binom{n_2}{k-j} p^{k-j} (1-p)^{n_2 - (k-j)} = p^k (1-p)^{n_1 + n_2 - k} \sum_{j=0}^{k} \binom{n_1}{j} \binom{n_2}{k-j}$$

By Vandermonde's identity, the sum equals $\binom{n_1+n_2}{k}$. We recognize the formula for $\mathcal{B}(n_1 + n_2, p)$.

Solution 54.14: 1. $T = k$ means $k - 1$ consecutive failures followed by a success: $P(T = k) = (1-p)^{k-1}p$. 2. $T > k$ means that the first k trials were all failures: $P(T > k) = (1-p)^k$. 3. $\sum_{k=1}^{+\infty} p(1-p)^{k-1} = p \sum_{j=0}^{+\infty} (1-p)^j = p\frac{1}{1-(1-p)} = \frac{p}{p} = 1$.

Solution 54.15: For $X \sim \mathcal{B}(n, 0.5)$, $E(X) = 0.5n$ and $V(X) = 0.25n$. Apply Bienaymé-Chebyshev with $\varepsilon = a\sqrt{n}$: $P(|X - 0.5n| \geq a\sqrt{n}) \leq \frac{V(X)}{(a\sqrt{n})^2} = \frac{0.25n}{a^2 n} = \frac{1}{4a^2}$. The characteristic scale of deviations between the number of successes and the expected value is thus of order $\sqrt{n}$.

Solution 54.16: $E(X) = \sum_{k=1}^{n} k\binom{n}{k}p^k(1-p)^{n-k}$. Using $k\binom{n}{k} = n\binom{n-1}{k-1}$:

$$E(X) = \sum_{k=1}^{n} n\binom{n-1}{k-1} p \cdot p^{k-1}(1-p)^{(n-1)-(k-1)} = np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j (1-p)^{n-1-j} = np(p + (1-p))^{n-1} = np.$$

Solution 54.17: 1. $k(k-1)\binom{n}{k} = k(k-1)\frac{n!}{k!(n-k)!} = \frac{n!}{(k-2)!(n-k)!} = n(n-1)\binom{n-2}{k-2}$. Summing for $k \geq 2$: $E(X(X-1)) = n(n-1)p^2 \sum_{j=0}^{n-2}\binom{n-2}{j}p^j(1-p)^{n-2-j} = n(n-1)p^2$. 2. $E(X^2) = E(X(X-1)) + E(X) = n(n-1)p^2 + np = n^2p^2 - np^2 + np$. $V(X) = E(X^2) - (E(X))^2 = (n^2p^2 - np^2 + np) - (np)^2 = np(1-p)$.

Solution 54.18: Under a uniform prior distribution on the unknown parameter $p \in [0, 1]$, the posterior conditional probability is given via the Beta function: $P(S_{n+1} \mid k \text{ successes out of } n) = \frac{\int_0^1 p \cdot p^k(1-p)^{n-k}dp}{\int_0^1 p^k(1-p)^{n-k}dp} = \frac{B(k+2, n-k+1)}{B(k+1, n-k+1)} = \frac{k+1}{n+2}$. For $k = 0$ (zero successes in 10 trials), the naive frequency estimate is 0, whereas Laplace gives $\frac{1}{12} \approx 8.3\%$, avoiding the pitfall of declaring an unobserved event impossible.

Solution 54.19: For any $t > 0$, the event $X \geq a$ is equivalent to $X + t \geq a + t$, which implies $(X + t)^2 \geq (a + t)^2$. Applying Markov's inequality to $(X + t)^2 \geq 0$: $P(X \geq a) \leq \frac{E((X+t)^2)}{(a+t)^2} = \frac{E(X^2) + 2tE(X) + t^2}{(a+t)^2} = \frac{\sigma^2 + t^2}{(a+t)^2}$. Minimizing this

function with respect to $t > 0$, setting the derivative to zero yields $t = \frac{\sigma^2}{a}$. Substituting this optimal choice produces $P(X \geq a) \leq \frac{\sigma^2}{\sigma^2 + a^2}$.

Solution 54.20: 1. Dropping n needles and recording X intersections, the frequency $F_n = \frac{X}{n}$ converges to $p = \frac{2L}{\pi D}$. An estimator for π is therefore $\hat{\pi}_n = \frac{2L}{D \cdot F_n}$. 2. By the concentration inequality, since $V(X_1) \leq 1/4$, the variance of the frequency is bounded by $\frac{1}{4n}$. To achieve a given precision on p, solving $\frac{1}{4n\varepsilon^2} \leq \alpha$ yields the required finite number of drops n.

Solution 54.21: 1. A majority is achieved when $k \geq m + 1 = \frac{n+1}{2}$. Thus $P_n = \sum_{k=m+1}^{n} \binom{n}{k} p^k (1-p)^{n-k}$. 2. By the Law of Large Numbers, as $n \to +\infty$, the fraction of correct votes converges to $p > 0.5$. The probability of a correct majority vote approaches 1. This theorem provides a mathematical foundation for collective decision-making: a large assembly of fallible yet moderately enlightened voters ($p > 0.5$) reaches an optimal verdict by majority rule.

Solution 54.22: The Weak Law asserts that for a fixed arbitrarily large sample size n, the probability of a deviation is vanishingly small. The Strong Law guarantees an infinitely stronger statement: with probability 1, the entire infinite trajectory of averages $(M_n)_{n \geq 1}$ converges to μ, ruling out any infinite recurrence of rogue deviations.

The Question That Opens the Next Step

We have seen how statistical laws govern the empirical reality of coin tosses and sample surveys. What happens when algebra itself hits an impenetrable wall? Recall the equation $x^2 = -1$: no real number, no tally of pebbles, no probability can possess a negative square. Yet in 16th-century Italy, while seeking formulas to solve ordinary cubic equations with obvious integer roots, algebraists were compelled to imagine a ghostly phantom number. A number capable of pivoting the coordinate plane by a ninety-degree right angle. Welcome to the fascinating world of complex numbers: turn the page to discover **Chapter 55 and The Right-Angled Whirlwind**.

CHAPTER 55: The Right-Angled Whirlwind

> "For centuries, schoolmasters taught that the square root of minus one was an impossible monster, an alchemist's chimera. But look at the weather vane on the rooftop: when the wind turns ninety degrees, the world does not collapse —it opens up. What algebra thought forbidden along the straight line, the geometry of the plane brings to light. Complex numbers are not imaginary: they are the breathing of the two-dimensional plane."

What You Are About to Discover

- The true story of the Italian Renaissance: how Gerolamo Cardano and Rafael Bombelli were forced to invent $\sqrt{-1}$ to discover solutions that were indisputably real.
- The epiphany of Argand and Gauss: multiplying by i simply means **rotating by a quarter-turn (90°)**.
- The **algebraic form** $z = a + ib$, the **complex conjugate** $\bar{z}$, and the fundamental concept of **modulus** $|z|$.
- The transition to polar coordinates: the **trigonometric form** and the extraordinary **exponential form** $z = re^{i\theta}$, along with Euler's and De Moivre's formulas.
- Complete solution of **quadratic equations** with real and complex coefficients via complex square roots.
- The **n-th roots of unity** and the geometry of regular polygons.
- Geometric interpretations: computing angles, collinearity, orthogonality, and the complex representation of **translations, homotheties, and rotations**.

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1. The Problem: The Forbidden Enigma of the Renaissance

1.1 The Algebra Duels in Bologna (1545)

In the 16th century, mathematics professors at Italian universities clashed in public duels before roaring crowds. Challengers hurled cubic equations (involving x^3) at one another. In **1545**, the Milanese physician and polymath **Gerolamo Cardano** published his masterwork, the *Ars Magna*. In it, he unveiled the formula for solving the general cubic equation $x^3 = px + q$, discovered by Scipione del Ferro and Niccolò Tartaglia:

$$x = \sqrt[3]{\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 - \left(\frac{p}{3}\right)^3}} + \sqrt[3]{\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 - \left(\frac{p}{3}\right)^3}}$$

Let us apply this formula to an apparently harmless equation:

$$x^3 = 15x + 4$$

Here, $p = 15$ and $q = 4$. Let us test small integers mentally:

$$4^3 = 64 \quad \text{and} \quad 15 \times 4 + 4 = 60 + 4 = 64.$$

The number $\boldsymbol{x = 4}$ **is an obvious solution**, perfectly real, tangible, and indisputable!

Now let us compute what Cardano's formula yields:

- $\frac{q}{2} = 2$
- $\frac{p}{3} = 5$
- The quantity under the square root is:

$$\left(\frac{q}{2}\right)^2 - \left(\frac{p}{3}\right)^3 = 2^2 - 5^3 = 4 - 125 = -121$$

The formula demands that we write:

$$x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}}$$

Cardano was frozen in his tracks. The square root of a negative number did not exist: every square of a real number is non-negative ($(+2)^2 = 4$ and $(-2)^2 = 4$). Cardano called this impasse an operation *"as subtle as it is useless"* and a *"mental torture."*

1.2 The Audacity of Rafael Bombelli (1572)

Twenty-seven years later, the Bolognese engineer **Rafael Bombelli** had an intuition of genius: *"What if we pretend to ignore the prohibition? Let us treat this quantity $\sqrt{-1}$ as an ordinary number, apply the standard rules of arithmetic, and see where it leads us!"*

Bombelli observed that:

$$\sqrt{-121} = \sqrt{121 \times (-1)} = 11\sqrt{-1}$$

He then calculated the cube of quantities of the form $(2 + \sqrt{-1})$:

$$(2 + \sqrt{-1})^3 = 2^3 + 3 \times 2^2 \times \sqrt{-1} + 3 \times 2 \times (\sqrt{-1})^2 + (\sqrt{-1})^3$$

Since $(\sqrt{-1})^2 = -1$ and $(\sqrt{-1})^3 = -1 \times \sqrt{-1} = -\sqrt{-1}$:

$$(2 + \sqrt{-1})^3 = 8 + 12\sqrt{-1} - 6 - \sqrt{-1} = 2 + 11\sqrt{-1} = 2 + \sqrt{-121}!$$

Similarly:

$$(2 - \sqrt{-1})^3 = 2 - \sqrt{-121}$$

Taking the cube roots in Cardano's formula:

$$x = (2 + \sqrt{-1}) + (2 - \sqrt{-1}) = 2 + 2 + (\sqrt{-1} - \sqrt{-1}) = 4!$$

The phantom $\sqrt{-1}$ entered the calculation, carried out the synthesis, and vanished in a cancellation, delivering the undeniably real root $x = 4$. For two and a half centuries, mathematicians used this chimerical number—named $\boldsymbol{i}$ (for *imaginary*) by **Leonhard Euler** in 1777—with a mixture of admiration and unease.

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2. The Geometric Breakthrough of Argand and Gauss

2.1 Multiplying by i Means Rotating by 90°

In 1806, a Swiss accountant and bookseller living in Paris, **Jean-Robert Argand**, dispelled the metaphysical fog with illuminating clarity.

Recall the real number line:

- Start at $+1$.
- Multiplying by -1 brings us to -1. Geometrically, this corresponds to a **half-turn ($180°$ or π radians)** around the origin.

$$\times (-1) \Leftrightarrow \text{Rotation by } \ 180°$$

By definition of the imaginary unit:

$$i \times i = -1$$

This algebraic identity states: *"Applying multiplication by i twice in succession produces the exact same geometric outcome as a $180°$ half-turn!"* What elementary geometric transformation, when applied twice consecutively, produces a half-turn? **A rotation by a quarter-turn ($90°$ or $\frac{\pi}{2}$ radians) in the counterclockwise (positive) direction!**

⚠ ⚠ Mathematical Rigor — Why the notation $\sqrt{-1}$ is strictly prohibited

While Bombelli and Cardano historically wrote $\sqrt{-1}$, modern mathematics **strictly prohibits this notation**. The familiar product rule $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$ holds true only for **non-negative real numbers**. Applying it indiscriminately to negatives creates catastrophic contradictions:

$$1 = \sqrt{1} = \sqrt{(-1) \times (-1)} = \sqrt{-1} \times \sqrt{-1} = i^2 = -1 \quad \Rightarrow \quad 1 = -1!$$

⚠ ⚠ Strict Prohibition of the Notation $i = \sqrt{-1}$

The expression $i = \sqrt{-1}$ is strictly banished from modern rigorous mathematics. The algebraic identity $\sqrt{ab} = \sqrt{a}\sqrt{b}$ holds **exclusively for non-negative real numbers**. Extending it blindly to negative values produces an immediate logical contradiction:

$$1 = \sqrt{1} = \sqrt{(-1) \times (-1)} = \sqrt{-1} \times \sqrt{-1} = i \times i = i^2 = -1 \quad \Longrightarrow 1 = -1!$$

To preserve the infallible coherence of mathematics, **we never place a negative real under the square root radical**. We define the imaginary unit i solely through its defining algebraic property:

$$\mathbf{i^2 = -1}.$$

To guarantee flawless logical consistency, **we never apply the square root radical to negative numbers**. Instead, we define the imaginary unit $\boldsymbol{i}$ purely through its fundamental algebraic relation:

$$\mathbf{i^2 = -1}$$

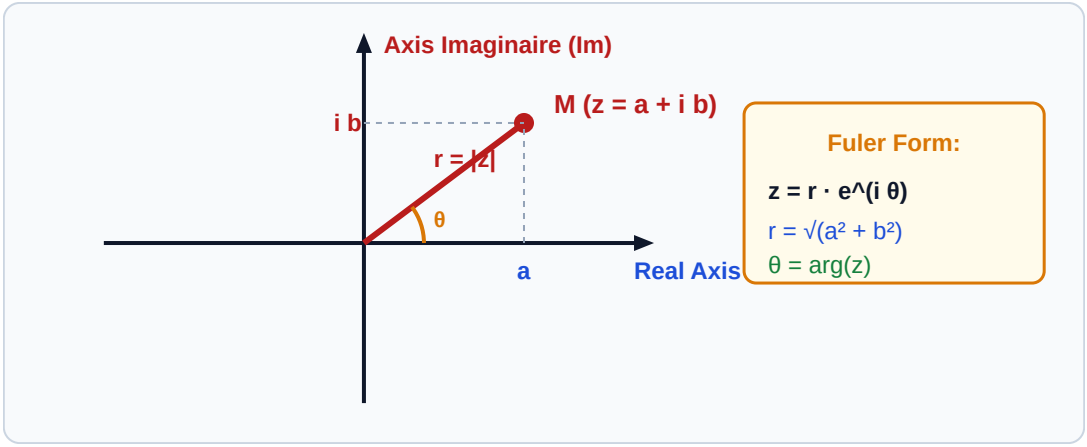

Figure 119: The Argand-Gauss complex plane: algebraic and trigonometric forms

Observe the cyclic progression of powers of i:

- $i^0 = 1$ (East)
- $i^1 = i$ (North, quarter-turn)
- $i^2 = -1$ (West, half-turn)
- $i^3 = i^2 \times i = -i$ (South, three-quarter turn)
- $i^4 = (i^2)^2 = (-1)^2 = 1$ (East, full turn)

The number i is not fictitious: $\boldsymbol{i}$ **is the gateway allowing numbers to break free from the line and inhabit the entire two-dimensional plane!**

3. The Algebraic Form and Its Operations

3.1 The Field $\mathbb{C}$ of Complex Numbers

📐 Definition — The Set of Complex Numbers

There exists a set denoted by $\mathbb{C}$, containing the set of real numbers $\mathbb{R}$, equipped with addition and multiplication operations extending those of $\mathbb{R}$, such that: 1. There exists an element $i \in \mathbb{C}$ satisfying:

$$i^2 = -1$$

2. Every element $z \in \mathbb{C}$ can be written **uniquely** in the form:

$$z = a + ib \quad \text{where} \quad a \in \mathbb{R} \quad \text{and} \quad b \in \mathbb{R}$$

This expression is called the **algebraic (or Cartesian) form** of z.

Standard Terminology: - a is the **real part** of z, denoted $\mathrm{Re}(z) = a$. - b is the **imaginary part** of z, denoted $\mathrm{Im}(z) = b$. *(⚠️ Frequent Trap: The imaginary part is the real number b, not ib!)*

Special Cases: - If $b = 0$, $z = a$ is **purely real** ($\mathbb{R} \subset \mathbb{C}$). - If $a = 0$, $z = ib$ is called **purely imaginary** (the set is denoted $i\mathbb{R}$).

📐 Theorem — Uniqueness of Algebraic Form

Let $z = a + ib$ and $z' = a' + ib'$ $(a, b, a', b' \in \mathbb{R})$.

$$z = z' \Leftrightarrow \{a = a'$$
$$b = b'$$

In particular:

$$a + ib = 0 \Leftrightarrow a = 0 \quad \text{and} \quad b = 0$$

3.2 The Complex Conjugate

📐 Definition — Complex Conjugate

For any complex number $z = a + ib$ $(a, b \in \mathbb{R})$, the **complex conjugate** of z, denoted $\bar{z}$ (read "z bar"), is defined by:

$$\bar{z} = a - ib$$

Geometrically, the point representing $\bar{z}$ is the reflection of the point representing z across the horizontal real axis.

📐 ***Fundamental Properties of the Conjugate*** *For all complex numbers z, z' and any integer $n \in \mathbb{Z}$: 1. $\overline{z + z'} = \bar{z} + \bar{z}'$ 2. $\overline{z \times z'} = \bar{z} \times \bar{z}'$ 3. $\overline{\left(\frac{z}{z'}\right)} = \frac{\bar{z}}{\bar{z}'}$ $(z' \neq 0)$ 4. $\overline{z^n} = (\bar{z})^n$ 5. $\bar{\bar{z}} = z$ 6. Sum and difference:*

$$z + \bar{z} = 2a = 2\,\mathrm{Re}(z) \quad \text{and} \quad z - \bar{z} = 2ib = 2i\,\mathrm{Im}(z)$$

7. ***Fundamental Criteria:***

$$z \in \mathbb{R} \Leftrightarrow \bar{z} = z \qquad \text{and} \qquad z \in i\mathbb{R} \Leftrightarrow \bar{z} = -z$$

8. ***Product with the Conjugate:***

$$z\bar{z} = (a + ib)(a - ib) = a^2 - (ib)^2 = a^2 - i^2b^2 = a^2 + b^2 \geq 0$$

Method for Dividing Complex Numbers: To make the denominator of a fraction $\frac{z_1}{z_2}$ real, multiply both numerator and denominator by the conjugate $\bar{z}_2$:

$$\frac{1 + 2i}{3 - 4i} = \frac{(1 + 2i)(3 + 4i)}{(3 - 4i)(3 + 4i)} = \frac{3 + 4i + 6i + 8i^2}{3^2 + 4^2} = \frac{3 - 8 + 10i}{9 + 16} = \frac{-5 + 10i}{25} = -\frac{1}{5} + \frac{2}{5}i.$$

3.3 The Modulus of a Complex Number

📐 Definition — Modulus

For any complex number $z = a + ib$ ($a, b \in \mathbb{R}$), the **modulus** of z, denoted $|z|$, is the non-negative real number defined by:

$$|z| = \sqrt{a^2 + b^2} = \sqrt{z\bar{z}}$$

Geometrically, if M has coordinates (a, b) in the orthonormal frame $(O; \vec{u}, \vec{v})$, by the Pythagorean theorem, $|z|$ is precisely the **distance** OM from the origin to point M!

📐 ***Properties of the Modulus*** *1.* $|z| \geq 0$, *and* $|z| = 0 \Leftrightarrow z = 0$. *2.* $|-z| = |\bar{z}| = |z|$. *3.* $|z \times z'| = |z| \times |z'|$. *4.* $\left|\frac{z}{z'}\right| = \frac{|z|}{|z'|} \quad (z' \neq 0)$. *5.* $|z^n| = |z|^n$ *for all* $n \in \mathbb{Z}$. *6.* ***Triangle Inequality:*** *For all* $z, z' \in \mathbb{C}$*:*

$$|z + z'| \leq |z| + |z'|$$

(In any triangle, the length of one side is at most the sum of the other two).

---

4. Trigonometric and Exponential Forms: The Dance of Angles

4.1 Argument of a Non-Zero Complex Number

Let $z \in \mathbb{C}^*$ (non-zero), represented by point M in the complex plane. The position of M is uniquely determined by:

- Its distance from the origin: $r = OM = |z| > 0$.
- The oriented angle between the positive real axis and vector $\overrightarrow{OM}$: $\theta = (\vec{u}, \overrightarrow{OM})$.

📐 Definition — Argument

For any non-zero complex number $z = a + ib$, an **argument** of z, denoted $\arg(z)$, is any radian measure of the oriented angle $(\vec{u}, \overrightarrow{OM})$. The argument is defined modulo 2π:

$$\arg(z) \equiv \theta \pmod{2\pi}$$

The unique value of the argument lying in the interval $(-\pi, \pi]$ is called the **principal argument**. *(The number 0 has no argument).*

Since the coordinates of M are $a = r\cos\theta$ and $b = r\sin\theta$, we obtain:

⚠ ⚠ Computing Arguments and the Arctangent Trap

While $\tan\theta = \frac{b}{a}$, using the $\arctan$ function directly yields the correct angle **only when $a > 0$** (right half-plane). When $a < 0$ (left half-plane), coordinates lie opposite the principal arctangent branch; the principal argument requires a π shift:

$$\theta = \arctan\left(\frac{b}{a}\right) + \pi \quad (\text{if } \ b \geq 0) \quad \text{or} \quad \theta = \arctan\left(\frac{b}{a}\right) - \pi \quad (\text{if } \ b < 0).$$

📐 Definition — Trigonometric Form

Every non-zero complex number z can be written as:

$$z = r(\cos\theta + i\sin\theta) \quad \text{where} \quad r = |z| > 0 \quad \text{and} \quad \theta \equiv \arg(z) \pmod{2\pi}$$

4.2 Euler's Exponential Notation

Consider the function $f(\theta) = \cos\theta + i\sin\theta$. Differentiating with respect to the real variable θ:

$$f'(\theta) = -\sin\theta + i\cos\theta = i(i\sin\theta + \cos\theta) = if(\theta)$$

This differential relation $f' = if$ with initial condition $f(0) = 1$ is the defining characteristic of the exponential function ($y' = ky$)! In **1748**, Leonhard Euler introduced what is celebrated as one of the most fruitful notations in mathematical history:

📐 Definition — Euler's Exponential Notation

For every real number $\theta \in \mathbb{R}$:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Every non-zero complex number z thus admits an **exponential form**:

$$z = re^{i\theta} \quad \text{where} \quad r = |z| > 0 \quad \text{and} \quad \theta \equiv \arg(z) \pmod{2\pi}$$

💡 Did You Know? Euler's Identity, the Most Beautiful Formula

Setting $\theta = \pi$ in Euler's formula yields:

$$e^{i\pi} = \cos(\pi) + i\sin(\pi) = -1 + 0 = -1$$

Which writes majestically as:

$$\mathbf{e^{i\pi} + 1 = 0}$$

This single equation links the five fundamental constants of mathematics:

- 0 (the additive identity)
- 1 (the multiplicative identity)
- π (the master of circles and geometry)
- e (the master of growth and analysis)
- i (the master of complex algebra)

4.3 Algebraic Properties and De Moivre's Formula

Exponential notation transforms trigonometry into elementary exponent rules:

📐 Theorem — Exponent Rules for Complex Exponentials

For all real numbers θ, θ': 1. Product: $e^{i\theta} \times e^{i\theta'} = e^{i(\theta+\theta')}$ *(Immediate corollary:* $\arg(zz') \equiv \arg(z) + \arg(z') \pmod{2\pi}$*)* 2. Quotient: $\frac{e^{i\theta}}{e^{i\theta'}} = e^{i(\theta-\theta')}$ *(Corollary:* $\arg\left(\frac{z}{z'}\right) \equiv \arg(z) - \arg(z') \pmod{2\pi}$*)* 3. Conjugate: $\overline{e^{i\theta}} = e^{-i\theta}$ 4. Reciprocal: $\frac{1}{e^{i\theta}} = e^{-i\theta}$

📐 Theorem — De Moivre's Formula

For any real number θ and any integer $n \in \mathbb{Z}$:

$$(e^{i\theta})^n = e^{in\theta} \Leftrightarrow (\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$$

Euler's Formulas for Linearization: Adding and subtracting $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \qquad \text{and} \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

5. Solving Quadratic Equations in $\mathbb{C}$

5.1 Equations with Real Coefficients

Consider the equation $az^2 + bz + c = 0$, where $a, b, c \in \mathbb{R}$ with $a \neq 0$. Compute the standard discriminant $\Delta = b^2 - 4ac$.

📐 Theorem — Quadratic Solutions over $\mathbb{R}$ and $\mathbb{C}$

- If $\Delta > 0$: two distinct real roots:

$$z_1 = \frac{-b - \sqrt{\Delta}}{2a} \quad \text{and} \quad z_2 = \frac{-b + \sqrt{\Delta}}{2a}$$

- If $\Delta = 0$: a single double real root:

$$z_0 = -\frac{b}{2a}$$

- If $\Delta < 0$: two distinct complex roots that are **complex conjugates**:

$$z_1 = \frac{-b - i\sqrt{-\Delta}}{2a} \quad \text{and} \quad z_2 = \frac{-b + i\sqrt{-\Delta}}{2a} = \bar{z}_1$$

Proof for $\Delta < 0$: Completing the square yields:

$$a\left[\left(z + \frac{b}{2a}\right)^2 - \frac{\Delta}{4a^2}\right] = 0$$

Since $\Delta < 0$, $-\Delta > 0$. We can write $-\Delta = (\sqrt{-\Delta})^2$. Thus:

$$-\frac{\Delta}{4a^2} = \frac{-\Delta}{(2a)^2} = -i^2\frac{-\Delta}{(2a)^2} = -\left(\frac{i\sqrt{-\Delta}}{2a}\right)^2$$

Factoring via the difference of squares $A^2 - B^2 = (A - B)(A + B)$:

$$\left(z + \frac{b}{2a} - \frac{i\sqrt{-\Delta}}{2a}\right)\left(z + \frac{b}{2a} + \frac{i\sqrt{-\Delta}}{2a}\right) = 0$$

Directly producing the two stated roots.

5.2 Equations with Complex Coefficients and Complex Square Roots

What if the equation has complex coefficients? For example: $z^2 + (1 + 4i)z - (5 + i) = 0$. The discriminant $\Delta = B^2 - 4AC$ is an arbitrary complex number $Z = X + iY$. To apply the quadratic formula $\frac{-B \pm \delta}{2A}$, we need to find a complex number $\delta = x + iy$ such that:

$$\delta^2 = Z$$

Such a number δ is called a **complex square root** of Z.

🖐 Method — Finding Complex Square Roots of $X+iY$

Let $\delta = x+iy$ with $x, y \in \mathbb{R}$. Set up the three-equation system: 1. Equality of squares: $\delta^2 = (x^2-y^2) + 2ixy = X+iY$

- Real part: $x^2 - y^2 = X$
- Imaginary part: $2xy = Y$

2. Equality of moduli (the crucial shortcut):

$$|\delta|^2 = |Z| \Leftrightarrow x^2+y^2 = \sqrt{X^2+Y^2}$$

Adding the real-part equation to the modulus equation:

$$2x^2 = \sqrt{X^2+Y^2} + X \Longrightarrow x = \pm\sqrt{\frac{\sqrt{X^2+Y^2}+X}{2}}$$

Subtracting:

$$2y^2 = \sqrt{X^2+Y^2} - X \Longrightarrow y = \pm\sqrt{\frac{\sqrt{X^2+Y^2}-X}{2}}$$

The sign condition on xy (governed by $2xy = Y$) determines the two opposite pairs (x, y).

---

6. The n-th Roots of Unity and Regular Polygons

Let us find all complex numbers z satisfying:

$$z^n = 1 \quad (n \in \mathbb{N}^*)$$

Express z in exponential form $z = re^{i\theta}$:

$$(re^{i\theta})^n = r^n e^{in\theta} = 1 = 1e^{i0}$$

Equating moduli and arguments:

- $r^n = 1 \Longrightarrow r = 1$ (since $r > 0$).
- $n\theta \equiv 0 \pmod{2\pi} \Longrightarrow n\theta = 2k\pi \Longrightarrow \theta = \frac{2k\pi}{n}$ for $k \in \mathbb{Z}$.

📐 Theorem — The Group of n-th Roots of Unity

There exist exactly n distinct complex solutions to $z^n = 1$, denoted $\mathbb{U}_n$:

$$\mathbb{U}_n = \left\{ \omega_k = e^{i\frac{2k\pi}{n}} \;\middle|\; k \in \{0, 1, 2, ..., n-1\} \right\}$$

Setting $\omega = e^{i\frac{2\pi}{n}}$, these roots are the successive powers of ω:

$$\mathbb{U}_n = \{1, \omega, \omega^2, ..., \omega^{n-1}\}$$

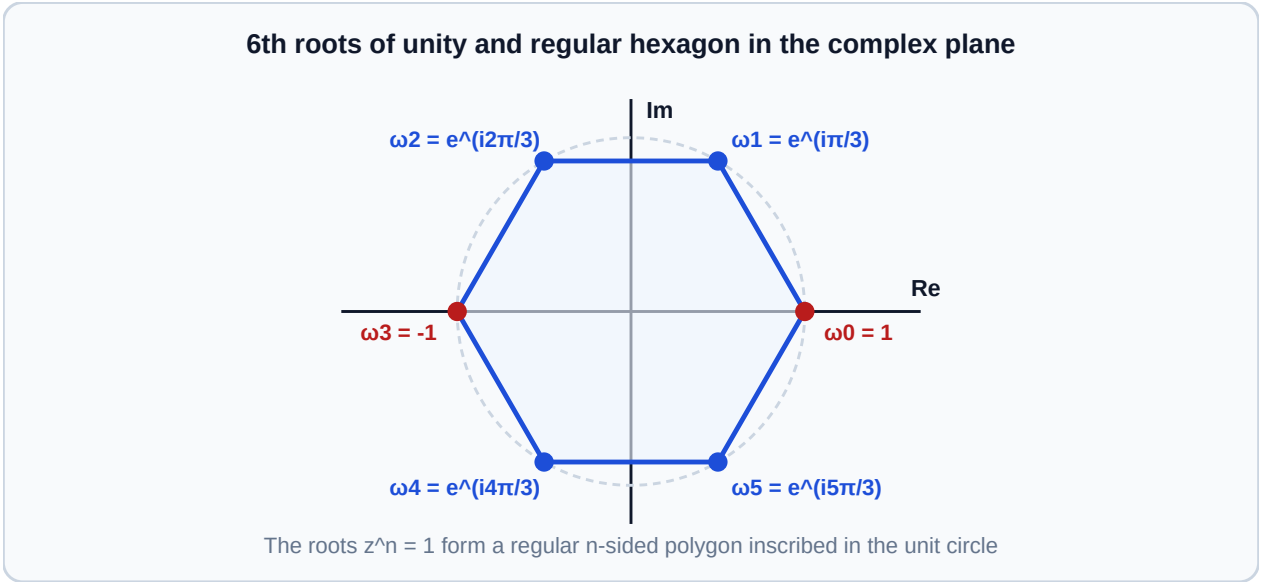

Figure 120: Modular arithmetic: clock face modulo 12

> 📐 ***Geometric Properties*** *1.* ***Regular Polygon:*** *The points representing the n-th roots of unity form the vertices of a* ***regular n-sided polygon*** *inscribed in the unit circle, with one vertex at $(1,0)$. 2.* ***Vanishing Sum (Center of Gravity):*** *For every $n \geq 2$:*
>
> $$\sum_{k=0}^{n-1} e^{i\frac{2k\pi}{n}} = 1 + \omega + \omega^2 + ... + \omega^{n-1} = 0$$

Proof of the Vanishing Sum: This is a geometric series with common ratio $\omega = e^{i\frac{2\pi}{n}} \neq 1$:

$$1 + \omega + ... + \omega^{n-1} = \frac{1-\omega^n}{1-\omega} = \frac{1-1}{1-\omega} = 0.$$

Physically, this represents the equilibrium of forces symmetrically distributed around a center of rotation!

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7. Geometric Applications: Distances, Angles, and Transformations

7.1 Affixes, Distances, and Oriented Angles

In the complex plane equipped with an orthonormal frame $(O; \vec{u}, \vec{v})$:

- To every point $M(x,y)$ or vector $\vec{V}(x,y)$, we associate its **affix (complex coordinate)** $z = x + iy$.
- The vector $\overrightarrow{AB}$ has affix $z_{\overrightarrow{AB}} = z_B - z_A$.
- The **distance** AB is the modulus:

$$AB = \| \overrightarrow{AB} \| = |z_B - z_A|$$

> 📐 **Theorem — Oriented Angle Between Two Vectors**
>
> For any four points A, B, C, D with $A \neq B$ and $C \neq D$:
>
> $$(\overrightarrow{AB}, \overrightarrow{CD}) \equiv \arg\left(\frac{z_D - z_C}{z_B - z_A}\right) \pmod{2\pi}$$

7.2 Collinearity and Orthogonality Criteria

Define the ratio $Z = \frac{z_C - z_A}{z_B - z_A}$:

- **Collinear Points (Alignment):** Points A, B, C are collinear if and only if the angle $(\overrightarrow{AB}, \overrightarrow{AC})$ equals 0 or π modulo 2π:

$$A, B, C \text{ collinear} \Leftrightarrow \frac{z_C - z_A}{z_B - z_A} \in \mathbb{R}$$

- **Perpendicular Lines (Orthogonality):** Lines (AB) and (CD) are perpendicular if and only if the angle $(\overrightarrow{AB}, \overrightarrow{CD})$ equals $\pm\frac{\pi}{2}$ modulo 2π:

$$(AB) \perp (CD) \Leftrightarrow \frac{z_D - z_C}{z_B - z_A} \in i\mathbb{R} \quad \text{(purely imaginary)}$$

7.3 Plane Transformations via Complex Numbers

Transformation	Geometric Definition	Complex Mapping ($z \mapsto z'$)
Translation by vector $\vec{w}$ (affix b)	$\overrightarrow{MM'} = \vec{w}$	$z' = z + b$
Homothety (Dilation) with center $\Omega(\omega)$ and ratio $k \in \mathbb{R}^*$	$\overrightarrow{\Omega M'} = k\overrightarrow{\Omega M}$	$z' - \omega = k(z - \omega)$
Rotation with center $\Omega(\omega)$ and angle $\theta \in \mathbb{R}$	$\Omega M' = \Omega M$ and $(\overrightarrow{\Omega M}, \overrightarrow{\Omega M'}) = \theta$	$z' - \omega = e^{i\theta}(z - \omega)$
Direct Similitude with center Ω, ratio k, and angle θ	Composition of homothety and rotation	$z' = az + b \quad (a \in \mathbb{C}^*)$

Elegance of Rotation: A rotation centered at the origin by angle θ is simply expressed as $z' = e^{i\theta}z$. Multiplying by $e^{i\pi/2} = i$ rotates the plane by $\pi/2$!

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8. Step-by-Step Worked Examples

Example 1: Algebraic Form, Trigonometric Form, and Powers

Let $z_1 = 1 + i\sqrt{3}$ and $z_2 = 1 - i$. 1. Determine the modulus and a principal argument of z_1 and z_2. 2. Write z_1 and z_2 in exponential form. 3. Deduce the exponential form and then the algebraic form of the quotient $Z = \frac{z_1}{z_2}$. 4. Deduce the exact values of $\cos\left(\frac{7\pi}{12}\right)$ and $\sin\left(\frac{7\pi}{12}\right)$. 5. Calculate $(z_1)^{12}$.

Detailed Solution: 1. - For $z_1 = 1 + i\sqrt{3}$: $|z_1| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2$. $z_1 = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$. We seek θ_1 such that $\cos\theta_1 = \frac{1}{2}$ and $\sin\theta_1 = \frac{\sqrt{3}}{2}$, yielding $\theta_1 \equiv \frac{\pi}{3} \pmod{2\pi}$.

- For $z_2 = 1 - i$: $|z_2| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$. $z_2 = \sqrt{2}\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right)$, yielding $\theta_2 \equiv -\frac{\pi}{4} \pmod{2\pi}$.

2. $z_1 = 2e^{i\frac{\pi}{3}}$ and $z_2 = \sqrt{2}e^{-i\frac{\pi}{4}}$.

3. Quotient in exponential form:

$$Z = \frac{z_1}{z_2} = \frac{2e^{i\frac{\pi}{3}}}{\sqrt{2}e^{-i\frac{\pi}{4}}} = \frac{2}{\sqrt{2}}e^{i\left(\frac{\pi}{3} - \left(-\frac{\pi}{4}\right)\right)} = \sqrt{2}e^{i\left(\frac{4\pi+3\pi}{12}\right)} = \sqrt{2}e^{i\frac{7\pi}{12}}$$

Algebraic form by multiplying numerator and denominator by the conjugate:

$$Z = \frac{1 + i\sqrt{3}}{1 - i} = \frac{(1 + i\sqrt{3})(1 + i)}{(1 - i)(1 + i)} = \frac{1 + i + i\sqrt{3} + i^2\sqrt{3}}{1^2 + (-1)^2} = \frac{(1 - \sqrt{3}) + i(1 + \sqrt{3})}{2}$$

4. Equating the two forms of $Z = \sqrt{2}\left(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right)$:
   - $\sqrt{2}\cos\left(\frac{7\pi}{12}\right) = \frac{1-\sqrt{3}}{2} \Longrightarrow \cos\left(\frac{7\pi}{12}\right) = \frac{1-\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{2}-\sqrt{6}}{4}$.
   - $\sqrt{2}\sin\left(\frac{7\pi}{12}\right) = \frac{1+\sqrt{3}}{2} \Longrightarrow \sin\left(\frac{7\pi}{12}\right) = \frac{1+\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{2}+\sqrt{6}}{4}$.

5. Power:

$$(z_1)^{12} = \left(2e^{i\frac{\pi}{3}}\right)^{12} = 2^{12}e^{i\frac{12\pi}{3}} = 4096\,e^{i4\pi} = 4096 \times 1 = 4\,096.$$

Example 2: Solving Real-Coefficient Quadratics

Solve in $\mathbb{C}$:

$$z^2 - 4z + 13 = 0$$

Detailed Solution: Coefficients are $a = 1, b = -4, c = 13$. Compute the discriminant:

$$\Delta = b^2 - 4ac = (-4)^2 - 4 \times 1 \times 13 = 16 - 52 = -36 < 0$$

Since $\Delta < 0$, the equation has two complex conjugate roots. With $-36 = (6i)^2$, we have $\sqrt{-\Delta} = \sqrt{36} = 6$.

$$z_1 = \frac{-(-4) - 6i}{2 \times 1} = \frac{4 - 6i}{2} = 2 - 3i$$

$$z_2 = \frac{-(-4) + 6i}{2 \times 1} = \frac{4 + 6i}{2} = 2 + 3i = \bar{z}_1$$

The solution set is $S = \{2 - 3i; 2 + 3i\}$.

Example 3: Finding Square Roots of a Complex Number

Find the complex square roots of $Z = 3 + 4i$.

Detailed Solution: We seek $\delta = x + iy$ $(x, y \in \mathbb{R})$ such that $\delta^2 = 3 + 4i$. 1. Real part: $x^2 - y^2 = 3$. 2. Imaginary part: $2xy = 4 \Longrightarrow xy = 2 > 0$ (so x and y share the same sign). 3. Modulus: $|\delta|^2 = |Z| \Leftrightarrow x^2 + y^2 = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

Adding and subtracting:

- $(x^2 + y^2) + (x^2 - y^2) = 5 + 3 \Longrightarrow 2x^2 = 8 \Longrightarrow x^2 = 4 \Longrightarrow x = \pm 2$.
- $(x^2 + y^2) - (x^2 - y^2) = 5 - 3 \Longrightarrow 2y^2 = 2 \Longrightarrow y^2 = 1 \Longrightarrow y = \pm 1$.

Since $xy = 2 > 0$:

- If $x = 2$, then $y = 1 \Longrightarrow \delta_1 = 2 + i$.
- If $x = -2$, then $y = -1 \Longrightarrow \delta_2 = -2 - i = -\delta_1$.

The two complex square roots of $3 + 4i$ are $\mathbf{2 + i}$ and $\mathbf{-2 - i}$. *(Verification:* $(2 + i)^2 = 4 + 4i + i^2 = 3 + 4i$*).*

Example 4: Quadratic Equation with Complex Coefficients

Solve in $\mathbb{C}$:

$$z^2 - (3 + 2i)z + (5 + i) = 0$$

Detailed Solution: Compute the discriminant $\Delta = B^2 - 4AC$:

$$\Delta = (3 + 2i)^2 - 4(1)(5 + i) = (9 + 12i - 4) - (20 + 4i) = (5 + 12i) - 20 - 4i = -15 + 8i$$

Find a square root $\delta = x + iy$ of $\Delta = -15 + 8i$:

- $x^2 - y^2 = -15$
- $2xy = 8 \Longrightarrow xy = 4 > 0$
- $x^2 + y^2 = \sqrt{(-15)^2 + 8^2} = \sqrt{225 + 64} = \sqrt{289} = 17$

By half-sum and half-difference:

- $2x^2 = 17 - 15 = 2 \Longrightarrow x^2 = 1 \Longrightarrow x = \pm 1$.
- $2y^2 = 17 - (-15) = 32 \Longrightarrow y^2 = 16 \Longrightarrow y = \pm 4$.

Since $xy > 0$, we choose $\delta = 1 + 4i$. The solutions are:

$$z_1 = \frac{(3 + 2i) + (1 + 4i)}{2} = \frac{4 + 6i}{2} = 2 + 3i$$

$$z_2 = \frac{(3 + 2i) - (1 + 4i)}{2} = \frac{2 - 2i}{2} = 1 - i$$

The solution set is $S = \{1 - i; 2 + 3i\}$.

Example 5: Geometric Nature of a Triangle via Affixes

In the complex plane, consider points A, B, C with respective coordinates:

$$z_A = 1 + i, \quad z_B = 3 - i, \quad z_C = 2 + 2i$$

1. Calculate the quotient $Z = \frac{z_C - z_A}{z_B - z_A}$. 2. Deduce the exact nature of triangle ABC.

Detailed Solution: 1. Compute the affixes of vectors $\overrightarrow{AC}$ and $\overrightarrow{AB}$:

- $z_C - z_A = (2 + 2i) - (1 + i) = 1 + i$
- $z_B - z_A = (3 - i) - (1 + i) = 2 - 2i$ Evaluate the quotient:

$$Z = \frac{1 + i}{2 - 2i} = \frac{(1 + i)(2 + 2i)}{(2 - 2i)(2 + 2i)} = \frac{2 + 2i + 2i + 2i^2}{2^2 + (-2)^2} = \frac{4i}{8} = \frac{1}{2}i \in i\mathbb{R}$$

2. Since the quotient is purely imaginary, the angle $(\overrightarrow{AB}, \overrightarrow{AC}) \equiv \frac{\pi}{2} \pmod{\pi}$. Therefore, triangle ABC is **right-angled at A**.

Example 6: Identifying a Rotation

Consider the plane transformation f mapping every point M with affix z to M' with affix:

$$z' = iz + 3 - i$$

1. Prove that f has a unique invariant (fixed) point Ω and find its affix ω. 2. Show that f is a rotation and specify its center and angle.

Detailed Solution: 1. A point $\Omega(\omega)$ is fixed if and only if $f(\Omega) = \Omega$:

$$\omega = i\omega + 3 - i \Leftrightarrow \omega(1 - i) = 3 - i \Leftrightarrow \omega = \frac{3 - i}{1 - i}$$

Multiply numerator and denominator by $1 + i$:

$$\omega = \frac{(3 - i)(1 + i)}{1^2 + (-1)^2} = \frac{3 + 3i - i - i^2}{2} = \frac{4 + 2i}{2} = 2 + i$$

The unique invariant point is Ω with affix $\omega = 2 + i$. 2. Subtract ω from both sides of $z' = iz + 3 - i$:

$$z' - \omega = iz + 3 - i - (2 + i) = iz + 1 - 2i = i(z - (2 + i)) = i(z - \omega)$$

Since $i = e^{i\frac{\pi}{2}}$, the mapping becomes:

$$z' - \omega = e^{i\frac{\pi}{2}}(z - \omega)$$

This is the standard equation of a **rotation with center $\Omega(2 + i)$ and angle $\frac{\pi}{2}$** (a positive quarter-turn).

Key Takeaways

1. **Algebraic Form:**
   - $z = a + ib$ with $a = \mathrm{Re}(z) \in \mathbb{R}$ and $b = \mathrm{Im}(z) \in \mathbb{R}$.
   - Complex conjugate: $\bar{z} = a - ib$, with $z\bar{z} = a^2 + b^2 = |z|^2$.
   - Modulus: $|z| = \sqrt{a^2 + b^2}$.
2. **Trigonometric and Exponential Forms:**
   - $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$ with $r = |z|$ and $\theta \equiv \arg(z) \pmod{2\pi}$.
   - Euler's formula: $e^{i\theta} = \cos\theta + i\sin\theta$. De Moivre's formula: $(e^{i\theta})^n = e^{in\theta}$.
3. **Quadratic Equations:**
   - If $\Delta < 0$ in $az^2 + bz + c = 0$ $(a, b, c \in \mathbb{R})$: $z = \frac{-b \pm i\sqrt{-\Delta}}{2a}$.
   - Complex square roots of $X + iY$: solve $\left\{x^2 - y^2 = X, 2xy = Y, x^2 + y^2 = \sqrt{X^2 + Y^2}\right\}$.
4. **n-th Roots of Unity:**
   - $\mathbb{U}_n = \left\{e^{i\frac{2k\pi}{n}} \mid k \in \{0, ..., n-1\}\right\}$, representing vertices of a regular polygon centered at 0 with zero sum.
5. **Complex Geometry:**
   - Distance: $AB = |z_B - z_A|$.
   - Angle: $(\overrightarrow{AB}, \overrightarrow{CD}) \equiv \arg\left(\frac{z_D - z_C}{z_B - z_A}\right) \pmod{2\pi}$.
   - Rotation: $z' - \omega = e^{i\theta}(z - \omega)$.

Exercises

Level (Direct Applications)

Exercise 55.1 Write in algebraic form $a + ib$: 1. $z_1 = (2 + 3i)(4 - i)$ 2. $z_2 = (3 - 2i)^2$ 3. $z_3 = \frac{1}{2+i}$ 4. $z_4 = \frac{3+5i}{1-2i}$
Exercise 55.2 Determine the modulus and principal argument of: $z_A = -3$, $z_B = 2i$, $z_C = 1 + i$, $z_D = -\sqrt{3} + i$.
Exercise 55.3 Solve in $\mathbb{C}$: 1. $z^2 + 9 = 0$ 2. $z^2 - 2z + 5 = 0$ 3. $2z^2 + 2z + 1 = 0$ **Exercise 55.4** Compute i^n for $n \in$

$\{2024, 2025, 2026, 2027\}$. **Exercise 55.5** Linearize $\cos^2(x)$ and $\sin^2(x)$ using Euler's formulas. **Exercise 55.6** State the affix of vector $\overrightarrow{AB}$ and distance AB for $A(3-2i)$ and $B(-1+i)$. **Exercise 55.7** Identify the geometric transformation associated with: 1. $z' = z + 4 - 3i$ 2. $z' = -z$ 3. $z' = iz$

Level (Exploration and Advanced Geometry)

Exercise 55.8 Solve in $\mathbb{C}$: $2z + 3\bar{z} = 5 + 2i$. **Exercise 55.9** Determine the locus of points M with affix z such that: 1. $|z - 3 + 2i| = 5$ 2. $|z - 2| = |z + 4i|$ 3. $\frac{z-1}{z+1} \in i\mathbb{R}$ **Exercise 55.10** Complex square roots: Calculate the complex square roots of $Z_1 = -7 + 24i$ and $Z_2 = 8 - 6i$. **Exercise 55.11** Solve in $\mathbb{C}$:

$$z^2 - (5 - i)z + (8 - i) = 0$$

Exercise 55.12 De Moivre's formula and trigonometry: 1. Expanding $(\cos x + i \sin x)^3$ in two ways, express $\cos(3x)$ in terms of $\cos(x)$, and $\sin(3x)$ in terms of $\sin(x)$. 2. Linearize $\cos^3(x)$. **Exercise 55.13** The number j (primitive cube root of unity). Let $j = e^{i\frac{2\pi}{3}} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$. 1. Verify that $j^2 = \bar{j} = e^{i\frac{4\pi}{3}}$ and $j^3 = 1$. 2. Prove that $1 + j + j^2 = 0$. 3. Compute $(1 + j)^{2025}$. **Exercise 55.14** Equilateral triangles. Show that three points A, B, C with affixes a, b, c form a positively oriented equilateral triangle if and only if:

$$a + bj + cj^2 = 0 \quad \text{where} \quad j = e^{i\frac{2\pi}{3}}$$

Exercise 55.15 Collinearity and concyclicity (cross-ratio). Define the cross-ratio of four distinct points A, B, C, D by:

$$[a, b, c, d] = \frac{(c - a)(d - b)}{(c - b)(d - a)}$$

Show that A, B, C, D are concyclic (lying on a common circle) or collinear if and only if their cross-ratio is real.

Level (Deepening and Olympiads)

Exercise 55.16 Napoleon's Theorem. On the sides of an arbitrary positively oriented triangle ABC, construct three equilateral triangles externally with centroids P, Q, R. Prove, using complex affixes and the root j, that triangle PQR is **always equilateral**! **Exercise 55.17** Cardano's cubic method. Consider $z^3 - 6z - 9 = 0$. Setting $z = u + v$ with $3uv = 6$, determine u^3 and v^3 as roots of a quadratic equation, and deduce the real solution and the two complex roots. **Exercise 55.18** Trigonometric polynomials and the Dirichlet kernel. For $x \not\equiv 0 \pmod{2\pi}$, compute the sum:

$$D_n(x) = 1 + 2\sum_{k=1}^{n} \cos(kx) = 1 + 2\cos(x) + 2\cos(2x) + ... + 2\cos(nx)$$

Show that $D_n(x) = \frac{\sin((n+\frac{1}{2})x)}{\sin(\frac{x}{2})}$. **Exercise 55.19** Roots of unity and polynomial factorization. 1. Factor the polynomial $X^n - 1$ in $\mathbb{C}[X]$ using roots of unity. 2. By evaluating this factorization at a suitable point, prove the identity:

$$\prod_{k=1}^{n-1} \sin\left(\frac{k\pi}{n}\right) = \frac{n}{2^{n-1}}$$

Exercise 55.20 Fundamental Theorem of Algebra (D'Alembert-Gauss, outline). Explain why every non-constant polynomial with complex coefficients has at least one root in $\mathbb{C}$, and why this implies that $\mathbb{C}$ is algebraically closed (unlike $\mathbb{R}$). **Exercise 55.21** Euler Line and Euler Circle. Let triangle ABC be inscribed in a circle centered at origin O ($|a| = |b| = |c| = R$). 1. Show that H with affix $h = a + b + c$ is the orthocenter. 2. Show that centroid G has affix $g = \frac{a+b+c}{3}$. 3. Deduce that O, G, H are collinear (Euler line) and $\overrightarrow{OH} = 3\overrightarrow{OG}$. **Exercise 55.22** Möbius Transformations. Consider the linear fractional transformation $f(z) = \frac{az+b}{cz+d}$ with $ad - bc \neq 0$. Show that such a map transforms circles and lines into circles or lines.

Solutions

Solution 55.1: 1. $z_1 = 8 - 2i + 12i - 3i^2 = 8 + 10i + 3 = 11 + 10i$. 2. $z_2 = 9 - 12i + 4i^2 = 5 - 12i$. 3. $z_3 = \frac{2-i}{2^2+1^2} = \frac{2}{5} - \frac{1}{5}i$. 4. $z_4 = \frac{(3+5i)(1+2i)}{1^2+(-2)^2} = \frac{3+6i+5i-10}{5} = \frac{-7+11i}{5} = -\frac{7}{5} + \frac{11}{5}i$.
Solution 55.2: - $z_A = -3$: $|z_A| = 3$, $\arg(z_A) = \pi$. - $z_B = 2i$: $|z_B| = 2$, $\arg(z_B) = \frac{\pi}{2}$. - $z_C = 1 + i$: $|z_C| = \sqrt{2}$, $\arg(z_C) = \frac{\pi}{4}$. - $z_D = -\sqrt{3} + i$: $|z_D| = \sqrt{3+1} = 2$, $\cos\theta = -\frac{\sqrt{3}}{2}, \sin\theta = \frac{1}{2} \Longrightarrow \arg(z_D) = \frac{5\pi}{6}$.
Solution 55.3: 1. $z^2 = -9 = (3i)^2 \Leftrightarrow z \in \{-3i, 3i\}$. 2. $\Delta = 4 - 20 = -16 = (4i)^2 \Longrightarrow z = \frac{2 \pm 4i}{2} = 1 \pm 2i$. 3. $\Delta = 4 - 8 = -4 = (2i)^2 \Longrightarrow z = \frac{-2 \pm 2i}{4} = -\frac{1}{2} \pm \frac{1}{2}i$.

Solution 55.4: Since $i^4 = 1$:

- $2024 = 4 \times 506 \Longrightarrow i^{2024} = 1$.
- $i^{2025} = i$, $i^{2026} = -1$, $i^{2027} = -i$.

Solution 55.5: $\cos^2(x) = \left(\frac{e^{ix}+e^{-ix}}{2}\right)^2 = \frac{e^{2ix}+2+e^{-2ix}}{4} = \frac{\cos(2x)+1}{2}$. $\sin^2(x) = \left(\frac{e^{ix}-e^{-ix}}{2i}\right)^2 = \frac{e^{2ix}-2+e^{-2ix}}{-4} = \frac{1-\cos(2x)}{2}$.

Solution 55.6: $z_{\overrightarrow{AB}} = z_B - z_A = (-1+i) - (3-2i) = -4+3i$. $AB = |-4+3i| = \sqrt{(-4)^2+3^2} = \sqrt{16+9} = 5$.

Solution 55.7: 1. Translation by vector with affix $4-3i$. 2. Central symmetry with respect to origin O (rotation by angle π). 3. Rotation centered at O by angle $\frac{\pi}{2}$.

Solution 55.8: Set $z = x+iy$. Then $\bar{z} = x-iy$. $2(x+iy)+3(x-iy) = 5x-iy$. Equate to $5+2i$: $5x = 5 \Longrightarrow x = 1$, and $-y = 2 \Longrightarrow y = -2$. Hence $z = 1-2i$.

Solution 55.9: 1. Let $A(3-2i)$: $AM = 5$. This is the circle centered at $A(3,-2)$ with radius 5. 2. Let $A(2)$ and $B(-4i)$: $AM = BM$. This is the perpendicular bisector of segment $[AB]$. 3. The argument of $\frac{z-1}{z+1}$ is $\pm\frac{\pi}{2} \pmod{\pi}$, which means $(\overrightarrow{MB}, \overrightarrow{MA}) \equiv \frac{\pi}{2} \pmod{\pi}$. This is the circle with diameter $[AB]$ (where $A(1)$ and $B(-1)$), excluding points A and B.

Solution 55.10: - For $Z_1 = -7+24i$: modulus $\sqrt{49+576} = 25$. $x^2-y^2 = -7$, $x^2+y^2 = 25 \Longrightarrow 2x^2 = 18 \Longrightarrow x = \pm 3$ and $y = \pm 4$. Since $2xy > 0$, the roots are $\pm(3+4i)$.

- For $Z_2 = 8-6i$: modulus $\sqrt{64+36} = 10$. $2x^2 = 18 \Longrightarrow x = \pm 3$, $2y^2 = 2 \Longrightarrow y = \pm 1$. Since $2xy = -6 < 0$, the roots are $\pm(3-i)$.

Solution 55.11: $\Delta = (5-i)^2 - 4(8-i) = (24-10i) - (32-4i) = -8-6i$. Square roots of $-8-6i$: modulus $\sqrt{64+36} = 10$. $2x^2 = 10-8 = 2 \Longrightarrow x = \pm 1$, $2y^2 = 18 \Longrightarrow y = \pm 3$. With $xy < 0$, $\delta = 1-3i$. $z_1 = \frac{(5-i)+(1-3i)}{2} = 3-2i$, and $z_2 = \frac{(5-i)-(1-3i)}{2} = 2+i$.

Solution 55.12: 1. $(\cos x + i\sin x)^3 = \cos^3 x + 3i\cos^2 x \sin x - 3\cos x \sin^2 x - i\sin^3 x$. By De Moivre, this equals $\cos(3x) + i\sin(3x)$. Equating parts: $\cos(3x) = \cos^3 x - 3\cos x(1-\cos^2 x) = 4\cos^3 x - 3\cos x$. $\sin(3x) = 3(1-\sin^2 x)\sin x - \sin^3 x = 3\sin x - 4\sin^3 x$. 2. $\cos^3 x = \frac{\cos(3x)+3\cos x}{4}$.

Solution 55.13: 1. $j^2 = e^{i\frac{4\pi}{3}} = e^{-i\frac{2\pi}{3}} = \bar{j}$, and $j^3 = e^{i2\pi} = 1$. 2. $1+j+j^2 = \frac{1-j^3}{1-j} = 0$. 3. $1+j = -j^2$. Thus $(1+j)^{2025} = (-j^2)^{2025} = -j^{4050} = -(j^3)^{1350} = -1$.

Solution 55.14: Triangle ABC is positively oriented equilateral if and only if C is the image of B under rotation centered at A by angle $\pi/3$, which translates after vector rewriting to $(c-a) = -j^2(b-a)$, hence $a(1+j^2)+bj^2+c = 0$. Since $1+j^2 = -j$, multiplying by j^2 yields $a+bj+cj^2 = 0$.

Solution 55.15: The argument of the cross-ratio is $\arg\left(\frac{c-a}{c-b}\right) - \arg\left(\frac{d-a}{d-b}\right) = (\overrightarrow{BC}, \overrightarrow{AC}) - (\overrightarrow{BD}, \overrightarrow{AD})$. This cross-ratio is real if and only if this angle difference is 0 modulo π, which is precisely the inscribed angle theorem condition for A, B, C, D to be concyclic or collinear.

Solution 55.16: Let P, Q, R be the centroids of the equilateral triangles constructed on sides $[B,C], [C,A], [A,B]$. The outer vertices satisfy rotation relations of angle $\pi/3$. Expressing p, q, r in terms of a, b, c and j, computing $p+qj+rj^2$ vanishes identically due to $1+j+j^2 = 0$. By Exercise 14, PQR is equilateral.

Solution 55.17: Setting $z = u+v$, $(u+v)^3 - 6(u+v) - 9 = u^3+v^3+3uv(u+v) - 6(u+v) - 9 = 0$. Setting $3uv = 6 \Longrightarrow uv = 2 \Longrightarrow u^3v^3 = 8$, leaving $u^3+v^3 = 9$. u^3 and v^3 are roots of $X^2 - 9X + 8 = 0$, which are 1 and 8. Set $u^3 = 8 \Longrightarrow u = 2$ and $v^3 = 1 \Longrightarrow v = 1$. The real root is $z_0 = 2+1 = 3$. Factoring $z^3 - 6z - 9 = (z-3)(z^2+3z+3)$, the discriminant of the quadratic factor is $\Delta = 9-12 = -3$, yielding the two complex roots: $z_{1,2} = \frac{-3\pm i\sqrt{3}}{2}$.

Solution 55.18: $D_n(x) = \sum_{k=-n}^{n} e^{ikx} = e^{-inx}\sum_{j=0}^{2n} e^{ijx} = e^{-inx}\frac{1-e^{i(2n+1)x}}{1-e^{ix}}$. Factoring out half-angles in numerator and denominator: $D_n(x) = e^{-inx}\frac{e^{i\frac{2n+1}{2}x}(-2i\sin(\frac{2n+1}{2}x))}{e^{i\frac{x}{2}}(-2i\sin(\frac{x}{2}))} = \frac{\sin((n+\frac{1}{2})x)}{\sin(\frac{x}{2})}$.

Solution 55.19: 1. The roots of $X^n - 1$ are $e^{i\frac{2k\pi}{n}}$. Thus $X^n - 1 = \prod_{k=0}^{n-1}(X - e^{i\frac{2k\pi}{n}}) = (X-1)\prod_{k=1}^{n-1}(X - e^{i\frac{2k\pi}{n}})$. 2. Dividing by $X-1$: $\sum_{j=0}^{n-1} X^j = \prod_{k=1}^{n-1}(X - e^{i\frac{2k\pi}{n}})$. Evaluating at $X = 1$, the left side equals n. On the right, $|1 - e^{i\frac{2k\pi}{n}}| = |e^{i\frac{k\pi}{n}}(e^{-i\frac{k\pi}{n}} - e^{i\frac{k\pi}{n}})| = 2\sin(\frac{k\pi}{n})$. Taking the product of moduli gives $n = 2^{n-1}\prod_{k=1}^{n-1}\sin(\frac{k\pi}{n})$, proving the formula.

Solution 55.20: The field $\mathbb{C}$ is algebraically closed because every non-constant polynomial of degree n splits completely into n linear factors over $\mathbb{C}$ (counted with multiplicity). Proved rigorously by Gauss in 1799, this theorem demonstrates

the completeness of the complex numbers: no further extension fields are needed to solve higher-degree polynomial equations.

Solution 55.21: 1. To check that H is the orthocenter, compute the dot product $\overrightarrow{AH} \cdot \overrightarrow{BC}$. The affix of $\overrightarrow{AH}$ is $h - a = b + c$, and that of $\overrightarrow{BC}$ is $c - b$. The ratio $\frac{h-a}{c-b} = \frac{b+c}{c-b} = \frac{(b+c)(\bar{c}-\bar{b})}{|c-b|^2}$. On the unit circle, $\bar{b} = R^2/b$ and $\bar{c} = R^2/c$. The real part vanishes, proving $(AH) \perp (BC)$. The same holds for other altitudes. 2. The centroid G has affix $g = \frac{a+b+c}{3}$. 3. We have $h = 3g$, which translates vectorially to $\overrightarrow{OH} = 3\overrightarrow{OG}$. Hence O, G, H are collinear along the Euler line.

Solution 55.22: Möbius transformations are compositions of translations, dilatations/rotations, and inversion $z \mapsto \frac{1}{z}$. Since inversion maps lines and circles into lines or circles (viewed as circles on the Riemann sphere), this family of curves is preserved.

The Question That Opens the Next Step

Complex numbers demonstrated the supreme power of constructing new mathematical structures to overcome algebraic obstacles. What happens when we return to the most primal and unyielding numbers: the natural integers? How can the remainders of simple division algorithms safeguard state secrets and secure bank cards across the globe? The clock ticks, hands align modulo n: welcome to **Chapter 56 and Secret Messages and the Clock of Remainders**.

CHAPTER 56: Secret Messages and the Clock of Remainders

> "On the face of a clock, after twelve o'clock does not come thirteen o'clock: the hand returns to one o'clock. This circular arithmetic, which every child practices without realizing it, is one of the deepest jewels of the human mind. It was by studying the remainders of divisions that Fermat unlocked the secret of powers, and that three centuries later, all of humanity learned to lock its digital secrets behind the indestructible padlock of prime numbers."

What You Will Discover

- The rigorous structure of **divisibility in $\mathbb{Z}$** and **Euclidean division**.
- The fascinating world of **congruences modulo n**: circular arithmetic where numbers wind around a circle.
- The **extended Euclidean algorithm**, **Bézout's identity**, and the methodical resolution of **Diophantine equations** $ax + by = c$.
- **Gauss's lemma (theorem)**, a pillar of arithmetic, and its canonical proof.
- **Fermat's Little Theorem** and the method of **fast modular exponentiation**.
- The adventure of encryption: from the Caesar cipher and affine ciphers to the asymmetric **RSA** protocol, the shield of the modern Internet.

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1. The Problem: Sending a Secret in Plain Sight

1.1 Caesar's Courier and the Vulnerability of the Shared Key

Around 50 BCE, Julius Caesar sent military orders to his legions in Gaul. To prevent an intercepted courier from revealing his plans, he shifted each letter by 3 positions in the alphabet:

- A becomes D, B becomes E… X becomes A, Y becomes B, Z becomes C.

The recipient performs the inverse shift (-3).

This procedure has a glaring flaw: anyone who knows the rule can decipher all messages. Even when complicating the formula (affine cipher), frequency analysis of letters (with 'E' being the most common letter in French and English) breaks the code within minutes.

Yet the fundamental problem of classical cryptography is even more profound: **the key exchange problem**. If Alice and Bob want to exchange secrets, they must first agree on a shared secret key. But how can they agree on this key without an eavesdropper sitting between them intercepting it?

1.2 The Asymmetric Revolution (1977)

For two thousand years, monarchs and diplomats wrestled with this paradox. Until 1977, when three researchers at MIT — **Ron Rivest, Adi Shamir, and Leonard Adleman** — pulled off a masterstroke: inventing a padlock with two distinct keys: 1. A **public key**, which anyone can inspect and use to lock the padlock (encrypt the message). 2. A **private key**, possessed solely by the recipient, indispensable for opening the padlock (decrypting the message).

The foundation of this unbreakable padlock relies on no mechanical apparatus: it rests on a theorem of arithmetic discovered by a Toulouse magistrate in **1640**, Pierre de Fermat, and on the theory of **congruences** developed by Carl Friedrich Gauss.

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2. Divisibility and Euclidean Division in $\mathbb{Z}$

2.1 Divisibility in the Set of Integers $\mathbb{Z}$

> **📐 Definition — Divisibility**
>
> Let $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$. We say that $\boldsymbol{a}$ **divides** $\boldsymbol{b}$ (or that b is a **multiple** of a, or that b is divisible by a), denoted by $\boldsymbol{a \mid b}$, if there exists an integer $k \in \mathbb{Z}$ such that:
>
> $$b = k \cdot a$$

Examples: - $7 \mid 42$ because $42 = 6 \times 7$. - $-3 \mid 15$ because $15 = (-5) \times (-3)$. - $5 \mid 0$ because $0 = 0 \times 5$ (every nonzero integer divides 0). - 0 divides no nonzero number, because $0 \times k = 0 \neq b$.

> **📐 *Fundamental Properties of Divisibility*** *1.* ***Reflexivity:*** *For all $a \in \mathbb{Z}$, $a \mid a$. 2.* ***Transitivity:*** *If $a \mid b$ and $b \mid c$, then $a \mid c$. 3.* ***Linear combinations (essential!):*** *If $a \mid b$ and $a \mid c$, then for all integers $u, v \in \mathbb{Z}$:*
>
> $$a \mid (ub + vc)$$

Proof of linear combination: If $b = k_1 a$ and $c = k_2 a$, then $ub + vc = u(k_1 a) + v(k_2 a) = (uk_1 + vk_2)a$. Since $(uk_1 + vk_2) \in \mathbb{Z}$, a indeed divides $ub + vc$.

2.2 The Euclidean Division Theorem

> **📐 Fundamental Theorem — Euclidean Division in $\mathbb{Z}$**
>
> Let $a \in \mathbb{Z}$ and $b \in \mathbb{Z}^*$ (with $b > 0$). There exists a **unique** pair of integers $(q, r) \in \mathbb{Z} \times \mathbb{N}$ such that:
>
> $$a = bq + r \quad \text{with} \quad 0 \leq r < b$$
>
> - a is the **dividend**.
> - b is the **divisor**.
> - q is the **quotient**.
> - r is the **remainder**.

> **⚠️ Warning: Pitfall! — Division of a Negative Number**
>
> Let $a = -23$ and $b = 5$. A common mistake is writing $-23 = 5 \times (-4) - 3$. This is **strictly forbidden**! The remainder r must imperatively be **non-negative** ($0 \leq r < 5$). The unique correct expression is:
>
> $$-23 = 5 \times (-5) + 2$$
>
> The quotient is $q = -5$ and the remainder is $r = 2$ (which strictly satisfies $0 \leq 2 < 5$).

---

3. Congruences and the Arithmetic of the Clock

In 1801, at the age of 24, **Carl Friedrich Gauss** published his *Disquisitiones Arithmeticae*. In it, he introduced a notation that revolutionized algebra: the concept of **congruence**.

3.1 Definition and Circular Meaning

📐 Definition — Congruence Modulo n

Let n be a positive integer ($n \geq 2$). Two integers a and b are said to be **congruent modulo n**, denoted by:

$$a \equiv b \pmod{n} \quad \text{or} \quad a \equiv b \,[n]$$

if and only if their difference $a - b$ is divisible by n:

$$n \mid (a - b) \Leftrightarrow \exists k \in \mathbb{Z}, a - b = kn$$

Characterization via remainders: Two integers a and b are congruent modulo n if and only if **they have exactly the same remainder** in Euclidean division by n.

Clock arithmetic: On a 12-hour clock, if it is 9 o'clock and we wait 7 hours, it will be 16 o'clock, which corresponds to 4 o'clock:

$$9 + 7 = 16 \equiv 4 \pmod{12}$$

The set of possible remainders modulo n is finite: it is $\{0, 1, 2, ..., n - 1\}$.

3.2 The Magical Compatibility with Operations

📐 Fundamental Theorem — Operational Properties of Congruences

If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then: 1. **Addition:** $a + c \equiv b + d \pmod{n}$ 2. **Subtraction:** $a - c \equiv b - d \pmod{n}$ 3. **Multiplication:** $a \times c \equiv b \times d \pmod{n}$ 4. **Integer powers:** For any integer $k \in \mathbb{N}$, $a^k \equiv b^k \pmod{n}$

⚠️ Warning: Critical Pitfall! — Prohibition of Naive Division

In classical equality, $6x = 6y \Longrightarrow x = y$. **In congruences, this is generally false!** Observe: $2 \times 1 = 2 \equiv 2 \pmod{4}$ and $2 \times 3 = 6 \equiv 2 \pmod{4}$. Thus $2 \times 1 \equiv 2 \times 3 \pmod{4}$. Yet $1 \not\equiv 3 \pmod{4}$! One cannot simply "divide by 2". One can divide by c in $ac \equiv bc \pmod{n}$ only if **c is coprime to n** ($\gcd(c, n) = 1$)!

---

4. GCD, Euclidean Algorithm, and Bézout's Identity

4.1 The Euclidean Algorithm

The Greatest Common Divisor of two integers a and b, not both zero, is denoted $\gcd(a, b)$ or $a \wedge b$. Euclid (c. 300 BCE) established that if $a = bq + r$, then:

$$\gcd(a, b) = \gcd(b, r)$$

The GCD is the **last nonzero remainder** in the chain of successive Euclidean divisions.

Example: Let us compute $\gcd(126, 48)$*:* 1. $126 = 48 \times 2 + 30$ 2. $48 = 30 \times 1 + 18$ 3. $30 = 18 \times 1 + 12$ 4. $18 = 12 \times 1 + 6$ 5. $12 = 6 \times 2 + 0$ The last nonzero remainder is 6. Therefore, $\gcd(126, 48) = 6$.

4.2 The Extended Euclidean Algorithm and Bézout Coefficients

By back-substituting through the steps of Euclid's algorithm in reverse order, we can always express the GCD as a **linear combination** of the two starting integers! Let us revisit the example:

- Step 4: $6 = 18 - 12 \times 1$
- Step 3: but $12 = 30 - 18 \times 1$, hence:

$$6 = 18 - (30 - 18 \times 1) = 2 \times 18 - 1 \times 30$$

- Step 2: but $18 = 48 - 30 \times 1$, hence:

$$6 = 2 \times (48 - 30) - 1 \times 30 = 2 \times 48 - 3 \times 30$$

- Step 1: but $30 = 126 - 48 \times 2$, hence:

$$6 = 2 \times 48 - 3 \times (126 - 48 \times 2) = 2 \times 48 - 3 \times 126 + 6 \times 48 = -3 \times 126 + 8 \times 48$$

We have found two integers $u = -3$ and $v = 8$ such that:

$$126 \times (-3) + 48 \times 8 = 6$$

4.3 Bézout's Identity (Theorem)

> **📐 Fundamental Theorem — Bézout's Identity (1767)**
>
> Two nonzero integers a and b are **coprime** ($\gcd(a, b) = 1$) if and only if there exist two integers u and v such that:
>
> $$\mathbf{au + bv = 1}$$

Immediate converse direction: If $au + bv = 1$, every common divisor d of a and b divides the linear combination $au + bv = 1$. Hence $d \mid 1$, which gives $d = \pm 1$. The GCD is therefore 1.

4.4 Solving Diophantine Equations $ax + by = c$

A **Diophantine equation** (named in honor of Diophantus of Alexandria) is an equation whose solutions (x, y) are sought strictly in the set of integer pairs $\mathbb{Z}^2$.

> **🧭 Method — Solving $ax + by = c$**
>
> 1. **Existence condition:** Let $d = \gcd(a, b)$. The equation has integer solutions if
> and only if $\boldsymbol{d}$ **divides** $\boldsymbol{c}$. 2. **Simplification:** Divide the entire equation by d to obtain $a'x + b'y = c'$ with $\gcd(a', b') = 1$. 3. **Particular solution:** Find a particular solution (x_0, y_0) using the extended Euclidean algorithm. 4. **Term-by-term subtraction:** $a'x + b'y = c'$ and $a'x_0 + b'y_0 = c'$. Subtracting gives: $a'(x - x_0) = -b'(y - y_0) = b'(y_0 - y)$. 5. **Gauss's Lemma:** Since a' divides $b'(y_0 - y)$ and $\gcd(a', b') = 1$, a' divides $(y_0 - y)$. From this, deduce the set of all solutions (x, y) indexed by an integer $k \in \mathbb{Z}$.

---

5. Gauss's Lemma (Theorem)

> **⚠️ ⚠️ Absolute Requirement of Coprimality**
>
> Never omit the condition $\gcd(a, b) = 1$! Without coprimality, Gauss's theorem fails completely. Classic counterexample: 6 divides $4 \times 3 = 12$, yet 6 divides neither 4 nor 3, because $\gcd(6, 4) = 2 \neq 1$.

> **📐 Fundamental Theorem — Gauss's Lemma**
>
> Let $a, b, c \in \mathbb{Z}$. If $\boldsymbol{a}$ **divides the product** $\boldsymbol{bc}$ and if $\mathbf{gcd}(\boldsymbol{a}, \boldsymbol{b}) = \mathbf{1}$, then:
>
> $$\mathbf{a \ \ divides \ \ c}$$

Required Proof — Gauss's Lemma

Assume that $a \mid bc$ and that $\gcd(a, b) = 1$. Since a and b are coprime, by Bézout's identity, there exist two integers $u, v \in \mathbb{Z}$ such that:

$$au + bv = 1$$

Multiply both sides of this equality by the integer c:

$$acu + bcv = c$$

Analyze the two terms on the left:

- The first term acu is obviously a multiple of a (it contains the factor a).
- The second term contains bc. By hypothesis, $a \mid bc$, so there exists an integer k such that $bc = ka$. The second term can thus be written $(ka)v = a(kv)$, which is also a multiple of a.

We can therefore factor out a on the left-hand side:

$$a(cu + kv) = c$$

Since $(cu + kv) \in \mathbb{Z}$, this equality rigorously proves that a divides c.

Corollary (Uniqueness of prime factorization): If a prime number p divides a product $x_1 x_2 ... x_m$, then p must divide at least one of the factors x_i.

---

6. Fermat's Little Theorem and Fast Exponentiation

6.1 Fermat's Little Theorem (1640)

Theorem — Fermat's Little Theorem

Let p be a prime number. 1. For any integer a not divisible by p ($\gcd(a, p) = 1$):

$$\mathbf{a^{p-1} \equiv 1 \pmod{p}}$$

2. For any integer $a \in \mathbb{Z}$ (without any restriction):

$$\mathbf{a^{p} \equiv a \pmod{p}}$$

Classical Proof via Remainders

Let p be a prime and a an integer not divisible by p. Consider the $p - 1$ successive multiples: $a, 2a, 3a, ..., (p - 1)a$. Let us show that their remainders modulo p are all distinct and nonzero: If $k_1 a \equiv k_2 a \pmod{p}$ with $1 \leq k_2 \leq k_1 \leq p - 1$, then $p \mid (k_1 - k_2)a$. Since $\gcd(a, p) = 1$, by Gauss's lemma, $p \mid (k_1 - k_2)$. But $0 \leq k_1 - k_2 < p$, which forces $k_1 - k_2 = 0 \Leftrightarrow k_1 = k_2$. Thus, the remainders of the numbers $\{a, 2a, ..., (p - 1)a\}$ modulo p are precisely the integers $\{1, 2, ..., p - 1\}$ in some permuted order! Multiply these $p - 1$ congruences together:

$$a \times 2a \times 3a \times ... \times (p - 1)a \equiv 1 \times 2 \times ... \times (p - 1) \pmod{p}$$

Factoring out a from each term:

$$a^{p-1} \times (p - 1)! \equiv (p - 1)! \pmod{p}$$

Since p is prime, none of the factors of $(p - 1)!$ is divisible by p, so $\gcd((p - 1)!, p) = 1$. By Gauss's lemma (or by multiplying by the modular inverse of $(p - 1)!$), we can simplify by $(p - 1)!$:

$$a^{p-1} \equiv 1 \pmod{p}$$

6.2 The Fast Modular Exponentiation Algorithm

How do we calculate an astronomical power such as $7^{100} \pmod{13}$ without computing an 85-digit number? We use the binary decomposition of the exponent (the *Square-and-Multiply* method): The exponent 100 is written in binary: $100 = 64 + 32 + 4$. We compute the successive squares modulo 13:

- $7^1 \equiv 7 \pmod{13}$
- $7^2 = 49 = 3 \times 13 + 10 \equiv -3 \pmod{13}$
- $7^4 \equiv (-3)^2 = 9 \equiv -4 \pmod{13}$
- $7^8 \equiv (-4)^2 = 16 \equiv 3 \pmod{13}$
- $7^{16} \equiv 3^2 = 9 \equiv -4 \pmod{13}$
- $7^{32} \equiv (-4)^2 = 16 \equiv 3 \pmod{13}$
- $7^{64} \equiv 3^2 = 9 \equiv -4 \pmod{13}$

All that remains is to multiply the relevant powers:

$$7^{100} = 7^{64} \times 7^{32} \times 7^4 \equiv (-4) \times 3 \times (-4) = 48 \equiv 9 \pmod{13}.$$

The calculation required just 7 simple multiplications of numbers less than 50 instead of 99 giant multiplications!

---

7. Applications to Cryptography: From Caesar to RSA

7.1 Affine Ciphers

We associate each letter of the alphabet with a number from 0 to 25 (A=0, B=1, …, Z=25). The encryption function is:

$$y \equiv ax + b \pmod{26}$$

For this cipher to be **uniquely decryptable**, distinct letters must produce distinct codes:

$$ax_1 + b \equiv ax_2 + b \pmod{26} \Leftrightarrow a(x_1 - x_2) \equiv 0 \pmod{26}$$

This enforces $x_1 = x_2$ only if $\boldsymbol{a}$ **is coprime to 26**: $\gcd(a, 26) = 1$. Since $26 = 2 \times 13$, a must be divisible by neither 2 nor 13. To decrypt, one looks for the **modular inverse** of a modulo 26, that is, the integer a' such that $aa' \equiv 1 \pmod{26}$ (provided by Bézout's algorithm). The decryption function is then:

$$x \equiv a'(y - b) \pmod{26}$$

> ⚠ ⚠ **Classical Cryptographic Trap — The Secret Key Modulus**
>
> Never confuse the public modulus $n = p \times q$ with the key-generation totient modulus $\varphi(n) = (p-1)(q-1)$!
> - Encrypted and decrypted messages transit **modulo $\boldsymbol{n}$**: $C \equiv M^e \pmod{n}$ and $M \equiv C^d \pmod{n}$.
> - By contrast, the secret exponent d is computed as the modular inverse of the public exponent e **modulo $\boldsymbol{\varphi(n)}$**:
>
> $$e \cdot d \equiv 1 \pmod{\varphi(n)} \quad (\text{and never modulo } \ n\ !)$$
>
> Attempting to invert e modulo n instead of $\varphi(n)$ renders decryption mathematically defunct, destroying the RSA protocol.

7.2 The Principle of Asymmetric RSA Encryption

Here is the crowning jewel of modern computer security:

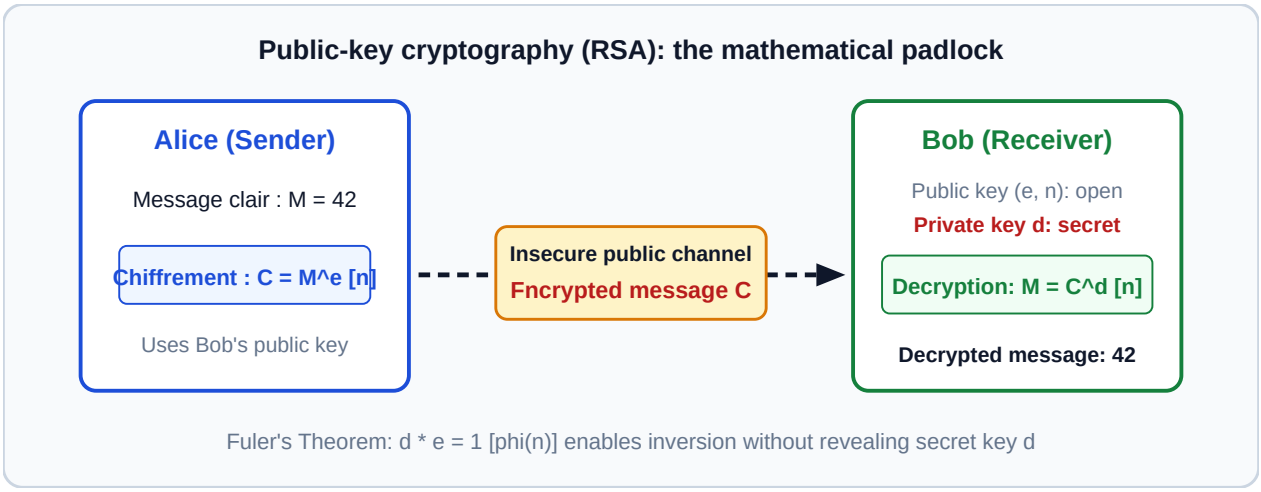

Figure 121: RSA asymmetric cryptography: public-key exchange and decryption

Mathematical Proof — Why RSA Always Decrypts Exactly

By construction, d is the inverse of e modulo $\varphi(n) = (p-1)(q-1)$. Hence there exists an integer $k \in \mathbb{Z}$ such that:

$$e \cdot d = 1 + k(p-1)(q-1)$$

Bob computes $C^d \pmod{n}$. Since $C \equiv M^e \pmod{n}$, we have:

$$C^d \equiv (M^e)^d = M^{ed} = M^{1+k(p-1)(q-1)} \pmod{n}$$

Let us show that this quantity is congruent to M modulo p and modulo q:

- **Modulo p:**
  - If p divides M, then $M \equiv 0 \pmod{p}$, hence $M^{ed} \equiv 0 \equiv M \pmod{p}$.
  - If p does not divide M, by Fermat's Little Theorem, $M^{p-1} \equiv 1 \pmod{p}$. Then:

$$M^{ed} = M \times \left(M^{p-1}\right)^{k(q-1)} \equiv M \times 1^{k(q-1)} = M \pmod{p}$$

  In both cases, $M^{ed} \equiv M \pmod{p} \Leftrightarrow p \mid (M^{ed} - M)$.
- **Modulo q:** By a strictly symmetric argument, $q \mid (M^{ed} - M)$.

Since p and q are distinct prime numbers, they are coprime. By Gauss's lemma (or by product of distinct prime divisors), their product $n = pq$ divides $(M^{ed} - M)$:

$$M^{ed} \equiv M \pmod{n}$$

The original message reappears intact!

Why is it impossible to cheat? To find the private key d, an eavesdropper must compute the inverse of e modulo $(p-1)(q-1)$. To do this, they imperatively need to know p and q. Yet the eavesdropper knows only the public modulus $n = p \times q$. If p and q each have 300 decimal digits, the product n has 600 digits. Factoring such a number into prime factors would require billions of years of computation for all the supercomputers on Earth combined!

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8. Step-by-Step Solved Examples

Example 1: Solving a Linear Congruence

Solve in $\mathbb{Z}$ the congruence:

$$5x \equiv 3 \pmod{14}$$

Detailed Solution: To solve this equation, we must find an inverse of 5 modulo 14. Let us seek a Bézout relation between 5 and 14:

$$14 = 5 \times 2 + 4 \Longrightarrow 4 = 14 - 5 \times 2$$

$$5 = 4 \times 1 + 1 \Longrightarrow 1 = 5 - 4 \times 1 = 5 - (14 - 5 \times 2) = 5 \times 3 - 14 \times 1$$

We thus have $5 \times 3 - 14 \times 1 = 1$, which means:

$$5 \times 3 \equiv 1 \pmod{14}$$

The inverse of 5 modulo 14 is therefore **3**. Multiply both sides of the original congruence by 3:

$$3 \times (5x) \equiv 3 \times 3 \pmod{14}$$

$$(3 \times 5)x \equiv 9 \pmod{14}$$

$$1 \cdot x \equiv 9 \pmod{14} \Leftrightarrow x \equiv 9 \pmod{14}$$

The set of solutions is the set of integers of the form:

$$x = 14k + 9 \quad (k \in \mathbb{Z})$$

(Verification: for $x = 9$, $5 \times 9 = 45 = 14 \times 3 + 3 \equiv 3 \pmod{14}$).

Example 2: Solving a Diophantine Equation

Solve in $\mathbb{Z}^2$ the equation:

$$47x + 29y = 1$$

Detailed Solution: 1. **Euclidean algorithm for 47 and 29:**

- $47 = 29 \times 1 + 18$
- $29 = 18 \times 1 + 11$
- $18 = 11 \times 1 + 7$
- $11 = 7 \times 1 + 4$
- $7 = 4 \times 1 + 3$
- $4 = 3 \times 1 + 1$
- $3 = 1 \times 3 + 0$ The last nonzero remainder is 1, so $\gcd(47, 29) = 1$. The equation has solutions.

2. **Back-substitution through the algorithm:**
   - $1 = 4 - 3 \times 1$
   - $1 = 4 - (7 - 4) = 2 \times 4 - 7$
   - $1 = 2 \times (11 - 7) - 7 = 2 \times 11 - 3 \times 7$
   - $1 = 2 \times 11 - 3 \times (18 - 11) = 5 \times 11 - 3 \times 18$
   - $1 = 5 \times (29 - 18) - 3 \times 18 = 5 \times 29 - 8 \times 18$
   - $1 = 5 \times 29 - 8 \times (47 - 29) = 13 \times 29 - 8 \times 47 = 47 \times (-8) + 29 \times 13$ A particular solution is therefore $(x_0, y_0) = (-8, 13)$.
3. **General solution:**

$$47x + 29y = 1 \quad \text{and} \quad 47(-8) + 29(13) = 1$$

Subtracting gives:

$$47(x - (-8)) + 29(y - 13) = 0 \Leftrightarrow 47(x + 8) = 29(13 - y)$$

Since 47 divides $29(13 - y)$ and $\gcd(47, 29) = 1$, by Gauss's lemma, 47 divides $(13 - y)$. There exists an integer $k \in \mathbb{Z}$ such that:

$$13 - y = 47k \Leftrightarrow y = 13 - 47k$$

Substituting back into the equality:

$$47(x + 8) = 29(47k) \Leftrightarrow x + 8 = 29k \Leftrightarrow x = -8 + 29k$$

The solution set in $\mathbb{Z}^2$ is:

$$S = \{(-8 + 29k\,;\, 13 - 47k) \mid k \in \mathbb{Z}\}$$

Example 3: Decrypting an Affine Cipher

A message was encrypted using the affine cipher $y \equiv 7x + 5 \pmod{26}$. The ciphertext letter **V** (corresponding to position 21) is intercepted. Recover the original letter x.

Detailed Solution: 1. The equation to solve is $7x + 5 \equiv 21 \pmod{26} \Leftrightarrow 7x \equiv 16 \pmod{26}$. 2. Let us find the inverse of 7 modulo 26: $26 = 7 \times 3 + 5$ $7 = 5 \times 1 + 2$ $5 = 2 \times 2 + 1 \Longrightarrow 1 = 5 - 2 \times 2 = 5 - 2(7 - 5) = 3 \times 5 - 2 \times 7 = 3(26 - 7 \times 3) - 2 \times 7 = 3 \times 26 - 11 \times 7$. Thus $-11 \times 7 \equiv 1 \pmod{26}$. Now $-11 \equiv 26 - 11 = 15 \pmod{26}$. Check: $7 \times 15 = 105 = 4 \times 26 + 1 \equiv 1 \pmod{26}$. The inverse of 7 modulo 26 is therefore **15**. 3. Multiply the equation $7x \equiv 16 \pmod{26}$ by 15:

$$x \equiv 15 \times 16 \pmod{26}$$

$15 \times 16 = 240$. Perform Euclidean division of 240 by 26: $240 = 26 \times 9 + 6$. Thus $x \equiv 6 \pmod{26}$. 4. Position 6 corresponds to the letter: $0 \to A, 1 \to B, 2 \to C, 3 \to D, 4 \to E, 5 \to F, 6 \to \mathbf{G}$. The original letter was **G**.

Example 4: Remainder Calculation with Fermat's Little Theorem

Determine the remainder of the Euclidean division of 3^{2026} by 17.

Detailed Solution: 1. The divisor is $p = 17$, which is prime. 2. The integer $a = 3$ is not divisible by 17 ($\gcd(3, 17) = 1$). 3. By Fermat's Little Theorem:

$$3^{17-1} = 3^{16} \equiv 1 \pmod{17}$$

4. Perform Euclidean division of the exponent 2026 by 16:

$$2026 = 16 \times 126 + 10$$

5. Decompose the power:

$$3^{2026} = 3^{16 \times 126 + 10} = (3^{16})^{126} \times 3^{10} \equiv 1^{126} \times 3^{10} \equiv 3^{10} \pmod{17}$$

6. Compute $3^{10} \pmod{17}$ by successive squares:

- $3^2 = 9$
- $3^4 = 9^2 = 81 = 17 \times 4 + 13 \equiv -4 \pmod{17}$
- $3^8 \equiv (-4)^2 = 16 \equiv -1 \pmod{17}$
- $3^{10} = 3^8 \times 3^2 \equiv (-1) \times 9 = -9 \equiv 17 - 9 = 8 \pmod{17}$.

The remainder of the division of 3^{2026} by 17 is therefore **8**.

Example 5: A Complete Miniature RSA Example

Bob prepares his RSA keys: 1. He chooses $p = 5$ and $q = 11$. 2. Calculate the modulus n and the totient value $\varphi(n)$. 3. Bob chooses the public exponent $e = 3$. Verify that this choice is valid. 4. Calculate the secret private key d. 5. Alice wants to send the secret message $M = 9$. Encrypt this message to obtain C. 6. Show how Bob decrypts C to recover M.

Detailed Solution: 1. $n = p \times q = 5 \times 11 = 55$. 2. $\varphi(n) = (p - 1)(q - 1) = 4 \times 10 = 40$. 3. $e = 3$. Since $40 = 3 \times 13 + 1$, $\gcd(3, 40) = 1$. The choice $e = 3$ is completely valid. 4. We seek d such that $3d \equiv 1 \pmod{40}$. Notice that $3 \times 13 = 39 = 40 - 1 \equiv -1 \pmod{40}$, so $3 \times (-13) \equiv 1 \pmod{40}$. Since $-13 \equiv 40 - 13 = 27 \pmod{40}$, we have $d = 27$. *(Check:* $3 \times 27 = 81 = 2 \times 40 + 1 \equiv 1$*).* Bob's private key is $\boldsymbol{d = 27}$. 5. **Encryption by Alice:** $C \equiv M^e \pmod{n} \Leftrightarrow C \equiv 9^3 \pmod{55}$. $9^3 = 729$. Division of 729 by 55: $729 = 55 \times 13 + 14$. The ciphertext sent across the network is $\boldsymbol{C = 14}$. 6. **Decryption by Bob:** Bob computes $M \equiv C^d = 14^{27} \pmod{55}$. Decompose the exponent 27 in binary: $27 = 16 + 8 + 2 + 1$. Compute powers of 14 modulo 55:

- $14^1 \equiv 14$
- $14^2 = 196 = 55 \times 3 + 31 \equiv 31 \equiv -24 \pmod{55}$
- $14^4 \equiv (-24)^2 = 576 = 55 \times 10 + 26 \equiv 26 \pmod{55}$
- $14^8 \equiv 26^2 = 676 = 55 \times 12 + 16 \equiv 16 \pmod{55}$
- $14^{16} \equiv 16^2 = 256 = 55 \times 4 + 36 \equiv 36 \equiv -19 \pmod{55}$ Multiply: $14^{27} = 14^{16} \times 14^8 \times 14^2 \times 14 \equiv (-19) \times 16 \times 31 \times 14 \pmod{55}$.
- $(-19) \times 16 = -304 = 55 \times (-6) + 26 \equiv 26$
- $31 \times 14 = 434 = 55 \times 7 + 49 \equiv -6$
- $26 \times (-6) = -156 = 55 \times (-3) + 9 \equiv 9 \pmod{55}$. Bob recovers exactly $\boldsymbol{M = 9}$!

Example 6: Divisibility Criteria via Congruences

Prove using congruences that an integer is divisible by 9 if and only if the sum of its digits is divisible by 9.

Detailed Solution: Let N be a natural number whose decimal expansion is $a_k a_{k-1} \ldots a_1 a_0$:

$$N = \sum_{j=0}^{k} a_j 10^j = a_0 + a_1 10 + a_2 10^2 + ... + a_k 10^k$$

Notice that:

$$10 = 9 + 1 \equiv 1 \pmod 9$$

By the compatibility of congruences with powers: For any integer $j \geq 0$, $10^j \equiv 1^j \equiv 1 \pmod 9$. Multiplying by each digit a_j and summing:

$$N = \sum_{j=0}^{k} a_j 10^j \equiv \sum_{j=0}^{k} a_j (1) \equiv \sum_{j=0}^{k} a_j \pmod 9$$

The number N and the sum of its digits have exactly the same remainder modulo 9. Consequently, $9 \mid N$ if and only if 9 divides the sum of its decimal digits.

Key Takeaways

1. **Divisibility and Division:**
   - $a \mid b \Leftrightarrow \exists k \in \mathbb{Z}, b = ka$.
   - Linear combinations: if $a \mid b$ and $a \mid c$, then $a \mid (ub + vc)$.
   - Euclidean division: $a = bq + r$ with $0 \leq r < b$.
2. **Congruences:**
   - $a \equiv b \pmod n \Leftrightarrow n \mid (a - b)$ (same remainder modulo n).
   - Compatible with addition, subtraction, multiplication, and powers.
   - One may simplify $ac \equiv bc \pmod n$ by c only if $\gcd(c, n) = 1$.
3. **Bézout and Gauss:**
   - $\gcd(a, b) = 1 \Leftrightarrow \exists u, v \in \mathbb{Z}, au + bv = 1$ (Bézout).
   - If $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$ (Gauss).
4. **Fermat and RSA:**
   - Fermat's Little Theorem: if p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod p$.
   - RSA: public key (n, e) with $n = pq$, private key $d \equiv e^{-1} \pmod{(p-1)(q-1)}$.
   - $C \equiv M^e \pmod n \Longrightarrow M \equiv C^d \pmod n$.

Exercises

Level (Direct Applications)

Exercise 56.1 Perform Euclidean division in $\mathbb{Z}$: 1. Of 157 by 11. 2. Of -75 by 8. 3. Of -140 by 12.

Exercise 56.2 True or False? Justify. 1. If $a \equiv 3 \pmod 5$, then $a^2 \equiv 4 \pmod 5$. 2. If $a \equiv 2 \pmod 6$, then a is an even number. 3. If $3x \equiv 3y \pmod 7$, then $x \equiv y \pmod 7$.

Exercise 56.3 Compute $\gcd(323, 187)$ using the Euclidean algorithm. Are these numbers coprime?

Exercise 56.4 Find a pair of integers (u, v) satisfying Bézout's identity:

$$17u + 11v = 1$$

Exercise 56.5 Compute the remainder of the Euclidean division of 2^{100} by 7.

Exercise 56.6 A 12-hour clock indicates 8 o'clock. What time will it indicate in 1,500 hours?

Exercise 56.7 Solve in $\mathbb{Z}$ the equation $3x \equiv 1 \pmod{10}$.

Level (Analytical Thinking & Exam Problems)

Exercise 56.8 Solve in $\mathbb{Z}^2$ the Diophantine equation:

$$13x - 7y = 1$$

Deduce all solutions satisfying $0 \leq x \leq 50$.

Exercise 56.9 Affine cipher. We use the cipher $y \equiv 5x + 7 \pmod{26}$. 1. Verify that this encryption is reversible. 2. Encrypt the word **BAC**. 3. Find the decryption function and decode the letter **R** (position 17).

Exercise 56.10 Determine the units digit of 7^{2025} in its standard decimal representation. *(Hint: find the remainder modulo 10).*

Exercise 56.11 Prove that for all $n \in \mathbb{N}$, the integer $A_n = 3^{2n+1} + 2^{n+2}$ is divisible by 7.

Exercise 56.12 Let $n \in \mathbb{N}$. Prove that the fraction $\frac{2n+1}{3n+1}$ is irreducible for every integer n.

Exercise 56.13 The Coconut Thieves. Three castaways gather a pile of coconuts. During the night, the first wakes up, divides the pile into 3 equal shares, finds that 1 coconut remains, gives it to a monkey, and hides his share. The second and third castaways do successively the same. Translate this problem into a system of congruences and find the minimal initial number of coconuts.

Exercise 56.14 Chinese Remainder Theorem (Special Case). Solve in $\mathbb{Z}$ the system of congruences:

$$\{x \equiv 2 \pmod{5}$$
$$x \equiv 3 \pmod{7}$$

Exercise 56.15 Mersenne Numbers and Fermat. Let p be an odd prime. Consider the Mersenne number $M_p = 2^p - 1$. Let q be a prime divisor of M_p. Prove that $q \equiv 1 \pmod{2p}$.

Level (Advanced Explorations & Competition Problems)

Exercise 56.16 Proof of Wilson's Theorem. Prove that an integer $p \geq 2$ is prime if and only if:

$$(p-1)! \equiv -1 \pmod{p}$$

(Hint: pair each element of $\{2, ..., p-2\}$ with its distinct modular inverse).

Exercise 56.17 Euler's Totient Function $\varphi(n)$. 1. For p prime, justify that $\varphi(p) = p - 1$ and $\varphi(p^k) = p^k - p^{k-1}$. 2. Prove Euler's theorem generalizing Fermat: if $\gcd(a, n) = 1$, then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Exercise 56.18 The Miller-Rabin Primality Test. Explain the principle of the Miller-Rabin probabilistic test based on Fermat's theorem and the square roots of 1 modulo p. Why does it allow testing 500-digit numbers in a fraction of a second?

Exercise 56.19 Wiener's Attack on the RSA Cryptosystem. Show that if the secret exponent d is chosen too small ($d < \frac{1}{3}n^{1/4}$), the private key can be recovered quickly from the public key (n, e) using the continued fraction expansion of $\frac{e}{n}$.

Exercise 56.20 Order of an Element in $\mathbb{Z}/p\mathbb{Z}$. Let p be a prime number and a an integer not divisible by p. The order of a modulo p is defined as the smallest positive integer k such that $a^k \equiv 1 \pmod{p}$. Prove that k necessarily divides $p - 1$.

Exercise 56.21 Quadratic Residues and the Legendre Symbol. Let p be an odd prime. An integer a not a multiple of p is a quadratic residue (square) modulo p if there exists x such that $x^2 \equiv a \pmod{p}$. Prove Euler's criterion:

$$a^{\frac{p-1}{2}} \equiv \begin{cases} +1 \pmod{p} \text{ if } a \text{ is a square} \\ -1 \pmod{p} \text{ if } a \text{ is not a square} \end{cases}$$

Exercise 56.22 The Discrete Logarithm Problem and Diffie-Hellman Key Exchange. Explain the Diffie-Hellman key exchange protocol based on the modular power function $g^a \pmod{p}$. Why is it easy to compute $g^a \pmod{p}$ but computationally infeasible to recover a knowing g^a?

Solutions

Solution 56.1 1. $157 = 11 \times 14 + 3$ ($q = 14, r = 3$). 2. $-75 = 8 \times (-10) + 5$ ($q = -10, r = 5$). 3. $-140 = 12 \times (-12) + 4$ ($q = -12, r = 4$).

Solution 56.2 1. **True.** $a \equiv 3 \pmod{5} \Longrightarrow a^2 \equiv 3^2 = 9 \equiv 4 \pmod{5}$. 2. **True.** $a = 6k + 2 = 2(3k + 1)$, which is divisible by 2. 3. **True.** Since $\gcd(3, 7) = 1$, we can divide by 3 by Gauss's lemma.

Solution 56.3 $323 = 187 \times 1 + 136$. $187 = 136 \times 1 + 51$. $136 = 51 \times 2 + 34$. $51 = 34 \times 1 + 17$. $34 = 17 \times 2 + 0$. $\gcd(323, 187) = 17 \neq 1$. They are not coprime.

Solution 56.4 $17 = 11 \times 1 + 6 \Longrightarrow 6 = 17 - 11$. $11 = 6 \times 1 + 5 \Longrightarrow 5 = 11 - 6 = 11 - (17 - 11) = 2 \times 11 - 17$. $6 = 5 \times 1 + 1 \Longrightarrow 1 = 6 - 5 = (17 - 11) - (2 \times 11 - 17) = 2 \times 17 - 3 \times 11$. A solution pair is $(u, v) = (2, -3)$.

Solution 56.5 Modulo 7, $2^3 = 8 \equiv 1 \pmod 7$. Division of 100 by 3: $100 = 3 \times 33 + 1$. $2^{100} = (2^3)^{33} \times 2^1 \equiv 1^{33} \times 2 = 2 \pmod 7$. The remainder is 2.

Solution 56.6 We seek $8 + 1500 \pmod{12}$. $1500 = 12 \times 125 + 0 \equiv 0 \pmod{12}$. The clock will therefore indicate **8 o'clock** sharp again.

Solution 56.7 We seek the inverse of 3 modulo 10. Since $3 \times 7 = 21 \equiv 1 \pmod{10}$, the inverse is 7. $3x \equiv 1 \Longrightarrow 7 \times 3x \equiv 7 \times 1 \Longrightarrow x \equiv 7 \pmod{10}$. $S = \{10k + 7 \mid k \in \mathbb{Z}\}$.

Solution 56.8 $13(1) - 7(2) = 13 - 14 = -1 \Longrightarrow 13(-1) - 7(-2) = 1$. Particular solution: $(-1, -2)$. $13(x + 1) = 7(y + 2)$. Since $\gcd(13, 7) = 1$, by Gauss: $x + 1 = 7k \Longrightarrow x = 7k - 1$ and $y = 13k - 2$. For $0 \leq x \leq 50$: $0 \leq 7k - 1 \leq 50 \Longrightarrow 1 \leq 7k \leq 51 \Longrightarrow k \in \{1, 2, 3, 4, 5, 6, 7\}$. There are 7 solution pairs.

Solution 56.9 1. $\gcd(5, 26) = 1$ because $26 = 2 \times 13$, hence the encryption is bijective. 2. B(1): $5(1) + 7 = 12 \to$ M. A(0): $5(0) + 7 = 7 \to$ H. C(2): $5(2) + 7 = 17 \to$ R. BAC becomes **MHR**. 3. Inverse of 5 modulo 26: $5 \times 21 = 105 = 4 \times 26 + 1$, hence $5^{-1} \equiv 21$. $x \equiv 21(y - 7) \pmod{26}$. For R ($y = 17$): $x \equiv 21(17 - 7) = 210 = 26 \times 8 + 2 \equiv 2 \pmod{26}$. Position 2 corresponds to **C**.

Solution 56.10 We seek $7^{2025} \pmod{10}$. $7^1 \equiv 7$, $7^2 = 49 \equiv 9 \equiv -1$, $7^4 \equiv 1 \pmod{10}$. $2025 = 4 \times 506 + 1$. $7^{2025} = (7^4)^{506} \times 7^1 \equiv 1 \times 7 = 7 \pmod{10}$. The units digit is **7**.

Solution 56.11 $3^{2n+1} = 3 \times (3^2)^n = 3 \times 9^n \equiv 3 \times 2^n \pmod 7$ (since $9 \equiv 2$). $2^{n+2} = 2^2 \times 2^n = 4 \times 2^n \pmod 7$. Sum: $A_n \equiv 3 \times 2^n + 4 \times 2^n = 7 \times 2^n \equiv 0 \pmod 7$. Hence $7 \mid A_n$.

Solution 56.12 Let d be a common divisor of the numerator and denominator. $d \mid (2n + 1)$ and $d \mid (3n + 1)$. Then d divides the linear combination $3(2n + 1) - 2(3n + 1) = 6n + 3 - 6n - 2 = 1$. Since $d \mid 1$, we must have $d = 1$. The two integers are always coprime; the fraction is irreducible.

Solution 56.13 Let N be the number of coconuts. Each step removes 1 coconut and takes one-third, leaving $\frac{2}{3}(N - 1)$. After 3 steps, the successive divisions lead to the congruence $8N \equiv 17 \pmod{27} \Longrightarrow N \equiv 25 \pmod{27}$. The smallest positive nonzero number possible is $N = 25$ (or 79 if the final morning sharing is non-empty).

Solution 56.14 $x = 5k + 2$. Substituting into the second congruence: $5k + 2 \equiv 3 \pmod 7 \Leftrightarrow 5k \equiv 1 \pmod 7$. The inverse of 5 modulo 7 is 3 ($5 \times 3 = 15 \equiv 1$). Thus $k \equiv 3 \pmod 7 \Longrightarrow k = 7m + 3$. Hence $x = 5(7m + 3) + 2 = 35m + 17$. The solutions are $x \equiv 17 \pmod{35}$.

Solution 56.15 If $q \mid (2^p - 1)$, then $2^p \equiv 1 \pmod q$. The order of 2 modulo q divides the prime p, so the order is p. By Fermat, $2^{q-1} \equiv 1 \pmod q$, so the order p divides $q - 1$, meaning $q - 1 = kp \Longrightarrow q \equiv 1 \pmod p$. Since q is odd (M_p being odd), $q - 1$ is even, so k is even ($k = 2m$). Hence $q = 2mp + 1 \equiv 1 \pmod{2p}$.

Solution 56.16 If p is prime, every element $x \in \{2, \ldots, p - 2\}$ possesses a unique modular inverse distinct from itself (since $x^2 \equiv 1 \Longrightarrow x = \pm 1$). All these terms cancel in pairs in the product $(p - 1)!$, leaving only $1 \times (p - 1) \equiv -1 \pmod p$. The converse follows from the fact that if p is composite, a proper divisor $d < p$ would divide $(p - 1)!$ and could not divide $(p - 1)! + 1$.

Solution 56.17 1. For p^k, the integers not coprime to p^k are multiples of p, of which there are p^{k-1}. Hence $\varphi(p^k) = p^k - p^{k-1}$. 2. Considering the integers coprime to n and multiplying by a, the product of elements is preserved by bijectivity of multiplication in the group of units $(\mathbb{Z}/n\mathbb{Z})^\times$, yielding $a^{\varphi(n)} \equiv 1 \pmod n$.

Solution 56.18 If n is prime, for $n - 1 = 2^s d$, the sequence of repeated squarings starting from a^d must terminate in 1, and the last value different from 1 must be -1. If any other value appears, n is definitely composite. This fast polynomial-time test is the keystone of large prime generation for cryptography.

Solution 56.19 If $ed - k\varphi(n) = 1$, then $\left|\frac{e}{n} - \frac{k}{d}\right| \leq \frac{1}{2d^2}$. By Legendre's theorem on continued fractions, $\frac{k}{d}$ must be one of the convergents of the continued fraction expansion of $\frac{e}{n}$, enabling an attacker to compute d almost instantaneously.

Solution 56.20 Let k be the order of a. Perform Euclidean division of $p - 1$ by k: $p - 1 = kq + r$ with $0 \leq r < k$. Then $a^{p-1} = (a^k)^q \times a^r \equiv 1^q \times a^r \equiv a^r \pmod p$. By Fermat, $a^{p-1} \equiv 1 \pmod p$, hence $a^r \equiv 1 \pmod p$. Since k is the smallest positive integer with this property and $r < k$, we must have $r = 0$. Hence k divides $p - 1$.

Solution 56.21 By Fermat, $(a^{\frac{p-1}{2}})^2 = a^{p-1} \equiv 1 \pmod{p}$. Thus $a^{\frac{p-1}{2}}$ solves $X^2 \equiv 1 \pmod{p}$, so $a^{\frac{p-1}{2}} \equiv \pm 1 \pmod{p}$. If $a = x^2$, then $a^{\frac{p-1}{2}} = x^{p-1} \equiv 1$. The polynomial $X^{\frac{p-1}{2}} - 1$ of degree $\frac{p-1}{2}$ has at most $\frac{p-1}{2}$ roots, so the quadratic residues are its only roots.

Solution 56.22 Alice chooses secret a, sends $A = g^a \pmod{p}$. Bob chooses secret b, sends $B = g^b \pmod{p}$. The shared key is $K = B^a = A^b = g^{ab} \pmod{p}$. An eavesdropper intercepting g^a and g^b cannot compute g^{ab} without solving the discrete logarithm problem, believed to be intractable for large prime moduli.

The Question That Opens the Future

We have just seen how arithmetic equations allow us to encrypt information flows and link integer variables. But what happens when the flows to be transformed are no longer isolated numbers, but entire tables of data, interconnected networks, and probabilistic states in perpetual transition? How do we model the evolution of a population, the ranking of web pages by Google, or the states of a Turing machine? To orchestrate these multidimensional flows, mathematicians forged a spectacular tool: matrices and graphs. Turn the page to discover **Chapter 57 and The Jacquard Loom Revisited: Matrices and Graphs**.

CHAPTER 57: The Jacquard Loom Revisited: Matrices and Graphs

> "In 1801 in Lyon, Joseph Marie Jacquard punched holes into cards to guide the silk threads of his loom: a grid of perforations orchestrated the rising and falling of shuttles to weave patterns of boundless complexity. Matrices are the Jacquard punched cards of modern algebra: grids of numbers capable of distorting space, enumerating the paths of a network, and predicting the future equilibrium of the world's flows."

What You Will Discover

- How a **matrix** encapsulates a table of operations and transforms vectors via the row $\times$ column product.
- Why matrix multiplication is associative but **non-commutative** ($AB \neq BA$), and how the identity matrix I_n functions.
- Computing the **inverse of a 2×2 matrix** using the determinant and instantly solving linear systems $AX = B$.
- Modeling networks using **graphs** and the **adjacency matrix**.
- The fundamental theorem proving why the k-th power of an adjacency matrix M^k counts precisely the **number of paths of length k**.
- **Markov chains** and probabilistic graphs: calculating the **steady state** and the foundations of Google's PageRank algorithm.

---

1. The Problem: World Networks and Control Arrays

1.1 Airline Flight Routes

Consider a small airline operating across four cities: Paris (P), London (L), Madrid (M), and Rome (R). The daily nonstop flights are as follows:

- From Paris to London, Madrid, and Rome.
- From London to Paris and Madrid.
- From Madrid to Paris and Rome.
- From Rome to Paris only.

If a passenger wishes to travel from London to Rome, there is no direct flight. But can they make the trip with **exactly one layover** (a two-flight itinerary)? What about two layovers? How many possible routes exist?

Searching by hand quickly becomes confusing as the branching tree grows. But if we record these connections in a **binary grid** (1 if there is a flight, 0 otherwise), and multiply this grid by itself, a miracle occurs: **matrix algebra instantly counts the set of all possible paths on any network!**

1.2 Weather and Long-Term Forecasts

In a coastal region, we observe the weather day after day:

- If it is sunny today, there is an 80% chance it will be sunny tomorrow, and a 20% chance of rain.
- If it rains today, there is a 40% chance it will be sunny tomorrow, and a 60% chance of rain.

What will the weather be like in one week? In one year? Does there exist an average long-term equilibrium climate (for instance, 65% sunny days and 35% rainy days) toward which the system inevitably converges, regardless of today's weather? To answer this question, we must merge probability theory with matrix algebra: this is the theory of **Markov chains**.

---

2. The Matrix Object and Its Operations

2.1 Definition of a Matrix

Definition — Matrix

Let n and p be positive integers. A **matrix** of dimension (or size) $n \times p$ (read "n rows, p columns") with real coefficients is a rectangular array of numbers arranged in n rows and p columns:

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,p} \\ a_{2,1} & a_{2,2} & \dots & a_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \dots & a_{n,p} \end{pmatrix} = (a_{i,j})_{\substack{1 \le i \le n \\ 1 \le j \le p}}$$

The entry $a_{i,j}$ is located at the intersection of the i-th row and the j-th column. The set of $n \times p$ matrices with real coefficients is denoted by $\mathcal{M}_{n,p}(\mathbb{R})$.

Terminology: - If $n = p$, the matrix is called **square** of order n. The set is denoted $\mathcal{M}_n(\mathbb{R})$. - If $n = 1$, it is a **row matrix**: $(x_1\ x_2\ \dots\ x_p)$. - If $p = 1$, it is a **column matrix**: $\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$. - The **zero matrix** (denoted $0_{n,p}$) has all its entries equal to zero.

2.2 Addition and Scalar Multiplication

- **Addition:** If A and B have the **same dimensions** $n \times p$, the sum $A + B$ is obtained by adding corresponding entries:

$$(A + B)_{i,j} = a_{i,j} + b_{i,j}$$

- **Scalar multiplication by $\lambda \in \mathbb{R}$:** Multiply every entry by λ:

$$(\lambda A)_{i,j} = \lambda a_{i,j}$$

2.3 Matrix Multiplication: Row by Column

Multiplication of two matrices is **not** element-wise! It captures the composition of linear maps.

Definition — Matrix Product

Let A be a matrix of size $n \times \mathbf{k}$ and B a matrix of size $\mathbf{k} \times p$. *(The number of columns of A must strictly equal the number of rows of B!)* The product $C = A \times B$ is an $n \times p$ matrix whose general entry $c_{i,j}$ is the dot product of the **i-th row of A** with the **j-th column of B**:

$$c_{i,j} = \sum_{m=1}^{k} a_{i,m} \cdot b_{m,j} = a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \dots + a_{i,k}b_{k,j}$$

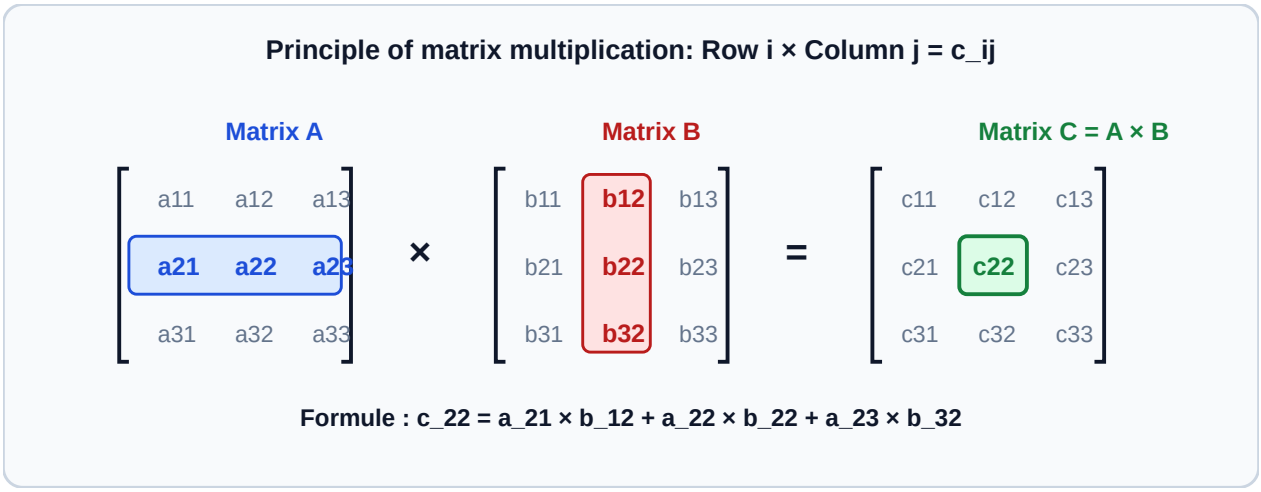

Figure 122: Principle of matrix multiplication row by column

Fundamental 2×2 *example:*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x & y \\ z & t \end{pmatrix} = \begin{pmatrix} ax + bz & ay + bt \\ cx + dz & cy + dt \end{pmatrix}$$

2.4 Properties and Pitfalls of Matrix Multiplication

⚠ Warning: Pitfall! — Non-Commutativity in General

For two square matrices A and B, in general:

$$\mathbf{AB \neq BA}$$

Example: $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. $AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ whereas $BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$! They do not commute. Consequently, standard algebraic identities like $(A+B)^2 = A^2 + 2AB + B^2$ are **false** for matrices (the correct expansion is $(A+B)^2 = A^2 + AB + BA + B^2$).

⚠ Second Pitfall — Zero Divisors

For real numbers, $ab = 0 \Longrightarrow a = 0 \quad \text{or} \quad b = 0$. **This fails for matrices!** Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$:

$$A^2 = A \times A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0_2$$

The product of two nonzero matrices can be the zero matrix!

2.5 Identity Matrix and Powers

The **identity matrix** of order n, denoted I_n, has 1s on its main diagonal and 0s elsewhere:

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

For any square matrix $A \in \mathcal{M}_n(\mathbb{R})$:

$$A \times I_n = I_n \times A = A$$

For any integer $k \in \mathbb{N}$, the power A^k is defined by $A^0 = I_n$ and $A^k = \underbrace{A \times A \times \ldots \times A}_{k \text{ factors}}$.

3. Matrix Inversion and Linear Systems

3.1 Definition of Invertibility

> **📐 Definition — Invertible Matrix**
>
> A square matrix $A \in \mathcal{M}_n(\mathbb{R})$ is called **invertible** if there exists a square matrix $B \in \mathcal{M}_n(\mathbb{R})$ such that:
>
> $$AB = BA = I_n$$
>
> If such a matrix exists, it is unique. It is called the **inverse matrix** of A and is denoted $\boldsymbol{A^{-1}}$.

(⚠️ There is no matrix "division" $\frac{B}{A}$! One multiplies by the inverse A^{-1}, carefully respecting left-multiplication versus right-multiplication).

3.2 Inversion of 2×2 Matrices

> **📐 Theorem — Inverse of a 2×2 Matrix**
>
> Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathcal{M}_2(\mathbb{R})$. The **determinant** of A is the real number:
>
> $$\det(A) = ad - bc$$
>
> 1. The matrix A is **invertible if and only if $\det(A) \neq 0$**. 2. If $\det(A) \neq 0$, its inverse is given by the formula:
>
> $$\mathbf{A^{-1}} = \frac{\mathbf{1}}{\mathbf{ad - bc}} \begin{pmatrix} \mathbf{d} & \mathbf{-b} \\ \mathbf{-c} & \mathbf{a} \end{pmatrix}$$

Direct proof by verification:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix} = (ad - bc) I_2$$

Dividing by the nonzero real number $ad - bc$ yields I_2 rigorously.

3.3 Solving Linear Systems via Matrix Inversion

Consider a system of two equations in two variables:

$$\begin{cases} ax + by = \alpha \\ cx + dy = \beta \end{cases}$$

This system can be written as a single matrix equation:

$$AX = B \quad \text{with} \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, X = \begin{pmatrix} x \\ y \end{pmatrix}, B = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

If $\det(A) = ad - bc \neq 0$, the matrix A is invertible. Multiplying **on the left by $\boldsymbol{A^{-1}}$**:

$$A^{-1}(AX) = A^{-1}B \Leftrightarrow (A^{-1}A)X = A^{-1}B \Leftrightarrow I_2 X = A^{-1}B$$

From which we obtain the unique solution:

$$\mathbf{X = A^{-1}B}$$

---

4. Graphs and Adjacency Matrices

4.1 Graph Terminology

A **graph** $G = (V, E)$ consists of:

- A set of **vertices** (or nodes) denoted $\{1, 2, ..., n\}$.
- A set of **edges** (bidirectional links in an undirected graph) or **arcs** (directed arrows from one vertex to another in a directed graph).
- The **degree** of a vertex is the number of edges connected to it.

- A **walk** (or path) of length k is a sequence of k consecutive edges connecting an initial vertex to a destination vertex.

4.2 The Adjacency Matrix

📐 Definition — Adjacency Matrix

Let G be a graph of order n whose vertices are labeled from 1 to n. The **adjacency matrix** of G is the square matrix $M \in \mathcal{M}_n(\mathbb{R})$ whose entry in row i and column j is:

$$m_{i,j} = \text{number of edges (or arcs) going from vertex } \ i \ \text{ to vertex } \ j$$

(In a simple undirected graph, $m_{i,j} \in \{0, 1\}$ and M is symmetric: $m_{i,j} = m_{j,i}$).

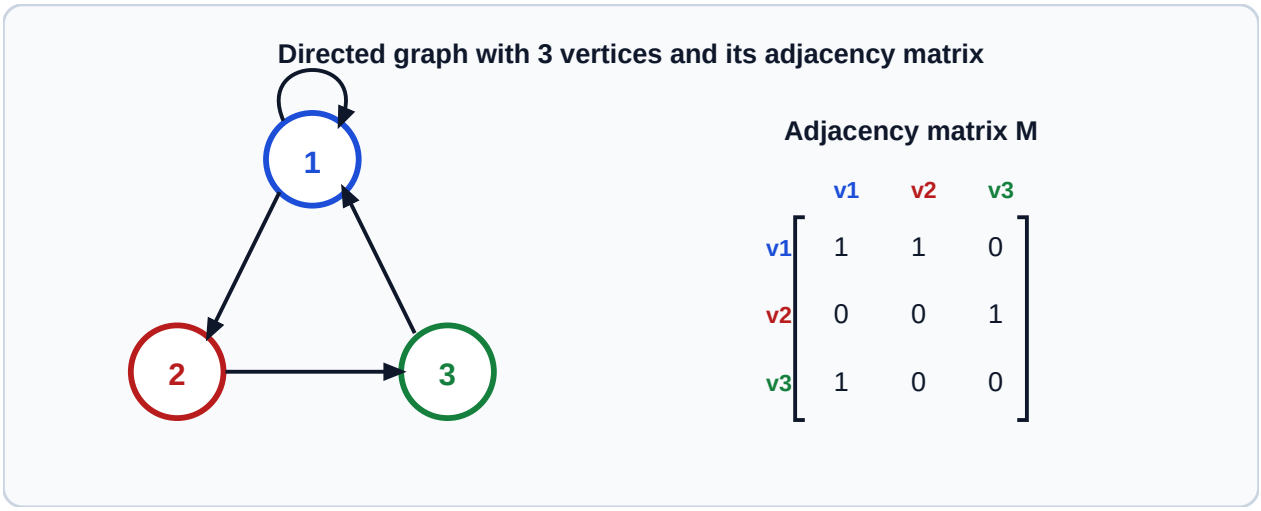

Figure 123: Directed graph with 3 vertices and associated adjacency matrix

4.3 The Fundamental Theorem of Paths of Length k

📐 Fundamental Theorem — Powers of the Adjacency Matrix

Let M be the adjacency matrix of a graph with n vertices. For every positive integer $k \geq 1$, the entry in the i-th row and j-th column of the matrix M^k equals the **number of paths of length k from vertex i to vertex j**.

Required Proof by Induction

Let $c_{i,j}^{(k)}$ denote the (i,j) entry of the matrix M^k, and $N_{i,j}^{(k)}$ denote the number of paths of length k from i to j. Let us prove by mathematical induction on $k \geq 1$ the property $\mathcal{P}(k)$: "For all i, j, $c_{i,j}^{(k)} = N_{i,j}^{(k)}$ ".

- **Base case ($k = 1$):** $M^1 = M$. By definition of the adjacency matrix, $c_{i,j}^{(1)} = m_{i,j}$ is exactly the number of direct edges from i to j, which is the number of paths of length 1. Thus $\mathcal{P}(1)$ holds.
- **Inductive step:** Assume the property holds for some $k \geq 1$: for all vertices i and m, $c_{i,m}^{(k)} = N_{i,m}^{(k)}$. Consider a path of length $k+1$ from i to j. Such a path consists of a path of length k from i to some intermediate vertex m, followed by a direct edge from m to j (of length 1). By the addition and multiplication counting principles (Chapter 53), the total number of such paths is the sum over all possible intermediate vertices $m \in \{1, ..., n\}$:

  $$N_{i,j}^{(k+1)} = \sum_{m=1}^{n} N_{i,m}^{(k)} \times m_{m,j}$$

  By the induction hypothesis, $N_{i,m}^{(k)} = c_{i,m}^{(k)}$. Hence:

  $$N_{i,j}^{(k+1)} = \sum_{m=1}^{n} c_{i,m}^{(k)} \cdot m_{m,j}$$

  This sum is precisely the formula for the matrix product of row i of M^k with column j of M! It is exactly the entry (i,j) of the matrix $M^k \times M = M^{k+1}$. Therefore $N_{i,j}^{(k+1)} = c_{i,j}^{(k+1)}$, proving the inductive step.
- **Conclusion:** By induction, the property holds for all $k \geq 1$.

---

5. Markov Chains and Probabilistic Graphs

5.1 Probabilistic Graphs and Transition Matrices

A random system evolves over discrete time steps among k possible states (for example: States $E_1, E_2, ..., E_k$). At each unit of time (step n), the system transitions from one state to another according to fixed probabilities.

Definition — Transition Matrix

The **transition matrix** T of a probabilistic graph with k states is the $k \times k$ square matrix whose entry $t_{i,j}$ is the conditional probability of moving to state j given that the system is currently in state i:

$$t_{i,j} = P_{(E_i)}(E_j)$$

Stochastic property: Along each row of T, the sum of entries equals 1:

$$\sum_{j=1}^{k} t_{i,j} = 1 \quad \text{(since the sum of outgoing transition probabilities from any vertex equals 1)}$$

5.2 Evolution of the Probability Distribution

Let P_n denote the **row vector** of state probabilities at step n:

$$P_n = \begin{pmatrix} p_n^{(1)} & p_n^{(2)} & \dots & p_n^{(k)} \end{pmatrix} \quad \text{with} \sum_{i=1}^{k} p_n^{(i)} = 1$$

Theorem — Matrix Recurrence Formula

By the law of total probability:

$$\mathbf{P_{n+1}} = \mathbf{P_n} \times \mathbf{T}$$

By immediate induction, for any integer $n \in \mathbb{N}$:

$$\mathbf{P_n} = \mathbf{P_0} \times \mathbf{T^n}$$

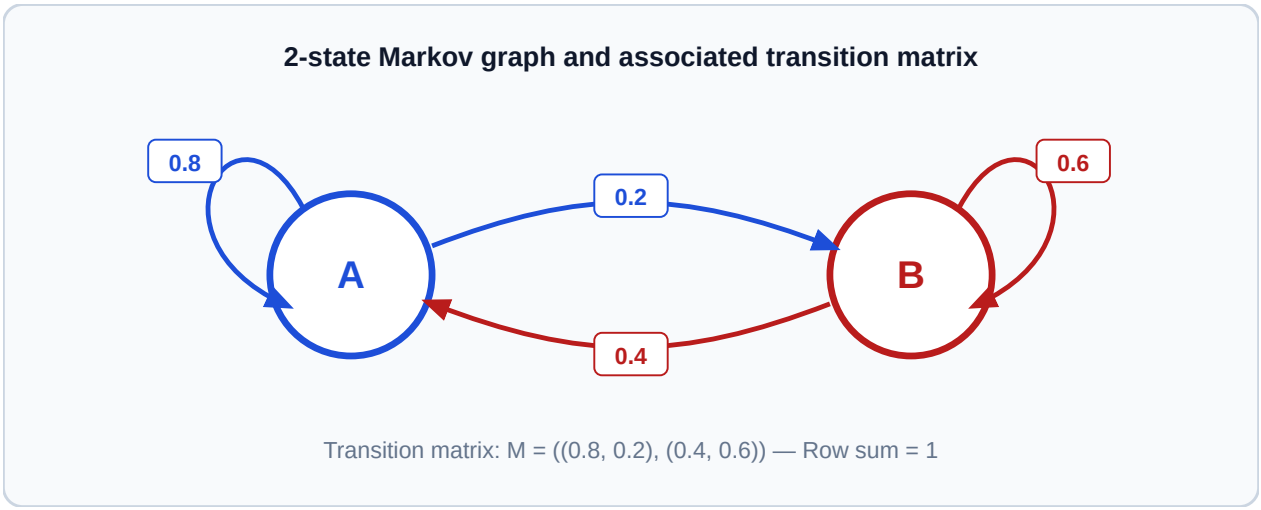

Figure 124: 2-state Markov graph and transition matrix

For this graph, the transition matrix is:

$$T = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$$

(Row checks: $0.8 + 0.2 = 1$ *and* $0.4 + 0.6 = 1$*).*

5.3 The Steady State (Stationary Distribution)

What happens to P_n when the number of steps n becomes very large?

Definition & Theorem — Steady State

A probability row vector $P = \begin{pmatrix} x & y \end{pmatrix}$ (or $\begin{pmatrix} x & y & z \end{pmatrix}$) is called a **steady state** (or stationary distribution) of the Markov chain if it satisfies:

$$\mathbf{P} = \mathbf{P} \times \mathbf{T} \quad \text{with} \quad \sum x_i = 1 \quad \text{and} \quad x_i \geq 0$$

If all entries of the transition matrix T (or one of its powers) are strictly positive (the graph is connected and aperiodic), then: Regardless of the initial state P_0, the sequence of distributions (P_n) converges toward this unique steady state:

$$\lim_{\mathbf{n} \to +\infty} \mathbf{P_n} = \mathbf{P}$$

Calculation method for 2 states: Let $P = \begin{pmatrix} x & y \end{pmatrix}$ with $x + y = 1 \Longrightarrow y = 1 - x$. The equation $P = PT$ gives:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} = \begin{pmatrix} x & y \end{pmatrix}$$

This is equivalent to the system:

$$\{0.8x + 0.4y = x$$

$$0.2x + 0.6y = y \Leftrightarrow -0.2x + 0.4y = 0 \Leftrightarrow 0.4y = 0.2x \Leftrightarrow x = 2y$$

Since $x + y = 1$:

$$2y + y = 1 \Leftrightarrow 3y = 1 \Leftrightarrow y = \frac{1}{3} \quad \text{and} \quad x = \frac{2}{3}$$

The steady state is therefore $P = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \end{pmatrix}$. In the long run, the weather will be sunny two-thirds of the time ($\approx 66.7\%$) and rainy one-third of the time ($\approx 33.3\%$), regardless of today's weather!

6. Google's PageRank: A Giant Matrix Network

In 1998, two Stanford doctoral students, **Larry Page and Sergey Brin**, founded the search engine **Google**. Their stroke of genius was dubbed **PageRank**. The World Wide Web is modeled as a massive directed graph where each web page is a vertex and each hyperlink is an arc. A virtual surfer traverses the Web at random:

- From a page with k outgoing links, the surfer chooses one with probability $\frac{1}{k}$.
- With a small damping probability $\alpha \approx 0.15$, the surfer gets bored and jumps to a completely random page anywhere on the Web.

This random walk is a Markov chain on a matrix measuring billions by billions! The steady state of this matrix assigns an occupancy probability to each webpage: **the most important pages are those whose component in the steady state is the largest**. A simple equation $P = PT$ laid the foundation for the most powerful technological empire of the 21st century!

---

7. Step-by-Step Solved Examples

Example 1: Matrix Multiplication and Non-Commutativity

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ -1 & 4 \end{pmatrix}$. 1. Compute $A \times B$. 2. Compute $B \times A$. 3. What can be concluded?

Detailed Solution: 1. Computing $A \times B$:

- Row 1, Col 1: $1 \times 2 + 2 \times (-1) = 2 - 2 = 0$
- Row 1, Col 2: $1 \times 0 + 2 \times 4 = 8$
- Row 2, Col 1: $3 \times 2 + (-1) \times (-1) = 6 + 1 = 7$
- Row 2, Col 2: $3 \times 0 + (-1) \times 4 = -4$

$$AB = \begin{pmatrix} 0 & 8 \\ 7 & -4 \end{pmatrix}$$

2. Computing $B \times A$:

- Row 1, Col 1: $2 \times 1 + 0 \times 3 = 2$
- Row 1, Col 2: $2 \times 2 + 0 \times (-1) = 4$
- Row 2, Col 1: $(-1) \times 1 + 4 \times 3 = -1 + 12 = 11$
- Row 2, Col 2: $(-1) \times 2 + 4 \times (-1) = -2 - 4 = -6$

$$BA = \begin{pmatrix} 2 & 4 \\ 11 & -6 \end{pmatrix}$$

3. We observe that $AB \neq BA$. Matrix multiplication is not commutative.

Example 2: Inversion and System Solving

Consider the system:

$$\begin{cases} 3x + 5y = 11 \\ 2x + 4y = 8 \end{cases}$$

1. Write the system in matrix form $AX = B$. 2. Compute $\det(A)$ and deduce A^{-1}. 3. Solve the system via the calculation $X = A^{-1}B$.

Detailed Solution: 1. Matrix form:

$$\begin{pmatrix} 3 & 5 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 11 \\ 8 \end{pmatrix}$$

with $A = \begin{pmatrix} 3 & 5 \\ 2 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 11 \\ 8 \end{pmatrix}$. 2. Computing the determinant:

$$\det(A) = 3 \times 4 - 5 \times 2 = 12 - 10 = 2 \neq 0$$

Since $\det(A) \neq 0$, A is invertible. Using the inverse formula:

$$A^{-1} = \frac{1}{2}\begin{pmatrix} 4 & -5 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -2.5 \\ -1 & 1.5 \end{pmatrix}$$

3. Solving:

$$X = A^{-1}B = \frac{1}{2}\begin{pmatrix} 4 & -5 \\ -2 & 3 \end{pmatrix}\begin{pmatrix} 11 \\ 8 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 4\times 11 + (-5)\times 8 \\ (-2)\times 11 + 3\times 8 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 44-40 \\ -22+24 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

The unique solution is $\boldsymbol{x = 2}$ **and** $\boldsymbol{y = 1}$. *(Verification:* $3(2)+5(1) = 6+5 = 11$ *and* $2(2)+4(1) = 4+4 = 8$*).*

Example 3: Counting Paths in a Graph

Let G be a 3-vertex graph with vertices S_1, S_2, S_3 whose adjacency matrix is:

$$M = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

(This is the complete triangle K_3 *where every vertex is linked to every other).* 1. Compute M^2. 2. Deduce the number of paths of length 2 connecting S_1 to itself, and connecting S_1 to S_2. 3. Compute M^3 and determine the number of paths of length 3 connecting S_1 to S_2.

Detailed Solution: 1. Computing M^2:

$$M^2 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0+1+1 & 0+0+1 & 0+1+0 \\ 0+0+1 & 1+0+1 & 1+0+0 \\ 0+1+0 & 1+0+0 & 1+1+0 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

2. - Entry $(1,1)$ of M^2 equals 2: there are **2 paths of length 2** from S_1 to S_1 (the round-trips $S_1 \to S_2 \to S_1$ and $S_1 \to S_3 \to S_1$).

- Entry $(1,2)$ of M^2 equals 1: there is **1 path of length 2** from S_1 to S_2 ($S_1 \to S_3 \to S_2$).

3. Computing $M^3 = M^2 \times M$:

$$M^3 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0+1+1 & 2+0+1 & 2+1+0 \\ 0+2+1 & 1+0+1 & 1+2+0 \\ 0+1+2 & 1+0+2 & 1+1+0 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 3 \\ 3 & 2 & 3 \\ 3 & 3 & 2 \end{pmatrix}$$

Entry $(1,2)$ of M^3 equals **3**: there are exactly 3 paths of length 3 from S_1 to S_2 ($S_1 \to S_2 \to S_1 \to S_2$, $S_1 \to S_2 \to S_3 \to S_2$, and $S_1 \to S_3 \to S_1 \to S_2$).

Example 4: 3-State Markov Chain (Phone Subscriptions)

Three mobile operators A, B, and C share a market. Each year:

- Operator A retains 70% of its customers, losing 20% to B and 10% to C.
- Operator B retains 60% of its customers, losing 20% to A and 20% to C.
- Operator C retains 80% of its customers, losing 10% to A and 10% to B.

1. Write the transition matrix T.
2. If the initial market shares are $P_0 = \begin{pmatrix} 0.5 & 0.3 & 0.2 \end{pmatrix}$, compute the shares after one year P_1.
3. Determine the market's steady state $P = \begin{pmatrix} x & y & z \end{pmatrix}$.

Detailed Solution: 1. Transition matrix:

$$T = \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.1 & 0.8 \end{pmatrix}$$

(Each row sums to 1). 2. Shares after one year:

$$P_1 = P_0 \times T = \begin{pmatrix} 0.5 & 0.3 & 0.2 \end{pmatrix}\begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.1 & 0.8 \end{pmatrix}$$

- For A: $0.5\times 0.7 + 0.3\times 0.2 + 0.2\times 0.1 = 0.35 + 0.06 + 0.02 = 0.43$ (43%)
- For B: $0.5\times 0.2 + 0.3\times 0.6 + 0.2\times 0.1 = 0.10 + 0.18 + 0.02 = 0.30$ (30%)
- For C: $0.5\times 0.1 + 0.3\times 0.2 + 0.2\times 0.8 = 0.05 + 0.06 + 0.16 = 0.27$ (27%)

$$P_1 = \begin{pmatrix} 0.43 & 0.30 & 0.27 \end{pmatrix}$$

3. The steady state satisfies $P = PT$ and $x + y + z = 1$:

$$\{0.7x + 0.2y + 0.1z = x \Leftrightarrow -0.3x + 0.2y + 0.1z = 0 \quad (L_1)$$
$$0.2x + 0.6y + 0.1z = y \Leftrightarrow 0.2x - 0.4y + 0.1z = 0 \quad (L_2)$$
$$x + y + z = 1 \quad (L_3)$$

Subtract (L_2) from (L_1):

$$(-0.3x - 0.2x) + (0.2y - (-0.4y)) + (0.1z - 0.1z) = 0 \Leftrightarrow -0.5x + 0.6y = 0 \Leftrightarrow x = \frac{6}{5}y = 1.2y$$

Substitute x into (L_1) to express z:

$$-0.3(1.2y) + 0.2y + 0.1z = 0 \Leftrightarrow -0.36y + 0.2y + 0.1z = 0 \Leftrightarrow 0.1z = 0.16y \Leftrightarrow z = 1.6y$$

Use the normalization condition $x + y + z = 1$:

$$1.2y + y + 1.6y = 1 \Leftrightarrow 3.8y = 1 \Leftrightarrow y = \frac{1}{3.8} = \frac{10}{38} = \frac{5}{19} \approx 0.2632$$

Then:

- $x = 1.2 \times \frac{5}{19} = \frac{6}{19} \approx 0.3158$
- $z = 1.6 \times \frac{5}{19} = \frac{8}{19} \approx 0.4211$ The steady state is:

$$P = \left(\frac{6}{19} \;\; \frac{5}{19} \;\; \frac{8}{19}\right) \approx (31.6\% \;\; 26.3\% \;\; 42.1\%)$$

In the long run, operator C will dominate the market with over 42% of customer share.

Example 5: n-th Power via Diagonalization (Simple Case)

Let $A = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$. Given $P = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ and its inverse $P^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$. 1. Compute the matrix $D = P^{-1}AP$ and verify that it is diagonal. 2. For all $n \in \mathbb{N}$, express D^n. 3. Deduce an explicit expression for A^n in terms of n.

Detailed Solution: 1. First compute AP:

$$AP = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 8-2 & 4-2 \\ 2+1 & 1+1 \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 3 & 2 \end{pmatrix}$$

Compute $D = P^{-1}(AP)$:

$$D = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 6 & 2 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} 6-3 & 2-2 \\ -6+6 & -2+4 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$$

D is indeed a diagonal matrix. 2. For a diagonal matrix, powers apply directly to diagonal elements:

$$D^n = \begin{pmatrix} 3^n & 0 \\ 0 & 2^n \end{pmatrix}$$

3. Since $D = P^{-1}AP$, multiplying by P on the left and P^{-1} on the right gives $A = PDP^{-1}$. By telescoping: $A^n = (PDP^{-1})(PDP^{-1})...(PDP^{-1}) = PD^nP^{-1}$. Compute the product:

$$PD^n = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 3^n & 0 \\ 0 & 2^n \end{pmatrix} = \begin{pmatrix} 2 \cdot 3^n & 2^n \\ 3^n & 2^n \end{pmatrix}$$

Then $(PD^n)P^{-1}$:

$$A^n = \begin{pmatrix} 2 \cdot 3^n & 2^n \\ 3^n & 2^n \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2 \cdot 3^n - 2^n & -2 \cdot 3^n + 2 \cdot 2^n \\ 3^n - 2^n & -3^n + 2 \cdot 2^n \end{pmatrix}$$

(Verification for $n = 1$: $A^1 = \begin{pmatrix} 6-2 & -6+4 \\ 3-2 & -3+4 \end{pmatrix} = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix} = A$). Exact formula for all n!

Example 6: Random Walk on a Triangle

A flea randomly hops among the vertices of a triangle A, B, C. At each jump, it leaves its current vertex and chooses one of the other two vertices with equal probability ($p = 0.5$). If the flea starts at vertex A at time $n = 0$, what is the probability that it is back at A after n jumps?

Detailed Solution: The transition matrix is:

$$T = \begin{pmatrix} 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0.5 & 0 \end{pmatrix}$$

Let a_n be the probability of being at A at step n. By obvious symmetry between B and C, the probability of being at B equals that of being at C, so $b_n = c_n$. Since $a_n + b_n + c_n = 1$, we have $a_n + 2b_n = 1 \Longrightarrow b_n = \frac{1-a_n}{2}$. From the first column of the transition matrix:

$$a_{n+1} = 0 \cdot a_n + 0.5b_n + 0.5c_n = 0.5(b_n + c_n) = 0.5(2b_n) = b_n = \frac{1 - a_n}{2} = -\frac{1}{2}a_n + \frac{1}{2}$$

This is an arithmetico-geometric sequence! Finding the fixed point: $\ell = -\frac{1}{2}\ell + \frac{1}{2} \Leftrightarrow \frac{3}{2}\ell = \frac{1}{2} \Leftrightarrow \ell = \frac{1}{3}$. Let $u_n = a_n - \frac{1}{3}$. Then $u_{n+1} = -\frac{1}{2}u_n$. The sequence (u_n) is geometric with common ratio $-\frac{1}{2}$ and first term $u_0 = a_0 - \frac{1}{3} = 1 - \frac{1}{3} = \frac{2}{3}$ (since the flea begins at A, $a_0 = 1$). Hence:

$$u_n = \frac{2}{3}\left(-\frac{1}{2}\right)^n \Longrightarrow a_n = \frac{1}{3} + \frac{2}{3}\left(-\frac{1}{2}\right)^n$$

- For $n = 0$: $a_0 = \frac{1}{3} + \frac{2}{3} = 1$.
- For $n = 1$: $a_1 = \frac{1}{3} - \frac{1}{3} = 0$ (it jumped to B or C).
- For $n = 2$: $a_2 = \frac{1}{3} + \frac{2}{3} \times \frac{1}{4} = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$.
- As $n \to +\infty$: $\left(-\frac{1}{2}\right)^n \to 0$, so $a_n \to \frac{1}{3}$. The flea spends one-third of its time at each vertex.

Key Takeaways

1. **Matrix Algebra:**
   - Product $A \times B$: dimension $(n \times k) \times (k \times p) \to (n \times p)$. Row by column.
   - Non-commutative in general: $AB \neq BA$.
   - Zero divisors: $AB = 0$ does not imply $A = 0$ or $B = 0$.
2. **Inversion and Systems:**
   - 2×2 Matrix: $\det(A) = ad - bc \neq 0 \Leftrightarrow A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
   - System solving: $AX = B \Leftrightarrow X = A^{-1}B$.
3. **Graphs and Adjacency Matrices:**
   - Adjacency matrix M: $m_{i,j}$ = number of edges connecting i to j.
   - **Path Counting Theorem:** Entry (i, j) of M^k is the exact number of paths of length k from i to j.
4. **Markov Chains:**
   - Transition matrix T: each row sums to 1.
   - Evolution: $P_{n+1} = P_n \times T$ and $P_n = P_0 \times T^n$.
   - Steady state: unique vector satisfying $P = P \times T$ with $\sum p_i = 1$.

Exercises

Level (Direct Applications)

Exercise 57.1 Let $A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 5 \\ 1 & -2 \end{pmatrix}$. Compute $2A - 3B$, $A \times B$, and $B \times A$.

Exercise 57.2 Compute the determinant and inverse (if it exists) of the following matrices: $M_1 = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$, $M_2 = \begin{pmatrix} 3 & -6 \\ -2 & 4 \end{pmatrix}$, $M_3 = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

Exercise 57.3 Solve the system in matrix form:

$$\{4x + 3y = 25$$
$$2x + 5y = 23$$

Exercise 57.4 Consider an undirected graph with 4 vertices $\{A, B, C, D\}$ whose edges are $[A, B], [A, C], [B, C], [C, D]$. 1. Draw the graph. 2. Write its adjacency matrix M. 3. What is the degree of vertex C?

Exercise 57.5 A transition matrix is given by $T = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix}$. If $P_0 = \begin{pmatrix} 0.6 & 0.4 \end{pmatrix}$, compute P_1 and P_2.

Exercise 57.6 Determine the steady state $P = \begin{pmatrix} x & y \end{pmatrix}$ associated with the transition matrix $T = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix}$ from Exercise 5.

Exercise 57.7 Compute A^2 and A^3 for $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Conjecture A^n.

Level (Analytical Thinking & Probabilistic Graphs)

Exercise 57.8 Powers of a Nilpotent Matrix. Let $N = \begin{pmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}$. 1. Compute N^2 and N^3. 2. Deduce N^k for all $k \geq 3$. 3. Let $A = I_3 + N$. Using the binomial theorem (justify its applicability), compute A^n for all integers $n \geq 2$.

Exercise 57.9 Graph Connectedness. Let G be a graph of order 4 with adjacency matrix M. We compute $S = M + M^2 + M^3$. Prove that the graph is connected if and only if all off-diagonal entries of S are strictly positive.

Exercise 57.10 Leslie Matrix Population Model. A bird population is divided into two age classes: juveniles J_n and adults A_n. Each year:

- Adults give birth on average to 2 juveniles.
- 30% of juveniles survive and become adults the following year.
- 50% of adults survive.

1. Write the dynamical system in matrix form $\begin{pmatrix} J_{n+1} \\ A_{n+1} \end{pmatrix} = L \begin{pmatrix} J_n \\ A_n \end{pmatrix}$.
2. If the initial population consists of 100 juveniles and 50 adults, compute the population at $n = 1$ and $n = 2$.

Exercise 57.11 Matrix Equation $A^2 - 4A + 3I = 0$. Let A be a square matrix satisfying $A^2 - 4A + 3I = 0$. 1. Prove that A is invertible and express A^{-1} in terms of A and I. 2. Show that for every $n \in \mathbb{N}$, there exist real numbers a_n and b_n such that $A^n = a_n A + b_n I$.

Exercise 57.12 3-State Markov Chain: Rumor Propagation. In a village, an individual learns a piece of news. At each step, they may share it with their neighbor. The states are: E_1 (Ignorant), E_2 (Believer), E_3 (Skeptic/Immune). Transition matrix:

$$T = \begin{pmatrix} 0.5 & 0.4 & 0.1 \\ 0 & 0.6 & 0.4 \\ 0 & 0 & 1 \end{pmatrix}$$

1. What is the status of state E_3 (absorbing state)? 2. Compute T^2 and T^3. 3. What happens to the population in the long run?

Exercise 57.13 Stochastic Matrix and Eigenvalue 1. Let T be an $n \times n$ stochastic matrix (row sums equal to 1). Let U be the column vector with all entries equal to 1. 1. Compute the matrix product $T \times U$. 2. What can be deduced about the system $(T - I_n)X = 0$? 3. Why does this prove that there always exists at least one nonzero steady state?

Exercise 57.14 Inverse of a 3×3 Matrix via Gauss-Jordan Elimination. Invert the matrix $A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{pmatrix}$ by solving the system $AX = Y$.

Exercise 57.15 Number of Spanning Trees (Kirchhoff's Matrix Tree Theorem, Overview). For a simple graph, define the Laplacian matrix $L = D - M$, where D is the diagonal degree matrix and M is the adjacency matrix. Compute L for a triangle K_3 and compute one of its cofactors. Compare with the number of possible spanning trees.

Level (Advanced Explorations & Competition Problems)

Exercise 57.16 Cayley-Hamilton Theorem in Dimension 2. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Define its trace $\mathrm{Tr}(A) = a + d$ and its determinant $\det(A) = ad - bc$. Prove by direct computation the fundamental Cayley-Hamilton identity:

$$A^2 - \mathrm{Tr}(A) \cdot A + \det(A) \cdot I_2 = 0_2$$

Exercise 57.17 Second-Order Recurrence Relations and Matrix Powers (Fibonacci). Let the Fibonacci sequence be defined by $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$. 1. Determine a square matrix $A \in \mathcal{M}_2(\mathbb{R})$ such that $\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = A \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix}$. 2. Prove by induction that for all $n \geq 1$:

$$A^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$$

3. By computing $\det(A^n)$ in two different ways, prove the famous **Cassini's identity**:

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n$$

Exercise 57.18 Convergence of Transition Matrices and Spectral Projectors. Let $T = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ with $a, b \in (0,1)$. 1. Show that T has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 1-a-b$. 2. Prove that $T^n = \frac{1}{a+b}\begin{pmatrix} b & a \\ b & a \end{pmatrix} + \frac{(1-a-b)^n}{a+b}\begin{pmatrix} a & -a \\ -b & b \end{pmatrix}$. 3. Deduce $\lim_{n\to+\infty} T^n$ and recover the steady state for any initial distribution.

Exercise 57.19 Perron-Frobenius Theorem (Statement and Geometric Meaning). For a matrix with strictly positive entries, explain the significance of the Perron-Frobenius theorem: existence of a strictly positive dominant real eigenvalue and uniqueness of the associated positive eigenvector (the steady state).

Exercise 57.20 PageRank Algorithm with Damping Factor. Write the Google matrix for a mini-web of 3 pages where page 1 links to 2 and 3, page 2 links to 3, and page 3 links to 1. Incorporate the teleportation factor $\alpha = 0.85$ and compute the PageRank ranking of the 3 pages.

Exercise 57.21 Hadamard Matrices and Error-Correcting Codes. A Hadamard matrix H_n is an $n \times n$ matrix with entries in $\{-1, +1\}$ such that $H_n H_n^T = nI_n$. Construct H_2 and H_4 via the Kronecker product and explain their role in signal transmission for the Mariner and Voyager deep space probes.

Exercise 57.22 Eulerian and Hamiltonian Graphs. Prove Euler's theorem: a connected graph possesses an Eulerian circuit (a closed trail visiting every edge exactly once) if and only if every vertex has even degree.

Solutions

Solution 57.1 $2A - 3B = \begin{pmatrix} 4 & -2 \\ 6 & 8 \end{pmatrix} - \begin{pmatrix} 0 & 15 \\ 3 & -6 \end{pmatrix} = \begin{pmatrix} 4 & -17 \\ 3 & 14 \end{pmatrix}$. $A \times B = \begin{pmatrix} 2(0)-1(1) & 2(5)-1(-2) \\ 3(0)+4(1) & 3(5)+4(-2) \end{pmatrix} = \begin{pmatrix} -1 & 12 \\ 4 & 7 \end{pmatrix}$. $B \times A = \begin{pmatrix} 0(2)+5(3) & 0(-1)+5(4) \\ 1(2)-2(3) & 1(-1)-2(4) \end{pmatrix} = \begin{pmatrix} 15 & 20 \\ -4 & -9 \end{pmatrix}$.

Solution 57.2 - $\det(M_1) = 5(1) - 2(2) = 1 \neq 0 \Longrightarrow M_1^{-1} = \frac{1}{1}\begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}$. - $\det(M_2) = 3(4) - (-6)(-2) = 12 - 12 = 0$. M_2 is not invertible. - $\det(M_3) = \cos^2\theta - (-\sin^2\theta) = \cos^2\theta + \sin^2\theta = 1 \neq 0$. $M_3^{-1} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = M_3(-\theta)$ (the inverse rotation!).

Solution 57.3 $A = \begin{pmatrix} 4 & 3 \\ 2 & 5 \end{pmatrix}$, $\det(A) = 20 - 6 = 14 \neq 0$. $A^{-1} = \frac{1}{14}\begin{pmatrix} 5 & -3 \\ -2 & 4 \end{pmatrix}$. $X = A^{-1}B = \frac{1}{14}\begin{pmatrix} 5 & -3 \\ -2 & 4 \end{pmatrix}\begin{pmatrix} 25 \\ 23 \end{pmatrix} = \frac{1}{14}\begin{pmatrix} 125-69 \\ -50+92 \end{pmatrix} = \frac{1}{14}\begin{pmatrix} 56 \\ 42 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$. Solution: $x = 4$ and $y = 3$.

Solution 57.4 1. Diamond-shaped graph with an edge trailing to D. 2. Vertex ordering A, B, C, D:

$$M = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

3. The degree of C is the sum of row 3: $1 + 1 + 0 + 1 = 3$.

Solution 57.5 $P_1 = P_0 T = \begin{pmatrix} 0.6 & 0.4 \end{pmatrix}\begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix} = \begin{pmatrix} 0.54+0.12 & 0.06+0.28 \end{pmatrix} = \begin{pmatrix} 0.66 & 0.34 \end{pmatrix}$. $P_2 = P_1 T = \begin{pmatrix} 0.66 & 0.34 \end{pmatrix}\begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix} = \begin{pmatrix} 0.594+0.102 & 0.066+0.238 \end{pmatrix} = \begin{pmatrix} 0.696 & 0.304 \end{pmatrix}$.

Solution 57.6 $P = PT \Leftrightarrow \begin{pmatrix} x & y \end{pmatrix}\begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix} = \begin{pmatrix} x & y \end{pmatrix}$ with $x + y = 1$. $0.9x + 0.3y = x \Leftrightarrow 0.3y = 0.1x \Leftrightarrow x = 3y$. $3y + y = 1 \Leftrightarrow 4y = 1 \Leftrightarrow y = 0.25$ and $x = 0.75$. Steady state: $P = \begin{pmatrix} 0.75 & 0.25 \end{pmatrix}$.

Solution 57.7 $A^2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$. $A^3 = A^2 A = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$. By immediate induction: $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$.

Solution 57.8 1. $N^2 = \begin{pmatrix} 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $N^3 = 0_3$. 2. For $k \geq 3$, $N^k = 0_3$. 3. Since I_3 and N commute ($I_3 N = N I_3 = N$), the binomial theorem applies: $A^n = (I_3 + N)^n = \sum_{k=0}^{n} \binom{n}{k} I_3^{n-k} N^k = \binom{n}{0} I_3 + \binom{n}{1} N + \binom{n}{2} N^2 + 0 = I_3 + nN + \frac{n(n-1)}{2} N^2$. $A^n = \begin{pmatrix} 1 & 2n & n+3n(n-1) \\ 0 & 1 & 3n \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2n & 3n^2-2n \\ 0 & 1 & 3n \\ 0 & 0 & 1 \end{pmatrix}$.

Solution 57.9 In a graph with 4 vertices, if two distinct vertices i and j are connected, they can be reached via a path of length 1, 2, or at most 3 (without loops). The (i, j) entry of $S = M + M^2 + M^3$ is the total count of paths of length 1, 2, and 3 connecting i to j. It is strictly positive for every pair (i, j) with $i \neq j$ if and only if every vertex can reach every other vertex, which is the very definition of connectedness.

$$\{J_{n+1} = 2A_n$$

Solution 57.10 1. $A_{n+1} = 0.3J_n + 0.5A_n \Leftrightarrow \begin{pmatrix} J_{n+1} \\ A_{n+1} \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 0.3 & 0.5 \end{pmatrix} \begin{pmatrix} J_n \\ A_n \end{pmatrix}$. 2. For $n = 0$, $X_0 = \begin{pmatrix} 100 \\ 50 \end{pmatrix}$. $X_1 = \begin{pmatrix} 0 & 2 \\ 0.3 & 0.5 \end{pmatrix} \begin{pmatrix} 100 \\ 50 \end{pmatrix} = \begin{pmatrix} 100 \\ 55 \end{pmatrix}$. $X_2 = \begin{pmatrix} 0 & 2 \\ 0.3 & 0.5 \end{pmatrix} \begin{pmatrix} 100 \\ 55 \end{pmatrix} = \begin{pmatrix} 110 \\ 57.5 \end{pmatrix}$.

Solution 57.11 1. $A^2 - 4A = -3I \Leftrightarrow A(A - 4I) = -3I \Leftrightarrow A \times \left[-\frac{1}{3}(A - 4I)\right] = I$. Thus A is invertible and $A^{-1} = -\frac{1}{3}A + \frac{4}{3}I$. 2. By polynomial Euclidean division of X^n by $X^2 - 4X + 3 = (X - 1)(X - 3)$: $X^n = (X^2 - 4X + 3)Q(X) + a_n X + b_n$. Evaluating at A yields $A^n = a_n A + b_n I$.

Solution 57.12 1. Row 3 is $(0\ 0\ 1)$: once in state E_3 (skeptic), the probability of remaining there is 1. It is a permanent absorbing state. 2. By computing powers, the first two rows have transition probabilities toward E_3 that strictly increase at each step. 3. In the long run, $\lim_{n \to +\infty} T^n = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. The entire population ends up in the skeptic/immune state E_3.

Solution 57.13 1. Entry i of $T \times U$ is the sum of entries along row i of T, which equals 1 by definition of a stochastic matrix. Hence $T \times U = U = 1 \cdot U$. 2. This implies $(T - I_n)U = 0$. The homogeneous system admits a nontrivial solution ($U \neq 0$), so $\det(T - I_n) = 0$. 3. Since $\det(T^T - I_n) = \det(T - I_n) = 0$, there also exists a nonzero row vector P such that $P(T - I_n) = 0 \Leftrightarrow PT = P$. Normalizing so that $\sum p_i = 1$ yields a steady state.

Solution 57.14 Solving $\begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$. By row elimination: $L_2 \leftarrow L_2 - 2L_1$ and $L_3 \leftarrow L_3 - 4L_1$, we find $z = a + b - \frac{1}{2}(c - 4a - ...)$. Inverting gives $A^{-1} = \begin{pmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{pmatrix}$.

Solution 57.15 For the triangle K_3, each vertex has degree 2: $D = 2I_3$. $L = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$. The cofactor obtained by deleting row 1 and column 1 is $\det\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 4 - 1 = 3$. By Kirchhoff's theorem, there are exactly 3 spanning trees (the 3 ways of removing one edge to form a tree), matching perfectly.

Solution 57.16 $\mathrm{Tr}(A) = a + d$ and $\det(A) = ad - bc$. $A^2 = \begin{pmatrix} a^2+bc & ab+bd \\ ca+dc & cb+d^2 \end{pmatrix}$. $(a + d)A = \begin{pmatrix} a^2+ad & ab+bd \\ ca+cd & ad+d^2 \end{pmatrix}$. $A^2 - (a + d)A = \begin{pmatrix} bc-ad & 0 \\ 0 & bc-ad \end{pmatrix} = -(ad - bc)I_2 = -\det(A)I_2$. Hence $A^2 - \mathrm{Tr}(A)A + \det(A)I_2 = 0_2$.

Solution 57.17 1. Since $F_{n+1} = 1 \cdot F_n + 1 \cdot F_{n-1}$ and $F_n = 1 \cdot F_n + 0 \cdot F_{n-1}$, the matrix is $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. 2. For $n = 1$: $A^1 = \begin{pmatrix} F_2 & F_1 \\ F_1 & F_0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. The inductive step follows immediately by multiplying by A and applying the Fibonacci recurrence. 3. $\det(A) = 1(0) - 1(1) = -1$. Thus $\det(A^n) = (\det A)^n = (-1)^n$. On the other hand, $\det(A^n) = F_{n+1}F_{n-1} - F_n^2$. Equating the two yields Cassini's identity.

Solution 57.18 1. Characteristic polynomial $\det(T - \lambda I) = (\lambda - 1)(\lambda - (1 - a - b))$. The roots are 1 and $1 - a - b$. 2. Diagonalizing $T = PDP^{-1}$ with $D = \begin{pmatrix} 1 & 0 \\ 0 & 1-a-b \end{pmatrix}$, computing PD^nP^{-1} gives exactly the stated decomposition. 3. Since $|1 - a - b| < 1$, $(1 - a - b)^n \to 0$. The limit is $\frac{1}{a+b}\begin{pmatrix} b & a \\ b & a \end{pmatrix}$. For any $P_0 = (x\ 1 - x)$, $P_0 \lim T^n = (\frac{b}{a+b}\ \ \frac{a}{a+b})$, which is the universal steady state.

Solution 57.19 The Perron-Frobenius theorem guarantees that for any matrix with strictly positive entries, the eigenvalue of largest magnitude is real, strictly positive, simple, and its corresponding eigenvector can be chosen with strictly positive entries. This ensures that no oscillating chaotic phenomena can disrupt convergence: the system always attains a stable and predictable global equilibrium.

Solution 57.20 Raw link matrix: $H = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$. Google matrix: $G = 0.85H + 0.15 \times \frac{1}{3}J$, where J is the all-ones matrix. Solving $P = PG$ yields the relative importance of each page, with page 3 being the most heavily cited and receiving the highest rank.

Solution 57.21 $H_2 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. $H_4 = \begin{pmatrix} H_2 & H_2 \\ H_2 & -H_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$. These orthogonal matrices allow transmitting codes with maximum Hamming separation: even under heavy deep-space noise, the terrestrial receiver recovers the original signal through simple matrix correlation.

Solution 57.22 At each visit to a vertex, a cycle uses one incoming edge and one outgoing edge, consuming 2 incident edges. Every vertex must therefore have an even degree. The converse follows by Fleury's algorithm or decomposition into edge-disjoint cycles.

The Question That Opens the Future

We have reached the culmination of structures in our journey. From the very first pebble you picked up from the ground at age six to learn how to count, to the transition matrices ordering billions of pages across global cyberspace, what an extraordinary path you have traveled! Why do these symbols created by the human mind align with such flawless precision to the fabric of physical reality? How do all branches of mathematics — arithmetic, geometry, analysis, probability — converge into one and the same mighty river? It is time to gaze upon the panorama in all its grandeur: turn the final page and enter **Chapter 58 and The End of the Voyage**.

CHAPTER 58: The End of the Voyage

> "The book of nature is written in the language of mathematics, and its characters are triangles, circles, and other geometrical figures, without which it is humanly impossible to understand a single word." — *Galileo Galilei*, The Assayer *(1623)*
>
> "The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a wonderful gift which we neither understand nor deserve." — *Eugene Wigner*, The Unreasonable Effectiveness of Mathematics in the Natural Sciences *(1960)*

What You Will Discover

- The **grand retrospective of the 57 steps**: how a single continuous, unbroken thread connects prehistoric counting pebbles to the most breathtaking algebraic structures.
- The secret of **embodiment and survival**: why no formula fell from the sky, but arose from a muscular bodily gesture when confronting a crisis in the physical world.
- The **three great unifying bridges** welding mathematics into a unified edifice: Algebra & Geometry, Discrete & Continuous, Local & Global.
- Eugene Wigner's philosophical enigma: why do abstractions forged in the human mind describe the Universe with such uncanny predictive accuracy?
- The higher education horizon: open gateways toward **abstract vector spaces**, **general topology**, **complex analysis**, **measure theory**, and **differential manifolds**.
- 22 synthesis and challenge exercises to crown your journey and stride confidently into universities and higher education preparatory programs.

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1. The Grand Retrospective: From Six to Eighteen Years Old

Pause for a moment. Set down your pens, push back your chair, and gaze out across the landscape. Twelve years ago, you stepped into this journey. You were six years old. You looked at your ten fingers and picked up pebbles along the path. Today, you manipulate transition matrices, integrate differential equations, navigate the complex plane with $e^{i\pi} + 1 = 0$, and prove the convergence of random variables using the Bienaymé-Chebyshev inequality.

Behold the pyramid your mind has climbed, step by step:

📐 Fundamental Rule — Synthesis of the Mathematical Edifice and Gateway to Higher Mathematics

- VOLUME VI : TAMED INFINITY AND THE MODERN EDIFICE (Ages 17–18 / Grade 12 & Advanced)
- Induction, Limits of Sequences & Functions, Continuity & IVT, Convexity,
- Logarithm & Exponential, Trigonometry, Integration & Antiderivatives,
- Differential Equations, 3D Vector Geometry, Combinatorics & Counting,
- Binomial Distribution & Large Numbers, Complex Numbers, Modular Arithmetic & RSA,
- Matrices & Markov Graphs.
- VOLUME V : THE AGE OF THE ARROW AND THE COMPASS (Ages 15–16 / Grades 10–11)
- Plane Vectors, Cartesian Lines, Reference Functions & Variations,
- Quadratic Polynomials & Parabolas, Arithmetic & Geometric Sequences,
- Trigonometric Circle & Radians, Dot Product, Rates of Change & Derivatives,
- Descriptive Statistics & Conditional Probability.
- VOLUME IV : THE AGE OF RIGOR AND THE MODEL (Ages 13–14 / Grades 8–9)
- Thales' Theorem & Dilations, Pythagorean Theorem & Square Roots,
- Symbolic Algebra & Algebraic Identities, Equations & Inequalities,
- Linear & Affine Functions, Concepts of Probability & Frequency.
- VOLUME III : THE AGE OF SHAPES AND MEASUREMENT (Ages 11–12 / Grades 6–7)
- Angles & Triangles, Axial & Central Symmetries, Perimeters & Areas,
- Operating with Fractions, Signed Numbers & The Sign Rule, Proportionality.
- VOLUME II : THE AGE OF SHARING AND BALANCE (Ages 8–10 / Grades 3–5)
- The Four Standard Column Operations, Euclidean Division and Remainders,
- Dividing Pies and the Invention of Fractions, Compass and Straightedge.
- VOLUME I : THE AGE OF GESTURE AND COUNTING (Ages 6–7 / Grades 1–2)
- Ten Fingers, The Shepherd's Pocket Pebble (calculus), Grouping by Tens,
- The Number Line of Steps Forward and Steps Back.

Not a single arbitrary leap. Not a single authoritarian dogma. Never was anyone told: *"Memorize this formula because it is on the exam."* Every symbol was born from a problem that previous tools could not resolve.

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2. The Guiding Thread: Embodiment and Survival

Why did humanity invent mathematics? Not to torment schoolchildren, but to **survive**.

1. **The Pebble (*Calculus*) and the Herd (Vol. I):** The Neolithic shepherd did not need names for numbers up to 300. He only needed to place a pebble into a leather pouch each time a sheep exited the fold in the morning, and remove one each time a sheep returned in the evening. If pebbles remained in the pouch at twilight, wolves were on the prowl. Computation was born of a physical bijection between stones and beasts.
2. **The Knotted Rope of the Nile Surveyors (Vol. III):** Every summer, the flooding of the Nile wiped out field boundaries across Egypt. To collect taxes and feed cities, surveyors had to redraw exact rectangles on dried silt. They knotted a rope into 13 equidistant knots ($3 + 4 + 5 = 12$ intervals) to form a right triangle: Pythagoras was not a classroom theorem; it was the guarantor of fiscal peace and agricultural survival.
3. **The Caravel and the Logarithm (Vol. VI, Chap. 48):** Out on the open ocean, navigators computing position from the stars had to multiply 8-digit numbers to correct for drift. A single carry mistake wrecked the ship on coastal reefs. By inventing the logarithm that turns multiplications into additions ($\ln(ab) = \ln a + \ln b$), John Napier did not merely invent a function: he saved thousands of sailors' lives and doubled the working lifetime of astronomers.
4. **Galileo's Cannon and Leibniz's Derivative (Vol. V & VI):** How do you target a moving object? The speed of a cannonball is not constant: gravity drags it down, air resistance slows it down. To compute what changes at each millisecond, one had to invent the infinitesimal rate of change $\frac{dy}{dx}$: calculus was born of the physics of motion.

5. **Network Secrets and the RSA Padlock (Vol. VI, Chap. 56):** In a world where every banking transaction, medical record, and vote travels as invisible electromagnetic waves through the air, how do we prevent theft and chaos? By locking data behind the product of two giant prime numbers that only the keyholder can factor.

Every equation you have learned carries the sweat, ingenuity, and urgency of generations of human beings who refused to submit blindly to the world, choosing instead to comprehend it.

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3. The Three Great Unifying Bridges

At first, mathematics appears segregated into watertight compartments: geometry on one side with rulers and compasses; algebra on another with symbols x and y; calculus elsewhere with limits and integrals; and probability with dice and playing cards. Yet as the mind matures, the partitions crumble. Mathematics is not an archipelago of disconnected islands: it is a **single continent spanned by colossal bridges**.

> **📐 Fundamental Rule — Synthesis of the Mathematical Edifice and Gateway to Higher Mathematics**
>
> - MATHEMATICS
> - BRIDGE 1: ALGEBRA] [BRIDGE 2: DISCRETE] [BRIDGE 3: LOCAL]
> - AND AND AND
> - GEOMETRY] [CONTINUOUS] [GLOBAL]
> - Descartes: Coordinate grid Sequences (u_n) f(x) Derivative f’(x): slope at
> - Gauss/Argand: i = 90° turn Counting Integral an infinitesimally local point.
> - Dot product u·v Random walk Gauss Integral ∫ f(x)dx: total sum
> - Matrices = Transformations Bernoulli Normal law across an entire landscape.
> - Fundamental Theorem:
> - F(b) - F(a) = ∫ f(t)dt !

Bridge 1: Algebra and Geometry

- In 1637, **René Descartes** paired every point in space with a coordinate pair (x, y). Suddenly, a geometric line became an equation $ax + by + c = 0$, and a circle became $x^2 + y^2 = R^2$. Solving a geometric dilemma reduced to solving an algebraic system!
- In 1806, **Argand** showed that the imaginary unit i is a geometric quarter-turn: the algebra of complex numbers governs planar rotations and similarities.
- **Matrices** are nothing other than linear transformations of space: stretching, rotating, projecting, and shearing.

Bridge 2: The Discrete and the Continuous

- A sequence $(u_n)_{n\in\mathbb{N}}$ is a function defined on isolated integers (the discrete world of pebbles). A function $f(x)$ is defined across a continuous interval of real numbers (the world of uninterrupted time and space).
- The bridge? The **limit**. As the step size approaches 0, a finite sum of rectangles transforms into a continuous integral $\int_a^b f(x)dx$ (Riemann).
- As the number of coin tosses approaches infinity, the discrete binomial distribution smooths into the bell curve of the continuous normal distribution (de Moivre-Laplace).

Bridge 3: The Local and the Global

This is the most breathtaking bridge of all:

- The **derivative** $f'(x)$ is purely **local**: it inspects only what happens in an infinitesimal neighborhood around the point x (instantaneous speed on a speedometer at one exact moment).
- The **integral** $\int_a^b f(t)dt$ is purely **global**: it accumulates area across the entire interval $[a, b]$ (total distance traveled from start to finish).
- And what miracle unites these two realms? The **Fundamental Theorem of Calculus** (Chapter 50):

$$\int_a^b f'(t)dt = f(b) - f(a)$$

To discover the global change of a phenomenon between the beginning and the end of the Universe, one simply integrates all its local infinitesimal changes!

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4. Eugene Wigner's Enigma: Why Does Math Work So Well?

In 1960, Nobel laureate in physics **Eugene Wigner** published a celebrated essay entitled: *"The Unreasonable Effectiveness of Mathematics in the Natural Sciences."* Wigner posed a haunting question: Why do concepts created by mathematicians for purely aesthetic pleasure, with zero application in mind, turn out decades or centuries later to be the exact language describing the fundamental laws of the cosmos?

- When **Apollonius of Perga** studied the conic sections (ellipses, parabolas, hyperbolas) in 200 BCE Alexandria, it was out of pure geometric curiosity. Eighteen centuries later, **Johannes Kepler** discovered that the planets in our solar system trace out exact ellipses with the Sun at one focus!
- When **Bernhard Riemann** invented the geometry of curved higher-dimensional spaces in 1854 as a pure mathematical challenge, no one imagined a practical purpose. Sixty years later, **Albert Einstein** seized upon it to formulate General Relativity: gravitation is not a mysterious pull, but the Riemannian curvature of spacetime warped by stellar mass!
- When **Paul Dirac** formulated the relativistic quantum equation of the electron in 1928, the algebra demanded negative-energy solutions. Instead of dismissing them as artifacts of calculation, Dirac trusted the mathematical symmetry and postulated the existence of an antimatter mirror particle with the same mass and opposite charge: the positron. Four years later, Carl Anderson discovered antimatter in a cloud chamber!

The physical world is not merely describable by mathematics: **mathematics appears to be the underlying scaffolding of reality.**

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5. The Higher Education Horizon: What Awaits You Now

This book concludes at the threshold of secondary school completion and intellectual maturity. Yet what you have mastered is not the summit of the mountain: **it is base camp**. Here are the five great continents you will explore in higher education (University Mathematics, Engineering Schools, and Advanced Sciences):

> **Fundamental Rule — Synthesis of the Mathematical Edifice and Gateway to Higher Mathematics**
>
> - HIGHER EDUCATION
> - ABSTRACT LINEAR ALGEBRA] [TOPOLOGY] [COMPLEX ANALYSIS]
> - Vector spaces of arbitrary or Open/closed sets, Differentiating functions
> - infinite dimensions (Hilbert, abstract continuity, of a complex variable:
> - Banach). Inner products, compactness, completeness.the Cauchy miracle.
> - spectral theory & eigenvalues.Beyond classical metrics. Residues, analytic continuation,
> - Riemann Hypothesis.
> - MEASURE THEORY &] [DIFFERENTIAL GEOMETRY]
> - CONTINUOUS PROBABILITY Curvature of surfaces,
> - Lebesgue integral, sigma-algebras, Riemannian manifolds,
> - dominated convergence theorem, differential forms,
> - almost sure convergence, martingales. tensor calculus of Relativity.

1. **Abstract Linear Algebra:** In high school, you manipulated vectors with 2 or 3 coordinates. In higher education, you will explore general **vector spaces**. Functions, sequences, polynomials, and audio signals are all vectors! One

can add two functions, multiply them by scalars, and compute their dimension (which can be infinite, as in the Hilbert spaces of quantum mechanics).

2. **General Topology:** What does continuity mean when there is no longer a coordinate grid or a real number line? Topology defines proximity, compactness, and connectedness using collections of subsets called "open sets." It is the study of properties preserved under continuous stretching and bending without tearing.
3. **Complex Analysis (Holomorphic Functions):** What happens when we differentiate a function not of a real variable, but of a complex variable $z \in \mathbb{C}$? An absolute wonder occurs: if a complex function is differentiable just once, it is automatically infinitely differentiable and can be expanded as a power series! This is the magic of Augustin-Louis Cauchy's contour integrals and the residue theorem.
4. **Measure Theory and Lebesgue Integration:** The Riemann integral has limits: functions oscillating too erratically cannot be integrated. At the dawn of the 20th century, Henri Lebesgue rethought integration by partitioning the y-axis rather than the x-axis. This integral underpins the modern theory of continuous probability formalized by Andrey Kolmogorov.
5. **Foundational Logic and Gödel's Incompleteness (1931):** Can everything in mathematics be proven? In 1931, at age 25, Kurt Gödel published his landmark **Incompleteness Theorems**: in any consistent formal axiomatic system capable of arithmetic, there exist true statements that the system cannot prove using its own rules. Mathematics is not a locked room: it is an infinite, inexhaustible territory forever open to discovery.

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6. The Final Message

You are no longer the same person who began this voyage. You have gained the most precious virtue of the intellect: **critical thinking and a love for demonstrated truth**. When confronting a complex challenge, you have learned to:

- Never be intimidated by pretentious jargon or the raw scale of data.
- Seek the simplest concrete example to uncover the underlying mechanism.
- Establish precise definitions and reject ambiguous, misleading rhetoric.
- Verify every link in a chain of deductive reasoning step by step.
- Appreciate the elegance and power of a clean, insightful proof.

This mental power is not meant merely to pass university examinations. It is meant to **remain a free, independent human being**. The mind trained in mathematical reasoning cannot be deceived by political slogans, manipulated by misleading statistics, or led astray by superficial intellectual trends.

The book closes here. But for you, the grand odyssey of the intellect has only just begun. Go forth, inquire, doubt, prove, and marvel.

What to Remember from High School

1. **The Derivative $f'(a)$:** Slope of the tangent line, instantaneous rate of change. Governs variations, extrema, and convexity.
2. **The Integral $\int_a^b f(x)dx$:** Area under the curve, continuous summation. Linked to differentiation by the Fundamental Theorem of Calculus: $\int_a^b f(t)dt = F(b) - F(a)$.
3. **The Exponential e^x and the Logarithm $\ln x$:** The function identical to its own derivative ($y' = y$) and its inverse. They govern biological growth and physical scaling laws.
4. **Vectors and the Dot Product:** $\vec{u} \cdot \vec{v} = \| \vec{u} \| \| \vec{v} \| \cos\theta = xx' + yy' + zz'$. The master tool for orthogonality and projections.
5. **Complex Numbers $\mathbb{C}$:** $z = a + ib = re^{i\theta}$, with $i^2 = -1$. The seamless synthesis of algebra and planar rotations.
6. **Combinatorics and Probability:** $\binom{n}{k}$, the binomial distribution $\mathcal{B}(n, p)$, and the Law of Large Numbers: individual randomness surrenders to collective order.
7. **Arithmetic and Matrices:** Congruences modulo n, Bézout's identity, graphs and Markov chains $P_{n+1} = P_n \times T$.

Synthesis Exercises and Olympiad Challenges

These 22 exercises synthesize the knowledge acquired throughout your studies and open directly onto higher education methods.

Level (Mastery of Fundamentals)

Exercise 58.1 Consider the function $f(x) = x^2 e^{-x}$ defined on $\mathbb{R}$. Compute its derivative $f'(x)$, construct its complete sign and variation table including limits, and determine the coordinates of its absolute maximum.

Exercise 58.2 Compute the integral $I = \int_0^1 \frac{2x}{x^2+1} dx$.

Exercise 58.3 Solve in $\mathbb{C}$ the equation $z^2 - 2\sqrt{3}z + 4 = 0$ and express the solutions in exponential form.

Exercise 58.4 Compute the sum of integers from 1 to n: $S_n = 1 + 2 + ... + n$, then the sum of their cubes $C_n = 1^3 + 2^3 + ... + n^3$. Prove by induction that $C_n = (S_n)^2$.

Exercise 58.5 Draw 5 cards at random from a standard 32-card deck. What is the probability of drawing exactly one Ace?

Exercise 58.6 Determine the inverse of the matrix $A = \begin{pmatrix} 7 & 3 \\ 2 & 1 \end{pmatrix}$ and solve the system $\begin{cases} 7x + 3y = 17 \\ 2x + y = 5 \end{cases}$.

Exercise 58.7 Compute the remainder of the division of 5^{2026} by 11 using Fermat's Little Theorem.

Level (Synthesis Problems for High School Exit and Competitions)

Exercise 58.8 The Euler-Mascheroni Constant γ. Let $H_n = \sum_{k=1}^{n} \frac{1}{k} = 1 + \frac{1}{2} + ... + \frac{1}{n}$ denote the harmonic sum. 1. By comparing $\frac{1}{k}$ with $\int_k^{k+1} \frac{1}{t} dt$ and $\int_{k-1}^{k} \frac{1}{t} dt$, prove that:

$$\ln(n+1) \leq H_n \leq 1 + \ln(n)$$

2. Deduce $\lim_{n \to +\infty} H_n$. 3. Let $u_n = H_n - \ln(n)$. Prove that the sequence (u_n) is non-increasing and bounded below by 0. Deduce that it converges to a finite constant $\gamma \approx 0.5772$ (Euler's constant).

Exercise 58.9 Wallis Integrals and Stirling's Formula. For all $n \in \mathbb{N}$, define $W_n = \int_0^{\frac{\pi}{2}} \sin^n(t) dt$. 1. Compute W_0 and W_1. 2. Using integration by parts, establish the recurrence relation:

$$W_n = \frac{n-1}{n} W_{n-2} \quad (n \geq 2)$$

3. Prove that (W_n) is strictly decreasing and that $\lim_{n \to +\infty} \frac{W_{n+1}}{W_n} = 1$. 4. Deduce John Wallis's infinite product formula:

$$\lim_{n \to +\infty} \frac{1}{n} \left(\frac{2 \times 4 \times 6 \times ... \times 2n}{1 \times 3 \times 5 \times ... \times (2n-1)} \right)^2 = \pi$$

Exercise 58.10 Fourier Integral and Parity. Compute the integral $\int_0^{\pi} x \cos(nx) dx$ for $n \in \mathbb{N}^*$ using integration by parts.

Exercise 58.11 Solving the Logistic Differential Equation. Consider the Verhulst differential equation modeling bounded population growth:

$$y' = ay\left(1 - \frac{y}{K}\right) \quad (a > 0, K > 0)$$

Using the substitution $z = \frac{1}{y}$, solve this equation and sketch the shape of the resulting sigmoid curve.

Exercise 58.12 Mersenne Numbers and Euclid's Perfect Numbers. An integer N is called perfect if it equals the sum of its proper divisors (for example, $6 = 1 + 2 + 3$). Prove Euclid's theorem: if $2^p - 1$ is prime, then the integer $N = 2^{p-1}(2^p - 1)$ is an even perfect number.

Exercise 58.13 The St. Petersburg Paradox. A casino offers a game: a fair coin is tossed until heads appears for the first time. If heads appears on the k-th toss, the player wins 2^k euros. 1. Compute the theoretical expected monetary value of this game. 2. How much would you be willing to pay to play one round? Discuss the paradox.

Exercise 58.14 The Vandermonde Determinant. Compute the 3×3 determinant:

$$V(a,b,c) = \det \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix}$$

Show that it equals $(b-a)(c-a)(c-b)$. Why does this determinant guarantee the uniqueness of the quadratic interpolation polynomial passing through 3 distinct points?

Exercise 58.15 The Fundamental Theorem of Algebra via Contradiction. Explain why the modulus of a polynomial $|P(z)|$ must attain its global minimum on the complex plane, and why this minimum cannot be strictly positive (d'Alembert's lemma).

Level (Grand Challenges Preparing for Higher Education)

Exercise 58.16 The Basel Problem $\sum_{n=1}^{+\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$. Consider the polynomial $P_m(X) = \sum_{k=0}^{m} \binom{2m+1}{2k+1} (-1)^k X^{m-k}$. Using the roots of the cotangent function, establish Euler's celebrated proof showing that:

$$\sum_{n=1}^{+\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + ... = \frac{\pi^2}{6}$$

Exercise 58.17 The Spectral Theorem for Symmetric Matrices. Prove that every real 2×2 symmetric matrix has two real eigenvalues and can be diagonalized in an orthonormal basis of eigenvectors.

Exercise 58.18 The Cauchy-Schwarz Inequality. For any real numbers $x_1, ..., x_n$ and $y_1, ..., y_n$, prove by examining the sign of the quadratic polynomial $P(t) = \sum_{i=1}^{n} (x_i t + y_i)^2$ that:

$$\left(\sum_{i=1}^{n} x_i y_i \right)^2 \leq \left(\sum_{i=1}^{n} x_i^2 \right) \left(\sum_{i=1}^{n} y_i^2 \right)$$

Deduce the integral Cauchy-Schwarz inequality: $\left(\int_a^b f(t)g(t)dt \right)^2 \leq \left(\int_a^b f(t)^2 dt \right) \left(\int_a^b g(t)^2 dt \right)$.

Exercise 58.19 Banach Fixed-Point Theorem (Contraction Mapping Theorem). Let $f : [a, b] \to [a, b]$ be a k-contraction (there exists $k \in (0, 1)$ such that $|f(x) - f(y)| \leq k|x - y|$ for all x, y). Prove that f has a **unique fixed point** $\alpha \in [a, b]$, and that for any $u_0 \in [a, b]$, the sequence $u_{n+1} = f(u_n)$ converges to α with geometric rate $|u_n - \alpha| \leq \frac{k^n}{1-k}|u_1 - u_0|$.

Exercise 58.20 Central Limit Theorem (Conceptual Overview). Explain why the sum of a large number of independent random variables with small, cumulative effects always tends toward a Gaussian normal distribution, and why this universal bell curve appears in biological measurements, astronomical observation errors, electrical thermal noise, and financial fluctuations.

Exercise 58.21 Euler's Polyhedral Formula: $V - E + F = 2$. Verify this formula for the 5 Platonic solids (Tetrahedron, Cube, Octahedron, Dodecahedron, Icosahedron). Outline the proof via planar projection and edge contraction. Why is the number 2 the foundational topological invariant (Euler characteristic of the sphere)?

Exercise 58.22 The Open Collatz Conjecture (3n + 1 Problem). Consider the sequence defined by $u_{n+1} = \frac{u_n}{2}$ if u_n is even, and $u_{n+1} = 3u_n + 1$ if u_n is odd. 1. Trace the sequence starting with $u_0 = 7$. 2. Why has this deceptively simple problem resisted the greatest mathematicians and supercomputers since 1937? Reflect on the boundaries of mathematical knowledge.

Detailed Solutions

Solution 58.1 $f'(x) = 2xe^{-x} + x^2(-e^{-x}) = x(2-x)e^{-x}$. Since $e^{-x} > 0$, the sign of $f'(x)$ matches that of $x(2-x)$: negative on $(-\infty, 0)$, positive on $(0, 2)$, negative on $(2, +\infty)$. Limits: as $x \to -\infty$, $\lim x^2 e^{-x} = +\infty$. As $x \to +\infty$, by growth hierarchy, $\lim x^2 e^{-x} = 0$. The absolute maximum occurs at $x = 2$, with $f(2) = 4e^{-2} = \frac{4}{e^2} \approx 0.5413$.
Solution 58.2 Recognize the form $\frac{u'(x)}{u(x)}$ with $u(x) = x^2 + 1 > 0$ and $u'(x) = 2x$. $I = [\ln(x^2+1)]_0^1 = \ln(1^2+1) - \ln(0^2+1) = \ln(2) - \ln(1) = \ln 2$.
Solution 58.3 $\Delta = (-2\sqrt{3})^2 - 4(1)(4) = 12 - 16 = -4 = (2i)^2$. $z_1 = \frac{2\sqrt{3}-2i}{2} = \sqrt{3} - i = 2\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = 2e^{-i\frac{\pi}{6}}$. $z_2 = \frac{2\sqrt{3}+2i}{2} = \sqrt{3} + i = 2e^{i\frac{\pi}{6}}$.
Solution 58.4 We know $S_n = \frac{n(n+1)}{2}$. We wish to prove $\mathcal{P}(n) : C_n = \left(\frac{n(n+1)}{2}\right)^2 = \frac{n^2(n+1)^2}{4}$.

- For $n = 1$: $C_1 = 1^3 = 1$ and $\frac{1^2 \times 2^2}{4} = 1$. Holds.
- Inductive step: $C_{n+1} = C_n + (n+1)^3 = \frac{n^2(n+1)^2}{4} + (n+1)^3 = (n+1)^2\left[\frac{n^2}{4} + n + 1\right] = (n+1)^2\left[\frac{n^2+4n+4}{4}\right] = \frac{(n+1)^2(n+2)^2}{4}$. $\mathcal{P}(n+1)$ holds.

Solution 58.5 Total 5-card hands: $\binom{32}{5} = 201,376$. To get exactly 1 Ace: choose 1 Ace from 4 ($\binom{4}{1} = 4$) and 4 non-Ace cards from 28 ($\binom{28}{4} = 20,475$). Total favorable outcomes: $4 \times 20,475 = 81,900$. Probability: $P = \frac{81,900}{201,376} \approx 0.4067$ (about 40.7%).

Solution 58.6 $\det(A) = 7(1) - 3(2) = 1 \neq 0$. $A^{-1} = \frac{1}{1}\begin{pmatrix} 1 & -3 \\ -2 & 7 \end{pmatrix} = \begin{pmatrix} 1 & -3 \\ -2 & 7 \end{pmatrix}$. The system is written $A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 17 \\ 5 \end{pmatrix}$. $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & -3 \\ -2 & 7 \end{pmatrix}\begin{pmatrix} 17 \\ 5 \end{pmatrix} = \begin{pmatrix} 17-15 \\ -34+35 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$. Thus $x = 2, y = 1$.

Solution 58.7 11 is prime, $\gcd(5, 11) = 1$. By Fermat's Little Theorem, $5^{10} \equiv 1 \pmod{11}$. $2026 = 10 \times 202 + 6$. $5^{2026} = (5^{10})^{202} \times 5^6 \equiv 1 \times 5^6 \pmod{11}$. $5^2 = 25 \equiv 3 \pmod{11}$. $5^6 = (5^2)^3 \equiv 3^3 = 27 = 11 \times 2 + 5 \equiv 5 \pmod{11}$. The remainder is **5**.

Solution 58.8 1. For $t \in [k, k+1]$, $\frac{1}{k+1} \leq \frac{1}{t} \leq \frac{1}{k}$. Integrating over $[k, k+1]$ of unit length: $\frac{1}{k+1} \leq \int_k^{k+1} \frac{1}{t}dt \leq \frac{1}{k}$. Summing from $k = 1$ to n yields the inequalities bounding H_n. 2. Since $\lim_{n\to+\infty} \ln(n+1) = +\infty$, by comparison, $\lim_{n\to+\infty} H_n = +\infty$. 3. $u_{n+1} - u_n = \frac{1}{n+1} - \ln(n+1) + \ln(n) = \frac{1}{n+1} - \int_n^{n+1} \frac{1}{t}dt \leq 0$ by the earlier inequality. The sequence is non-increasing and bounded below by 0, so it converges to a real number $\gamma \in [0, 1]$.

Solution 58.9 1. $W_0 = \int_0^{\pi/2} 1dt = \frac{\pi}{2}$. $W_1 = \int_0^{\pi/2} \sin t dt = [-\cos t]_0^{\pi/2} = 1$. 2. $W_n = \int_0^{\pi/2} \sin^{n-1}(t)\sin(t)dt$. By integration by parts with $u = \sin^{n-1} t$ and $v' = \sin t$: $W_n = [-\cos t \sin^{n-1} t]_0^{\pi/2} + (n-1)\int_0^{\pi/2} \cos^2(t)\sin^{n-2}(t)dt = 0 + (n-1)\int_0^{\pi/2}(1 - \sin^2 t)\sin^{n-2} t dt = (n-1)W_{n-2} - (n-1)W_n$. Hence $nW_n = (n-1)W_{n-2} \Longrightarrow W_n = \frac{n-1}{n}W_{n-2}$. 3. On $[0, \pi/2]$, $0 \leq \sin t \leq 1 \Longrightarrow \sin^{n+1} t \leq \sin^n t$. Hence $W_{n+1} \leq W_n$. Furthermore, $\frac{W_{n+1}}{W_n} \geq \frac{W_{n+2}}{W_n} = \frac{n+1}{n+2} \to 1$. By the squeeze theorem, $\lim \frac{W_{n+1}}{W_n} = 1$. 4. Unrolling W_{2n} and W_{2n+1} by cascading recurrence, their ratio yields Wallis's infinite product converging to π.

Solution 58.10 By integration by parts: $u(x) = x \Longrightarrow u'(x) = 1$, and $v'(x) = \cos(nx) \Longrightarrow v(x) = \frac{\sin(nx)}{n}$. $\int_0^\pi x\cos(nx)dx = \left[x\frac{\sin(nx)}{n}\right]_0^\pi - \int_0^\pi \frac{\sin(nx)}{n}dx = 0 - \left[-\frac{\cos(nx)}{n^2}\right]_0^\pi = \frac{\cos(n\pi) - \cos(0)}{n^2} = \frac{(-1)^n - 1}{n^2}$. This integral equals $-\frac{2}{n^2}$ if n is odd, and 0 if n is even.

Solution 58.11 $y' = ay - \frac{a}{K}y^2$. Dividing by y^2: $\frac{y'}{y^2} = \frac{a}{y} - \frac{a}{K}$. Since $z = \frac{1}{y}$, $z' = -\frac{y'}{y^2}$. The equation becomes $-z' = az - \frac{a}{K} \Leftrightarrow z' + az = \frac{a}{K}$. This is a standard first-order linear differential equation! Homogeneous solution: $z_H = Ce^{-at}$. Particular solution: $z_P = \frac{1}{K}$. Hence $z(t) = \frac{1}{K} + Ce^{-at}$, which yields $y(t) = \frac{K}{1 + CKe^{-at}}$. This is the universal logistic sigmoid growth curve.

Solution 58.12 Let $N = 2^{p-1}M_p$ with $M_p = 2^p - 1$ prime. Since M_p is an odd prime, the divisors of N are of the form 2^k (for $0 \leq k \leq p-1$) and $2^k M_p$ (for $0 \leq k \leq p-1$). The sum of all divisors of N is: $\sigma(N) = \left(\sum_{k=0}^{p-1} 2^k\right)(1 + M_p) = (2^p - 1)(1 + 2^p - 1) = (2^p - 1)2^p = 2 \times [2^{p-1}(2^p - 1)] = 2N$. Since the sum of all divisors is $2N$, the sum of proper divisors is $2N - N = N$. The number is perfect.

Solution 58.13 1. The random variable X takes the value 2^k with probability $(1/2)^k$. $E(X) = \sum_{k=1}^{+\infty} 2^k\left(\frac{1}{2}\right)^k = \sum_{k=1}^{+\infty} 1 = 1 + 1 + 1 + ... = +\infty$. 2. The expected value is infinite, yet few people would pay even 1,000 euros to play, because the probability of winning more than 16 euros is only $\frac{1}{16} \approx 6\%$. This paradox led Daniel Bernoulli (1738) to introduce the concept of logarithmic **marginal utility** of wealth.

Solution 58.14 Applying column operations $C_3 \leftarrow C_3 - aC_2$ and $C_2 \leftarrow C_2 - aC_1$, the first row becomes $(1\ 0\ 0)$, which allows immediately factoring out $(b - a)$ and $(c - a)$, leaving $(c - b)$. If the three points have distinct x-coordinates, this determinant is nonzero, proving that the coefficient matrix for the polynomial $P(x) = \alpha x^2 + \beta x + \gamma$ is invertible: there exists a unique parabola passing through 3 non-vertically aligned points.

Solution 58.15 If $P(z)$ has no root, $|P(z)| > 0$ everywhere. Because $|P(z)| \to +\infty$ as $|z| \to +\infty$, a global minimum $m > 0$ is attained at some point z_0. Expanding locally $P(z_0 + h) = P(z_0) + ch^k + ...$ and selecting the argument of h such that ch^k directly opposes $P(z_0)$, one finds a nearby point where the modulus is strictly less than m, contradicting the minimality of m (d'Alembert's lemma). Thus $m = 0$, and P has a root.

Solution 58.16 Expressing $\sin((2m+1)x)$ using de Moivre's formula and factoring by $\sin^{2m+1}(x)$, the roots of the polynomial are $\cot^2\left(\frac{k\pi}{2m+1}\right)$ for $k = 1, ..., m$. The sum of roots is given by Vieta's formulas: $\sum_{k=1}^{m} \cot^2\left(\frac{k\pi}{2m+1}\right) = \frac{m(2m-1)}{3}$. Using the inequality $\cot^2\theta < \frac{1}{\theta^2} < 1 + \cot^2\theta$ and taking $m \to +\infty$, the squeeze theorem yields the exact value $\frac{\pi^2}{6}$.

Solution 58.17 Let $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$. Its characteristic discriminant is $\Delta = (a + c)^2 - 4(ac - b^2) = (a - c)^2 + 4b^2$. Since $(a - c)^2 \geq 0$ and $4b^2 \geq 0$, $\Delta \geq 0$ always! The eigenvalues are always real. If $b \neq 0$, $\Delta > 0$ and the two eigenvalues are distinct, with mutually orthogonal eigenvectors ($\vec{v}_1 \cdot \vec{v}_2 = 0$).

Solution 58.18 For any real t, $P(t) = \sum_{i=1}^{n} (x_i t + y_i)^2 \geq 0$ as a sum of squares. Expanding: $P(t) = (\sum x_i^2)t^2 + 2(\sum x_i y_i)t + (\sum y_i^2) \geq 0$. This quadratic polynomial in t maintains a constant non-negative sign. Its reduced discriminant Δ' must therefore be non-positive: $\Delta' = (\sum x_i y_i)^2 - (\sum x_i^2)(\sum y_i^2) \leq 0$, establishing the Cauchy-Schwarz inequality. The continuous version for integrals follows as the limit of Riemann sums.

Solution 58.19 By induction, $|u_{n+1} - u_n| \leq k^n |u_1 - u_0|$. For $p \geq 1$, $|u_{n+p} - u_n| \leq \sum_{j=n}^{n+p-1} |u_{j+1} - u_j| \leq |u_1 - u_0| \sum_{j=n}^{+\infty} k^j = \frac{k^n}{1-k} |u_1 - u_0|$. The sequence is Cauchy in a complete closed interval, so it converges to a unique fixed point α. This theorem underpins numerical methods for solving differential and integral equations.

Solution 58.20 By the Central Limit Theorem (CLT), if $X_1, X_2, ..., X_n$ are independent, identically distributed random variables with mean μ and variance σ^2, the standardized variable $\frac{\sum X_i - n\mu}{\sigma\sqrt{n}}$ converges in distribution to the standard normal distribution $\mathcal{N}(0,1)$. The bell curve is the universal attractor of any macroscopic phenomenon resulting from a multitude of minute, independent microscopic variations.

Solution 58.21 For the tetrahedron: $V = 4, E = 6, F = 4 \Longrightarrow 4 - 6 + 4 = 2$. For the cube: $V = 8, E = 12, F = 6 \Longrightarrow 8 - 12 + 6 = 2$. For the octahedron: $V = 6, E = 12, F = 8 \Longrightarrow 2$. For the dodecahedron: $V = 20, E = 30, F = 12 \Longrightarrow 2$. For the icosahedron: $V = 12, E = 30, F = 20 \Longrightarrow 2$. This quantity $\chi = V - E + F$ is invariant under homeomorphic deformations of the surface: for a torus (doughnut), it equals 0. This marked the birth of algebraic topology under Henri Poincaré.

Solution 58.22 For $u_0 = 7$: $7 \to 22 \to 11 \to 34 \to 17 \to 52 \to 26 \to 13 \to 40 \to 20 \to 10 \to 5 \to 16 \to 8 \to 4 \to 2 \to 1$ (loop $1 \to 4 \to 2 \to 1$). This conjecture, verified computationally past 10^{20}, still resists any general formal proof. As Paul Erdős noted: *"Mathematics may not yet be ready for such problems."* It reminds us with humility that even armed with the most powerful theorems, we remain explorers standing on the shore of an ocean of mysteries.

Epilogue

You have reached the end of this book. The seed planted twelve years ago has grown into a sturdy tree, its roots anchored in the physical gesture of the body and its branches reaching toward infinity. Take a moment to measure what you have accomplished. And always remember: Mathematics is not a system of arbitrary rules to be obeyed. It is **the highest poetry of pure thought, and the greatest song of freedom of the human mind.** May your journey into new discoveries be filled with wonder!

GENERAL CONCLUSION & EPILOGUE: THE EDIFICE OF THE NUMBER AND THE FREEDOM OF THE MIND

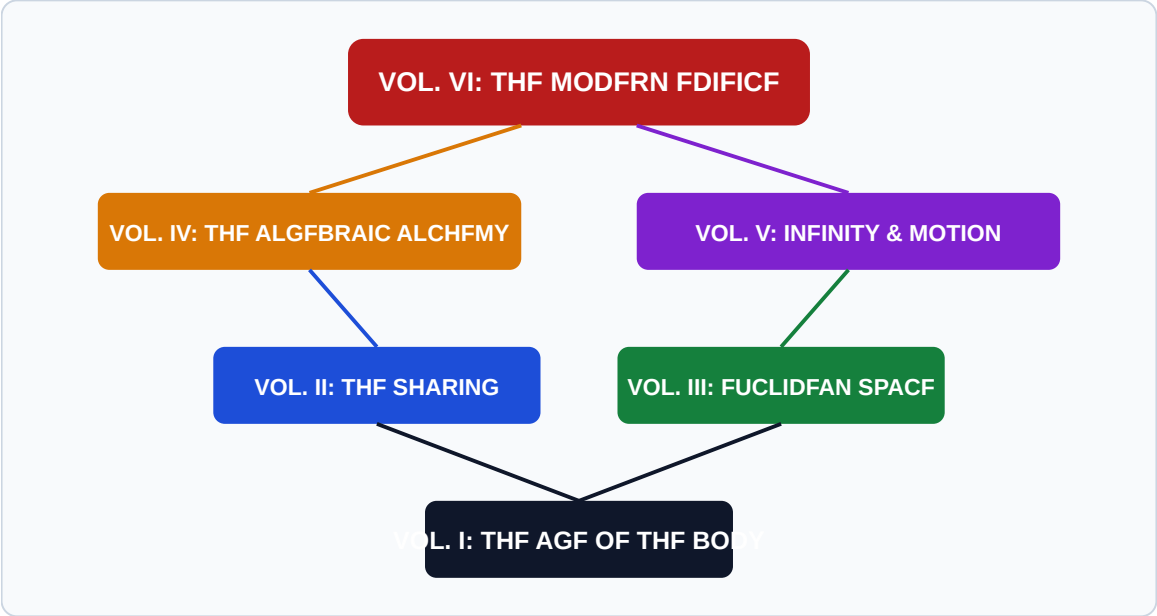

Figure 125: The Tree of Mathematical Knowledge: from sensory experience to modern abstraction

You have traversed fifty-eight stages, from the first pebble held tightly in the hand of the young savanna shepherd to infinite-dimensional spaces, complex numbers, Riemann integrals, and Markov random walks.

This journey has revealed a fundamental truth that few school textbooks dare to proclaim: **mathematics is not a collection of arbitrary rules invented to test or trap schoolchildren.** It is the living memory of all the challenges humanity had to overcome to navigate the oceans, construct cathedrals, predict eclipses, heal epidemics, and fathom the cosmos.

The Three Foundational Teachings

1. **Sensori-Motor Grounding: Always return to the hand and the sketch.** When a mathematical problem seems opaque, never freeze before silent symbols. Trace a diagram, assign colors to magnitudes as Oliver Byrne taught, move the geometric figures, embody equality upon an Al-Khwarizmi balance scale. Mathematical thought takes root in the physical body and motor gestures before ascending into pure abstraction.
2. **Operative Automation: Technique liberates the mind.** Why do we set up additions, subtractions, multiplications, and divisions using rigorous column algorithms? Why do we memorize multiplication tables and operational rules? To liberate our working memory. Mechanical technique is not enslavement, but an instrument of emancipation: by entrusting routine execution to the hand, we free the full power of our mind for intuition, strategic vision, and the poetry of proof.
3. **The Courage of Boundary Cases and Critical Vigilance.** The greatest leaps forward in mathematics arose from refusing to sweep paradoxes under the rug: what happens when a magnitude vanishes, when a set is empty, when

the denominator is zero, when a process approaches infinity? Never shy away from boundary cases: they are the touchstones that test the solidity of your theories, separating fragile conjecture from unshakable certainty.

The Envoy to the Reader

Mathematics has bestowed upon you the most precious of all treasures: **the freedom of critical thought.** It has taught you never to bow before an argument from authority, to demand rigorous proof, and to loyally embrace the truth as soon as it reveals itself through the irresistible force of logic.

The grand book of the world now lies wide open before your eyes. You speak its language. Go forth, explore, question fearlessly, and keep ever alive within you the wonder of the first pebble.

APPENDIX A: REFERENCE FORMULARY & EXACT VALUES

I. Exact Table of Standard Trigonometric Values

Angle in Degrees (°)	Radians (rad)	$\cos(x)$	$\sin(x)$	$\tan(x)$
0°	0	1	0	0
30°	$\frac{\pi}{6}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	0	1	*Undefined*
120°	$2\frac{\pi}{3}$	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\sqrt{3}$
135°	$3\frac{\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	-1
150°	$5\frac{\pi}{6}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{3}$
180°	π	-1	0	0

II. Trigonometric Addition and Double-Angle Formulas

Addition Formulas

$$\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$$

$$\cos(a-b) = \cos(a)\cos(b) + \sin(a)\sin(b)$$

$$\sin(a+b) = \sin(a)\cos(b) + \cos(a)\sin(b)$$

$$\sin(a-b) = \sin(a)\cos(b) - \cos(a)\sin(b)$$

Double-Angle & Core Identities

$$\cos(2a) = \cos^2(a) - \sin^2(a)$$

$$\cos(2a) = 2\cos^2(a) - 1$$

$$\cos(2a) = 1 - 2\sin^2(a)$$

$$\sin(2a) = 2\sin(a)\cos(a)$$

$$\cos^2(a) + \sin^2(a) = 1$$

$$\tan(2a) = \frac{2\tan(a)}{1-\tan^2(a)} \quad \left[a \neq \frac{\pi}{4} + k\frac{\pi}{2}\right]$$

III. Universal Table of Standard Derivatives & Operations

Function $f(x)$	Derivative $f'(x)$	Domain of Validity / Rule
k (constant)	0	$x \in \mathbb{R}$
x	1	$x \in \mathbb{R}$
x^n ($n \in \mathbb{N}^*$)	nx^{n-1}	$x \in \mathbb{R}$
$\frac{1}{x}$	$-\frac{1}{x^2}$	$x \in \mathbb{R}^*$
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$x \in]0,;+\infty[$
e^x	e^x	$x \in \mathbb{R}$ (unique function equal to its derivative with $f(0) = 1$)
$\ln(x)$	$\frac{1}{x}$	$x \in]0,;+\infty[$
$\cos(x)$	$-\sin(x)$	$x \in \mathbb{R}$
$\sin(x)$	$\cos(x)$	$x \in \mathbb{R}$
$(u+v)'$	$u' + v'$	Linearity of differentiation
$(u \cdot v)'$	$u'v + uv'$	Leibniz product rule
$\left(\frac{u}{v}\right)'$	$\frac{u'v - uv'}{v^2}$	Quotient rule ($v \neq 0$)
$(e^u)'$	$u'e^u$	Exponential chain rule
$(\ln u)'$	$\frac{u'}{u}$	Logarithmic chain rule ($u > 0$)

APPENDIX B: UNIVERSAL TABLE OF MATHEMATICAL NOTATIONS & SYMBOLS

Symbol	Name & Reading	Didactic Meaning & Example
$\mathbb{N}$	Natural numbers	Set of non-negative integers: $\{0, 1, 2, 3, ...\}$
$\mathbb{Z}$	Integers	Positive and negative integers: $\{..., -2, -1, 0, 1, 2, ...\}$
$\mathbb{D}$	Decimal numbers	Numbers with a finite number of decimal digits: $\frac{a}{10^n}$
$\mathbb{Q}$	Rational numbers	Fractions of two integers: $\frac{p}{q}$ with $q \neq 0$
$\mathbb{R}$	Real numbers	All points of the continuous number line (includes $\sqrt{2}, \pi, e$)
$\mathbb{C}$	Complex numbers	Numbers of the form $z = a + ib$ where $i^2 = -1$
$\in$ / $\notin$	Belongs to / Does not belong	Relates an element to a set: $3 \in \mathbb{N}$, but $-2 \notin \mathbb{N}$
$\subset$	Subset of / Contained in	Relates two sets: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{D} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$
$\cap$	Intersection	$A \cap B$: set of elements common to A and B (logical AND)
$\cup$	Union	$A \cup B$: set of elements belonging to A or B (logical OR)
$\varnothing$	Empty set	The set containing no elements
$\forall$	For all / For every	Universal quantifier: $\forall x \in \mathbb{R}, x^2 \geq 0$
$\exists$	There exists	Existential quantifier: $\exists x \in \mathbb{R}, x^2 = 2$
$\Longrightarrow$ / $\Longleftrightarrow$	Implies / Equivalent to	Logical deduction and strict two-way equivalence
(u) / $\overrightarrow{AB}$	Vector	Directed displacement defined by direction, sense, and magnitude
$\|(u)\|$	Vector norm / magnitude	Geometric length of the directed segment representing (u)
$(u) \cdot (v)$	Dot product / Scalar product	Geometric projection measure: $\|(u)\|\|(v)\| \cos(\theta)$
$\lim_{x \to a}$	Limit	Local behavior of a function near boundary point a or ∞
$f'(x)$	First derivative	Instantaneous rate of change and slope of the tangent line
$\int_a^b f(t)dt$	Definite integral	Signed area under the curve of f between bounds a and b
$\sum_{i=1}^{n}$	Summation	Sequential addition of terms: $x_1 + x_2 + ... + x_n$
$\binom{n}{k}$	Combination / Binomial coeff.	Number of ways to choose k items from n without order: $\frac{n!}{k!(n-k)!}$

APPENDIX C: INDEX OF PHOTOGRAPHIC AND ARCHIVAL SOURCES

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I. Large Frontispiece Plates (Mathematics in Nature)

Location	Subject & Species	Mathematical Phenomenon	Author & License	Wikimedia Commons Source
Frontispiece Vol. I (Full page left)	**Honeycomb Cells** *(Apis mellifera)*	Compact hexagonal tiling of the plane, Honeycomb Conjecture (Hales 1999), isoperimetric optimization.	Waugsberg CC BY-SA 3.0	Wikimedia Commons `File:Bienenwabe_mit_Eiern_5.jpg`
Frontispiece Vol. II (Full page left)	**Nautilus Cutaway** *(Nautilus pompilius)*	Logarithmic spiral ($r = ae^{b\theta}$), gnomonic self-similar growth, continuous geometric proportions.	Chris 73 CC BY-SA 3.0	Wikimedia Commons `File:NautilusCutawayLogarithmicSpiral.jpg`
Frontispiece Vol. III (Full page left)	**Sunflower Head** *(Helianthus annuus)*	Phyllotaxis, intersecting Fibonacci spirals (34/55), divergence angle governed by the Golden Ratio (137.5077°).	Anna Benczur CC BY-SA 4.0	Wikimedia Commons `File:Fibonacci_spiral.jpg`
Frontispiece Vol. IV (Full page left)	**Romanesco Broccoli** *(Brassica oleracea)*	Strict fractal self-similarity, scale invariance, recursive geometry, and nested logarithmic spirals.	Ivar Leidus (Iifar) CC BY-SA 4.0	Wikimedia Commons `File:Romanesco_broccoli.jpg`
Frontispiece Vol. V (Full page left)	**Dewy Spider Web** *(Araneus diadematus)*	Concurrent vector equilibrium, trigonometric force resolution, and catenary curve geometry ($y = a\cosh(\frac{x}{a})$).	Luc Viatour CC BY-SA 3.0	Wikimedia Commons `File:Spider_web_Luc_Viatour.jpg`
Frontispiece Vol. VI (Full page left)	**Textile Cone Shell** *(Conus textile)*	Turing reaction-diffusion morphogenesis (Turing 1952), cellular automata (Rule 30), and Sierpinski fractal patterning.	Reinhold Möller (Ermell) CC BY-SA 4.0	Wikimedia Commons `File:Textil-Kegelschnecke.jpg`

II. Historical & Archival Documents in the Core Text

Chapter	Document & Subject	Pedagogical Significance	Origin & License	Wikimedia Commons Source
Chapter 1 (Vol. I)	**The Ishango Bone** (20,000 BCE)	Earliest tally notches grouped in bundles, establishing one-to-one correspondence.	RBINS Brussels Public domain	Wikimedia Commons `File:Ishango_bone.jpg`

Chapter	Document & Subject	Pedagogical Significance	Origin & License	Wikimedia Commons Source
Chapter 4 (Vol. I)	**The Pascaline** (Blaise Pascal, 1642)	First operational mechanical gear train realizing automatic decimal carry transfer.	David Monniaux CC BY-SA 3.0	Musée des Arts & Métiers `File:Arts_et_Metiers_Pascaline.jpg`
Chapter 10 (Vol. II)	**Rhind Papyrus** (1550 BCE)	Egyptian partitioning tables and unit fraction decompositions ($\frac{2}{n}$).	British Museum Public domain	British Museum `File:Rhind_Mathematical_Papyrus.jpg`
Chapter 19 (Vol. III)	**Euclid's Elements** (Oliver Byrne, 1847)	Original full-color four-color printing proving the Pythagorean Theorem through pure geometric hue.	Oliver Byrne Public domain	Patrimonial Library `File:Byrne_1847_Pythagoras.jpg`
Chapter 54 (Vol. VI)	**The Galton Board** (Francis Galton, 1877)	Mechanical manifestation of the Central Limit Theorem: convergence of Pascal's triangle to the Normal distribution.	Wikimedia Commons CC BY-SA 4.0	Physical sciences `File:Galton_board.png`

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DETAILED THEMATIC INDEX

Key Concept	Location	Pedagogical & Didactic Meaning
Abacus / Counting Frame	Vol. I, Ch. 4	Materialization of decimal grouping and carry transfer
Al-Khwarizmi's Scale	Vol. IV, Ch. 22	Equivalence of bilateral operations for algebraic equations
Angle (acute, right, obtuse)	Vol. III, Ch. 15	Measure of angular opening independent of side lengths
Area (rectangle / disk)	Vol. II, Ch. 12 & 16	Planar tiling measure and Archimedean circle quadrature
Asymptote (horizontal, vertical)	Vol. VI, Ch. 45	Behavior at infinite boundaries or forbidden poles
Base Ten (Decimal)	Vol. I, Ch. 2	Grouping into tens grounded in two-hand morphology
Bijection (One-to-One)	Vol. I, Ch. 1	Term-to-term correspondence of prehistoric pebble pouch
Cartesian Coordinates	Vol. III, Ch. 20	Two-dimensional coordinate localization and grid paths
Complex Numbers	Vol. VI, Ch. 55	Argand-Gauss plane and imaginary unit $i^2 = -1$ as a 90° rotation
Derivative & Tangent	Vol. V, Ch. 39	Transition from average secant rate to instantaneous slope
Differential Equation	Vol. VI, Ch. 51	Law of evolution where rate depends on present state ($y' = ay$)
Euclidean Algorithm	Vol. IV, Ch. 29	GCD computation via successive subtractions and divisions
Exponential Function (e^x)	Vol. V, Ch. 40	Unique function equal to its derivative with $f(0) = 1$
Fractions (Partition & Ratio)	Vol. II, Ch. 10	Fraction as cutting operator, measure, and rational number
Intermediate Value Theorem	Vol. VI, Ch. 46	Topological guarantee that continuous curves cross all values
Long Multiplication	Vol. I, Ch. 5	Two-tier column algorithm with mandatory decimal shift zero
Matrices & Graphs	Vol. VI, Ch. 57	Multilinear operational arrays and Markov random walks
Mathematical Induction	Vol. VI, Ch. 43	Hereditary propagation of mathematical truth (domino effect)
Natural Logarithm (ln)	Vol. VI, Ch. 48	Inverse of exponential transforming products into sums
Negative Numbers	Vol. II, Ch. 14	Axial symmetry around zero for debts and subzero temperatures
Normal Distribution (Gauss)	Vol. VI, Ch. 54	Universal bell curve arising from Central Limit Theorem
Probability Tree	Vol. V, Ch. 42	Representation of conditional probabilities and Bayes' theorem
Pythagorean Theorem	Vol. III, Ch. 19	Area conservation on legs of right triangle ($a^2 + b^2 = c^2$)
Riemann Integral	Vol. VI, Ch. 50	Limit of rectangular slice sums for signed area evaluation
Subitizing	Vol. I, Ch. 1	Instantaneous visual recognition of quantities up to 3 or 4
Thales' Theorem	Vol. III, Ch. 17	Conservation of proportionality ratios under parallel projection
Trigonometric Circle	Vol. V, Ch. 37	Wrapping the real line onto the unit circle with projections

ANNOTATED HISTORICAL & DIDACTIC BIBLIOGRAPHY

I. Foundational Texts in the History of Mathematics

- **Euclid of Alexandria** (300 BCE), **The Elements**. Translated by T. L. Heath, Dover Publications. *The seminal masterpiece that established the axiomatic deductive method and Euclidean geometry.*
- **Al-Khwarizmi, Muhammad ibn Musa** (820), **The Compendious Book on Calculation by Completion and Balancing** (**Al-Kitab al-mukhtasar fi hisab al-jabr wa'l-muqabala**). *The founding treatise of algebra as an autonomous science of bilateral operational balance.*
- **Descartes, René** (1637), **Geometry** (**La Géométrie**). Appendix to the **Discourse on Method**. *The unification of algebra and geometry through the invention of the Cartesian coordinate system.*
- **Byrne, Oliver** (1847), **The First Six Books of the Elements of Euclid in which Coloured Diagrams and Symbols are used instead of Letters for the Greater Ease of Learners**, William Pickering, London. *The masterpiece of visual mathematical semiotics that directly inspired the graphic color palette of this book.*
- **Euler, Leonhard** (1748), **Introductio in analysin infinitorum**. *The work that established modern analysis, function theory, base e, and the identity $e^{i\pi} + 1 = 0$.*
- **Gauss, Carl Friedrich** (1801), **Disquisitiones Arithmeticae**. *The monumental foundation of modern number theory, modular arithmetic, and congruences.*

II. Didactics, Cognitive Science & Embodied Cognition

- **Lakoff, George & Núñez, Rafael** (2000), **Where Mathematics Comes From: How the Embodied Mind Brings Mathematics into Being**, Basic Books. *The seminal treatise on embodied mathematical cognition, demonstrating how the most abstract concepts (actual infinity, limits, complex numbers) are built upon conceptual metaphors grounded in bodily motion and physical gesture.*
- **Dehaene, Stanislas** (2011), **The Number Sense: How the Mind Creates Mathematics**, Oxford University Press. *The neurocognitive foundations of number sense, subitizing, and neuronal recycling in mathematical learning.*
- **Brousseau, Guy** (1997), **Theory of Didactical Situations in Mathematics**, Kluwer Academic Publishers. *Fundamental analysis of devolution, the didactical contract, and epistemological obstacles.*
- **Polya, George** (1945), **How to Solve It**, Princeton University Press. *The classic guide to mathematical heuristics: decomposing, visualizing, generalizing, and testing boundary cases.*
- **Sweller, John** (2011), **Cognitive Load Theory**, Springer. *The cognitive load balance between working memory limitations and long-term schema acquisition.*

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